*TM 1-1520-237-23-3

TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS

CHAPTER 3 TROUBLESHOOTING PROCEDURES

*This manual together with TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10, TM 1-1520-237-23-11, TM 1-1520-237-23-12, TM 1-1520-237-23-13, TM 1-1520-237-23-14, TM 1-1520-237-23-15, TM 1-1520-237-23-16, TM 1-1520-237-23-17, TM 1-1520-237-23-18, TM 1-1520-237-23-19 and TM 1-1520-237-23-20, dated 17 April 2006, supersedes TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10-1, TM 1-1520-237-23-10-2, TM 1-1520-237-23-10-3, TM 1-1520-237-23-11, dated 1 May 2003 including all changes.

This information is furnished upon the condition that it will not be released to another nation without the specific authority of the Department of the Army of the United States, that it will be used for military purposes only, that individual or corporate rights originating in the information, whether patented or not, will be respected, that the recipient will report promptly to the United States, any known or suspected compromise, and that the information will be provided substantially the same degree of security afforded it by the Department of Defense of the United States. Also, regardless of any other markings on the document, it will not be downgraded or declassified without written approval of the originating United States agency.

DISTRIBUTION STATEMENT D - Distribution authorized to the DoD and DoD contractors only due to Critical Technology effective as of 15 June 2003. Other requests must be referred to Commander, US Army Aviation and Missile Command, ATTN: SFAE-AV-UH/L, Redstone Arsenal, AL 35898-5000.

WARNING - This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C., Sec 2751, et. seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et. seq. Violations of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.

DESTRUCTION NOTICE - Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

HEADQUARTERS, DEPARTMENT OF THE ARMY

17 April 2006

WARNING SUMMARY

Personnel performing operations, procedures, and practices which are included or implied in this technical manual shall observe the following warnings. To disregard these warnings and precautionary information can cause serious injury, death, or an aborted mission.

WARNING

FIRST AID - refer to FM 4-25-11.

AC POWER - before applying ac power to the helicopter, make sure that the area around the stabilator is clear of personnel and equipment. Should the stabilator be in any position other than trailing edge fully down, it may reposition automatically to fully down when ac power is applied.

CARBON MONOXIDE - during cold weather operations when preheating is necessary, all personnel shall become acquainted with the various types of heaters and where they are to be used. Be careful of accumulations of carbon monoxide and damage to heat-sensitive equipment.

CONSUMABLE MATERIALS - observe all cautions and warnings on the containers when using consumables. When applicable wear necessary protective gear during handling and use. If a consumable is flammable or explosive, MAKE SURE consumable and its vapors are kept away from heat, spark, and flame. MAKE SURE helicopter is properly grounded and firefighting equipment is readily available for use.

SAFETY PINS - make sure all safety pins are installed in ejector racks prior to performing any maintenance. To disregard this warning can cause serious injury, death, or an aborted mission.

ELECTRICAL GROUNDING OF THE HELICOPTER - army regulations require the grounding of the helicopter when doing all fueling and defueling operations. Do not operate helicopter electrical switches, except those essential for servicing during fueling and defueling. Do not smoke or use flame during fueling and defueling operations.

HIGH VOLTAGE - there are dangerous voltages in the helicopter. Use extreme care when working with equipment having these voltages.

HYDRAULIC SYSTEMS - there are dangerous high hydraulic system pressures in the helicopter. Use extreme care when working on systems having this pressure.

USE OF FIRE EXTINGUISHER - when fire extinguishers are used in a confined area, ventilate immediately. Serious injury or death could result if the area is not ventilated or the user does not use a self-contained breathing device.

MAIN ROTOR PYLON SLIDING COVER - when opening and closing main pylon sliding cover, keep hands away from sharp edges near track, mating end, and especially ventilation blower inlet hole.

TECHNICAL ACETONE - technical Acetone is flammable and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well-ventilated areas (or use approved respirator). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

ALKALINE AQUEOUS CLEANER - alkaline cleaner is harmful to skin and eyes. Use adequate ventilation. Operators should wear protective gloves, aprons, and goggles. Upon contact with cleaning compound, wash affected area for at least 20 minutes with clean water. Seek medical attention.

CLEANING COMPOUND SOLVENT - cleaning Compound Solvent is combustible and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well-ventilated areas (or use approved respirator). Keep away from open flames or other sources of ignition.

ELECTRON - use electron in well ventilated area only. Avoid prolonged breathing of vapors. Avoid bodily contact. The use of chemical gloves and chemical splash goggles are required. Do not use near heat, spark or flame. This solvent is reactive with acids and oxidizers; do not mix or cross-apply with other chemicals. Organic vapor respirator with dust and mist filter is recommended when solvent is spray applied. Keep containers closed between applications.

- Provide mechanical ventilation if used in confined spaces. Coordinate the use of this material with your supporting industrial hygiene and safety offices. Ensure you read and understand the material safety data sheet (MSDS) for electron prior to use.

ISOPROPYL ALCOHOL - isopropyl alcohol is a volatile and flammable liquid which must be kept away from flame and other ignition sources. It must be used only in well ventilated area. Personnel must wear protective gloves and eye protection. Excessive inhalation of vapors or contact with skin must be avoided. If contact occurs with eyes or skin, flush area thoroughly with water. If ingestion occurs, induce vomiting and call a doctor. Remove to fresh air if overexposed to vapor.

DS-108 - use in well ventilated area. Avoid breathing vapors. Organic vapor respirator with dust and mist filter is recommended. May be harmful by inhalation, ingestion, or skin absorption. Avoid bodily contact. Chemical safety glasses/ goggles and chemical resistant gloves are required. Combustible, keep away from heat, spark, or flame. Vapors are heavier than air and may travel over a distance to an ignition source and then flash back. Do not mix or cross apply with other chemicals.

LOCKWIRE - in critical applications use lockwire instead of safety cable which is specifically prohibited from use in critical applications. The use of safety cable is restricted in these specific applications with a warning.

PURPOSE OF A WARNING - alerts personal to operation, procedure, practice, etc., which if not strictly observed, could result in personal injury or loss of life.

CHANGE
NO. 2

HEADQUARTERS DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 15 December 2007

TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS

TM 1-1520-237-23-3, 17 April 2006, is updated as follows:

1. File this sheet in front of the manual for reference.
2. This change results from updated procedures for maintenance, checks and services, and/or updated lists of expendable/durable supplies and materials.
3. New or updated text is indicated by a vertical bar in the outer margin of the page.
4. Added illustrations are indicated by a vertical bar adjacent to the figure number. Changed illustrations are indicated by a miniature pointing hand adjacent to the updated area or a MAJOR CHANGE symbol and a vertical bar adjacent to the figure number.
5. Remove old pages and insert new pages as indicated below.

Remove Pages	Insert Pages
None	a and b
A/B Blank	A/B Blank

6. Replace the following work packages with their revised version.

Work Package Number
WP 0101 00
WP 0112 00
WP 0113 00
WP 0114 00
WP 0122 00

Change
NO. 2

By Order of the Secretary of the Army:

GEORGE W. CASEY, JR.
General, United States Army
Chief of Staff

Official:

JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army
0733117

DISTRIBUTION:

To be distributed in accordance with Initial Distribution Number (IDN) 311286, requirements for TM 1-1520-237-23-3.

CHANGE
NO. 1

HEADQUARTERS
DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 21 March 2007

TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE

FOR
ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A, AND HH-60L HELICOPTERS

WARNING - This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C. Sec. 2751 et seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et seq. Violation of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.

DISTRIBUTION STATEMENT D. Distribution authorized to the DOD and DOD contractors only due to Critical Technology effective as of 15 June 2003. Other requests must be referred to Commander, US Army Aviation and Missile Command, ATTN: SFAE-AV-UH/L, Redstone Arsenal, AL 35898-5230.

DESTRUCTION NOTICE - Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

TM 1-1520-237-23-3, dated 17 April 2006, is changed as follows:

1. File these sheets in front of the manual for reference.
2. This change results from updated procedures for maintenance, checks and services, and/or updated lists of expendable/durable supplies and materials.
3. New or updated text is indicated by a vertical bar in the outer margin of the page.
4. Added illustrations are indicated by a vertical bar adjacent to the figure number. Changed illustrations are indicated by a miniature pointing hand adjacent to the updated area or a MAJOR CHANGE symbol and a vertical bar adjacent to the figure number.
5. Remove old pages and insert new pages as indicated below.

Remove pages	Insert pages
A/B Blank	A/ B Blank
i and ii	i and ii
Chapter 3 Title and Blank	Chapter 3 Title and Blank
Cover	Cover

6. Replace the following work packages with their revised version.

Work Package Number

WP 0108 00
WP 0112 00
WP 0114 00
WP 0116 00
WP 0122 00

<u>Work Package Number</u>

WP 0123 00
WP 0124 00
WP 0125 00
WP 0126 00
WP 0127 00
WP 0128 00
WP 0129 00
WP 0131 00
WP 0132 00
WP 0134 00
WP 0135 00
WP 0136 00

.

By Order of the Secretary of the Army:

PETER J. SCHOOMAKER
General, United States
Army
Chief of Staff

Official:

JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army
0704003

DISTRIBUTION:
To be distributed in accordance with Initial Distribution Number (IDN) 311286, requirements for TM 1-1520-237-23-3.

INSERT LATEST CHANGED PAGES/WORK PACKAGES. DESTROY SUPERSEDED DATA.

LIST OF EFFECTIVE PAGES/WORK PACKAGES

Dates of issue for original and changed pages / work packages are:

Original 0 17 April 2006

Change 1 21 March 2007

Change 2 15 December 2007

TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS 8 AND TOTAL NUMBER OF WORK PACKAGES IS 40 CONSISTING OF THE FOLLOWING:

Page / WP No.	*Change No.
Volume 3 Title	1
a - b	2
A	2
B Blank	2
i - ii	1
Chapter 3 Title Page	1
WP 0101 00 (150 pgs)	2
WP 0102 00 (70 pgs)	0
WP 0103 00 (8 pgs)	0
WP 0104 00 (2 pgs)	0
WP 0105 00 (28 pgs)	0
WP 0106 00 (52 pgs)	0
WP 0107 00 (6 pgs)	0
WP 0108 00 (8 pgs)	1
WP 0109 00 (10 pgs)	0
WP 0110 00 (26 pgs)	0
WP 0111 00 (14 pgs)	0
WP 0112 00 (20 pgs)	2
WP 0113 00 (16 pgs)	2
WP 0114 00 (12 pgs)	2
WP 0115 00 (10 pgs)	0
WP 0116 00 (8 pgs)	1
WP 0117 00 (18 pgs)	0
WP 0118 00 (8 pgs)	0
WP 0119 00 (6 pgs)	0
WP 0120 00 (20 pgs)	0
WP 0121 00 (22 pgs)	0
WP 0122 00 (78 pgs)	2
WP 0123 00 (48 pgs)	1
WP 0124 00 (22 pgs)	1
WP 0125 00 (18 pgs)	1
WP 0126 00 (24 pgs)	1
WP 0127 00 (12 pgs)	1
WP 0128 00 (30 pgs)	1
WP 0129 00 (18 pgs)	1
WP 0130 00 (8 pgs)	0
WP 0131 00 (26 pgs)	1
WP 0132 00 (24 pgs)	1
WP 0133 00 (12 pgs)	0
WP 0134 00 (14 pgs)	1
WP 0135 00 (18 pgs)	1
WP 0136 00 (44 pgs)	1
WP 0137 00 (8 pgs)	0
WP 0138 00 (8 pgs)	0
WP 0139 00 (8 pgs)	0
WP 0140 00 (8 pgs)	0

* Zero in this column indicates an original page or work package.

HEADQUARTERS,
DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 17 April 2006

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

Army Models UH-60A, UH-60L, EH-60A, HH-60A, and HH-60L Helicopters

REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS.

You can help improve this manual. If you find any mistakes or if you know of a way to improve these procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank Forms) located in the back of this manual directly to: Commander, U.S. Army Aviation and Missile Command, ATTN: AMSAM-MMC-MA-NP, Redstone Arsenal, AL 35898-5000. A reply will be furnished to you. You may also provide DA Form 2028 information to AMCOM via e-mail, fax, or the World Wide Web. Our fax number is: DSN 788-6546 or Commercial 256-842-6546. Our e-mail address is: 2028@redstone.army.mil. Instructions for sending an electronic 2028 may be found at the back of this manual immediately preceding the hard copy 2028. For the World Wide Web use: https://amcom2028.redstone.army.mil.

TABLE OF CONTENTS

*This manual together with TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10, TM 1-1520-237-23-11, TM 1-1520-237-23-12, TM 1-1520-237-23-13, TM 1-1520-237-23-14, TM 1-1520-237-23-15, TM 1-1520-237-23-16, TM 1-1520-237-23-17, TM 1-1520-237-23-18, TM 1-1520-237-23-19, and TM 1-1520-237-23-20, dated 17 April 2006, supersedes TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10, TM 1-1520-237-23-10-1, TM 1-1520-237-23-10-2, TM 1-1520-237-23-10-3, TM 1-1520-237-23-11, dated 1 May 2003 including all changes.

TABLE OF CONTENTS - Continued

CHAPTER 3

TROUBLESHOOTING PROCEDURES

FOR

ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A, AND HH-60L

HELICOPTERS

UNIT LEVEL

ELECTRICAL SYSTEMS

AC ELECTRICAL SYSTEM

This WP supersedes WP 0101 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A
Power System Analyzer, 60B63-5-A

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer Toolkit, SC 5180-99-B06
Locally-Made Drag Beam Wedge Figure 167, WP 1805 00

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-237-10
WP 0086 00
WP 0090 00
WP 0102 00
WP 0104 00
WP 0145 00
WP 0443 00
WP 0785 00
WP 0824 00
WP 0825 00
WP 0827 00
WP 0828 00
WP 0833 00
WP 0834 00
WP 0835 00
WP 0836 00
WP 0837 00
WP 0838 00
WP 0839 00
WP 0840 00
WP 0843 00
WP 0844 00
WP 0845 00
WP 0846 00
WP 0847 00
WP 0849 00
WP 0850 00
WP 0851 00
WP 0925 00
WP 1227 00
WP 1685 00
WP 1747 00
WP 1805 00

Equipment Condition

External Electrical Power Available, WP 1685 00 and APU Operational, TM 1-1520-237-23

SPECIAL INSTRUCTIONS

NOTE

See UH-60A UH-60L Figure 1 and EH-60A Figure 2 for component location diagrams, UH-60A UH-60L Figure 3 and EH-60A Figure 4 for schematic diagrams, UH-60A UH-60L 89-26149-96-26722 Figure 5 , UH-60L 96-26723-SUBQ Figure 6 , and EH-60A Figure 7 for distribution diagrams as an aid in operational/troubleshooting.

The AC electrical system operational/troubleshooting procedure is subdivided as follows:

- APU Generator System
- No. 1 Generator System

SPECIAL INSTRUCTIONS - Continued

- No. 2 Generator System
- AC External Power System
- Mission Interface Panel Ground Power **EH-60A**
- Mission Interface Panel Single Generator Power **EH-60A**
- Mission Interface Panel Dual Generators Power **EH-60A**
- Mission Interface Panel Auxiliary Power **EH-60A**
- 26 VAC Power System
- Cabin UTIL RCPT 115 VAC 3ø 400 Hz Receptacle
- 60 Hz Converter Receptacle
- Generator and GCU Troubleshooting

CAUTION

In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.

Table 1. Temperature Requirements.

AIR TEMPERATURE F° (C°)	OPERATING TIME (MINUTES)	COOLDOWN TIME, (PUMP OFF) (MINUTES)
90° (32°) max	Unlimited	-
91° (33°) to 100° (38°)	24	72
101° (39°) to 125° (52°)	16	48

APU GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
UH-60A UH-60L	
AC ESNTL BUS SPLY	Pilot’s
AC ESNTL BUS SPLY	Copilot’s
AC ESNTL BUS WARN	Pilot’s
APU CONTR INST	Lower Console
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot’s
DC ESNTL BUS SPLY	Copilot’s

APU GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
ESNTL BUS AC & CONV WARN	Lower Console
ESNTL BUS DC SPLY	Lower Console
J207 APU-AC POWER SYSTEM TEST RECEPTACLES	No. 2 Junction Box
LIGHTS CAUT ADVSY	Copilot's
NO. 1 CONVERTER	Copilot's
NO. 2 CONVERTER	Pilot's
EH-60A	
AC ESNTL BUS SPLY	Pilot's
AC ESNTL BUS SPLY	Copilot's
AC ESNTL BUS WARN	Pilot's
APU CONTR INST	Lower Console
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot's
DC ESNTL BUS SPLY	Copilot's
ESNTL BUS AC & CONV WARN	Lower Console
ESNTL BUS DC SPLY	Lower Console
J207 APU-AC POWER SYSTEM TEST RECEPTACLES	No. 2 Junction Box
LIGHTS CAUT ADVSY	Copilot's
NO. 1 CONVERTER	Copilot's
NO. 2 CONVERTER	Pilot's

2. Place upper console BATT switch to ON.

APU GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

a. AC ESS BUS OFF capsule shall go on.

b. #1 CONV capsule shall go on.

c. #2 CONV capsules shall go on.

d. DC ESS BUS OFF capsule shall be off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to AC ESS BUS OFF CAPSULE DOES NOT GO ON WITHOUT POWER ON AC ESNTL BUS, in this work package.
2. If Indication/Condition a. or b. is not as specified, troubleshoot DC electrical system (WP 0102 00).

Step

3. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and APU switches are in OFF/RESET.
4. Start APU (TM 1-1520-237-10).
5. Place upper console GENERATORS APU switch to ON.

Indication/Condition

a. APU GEN ON capsule shall go on.

b. #1 CONV capsule shall go off.

c. #2 CONV capsule shall go off.

d. AC ESS BUS OFF capsule shall go off.

e. #1 GEN and #2 GEN capsules shall go on.

Corrective Action

1. If Indication/Conditions a., b., and c. are not as specified, go to WITH GENERATORS APU SWITCH ON, APU GEN ON CAPSULE DOES NOT GO ON AND #1 CONV AND #2 CONV CAPSULES REMAIN ON (PRIMARY AC BUSES NOT ENERGIZED), in this work package.
2. If Indication/Condition a. is not as specified, go to APU GEN ON CAPSULE IS OFF WITH APU GENERATOR SUPPLYING BUSES (#1 CONV AND #2 CONV CAPSULES OFF), in this work package.
3. If Indication/Condition b. is not as specified, go to APU GENERATOR OR AC EXTERNAL POWER DOES NOT SUPPLY NO. 1 AC PRI BUS (#1 CONV CAPSULE ON), in this work package.
4. If Indication/Condition c. is not as specified, check wiring from NO. 2 CONVERTER circuit breaker terminals A2, B2, and C2 to NO. 2 GEN CNTOR contactor terminals A2, B2, and C2.
   a. If continuity is not present, repair/replace wiring as required (WP 1747 00).
   b. If continuity is present, replace NO. 2 GEN CNTOR (WP 0835 00).
5. If Indication/Condition d. is not as specified, go to AC ESS BUS OFF CAPSULE DOES NOT GO OFF WHEN NO. 1 AND NO. 2 AC PRIMARY BUSES ARE ENERGIZED, in this work package.
6. If Indication/Condition e. is not as specified, do NO. 1 GENERATOR SYSTEM or NO. 2 GENERATOR SYSTEM, as required, in this work package.

Step

6. Place upper console GENERATORS APU switch to OFF/RESET.

APU GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

7. Shut down APU (TM 1-1520-237-10).
8. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

NO. 1 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
UH-60A UH-60L	
AC ESNTL BUS SPLY	Pilot’s
AC ESNTL BUS SPLY	Copilot’s
AC ESNTL BUS WARN	Pilot’s

NO. 1 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
APU CONTR INST	Lower Console
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot's
DC ESNTL BUS SPLY	Copilot's
ESNTL BUS AC & CONV WARN	Lower Console
ESNTL BUS DC SPLY	Lower Console
LIGHTS CAUT ADVSY	Copilot's
NO. 1 CONVERTER	Copilot's
NO. 2 CONVERTER	Pilot's
NO. 1 GEN WARN	Copilot's
NO. 2 GEN WARN	Pilot's
PWR SYS TEST RECP A-B-C 115 VAC D GROUND E 28 VDC J211	No. 1 Junction Box
EH-60A	
AC ESNTL BUS SPLY	Pilot's
AC ESNTL BUS SPLY	Copilot's
AC ESNTL BUS WARN	Pilot's
APU CONTR INST	Lower Console
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot's
DC ESNTL BUS SPLY	Copilot's

NO. 1 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
ESNTL BUS AC & CONV WARN	Lower Console
ESNTL BUS DC SPLY	Lower Console
LIGHTS CAUT ADVSY	Copilot's
NO. 1 CONVERTER	Copilot's
NO. 2 CONVERTER	Pilot's
NO. 1 GEN WARN	Copilot's
NO. 2 GEN WARN	Pilot's
PWR SYS TEST RECP A-B-C 115 VAC D GROUND E 28 VDC J211	No. 1 Junction Box

2. Place upper console BATT switch to ON.

Indication/Condition

#1 GEN BRG capsule on caution/advisory panel shall not go on.

Corrective Action

If Indication/Condition is not as specified, go to #1 GEN BRG AND/OR #2 GEN BRG CAPSULE GOES ON, in this work package.

Step

3. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and APU switches are placed to OFF/RESET.
4. Start APU (TM 1-1520-237-10).
5. Place upper console GENERATOR APU switch to ON.

Indication/Condition

a. #1 GEN capsule shall go on.

b. #2 GEN capsule shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #1 GEN CAPSULE DOES NOT GO ON WITH NO. 1 GENERATOR OFF, in this work package.
2. If Indication/Condition b. is not as specified, do NO. 2 GENERATOR SYSTEM, in this work package.

Step

6. With qualified personnel at the controls, start No. 1 engine and operate with main rotor turning at 100% (TM 1-1520-237-10).
7. Place upper console GENERATORS NO. 1 switch to ON.

NO. 1 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

a. #1 GEN capsule shall go off.

b. #2 GEN capsule shall remain on.

c. #1 CONV capsule shall go off.

d. #2 CONV capsule shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #1 GEN CAPSULE IS ON WITH GENERATORS NO. 1 SWITCH ON, in this work package.
2. If Indication/Condition b. is not as specified, do NO. 2 GENERATOR SYSTEM, in this work package.
3. If Indication/Condition c. is not as specified, go to #1 CONV CAPSULE STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 1 AC PRIMARY BUS, in this work package.
4. If Indication/Condition d. is not as specified, go to #2 CONV CAPSULE STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 2 AC PRIMARY BUS, in this work package.
5. If No. 1 generator is disabled due to underfrequency, go to NO. 1 AND/OR NO. 2 GENERATOR DISABLED BY UNDERFREQUENCY CONDITION DURING AUTO ROTATIVE DESCENT, in this work package.

Step

8. Place upper console GENERATORS NO. 1 switch to OFF/RESET.
9. Shut down No. 1 engine and APU (TM 1-1520-237-10).
10. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

NO. 2 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
UH-60A UH-60L	
AC ESNTL BUS SPLY	Pilot’s
AC ESNTL BUS SPLY	Copilot’s
AC ESNTL BUS WARN	Pilot’s
APU CONTR INST	Lower Console
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot’s

NO. 2 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
DC ESNTL BUS SPLY	Copilot's
ESNTL BUS AC & CONV WARN	Lower Console
ESNTL BUS DC SPLY	Lower Console
J208 NO. 2-AC/DC POWER SYSTEM TEST RECEPTACLES	No. 1 Junction Box
LIGHTS CAUT ADVSY	Copilot's
NO. 1 CONVERTER	Copilot's
NO. 2 CONVERTER	Pilot's
NO. 1 GEN WARN	Copilot's
NO. 2 GEN WARN	Pilot's
EH-60A	
AC ESNTL BUS SPLY	Pilot's
AC ESNTL BUS SPLY	Copilot's
AC ESNTL BUS WARN	Pilot's
APU CONTR INST	Lower Console
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot's
DC ESNTL BUS SPLY	Copilot's
ESNTL BUS AC & CONV WARN	Lower Console
ESNTL BUS DC SPLY	Lower Console
J208 NO. 2-AC/DC POWER SYSTEM TEST RECEPTACLES	No. 1 Junction Box
LIGHTS CAUT ADVSY	Copilot's

NO. 2 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
NO. 1 CONVERTER	Copilot's
NO. 2 CONVERTER	Pilot's
NO. 1 GEN WARN	Copilot's
NO. 2 GEN WARN	Pilot's

2. Place upper console BATT switch to ON.

Indication/Condition

#2 GEN BRG capsule on caution/advisory panel shall not go on.

Corrective Action

If Indication/Condition is not as specified, go to #1 GEN BRG AND/OR #2 GEN BRG CAPSULE GOES ON, in this work package.

Step

3. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and APU switches are placed to OFF/RESET.
4. Start APU (TM 1-1520-237-10).
5. Place upper console GENERATORS APU switch to ON.

Indication/Condition

a. #1 GEN capsule shall go on.

b. #2 GEN capsule shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, do NO. 1 GENERATOR SYSTEM, in this work package.
2. If Indication/Condition b. is not as specified, go to #2 GEN CAPSULE IS OFF WITH NO. 2 GENERATOR OFF, in this work package.

Step

6. With qualified personnel at the controls, start No. 2 engine and operate with main rotor turning at 100% (TM 1-1520-237-10).
7. Place upper console GENERATORS NO. 2 switch to ON.

Indication/Condition

a. #2 GEN capsule shall go off.

b. #1 GEN capsule shall remain on.

c. #1 CONV and #2 CONV capsules shall go off.

NO. 2 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition a. is not as specified, go to #2 GEN CAPSULE IS ON WITH GENERATORS NO. 2 SWITCH ON, in this work package.
2. If Indication/Condition b. is not as specified, do NO. 1 GENERATOR SYSTEM, in this work package.
3. If Indication/Condition c. is not as specified, go to #1 CONV CAPSULE STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 1 AC PRIMARY BUS or #2 CONV CAPSULE STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 2 AC PRIMARY BUS, as required, in this work package.
4. If No. 2 generator is disabled due to underfrequency, go to NO. 1 AND/OR NO. 2 GENERATOR DISABLED BY UNDERFREQUENCY CONDITION DURING AUTO ROTATIVE DESCENT, in this work package.

Step

8. Place upper console GENERATORS NO. 2 switch to OFF/RESET.
9. Shut down No. 2 engine and APU (TM 1-1520-237-10).
10. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

AC EXTERNAL POWER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
AC ESNTL BUS SPLY	Pilot’s
AC ESNTL BUS SPLY	Copilot’s
BATT & ESNTL DC WARN EXT PWR CONTR	Lower Console
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot’s
DC ESNTL BUS SPLY	Copilot’s
ESNTL BUS AC & CONV WARN	Lower Console

AC EXTERNAL POWER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
ESNTL BUS DC SPLY	Lower Console
LIGHTS CAUT ADVSY	Copilot’s
NO. 1 CONVERTER	Copilot’s
NO. 2 CONVERTER	Pilot’s

2. Place upper console BATT switch to ON.
3. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and APU switches are in OFF/RESET.
4. Turn on external electrical power.

Indication/Condition

a. EXT PWR CONNECTED capsule shall go on.

b. #1 CONV and #2 CONV capsules shall remain on.

c. AC ESS BUS OFF capsule shall remain on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to EXT PWR CONNECTED CAPSULE IS OFF WITH A SOURCE OF EXTERNAL POWER CONNECTED, in this work package.
2. If Indication/Condition b. is not as specified, troubleshoot DC electrical system (WP 0102 00).
3. If Indication/Condition c. is not as specified, go to AC ESS BUS OFF CAPSULE DOES NOT GO ON WITHOUT POWER ON AC ESNTL BUS, in this work package.

Step

5. Momentarily place upper console EXT PWR switch to RESET and then to ON.

Indication/Condition

a. #1 CONV capsule shall go off.

b. #2 CONV capsule shall go off.

c. #1 GEN and #2 GEN capsules shall go on.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, go to EXTERNAL POWER DOES NOT SUPPLY AC BUSES (#1 CONV AND #2 CONV CAPSULES ON), in this work package.
2. If Indication/Condition a. is not as specified, go to APU GENERATOR OR AC EXTERNAL POWER DOES NOT SUPPLY NO. 1 AC PRI BUS (#1 CONV CAPSULE ON), in this work package.
3. If Indication/Condition c. is not as specified, do NO. 1 GENERATOR SYSTEM or NO. 2 GENERATOR SYSTEM, as required, in this work package.

Step

6. Turn off external electrical power.

AC EXTERNAL POWER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

7. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

MISSION INTERFACE PANEL GROUND POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
BACKUP HYD CONTR	Upper Console
BACKUP PUMP PWR	Copilot’s
BATT & ESNTL DC WARN EXT PWR CONTR	Lower Console
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console

MISSION INTERFACE PANEL GROUND POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot’s
DC ESNTL BUS SPLY	Copilot’s
ESNTL BUS DC SPLY	Lower Console
LIGHTS CAUT ADVSY	Copilot’s
NO. 1 CONVERTER	Copilot’s
NO. 2 CONVERTER	Pilot’s
NO. 2 SERVO WARN	Pilot’s

2. Place upper console BATT switch to ON.
3. Turn on external electrical power.
4. Pull out SEC MON BUS CONTR circuit breaker on pilot’s circuit breaker panel.
5. Make sure upper console BACKUP HYD PUMP switch is placed to OFF.
6. Place upper console EXT PWR switch first to RESET and then to ON.

Indication/Condition

#1 CONV and #2 CONV capsules shall be off.

Corrective Action

If Indication/Condition is not as specified, do APU GENERATOR SYSTEM or AC EXTERNAL POWER SYSTEM, as required, in this work package.

Step

NOTE

Backup hydraulic pump may run to charge APU accumulator. AC secondary bus will not energize until backup pump stops.

7. Check for 115 vac between the following points on the mission interface panel:
   - J511-A and J511-G
   - J511-A and J511-H
   - J511-B and J511-G
   - J511-B and J511-H
   - J511-C and J511-G
   - J511-C and J511-H

MISSION INTERFACE PANEL GROUND POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A - Continued

Indication/Condition

Multimeter shall indicate 115 vac between specified points.

Corrective Action

If Indication/Condition is not as specified, go to NO POWER AT MISSION INTERFACE PANEL WITH EXTERNAL POWER APPLIED, in this work package.

Step

8. Place upper console BACKUP HYD PUMP switch to ON.
9. Check voltage between J511-C and ground.

Indication/Condition

a. Multimeter shall indicate 0 vac.

b. BACK-UP PUMP ON capsule shall be on.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, troubleshoot hydraulic system (WP 0086 00).
2. If Indication/Condition a. is not as specified, go to 115 VAC AT MISSION INTERFACE PANEL WITH EXTERNAL POWER CONNECTED AND BACK-UP PUMP RUNNING, in this work package.

Step

10. Place upper console BACKUP HYD PUMP switch to OFF.
11. Turn off external electrical power.
12. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

MISSION INTERFACE PANEL SINGLE GENERATOR POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
BACKUP HYD CONTR	Upper Console
BACKUP PUMP PWR	Copilot’s
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot’s
DC ESNTL BUS SPLY	Copilot’s
ESNTL BUS AC & CONV WARN	Lower Console
ESNTL BUS DC SPLY	Lower Console

MISSION INTERFACE PANEL SINGLE GENERATOR POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
LIGHTS CAUT ADVSY	Copilot's
NO. 1 CONVERTER	Copilot's
NO. 2 CONVERTER	Pilot's
NO. 1 GEN WARN	Copilot's
NO. 2 GEN WARN	Pilot's
NO. 2 SERVO WARN	Pilot's
SEC MON BUS CONTR	Pilot's

2. Make sure upper console BACKUP HYD PUMP switch is OFF.
3. Start APU (TM 1-1520-237-10).
4. Place upper console GENERATORS NO. 1 switch to ON and GENERATORS NO. 2 and GENERATORS APU switches to OFF/RESET.
5. Place upper console BATT switch to ON.

Indication/Condition

#1 GEN BRG or #2 GEN BRG capsule on caution/advisory panel shall not go on.

Corrective Action

If Indication/Condition is not as specified, go to #1 GEN BRG AND/OR #2 GEN BRG CAPSULE GOES ON, in this work package.

Step

6. With qualified personnel at the controls, start No. 1 engine and operate with main rotor turning at 100% (TM 1-1520-237-10).

Indication/Condition

#1 CONV and #2 CONV capsules shall be off.

Corrective Action

If Indication/Condition is not as specified, do NO. 2 GENERATOR SYSTEM, in this work package.

Step

NOTE

Backup hydraulic pump may run to charge APU accumulator. AC secondary bus will not operate until backup pump stops.

7. Check for 115 vac between the following points on the mission interface panel:

- J511-A and J511-G

MISSION INTERFACE PANEL SINGLE GENERATOR POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A - Continued

- J511-A and J511-H
- J511-B and J511-G
- J511-B and J511-H
- J511-C and J511-G
- J511-C and J511-H

Indication/Condition

Multimeter shall indicate 115 vac between specified points.

Corrective Action

If Indication/Condition is not as specified, go to NO POWER AT MISSION INTERFACE PANEL WITH NO. 1 OR NO. 2 GENERATOR ON, in this work package.

Step

8. Place upper console BACKUP HYD PUMP switch to ON.
9. Check voltage between J511-C and ground.

Indication/Condition

a. Multimeter shall indicate 0 vac.

b. BACK-UP PUMP ON capsule shall be on.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, troubleshoot hydraulic system (WP 0086 00).
2. If Indication/Condition a. is not as specified, go to 115 VAC AT MISSION INTERFACE PANEL WITH EXTERNAL POWER CONNECTED AND BACK-UP PUMP RUNNING, in this work package.

Step

10. Place upper console BACKUP HYD PUMP switch to OFF.
11. Shut down No. 1 engine (TM 1-1520-237-10).
12. Place upper console GENERATORS NO. 1 switch to OFF.
13. Place upper console BATT switch to OFF.
14. Repeat steps 4. through 13. substituting No. 2 engine for No. 1 engine and GENERATORS NO. 2 switch for GENERATORS NO. 1 switch.
15. Shut down APU (TM 1-1520-237-10).

Indication/Condition

APU shall be shut down.

Corrective Action

None Required

MISSION INTERFACE PANEL DUAL GENERATORS POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
BACKUP HYD CONTR	Upper Console
BACKUP PUMP PWR	Copilot’s
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot’s
DC ESNTL BUS SPLY	Copilot’s
ESNTL BUS AC & CONV WARN	Lower Console
ESNTL BUS DC SPLY	Lower Console

MISSION INTERFACE PANEL DUAL GENERATORS POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
LIGHTS CAUT ADVSY	Copilot's
NO. 1 CONVERTER	Copilot's
NO. 2 CONVERTER	Pilot's
NO. 1 GEN WARN	Copilot's
NO. 2 GEN WARN	Pilot's
NO. 2 SERVO WARN	Pilot's
SEC MON BUS CONTR	Pilot's

2. Place upper console BATT switch to ON.

Indication/Condition

#1 GEN BRG or #2 GEN BRG capsule on caution/advisory panel shall not go on.

Corrective Action

If Indication/Condition is not as specified, go to #1 GEN BRG AND/OR #2 GEN BRG CAPSULE GOES ON, in this work package.

Step

3. Start APU (TM 1-1520-237-10).
4. Place upper console GENERATORS NO. 1 and NO. 2 switches to ON and GENERATORS APU switch to OFF/RESET.
5. With qualified personnel at the controls, start No. 1 and No. 2 engines and operate with main rotor turning at 100% (TM 1-1520-237-10).

Indication/Condition

#1 GEN and #2 GEN capsules shall be off.

Corrective Action

If Indication/Condition is not as specified, do NO. 1 GENERATOR SYSTEM or NO. 2 GENERATOR SYSTEM, as required, in this work package.

Step

6. Place upper console BACKUP HYD PUMP switch to ON.
7. Check for 115 vac between the following points on the mission interface panel:
   - J511-A and J511-G
   - J511-A and J511-H
   - J511-B and J511-G

MISSION INTERFACE PANEL DUAL GENERATORS POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A - Continued

- J511-B and J511-H
- J511-C and J511-G
- J511-C and J511-H

Indication/Condition

a. Multimeter shall indicate 115 vac between specified points.

b. BACK-UP PUMP ON capsule shall be on.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, go to NO POWER AT MISSION INTERFACE PANEL WITH NO. 1 AND NO. 2 GENERATORS ON, AND BACK-UP PUMP RUNNING, in this work package.
2. If Indication/Condition b. is not as specified, troubleshoot hydraulic system (WP 0086 00).

Step

8. Place upper console BACKUP HYD PUMP switch to OFF.
9. Place upper console GENERATORS NO. 1 and NO. 2 switches to OFF/RESET.
10. Shut down No. 1 and No. 2 engines (TM 1-1520-237-10).
11. Shut down APU (TM 1-1520-237-10).
12. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

MISSION INTERFACE PANEL AUXILIARY POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
APU CONTR INST	Lower Console
BACKUP HYD CONTR	Upper Console
BACKUP PUMP PWR	Copilot’s
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot’s
DC ESNTL BUS SPLY	Copilot’s
DE-ICE CONTRLR	Copilot’s

MISSION INTERFACE PANEL AUXILIARY POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
ESNTL BUS AC & CONV WARN	Lower Console
ESNTL BUS DC SPLY	Lower Console
LIGHTS CAUT ADVSY	Copilot's
NO. 1 CONVERTER	Copilot's
NO. 2 CONVERTER	Pilot's
NO. 1 GEN WARN	Copilot's
NO. 2 GEN WARN	Pilot's
NO. 2 SERVO WARN	Pilot's
SEC MON BUS CONTR	Pilot's

2. Place upper console BATT switch to ON.

Indication/Condition

#1 GEN BRG capsule on caution/advisory panel shall not go on.

Corrective Action

If Indication/Condition is not as specified, go to #1 GEN BRG AND/OR #2 GEN BRG CAPSULE GOES ON, in this work package.

Step

3. Start APU (TM 1-1520-237-10).
4. Place upper console GENERATORS NO. 1 and GENERATORS NO. 2 switches to ON and GENERATORS APU switch to OFF/RESET.
5. Place upper console GENERATORS APU switch to ON.

Indication/Condition

APU GEN ON capsule shall be on, and #1 CONV and #2 CONV capsules shall be off.

Corrective Action

If Indication/Condition is not as specified, do APU GENERATOR SYSTEM, in this work package.

Step

6. Place upper console GENERATORS APU switch to OFF/RESET.
7. With qualified personnel at the controls, start No. 1 engine and operate with main rotor turning at 100% (TM 1-1520-237-10).
8. Place upper console GENERATORS NO. 1 switch to ON.

MISSION INTERFACE PANEL AUXILIARY POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A - Continued

Indication/Condition

#1 GEN, #1 CONV, and #2 CONV capsules shall be off.

Corrective Action

If Indication/Condition is not as specified, do NO. 1 GENERATOR SYSTEM, in this work package.

Step

9. Place upper console GENERATORS APU switch to ON.
10. Place upper console BACKUP HYD PUMP switch to ON.

Indication/Condition

BACK-UP PUMP ON capsule on caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot hydraulic system (WP 0086 00).

Step

11. Check for 28 vdc between contactor terminals K11-Y1 and K11-COM.

Indication/Condition

Multimeter shall indicate 28 vdc.

Corrective Action

If Indication/Condition is not as specified, go to NO EMERGENCY AUXILIARY POWER AT MISSION INTERFACE PANEL, in this work package.

Step

12. Check for 115 vac between the following points on the mission interface panel:
    - J511-A and J511-G
    - J511-A and J511-H
    - J511-B and J511-G
    - J511-B and J511-H
    - J511-C and J511-G
    - J511-C and J511-H

Indication/Condition

Multimeter shall indicate 115 vac between specified points.

Corrective Action

1. If Indication/Condition is not as specified, repair/replace wiring to mission interface panel connector J511 as required (WP 1747 00).
   a. If trouble remains, replace contactor K11 (WP 0847 00).

MISSION INTERFACE PANEL AUXILIARY POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A - Continued

Step

13. Place blade de-ice control panel POWER switch to ON.
14. Check for 0 vac between J511-C and ground.

Indication/Condition

Multimeter shall indicate 0 vac.

Corrective Action

If Indication/Condition is not as specified, go to BLADE DE-ICE SYSTEM DOES NOT DISABLE AUXILIARY POWER WITH BLADE DE-ICE CONTROL PANEL POWER SWITCH ON AND/OR TEST, in this work package.

Step

15. Place blade de-ice control panel POWER switch to TEST.
16. Check for 0 vac between J511-C and ground.

Indication/Condition

Multimeter shall indicate 0 vac.

Corrective Action

If Indication/Condition is not as specified, go to BLADE DE-ICE SYSTEM DOES NOT DISABLE AUXILIARY POWER WITH BLADE DE-ICE CONTROL PANEL POWER SWITCH ON AND/OR TEST, in this work package.

Step

17. Place blade de-ice control panel POWER switch to OFF.
18. Repeat steps 7. through 17. substituting No. 2 engine for No. 1 engine and GENERATORS NO. 2 switch for GENERATORS NO. 1 switch.
19. Place upper console GENERATORS NO. 1 switch to OFF.
20. Using locally-made drag beam wedge (WP 1805 00), push in plunger of left drag beam switch to simulate in-flight condition.
21. Check for 115 vac between the following points on the mission interface panel:
    - J511-A and J511-G
    - J511-A and J511-H
    - J511-B and J511-G
    - J511-B and J511-H
    - J511-C and J511-G
    - J511-C and J511-H

Indication/Condition

Multimeter shall indicate 115 vac between specified points.

MISSION INTERFACE PANEL AUXILIARY POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE EH-60A - Continued

Corrective Action

If Indication/Condition is not as specified, go to 115 VAC AT MISSION INTERFACE PANEL WITH NO. 1 AND NO. 2 GENERATORS OFF, AND SIMULATED WEIGHT-OFF-WHEELS, in this work package.

Step

22. Remove locally-made drag beam wedge from drag beam switch.
23. Place upper console BACKUP HYD PUMP switch to OFF.
24. Place upper console GENERATORS APU and GENERATORS NO. 1 switches to OFF/RESET.
25. Shut down No. 1 engine (TM 1-1520-237-10).
26. Shut down APU (TM 1-1520-237-10).
27. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

26 VAC POWER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
UH-60A UH-60L	
AUTO XFMR	Pilot’s
BATT & ESNTL DC WARN EXT PWR CONTR	Lower Console
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot’s
DC ESNTL BUS SPLY	Copilot’s
ESNTL BUS AC & CONV WARN	Lower Console
ESNTL BUS DC SPLY	Lower Console

26 VAC POWER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
LIGHTS CAUT ADVSY	Copilot's
NO. 1 CONVERTER	Copilot's
NO. 2 CONVERTER	Pilot's
EH-60A	
AUTO XFMR	Pilot's
BATT & ESNTL DC WARN EXT PWR CONTR	Lower Console
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
CAUT/ADVSY PNL	Upper Console
DC ESNTL BUS SPLY	Pilot's
DC ESNTL BUS SPLY	Copilot's
ESNTL BUS AC & CONV WARN	Lower Console
ESNTL BUS DC SPLY	Lower Console
LIGHTS CAUT ADVSY	Copilot's
NO. 1 CONVERTER	Copilot's
NO. 2 CONVERTER	Pilot's

2. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and APU GENERATORS switches are set to OFF/RESET.
3. Place upper console BATT switch to ON.
4. Turn on external electrical power.
5. Check for 26 vac between terminal 2 of STAB IND, INST, and DPLR circuit breakers and ground.

Indication/Condition

Multimeter shall indicate 26 vac.

Corrective Action

If Indication/Condition is not as specified, go to NO 26 VAC POWER ON AC ESNTL BUS, in this work package.

Step

6. Turn off external electrical power.

26 VAC POWER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

7. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

UTIL RCPT 115 VAC 3Ø 400 HZ OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure UTIL RECP circuit breaker on pilot’s circuit breaker panel is pushed in.
2. Place upper console BATT switch to ON.
3. Turn on external electrical power.
4. Check for 115 vac between the following:
   - J257-A and ground
   - J257-B and ground
   - J257-C and ground

Indication/Condition

Multimeter shall indicate 115 vac.

Corrective Action

1. If Indication/Condition is not as specified, check for 115 vac between terminals A2, B2, and C2 of UTIL RECP circuit breaker and ground.

UTIL RCPT 115 VAC 3Ø 400 HZ OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

a. If voltage is as specified, repair/replace wiring between circuit breaker and J257 as required (WP 1747 00).

b. If voltage is not as specified, replace circuit breaker (WP 0824 00).

Step

5. Turn off external electrical power.
6. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

60 HZ AC CONVERTER RECEPTACLE OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure 60 HZ AC CONVERTER circuit breaker on copilot's circuit breaker panel is pushed in.
2. Place upper console BATT switch to ON.
3. Turn on external electrical power.
4. Check for 115 vac between the following:
   - J276-A and ground
   - J276-B and ground
   - J276-C and ground

60 HZ AC CONVERTER RECEPTACLE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Multimeter shall indicate 115 vac.

Corrective Action

1. If Indication/Condition is not as specified, check for 115 vac between terminals A1, B1, and C1 of 60 HZ AC CONVERTER circuit breaker and ground.
   a. If voltage is as specified, repair/replace wiring between circuit breaker and P276 as required (WP 1747 00).
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00).

Step

5. Turn off external electrical power.
6. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

GENERATOR AND GCU OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).
- If generator control unit is removed from its mounting at any time during system troubleshooting, the generator control unit case must be grounded to airframe ground.
- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Plug ground cable (supplied with analyzer) into ground test jack on analyzer, and attach alligator clip to ground.
2. Place upper console GENERATORS NO. 1, GENERATORS NO. 2, or GENERATORS APU switch to OFF/RESET, as required.
3. Place power system analyzer GEN, CUR LIM, and FF toggle switches to OFF, and all other toggle switches down.
4. Turn power system analyzer TEST SELECT switch to START TEST.
5. Connect P1 of adapter cable (supplied with analyzer) to J1 AIRCRAFT connector on analyzer.

CAUTION

Always shut down APU or No. 1 engine (as applicable) before disconnecting generator control unit.

6. Disconnect P214, P215, or P216 from generator control unit and connect to P3 of analyzer adapter cable, as required.
7. Connect P2 of analyzer adapter cable to generator control unit.
8. Make sure APU CONTR INST circuit breakers on lower console circuit breaker panel are pushed in.
9. Start APU (TM 1-1520-237-10).

GENERATOR AND GCU OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

10. With qualified personnel at controls, start No. 1 engine and operate with main rotor turning at 100% rpm (for No. 1 or No. 2 main generator) (TM 1-1520-237-10).

CAUTION

All tests with the system analyzer must be done with the GENERATORS NO. 1, GENERATORS NO. 2, or GENERATORS APU switch set to OFF/RESET to prevent applying a simulated fault to any helicopter load.

11. Turn power system analyzer GEN/PMG VOLTS meter switch to A-B, B-C, and C-A (PMG 30V RANGE) and check pmg voltage on meter for each switch position.

Indication/Condition

Each voltage indication shall be more than 20.5 volts.

Corrective Action

If Indication/Condition is not as specified, replace generator (WP 0839 00).

Step

12. On power system analyzer, hold TEST SET switch at RESET and GEN switch to ON; then release TEST SET switch.
13. Turn power system analyzer GEN/PMG VOLTS meter switch to T-1, T-2, and T-3 (GEN 140V RANGE) and check generator output voltage on meter for each switch position.

Indication/Condition

Each voltage indication shall be between 112.5 and 117.5 vac.

Corrective Action

If Indication/Condition is not as specified, replace generator (WP 0839 00).

Step

14. Check field current on power system analyzer FIELD AMPS/REG VOLTS meter.

Indication/Condition

Field current shall be 1.0 ampere maximum.

Corrective Action

If Indication/Condition is not as specified, replace generator (WP 0839 00).

Step

15. Check power system analyzer FREQUENCY meter indication.

Indication/Condition

FREQUENCY meter shall indicate between 380 and 420 Hz and ø SEQ T1-2-3 lamp shall be on.

Corrective Action

If Indication/Condition is not as specified, replace generator (WP 0839 00).

Step

16. Check power system analyzer GCR RESET, CCR RESET, CCR TRIP, SIM CONT, and TEST RELAY lamps.

GENERATOR AND GCU OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

GCR RESET, CCR RESET, SIM CONT, and TEST RELAY lamps shall be on and CCR TRIP lamp shall be off.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

17. Turn power system analyzer TEST SELECT switch to NL REG.
18. Place power system analyzer FIELD AMPS/REG VOLTS meter switch to READ REG BUS VOLTS and check voltage indication on meter.

Indication/Condition

Voltage indication shall be between 21 and 27 volts.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

19. Place power system analyzer FIELD AMPS/REG VOLTS meter switch to FLD CUR and GEN switch to OFF. Check GCR RESET, CCR RESET, CCR TRIP, SIM CONT, SIM WARN, and TEST RELAY lamps.

Indication/Condition

CCR TRIP, SIM WARN, and TEST RELAY lamps shall be on and GCR RESET, CCR RESET, and SIM CONT lamps shall be off.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

20. Hold power system analyzer TEST SET switch at RESET and hold GEN switch to TEST; then only release TEST SET switch.
21. Check power system analyzer GCR RESET, CCR RESET, CCR TRIP, SIM CONT, SIM WARN, and TEST RELAY lamps.

Indication/Condition

GCR RESET and CCR RESET lamps shall be on and CCR TRIP, SIM CONT, SIM WARN, and TEST RELAY lamps shall be off.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

22. Hold power system analyzer TEST SET switch to RESET and only release GEN switch to OFF, then place to ON; then only release TEST SET switch.
23. Turn power system analyzer GEN/PMG VOLTS meter switch to T-1, T-2, and T-3 (GEN 140V RANGE) and record generator output voltage on meter for each switch position.

GENERATOR AND GCU OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Each voltage indication shall be between 112.5 and 117.5 vac.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

24. Momentarily place power system analyzer FLYWHEEL DIODE switch to TEST.

Indication/Condition

FLYWHEEL DIODE OPEN lamp shall be off.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

25. Turn power system analyzer TEST SELECT switch to FL REG.
26. Turn power system analyzer GEN/PMG VOLTS meter switch to A-B, B-C, and C-A (PMG 30V RANGE) and check pmg voltage on meter for each switch position.

Indication/Condition

Each voltage indication shall be at least 19 volts.

Corrective Action

If Indication/Condition is not as specified, replace generator (WP 0839 00).

Step

27. Turn power system analyzer GEN/PMG VOLTS meter switch to T-1, T-2, and T-3 (GEN 140V RANGE) and check generator output voltage on meter for each switch position.

Indication/Condition

Each voltage indication shall not change more than 2.5 vac from indication recorded in step 23.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

28. Turn power system analyzer GEN/PMG VOLTS meter switch to T-1 (GEN 140V RANGE).
29. Place power system analyzer CUR LIM switch to 20 KVA.

NOTE

Voltage checks in step 30. must be done within 4 seconds of selection of each switch position to prevent an undervoltage trip. If an undervoltage trip occurs, turn TEST SELECT switch to FL REG and place GEN switch OFF. Then, with TEST SET switch held at RESET, place GEN switch ON and repeat voltage checks.

30. Turn power system analyzer TEST SELECT switch to A ø CUR LIM, B ø CUR LIM, and C ø CUR LIM and check GEN/PMG VOLTS meter T-1 indication for each switch position.

GENERATOR AND GCU OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Each voltage indication shall be between 90 and 110 volts.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

31. Place power system analyzer CUR LIM switch to 30 KVA and repeat step 30.
32. Place power system analyzer CUR LIM switch to OFF.
33. Turn power system analyzer TEST SELECT switch to A ø GEN CT, press CT CUR switch to READ, and check CT TEST CURRENT meter indication.

Indication/Condition

NOTE

It is normal for the 60B63-5A Power System Analyzer to indicate a fault when it is switched to test current transformers and contactors.

Current indication shall be between 0.70 and 1.70 mA.

Corrective Action

If Indication/Condition is not as specified, replace generator (WP 0839 00).

Step

34. Repeat step 33. with power system analyzer TEST SELECT switch in B ø GEN CT and C ø GEN CT positions.
35. Disconnect P333 from auxiliary AC junction box.
36. Turn power system analyzer TEST SELECT switch to A ø DP CT, press CT CUR switch to READ, and check CT TEST CURRENT meter indication.

Indication/Condition

Current indication shall be between 0.6 and 2.0 mA.

Corrective Action

If Indication/Condition is not as specified, replace current transformer (WP 0837 00).

Step

37. Repeat step 36. with power system analyzer TEST SELECT switch at B ø DP CT and C ø DP CT.
38. Place power system analyzer GEN switch OFF and turn TEST SELECT switch to START TEST.
39. Check continuity between the following:
    - J333-C and J333-D
    - J333-E and J333-F
    - J333-G and J333-H

Indication/Condition

Continuity shall be present.

GENERATOR AND GCU OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

If Indication/Condition is not as specified, replace auxiliary AC junction box (WP 1227 00).

Step

40. Disconnect P202 from No. 2 junction box.
41. Check continuity between the following:
    - P333-C and P333-D
    - P333-E and P333-F
    - P333-G and P333-H

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, replace No. 2 junction box transformer T3 (WP 0837 00).

Step

42. Install P202 and P333.
43. Shut down APU and/or No. 1 engine, as applicable (TM 1-1520-237-10).
44. Disconnect P214, P215, or P216 (as applicable) from P3 of analyzer adapter cable and connect it to generator control unit.
45. Disconnect adapter cable from analyzer.

Indication/Condition

Adapter cable shall be disconnected from analyzer.

Corrective Action

None Required

AC ESS BUS OFF CAPSULE DOES NOT GO ON WITHOUT POWER ON AC ESNTL BUS

SYMPTOM

AC ESS BUS OFF capsule does not go on without power on AC ESNTL bus.

MALFUNCTION

AC ESS BUS OFF capsule does not go on without power on AC ESNTL bus.

CORRECTIVE ACTION

1. Check helicopter configuration.

a. **EMEP** , go to **2.**

b. **W/O EMEP** , go to **4.**

2. EMEP Remove and check pin filtered adapter from P117 (WP 0925 00).

a. If pin filtered adapter is good, go to **3.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **8.**

AC ESS BUS OFF CAPSULE DOES NOT GO ON WITHOUT POWER ON AC ESNTL BUS - Continued

3. **EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**
4. **Check for 28 vdc between P117-20 and ground.**
   a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00). Go to **8.**
   b. If voltage is not as specified, go to **5.**
5. **Check for 28 vdc between K13 AC ESNTL BUS FAIL relay terminal A3 and ground.**
   a. If voltage is as specified, repair/replace wiring between K13 AC ESNTL BUS FAIL relay terminal A3 and P117-20 (WP 1747 00). Go to **8.**
   b. If voltage is not as specified, go to **6.**
6. **Check for 28 vdc between K13 AC ESNTL BUS FAIL relay terminal A2 and ground.**
   a. If voltage is as specified, replace relay (WP 0845 00). Go to **8.**
   b. If voltage is not as specified, go to **7.**
7. **Check for 28 vdc between ESNTL BUS AC & CONV WARN circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, repair/replace wiring between circuit breaker and relay terminal A2 (WP 1747 00). Go to **8.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8.**
8. **Procedure completed.**

WITH GENERATORS APU SWITCH ON, APU GEN ON CAPSULE DOES NOT GO ON AND #1 CONV AND #2 CONV CAPSULES REMAIN ON (PRIMARY AC BUSES NOT ENERGIZED)

SYMPTOM

With GENERATORS APU switch ON, APU GEN ON capsule does not go on and #1 CONV and #2 CONV capsules remain on (primary ac buses not energized).

MALFUNCTION

With GENERATORS APU switch ON, APU GEN ON capsule does not go on and #1 CONV and #2 CONV capsules remain on (primary ac buses not energized).

CORRECTIVE ACTION

1. **Check for 115 vac between:**
   - **J207-A and J207-D**
   - **J207-B and J207-D**
   - **J207-C and J207-D**

   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, go to **25.**
2. **Check helicopter configuration.**
   a. **EMEP** , go to **3.**
   b. **W/O EMEP** , go to **5.**

WITH GENERATORS APU SWITCH ON, APU GEN ON CAPSULE DOES NOT GO ON AND #1 CONV AND #2 CONV CAPSULES REMAIN ON (PRIMARY AC BUSES NOT ENERGIZED) - Continued

3. EMEP **Remove and check pin filtered adapters from P118 and P216 (WP 0925 00).**
   a. If pin filtered adapters are good, go to **4.**
   b. If pin filtered adapters are not good, replace pin filtered adapters (WP 0925 00). Go to **26.**
4. EMEP **Install pin filtered adapters (WP 0925 00). Go to 5.**
5. **Check for 28 vdc between GENERATORS APU switch terminal 2 and ground.**
   a. If voltage is as specified, go to **6.**
   b. If voltage is not as specified, go to **24.**
6. **Check for 28 vdc between GENERATORS APU switch terminal 5 and ground.**
   a. If voltage is as specified, go to **7.**
   b. If voltage is not as specified, go to **23.**
7. **Place GENERATORS APU switch to TEST and check that APU GEN ON capsule goes on.**
   a. If APU GEN ON capsule is on, go to **8.**
   b. If APU GEN ON capsule is not on, go to **21.**
8. **Place GENERATORS APU switch to OFF/RESET and then ON. Go to 9.**
9. **Check for 28 vdc between GENERATORS APU switch terminal 6 and ground.**
   a. If voltage is as specified, go to **10.**
   b. If voltage is not as specified, replace GENERATORS APU switch (WP 0828 00). Go to **26.**
10. **Check for 28 vdc between P205-P and P205-N.**
    a. If voltage is as specified, go to **11.**
    b. If voltage is not as specified, go to **20.**
11. **Check for 28 vdc between P206-C and ground.**
    a. If voltage is as specified, go to **12.**
    b. If voltage is not as specified, replace K3 APU/EXT PWR CNTOR (WP 0840 00). Go to **26.**
12. **Check for 28 vdc between P241-J and ground.**
    a. If voltage is as specified, go to **13.**
    b. If voltage is not as specified, repair/replace wiring between P230-r and P241-J (WP 1747 00). Go to **26.**
13. **Check for 28 vdc between P203-S and ground.**
    a. If voltage is as specified, go to **14.**
    b. If voltage is not as specified, go to **19.**
14. **Check for 28 vdc between P210-U and ground.**
    a. If voltage is as specified, go to **15.**
    b. If voltage is not as specified, go to **18.**
15. **Check for 28 vdc between P206-P and P206-N.**

WITH GENERATORS APU SWITCH ON, APU GEN ON CAPSULE DOES NOT GO ON AND #1 CONV AND #2 CONV CAPSULES REMAIN ON (PRIMARY AC BUSES NOT ENERGIZED) - Continued

a. If voltage is as specified, go to **16.**

b. If voltage is not as specified, go to **17.**

16. Check for 28 vdc between P118-E and ground.

a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00). Go to **26.**

b. If voltage is not as specified, replace K4 AC BUS TIE CNTOR (WP 0844 00). Go to **26.**

17. Check continuity between P206-N and ground.

a. If continuity is present, replace No. 1 generator contactor (WP 0834 00). Go to **26.**

b. If continuity is not present, repair/replace wiring to ground (WP 1747 00). Go to **26.**

18. Check continuity between P210-U and P203-P.

a. If continuity is present, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **26.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **26.**

19. Check continuity between P203-S and P241-P.

a. If continuity is present, replace right relay panel (WP 0825 00). Go to **26.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **26.**

20. Check continuity between P205-N and ground.

a. If continuity is present, repair/replace wiring between P205-P and GENERATORS APU switch terminal 6 (WP 1747 00). Go to **26.**

b. If continuity is not present, repair/replace wiring between P205-N and ground (WP 1747 00). Go to **26.**

21. Check for 28 vdc between GENERATORS APU switch terminal 4 and ground.

a. If voltage is as specified, go to **22.**

b. If voltage is not as specified, replace GENERATORS APU switch (WP 0828 00). Go to **26.**

22. Check for 28 vdc between P118-E and ground.

a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00). Go to **26.**

b. If voltage is not as specified, repair/replace wiring between GENERATORS APU switch terminal 4 and P118-E (WP 1747 00). Go to **26.**

23. Check continuity between GENERATORS APU switch terminal 5 and P216-e.

a. If continuity is present, replace generator control unit (WP 0838 00). Go to **26.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **26.**

24. Check continuity between P230-a and P216-c.

a. If continuity is present, replace generator control unit (WP 0838 00). Go to **26.**

b. If continuity is present, repair/replace wiring (WP 1747 00). Go to **26.**

25. Check continuity between the following for APU GENERATOR:

- **Terminal T1 and J207-A**

WITH GENERATORS APU SWITCH ON, APU GEN ON CAPSULE DOES NOT GO ON AND #1 CONV AND #2 CONV CAPSULES REMAIN ON (PRIMARY AC BUSES NOT ENERGIZED) - Continued

- **Terminal T2 and J207-B**
- **Terminal T3 and J207-C**

a. If continuity is present, troubleshoot generator and GCU using system analyzer. Go to **26.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **26.**

26. Procedure completed.

APU GEN ON CAPSULE IS OFF WITH APU GENERATOR SUPPLYING BUSES (#1 CONV AND #2 CONV CAPSULES OFF)

SYMPTOM

APU GEN ON capsule is off with APU generator supplying buses (#1 CONV and #2 CONV capsules off).

MALFUNCTION

APU GEN ON capsule is off with APU generator supplying buses (#1 CONV and #2 CONV capsules off).

CORRECTIVE ACTION

1. Momentarily hold caution/advisory panel BRT/DIM-TEST switch to TEST and check that APU GEN ON capsule goes on.

a. If APU GEN ON capsule is on, go to **2.**

b. If APU GEN ON capsule is not on, troubleshoot caution/advisory warning system (WP 0090 00). Go to **14.**

2. Check helicopter configuration.

a. **EMEP** , go to **3.**

b. **W/O EMEP** , go to **5.**

3. EMEP Remove and check pin filtered adapter from P118 (WP 0925 00).

a. If pin filtered adapter is good, go to **4.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **14.**

4. EMEP Install pin filtered adapter (WP 0925 00). Go to 5.

5. Check for 28 vdc between P118-E and ground.

a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00). Go to **14.**

b. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between P205-K and ground.

a. If voltage is as specified, go to **7.**

b. If voltage is not as specified, repair/replace wiring between P205-K and J203-T (WP 1747 00). Go to **14.**

7. Check for 28 vdc between P206-C and ground.

a. If voltage is as specified, go to **8.**

b. If voltage is not as specified, go to **13.**

8. Check helicopter configuration.

APU GEN ON CAPSULE IS OFF WITH APU GENERATOR SUPPLYING BUSES (#1 CONV AND #2 CONV CAPSULES OFF) - Continued

a. EH-60A , go to **9.**

b. UH-60A UH-60L , go to **12.**

9. EH-60A Check for 28 vdc between P398-Y and ground.

a. If voltage is as specified, go to **10.**

b. If voltage is not as specified, go to **11.**

10. EH-60A Check continuity between P118-E and P398-Z.

a. If continuity is present, replace No. 3 relay panel (WP 0827 00). Go to **14.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**

11. EH-60A Check continuity between P398-Y and P206-B.

a. If continuity is present, replace K4 AC BUS TIE CNTOR (WP 0844 00). Go to **14.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**

12. UH-60A UH-60L Check continuity between P118-E and P206-B.

a. If continuity is present, replace K4 AC BUS TIE CNTOR (WP 0844 00). Go to **14.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**

13. Check continuity between P206-C and P205-L.

a. If continuity is present, replace K3 APU/EXT PWR CNTOR (WP 0840 00). Go to **14.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**

14. Procedure completed.

APU GENERATOR OR AC EXTERNAL POWER DOES NOT SUPPLY NO. 1 AC PRI BUS (#1 CONV CAPSULE ON)

SYMPTOM

APU generator or ac external power does not supply No. 1 AC PRI bus (#1 CONV capsule on).

MALFUNCTION

APU generator or ac external power does not supply No. 1 AC PRI bus (#1 CONV capsule on).

CORRECTIVE ACTION

1. Check for 115 vac between K1 NO. 1 GEN CNTOR:

- **Terminal A2 and ground**
- **Terminal B2 and ground**
- **Terminal C2 and ground**

a. If voltage is as specified, repair/replace wiring between K1 NO. 1 GEN CNTOR and NO. 1 CONVERTER circuit breaker (WP 1747 00). Go to **4.**

b. If voltage is not as specified, go to **2.**

2. Check for 115 vac between K1 NO. 1 GEN CNTOR:

- **Terminal A3 and ground**
- **Terminal B3 and ground**

APU GENERATOR OR AC EXTERNAL POWER DOES NOT SUPPLY NO. 1 AC PRI BUS (#1 CONV CAPSULE ON) - Continued

- **Terminal C3 and ground**

a. If voltage is as specified, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **4.**

b. If voltage is not as specified, go to **3.**

3. Check for open AC current limiters CL-1, CL-2, and CL-3.

a. If AC current limiters are open, replace ac current limiters as required (WP 0836 00). Go to **4.**

b. If AC current limiters are not open, repair/replace wiring between K1 NO. 1 GEN CNTOR and K2 NO. 2 GEN CNTOR as required (WP 1747 00). Go to **4.**

4. Procedure completed.

AC ESS BUS OFF CAPSULE DOES NOT GO OFF WHEN NO. 1 AND NO. 2 AC PRIMARY BUSES ARE ENERGIZED

SYMPTOM

AC ESS BUS OFF capsule does not go off when No. 1 and No. 2 ac primary buses are energized.

MALFUNCTION

AC ESS BUS OFF capsule does not go off when No. 1 and No. 2 ac primary buses are energized.

CORRECTIVE ACTION

1. Check for 115 vac between K13 AC ESNTL BUS FAIL relay terminals X1 and X2.

a. If voltage is as specified, replace relay (WP 0845 00). Go to **15.**

b. If voltage is not as specified, go to **2.**

2. Check for 115 vac between K13 AC ESNTL BUS FAIL relay terminal X1 and ground.

a. If voltage is as specified, repair/replace wiring between K13 AC ESNTL BUS FAIL relay terminal X2 and ground (WP 1747 00). Go to **15.**

b. If voltage is not as specified, go to **3.**

3. Check for 115 vac between AC ESNTL BUS WARN circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between AC ESNTL BUS WARN circuit breaker terminal 1 and K13 AC ESNTL BUS FAIL relay terminal X1 (WP 1747 00). Go to **15.**

b. If voltage is not as specified, replace AC ESNTL BUS WARN circuit breaker (WP 0824 00). Go to **15.**

c. If trouble remains, go to **4.**

4. Check for 115 vac between K8 AC ESNTL BUS XFER relay terminal A2 and ground.

a. If voltage is as specified, go to **5.**

b. If voltage is not as specified, go to **13.**

5. Pull out AC ESNTL BUS SPLY circuit breaker (copilot's circuit breaker panel). Go to 6.

6. Check for 115 vac between K8 AC ESNTL BUS XFER relay terminal A2 and ground.

AC ESS BUS OFF CAPSULE DOES NOT GO OFF WHEN NO. 1 AND NO. 2 AC PRIMARY BUSES ARE ENERGIZED - Continued

a. If voltage is as specified, go to **7.**

b. If voltage is not as specified, go to **9.**

7. Check for 115 vac between K8 AC ESNTL BUS XFER relay terminal A3 and ground.

a. If voltage is as specified, replace K8 AC ESNTL BUS XFER relay (WP 0845 00). Go to **15.**

b. If voltage is not as specified, go to **8.**

8. Check for 115 vac between AC ESNTL BUS SPLY circuit breaker (pilot's circuit breaker panel) terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between AC ESNTL BUS SPLY circuit breaker terminal 1 and K8 AC ESNTL BUS XFER relay terminal A3 (WP 1747 00). Go to **15.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **15.**

9. Push in AC ESNTL BUS SPLY circuit breaker (copilot's circuit breaker panel). Go to 10.

10. Check for 115 vac between K8 AC ESNTL BUS XFER relay terminal A2 and ground.

a. If voltage is as specified, repair/replace wiring between K8 AC ESNTL BUS XFER relay terminal A2 and AC ESNTL BUS WARN circuit breaker terminal 2 (WP 1747 00). Go to **15.**

b. If voltage is not as specified, go to **11.**

11. Check for 115 vac between K8 AC ESNTL BUS XFER relay terminal A1 and ground.

a. If voltage is as specified, repair/replace wiring between relay terminal X2 and ground (WP 1747 00). Go to **15.**

b. If voltage is not as specified, go to **12.**

12. Check for 115 vac between AC ESNTL BUS SPLY circuit breaker (copilot's circuit breaker panel) terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and K8 AC ESNTL BUS XFER relay terminal A1 (WP 1747 00). Go to **15.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **15.**

13. Check for 115 vac between K8 AC ESNTL BUS XFER relay terminal A1 and ground.

a. If voltage is as specified, repair/replace wiring between relay terminal X2 and ground (WP 1747 00). Go to **15.**

b. If voltage is not as specified, go to **14.**

14. Check for 115 vac between AC ESNTL BUS SPLY circuit breaker (copilot's circuit breaker panel) terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and K8 AC ESNTL BUS XFER relay terminal A1 (WP 1747 00). Go to **15.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **15.**

15. Procedure completed.

#1 GEN BRG AND/OR #2 GEN BRG CAPSULE GOES ON

SYMPTOM

#1 GEN BRG and/or #2 GEN BRG capsule goes on.

MALFUNCTION

#1 GEN BRG and/or #2 GEN BRG capsule goes on.

CORRECTIVE ACTION

1. **Are both #1 GEN BRG and #2 GEN BRG capsules on?**
   a. If both capsules are on, troubleshoot both #1 GEN BRG and #2 GEN BRG capsule circuits (WP 0090 00). Go to **17.**
   b. If both capsules are not on, go to **2.**
2. **Is #1 GEN BRG capsule on?**
   a. If capsule is on, go to **3.**
   b. If capsule is not on, go to **10.**
3. **Did #1 GEN BRG capsule go on steady for more than 1 minute?**
   a. If capsule was steady, go to **4.**
   b. If capsule was not steady, no maintenance required. Go to **17.**
4. **Check helicopter configuration.**
   a. **EMEP** , go to **5.**
   b. **W/O EMEP** , go to **7.**
5. **EMEP Remove and check pin filtered adapter from P117 (WP 0925 00).**
   a. If pin filtered adapter is good, go to **6.**
   b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **17.**
6. **EMEP Install pin filtered adapter (WP 0925 00). Go to 7.**
7. **Pull out, then push in NO. 1 GEN WARN circuit breaker. Go to 8.**
8. **Did #1 GEN BRG capsule go off and remain off?**
   a. If capsule did go off and remain off, no maintenance required. Go to **17.**
   b. If capsule did not go off and remain off, go to **9.**
9. **Check for less than 50 ohms between P117-18 and ground.**
   a. If resistance is as specified, replace No. 1 generator (WP 0839 00). Go to **17.**
   b. If resistance is not as specified, replace caution/advisory panel (WP 0785 00). Go to **17.**
10. **Did #2 BRG capsule go on steady for more than 1 minute?**
    a. If capsule was steady, go to **11.**
    b. If capsule was not steady, no maintenance required. Go to **17.**
11. **Check helicopter configuration.**
    a. **EMEP** , go to **12.**

#1 GEN BRG AND/OR #2 GEN BRG CAPSULE GOES ON - Continued

b. **W/O EMEP** , go to **14.**

12. **EMEP Remove and check pin filtered adapter from P117 (WP 0925 00).**

    a. If pin filtered adapter is good, go to **13.**

    b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **17.**

13. **EMEP Install pin filtered adapter (WP 0925 00). Go to 14.**

14. **Pull out then push in NO. 2 GEN WARN circuit breaker. Go to 15.**

15. **Did #2 GEN BRG capsule go off and remain off?**

    a. If capsule went off and remained off, no maintenance required. Go to **17.**

    b. If capsule did not go off and remain off, go to **16.**

16. **Check for less than 50 ohms between P117-34 and ground.**

    a. If resistance is as specified, replace No. 2 generator (WP 0839 00). Go to **17.**

    b. If resistance is not as specified, replace caution/advisory panel (WP 0785 00). Go to **17.**

17. **Procedure completed.**

#1 GEN CAPSULE DOES NOT GO ON WITH NO. 1 GENERATOR OFF

SYMPTOM

#1 GEN capsule does not go on with No. 1 generator off.

MALFUNCTION

#1 GEN capsule does not go on with No. 1 generator off.

CORRECTIVE ACTION

1. **Momentarily hold caution/advisory panel BRT/DIM-TEST switch to TEST and check that #1 GEN capsule goes on.**

    a. If #1 GEN capsule goes on, go to **2.**

    b. If #1 GEN capsule does not go on, troubleshoot caution/advisory warning system (WP 0090 00). Go to **12.**

2. **Check helicopter configuration.**

    a. **EMEP** , go to **3.**

    b. **W/O EMEP** , go to **5.**

3. **EMEP Remove and check pin filtered adapter from P117 (WP 0925 00).**

    a. If pin filtered adapter is good, go to **4.**

    b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **12.**

4. **EMEP Install pin filtered adapter (WP 0925 00). Go to 5.**

5. **Check for 28 vdc between P117-17 and ground.**

    a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00). Go to **12.**

    b. If voltage is not as specified, go to **6.**

#1 GEN CAPSULE DOES NOT GO ON WITH NO. 1 GENERATOR OFF - Continued

6. **Check for 28 vdc between P212-B and ground.**

   a. If voltage is as specified, go to **7.**

   b. If voltage is not as specified, go to **8.**

7. **Check continuity between P117-17 and P212-A.**

   a. If continuity is present, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **12.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12.**

8. **Check continuity between P212-B and P215-Z.**

   a. If continuity is present, go to **9.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12.**

9. **Check continuity between P212-A and P215-f.**

   a. If continuity is present, go to **10.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12.**

10. **Check for 28 vdc between P215-Y and ground.**

    a. If voltage is as specified, replace No. 1 generator control unit (WP 0838 00). Go to **12.**

    b. If voltage is not as specified, go to **11.**

11. **Check for 28 vdc between NO. 1 GEN WARN circuit breaker terminal 1 and ground.**

    a. If voltage is as specified, repair/replace wiring between circuit breaker and P215-Y (WP 1747 00). Go to **12.**

    b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **12.**

12. **Procedure completed.**

#1 GEN CAPSULE IS ON WITH GENERATORS NO. 1 SWITCH ON

SYMPTOM

#1 GEN capsule is on with GENERATORS NO. 1 switch ON.

MALFUNCTION

#1 GEN capsule is on with GENERATORS NO. 1 switch ON.

CORRECTIVE ACTION

1. **Shut down No. 1 engine and APU (TM 1-1520-237-10). Go to 2.**

2. **Inspect connector P251-4 and P251-5 for scorched sockets at the front and back ends of the connector.**

   a. If pins are sorched, clean or replace connector as required. Go to **23.**

   b. If pins are not sorched, go to **3.**

3. **With qualified personnel at the controls, start No. 1 engine and operate with main rotor turning at 100% (TM 1-1520-237-10). Go to 4.**

4. **Check for 115 vac between:**

   - **J211-A and J211-D**

#1 GEN CAPSULE IS ON WITH GENERATORS NO. 1 SWITCH ON - Continued

- **J211-B and J211-D**
- **J211-C and J211-D**

a. If voltage is as specified, go to **5.**

b. If voltage is not as specified, go to **22.**

5. **Place and hold GENERATORS NO. 1 switch to TEST. Go to 6.**
6. **Check that NO. 1 GEN capsule goes off.**

a. If NO. 1 GEN capsule goes off, go to **7.**

b. If NO. 1 GEN capsule does not go off, go to **11.**

7. **Place GENERATORS NO. 1 switch to ON. Go to 8.**
8. **Check for 28 vdc between P212-P and P212-N.**

a. If voltage is as specified, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **23.**

b. If voltage is not as specified, go to **9.**

9. **Check for 28 vdc between P212-P and ground.**

a. If voltage is as specified, repair/replace wiring between P212-N and ground (WP 1747 00). Go to **23.**

b. If voltage is not as specified, go to **10.**

10. **Check continuity between P212-P and GENERATORS NO. 1 switch terminal 6.**

a. If continuity is present, replace switch (WP 0828 00). Go to **23.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **23.**

11. **Check helicopter configuration.**

a. **EMEP** , go to **12.**

b. **W/O EMEP** , go to **14.**

12. **EMEP Remove and check pin filtered adapter from P118 (WP 0925 00).**

a. If pin filtered adapter is good, go to **13.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **23.**

13. **EMEP Install pin filtered adapter (WP 0925 00). Go to 14.**
14. **Check for 28 vdc between GENERATORS NO. 1 switch terminal 5 and ground.**

a. If voltage is as specified, go to **15.**

b. If voltage is not as specified, go to **17.**

15. **Check for 28 vdc between GENERATORS NO. 1 switch terminal 4 and ground.**

a. If voltage is as specified, go to **16.**

b. If voltage is not as specified, replace switch (WP 0828 00). Go to **23.**

16. **Check continuity between GENERATORS NO. 1 switch terminal 4 and P215-d.**

a. If continuity is present, replace No. 1 generator control unit (WP 0838 00). Go to **23.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **23.**

#1 GEN CAPSULE IS ON WITH GENERATORS NO. 1 SWITCH ON - Continued

17. **Check continuity between GENERATORS NO. 1 switch terminal 5 and P215-e.**
    a. If continuity is present, go to **18.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **23.**
18. **Check for 28 vdc between GENERATORS NO. 1 switch terminal 1 and ground.**
    a. If voltage is as specified, go to **19.**
    b. If voltage is not as specified, go to **21.**
19. **Check for 28 vdc between GENERATORS NO. 1 switch terminal 1 and ground.**
    a. If voltage is as specified, go to **20.**
    b. If voltage is not as specified, replace switch (WP 0828 00). Go to **23.**
20. **Check continuity between GENERATORS NO. 1 switch terminal 1 and P215-a.**
    a. If continuity is present, replace No. 1 generator control unit (WP 0838 00). Go to **23.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **23.**
21. **Check continuity between GENERATORS NO. 1 switch terminal 2 and P215-c.**
    a. If continuity is present, replace No. 1 generator control unit (WP 0838 00). Go to **23.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **23.**
22. **Check continuity between No. 1 generator:**
    - **Terminal T1 and J211-A**
    - **Terminal T2 and J211-B**
    - **Terminal T3 and J211-C**

    a. If continuity is present, do GENERATOR AND GCU, in this work package, using system analyzer. Go to **23.**
    b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **23.**
23. **Procedure completed.**

#1 CONV CAPSULE STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 1 AC PRIMARY BUS

SYMPTOM

#1 CONV capsule stays on when either generator is supplying power to No. 1 ac primary bus.

MALFUNCTION

#1 CONV capsule stays on when either generator is supplying power to No. 1 ac primary bus.

CORRECTIVE ACTION

1. **Check for 115 vac between NO. 1 CONVERTER circuit breaker:**
    - **Terminal A2 and ground**
    - **Terminal B2 and ground**
    - **Terminal C2 and ground**

#1 CONV CAPSULE STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 1 AC PRIMARY BUS - Continued

a. If voltage is as specified, troubleshoot DC electrical system (WP 0102 00). Go to **4.**

b. If voltage is not as specified, go to **2.**

2. **Check for 115 vac between NO. 1 CONVERTER circuit breaker:**
   - **Terminal A1 and ground**
   - **Terminal B1 and ground**
   - **Terminal C1 and ground**

a. If voltage is as specified, replace circuit breaker (WP 0824 00). Go to **4.**

b. If voltage is not as specified, go to **3.**

3. **Check for 115 vac between K1 NO. 1 GEN CNTOR:**
   - **Terminal A2 and ground**
   - **Terminal B2 and ground**
   - **Terminal C2 and ground**

a. If voltage is as specified, repair/replace wiring between K1 NO. 1 GEN CNTOR and NO. 1 CONVERTER circuit breaker (WP 1747 00). Go to **4.**

b. If voltage is not as specified, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **4.**

4. **Procedure completed.**

#2 CONV CAPSULE STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 2 AC PRIMARY BUS

SYMPTOM

#2 CONV capsule stays on when either generator is supplying power to No. 2 ac primary bus.

MALFUNCTION

#2 CONV capsule stays on when either generator is supplying power to No. 2 ac primary bus.

CORRECTIVE ACTION

1. **Check for 115 vac between NO. 2 CONVERTER circuit breaker:**
   - **Terminal A1 and ground**
   - **Terminal B1 and ground**
   - **Terminal C1 and ground**

a. If voltage is as specified, troubleshoot DC electrical system (WP 0102 00). Go to **6.**

b. If voltage is not as specified, go to **2.**

2. **Check for 115 vac between NO. 2 CONVERTER circuit breaker:**
   - **Terminal A2 and ground**
   - **Terminal B2 and ground**
   - **Terminal C2 and ground**

a. If voltage is as specified, replace circuit breaker (WP 0824 00). Go to **6.**

#2 CONV CAPSULE STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 2 AC PRIMARY BUS - Continued

b. If voltage is not as specified, go to **3.**

3. Check for 115 vac between K2 NO. 2 GEN CNTOR:

- **Terminal A2 and ground**
- **Terminal B2 and ground**
- **Terminal C2 and ground**

a. If voltage is as specified, repair/replace wiring between K2 NO. 2 GEN CNTOR and NO. 2 CONVERTER circuit breaker (WP 1747 00). Go to **6.**

b. If voltage is not as specified, go to **4.**

4. Check for 115 vac between K1 NO. 1 GEN CNTOR:

- **Terminal A2 and ground**
- **Terminal B2 and ground**
- **Terminal C2 and ground**

a. If voltage is as specified, go to **5.**

b. If voltage is not as specified, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **6.**

5. Check for open current limiters CL4, CL5, and CL6.

a. If limiters are open, replace open AC current limiters as required (WP 0836 00). Go to **6.**

b. If limiters are not open, repair/replace wiring between K1 NO. 1 GEN CNTOR and K2 NO. 2 GEN CNTOR (WP 1747 00).

c. If trouble remains, replace K4 AC BUS TIE CNTOR (WP 0844 00). Go to **6.**

6. Procedure completed.

NO. 1 AND/OR NO. 2 GENERATOR DISABLED BY UNDERFREQUENCY CONDITION DURING AUTO ROTATIVE DESCENT

SYMPTOM

No. 1 and/or No. 2 generator disabled by underfrequency condition during auto rotative descent.

MALFUNCTION

No. 1 and/or No. 2 generator disabled by underfrequency condition during auto rotative descent.

CORRECTIVE ACTION

1. Are both generators disabled?

a. If both are disabled, go to **2**

b. If both are not disabled, go to **5**

2. Check helicopter configuration.

a. **MOD** , go to **3.**

b. **W/O MOD** , go to **4.**

3. W/O MOD Is left drag beam switch in proper adjustment?

a. If switch is in proper adjustment, replace switch (WP 0443 00). Go to **20.**

NO. 1 AND/OR NO. 2 GENERATOR DISABLED BY UNDERFREQUENCY CONDITION DURING AUTO ROTATIVE DESCENT - Continued

b. If switch is not in proper adjustment, adjust switch (WP 0443 00). Go to **20.**

4. **MOD Is right drag beam switch in proper adjustment?**

   a. If switch is in proper adjustment, replace switch (WP 0443 00). Go to **20.**

   b. If switch is not in proper adjustment, adjust switch (WP 0443 00). Go to **20.**

5. **Check helicopter configuration.**

   a. **EMEP** , go to **6.**

   b. **W/O EMEP** , go to **13.**

6. **EMEP Remove and check pin filtered adapter from P214 (WP 0925 00).**

   a. If pin filtered adapter is good, go to **7.**

   b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **20.**

7. **EMEP Install pin filtered adapter on P214 (WP 0925 00). Go to 8.**

8. **EMEP Is No. 1 generator disabled?**

   a. If No. 1 generator is disabled, go to **9.**

   b. If No. 1 generator is not disabled, go to **12.**

9. **EMEP Remove and check pin filtered adapter from P215 (WP 0925 00).**

   a. If pin filtered adapter is good, go to **10.**

   b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **20.**

10. **EMEP Install pin filtered adapter on P215 (WP 0925 00). Go to 11.**

11. **EMEP Check continuity between P163-F and P215-F.**

    a. If continuity is present, replace No. 1 generator control unit (WP 0838 00). Go to **20.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20.**

12. **EMEP Check continuity between P163-F and P214-F.**

    a. If continuity is present, replace No. 2 generator control unit (WP 0838 00). Go to **20.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20.**

13. **W/O EMEP Is No. 1 generator disabled?**

    a. If generator is disabled, go to **14.**

    b. If generator is not disabled, go to **17.**

14. **W/O EMEP Check helicopter configuration.**

    a. **W/O MOD** , go to **15.**

    b. **MOD** , go to **16.**

15. **W/O EMEP MOD Check continuity between J163-F and P215-F.**

    a. If continuity is present, replace No. 1 generator control unit (WP 0838 00). Go to **20.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20.**

16. **W/O EMEP MOD Check continuity between J164-C and P215-F.**

    a. If continuity is present, replace No. 1 generator control unit (WP 0838 00). Go to **20.**

NO. 1 AND/OR NO. 2 GENERATOR DISABLED BY UNDERFREQUENCY CONDITION DURING AUTO ROTATIVE DESCENT - Continued

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20.**

17. **W/O EMEP Check helicopter configuration.**

a. MOD , go to **18.**

b. W/O MOD , go to **19.**

18. **W/O EMEP MOD Check continuity between J164-C and P214-F.**

a. If continuity is present, replace No. 2 generator control unit (WP 0838 00). Go to **20.**

b. If continuity is not present, repair/replace wiring (WP 0838 00). Go to **20.**

19. **W/O EMEP W/O MOD Check continuity between J164-C and P214-F.**

a. If continuity is present, replace No. 2 generator control unit (WP 0838 00). Go to **20.**

b. If continuity is not present, repair/replace wiring (WP 0838 00). Go to **20.**

20. **Procedure completed.**

#2 GEN CAPSULE IS OFF WITH NO. 2 GENERATOR OFF

SYMPTOM

#2 GEN capsule is off with No. 2 generator off.

MALFUNCTION

#2 GEN capsule is off with No. 2 generator off.

CORRECTIVE ACTION

1. **Check helicopter configuration.**

a. EMEP , go to **2.**

b. W/O EMEP , go to **4.**

2. **EMEP Remove and check pin filtered adapters from P118 and P214 (WP 0925 00).**

a. If pin filtered adapter is good, go to **3.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **13.**

3. **EMEP Install pin filtered adapters (WP 0925 00). Go to 4.**

4. **Momentarily hold caution/advisory panel BRT/DIM-TEST switch to TEST and check that #2 GEN capsule goes on.**

a. If capsule goes on, go to **5.**

b. If capsule does not go on, troubleshoot caution/advisory warning system (WP 0090 00). Go to **13.**

5. **Check for 28 vdc between P117-33 and ground.**

a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00). Go to **13.**

b. If voltage is not as specified, go to **6.**

6. **Check for 28 vdc between P204-B and ground.**

a. If voltage is as specified, go to **7.**

#2 GEN CAPSULE IS OFF WITH NO. 2 GENERATOR OFF - Continued

b. If voltage is not as specified, go to **8.**

7. **Check continuity between P204-A and P117-33.**

   a. If continuity is present, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **13.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13.**

8. **Check continuity between P204-B and P214-Z.**

   a. If continuity is present, go to **9.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13.**

9. **Check continuity between P204-A and P214-f.**

   a. If continuity is present, go to **10.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13.**

10. **Check for 28 vdc between P214-Y and ground.**

    a. If voltage is as specified, replace No. 2 generator control unit (WP 0838 00). Go to **13.**

    b. If voltage is not as specified, go to **11.**

11. **Check for 28 vdc between P117-66 and ground.**

    a. If voltage is as specified, repair/replace wiring between NO. 2 GEN WARN circuit breaker and P214-Y (WP 1747 00). Go to **13.**

    b. If voltage is not as specified, go to **12.**

12. **Check for 28 vdc between NO. 2 GEN WARN circuit breaker terminal 1 and ground.**

    a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **13.**

    b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **13.**

13. **Procedure completed.**

#2 GEN CAPSULE IS ON WITH GENERATORS NO. 2 SWITCH ON

SYMPTOM

#2 GEN capsule is on with GENERATORS NO. 2 switch ON.

MALFUNCTION

#2 GEN capsule is on with GENERATORS NO. 2 switch ON.

CORRECTIVE ACTION

1. **Shut down No. 1 engine and APU (TM 1-1520-237-10). Go to 2.**

2. **Inspect connector P250-4 and P250-5 for scorched sockets at the front and back ends of the connector.**

   a. If pins are sorched, clean or replace connector as required. Go to **22.**

   b. If pins are not sorched, go to **3.**

3. **With qualified personnel at the controls, start No. 1 engine and operate with main rotor turning at 100% (TM 1-1520-237-10). Go to 4.**

4. **Check for 115 vac between:**

#2 GEN CAPSULE IS ON WITH GENERATORS NO. 2 SWITCH ON - Continued

- **J208-A and J208-D**
- **J208-B and J208-D**
- **J208-C and J208-D**

a. If voltage is as specified, go to **5.**

b. If voltage is not as specified, go to **21.**

5. **Place and hold GENERATORS NO. 2 switch to TEST and check that NO. 2 GEN capsule goes off.**

a. If NO. 2 GEN capsule goes off, go to **6.**

b. If NO. 2 GEN capsule does not go off, go to **10.**

6. **Place GENERATORS NO. 2 switch to ON. Go to 7.**

7. **Check for 28 vdc between P204-P and P204-N.**

a. If voltage is as specified, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **22.**

b. If voltage is not as specified, go to **8.**

8. **Check for 28 vdc between P204-P and ground.**

a. If voltage is as specified, repair/replace wiring between P204-N and ground (WP 1747 00). Go to **22.**

b. If voltage is not as specified, go to **9.**

9. **Check continuity between P204-P and GENERATORS NO. 2 switch terminal 6.**

a. If continuity is present, replace switch (WP 0828 00). Go to **22.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **22.**

10. **Check helicopter configuration.**

a. **EMEP** , go to **11.**

b. **W/O EMEP** , go to **13.**

11. **EMEP Remove and check pin filtered adapter from P214 (WP 0925 00).**

a. If pin filtered adapter is good, go to **12.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **22.**

12. **EMEP Install pin filtered adapter (WP 0925 00). Go to 13.**

13. **Check for 28 vdc between GENERATORS NO. 2 switch terminal 5 and ground.**

a. If voltage is as specified, go to **14.**

b. If voltage is not as specified, go to **16.**

14. **Check for 28 vdc between GENERATORS NO. 2 switch terminal 4 and ground.**

a. If voltage is as specified, go to **15.**

b. If voltage is not as specified, replace switch (WP 0828 00). Go to **22.**

15. **Check continuity between GENERATORS NO. 2 switch terminal 4 and P214-d.**

a. If continuity is present, replace switch (WP 0828 00). Go to **22.**

#2 GEN CAPSULE IS ON WITH GENERATORS NO. 2 SWITCH ON - Continued

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **22.**

16. Check continuity between GENERATORS NO. 2 switch terminal 5 and P214-e.

a. If continuity is present, go to **17.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **22.**

17. Check for 28 vdc between GENERATORS NO. 2 switch terminal 2 and ground.

a. If voltage is as specified, go to **18.**

b. If voltage is not as specified, go to **20.**

18. Check for 28 vdc between GENERATORS NO. 2 switch terminal 1 and ground.

a. If voltage is as specified, go to **19.**

b. If voltage is not as specified, replace switch (WP 0828 00). Go to **22.**

19. Check continuity between GENERATORS NO. 2 switch terminal 1 and P214-a.

a. If continuity is present, replace No. 2 generator control unit (WP 0838 00). Go to **22.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **22.**

20. Check continuity between switch terminal 2 and P214-c.

a. If continuity is present, replace No. 2 generator control unit (WP 0838 00). Go to **22.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **22.**

21. Check continuity between No. 2 generator:

- **Terminal T1 and J208-A**
- **Terminal T2 and J208-B**
- **Terminal T3 and J208-C**

a. If continuity is present, do GENERATOR AND GCU TROUBLESHOOTING, in this work package using system analyzer. Go to **22.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **22.**

22. Procedure completed.

EXT PWR CONNECTED CAPSULE IS OFF WITH A SOURCE OF EXTERNAL POWER CONNECTED

SYMPTOM

EXT PWR CONNECTED capsule is off with a source of external power connected.

MALFUNCTION

EXT PWR CONNECTED capsule is off with a source of external power connected.

CORRECTIVE ACTION

1. Check helicopter configuration.

a. **EMEP** , go to **2.**

b. **W/O EMEP** , go to **4.**

2. EMEP Remove and check pin filtered adapter from P118 (WP 0925 00).

EXT PWR CONNECTED CAPSULE IS OFF WITH A SOURCE OF EXTERNAL POWER CONNECTED - Continued

a. If pin filtered adapter is good, go to **3.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9.**

3. **EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**

4. **Momentarily hold caution/advisory panel BRT/DIM-TEST switch to TEST and check that EXT PWR CONNECTED capsule goes on.**

a. If EXT PWR CONNECTED capsule goes on, go to **5.**

b. If EXT PWR CONNECTED capsule does not go on, troubleshoot caution/advisory warning system (WP 0090 00). Go to **9.**

5. **Check for 28 vdc between J136-F and J136-N.**

a. If voltage is as specified, go to **6.**

b. If voltage is not as specified, go to **7.**

6. **Check for 28 vdc between P118-N and ground.**

a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00). Go to **9.**

b. If voltage is not as specified, repair/replace wiring between J136-E and P118-N (WP 1747 00). Go to **9.**

7. **Check continuity between J136-N and ground.**

a. If continuity is present, go to **8.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

8. **Check for 28 vdc between BATT & ESNTL DC WARN EXT PWR CONTR circuit breaker terminal 1 and ground.**

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J136-F (WP 1747 00). Go to **9.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9.**

9. **Procedure completed.**

EXTERNAL POWER DOES NOT SUPPLY AC BUSES (#1 CONV AND #2 CONV CAPSULES ON)

SYMPTOM

External power does not supply ac buses (#1 CONV and #2 CONV capsules on).

MALFUNCTION

External power does not supply ac buses (#1 CONV and #2 CONV capsules on).

CORRECTIVE ACTION

1. **Check for 115 vac between K2 NO. 2 GEN CNTOR:**
   - **Terminal A3 and ground**
   - **Terminal B3 and ground**
   - **Terminal C3 and ground**

a. If voltage is as specified, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **18.**

EXTERNAL POWER DOES NOT SUPPLY AC BUSES (#1 CONV AND #2 CONV CAPSULES ON) - Continued

b. If voltage is not as specified, go to **2.**

2. Check for 115 vac between K4 AC BUS TIE CNTOR:

- **Terminal A2 and ground**
- **Terminal B2 and ground**
- **Terminal C3 and ground**

a. If voltage is as specified, repair/replace wiring between K2 NO. 2 GEN CNTOR and K4 AC BUS TIE CNTOR as required (WP 1747 00). Go to **18.**

b. If voltage is not as specified, go to **3.**

3. Check for 115 vac between K4 AC BUS TIE CNTOR:

- **Terminal A1 and ground**
- **Terminal B1 and ground**
- **Terminal C1 and ground**

a. If voltage is as specified, go to **4.**

b. If voltage is not as specified, go to **16.**

4. Check for 28 vdc between:

- **P206-P and ground**
- **P206-N and ground**

a. If voltage is as specified, replace K4 AC BUS TIE CNTOR (WP 0844 00). Go to **18.**

b. If voltage is not as specified, go to **5.**

5. Check for 28 vdc between P206-P and ground.

a. If voltage is as specified, repair/replace wiring between P206-N and ground (WP 1747 00). Go to **18.**

b. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between P212-L and ground.

a. If voltage is as specified, go to **7.**

b. If voltage is not as specified, go to **8.**

7. Check continuity between P212-M and P206-P.

a. If continuity is present, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **18.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

8. Check for 28 vdc between P204-M and ground.

a. If voltage is as specified, go to **9.**

b. If voltage is not as specified, go to **10.**

9. Check continuity between P204-L and P212-L.

a. If continuity is present, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **18.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

EXTERNAL POWER DOES NOT SUPPLY AC BUSES (#1 CONV AND #2 CONV CAPSULES ON) - Continued

10. Check for 28 vdc between P241-H and ground.

a. If voltage is as specified, go to **11.**

b. If voltage is not as specified, go to **12.**

11. Check continuity between P241-P and P204-M.

a. If continuity is present, replace right relay panel (WP 0825 00). Go to **18.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

12. Check for 28 vdc between EXT PWR switch terminal 2 and ground.

a. If voltage is as specified, repair/replace wiring between switch terminal 2 and P241-H (WP 1747 00). Go to **18.**

b. If voltage is not as specified, go to **13.**

13. Check for 28 vdc between EXT PWR switch terminal 3 and ground.

a. If voltage is as specified, replace switch (WP 0828 00). Go to **18.**

b. If voltage is not as specified, go to **14.**

14. Check continuity between EXT PWR switch terminal 3 and P213-5.

a. If continuity is present, go to **15.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

15. Check continuity between K3 APU/EXT PWR CNTOR:

- **Terminal A3 and P213-3**
- **Terminal B3 and P213-2**
- **Terminal C3 and P213-1**

a. If continuity is present, replace external power monitor panel (WP 0843 00). Go to **18.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **18.**

16. Check for 115 vac between K3 APU/EXT PWR CNTOR:

- **Terminal A2 and ground**
- **Terminal B2 and ground**
- **Terminal C2 and ground**

a. If voltage is as specified, repair/replace wiring between K3 APU/EXT PWR CNTOR and K4 AC BUS TIE CNTOR as required (WP 1747 00). Go to **18.**

b. If voltage is not as specified, go to **17.**

17. Check for 115 vac between K3 APU/EXT PWR CNTOR:

- **Terminal A3 and ground**
- **Terminal B3 and ground**
- **Terminal C3 and ground**

a. If voltage is as specified, replace K3 APU/EXT PWR CNTOR (WP 0840 00). Go to **18.**

EXTERNAL POWER DOES NOT SUPPLY AC BUSES (#1 CONV AND #2 CONV CAPSULES ON) - Continued

b. If voltage is not as specified, repair/replace wiring between K3 APU/EXT PWR CNTOR and external receptacle as required (WP 1747 00). Go to **18.**

18. Procedure completed.

NO POWER AT MISSION INTERFACE PANEL WITH EXTERNAL POWER APPLIED

SYMPTOM

No power at mission interface panel with external power applied.

MALFUNCTION

No power at mission interface panel with external power applied.

CORRECTIVE ACTION

1. Check continuity between J511-H and ground.

a. If continuity is present, go to **2.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13.**

2. Check for 115 vac between K11 AC SEC BUS CNTOR:

- **Terminal A2 and ground**
- **Terminal B2 and ground**
- **Terminal C2 and ground**

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, go to **4.**

3. Check for open secondary bus current limiters CL-16, CL-17, and CL-18.

a. If secondary bus current limiters are open, replace secondary bus current limiters as required (WP 0846 00). Go to **13.**

b. If secondary bus current limiters are not open, repair/replace wiring between K11 AC SEC BUS CNTOR and J511 as required (WP 1747 00). Go to **13.**

4. Check for 115 vac between K11 AC SEC BUS CNTOR:

- **Terminal A1 and ground**
- **Terminal B1 and ground**
- **Terminal C1 and ground**

a. If voltage is as specified, go to **5.**

b. If voltage is not as specified, repair/replace wiring between K1 NO. 1 GEN CNTOR and K11 AC SEC BUS CNTOR as required (WP 1747 00). Go to **13.**

5. Check for 28 vdc between K11 AC SEC BUS CNTOR terminals X1 and COM.

a. If voltage is as specified, replace K11 AC SEC BUS CNTOR (WP 0847 00). Go to **13.**

b. If voltage is not as specified, go to **6.**

6. Check continuity between K11 AC SEC BUS CNTOR terminal COM and ground.

a. If continuity is present, go to **7.**

NO POWER AT MISSION INTERFACE PANEL WITH EXTERNAL POWER APPLIED - Continued

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13.**

7. Check for 28 vdc between K80 BACK-UP PUMP INTERLK relay terminal A3 and ground.

a. If voltage is as specified, repair/replace wiring between K80 BACK-UP PUMP INTERLK relay terminal A3 and K11 AC SEC BUS CNTOR terminal X1 (WP 1747 00). Go to **13.**

b. If voltage is not as specified, go to **8.**

8. Check for 28 vdc between K80 BACK-UP PUMP INTERLK relay terminal A2 and ground.

a. If voltage is as specified, replace relay (WP 0851 00). Go to **13.**

b. If voltage is not as specified, go to **9.**

9. Check for 28 vdc between diode CR19 terminal E5 and ground.

a. If voltage is as specified, repair/replace wiring between diode CR19 terminal E5 and K80 BACK-UP PUMP INTERLK relay terminal A2 (WP 1747 00). Go to **13.**

b. If voltage is not as specified, go to **10.**

10. Check for 28 vdc between diode CR19 terminal E6 and ground.

a. If voltage is as specified, replace diode (WP 0849 00). Go to **13.**

b. If voltage is not as specified, go to **11.**

11. Check for 28 vdc between K82 SEC BUS GND PWR CONT RLY terminal B3 and ground.

a. If voltage is as specified, repair/replace wiring between relay terminal B3 and diode CR19 terminal E6 (WP 1747 00). Go to **13.**

b. If voltage is not as specified, go to **12.**

12. Check for 28 vdc between K82 SEC BUS GND PWR CONT RLY terminal B2 and ground.

a. If voltage is as specified, replace relay (WP 0851 00). Go to **13.**

b. If voltage is not as specified, repair/replace wiring between relay terminal B2 and P212-M (WP 1747 00). Go to **13.**

c. If trouble remains, go to AC ESS BUS OFF CAPSULE DOES NOT GO ON WITHOUT POWER ON AC ESNTL BUS and WITH GENERATORS APU SWITCH ON, APU GEN ON CAPSULE DOES NOT GO ON AND #1 CONV AND #2 CONV CAPSULES REMAIN ON (PRIMARY AC BUSES NOT ENERGIZED), in this work package.

13. Procedure completed.

115 VAC AT MISSION INTERFACE PANEL WITH EXTERNAL POWER CONNECTED AND BACK-UP PUMP RUNNING

SYMPTOM

115 vac at mission interface panel with external power connected and back-up pump running.

MALFUNCTION

115 vac at mission interface panel with external power connected and back-up pump running.

CORRECTIVE ACTION

1. **Check for 28 vdc between K80 BACK-UP PUMP INTERLK relay terminals X1 and X2.**
   a. If voltage is as specified, replace relay (WP 0851 00). Go to **5.**
   b. If voltage is not as specified, go to **2.**
2. **Check for 28 vdc between K80 BACK-UP PUMP INTERLK relay terminal X1 and ground.**
   a. If voltage is as specified, repair/replace wiring between relay terminal X2 and ground (WP 1747 00). Go to **5.**
   b. If voltage is not as specified, go to **3.**
3. **Check for 28 vdc between K19 NO. 3 HYD PWR PUMP relay terminal X1 and ground.**
   a. If voltage is as specified, repair/replace wiring between K80 BACK-UP PUMP INTERLK relay terminal X1 and K19 NO. 3 HYD PWR PUMP relay (WP 1747 00). Go to **5.**
   b. If voltage is not as specified, go to **4.**
4. **Check continuity between K19 NO. 3 HYD PWR PUMP relay terminal X1 and P244-A.**
   a. If continuity is present, replace right relay panel (WP 0825 00). Go to **5.**
   b. If trouble remains, troubleshoot hydraulic system (WP 0086 00). Go to **5.**
   c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5.**
5. **Procedure completed.**

NO POWER AT MISSION INTERFACE PANEL WITH NO. 1 OR NO. 2 GENERATOR ON

SYMPTOM

No power at mission interface panel with No. 1 or No. 2 generator on.

MALFUNCTION

No power at mission interface panel with No. 1 or No. 2 generator on.

CORRECTIVE ACTION

1. **Check for 115 vac between K11 AC SEC BUS CNTOR:**
   - **Terminal A2 and ground**
   - **Terminal B2 and ground**

NO POWER AT MISSION INTERFACE PANEL WITH NO. 1 OR NO. 2 GENERATOR ON - Continued

- **Terminal C2 and ground**

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **3.**

2. Check for open secondary bus current limiters CL-16, CL-17, and CL-18.

a. If secondary bus current limiters are open, replace current limiter as required (WP 0846 00). Go to **12.**

b. If secondary bus current limiters are not open, repair/replace wiring between the following (WP 1747 00), then go to **12:**

- Terminal A2 and J511-A
- Terminal B2 and J511-B
- Terminal C2 and J511-C

3. Check for 115 vac between K11 AC SEC BUS CNTOR:

- **Terminal A1 and ground**
- **Terminal B1 and ground**
- **Terminal C1 and ground**

a. If voltage is as specified, go to **4.**

b. If voltage is not as specified, repair/replace wiring between K11 AC SEC BUS CNTOR and K1 NO. 1 and K2 NO. 2 GEN CNTOR as required (WP 1747 00). Go to **12.**

4. Check for 28 vdc between K11 AC SEC BUS CNTOR terminals X1 and COM.

a. If voltage is as specified, replace K11 AC SEC BUS CNTOR (WP 0847 00). Go to **12.**

b. If voltage is not as specified, go to **5.**

5. Check for 28 vdc between K80 BACK-UP PUMP INTERLK relay terminal A3 and ground.

a. If voltage is as specified, repair/replace wiring between K80 BACK-UP PUMP INTERLK relay terminal A3 and K11 AC SEC BUS CNTOR terminal X1 (WP 1747 00). Go to **12.**

b. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between K80 BACK-UP PUMP INTERLK relay terminal A2 and ground.

a. If voltage is as specified, replace relay (WP 0851 00). Go to **12.**

b. If voltage is not as specified, go to **7.**

7. Check for 28 vdc between diode CR18 terminal E3 and ground.

a. If voltage is as specified, repair/replace wiring between diode CR18 terminal E3 and K80 BACK-UP PUMP INTERLK relay terminal A2 (WP 1747 00). Go to **12.**

b. If voltage is not as specified, go to **8.**

8. Check for 28 vdc between diode CR18 terminal E4 and ground.

a. If voltage is as specified, replace diode (WP 0849 00). Go to **12.**

NO POWER AT MISSION INTERFACE PANEL WITH NO. 1 OR NO. 2 GENERATOR ON - Continued

b. If voltage is not as specified, go to **9.**

9. Check for 28 vdc between P205-D and ground.

a. If voltage is as specified, go to **10.**

b. If voltage is not as specified, go to **11.**

10. Check continuity between P205-E and diode CR18 terminal E4.

a. If continuity is present, replace K3 APU/EXT PWR CNTOR (WP 0840 00). Go to **12.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12.**

11. Check for 28 vdc between SEC MON BUS CONTR circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P205-D (WP 1747 00). Go to **12.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **12.**

12. Procedure completed.

NO POWER AT MISSION INTERFACE PANEL WITH NO. 1 AND NO. 2 GENERATORS ON, AND BACK-UP PUMP RUNNING

SYMPTOM

No power at mission interface panel with No. 1 and No. 2 generators on, and back-up pump running.

MALFUNCTION

No power at mission interface panel with No. 1 and No. 2 generators on, and back-up pump running.

CORRECTIVE ACTION

1. Check for 115 vac between K11 AC SEC BUS CNTOR:

- **Terminal B2 and ground**
- **Terminal A2 and ground**
- **Terminal C2 and ground**

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **3.**

2. Check for open secondary bus current limiters CL-16, CL-17, and CL-18.

a. If secondary bus current limiters are open, replace current limiter as required (WP 0846 00). Go to **15.**

b. If secondary bus current limiters are not open, repair/replace wiring between the following for K11 AC SEC BUS CNTOR (WP 1747 00), then go to **15:**

- Terminal A2 and J511-A
- Terminal B2 and J511-B
- Terminal C2 and J511-C

3. Check for 115 vac between K11 AC SEC BUS CNTOR:

NO POWER AT MISSION INTERFACE PANEL WITH NO. 1 AND NO. 2 GENERATORS ON, AND BACK-UP PUMP RUNNING - Continued

- **Terminal A1 and ground**
- **Terminal B1 and ground**
- **Terminal C1 and ground**

a. If voltage is as specified, go to **4.**

b. If voltage is not as specified, repair/replace wiring between K11 AC SEC BUS CNTOR, K1 NO. 1 GEN CNTOR, and K2 NO. 2 GEN CNTOR as required (WP 1747 00). Go to **15.**

4. Check for 28 vdc between K11 AC SEC BUS CNTOR terminals X1 and COM.

a. If voltage is as specified, replace K11 AC SEC BUS CNTOR (WP 0847 00). Go to **15.**

b. If voltage is not as specified, go to **5.**

5. Check for 28 vdc between K11 AC SEC BUS CNTOR terminal X1 and ground.

a. If voltage is as specified, repair/replace wiring between K11 AC SEC BUS CNTOR terminal COM and ground (WP 1747 00). Go to **15.**

b. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between diode CR17 terminal E1 and ground.

a. If voltage is as specified, repair/replace wiring between diode CR17 terminal E1 and contactor terminal X1 (WP 1747 00). Go to **15.**

b. If voltage is not as specified, go to **7.**

7. Check for 28 vdc between diode CR17 terminal E2 and ground.

a. If voltage is as specified, replace diode (WP 0849 00). Go to **15.**

b. If voltage is not as specified, go to **8.**

8. Check for 28 vdc between P204-L and ground.

a. If voltage is as specified, go to **9.**

b. If voltage is not as specified, go to **10.**

9. Check continuity between P204-K and diode CR17 terminal E2.

a. If continuity is present, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **15.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15.**

10. Check for 28 vdc between P212-K and ground.

a. If voltage is as specified, go to **11.**

b. If voltage is not as specified, go to **12.**

11. Check continuity between P212-L and P204-L.

a. If continuity is present, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **15.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15.**

12. Check for 28 vdc between P244-V and ground.

a. If voltage is as specified, go to **13.**

b. If voltage is not as specified, go to **14.**

13. Check continuity between P212-K and P244-L.

NO POWER AT MISSION INTERFACE PANEL WITH NO. 1 AND NO. 2 GENERATORS ON, AND BACK-UP PUMP RUNNING - Continued

a. If continuity is present, replace right relay panel (WP 0825 00). Go to **15.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15.**

14. Check for 28 vdc between SEC MON BUS CONTR circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P244-V (WP 1747 00). Go to **15.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **15.**

15. Procedure completed.

NO EMERGENCY AUXILIARY POWER AT MISSION INTERFACE PANEL

SYMPTOM

No emergency auxiliary power at mission interface panel.

MALFUNCTION

No emergency auxiliary power at mission interface panel.

CORRECTIVE ACTION

1. Check for 115 vac between K11 AC SEC BUS CNTOR:

- **Terminal A3 and ground**
- **Terminal B3 and ground**
- **Terminal C3 and ground**

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, repair/replace wiring between the following for K11 AC SEC BUS CNTOR and K3 APU/EXT PWR CNTOR (WP 1747 00), then go to **19:**

- Terminal A3 and Terminal A1
- Terminal B3 and Terminal B1
- Terminal C3 and Terminal C1

2. Check for 28 vdc between K11 AC SEC BUS CNTOR terminals Y1 and COM.

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, go to **4.**

3. Check continuity between K11 AC SEC BUS CNTOR terminal COM and ground.

a. If continuity is present, replace K11 AC SEC BUS CNTOR (WP 0847 00). Go to **19.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19.**

4. Check for 28 vdc between K12 relay terminal A3 and ground.

a. If voltage is as specified, repair/replace wiring between K12 relay terminal A3 and K11 AC SEC BUS CNTOR terminal Y1 (WP 1747 00). Go to **19.**

b. If voltage is not as specified, go to **5.**

5. Check for 28 vdc between K12 relay terminal A2 and ground.

NO EMERGENCY AUXILIARY POWER AT MISSION INTERFACE PANEL - Continued

a. If voltage is as specified, go to **6.**

b. If voltage is not as specified, go to **7.**

6. Check for 28 vdc between K12 relay terminals X1 and X2.

a. If voltage is as specified, troubleshoot blade de-ice system (WP 0145 00). Go to **19.**

b. If voltage is not as specified, replace relay (WP 0851 00). Go to **19.**

7. Check for 28 vdc between K80 BACK-UP PUMP INTERLK relay terminal B1 and ground.

a. If voltage is as specified, go to **8.**

b. If voltage is not as specified, repair/replace wiring between K80 BACK-UP PUMP INTERLK relay terminal B1 and K12 relay terminal A2 (WP 1747 00). Go to **19.**

8. Check for 28 vdc between K80 BACK-UP PUMP INTERLK relay terminal B2 and ground.

a. If voltage is as specified, go to **9.**

b. If voltage is not as specified, go to NO. 1 AND/OR NO. 2 GENERATOR DISABLED BY UNDERFREQUENCY CONDITION DURING AUTO ROTATIVE DESCENT, in this work package.

9. Check for 28 vdc between K81 GENS ON RLY terminal A3 and ground.

a. If voltage is as specified, repair/replace wiring between K80 BACK-UP PUMP INTERLK relay terminal B2 and K81 GENS ON RLY terminal A3 (WP 1747 00). Go to **19.**

b. If voltage is not as specified, go to **10.**

10. Check for 28 vdc between K81 GENS ON RLY terminal A2 and ground.

a. If voltage is as specified, go to **11.**

b. If voltage is not as specified, go to **13.**

11. Check for 28 vdc between K81 GENS ON RLY terminalsX1 and X2.

a. If voltage is as specified, go to **12.**

b. If voltage is not as specified, replace relay (WP 0851 00). Go to **19.**

12. Is operational check being performed with both No. 1 and APU generators on?

a. If both No. 1 and APU generators are on, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **19.**

b. If both No. 1 and APU generators are not on, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **19.**

13. Check for 28 vdc between diode CR20 terminal E8 and ground.

a. If voltage is as specified, repair/replace wiring between diode CR20 terminal E8 and relay terminal A2 (WP 1747 00). Go to **19.**

b. If voltage is not as specified, go to **14.**

14. Check for 28 vdc between P205-E and ground.

a. If voltage is as specified, go to **15.**

b. If voltage is not as specified, go to **16.**

NO EMERGENCY AUXILIARY POWER AT MISSION INTERFACE PANEL - Continued

15. **Check continuity between P205-F and diode CR20 terminal E8.**
    a. If continuity is present, replace K3 APU/EXT PWR CNTOR (WP 0840 00). Go to **19.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19.**
16. **Check for 28 vdc between P206-M and ground.**
    a. If voltage is as specified, go to **17.**
    b. If voltage is not as specified, go to **18.**
17. **Check continuity between P206-L and P205-E.**
    a. If continuity is present, replace K4 AC BUS TIE CNTOR (WP 0844 00). Go to **19.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19.**
18. **Check for 28 vdc between SEC MON BUS CNTOR circuit breaker terminal 1 and ground.**
    a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P206-M (WP 1747 00). Go to **19.**
    b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **19.**
19. **Procedure completed.**

BLADE DE-ICE SYSTEM DOES NOT DISABLE AUXILIARY POWER WITH BLADE DE-ICE CONTROL PANEL POWER SWITCH ON AND/OR TEST

SYMPTOM

Blade de-ice system does not disable auxiliary power with blade de-ice control panel power switch ON and/or TEST.

MALFUNCTION

Blade de-ice system does not disable auxiliary power with blade de-ice control panel power switch ON and/or TEST.

CORRECTIVE ACTION

1. **Check for 26 vdc between K12 relay terminals X1 and X2.**
    a. If voltage is as specified, replace relay (WP 0851 00). Go to **8.**
    b. If voltage is not as specified, go to **2.**
2. **Check for 28 vdc between K12 relay terminals X1 and ground.**
    a. If voltage is as specified, repair/replace wiring between relay terminal X2 and ground (WP 1747 00). Go to **8.**
    b. If voltage is not as specified, go to **3.**
3. **Check for 28 vdc between diode CR22 terminal E3 and ground.**
    a. If voltage is as specified, repair/replace wiring between diode terminal E3 and K12 relay terminal X1 (WP 1747 00). Go to **8.**
    b. If voltage is not as specified, go to **4.**
4. **Is BLADE DE-ICE control panel power switch in ON position?**
    a. If switch is on, go to **5.**

BLADE DE-ICE SYSTEM DOES NOT DISABLE AUXILIARY POWER WITH BLADE DE-ICE CONTROL PANEL POWER SWITCH ON AND/OR TEST - Continued

b. If switch is not on, go to **6.**

5. Check for 28 vdc between diode CR22 terminal E4 and ground.

a. If voltage is as specified, replace diode (WP 0850 00). Go to **8.**

b. If voltage is not as specified, troubleshoot blade de-ice system (WP 0145 00). Go to **8.**

6. Check for 28 vdc between diode CR23 terminal E5 and ground.

a. If voltage is as specified, repair/replace wiring between diode terminal E5 and diode CR22 terminal E3 (WP 1747 00). Go to **8.**

b. If voltage is not as specified, go to **7.**

7. Check for 28 vdc between diode CR23 terminal E6 and ground.

a. If voltage is as specified, replace diode (WP 0850 00). Go to **8.**

b. If voltage is not as specified, troubleshoot blade de-ice system (WP 0145 00). Go to **8.**

8. Procedure completed.

115 VAC AT MISSION INTERFACE PANEL WITH NO. 1 AND NO. 2 GENERATORS OFF, AND SIMULATED WEIGHT-OFF-WHEELS

SYMPTOM

115 vac at mission interface panel with No. 1 and No. 2 generators off, and simulated weight-off-wheels.

MALFUNCTION

115 vac at mission interface panel with No. 1 and No. 2 generators off, and simulated weight-off-wheels.

CORRECTIVE ACTION

1. Check for 28 vdc between K82 SEC BUS GND PWR CONT relay terminals X1 and X2.

a. If voltage is as specified, replace K82 SEC BUS GND PWR CONT relay (WP 0851 00). Go to **7.**

b. If voltage is not as specified, go to **2.**

2. Check for 28 vdc between K82 SEC BUS GND PWR CONT relay terminal X1 and ground.

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, go to **6.**

3. Check for 28 vdc between J163-C and ground.

a. If voltage is as specified, go to **4.**

b. If voltage is not as specified, repair/replace wiring between relay terminal X2 andJ163-C (WP 1747 00). Go to **7.**

4. Check continuity between J163-A and ground.

a. If continuity is present, go to **5.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7.**

5. Is left drag beam switch properly adjusted?

115 VAC AT MISSION INTERFACE PANEL WITH NO. 1 AND NO. 2 GENERATORS OFF, AND SIMULATED WEIGHT-OFF-WHEELS - Continued

a. If switch is properly adjusted, replace switch (WP 0443 00). Go to **7.**

b. If switch is not properly adjusted, adjust switch (WP 0443 00). Go to **7.**

6. Check for 28 vdc between SEC MON BUS CONTR circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and K82 SEC BUS GND PWR CONT relay terminal X1 (WP 1747 00). Go to **7.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **7.**

7. Procedure completed.

NO 26 VAC POWER ON AC ESNTL BUS

SYMPTOM

No 26 vac power on AC ESNTL bus.

MALFUNCTION

No 26 vac power on AC ESNTL bus.

CORRECTIVE ACTION

1. Check for 115 vac between T12 AUTO XFMR terminal HV and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **3.**

2. Check for 26 vac between T12 AUTO XFMR terminal LV and ground.

a. If voltage is as specified, repair/replace wiring between transformer terminal LV and AC ESNTL BUS 26 VAC circuit breaker (WP 1747 00). Go to **5.**

b. If voltage is not as specified, replace transformer (WP 0833 00). Go to **5.**

3. Check for 115 vac between T12 AUTO XFMR terminal HV and ground.

a. If voltage is as specified, repair/replace wiring between T12 transformer terminal GND and ground (WP 1747 00). Go to **5.**

b. If voltage is not as specified, go to **4.**

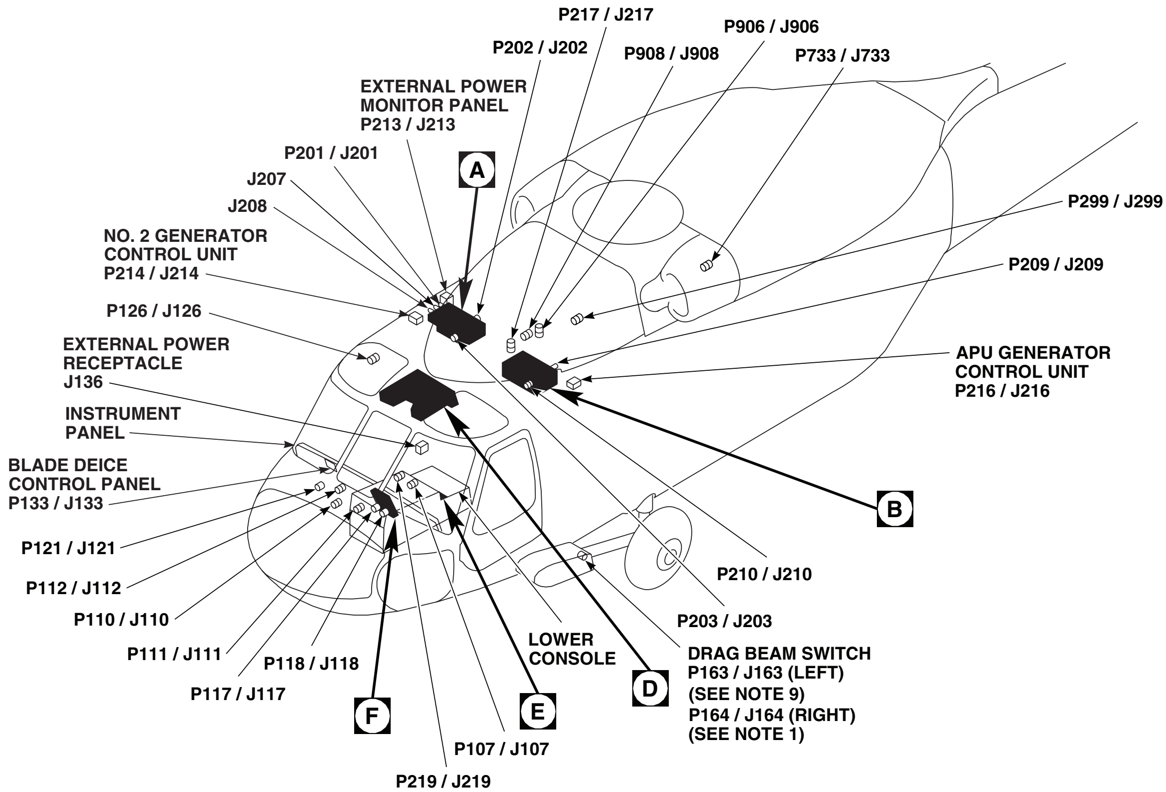
4. Check for 115 vac between AUTO XFMR circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T12 transformer terminal HV (WP 1747 00). Go to **5.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **5.**

5. Procedure completed.

LOCATION



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
J136	EXTERNAL POWER RECEPTACLE
J207	NO. 2 JUNCTION BOX
J208	NO. 2 JUNCTION BOX
J211	NO. 1 JUNCTION BOX
J257	UTILITY RECEPTACLE 115 V 400 Hz
P107 / J107	COCKPIT, BL 0 LH, STA 236
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P112 / J112	COCKPIT, BL 6 RH, STA 197
P117 / J117	CAUTION / ADVISORY PANEL
P118 / J118	CAUTION / ADVISORY PANEL
P121 / J121	COCKPIT, BL 10 RH, STA 195
P126 / J126	COCKPIT CEILING BL 28 RH, STA 243
P127 / J127	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 28 LH, STA 243
P133 / J133	BLADE DEICE CONTROL PANEL

EFFECTIVITY

UH60A UH60L

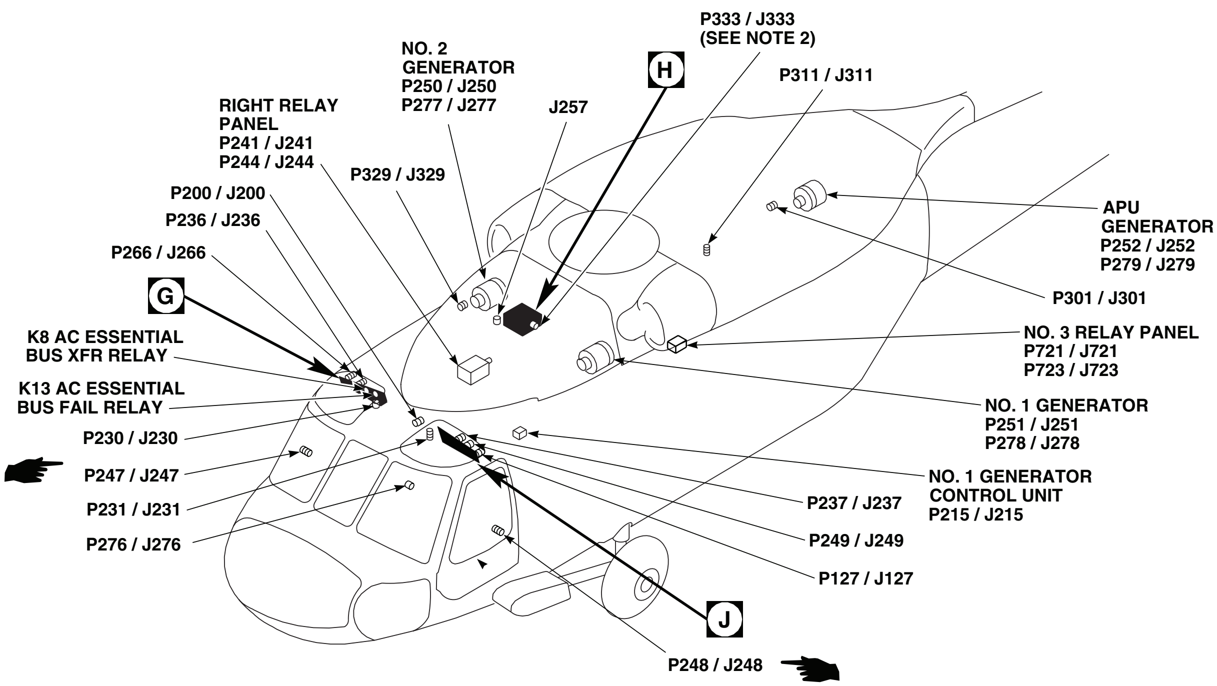
NOTES

1. UH60L MOD
2. HELICOPTERS EQUIPPED WITH BLADE DEICING SYSTEM.
3. W/O HCW
4. HCW
5. UH60A WITH AUXILIARY HEATER.
6. HELICOPTERS WITH AUXILIARY CABIN HEATER INSTALLED, CIRCUIT BREAKER CB1A REIDENTIFIED AS AUS HTR BLOWER.
7. 77–27714 – 97–26743
8. UH60L 96–26723 – SUBQ
   CIRCUIT BREAKER IS LABELED DPLR/GPS.
9. W/O MOD

AA2696_1C
SA

Figure 1. AC Electrical System Location Diagram UH-60A UH-60L . (Sheet 1 of 8)

LOCATION - Continued



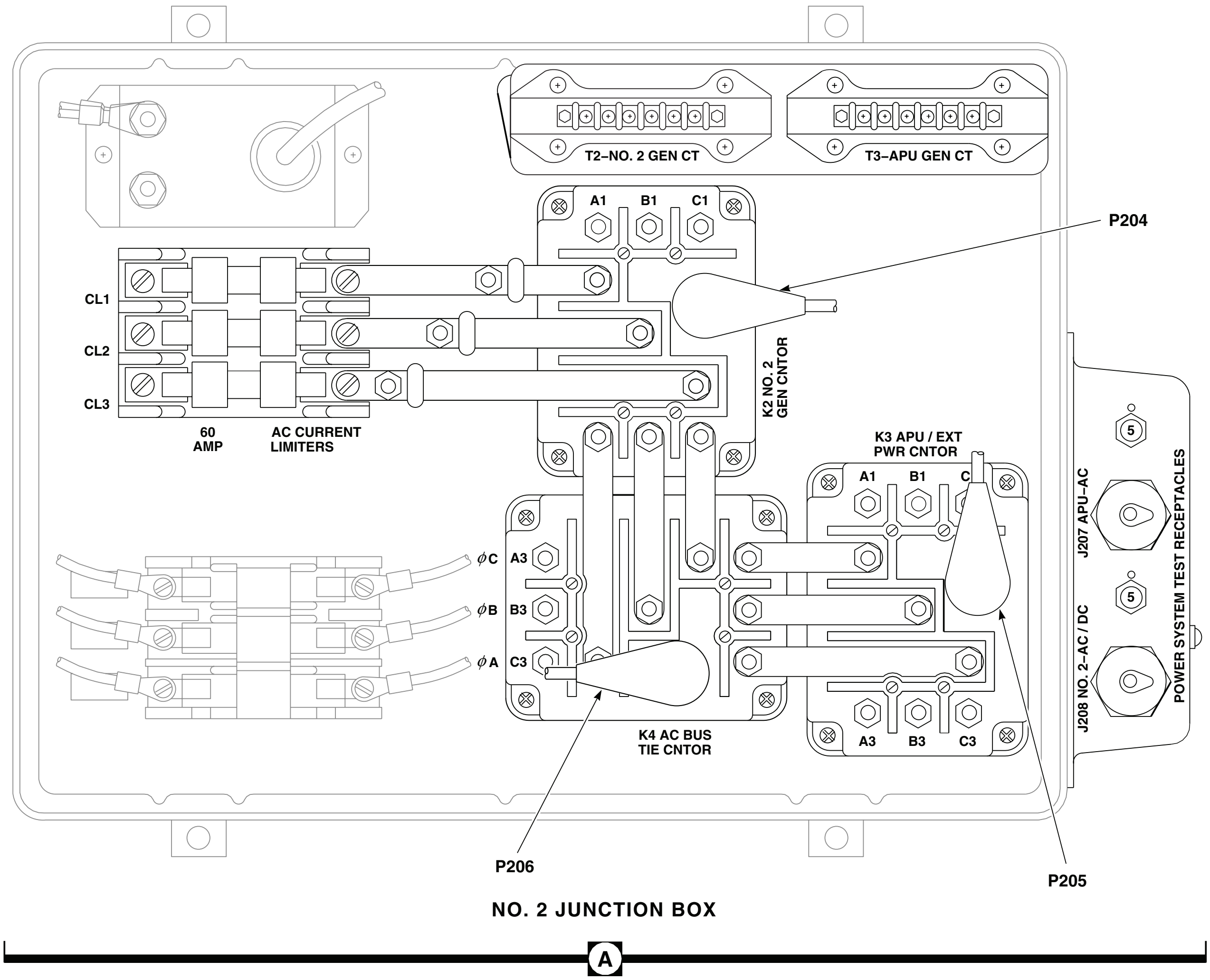
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P163 / J163 (SEE NOTE 9)	LEFT DRAG BEAM SWITCH
P164 / J164 (SEE NOTE 1)	RIGHT DRAG BEAM SWITCH
P200 / J200	COCKPIT CEILING BL 13 LH, STA 246
P201 / J201	NO. 2 JUNCTION BOX
P202 / J202	NO. 2 JUNCTION BOX
P203 / J203	NO. 2 JUNCTION BOX
P204	NO. 2 JUNCTION BOX
P205	NO. 2 JUNCTION BOX
P206	NO. 2 JUNCTION BOX
P209 / J209	NO. 1 JUNCTION BOX
P210 / J210	NO. 1 JUNCTION BOX
P212	NO. 1 JUNCTION BOX
P213 / J213	EXTERNAL POWER MONITOR PANEL
P214 / J214	NO. 2 GENERATOR CONTROL UNIT
P215 / J215	NO. 1 GENERATOR CONTROL UNIT
P216 / J216	APU GENERATOR CONTROL UNIT

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P231/ J231	CABIN CEILING BL 9 LH, STA 247
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 23 RH
P237 / J237	BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 25 LH
P241 / J241	RIGHT RELAY PANEL
P244 / J244	RIGHT RELAY PANEL
P247 / J247 (SEE NOTE 1)	CABIN, BL 40 RH, STA 248
P248 / J248 (SEE NOTE 9)	CABIN, BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P250 / J250	NO. 2 GENERATOR
P251 / J251	NO. 1 GENERATOR
P252 / J252	APU GENERATOR

AA2696_2B
SA

Figure 1. AC Electrical System Location Diagram UH-60A UH-60L . (Sheet 2 of 8)

LOCATION - Continued



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH
P276 / J276	60 Hz CONVERTER (PROVISIONAL)
P277 / J277	NO. 2 GENERATOR
P278 / J278	NO. 1 GENERATOR
P279 / J279	APU GENERATOR
P299 / J299 (SEE NOTE 5)	RANGE EXTENSION PROVISION CONTROL PANEL, STA 295, WL 265
P301 / J301	TRANSITION DECK BL 12.88 LH, STA 415
P311 / J311	CABIN CEILING BL 16.5 LH, STA 383
P329 / J329	MAIN BLADE DE–ICE JUNCTION BOX

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P333 / J333 (SEE NOTE 2)	AUXILIARY AC JUNCTION BOX
P721 / J721	NO. 3 RELAY PANEL
P723 / J723	NO. 3 RELAY PANEL
P733 / J733	STA 360, BL 24.71, WL 265.30
P906 / J906	CABIN CEILING BL 0, STA 284
P908 / J908	CABIN CEILING BL 0, STA 279

AA2696_3A
SA

Figure 1. AC Electrical System Location Diagram **UH-60A UH-60L** . (Sheet 3 of 8)

LOCATION - Continued

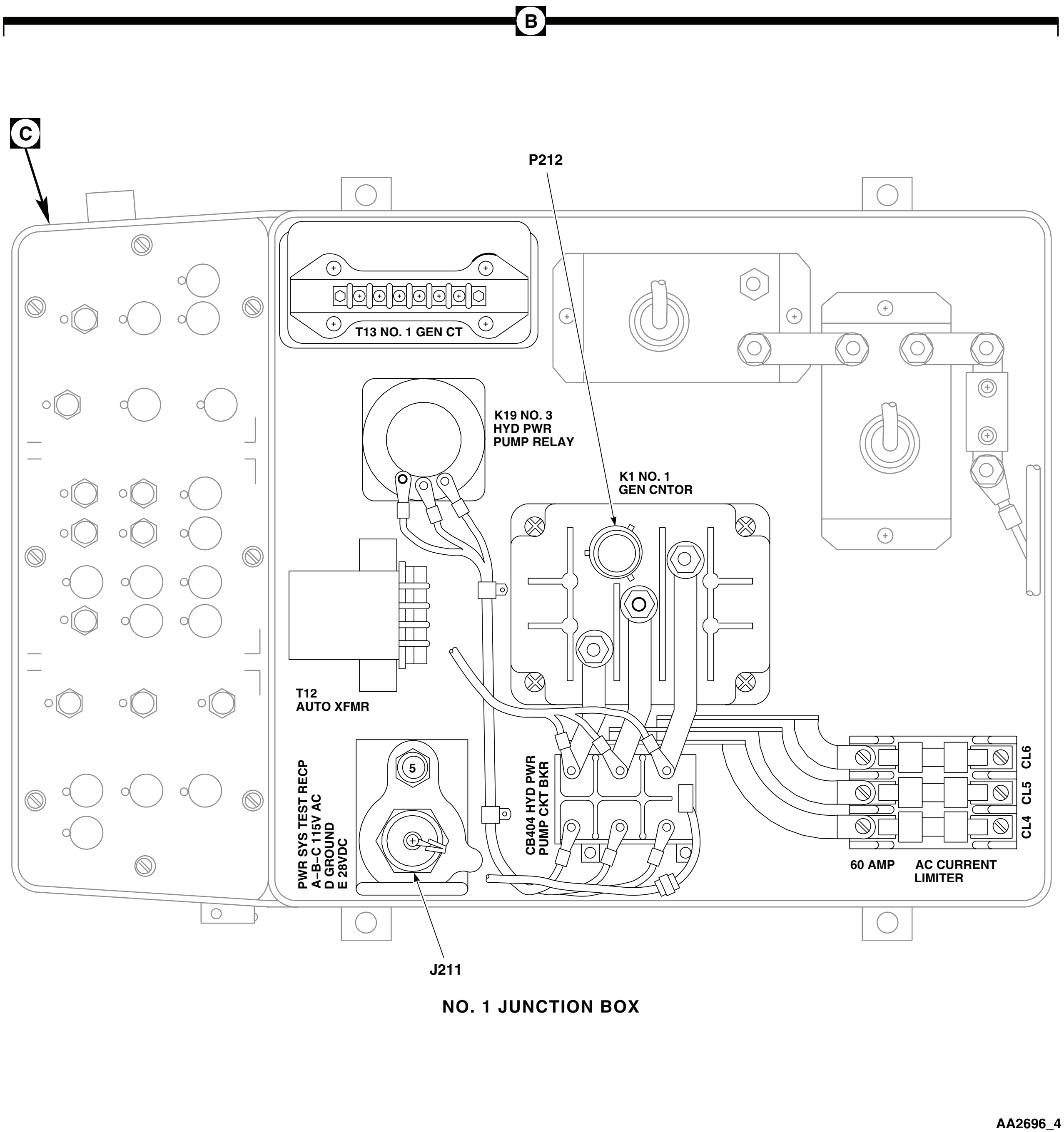


Figure 1. AC Electrical System Location Diagram UH-60A UH-60L . (Sheet 4 of 8)

LOCATION - Continued

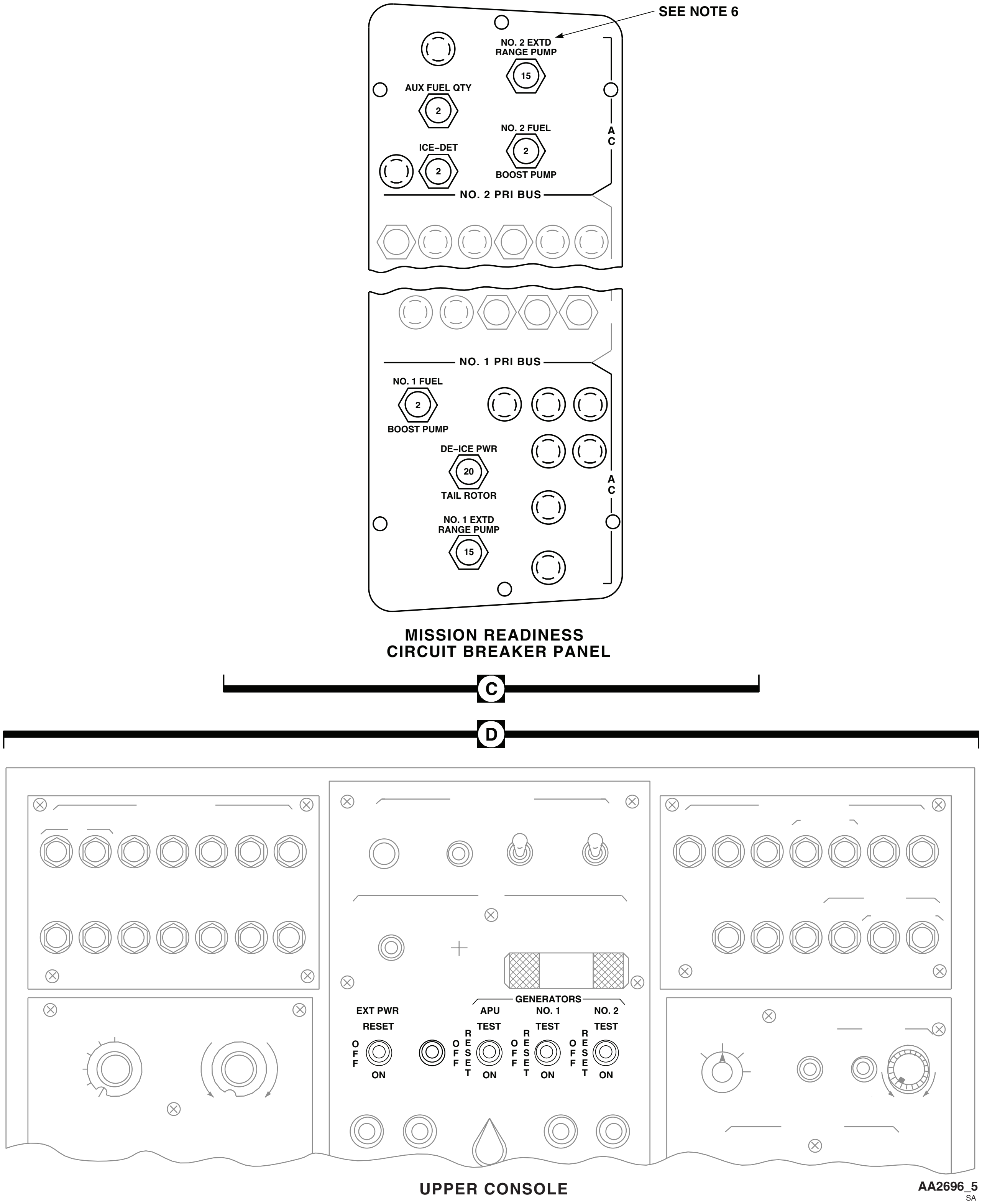
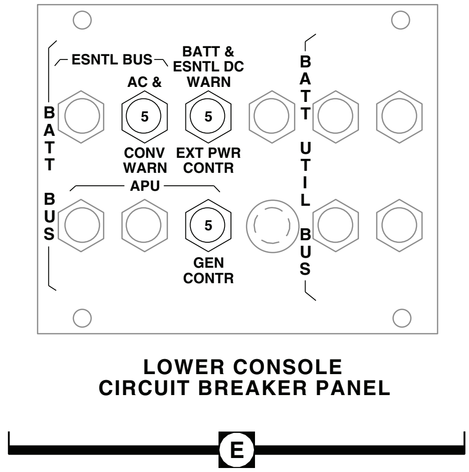
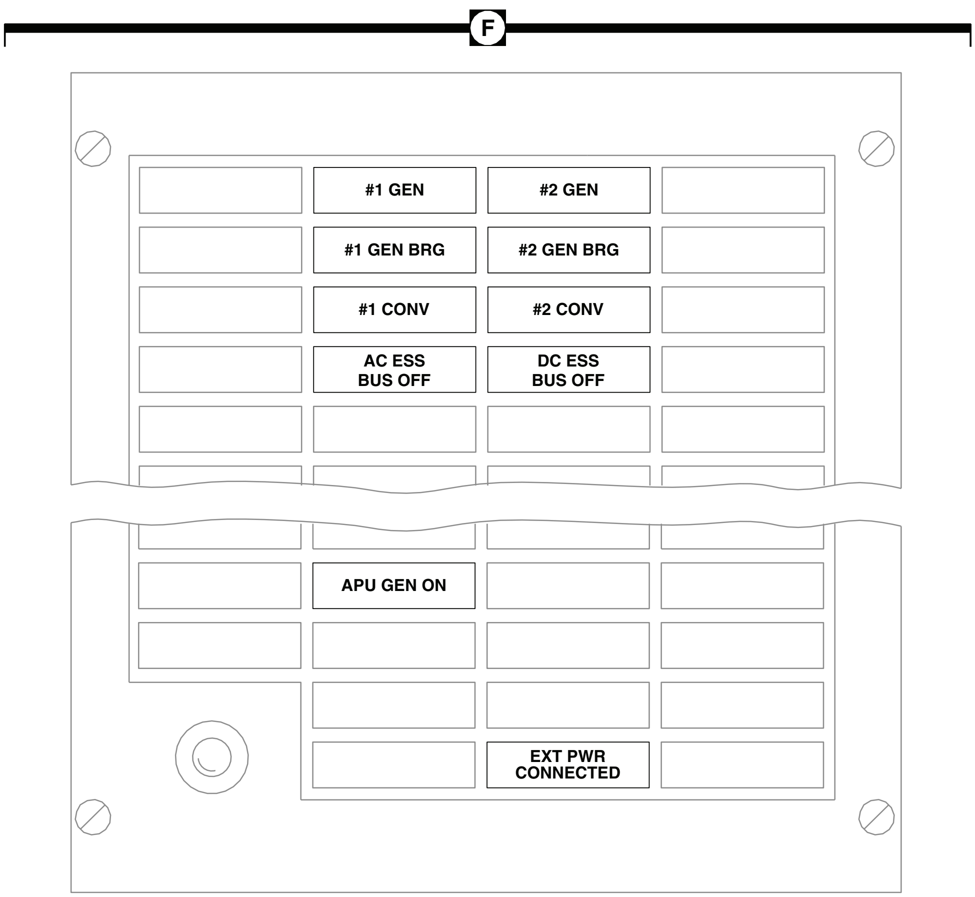


Figure 1. AC Electrical System Location Diagram **UH-60A UH-60L** . (Sheet 5 of 8)

LOCATION - Continued



LOWER CONSOLE
CIRCUIT BREAKER PANEL

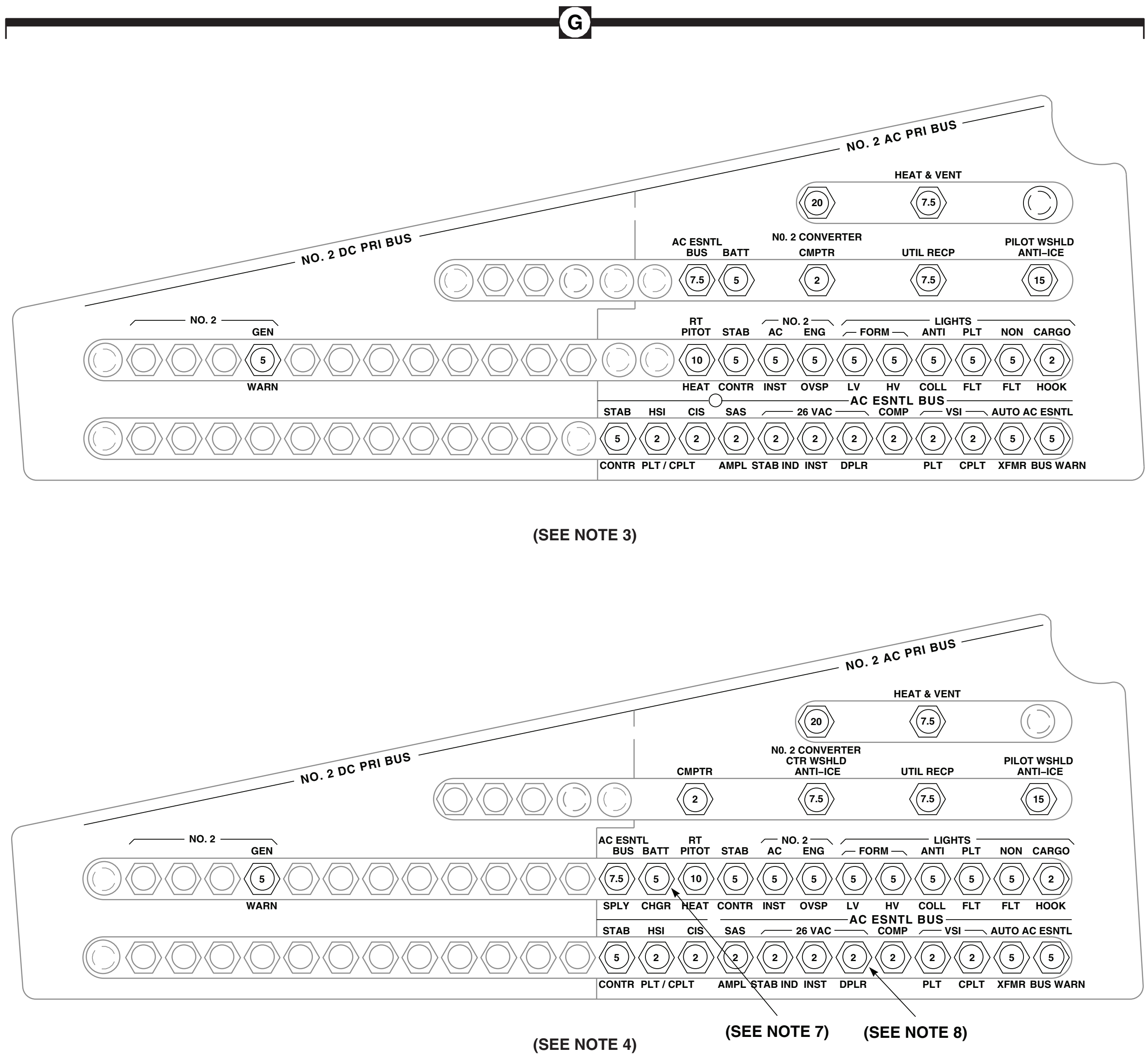


CAUTION / ADVISORY PANEL

AA2696_6
SA

Figure 1. AC Electrical System Location Diagram UH-60A UH-60L . (Sheet 6 of 8)

LOCATION - Continued



PILOT'S CIRCUIT BREAKER PANEL

AA2696_7B
SA

Figure 1. AC Electrical System Location Diagram UH-60A UH-60L . (Sheet 7 of 8)

LOCATION - Continued

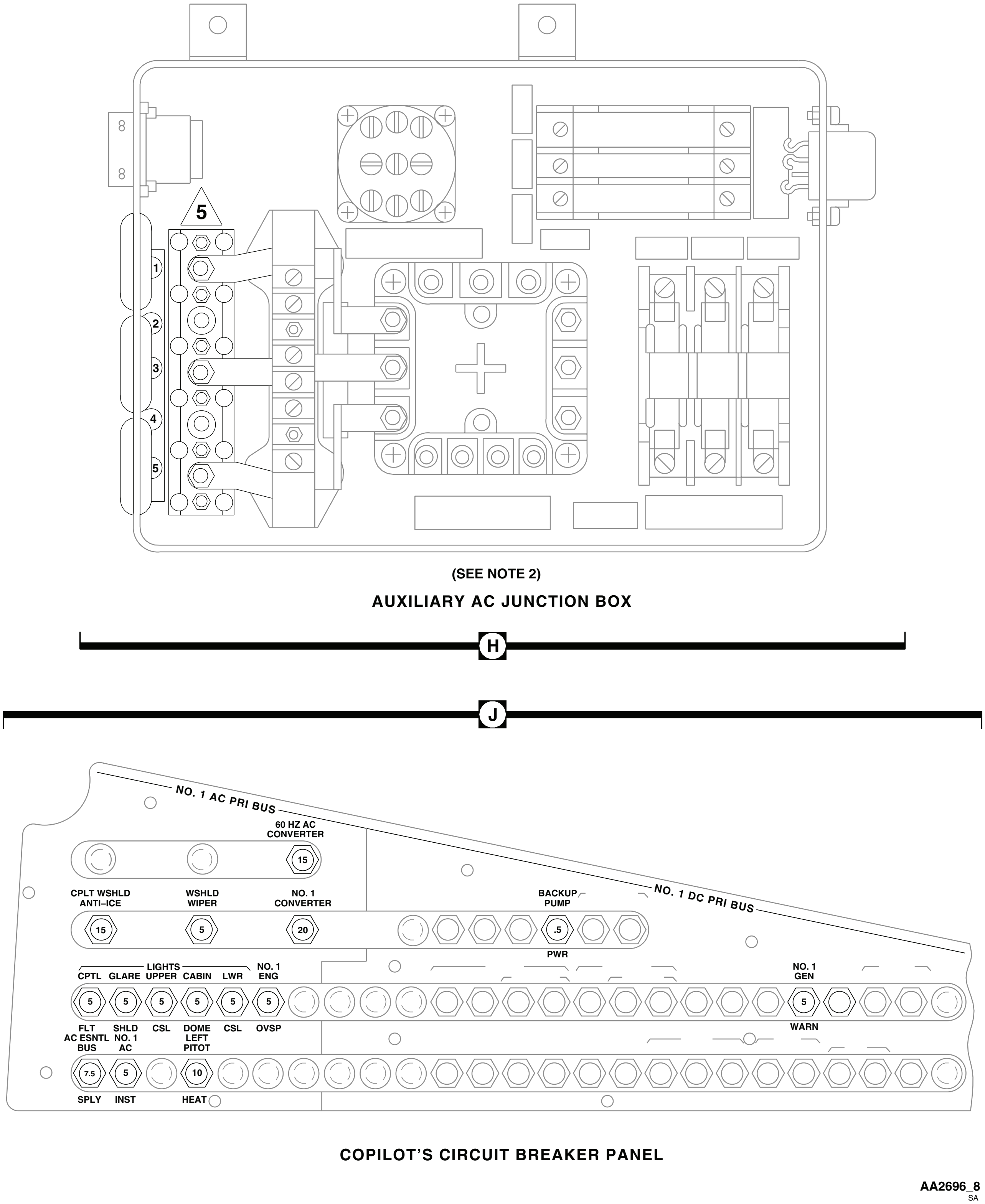
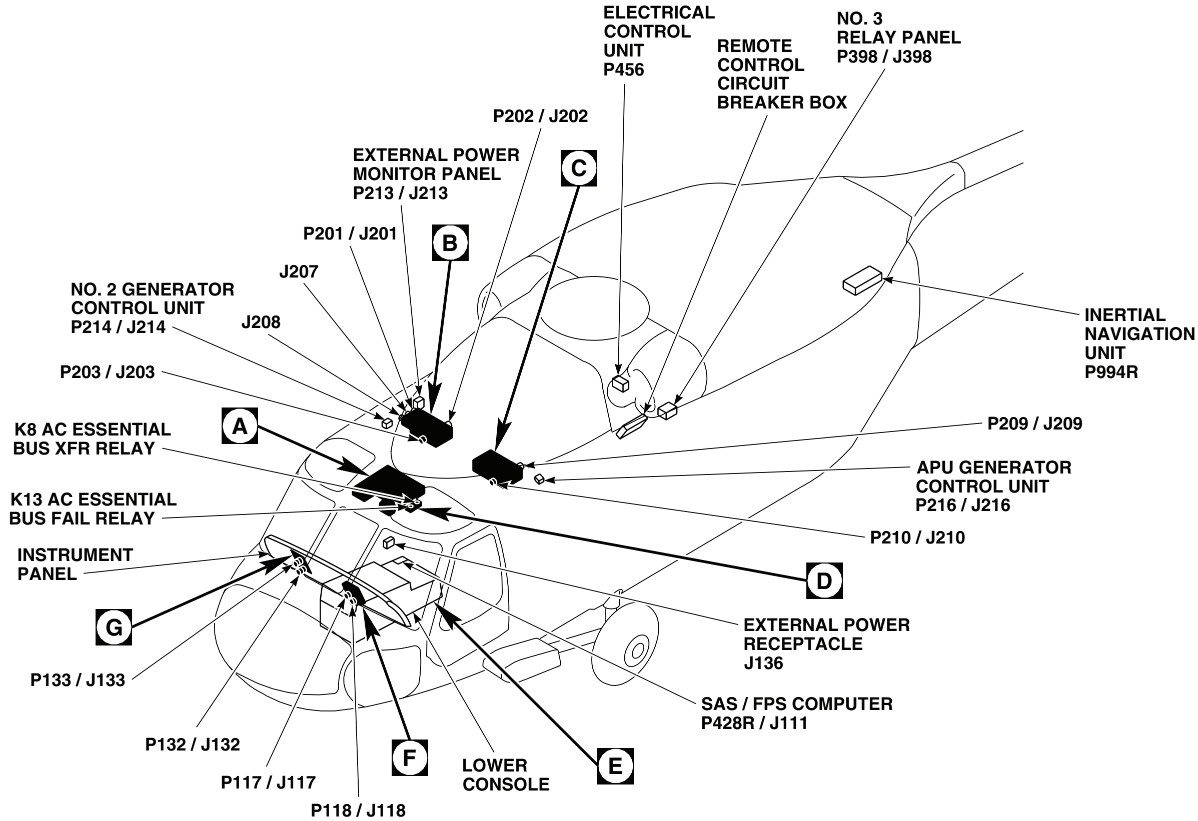


Figure 1. AC Electrical System Location Diagram UH-60A UH-60L . (Sheet 8 of 8)

LOCATION - Continued



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
J136	EXTERNAL POWER RECEPTACLE
J207	NO. 2 JUNCTION BOX
J208	NO. 2 JUNCTION BOX
J211	NO. 1 JUNCTION BOX
J257	UTILITY RECEPTACLE 115V 400 Hz
J511	MISSION INTERFACE PANEL
P1R / J1R	COCKPIT BL 6, STA 210
P3R / J3R	STABILATOR CONTROLS / AUTO FLIGHT CONTROL PANEL
P21R / J21R	BL 40 LH, STA 247
P107 / J107	COCKPIT BL 9 RH, STA 236
P110 / J110	COCKPIT BL 7.5 RH, STA 197

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P112 / J112	COCKPIT BL 6 RH, STA 197
P117 / J117	CAUTION / ADVISORY PANEL
P118 / J118	CAUTION / ADVISORY PANEL
P121 / J121	COCKPIT BL 10 RH, STA 195
P126 / J126	COCKPIT CEILING BL 28 RH, STA 243
P132 / J132	ICING RATE METER
P133 / J133	BLADE DEICE CONTROL PANEL
P142 / J142	COCKPIT BL 3 RH, STA 210
P149R / J149R	RADIO RECEIVER

NOTE

HELICOPTERS EQUIPPED WITH BLADE DEICING SYSTEM.

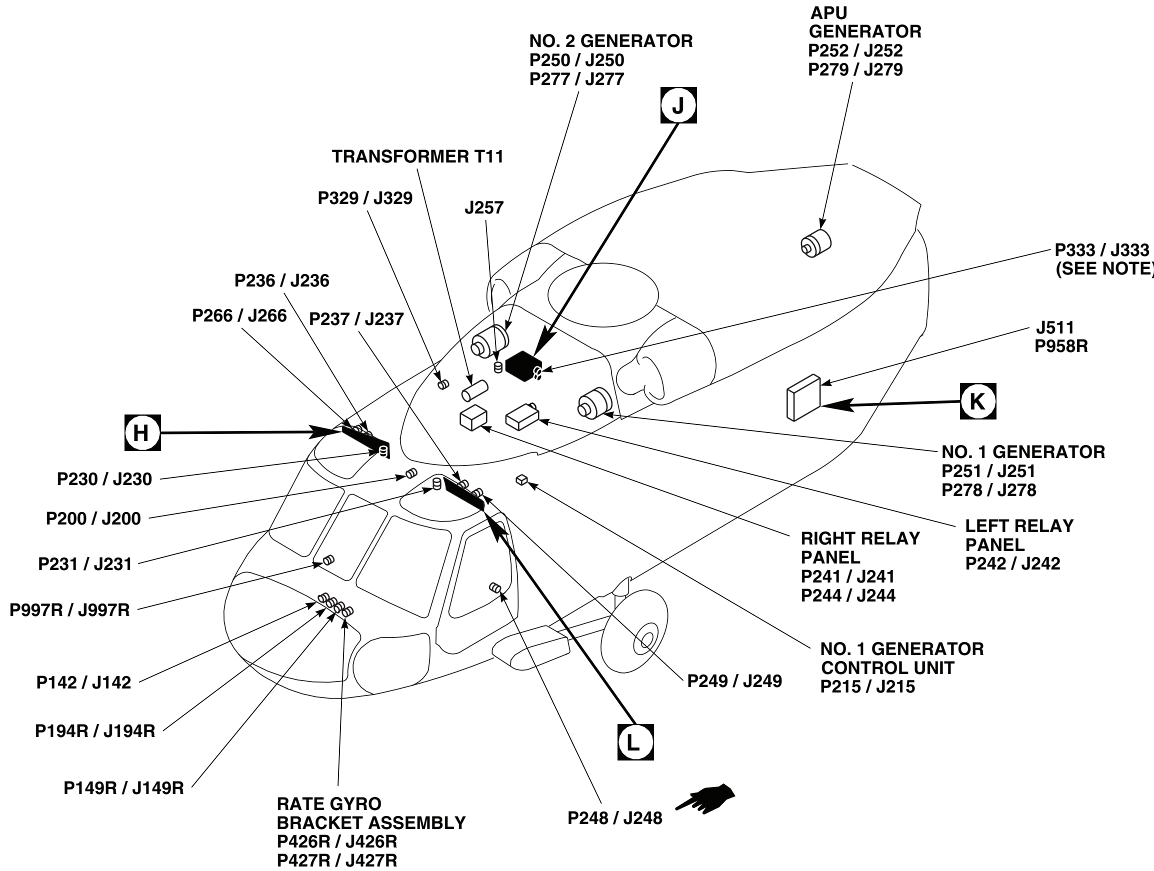
EFFECTIVITY

EH60A

AA2698_1
SA

Figure 2. AC Electrical System Location Diagram EH-60A . (Sheet 1 of 9)

LOCATION - Continued



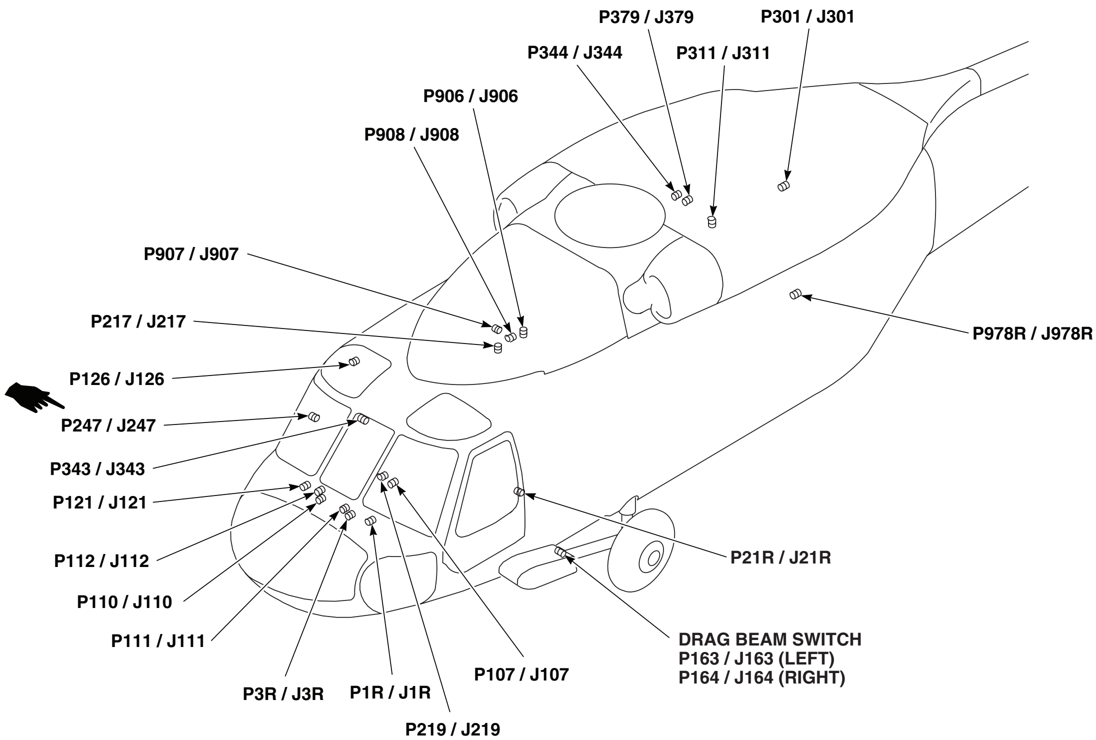
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P163 / J163	LEFT DRAG BEAM SWITCH
P164 / J164	RIGHT DRAG BEAM SWITCH
P194R / J194R	RADIO RECEIVER
P200 / J200	COCKPIT CEILING BL 13 LH, STA 246
P201 / J201	NO. 2 JUNCTION BOX
P202 / J202	NO. 2 JUNCTION BOX
P203 / J203	NO. 2 JUNCTION BOX
P204	NO. 2 JUNCTION BOX
P205	NO. 2 JUNCTION BOX
P206	NO. 2 JUNCTION BOX
P209 / J209	NO. 1 JUNCTION BOX
P210 / J210	NO. 1 JUNCTION BOX
P212	NO. 1 JUNCTION BOX
P213 / J213	EXTERNAL POWER MONITOR PANEL
P214 / J214	NO. 2 GENERATOR CONTROL UNIT
P215 / J215	NO. 1 GENERATOR CONTROL UNIT

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P216 / J216	APU GENERATOR CONTROL UNIT
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P231 / J231	CABIN CEILING BL 9 LH, STA 247
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 23 RH
P237 J237	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 25 LH
P241 / J241	RIGHT RELAY PANEL
P242 / J242	LEFT RELAY PANEL
P244 / J244	RIGHT RELAY PANEL
P247 / J247	CABIN BL 40 RH, STA 248
P248 / J248	CABIN BL 40 LH, STA 248

AA2698_2A
SA

Figure 2. AC Electrical System Location Diagram EH-60A . (Sheet 2 of 9)

LOCATION - Continued



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 23 LH
P250 / J250	NO. 2 GENERATOR
P251 / J251	NO. 1 GENERATOR
P252 / J252	APU GENERATOR
P266 / J266	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 25 RH
P277 / J277	NO. 2 GENERATOR
P278 / J278	NO. 1 GENERATOR
P279 / J279	APU GENERATOR
P301 / J301	TRANSITION DECK BL 12.88 LH, STA 415
P311 / J311	CABIN CEILING BL 16.5 LH, STA 383
P329 / J329	MAIN ROTOR BLADE DE–ICE JUNCTION BOX
P333 (SEE NOTE)	AUXILIARY AC JUNCTION BOX
P343 / J343	BL 30 RH, STA 245
P344 / J344	BL 3.5 RH, STA 397

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P379 / J379	CABIN CEILING BL 5 LH, STA 379
P398 / J398	NO. 3 RELAY PANEL
P426R / J426R	RATE GYRO BRACKET ASSEMBLY
P427R / J427R	RATE GYRO BRACKET ASSEMBLY
P428R / J111	SAS / FPS COMPUTER
P456	ELECTRICAL CONTROL UNIT
P906 / J906	CABIN CEILING BL 0, STA 284
P907 / J907	CABIN CEILING BL 0, STA 289
P908 / J908	CABIN CEILING BL 0, STA 279
P958R	MISSION INTERFACE PANEL
P978R / J978R	STA 398, WL 238
P994R	INTERTIAL NAVIGATION UNIT
P997R / J997R	BL 30 RH, STA 245

AA2698_3A
SA

Figure 2. AC Electrical System Location Diagram EH-60A . (Sheet 3 of 9)

LOCATION - Continued

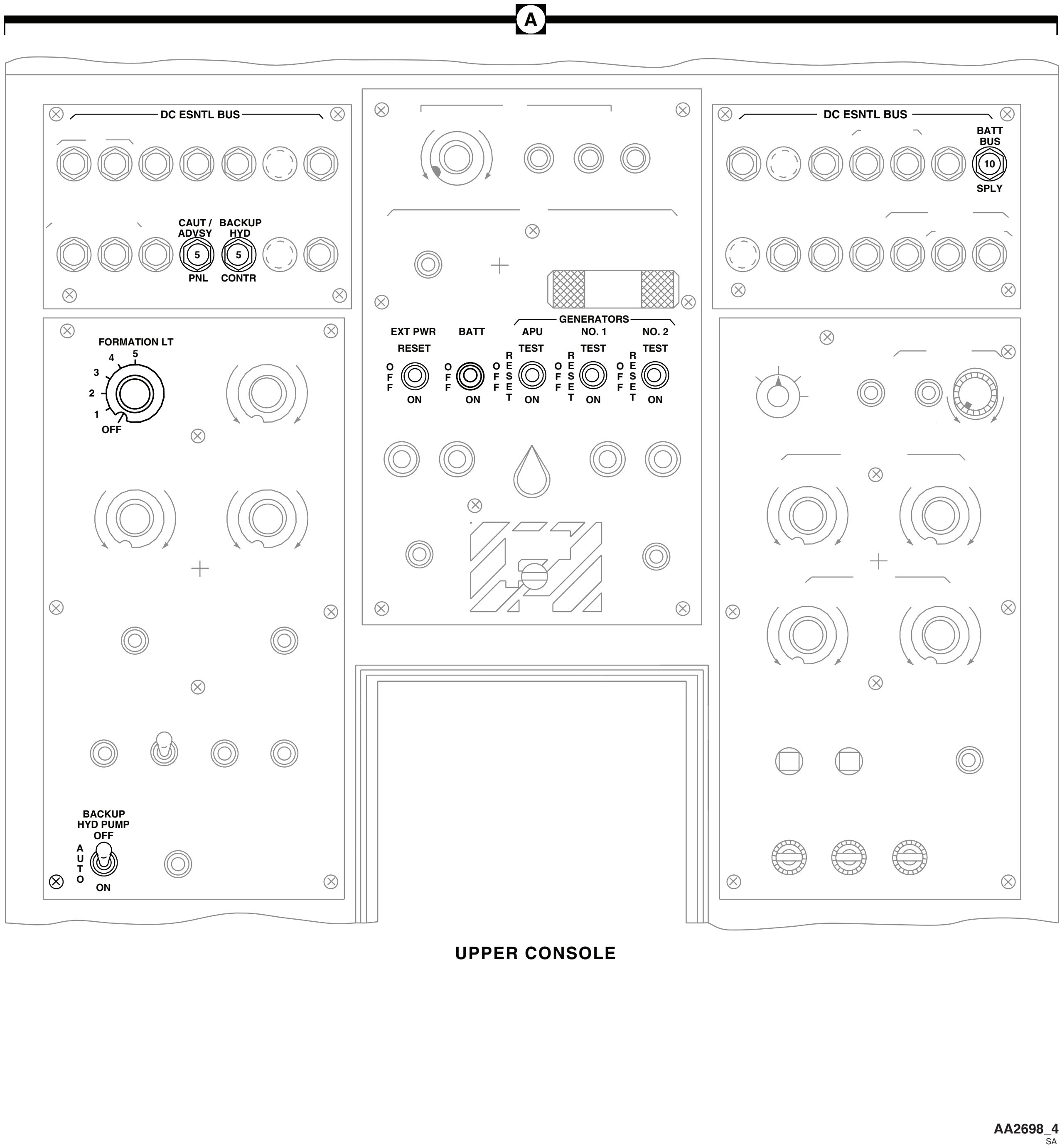


Figure 2. AC Electrical System Location Diagram EH-60A . (Sheet 4 of 9)

LOCATION - Continued

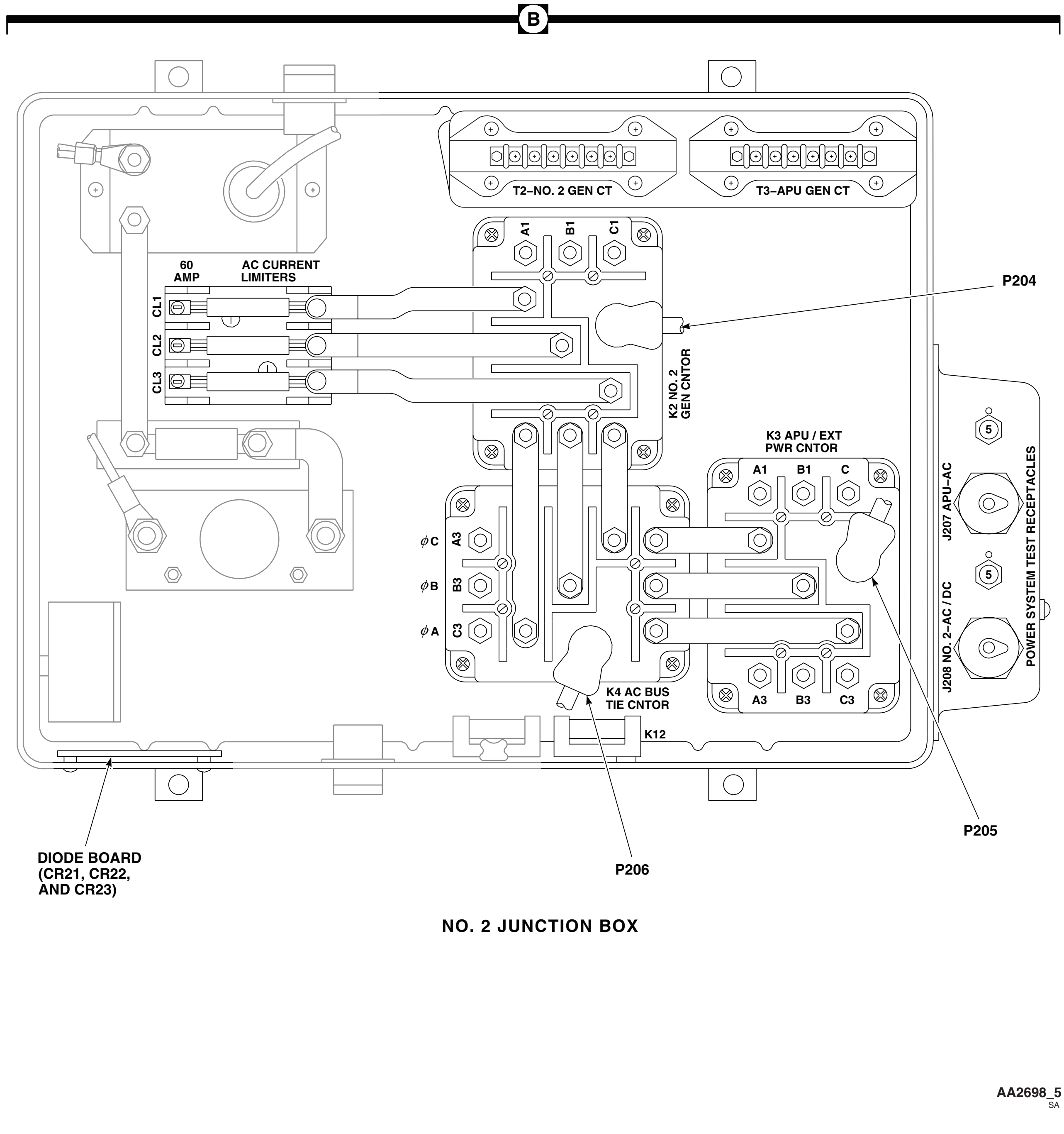


Figure 2. AC Electrical System Location Diagram EH-60A . (Sheet 5 of 9)

LOCATION - Continued

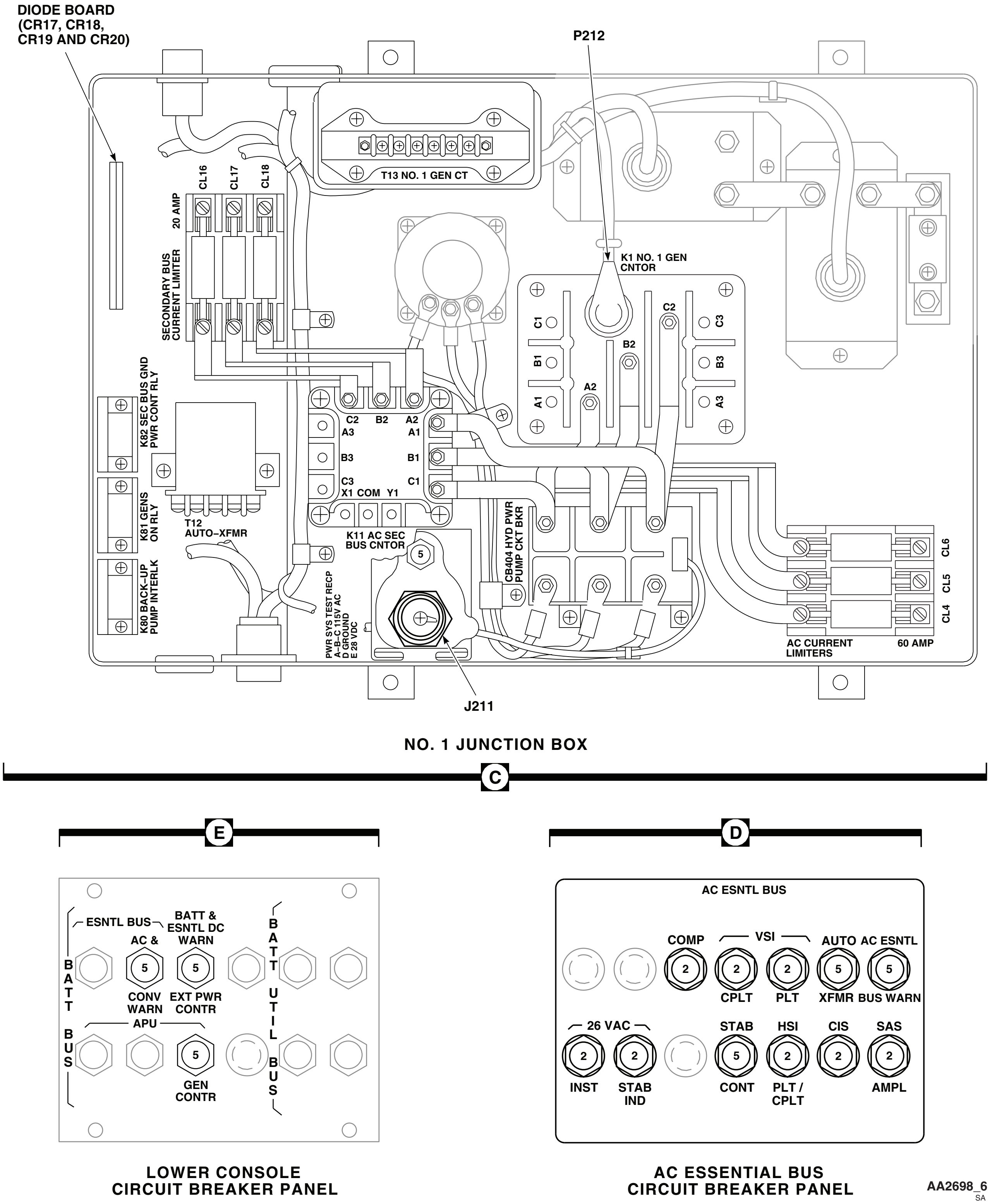


Figure 2. AC Electrical System Location Diagram EH-60A . (Sheet 6 of 9)

LOCATION - Continued

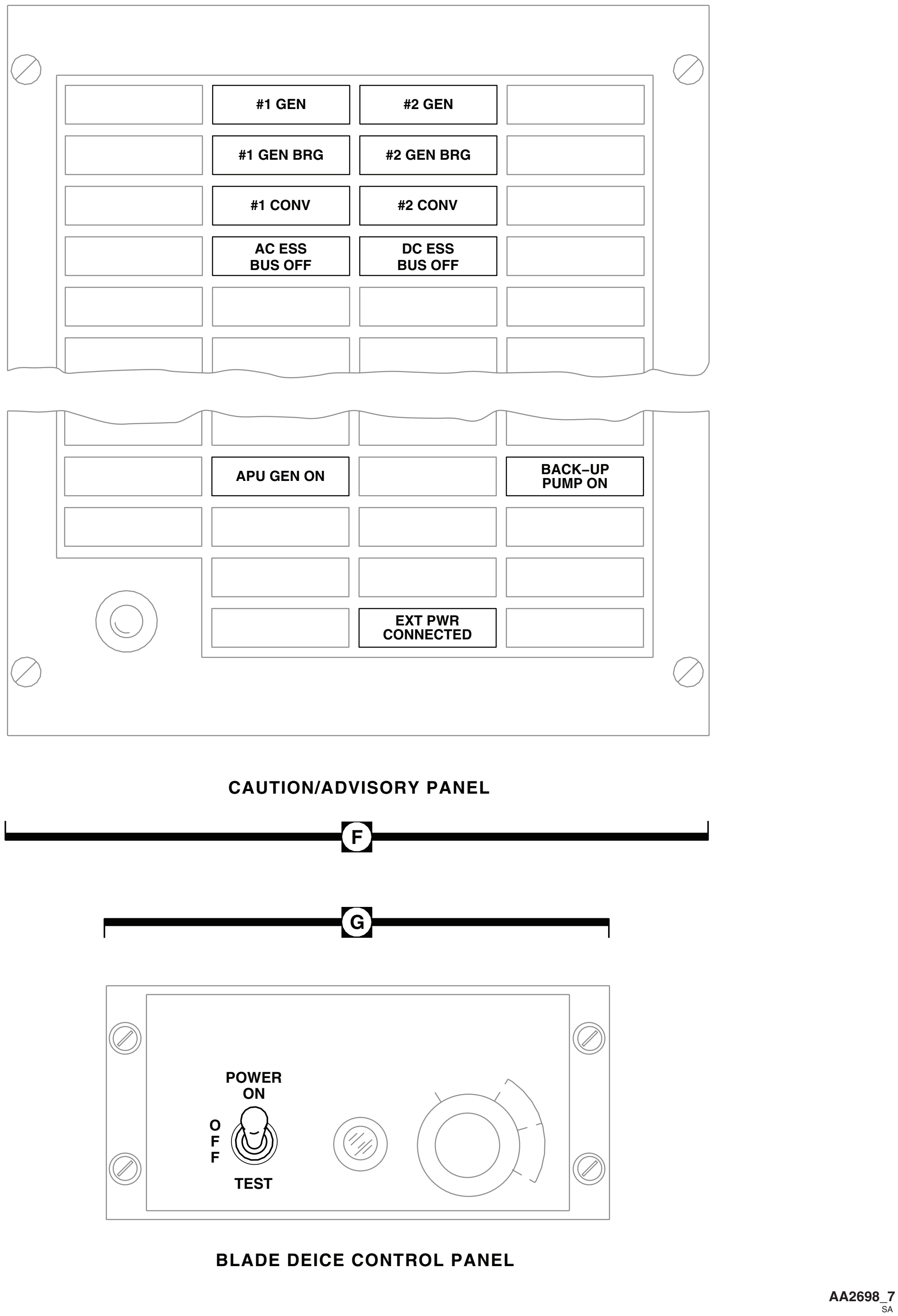
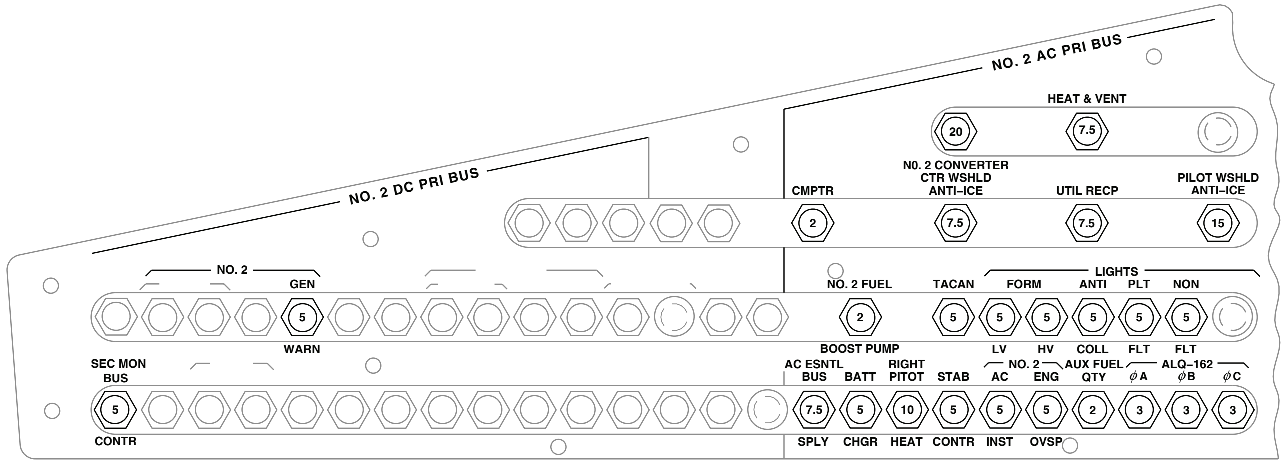


Figure 2. AC Electrical System Location Diagram EH-60A . (Sheet 7 of 9)

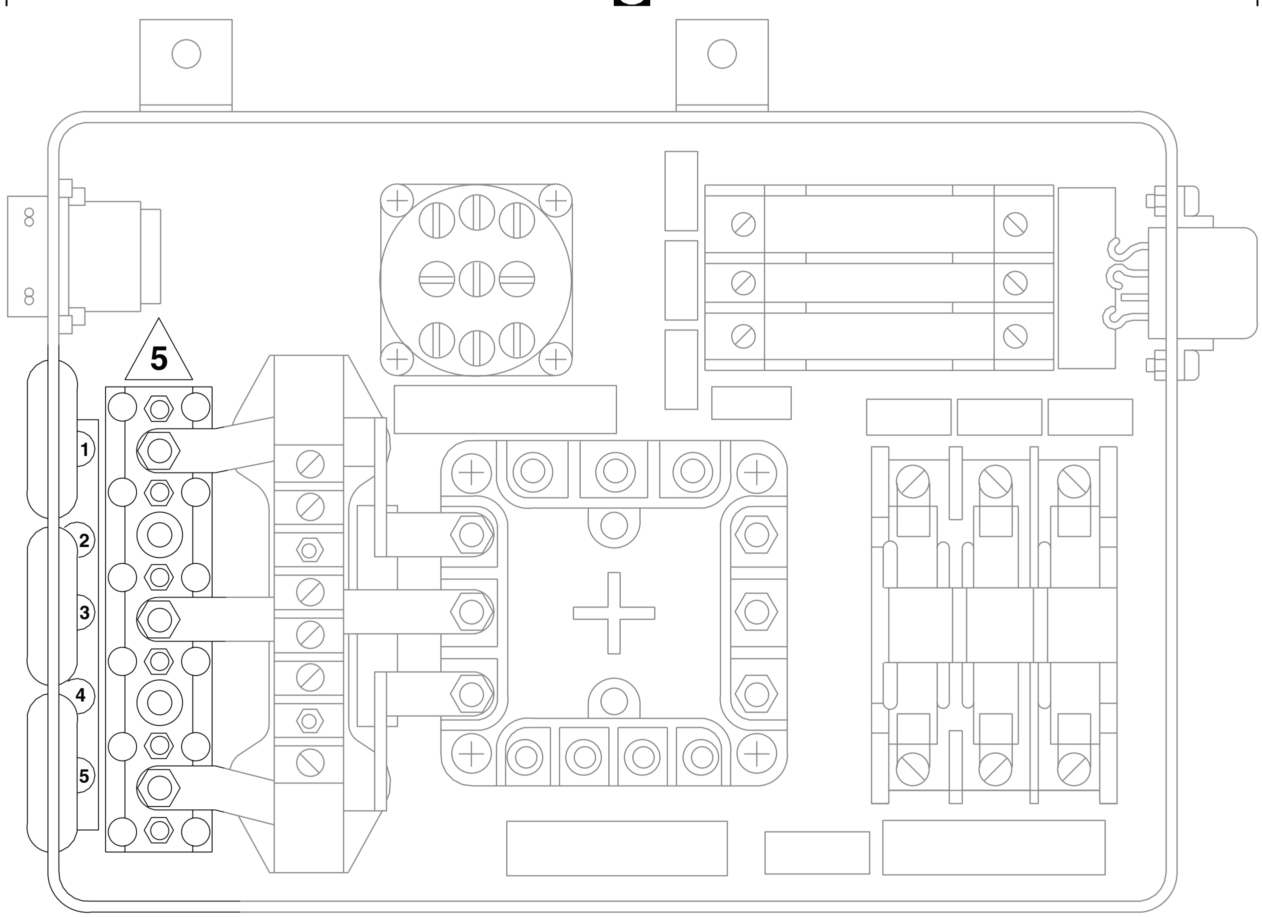
LOCATION - Continued



PILOT'S CIRCUIT BREAKER PANEL

H

J



AUXILIARY AC JUNCTION BOX

AA2698_8
SA

Figure 2. AC Electrical System Location Diagram EH-60A . (Sheet 8 of 9)

LOCATION - Continued

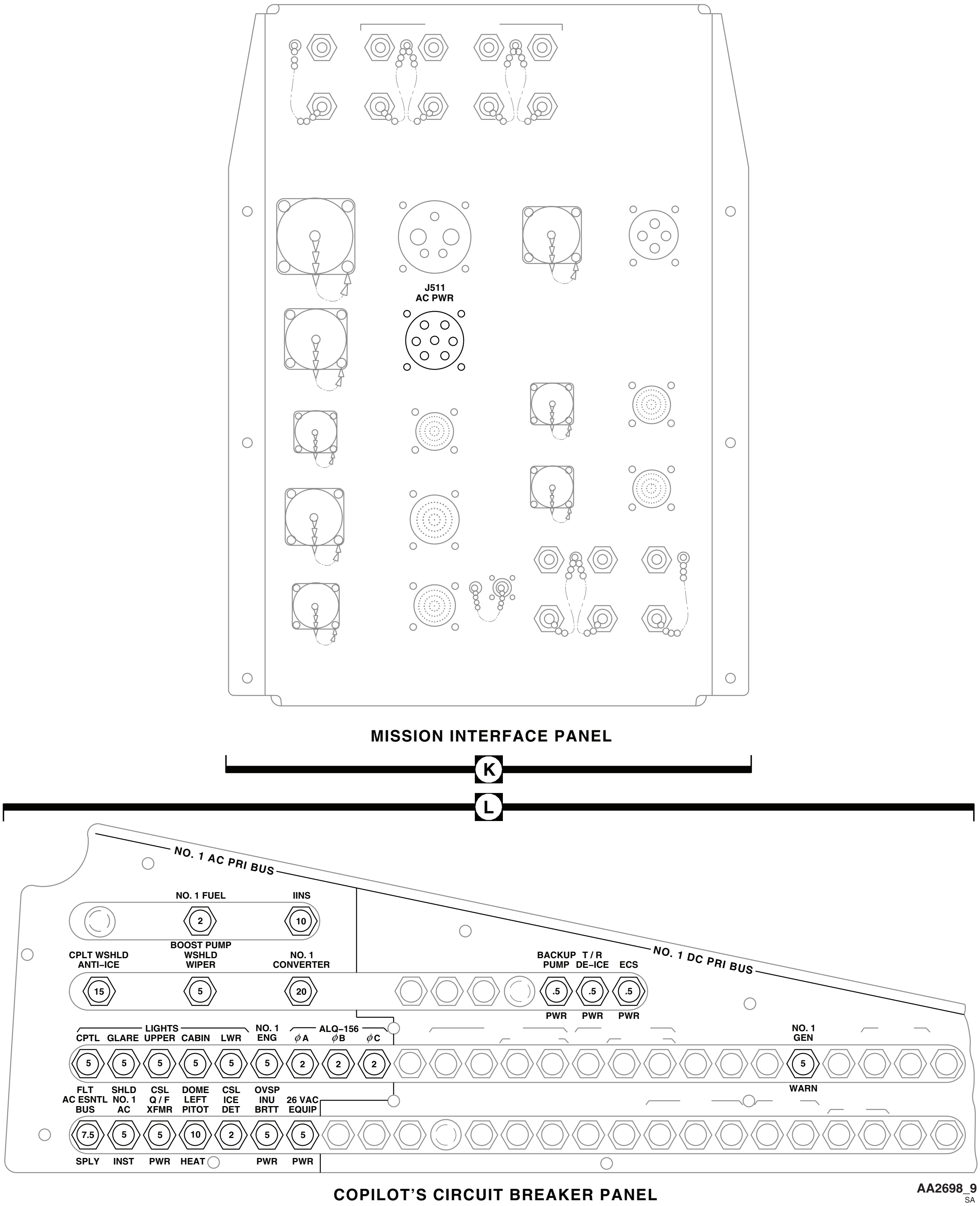


Figure 2. AC Electrical System Location Diagram EH-60A . (Sheet 9 of 9)

SCHEMATICS

NOTES

1. W/O EMEP PIN FILTERED ADAPTERS ARE NOT INSTALLED.
2. PIN A OF P215 SHOWS TYPICAL CONFIGURATION FOR ALL PIN FILTERED ADAPTERS.
3. UH60A WITH AUXILIARY HEATER.
4. W/O HCW AC ESNTL BUS SPLY IDENTIFIED AS CB257.
5. W/O MOD
6. UH60A 77-22714 - 80-23422
7. MOD UH60A

EFFECTIVITY

UH60A UH60L

LEGEND

POWER SOURCE

A=NO. 1 MAIN GENERATOR
B=NO. 2 MAIN GENERATOR
C=APU GENERATOR
D=EXTERNAL AC POWER
1=TURNED ON AND OPERATING
0=TURNED OFF OR FAILED

CONTACTORS/RELAYS

K1=NO. 1 GENERATOR CONTACTOR
K2=NO. 2 GENERATOR CONTACTOR
K3=APU / EXT PWR CONTACTOR
K4=AC BUS TIE CONTACTOR
K8=AC ESNTL BUS XFR RELAY
K13=AC ESNTL BUS FAIL RELAY
X=ENERGIZED
O=DE-ENERGIZED

AC POWER SOURCE				CONDITION	AC SYSTEM CONTACTORS/ RELAYS					
A	B	C	D		K1	K2	K3	K4	K8	K13
0	0	0	0	1	0	0	0	0	0	0
1	0	0	0	2	X	0	0	0	X	X
0	1	0	0	3	0	X	0	0	X	X
1	1	0	0	4	X	X	0	0	X	X
0	0	1	0	5	0	0	X	X	X	X
1	0	1	0	6	X	0	X	0	X	X
0	1	1	0	7	0	X	X	0	X	X
1	1	1	0	8	X	X	X	0	X	X
0	0	0	1	9	0	0	0	X	X	X
1	0	0	1	10	X	0	0	0	X	X
0	1	0	1	11	0	X	0	0	X	X
1	1	0	1	12	X	X	0	0	X	X
0	0	1	1	13	0	0	X	X	X	X
1	0	1	1	14	X	0	X	0	X	X
0	1	1	1	15	0	X	X	0	X	X
1	1	1	1	16	X	X	X	0	X	X

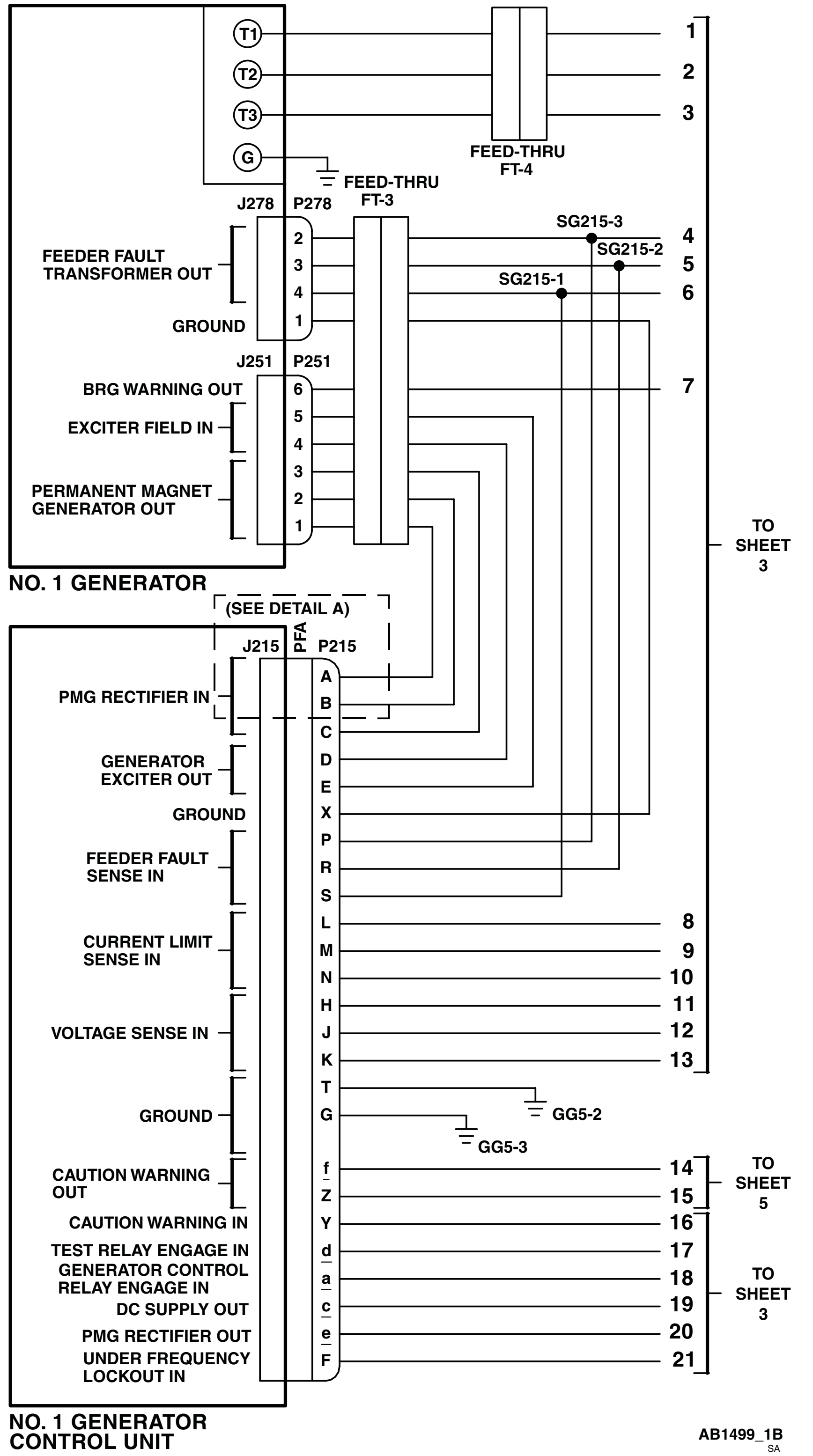


Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 1 of 22)

SCHEMATICS - Continued

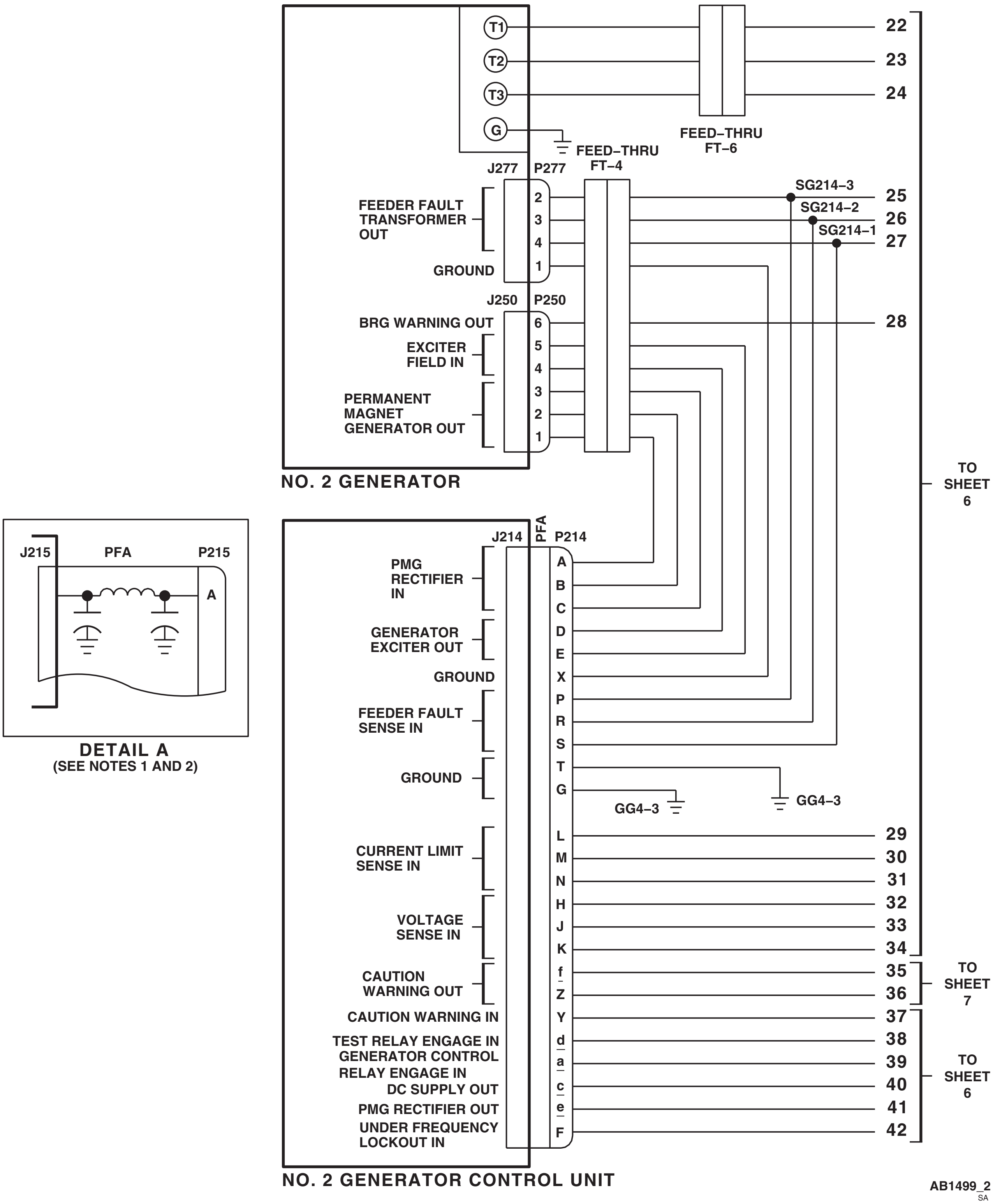


Figure 3. AC Electrical System Schematic Diagram **UH-60A UH-60L** . (Sheet 2 of 22)

SCHEMATICS - Continued

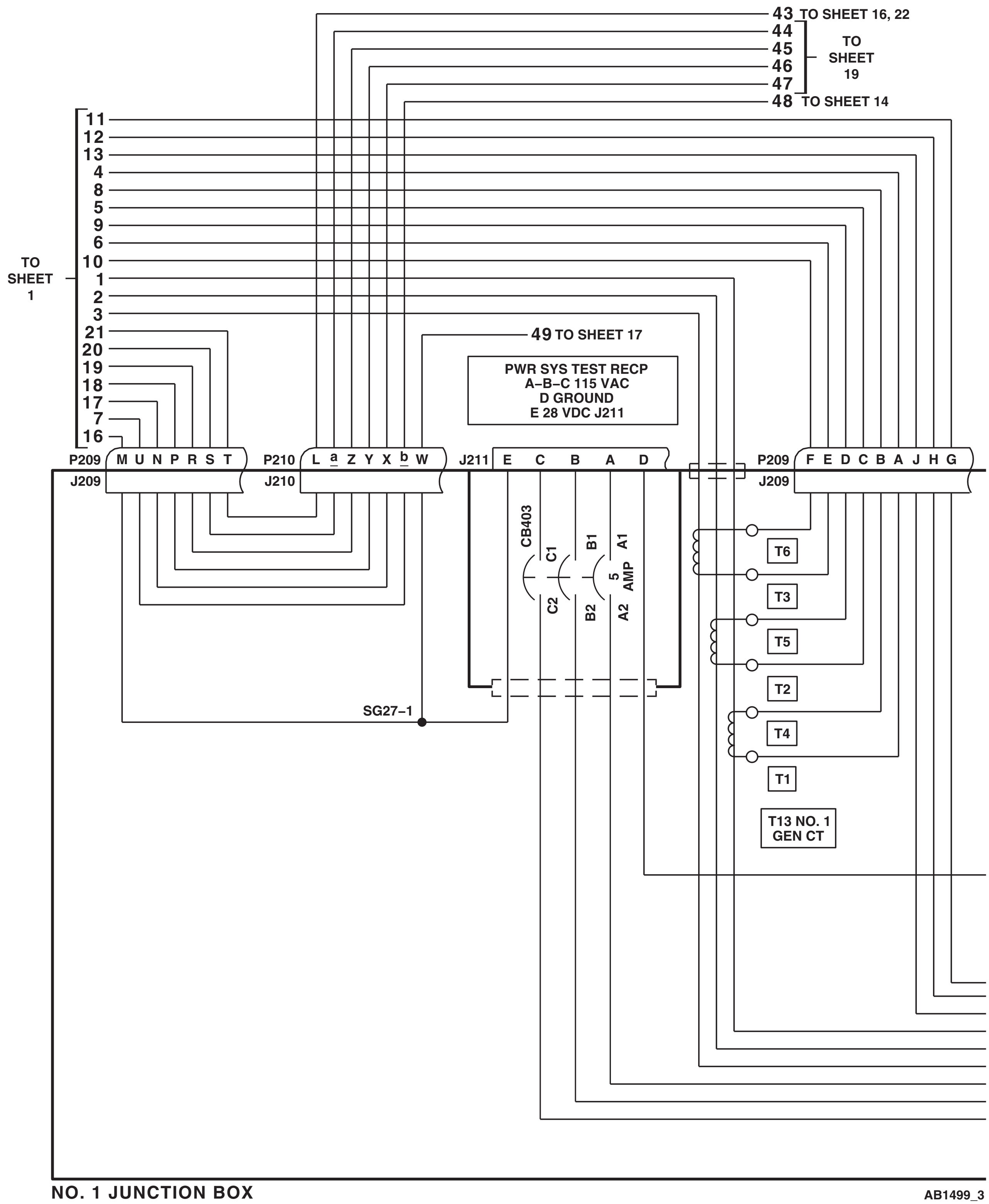


Figure 3. AC Electrical System Schematic Diagram **UH-60A UH-60L** . (Sheet 3 of 22)

SCHEMATICS - Continued

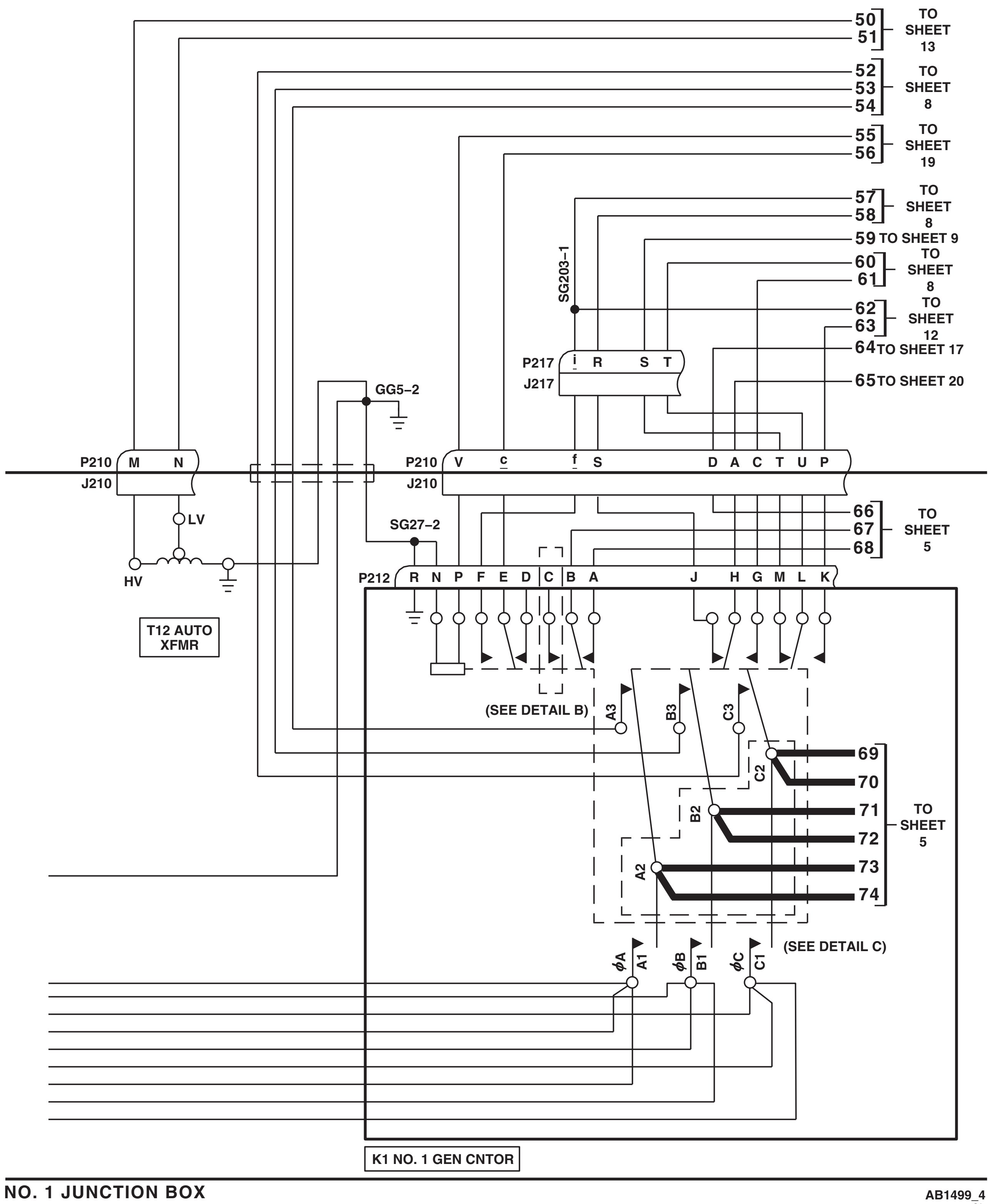


Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 4 of 22)

SCHEMATICS - Continued

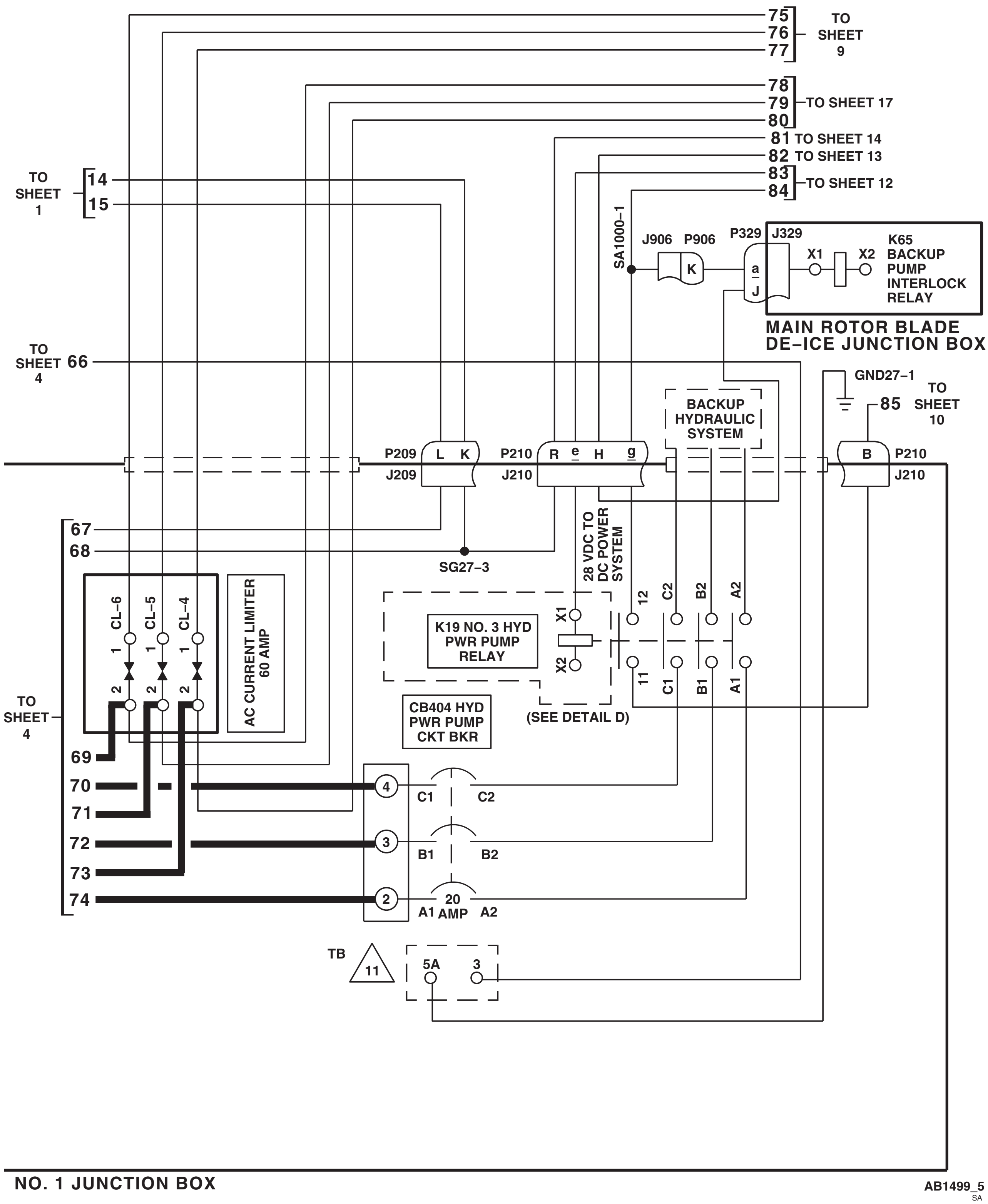


Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 5 of 22)

SCHEMATICS - Continued

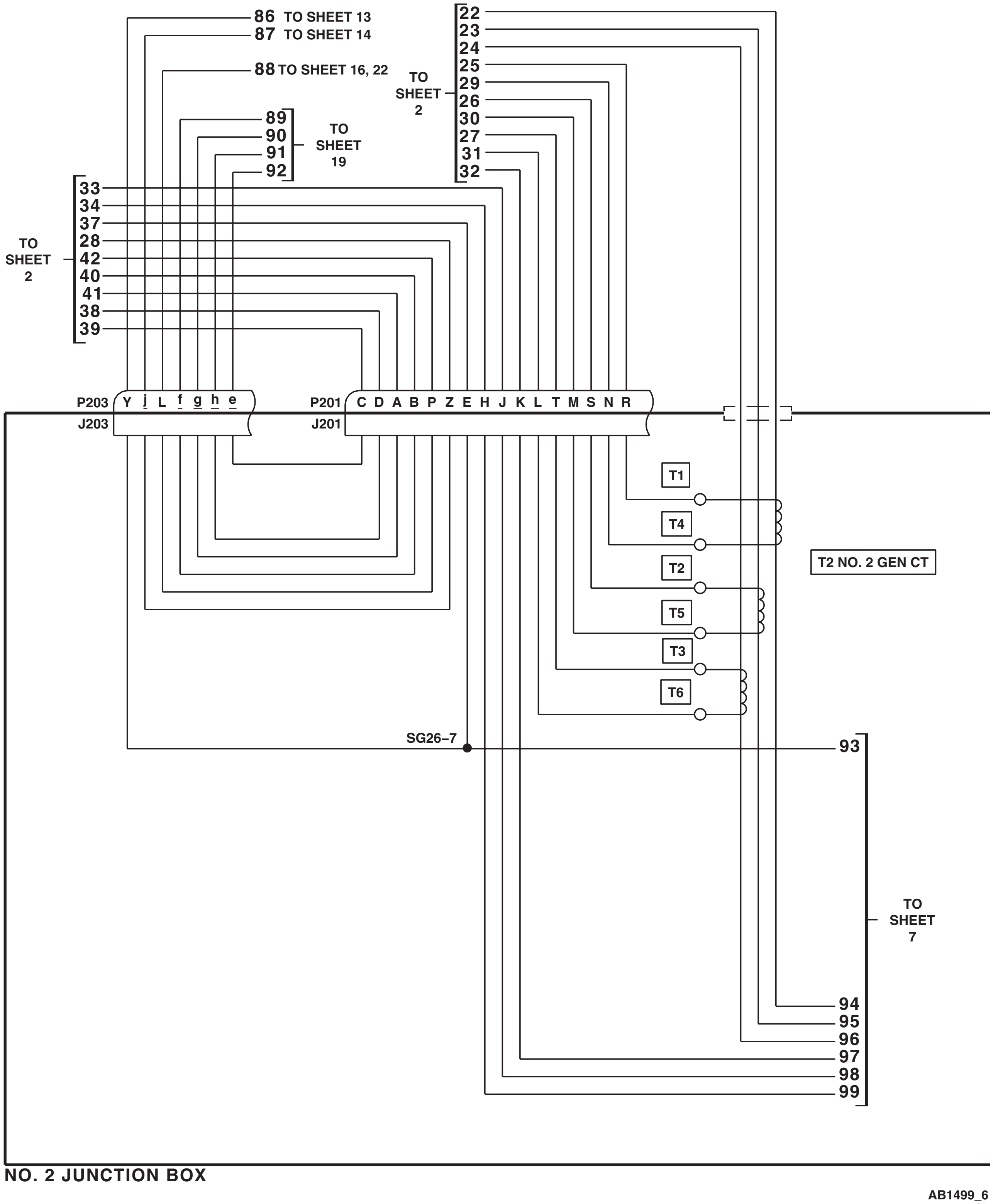


Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 6 of 22)

SCHEMATICS - Continued

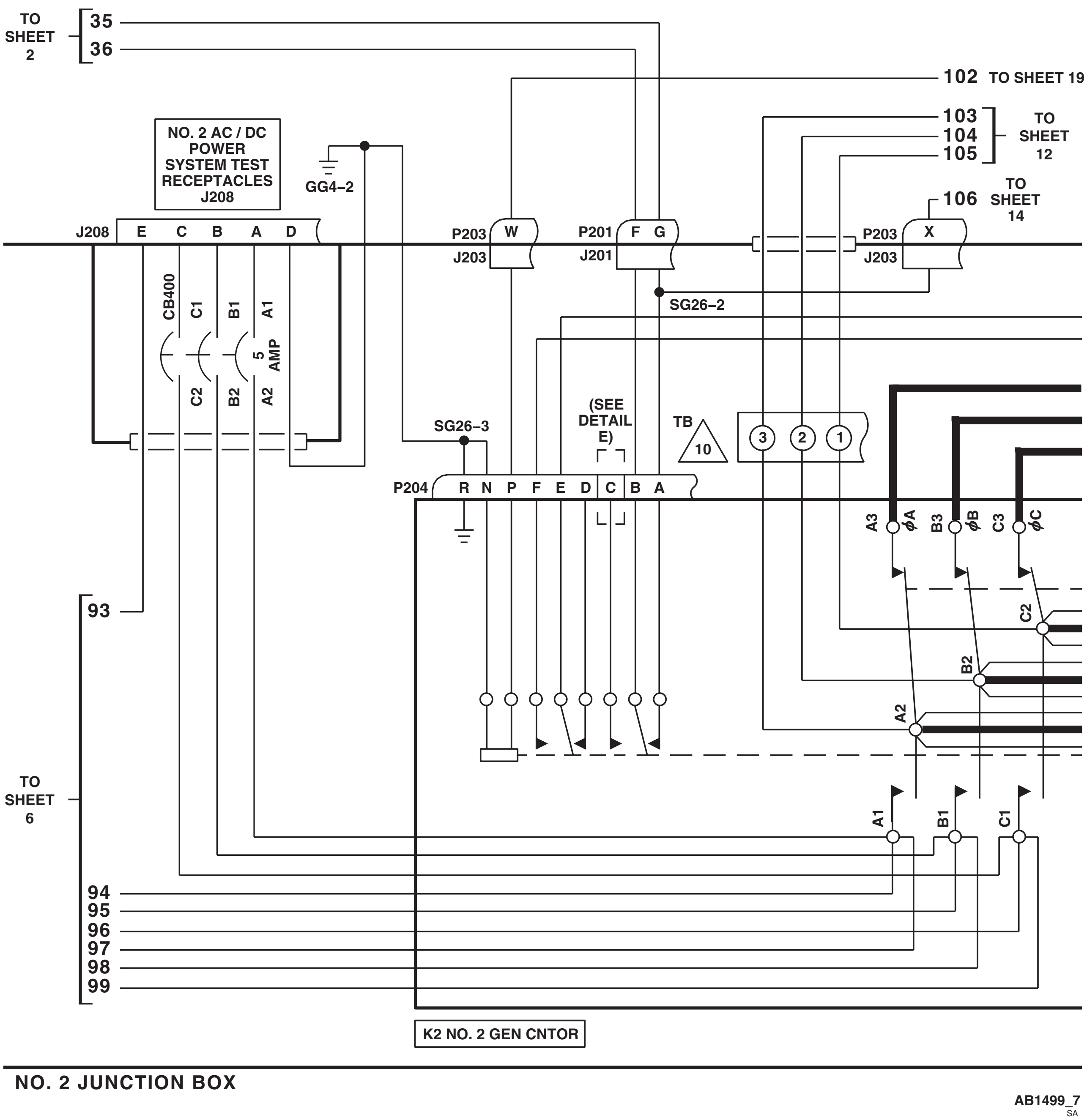


Figure 3. AC Electrical System Schematic Diagram **UH-60A UH-60L** . (Sheet 7 of 22)

SCHEMATICS - Continued

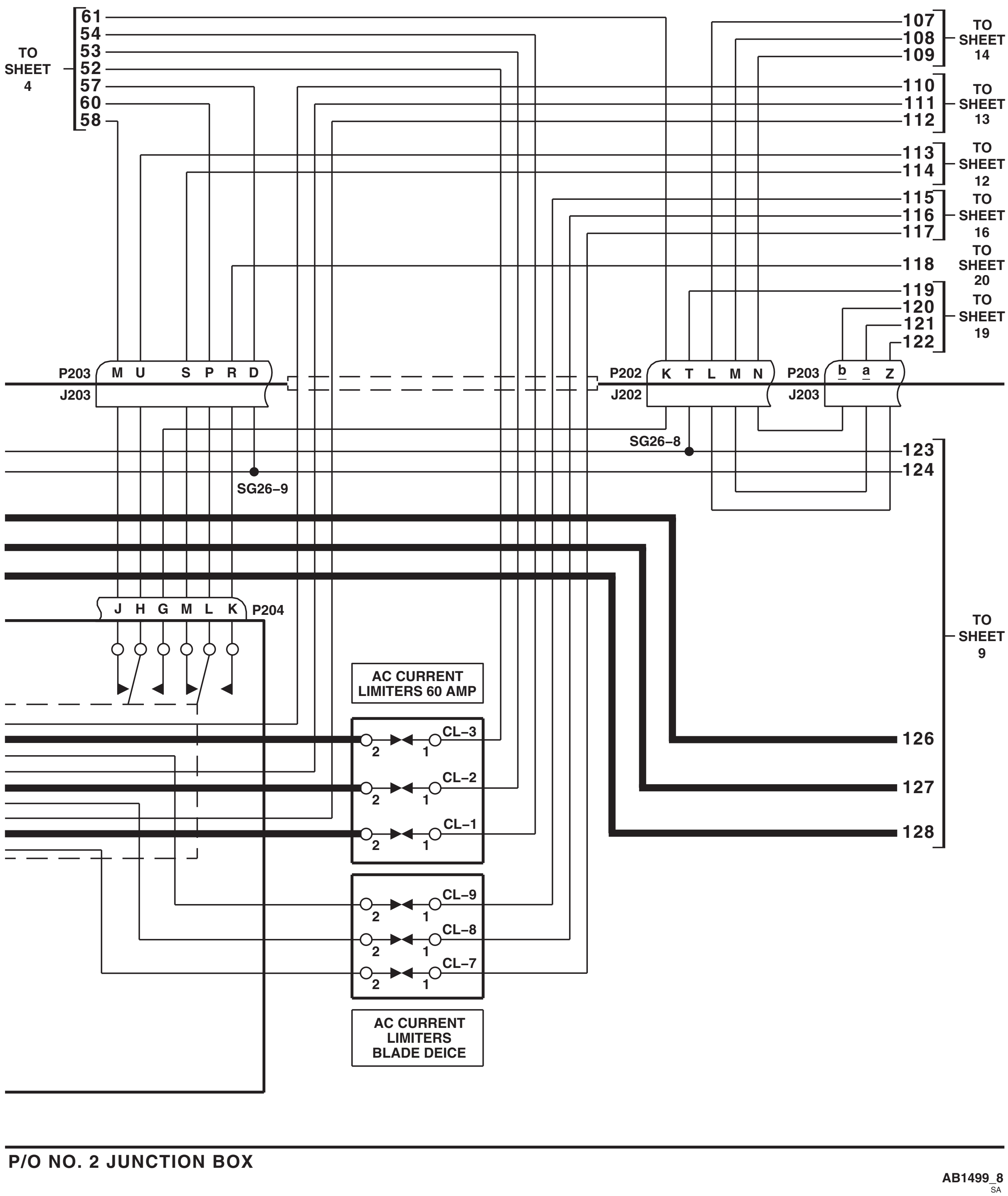


Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 8 of 22)

SCHEMATICS - Continued

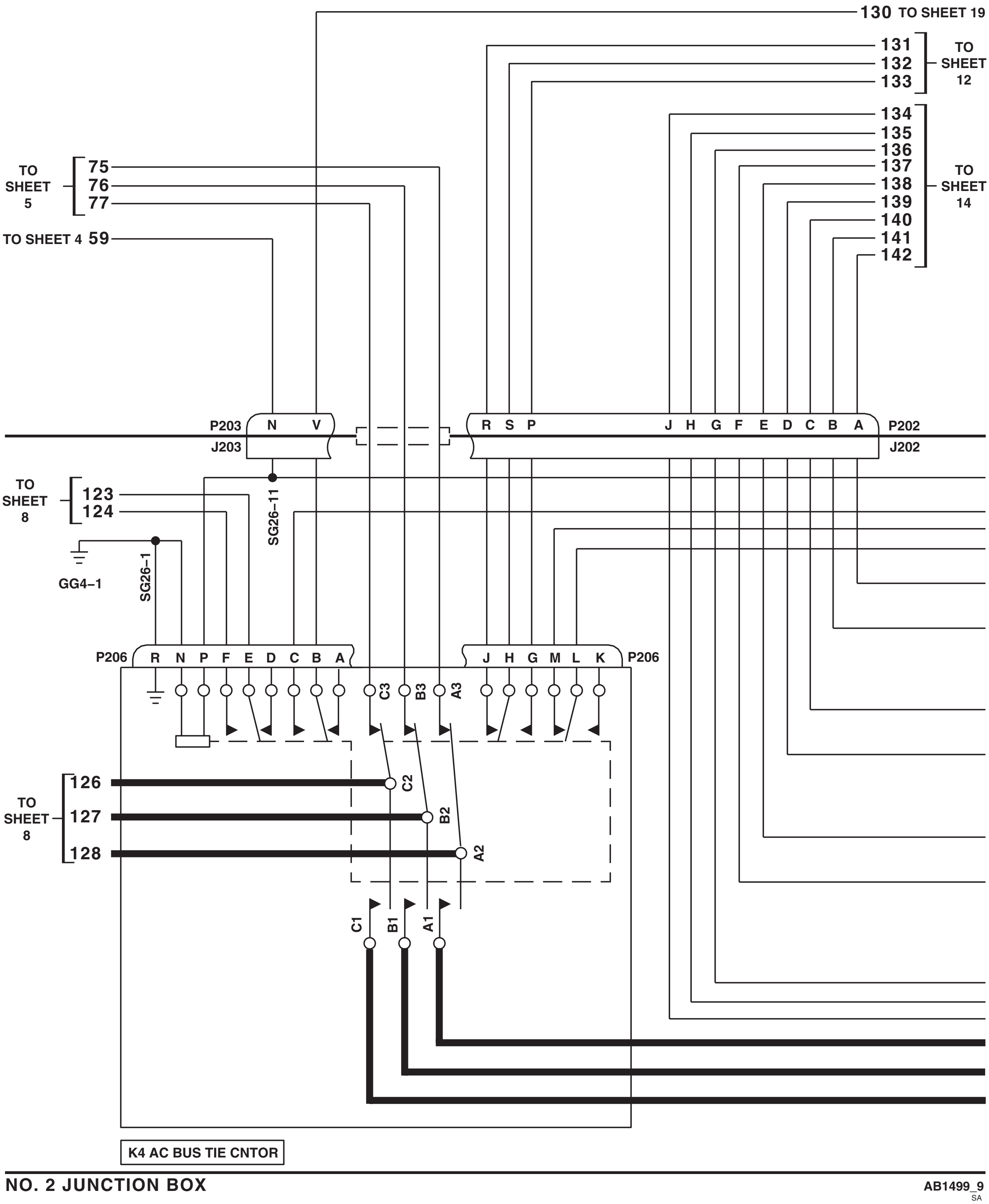


Figure 3. AC Electrical System Schematic Diagram **UH-60A UH-60L** . (Sheet 9 of 22)

SCHEMATICS - Continued

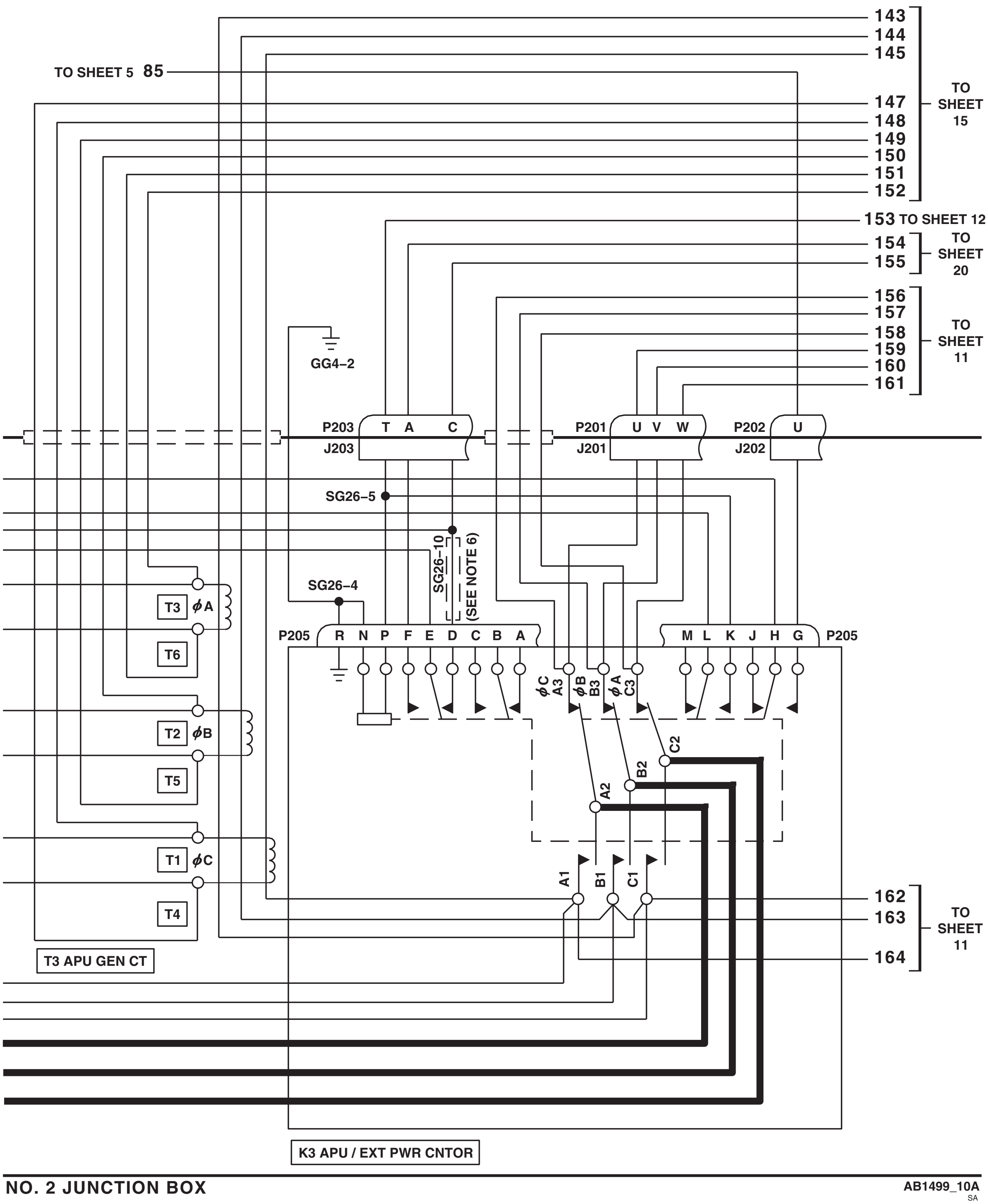


Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 10 of 22)

SCHEMATICS - Continued

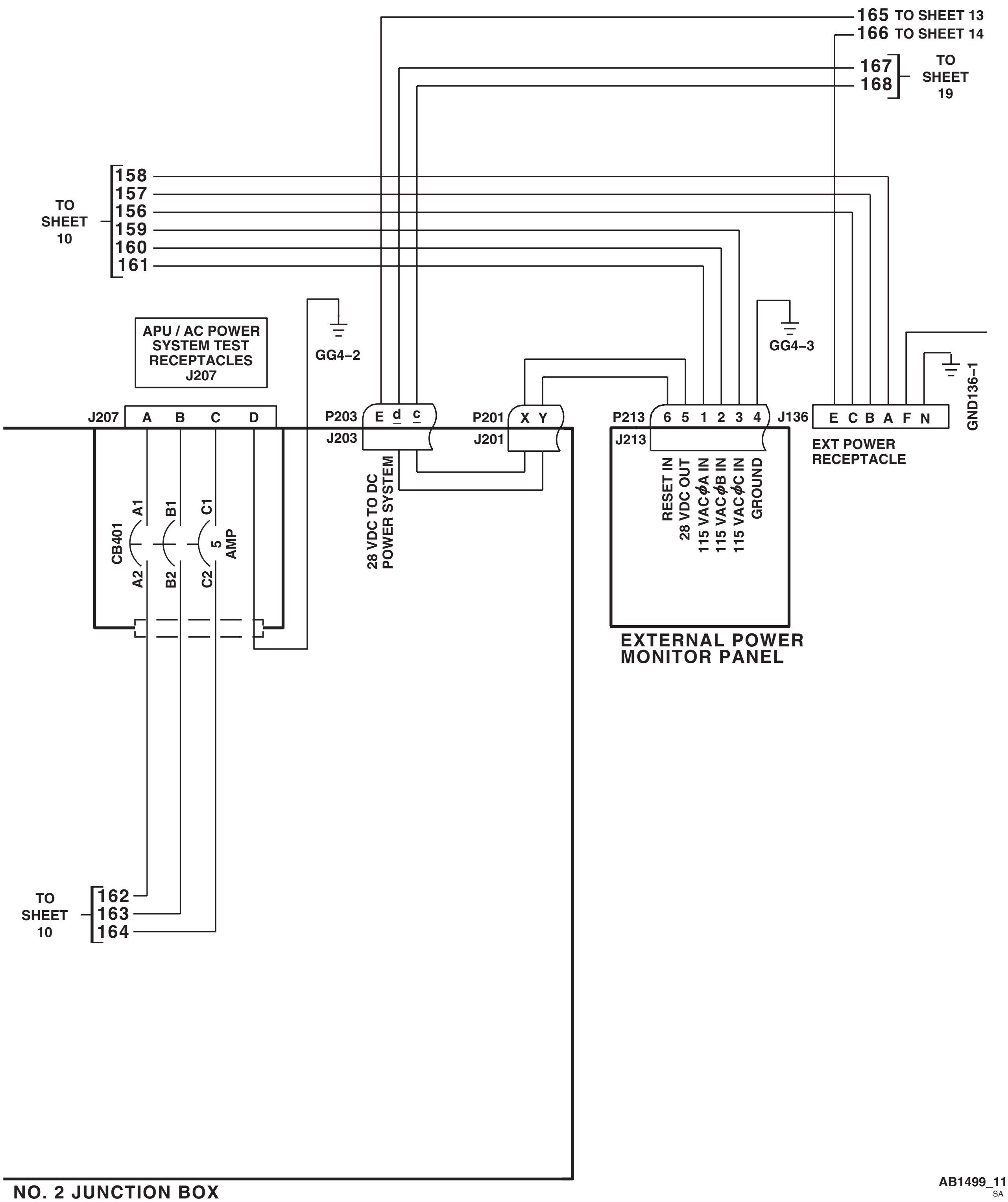


Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 11 of 22)

SCHEMATICS - Continued

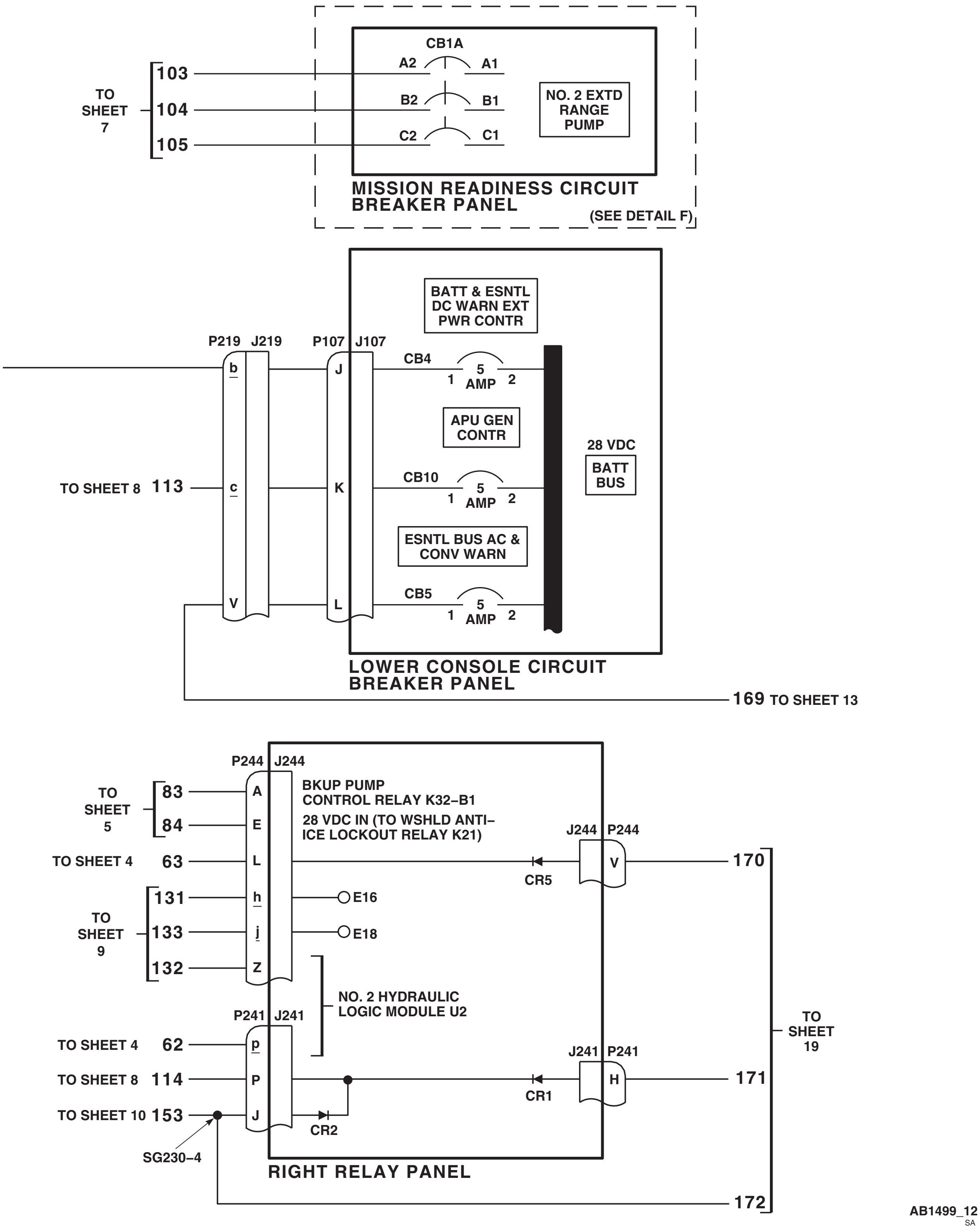


Figure 3. AC Electrical System Schematic Diagram **UH-60A UH-60L** . (Sheet 12 of 22)

SCHEMATICS - Continued

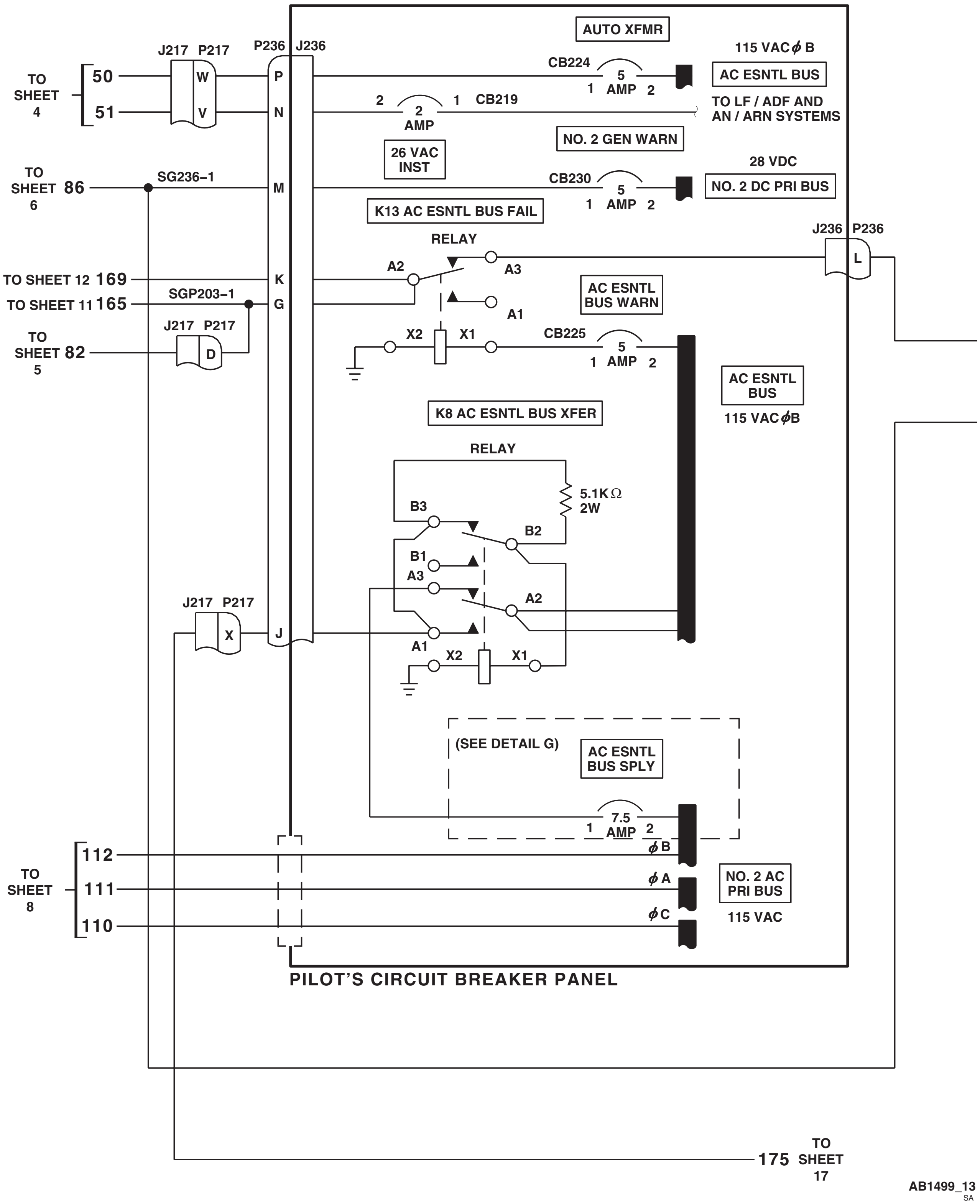


Figure 3. AC Electrical System Schematic Diagram **UH-60A UH-60L** . (Sheet 13 of 22)

SCHEMATICS - Continued

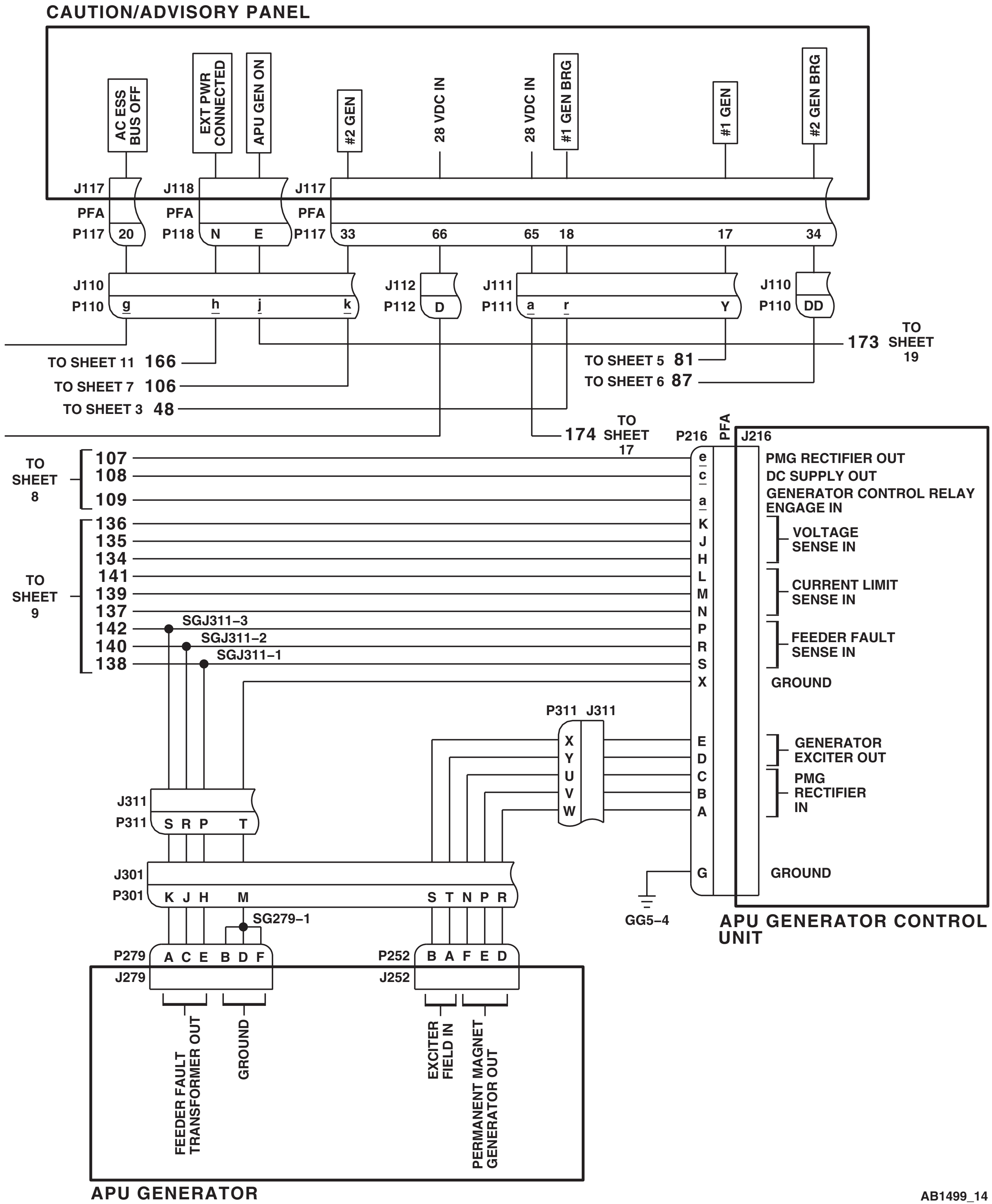
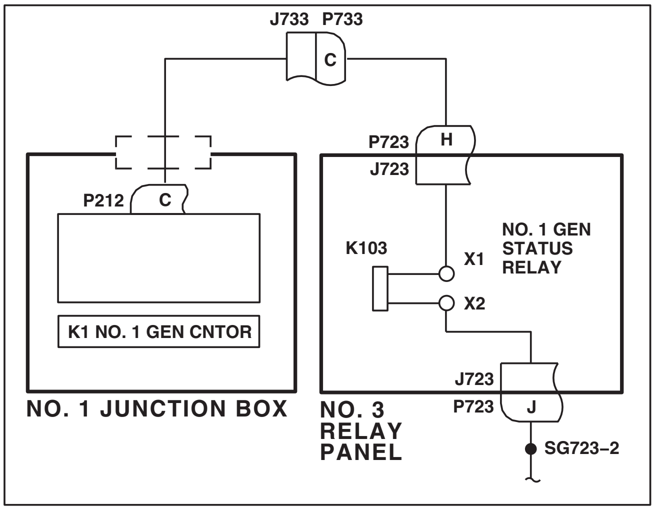
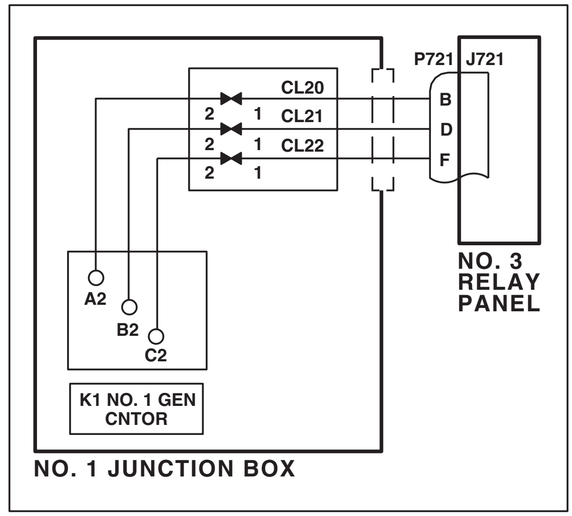


Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 14 of 22)

SCHEMATICS - Continued



DETAIL B
(SEE NOTE 3)



DETAIL C
(SEE NOTE 3)

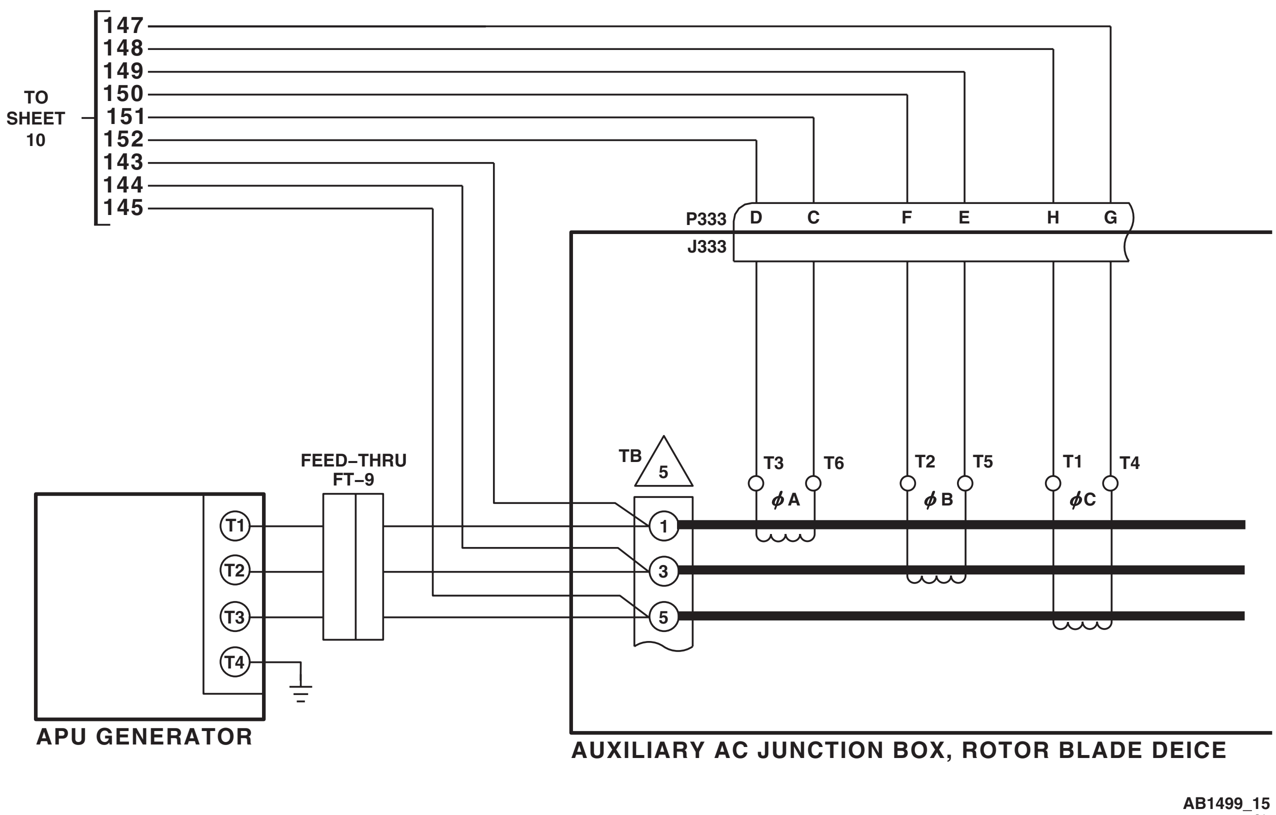


Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 15 of 22)

SCHEMATICS - Continued

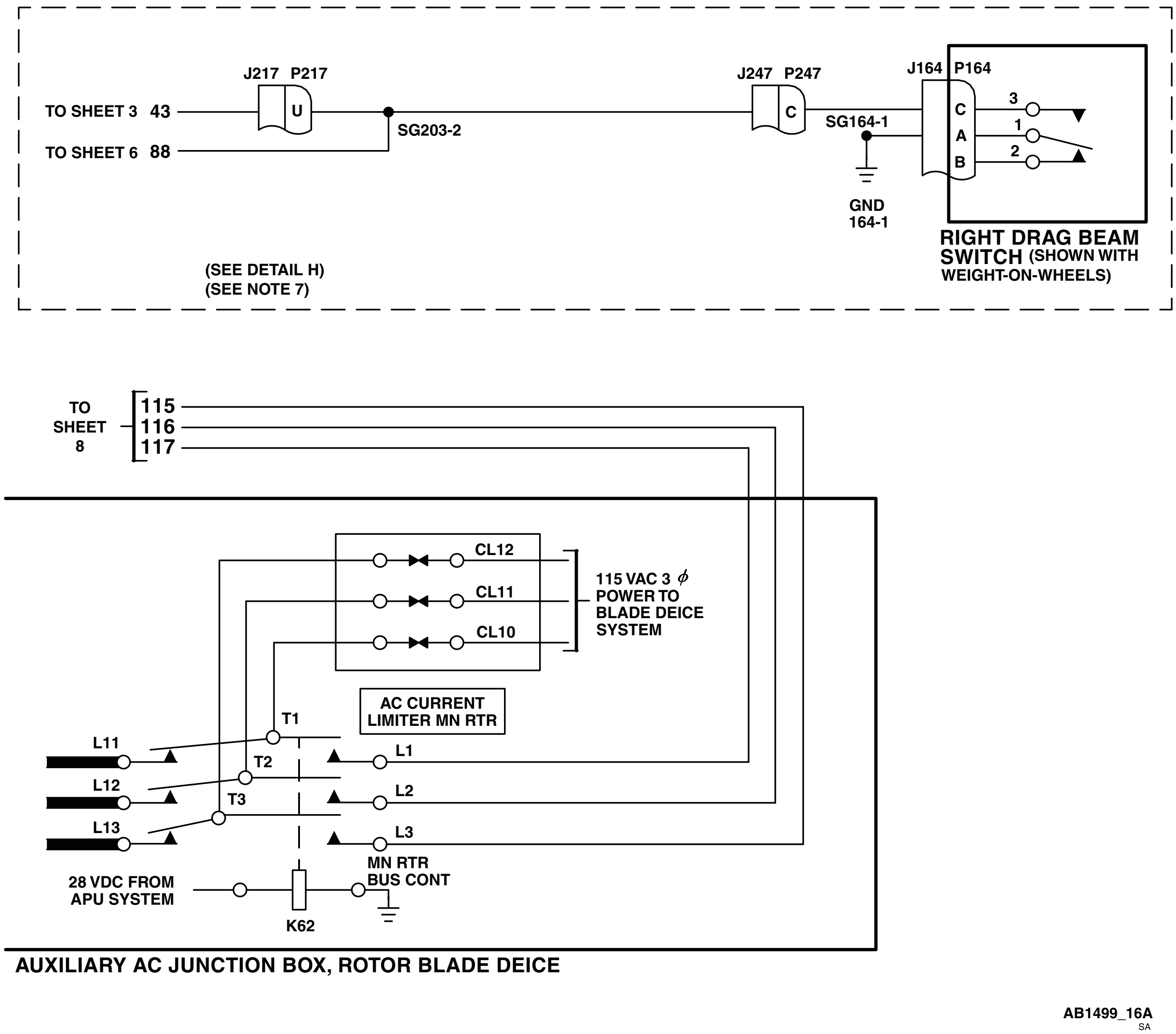


Figure 3. AC Electrical System Schematic Diagram **UH-60A UH-60L**. (Sheet 16 of 22)

SCHEMATICS - Continued

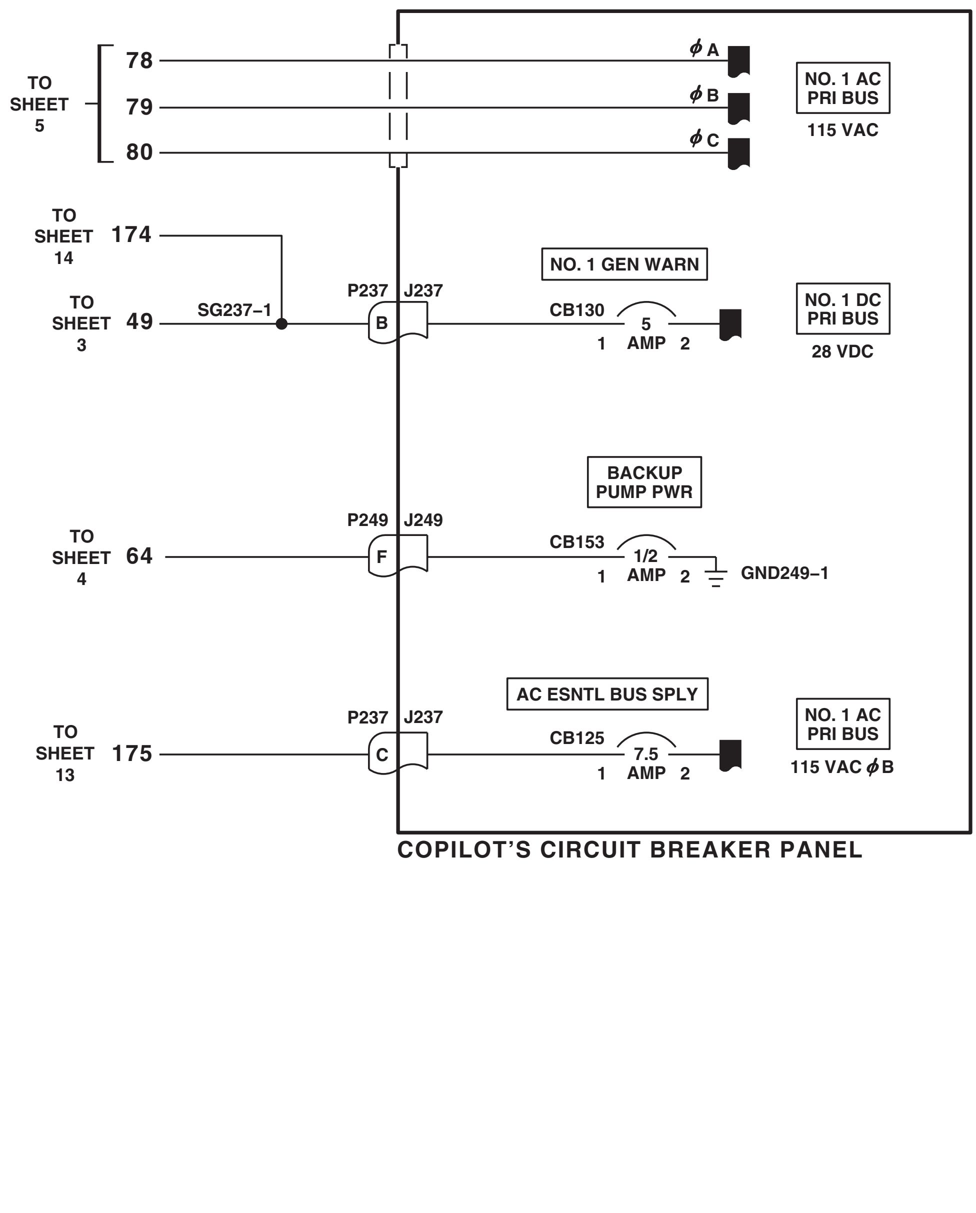
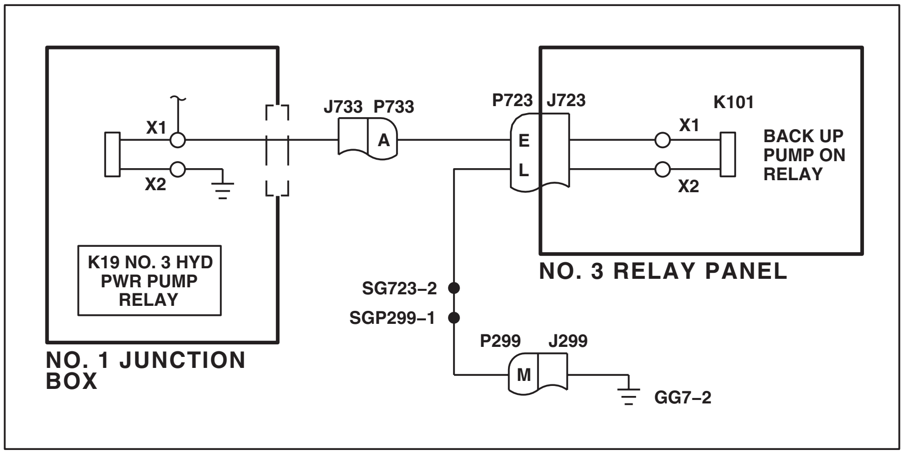
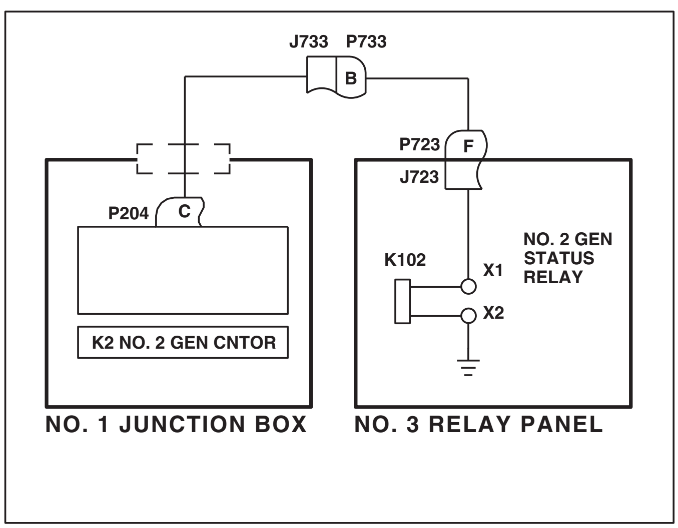


Figure 3. AC Electrical System Schematic Diagram **UH-60A UH-60L** . (Sheet 17 of 22)

SCHEMATICS - Continued



DETAIL D
(SEE NOTE 3)



DETAIL E
(SEE NOTE 3)

AB1499_18
SA

Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 18 of 22)

SCHEMATICS - Continued

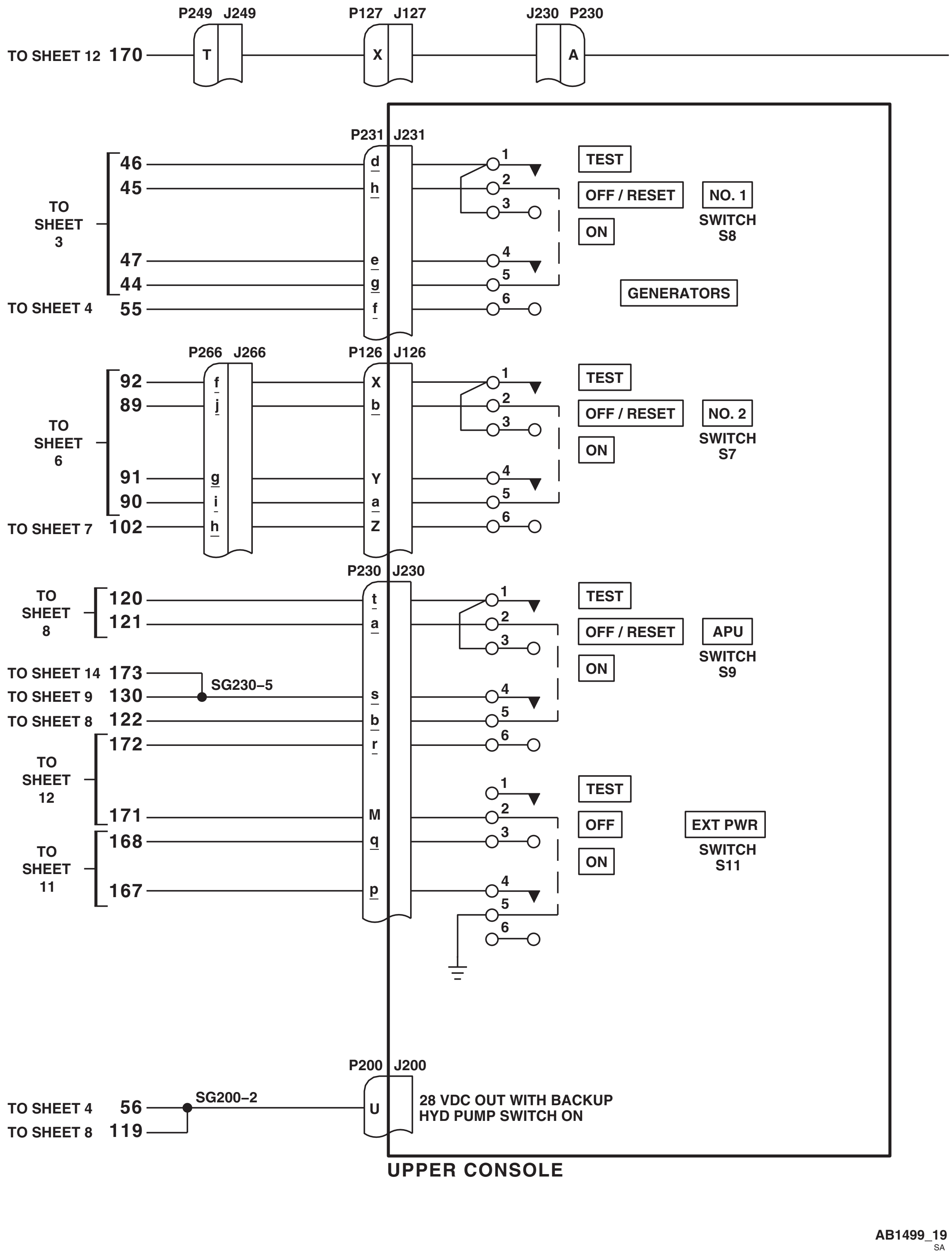


Figure 3. AC Electrical System Schematic Diagram **UH-60A UH-60L** . (Sheet 19 of 22)

SCHEMATICS - Continued

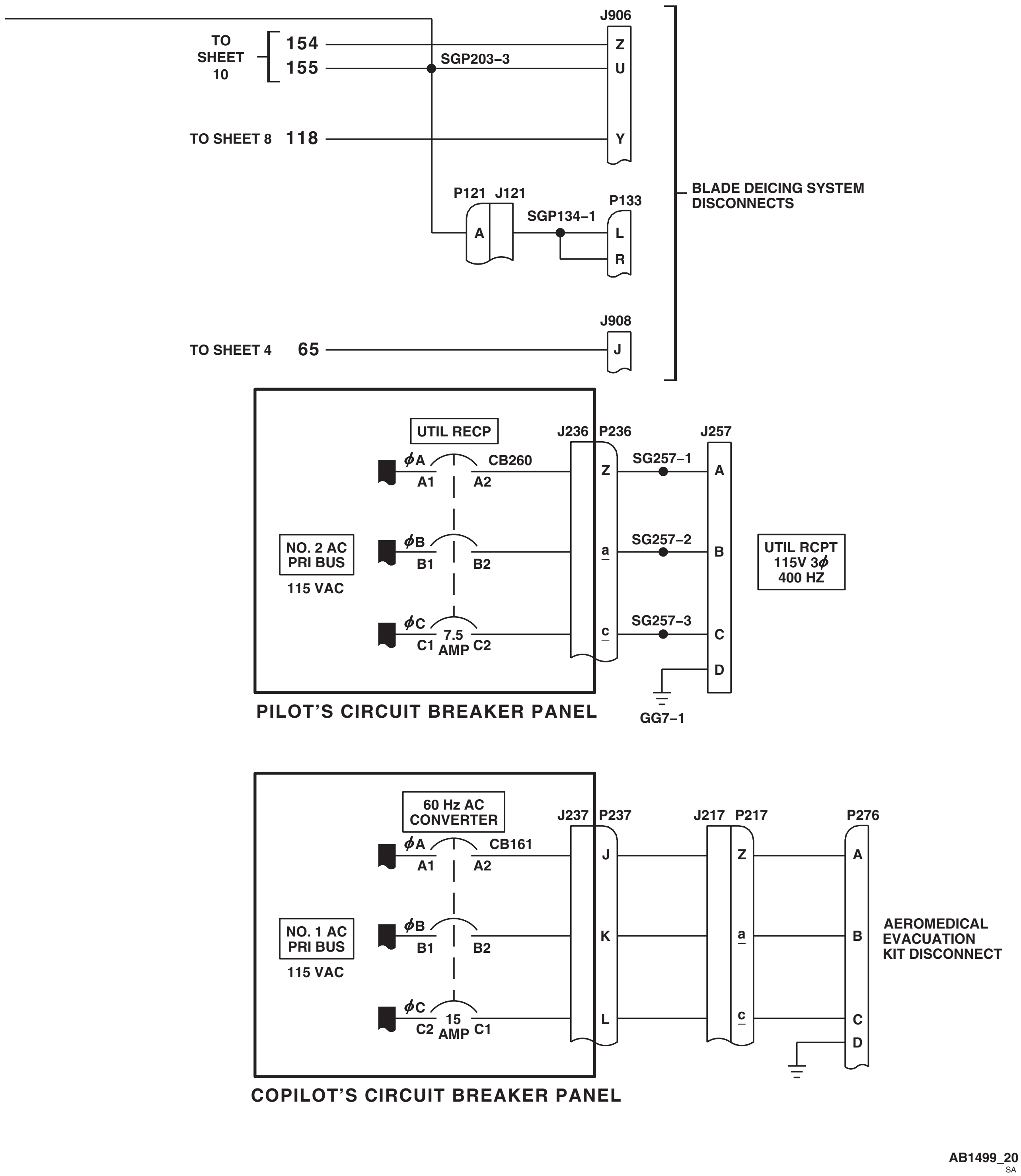
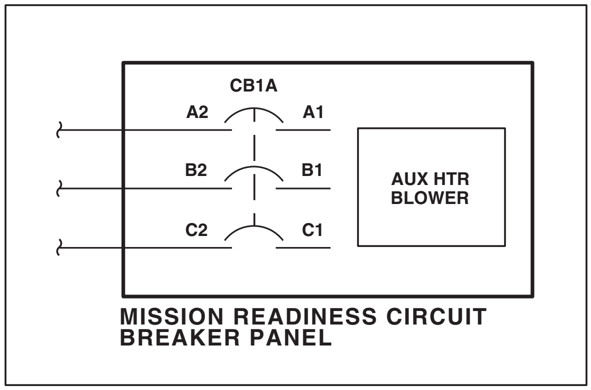
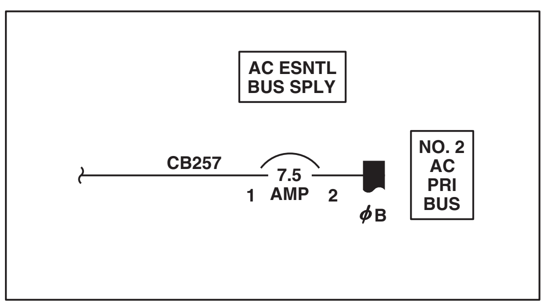


Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 20 of 22)

SCHEMATICS - Continued



DETAIL F
(SEE NOTE 3)



DETAIL G
(SEE NOTE 4)

AB1499_21
SA

Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 21 of 22)

SCHEMATICS - Continued

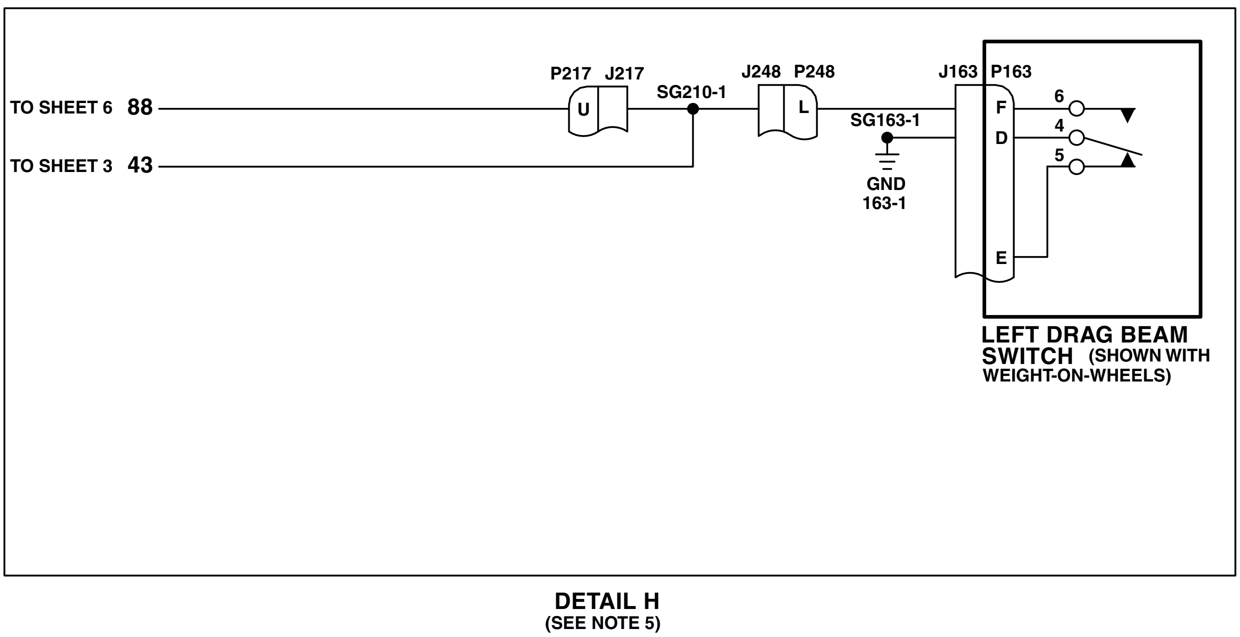


Figure 3. AC Electrical System Schematic Diagram UH-60A UH-60L . (Sheet 22 of 22)

SCHEMATICS - Continued

AC POWER SOURCE				CONDITION	AC SYSTEM CONTACTORS/RELAYS												
										K11					K80	K81	K82
A	B	C	D		K1	K2	K3	K4	K8	X1		Y1		K13	SEE NOTE 6		SEE NOTE 7
0	0	0	0	1	0	0	0	0	0	0		0		0	X	0	X
1	0	0	0	2	X	0	0	0	X	X	SEE NOTE 3	0		X	X	0	X
0	1	0	0	3	0	X	0	0	X	X	SEE NOTE 3	0		X	X	0	X
1	1	0	0	4	X	X	0	0	X	X		0		X	X	X	X
0	0	1	0	5	0	0	X	X	X	X	SEE NOTE 4	0		X	X	0	X
1	0	1	0	6	X	0	X	0	X	X	SEE NOTE 3	X	SEE NOTE 5	X	X	0	X
0	1	1	0	7	0	X	X	0	X	X	SEE NOTE 3	X	SEE NOTE 5	X	X	0	X
1	1	1	0	8	X	X	X	0	X	X		0		X	X	X	X
0	0	0	1	9	0	0	0	X	X	X	SEE NOTE 4	0		X	X	0	X
1	0	0	1	10	X	0	0	0	X	X	SEE NOTE 3	0		X	X	0	X
0	1	0	1	11	0	X	0	0	X	X	SEE NOTE 3	0		X	X	0	X
1	1	0	1	12	X	X	0	0	X	X		0		X	X	X	X
0	0	1	1	13	0	0	X	X	X	X	SEE NOTE 4	0		X	X	0	X
1	0	1	1	14	X	0	X	0	X	X	SEE NOTE 3	X	SEE NOTE 5	X	X	0	X
0	1	1	1	15	0	X	X	0	X	X	SEE NOTE 3	X	SEE NOTE 5	X	X	0	X
1	1	1	1	16	X	X	X	0	X	X		0		X	X	X	X

EFFECTIVITY

EH60A

NOTES

1. W/O EMEP PIN FILTERED ADAPTERS (PFA) ARE NOT INSTALLED.
2. PIN A OF P215 SHOWS TYPICAL CONFIGURATION FOR ALL PIN FILTERED ADAPTERS.
3. X INDICATES ENERGIZED WITH BACKUP PUMP OFF.
4. X INDICATES ENERGIZED WITH WEIGHT–ON–WHEELS AND BACKUP PUMP OFF.
5. X INDICATES ENERGIZED WITH BACKUP PUMP ON AND BLADE DEICE OFF.
6. X INDICATES ENERGIZED WHEN AIRBORNE AND HYD PRESSURE IS LESS THAN 2000 PSI, OF WHEN WEIGHT–ON–WHEELS AND BACKUP PUMP IS ON.
7. INDICATES ENERGIZED WHEN WEIGHT–OFF–WHEELS.
8. EH60A 84–24017 – 86–24567
9. EH60A 84–24017 – 86–24560
10. GROUND IS APPLIED TO X2 OF K19 WHEN WEIGHT–ON–WHEELS AND BACKUP PUMP IS NOT OVERHEATED, OR WHEN HELICOPTER IS AIRBORNE.
11. EH60A 84–24017 – 86–24476
12. EH60A 84–24017 – 86–24663

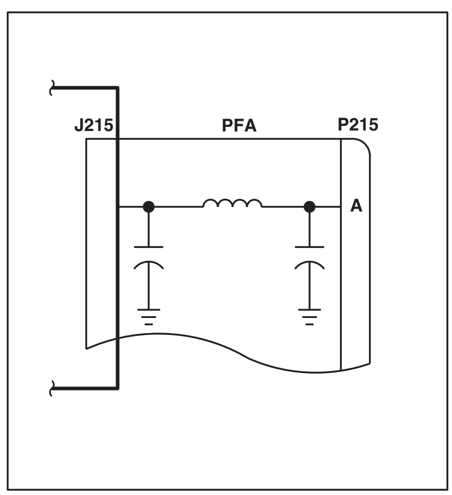
LEGEND

POWER SOURCE

A=NO. 1 MAIN GENERATOR
B=NO. 2 MAIN GENERATOR
C=APU GENERATOR
D=EXTERNAL AC POWER
1=TURNED ON AND OPERATING
0=TURNED OFF OR FAILED

CONTACTORS/RELAYS

K1=NO. 1 GENERATOR CONTACTOR
K2=NO. 2 GENERATOR CONTACTOR
K3=APU / EXT PWR CONTACTOR
K4=AC BUS TIE CONTACTOR
K8=AC ESNTL BUS XFR RELAY
K11=AC SECONDARY BUS CONTACTOR
K13=AC ESNTL BUS FAIL RELAY
K80=BACK–UP PUMP INTERLOCK
K81=GENERATORS ON RELAY
K82=SECONDARY BUS GND PWR CONTROL RELAY
X=ENERGIZED
0=DE–ENERGIZED



DETAIL A
(SEE NOTES 1 AND 2)

AB1500_1A
SA

Figure 4. AC Electrical System Schematic Diagram EH-60A . (Sheet 1 of 22)

SCHEMATICS - Continued

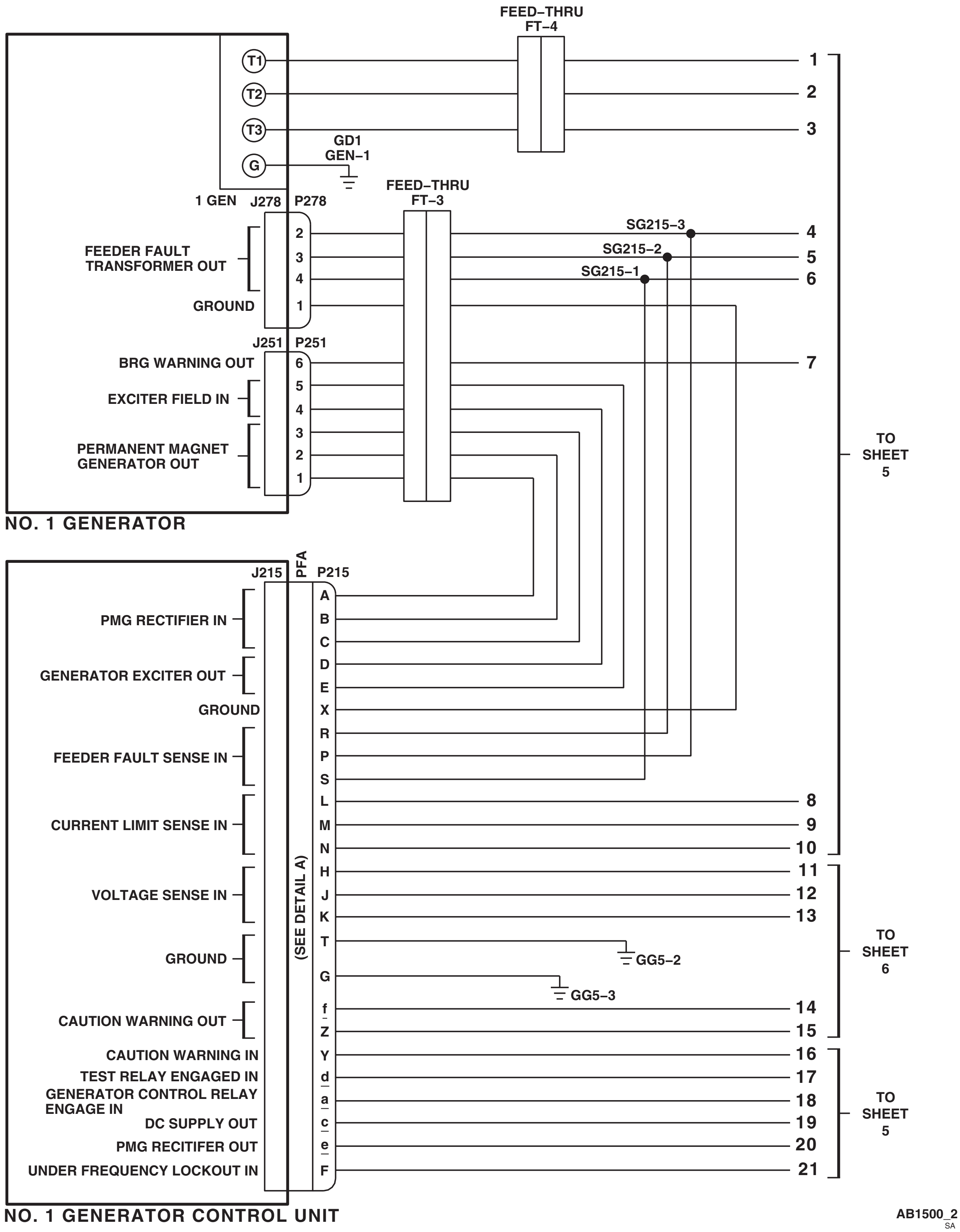


Figure 4. AC Electrical System Schematic Diagram EH-60A . (Sheet 2 of 22)

SCHEMATICS - Continued

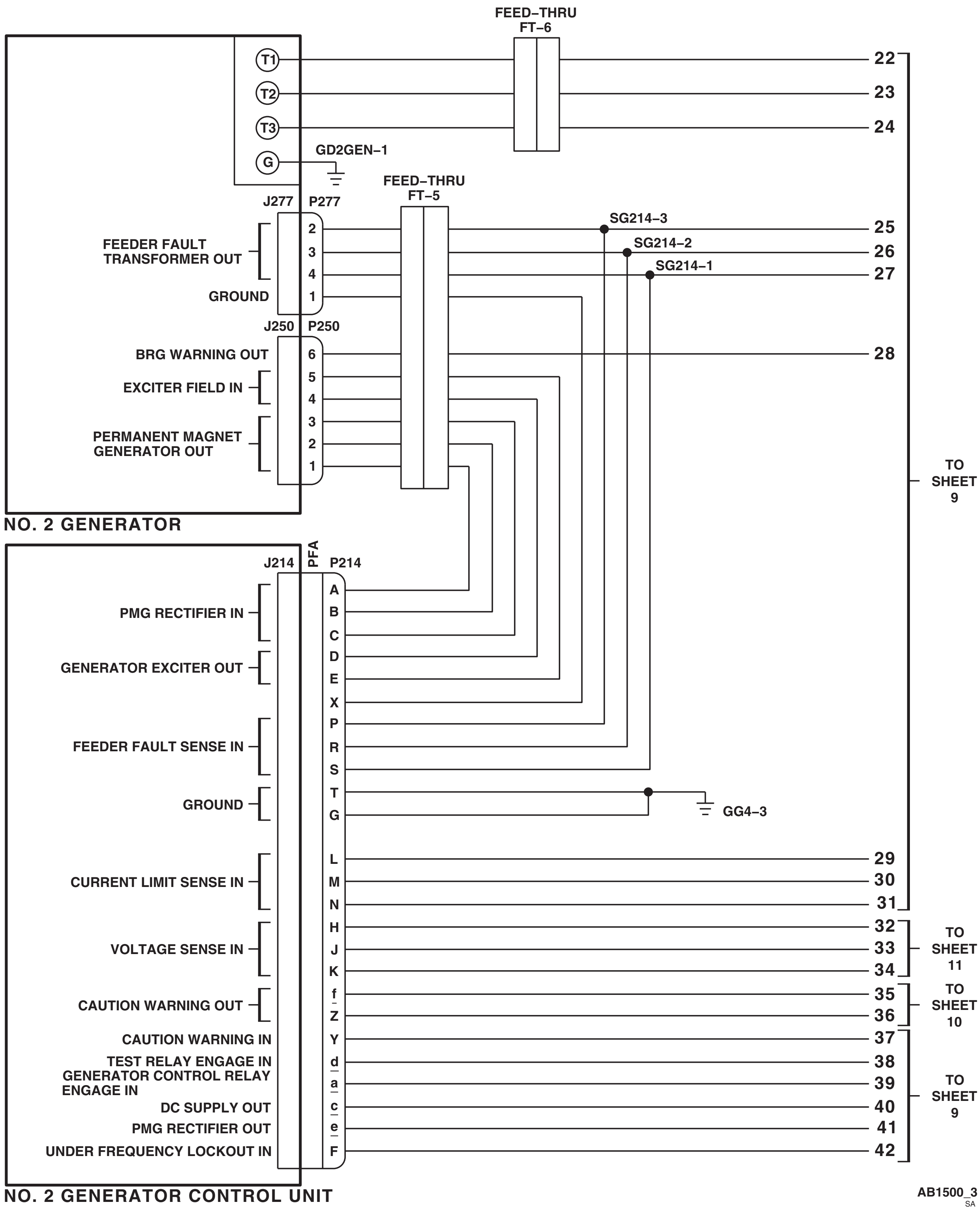


Figure 4. AC Electrical System Schematic Diagram EH-60A . (Sheet 3 of 22)

SCHEMATICS - Continued

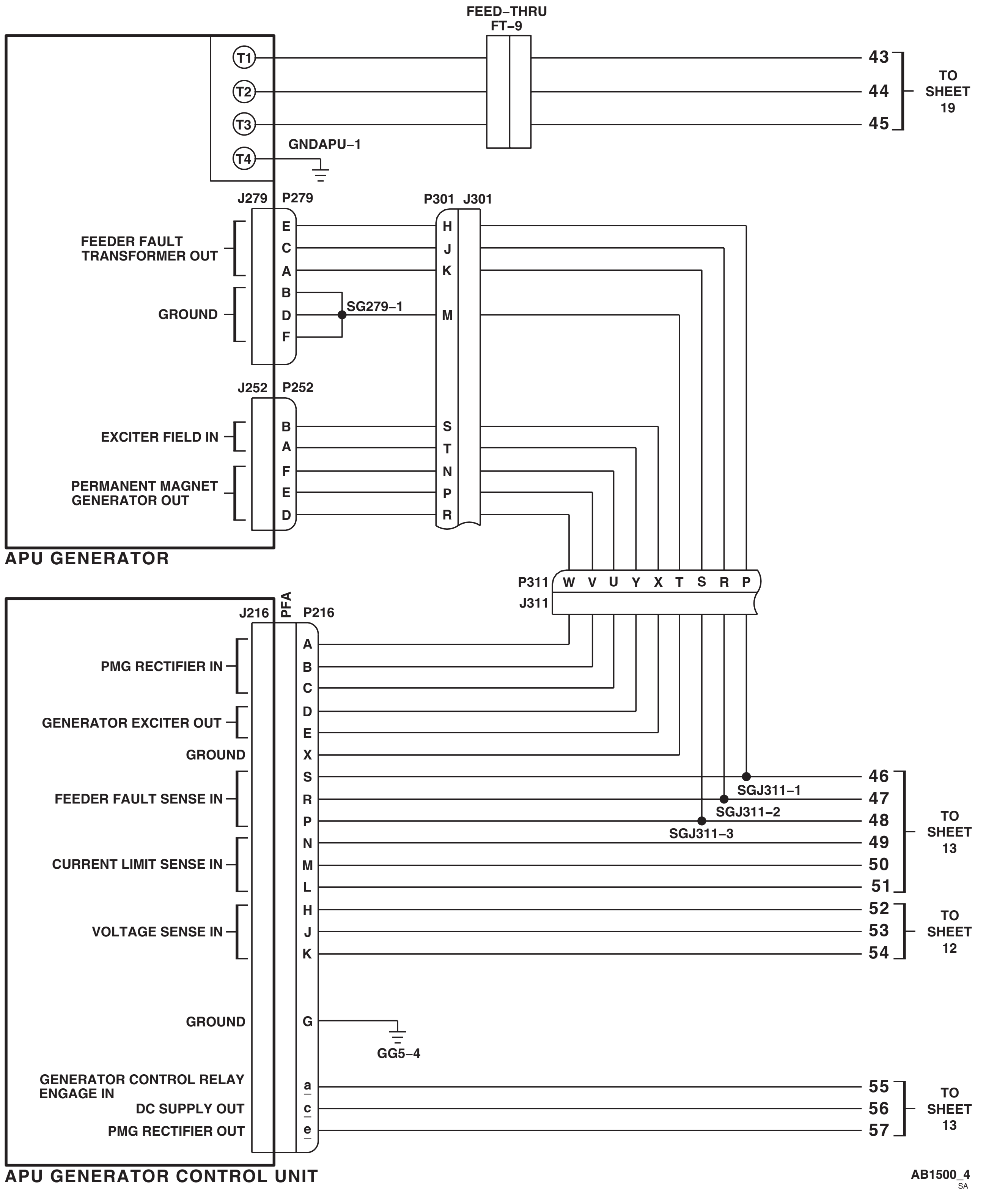


Figure 4. AC Electrical System Schematic Diagram **EH-60A** . (Sheet 4 of 22)

SCHEMATICS - Continued

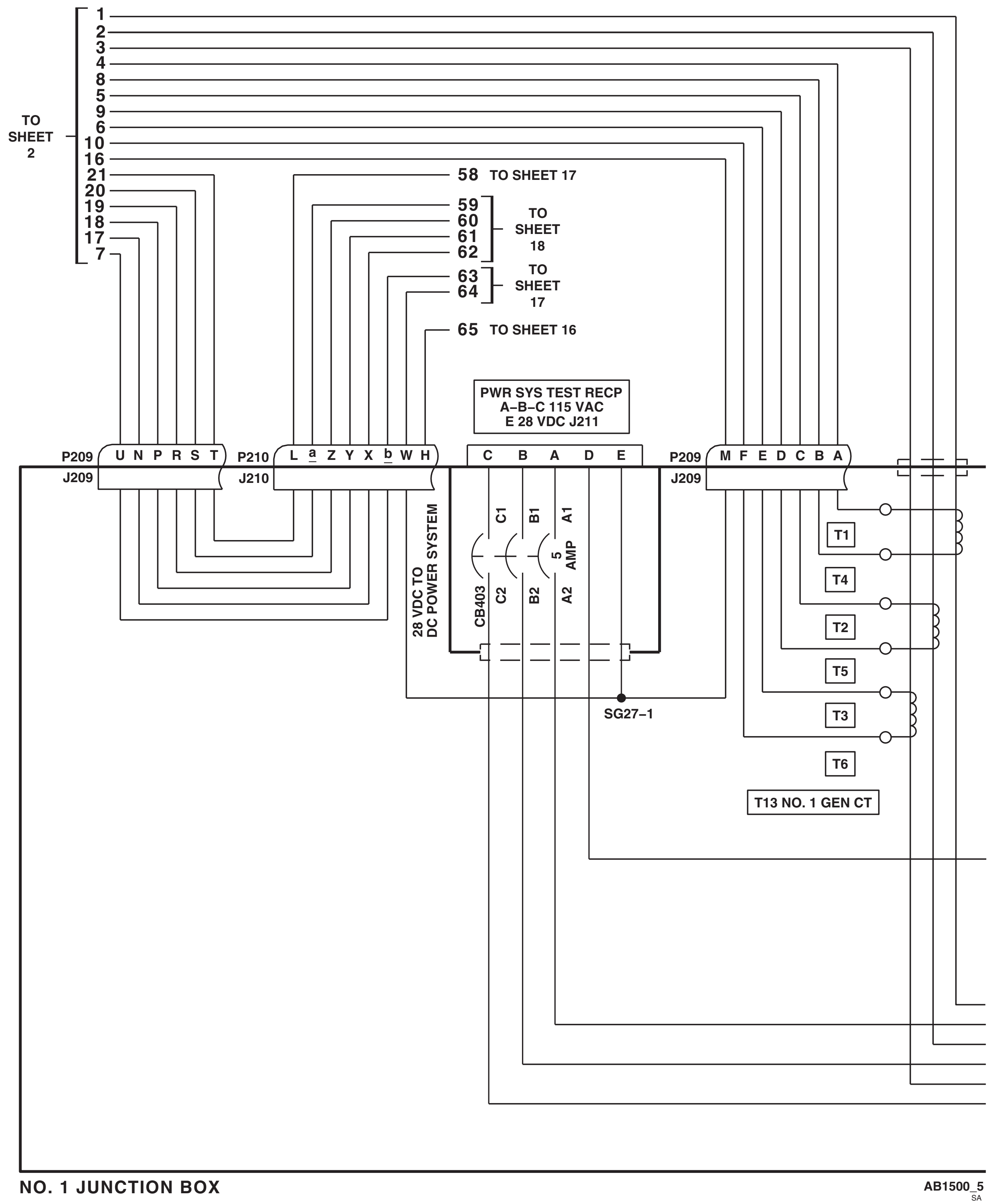


Figure 4. AC Electrical System Schematic Diagram EH-60A . (Sheet 5 of 22)

SCHEMATICS - Continued

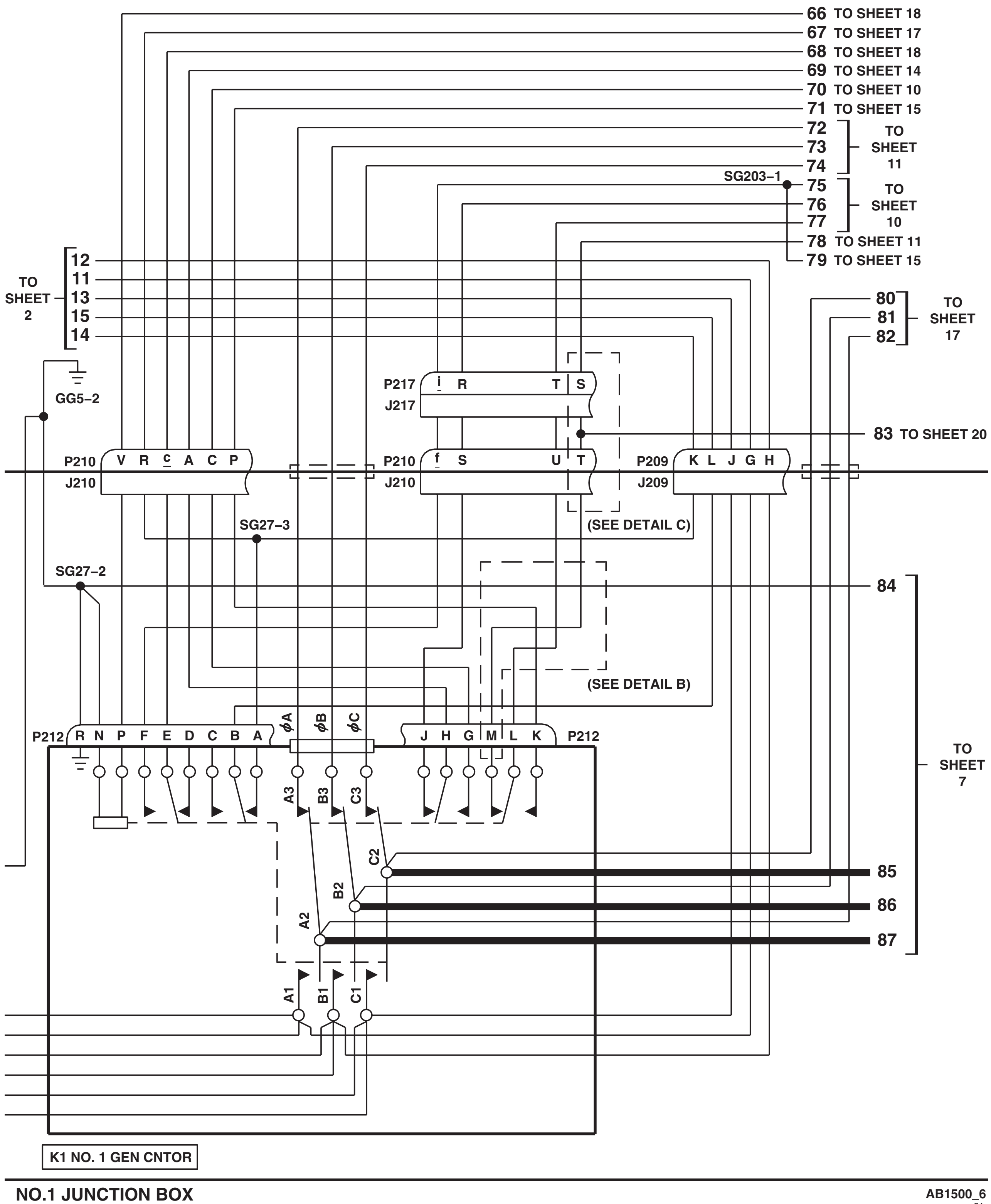


Figure 4. AC Electrical System Schematic Diagram EH-60A . (Sheet 6 of 22)

SCHEMATICS - Continued

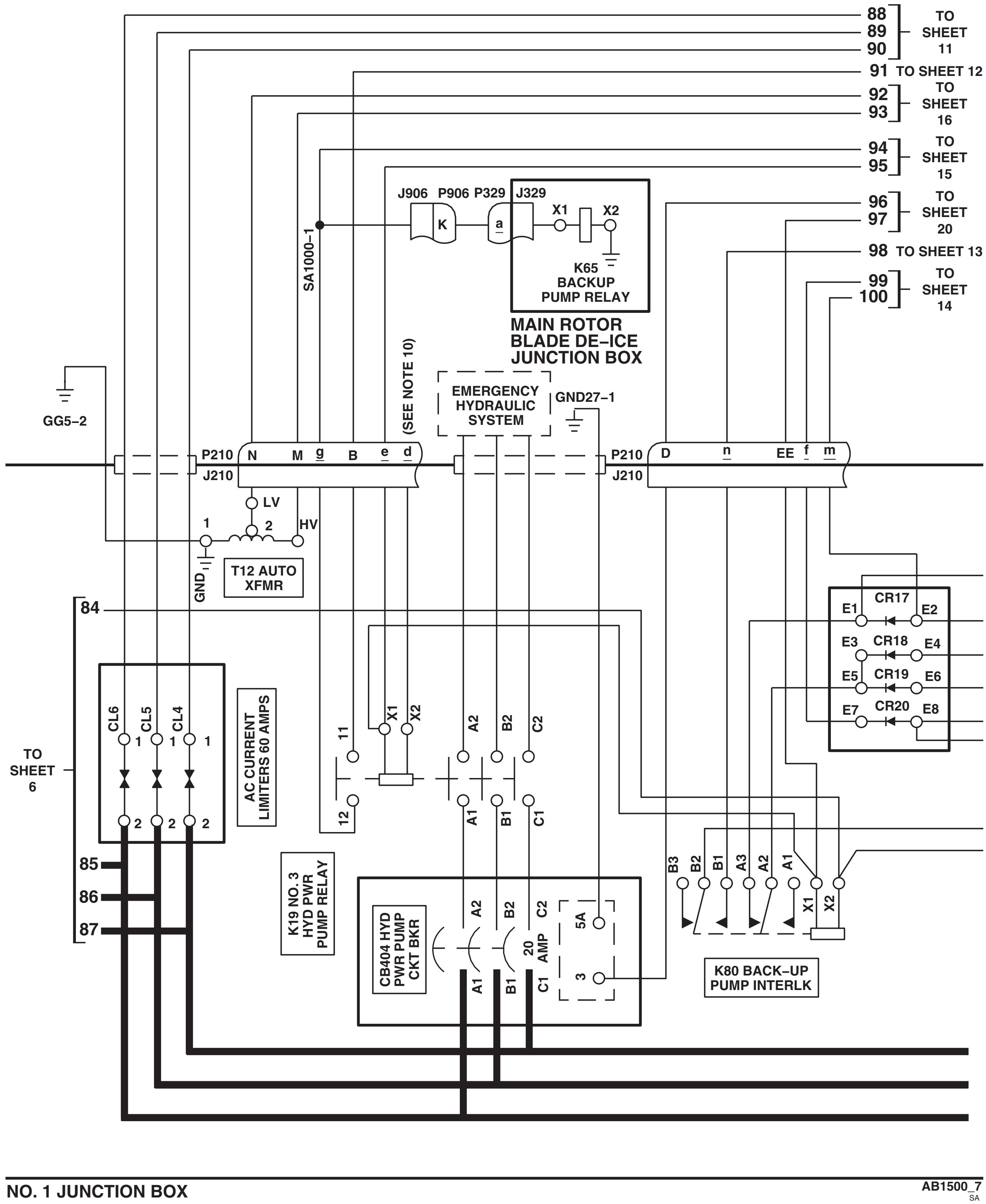


Figure 4. AC Electrical System Schematic Diagram **EH-60A** . (Sheet 7 of 22)

SCHEMATICS - Continued

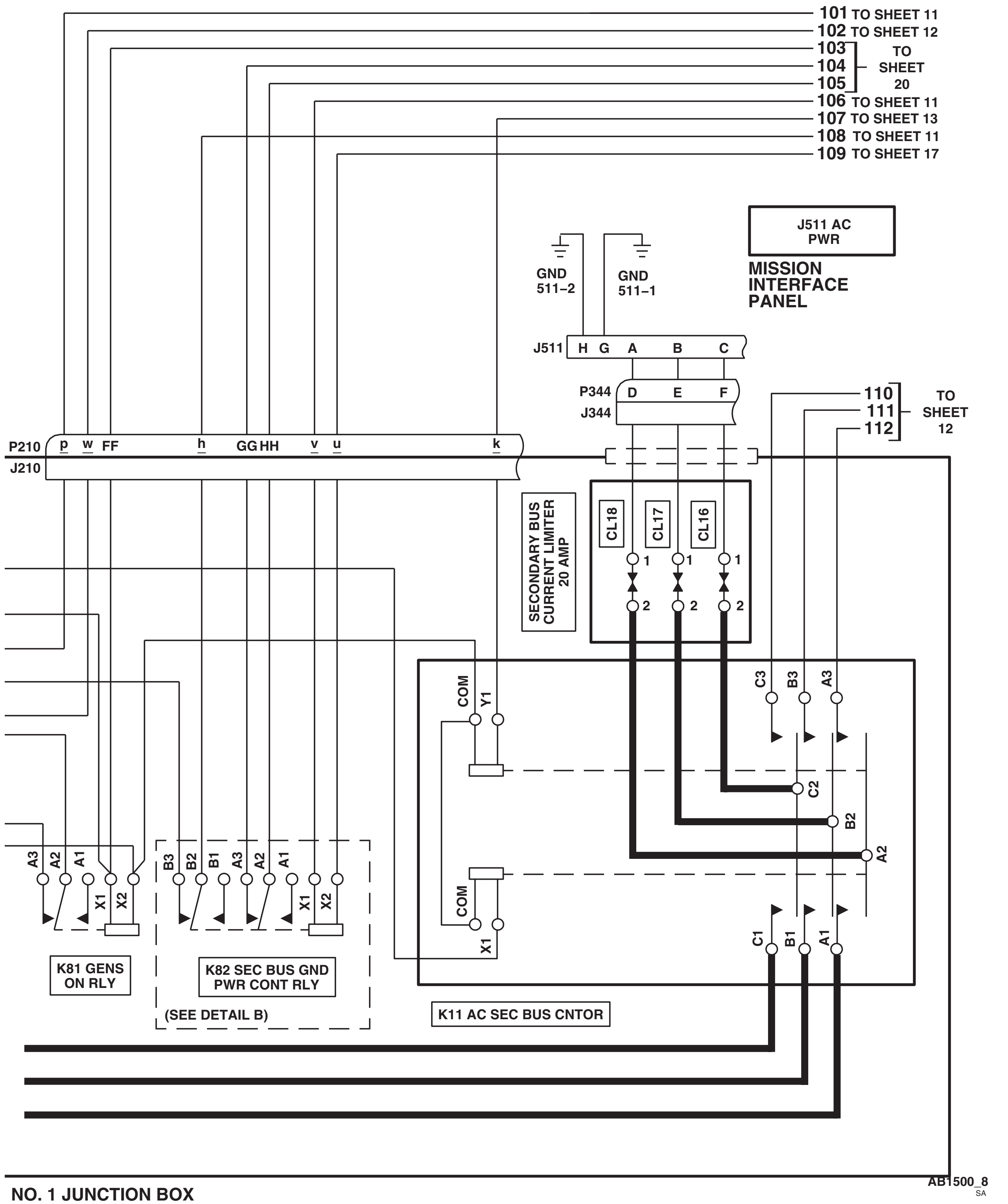


Figure 4. AC Electrical System Schematic Diagram **EH-60A** . (Sheet 8 of 22)

SCHEMATICS - Continued

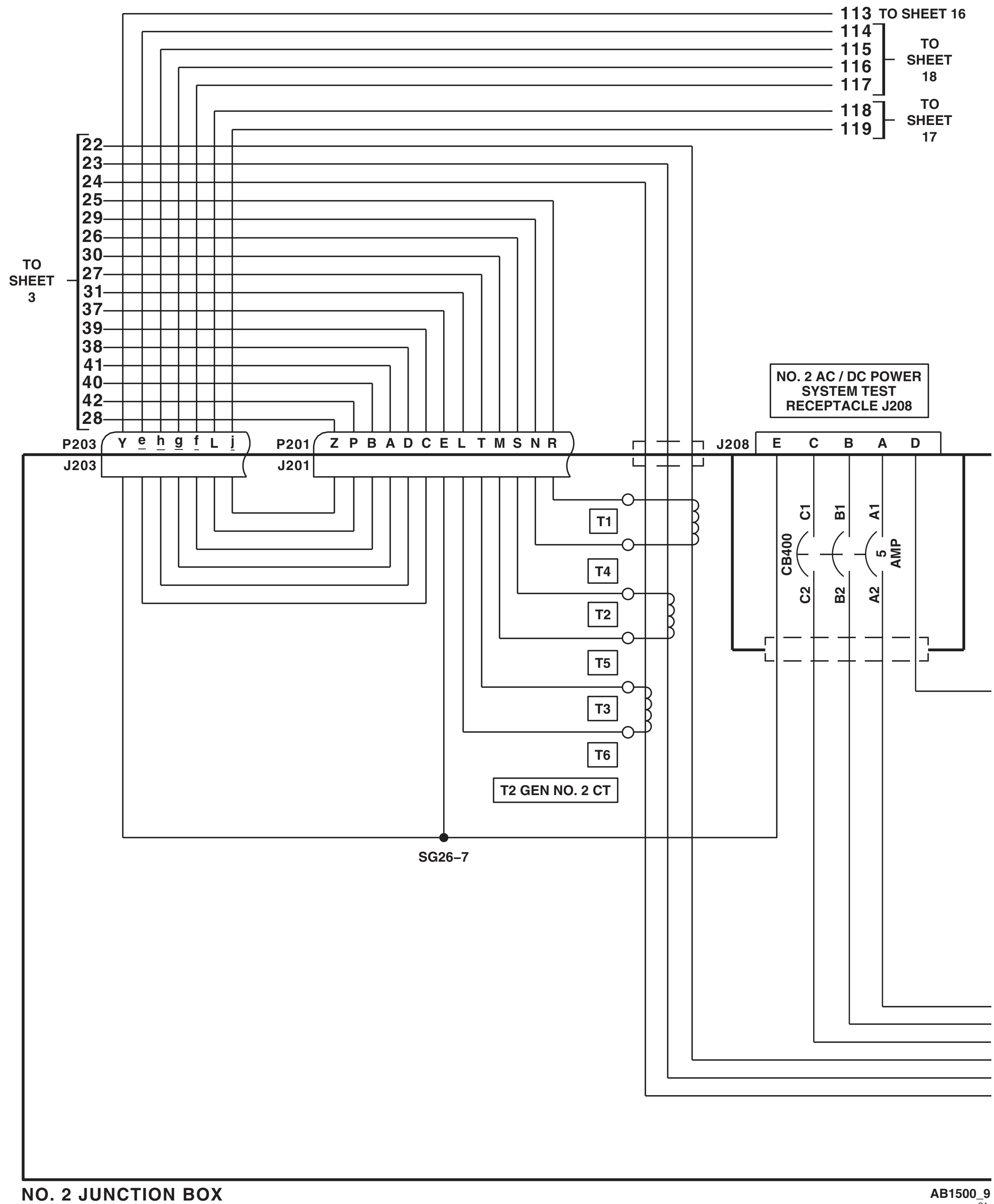


Figure 4. AC Electrical System Schematic Diagram EH-60A . (Sheet 9 of 22)

SCHEMATICS - Continued

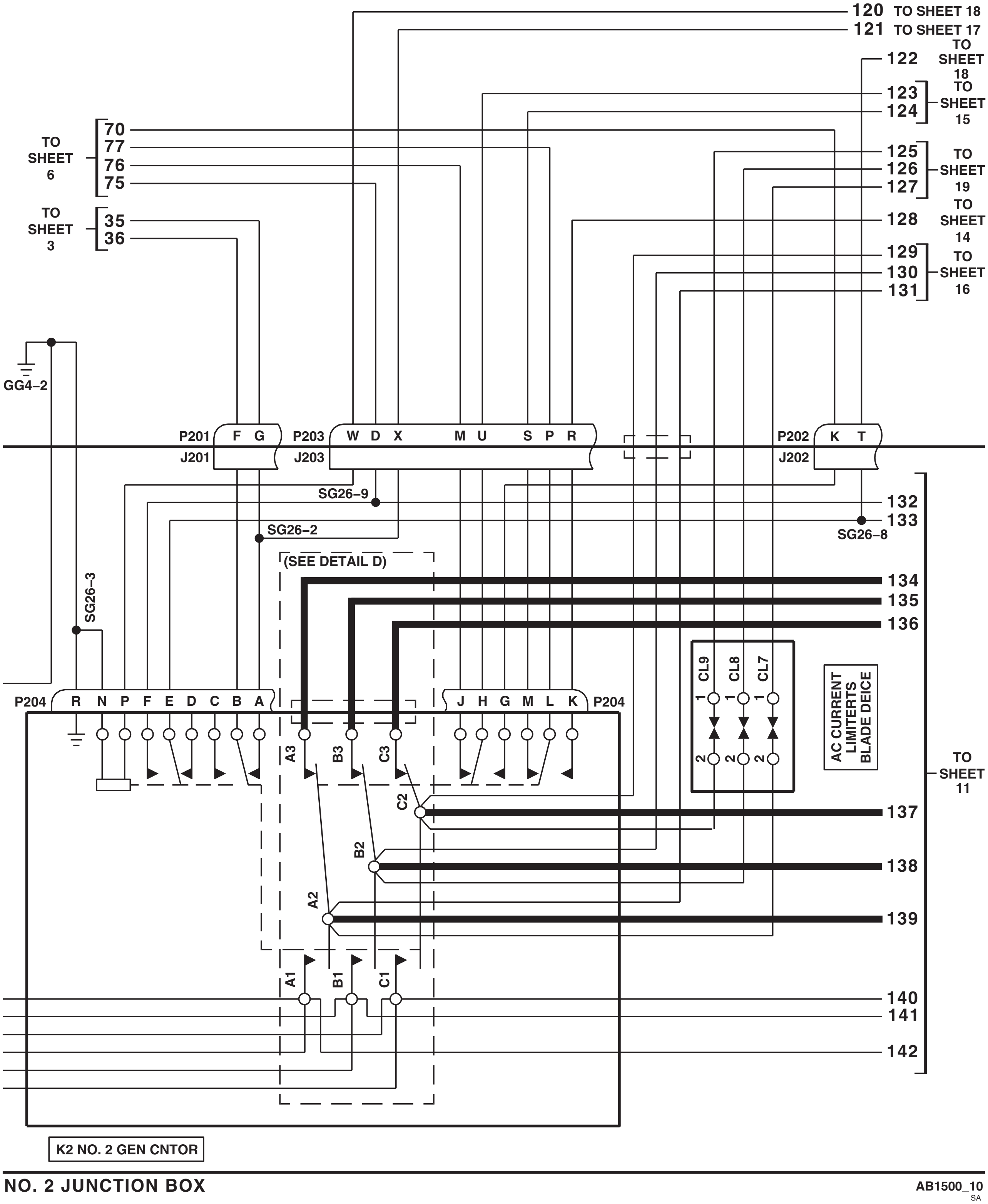


Figure 4. AC Electrical System Schematic Diagram EH-60A . (Sheet 10 of 22)

SCHEMATICS - Continued

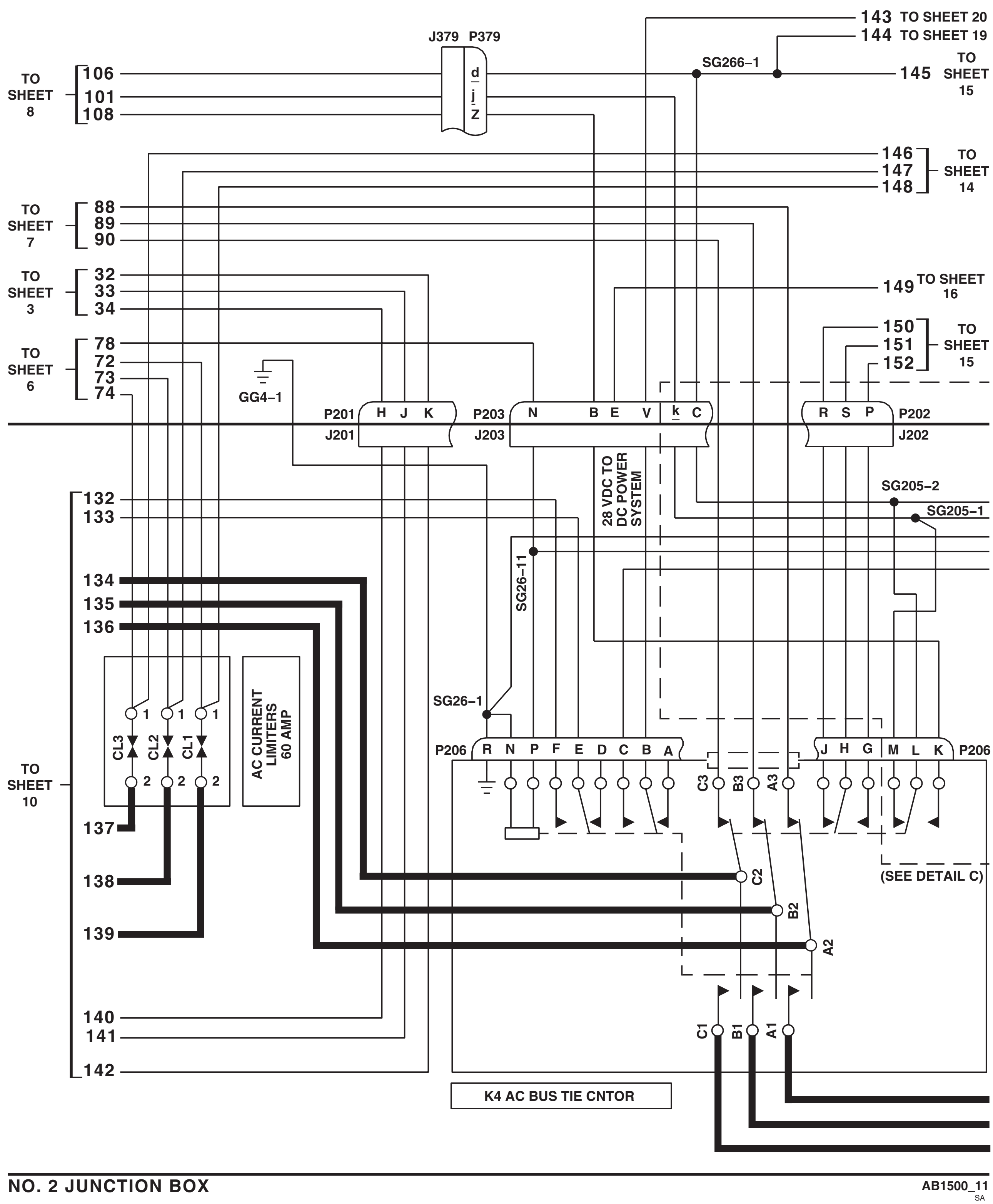


Figure 4. AC Electrical System Schematic Diagram **EH-60A** . (Sheet 11 of 22)

SCHEMATICS - Continued

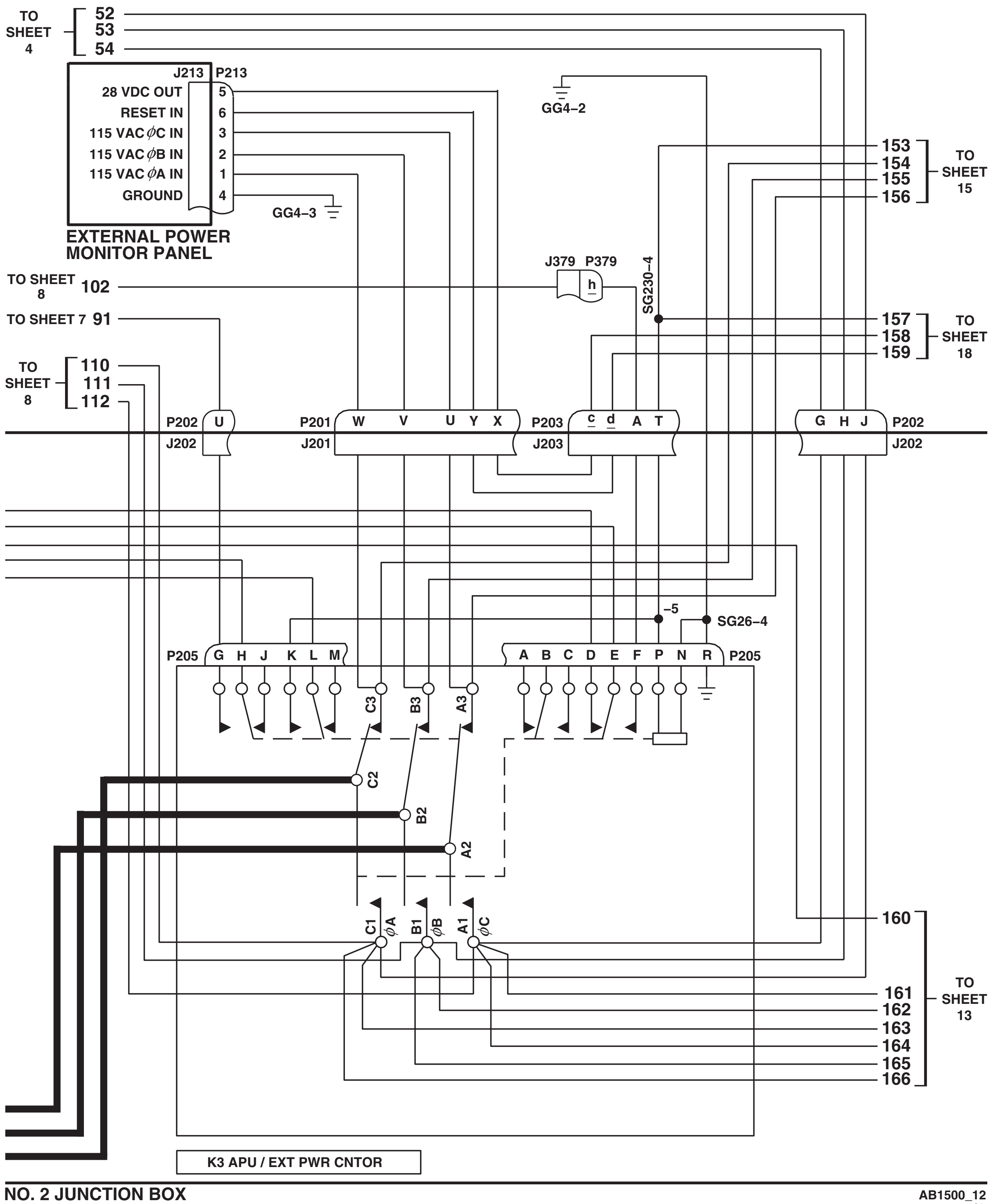


Figure 4. AC Electrical System Schematic Diagram **EH-60A** . (Sheet 12 of 22)

SCHEMATICS - Continued

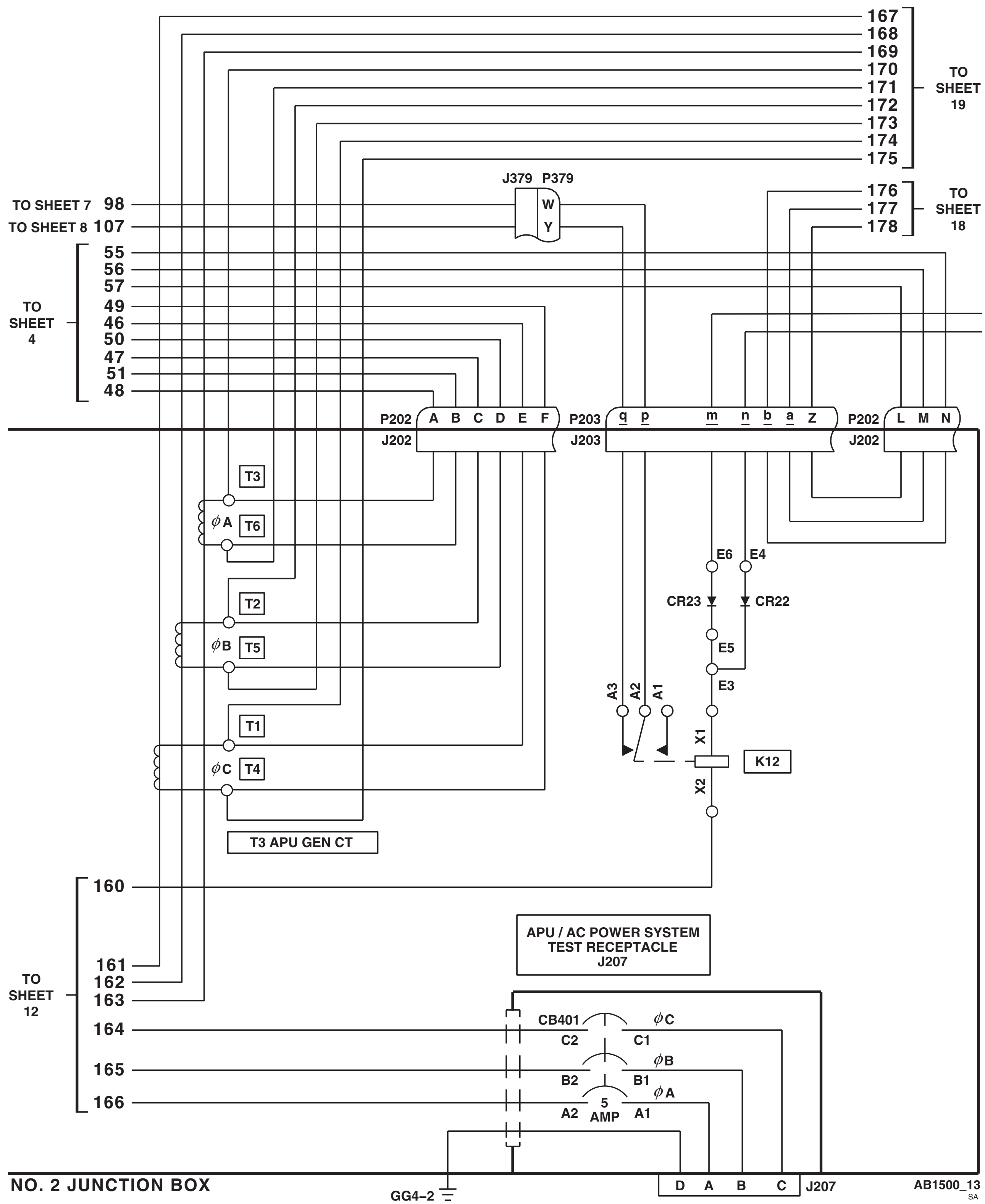


Figure 4. AC Electrical System Schematic Diagram EH-60A . (Sheet 13 of 22)

SCHEMATICS - Continued

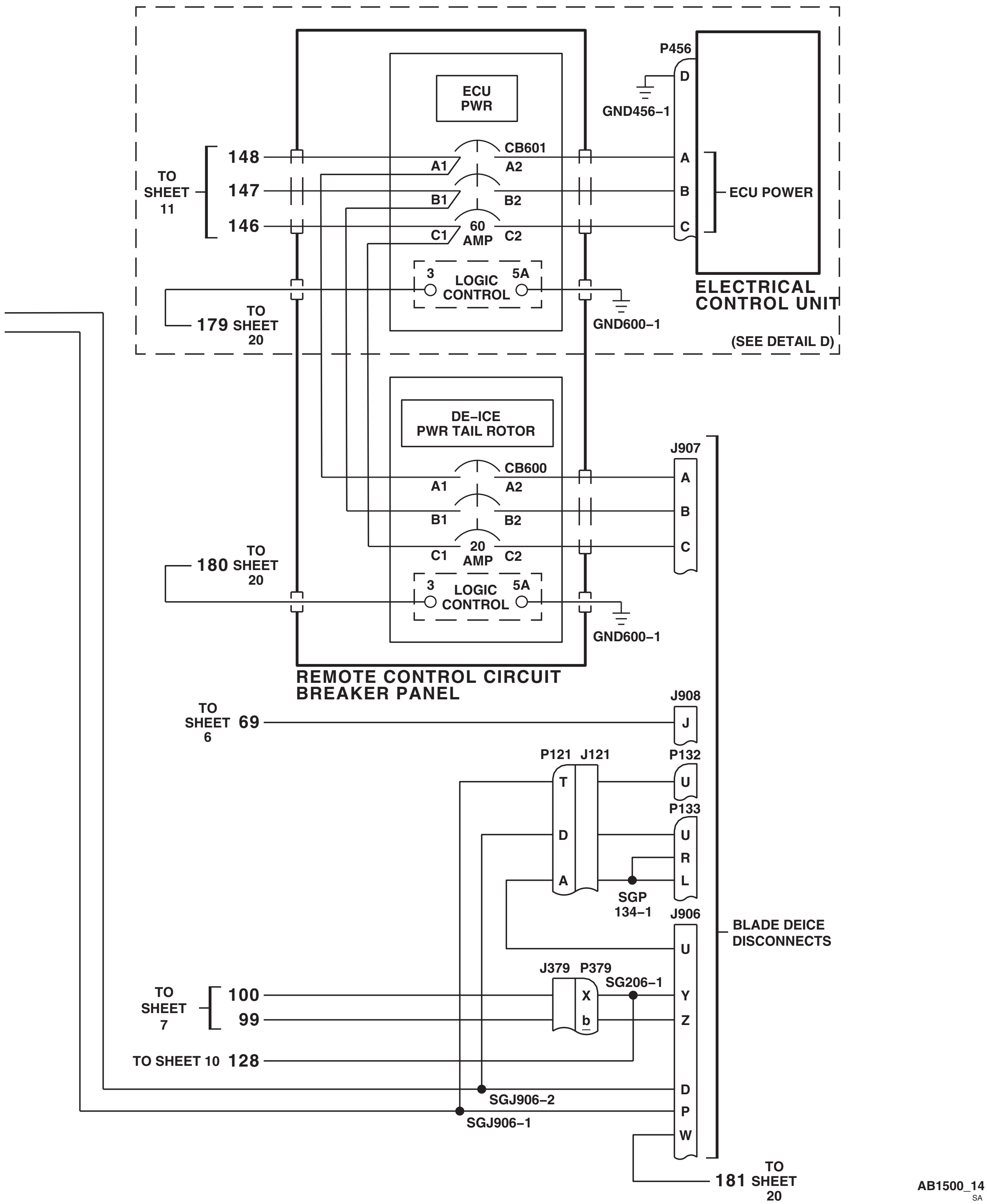


Figure 4. AC Electrical System Schematic Diagram **EH-60A** . (Sheet 14 of 22)

SCHEMATICS - Continued

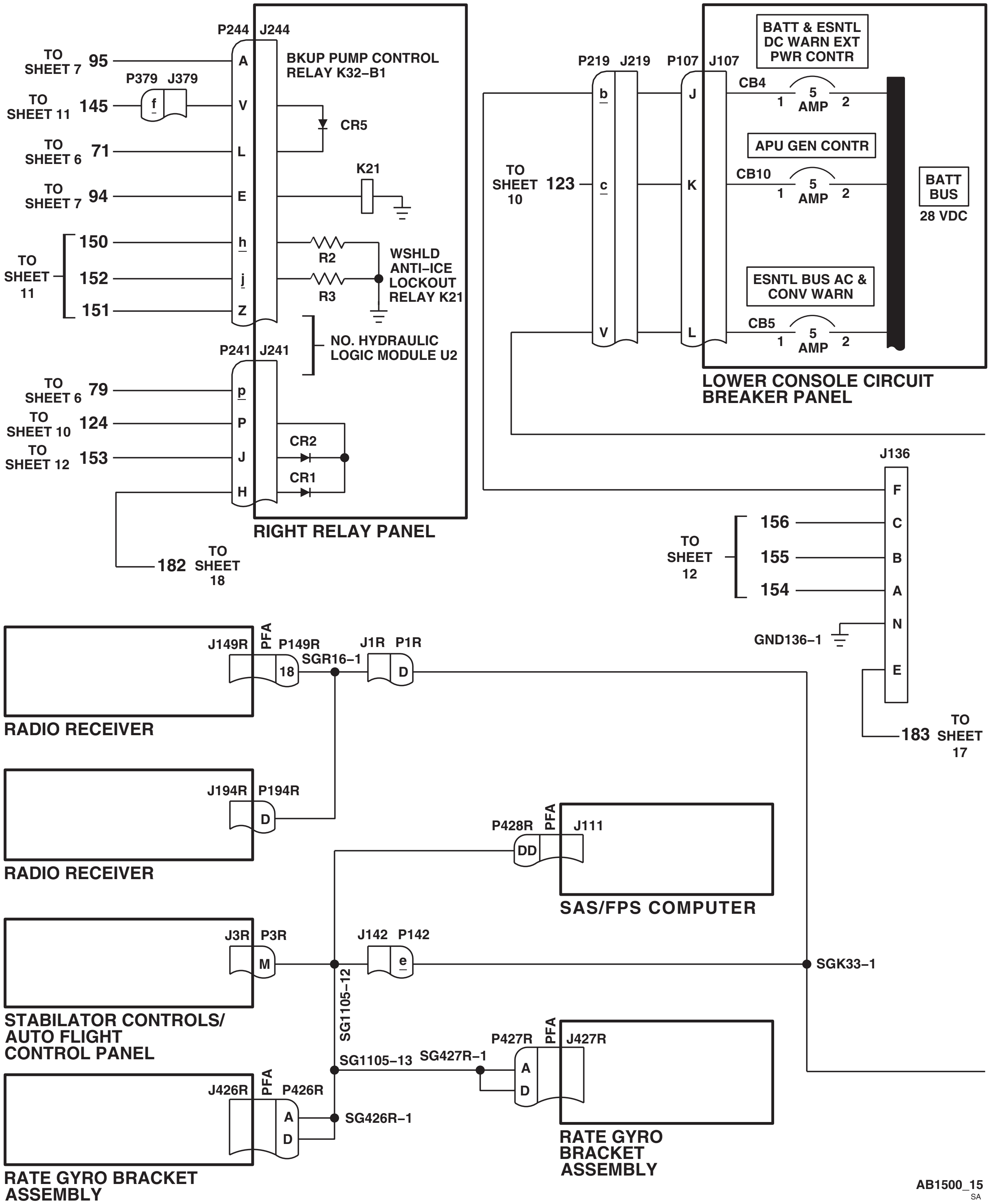


Figure 4. AC Electrical System Schematic Diagram **EH-60A** . (Sheet 15 of 22)

SCHEMATICS - Continued

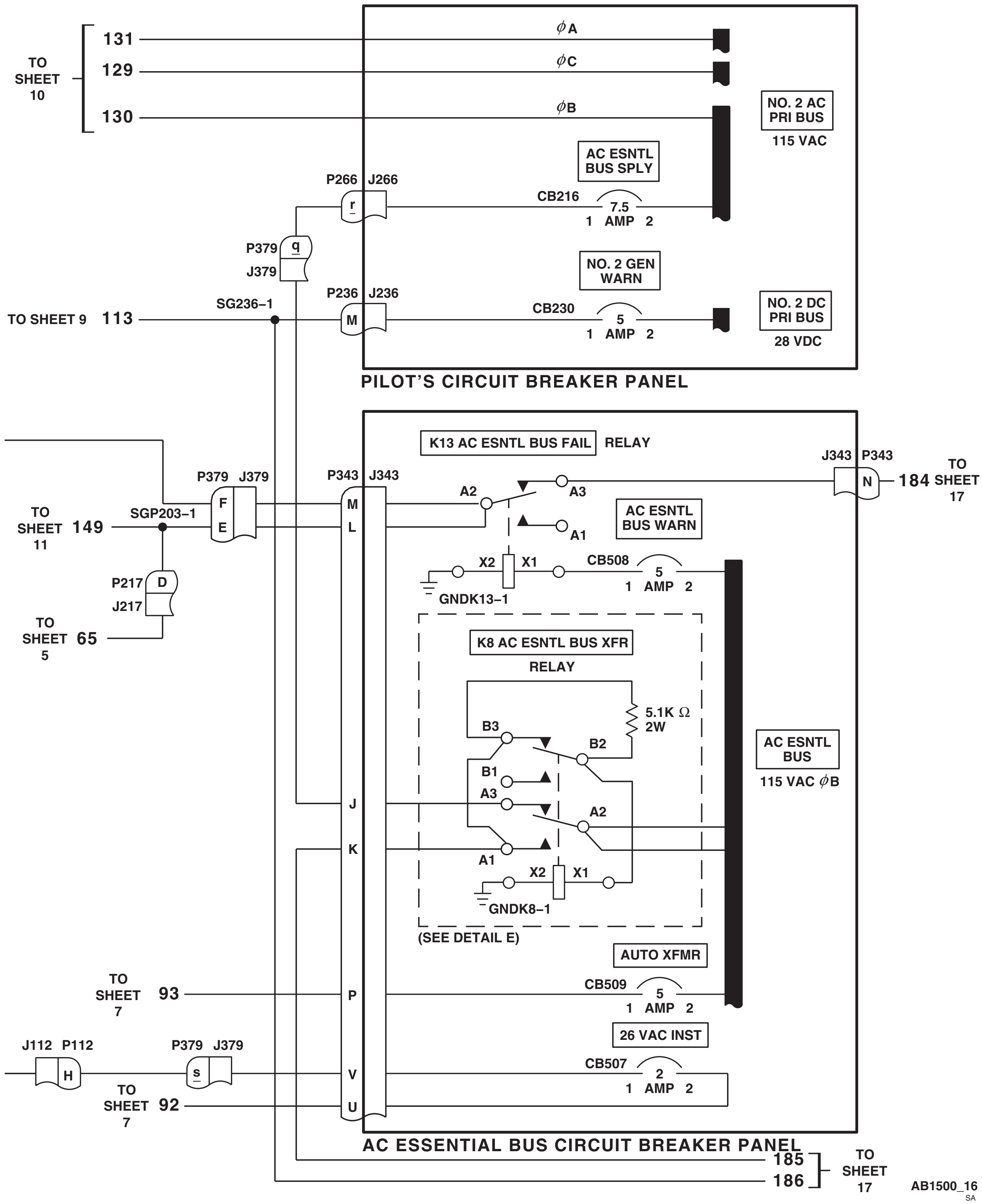


Figure 4. AC Electrical System Schematic Diagram EH-60A . (Sheet 16 of 22)

SCHEMATICS - Continued

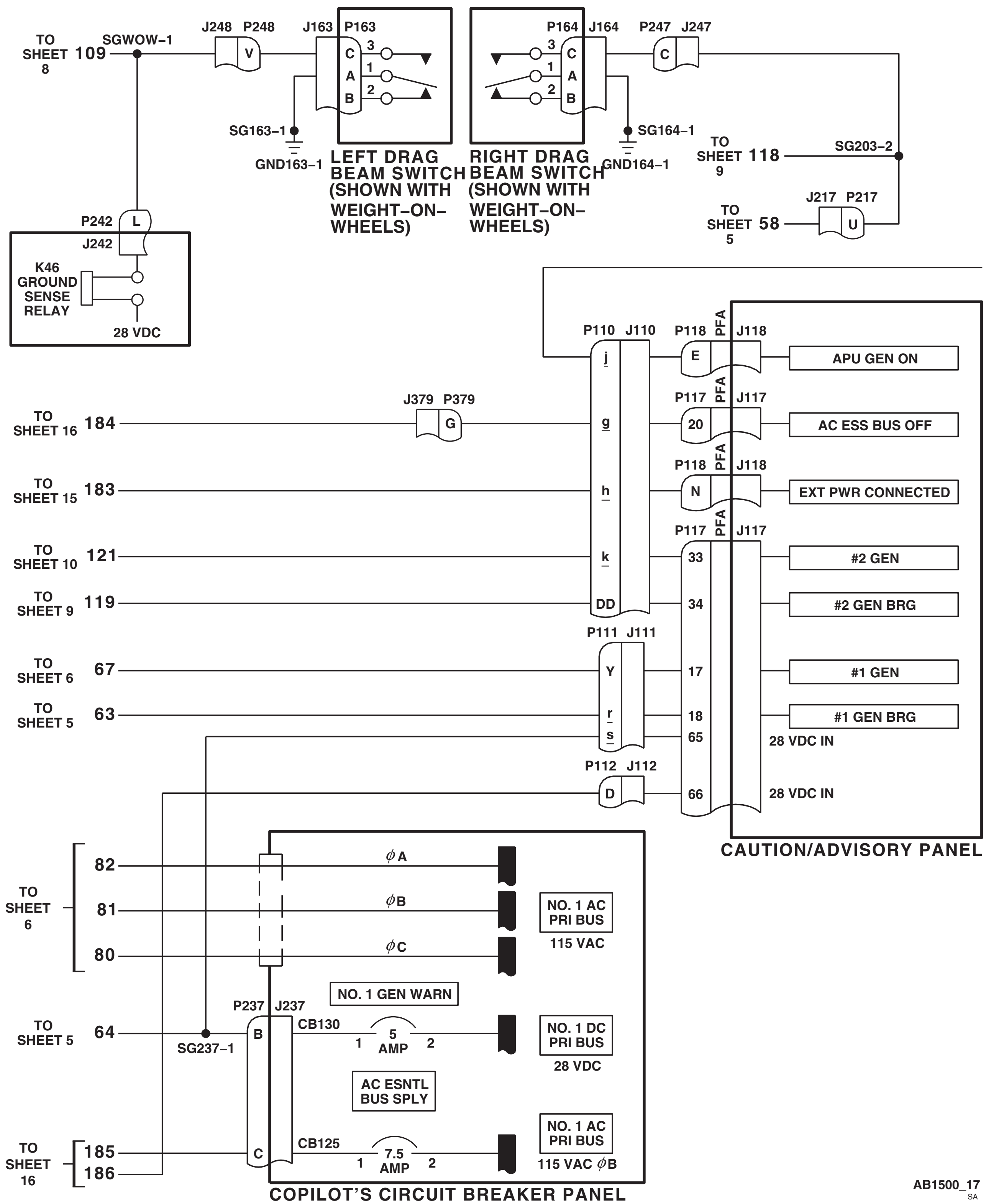


Figure 4. AC Electrical System Schematic Diagram **EH-60A** . (Sheet 17 of 22)

SCHEMATICS - Continued

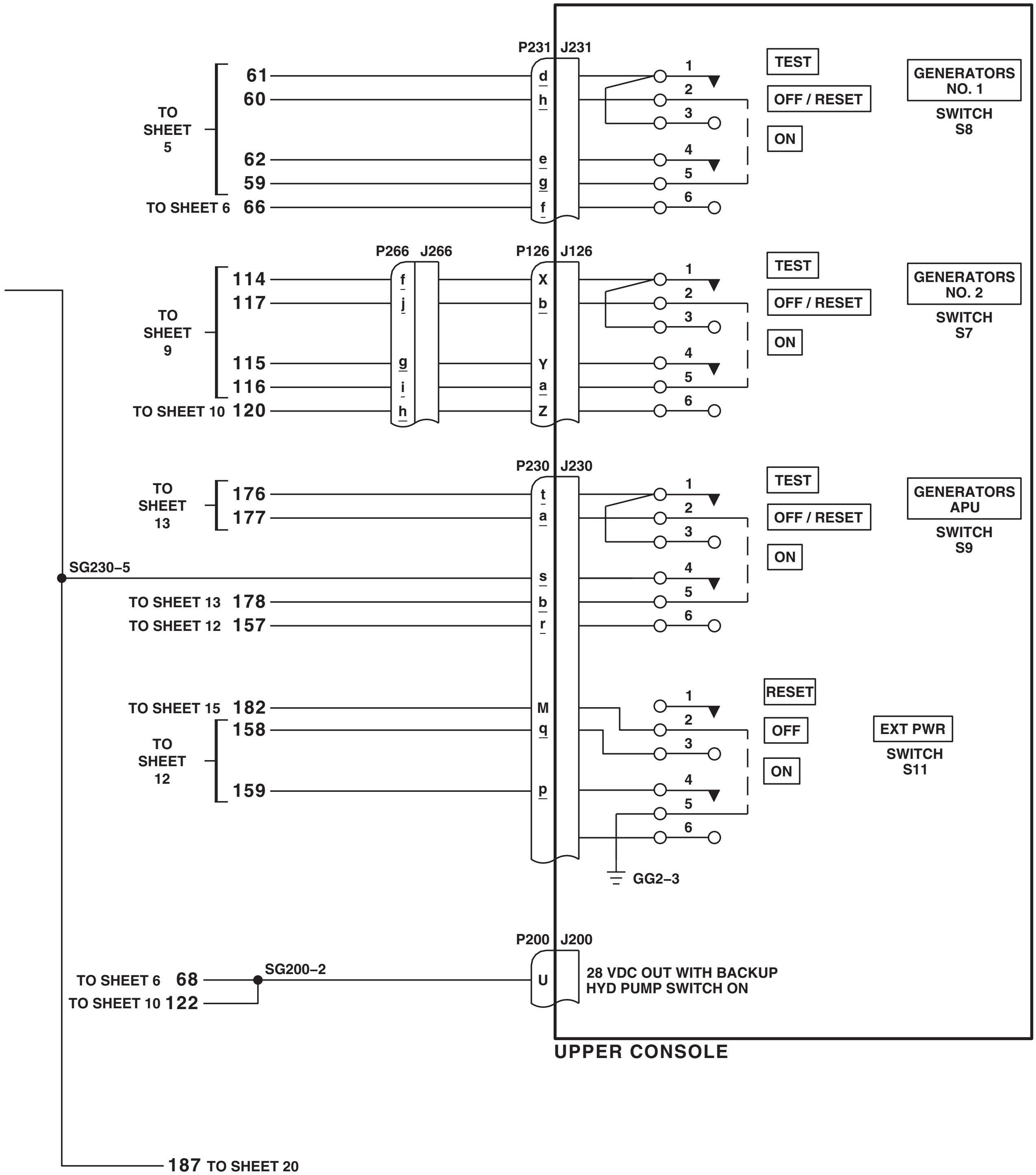


Figure 4. AC Electrical System Schematic Diagram EH-60A . (Sheet 18 of 22)

SCHEMATICS - Continued

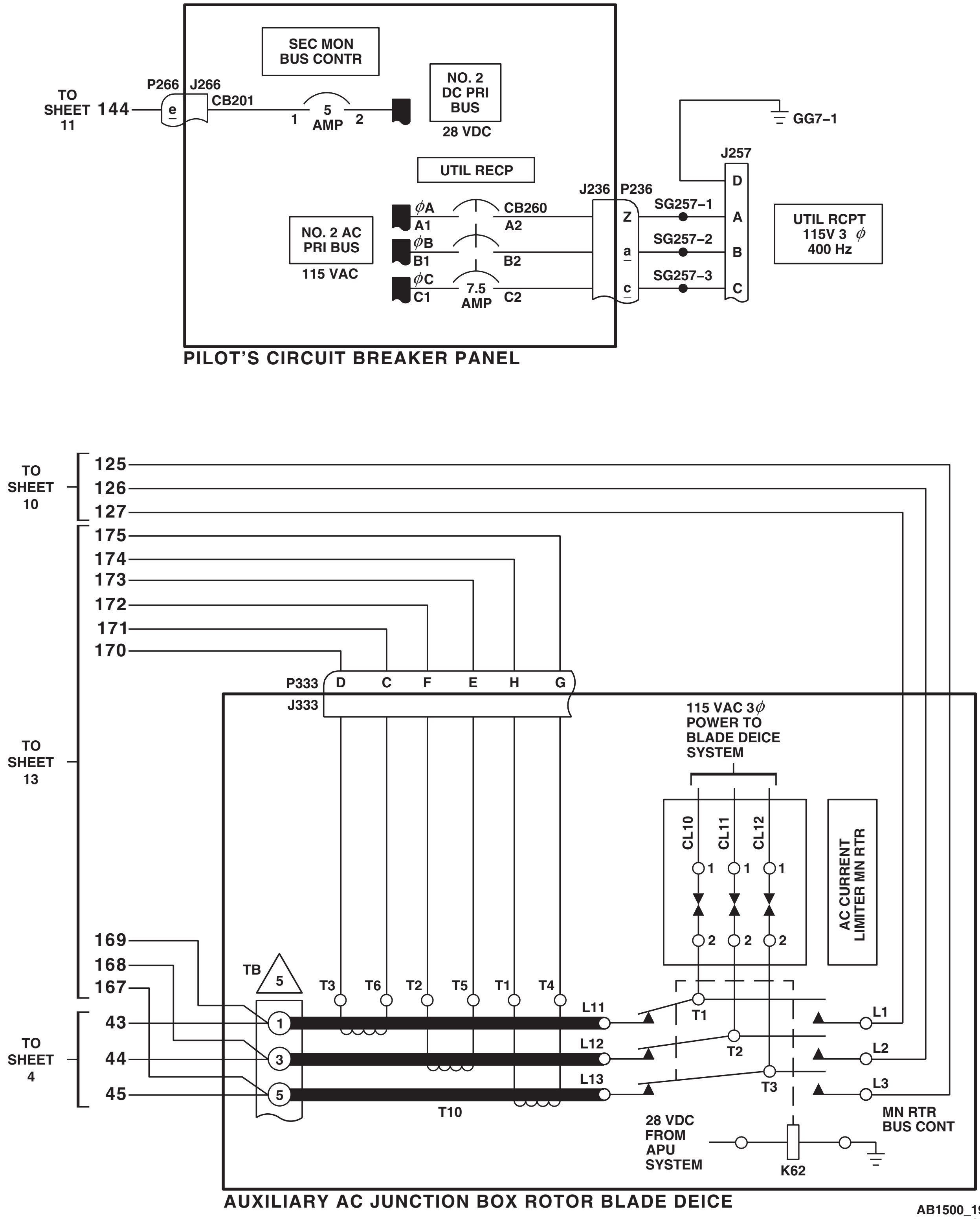


Figure 4. AC Electrical System Schematic Diagram **EH-60A** . (Sheet 19 of 22)

SCHEMATICS - Continued

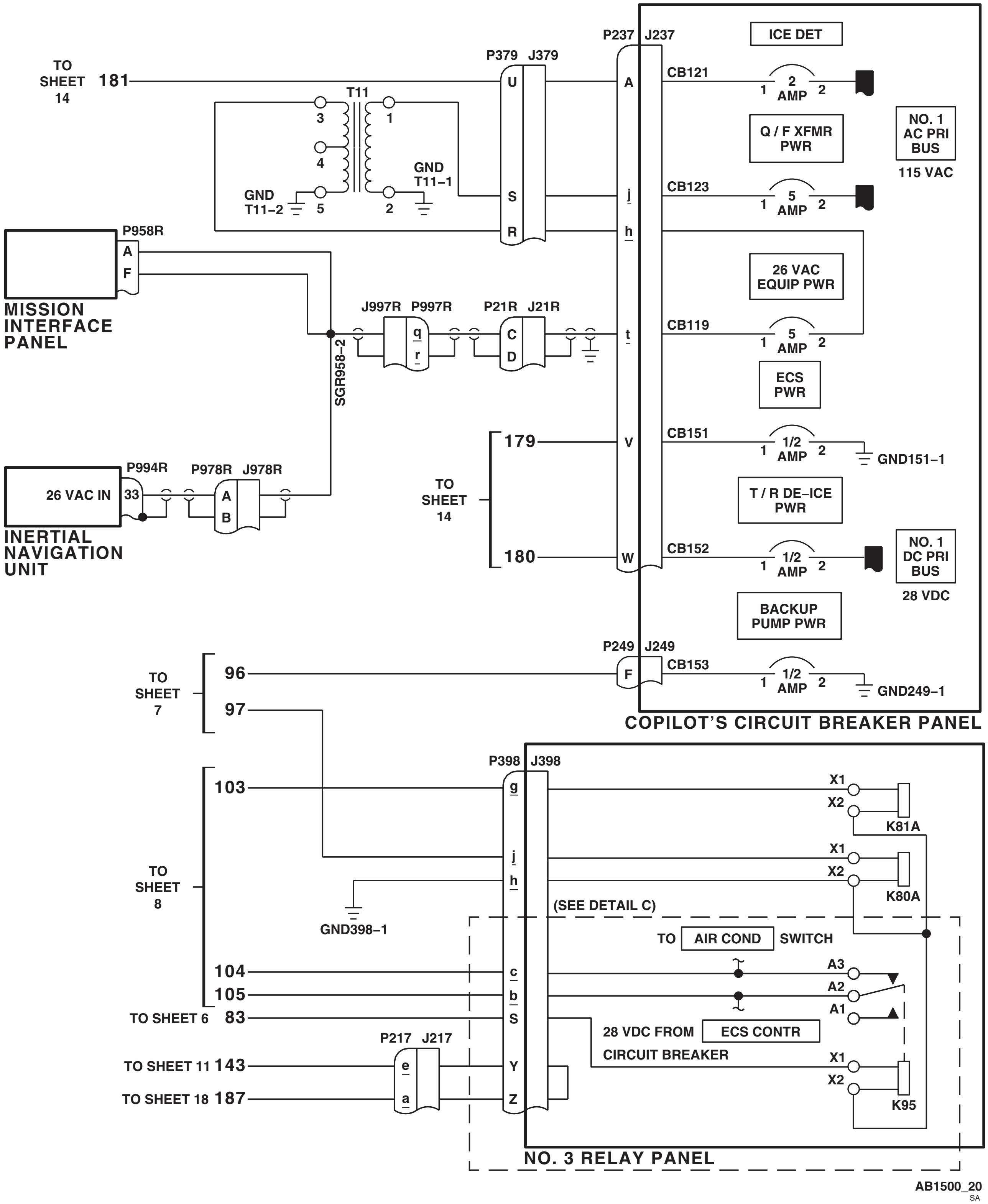
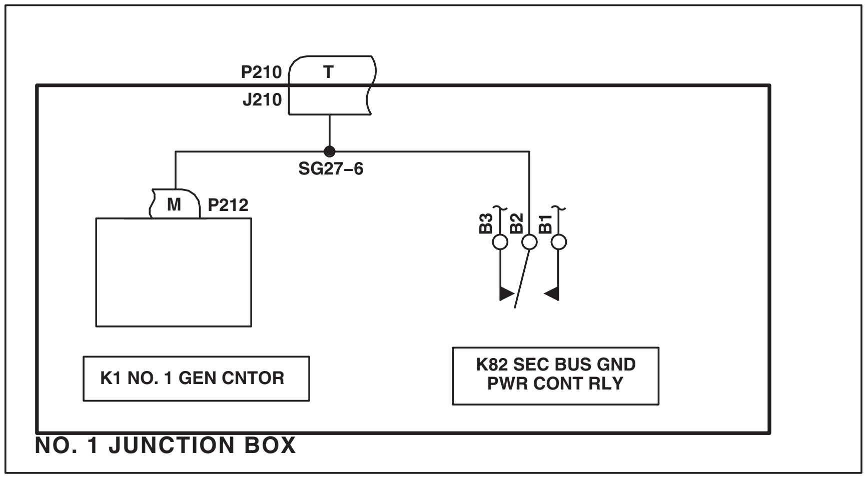
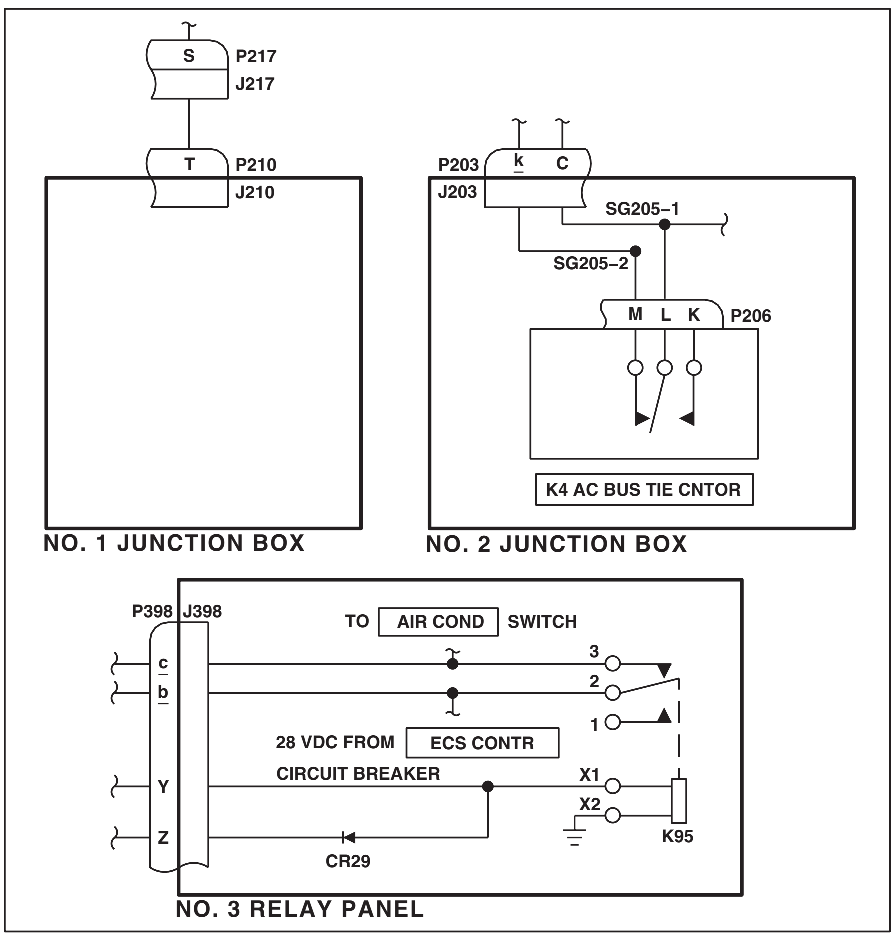


Figure 4. AC Electrical System Schematic Diagram EH-60A . (Sheet 20 of 22)

SCHEMATICS - Continued



DETAIL B
(SEE NOTE 8)

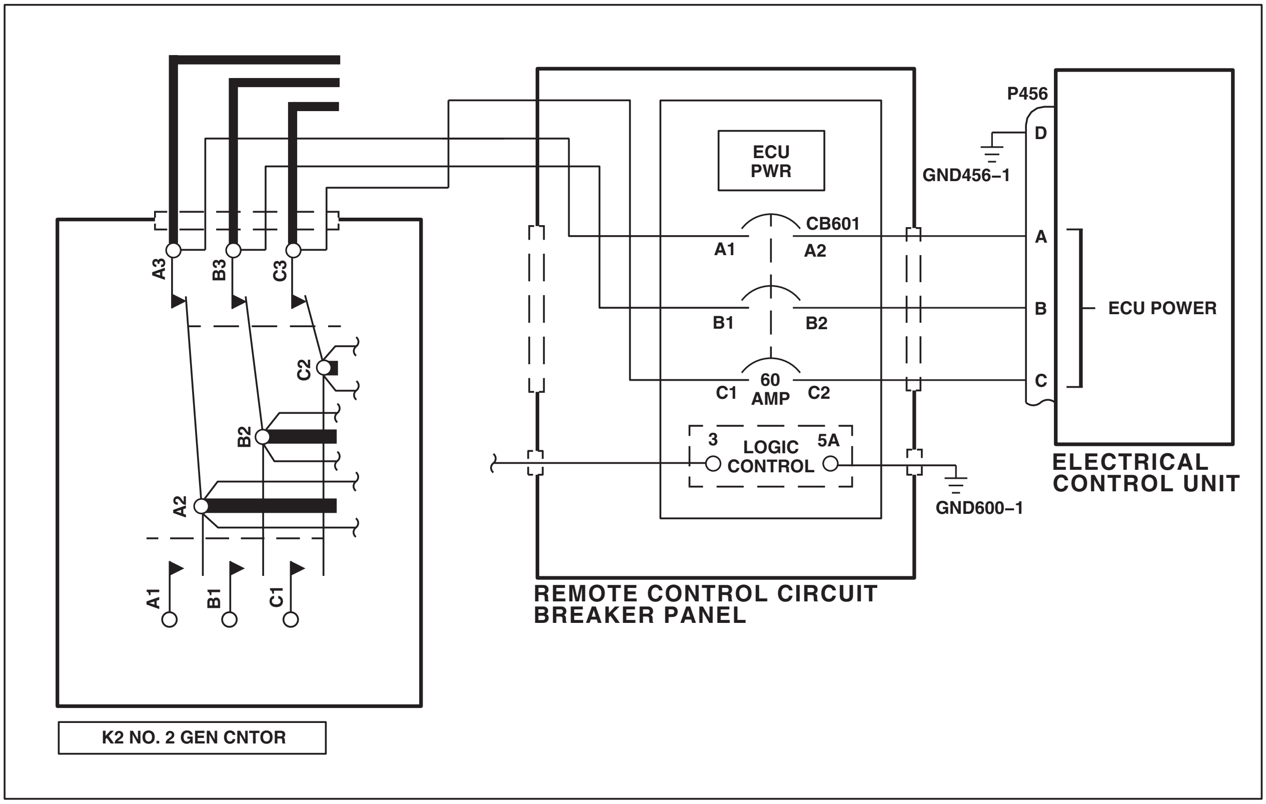


DETAIL C
(SEE NOTE 9)

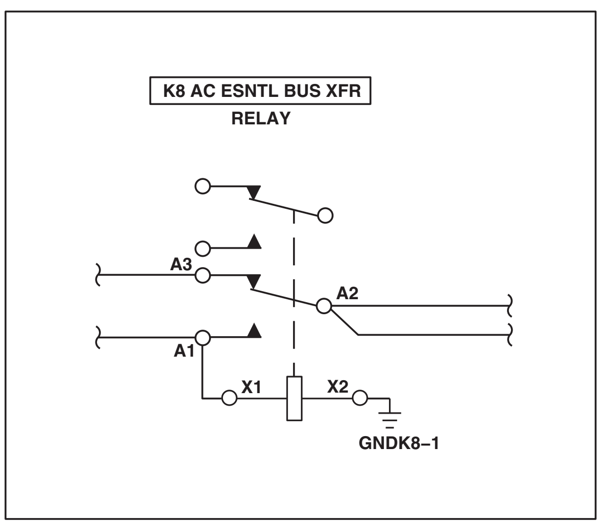
AB1500_21
SA

Figure 4. AC Electrical System Schematic Diagram **EH-60A** . (Sheet 21 of 22)

SCHEMATICS - Continued



DETAIL D
(SEE NOTE 11)

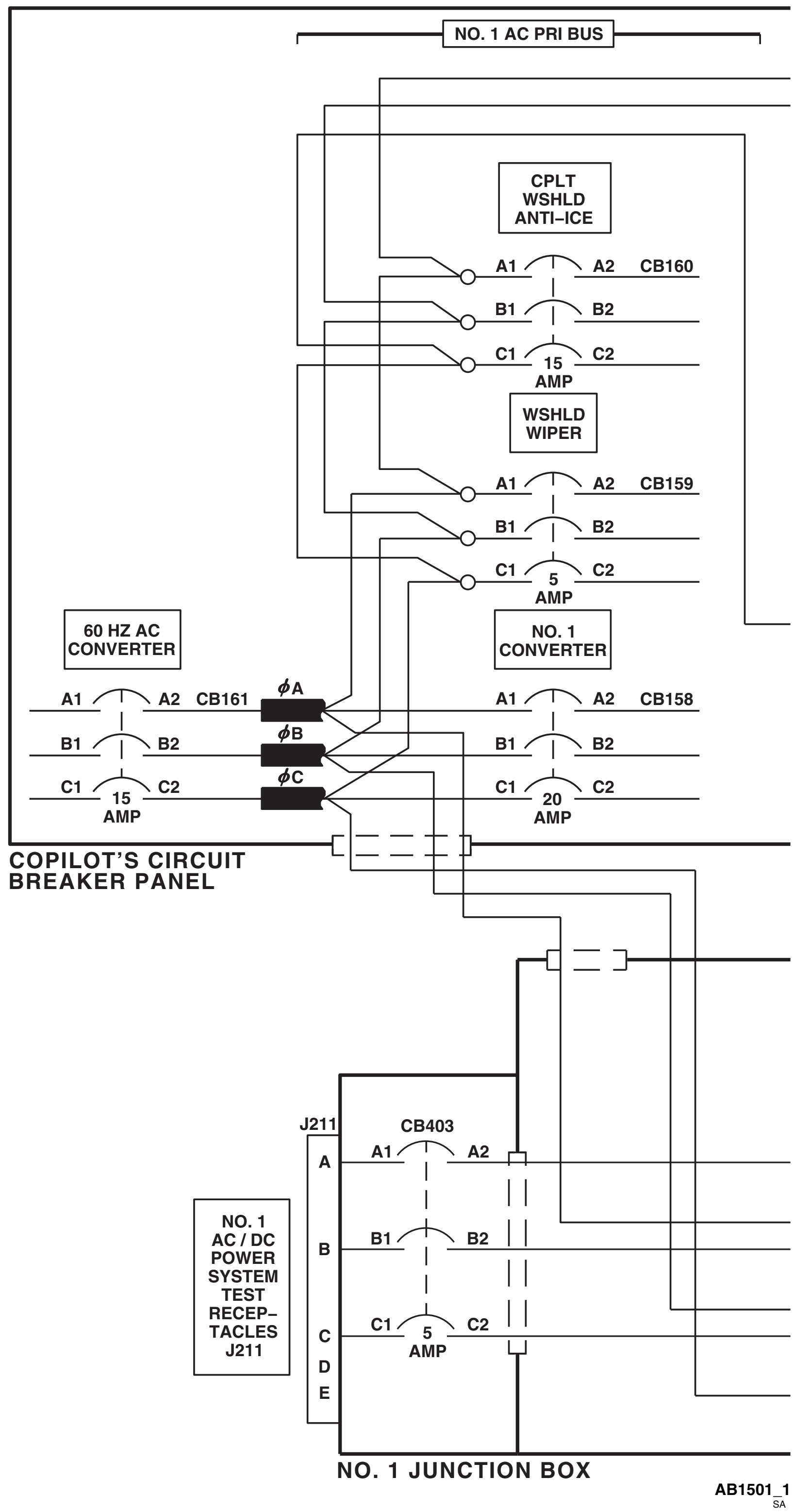


DETAIL E
(SEE NOTE 12)

AB1500_22
SA

Figure 4. AC Electrical System Schematic Diagram EH-60A . (Sheet 22 of 22)

SCHEMATICS - Continued



EFFECTIVITY

UH60A

UH60L 89–26149 – 96–26722

NOTES

1. HELICOPTERS WITH AUXILIARY CABIN HEATER INSTALLED, CIRCUIT BREAKER CB1A REIDENTIFIED AS AUX HTR BLOWER.
2. HCW
3. UH60A 77–22714 – 77–22720

   W/O MOD NO. 2 CONVERTER CIRCUIT BREAKER IS LOCATED / IDENTIFIED AS CB264.
4. W/O HCW CMPTR CIRCUIT BREAK– ER IS LOCATED / IDENTIFIED AS CB259.
5. W/O HCW AC ESNTL BUS SPLY CIRCUIT BREAKER IS LOCATED / IDENTIFIED AS CB257.
6. W/O HCW BATT CHGR CIRCUIT BREAKER IS LOCATED / IDENTIFIED AS CB258.

Figure 5. AC Power Distribution Diagram UH-60A UH-60L 89-26149-96-26722 . (Sheet 1 of 5)

SCHEMATICS - Continued

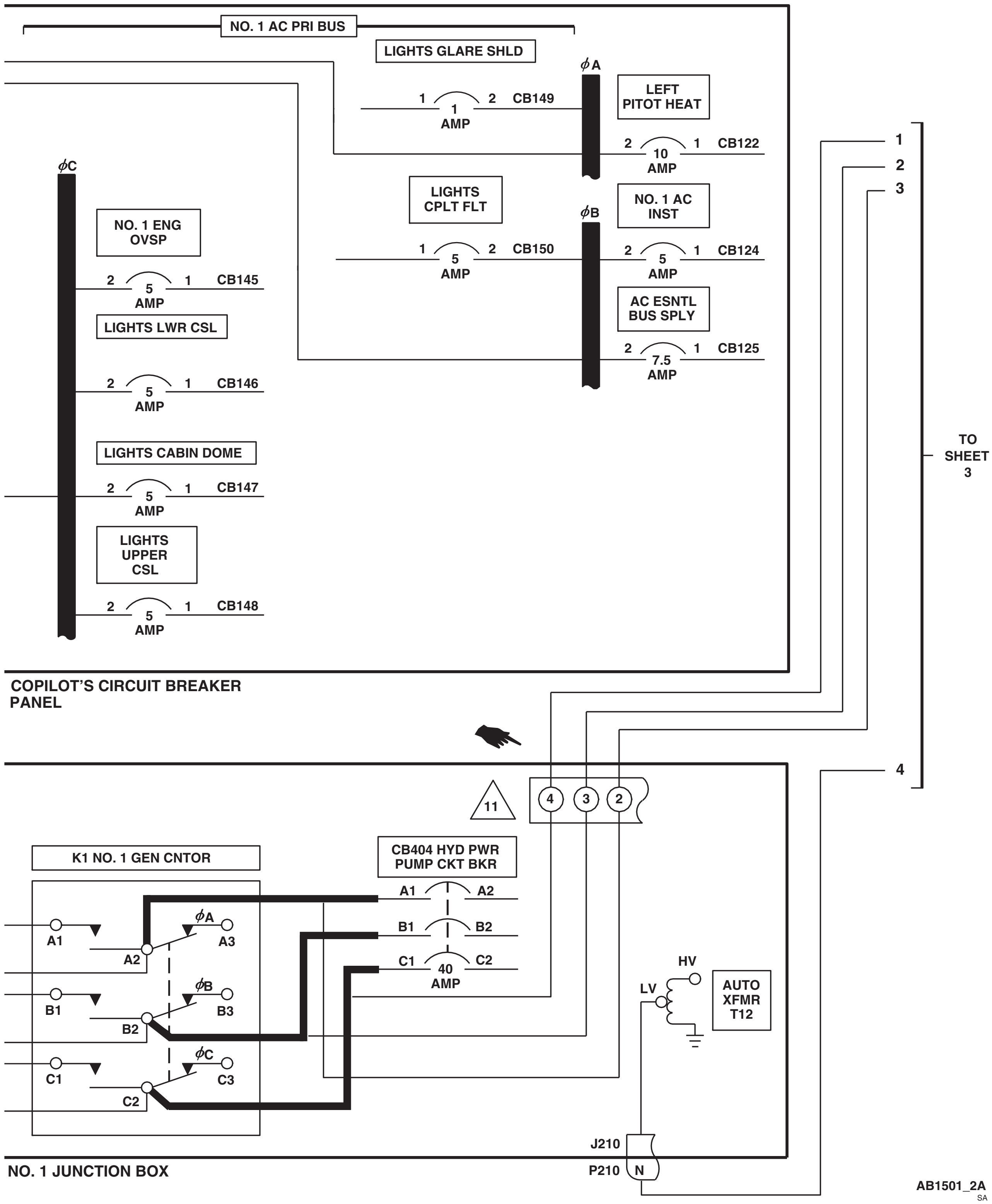


Figure 5. AC Power Distribution Diagram UH-60A UH-60L 89-26149-96-26722 . (Sheet 2 of 5)

SCHEMATICS - Continued

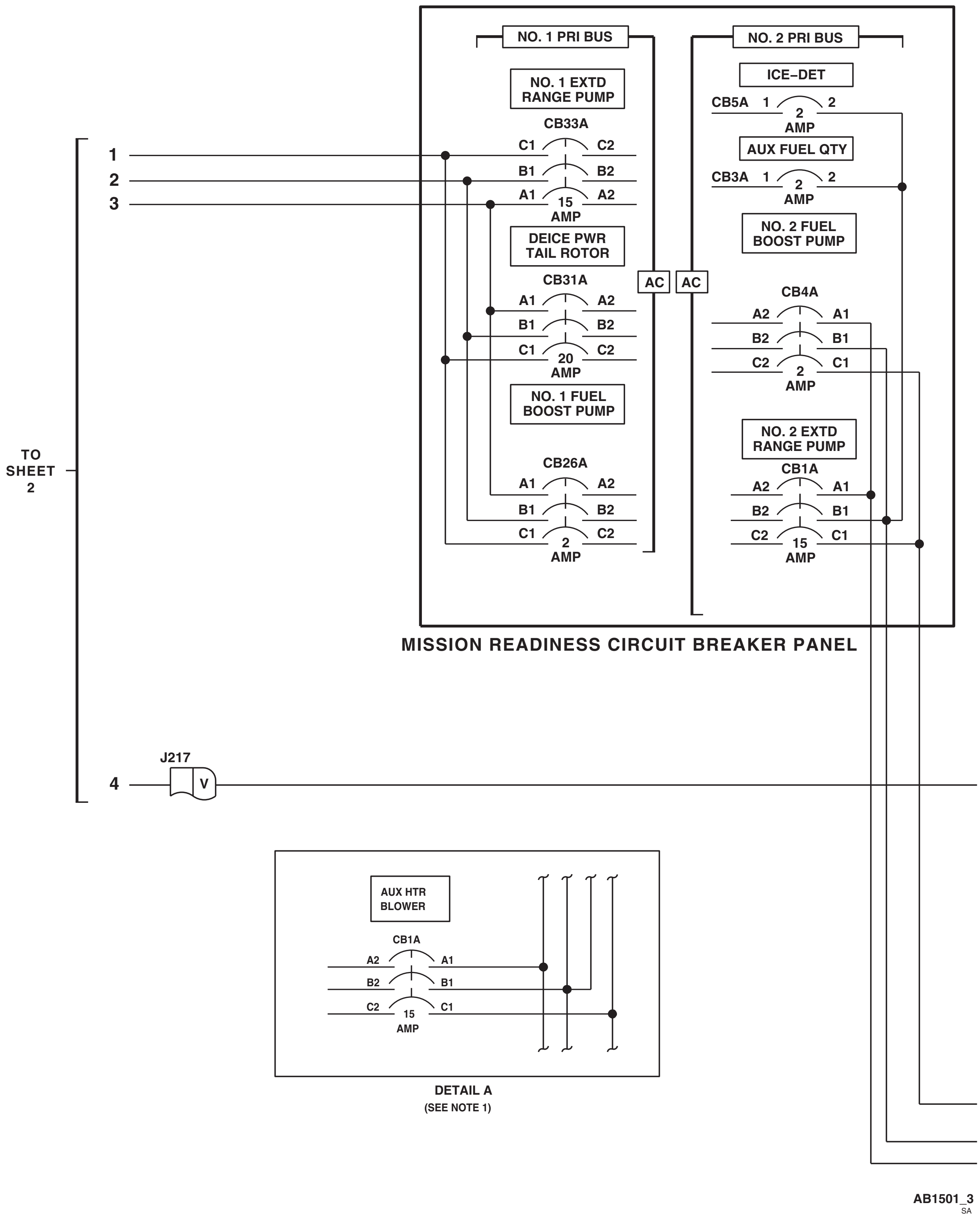


Figure 5. AC Power Distribution Diagram UH-60A UH-60L 89-26149-96-26722 . (Sheet 3 of 5)

SCHEMATICS - Continued

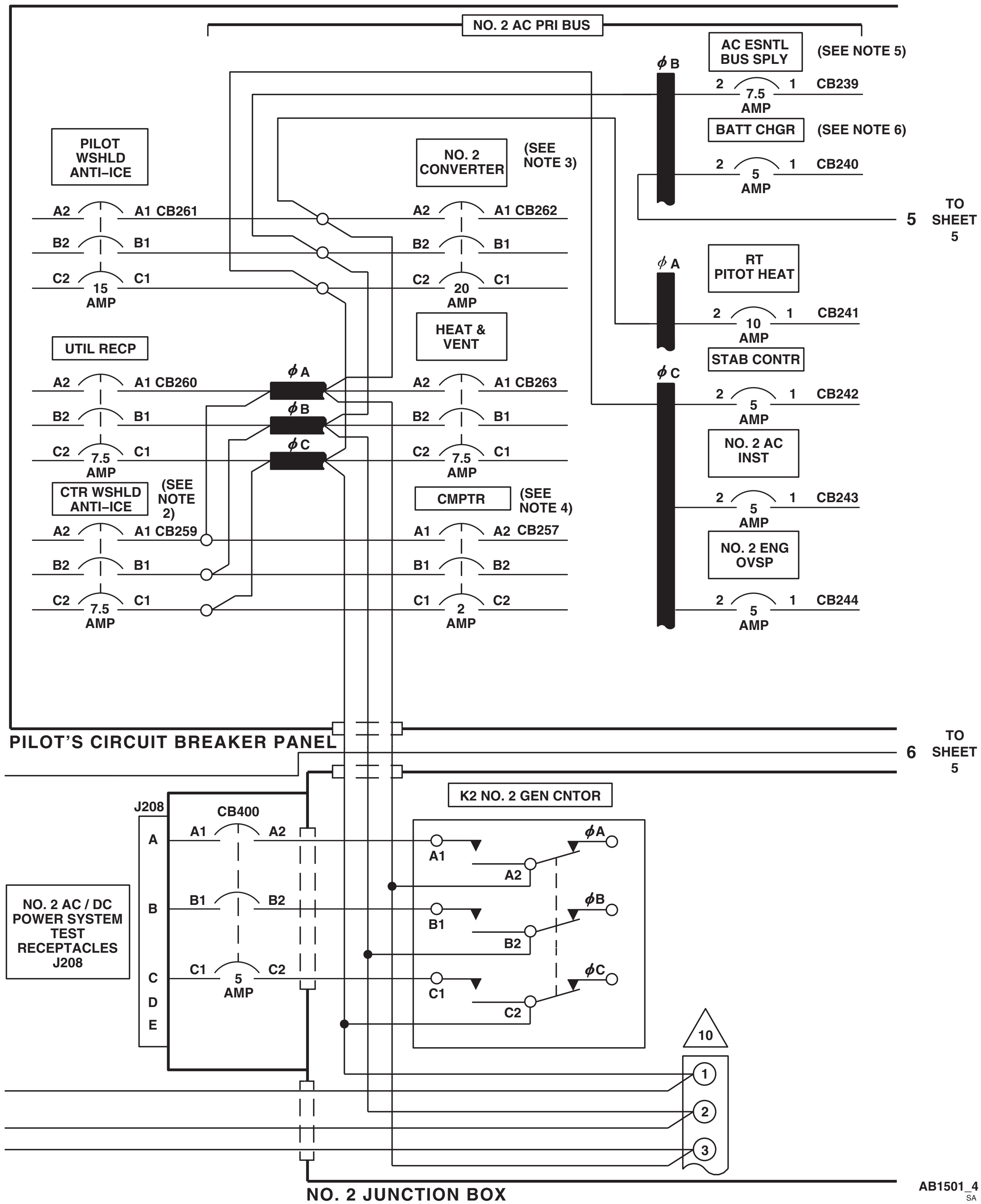


Figure 5. AC Power Distribution Diagram UH-60A UH-60L 89-26149-96-26722 . (Sheet 4 of 5)

SCHEMATICS - Continued

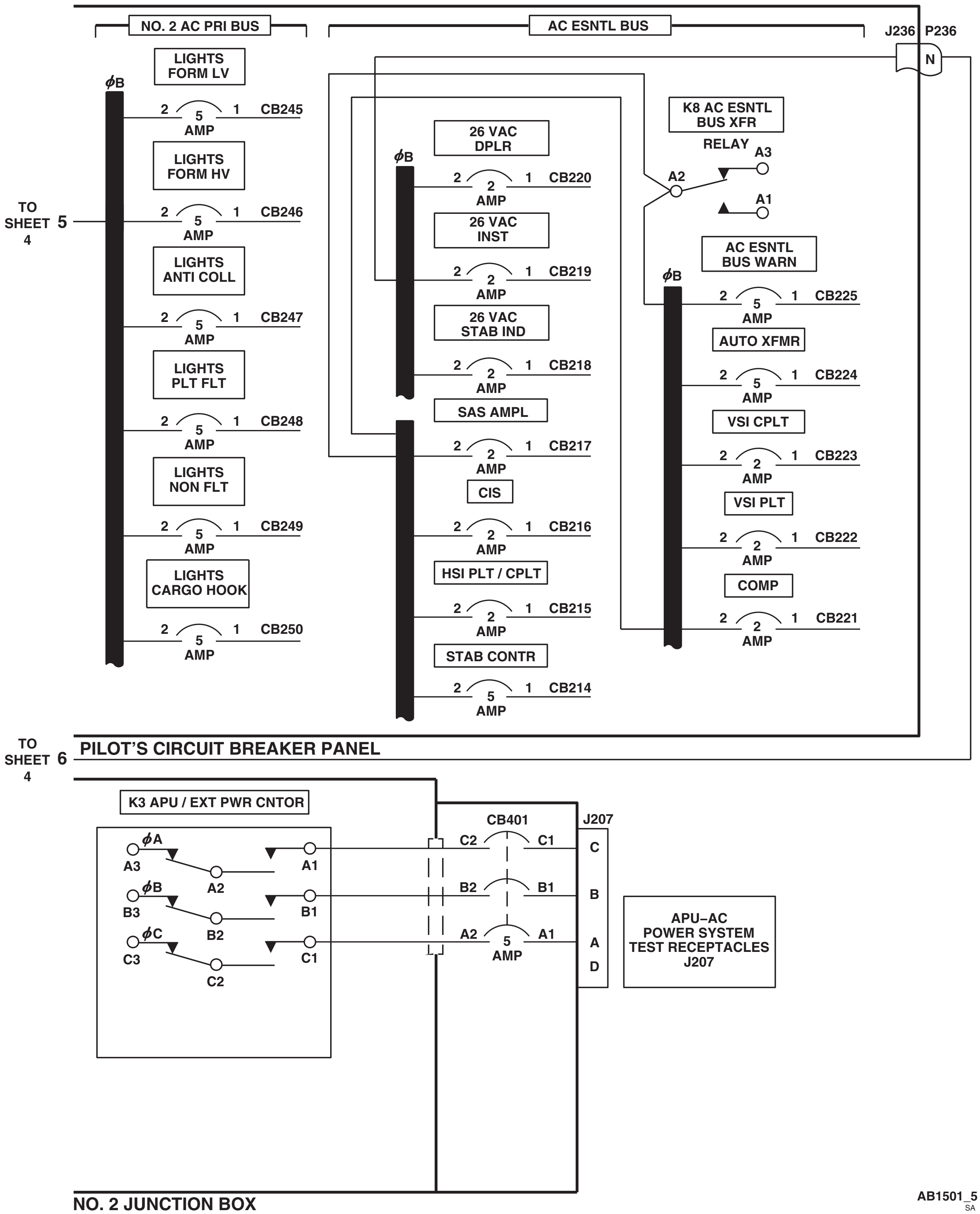


Figure 5. AC Power Distribution Diagram UH-60A UH-60L 89-26149-96-26722 . (Sheet 5 of 5)

SCHEMATICS - Continued

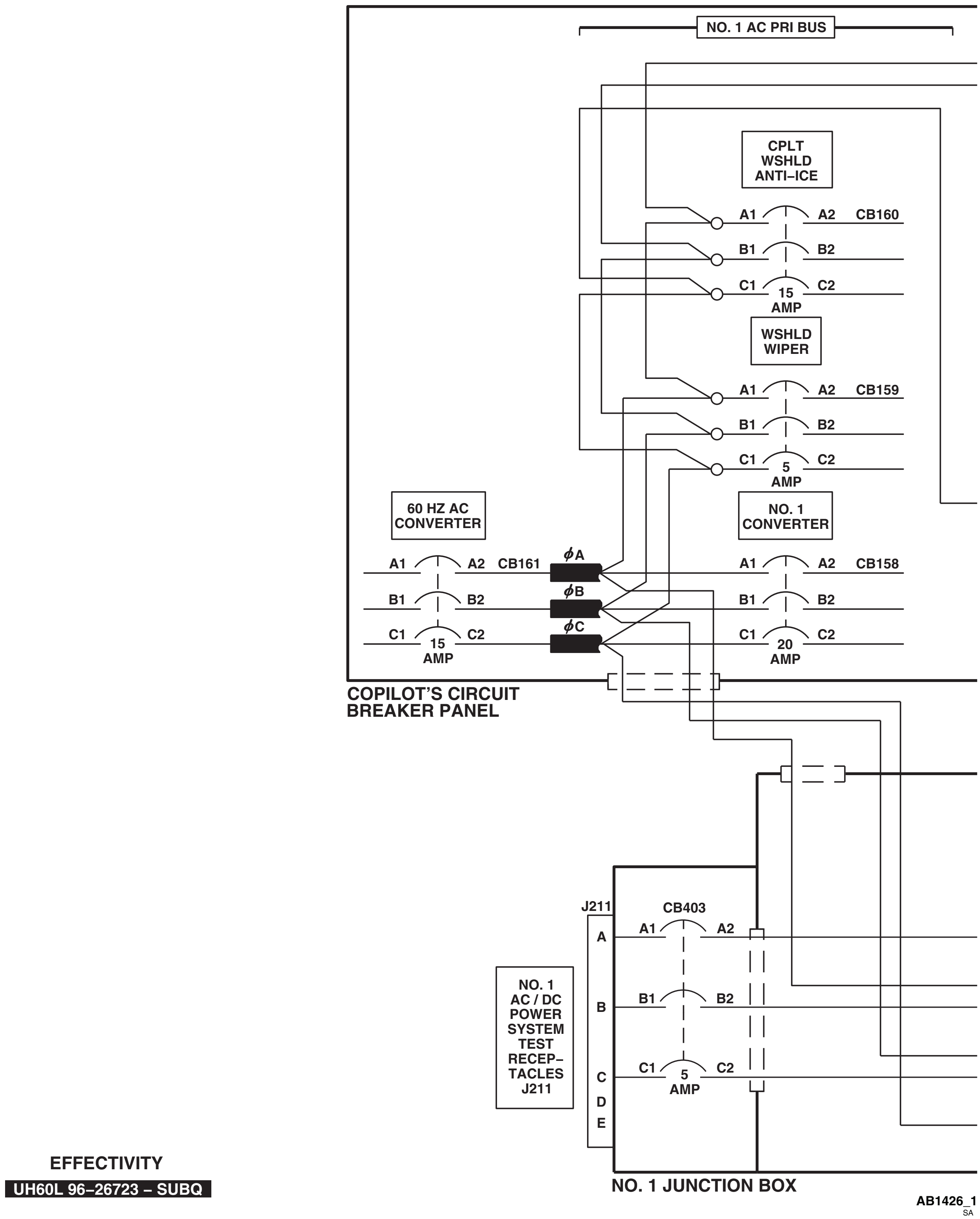


Figure 6. AC Power Distribution Diagram UH-60L 96-26723-SUBQ . (Sheet 1 of 5)

SCHEMATICS - Continued

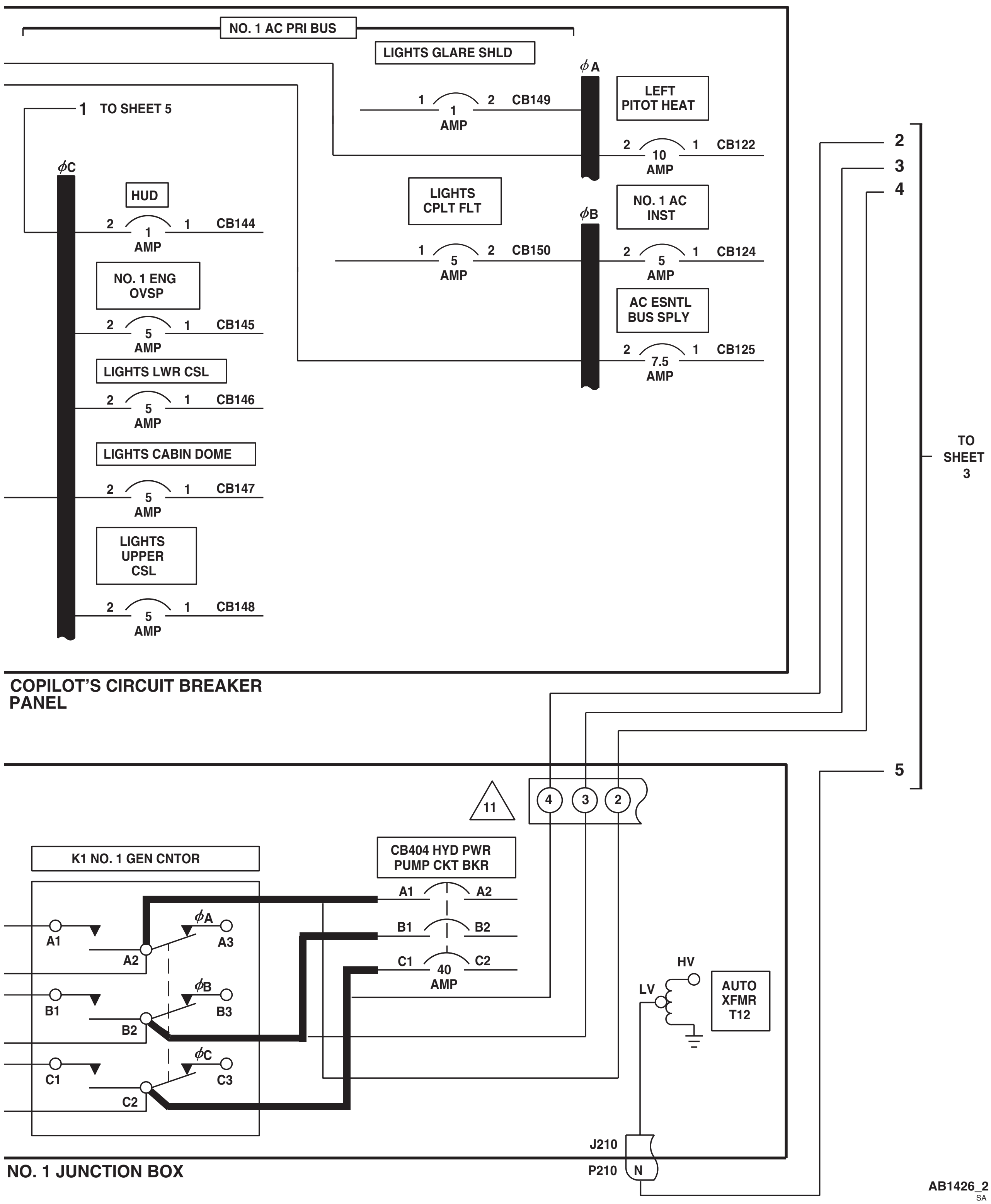


Figure 6. AC Power Distribution Diagram UH-60L 96-26723-SUBQ . (Sheet 2 of 5)

SCHEMATICS - Continued

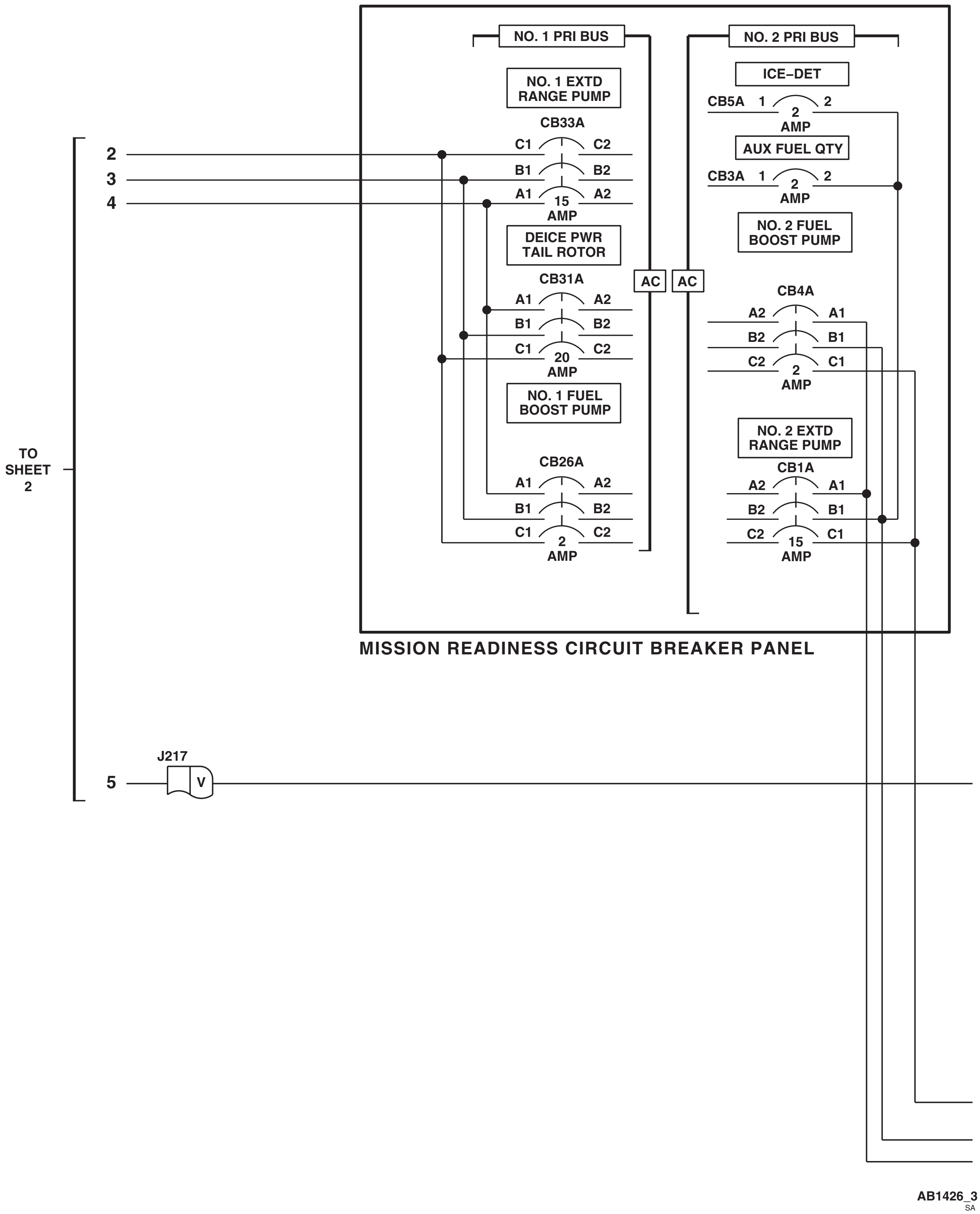


Figure 6. AC Power Distribution Diagram UH-60L 96-26723-SUBQ . (Sheet 3 of 5)

SCHEMATICS - Continued

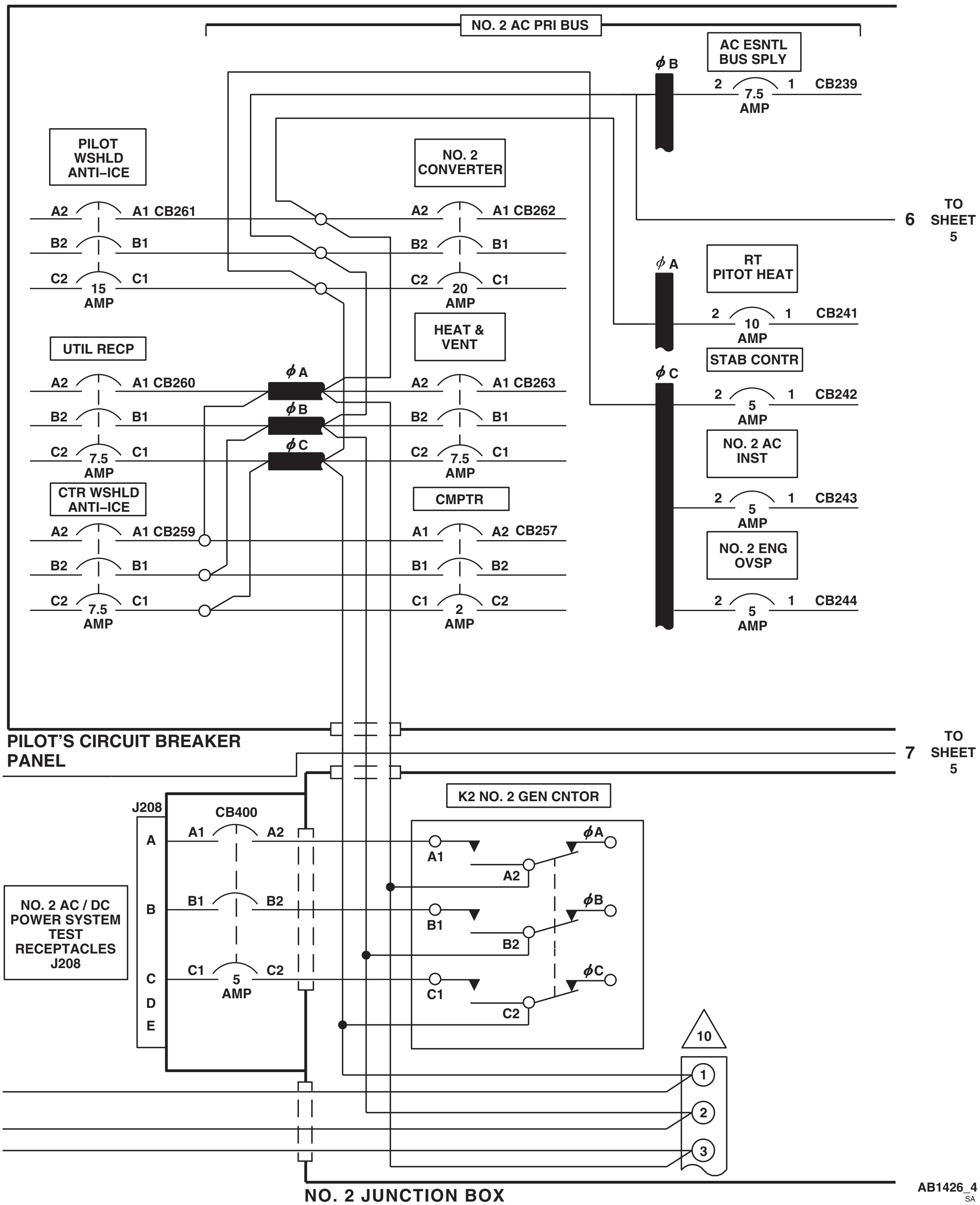


Figure 6. AC Power Distribution Diagram UH-60L 96-26723-SUBQ . (Sheet 4 of 5)

SCHEMATICS - Continued

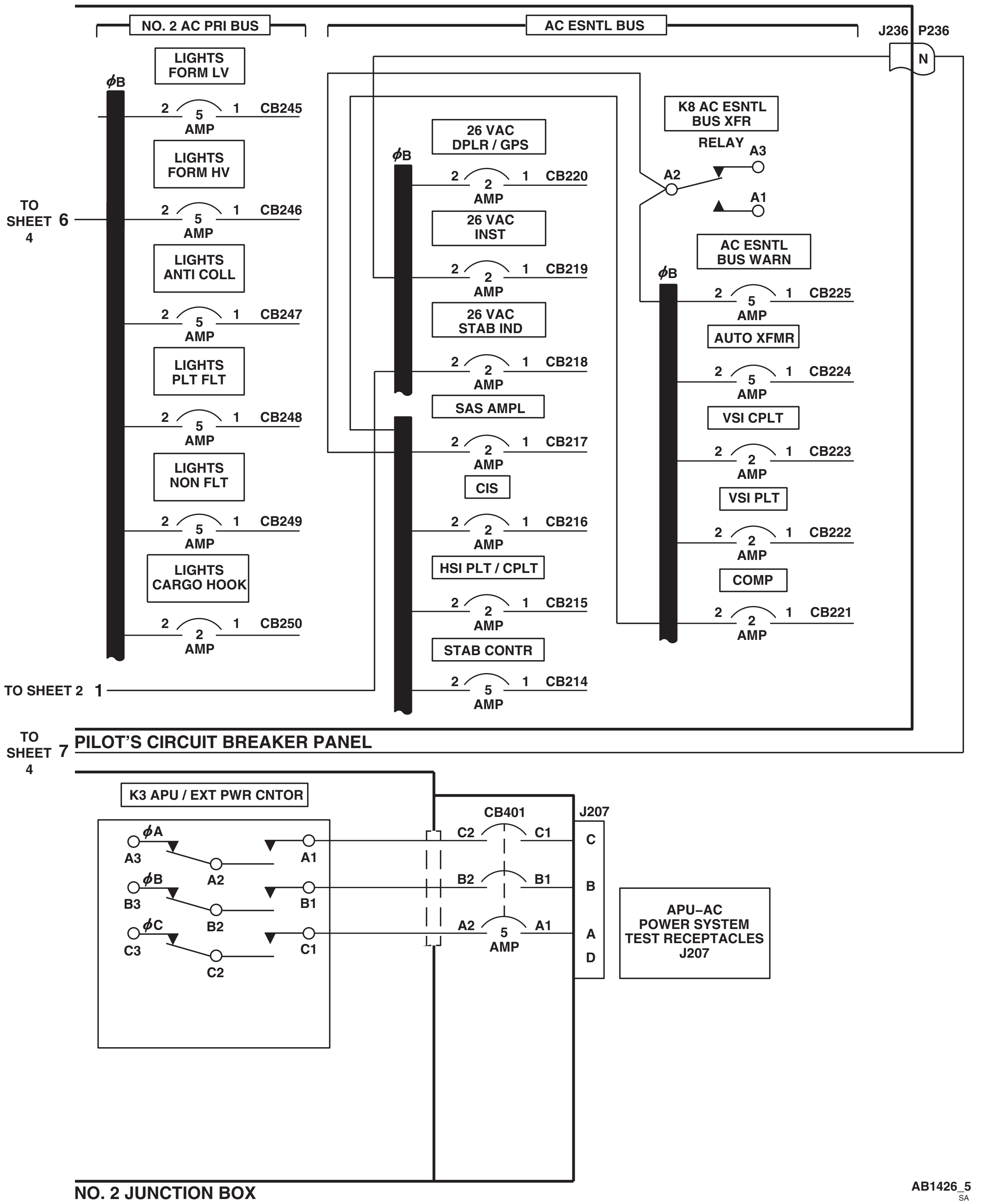


Figure 6. AC Power Distribution Diagram **UH-60L 96-26723-SUBQ** . (Sheet 5 of 5)

SCHEMATICS - Continued

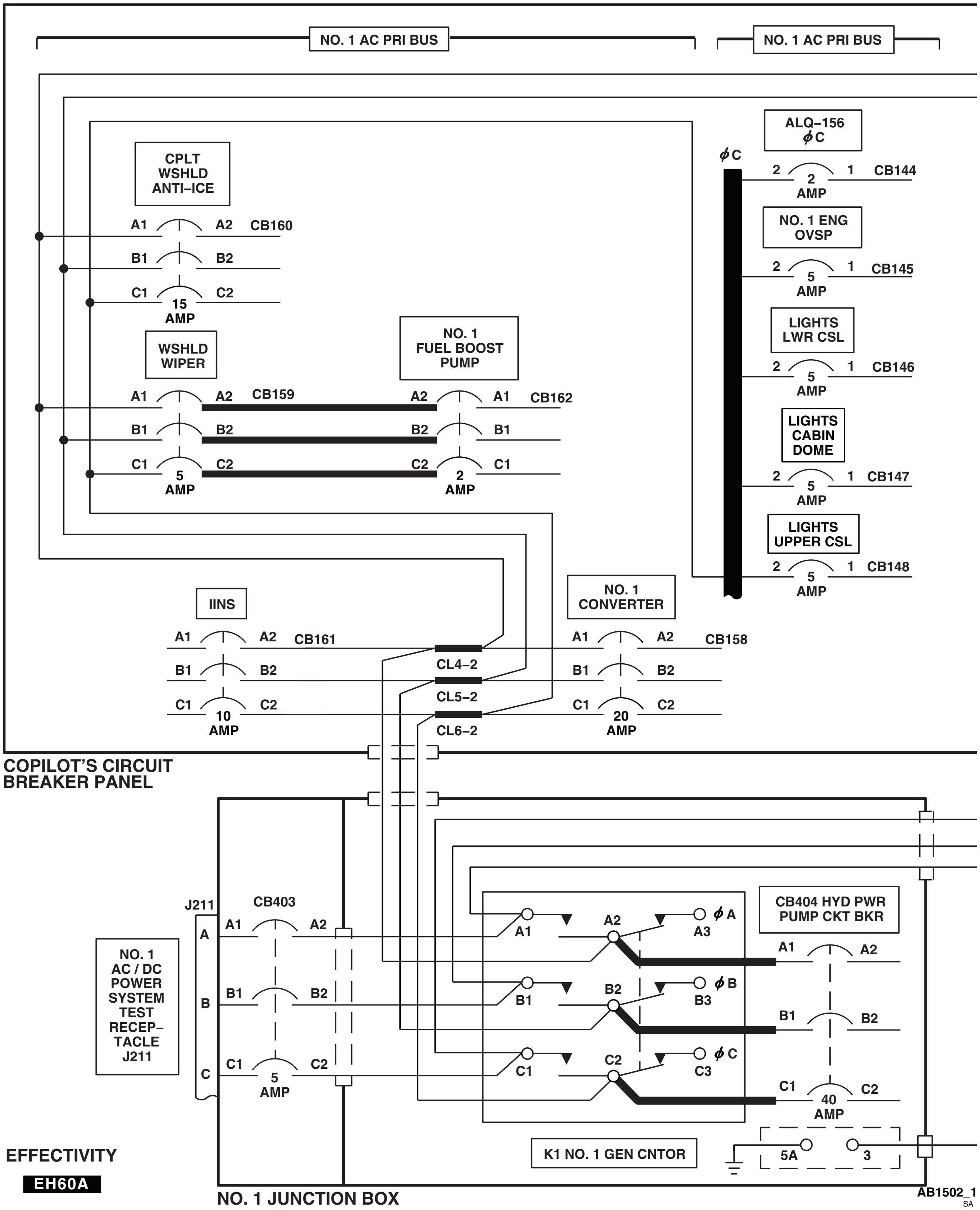


Figure 7. AC Power Distribution Diagram EH-60A . (Sheet 1 of 6)

SCHEMATICS - Continued

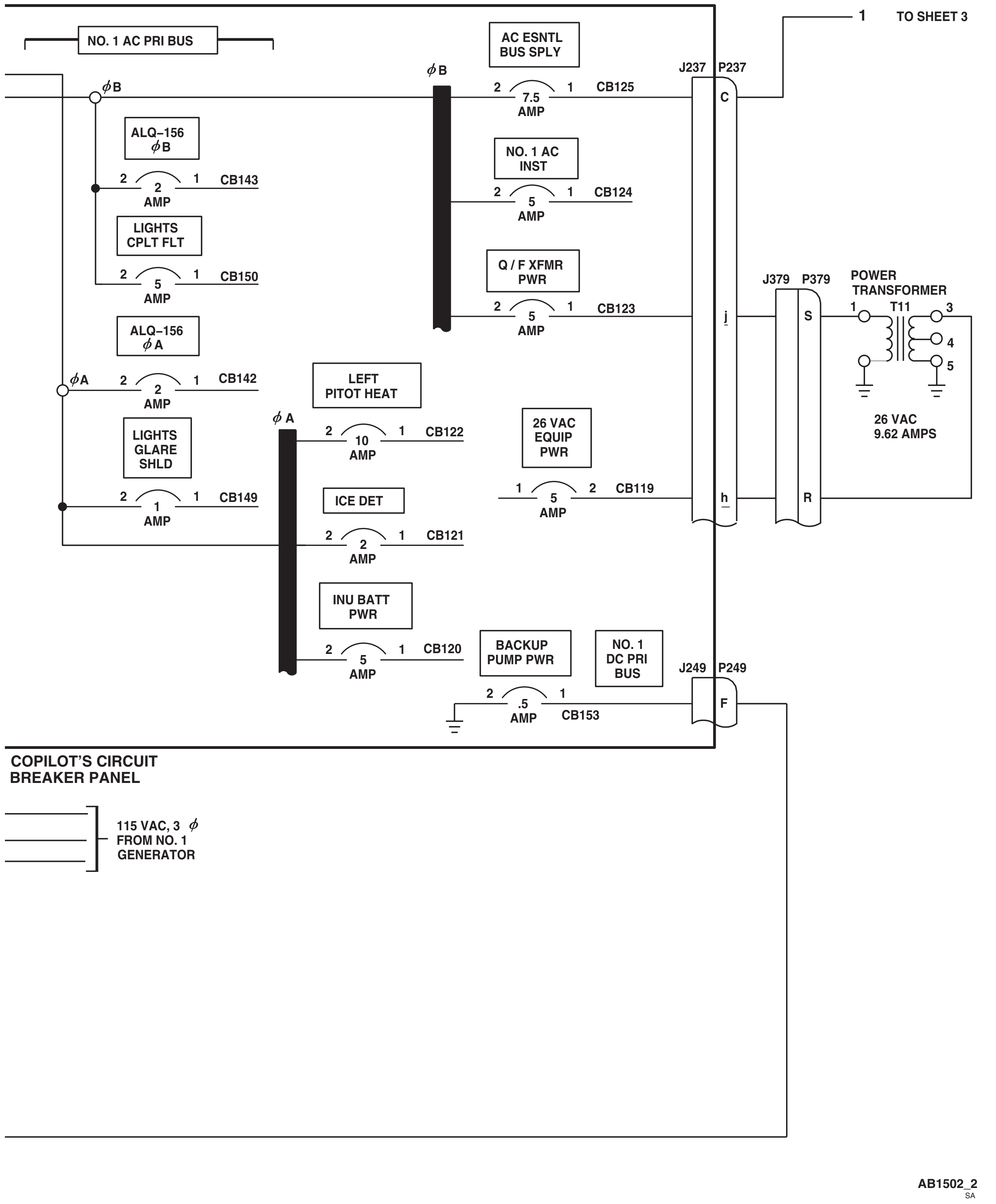


Figure 7. AC Power Distribution Diagram EH-60A . (Sheet 2 of 6)

SCHEMATICS - Continued

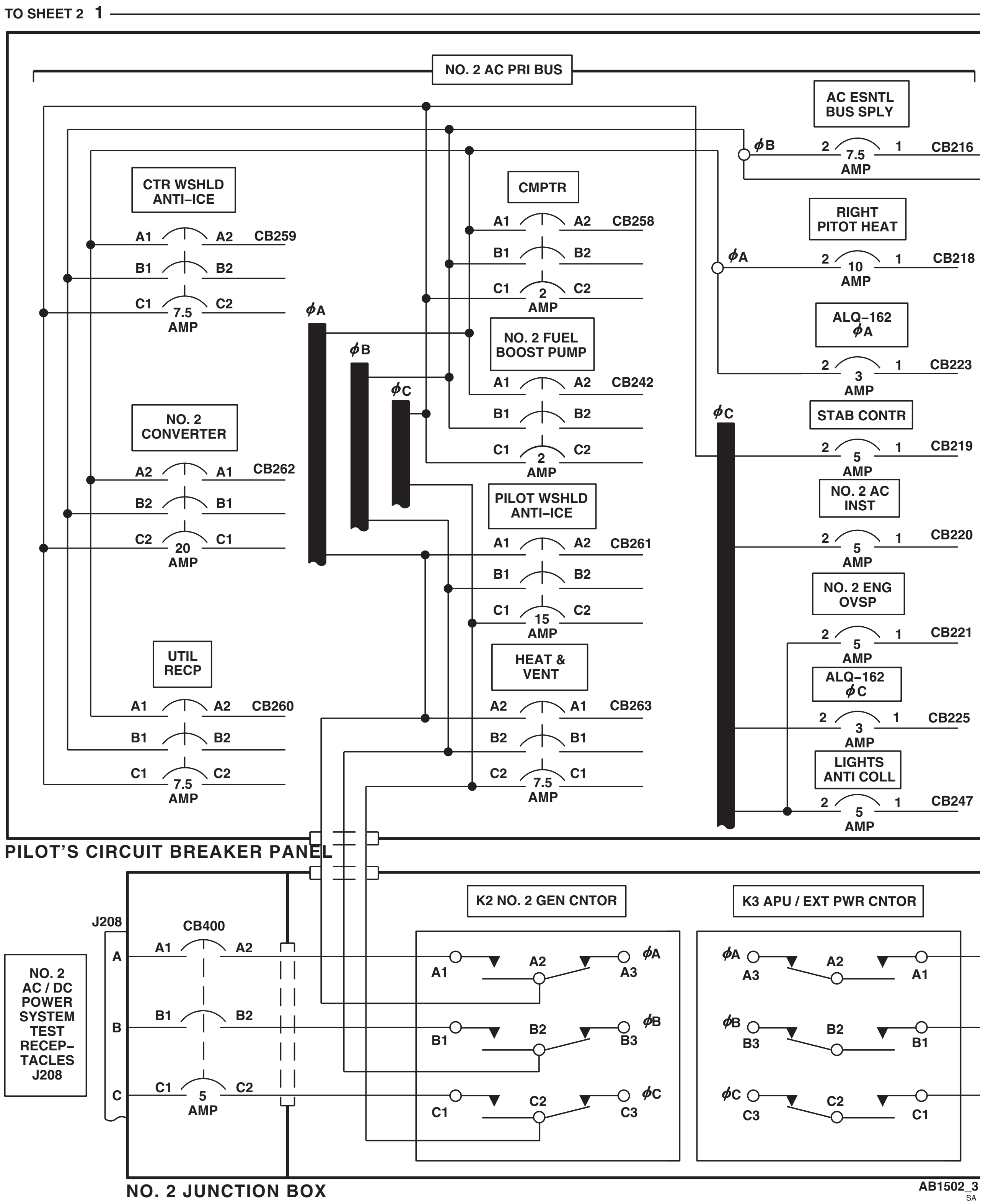


Figure 7. AC Power Distribution Diagram EH-60A . (Sheet 3 of 6)

SCHEMATICS - Continued

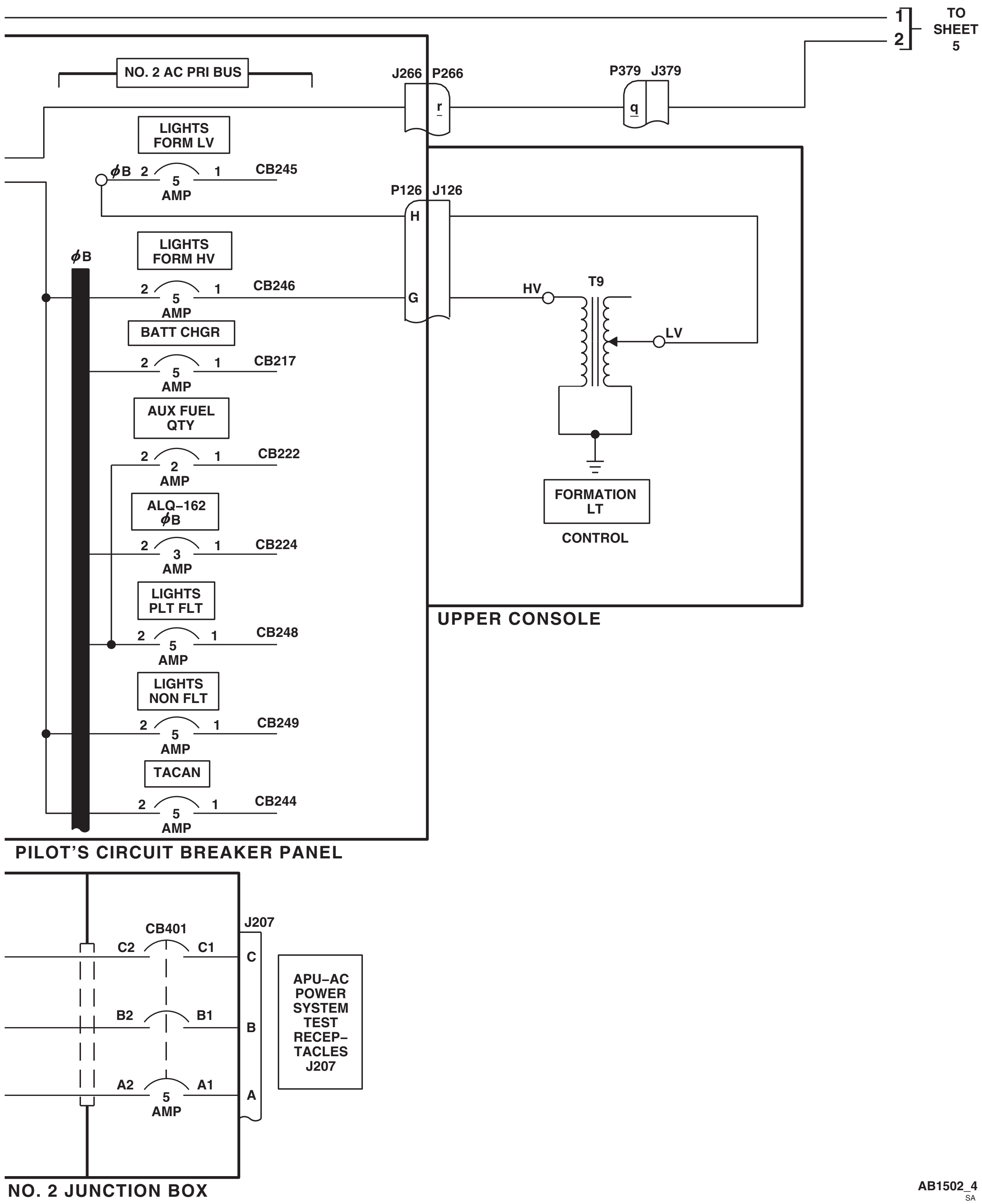


Figure 7. AC Power Distribution Diagram EH-60A . (Sheet 4 of 6)

SCHEMATICS - Continued

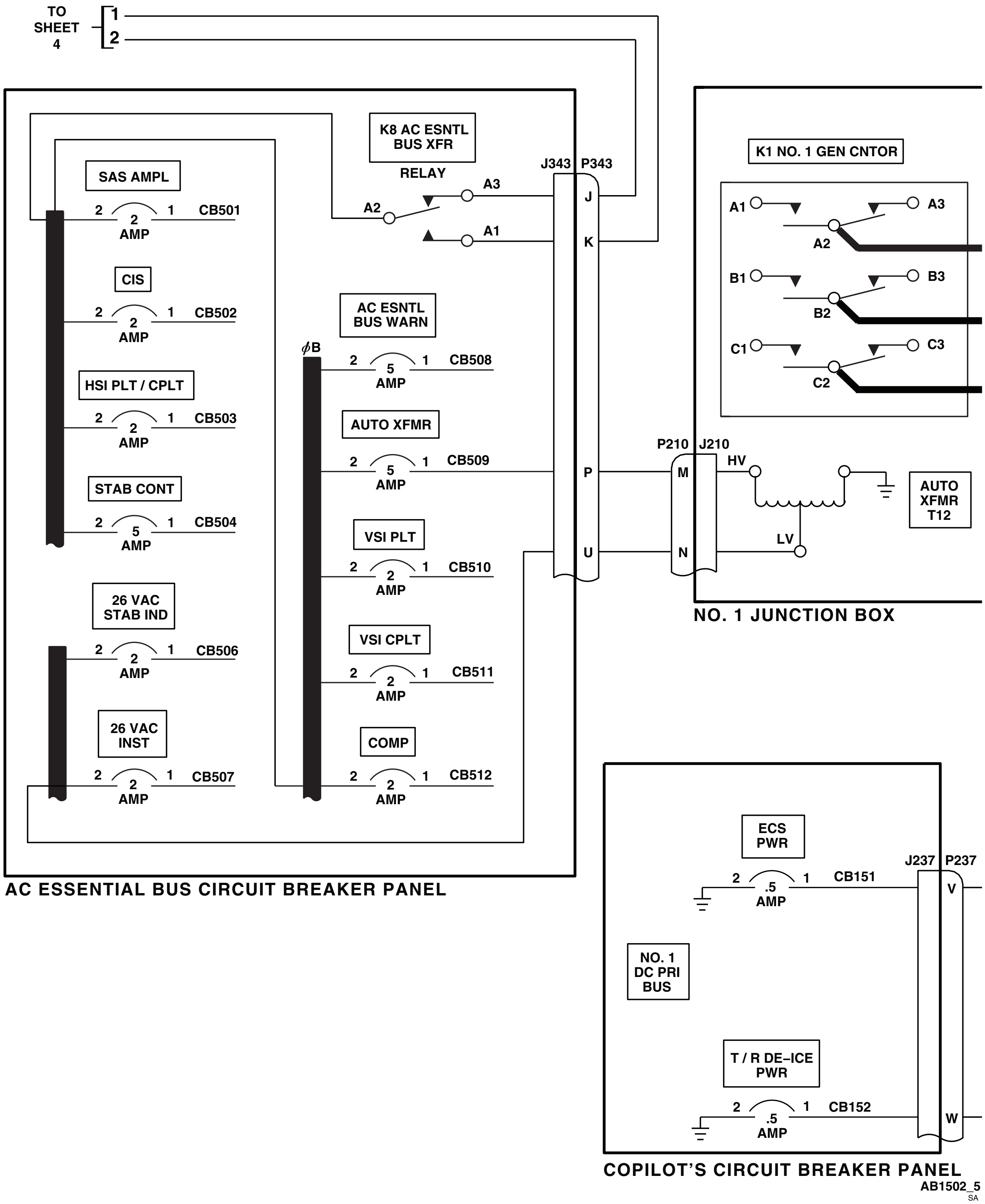


Figure 7. AC Power Distribution Diagram **EH-60A** . (Sheet 5 of 6)

SCHEMATICS - Continued

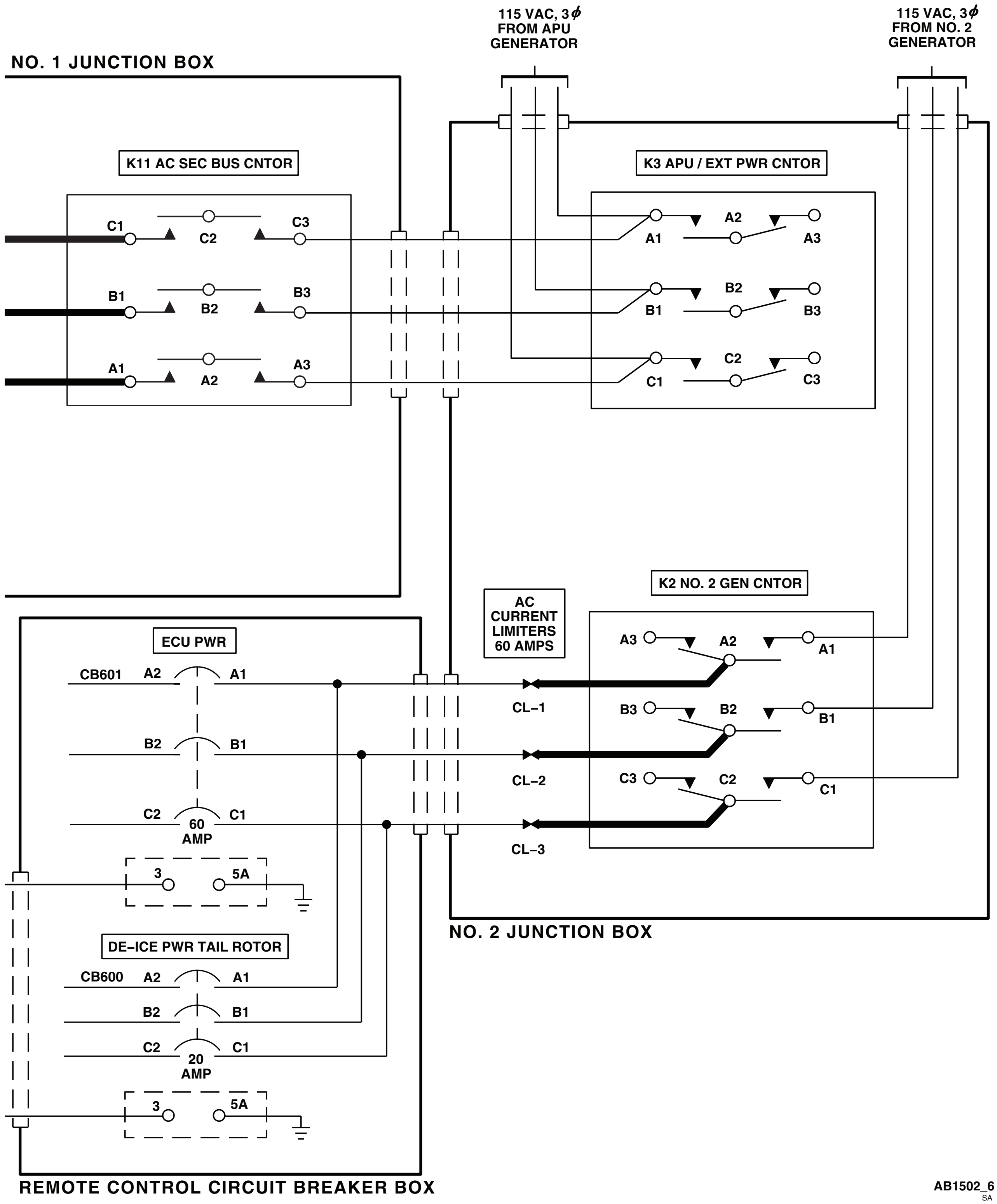


Figure 7. AC Power Distribution Diagram **EH-60A** . (Sheet 6 of 6)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

DC ELECTRICAL SYSTEM

INITIAL SETUP:

Test Equipment

Battery Sensor Relay Tester Figure 258, WP 1805 00
Multimeter, AS/PSM-45A

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required

Aircraft Electrician MOS 15F (1)

References

WP 0828 00
WP 0090 00
WP 0104 00
WP 0824 00
WP 0825 00
WP 0853 00
WP 0854 00
WP 0855 00
WP 0856 00
WP 0857 00
WP 0858 00
WP 0859 00
WP 0861 00
WP 0862 00
WP 0863 00
WP 0864 00
WP 0925 00
WP 0931 00
WP 0936 00
WP 1685 00
WP 1747 00
WP 1805 00

Equipment Condition

External Power Available, WP 1685 00
or APU Operational,
TM 1-1520-237-10TM 1-1520-237-10

SPECIAL INSTRUCTIONS

NOTE

See **UH60A UH60L** Figure 1 and **EH60A** Figure 2 for component location diagrams, **UH60L 96-26723-SUBQ MWO 50-77** Figure 3 , **UH60A UH60L** Figure 4 and **EH60A** Figure 5 for schematic diagrams, and **UH60A UH60L** Figure 6 and **EH60A** Figure 7 for distribution diagrams as an aid in operational/troubleshooting.

The dc electrical system operational/troubleshooting procedure is subdivided as follows:

- Battery
- No. 1 Converter
- No. 2 Converter
- Utility DC **UH60A UH60L**
- Mission Interface Power **EH60A**
- Battery Low Sensing **UH60L 96-26723 - SUBQ**
- Shutdown

SPECIAL INSTRUCTIONS - Continued

The battery sensor relay tester may be used at the AVIM level to check the battery low sensor relay. The following preset conditions must be met prior to applying power from a bench power supply:

- S2 must be set to the on position on the battery sensor relay tester.
- The rheostat R1 on the battery sensor relay tester must be set to approximately five (5) ohms to prevent damage to the zenner diode.

BATTERY OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- Do not start APU or connect external electrical power for battery system checkout.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Place upper console BATT switch to ON.

Indication/Condition

a. BATT LOW CHARGE capsule shall be off.

b. 77-27714-96-26722 BATTERY FAULT capsule shall be off.

c. DC ESS BUS OFF capsule shall be off.

d. SAS OFF, STABILATOR, AC ESS BUS OFF, and BOOST SERVO OFF capsules shall be on.

e. #1 CONV capsule shall be on.

f. #2 CONV capsule shall be on.

Corrective Action

1. 77-27714-96-26722 If no caution/advisory capsules go on, go to NO CAUTION/ADVISORY CAPSULES GO ON WHEN BATT SWITCH IS PLACES TO ON, in this work package.
2. UH60L 96-26723-SUBQ If no caution/advisory capsules go on, go to NO CAUTION/ADVISORY CAPSULES GO ON WHEN BATT SWITCH IS PLACED TO ON, in this work package.
3. 77-27714-96-26722 If Indication/Condition a. is not as specified, go to BATT LOW CHARGE CAPSULE IS ON WHEN BATT SWITCH IS SET TO ON, in this work package.
4. UH60L 96-26723-SUBQ If Indication/Condition a. is not as specified, go to BATT LOW CHARGE CAPSULE IS ON WHEN BATT SWITCH IS SET TO ON, in this work package.
5. 77-27714-96-26722 EMEP If Indication/Condition b. is not as specified, place BATT switch to OFF, remove pin filtered adapter from P123, and do pin filtered adapter check (WP 0925 00).

   a. If pin filtered adapter is good, check continuity between P123 and P141.

      (1) If continuity is present, replace battery or battery conditioner/analyzer as required (WP 0857 00) or (WP 0858 00).

      (2) If continuity is not present, repair/replace wiring as required (WP 1747 00).

BATTERY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00).

c. Install pin filtered adapter and do EME check (WP 0925 00).

6. **77-27714-96-26722** **W/O EMEP** If Indication/Condition b. is not as specified, place BATT switch to OFF and check continuity between P123 and P141.

   a. If continuity is present, replace battery or battery conditioner/analyzer as required (WP 0857 00) or (WP 0858 00).

   b. If continuity is not present, repair/replace wiring as required (WP 1747 00).

7. If Indication/Condition c. is not as specified, go to DC ESS BUS OFF CAPSULE IS ON WHEN BATT SWITCH IS SET TO ON, in this work package.

8. If both #1 and #2 CONV capsules do not go on, go to #1 AND #2 CONV CAPSULES ARE NOT ON, in this work package.

9. If Indication/Condition e. is not as specified, go to #1 CONV CAPSULE IS NOT ON, in this work package.

10. If Indication/Condition f. is not as specified, go to #2 CONV CAPSULE IS NOT ON, in this work package.

Step

3. Pull out ESNTL BUS DC SPLY circuit breaker on BATT BUS (lower console circuit breaker panel).

Indication/Condition

DC ESS BUS OFF capsule shall be on.

Corrective Action

If Indication/Condition is not as specified, replace right relay panel (WP 0825 00).

Step

4. Push in ESNTL BUS DC SPLY circuit breaker (lower console circuit breaker panel).
5. Place upper console BATT switch to OFF.
6. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 1 CONVERTER OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Turn on electrical power, if not already done.

NO. 1 CONVERTER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

3. Make sure upper console BATT switch is OFF.
4. Pull out NO. 2 CONVERTER circuit breaker (pilot's circuit breaker panel).

Indication/Condition

a. #1 CONV capsule shall be off.

b. #2 CONV capsule shall be on.

c. DC ESS BUS OFF capsule shall be off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #1 CONV CAPSULE IS ON, in this work package.
2. If Indication/Condition b. is not as specified, go to #2 CONV CAPSULE IS ON, in this work package.
3. If Indication/Condition c. is not as specified, go to DC ESS BUS OFF CAPSULE IS ON WHEN BATT SWITCH IS SET TO ON, in this work package.

Step

5. Pull out ESNTL DC SENSE circuit breaker on DC ESNTL BUS (pilot's side upper console).

Indication/Condition

DC ESS BUS OFF capsule shall go on.

Corrective Action

If Indication/Condition is not as specified, go to DC ESS BUS OFF CAPSULE IS OFF WITH ESNTL DC SENSE CIRCUIT BREAKER PULLED, in this work package.

Step

6. Push in ESNTL DC SENSE circuit breaker (pilot's side upper console).
7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 2 CONVERTER OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Turn on electrical power, if not already done.
3. Make sure upper console BATT switch is OFF.

NO. 2 CONVERTER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

4. Pull out NO. 1 CONVERTER circuit breaker (copilot's circuit breaker panel).

Indication/Condition

a. #2 CONV capsule shall be off.

b. #1 CONV capsule shall be on.

c. DC ESS BUS OFF capsule shall be off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #2 CONV CAPSULE IS ON, in this work package.
2. If Indication/Condition b. is not as specified, go to #1 CONV CAPSULE IS NOT ON, in this work package.
3. If Indication/Condition c. is not as specified, go to DC ESS BUS OFF CAPSULE IS ON WHEN BATT SWITCH IS SET TO ON, in this work package.

Step

5. Pull out ESNTL DC SENSE circuit breaker on DC ESNTL BUS (pilot's side upper console).

Indication/Condition

DC ESS BUS OFF capsule shall go on.

Corrective Action

If Indication/Condition is not as specified, replace K9 NO. 2 DC ESNTL BUS SPLY relay (WP 0855 00).

Step

6. Push in ESNTL DC SENSE circuit breaker (pilot's side upper console).
7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

UTILITY DC UH60A UH60L OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Make sure upper console BATT switch is OFF.
3. Turn on electrical power, if not already done.

UTILITY DC **UH60A UH60L** **Operational/Troubleshooting Procedure** - Continued

Indication/Condition

DC ESS BUS OFF capsule shall be off.

Corrective Action

If Indication/Condition is not as specified, go to DC ESS BUS OFF CAPSULE IS ON WHEN BATT SWITCH IS SET TO ON, in this work package.

Step

4. Pull out ESNTL DC SENSE circuit breaker (upper console circuit breaker panel).

Indication/Condition

DC ESS BUS OFF capsule shall go on.

Corrective Action

1. If Indication/Condition is not as specified, check helicopter configuration, go to EMEP or W/O EMEP.
   a. EMEP Remove pin filtered adapter from P117, and do pin filtered adapter check (WP 0925 00).
      (1) EMEP If pin filtered adapter is good, install pin filtered adapter (WP 0925 00).
      (2) EMEP If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00).
   b. W/O EMEP Check for 28 vdc between P117-36 and ground.
      (1) If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00).
      (2) If voltage is not as specified, check continuity between P117-36 and P900-G.
         (a) If continuity is present, replace right relay panel (WP 0825 00).
         (b) If continuity is not present, repair/replace wiring (WP 1747 00).

Step

5. Push in ESNTL DC SENSE circuit breaker (upper console circuit breaker panel).
6. Check for 28 vdc between J258-B and J258-A.

Indication/Condition

Multimeter shall indicate 28 vdc.

Corrective Action

1. If Indication/Condition is not as specified, check for 28 vdc between UTIL RECP CABIN circuit breaker terminal 1 (copilot's circuit breaker panel) and ground.
   a. If voltage is not as specified, replace circuit breaker (WP 0824 00).
   b. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J258-B, or between J258-A and ground, as required (WP 1747 00).

Step

7. Check for 28 vdc between J265-B and J265-A.

Indication/Condition

Multimeter shall indicate 28 vdc.

UTILITY DC UH60A UH60L Operational/Troubleshooting Procedure - Continued

Corrective Action

1. If Indication/Condition is not as specified, check for 28 vdc between CMD CSL SET circuit breaker terminal 1 (copilot's circuit breaker panel) and ground.
   a. If voltage is not as specified, replace circuit breaker (WP 0824 00).
   b. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J265-B, or between J265-A and ground, as required (WP 1747 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

MISSION INTERFACE POWER EH60A OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Turn on electrical power, if not already done.
3. Check for 28 vdc between J258-B and J258-A of UTIL RECP +28 VDC.

Indication/Condition

Multimeter shall indicate 28 vdc.

Corrective Action

If Indication/Condition is not as specified, go to NO 28 VDC INDICATION BETWEEN J258-A AND J258-B, in this work package.

Step

4. Check for 28 vdc between J510-A and J510-B.

Indication/Condition

Multimeter shall indicate 28 vdc.

Corrective Action

1. If Indication/Condition is not as specified, check for 28 vdc between Q/F EQUIP PWR circuit breaker terminal 1 (copilot's circuit breaker panel) and ground.
   a. If voltage is not as specified, replace circuit breaker (WP 0824 00).

MISSION INTERFACE POWER EH60A Operational/Troubleshooting Procedure - Continued

b. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J510-B, or between J510-A and ground, as required (WP 1747 00).

Step

5. Place upper console Q/F PWR switch to ON.
6. Check for 28 vdc between the following points:
   - J512-E and J512-A
   - J512-E and J512-B
   - J512-E and J512-D

Indication/Condition

Multimeter shall indicate 28 vdc between the specified points.

Corrective Action

1. If Indication/Condition is not as specified for any points, repair/replace wiring as required (WP 1747 00).
2. If Indication/Condition is not as specified for all points, go to NO VOLTAGE ON J512, in this work package.

Step

7. Place upper console Q/F PWR switch to OFF.
8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

BATTERY LOW SENSING UH60L 96-26723 - SUBQ OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Turn off electrical power.
2. Make sure upper console BATT switch is OFF.
3. Disconnect electrical connector P140 from battery.
4. Connect P140 to TP3 and TP4 of battery sensor relay tester (tester) (WP 1805 00) as follows:
   - P140 (+) to red alligator clip.
   - P140 (-) to black alligator clip.

BATTERY LOW SENSING UH60L 96-26723 - SUBQ Operational/Troubleshooting Procedure - Continued

- Wire with red banana plug to TP3 on tester.
- Wire with black banana plug to TP4 on tester.

5. Using battery connector P1 with pigtail wires, connect battery to tester as follows:
   - Wire with black banana plug to TP2 of tester.
   - Wire with red banana plug to TP1 of tester.
   - Connector P1 to battery.

NOTE

Make sure R1 is at the minimum resistance, 0 ohms, and S2 is in the off position before applying power to the tester when use on the aircraft.

6. Connect multimeter to TP3 and TP4 on tester.
7. Place upper console BATT switch ON.
8. On tester, push in circuit breaker CB1 and place switches S1 and S2 to on.
9. On tester, turn control R1 until 24 vdc or higher is observed on multimeter.

Indication/Condition

BATT LOW CHARGE capsule shall be off.

Corrective Action

If Indication/Condition is not as specified, go to BATT LOW CHARGE CAPSULE IS ON WHEN BATTERY VOLTAGE IS 24 VDC, in this work package.

Step

10. On tester, turn R1 slowly clockwise until battery voltage decreases to 22 vdc.

Indication/Condition

BATT LOW CHARGE capsule shall go on.

Corrective Action

If Indication/Condition is not as specified, go to BATT LOW CHARGE CAPSULE IS OFF WHEN BATTERY VOLTAGE IS 22 VDC, in this work package.

Step

11. Place upper console BATT switch OFF.
12. Disconnect all wiring from tester to electrical connector P140 and battery.
13. Reconnect P140 to battery.
14. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

NO CAUTION/ADVISORY CAPSULES GO ON WHEN BATT SWITCH IS PLACED TO ON 77-27714-96-26722

SYMPTOM

No caution/advisory capsules go on when BATT switch is placed to ON.

MALFUNCTION

No caution/advisory capsules go on when BATT switch is placed to ON.

CORRECTIVE ACTION

1. Check for greater than 23 vdc between K10-A3 and ground.

a. If voltage is as specified, repair/replace wiring between K10-A3 and DC ESNTL BUS (WP 1747 00). Go to **37.**

b. If voltage is not as specified, go to **2.**

2. Check for greater than 23 vdc between K10-A4 and ground.

a. If voltage is as specified, replace relay K10 (WP 0855 00). Go to **37.**

b. If voltage is not as specified, go to **3.**

3. Check for greater than 23 vdc between K9-A3 and ground.

a. If voltage is as specified, repair/replace wiring between K9-A3 and K10-A4 (WP 1747 00). Go to **37.**

b. If voltage is not as specified, go to **4.**

4. Check for greater than 23 vdc between K9-A1 and ground.

a. If voltage is as specified, repair/replace wiring between K9-A3 and K9-A1 (WP 1747 00). Go to **37.**

b. If voltage is not as specified, go to **5.**

5. Check for greater than 23 vdc between K9-A2 and ground.

a. If voltage is as specified, go to **6.**

b. If voltage is not as specified, go to **20.**

6. Check for greater than 23 vdc between K9-X1 and ground.

a. If voltage is as specified, go to **7.**

NO CAUTION/ADVISORY CAPSULES GO ON WHEN BATT SWITCH IS PLACED TO ON 77-27714-96-26722 - Continued

b. If voltage is not as specified, go to **8.**

7. **Check continuity between K9-X2 and ground.**

   a. If continuity is present, replace relay K9 (WP 0855 00). Go to **37.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **37.**

8. **Check continuity between P123-N and K9-X1.**

   a. If continuity is present, go to **9.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **37.**

9. **Check for greater than 23 vdc between P123-M and ground.**

   a. If voltage is as specified, go to **10.**

   b. If voltage is not as specified, repair/replace wiring between P123-M and P238-G (WP 1747 00). Go to **37.**

10. **Check for greater than 23 vdc between P123-R and ground.**

    a. If voltage is as specified, go to **11.**

    b. If voltage is not as specified, go to **18.**

11. **Check continuity between P123-P and ground.**

    a. If continuity is present, go to **12.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **37.**

12. **Check continuity between P123-D and ground.**

    a. If continuity is present, go to **13.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **37.**

13. **Check for less than 0.2 vdc between P123-C and P123-V.**

    a. If voltage is as specified, go to **14.**

    b. If voltage is not as specified, go to **15.**

14. **Check for less than 0.2 vdc between J141-C and J141-H.**

    a. If voltage is as specified, replace battery (WP 0857 00). Go to **37.**

    b. If voltage is not as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **37:**

       - P123-V and P141-H
       - P123-C and P141-C

15. **Check for voltage between 0.2 to 1.03 vdc between P123-C and P123-V.**

    a. If voltage is as specified, go to **16.**

    b. If voltage is not as specified, check for less than 23 vdc between P117-21 and ground.

    c. If condition exists, replace battery conditioner/analyzer (WP 0858 00). Go to **37.**

16. **Check for greater than 23 vdc between P117-21 and ground.**

    a. If voltage is as specified, replace battery (WP 0857 00). Go to **37.**

NO CAUTION/ADVISORY CAPSULES GO ON WHEN BATT SWITCH IS PLACED TO ON 77-27714-96-26722 - Continued

b. If voltage is not as specified, go to **17.**

17. Check continuity between P123-K and P117-21.

a. If continuity is present, replace battery conditioner/analyzer (WP 0858 00). Go to **37.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **37.**

18. Check for greater than 23 vdc between S10-6 and ground.

a. If voltage is as specified, repair/replace wiring between P123-R and S10-6 (WP 1747 00). Go to **37.**

b. If voltage is not as specified, go to **19.**

19. Check for greater than 23 vdc between S10-5 and ground.

a. If voltage is as specified, replace BATT switch S10 (WP 0861 00). Go to **37.**

b. If voltage is not as specified, repair/replace wiring between S10-5 and BATT BUS CONTR circuit breaker terminal 1 (WP 1747 00). Go to **37.**

20. Check for greater than 23 vdc between ESNTL BUS DC SPLY circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker and K9-A2 (WP 1747 00). Go to **37.**

b. If voltage is not as specified, go to **21.**

21. Check for greater than 23 vdc between ESNTL BUS DC SPLY circuit breaker terminal 2 and ground.

a. If voltage is as specified, replace circuit breaker (WP 0824 00). Go to **37.**

b. If voltage is not as specified, go to **22.**

22. Check for greater than 23 vdc between lower console circuit breaker panel terminal E1 and ground.

a. If voltage is as specified, repair/replace wiring between terminal E1 and BATT bus (WP 1747 00). Go to **37.**

b. If voltage is not as specified, go to **23.**

23. Check for greater than 23 vdc between K7-A1 and ground.

a. If voltage is as specified, repair/replace wiring between relay K7 and terminal E1 (WP 1747 00). Go to **37.**

b. If voltage is not as specified, go to **24.**

24. Check for greater than 23 vdc between K7-A2 and ground.

a. If voltage is as specified, go to **25.**

b. If voltage is not as specified, go to **35.**

25. Check for greater than 23 vdc between K7-X1 and K7-X2.

a. If voltage is as specified, replace battery relay K7 (WP 0856 00). Go to **37.**

b. If voltage is not as specified, go to **26.**

26. Check continuity between K7-X2 and ground.

NO CAUTION/ADVISORY CAPSULES GO ON WHEN BATT SWITCH IS PLACED TO ON 77-27714-96-26722 - Continued

a. If continuity is present, go to **27.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **37.**

27. Check for greater than 23 vdc between BATT switch terminal 3 and ground.

a. If voltage is as specified, repair/replace wiring between BATT switch terminal 3 and K7-X1 (WP 1747 00). Go to **37.**

b. If voltage is not as specified, go to **28.**

28. Check for greater than 23 vdc between BATT switch terminal 2 and ground.

a. If voltage is as specified, replace BATT switch (WP 0861 00). Go to **37.**

b. If voltage is not as specified, go to **29.**

29. Check continuity between BATT switch terminal 2 and P238-G.

a. If continuity is present, go to **30.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **37.**

30. Check for greater than 23 vdc between P238-H and ground.

a. If voltage is as specified, replace K6 NO. 2 DC PRI BUS CNTOR (WP 0853 00). Go to **37.**

b. If voltage is not as specified, go to **31.**

31. Check continuity between P238-H and P239-H.

a. If continuity is present, go to **32.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **37.**

32. Check for greater than 23 vdc between P239-G and ground.

a. If voltage is as specified, replace K16 NO. 1 DC PRI BUS CNTOR (WP 0853 00). Go to **37.**

b. If voltage is not as specified, go to **33.**

33. Check for greater than 23 vdc between BATT BUS CONTR circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P239-G (WP 1747 00). Go to **37.**

b. If voltage is not as specified, go to **34.**

34. Check for greater than 23 vdc between BATT BUS CONTR circuit breaker terminal 2 and ground.

a. If voltage is as specified, replace circuit breaker (WP 0824 00). Go to **37.**

b. If voltage is not as specified, repair/replace wiring between circuit breaker terminal 2 and K7-A2 (WP 1747 00). Go to **37.**

35. Check continuity between:

- **K7-A2 and P140 (+)**
- **K7-A2 and Terminal E2 (lower console circuit breaker panel)**

a. If continuity is present, go to **36.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **37.**

NO CAUTION/ADVISORY CAPSULES GO ON WHEN BATT SWITCH IS PLACED TO ON 77-27714-96-26722 - Continued

36. Check continuity between P140 (-) and ground.

a. If continuity is present, replace battery (WP 0857 00). Go to **37.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **37.**

37. Procedure completed.

NO CAUTION/ADVISORY CAPSULES GO ON WHEN BATT SWITCH IS PLACED TO ON UH60L 96-26723-SUBQ

SYMPTOM

No caution/advisory capsules go on when BATT switch is placed to ON.

MALFUNCTION

No caution/advisory capsules go on when BATT switch is placed to ON.

CORRECTIVE ACTION

1. Check for greater than 23 vdc between K9-X1 and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **3.**

2. Check continuity between K9-X2 and ground.

a. If continuity is present, go to **4.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20.**

3. Check for greater than 23 vdc between BATT switch terminal 3 and ground.

a. If voltage is as specified, repair/replace wiring between BATT switch terminal 3 and K9-X1 (WP 1747 00). Go to **20.**

b. If voltage is not as specified, go to **15.**

4. Check for greater than 23 vdc between K10-A3 and ground.

a. If voltage is as specified, repair/replace wiring between K10-A3 and DC ESNTL Bus (WP 1747 00). Go to **20.**

b. If voltage is not as specified, go to **5.**

5. Check for greater than 23 vdc between K10-A4 and ground.

a. If voltage is as specified, replace K10 (WP 0855 00). Go to **20.**

b. If voltage is not as specified, go to **6.**

6. Check for greater than 23 vdc between K9-A3 and ground.

a. If voltage is as specified, repair/replace wiring between K9-A3 and K10-A4 (WP 1747 00). Go to **20.**

b. If voltage is not as specified, go to **7.**

7. Check for greater than 23 vdc between K9-A2 and ground.

a. If voltage is as specified, replace relay K9 (WP 0855 00). Go to **20.**

b. If voltage is not as specified, go to **8.**

NO CAUTION/ADVISORY CAPSULES GO ON WHEN BATT SWITCH IS PLACED TO ON UH60L 96-26723-SUBQ - Continued

8. **Check for greater than 23 vdc between terminal 1 of ESNTL BUS DC SPLY circuit breaker and ground.**

   a. If voltage is as specified, repair/replace wiring between terminal 1 of ESNTL BUS DC SPLY circuit breaker and K9-A2 (WP 1747 00). Go to **20.**

   b. If voltage is not as specified, go to **9.**

9. **Check for greater than 23 vdc between terminal 2 of ESNTL BUS DC SPLY circuit breaker and ground.**

   a. If voltage is as specified, replace ESNTL BUS DC SPLY circuit breaker (WP 0824 00). Go to **20.**

   b. If voltage is not as specified, go to **10.**

10. **Check for greater than 23 vdc between K7-X1 and ground.**

    a. If voltage is as specified, go to **11.**

    b. If voltage is not as specified, repair/replace wiring between BATT switch terminal 3 and K7-X1 (WP 1747 00). Go to **20.**

11. **Check continuity between:**

    - **K7-X2 and ground**
    - **K7-A1 and K7-A3**

    a. If continuity is present, go to **12.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20.**

12. **Check for greater than 23 vdc between K7-A3 and ground.**

    a. If voltage is as specified, repair/replace wiring between K7-A3and BATT BUS (WP 1747 00). Go to **20.**

    b. If voltage is not as specified, go to **13.**

13. **Check for greater than 23 vdc between K7-A2 and ground.**

    a. If voltage is as specified, replace K7 (WP 0856 00). Go to **20.**

    b. If voltage is not as specified, go to **14.**

14. **Check continuity between:**

    - **P140 + and K7-A2**
    - **P140 - and ground**

    a. If continuity is present, replace battery (WP 0857 00). Go to **20.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20.**

15. **Check for greater than 23 vdc between BATT switch terminal 2 and ground.**

    a. If voltage is as specified, replace BATT switch (WP 0861 00). Go to **20.**

    b. If voltage is not as specified, go to **16.**

16. **Check for greater than 23 vdc between BATT BUS CONTR circuit breaker terminal 1 and ground.**

    a. If voltage is as specified, go to **17.**

NO CAUTION/ADVISORY CAPSULES GO ON WHEN BATT SWITCH IS PLACED TO ON UH60L 96-26723-SUBQ - Continued

b. If voltage is not as specified, go to **19.**

17. Check continuity between terminals 7 and 9 of contactor K16.

a. If continuity is present, go to **18.**

b. If continuity is not present, replace K16 (WP 0853 00). Go to **20.**

18. Check continuity between terminals 7 and 9 of contactor K6.

a. If continuity is present, repair/replace wiring between the following as required (WP 1747 00), then go to **20:**

- BATT BUS CONTR terminal 1 and K16-9
- K16-7 and K6-7
- K6-9 and BATT switch terminal 2

b. If continuity is not present, replace K16 (WP 0853 00). Go to **20.**

19. Check for greater than 23 vdc between BATT BUS CONTR circuit breaker terminal 2 and ground.

a. If voltage is as specified, replace BATT BUS CONTR circuit breaker (WP 0824 00. Go to **20.**

b. If voltage is not as specified, repair/replace wiring (WP 1747 00). Go to **20.**

20. Procedure completed.

BATT LOW CHARGE CAPSULE IS ON WHEN BATT SWITCH IS SET TO ON 77-27714-96-26722

SYMPTOM

BATT LOW CHARGE capsule is on when BATT switch is set to ON.

MALFUNCTION

BATT LOW CHARGE capsule is on when BATT switch is set to ON.

CORRECTIVE ACTION

1. Turn on electrical power. Go to 2.

2. Place BATT switch to OFF. Go to 3.

3. Check helicopter configuration.

a. W/O EMEP , go to **6.**

b. EMEP , go to **4.**

4. EMEP Remove and check pin filtered adapter from P123 (WP 0925 00).

a. If pin filtered adapter is good, go to **5.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **14.**

5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.

6. Check for greater than 23 vdc between P123-S and ground.

a. If voltage is as specified, go to **7.**

b. If voltage is not as specified, go to **11.**

BATT LOW CHARGE CAPSULE IS ON WHEN BATT SWITCH IS SET TO ON 77-27714-96-26722 - Continued

7. **Check for 115 vac between P123-H and ground.**

   a. If voltage is as specified, go to **8.**

   b. If voltage is not as specified, go to **10.**

8. **Check for greater than 23 vdc between P123-J and ground.**

   a. If voltage is as specified, replace battery condition/analyzer (WP 0858 00). Go to **14.**

   b. If voltage is not as specified, go to **9.**

9. **Check continuity between P123-J and DC BATT CHRG circuit breaker terminal 1.**

   a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **14.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**

10. **Check for 115 vac between AC BATT CHGR circuit breaker terminal 1 and ground.**

    a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P123-H (WP 1747 00). Go to **14.**

    b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **14.**

11. **Check for greater than 23 vdc between K7-X1 and K7-X2.**

    a. If voltage is as specified, replace BATT switch S10 (WP 0861 00). Go to **14.**

    b. If voltage is not as specified, go to **12.**

12. **Check continuity between P123-S and K7-13.**

    a. If continuity is present, go to **13.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**

13. **Check continuity between K7-12 and K7-A2.**

    a. If continuity is present, replace battery relay K7 (WP 0856 00). Go to **14.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**

14. **Procedure completed.**

BATT LOW CHARGE CAPSULE IS ON WHEN BATT SWITCH IS SET TO ON UH60L 96-26723-SUBQ

SYMPTOM

BATT LOW CHARGE capsule is on when BATT switch is set to ON.

MALFUNCTION

BATT LOW CHARGE capsule is on when BATT switch is set to ON.

CORRECTIVE ACTION

1. **Check for 28 vdc between K201-X1 and K201-X2.**

   a. If voltage is as specified, replace K201 (WP 0936 00). Go to **4.**

   b. If voltage is not as specified, go to **2.**

2. **Check for 28 vdc between K201-X1 and ground.**

   a. If voltage is as specified, repair/repair wiring (WP 1747 00). Go to **4.**

   b. If voltage is not as specified, go to **3**.

BATT LOW CHARGE CAPSULE IS ON WHEN BATT SWITCH IS SET TO ON UH60L 96-26723-SUBQ - Continued

3. **Check continuity between:**
   - **K201-X1 and BATT switch terminal 6**
   - **BATT switch terminal 5 and P107-H**

   a. If continuity is present, replace BATT switch (WP 0861 00). Go to **4.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**

4. **Procedure completed.**

DC ESS BUS OFF CAPSULE IS ON WHEN BATT SWITCH IS SET TO ON

SYMPTOM

DC ESS BUS OFF capsule is on when BATT switch is set to ON.

MALFUNCTION

DC ESS BUS OFF capsule is on when BATT switch is set to ON.

CORRECTIVE ACTION

1. **Check for greater than 23 vdc between P900-J and ground.**

   a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **3.**

   b. If voltage is not as specified, go to **2.**

2. **Check for greater than 23 vdc between ESNTL DC SENSE circuit breaker terminal 1 and ground.**

   a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P900-J (WP 1747 00). Go to **3.**

   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **3.**

3. **Procedure completed.**

#1 AND #2 CONV CAPSULES ARE NOT ON

SYMPTOM

#1 and #2 CONV capsules are not on.

MALFUNCTION

#1 and #2 CONV capsules are not on.

CORRECTIVE ACTION

1. **Check for 28 vdc between K13 AC ESNTL BUS FAIL relay terminal A2 and ground.**

   a. If voltage is as specified, repair/replace wiring between terminal A2 and SGP203-1 (WP 1747 00). Go to **3.**

   b. If voltage is not as specified, go to **2.**

2. **Check for 28 vdc between ESNTL BUS AC & CONV WARN circuit breaker terminal 1 and ground.**

   a. If voltage is as specified, repair/replace wiring between terminal 1 and K13 AC ESNTL BUS FAIL relay terminal A2 (WP 1747 00). Go to **3.**

   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **3.**

#1 AND #2 CONV CAPSULES ARE NOT ON - Continued

3. Procedure completed.

#1 CONV CAPSULE IS NOT ON

SYMPTOM

#1 CONV capsule is not on.

MALFUNCTION

#1 CONV capsule is not on.

CORRECTIVE ACTION

1. Hold caution/advisory panel BRT/DIM-TEST switch to TEST.

a. If #1 CONV capsule goes on, go to **2.**

b. If #1 CONV capsule does not go on, troubleshoot caution/advisory warning system (WP 0090 00). Go to **8.**

2. Check helicopter configuration.

a. **EMEP** , go to **3.**

b. **W/O EMEP** , go to **5.**

3. EMEP Remove and check pin filtered adapter from P117 (WP 0925 00).

a. If pin filtered adapter is good, go to **4.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **8.**

4. EMEP Install pin filtered adapter (WP 0925 00). Go to 5.

5. Check for 28 vdc between P117-19 and ground.

a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00). Go to **8.**

b. If voltage is not as specified, go to **6.**

6. Check continuity between P117-19 and P239-E.

a. If continuity is present, go to **7.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

7. Check for 28 vdc between P239-J and ground.

a. If voltage is as specified, replace K16 NO. 1 DC PRI BUS CNTOR (WP 0853 00). Go to **8.**

b. If voltage is not as specified, repair/replace wiring between K13 AC ESNTL BUS FAIL relay terminal A2 and P239-J (WP 1747 00). Go to **8.**

8. Procedure completed.

#2 CONV CAPSULE IS NOT ON

SYMPTOM

#2 CONV capsule is not on.

MALFUNCTION

#2 CONV capsule is not on.

CORRECTIVE ACTION

1. **Hold caution/advisory panel BRT/DIM-TEST switch to TEST.**
   a. If #2 CONV capsule goes on, go to **2.**
   b. If #2 CONV capsule does not go on, troubleshoot caution/advisory warning system (WP 0090 00). Go to **8.**
2. **Check helicopter configuration.**
   a. **EMEP** , go to **3.**
   b. **W/O EMEP** , go to **5.**
3. **EMEP Remove and check pin filtered adapter from P117 (WP 0925 00).**
   a. If pin filtered adapter is good, go to **4.**
   b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **8.**
4. **EMEP Install pin filtered adapter (WP 0925 00). Go to 5.**
5. **Check for 28 vdc between P117-35 and ground.**
   a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00). Go to **8.**
   b. If voltage is not as specified, go to **6.**
6. **Check continuity between P117-35 and P238-C.**
   a. If continuity is present, go to **7.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**
7. **Check for 28 vdc between P238-K and ground.**
   a. If voltage is as specified, replace K6 NO. 2 DC PRI BUS CNTOR (WP 0853 00). Go to **8.**
   b. If voltage is not as specified, repair/replace wiring between K13 AC ESNTL BUS FAIL relay terminal A2 and P238-K (WP 1747 00). Go to **8.**
8. **Procedure completed.**

#1 CONV CAPSULE IS ON

SYMPTOM

#1 CONV capsule is on.

MALFUNCTION

#1 CONV capsule is on.

CORRECTIVE ACTION

1. **Check for 28 vdc between P239-A and P239-B.**

   a. If voltage is as specified, replace K16 NO. 1 DC PRI BUS CNTOR (WP 0853 00). Go to **8.**

   b. If voltage is not as specified, go to **2.**

2. **Check for 28 vdc between P239-A and ground.**

   a. If voltage is as specified, repair/replace wiring between P239-B and ground (WP 1747 00). Go to **8.**

   b. If voltage is not as specified, go to **3.**

3. **Check for 28 vdc between K16 NO. 1 DC PRI BUS CNTOR terminal A1 and ground.**

   a. If voltage is as specified, repair/replace wiring between contactor terminal A1 and P239-A (WP 1747 00). Go to **8.**

   b. If voltage is not as specified, go to **4.**

4. **Check continuity between No. 1 converter and K16 NO. 1 DC PRI BUS CNTOR terminal A1.**

   a. If continuity is present, go to **5.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

5. **Check for 115 vac between:**

   - **P903-A and P903-D**
   - **P903-B and P903-D**
   - **P903-C and P903-D**

   a. If voltage is as specified, replace No. 1 converter (WP 0859 00). Go to **8.**

   b. If voltage is not as specified, go to **6.**

6. **Check continuity between P903-D and ground.**

   a. If continuity is present, go to **7.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

7. **Check for 115 vac between NO. 1 CONVERTER circuit breaker:**

   - **Terminal A1 and ground**
   - **Terminal B1 and ground**
   - **Terminal C1 and ground**

#1 CONV CAPSULE IS ON - Continued

a. If voltage is as specified, repair/replace wiring between circuit breaker and No. 1 converter (WP 1747 00). Go to **8.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8.**

8. **Procedure completed.**

DC ESS BUS OFF CAPSULE IS OFF WITH ESNTL DC SENSE CIRCUIT BREAKER PULLED

SYMPTOM

DC ESS BUS OFF capsule is off with ESNTL DC SENSE circuit breaker pulled.

MALFUNCTION

DC ESS BUS OFF capsule is off with ESNTL DC SENSE circuit breaker pulled.

CORRECTIVE ACTION

1. **Check for 28 vdc between K10 NO. 1 DC ESNTL BUS SPLY relay terminals X1 and X2.**

   a. If voltage is as specified, replace relay (WP 0855 00). Go to **5.**

   b. If voltage is not as specified, go to **2.**

2. **Check for 28 vdc between K10 NO. 1 DC ESNTL BUS SPLY relay terminal X1 and ground.**

   a. If voltage is as specified, repair/replace wiring between relay terminal X2 and ground (WP 1747 00). Go to **5.**

   b. If voltage is not as specified, go to **3.**

3. **Check for 28 vdc between K10 NO. 1 DC ESNTL BUS SPLY relay terminal A2 and ground.**

   a. If voltage is as specified, repair/replace wiring between relay terminal A2 and X1 (WP 1747 00). Go to **5.**

   b. If voltage is not as specified, go to **4.**

4. **Check for 28 vdc between NO. 1 DC ESNTL BUS SPLY circuit breaker terminal 1 and ground.**

   a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and K10 NO. 1 DC ESNTL BUS SPLY relay terminal A2 (WP 1747 00). Go to **5.**

   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **5.**

5. **Procedure completed.**

#2 CONV CAPSULE IS ON

SYMPTOM

#2 CONV capsule is on.

MALFUNCTION

#2 CONV capsule is on.

CORRECTIVE ACTION

1. **Check for 28 vdc between P238-A and P238-B.**

#2 CONV CAPSULE IS ON - Continued

a. If voltage is as specified, replace K6 NO. 2 DC PRI BUS CNTOR (WP 0853 00). Go to **8.**

b. If voltage is not as specified, go to **2.**

2. **Check for 28 vdc between P238-A and ground.**

a. If voltage is as specified, repair/replace wiring between P238-B and ground (WP 1747 00). Go to **8.**

b. If voltage is not as specified, go to **3.**

3. **Check for 28 vdc between K6 NO. 2 DC PRI BUS CNTOR terminal A2 and ground.**

a. If voltage is as specified, repair/replace wiring between K6 NO. 2 DC PRI BUS CNTOR terminal A2 and P238-A (WP 1747 00). Go to **8.**

b. If voltage is not as specified, go to **4.**

4. **Check continuity between K6 NO. 2 DC PRI BUS CNTOR terminal A2 and No. 2 converter.**

a. If continuity is present, go to **5.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

5. **Check for 115 vac between:**

- **P904-A and P904-D**
- **P904-B and P904-D**
- **P904-C and P904-D**

a. If voltage is as specified, replace converter (WP 0859 00). Go to **8.**

b. If voltage is not as specified, go to **6.**

6. **Check continuity between P904-D and ground.**

a. If continuity is present, go to **7.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

7. **Check for 115 vac between NO. 2 CONVERTER circuit breaker:**

- **Terminal A1 and ground**
- **Terminal B1 and ground**
- **Terminal C1 and ground**

a. If voltage is as specified, repair/replace wiring between circuit breaker and No. 2 converter (WP 1747 00). Go to **8.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8.**

8. **Procedure completed.**

NO 28 VDC INDICATION BETWEEN J258-A AND J258-B EH60A

SYMPTOM

No 28 vdc indication between J258-A and J258-B.

MALFUNCTION

No 28 vdc indication between J258-A and J258-B.

CORRECTIVE ACTION

1. Check continuity between J258-A and ground.

a. If continuity is present, go to **2.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12.**

2. Check for 28 vdc between UTIL RECEPT CABIN circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J258-B (WP 1747 00). Go to **12.**

b. If voltage is not as specified, go to **3.**

3. Check for 28 vdc between UTIL RECEPT CABIN circuit breaker terminal 2 and ground.

a. If voltage is as specified, replace circuit breaker (WP 0824 00). Go to **12.**

b. If voltage is not as specified, go to **4.**

4. Check for 28 vdc between K5-A2 and ground.

a. If voltage is as specified, repair/replace wiring between K5-A2 and UTIL RECEPT CABIN circuit breaker terminal 2 (WP 1747 00). Go to **12.**

b. If voltage is not as specified, go to **5.**

5. Check for 28 vdc between K5-A1 and ground.

a. If voltage is as specified, go to **6.**

b. If voltage is not as specified, go to **11.**

6. Check for 28 vdc between K5-X1 and K5-X2.

a. If voltage is as specified, replace relay K5 (WP 0864 00). Go to **12.**

b. If voltage is not as specified, go to **7.**

7. Check for 28 vdc between K5-X1 and ground.

a. If voltage is as specified, repair/replace wiring between K5-X2 and ground (WP 1747 00). Go to **12.**

b. If voltage is not as specified, go to **8.**

8. Check continuity between K5-X1 and P240-K.

a. If continuity is present, go to **9.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12.**

9. Check for 28 vdc between P240-C and ground.

NO 28 VDC INDICATION BETWEEN J258-A AND J258-B EH60A - Continued

a. If voltage is as specified, replace K15 DC BUS TIE CNTOR (WP 0854 00). Go to **12.**

b. If voltage is not as specified, go to **10.**

10. Check for 28 vdc between NO. 2 BUS TIE CNTOR circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P240-C (WP 1747 00). Go to **12.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **12.**

11. Check for 28 vdc between DC MON BUS SPLY circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and K5-A1 (WP 1747 00). Go to **12.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **12.**

12. Procedure completed.

NO VOLTAGE ON J512 EH60A

SYMPTOM

No voltage on J512.

MALFUNCTION

No voltage on J512.

CORRECTIVE ACTION

1. Check for 28 vdc between K83 QUICK FIX POWER CNTOR terminal A1 and ground.

a. If voltage is as specified, repair/replace wiring between contactor terminal A1 and J512-E (WP 1747 00). Go to **7.**

b. If voltage is not as specified, go to **2.**

2. Check for 28 vdc between K83 QUICK FIX POWER CNTOR terminal A2 and ground.

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, replace QUICK FIX POWER CL10 (WP 0863 00). Go to **7.**

3. Check for 28 vdc between K83 QUICK FIX POWER CNTOR terminals X1 and X2.

a. If voltage is as specified, replace contactor (WP 0862 00). Go to **7.**

b. If voltage is not as specified, go to **4.**

4. Check for 28 vdc between K83 QUICK FIX POWER CNTOR terminal X1 and ground.

a. If voltage is as specified, repair/replace wiring between contactor terminal X2 and ground (WP 1747 00). Go to **7.**

b. If voltage is not as specified, go to **5.**

5. Check for 28 vdc between Q/F PWR switch terminal 3 and ground.

a. If voltage is as specified, repair/replace wiring between switch terminal 3 and K83 QUICK FIX POWER CNTOR terminal X1 (WP 1747 00). Go to **7.**

b. If voltage is not as specified, go to **6.**

NO VOLTAGE ON J512 EH60A - Continued

6. **Check for 28 vdc between Q/F PWR switch terminal 2 and ground.**
   a. If voltage is as specified, replace switch (WP 0861 00). Go to **7.**
   b. If voltage is not as specified, repair/replace wiring between switch terminal 2 and K5-X1 (WP 1747 00). Go to **7.**
7. **Procedure completed.**

BATT LOW CHARGE CAPSULE IS ON WHEN BATTERY VOLTAGE IS 24 VDC UH60L 96-26723-SUBQ

SYMPTOM

BATT LOW CHARGE capsule is on when battery voltage is 24 vdc.

MALFUNCTION

BATT LOW CHARGE capsule is off when battery voltage is 24 vdc.

CORRECTIVE ACTION

1. **Check continuity between:**
   - **P230-g and K201-X1**
   - **P230-g and K201-Y1**
   - **K201-X2 and ground**

   a. If continuity is present, go to **2.**
   b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7.**
2. **With BATT switch ON, check for 24 vdc or more between K201-X1 and ground.**
   a. If voltage is as specified, go to **3.**
   b. If voltage is not as specified, go to **4.**
3. **Check for 24 vdc or more between K201-Y1 and ground.**
   a. If voltage is as specified, replace relay K201 (WP 0936 00). Go to **7.**
   b. If voltage is not as specified, repair/replace wiring between (WP 1747 00), then go to **7:**
      - K201-Y1 and BATT switch terminal 6
      - K201-X1 and BATT switch terminal 6
4. **Check continuity between CB2-1 and CB2-2.**
   a. If continuity is present, go to **5.**
   b. If continuity is not present, replace circuit breaker (WP 0824 00). Go to **7.**
5. **Check for 24 vdc or more between BATT switch terminal 5 and ground.**
   a. If voltage is as specified, replace BATT switch (WP 0861 00). Go to **7.**
   b. If voltage is not as specified, go to **6.**
6. **Check continuity between:**
   - **BATT switch terminal 5 and P140 (+)**
   - **P140 (-) and ground**

BATT LOW CHARGE CAPSULE IS ON WHEN BATTERY VOLTAGE IS 24 VDC UH60L 96-26723-SUBQ - Continued

a. If continuity is present, replace battery (WP 0931 00). Go to **7.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7.**

7. Procedure completed.

BATT LOW CHARGE CAPSULE IS OFF WHEN BATTERY VOLTAGE IS 22 VDC UH60L 96-26723-SUBQ

SYMPTOM

BATT LOW CHARGE capsule is off when battery voltage is 22 vdc.

MALFUNCTION

BATT LOW CHARGE capsule is off when battery voltage is 22 vdc.

CORRECTIVE ACTION

1. Check continuity between:

- **P117-21 and K201-B3**
- **K201-X2 and ground**

a. If continuity is present, go to **2.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7.**

2. With BATT switch ON, check for 22 vdc or less between K201-X1 and ground.

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, go to **4.**

3. Check for 22 vdc or less between K201-Y1 and ground.

a. If voltage is as specified, replace relay K201 (WP 0936 00). Go to **7.**

b. If voltage is not as specified, repair/replace wiring between (WP 1747 00), then go to **7:**

- K201-Y1 and BATT switch terminal 6
- K201-X1 and BATT switch terminal 6

4. Check continuity between CB2-1 and CB2-2.

a. If continuity is present, go to **5.**

b. If continuity is not present, replace circuit breaker (WP 0824 00). Go to **7.**

5. Check for 22 vdc or less between BATT switch terminal 5 and ground.

a. If voltage is as specified, replace BATT switch (WP 0861 00). Go to **7.**

b. If voltage is not as specified, go to **6.**

6. Check continuity between:

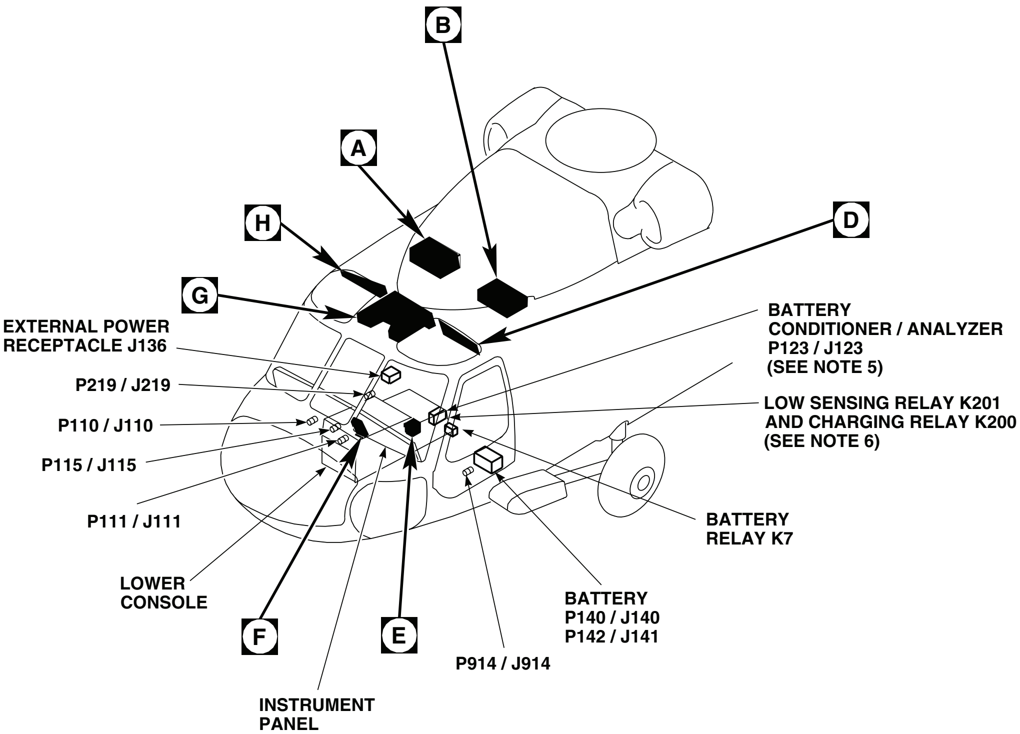
- **BATT switch terminal 5 and P140 (+)**
- **P140 (-) and ground**

a. If continuity is present, replace battery (WP 0931 00). Go to **7.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7.**

7. Procedure completed.

LOCATION



EFFECTIVITY

UH60A UH60L

NOTES

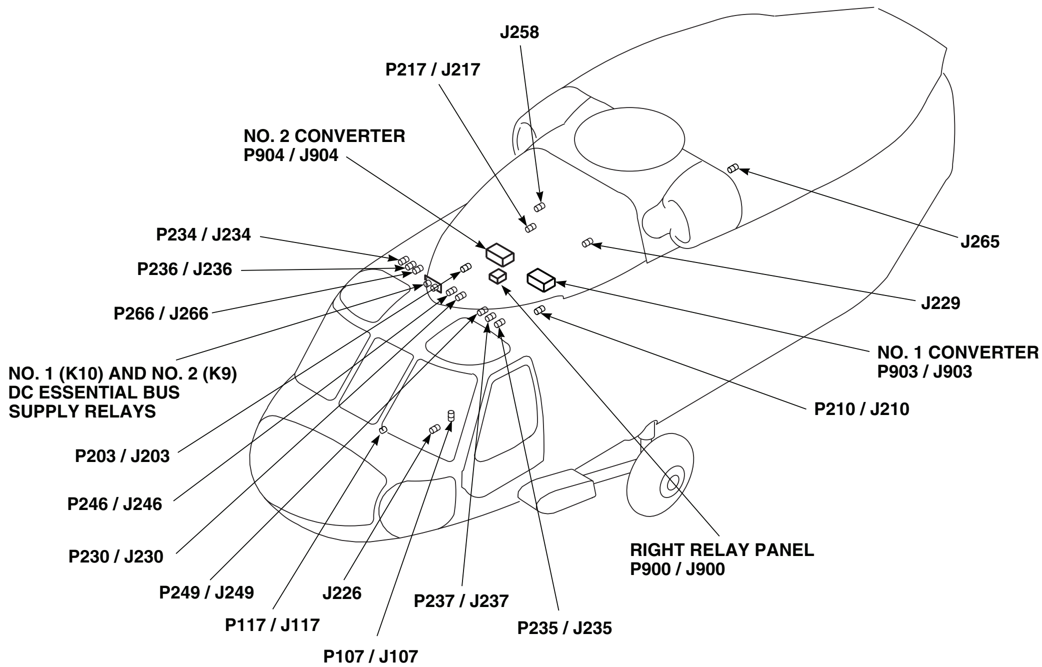
1. W/O HCW
2. HCW
3. ESSS
4. HELICOPTERS WITH AUXILIARY CABIN HEATER INSTALLED, CIRCUIT BREAKER 18A REIDENTIFIED AS AUX HTR CONTROL.
5. UH60A
   UH60L 89-26149 - 96-26722
6. UH60L 96-26723 - SUBQ
7. UH60L 96-26723 - SUBQ
   CIRCUIT BREAKER IS LABELED DPLR/GPS.
8. MWO 50-76 HF RADIO SET (AN/ARC-220(V)1).
9. MWO 50-75
10. UH60L 96-26723 - SUBQ
    MWO 50-77
11. MWO 50-82

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
J136	EXTERNAL POWER RECEPTACLE
J226	RESCUE HOIST KIT DISCONNECTS
J229	RESCUE HOIST KIT DISCONNECTS
J258	UTILITY RECEPTACLE +28 VDC
J265	CMD CSL POWER RECPT +28 VDC
P107 / J107	COCKPIT BL 9 LH, STA 236
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P115 / J115	COCKPIT BL 6 RH, STA 210

AA2753_1D
SA

Figure 1. DC Electrical System Location Diagram UH60A UH60L . (Sheet 1 of 6)

LOCATION - Continued



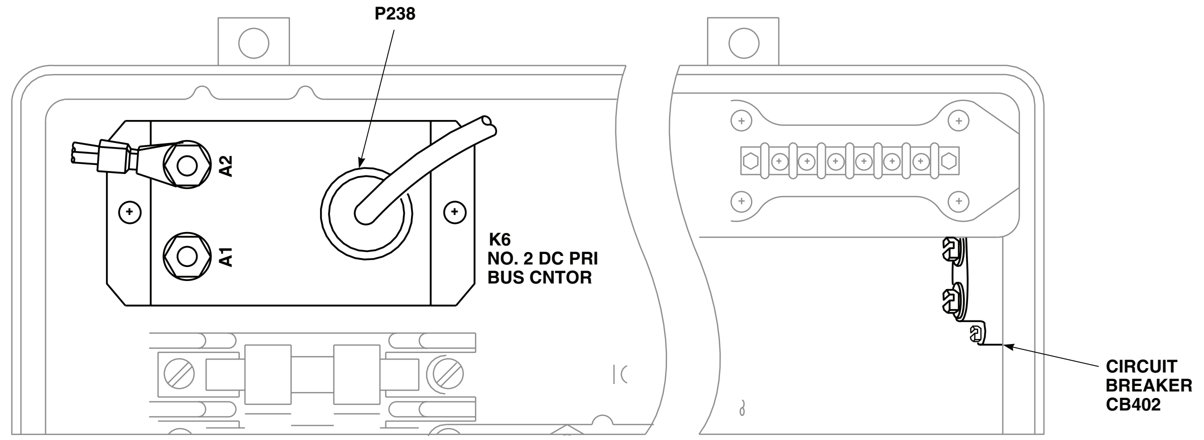
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P117 / J117	CAUTION / ADVISORY PANEL
P123 / J123	BATTERY CONDITIONER / ANALYZER (SEE NOTE 5)
P140 / J140	BATTERY
P141 / J141	BATTERY
P203 / J203	NO. 2 JUNCTION BOX
P210 / J210	NO. 1 JUNCTION BOX
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P234 / J234	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 27 RH
P235 / J235	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 20 LH

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 23 RH
P237 / J237	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 25 LH
P238	NO. 2 JUNCTION BOX
P239	NO. 1 JUNCTION BOX
P240	NO. 1 JUNCTION BOX
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH
P900 / J900	RIGHT RELAY PANEL
P903 / J903	NO. 1 CONVERTER
P904 / J904	NO. 2 CONVERTER
P914 / J914	COCKPIT BL 24 LH, STA 247

AA2753_2A
SA

Figure 1. DC Electrical System Location Diagram UH60A UH60L . (Sheet 2 of 6)

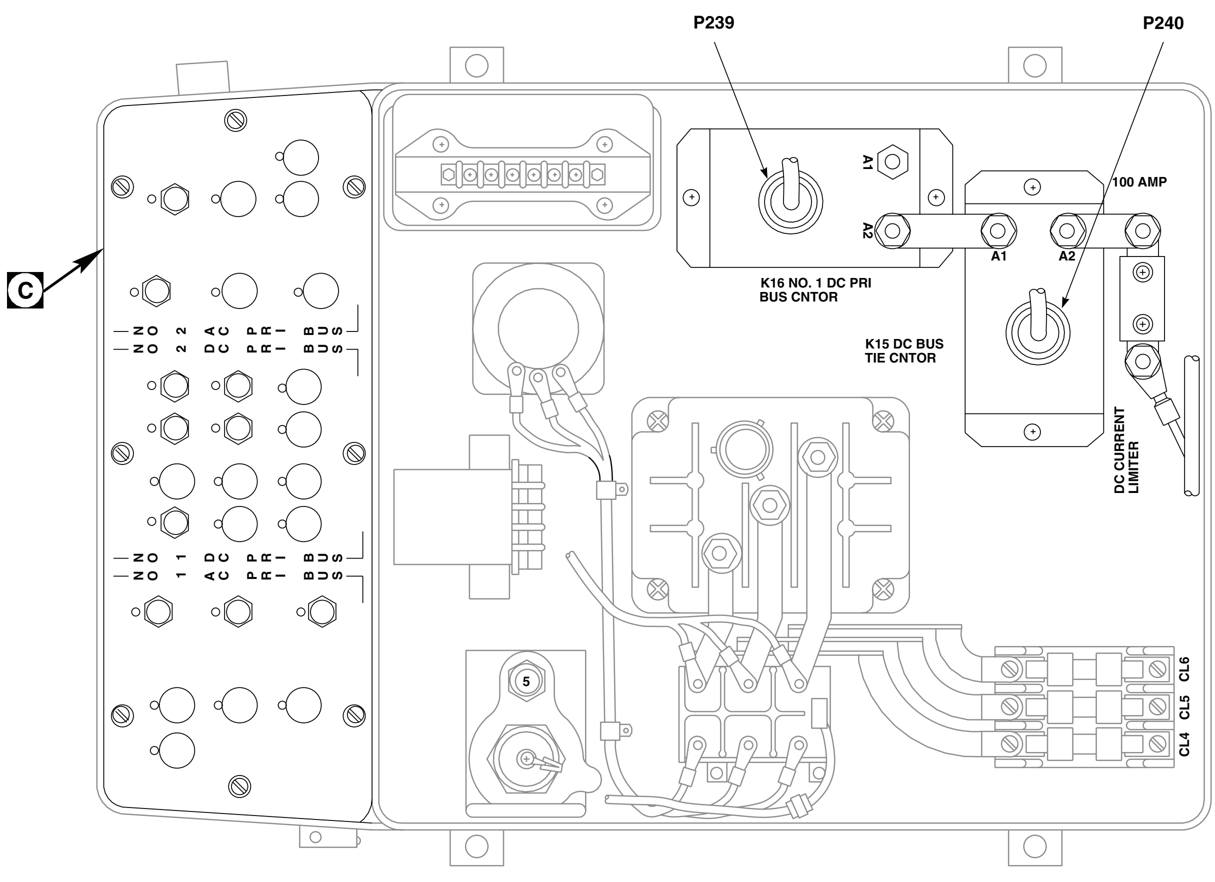
LOCATION - Continued



NO. 2 JUNCTION BOX

A

B



NO. 1 JUNCTION BOX

AA2753_3
SA

Figure 1. DC Electrical System Location Diagram **UH60A UH60L** . (Sheet 3 of 6)

LOCATION - Continued

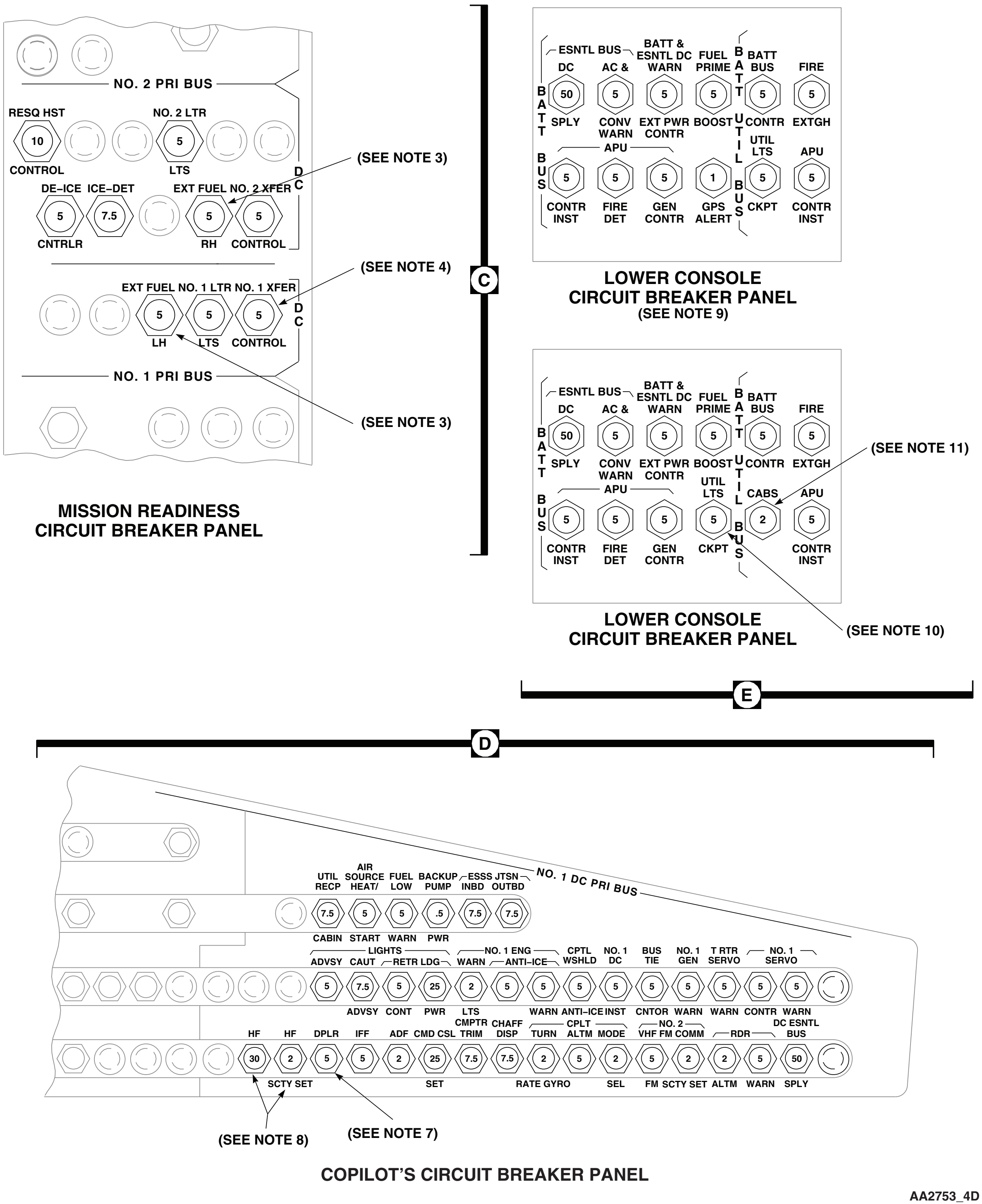
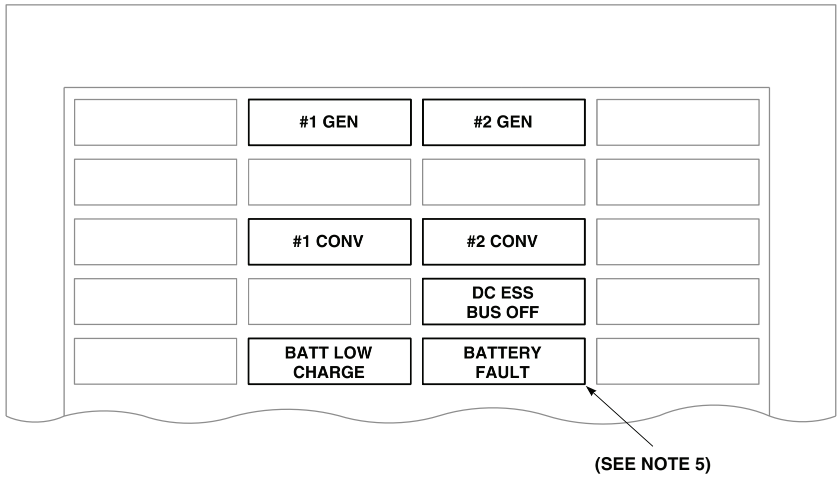
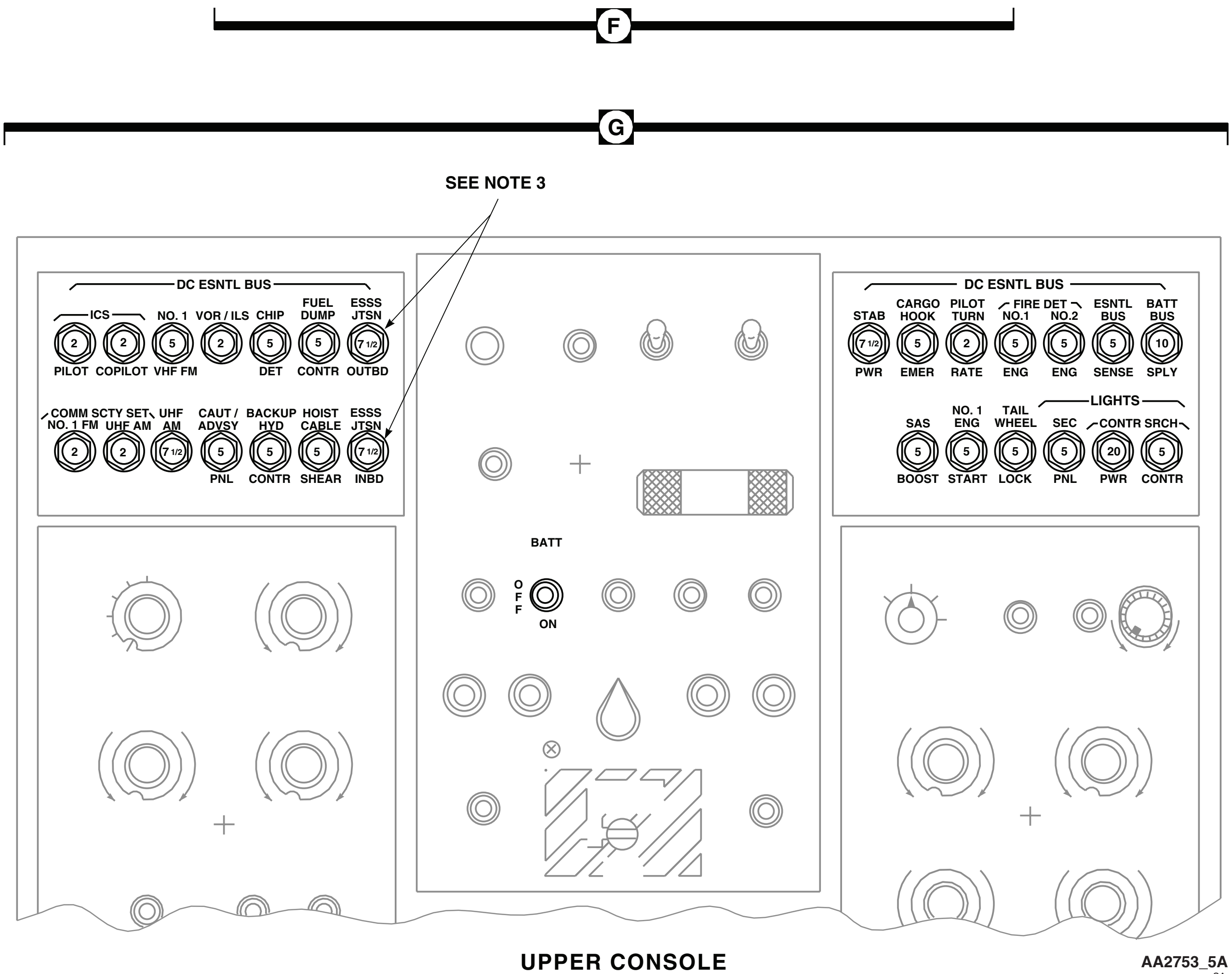


Figure 1. DC Electrical System Location Diagram UH60A UH60L . (Sheet 4 of 6)

LOCATION - Continued



CAUTION/ADVISORY PANEL

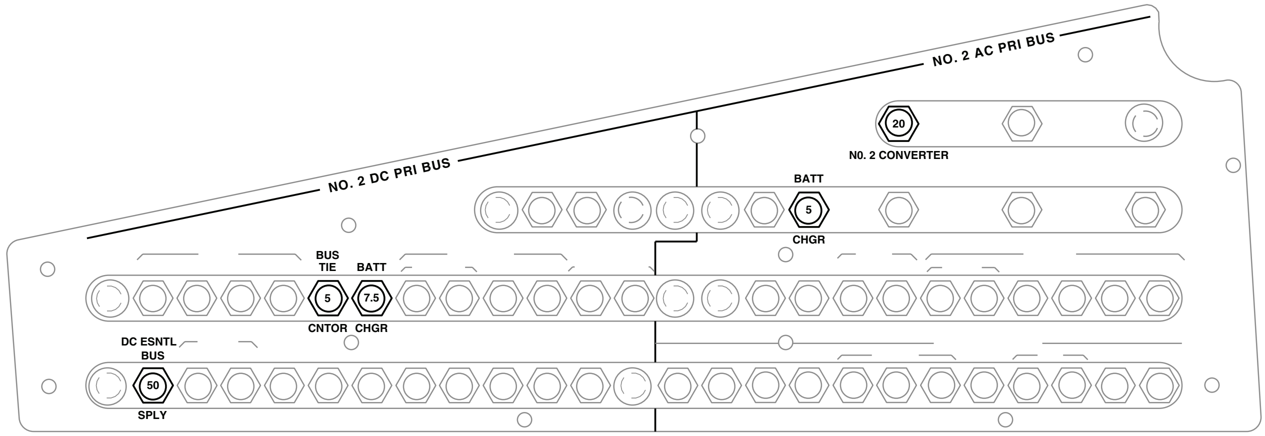


UPPER CONSOLE

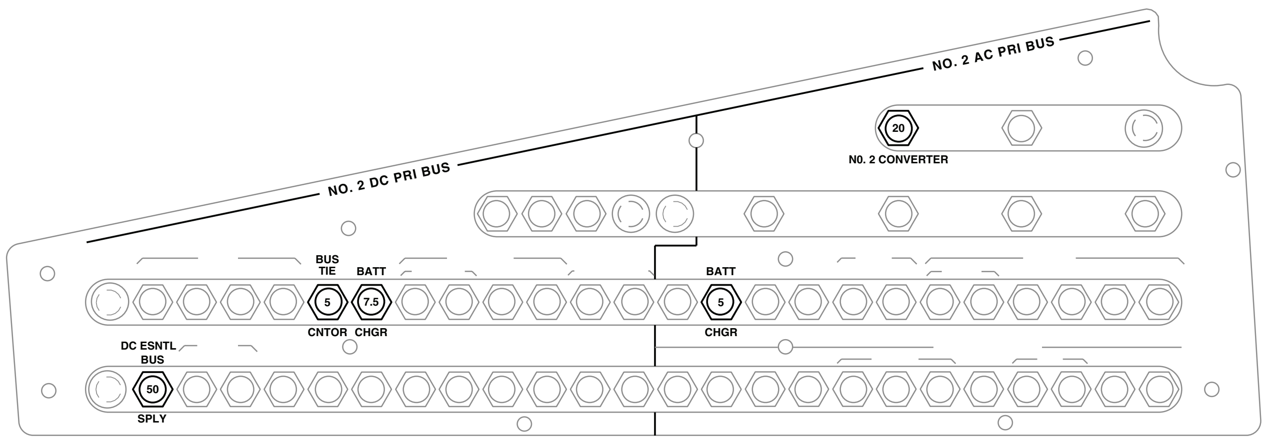
AA2753_5A
SA

Figure 1. DC Electrical System Location Diagram UH60A UH60L . (Sheet 5 of 6)

LOCATION - Continued



(SEE NOTE 1)



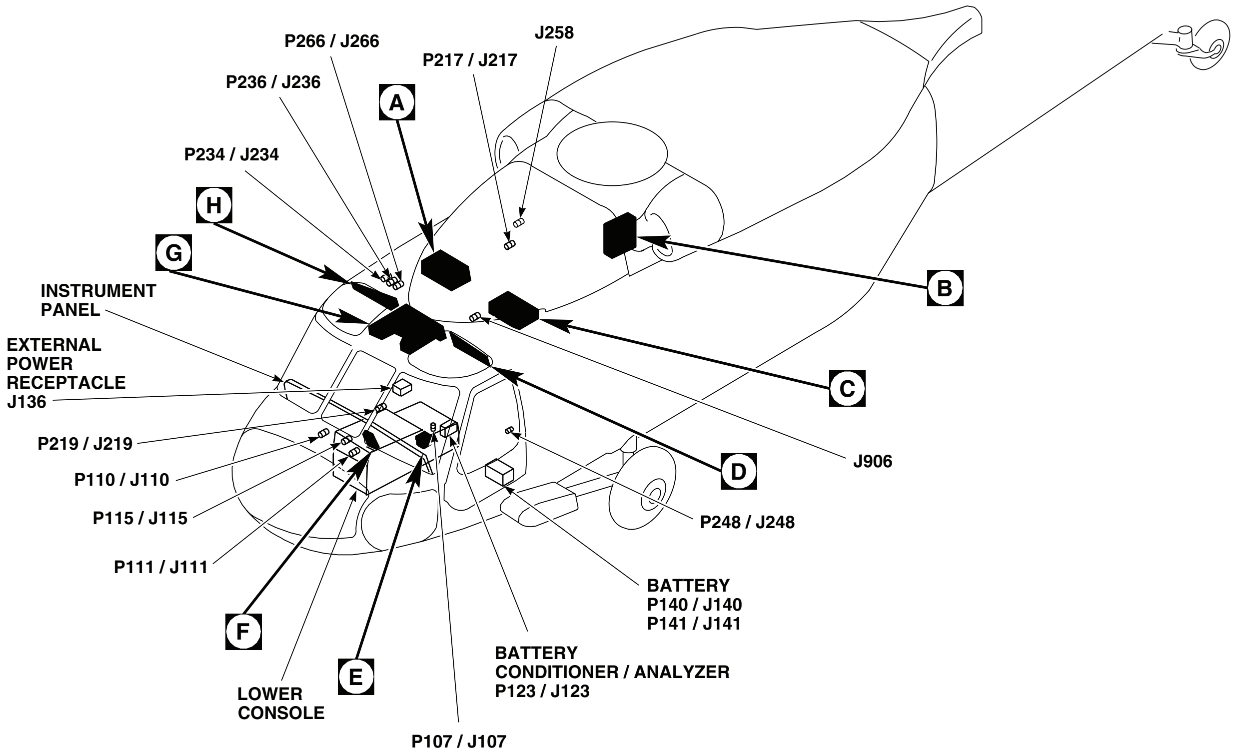
(SEE NOTE 2)

PILOT'S CIRCUIT BREAKER PANEL

AA2753_6
SA

Figure 1. DC Electrical System Location Diagram UH60A UH60L . (Sheet 6 of 6)

LOCATION - Continued



EFFECTIVITY

EH60A

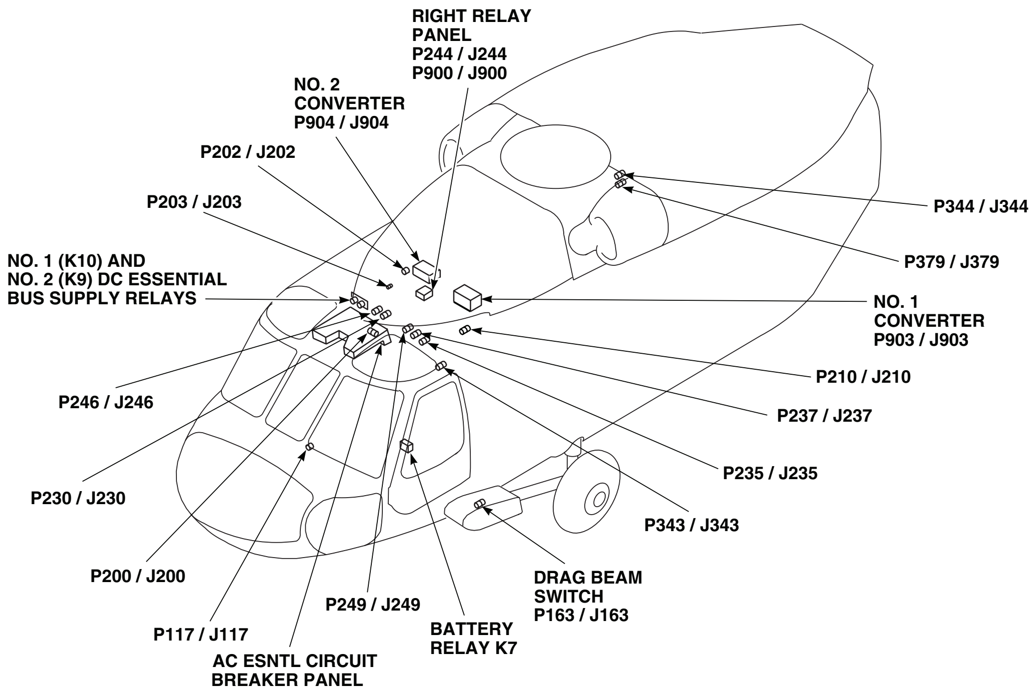
TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
J136	EXTERNAL POWER RECEPTACLE
J258	UTILITY RECEPTACLE +28 VDC
J510	MISSION INTERFACE PANEL
J512	MISSION INTERFACE PANEL
J906	CABIN CEILING BL 0, STA 284
P107 / J107	COCKPIT BL 9 LH, STA 236
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P111 / J111	COCKPIT BL 7.5 LH, STA 197

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P115 / J115	COCKPIT BL 6 RH, STA 210
P117 / J117	CAUTION / ADVISORY PANEL
P123 / J123	BATTERY CONDITIONER / ANALYZER
P140 / J140	BATTERY
P141 / J141	BATTERY
P163 / J163	DRAG BEAM SWITCH
P200 / J200	COCKPIT CEILING BL 13 RH, STA 246
P202 / J202	NO. 2 JUNCTION BOX
P203 / J203	NO. 2 JUNCTION BOX
P205	NO. 2 JUNCTION BOX
P206	NO. 2 JUNCTION BOX
P210 / J210	NO. 1 JUNCTION BOX
P212	NO. 1 JUNCTION BOX
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P230 / J230	CABIN CEILING BL 8 RH, STA 247

AA2754_1
SA

Figure 2. DC Electrical System Location Diagram EH60A . (Sheet 1 of 7)

LOCATION - Continued



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P234 / J234	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 27 RH
P235 / J235	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 20 LH
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 23 RH
P237 / J237	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 25 LH
P238	NO. 2 JUNCTION BOX
P239	NO. 1 JUNCTION BOX
P240	NO. 1 JUNCTION BOX
P244 / J244	RIGHT RELAY PANEL
P246 / J246	CABIN CEILING BL 5 RH, STA 247

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P248 / J248	CABIN BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH
P343 / J343	CABIN CEILING BL 30 LH, STA 245
P344 / J344	CABIN CEILING BL 3.5 RH, STA 379
P379 / J379	CABIN CEILING BL 3.5 RH, STA 379
P900 / J900	RIGHT RELAY PANEL
P903 / J903	NO. 1 CONVERTER
P904 / J904	NO. 2 CONVERTER

AA2754_2A
SA

Figure 2. DC Electrical System Location Diagram EH60A . (Sheet 2 of 7)

LOCATION - Continued

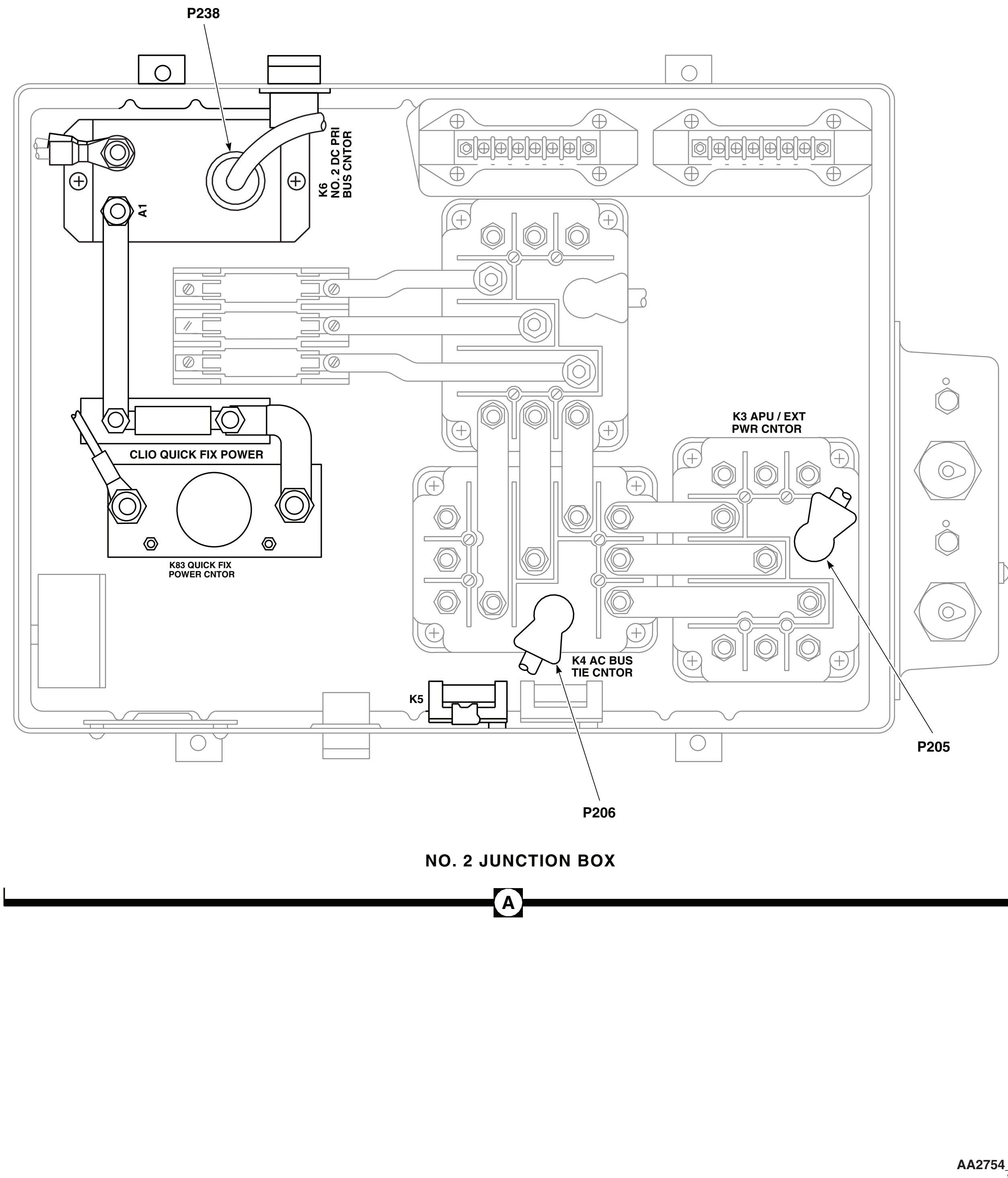


Figure 2. DC Electrical System Location Diagram EH60A . (Sheet 3 of 7)

LOCATION - Continued

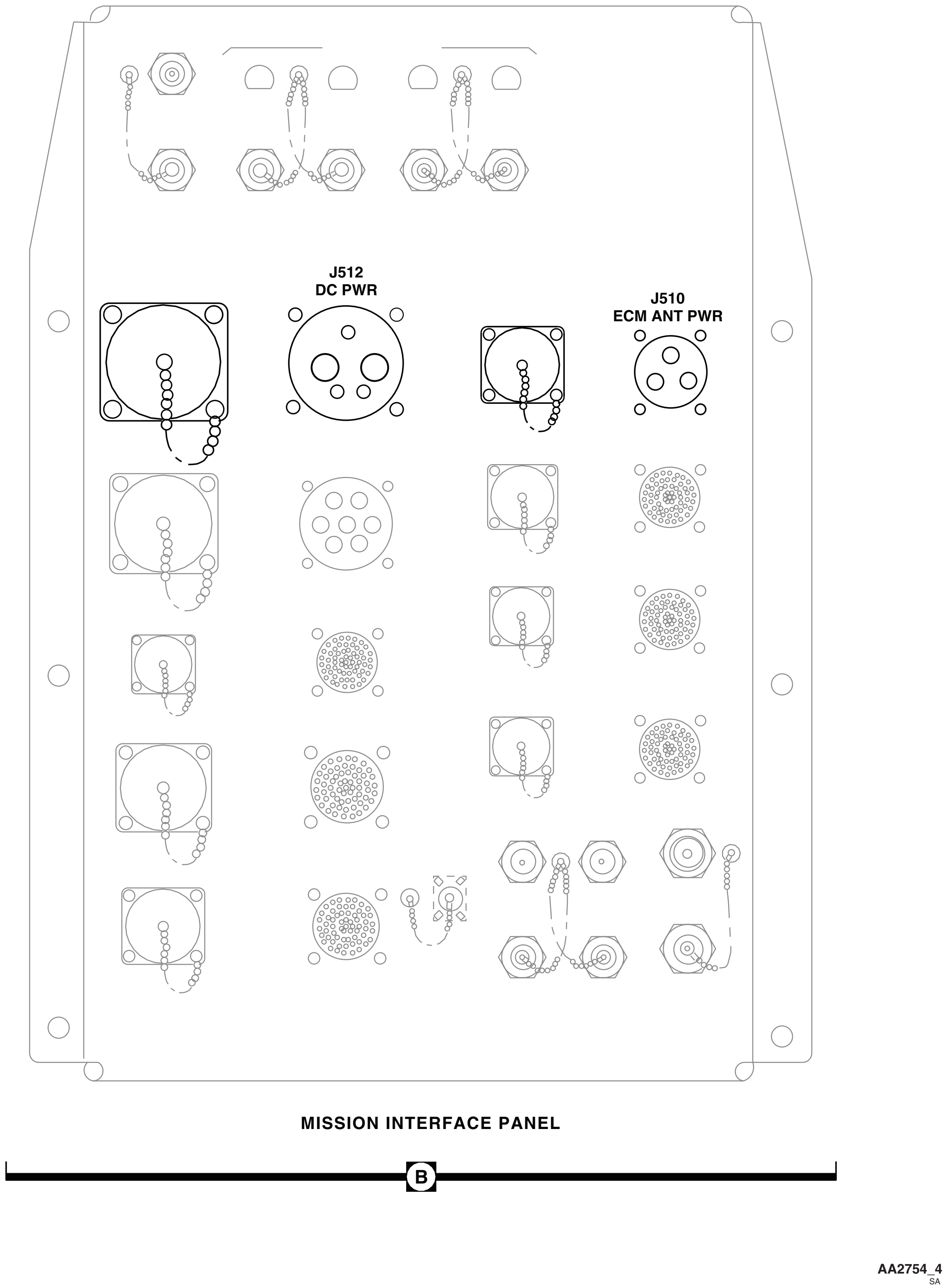


Figure 2. DC Electrical System Location Diagram EH60A . (Sheet 4 of 7)

LOCATION - Continued

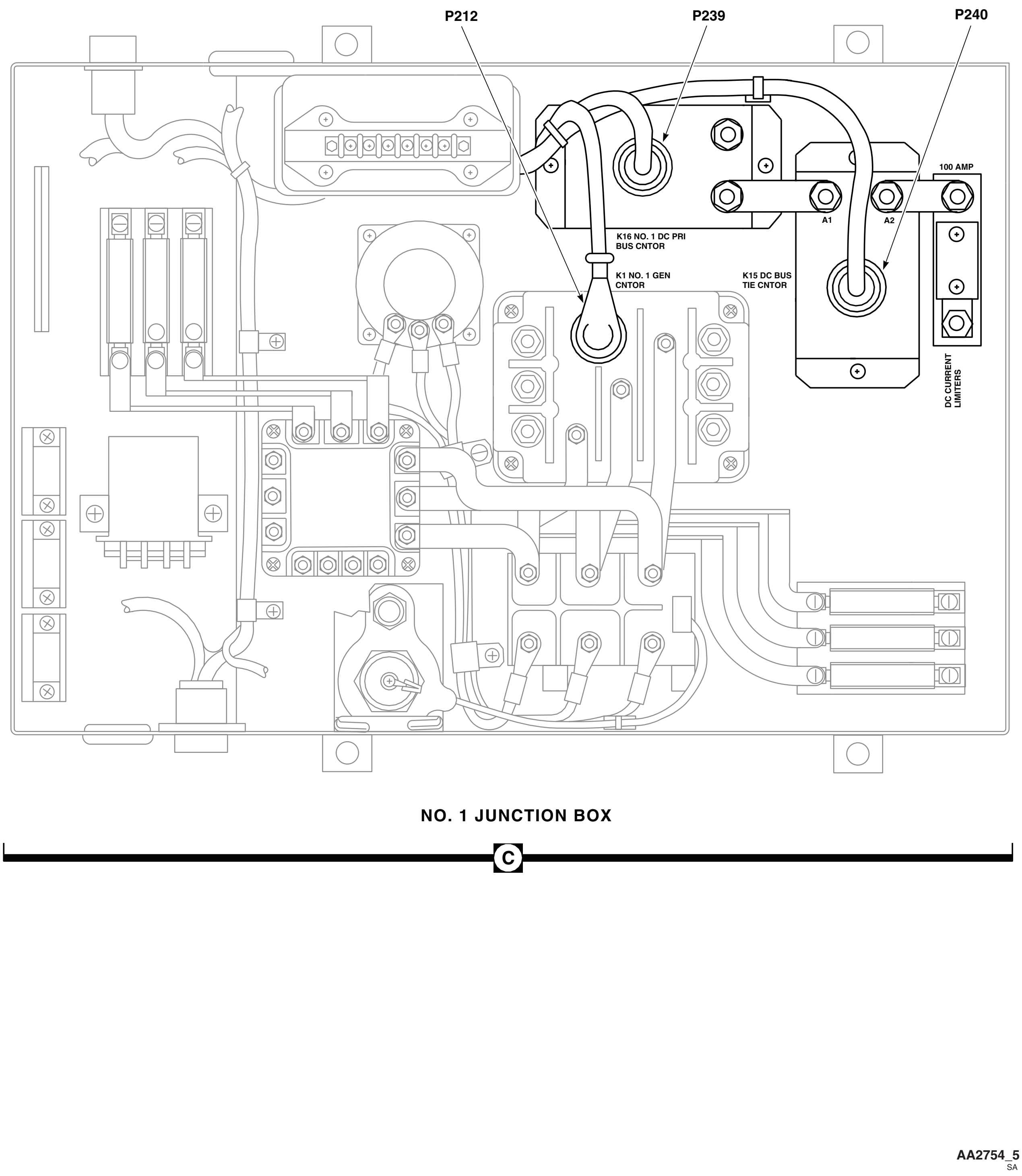


Figure 2. DC Electrical System Location Diagram EH60A . (Sheet 5 of 7)

LOCATION - Continued

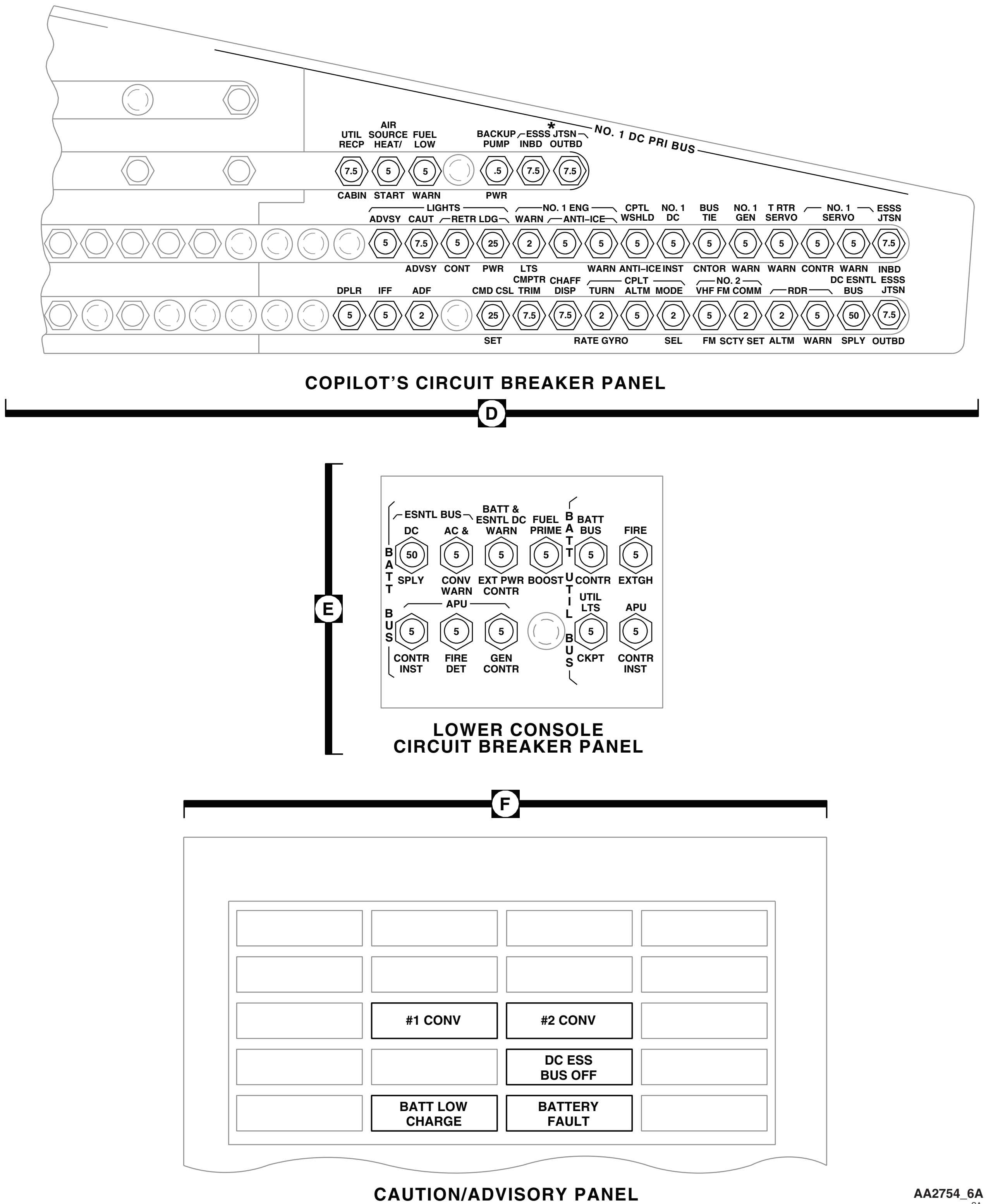
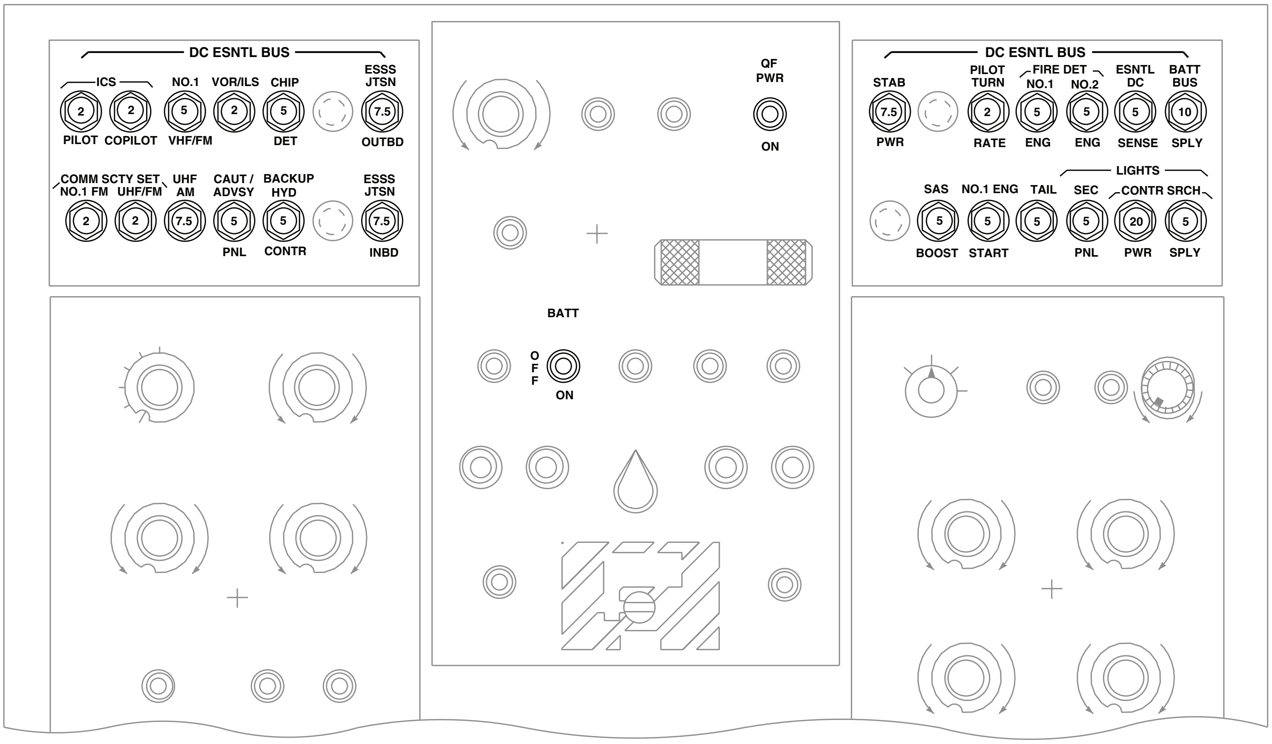


Figure 2. DC Electrical System Location Diagram EH60A . (Sheet 6 of 7)

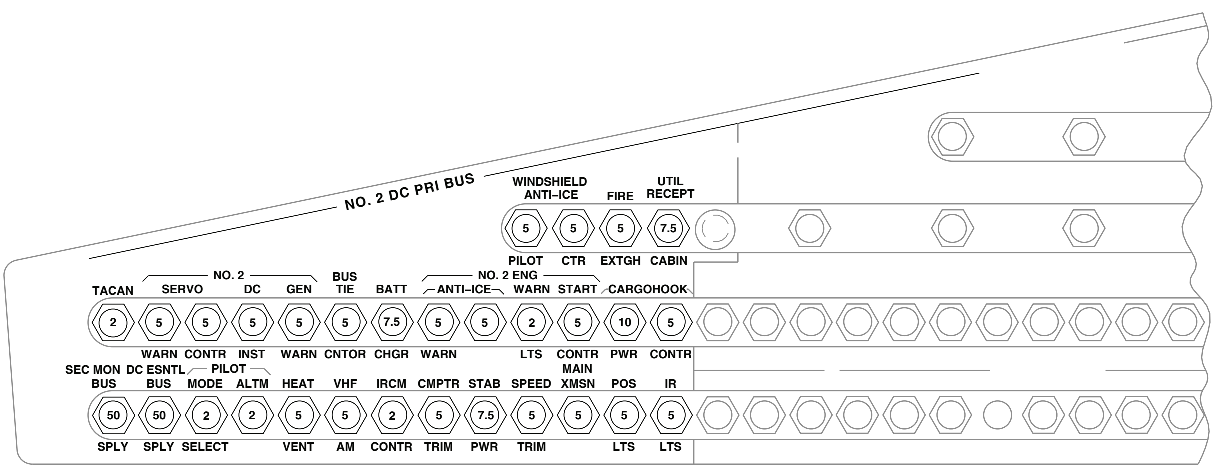
LOCATION - Continued



UPPER CONSOLE

G

H



PILOT'S CIRCUIT BREAKER PANEL

AA2754_7A
SA

Figure 2. DC Electrical System Location Diagram EH60A . (Sheet 7 of 7)

SCHEMATICS

EFFECTIVITY

UH60L 96–26723 – SUBQ

MWO 50–77

NOTES

1. W/O EMEP
   PIN FILTERED ADAPTERS (PFA) ARE NOT INSTALLED.
2. PIN 21 OF P117 SHOWS TYPICAL CONFIGURATION FOR ALL PIN FILTERED ADAPTERS.
3. HELICOPTERS WITH AUXILIARY HEATER INSTALLED, CIRCUIT BREAKER CB18A REIDENTIFIED AS **AUX HTR CONTROL**.

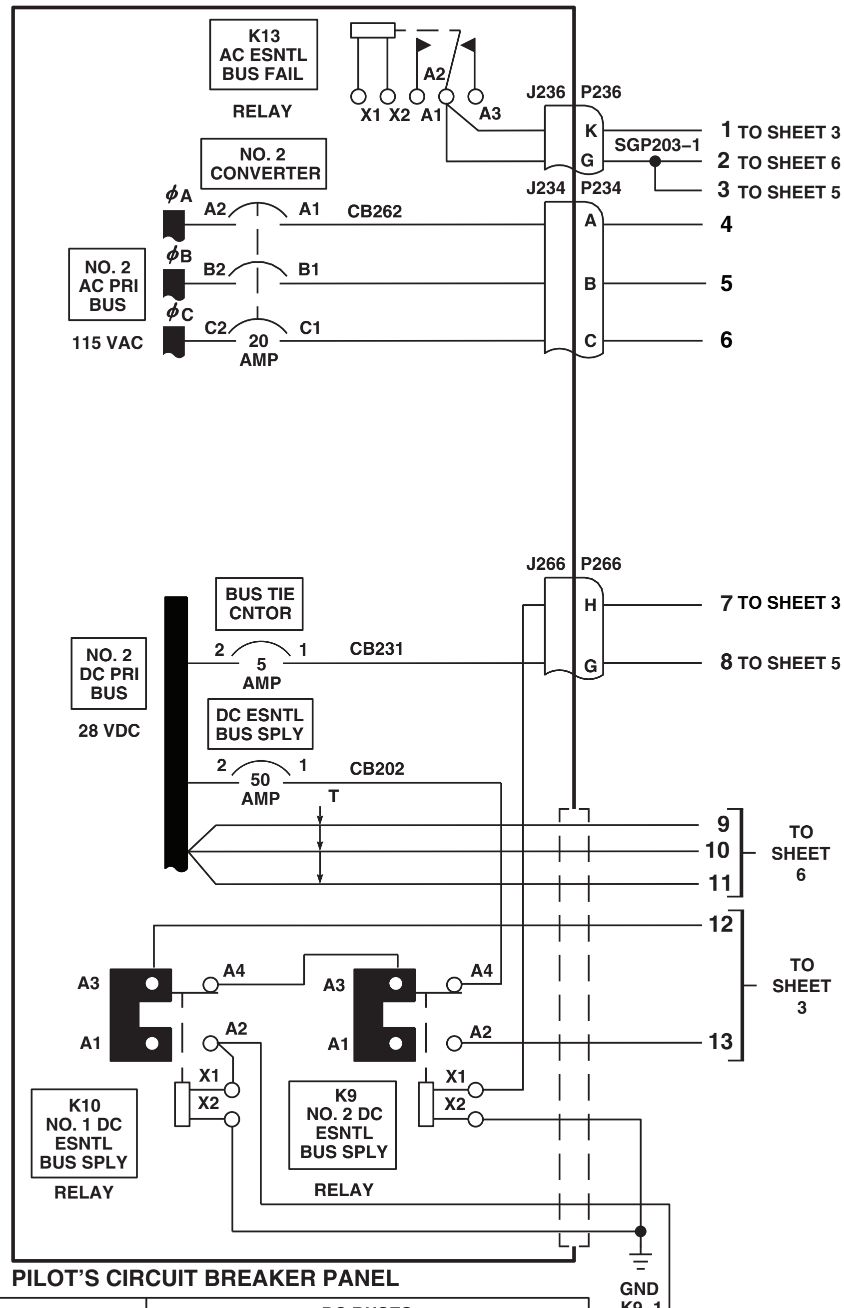
LEGEND

POWER SOURCE

A = NO. 1 CONVERTER
B = NO. 2 CONVERTER
C = BATTERY (GREATER THAN 35% CHARGED)
1 = TURNED ON AND OPERATING
0 = TURNED OFF OR FAILED

CONTACTORS/RELAYS/BUSES

K7 = BATTERY RELAY
K16 = NO. 1 DC PRI BUS CONTACTOR
K6 = NO. 2 DC PRI BUS CONTACTOR
K15 = DC BUS TIE CONTACTOR
K10 = NO. 1 DC ESNTL BUS SPLY RELAY
K9 = NO. 2 ESNTL BUS SPLY RELAY
K20 = DC ESNTL BUS FAIL RELAY
X = ENERGIZED
O = DE–ENERGIZED
= SIGNAL OR POWER GROUND
= CASE GROUND



DC POWER SOURCE			CONDITION	DC SYSTEM CONTACTORS / RELAYS							DC BUSES				
A	B	C		K7	K16	K6	K15	K10	K9	K20	NO. 1 DC PRI	NO. 2 DC PRI	DC ESNTL	BATT UTIL	BATT
0	0	0	1	O	O	O	O	O	O	O	O	O	O	O	O
1	0	0	2	O	X	O	X	X	O	X	X	X	X	X	X
0	1	0	3	O	O	X	X	X	O	X	X	X	X	X	X
1	1	0	4	O	X	X	O	X	O	X	X	X	X	X	X
0	0	1	5	X	O	O	O	O	X	X	O	O	X	X	X
1	0	1	6	O	X	O	X	X	O	X	X	X	X	X	X
0	1	1	7	O	O	X	X	X	O	X	X	X	X	X	X
1	1	1	8	O	X	X	O	X	O	X	X	X	X	X	X

AB1438_1A
SA

Figure 3. DC Electrical System Schematic Diagram UH60L 96-26723-SUBQ MWO 50-77 . (Sheet 1 of 6)

SCHEMATICS - Continued

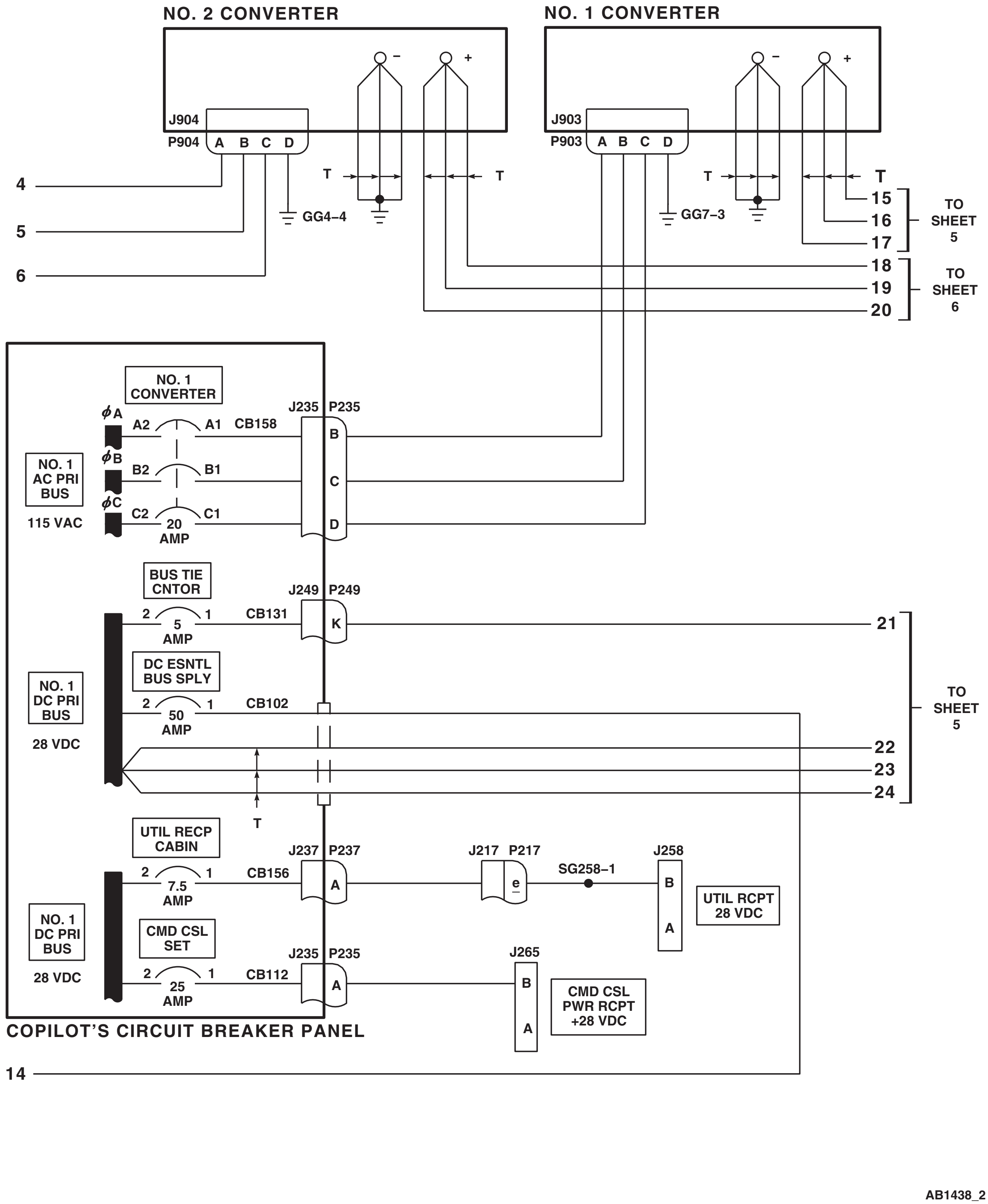


Figure 3. DC Electrical System Schematic Diagram UH60L 96-26723-SUBQ MWO 50-77 . (Sheet 2 of 6)

SCHEMATICS - Continued

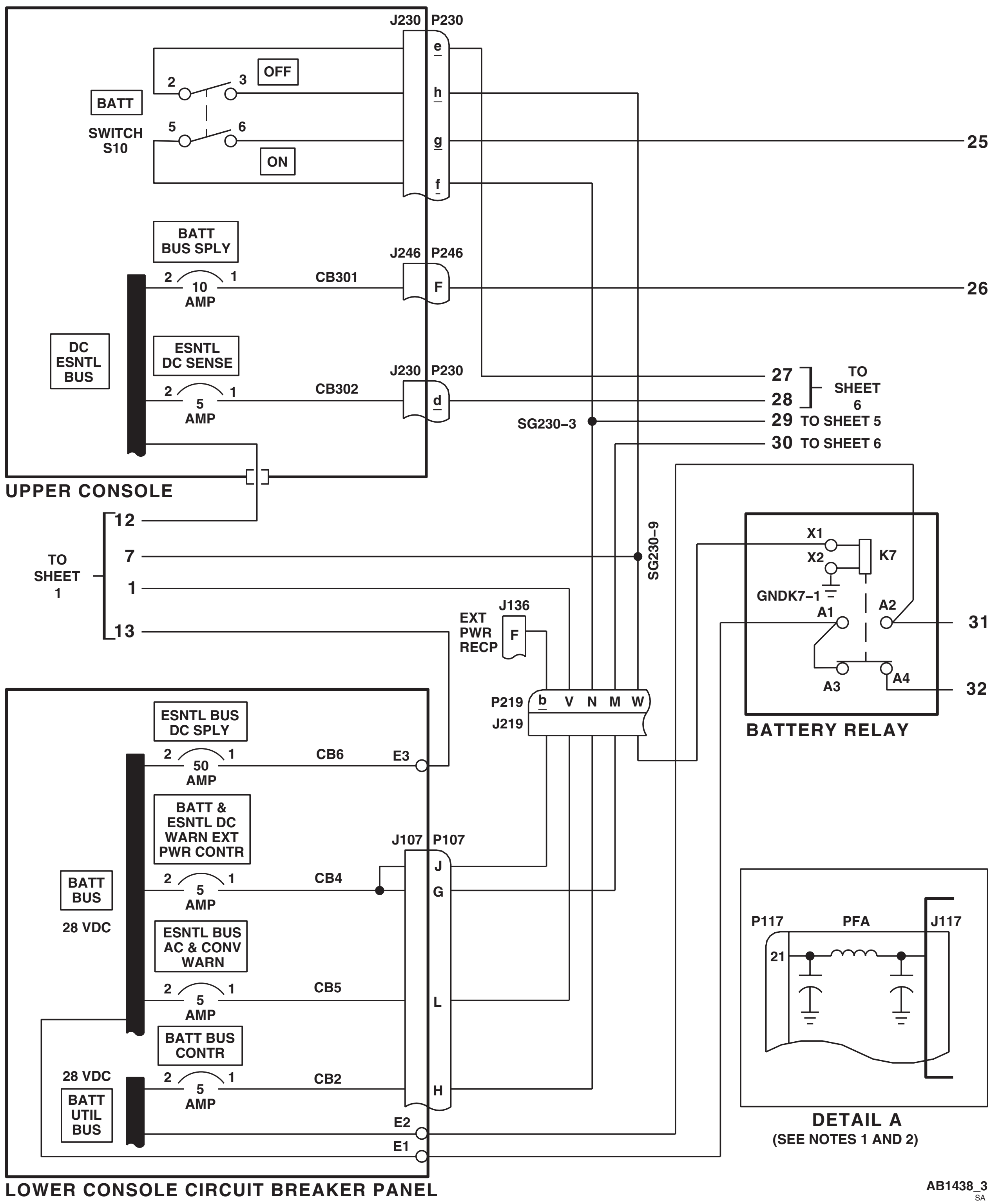


Figure 3. DC Electrical System Schematic Diagram UH60L 96-26723-SUBQ MWO 50-77 . (Sheet 3 of 6)

SCHEMATICS - Continued

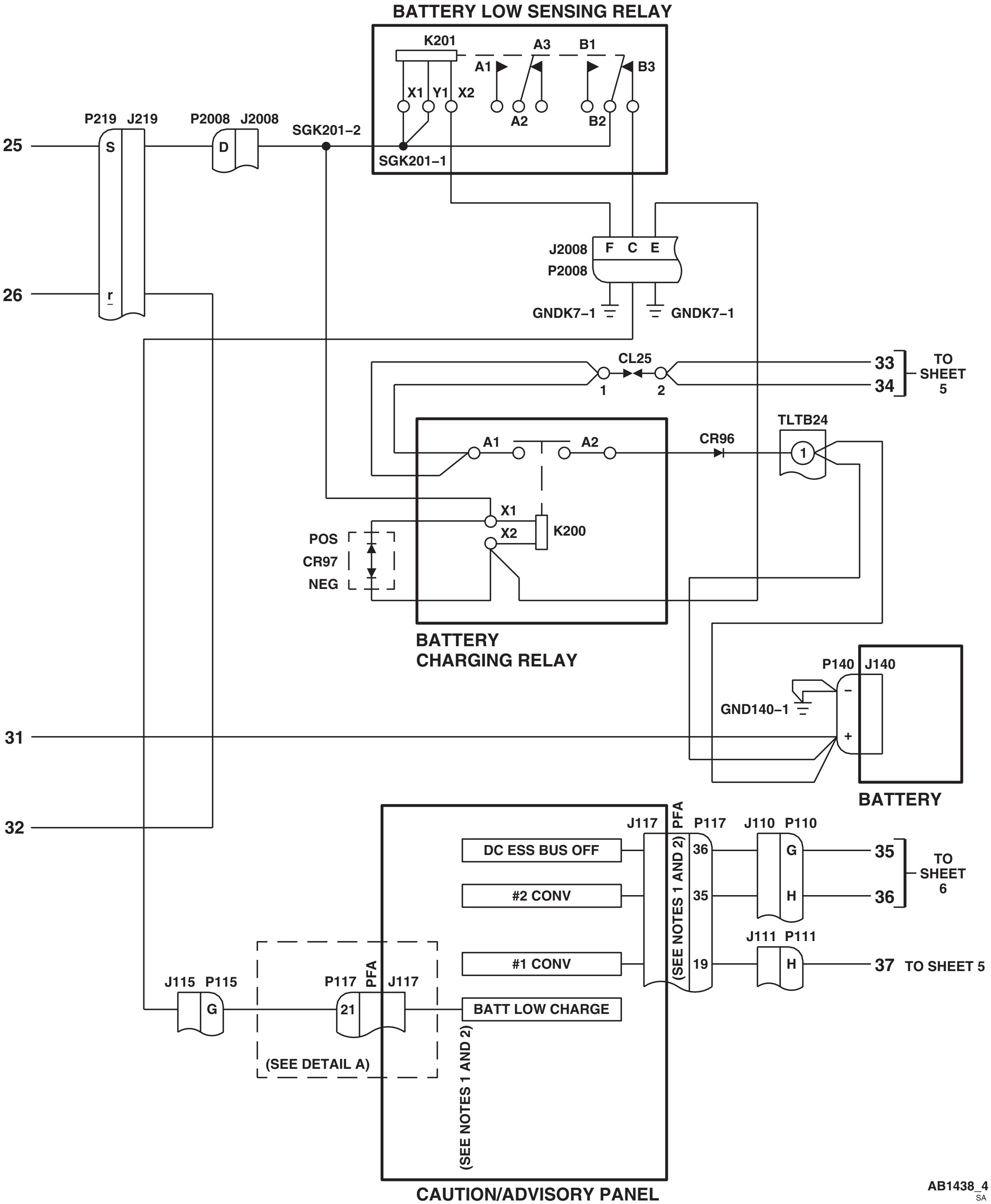


Figure 3. DC Electrical System Schematic Diagram UH60L 96-26723-SUBQ MWO 50-77 . (Sheet 4 of 6)

SCHEMATICS - Continued

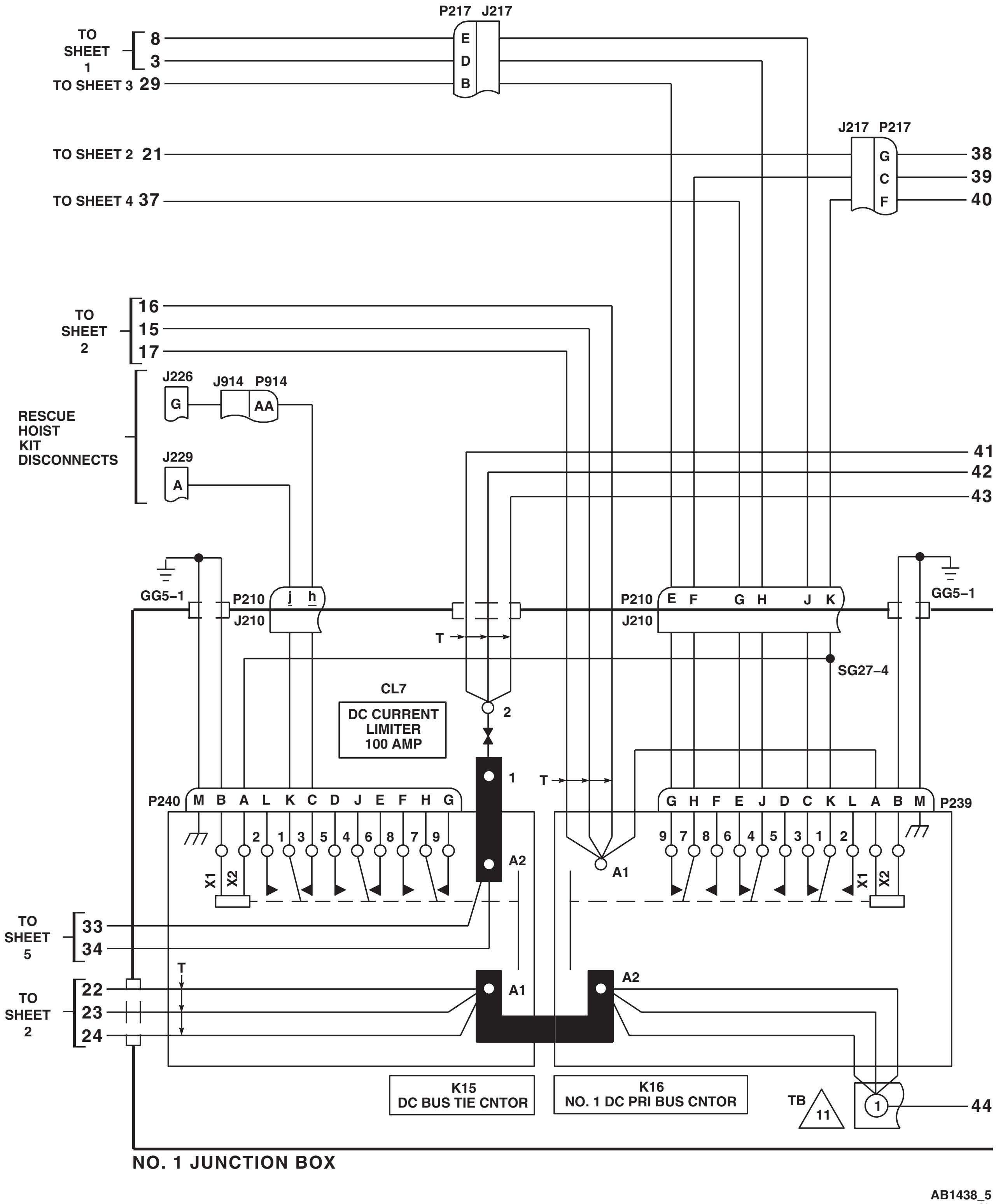


Figure 3. DC Electrical System Schematic Diagram **UH60L 96-26723-SUBQ MWO 50-77** . (Sheet 5 of 6)

SCHEMATICS - Continued

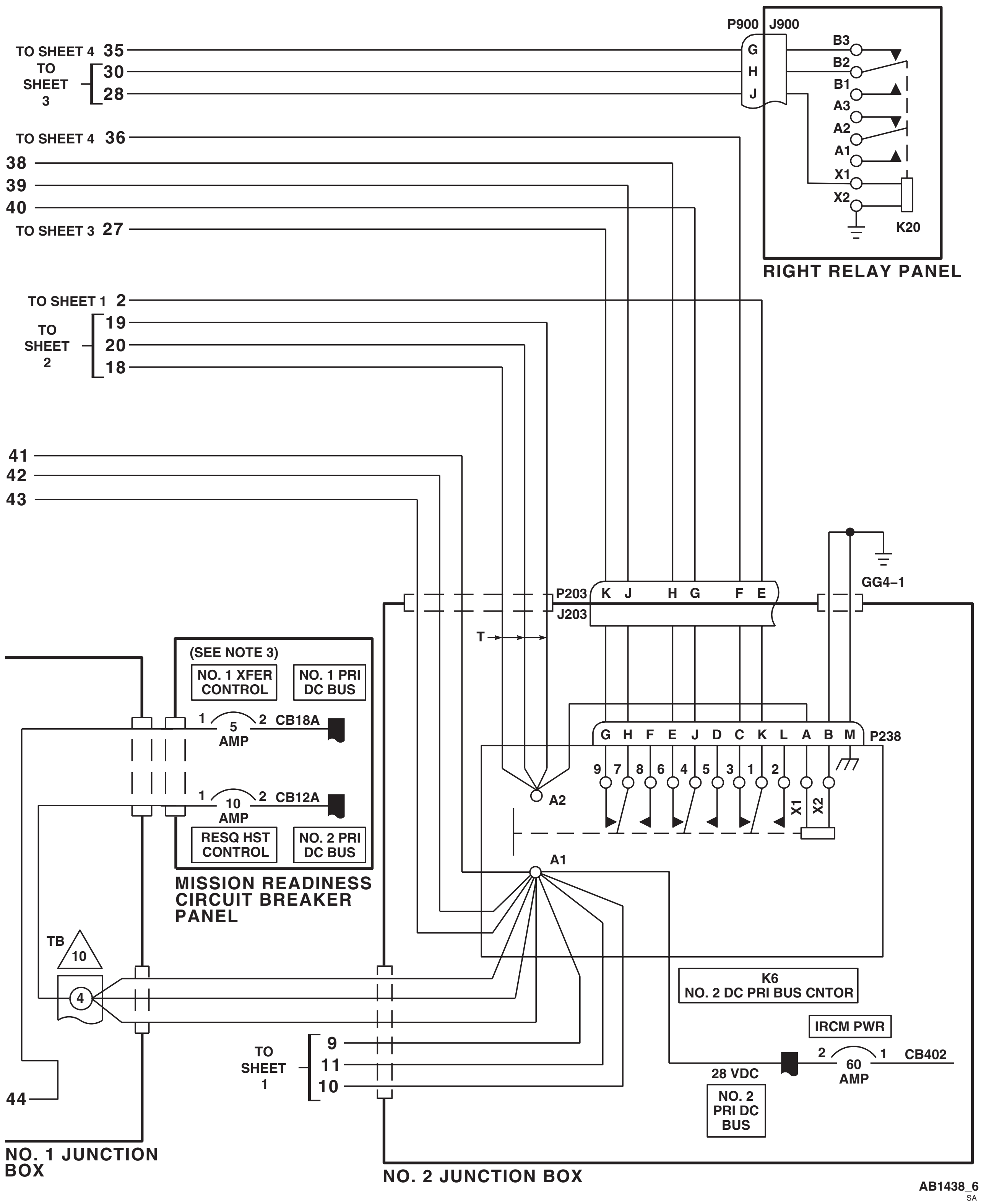
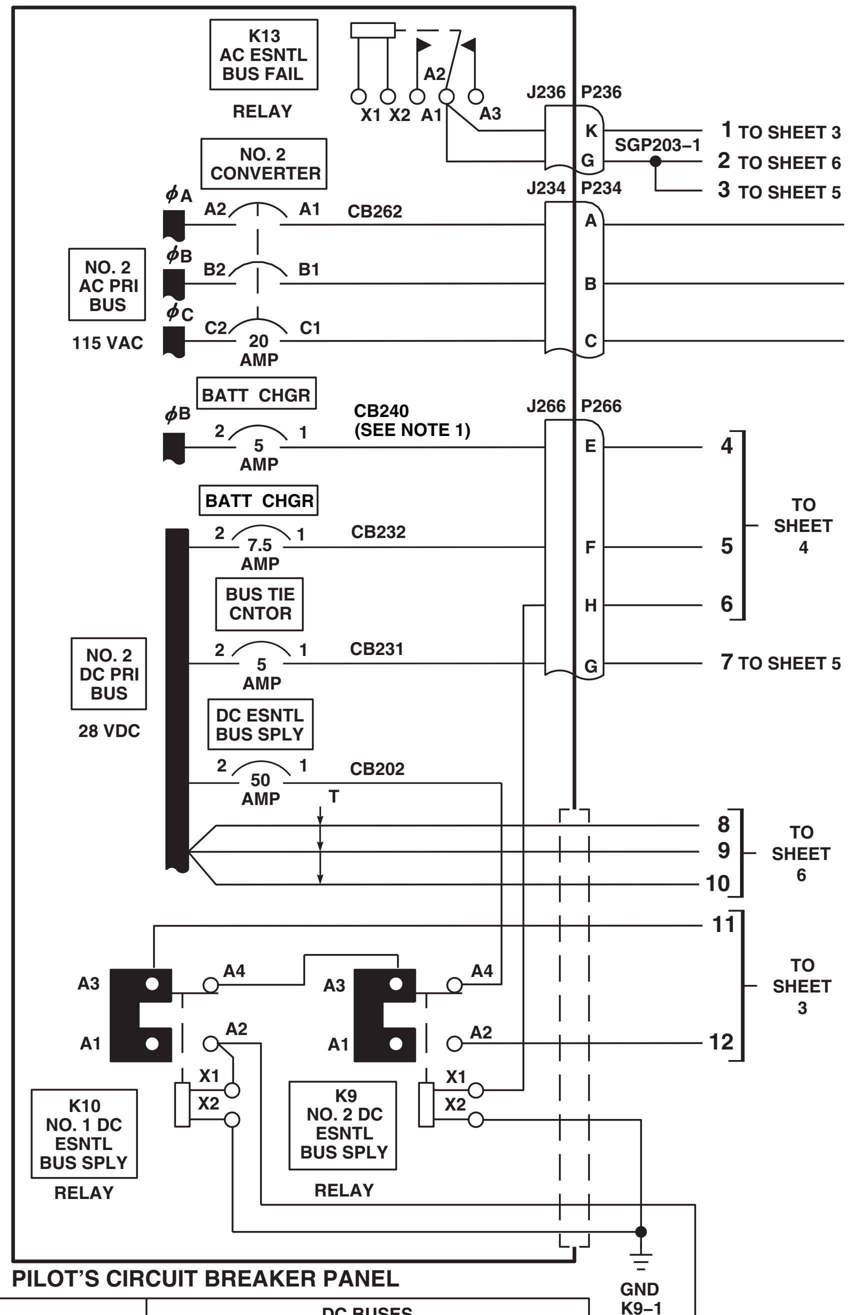


Figure 3. DC Electrical System Schematic Diagram **UH60L 96-26723-SUBQ MWO 50-77**. (Sheet 6 of 6)

SCHEMATICS - Continued



EFFECTIVITY

UH60A UH60L

NOTES

1. W/O HCW
   THIS CIRCUIT BREAKER RELOCATED / IDENTIFIED AS CB258.
2. W/O EMEP
   PIN FILTERED ADAPTERS (PFA) ARE NOT INSTALLED.
3. PIN S OF P123 SHOWS TYPICAL CONFIGURATION FOR ALL PIN FILTERED ADAPTERS.
4. HELICOPTERS WITH AUXILIARY HEATER INSTALLED, CIRCUIT BREAKER CB18A REIDENTIFIED AS AUX HTR CONTROL.

LEGEND

POWER SOURCE

A = NO. 1 CONVERTER
B = NO. 2 CONVERTER
C = BATTERY (GREATER THAN 35% CHARGED)
1 = TURNED ON AND OPERATING
0 = TURNED OFF OR FAILED

CONTACTORS/RELAYS/BUSES

K7 = BATTERY RELAY
K16 = NO. 1 DC PRI BUS CONTACTOR
K6 = NO. 2 DC PRI BUS CONTACTOR
K15 = DC BUS TIE CONTACTOR
K10 = NO. 1 DC ESNTL BUS SPLY RELAY
K9 = NO. 2 ESNTL BUS SPLY RELAY
K20 = DC ESNTL BUS FAIL RELAY
X = ENERGIZED
O = DE–ENERGIZED
= SIGNAL OR POWER GROUND
= CASE GROUND

DC POWER SOURCE			CONDITION	DC SYSTEM CONTACTORS / RELAYS							DC BUSES				
A	B	C		K7	K16	K6	K15	K10	K9	K20	NO. 1 DC PRI	NO. 2 DC PRI	DC ESNTL	BATT UTIL	BATT
0	0	0	1	O	O	O	O	O	O	O	O	O	O	O	O
1	0	0	2	O	X	O	X	X	O	X	X	X	X	X	X
0	1	0	3	O	O	X	X	X	O	X	X	X	X	X	X
1	1	0	4	O	X	X	O	X	O	X	X	X	X	X	X
0	0	1	5	X	O	O	O	O	X	X	O	O	X	X	X
1	0	1	6	O	X	O	X	X	O	X	X	X	X	X	X
0	1	1	7	O	O	X	X	X	O	X	X	X	X	X	X
1	1	1	8	O	X	X	O	X	O	X	X	X	X	X	X

AB1503_1
SA

Figure 4. DC Electrical System Schematic Diagram UH60A UH60L . (Sheet 1 of 6)

SCHEMATICS - Continued

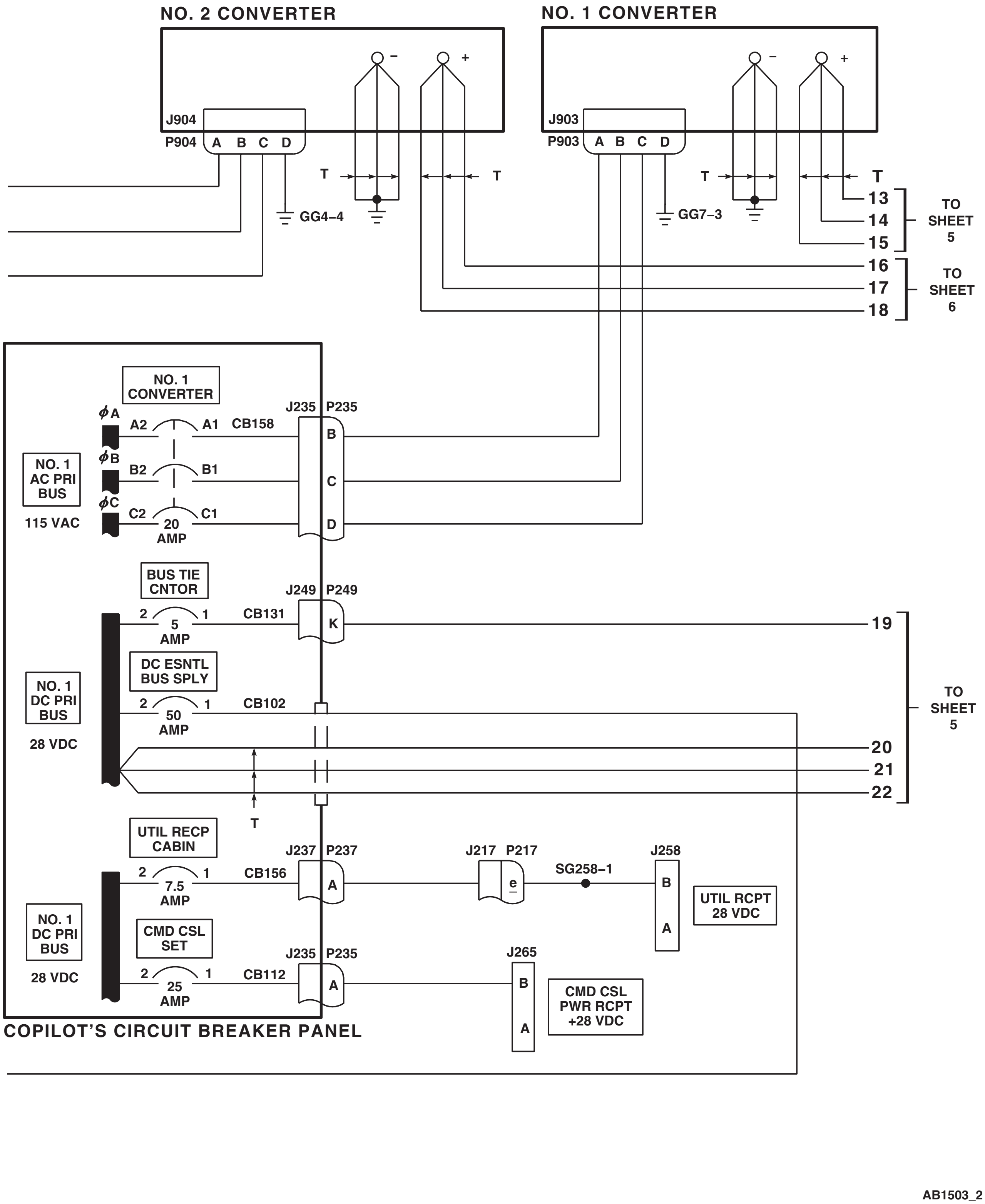


Figure 4. DC Electrical System Schematic Diagram **UH60A UH60L** . (Sheet 2 of 6)

SCHEMATICS - Continued

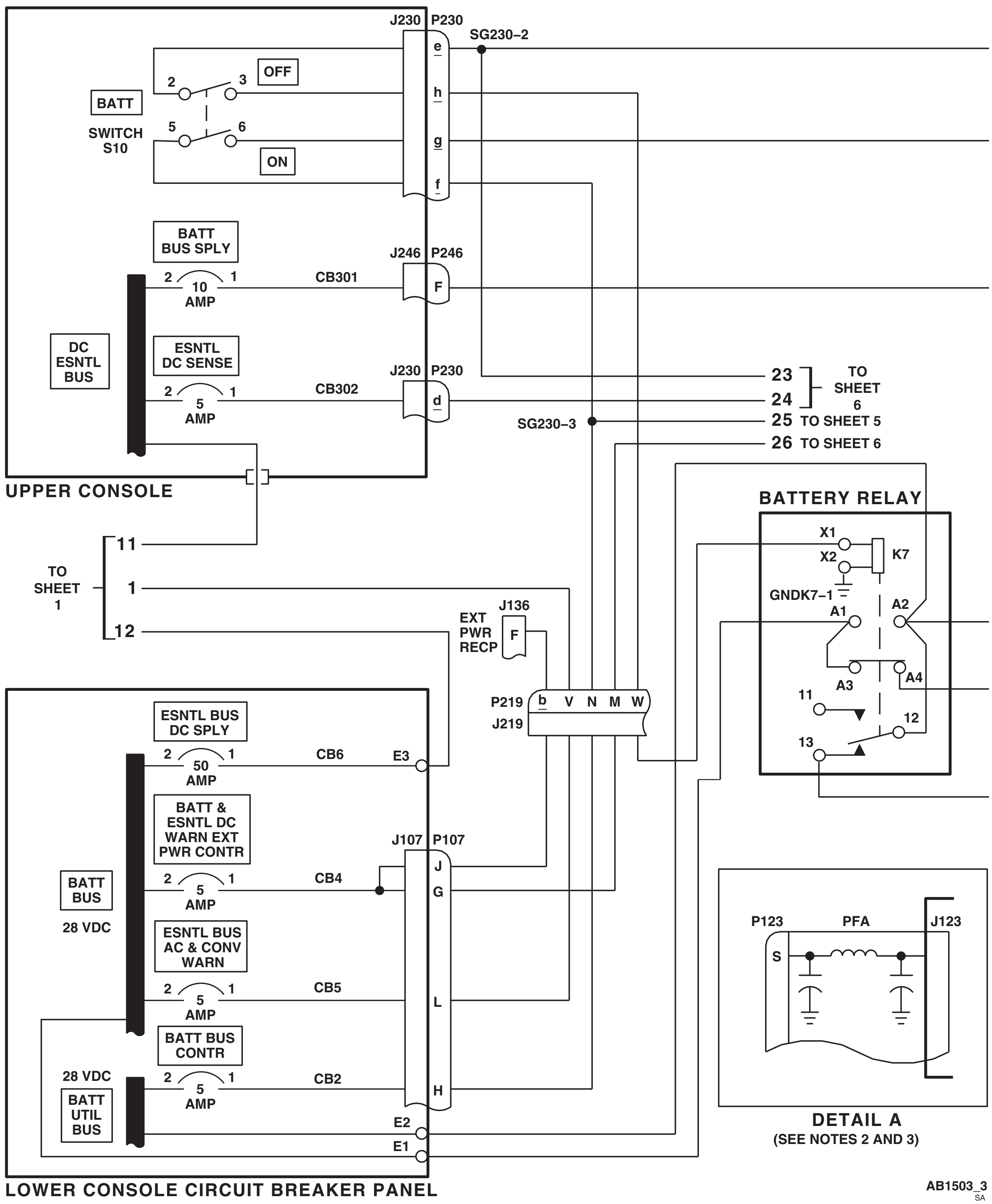


Figure 4. DC Electrical System Schematic Diagram UH60A UH60L . (Sheet 3 of 6)

SCHEMATICS - Continued

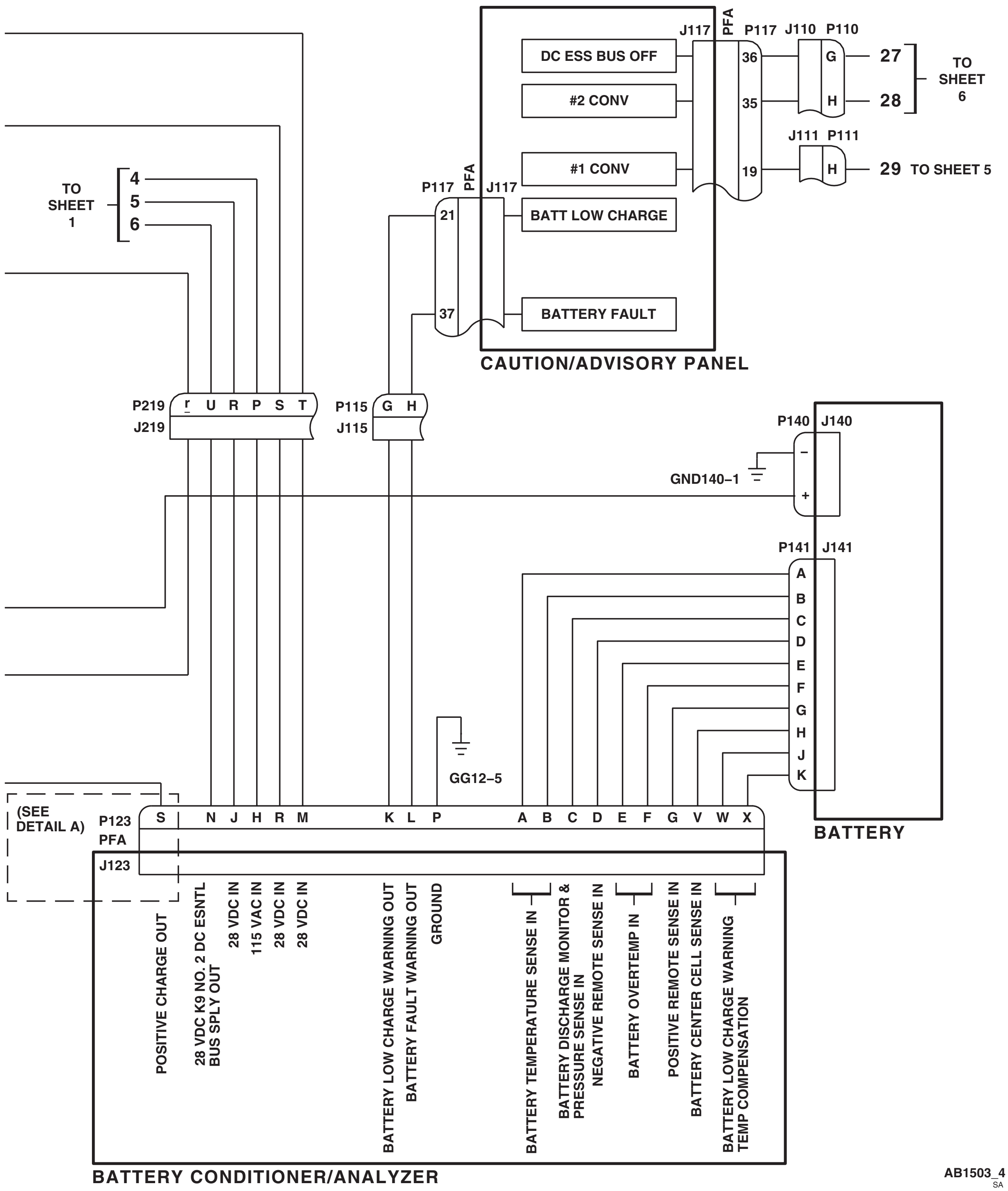


Figure 4. DC Electrical System Schematic Diagram **UH60A UH60L**. (Sheet 4 of 6)

SCHEMATICS - Continued

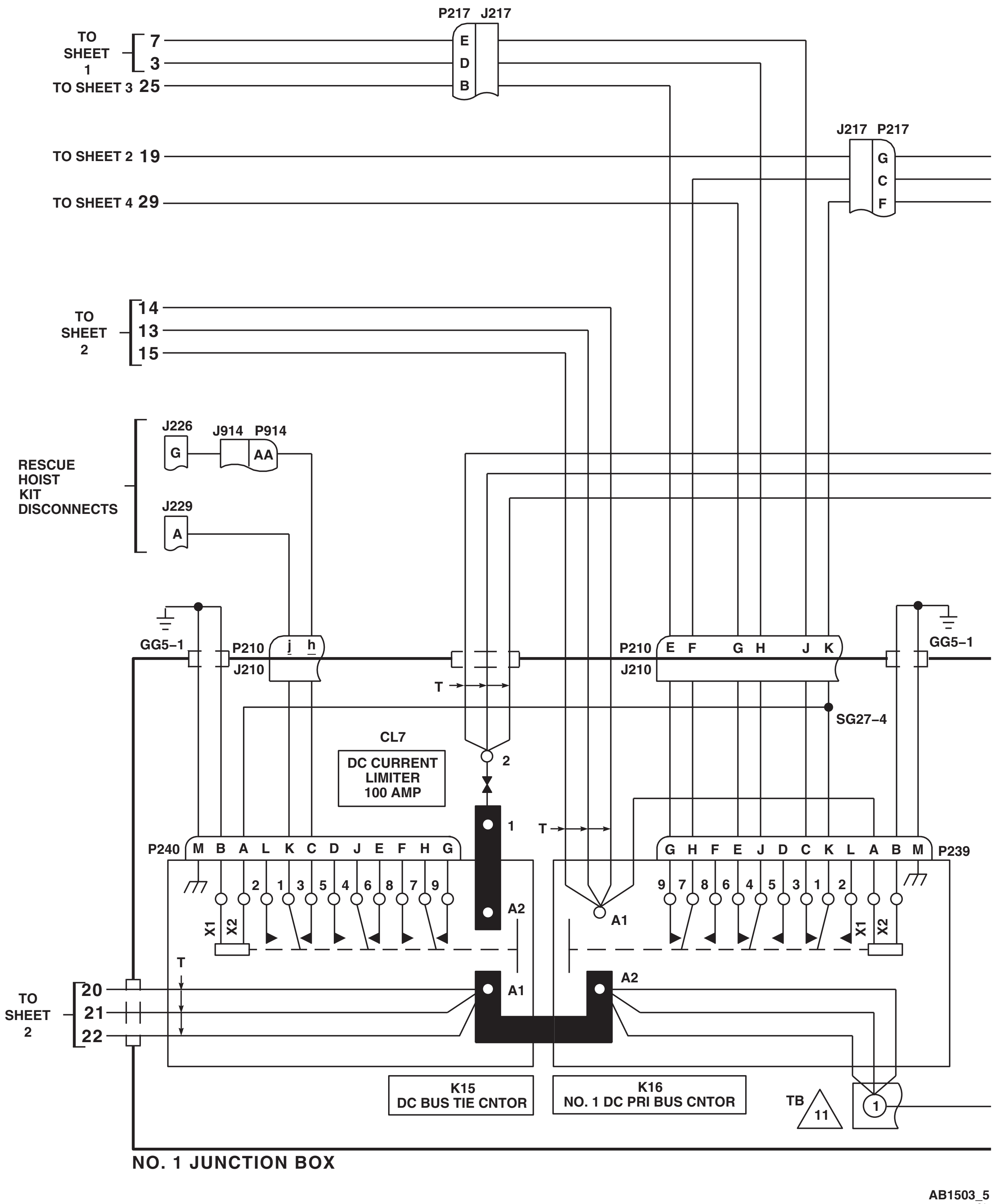


Figure 4. DC Electrical System Schematic Diagram UH60A UH60L . (Sheet 5 of 6)

SCHEMATICS - Continued

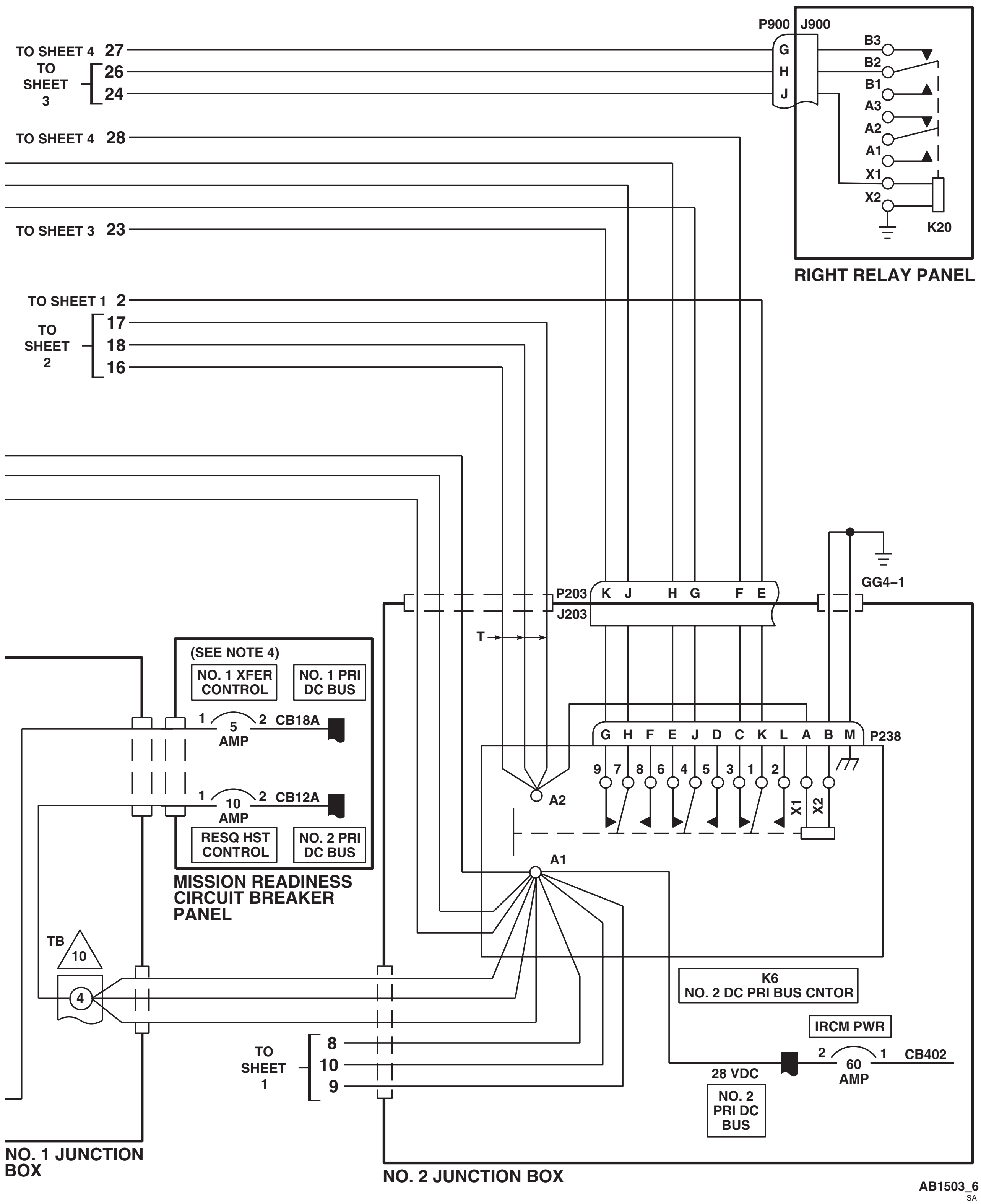


Figure 4. DC Electrical System Schematic Diagram UH60A UH60L . (Sheet 6 of 6)

SCHEMATICS - Continued

EFFECTIVITY

EH60A

NOTES

1. SEE AC ELECTRICAL SYSTEM WP 0101 00.
2. X INDICATES ENERGIZED WHEN QUICK FIX PWR SWITCH IS ON.
3. X INDICATES OPERATIONAL WHEN QUICK FIX PWR SWITCH IS ON.
4. EH60A 84–24017 – 86–24567
5. W/O EMEP PIN FILTERED ADAPTERS (PFA) ARE NOT INSTALLED.
6. PIN S OF P123 SHOWS TYPICAL CONFIGURATION FOR ALL PIN FILTERED ADAPTERS.

LEGEND

POWER SOURCE

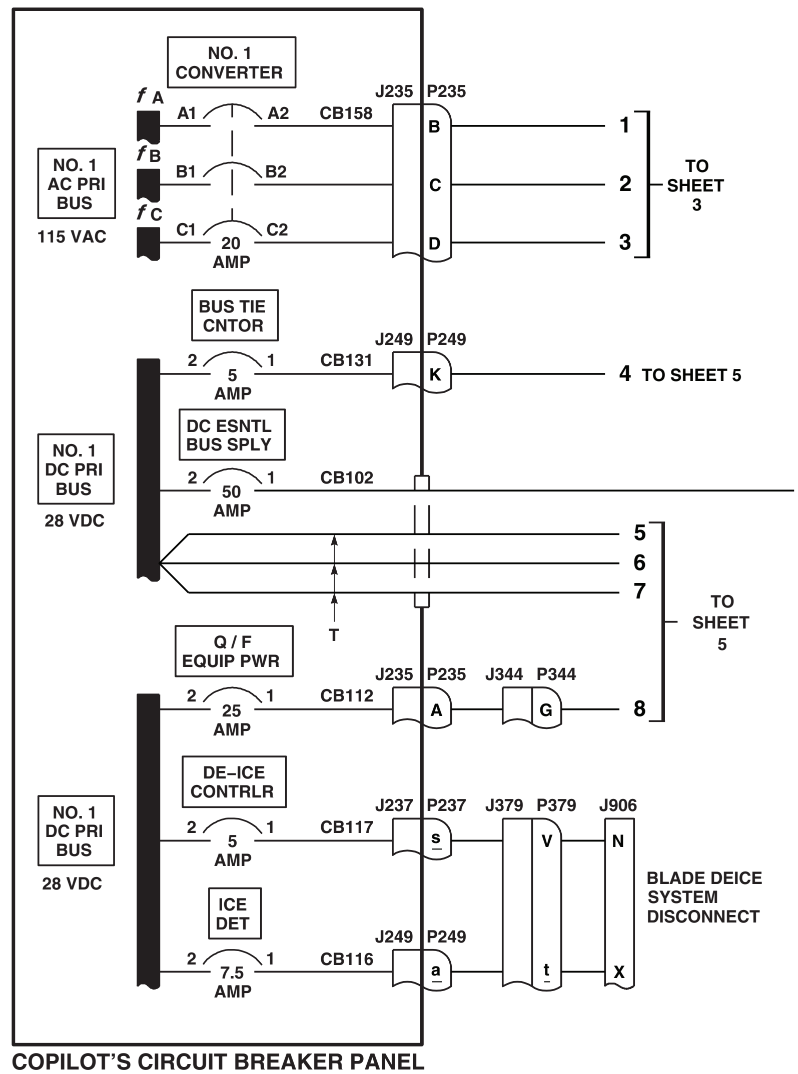
A = NO. 1 CONVERTER
B = NO. 2 CONVERTER
C = BATTERY (GREATER THAN 35% CHARGED)
1 = TURNED ON AND OPERATING
0 = TURNED OFF OR FAILED

CONTACTORS/RELAYS/BUSES

K5 = DC MONITOR BUS RELAY
K6 = NO. 2 DC PRI BUS CONTACTOR
K7 = BATTERY RELAY
K9 = NO. 2 DC ESNTL BUS SPLY RELAY
K10 = NO. 1 DC ESNTL BUS SPLY RELAY
K15 = DC BUS TIE CONTACTOR
K16 = NO. 1 DC PRI BUS CONTACTOR
K20 = DC ESNTL BUS FAIL RELAY
K83 = QUICK FIX POWER CONTACTOR
X = ENERGIZED
O = DE–ENERGIZED

= SIGNAL OR POWER GROUND

= CASE GROUND



DC POWER SOURCE			CONDITION	DC SYSTEM CONTACTORS / RELAYS									DC BUSES						
A	B	C		K5	K6	K7	K9	K10	K15	K16	K20	K83	NO. 1 DC PRI	NO. 2 DC PRI	DC ESNTL	BATT UTIL	BATT	DC MON	QUICK FIX PWR
0	0	0	1	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O
1	0	0	2	O	O	O	O	X	X	X	X	O	X	X	X	X	X	O	O
0	1	0	3	O	X	O	O	X	X	O	X	O	X	X	X	X	X	O	O
1	1	0	4	X	X	O	O	X	O	X	X	X SEE NOTE 2	X	X	X	X	X	X	X SEE NOTE 3
0	0	1	5	O	O	X	X	O	O	O	X	O	O	O	X	X	X	O	O
1	0	1	6	O	O	O	O	X	X	X	X	O	X	X	X	X	X	O	O
0	1	1	7	O	X	O	O	X	X	O	X	O	X	X	X	X	X	O	O
1	1	1	8	X	X	O	O	X	O	X	X	X SEE NOTE 2	X	X	X	X	X	X	X SEE NOTE 3

AB1504_1A
SA

Figure 5. DC Electrical System Schematic Diagram EH60A . (Sheet 1 of 7)

SCHEMATICS - Continued

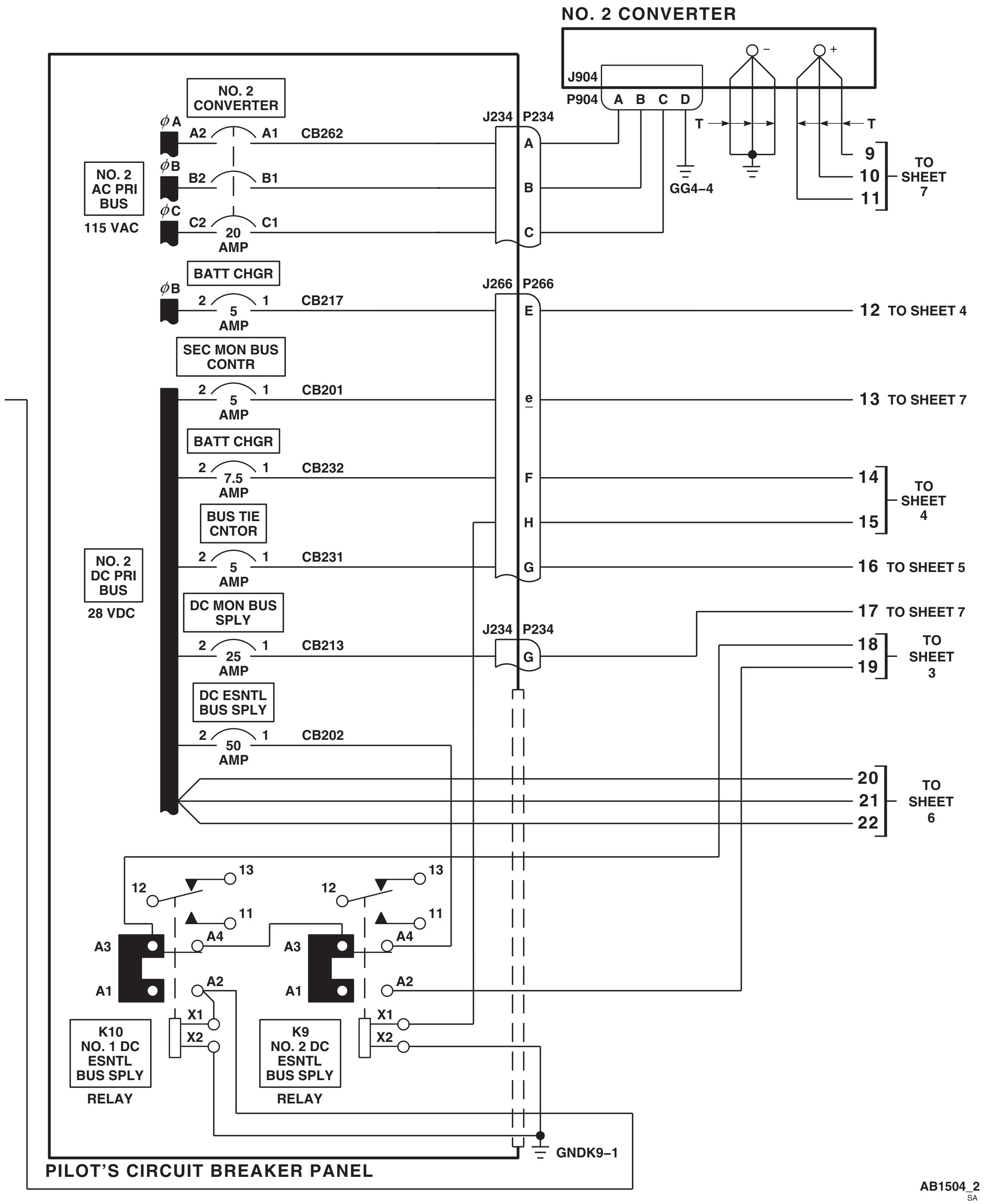


Figure 5. DC Electrical System Schematic Diagram EH60A . (Sheet 2 of 7)

SCHEMATICS - Continued

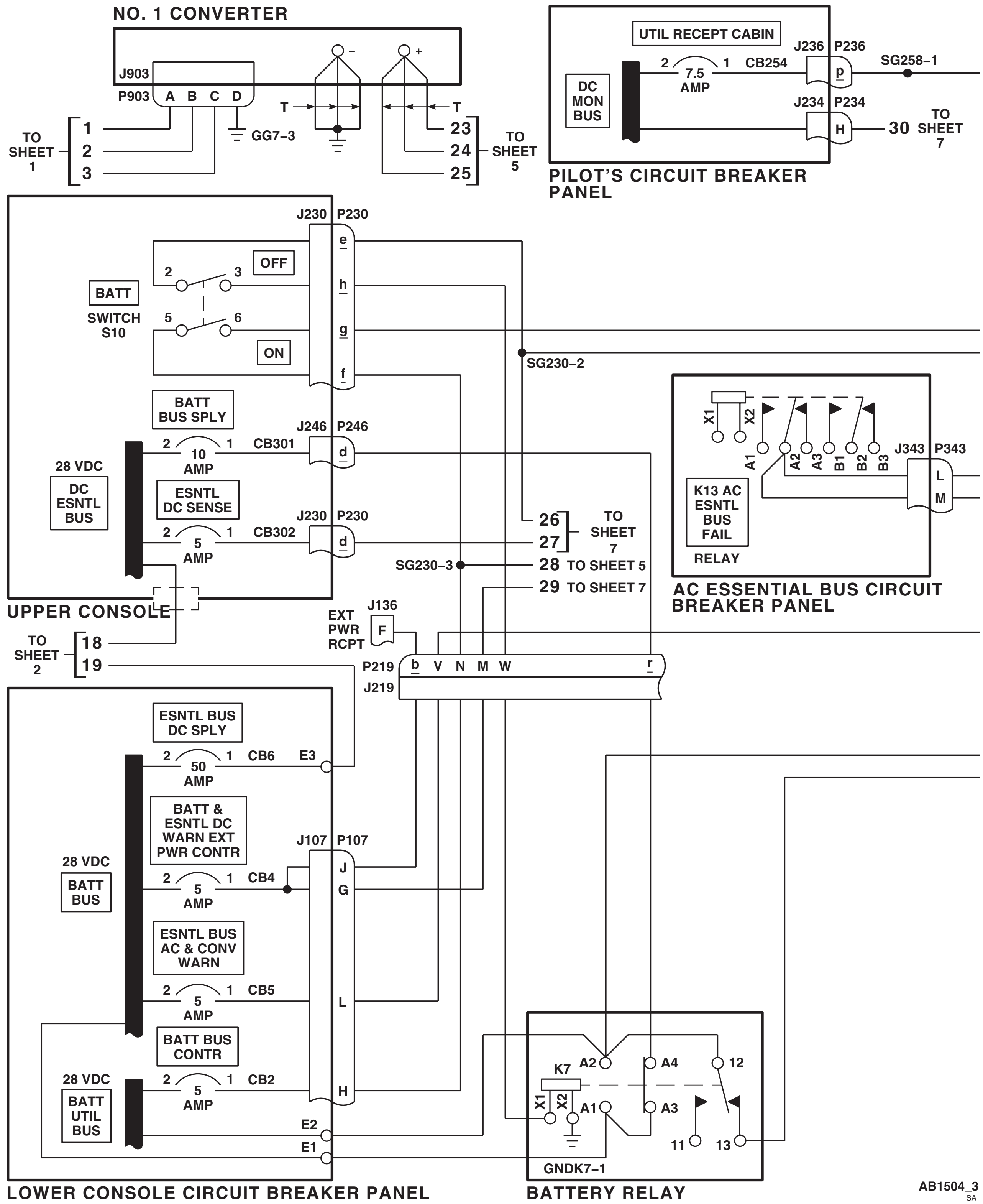


Figure 5. DC Electrical System Schematic Diagram **EH60A**. (Sheet 3 of 7)

SCHEMATICS - Continued

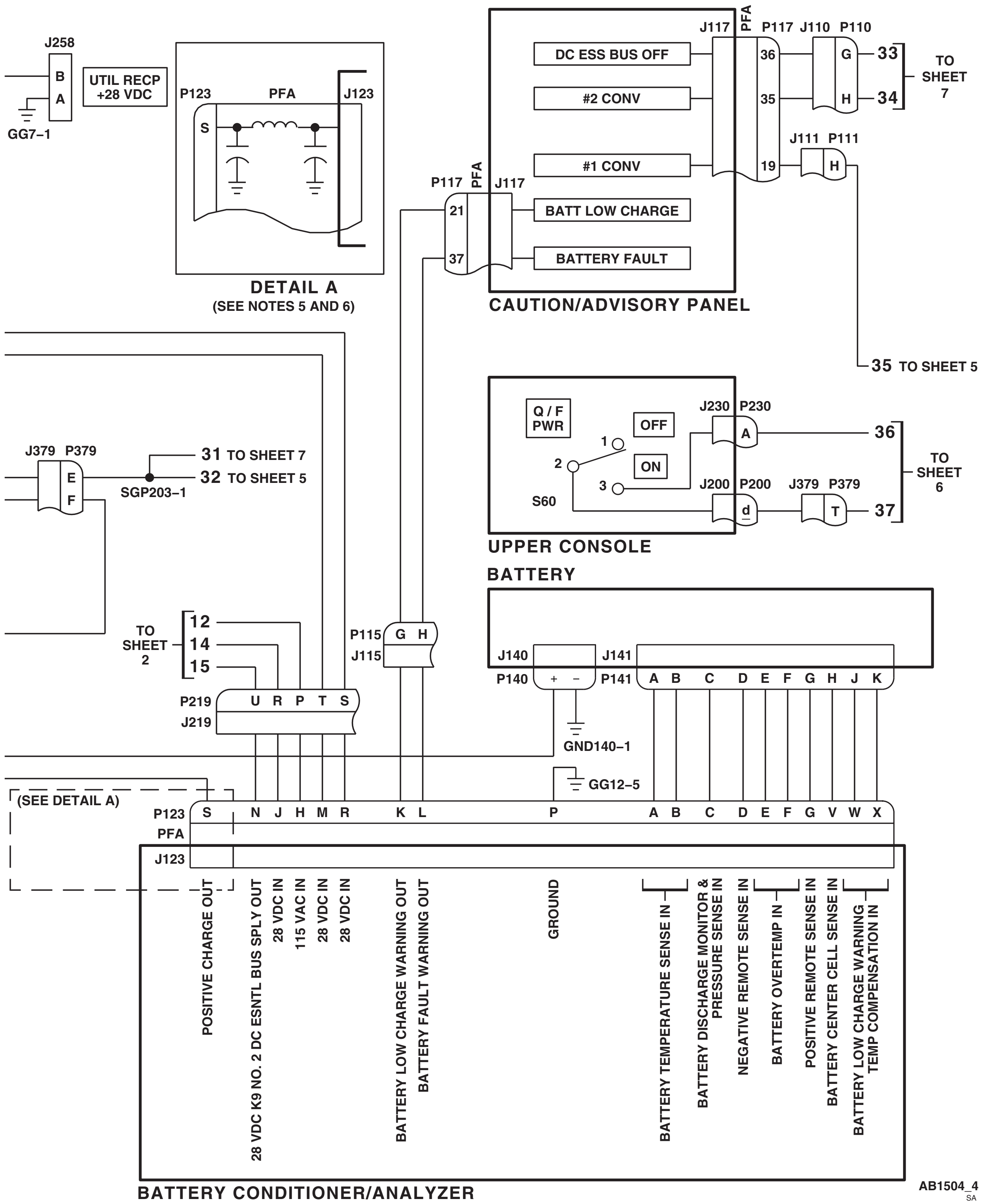


Figure 5. DC Electrical System Schematic Diagram EH60A . (Sheet 4 of 7)

SCHEMATICS - Continued

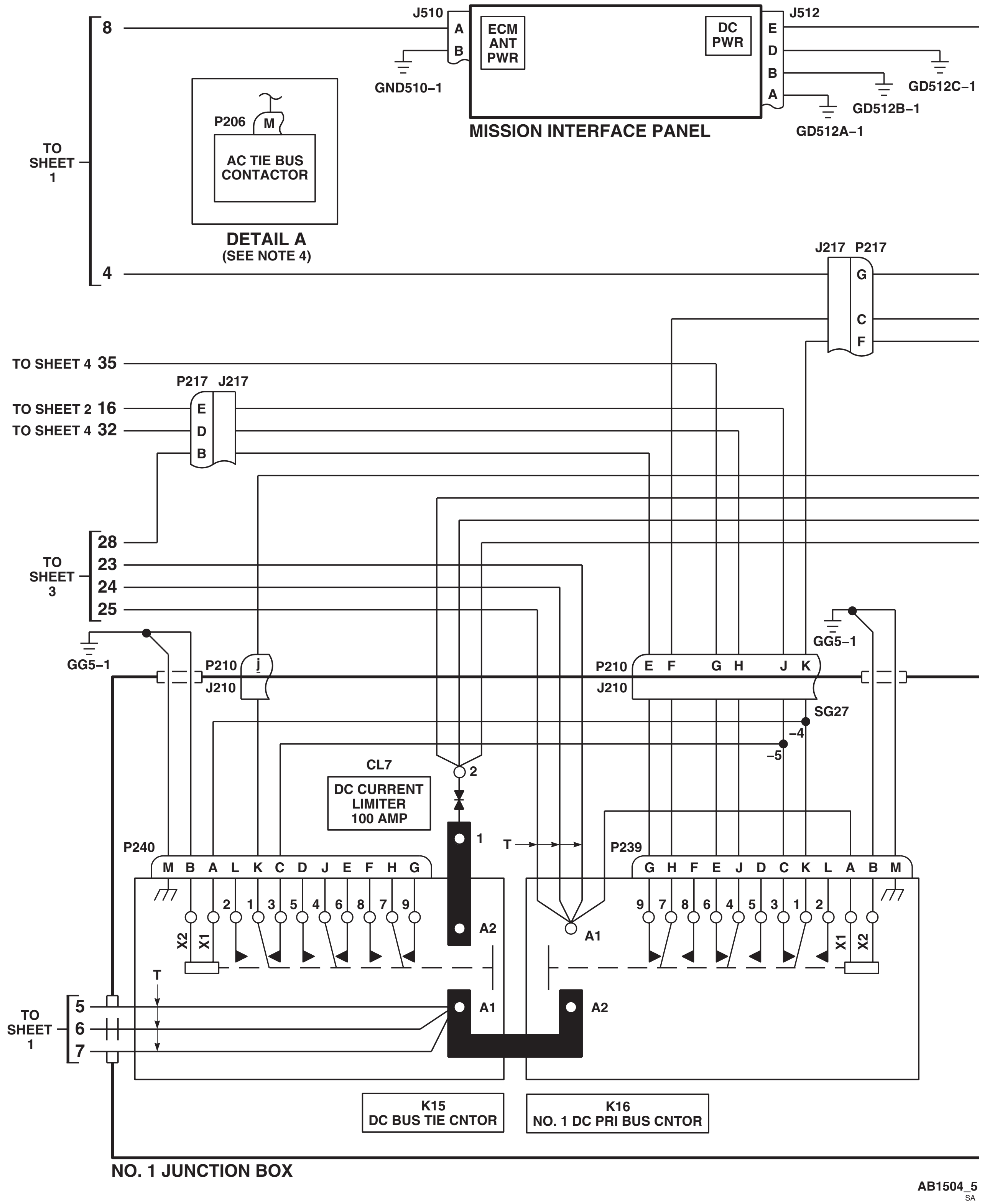


Figure 5. DC Electrical System Schematic Diagram **EH60A** . (Sheet 5 of 7)

SCHEMATICS - Continued

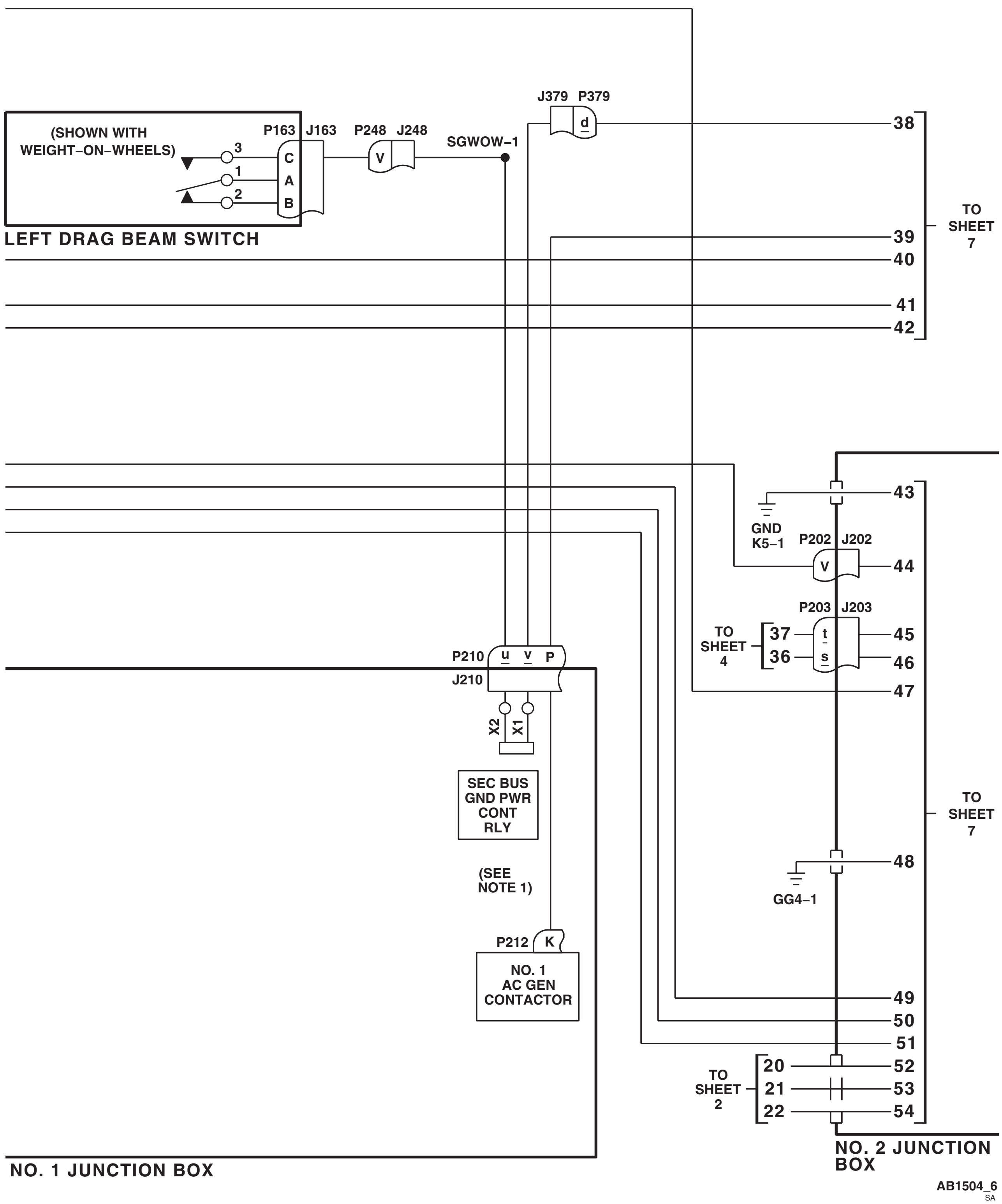


Figure 5. DC Electrical System Schematic Diagram EH60A . (Sheet 6 of 7)

SCHEMATICS - Continued

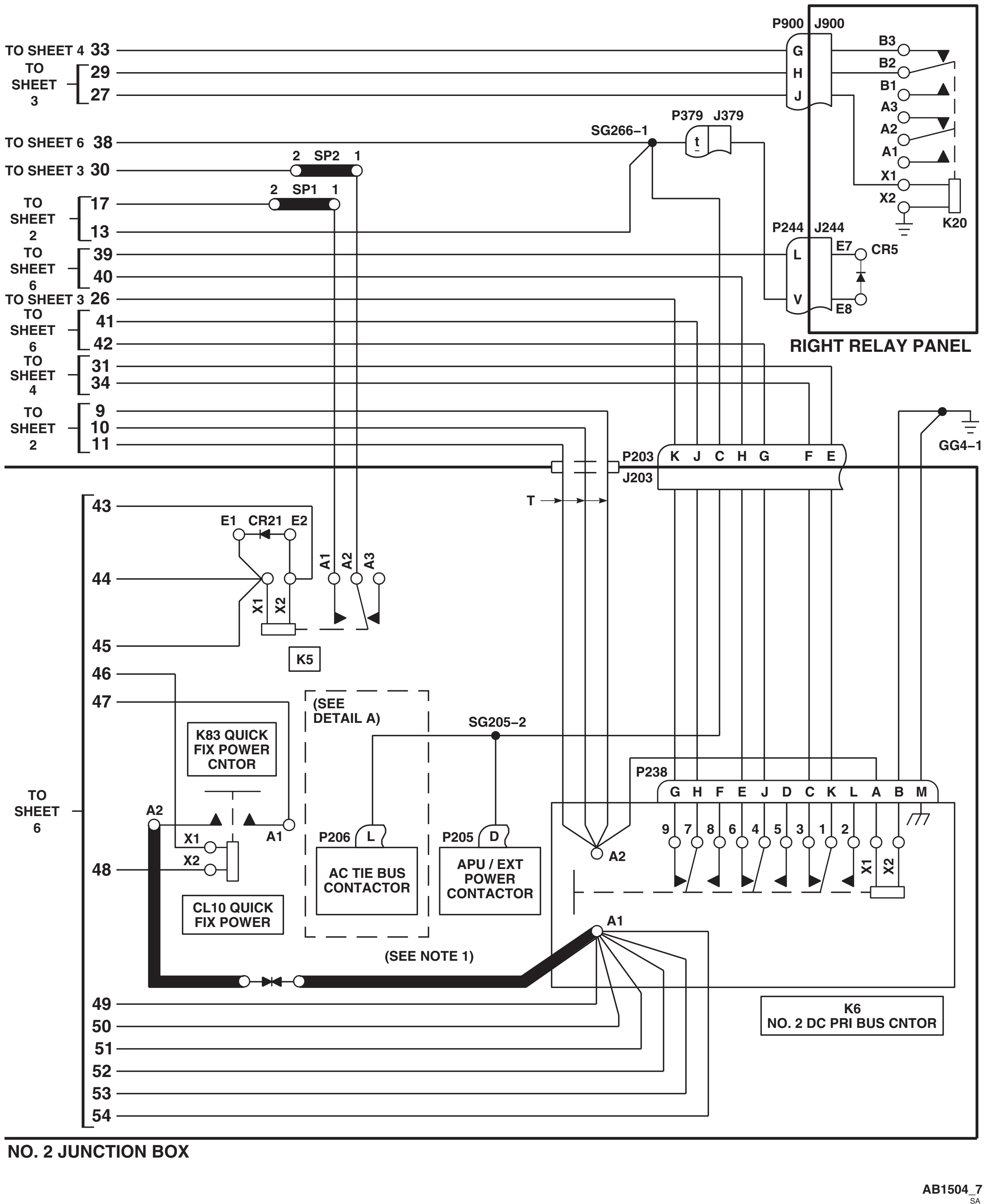
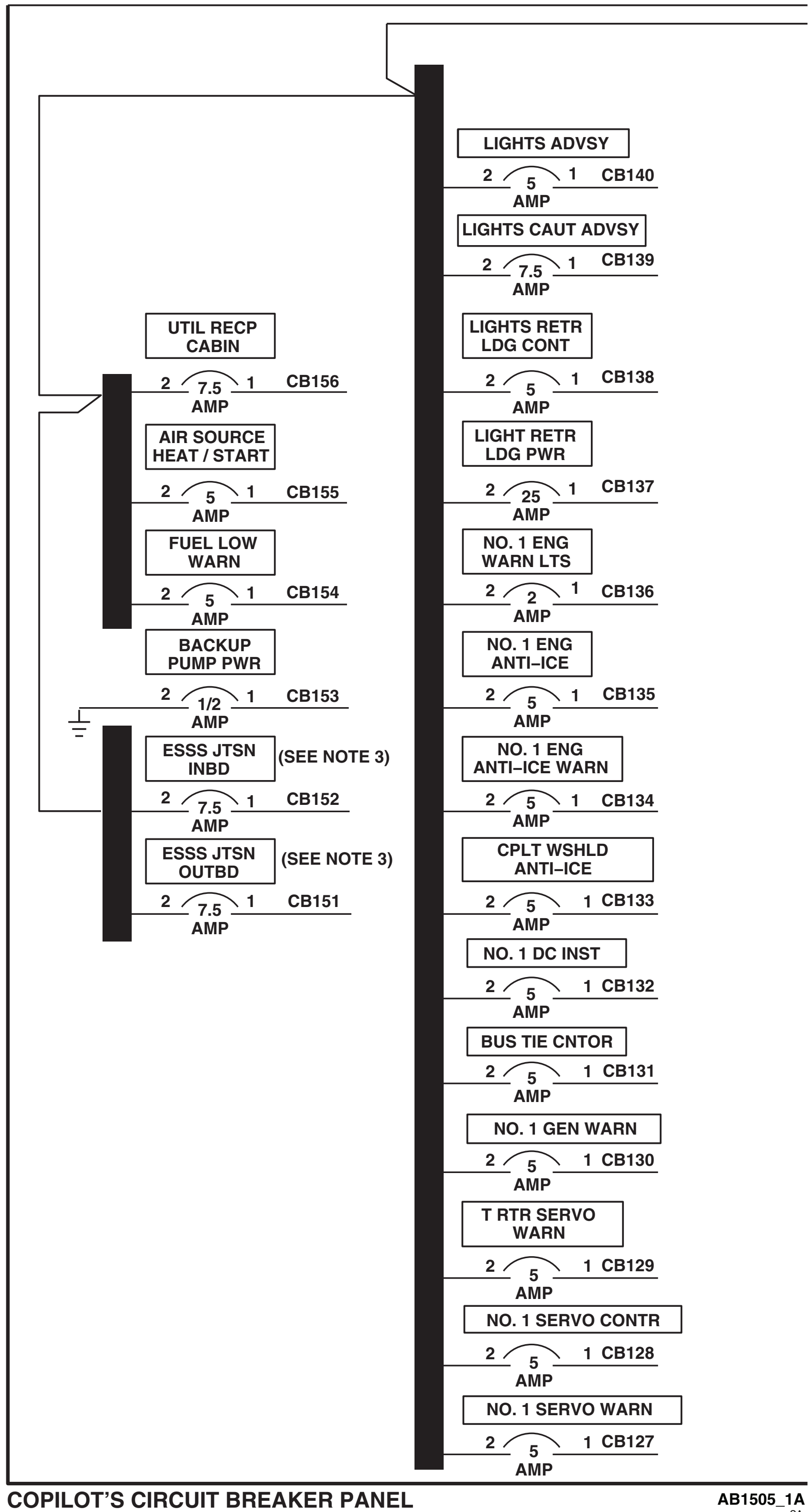


Figure 5. DC Electrical System Schematic Diagram **EH60A** . (Sheet 7 of 7)

SCHEMATICS - Continued



EFFECTIVITY

UH60A UH60L

NOTES

1. HCW
2. HELICOPTERS WITH AUXILIARY CABIN HEATER INSTALLED.
3. ESSS
4. W/O HCW CIRCUIT BREAKER IDENTIFIED AS PILOT WSHLD CTR.
5. UH60L 96–26723–SUBQ
   MWO 50–77
6. MWO 50–75
7. UH60L 96–26723 – SUBQ CIRCUIT BREAKER IS LABELED DPLR/GPS.
8. MWO 50–76 HF RADIO SET (AN/ARC–220(V)1).
9. MWO 50–82

Figure 6. DC Power Distribution Diagram UH60A UH60L . (Sheet 1 of 7)

SCHEMATICS - Continued

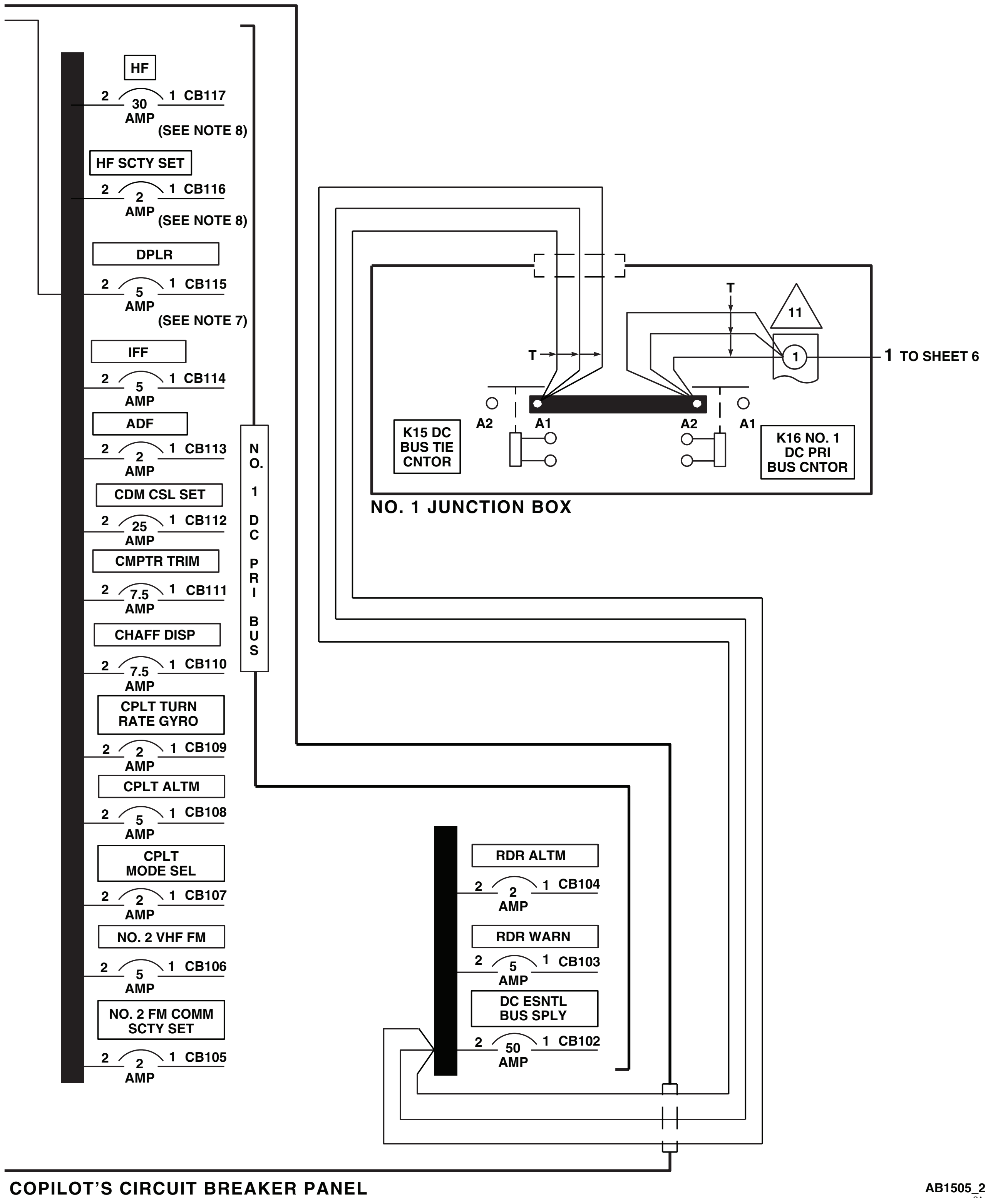


Figure 6. DC Power Distribution Diagram UH60A UH60L . (Sheet 2 of 7)

SCHEMATICS - Continued

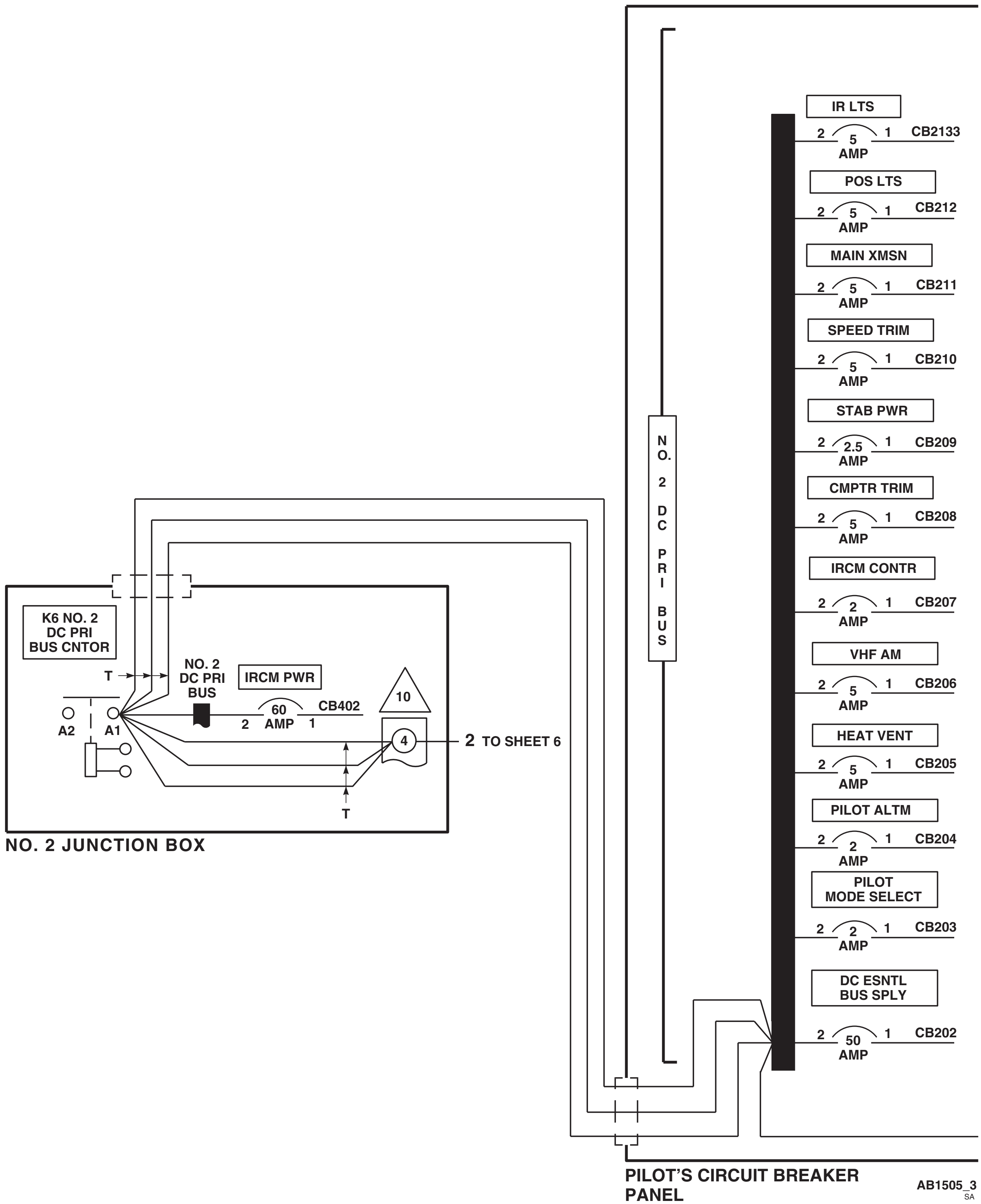


Figure 6. DC Power Distribution Diagram **UH60A UH60L** . (Sheet 3 of 7)

SCHEMATICS - Continued

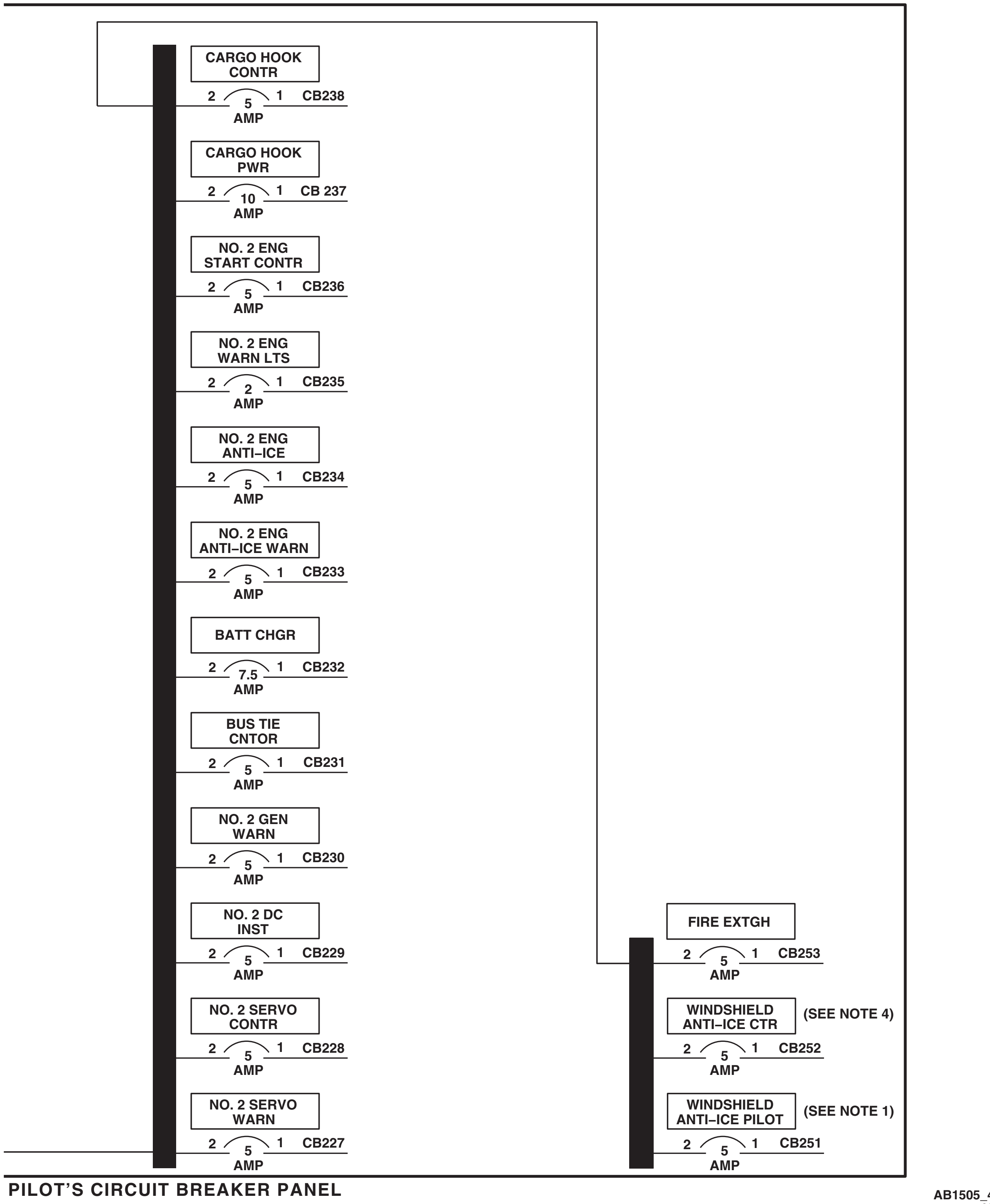


Figure 6. DC Power Distribution Diagram UH60A UH60L . (Sheet 4 of 7)

SCHEMATICS - Continued

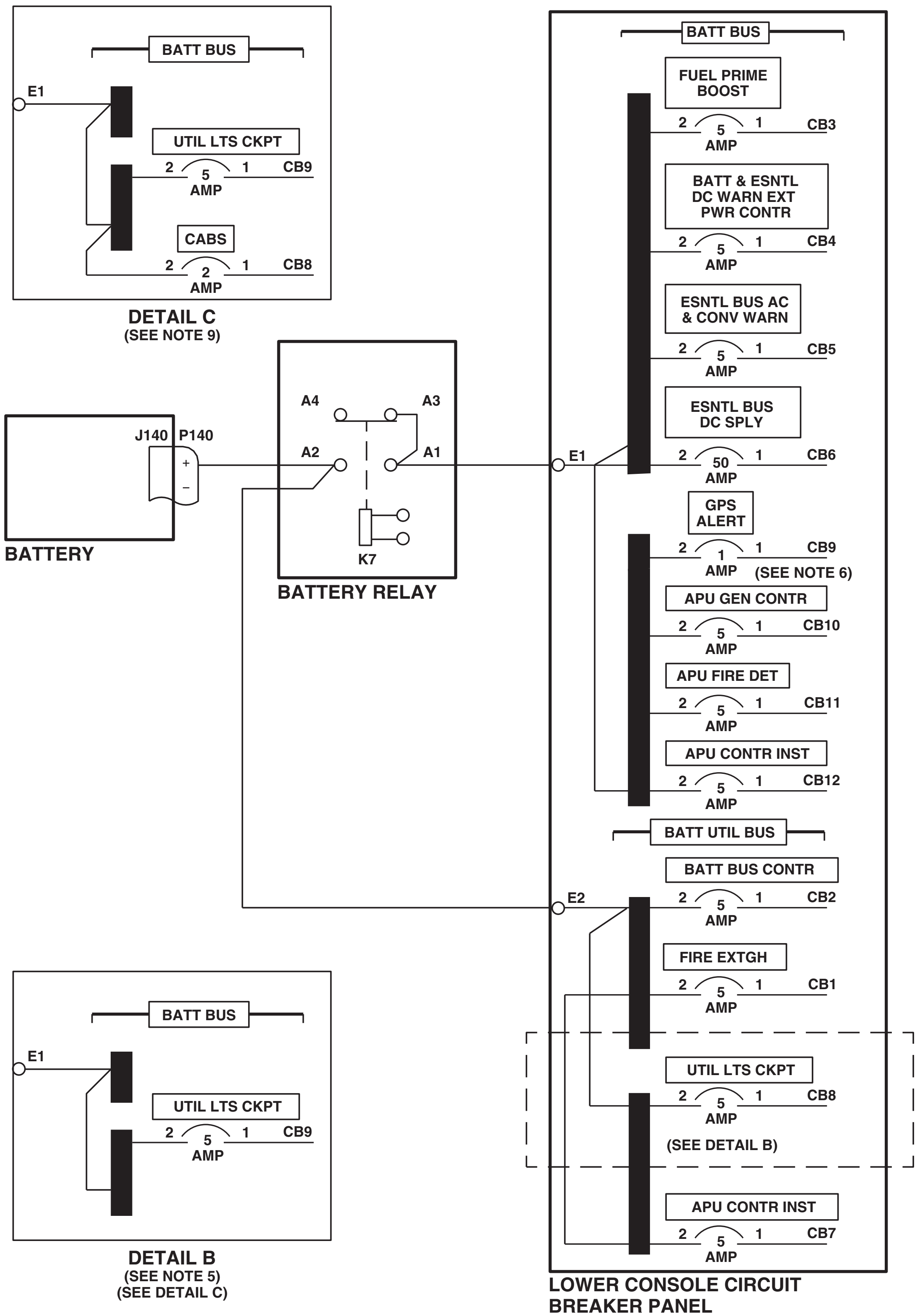


Figure 6. DC Power Distribution Diagram UH60A UH60L . (Sheet 5 of 7)

SCHEMATICS - Continued

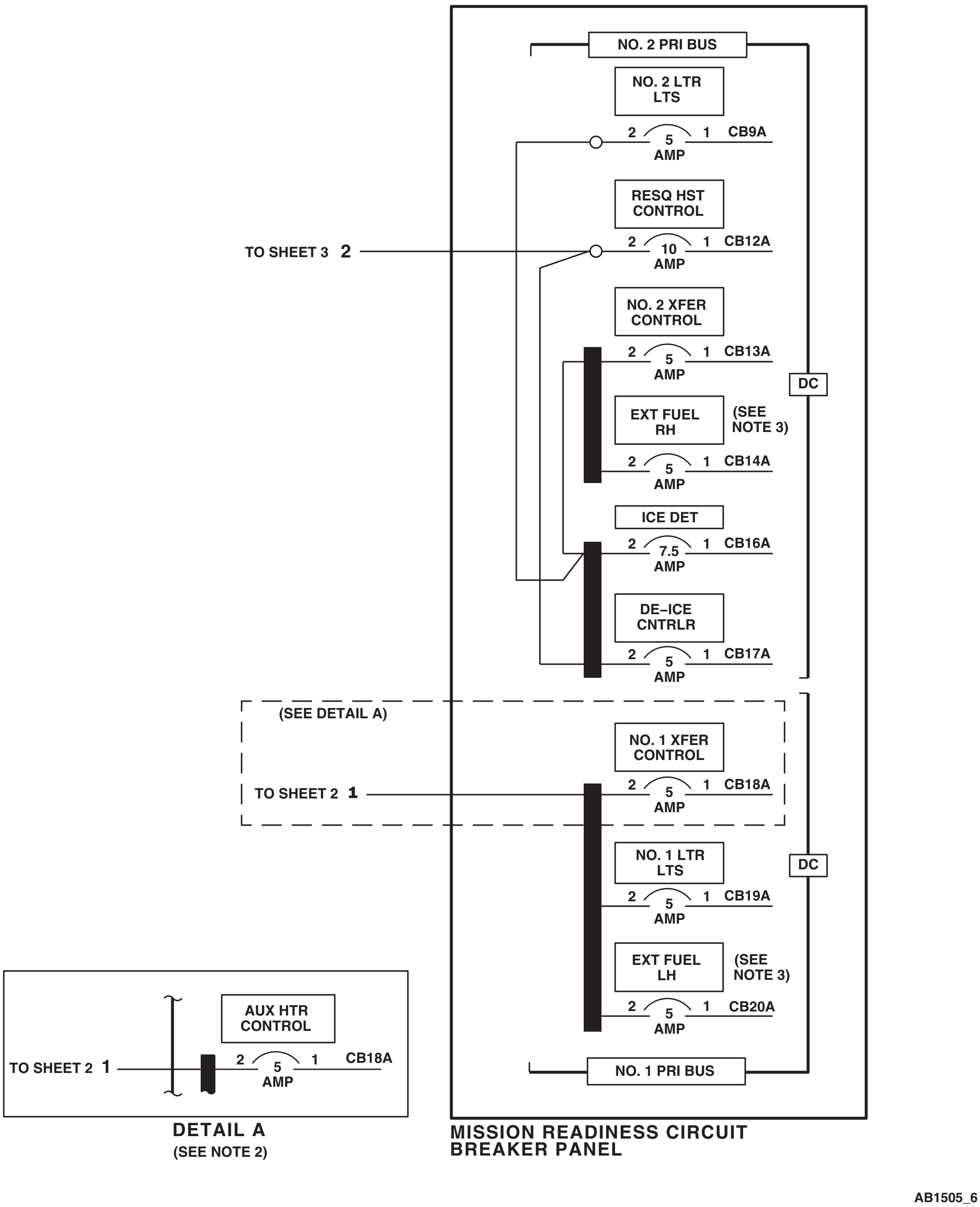


Figure 6. DC Power Distribution Diagram UH60A UH60L . (Sheet 6 of 7)

SCHEMATICS - Continued

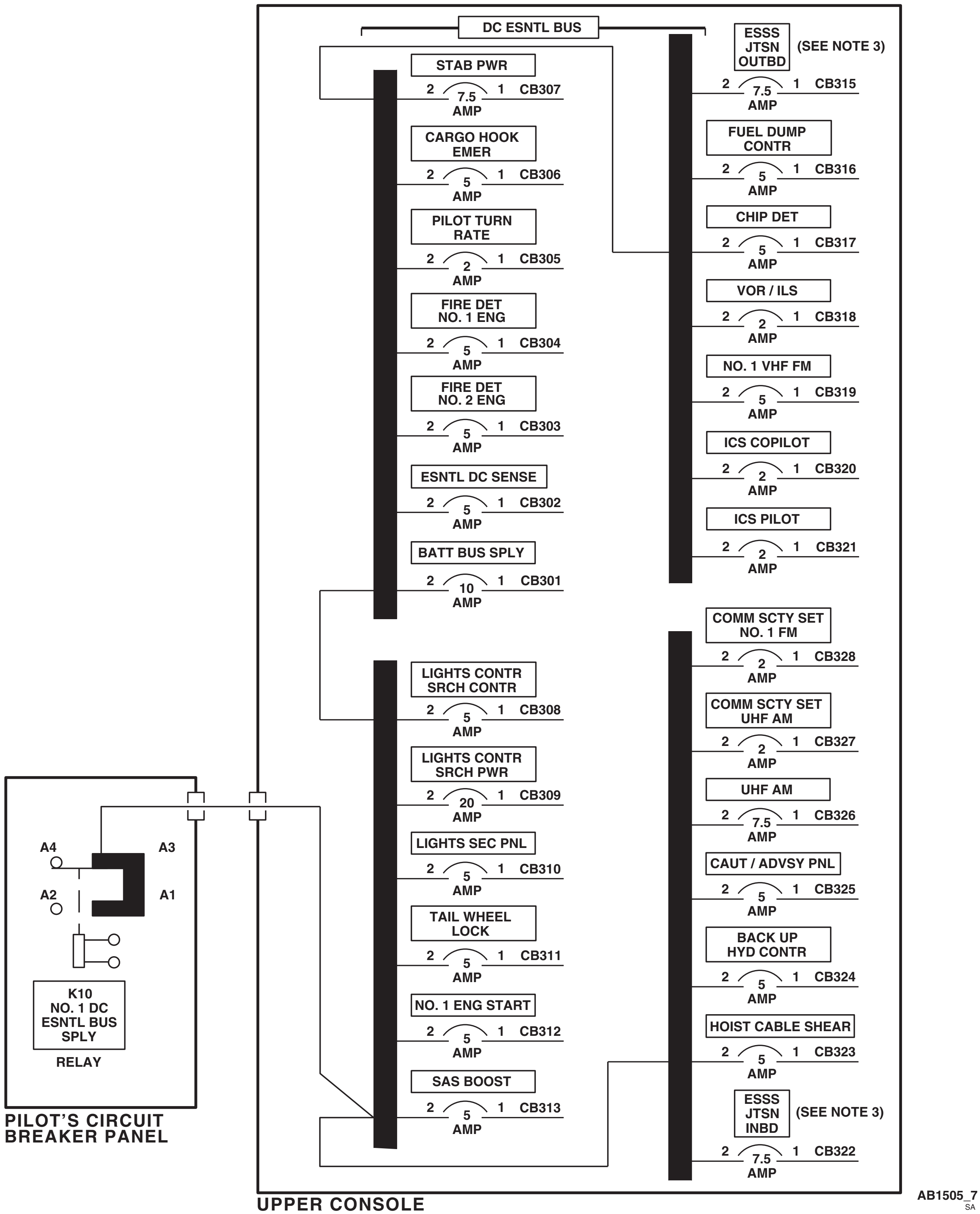


Figure 6. DC Power Distribution Diagram UH60A UH60L . (Sheet 7 of 7)

SCHEMATICS - Continued

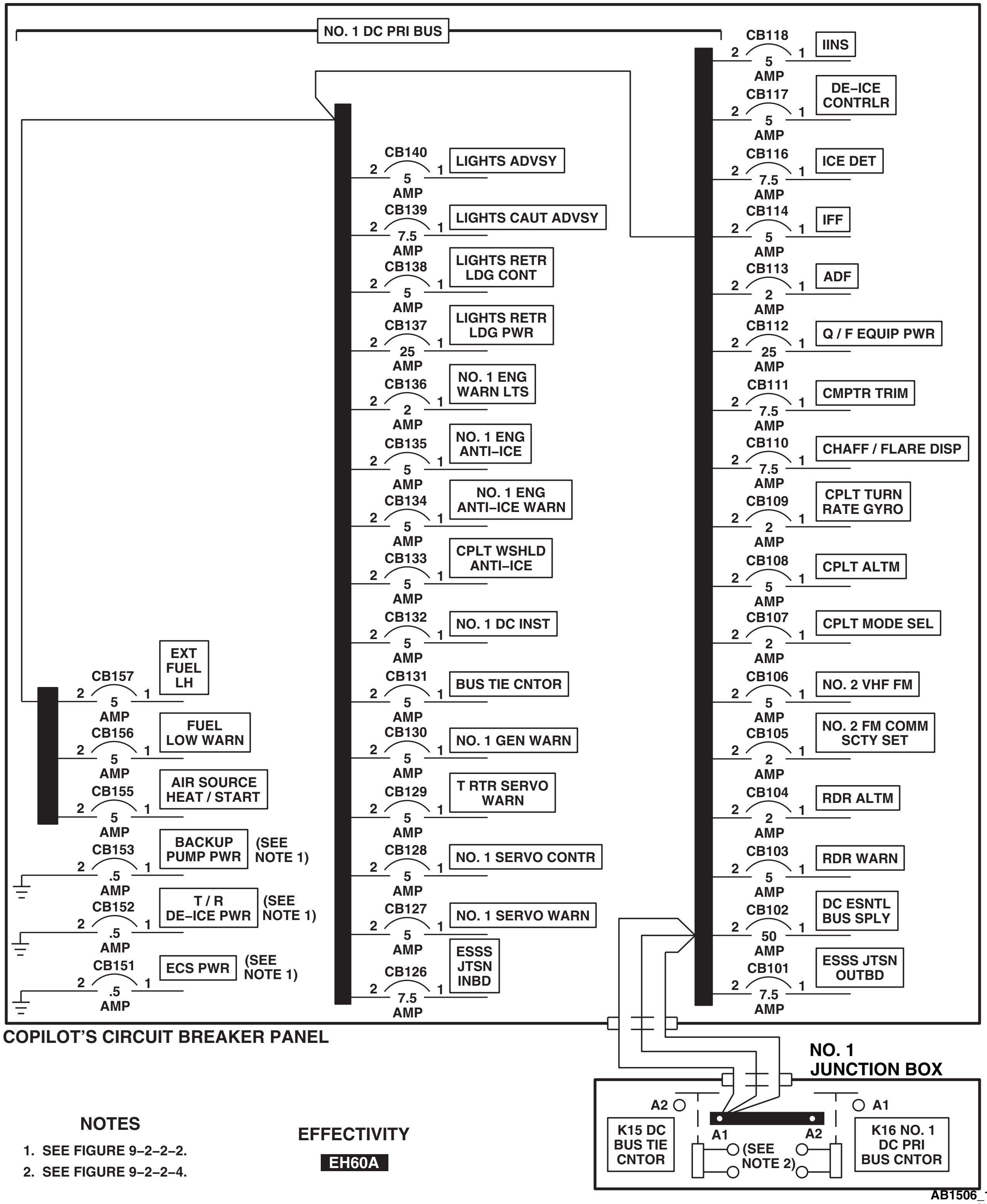


Figure 7. DC Power Distribution Diagram EH60A . (Sheet 1 of 3)

SCHEMATICS - Continued

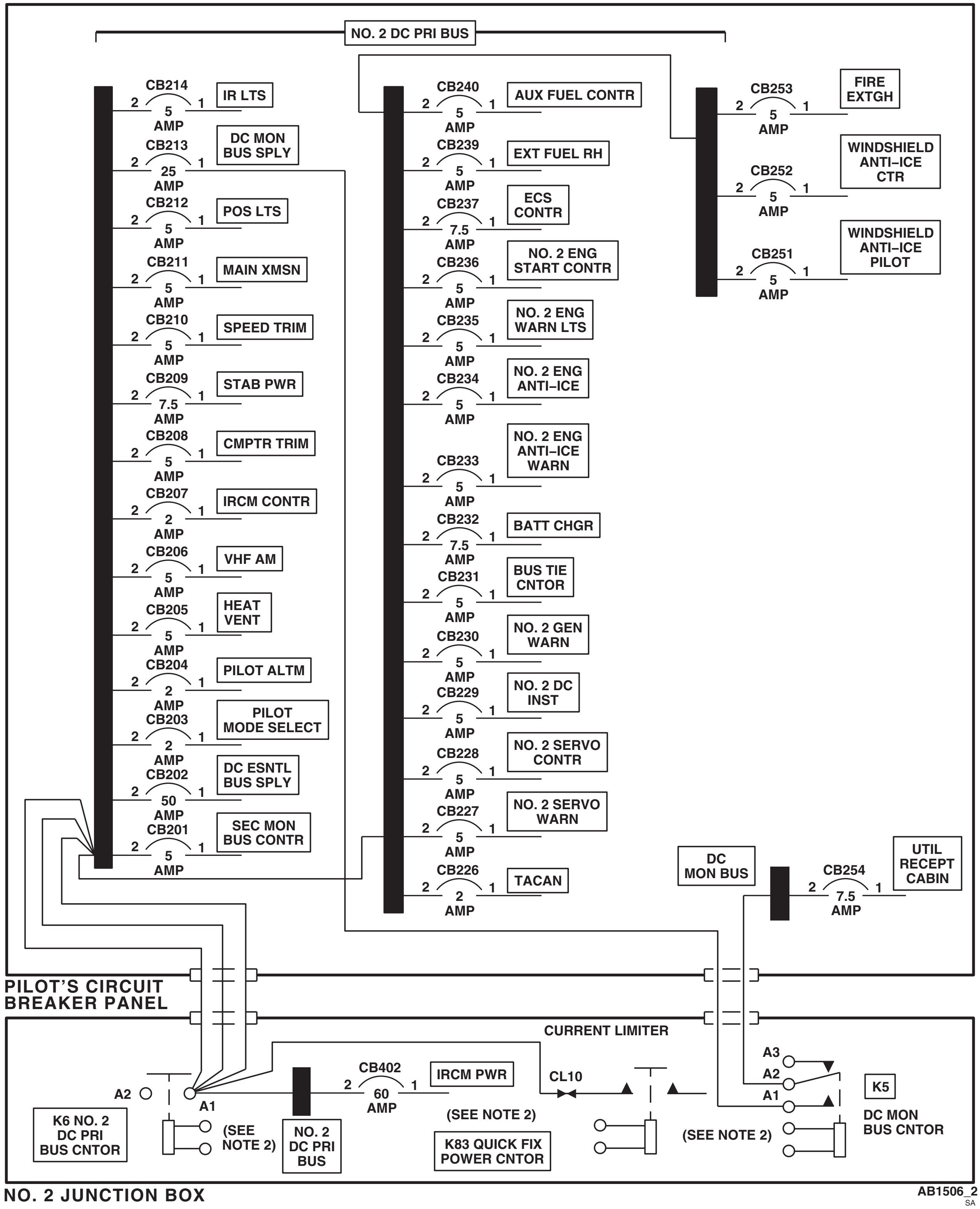


Figure 7. DC Power Distribution Diagram **EH60A** . (Sheet 2 of 3)

SCHEMATICS - Continued

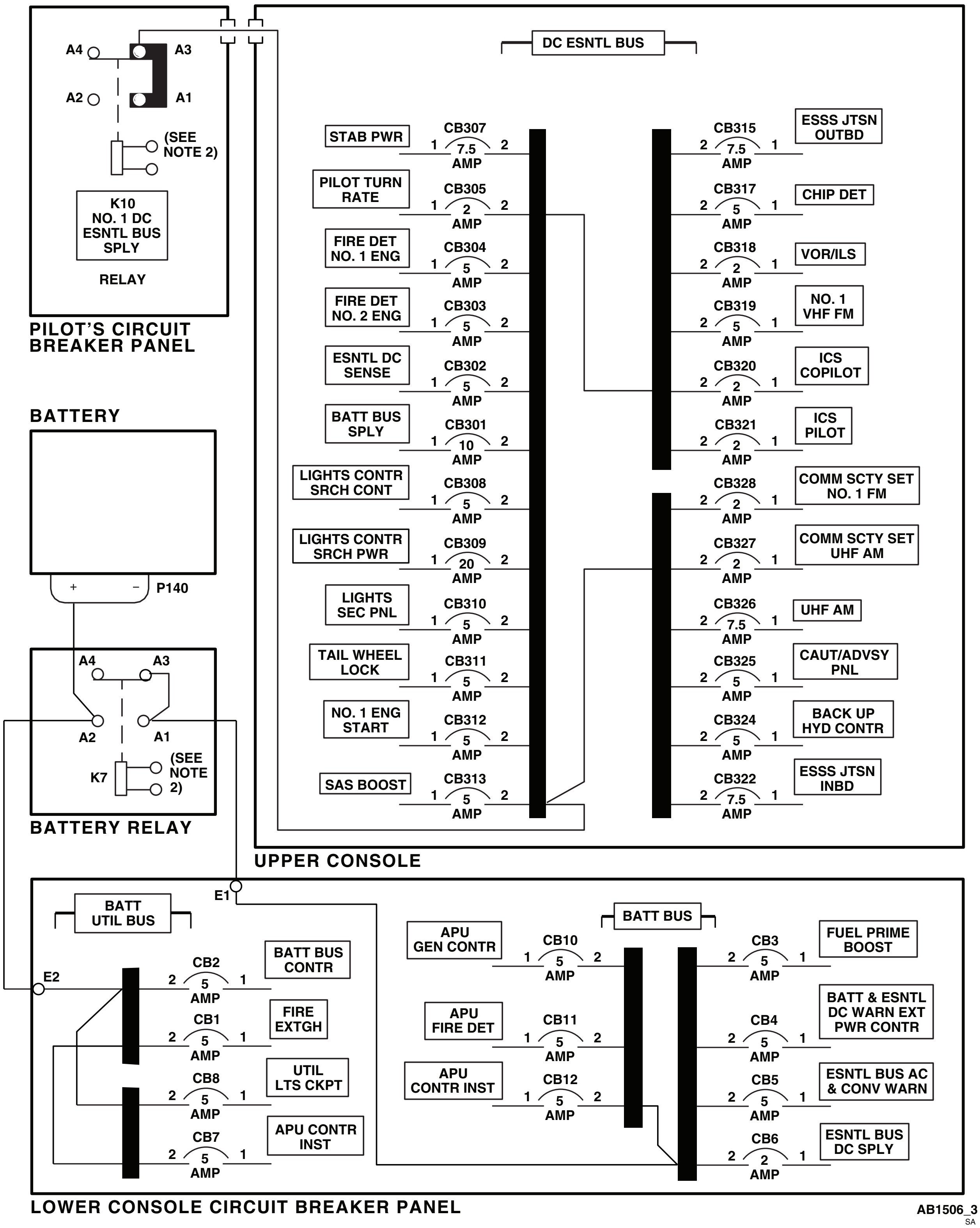


Figure 7. DC Power Distribution Diagram **EH60A** . (Sheet 3 of 3)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

MISSION ELECTRICAL INTERFACE EH60A

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
WP 0101 00
WP 0102 00
WP 0104 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Connected, WP 1685 00 or APU Operational, TM 1-1520-237-10

CAUTION

In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.

NOTE

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

Table 1. Temperature Requirements.

AIR TEMPERATURE F° (C°)	OPERATING TIME (MINUTES)	COOLDOWN TIME, (PUMP OFF) (MINUTES)
90° (32°) max	Unlimited	-
91° (33°) to 100° (38°)	24	72
101° (39°) to 125° (52°)	16	48

MISSION ELECTRICAL INTERFACE OPERATIONAL/TROUBLESHOOTING PROCEDURE EH60A

PROCEDURE

Step

NOTE

If a circuit breaker pops out during operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
BACKUP HYD CONTR	Upper Console
BACKUP PUMP PWR	Copilot’s
BUS TIE CNTOR	Copilot’s
NO 1. CONVERTER	Copilot’s
NO 2. CONVERTER	Pilot’s

2. If cables from interface panel J511 and J512 are connected, disconnect them.
3. Turn on electrical power.
4. Place upper console BACKUP HYD pump switch to OFF.

NOTE

Backup hydraulic pump may run to charge APU accumulator. The AC secondary bus will not operate until backup pump stops.

5. Check for 115 vac between the following points:
   - J511-C and J511-H
   - J511-B and J511-H
   - J511-A and J511-G

Indication/Condition

Multimeter shall indicate 115 vac.

Corrective Action

If Indication/Condition is not as specified, troubleshoot AC electrical system (WP 0101 00).

Step

6. Place upper console Q/F PWR switch to ON.
7. Check for 28 vdc between the following points:

MISSION ELECTRICAL INTERFACE OPERATIONAL/TROUBLESHOOTING PROCEDURE EH60A - Continued

- J512-E and J512-A
- J512-E and J512-B
- J512-E and J512-D

Indication/Condition

Multimeter shall indicate 28 vdc.

Corrective Action

1. If Indication/Condition is not as specified for some points, repair/replace wiring, as required (WP 1747 00).
2. If Indication/Condition is not as specified for all points, troubleshoot DC electrical system (WP 0102 00).

Step

8. Place upper console Q/F PWR switch to OFF.
9. Turn off electrical power.
10. Connect cables J511 and J512 to interface panel.

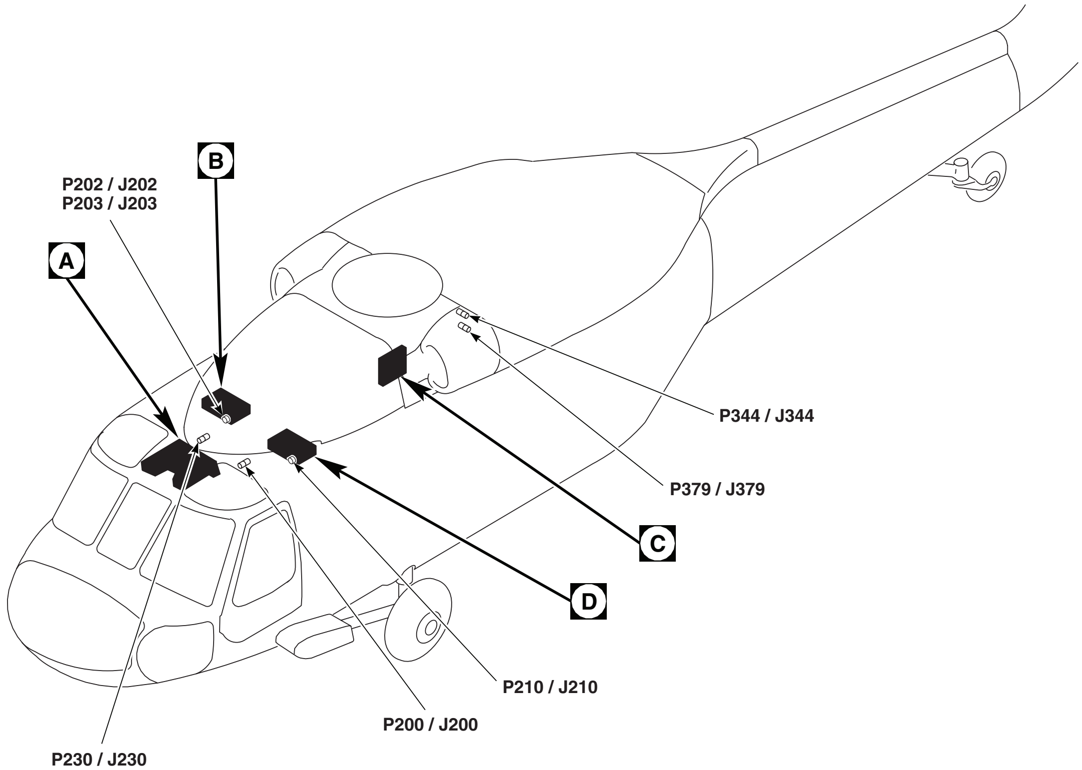
Indication/Condition

Cables shall be connected to the interface panel.

Corrective Action

None Required

LOCATION



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J511	MISSION INTERFACE PANEL
J512	MISSION INTERFACE PANEL
P200 / J200	COCKPIT CEILING BL 13 LH, STA 246
P202 / J202	NO. 2 JUNCTION BOX
P203 / J203	NO. 2 JUNCTION BOX
P210 / J210	NO. 1 JUNCTION BOX
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P344 / J344	CABIN CEILING, BL 5 RH, STA 379
P379 / J379	CABIN CEILING, BL 5 RH, STA 379

AA2683_1
SA

Figure 1. Mission Electrical Interface Location Diagram. (Sheet 1 of 4)

LOCATION - Continued

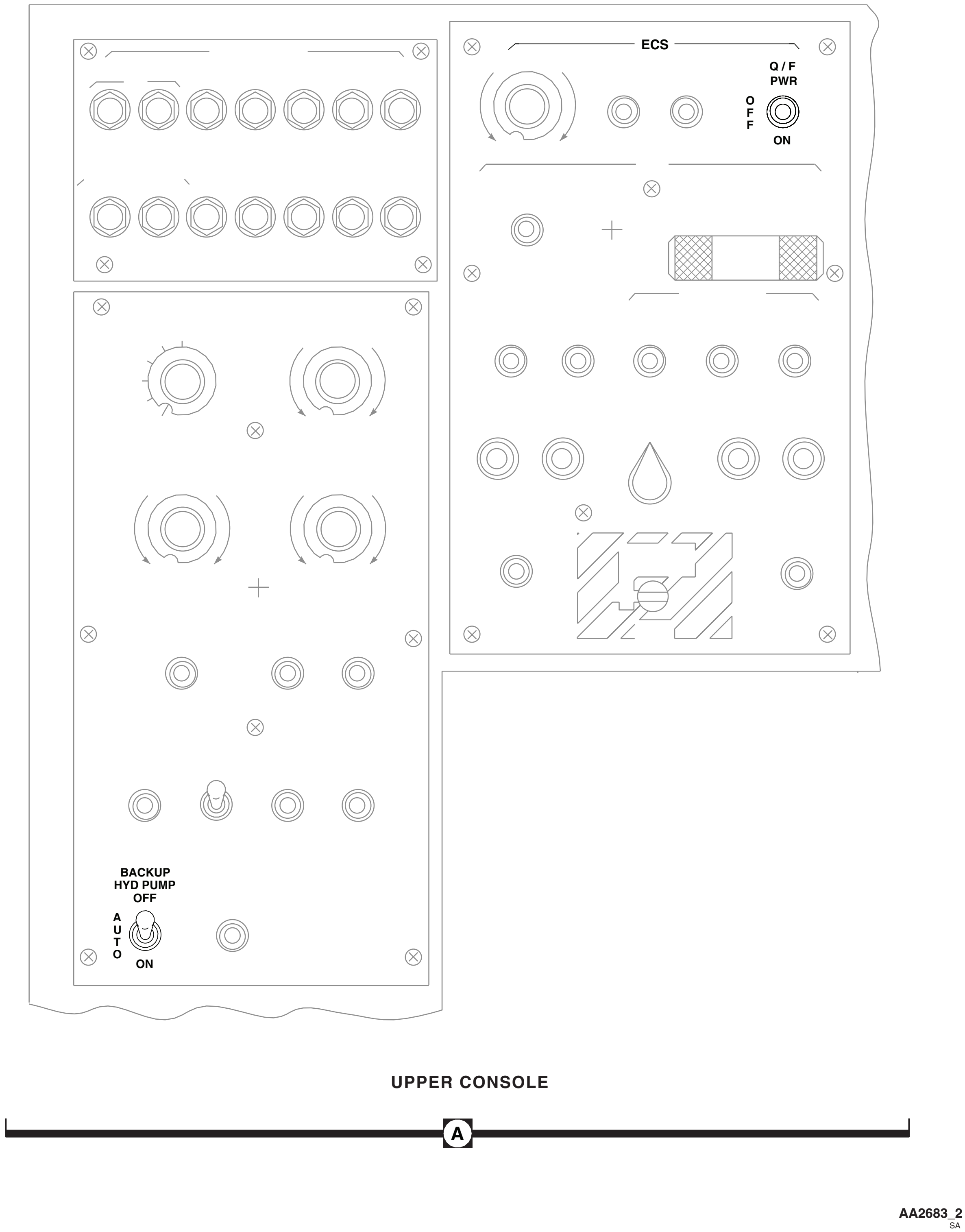
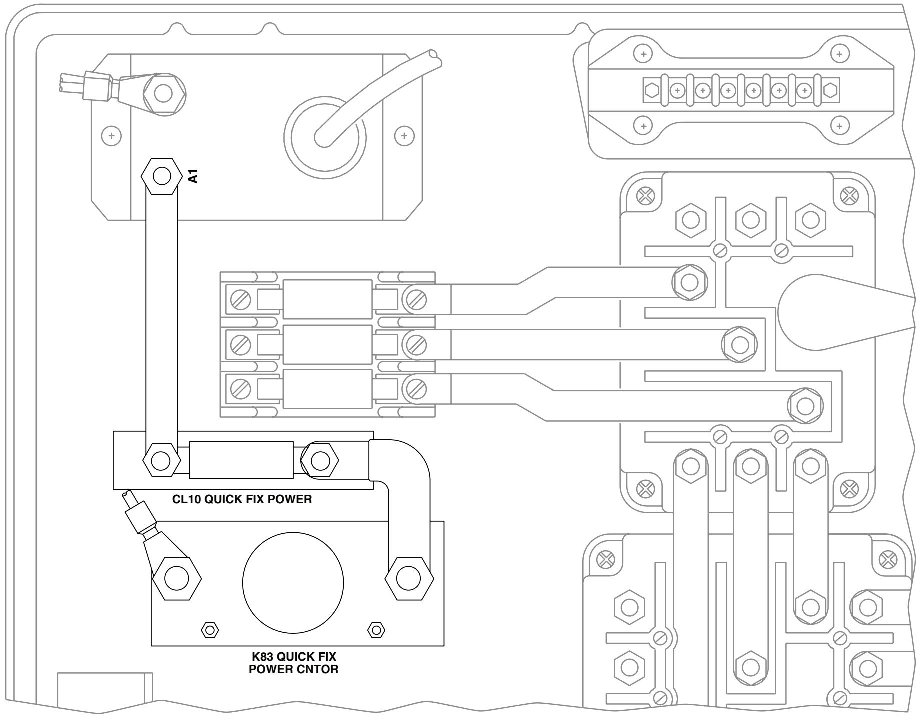
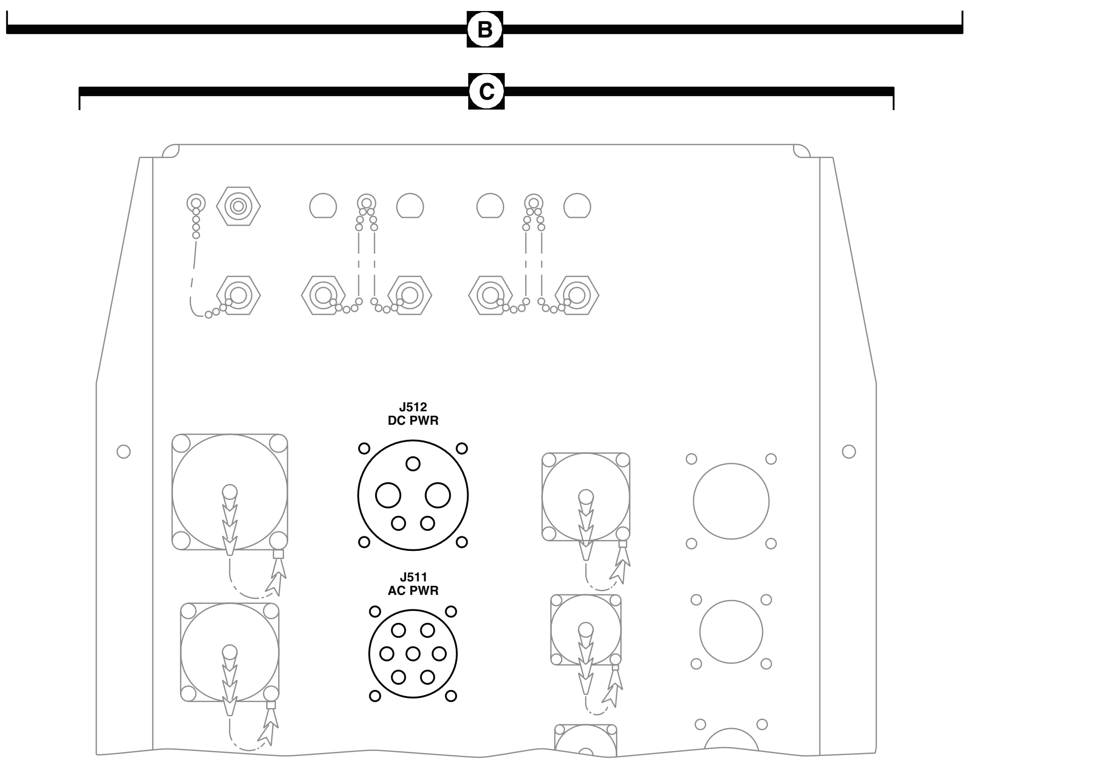


Figure 1. Mission Electrical Interface Location Diagram. (Sheet 2 of 4)

LOCATION - Continued



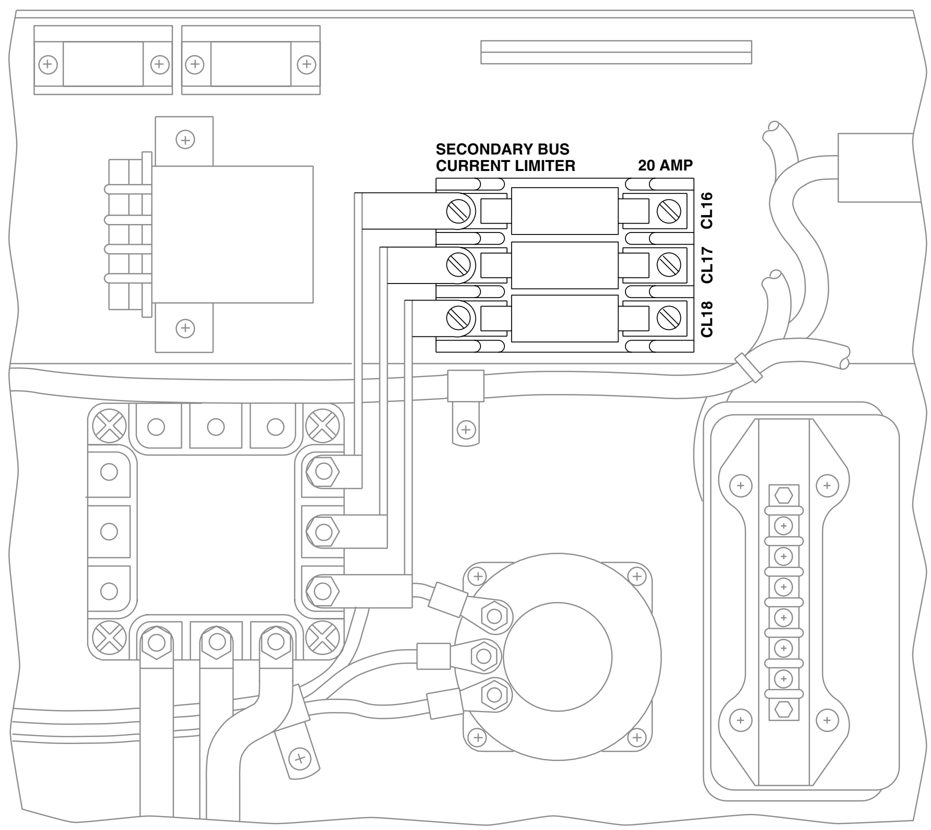
NO. 2 JUNCTION BOX



MISSION INTERFACE PANEL

Figure 1. Mission Electrical Interface Location Diagram. (Sheet 3 of 4)

LOCATION - Continued



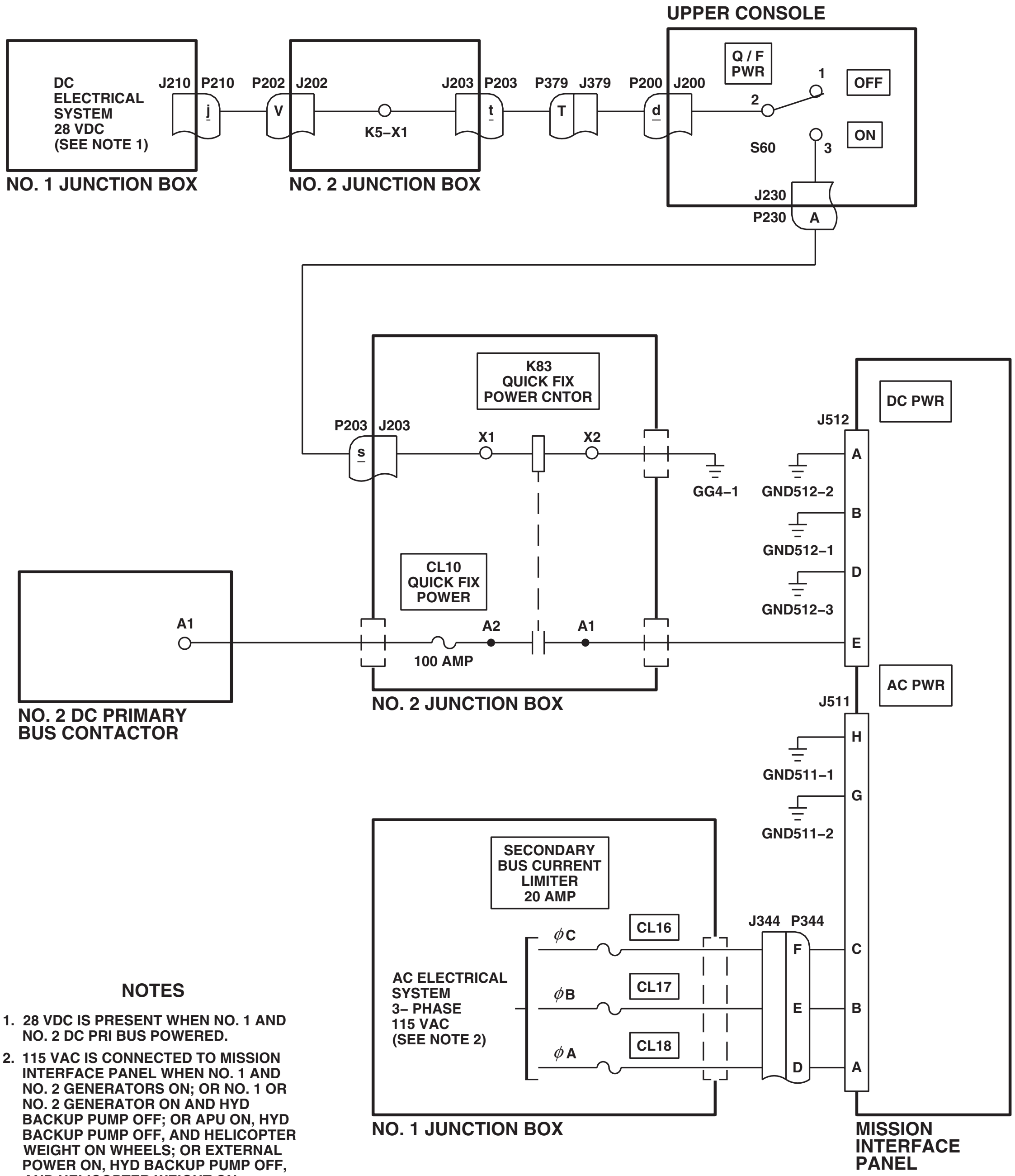
NO. 1 JUNCTION BOX

D

AA2683_4
SA

Figure 1. Mission Electrical Interface Location Diagram. (Sheet 4 of 4)

SCHEMATICS



NOTES

1. 28 VDC IS PRESENT WHEN NO. 1 AND NO. 2 DC PRI BUS POWERED.
2. 115 VAC IS CONNECTED TO MISSION INTERFACE PANEL WHEN NO. 1 AND NO. 2 GENERATORS ON; OR NO. 1 OR NO. 2 GENERATOR ON AND HYD BACKUP PUMP OFF; OR APU ON, HYD BACKUP PUMP OFF, AND HELICOPTER WEIGHT ON WHEELS; OR EXTERNAL POWER ON, HYD BACKUP PUMP OFF, AND HELICOPTER WEIGHT ON WHEELS; OR NO. 1 OR NO. 2 GENERATOR ON, APU ON, AND BLADE DEICE OFF.

Figure 2. Mission Electrical Interface Schematic Diagram.

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

SYSTEM CIRCUIT BREAKERS

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A
Multimeter, Simpson 260

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 1747 00

Equipment Condition
28 Vdc or 115 Vac (as applicable), Available at System Circuit Breaker

SYSTEM CIRCUIT BREAKERS OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

The following applies to any circuit breaker which pops out during system operation.

1. Disconnect electrical connectors from all system components and check for signs of overheating, bent connector pins, or improper connection. Leave connectors off.
2. Push in circuit breaker.

Indication/Condition

Reference Action.

Corrective Action

1. If circuit breaker stays in, component malfunction is indicated. Isolate malfunctioning component as follows:
   - Pull out circuit breaker.
   - Sequentially connect all components serviced by circuit breaker, one component at a time.
   - Push in circuit breaker after each connection. When circuit breaker pops out, last component connected is then known to be malfunctioning.
2. If circuit breaker pops, short circuit in helicopter wiring is indicated. Check for short circuit between helicopter ground and system wiring using multimeter. Repair/replace wiring as required (WP 1747 00).

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

INSTRUMENT PANEL LIGHTS

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
WP 0104 00
WP 0770 00
WP 0824 00
WP 0875 00
WP 0881 00
WP 1061 00
WP 1062 00
WP 1066 00
WP 1221 00
WP 1549 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

INSTRUMENT PANEL LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

To prevent damage to lighting, do not measure voltage or continuity across transformers T5, T6, and T7.

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See **UH60A UH60L** Figure 1 and **EH60A** Figure 2 for component location diagrams and Figure 3 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
LIGHTS SEC	Copilot’s
LIGHTS NON FLT	Pilot’s
LIGHTS PNL FLT	Pilot’s

INSTRUMENT PANEL LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
LIGHTS CPLT FLT	Copilot's

2. Turn on electrical power.
3. Turn CPLT FLT INST LTS control to OFF.
4. Turn INSTR LT PILOT FLT control to OFF.
5. Turn INSTR LT NON FLT control to BRT.

Indication/Condition

a. All nonflight instrument panel lights shall go on.

b. Pilot's and copilot's collective stick grip lights shall go on.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, go to NO NONFLIGHT INSTRUMENT PANEL LIGHTS GO ON, in this work package.
2. If only icing rate meter does not go on, go to ICING RATE METER LIGHT DOES NOT GO ON, in this work package.
3. If only auxiliary fuel management control panel lights do not go on, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL LIGHTS DO NOT GO ON, in this work package.
4. If Indication/Condition b. is not as specified, go to PILOT'S AND/OR COPILOT'S COLLECTIVE STICK GRIP LIGHTING DOES NOT GO ON, in this work package.
5. If lights on individual nonflight instruments do not go on, check continuity between P280-L and affected instrument.
   a. If continuity is present, replace instrument or lighted bezel assembly (WP 0770 00).
   b. If continuity is not present, repair/replace wiring (WP 1747 00).

Step

6. Turn INSTR LT NON FLT control to OFF.

Indication/Condition

All nonflight instrument panel lights shall dim and go off.

Corrective Action

If Indication/Condition is not as specified, replace INSTR LT NON FLT dimmer control T7 (WP 0875 00).

Step

7. Turn INSTR LT PILOT FLT control to BRT.

Indication/Condition

All pilot flight instrument panel lights shall go on.

INSTRUMENT PANEL LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition is not as specified, go to NO PILOT'S FLIGHT INSTRUMENT PANEL LIGHTS GO ON, in this work package.
2. If lights on pilot's stabilator position placard and pilot's HSI/VSI mode select panel do not go on, go to LIGHTS ON PILOT'S STABILATOR POSITION PLACARD AND PILOT'S HSI/VSI MODE SELECT PANEL DO NOT GO ON, in this work package.
3. If individual pilot flight instrument lights do not go on, check continuity between instrument panel J112 and affected instrument light.
   a. If continuity is present, replace lighted bezel assembly (WP 0770 00).
   b. If continuity is not present, repair/replace wiring (WP 1747 00).

Step

8. Turn INSTR LT PILOT FLT control to OFF.

Indication/Condition

All pilot flight instrument panel lights shall dim and go off.

Corrective Action

If Indication/Condition is not as specified, replace INSTR LT PILOT FLT dimmer control T6 (WP 0875 00).

Step

9. Turn CPLT FLT INST LTS control to BRT.

Indication/Condition

All copilot flight instrument panel lights shall go on.

Corrective Action

1. If Indication/Condition is not as specified, go to NO COPILOT'S FLIGHT INSTRUMENT PANEL LIGHTS GO ON, in this work package.
2. If lights on copilot's stabilator position placard and copilot's HSI/VSI mode select panel do not go on, go to LIGHTS ON COPILOT'S STABILATOR POSITION PLACARD AND COPILOT'S HSI/VSI MODE SELECT PANEL DO NOT GO ON, in this work package.
3. If lights on individual copilot flight instruments do not go on, check continuity between instrument panel J113 and affected instrument light.
   a. If continuity is present, replace lighted bezel assembly (WP 0770 00).
   b. If continuity is not present, repair/replace wiring (WP 1747 00).

Step

10. Turn CPLT FLT INST LTS control to OFF.

Indication/Condition

All copilot flight instrument panel lights shall dim and go off.

Corrective Action

If Indication/Condition is not as specified, replace CPLT FLT INST LTS dimmer control T5 (WP 0875 00).

INSTRUMENT PANEL LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

11. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

NO NONFLIGHT INSTRUMENT PANEL LIGHTS GO ON

SYMPTOM

No nonflight instrument panel lights go on.

MALFUNCTION

No nonflight instrument panel lights go on.

CORRECTIVE ACTION

1. Do all nonflight instrument lights fail to go on?

a. If nonflight instrument lights fail to go on, go to **2.**

b. If nonflight instrument lights do not fail to go on, go to **5.**

2. Check continuity between T6-G and GG2-4.

a. If continuity is present, go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13.**

3. Check for 115 vac between T7-HV and T7-G.

a. If voltage is as specified, replace T7 (WP 0875 00). Go to **13.**

b. If voltage is not as specified, go to **4.**

4. Check for 115 vac between LIGHTS NON FLT circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T7-HV (WP 1747 00). Go to **13.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **13.**

5. Do the nonflight instrument panel lights go on?

a. If nonflight instrument panel lights go on, go to **6.**

b. If nonflight instrument panel lights do not go on, go to **12.**

6. Do pilot's or copilot's collective stick grip lights go on?

a. If collective stick grip lights go on, go to **7.**

b. If collective stick grip lights do not go on, repair/replace wiring, as required, between (WP 1747 00), then go to **13:**

- J103-T and T7-LV
- J104-T and T7-LV

NO NONFLIGHT INSTRUMENT PANEL LIGHTS GO ON - Continued

7. **Do pilot's collective stick grip lights go on?**
   a. If lights go on, go to **8.**
   b. If lights do not go on, go to **10.**
8. **Check continuity between J103-V and ground.**
   a. If continuity is present, go to **9.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13.**
9. **Check continuity between J103-T and J104-T.**
   a. If continuity is present, replace copilot's collective stick (WP 1062 00). Go to **13.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13.**
10. **Check continuity between J104-V and ground.**
    a. If continuity is present, go to **11.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13.**
11. **Check continuity between J103-T and J104-T.**
    a. If continuity is present, replace pilot's collective stick (WP 1061 00). Go to **13.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13.**
12. **Check continuity between P133-A and P388-C.**
    a. If continuity is present, repair/replace wiring between T7-LV and P388-C (WP 1747 00). Go to **13.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13.**
13. **Procedure completed.**

ICING RATE METER LIGHT DOES NOT GO ON

SYMPTOM
Icing rate meter light does not go on.

MALFUNCTION
Icing rate meter light does not go on.

CORRECTIVE ACTION

1. **Check for 115 vac between P388-C and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, repair/replace wiring between P280-L and P388-C (WP 1747 00). Go to **4.**
2. **Check for 5 vac between P132-M and ground.**
   a. If voltage is as specified, replace icing rate meter (WP 1221 00). Go to **4.**
   b. If voltage is not as specified, go to **3.**
3. **Check continuity between P132-M and P388-A.**

ICING RATE METER LIGHT DOES NOT GO ON - Continued

a. If continuity is present, replace nonflight instrument lights 5v/115v transformer (WP 0881 00). Go to **4.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**

4. Procedure completed.

PILOT'S AND/OR COPILOT'S COLLECTIVE STICK GRIP LIGHTING DOES NOT GO ON

SYMPTOM

Pilot's and/or copilot's collective stick grip lighting does not go on.

MALFUNCTION

Pilot's and/or copilot's collective stick grip lighting does not go on.

CORRECTIVE ACTION

1. Does either pilot's or copilot's collective stick grip lighting go on?

a. If lighting goes on, go to **2.**

b. If lighting does not go on, go to **5.**

2. Does pilot's collective stick grip lighting go on?

a. If lighting goes on, go to **3.**

b. If lighting does not go on, go to **4.**

3. Check for 115 vac between J103-T and J103-V.

a. If voltage is as specified, replace copilot's collective stick grip information plate (WP 1066 00). Go to **6.**

b. If voltage is not as specified, repair/replace wiring between P231-U and J103-T (WP 1747 00). Go to **6.**

4. Check for 115 vac between J104-T and J104-V.

a. If voltage is as specified, replace pilot's collective stick grip information plate (WP 1066 00). Go to **6.**

b. If voltage is not as specified, repair/replace wiring between P231-U and J104-T (WP 1747 00). Go to **6.**

5. Check for 115 vac between J231-U and ground.

a. If voltage is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **6:**

- P231-U and J103-T
- P231-U and J104-T

b. If voltage is not as specified, repair/replace wiring between P231-U and T7-LV (WP 1747 00). Go to **6.**

6. Procedure completed.

NO PILOT'S FLIGHT INSTRUMENT PANEL LIGHTS GO ON

SYMPTOM

No pilot's flight instrument panel lights go on.

MALFUNCTION

No pilot's flight instrument panel lights go on.

CORRECTIVE ACTION

1. Do all pilot flight instrument lights fail to go on?

a. If lights fail to go on, go to **2.**

b. If lights do not fail to go on, go to **5.**

2. Check continuity between T6-G and GG2-4.

a. If continuity is present, go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

3. Check for 115 vac between T6-HV and T6-G.

a. If voltage is as specified, replace T6 (WP 0875 00). Go to **9.**

b. If voltage is not as specified, go to **4.**

4. Check for 115 vac between LIGHTS PLT FLT circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T6-HV (WP 1747 00). Go to **9.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9.**

5. Do pilot HSI/VSI mode select panel lights go on?

a. If lights go on, repair/replace wiring, as required, between (WP 1747 00), then go to **9:**

- T6-LV and P187-A
- T6-LV and P181-A

b. If lights do not go on, go to **6.**

6. Check continuity between P317R-36 and P386-C.

a. If continuity is present, go to **7.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

7. Check continuity between:

- **P386-B and ground**
- **P386-D and ground**
- **P386-E and ground**

a. If continuity is present, go to **8.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **9.**

8. Check continuity between T6-LV and P386-A.

NO PILOT'S FLIGHT INSTRUMENT PANEL LIGHTS GO ON - Continued

a. If continuity is present, replace pilot's flight instrument lights 5v/115v transformer (WP 0881 00). Go to **9.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

9. Procedure completed.

LIGHTS ON PILOT'S STABILATOR POSITION PLACARD AND PILOT'S HSI/VSI MODE SELECT PANEL DO NOT GO ON

SYMPTOM

Lights on pilot's stabilator position placard and pilot's HSI/VSI mode select panel do not go on.

MALFUNCTION

Lights on pilot's stabilator position placard and pilot's HSI/VSI mode select panel do not go on.

CORRECTIVE ACTION

1. Check for 5 vac between J246-a and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, repair/replace wiring between J246-a and T6-LV (WP 1747 00). Go to **4.**

2. Check for 5 vac between P386-A and P386-B.

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **4:**

- P386-A and P246-a
- P386-B and GND386-1

3. Check for 115 vac between P137-FF and ground.

a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **4.**

b. If voltage is not as specified, replace pilot's flight instrument lights 5v/115v transformer (WP 0881 00). Go to **4.**

4. Procedure completed.

NO COPILOT'S FLIGHT INSTRUMENT PANEL LIGHTS GO ON

SYMPTOM

No copilot's flight instrument panel lights go on.

MALFUNCTION

No copilot's flight instrument panel lights go on.

CORRECTIVE ACTION

1. Do all copilot flight instrument lights fail to go on?

a. If lights fail to go on, go to **2.**

b. If lights do not fail to go on, go to **5.**

NO COPILOT'S FLIGHT INSTRUMENT PANEL LIGHTS GO ON - Continued

2. **Check continuity between T5-G and GG2-4.**

   a. If continuity is present, go to **3.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

3. **Check for 115 vac between T5-HV and T5-G.**

   a. If voltage is as specified, replace T5 (WP 0875 00). Go to **9.**

   b. If voltage is not as specified, go to **4.**

4. **Check for 115 vac between LIGHTS CPLT FLT circuit breaker terminal 1 and ground.**

   a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T5-HV (WP 1747 00). Go to **9.**

   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9.**

5. **Do pilot HSI/VSI mode select panel lights go on?**

   a. If lights go on, repair/replace wiring, as required, between (WP 1747 00), then go to **9:**

      - T5-LV and P189-A
      - T5-LV and P191-A

   b. If lights do not go on, go to **6.**

6. **Check continuity between P300R-36 and P387-C.**

   a. If continuity is present, go to **7.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

7. **Check continuity between:**

   - **P387-B and ground**
   - **P387-D and ground**
   - **P387-E and ground**

   a. If continuity is present, go to **8.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

8. **Check continuity between T5-LV and P387-A.**

   a. If continuity is present, replace copilot's flight instruments lights 5v/115v transformer (WP 0881 00). Go to **9.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

9. **Procedure completed.**

LIGHTS ON COPILOT'S STABILATOR POSITION PLACARD AND COPILOT'S HSI/VSI MODE SELECT PANEL DO NOT GO ON

SYMPTOM

Lights on copilot's stabilator position placard and copilot's HSI/VSI mode select panel do not go on.

MALFUNCTION

Lights on copilot's stabilator position placard and copilot's HSI/VSI mode select panel do not go on.

CORRECTIVE ACTION

1. **Check for 5 vac between J246-g and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, repair/replace wiring between J246-g and T5-LV (WP 1747 00). Go to **4.**
2. **Check for 5 vac between P387-A and ground.**
   a. If voltage is as specified, go to **3.**
   b. If voltage is not as specified, repair/replace wiring between P387-A and P246-g (WP 1747 00). Go to **4.**
3. **Check for 115 vac between P138-FF and ground.**
   a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **4.**
   b. If voltage is not as specified, replace copilot's flight instrument lights 5v/115v transformer (WP 0881 00). Go to **4.**
4. **Procedure completed.**

AUXILIARY FUEL MANAGEMENT CONTROL PANEL LIGHTS DO NOT GO ON

SYMPTOM

Auxiliary fuel management control panel lights do not go on.

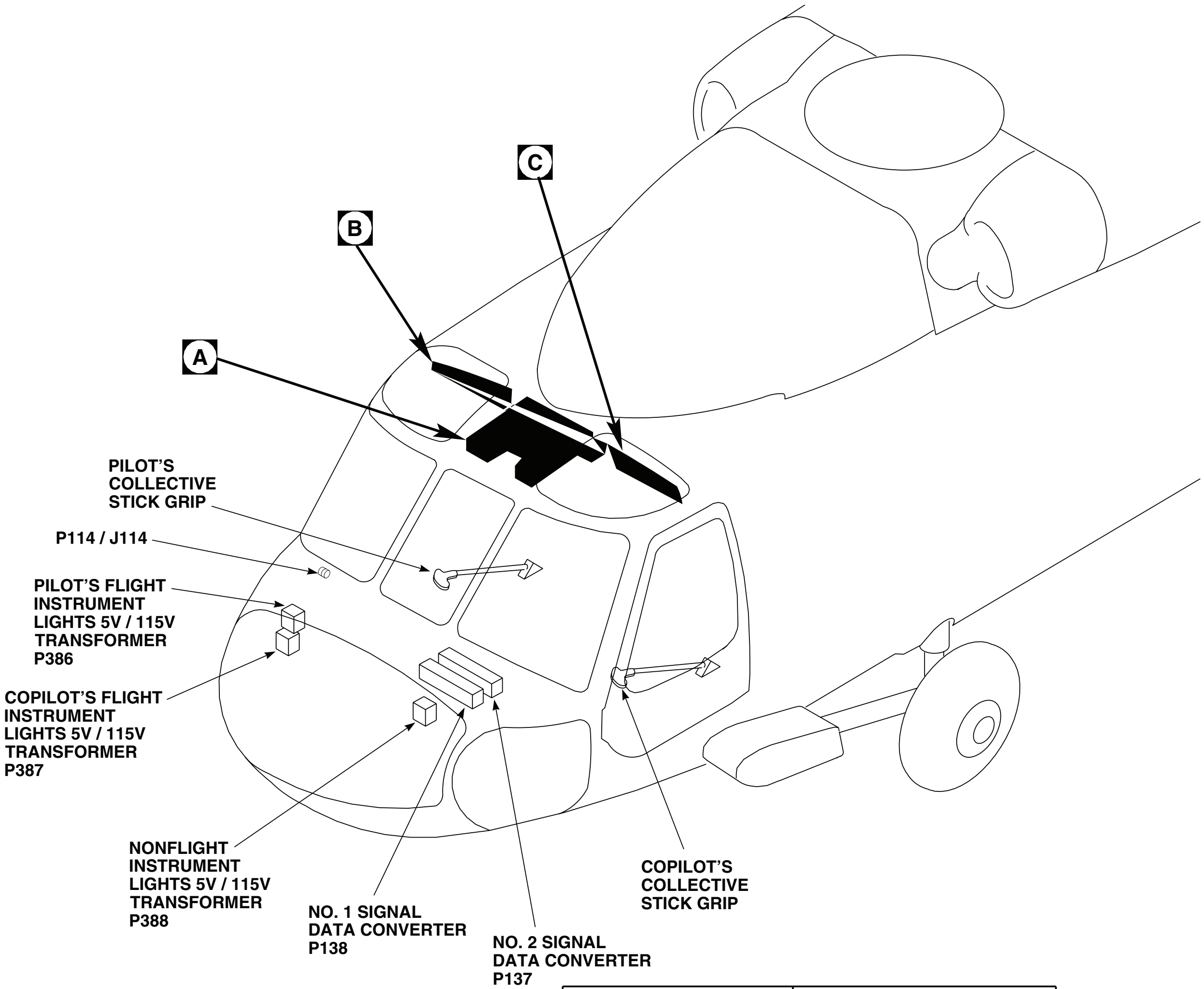
MALFUNCTION

Auxiliary fuel management control panel lights do not go on.

CORRECTIVE ACTION

1. **Check for 5 vac between P857-36 and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, repair/replace wiring between P857-36 and ground (WP 1747 00). Go to **3.**
2. **Check continuity between P857-36 and P388-A.**
   a. If continuity is present, replace auxiliary fuel management control panel (WP 1549 00). Go to **3.**
   b. If continuity is not present, repair/replace wiring between P857-36 and P388-A (WP 1747 00). Go to **3.**
3. **Procedure completed.**

LOCATION



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P103 / J103	COCKPIT TUB, BL 32 LH, STA 247
P104 / J104	COCKPIT TUB, BL 15 RH, STA 247
P112 / J112	COCKPIT, BL 6 RH, STA 197
P113 / J113	COCKPIT, BL 6 LH, STA 197
P114 / J114	INSTRUMENT PANEL LIGHTS
P118 / J118	CAUTION / ADVISORY PANEL
P126 / J126	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 28 RH, STA 243
P127 / J127	BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 28 LH, STA 243
P132	ICING RATE METER
P133	BLADE DE-ICE CONTROL PANEL

EFFECTIVITY

UH60A UH60L

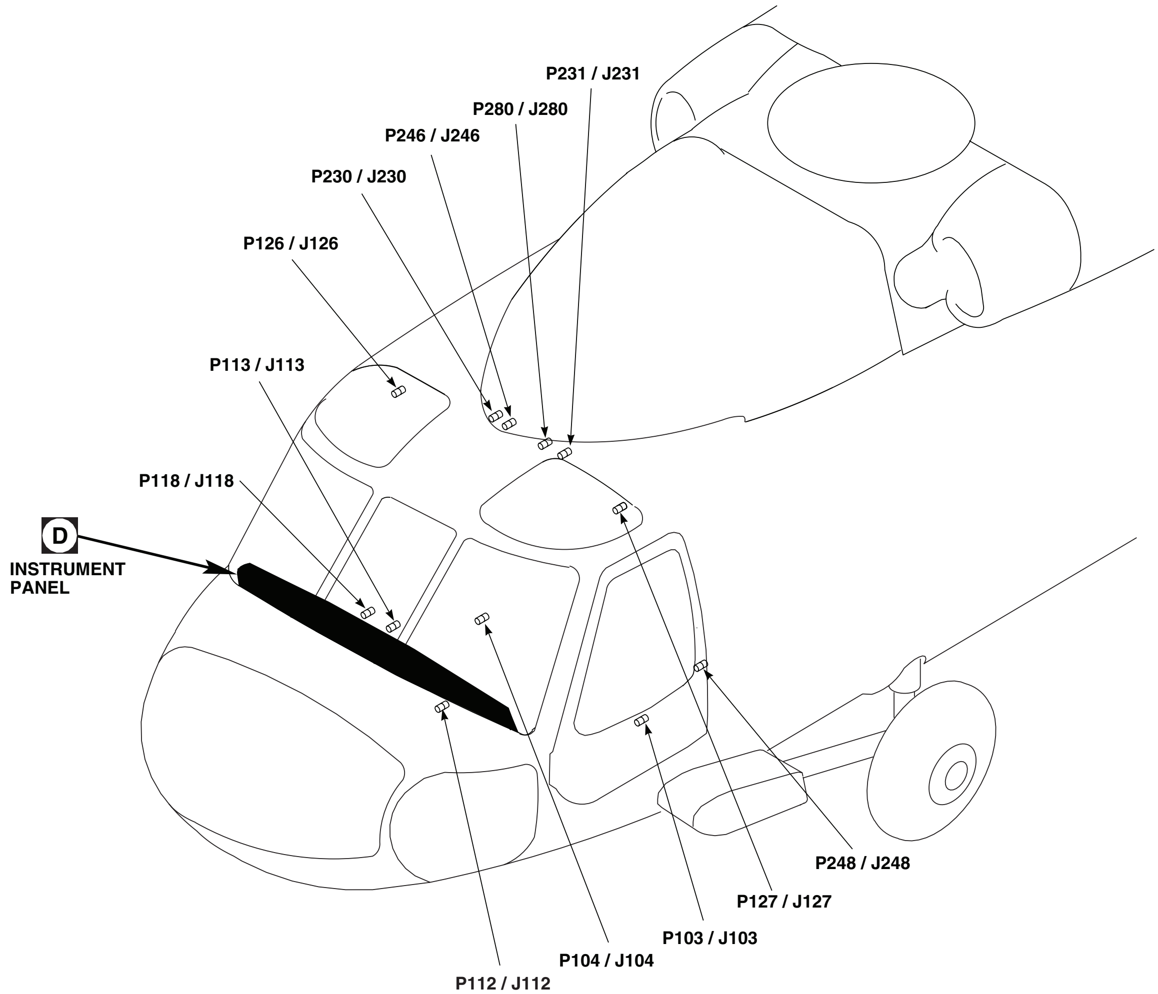
NOTE

MWO 50-78

AA2612_1B
SA

Figure 1. Instrument Panel Lights Location Diagram UH60A UH60L . (Sheet 1 of 6)

LOCATION - Continued



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P134	BLADE DE–ICE TEST PANEL
P137	NO. 2 SIGNAL DATA CONVERTER
P138	NO. 1 SIGNAL DATA CONVERTER
P181	PILOT'S VERTICAL SPEED INDICATOR LIGHTED BEZEL ASSEMBLY
P182	PILOT'S RADAR ALTIMETER LIGHTED BEZEL ASSEMBLY
P183	PILOT'S BAROMETRIC ALTIMETER LIGHTED BEZEL ASSEMBLY
P184	PILOT'S VERTICAL SITUATION INDICATOR LIGHTED BEZEL ASSEMBLY

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P185	PILOT'S HORIZONTAL SITUATION INDICATOR LIGHTED BEZEL ASSEMBLY
P186	PILOT'S CLOCK LIGHTED BEZEL ASSEMBLY
P187	PILOT'S AIRSPEED INDICATOR LIGHTED BEZEL ASSEMBLY
P188	PILOT'S STABILATOR INDICATOR LIGHTED BEZEL ASSEMBLY
P189	COPILOT'S AIRSPEED INDICATOR LIGHTED BEZEL ASSEMBLY
P190	COPILOT'S VERTICAL SITUATION INDICATOR LIGHTED BEZEL ASSEMBLY

AA2612_2
SA

Figure 1. Instrument Panel Lights Location Diagram UH60A UH60L . (Sheet 2 of 6)

LOCATION - Continued

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P191	COPILOT'S HORIZONTAL SITUATION INDICATOR LIGHTED BEZEL ASSEMBLY
P192	COPILOT'S RADAR ALTIMETER LIGHTED BEZEL ASSEMBLY
P193	COPILOT'S BAROMETRIC ALTIMETER LIGHTED BEZEL ASSEMBLY
P194	COPILOT'S VERTICAL SPPED INDICATOR LIGHTED BEZEL ASSEMBLY
P195	COPILOT'S CLOCK LIGHTED BEZEL ASSEMBLY
P196	COPILOT'S STABILATOR INDICATOR LIGHTED BEZEL ASSEMBLY
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P231 / J231	CABIN CEILING, BL 9 LH, STA 247
P246 / J246	CABIN CEILING, BL 5 RH, STA 247
P248 / J248	CABIN, BL 40 LH, STA 248
P280 / J280	CABIN CEILING, BL 6 LH, STA 247
P300R	COPILOT'S HSI / VSI MODE SELECT PANEL
P317R	PILOT'S HSI / VSI MODE SELECT PANEL
P346 / J346	DISCONNECT
P386	PILOT'S FLIGHT INSTRUMENT LIGHTS 5V / 115V TRANSFORMER
P387	COPILOT'S FLIGHT INSTRUMENT LIGHTS 5V / 115V TRANSFORMER
P388	NONFLIGHT INSTRUMENT LIGHTS 5V / 115V TRANSFORMER
P419R / J419R	DISCONNECT
P857 / J857 (SEE NOTE)	AUXILIARY FUEL MANAGEMENT PANEL

AA2612_3A
SA

Figure 1. Instrument Panel Lights Location Diagram **UH60A UH60L** . (Sheet 3 of 6)

LOCATION - Continued

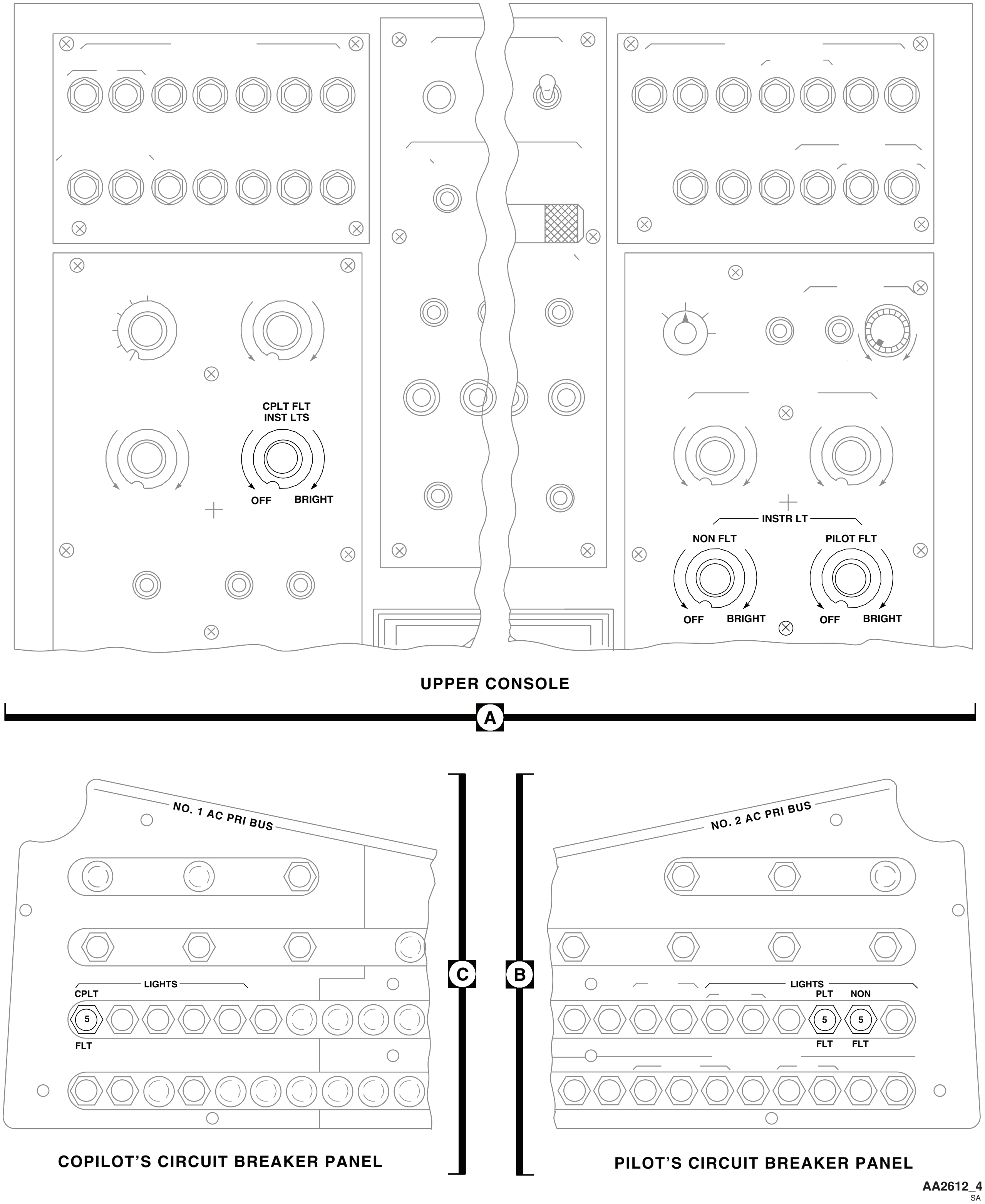


Figure 1. Instrument Panel Lights Location Diagram UH60A UH60L . (Sheet 4 of 6)

LOCATION - Continued

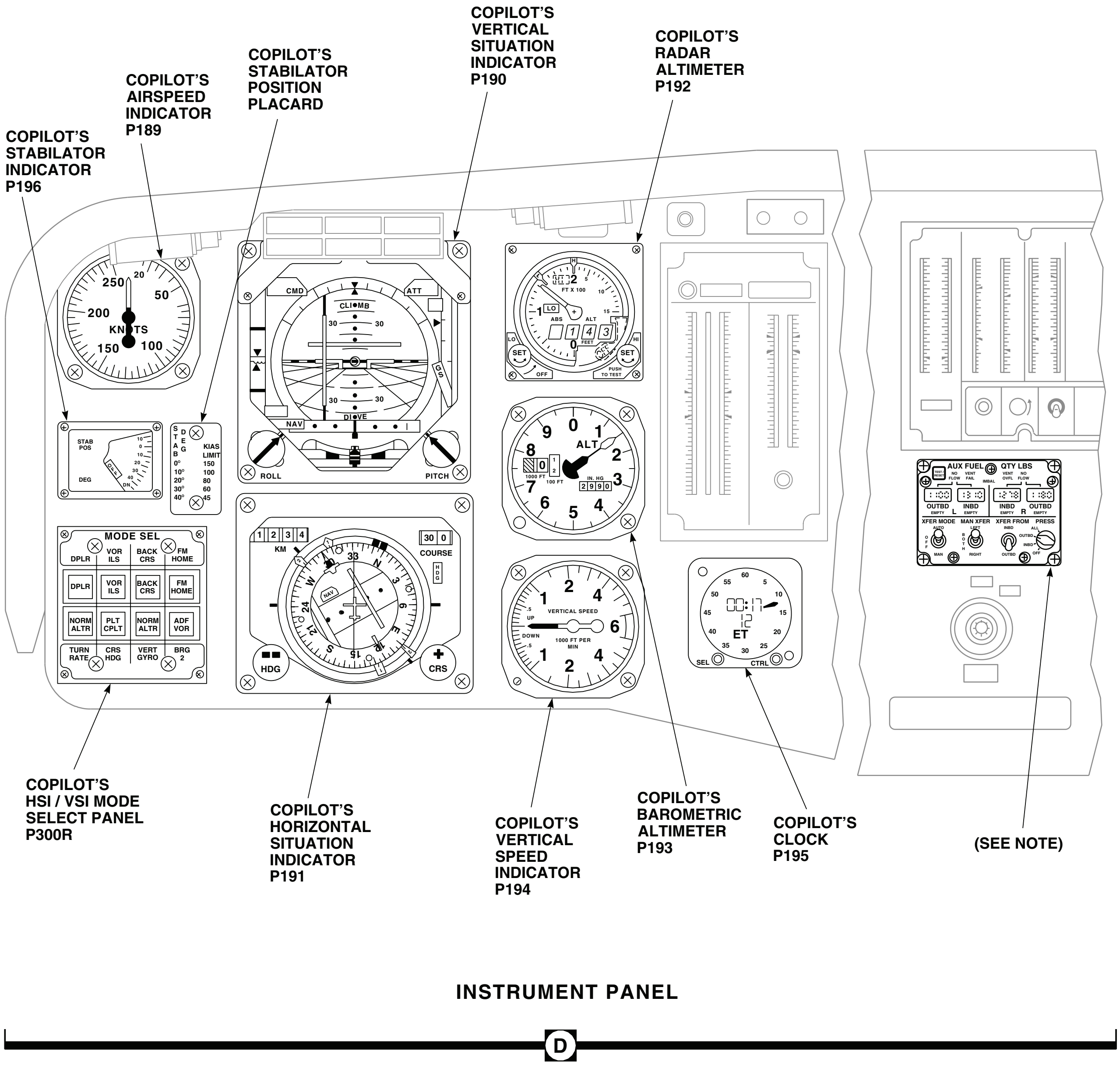


Figure 1. Instrument Panel Lights Location Diagram UH60A UH60L . (Sheet 5 of 6)

LOCATION - Continued

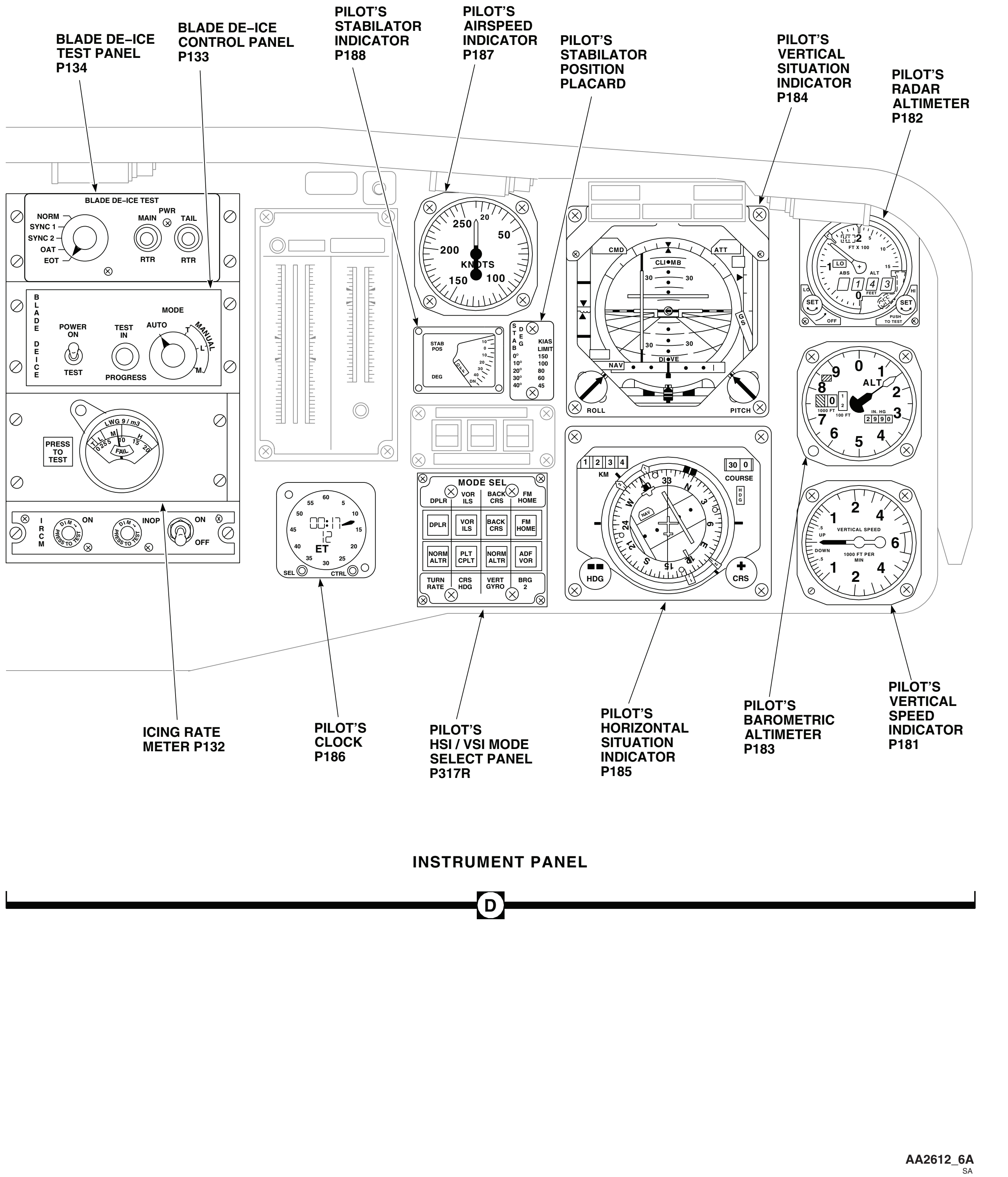
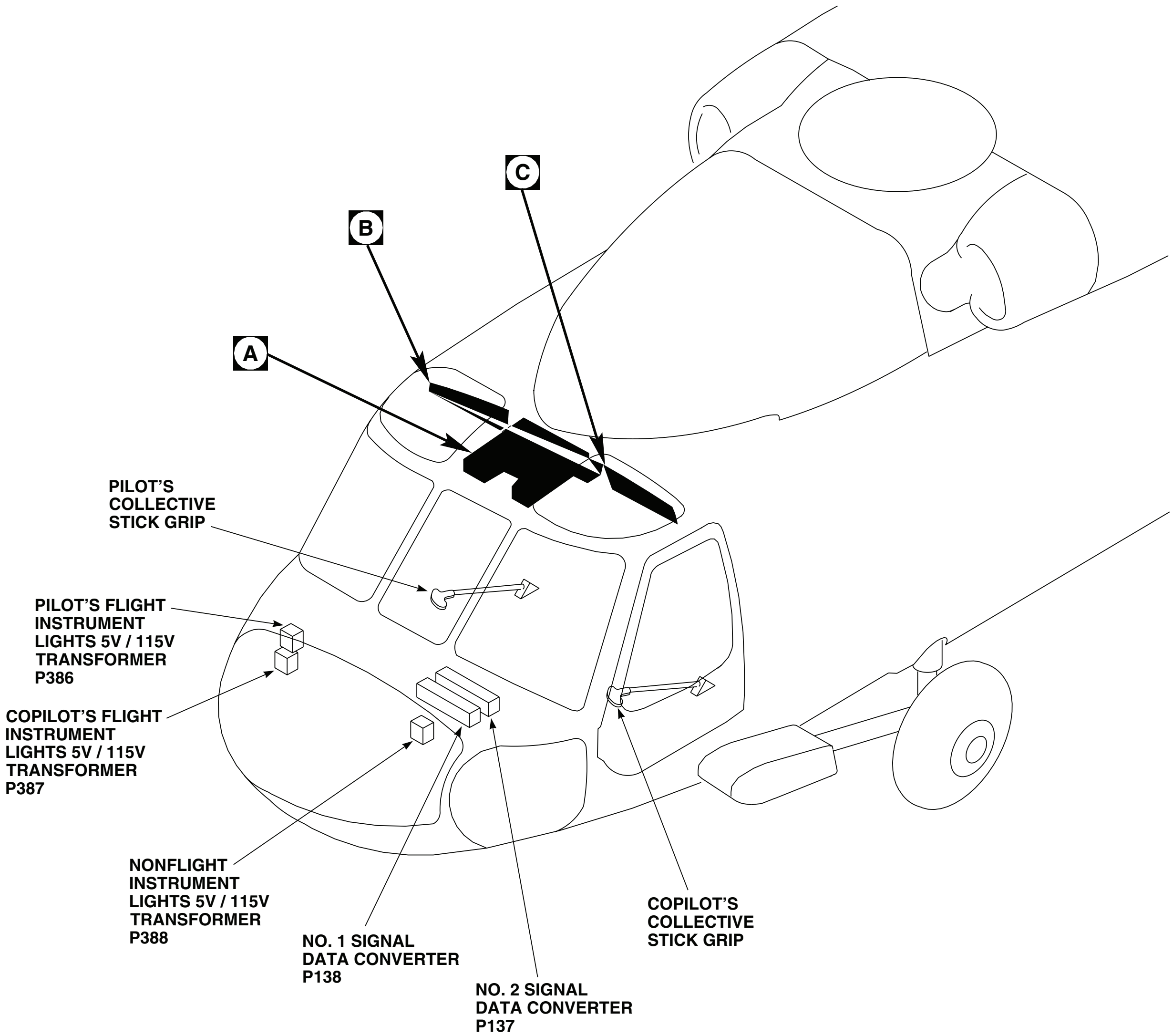


Figure 1. Instrument Panel Lights Location Diagram UH60A UH60L . (Sheet 6 of 6)

LOCATION - Continued



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P1R	COCKPIT, BL 6 LH, STA 210
P2	ALQ-156 CONTROL INDICATOR PANEL
P103 / J103	COCKPIT TUB, BL 32 LH, STA 247
P104 / J104	COCKPIT TUB, BL 15 RH, STA 247
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P112 / J112	COCKPIT, BL 6 RH, STA 197
P113 / J113	COCKPIT, BL 6 LH, STA 197
P118 / J118	CAUTION / ADVISORY PANEL

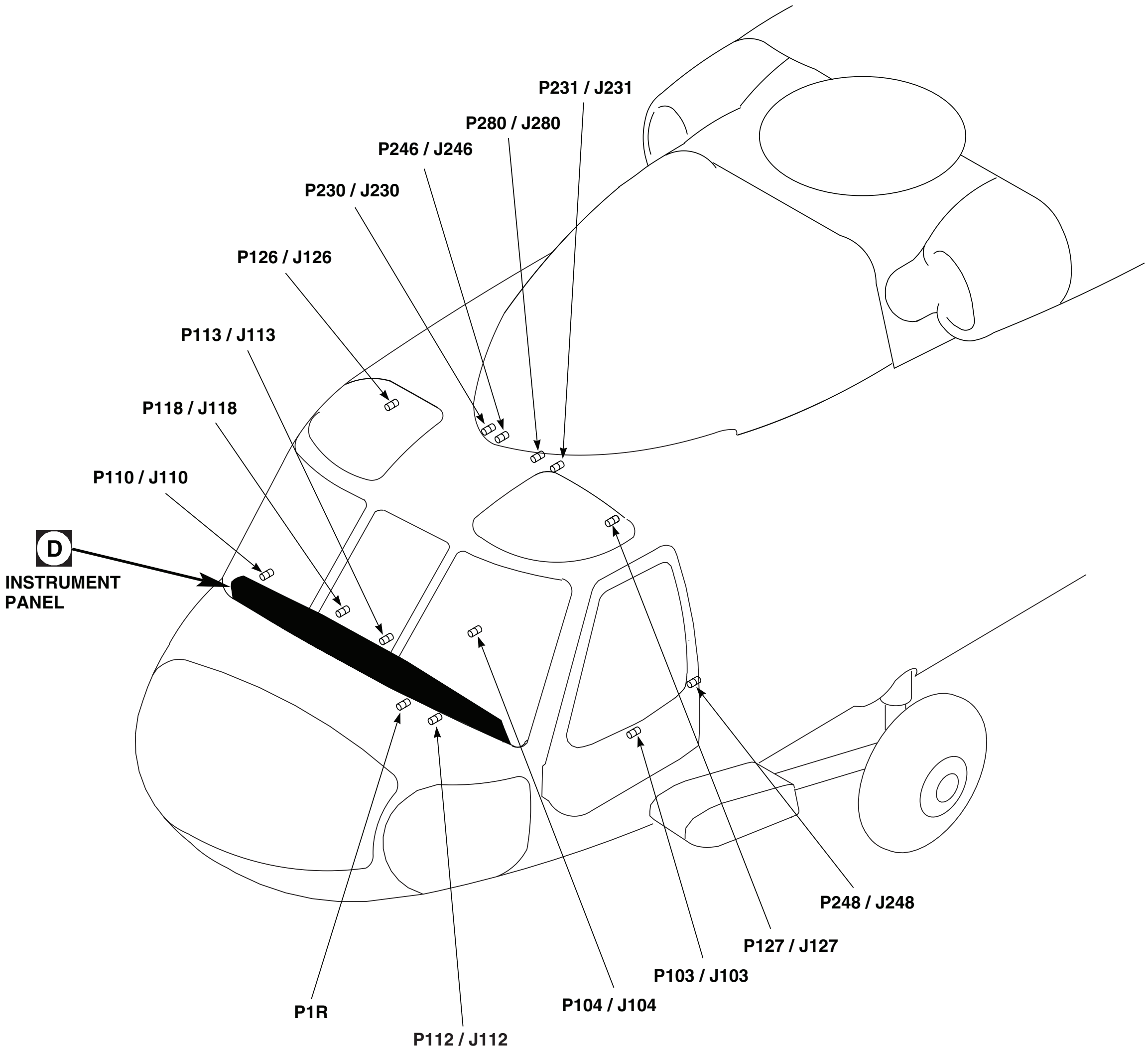
EFFECTIVITY

EH60A

AA2613_1
SA

Figure 2. Instrument Panel Lights Location Diagram EH60A . (Sheet 1 of 6)

LOCATION - Continued



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P126 / J126	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 28 RH, STA 243
P127 / J127	BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 28 LH, STA 243
P132	ICING RATE METER
P133	BLADE DE–ICE CONTROL PANEL
P134	BLADE DE–ICE TEST PANEL
P137	NO. 2 SIGNAL DATA CONVERTER

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P138	NO. 1 SIGNAL DATA CONVERTER
P181	PILOT'S VERTICAL SPEED INDICATOR LIGHTED BEZEL ASSEMBLY
P182	PILOT'S RADAR ALTIMETER LIGHTED BEZEL ASSEMBLY
P183	PILOT'S BAROMETRIC ALTIMETER LIGHTED BEZEL ASSEMBLY
P184	PILOT'S VERTICAL SITUATION INDICATOR LIGHTED BEZEL ASSEMBLY

AA2613_2
SA

Figure 2. Instrument Panel Lights Location Diagram EH60A . (Sheet 2 of 6)

LOCATION - Continued

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P185	PILOT'S HORIZONTAL SITUATION INDICATOR LIGHTED BEZEL ASSEMBLY
P186	PILOT'S CLOCK LIGHTED BEZEL ASSEMBLY
P187	PILOT'S AIRSPEED INDICATOR LIGHTED BEZEL ASSEMBLY
P188	PILOT'S STABILATOR INDICATOR LIGHTED BEZEL ASSEMBLY
P189	COPILOT'S AIRSPEED INDICATOR LIGHTED BEZEL ASSEMBLY
P190	COPILOT'S VERTICAL SITUATION INDICATOR LIGHTED BEZEL ASSEMBLY
P191	COPILOT'S HORIZONTAL SITUATION INDICATOR LIGHTED BEZEL ASSEMBLY
P192	COPILOT'S RADAR ALTIMETER LIGHTED BEZEL ASSEMBLY
P193	COPILOT'S BAROMETRIC ALTIMETER LIGHTED BEZEL ASSEMBLY
P194	COPILOT'S VERTICAL SPEED INDICATOR LIGHTED BEZEL ASSEMBLY
P195	COPILOT'S CLOCK LIGHTED BEZEL ASSEMBLY

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P196	COPILOT'S STABILATOR INDICATOR LIGHTED BEZEL ASSEMBLY
P197	COPILOT'S BEARING DISTANCE HEADING INDICATOR (BDHI)
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P231 / J231	CABIN CEILING, BL 9 LH, STA 247
P246 / J246	CABIN CEILING, BL 5 RH, STA 247
P248 / J248	CABIN, BL 40 LH, STA 248
P280 / J280	CABIN CEILING, BL 6 RH, STA 247
P300R	COPILOT'S VSI / HSI MODE SELECT PANEL
P317R	PILOT'S HSI / VSI MODE SELECT PANEL
P386	PILOT'S FLIGHT INSTRUMENT LIGHTS 5V / 115 V TRANSFORMER
P387	COPILOT'S FLIGHT INSTRUMENT LIGHTS 5V /115V TRANSFORMER
P388	NONFLIGHT INSTRUMENT LIGHTS 5V / 115V TRANSFORMER
P935R	SYSTEM SELECT PANEL

AA2613_3
SA

Figure 2. Instrument Panel Lights Location Diagram EH60A . (Sheet 3 of 6)

LOCATION - Continued

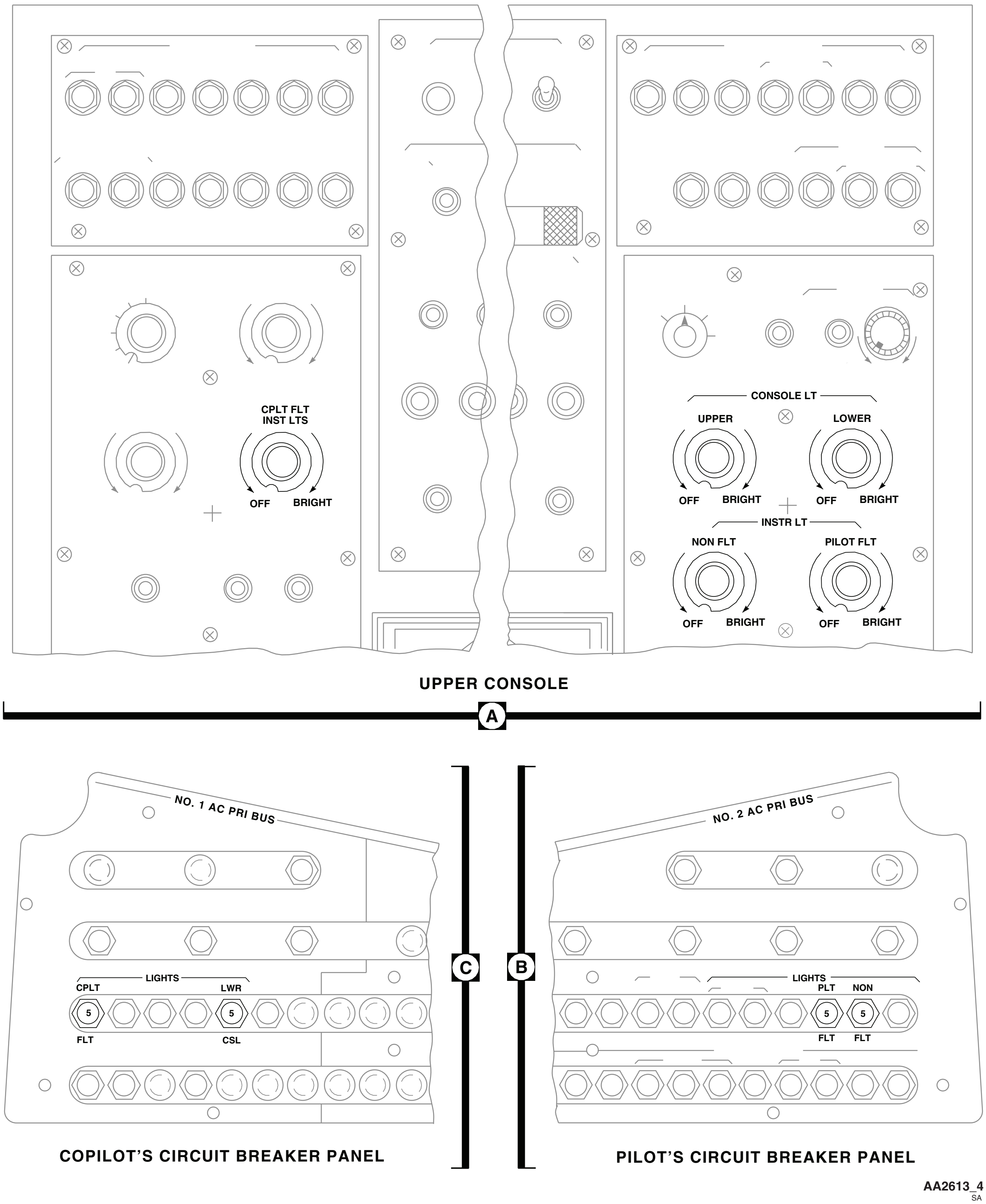


Figure 2. Instrument Panel Lights Location Diagram **EH60A** . (Sheet 4 of 6)

LOCATION - Continued

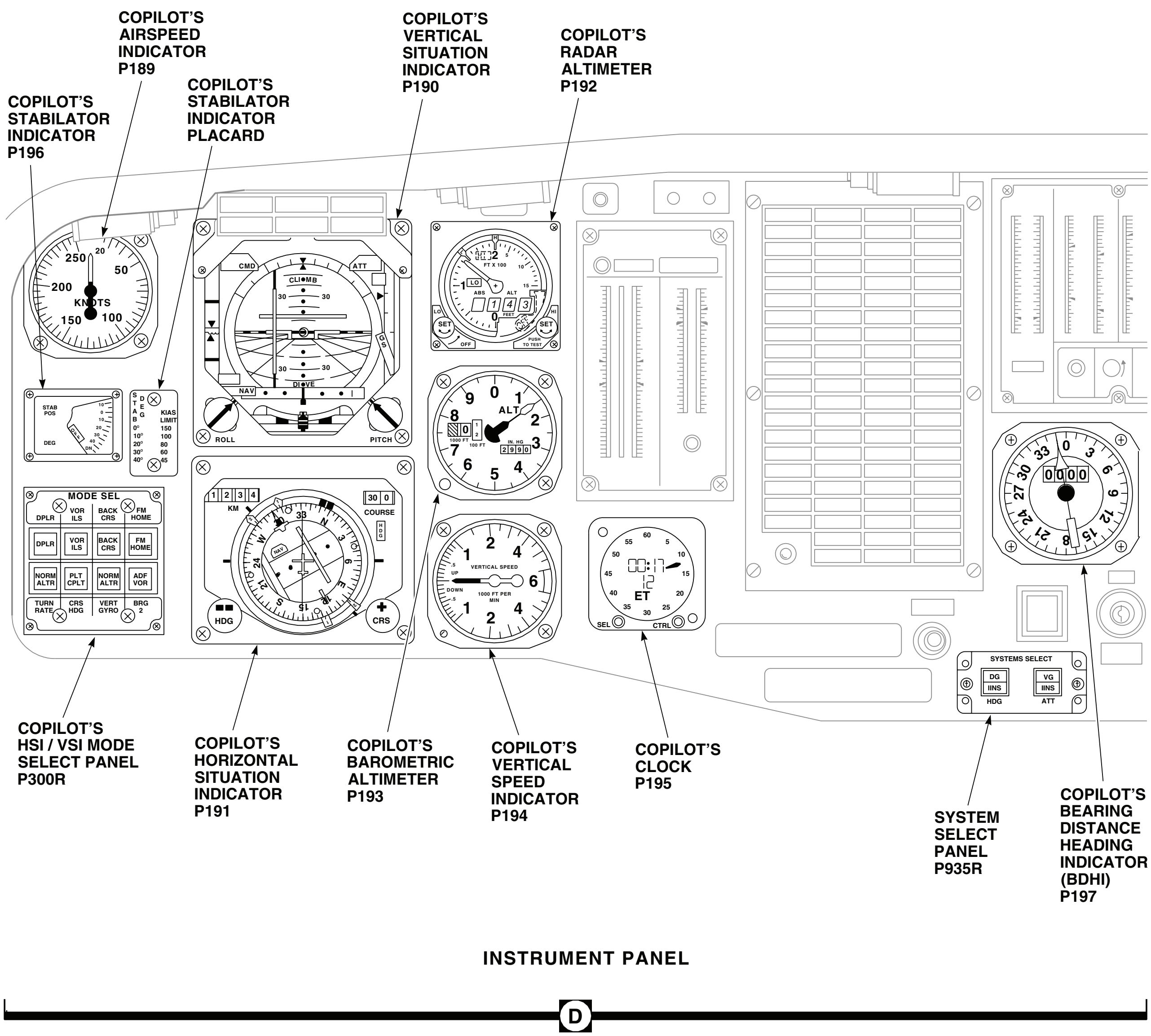


Figure 2. Instrument Panel Lights Location Diagram EH60A . (Sheet 5 of 6)

LOCATION - Continued

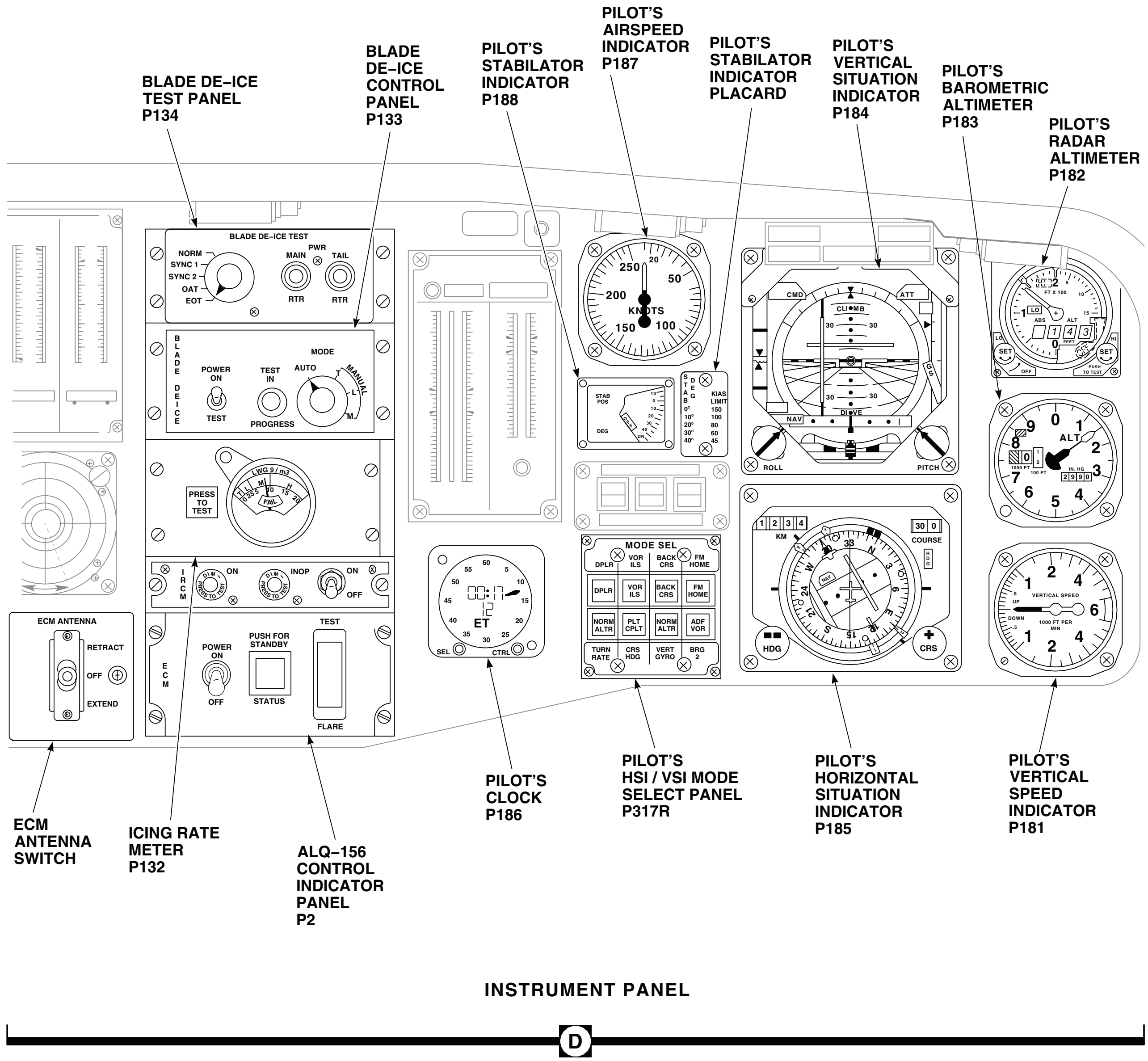
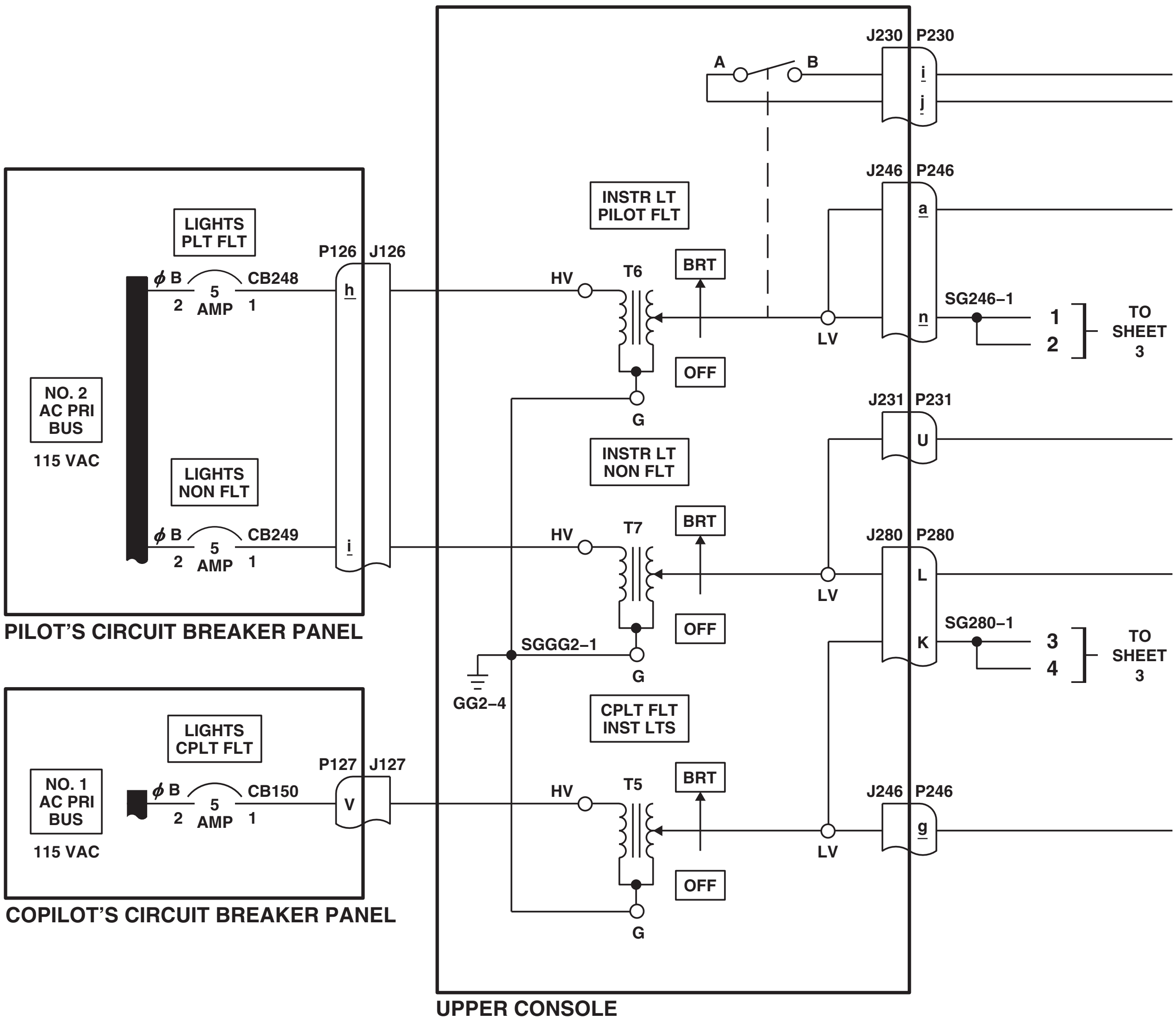


Figure 2. Instrument Panel Lights Location Diagram EH60A . (Sheet 6 of 6)

SCHEMATICS



NOTES

1. EH60A
2. W/O EMEP PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.
3. HELICOPTERS WITH DIGITAL CLOCK.
4. HELICOPTERS WITH ANALOG CLOCK.
5. UH60L 96–26723 – SUBQ
6. UH60A UH60L MWO 50–78

AB1470_1
SA

Figure 3. Instrument Panel Lights Schematic Diagram. (Sheet 1 of 6)

SCHEMATICS - Continued

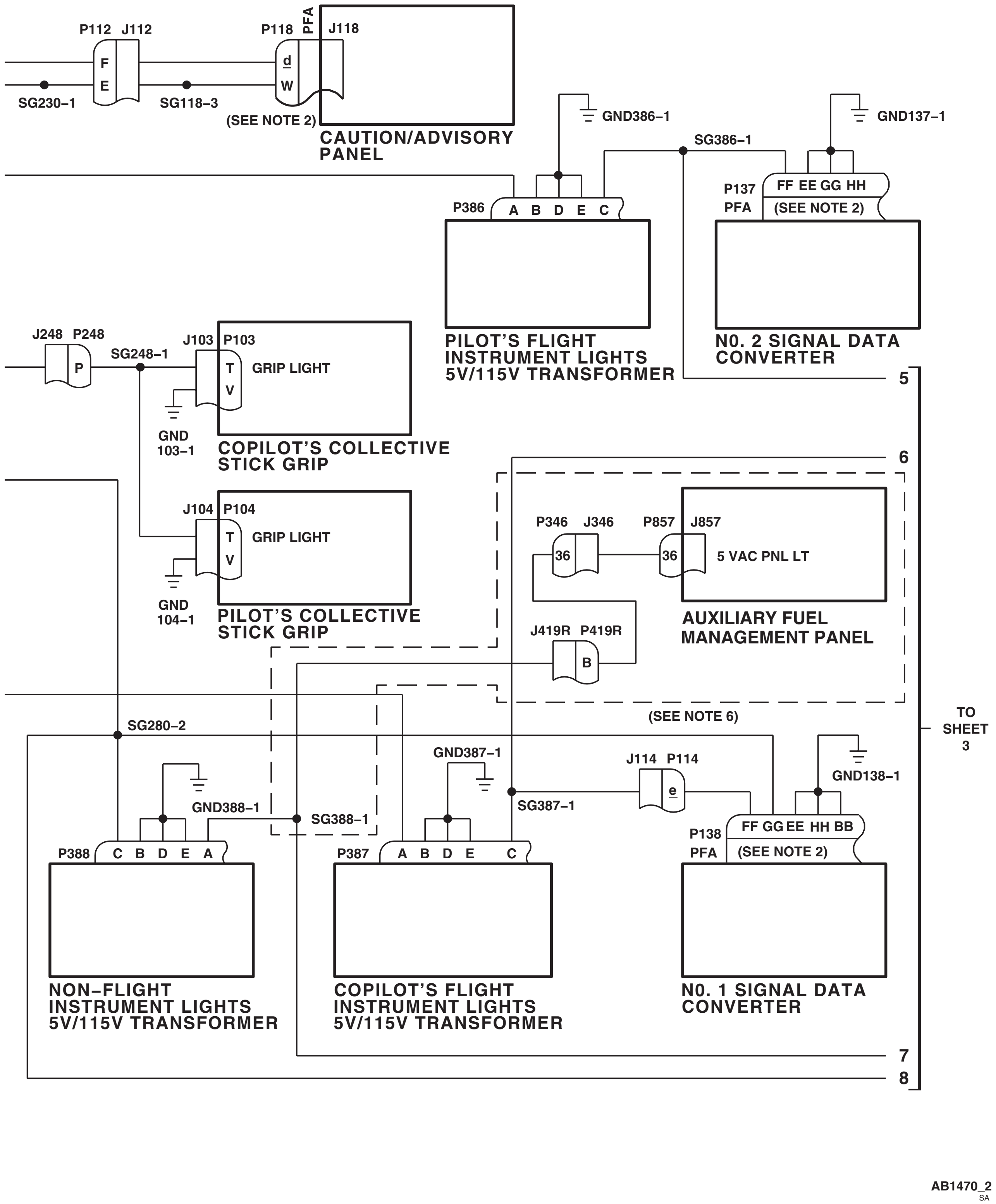


Figure 3. Instrument Panel Lights Schematic Diagram. (Sheet 2 of 6)

SCHEMATICS - Continued

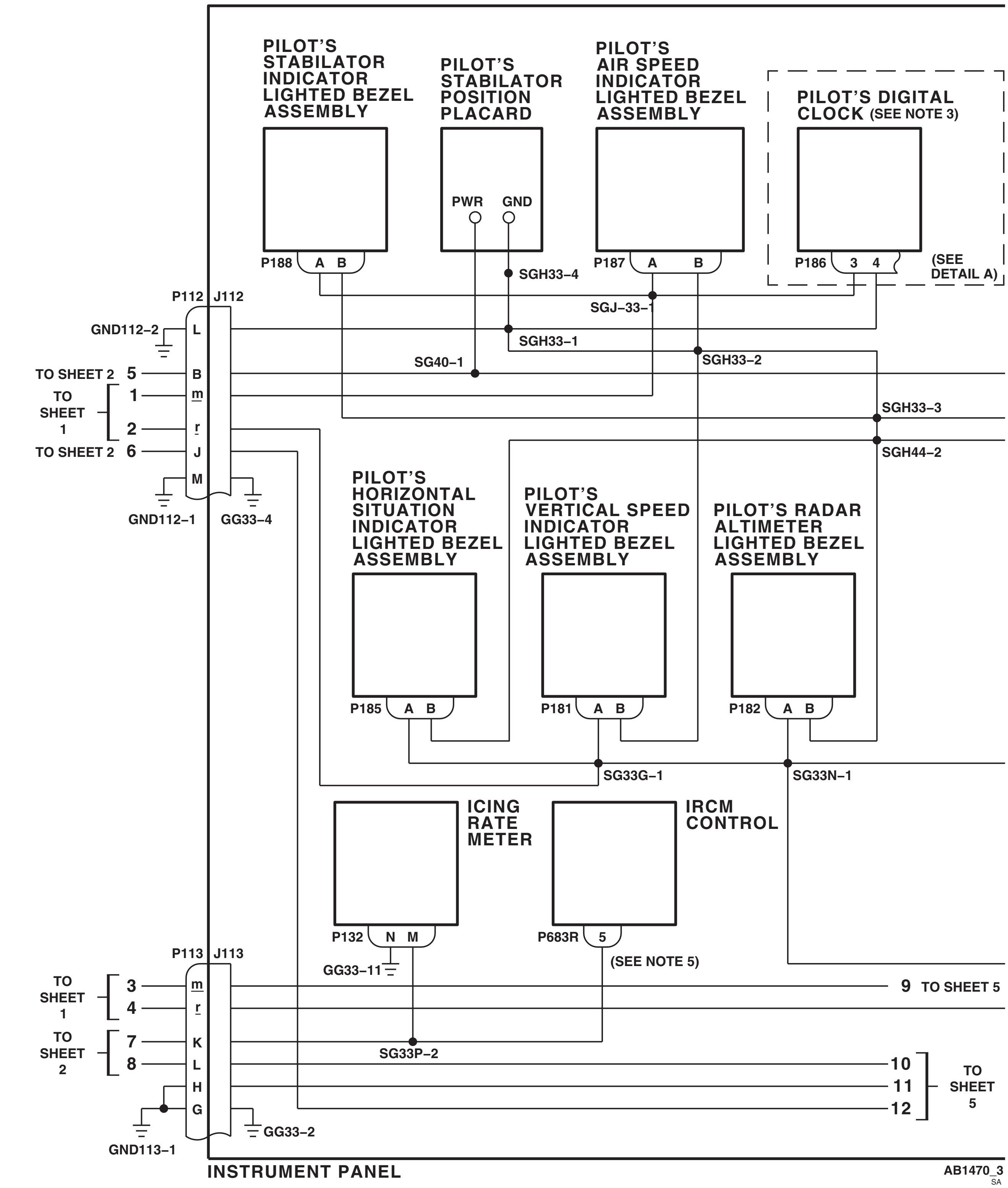


Figure 3. Instrument Panel Lights Schematic Diagram. (Sheet 3 of 6)

SCHEMATICS - Continued

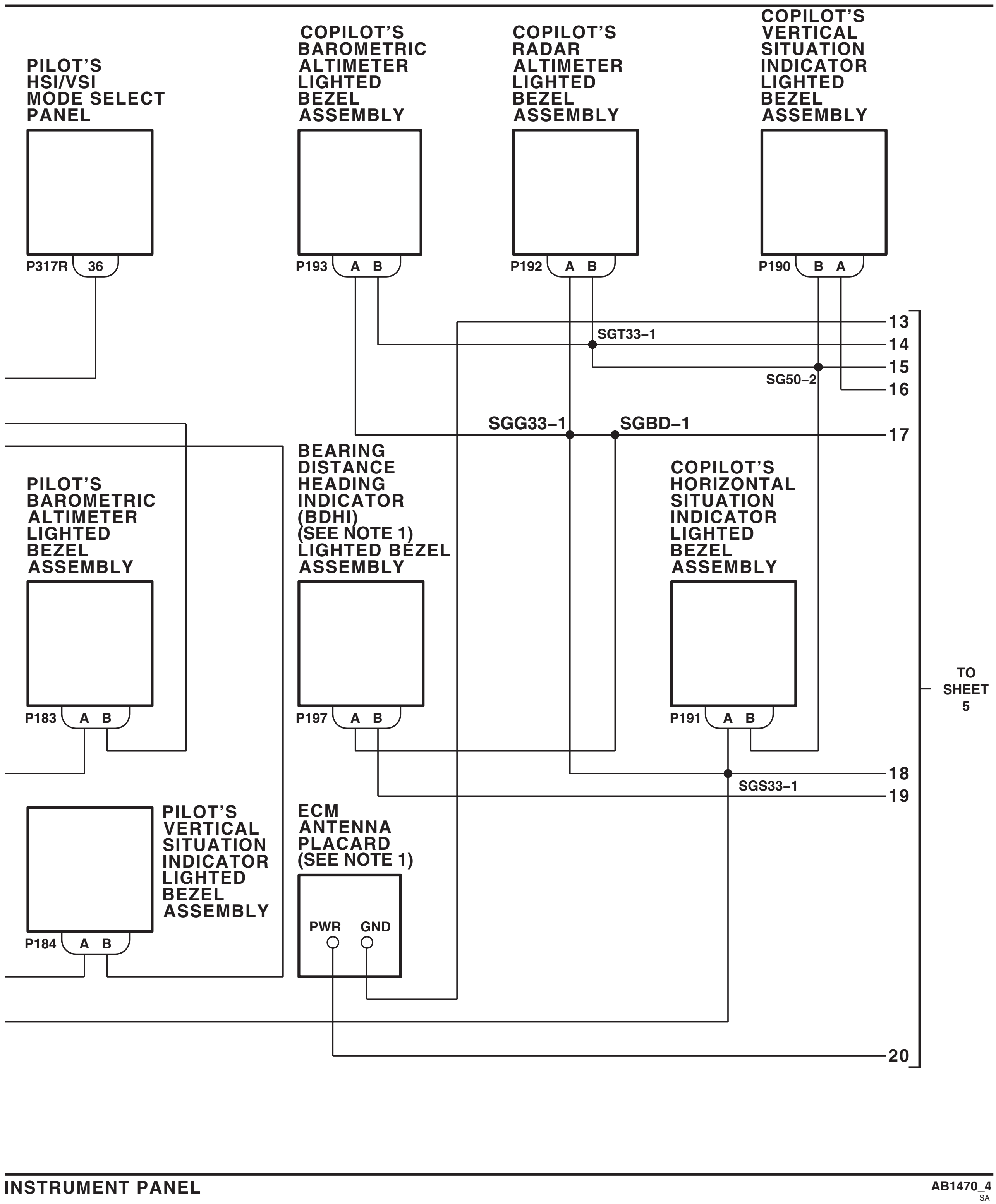


Figure 3. Instrument Panel Lights Schematic Diagram. (Sheet 4 of 6)

SCHEMATICS - Continued

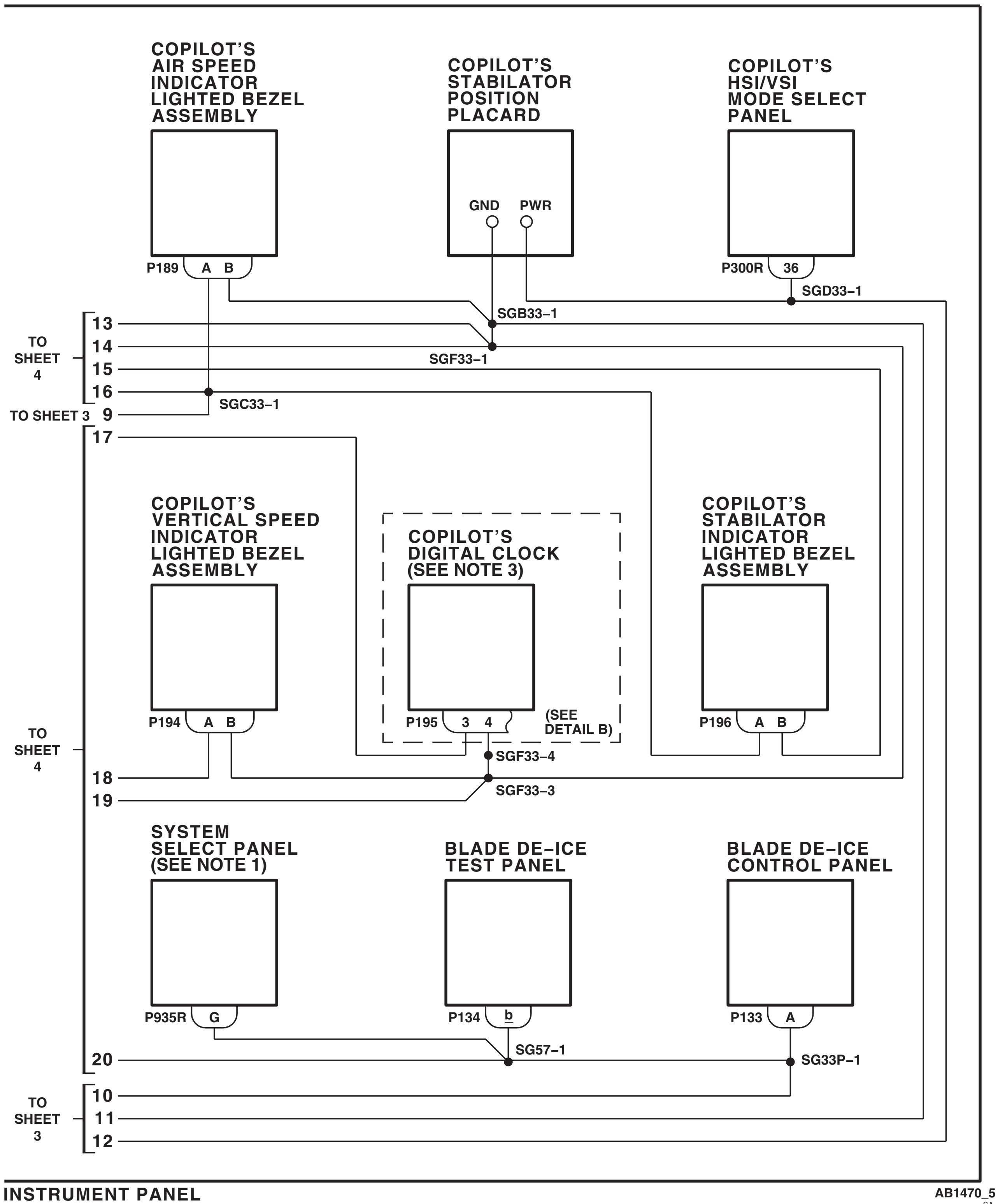
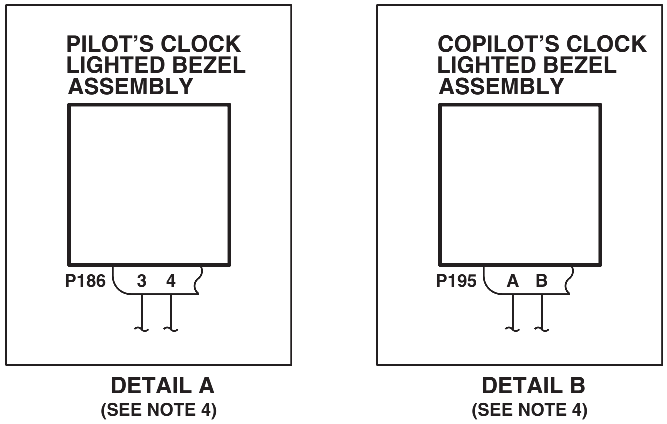


Figure 3. Instrument Panel Lights Schematic Diagram. (Sheet 5 of 6)

SCHEMATICS - Continued



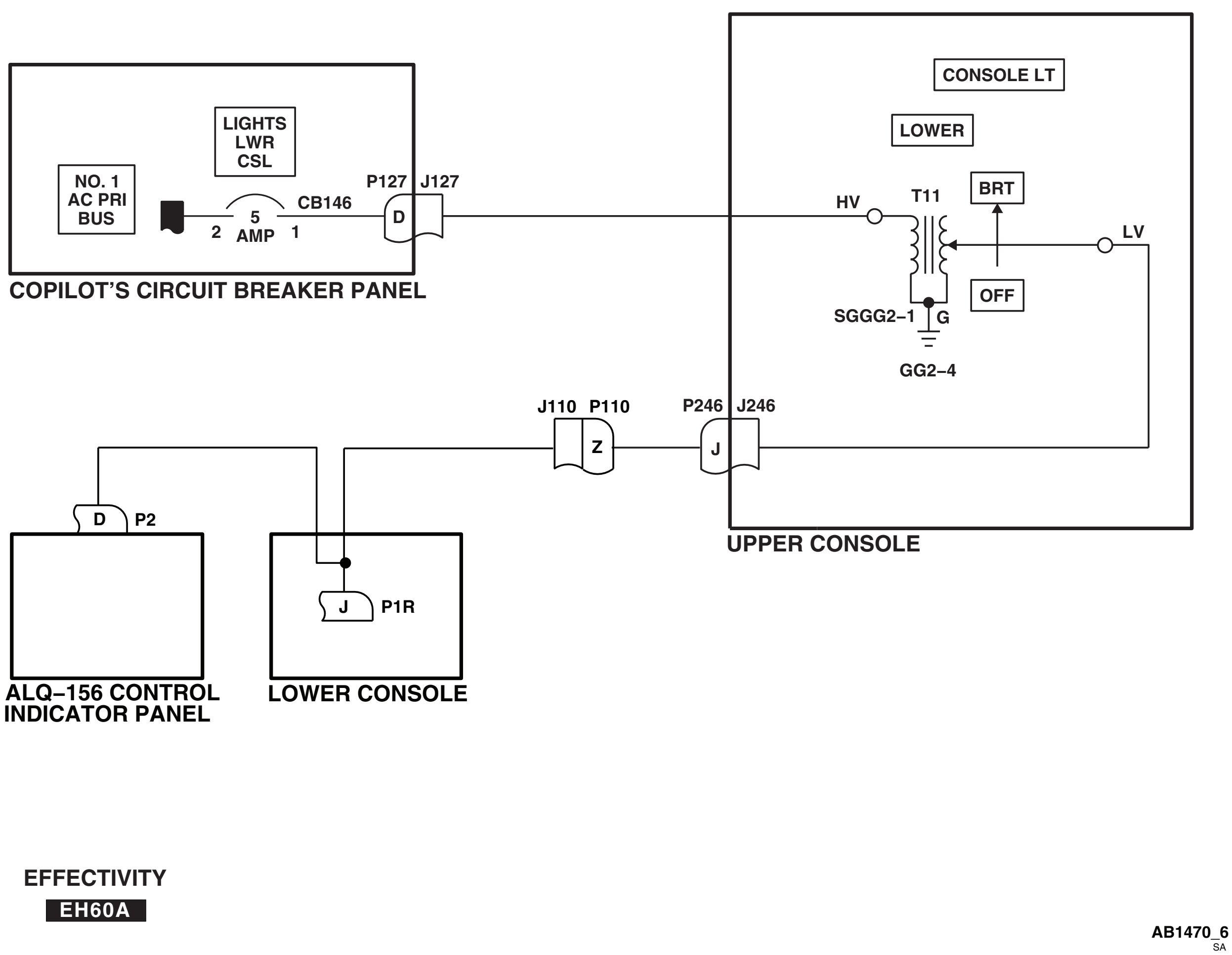


Figure 3. Instrument Panel Lights Schematic Diagram. (Sheet 6 of 6)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

UPPER AND LOWER CONSOLE LIGHTS

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
TM 1-1520-237-10
TM 11-1520-237-23
WP 0104 00
WP 0824 00
WP 0875 00
WP 0930 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational,
TM 1-1520-237-10TM 1-1520-237-10

UPPER AND LOWER CONSOLE LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- RIS modified helicopter's have the lower console lighting limited to 65 - 70 vac by a mechanical limit attachment to the lower console dimmer control.
- See Figure 1 for component location diagram and EH-60A UH-60A UH-60L 89-26149-96-26722 W/O RIS W/O MWO 50-78 Figure 2 , UH-60L 96-26723-SUBQ W/O RIS NVG Figure 3 , EH-60A UH-60A UH-60L W/0 MWO 50-56 RIS Figure 4 , EH-60A UH-60A UH-60L MWO 50-56 AN/ARC 186 CONFIGURATION Figure 5 , and EH-60A UH-60A UH-60L MWO 50-56 AN/ARC 201 CONFIGURATION Figure 6 for schematic diagrams as an aid in operational/troubleshooting.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance.
2. Turn on electrical power.
3. On upper console, turn CONSOLE LT LOWER BRT/OFF control to BRT.

Indication/Condition

All lower console control panel lights, troop commander's, left gunner's and right gunner's ICS control panel floodlights shall go from dim to bright as control is turned clockwise.

UPPER AND LOWER CONSOLE LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If none of the lower console control panel lighting, troop commander's, left gunner's and right gunner's ICS control panel floodlights goes from dim to bright, go to LOWER CONSOLE CONTROL PANEL LIGHTS, TROOP COMMANDER'S, LEFT GUNNER'S AND RIGHT GUNNER'S ICS CONTROL PANEL FLOODLIGHTS DO NOT GO TO BRIGHT, in this work package.
2. **W/O MWO 50-56** If RESCUE HOIST CONTROL panel, range extension kit control panel, STABILATOR CONTROL panel, and FUEL BOOST PUMP CONTROL panel goes from dim to bright while the other lower console control panel's lighting, troop commander's, left gunner's and right gunner's ICS control panel floodlights stay dim, go to RESCUE HOIST CONTROL PANEL, RANGE EXTENSION KIT CONTROL PANEL, STABILATOR CONTROL PANEL, AND FUEL BOOST PUMP CONTROL PANEL GOES FROM DIM TO BRIGHT WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING, TROOP COMMANDER'S, LEFT GUNNER'S AND RIGHT GUNNER'S ICS CONTROL PANEL FLOODLIGHTS STAY DIM, in this work package.
3. **W/O MWO 50-56** **EH-60A** If No. 1 and No. 2 VHF/FM Radio Set Controls (AN/ARC-186) or No. 1 VHF/FM Radio Set Control (C-11188A/ARC) (AN/ARC-186) or No. 1 and No. 2 VHF/FM Radio Set Controls (AN/ARC-201) **UH-60L** No. 1 and No. 2 VHF/FM Radio Set Controls (AN/ARC-201) **UH-60A** No. 1 and No. 2 VHF/FM Radio Set Controls (AN/ARC-186), **UH-60A 77-22714 - 84-23952 UH-60A 84-23954 - 84-23961** VHF/AM Radio Set Control (AN/ARC 186(V)) or No. 1 and No. 2 VHF/FM Radio Set Controls (AN/ARC-201) do not go from dim to bright, go to COMMUNICATION SYSTEM CONTROL PANELS LIGHTING DOES NOT GO FROM DIM TO BRIGHT, in this work package.
4. **MWO 50-56** If RESCUE HOIST CONTROL panel, range extension kit control panel, STABILATOR CONTROL panel, MISC SW control panel, radio retransmission control panel, COMPASS system control panel, FUEL BOOST PUMP CONTROL panel, cabin dome light dimmer panel, No. 1 VHF/FM radio set control panel and No. 2 VHF/FM radio set control panel lighting goes from dim to bright while the other lower console control panel's lighting, troop commander's, left gunner's and right gunner's ICS control panel floodlights stays dim, go to RESCUE HOIST, RANGE EXTENSION, STABILATOR, MISC SW, RADIO RETRANSMISSION, COMPASS SYSTEM, FUEL BOOST PUMP, BOTH VHF/FM RADIO SET PANELS LIGHTING GOES FROM DIM TO BRIGHT, in this work package.
5. **MWO 50-56** If MISC SW control panel, radio retransmission control panel, COMPASS system control panel, No. 1 and No. 2 VHF/FM radio set control (AN/ARC-201) lighting stays dim while the other lower console panel lighting goes from dim to bright, go to MISC SW CONTROL PANEL, RADIO RETRANSMISSION CONTROL PANEL, COMPASS SYSTEM CONTROL PANEL, NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE PANEL LIGHTING GOES FROM BRIGHT TO DIM, in this work package.
6. **MWO 50-56** If No. 1 and No. 2 VHF/FM radio set control (AN/ARC-201) lighting stays dim while the other lower console control panel's lighting goes from dim to bright, go to NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING GOES FROM DIM TO BRIGHT, in this work package.
7. If any individual lower console control panel's lighting, troop commander's, left gunner's and right gunner's ICS control panel floodlights do not go from dim to bright, go to **W/O MWO 50-56** INDIVIDUAL LOWER CONSOLE CONTROL PANEL LIGHTING CHECK, in this work package **MWO 50-56** INDIVIDUAL LOWER CONSOLE CONTROL PANEL LIGHTING CHECK, in this work package.
8. If lower console control panel lighting, troop commander's, left gunner's and right gunner's ICS control panel floodlights flickers when control is turned clockwise, replace CONSOLE LT LOWER dimming control T11 (WP 0875 00).

UPPER AND LOWER CONSOLE LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

4. On upper console, turn CONSOLE LT LOWER BRT/OFF control to OFF.

Indication/Condition

Lower console control panels lighting, troop commander's, left gunner's and right gunner's ICS control panel floodlights shall go from bright to dim, then off as control is turned counterclockwise.

Corrective Action

If Indication/Condition is not as specified, replace CONSOLE LT LOWER dimmer control T11 (WP 0875 00).

Step

5. On upper console, turn CONSOLE LT UPPER BRT/OFF control to BRT.

Indication/Condition

The following lighting shall go from dim to bright as control is turned clockwise:

- Upper console left side control panel
- Upper console center control panel
- Upper console right side control panel
- Cockpit flood and secondary lights panel
- Engine control left quadrant
- Engine control center quadrant
- Engine control right quadrant

Corrective Action

1. If none of the upper console lights go on, go to UPPER CONSOLE LIGHTING DOES NOT GO ON, in this work package.
2. If any individual upper console panel lighting does not go from dim to bright, go to INDIVIDUAL UPPER CONSOLE PANEL LIGHTING CHECK, in this work package.
3. If upper console lighting flickers when control is turned clockwise, replace CONSOLE LT UPPER dimmer control T3 (WP 0875 00).

Step

6. On upper console, turn CONSOLE LT UPPER BRT/OFF control to OFF.

Indication/Condition

The following lights shall go from bright to dim when the control is turned counterclockwise:

- Upper console left side control panel
- Upper console center control panel
- Upper console right side control panel
- Cockpit flood and secondary lights panel
- Engine control left quadrant

UPPER AND LOWER CONSOLE LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- Engine control center quadrant
- Engine control right quadrant

Corrective Action

If Indication/Conditions a. through g. are not as specified, replace CONSOLE LT UPPER dimmer control T3 (WP 0875 00).

Step

7. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

LOWER CONSOLE CONTROL PANEL LIGHTS, TROOP COMMANDER'S, LEFT GUNNER'S AND RIGHT GUNNER'S ICS CONTROL PANEL FLOODLIGHTS DO NOT GO TO BRIGHT

SYMPTOM

Lower console control panel lights, troop commander's, left gunner's and right gunner's ICS control panel floodlights do not go to bright.

MALFUNCTION

Lower console control panel lights, troop commander's, left gunner's and right gunner's ICS control panel floodlights do not go to bright.

CORRECTIVE ACTION

1. **Check continuity between CONSOLE LT LOWER dimming control T11-G and ground.**
   a. If continuity is present, go to **2.**
   b. If continuity is not present, repair/replace between wiring T11-G and GG2-4 (WP 1747 00). Go to **5.**
2. **Check for 115 vac between T11-HV and T11-G.**
   a. If voltage is as specified, go to **3.**
   b. If voltage is not as specified, go to **4.**
3. **With CONSOLE LTS LOWER OFF/BRT control in BRT, check for RIS 65 - 70 vac W/O RIS 115 vac between T11-LV and ground.**
   a. If voltage is as specified, repair/replace wiring between EH-60A J246-J UH-60A UH-60L J246-Z and T11-LV (WP 1747 00). Go to **5.**
   b. If voltage is not as specified, replace CONSOLE LT LOWER OFF/BRT dimming control T11 (WP 0875 00). Go to **5.**
4. **Check for 115 vac between terminal 1 of LIGHTS LWR CSL circuit breaker CB146 and ground.**

LOWER CONSOLE CONTROL PANEL LIGHTS, TROOP COMMANDER'S, LEFT GUNNER'S AND RIGHT GUNNER'S ICS CONTROL PANEL FLOODLIGHTS DO NOT GO TO BRIGHT - Continued

a. If voltage is as specified, repair/replace wiring between T11-HV and terminal 1 of LIGHTS LWR CSL circuit breaker CB146 (WP 1747 00). Go to **5.**

b. If voltage is not as specified, replace LIGHTS LWR CSL circuit breaker CB146 (WP 0824 00). Go to **5.**

5. Procedure completed.

RESCUE HOIST CONTROL PANEL, RANGE EXTENSION KIT CONTROL PANEL, STABILATOR CONTROL PANEL, AND FUEL BOOST PUMP CONTROL PANEL GOES FROM DIM TO BRIGHT WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING, TROOP COMMANDER'S, LEFT GUNNER'S AND RIGHT GUNNER'S ICS CONTROL PANEL FLOODLIGHTS STAY DIM

SYMPTOM

Rescue hoist control panel, range extension kit control panel, stabilator control panel, and fuel boost pump control panel goes from dim to bright while the other lower console control panel's lighting, troop commander's, left gunner's and right gunner's ICS control panel floodlights stay dim.

MALFUNCTION

Rescue hoist control panel, range extension kit control panel, stabilator control panel, and fuel boost pump control panel goes from dim to bright while the other lower console control panel's lighting, troop commander's, left gunner's and right gunner's ICS control panel floodlights stay dim.

CORRECTIVE ACTION

1. Check for RIS 65 - 70 vac W/O RIS 115 vac between P657R-Z and ground.

a. If voltage is a specified, replace audio junction box assembly (TM 11-1520-237-23). Go to **2.**

b. If voltage is not as specified, repair/replace wiring between P657R-Z and UH-60A UH-60L P246-Z EH-60A P246-J (WP 1747 00). Go to **2.**

2. Procedure completed.

COMMUNICATION SYSTEM CONTROL PANELS LIGHTING DOES NOT GO FROM DIM TO BRIGHT

SYMPTOM

Communication system control panels lighting does not go from dim to bright.

MALFUNCTION

Communication system control panels lighting does not go from dim to bright.

CORRECTIVE ACTION

1. With RIS TJ1-1 W/O RIS TJ1-2 terminal lug E disconnected, check continuity between disconnected terminal lug E from RIS TJ1-1 W/O RIS TJ1-2 and ground.

a. If continuity is present, reconnect terminal lug E to TJ1-1 and go to **2.**

b. If continuity is not present, repair/replace wiring between terminal lug E and UH-60A UH-60L GND16-1 EH-60A GG12-3 (WP 1747 00). Go to **4.**

2. Check for 115 vac between RIS TJ1-1 W/O RIS TJ1-2 terminal K and RIS TJ1-1 W/O RIS TJ1-2 terminal E.

a. If voltage is a specified, replace terminal board RIS TJ1-1 W/O RIS TJ1-2 (, WP 0872 00). Go to **4.**

COMMUNICATION SYSTEM CONTROL PANELS LIGHTING DOES NOT GO FROM DIM TO BRIGHT - Continued

b. If voltage is not as specified, go to **3.**

3. Check continuity between RIS TJ1-1 W/O RIS TJ1-2 terminal K and P657R-C.

a. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). Go to **4.**

b. If continuity is not present, repair/replace wiring between RIS TJ1-1 W/O RIS TJ1-2 terminal K and P657R-C (WP 1747 00). Go to **4.**

4. Procedure completed.

RESCUE HOIST, RANGE EXTENSION, STABILATOR, MISC SW, RADIO RETRANSMISSION, COMPASS SYSTEM, FUEL BOOST PUMP, BOTH VHF/FM RADIO SET PANELS LIGHTING GOES FROM DIM TO BRIGHT WHILE OTHER LOWER CONSOLE CONTROL PANEL LIGHTING, TROOP COMMANDER, LEFT/RIGHT GUNNER ICS FLOODLIGHTS STAYS DIM

SYMPTOM

Rescue hoist, range extension, stabilator, MISC SW, radio retransmission, compass system, fuel boost pump, both VHF/FM radio set panels lighting goes from dim to bright while other lower console control panel lighting, troop commander, left/right gunner ICS stays dim.

MALFUNCTION

Rescue hoist, range extension, stabilator, MISC SW, radio retransmission, compass system, fuel boost pump, both VHF/FM radio set panels lighting goes from dim to bright while other lower console control panel lighting, troop commander, left/right gunner ICS stays dim.

CORRECTIVE ACTION

1. Check for 28 vdc between P657R-Z and ground.

a. If voltage is as specified, replace audio junction box assembly (TM 11-1520-237-23). Go to **5.**

b. If voltage is not as specified, go to **2.**

2. Check continuity between P657R-Z and P168-D.

a. If continuity is present, go to **3.**

b. If continuity is not present, repair/replace wiring between P657R-Z and P168-D (WP 1747 00). Go to **5.**

3. Check for 115 vac between P168-F and ground.

a. If voltage is as specified, go to **4.**

b. If voltage is not as specified, repair/replace wiring between P168-F and UH-60L UH-60A P246-Z EH-60A P246-J (WP 1747 00). Go to **5.**

4. Check continuity between:

- **P168-K and ground**
- **P168-L and ground**
- **P168-E and ground**

a. If continuity is present, replace low voltage power supply (, WP 0927 00). Go to **5.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **5.**

5. Procedure completed.

IF MISC SW CONTROL PANEL, RADIO RETRANSMISSION CONTROL PANEL, COMPASS SYSTEM CONTROL PANEL, NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE PANEL LIGHTING GOES FROM DIM TO BRIGHT

SYMPTOM

If MISC SW control panel, radio retransmission control panel, compass system control panel, No. 1 and No. 2 VHF/FM radio set control (AN/ARC-201) lighting stays dim while the other lower console panel lighting goes from dim to bright.

MALFUNCTION

If MISC SW control panel, radio retransmission control panel, compass system control panel, No. 1 and No. 2 VHF/FM radio set control (AN/ARC-201) lighting stays dim while the other lower console panel lighting goes from dim to bright.

CORRECTIVE ACTION

1. **Check for 115 vac between TB2A-A and ground.**

   a. If voltage is a specified, replace terminal board TB2A. Go to **2.**

   b. If voltage is not as specified, repair/replace wiring between TB2A-A and **UH-60L UH-60A** P246-Z **EH-60A** P246-J (WP 1747 00). Go to **2.**

2. **Procedure completed.**

NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING GOES FROM DIM TO BRIGHT

SYMPTOM

No. 1 and No. 2 VHF/FM radio set control (AN/ARC-201) lighting stays dim while the other lower console control panel's lighting goes from dim to bright.

MALFUNCTION

No. 1 and No. 2 VHF/FM radio set control (AN/ARC-201) lighting stays dim while the other lower console control panel's lighting goes from dim to bright.

CORRECTIVE ACTION

1. **With TJ1-2 terminal lug E disconnected, check continuity between disconnected terminal lug E from TJ1-2 and ground.**

   a. If continuity is present, reconnect terminal lug E to TJ1-2 and go to **2.**

   b. If continuity is not present, repair/replace wiring between terminal lug E and **UH-60L UH-60A** GND16-1 **EH-60A** GG12-3 (WP 1747 00). Go to **6.**

2. **Check for 70 - 75 vac between TJ1-2 terminal K and TJ1-2 terminal E.**

   a. If voltage is a specified, replace terminal board TJ1-2. Go to **6.**

   b. If voltage is not as specified, go to **3.**

3. **Check continuity between TJ1-2 terminal K and EL BALANCE ASSEMBLY T110-LV.**

   a. If continuity is present, go to **4.**

   b. If continuity is not present, repair/replace wiring between TJ1-2 terminal K and EL BALANCE ASSEMBLY T110-LV (WP 1747 00). Go to **6.**

4. **Check for 115 vac between EL BALANCE ASSEMBLY T110-HV and ground.**

NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING GOES FROM DIM TO BRIGHT - Continued

a. If voltage is as specified, replace EL BALANCE ASSEMBLY. Go to **6.**

b. If voltage is not as specified, go to **5.**

5. Check continuity between EL BALANCE ASSEMBLY T110-HV and terminal G of TB2A.

a. If continuity is present, replace terminal board TB2A. Go to **6.**

b. If continuity is not present, repair/replace wiring between T110-HV and terminal G of TB2A (WP 1747 00). Go to **6.**

6. Procedure completed.

Table 1. Individual Lower Console Control Panel Lighting Check. W/O MWO 50-56

COMPONENT	CHECK FOR W/O RIS 115 vac RIS 65 - 70 vac BETWEEN THESE POINTS:	IF YES	IF NO
UH-60A UH-60L Rescue Hoist Control Panel	P228-K and ground	Replace Panel	Repair/replace wiring between P228-K and P246-Z (WP 1747 00).
UH-60A Range Extension Kit Control Panel	P170-u and ground	Replace Panel	Repair/replace wiring between P170-u and P246-Z (WP 1747 00).
UH-60L AUX CABIN HEATER Control Panel	P724-D and ground	Replace Panel	Repair/replace wiring between P724-D and P246-Z (WP 1747 00).
Stabilator Control/Auto Flight Control Panel	P4R-e and ground	Replace Panel	Repair/replace wiring between P4R-e and **EH-60A P246-J UH-60A UH-60L P246-Z (WP 1747 00).**
MISC SW Panel	P659R-G and ground	Replace Panel	Check continuity between P659R-G and P657R-L. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P659R-G and P657R-L.
RADIO RETRANS Control Panel	P116R-J and ground	Replace Panel	Check continuity between P116R-J and P657R-U. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P116R-J and P657R-U (WP 1747 00).

NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING GOES FROM DIM TO BRIGHT - Continued

Table 1. Individual Lower Console Control Panel Lighting Check. W/O MWO 50-56 - Continued

COMPONENT	CHECK FOR W/O RIS 115 vac RIS 65 - 70 vac BETWEEN THESE POINTS:	IF YES	IF NO
Compass Control Panel	P134R-U and ground	Replace Panel	Check continuity between P134R-U and P657R-J. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P134R-U and P657R-J (WP 1747 00).
Fuel Boost Pump Control Panel	P139-L and ground	Replace Panel	Repair/replace wiring between P139-L and **EH-60A P246-J UH-60A UH-60L P246-Z (WP 1747 00).**
UH-60A 84-23953, 84-23962-SUBQ VHF/AM Radio Set Control (ARC-186(V))	Check for 28 vdc between P814R-a and ground	Replace Panel	Check continuity between P814R-a and P657R-X. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P814R-a and P657R-X (WP 1747 00).
UH-60A 77-22714 - 84-23952 VHF/AM Radio Set Control (ARC-186(V))	Check for 28 vdc between P118R-C and ground	Replace Panel	Check continuity between P118R-C and P657R-S. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P814R-a and P657R-X (WP 1747 00).
No. 1 VHF/FM Radio Set Control Panel (AN/ARC-201)	P820R-a and P820R-W	Replace Panel	Check continuity between: P820R-a and TJ1-1 terminal G P820R-W and TJ1-1 terminal B. If continuity is present, replace TJ1-1. If continuity is not present, repair/replace wiring as required (WP 1747 00).
No. 2 VHF/FM Radio Set Control Panel (AN/ARC-201)	P817R-a and P817R-W	Replace Panel	Check continuity between: P817R-a and TJ1-1 terminal H P817R-W and TJ1-1 terminal C. If continuity is present, replace TJ1-1. If continuity is not present, repair/replace wiring as required (WP 1747 00).

NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING GOES FROM DIM TO BRIGHT - Continued

Table 1. Individual Lower Console Control Panel Lighting Check. W/O MWO 50-56 - Continued

COMPONENT	CHECK FOR W/O RIS 115 vac RIS 65 - 70 vac BETWEEN THESE POINTS:	IF YES	IF NO
No. 1 VHF/FM Radio Set Control (AN/ARC-186)	Check for 28 vdc between P820R-a and P820R-W	Replace Panel	Check continuity between: P820R-a and TJ1-2 terminal G P820R-$\overline{W}$ and TJ1-2 terminal B. If continuity is present, replace TJ1-2. If continuity is not present, repair/replace wiring as required (WP 1747 00).
No. 2 VHF/FM Radio Set Control (AN/ARC-186)	Check 28 vdc between P817R-a and P817R-W	Replace Panel	Check continuity between: P817R-a and TJ1-2 terminal H P817R-$\overline{W}$ and TJ1-2 terminal C. If continuity is present, replace TJ1-2. If continuity is not present, repair/replace wiring as required (WP 1747 00).
EH-60A IFM Control Panel (C-11188A/ARC) (AN/ARC-186)	P1034-G and P1034-H	Replace Panel	Check continuity between: P1034-G and TJ1-1 terminal A P1034-H and TJ1-1 terminal F. If continuity is present, replace TJ1-1. If continuity is not present, repair/replace wiring as required (WP 1747 00).
EH-60A No. 1 VHF/FM Radio Set Control (AN/ARC-186)	P820R-a and P820R-W	Replace Panel	Check continuity between: P820R-a and TJ1-3 terminal G P820R-$\overline{W}$ and TJ1-3 terminal B. If continuity is present, replace TJ1-3. If continuity is not present, repair/replace wiring as required (WP 1747 00).
EH-60A No. 2 VHF/FM Radio Set Control (AN/ARC-186)	P817R-a and P817R-W	Replace Panel	Check continuity between: P817R-a and TJ1-3 terminal H P817R-$\overline{W}$ and TJ1-3 terminal C. If continuity is present, replace TJ1-3. If continuity is not present, repair/replace wiring as required (WP 1747 00).
Cabin Dome Light Dimmer Panel	P455-D and P455-E	Replace Panel	Check continuity between: P455-D and P657R-E P455-E and ground. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring as required (WP 1747 00).

NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING GOES FROM DIM TO BRIGHT - Continued

Table 1. Individual Lower Console Control Panel Lighting Check. W/O MWO 50-56 - Continued

COMPONENT	CHECK FOR W/O RIS 115 vac RIS 65 - 70 vac BETWEEN THESE POINTS:	IF YES	IF NO
Crew Chief/Right Gunner's ICS Control Panel Floodlight	SA21R-1 and SA21R-2	Replace Floodlight	Check continuity between: SA21R-1 and P653R-F SA21R-2 and ground. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring as required (WP 1747 00).
Left Gunner's ICS Control Panel Floodlight	SA23R-1 and SA23R-2	Replace Floodlight	Check continuity between: SA23R-1 and P655R-R SA23R-2 and ground. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring as required (WP 1747 00).
Troop Commander's ICS Control Panel Floodlight	SA20R-1 and SA20R-2	Replace Floodlight	Check continuity between: SA20R-1 and P653R-F SA20R-2 and ground. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring as required (WP 1747 00).

Table 2. Individual Lower Console Control Panel Lighting Check. MWO 50-56

COMPONENT	CHECK FOR SPECIFIED VOLTAGE BETWEEN THESE POINTS:	IF YES	IF NO
UH-60A UH-60L Rescue Hoist Control Panel	Check for 115 vac between P228-K and ground	Replace Panel	Repair/replace wiring between P228-K and P246-Z (WP 1747 00).
UH-60A Range Extension Kit Control Panel	Check for 115 vac between P170-u and ground	Replace Panel	Repair/replace wiring between P170-u and P246-Z (WP 1747 00).
UH-60L AUX CABIN HEATER Control Panel	Check for 115 vac between P724-D and ground	Replace Panel	Repair/replace wiring between P724-D and P246-Z (WP 1747 00).

NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING GOES FROM DIM TO BRIGHT - Continued

Table 2. Individual Lower Console Control Panel Lighting Check. MWO 50-56 - Continued

COMPONENT	CHECK FOR SPECIFIED VOLTAGE BETWEEN THESE POINTS:	IF YES	IF NO
Stabilator Control/Auto Flight Control Panel	Check for 115 vac between P4R-e and ground	Replace Panel	Repair/replace wiring between P4R-e and **EH-60A P246-J (WP 1747 00)** [redacted] **UH-60A UH-60L P246-Z** [redacted] **.** **UH-60A UH-60L P246-Z** [redacted] **(WP 1747 00).**
MISC SW Panel	Check for 115 vac between P659R-G and ground	Replace Panel	Check continuity between P659R-G and TB2A terminal B. If continuity is present, replace TB2A. If continuity is not present, repair/replace wiring between P659R-G and TB2A terminal B (WP 1747 00).
RADIO RETRANS Control Panel	Check for 115 vac between P116R-J and ground	Replace Panel	Check continuity between P116R-J and TB2A terminal C. If continuity is present, replace TB2A. If continuity is not present, repair/replace wiring between P116R-J and TB2A terminal C (WP 1747 00).
Compass Control Panel	Check for 115 vac between P134R-U and ground	Replace Panel	Check continuity between P134R-U and TB2A terminal F. If continuity is present, replace TB2A. If continuity is not present, repair/replace wiring between P134R-U and TB2A terminal F (WP 1747 00).
Fuel Boost Pump Control Panel	Check for 115 vac between P139-L and ground	Replace Panel	Repair/replace wiring between P139-L and **EH-60A P246-J** [redacted] **.** **UH-60A UH-60L P246-Z** [redacted] **(WP 1747 00).**

NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING GOES FROM DIM TO BRIGHT - Continued

Table 2. Individual Lower Console Control Panel Lighting Check. MWO 50-56 - Continued

COMPONENT	CHECK FOR SPECIFIED VOLTAGE BETWEEN THESE POINTS:	IF YES	IF NO
No. 1 VHF/FM Radio Set Control Panel (AN/ARC-201)	Check for 70 - 75 vac between P820R-a and P820R-W	Replace Panel	Check continuity between: P820R-a and TJ1-1 terminal G P820R-W and TJ1-1 terminal B. If continuity is present, replace TJ1-1. If continuity is not present, repair/replace wiring as required (WP 1747 00).
No. 2 VHF/FM Radio Set Control Panel (AN/ARC-201)	Check for 70 - 75 vac between P817R-a and P817R-W	Replace Panel	Check continuity between: P817R-a and TJ1-1 terminal H P817R-W and TJ1-1 terminal C. If continuity is present, replace TJ1-1. If continuity is not present, repair/replace wiring as required (WP 1747 00).
VHF/AM Radio Set Control (ARC-186(V))	Check for 28 vdc between P814R-a and ground	Replace Panel	Check continuity between P814R-a and P657R-X (for AN/ARC-186 configuration) or P657R-S (for AN/ARC-201 configuration). If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P814R-a and P657R-X (for AN/ARC-186 configuration) or P657R-S (for AN/ARC-201 configuration) (WP 1747 00).
No. 1 VHF/FM Radio Set Control (AN/ARC-186)	Check for 28 vdc between P820R-a and P820R-W	Replace Panel	Check continuity between: P820R-a and TJ1-2 terminal G P820R-W and TJ1-2 terminal B. If continuity is present, replace TJ1-2. If continuity is not present, repair/replace wiring as required (WP 1747 00). If trouble still remains, repair/replace wiring between TJ1-2 and P657R-S (WP 1747 00).

NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING GOES FROM DIM TO BRIGHT - Continued

Table 2. Individual Lower Console Control Panel Lighting Check. MWO 50-56 - Continued

COMPONENT	CHECK FOR SPECIFIED VOLTAGE BETWEEN THESE POINTS:	IF YES	IF NO
No. 2 VHF/FM Radio Set Control (AN/ARC-186)	Check for 28 vdc between P817R-a and P817R-W	Replace Panel	Check continuity between: P817R-a and TJ1-2 terminal H P817R-W and TJ1-2 terminal C. If continuity is present, replace TJ1-2. If continuity is not present, repair/replace wiring as required (WP 1747 00). If trouble still remains, repair/replace wiring between TJ1-2 and P657R-S (WP 1747 00).
Doppler Navigation Display Control Panel	Check for 5 vac between P687R-K and ground	Replace Panel	Check continuity between P687R-K and P168-A. If continuity is present, replace NVG power supply (WP 0930 00). If continuity is not present, repair/replace wiring between P687R-K and P168-A.
IFF Receiver-Transmitter Control Panel	Check for 5 vac between P664R-15 and ground	Replace Panel	Check continuity between P664R-15 and P168-B. If continuity is present, replace NVG power supply (WP 0930 00). If continuity is not present, repair/replace wiring between P664R-15 and P168-B.
Cabin Dome Light Dimmer Panel	Check for 115 vac between P455-D and P455-E	Replace Panel	Check continuity between: P455-D and TB2A terminal D P455-E and ground. If continuity is present, replace TB2A. If continuity is not present, repair/replace wiring as required (WP 1747 00).
Crew Chief/Right Gunner's ICS Control Panel Floodlight	Check for 28 vdc between P126R-U and ground	Replace Floodlight	Check continuity between P126R-U and P653R-F. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring as required (WP 1747 00).

NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING GOES FROM DIM TO BRIGHT - Continued

Table 2. Individual Lower Console Control Panel Lighting Check. MWO 50-56 - Continued

COMPONENT	CHECK FOR SPECIFIED VOLTAGE BETWEEN THESE POINTS:	IF YES	IF NO
Left Gunner's ICS Control Panel Floodlight	Check for 28 vdc between P127R-U and ground	Replace Floodlight	Check continuity between P127R-U and P655R-R. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring as required (WP 1747 00).
Troop Commander's ICS Control Panel Floodlight	Check for 28 vdc between P200R-U and ground	Replace Floodlight	Check continuity between P200R-U and P653R-F. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring as required (WP 1747 00).
LF/ADF Radio Set Control Panel (ARN-89)	Check for 28 vdc between P194R-M and ground	Replace Panel	Check continuity between P194R-M and P657R-Y. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P194R-M and P57R-Y (WP 1747 00).
Civil Navigation Receiver Control Panel (ARN-123)	Check for 28 vdc between P107R-21 and ground	Replace Panel	Check continuity between P107R-21 and P657R-K. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P107R-21 and P657R-K (WP 1747 00).
No. 2 COMSEC Control Panel (KY-58)	Check for 28 vdc between P112R-H and ground	Replace Panel	Check continuity between P112R-H and P657R-R. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P112R-H and P657R-R (WP 1747 00).

NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING GOES FROM DIM TO BRIGHT - Continued

Table 2. Individual Lower Console Control Panel Lighting Check. MWO 50-56 - Continued

COMPONENT	CHECK FOR SPECIFIED VOLTAGE BETWEEN THESE POINTS:	IF YES	IF NO
Detecting Set Control Panel (APR-39A)	Check for 28 vdc between P690R-13 and ground	Replace Panel	Check continuity between P690R-13 and P657R-P. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P690R-13 and P657R-P (WP 1747 00).
Chaff Dispenser Control Panel	Check for 28 vdc between P669R-G and ground	Replace Panel	Check continuity between P669R-G and P657R-T. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P669R-G and P657R-T (WP 1747 00).
No. 1 COMSEC Control Panel (KY-28)	Check for 28 vdc between P143R-H and ground	Replace Panel	Check continuity between P143R-H and P657R-W. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P143R-H and P657R-W (WP 1747 00).
UHF/AM Radio Set Control Panel (ARC-164(V))	Check for 28 vdc between P120R-C and ground	Replace Panel	Check continuity between P120R-C and P657R-N. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P120R-C and P657R-N (WP 1747 00).
No. 3 COMSEC Control Panel (KY-28)	Check for 28 vdc between P122R-H and ground	Replace Panel	Check continuity between P122R-H and P657R-M. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P122R-H and P657R-M (WP 1747 00).

NO. 1 AND NO. 2 VHF/FM RADIO SET CONTROL (AN/ARC-201) LIGHTING STAYS DIM WHILE THE OTHER LOWER CONSOLE CONTROL PANEL'S LIGHTING GOES FROM DIM TO BRIGHT - Continued

Table 2. Individual Lower Console Control Panel Lighting Check. MWO 50-56 - Continued

COMPONENT	CHECK FOR SPECIFIED VOLTAGE BETWEEN THESE POINTS:	IF YES	IF NO
Copilot ICS Control Panel	Check for 28 vdc between P125R-U and ground	Replace Panel	Check continuity between P125R-U and P651R-AA. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P125R-U and P651R-AA (WP 1747 00).
Pilot ICS Control Panel	Check for 28 vdc between P124R-U and ground	Replace Panel	Check continuity between P124R-U and P651R-AA. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). If continuity is not present, repair/replace wiring between P124R-U and P651R-AA (WP 1747 00).

UPPER CONSOLE LIGHTING DOES NOT GO ON

SYMPTOM

Upper console lighting does not go on.

MALFUNCTION

Upper console lighting does not go on.

CORRECTIVE ACTION

1. **Check continuity between CONSOLE LT UPPER dimming control T3-G and ground.**
   a. If continuity is present, go to **2.**
   b. If continuity is not present, repair/replace wiring between T3-G and GG2-4 (WP 1747 00). Go to **5.**
2. **Check for 115 vac between T3-HV and T3-G.**
   a. If voltage is as specified, replace CONSOLE LT UPPER dimmer control T3 (WP 0875 00). Go to **5.**
   b. If voltage is not as specified, go to **3.**
3. **Turn CONSOLE LT UPPER BRT/OFF control to BRT and check for 65-70 vac between T3-LV and T3-G.**
   a. If voltage is as specified, repair/replace wiring between T3-LV and upper console lights (WP 1747 00). Go to **5.**
   b. If voltage is not as specified, replace dimming control (WP 0875 00). Go to **5.**
4. **Check for 115 vac between terminal 1 of LIGHTS UPPER CSL circuit breaker CB148 and ground.**

UPPER CONSOLE LIGHTING DOES NOT GO ON - Continued

a. If voltage is as specified, repair/replace wiring between T3-HV and terminal 1 of LIGHTS UPPER CSL circuit breaker CB148 (WP 1747 00). Go to **5.**

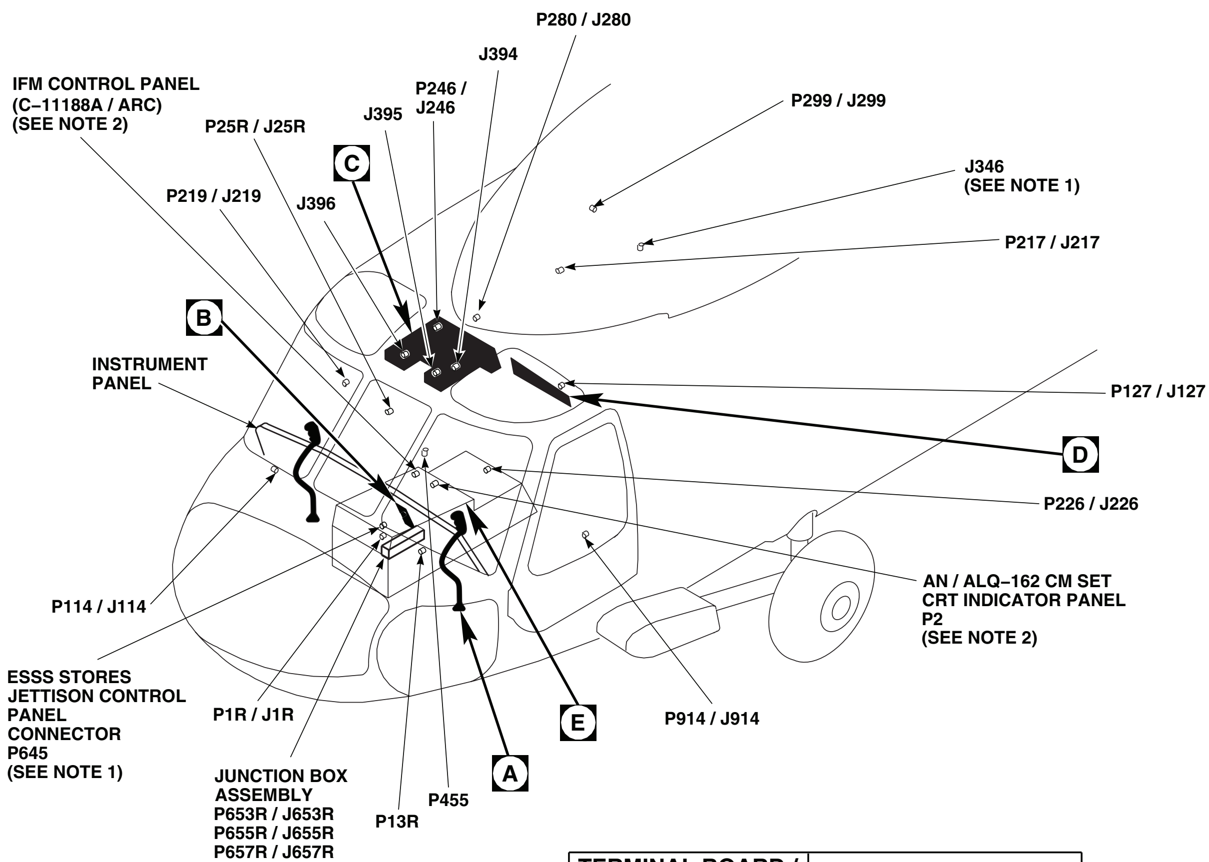
b. If voltage is not as specified, replace LIGHTS UPPER CSL circuit breaker CB148 (WP 0824 00). Go to **5.**

5. Procedure completed.

Table 3. Individual Upper Console Panel Lighting Checks.

INFORMATION PANEL	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
Engine Control Left Quadrant	J394-1 and T3-LV J394-2 and ground	Replace panel	Repair/replace wiring (WP 1747 00).
Engine Control Center Quadrant	J395-1 and T3-LV J395-2 and ground	Replace panel	Repair/replace wiring (WP 1747 00).
Engine Control Right Quadrant	J396-1 and T3-LV J396-2 and ground	Replace panel	Repair/replace wiring (WP 1747 00).
Cockpit Flood And Secondary Lights Panel	P288-E and T3-LV	Troubleshoot Cockpit Flood And Secondary Lights (, WP 0107 00)	Repair/replace wiring (WP 1747 00).
Upper Console Left Side Control Panel	1J2-PWR and T3-LV 1J2-GND and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
Upper Console Center Side Control Panel	1J1-PWR and 1J3-LV 1J1-PWR AND 1J2-PWR	Replace Panel	Repair/replace wiring (WP 1747 00).
Upper Console Right Side Control Panel	1J3-PWR and T3-LV 1J3-GND and ground	Replace Panel	Repair/replace wiring (WP 1747 00).

LOCATION



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
J346	CABIN CEILING, BL 18 LH, STA 301 (SEE NOTE 1)
J394	COCKPIT CEILING, BL 9 LH, STA 233
J395	COCKPIT CEILING, BL 4 LH, STA 229
J396	COCKPIT CEILING, BL 4 RH, STA 229
P2	AN / ALQ–162 CM SET CRT INDICATOR PANEL
P4R	STABILATOR CONTROLS / AUTO FLIGHT CONTROL PANEL
P13R / J13R	CABIN, BL 20 LH, STA 247
P25R / J25R	CABIN, BL 40 RH, STA 282
P107R	CIVIL NAVIGATION RECEIVER CONTROL PANEL (ARN–23) (SEE NOTE 4)
P112R	NO. 2 COMSEC CONTROL PANEL (KY–58) (SEE NOTE 4)
P120R	UHF / AM RADIO SET CONTROL PANEL (ARC–164(V)) (SEE NOTE 4)

NOTES

1. ESSS
2. EH60A
3. UH60A AUXILIARY CABIN HEATER INSTALLED, AUX CABIN HEATER CONTROL PANEL REPLACES RADIO RETRANSMISSION CONTROL PANEL.
4. MWO 50–56
5. W / O MWO 50–56
6. IF INSTALLED.
7. 84–23953 84–23962–SUBQ
   RIS MOD W/O RIS
8. 77–22714–84–23952
   84–23954–84–23961
   W/O MOD W/O RIS

AB2375_1A
SA

Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 1 of 9)

LOCATION - Continued

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P114 / J114	COCKPIT, BL 8 RH, STA 200
P116R	RADIO RETRANSMISSION CONTROL PANEL
P118R	VHF / AM RADIO SET CONTROL PANEL (AN / ARC–186(V)) (SEE NOTE 8)
P122R	NO. 3 COMSEC CONTROL PANEL (KY–28) (SEE NOTE 4)
P124R	PILOT ICS CONTROL PANEL (SEE NOTE 4)
P125R	COPILOT ICS CONTROL PANEL (SEE NOTE 4)
P126R	CREW CHIEF / RIGHT GUNNER'S ICS CONTROL PANEL FLOODLIGHT (SEE NOTE 4)
P127 / J127	BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 28 LH, STA 243
P127R	LEFT GUNNER'S ICS CONTROL PANEL FLOODLIGHT (SEE NOTE 4)
P134R	COMPASS SYSTEM CONTROL PANEL
P139	FUEL BOOST PUMP CONTROL PANEL (SEE NOTE 1)
P143R	NO. 1 COMSEC CONTROL PANEL (KY–28) (SEE NOTE 4)
P194R	LF / ADF RADIO SET CONTROL (ARN–89) (SEE NOTE 4)
P200R	TROOP COMMANDER'S ICS CONTROL PANEL FLOODLIGHT (SEE NOTE 4)
P217 / J217	MAIN ROTOR PYLON DECK, BL 0, STA 284
P219 / J219	CABIN FLOOR, BL 25 RH, STA 247
P226 / J226	COCKPIT, BL 10 LH, STA 228
P228	RESCUE HOIST CONTROL PANEL (WHEN INSTALLED)
P246 / J246	CABIN CEILING, BL 5 RH, STA 247

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P280 / J280	CABIN CEILING, BL 6 LH, STA 247
P288	COCKPIT FLOOD AND SECONDARY LIGHTS PANEL
P299 / J299	CABIN CEILING, BL 10 LH, STA 302
P455	CABIN DOME LT DIMMER, BEHIND PILOT'S SEAT, BL 11 RH, STA 245
P645	ESSS STORES JETTISON CONTROL PANEL CONNECTOR (SEE NOTE 1)
P653R / J653R	JUNCTION BOX ASSEMBLY
P655R / J655R	JUNCTION BOX ASSEMBLY
P657R / J657R	JUNCTION BOX ASSEMBLY
P659R	MISCELLANEOUS SWITCH PANEL
P664R	IFF RECEIVER TRANSMITTER CONTROL PANEL (SEE NOTE 4)
P687R	DOPPLER NAVIGATION DISPLAY CONTROL PANEL (AN / ASN–128) (SEE NOTE 4)
P690R	DETECTING SET CONTROL PANEL (APR–39A) (SEE NOTE 4)
P724 / J724	AUX CABIN HEATER CONTROL PANEL (SEE NOTE 3)
P814R	VHF / AM RADIO SET CONTROL PANEL (AN / ARC–186(V)) (SEE NOTES 4 AND 7)
P817R	NO. 2 VHF / FM RADIO SET CONTROL
P820R	NO. 1 VHF / FM RADIO SET CONTROL
P914 / J914	COCKPIT BL 24 LH, STA 247
P1034	IFM CONTROL PANEL (C–11188A / ARC) (SEE NOTE 2)

AB2375_2A
SA

Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 2 of 9)

LOCATION - Continued

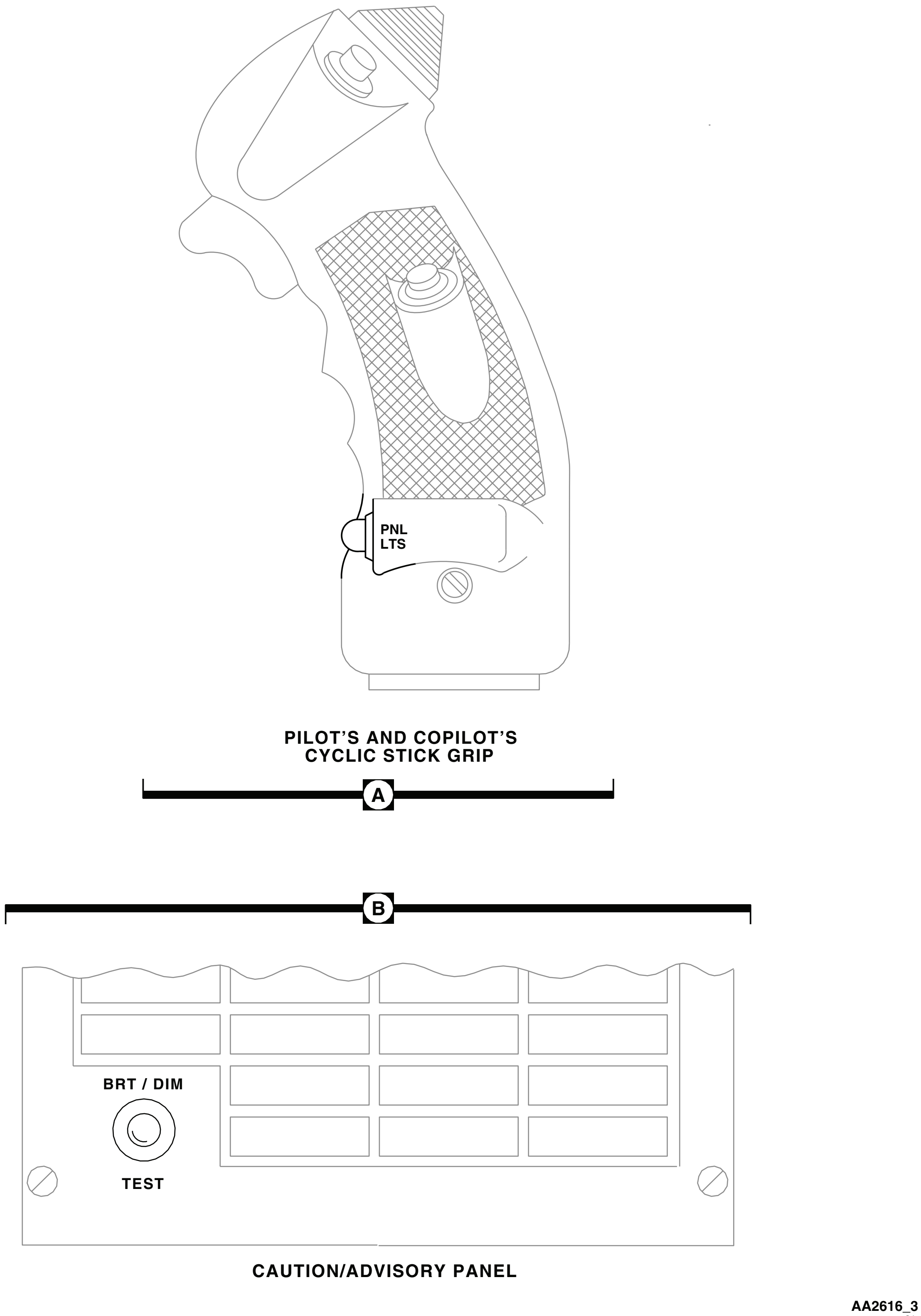


Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 3 of 9)

LOCATION - Continued

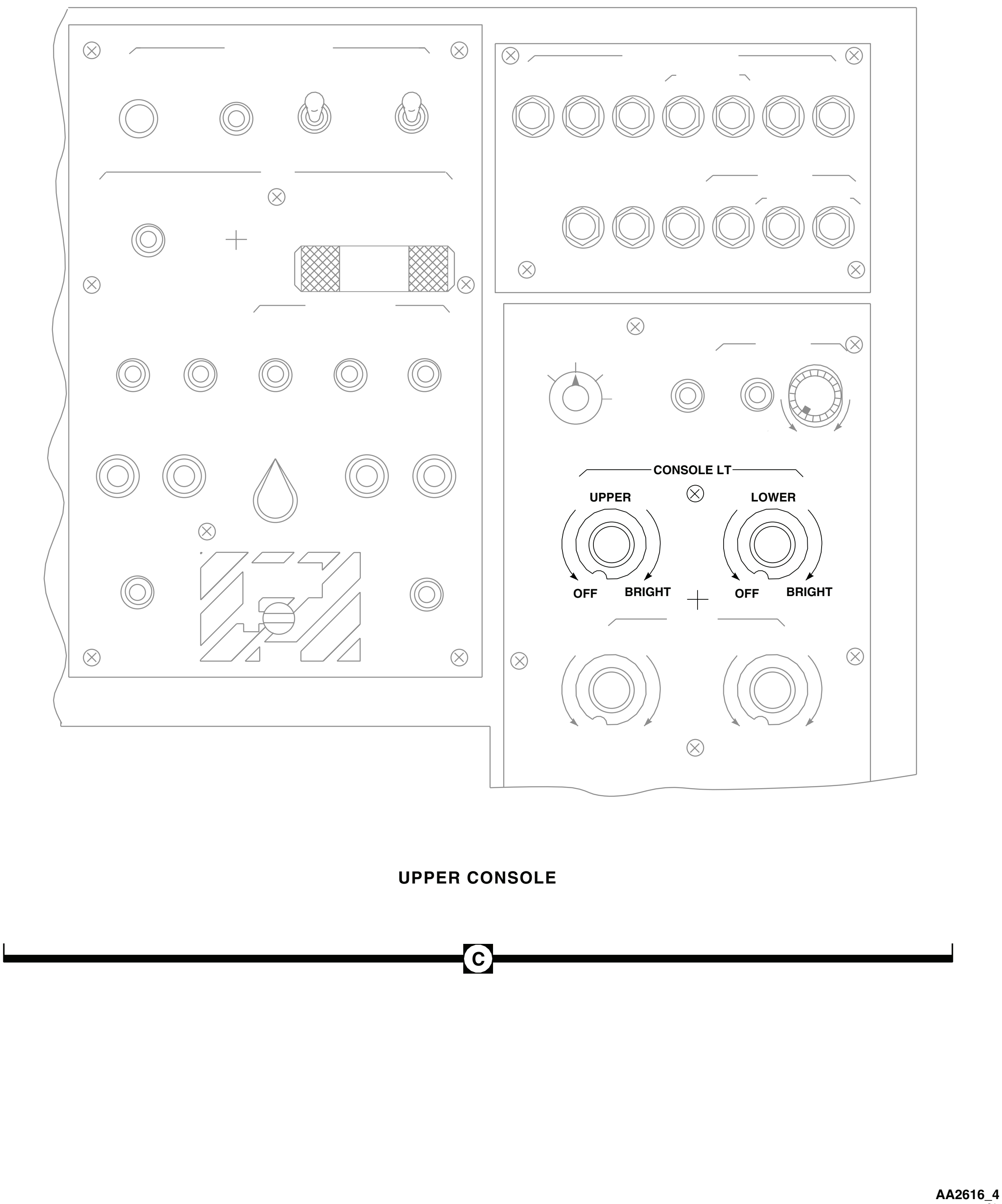


Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 4 of 9)

LOCATION - Continued

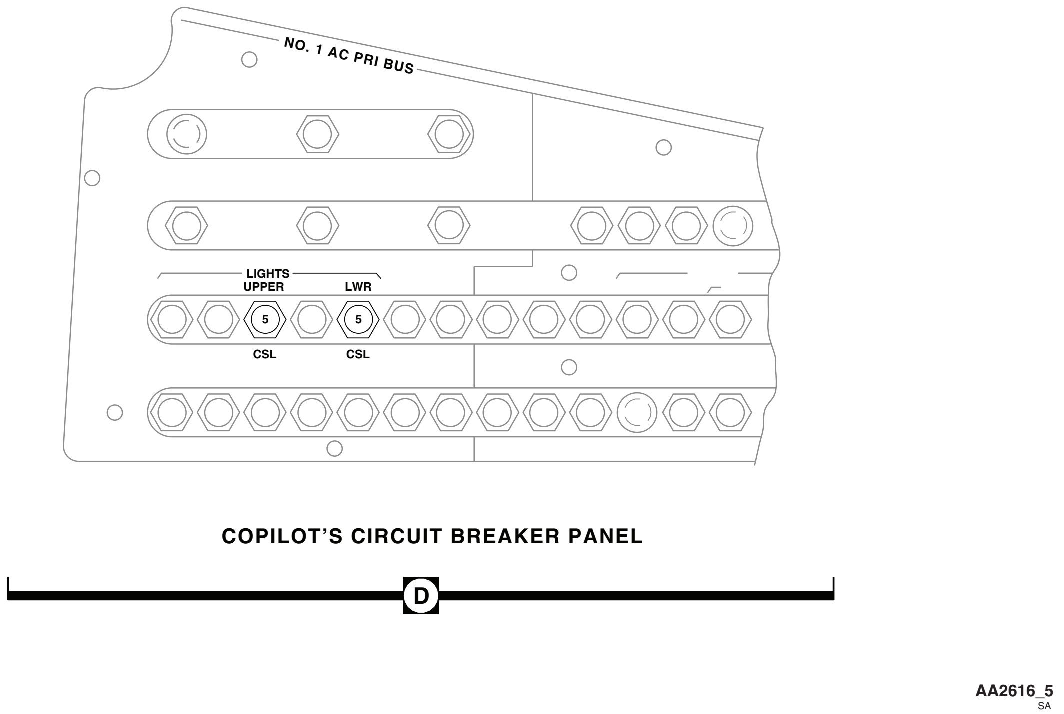


Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 5 of 9)

LOCATION - Continued

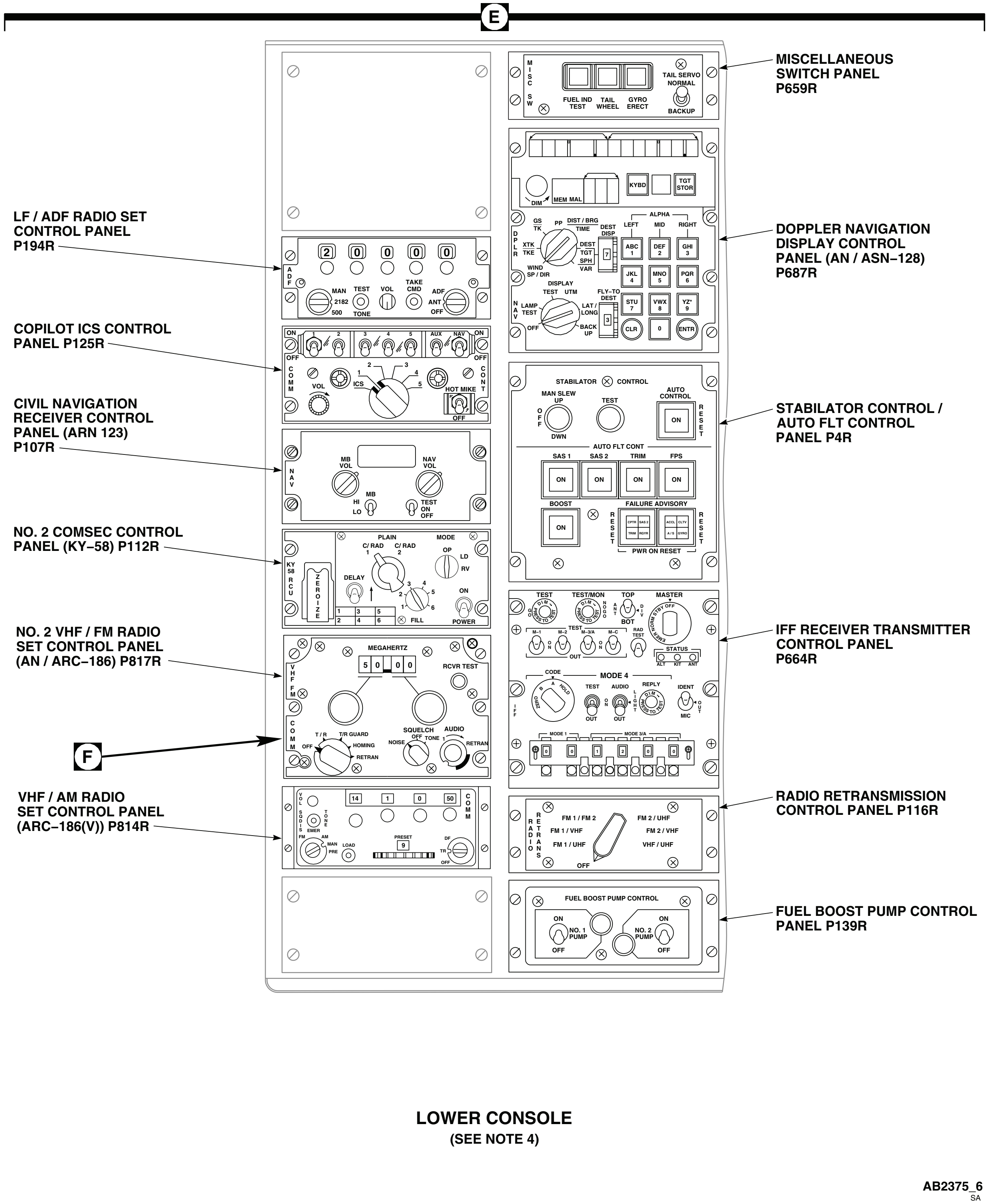


Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 6 of 9)

LOCATION - Continued

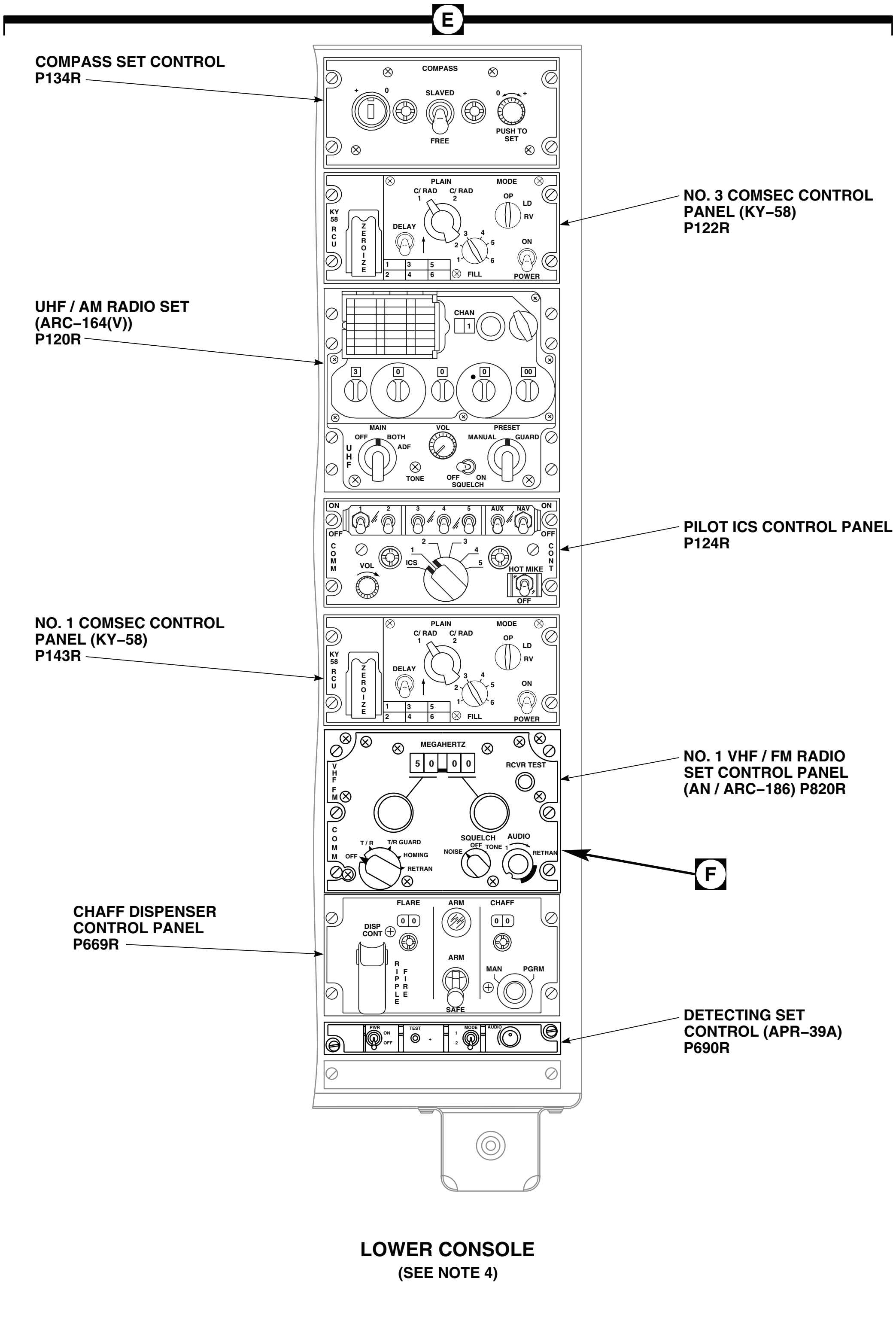


Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 7 of 9)

LOCATION - Continued

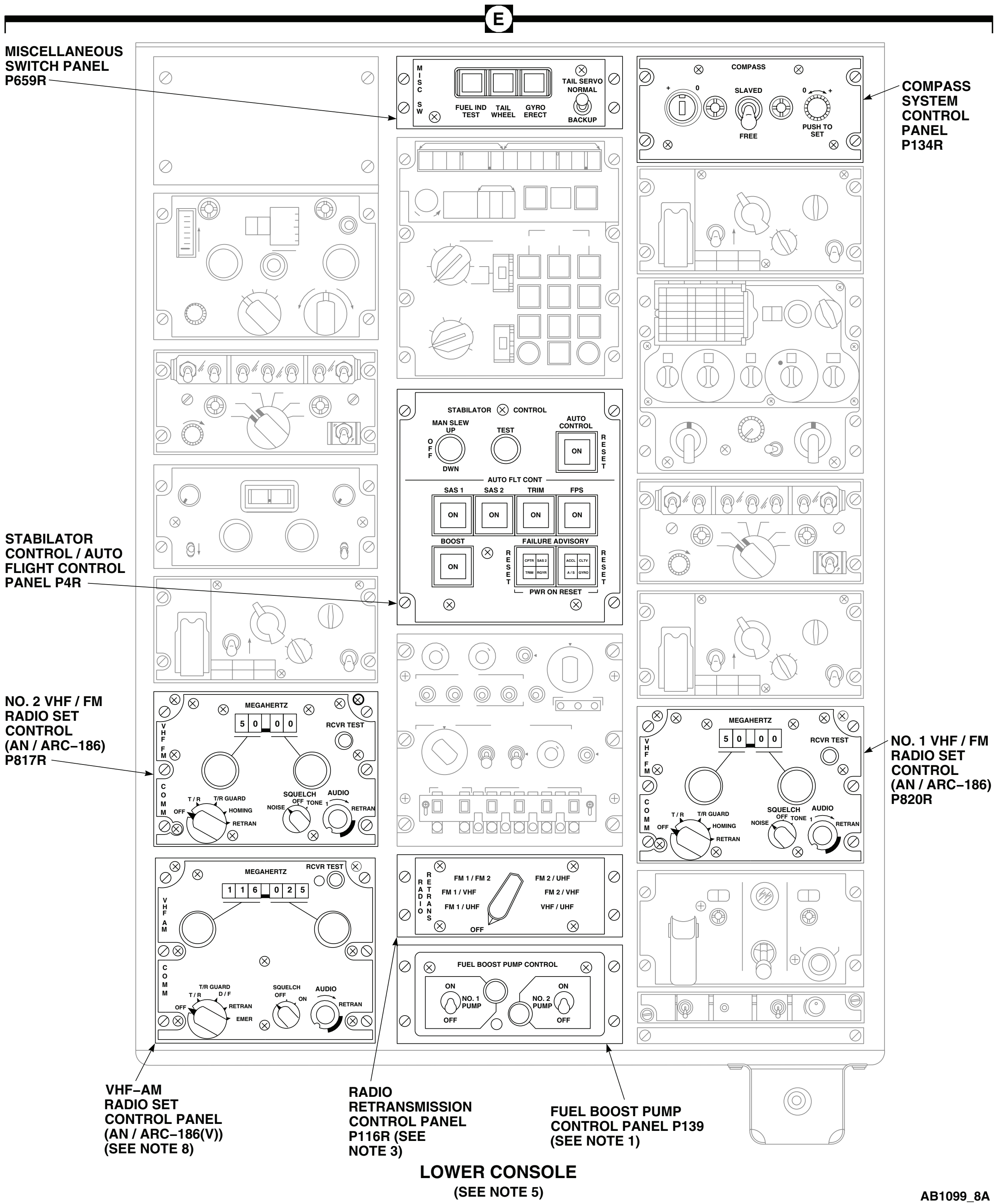
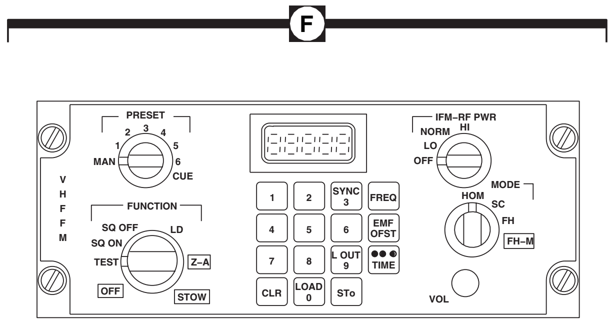


Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 8 of 9)

LOCATION - Continued



VHF/FM RADIO SET CONTROL PANEL
AN/ARC–201
(SEE NOTES 4 AND 6)

AB1099_9
SA

Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 9 of 9)

SCHEMATICS

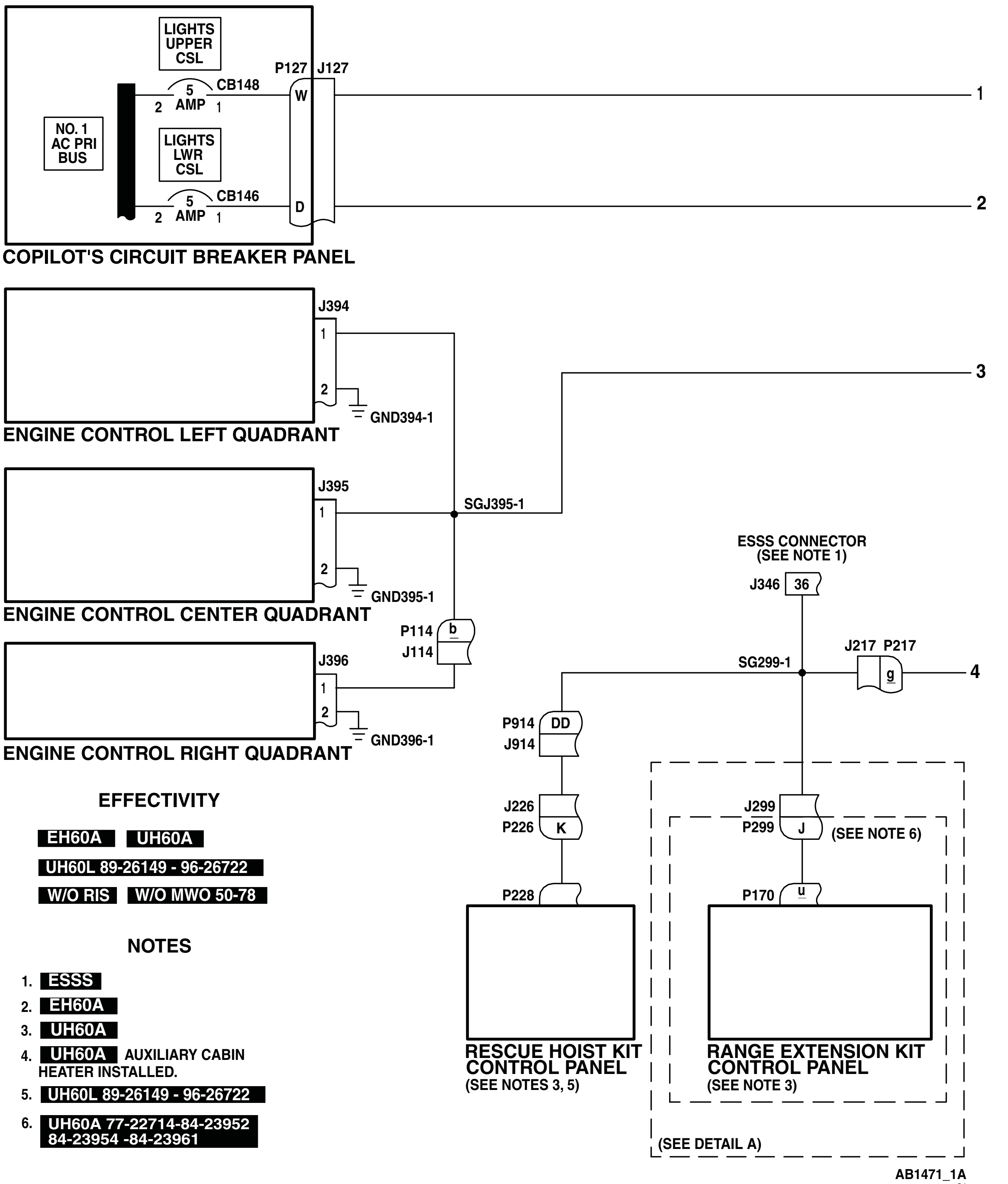


Figure 2. Upper and Lower Console Lights Schematic Diagram
EH-60A UH-60A UH-60L 89-26149-96-26722 W/O RIS W/O MWO 50-78 . (Sheet 1 of 4)

SCHEMATICS - Continued

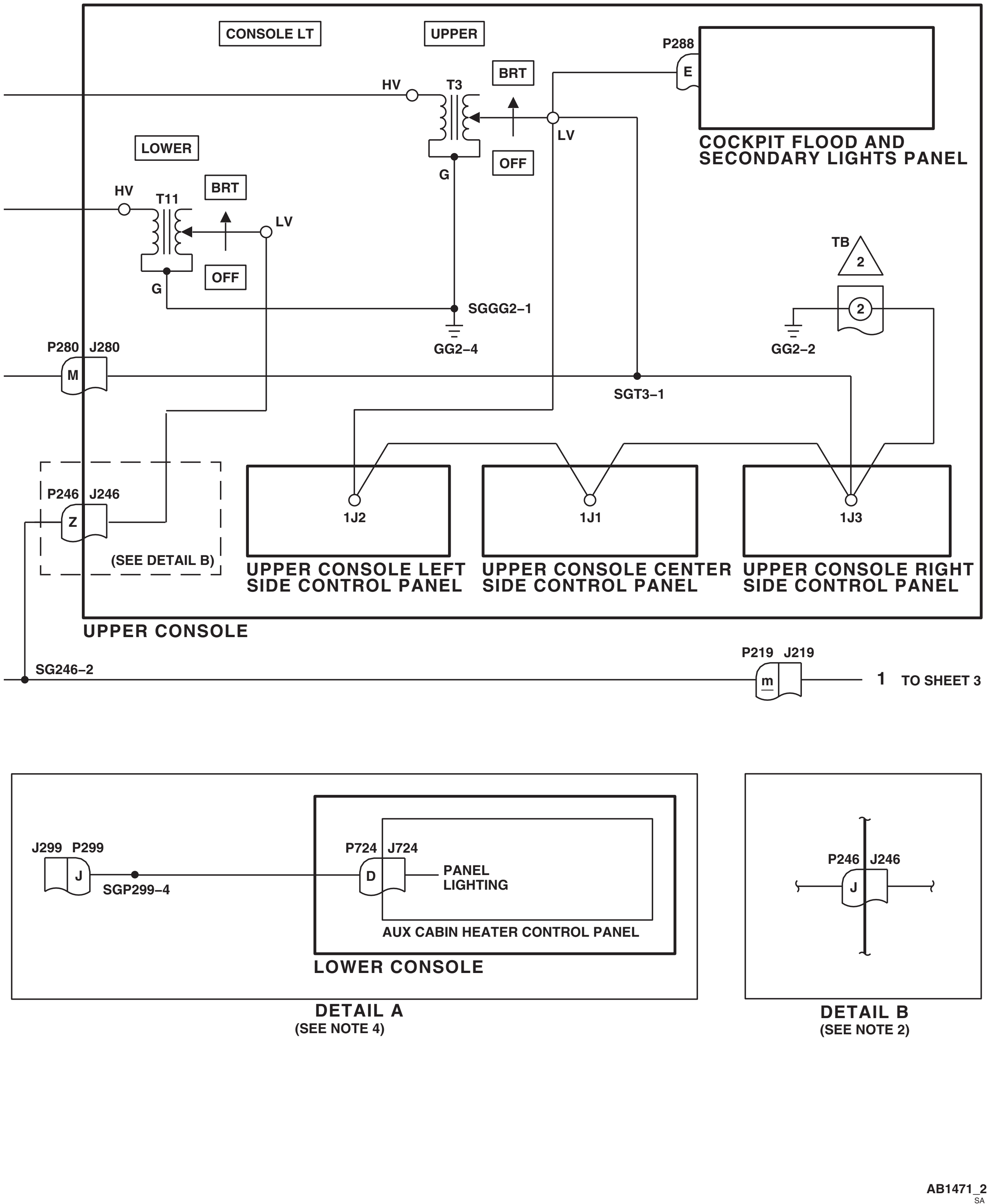


Figure 2. Upper and Lower Console Lights Schematic Diagram

EH-60A UH-60A UH-60L 89-26149-96-26722 W/O RIS W/O MWO 50-78 . (Sheet 2 of 4)

SCHEMATICS - Continued

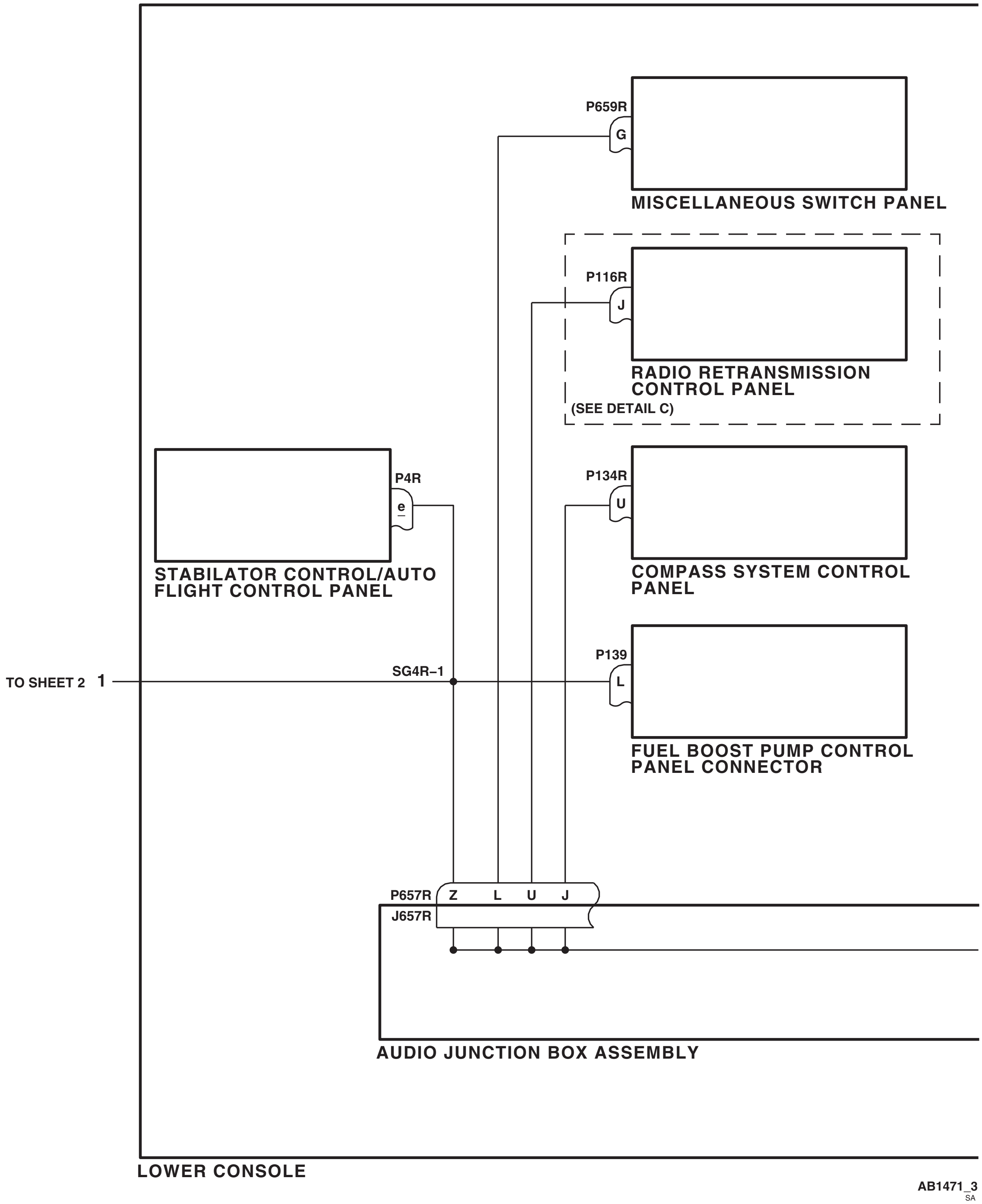


Figure 2. Upper and Lower Console Lights Schematic Diagram

EH-60A UH-60A UH-60L 89-26149-96-26722 W/O RIS W/O MWO 50-78 . (Sheet 3 of 4)

SCHEMATICS - Continued

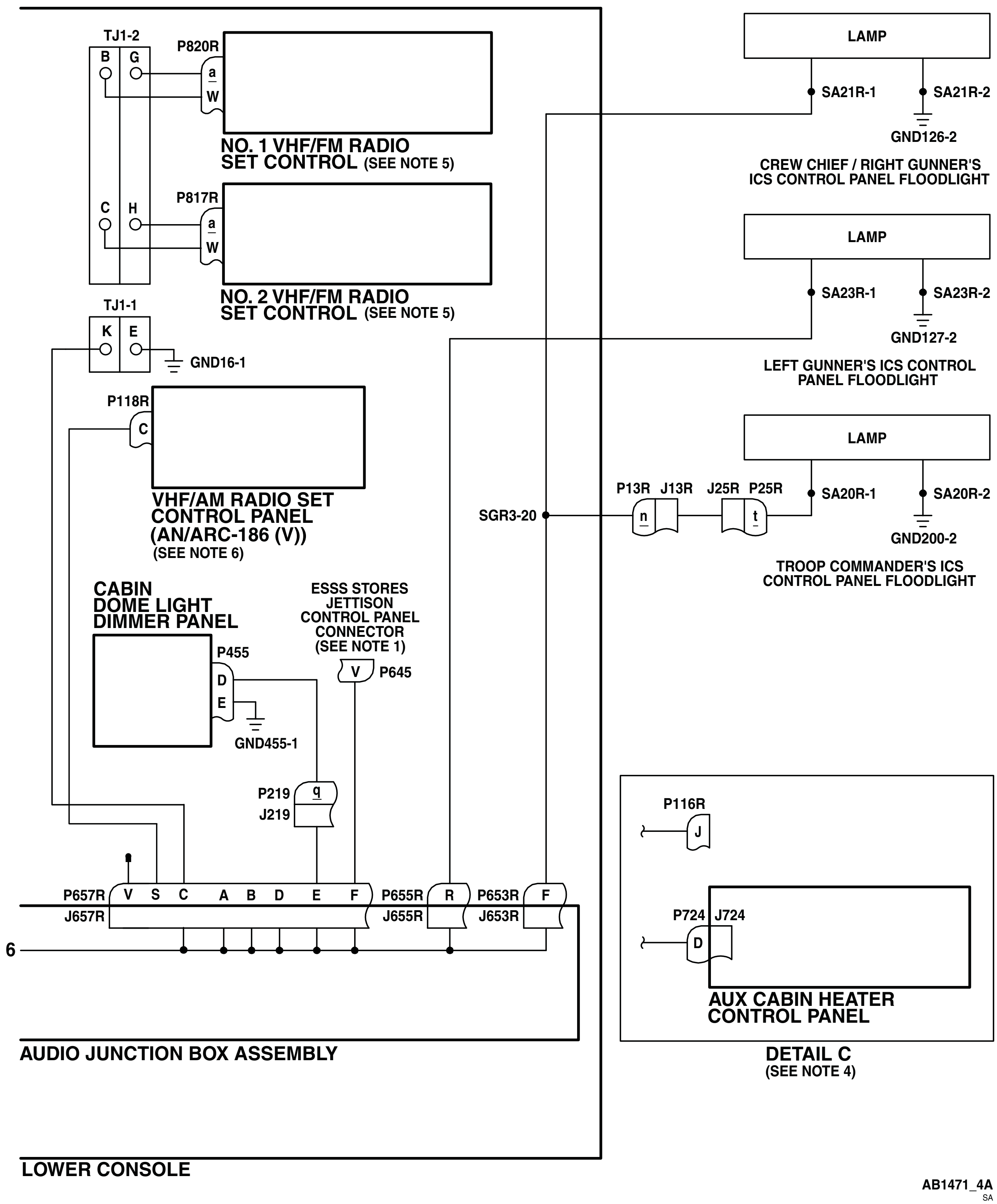


Figure 2. Upper and Lower Console Lights Schematic Diagram

EH-60A UH-60A UH-60L 89-26149-96-26722 W/O RIS W/O MWO 50-78 . (Sheet 4 of 4)

SCHEMATICS - Continued

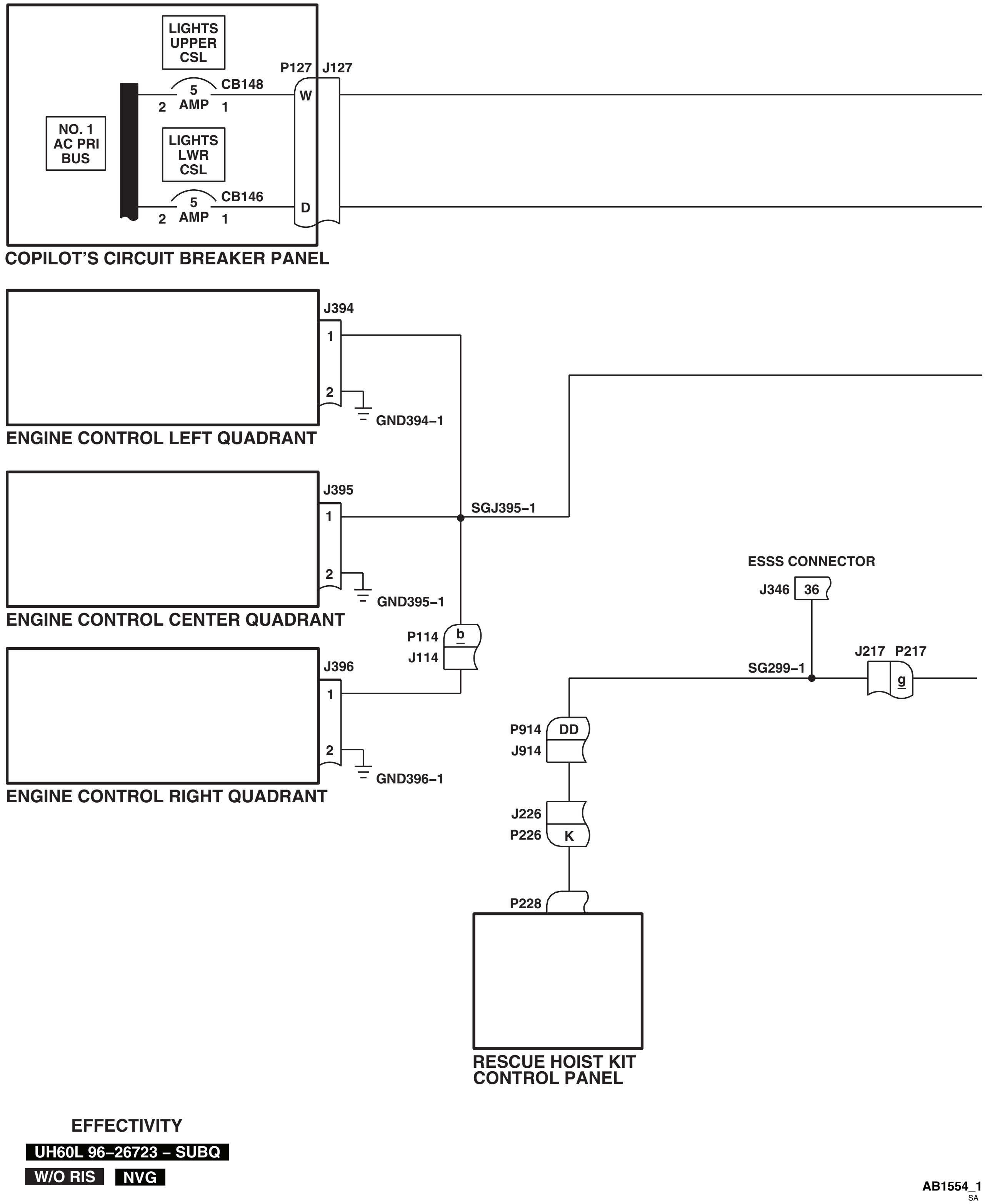


Figure 3. Upper and Lower Console Lights Schematic Diagram UH-60L 96-26723-SUBQ W/O RIS NVG . (Sheet 1 of 4)

SCHEMATICS - Continued

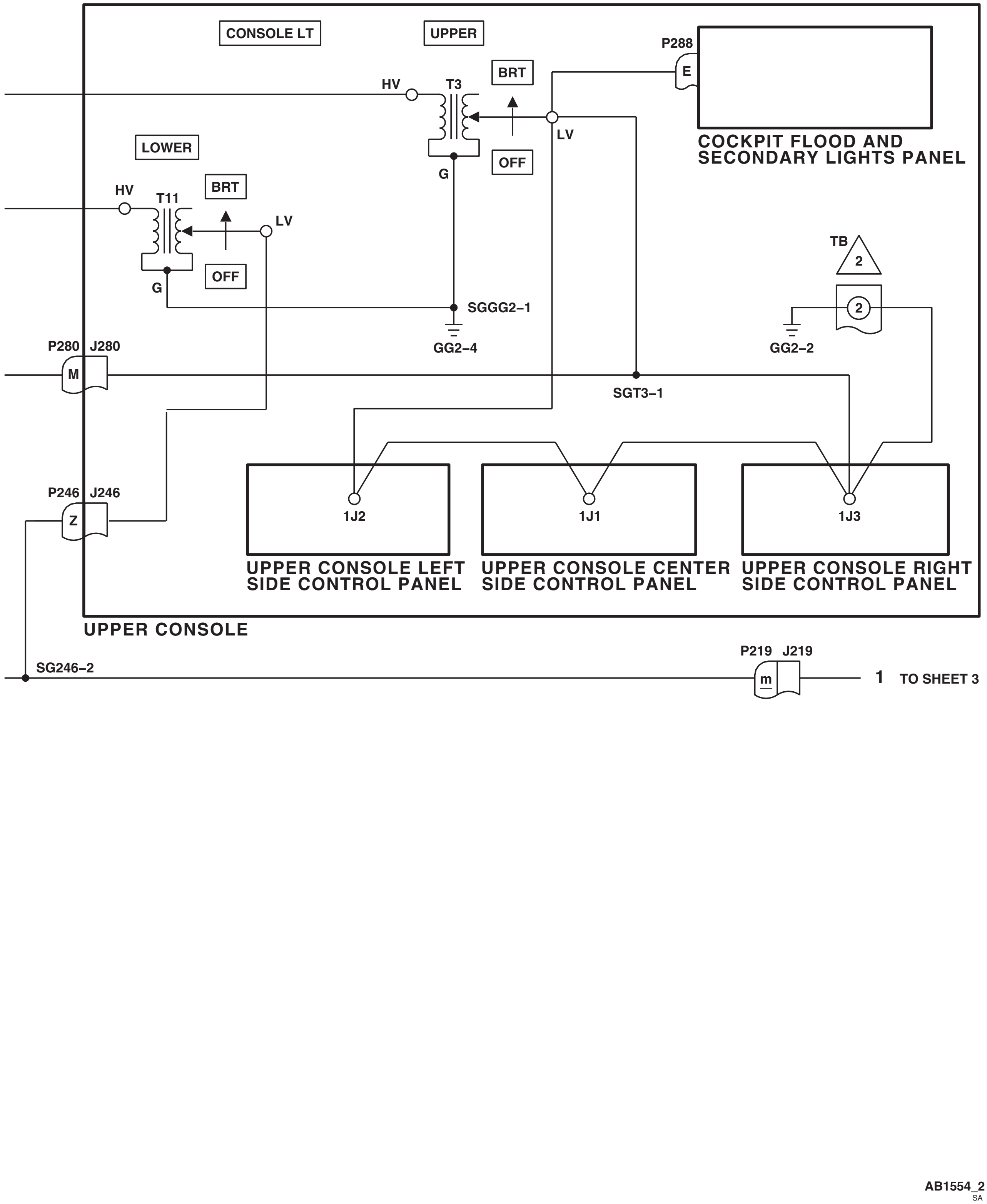


Figure 3. Upper and Lower Console Lights Schematic Diagram UH-60L 96-26723-SUBQ W/O RIS NVG . (Sheet 2 of 4)

SCHEMATICS - Continued

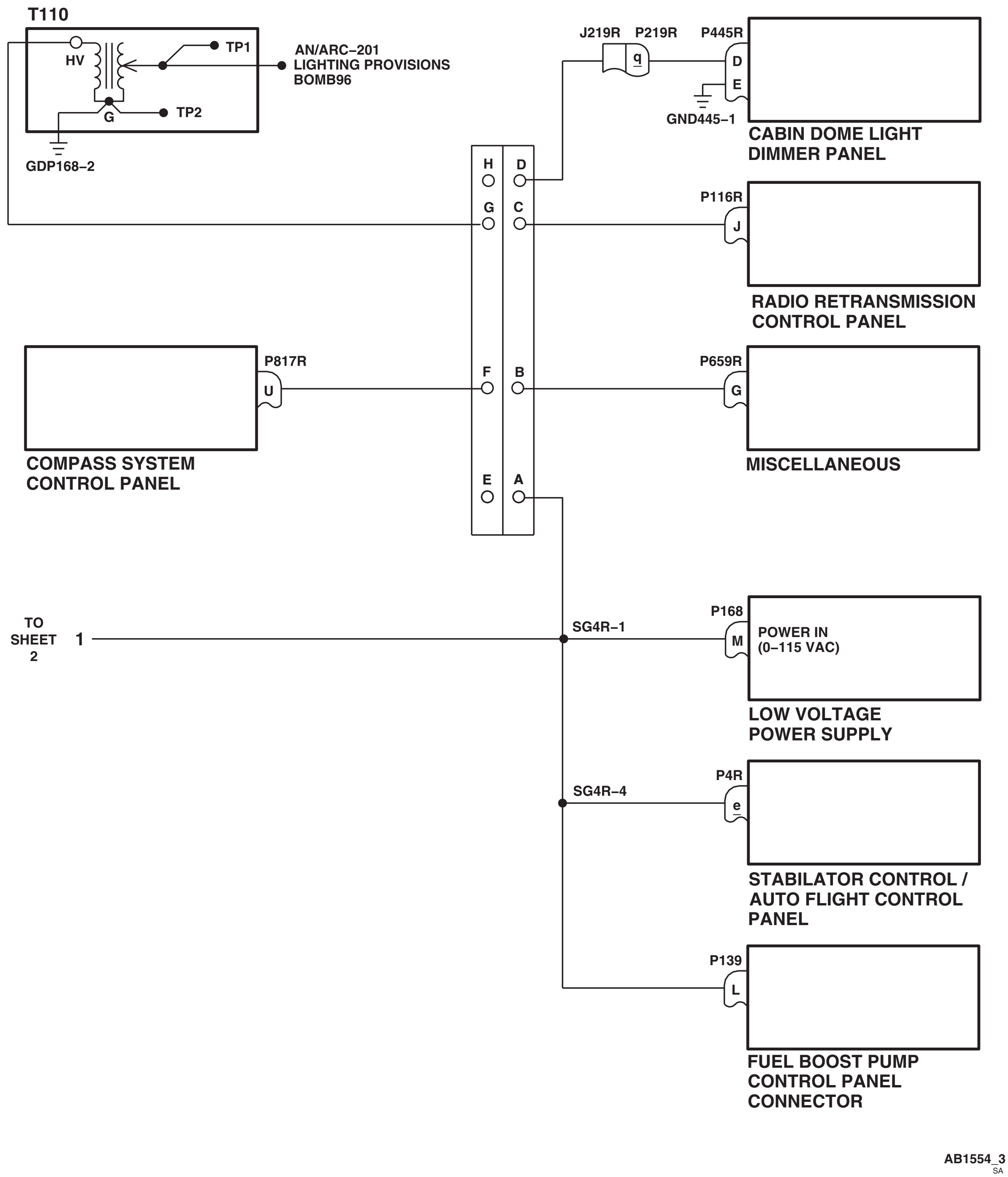


Figure 3. Upper and Lower Console Lights Schematic Diagram UH-60L 96-26723-SUBQ W/O RIS NVG . (Sheet 3 of 4)

SCHEMATICS - Continued

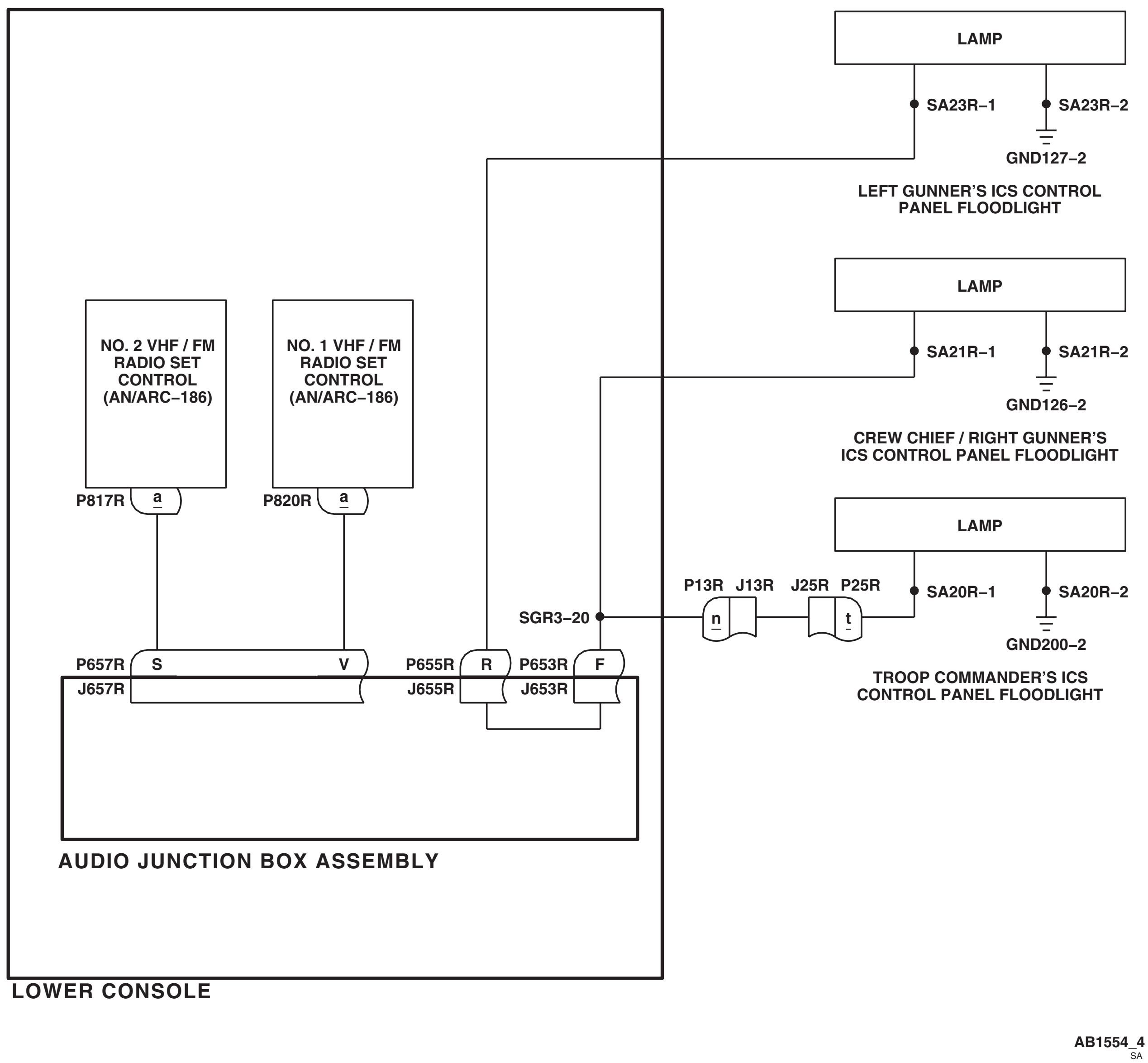


Figure 3. Upper and Lower Console Lights Schematic Diagram **UH-60L 96-26723-SUBQ W/O RIS NVG** . (Sheet 4 of 4)

SCHEMATICS - Continued

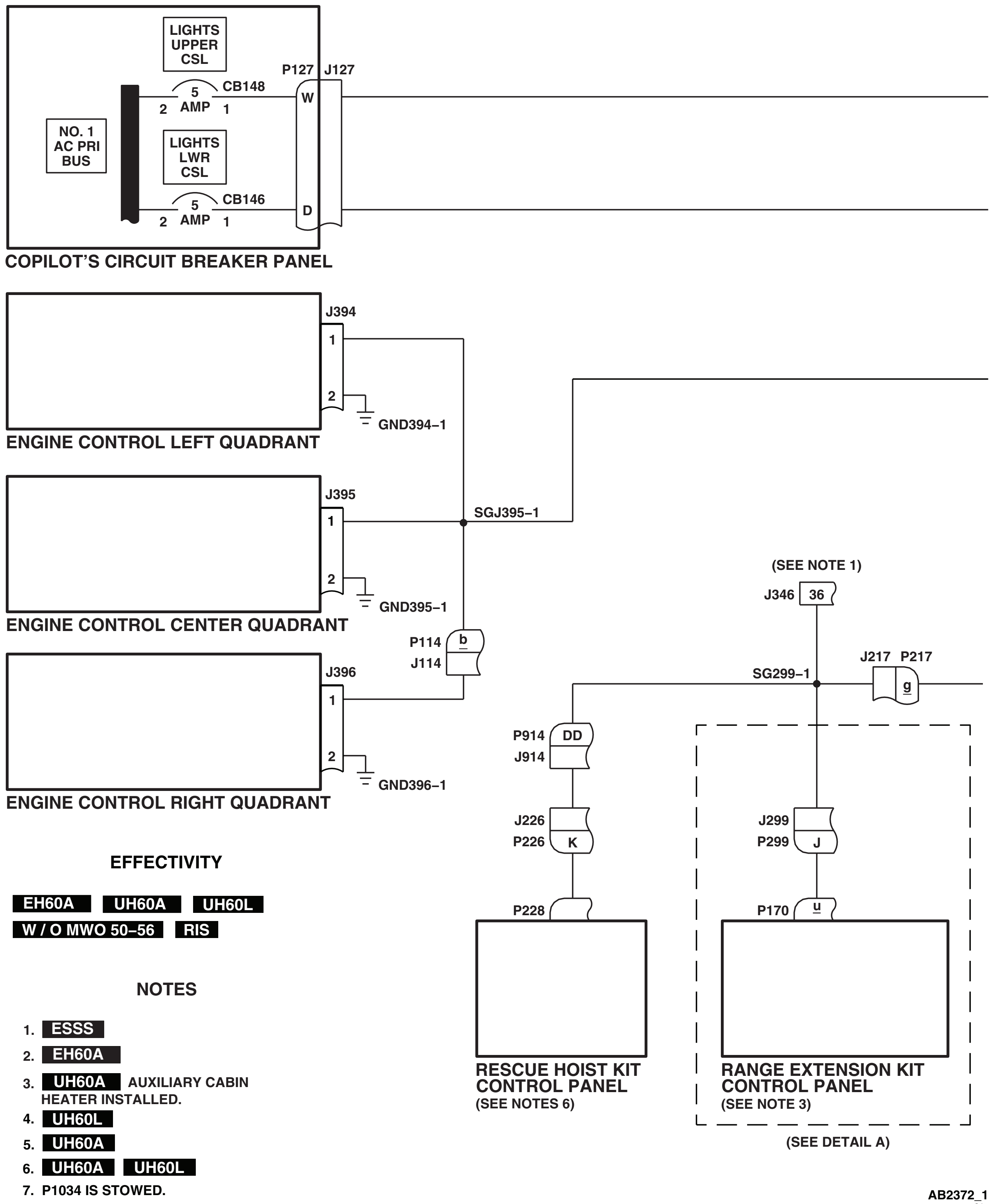
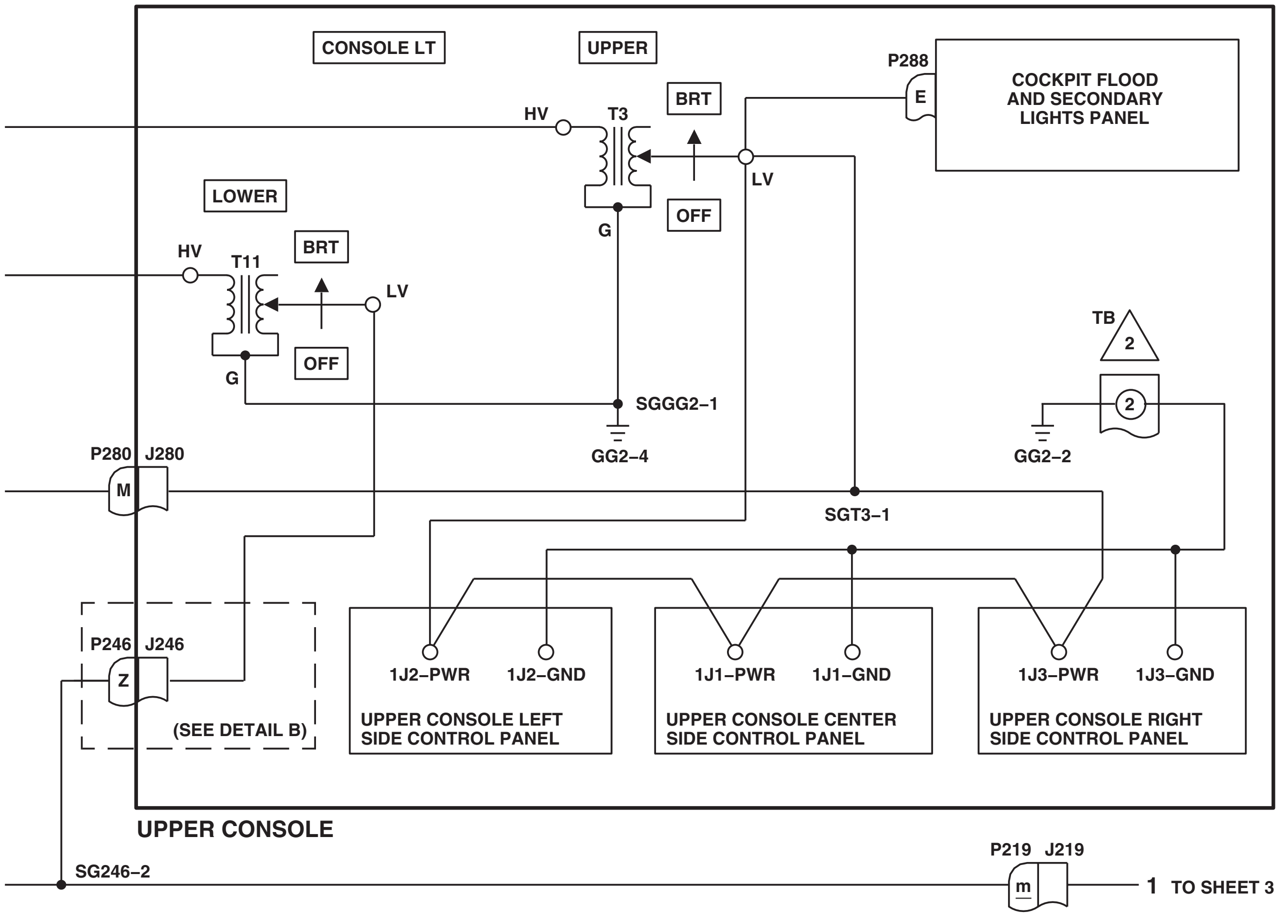
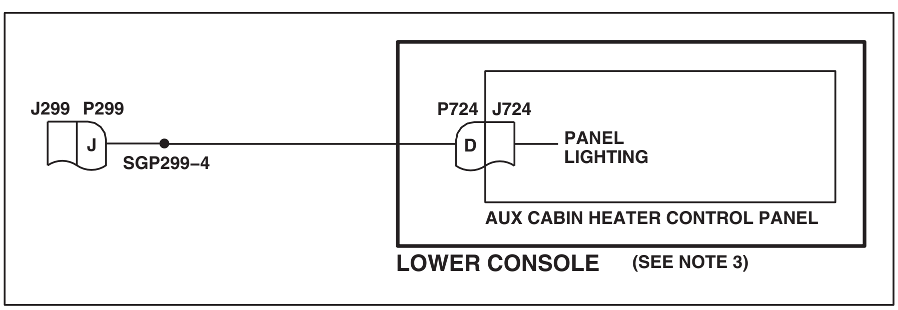


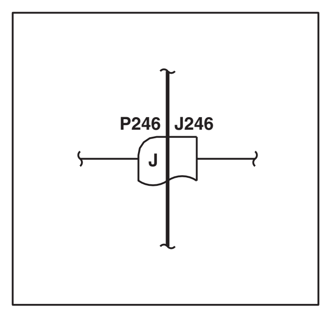
Figure 4. Upper and Lower Console Lights Schematic Diagram EH-60A UH-60A UH-60L W/0 MWO 50-56 RIS .
(Sheet 1 of 7)

SCHEMATICS - Continued





DETAIL A
(SEE NOTE 4)



DETAIL B
(SEE NOTE 2)

AB1555_2
SA

Figure 4. Upper and Lower Console Lights Schematic Diagram **EH-60A UH-60A UH-60L W/0 MWO 50-56 RIS** .
(Sheet 2 of 7)

SCHEMATICS - Continued

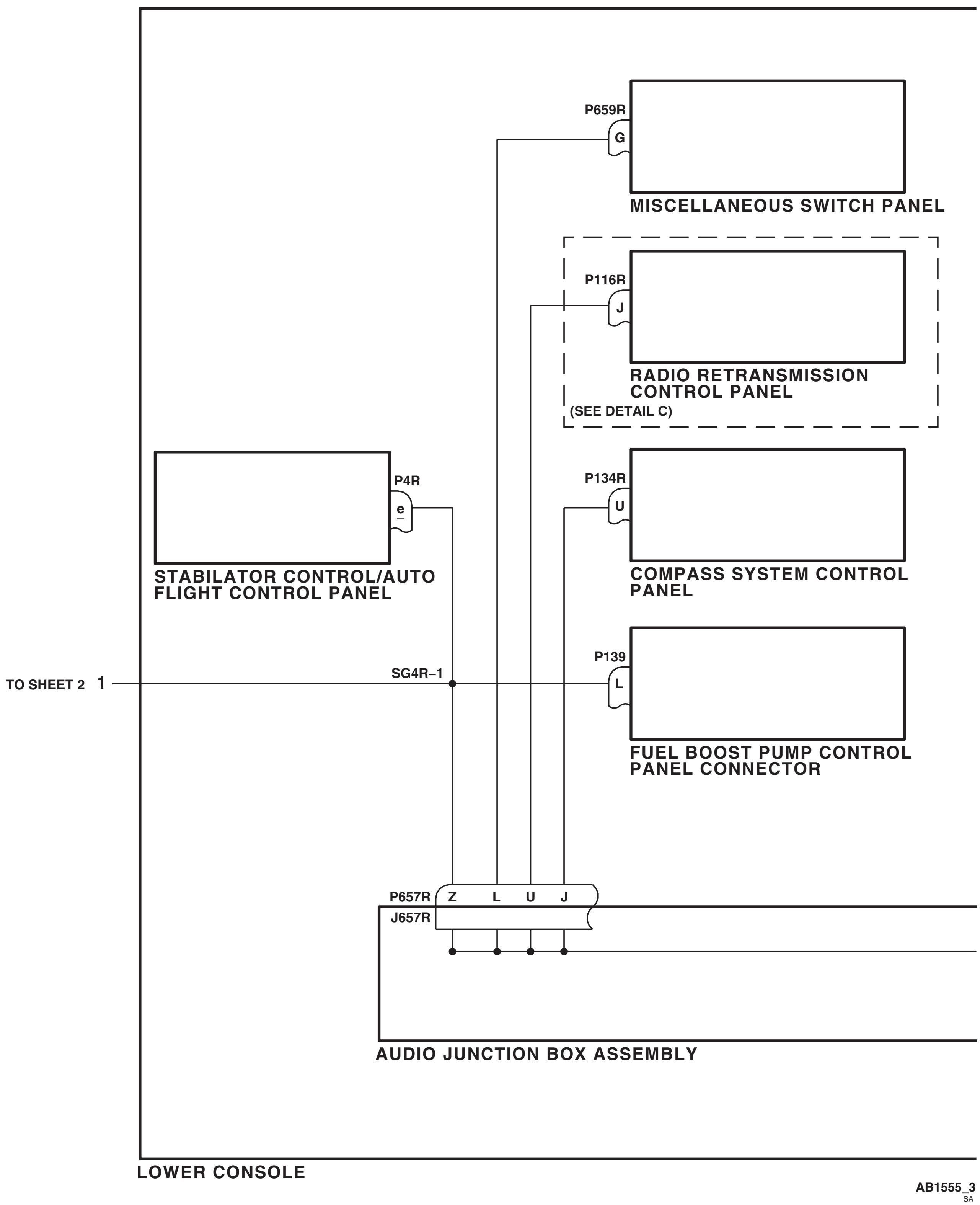


Figure 4. Upper and Lower Console Lights Schematic Diagram **EH-60A UH-60A UH-60L W/0 MWO 50-56 RIS** .
(Sheet 3 of 7)

SCHEMATICS - Continued

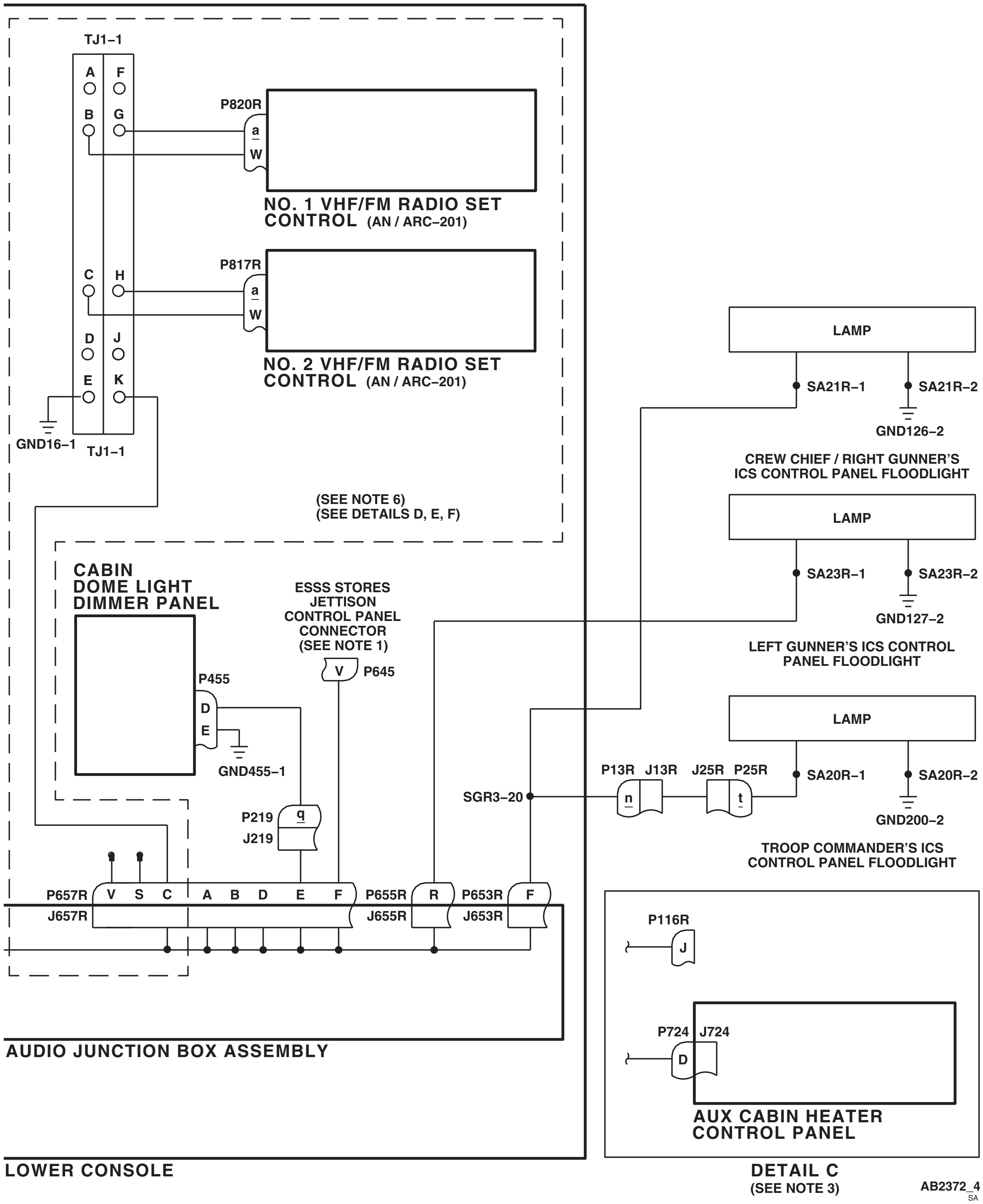
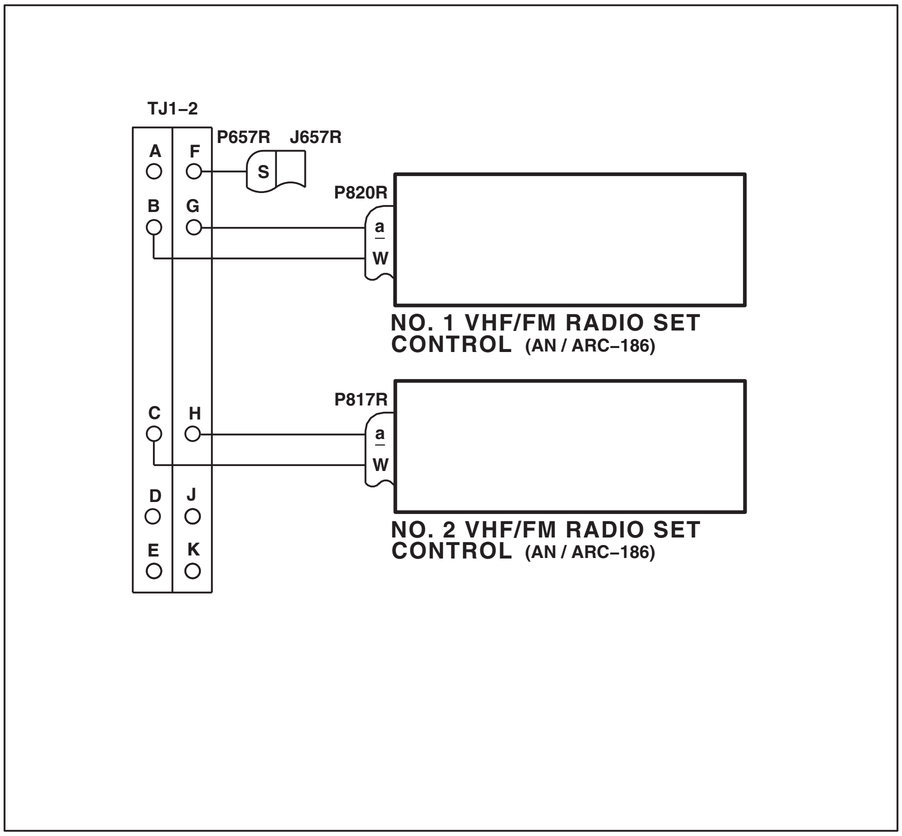


Figure 4. Upper and Lower Console Lights Schematic Diagram **EH-60A UH-60A UH-60L W/0 MWO 50-56 RIS** .
(Sheet 4 of 7)

SCHEMATICS - Continued



DETAIL D
(SEE NOTE 5)

AB2372_5A
SA

Figure 4. Upper and Lower Console Lights Schematic Diagram EH-60A UH-60A UH-60L W/0 MWO 50-56 RIS .
(Sheet 5 of 7)

SCHEMATICS - Continued

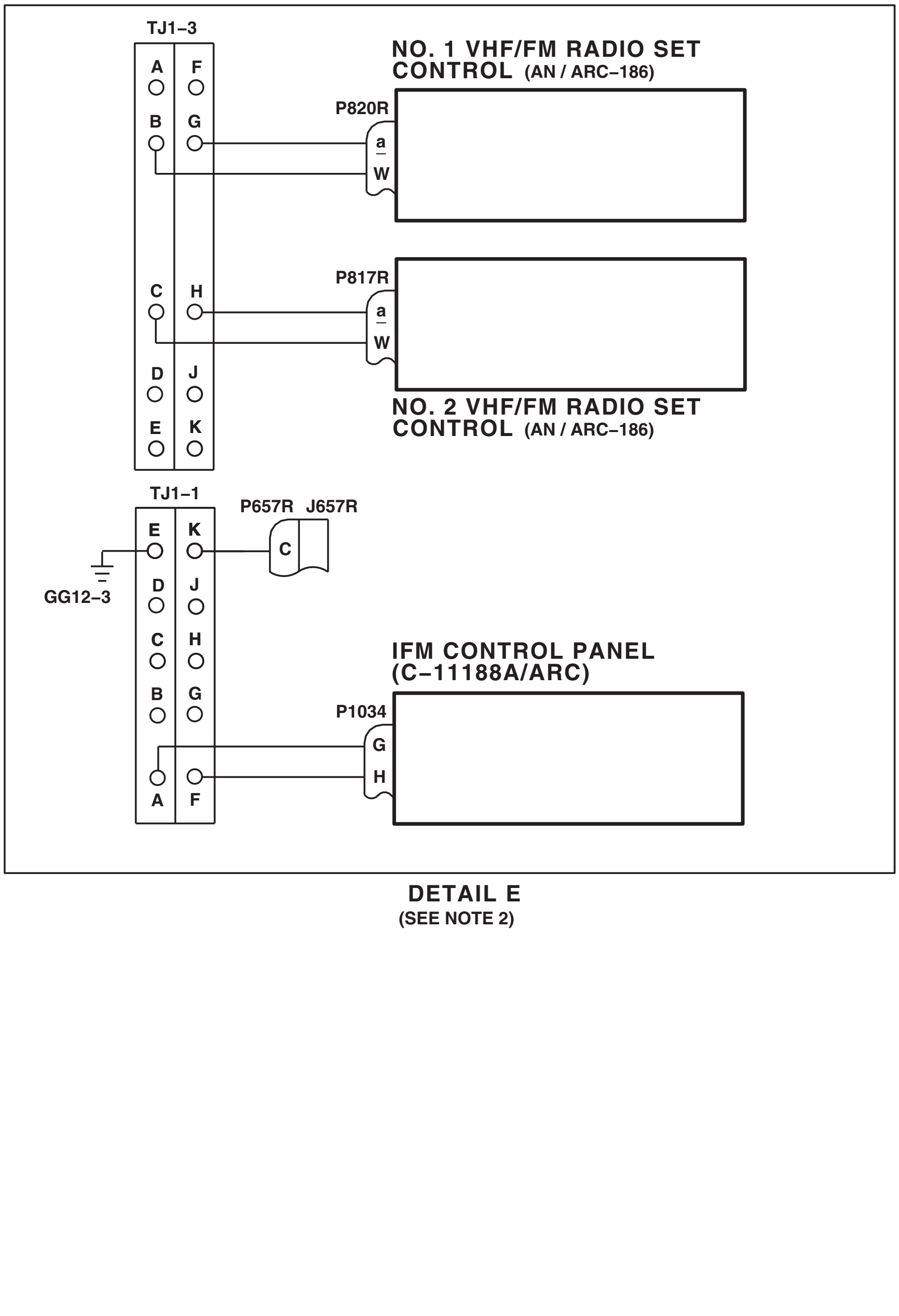


Figure 4. Upper and Lower Console Lights Schematic Diagram EH-60A UH-60A UH-60L W/0 MWO 50-56 RIS .
(Sheet 6 of 7)

SCHEMATICS - Continued

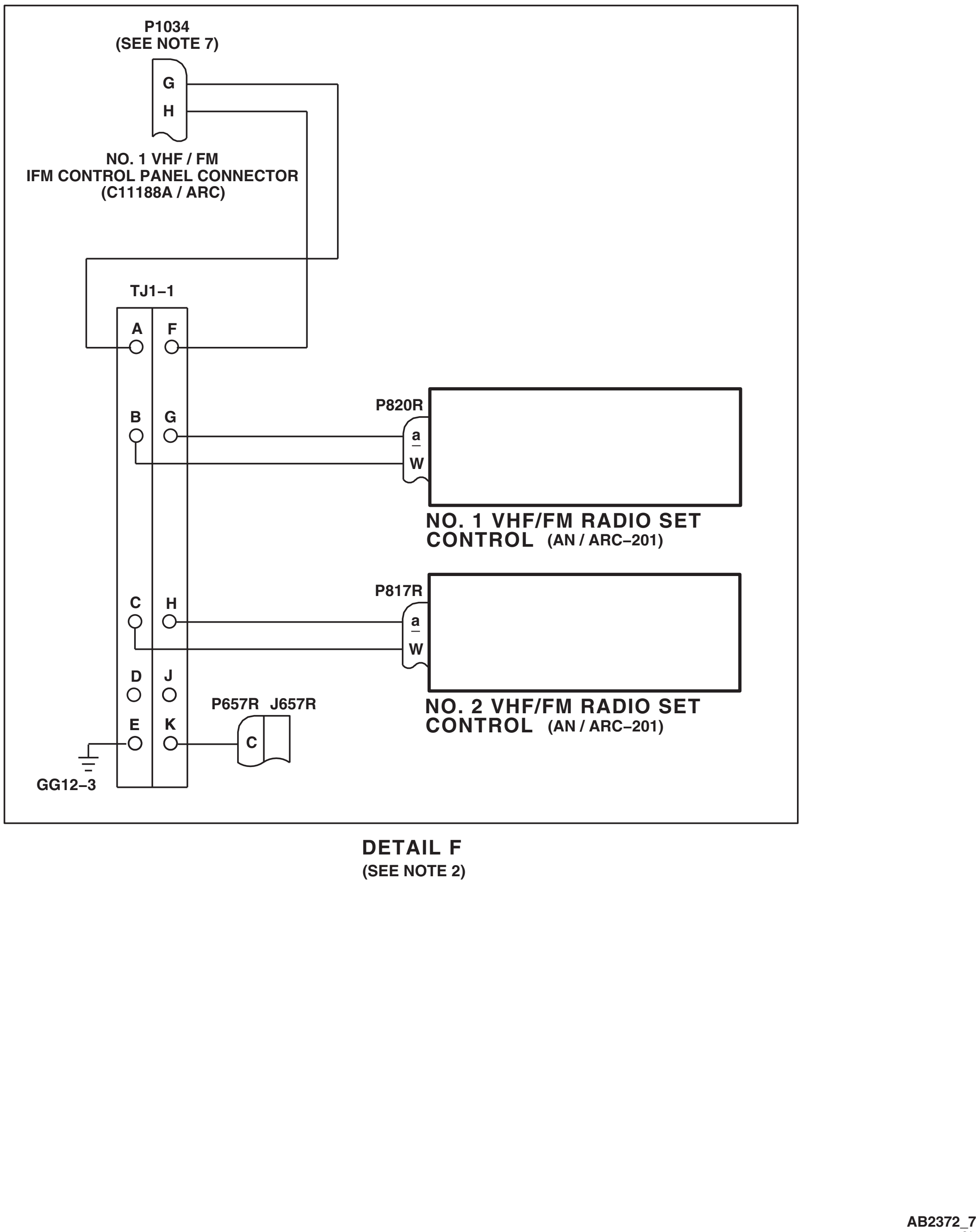


Figure 4. Upper and Lower Console Lights Schematic Diagram EH-60A UH-60A UH-60L W/0 MWO 50-56 RIS .
(Sheet 7 of 7)

SCHEMATICS - Continued

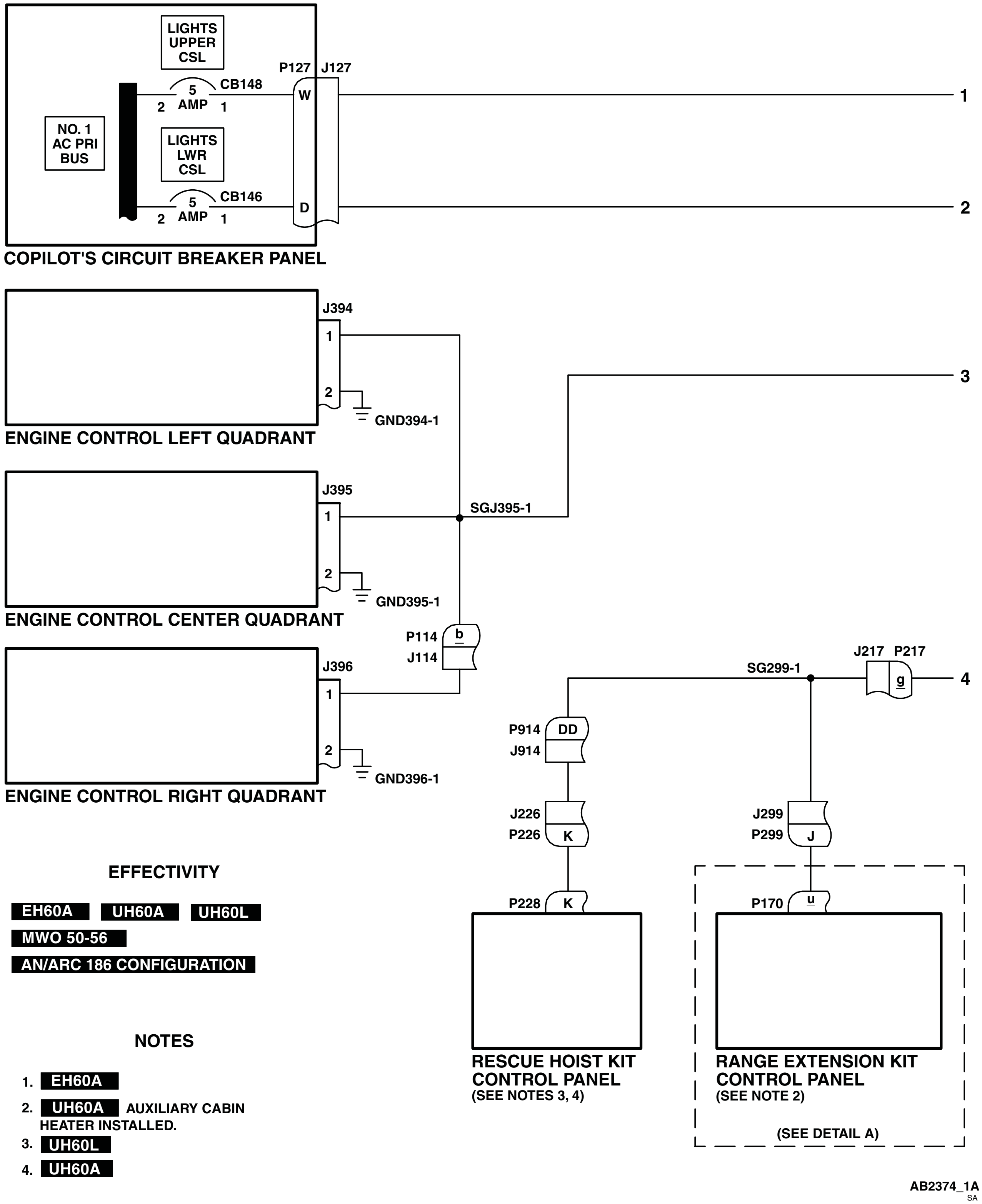
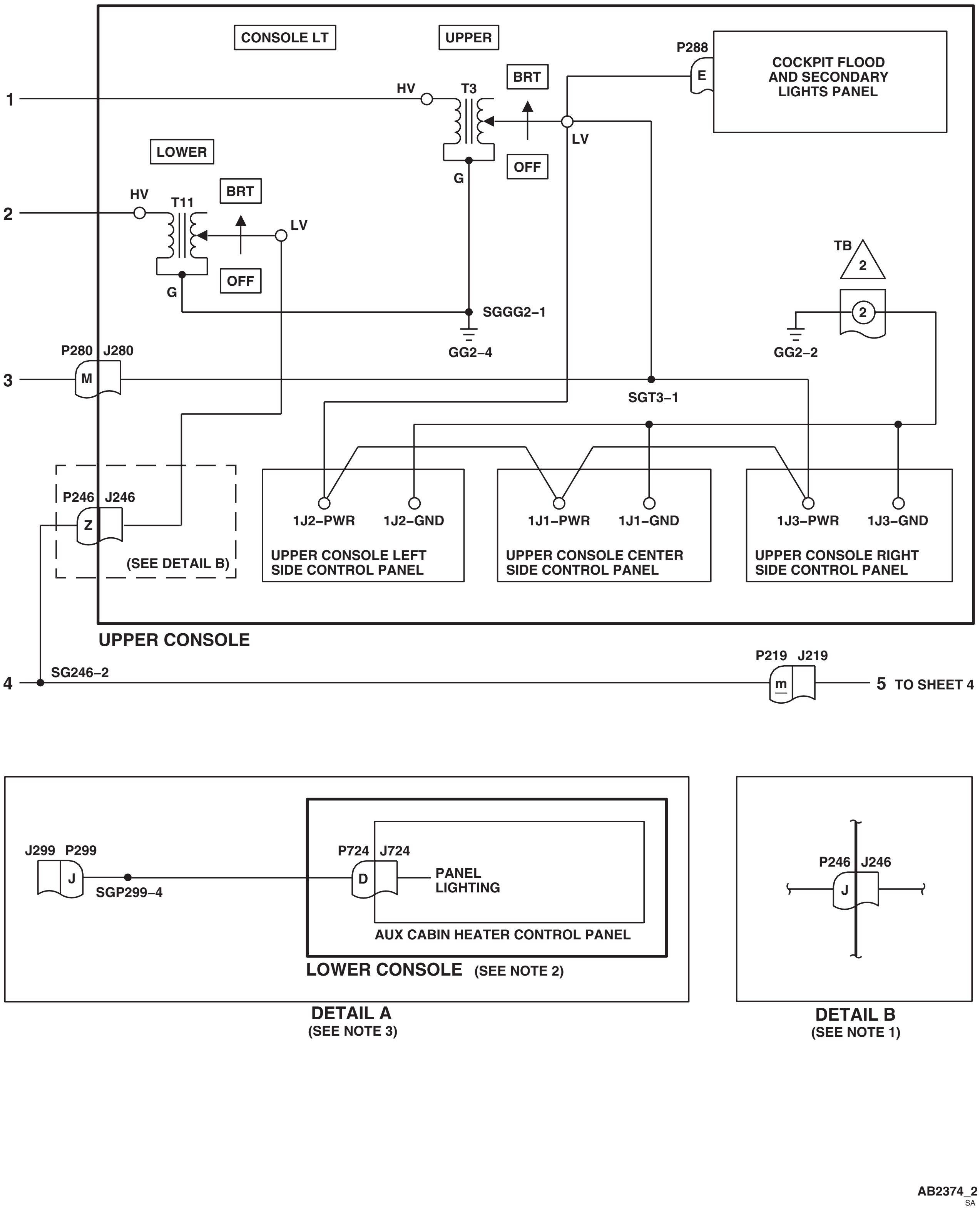


Figure 5. Upper and Lower Console Lights Schematic Diagram
EH-60A UH-60A UH-60L MWO 50-56 AN/ARC 186 CONFIGURATION. (Sheet 1 of 5)

SCHEMATICS - Continued



DETAIL A
(SEE NOTE 3)

DETAIL B
(SEE NOTE 1)

Figure 5. Upper and Lower Console Lights Schematic Diagram
EH-60A UH-60A UH-60L MWO 50-56 AN/ARC 186 CONFIGURATION . (Sheet 2 of 5)

SCHEMATICS - Continued

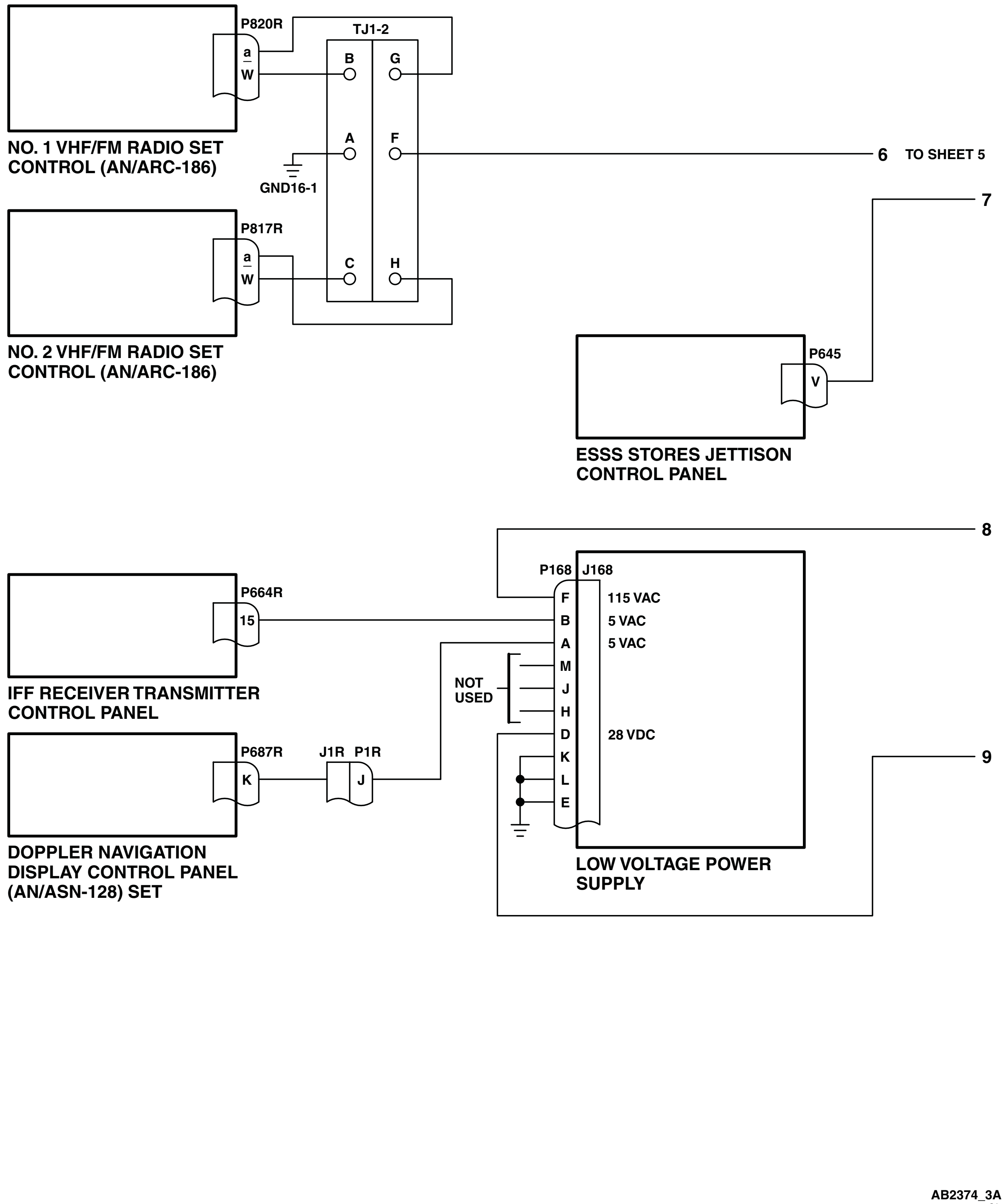


Figure 5. Upper and Lower Console Lights Schematic Diagram
EH-60A UH-60A UH-60L MWO 50-56 AN/ARC 186 CONFIGURATION . (Sheet 3 of 5)

SCHEMATICS - Continued

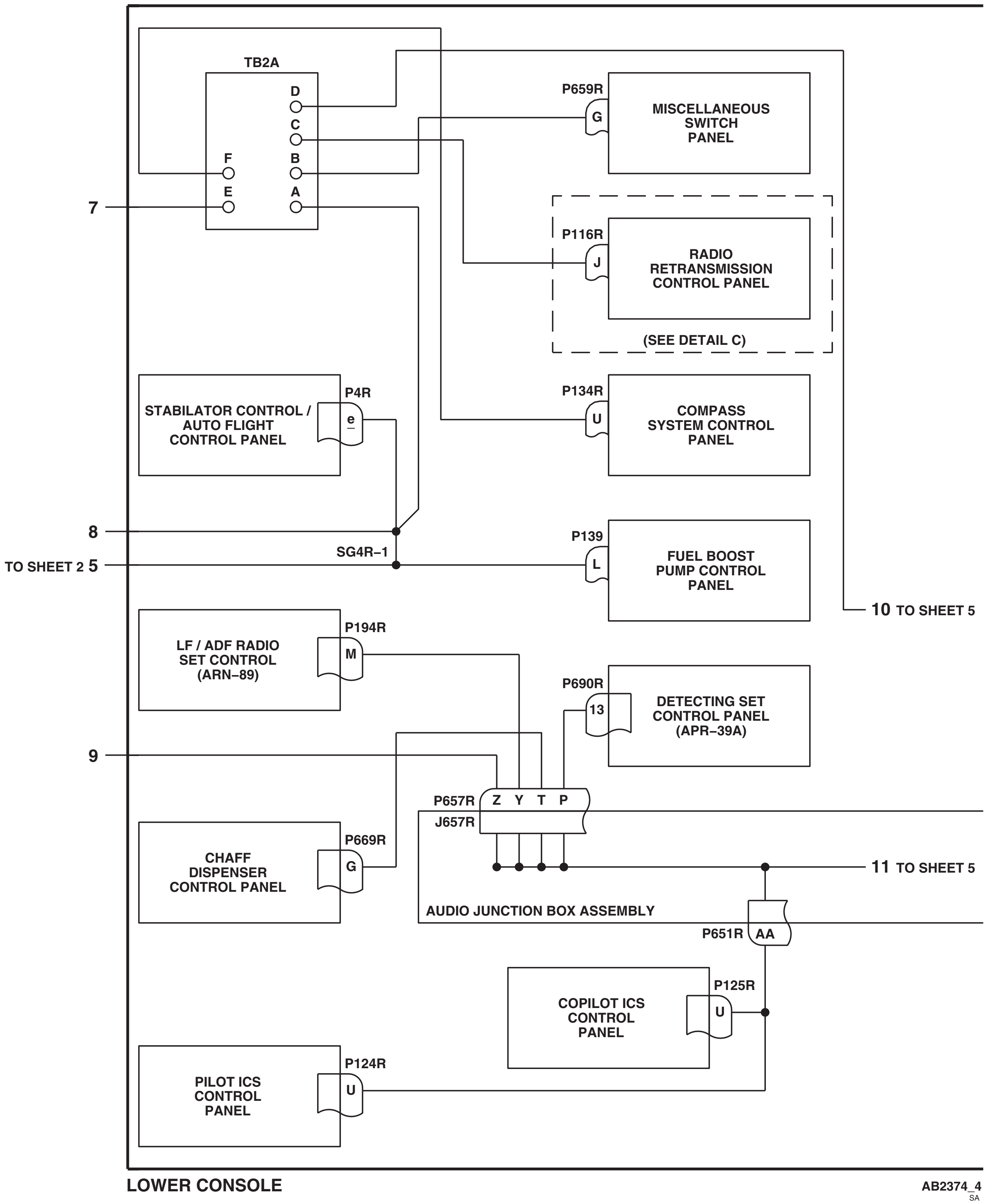


Figure 5. Upper and Lower Console Lights Schematic Diagram

EH-60A UH-60A UH-60L MWO 50-56 AN/ARC 186 CONFIGURATION . (Sheet 4 of 5)

SCHEMATICS - Continued

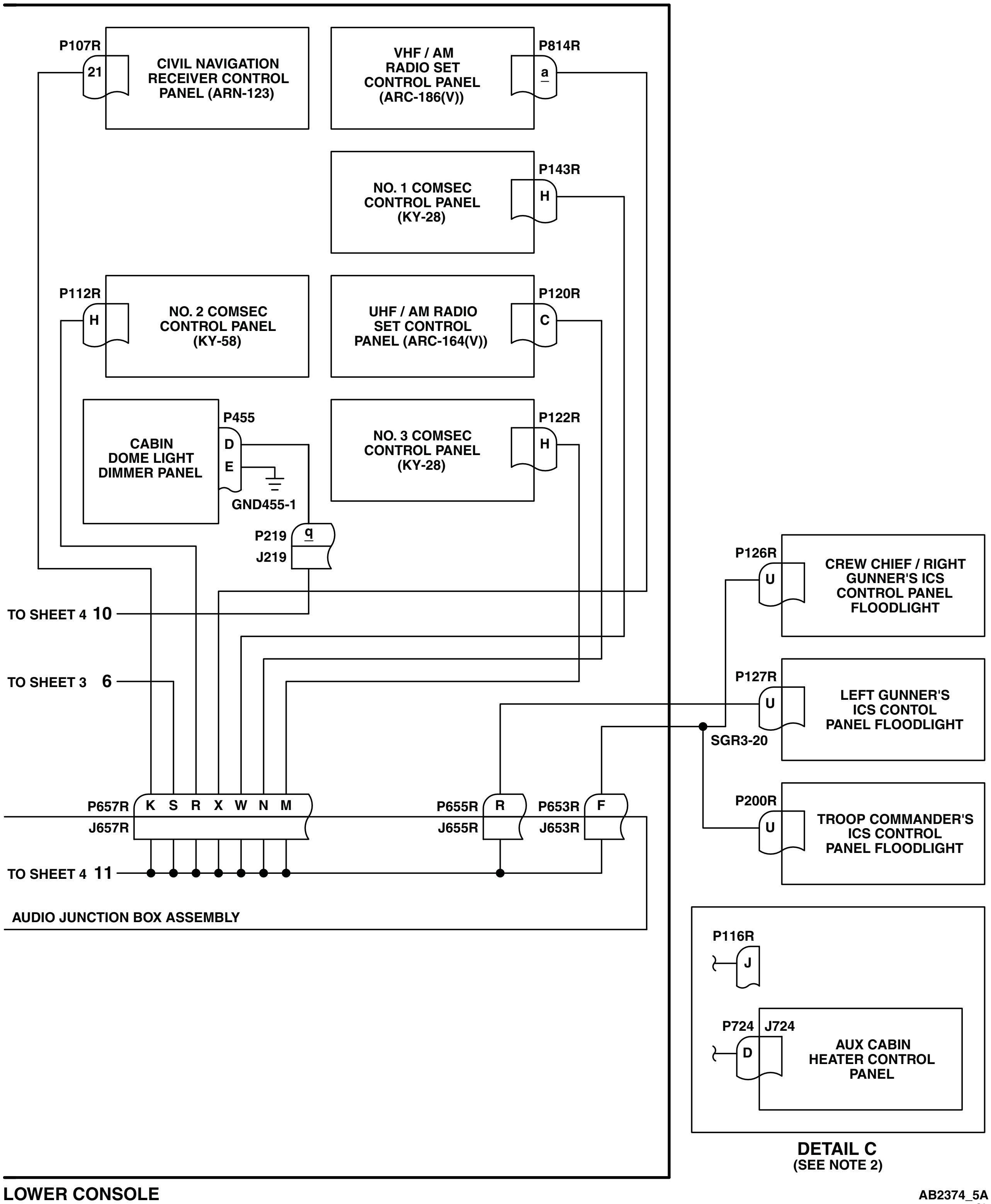


Figure 5. Upper and Lower Console Lights Schematic Diagram

EH-60A UH-60A UH-60L MWO 50-56 AN/ARC 186 CONFIGURATION . (Sheet 5 of 5)

SCHEMATICS - Continued

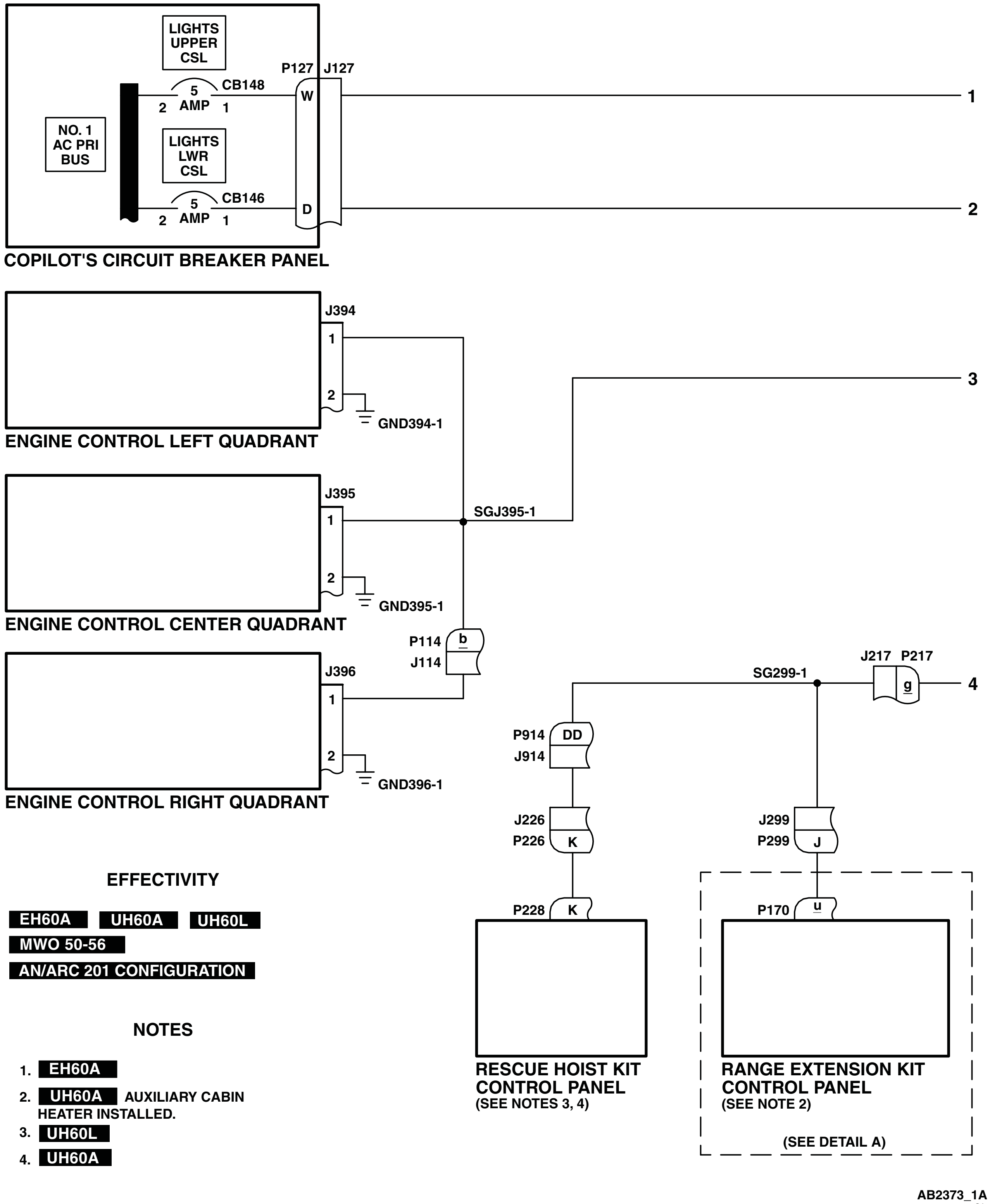


Figure 6. Upper and Lower Console Lights Schematic Diagram
EH-60A UH-60A UH-60L MWO 50-56 AN/ARC 201 CONFIGURATION . (Sheet 1 of 5)

SCHEMATICS - Continued

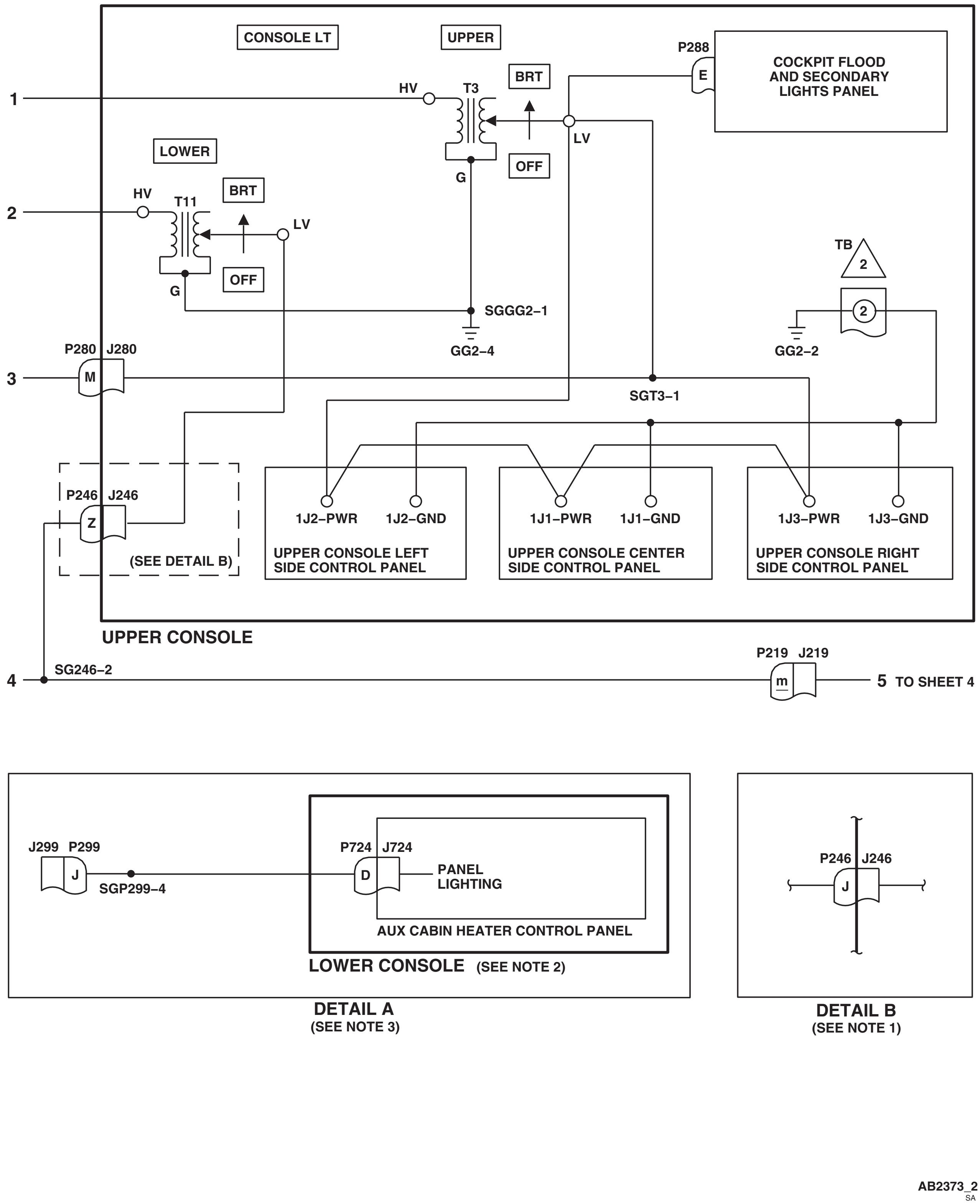


Figure 6. Upper and Lower Console Lights Schematic Diagram

EH-60A UH-60A UH-60L MWO 50-56 AN/ARC 201 CONFIGURATION . (Sheet 2 of 5)

SCHEMATICS - Continued

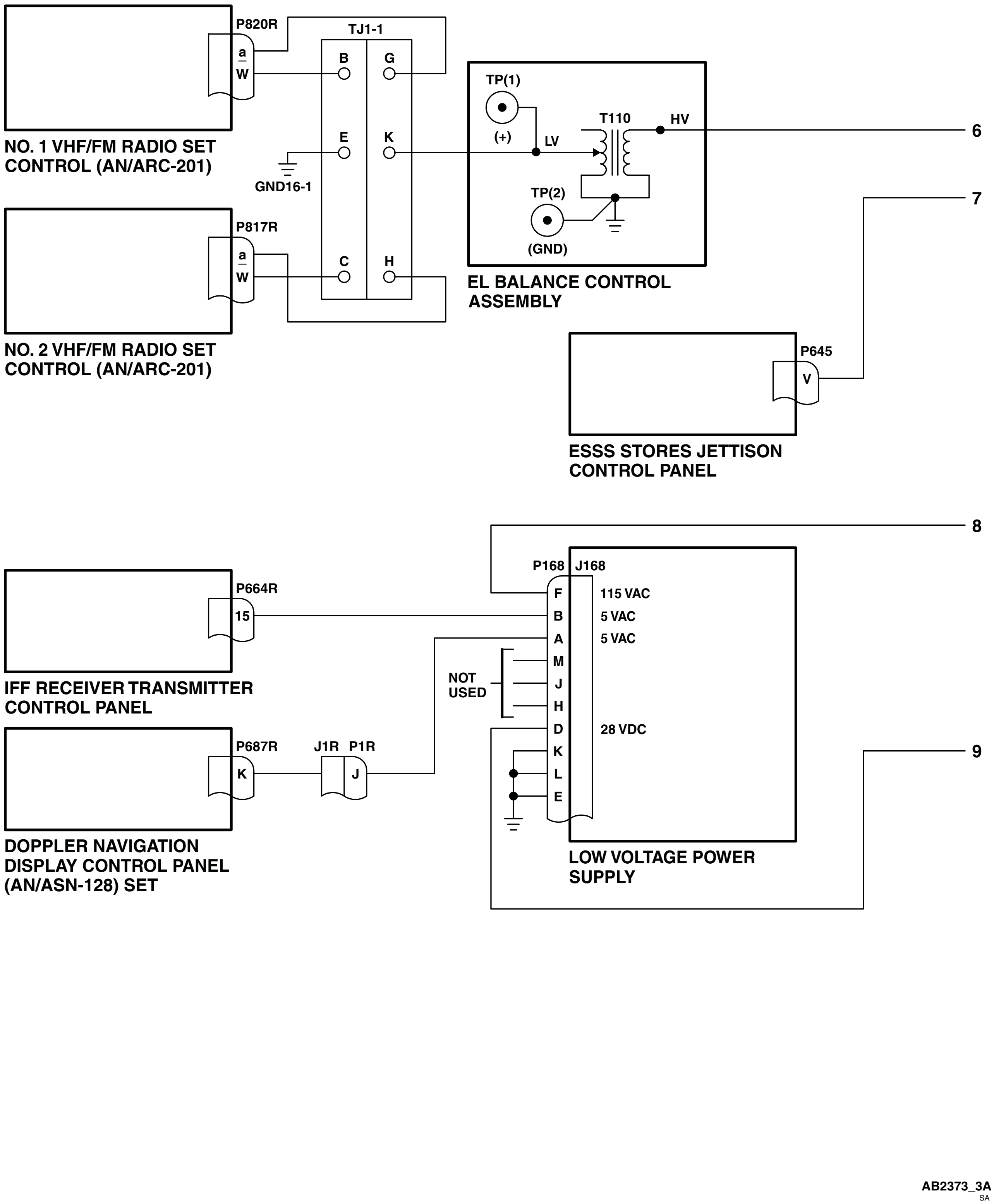


Figure 6. Upper and Lower Console Lights Schematic Diagram

EH-60A UH-60A UH-60L MWO 50-56 AN/ARC 201 CONFIGURATION . (Sheet 3 of 5)

SCHEMATICS - Continued

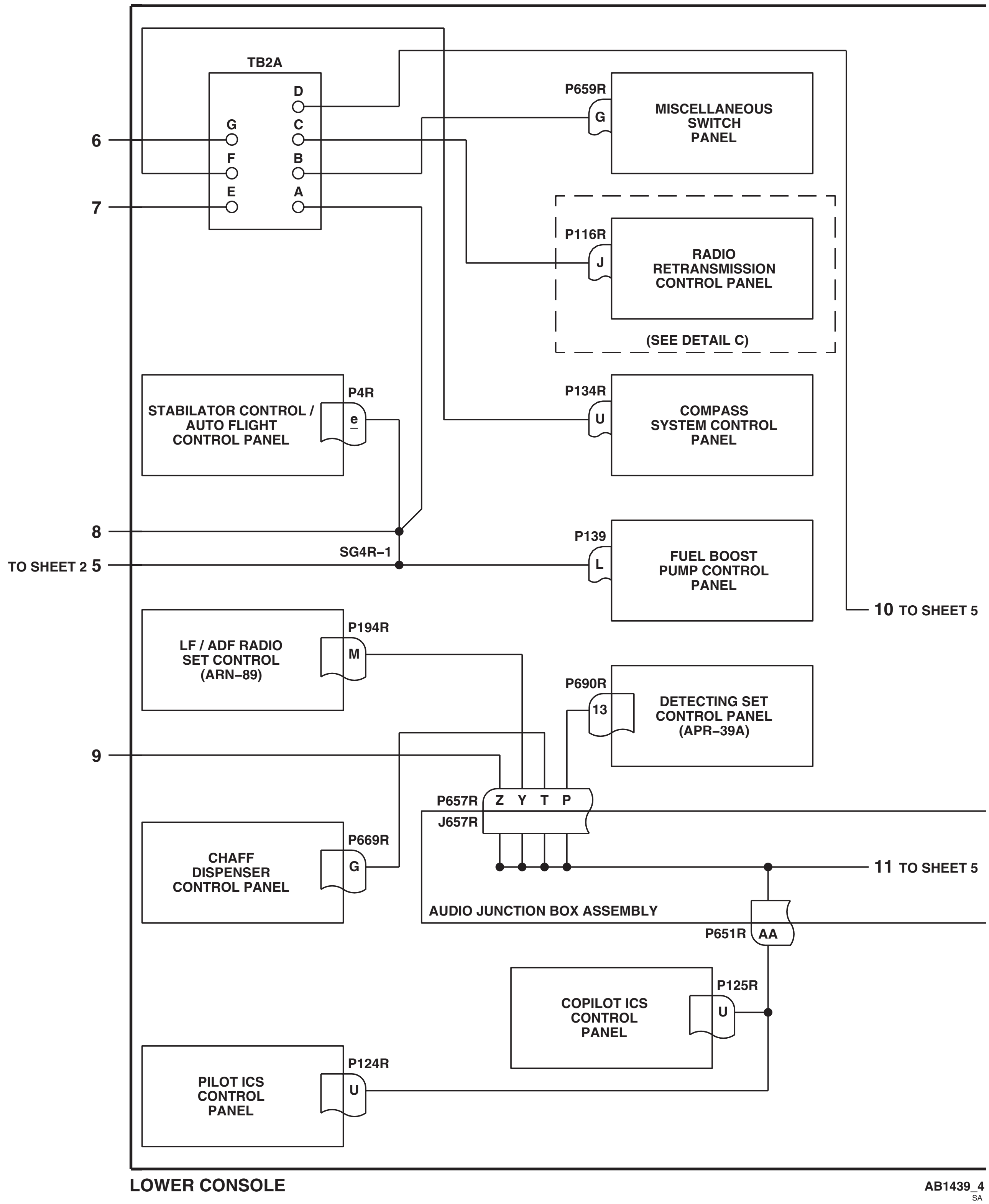


Figure 6. Upper and Lower Console Lights Schematic Diagram

EH-60A UH-60A UH-60L MWO 50-56 AN/ARC 201 CONFIGURATION . (Sheet 4 of 5)

SCHEMATICS - Continued

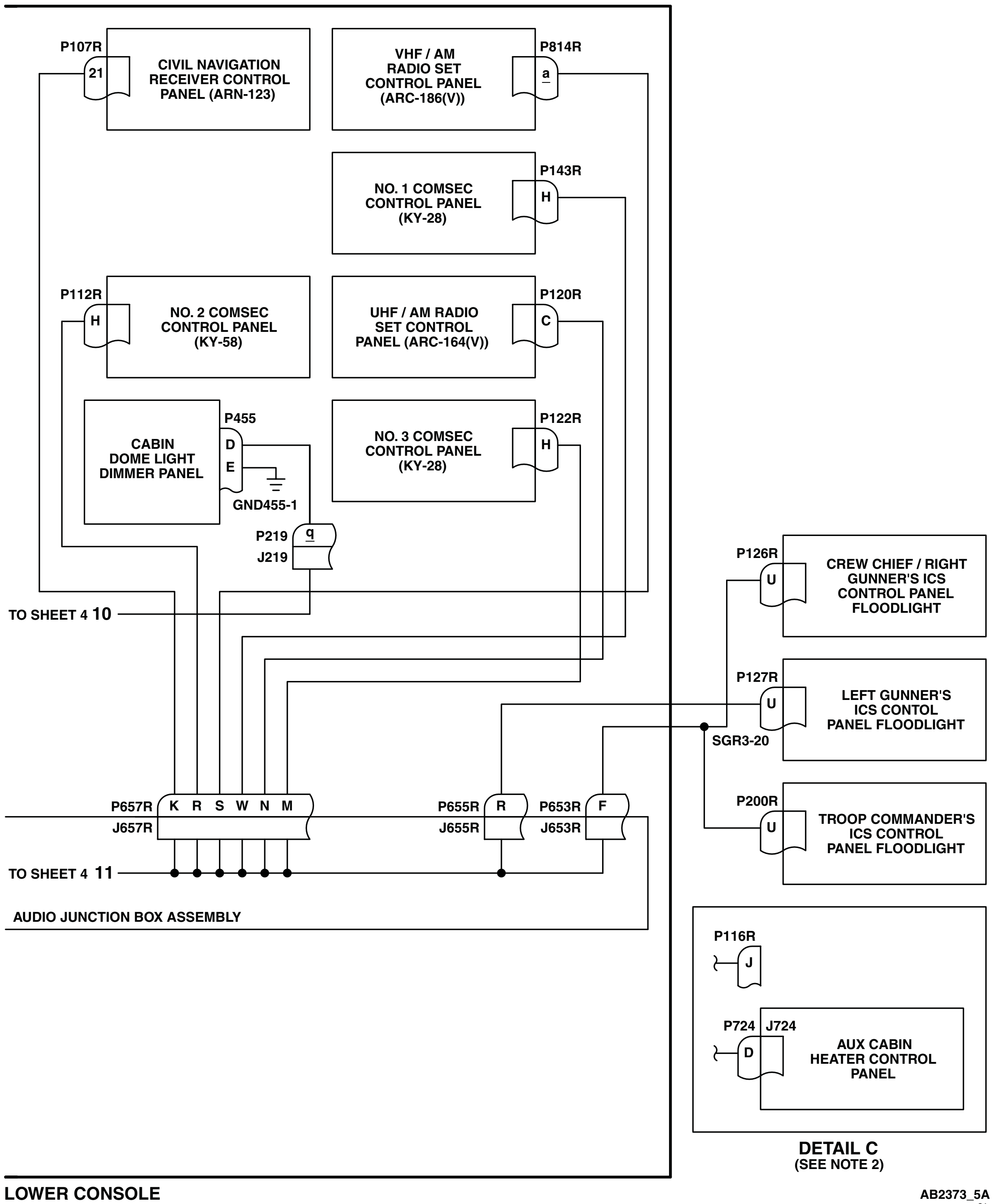


Figure 6. Upper and Lower Console Lights Schematic Diagram
EH-60A UH-60A UH-60L MWO 50-56 AN/ARC 201 CONFIGURATION . (Sheet 5 of 5)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

CABIN DOME LIGHTS

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
WP 0104 00
WP 0824 00
WP 0828 00
WP 0876 00
WP 0887 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

CABIN DOME LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- RIS modified helicopter's have the lower console lighting limited to 65 - 70 vac by a mechanical limit attachment to the lower console dimmer control.
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure LIGHTS CABIN DOME circuit breaker, on copilot's circuit breaker panel, is pushed in.
2. Turn on electrical power.
3. Turn CABIN DOME LT BRT/OFF control to BRT.
4. Place upper console CABIN DOME LT WHITE/OFF/BLUE switch to WHITE.

Indication/Condition

All white cabin dome lights shall go on.

Corrective Action

1. If Indication/Condition is not as specified, go to NO WHITE CABIN DOME LIGHTS GO ON, in this work package.
2. If one white cabin dome light does not go on, check lamp.
   a. If lamp is not good, replace lamp (WP 0887 00).

CABIN DOME LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

b. If lamp is good, repair/replace wiring between CABIN DOME LT switch terminal 1 and the following, as required (WP 1747 00):

- J262-A
- J263-A
- J264-A

Step

5. Place upper console CABIN DOME LT WHITE/OFF/BLUE switch to BLUE.

Indication/Condition

All blue cabin dome lights shall go on.

Corrective Action

1. If Indication/Condition is not as specified, replace CABIN DOME LT switch (WP 0828 00).
2. If one blue cabin dome light does not go on, check lamp.
   a. If lamp is not good, replace lamp (WP 0887 00).
   b. If lamp is good, repair/replace wiring between CABIN DOME LT switch terminal 3 and the following, as required (WP 1747 00):
      - J262-B
      - J263-B
      - J264-B

Step

6. Turn CABIN DOME LT BRT/OFF control to OFF.

Indication/Condition

Lights shall dim and go off.

Corrective Action

1. If Indication/Condition is not as specified, replace CABIN DOME LT BRT/OFF dimmer control (WP 0876 00).

Step

7. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

NO WHITE CABIN DOME LIGHTS GO ON

SYMPTOM

No white cabin dome lights go on.

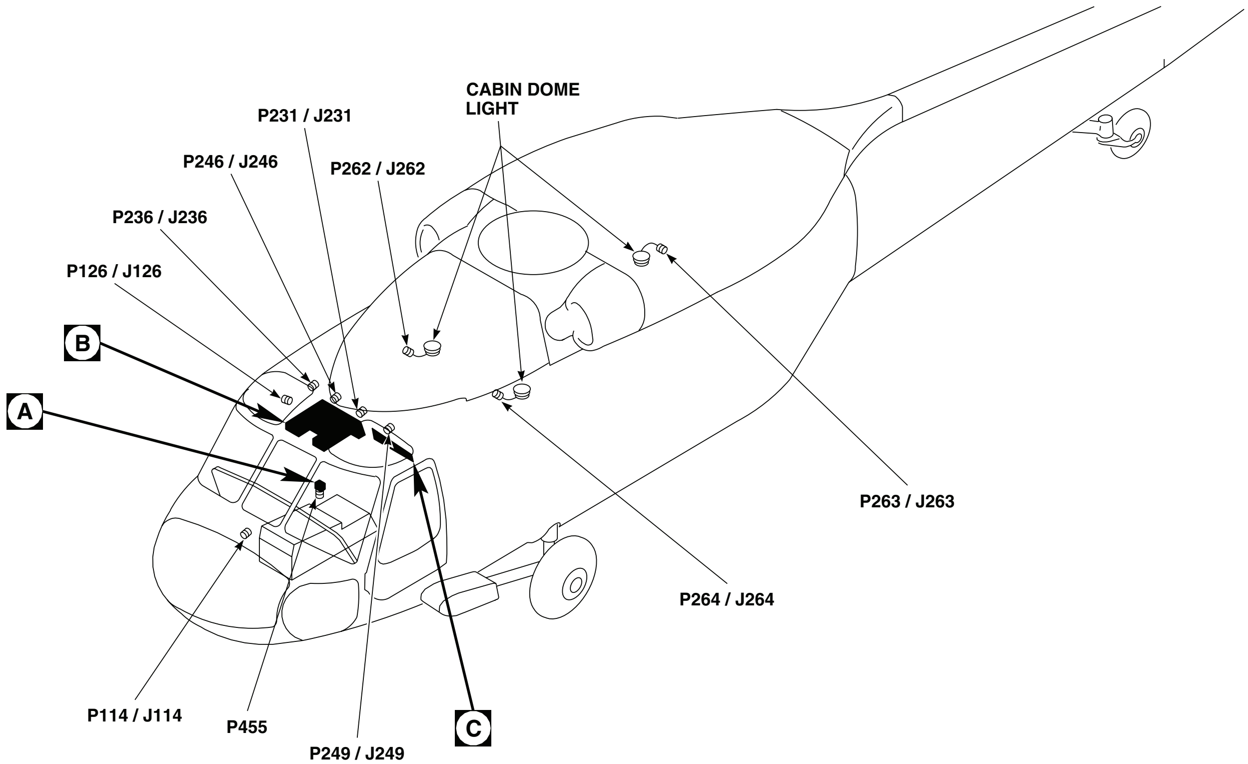
MALFUNCTION

No white cabin dome lights go on.

CORRECTIVE ACTION

1. **Check for 115 vac between CABIN DOME LT dimming control T1-HV and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, go to **4.**
2. **Check for 28 vac between T1-LV and ground.**
   a. If voltage is as specified, go to **3.**
   b. If voltage is not as specified, replace dimming control (WP 0876 00). Go to **5.**
3. **Check for 28 vac between CABIN DOME LT switch S12-2 and ground.**
   a. If voltage is as specified, replace switch S12 (WP 0828 00). Go to **5.**
   b. If voltage is not as specified, repair/replace wiring between S12-2 and T1- LV (WP 1747 00). Go to **5.**
4. **Check for 115 vac between LIGHTS CABIN DOME circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T1-HV (WP 1747 00). Go to **5.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **5.**
5. **Procedure completed.**

LOCATION



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P114 / J114	COCKPIT, BL 8 RH, STA 200
P126 / J126	COCKPIT CEILING BL 28 RH, STA 243
P231 / J231	CABIN CEILING, BL 9 LH, STA 247
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 23 RH, STA 247
P246 / J246	CABIN CEILING, BL 5 RH, STA 247

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH, STA 247
P262 / J262	FORWARD RIGHT CABIN DOME LIGHT
P263 / J263	AFT RIGHT CABIN DOME LIGHT
P264 / J264	FORWARD LEFT CABIN DOME LIGHT
P455	CABIN DOME LT DIMMER, BEHIND PILOT'S SEAT, BL 11 RH, STA 245

AA2766_1
SA

Figure 1. Cabin Dome Lights Location Diagram. (Sheet 1 of 2)

LOCATION - Continued

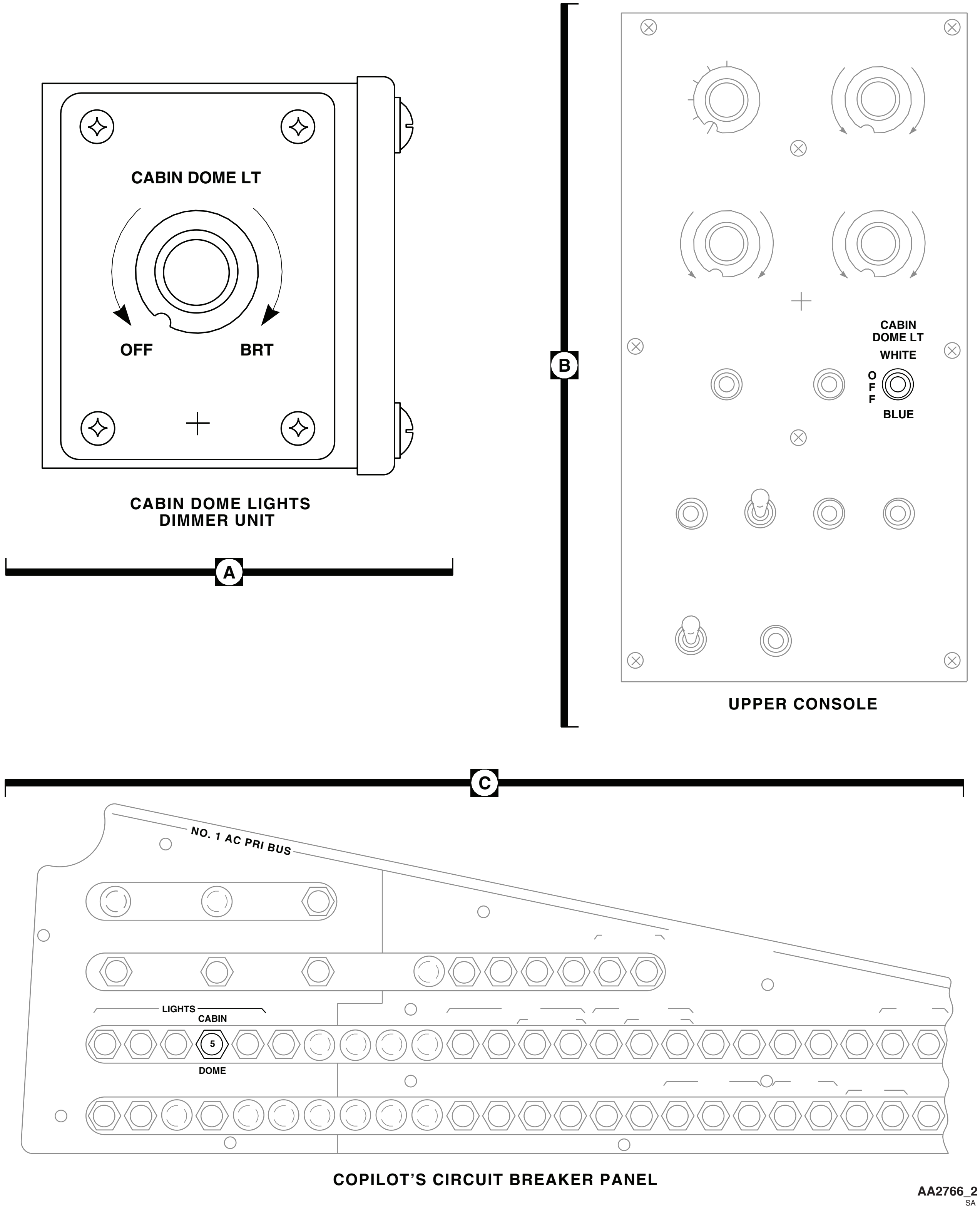


Figure 1. Cabin Dome Lights Location Diagram. (Sheet 2 of 2)

SCHEMATICS

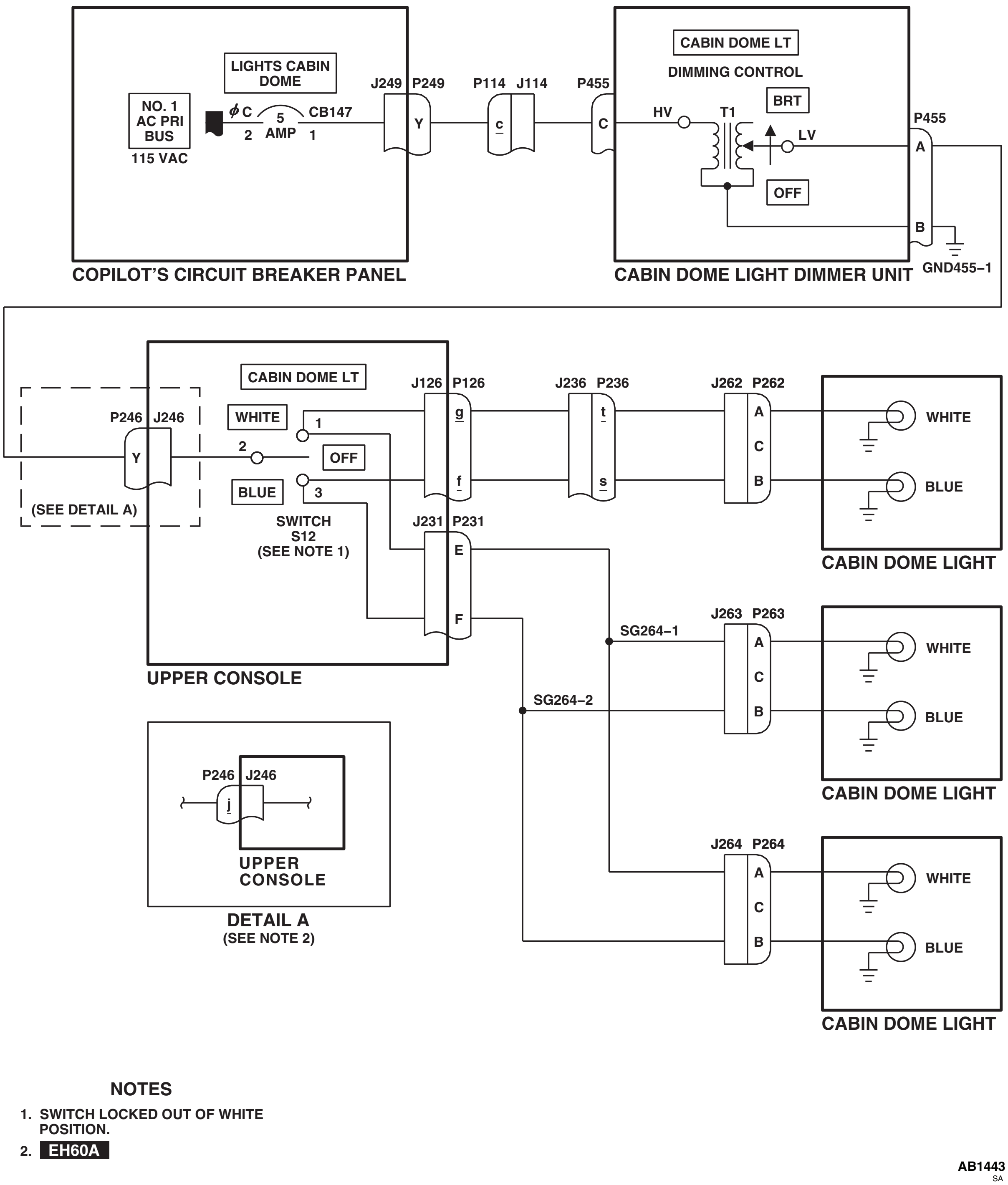


Figure 2. Cabin Dome Lights Schematic Diagram.

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

COCKPIT FLOOD AND SECONDARY LIGHTS

This WP supersedes WP 0108 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
WP 0104 00
WP 0773 00
WP 0775 00
WP 0824 00
WP 0875 00
WP 0892 00
WP 0893 00
WP 0894 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

COCKPIT FLOOD AND SECONDARY LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure LIGHTS GLARE SHLD circuit breaker, on copilot's circuit breaker panel, is pushed in.
2. Make sure GLARESHIELD LIGHTS dimmer control is in OFF position.
3. Turn on electrical power.
4. Place BLUE/OFF/WHITE switch, on the cockpit flood and secondary lights panel, to OFF.
5. Turn GLARESHIELD LIGHTS control to BRT.

Indication/Condition

Six instrument panel glare shield lights shall go on and increase in brilliance.

Corrective Action

1. If Indication/Condition is not as specified, go to GLARE SHIELD LIGHTS DO NOT GO ON WITH GLARESHIELD LIGHTS CONTROL AT BRT, in this work package.
2. If one glare shield light does not go on, replace lamp or socket (WP 0892 00).

COCKPIT FLOOD AND SECONDARY LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

6. Turn GLARESHIELD LIGHTS control to OFF.

Indication/Condition

Six instrument panel glare shield lights shall decrease in brilliance and go off.

Corrective Action

If Indication/Condition is not as specified, replace GLARESHIELD LIGHTS dimmer control T4 (WP 0875 00).

Step

7. Place BLUE/OFF/WHITE switch, on cockpit flood and secondary lights panel, to WHITE.

Indication/Condition

Cockpit flood and secondary lights panel white flood lights shall go on.

Corrective Action

1. If Indication/Condition is not as specified, go to NEITHER COCKPIT WHITE FLOOD LIGHT GOES ON, in this work package.
2. If one cockpit white flood light does not go on, replace flood light lamp (WP 0894 00).

Step

8. Place BLUE/OFF/WHITE switch, on cockpit flood and secondary light panel, to BLUE.

Indication/Condition

Both white flood lights shall go off and both blue lights shall go on.

Corrective Action

1. If Indication/Condition is not as specified, replace cockpit flood and secondary lights panel (WP 0894 00).
2. If one blue secondary light does not go on, replace lamp (WP 0894 00).

Step

9. Place BLUE/OFF/WHITE switch, on cockpit flood and secondary light panel, to OFF.

Indication/Condition

All secondary overhead lights shall go off.

Corrective Action

If Indication/Condition is not as specified, replace secondary light panel (WP 0894 00).

Step

10. Place standby magnetic compass ON/OFF light switch to ON.

Indication/Condition

Light on standby magnetic compass shall go on.

Corrective Action

If Indication/Condition is not as specified, go to LIGHT ON STANDBY MAGNETIC COMPASS DOES NOT GO ON, in this work package.

COCKPIT FLOOD AND SECONDARY LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

11. Place standby magnetic compass ON/OFF light switch to OFF.
12. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

GLARE SHIELD LIGHTS DO NOT GO ON WITH GLARESHIELD LIGHTS CONTROL AT BRT

SYMPTOM

Glare shield lights do not go on with GLARESHIELD LIGHTS control at BRT.

MALFUNCTION

Glare shield lights do not go on with GLARESHIELD LIGHTS control at BRT.

CORRECTIVE ACTION

1. **Do either right or left glare shield lights go on?**
   a. If lights go on, go to **2.**
   b. If lights do not go on, repair/replace wiring between J162-A and dimming control T4-LV (WP 1747 00). Go to **3.**
2. **Check continuity between J110-E and associated pin of malfunctioning glare shield lights, and between GG33-1 and associated pin of malfunctioning glare shield lights.**
   a. If continuity is present, replace glare shield harness (WP 0893 00). Go to **3.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3.**
3. **Procedure completed.**

NEITHER COCKPIT WHITE FLOOD LIGHT GOES ON

SYMPTOM

Neither cockpit white flood light goes on.

MALFUNCTION

Neither cockpit white flood light goes on.

CORRECTIVE ACTION

1. **Check for 28 vdc between P288-C and ground.**
   a. If voltage is as specified, replace cockpit flood and secondary lights panel (WP 0894 00). Go to **3.**
   b. If voltage is not as specified, go to **2.**
2. **Check continuity between P288-C and LIGHTS SEC PNL circuit breaker terminal 1.**
   a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **3.**

NEITHER COCKPIT WHITE FLOOD LIGHT GOES ON - Continued

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3.**

3. Procedure completed.

LIGHT ON STANDBY MAGNETIC COMPASS DOES NOT GO ON

SYMPTOM

Light on standby magnetic compass does not go on.

MALFUNCTION

Light on standby magnetic compass does not go on.

CORRECTIVE ACTION

1. Check for 28 vdc between compass post light and ground.

a. If voltage is as specified, replace standby compass (WP 0773 00). Go to **3.**

b. If voltage is not as specified, go to **2.**

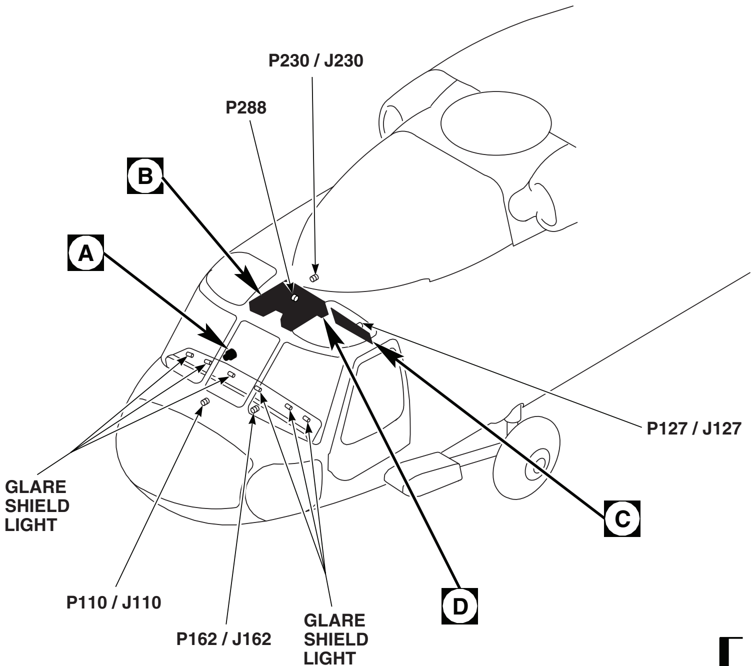
2. Check continuity across standby compass switch.

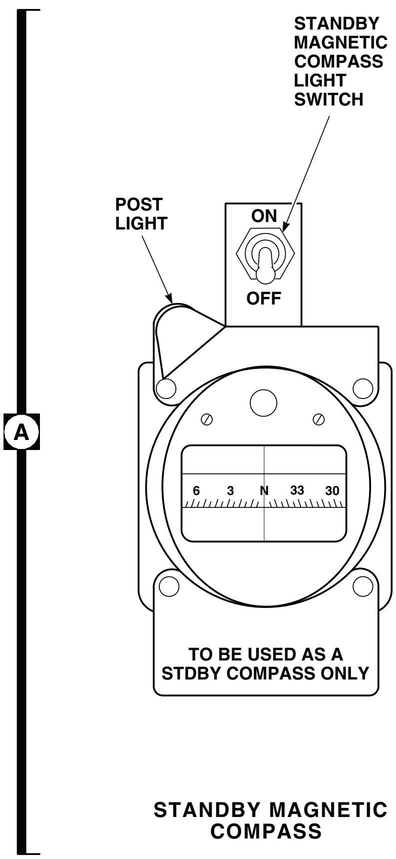
a. If continuity is present, repair/replace wiring between switch terminal 2 and LIGHTS SEC PNL circuit breaker terminal 1 (WP 1747 00). Go to **3.**

b. If continuity is not present, replace standby compass switch (WP 0775 00). Go to **3.**

3. Procedure completed.

LOCATION





TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P127 / J127	COCKPIT CEILING, BL 28 LH, STA 243
P162 / J162	BACK OF INSTRUMENT PANEL, BL 12 LH
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P288	COCKPIT FLOOD AND SECONDARY LIGHTS PANEL

AA2767_1
SA

Figure 1. Cockpit Flood and Secondary Lights Location Diagram. (Sheet 1 of 2)

LOCATION - Continued

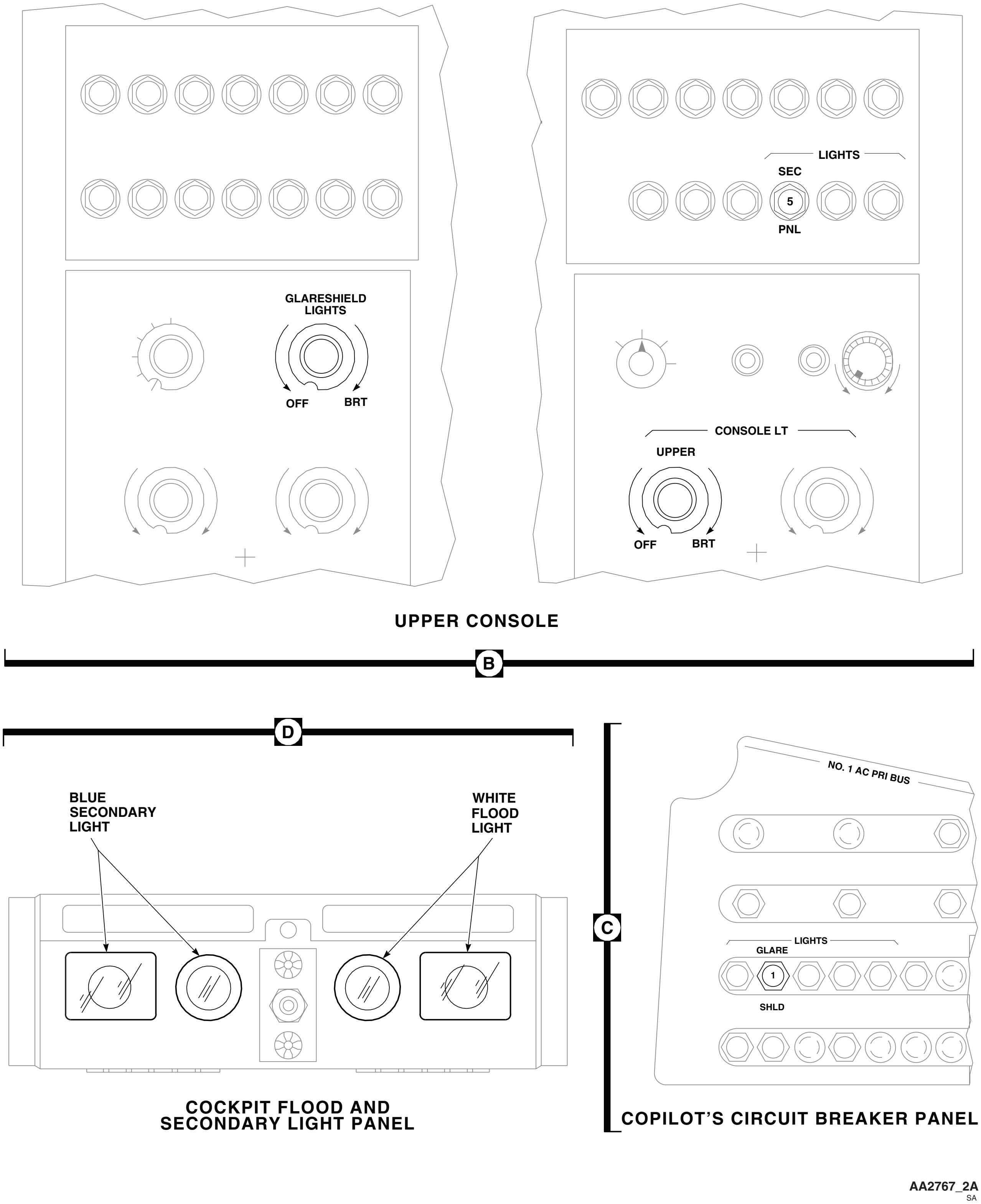


Figure 1. Cockpit Flood and Secondary Lights Location Diagram. (Sheet 2 of 2)

SCHEMATICS

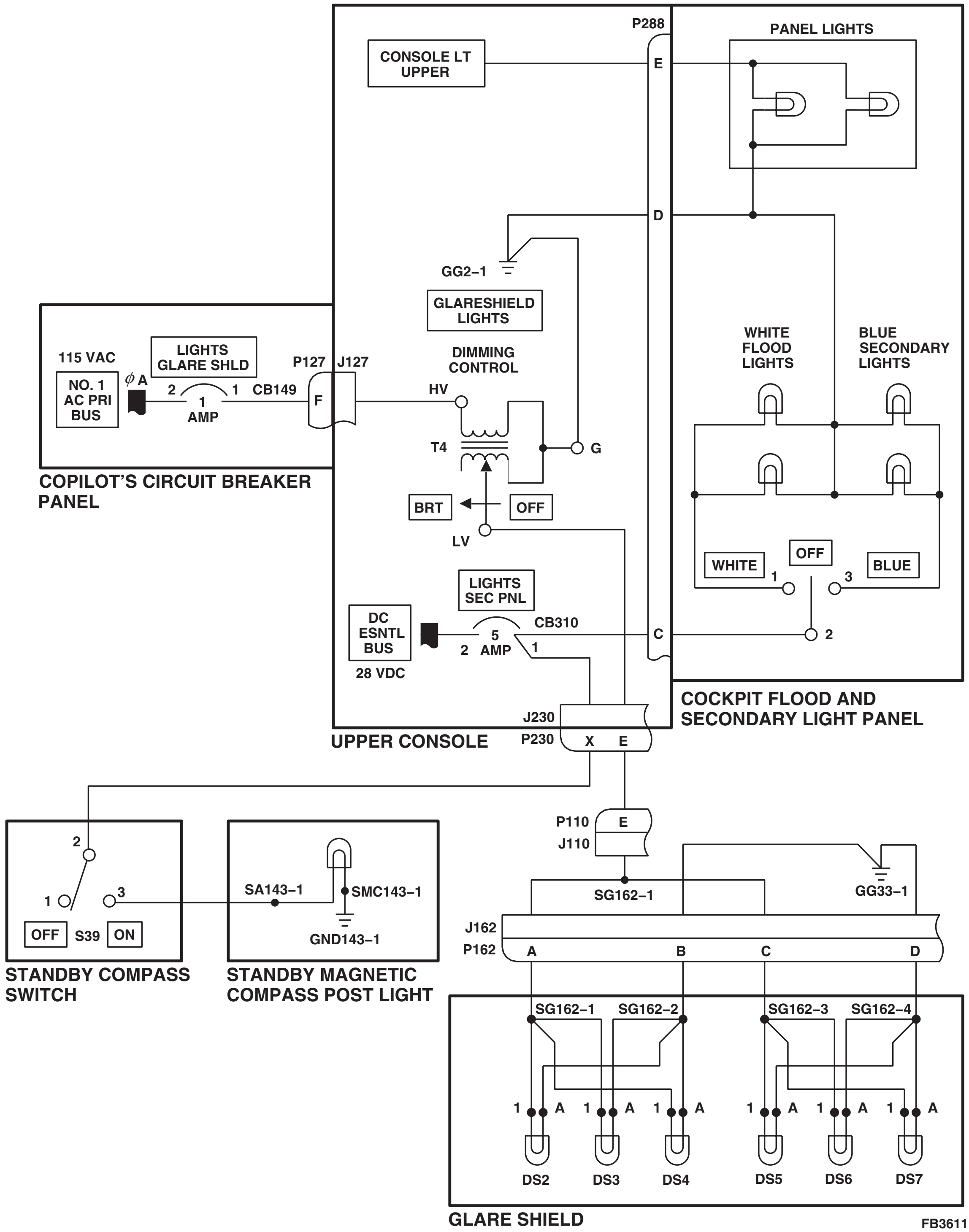


Figure 2. Cockpit Flood and Secondary Lights Schematic Diagram.

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

UTILITY AND MAINTENANCE LIGHTS

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0104 00
WP 0824 00
WP 0889 00
WP 0890 00
WP 0891 00
WP 1747 00

Equipment Condition
Battery at Least 35% Charged

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

- **UH60A UH60L** Repeat steps 1. through 9. for pilot's, copilot's, and auxiliary utility lights. Then do steps 10. and 11.
- **EH60A** Repeat steps 1. through 9. for pilot's, copilot's, and auxiliary utility lights. Then do steps 10., 11., and 12.

UTILITY AND MAINTENANCE LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make certain that UTIL LTS CKPT circuit breaker on LOWER CONSOLE circuit breaker panel is pushed in.
2. Press and hold button in center of rheostat knob on back of utility light assembly.

Indication/Condition

Utility light shall go on to full brightness.

Corrective Action

If Indication/Condition is not as specified, go to UTILITY LIGHT DOES NOT GO ON, in this work package.

UTILITY AND MAINTENANCE LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

3. Release button in center of rheostat knob.

Indication/Condition

Utility light shall go off.

Corrective Action

If Indication/Condition is not as specified, replace utility light assembly (WP 0890 00).

Step

4. Turn rheostat on back of utility light assembly fully clockwise.

Indication/Condition

Utility light shall go to full brightness.

Corrective Action

If Indication/Condition is not as specified, replace utility light assembly (WP 0890 00).

Step

5. Pull collar on light assembly forward, over lens.
6. Press lock button and turn collar so index marks on collar and light assembly line up.

Indication/Condition

Light output shall be bright, white, spotlight beam.

Corrective Action

If Indication/Condition is not as specified, replace utility light assembly (WP 0890 00).

Step

7. Push collar backwards, away from lens.
8. Press lock button and turn collar to obtain blue lens.

Indication/Condition

Light output shall be bright blue.

Corrective Action

If Indication/Condition is not as specified, replace utility light assembly (WP 0890 00).

Step

9. Turn rheostat on back of utility light assembly fully counterclockwise to OFF.

Indication/Condition

Light shall dim and go off.

Corrective Action

If Indication/Condition is not as specified, replace utility light assembly (WP 0890 00).

UTILITY AND MAINTENANCE LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

10. Connect maintenance light assembly to forward maintenance light receptacle J259.

Indication/Condition

Maintenance light shall go on.

Corrective Action

If Indication/Condition is not as specified, go to MAINTENANCE LIGHT DOES NOT GO ON WHEN CONNECTED TO ONE MAINTENANCE RECEPTACLE, in this work package.

Step

11. Connect maintenance light assembly to rear maintenance receptacle J509.

Indication/Condition

Maintenance light shall go on.

Corrective Action

If Indication/Condition is not as specified, repair/replace wiring to J509-A or J509-B as required (WP 1747 00).

Step

12. **EH60A** Connect maintenance light assembly to transition maintenance receptacle J517.

Indication/Condition

Maintenance light shall go on.

Corrective Action

If Indication/Condition is not as specified, repair/replace wiring to J517-A or J517-B as required (WP 1747 00).

UTILITY LIGHT DOES NOT GO ON

SYMPTOM

Utility light does not go on.

MALFUNCTION

Utility light does not go on.

CORRECTIVE ACTION

1. **Does any utility light go on?**
   a. If any utility light is on, go to **2.**
   b. If any utility light is not on, go to **11.**
2. **Does pilot's utility light not go on?**
   a. If light is not on, go to **3.**
   b. If light is on, go to **5.**
3. **Check lamp.**
   a. If lamp is good, go to **4.**

UTILITY LIGHT DOES NOT GO ON - Continued

b. If lamp is not good, replace lamp (WP 0889 00). Go to **12.**

4. **Check for 28 vdc between TB2-1 and TB2-2.**

a. If voltage is as specified, replace pilot's utility light assembly (WP 0890 00). Go to **12.**

b. If voltage is not as specified, repair/replace wiring to TB2-1 or TB2-2 (WP 1747 00). Go to **12.**

5. **Does copilot's utility light not go on?**

a. If copilot's light is not on, go to **6.**

b. If copilot's light is on, go to **8.**

6. **Check lamp.**

a. If lamp is good, go to **7.**

b. If lamp is not good, replace lamp (WP 0889 00). Go to **12.**

7. **Check for 28 vdc between TB1-1 and TB1-2.**

a. If voltage is as specified, replace copilot's utility light assembly (WP 0890 00). Go to **12.**

b. If voltage is not as specified, repair/replace wiring to TB1-1 or TB1-2 (WP 1747 00). Go to **12.**

8. **Does auxiliary utility light not go on?**

a. If auxiliary utility light is not on, go to **9.**

9. **Check lamp.**

a. If lamp is good, go to **10.**

b. If lamp is not good, replace lamp (WP 0889 00). Go to **12.**

10. **Check for 28 vdc between TB6-1 and TB6-2.**

a. If voltage is as specified, replace auxiliary utility light assembly (WP 0890 00). Go to **12.**

b. If voltage is not as specified, repair/replace wiring to TB6-1 or TB6-2 (WP 1747 00). Go to **12.**

11. **Check for 28 vdc between UTIL LTS CKPT circuit breaker terminal 1 and ground.**

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and TB1-1 (WP 1747 00). Go to **12.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **12.**

12. **Procedure completed.**

MAINTENANCE LIGHT DOES NOT GO ON WHEN CONNECTED TO ONE MAINTENANCE RECEPTACLE

SYMPTOM

Maintenance light does not go on when connected to one maintenance receptacle.

MALFUNCTION

Maintenance light does not go on when connected to one maintenance receptacle.

CORRECTIVE ACTION

1. **Check helicopter configuration.**
   a. EH60A , go to **2.**
   b. UH60A UH60L , go to **4.**
2. EH60A **Check for 28 vdc between:**
   - **J259-A and J259-B**
   - **J509-A and J509-B**
   - **J517-A and J517-B**

   a. If voltage is as specified, repair maintenance light (WP 0891 00). Go to **6.**
   b. If voltage is not as specified, go to **3.**
3. EH60A **Check continuity between:**
   - **J259-B and ground**
   - **J509-B and ground**
   - **J517-B and ground**

   a. If continuity is present, repair/replace wiring, as required, between (WP 1747 00), then go to **6:**
      - UTIL LTS CKPT terminal 1 and J259-A
      - UTIL LTS CKPT terminal 1 and J509-A
      - UTIL LTS CKPT terminal 1 and J517-A

   b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**
4. UH60A UH60L **Check for 28 vdc between:**
   - **J259-A and J259-B**
   - **J509-A and J509-B**

   a. If voltage is as specified, repair maintenance light (WP 0891 00). Go to **6.**
   b. If voltage is not as specified, go to **5.**
5. **Check continuity between:**
   - **J259-B and ground**
   - **J509-B and ground**

MAINTENANCE LIGHT DOES NOT GO ON WHEN CONNECTED TO ONE MAINTENANCE RECEPTACLE - Continued

a. If continuity is present, repair/replace wiring, as required, between (WP 1747 00), then go to **6:**

- UTIL LTS CKPT terminal 1 and J259-A
- UTIL LTS CKPT terminal 1 and J509-A

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**

6. **Procedure completed.**

LOCATION

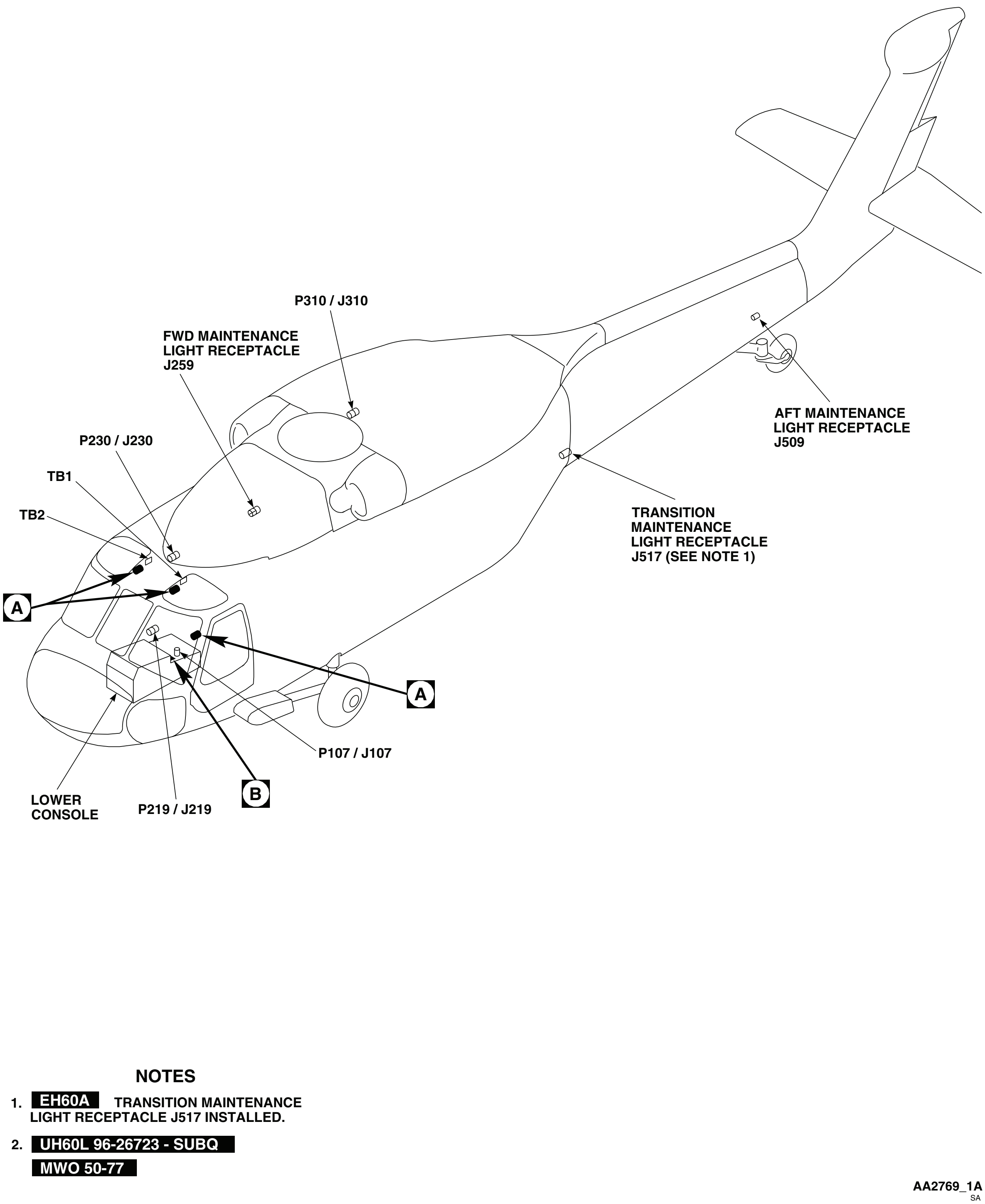
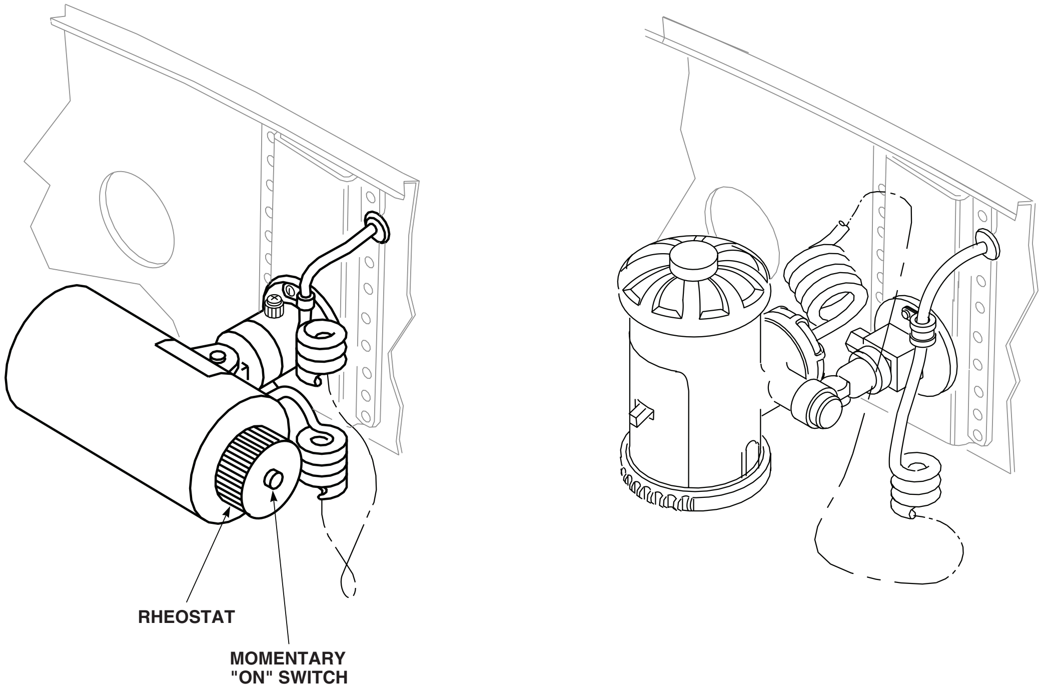


Figure 1. Utility and Maintenance Lights Location Diagram. (Sheet 1 of 2)

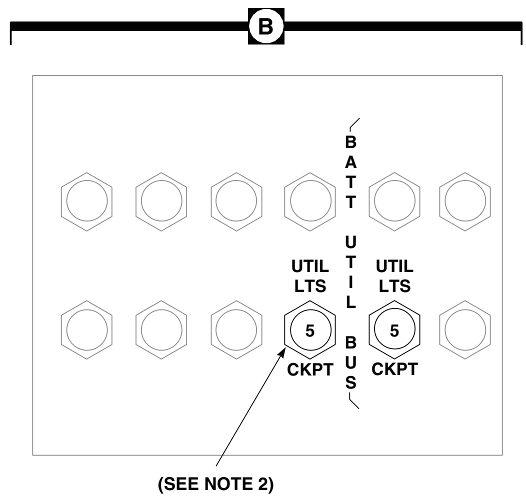
LOCATION - Continued



PILOT'S, COPILOT'S AND AUXILIARY UTILITY LIGHT

A

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J259	FWD MAINTENANCE LIGHT RECEPTACLE
J509	AFT MAINTENANCE LIGHT RECEPTACLE
J517 (SEE NOTE)	TRANSITION MAINTENANCE LIGHT RECEPTACLE
P107 / J107	COCKPIT, BL 9 LH, STA 236
P219 / J219	CABIN FLOOR, BL 25 RH, STA 247
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P310 / J310	CABIN CEILING, BL 17 RH, STA 387
TB1	COCKPIT CEILING, BL 9 LH, STA 236
TB2	COCKPIT CEILING, BL 9 RH, STA 236
TB6	BEHIND COPILOT'S SEAT BL 11 LH, STA 245



CIRCUIT BREAKER PANEL

AA2769_2A
SA

Figure 1. Utility and Maintenance Lights Location Diagram. (Sheet 2 of 2)

SCHEMATICS

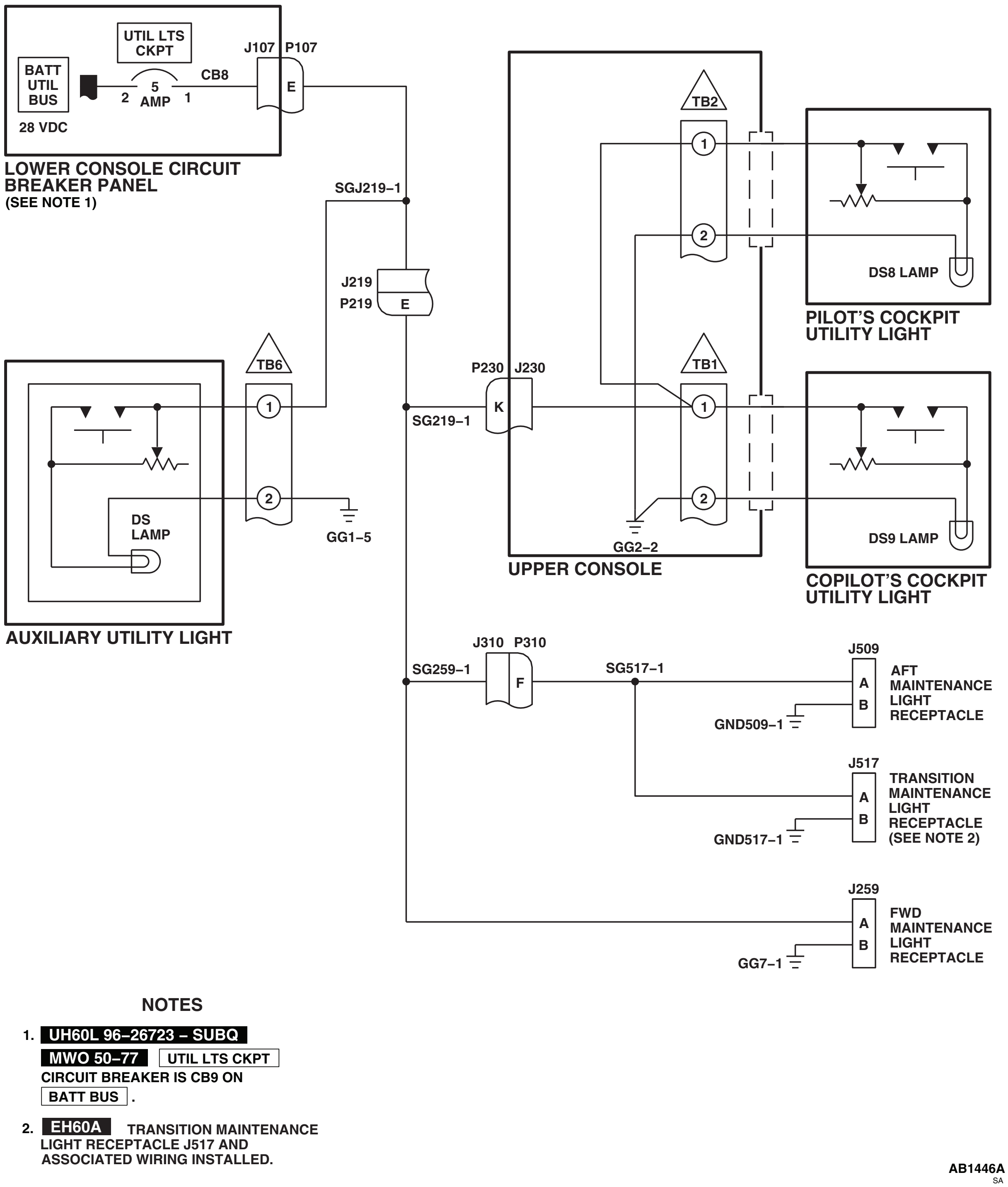


Figure 2. Utility and Maintenance Lights Schematic Diagram.

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
TM 11-1520-237-23
TM 11-1520-249-23-2
WP 0071 00
WP 0090 00
WP 0104 00
WP 0141 00
WP 0785 00
WP 0824 00
WP 0825 00
WP 0828 00
WP 0829 00
WP 0830 00
WP 0832 00
WP 0878 00
WP 0879 00
WP 0925 00
WP 0954 00
WP 1549 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING OPERATIONAL/ TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure that the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
APU FIRE DET	Lower console
CAUT/ADVSY PNL	Upper console
CPLT MODE SEL	Copilot’s

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
DPLR	Copilot's
FIRE DET NO. 1 ENG	Upper console
FIRE DET NO. 2 ENG	Upper console
LIGHTS ADVSY	Copilot's
LIGHTS CAUT ADVSY	Copilot's
NO. 1 DC PRI BUS CMPTR TRIM	Copilot's
NO. 2 DC PRI BUS CMPTR TRIM	Pilot's
PILOT MODE SELECT	Pilot's
SAS BOOST	Upper console
TAIL WHEEL LOCK	Upper console

2. Make sure INST LT PILOT FLT and LIGHTED SWITCHES controls are turned OFF.
3. Make sure the pilot's and copilot's RAD ALT DIMMING controls are turned fully counterclockwise.
4. Turn on electrical power.
5. Test the following press-to-test indicators:
   - Blade de-ice control panel TEST IN PROGRESS lamp.
   - Blade de-ice test panel PWR MAIN ROTOR and TAIL ROTOR lamps.
   - **UH60A UH60L** Rescue hoist control panel SQUIB IND lamp (when installed).
   - **UH60A UH60L** Upper console CARGO HOOK EMER REL TEST lamp.
   - Fuel boost pump control panel NO. 1 PUMP and NO. 2 PUMP lamps.
   - Auxiliary cabin heater control panel HTR ON and HTR INOP lamps.

Indication/Condition

When pressed, each indicator shall light brightly.

Corrective Action

1. If Indication/Condition is not as specified, go to INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DO NOT GO ON BRIGHT, in this work package.
2. If one press-to-test indicator does not go on, replace lamp (WP 0829 00).
   a. If trouble remains, check continuity of affected indicator wiring.
      (1) If continuity is present, replace indicator (WP 0954 00).

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

(2) If continuity is not present, repair/replace wiring as required (WP 1747 00).

(3) If trouble remains, replace left relay panel (WP 0825 00).

Step

6. Place and hold the caution/advisory panel BRT/DIM-TEST switch to TEST.

Indication/Condition

a. All capsules on caution/advisory panel and pilot's and copilot's master warning panels shall go on bright.

b. Master warning panel LOW ROTOR RPM capsules shall flash.

c. All switch legends on CIS mode select panel and pilot's and copilot's HSI/VSI mode select panels shall go on bright.

d. CPTR, SAS 2, TRIM, RGYR, ACCL, CLTV, A/S, and GYRO failure indications on stabilator controls/auto flight control panel shall go on bright.

e. GA, DH, and MB indicators on pilot's and copilot's vertical situation indicators shall go on bright.

Corrective Action

1. If Indication/Condition a. or b. is not as specified, troubleshoot caution/advisory warning system (WP 0090 00).
2. If Indication/Condition c., d., or e. is not as specified, troubleshoot command instrument system (TM 11-1520-237-23 or TM 11-1520-249-23-2).
3. If one switch legend or failure indication of Indication/Condition c. or d. is not as specified, replace associated lamp (TM 11-1520- 237-23 or TM 11-1520-249-23-2).

   a. If trouble remains, troubleshoot panel (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step

7. Place radar altimeter system on by turning pilot's and copilot's radar altimeter LO/SET/OFF control from OFF.

Indication/Condition

a. Pilot's and copilot's OFF flags shall go out of view.

b. Pilot's and copilot's digital displays shall light brightly.

Corrective Action

1. If Indication/Condition a. is not as specified, troubleshoot radar altimeter (AN/APN-209) system (TM 11-1520-237-23 or TM 11-1520-249-23-2).
2. If Indication/Condition b. is not as specified, go to PILOT'S AND COPILOT'S RADAR ALTIMETER DIGITAL DISPLAYS DO NOT GO ON BRIGHT, in this work package.

Step

8. Press miscellaneous switch panel TAIL WHEEL switch. Observe TAIL WHEEL switch legends.

Indication/Condition

When pressed, TAIL WHEEL switch shall alternately indicate LOCK or UNLK.

Corrective Action

1. If Indication/Condition is not as specified, replace lamp (WP 0832 00).

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

a. If trouble remains, troubleshoot tailwheel lock system (WP 0071 00).

b. If trouble still remains, replace miscellaneous switch panel (WP 0830 00).

Step

9. **UH60A UH60L** Place doppler computer display MODE switch to LAMP TEST.

Indication/Condition

a. MEM and MAL failure indicators shall go on bright.

b. Digital displays shall vary according to DIM control.

Corrective Action

1. If Indication/Condition a. is not as specified, repair/replace wiring (WP 1747 00) or replace doppler computer display (TM 11-1520-237-23), as required.
2. If one indicator in Indication/Condition a. is not as specified, replace lamp (TM 11-1520-237-23).

   a. If trouble remains, replace doppler computer display (TM 11-1520-237-23).

Step

10. Place doppler computer display MODE switch to OFF.

NOTE

EH60A Do steps 11. through 13. only if mission equipment is installed.

11. Press system select panel HDG switch.

Indication/Condition

When pressed, HDG switch shall alternately display DG or IINS legends.

Corrective Action

1. If Indication/Condition is not as specified, troubleshoot gyromagnetic compass (AN/ASN-43) system (TM 11-1520-249-23-2).

   a. If trouble remains, repair/replace wiring as required (WP 1747 00).

Step

12. Press system select panel ATT switch.

Indication/Condition

When pressed, ATT switch shall alternately display VG or IINS legends.

Corrective Action

1. If Indication/Condition is not as specified, troubleshoot attitude indicating system (TM 11-1520-249-23-2).

   a. If trouble remains, repair/replace wiring as required (WP 1747 00).

Step

13. Press cabin mounted CREW CALL switch to light instrument panel CREW CALL switch.

Indication/Condition

CREW CALL switch on instrument panel shall light.

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition is not as specified, troubleshoot intercommunication system (TM 11-1520-249-23-2).
   a. If trouble remains, repair/replace wiring as required (WP 1747 00).

Step

14. Turn upper console INST LT PILOT FLT control one quarter turn from OFF.
15. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM to select dim mode.

Indication/Condition

a. Any lit caution/advisory panel and master warning panel capsules shall dim.

b. Switch legends on pilot's and copilot's HSI/VSI mode select panels and TAILWHEEL LOCK switch shall go off.

c. Digital displays on pilot's and copilot's radar altimeters shall go off.

d. **EH60A** With mission equipment installed, switch legends on system select panel and CREW CALL switch shall go off.

e. With caution/advisory panel BRT/DIM-TEST switch placed to TEST, all VSI advisory legends shall be dim.

Corrective Action

1. If Indication/Conditions are not as specified, troubleshoot caution/advisory warning system (WP 0090 00).
   a. If trouble remains, replace left relay panel (WP 0825 00).

Step

16. Turn upper console LIGHTED SWITCHES dimmer control from OFF to BRT.

Indication/Condition

a. Selected legends on miscellaneous switch panel TAILWHEEL LOCK switch and bottom row of pilot's and copilot's HSI/VSI mode select panels shall vary smoothly from off to bright.

b. Displays and annunciators on auxiliary fuel management panel shall vary smoothly from off to bright.

Corrective Action

1. If Indication/Conditions a. and b. lights do not vary smoothly from dim to bright, troubleshoot LIGHTED SWITCHES dimmer control resistance as follows:
   a. Remove LIGHTED SWITCHES dimmer control (WP 0828 00).
   b. Remove three screws from hinged cover of LIGHTED SWITCHES dimmer control and open hinged cover.
   c. Measure resistance of variable resistor as follows:

CONTROL POSITION	PINS	RESISTANCE + OR - 10%
OFF (Counterclockwise)	1 - 2 2 - 3	0.3 ohms 4.41K ohms

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

CONTROL POSITION	PINS	RESISTANCE + OR - 10%
BRT (Clockwise)	1 - 2 2 - 3	4.62K ohms 0.2 ohms

2. If LIGHTED SWITCHES dimmer control variable resistor’s resistance is not as specfied, replace LIGHTED SWITCHES dimmer control variable resistor (WP 0828 00).
3. If LIGHT SWITCHES dimmer control variable resistor resistance is as specified, replace LIGHTED SWITCHES dimmer control (WP 0828 00).
4. If Indication/Conditions a. and b. do not go on as specified, go to LIGHTED SWITCHES ON SELECTED PANELS, OR DISPLAYS AND ANNUNCIATORS ON AUXILIARY FUEL MANAGEMENT PANEL DO NOT DIM, in this work package.

Step

17. Press blade de-ice control panel TEST IN PROGRESS press-to-test indicator.

Indication/Condition

TEST IN PROGRESS lamp shall light with decreased intensity from that noted in step 5.

Corrective Action

If Indication/Condition is not as specified, go to INDICATOR LIGHTS DO NOT DIM, in this work package.

Step

18. Turn pilot’s and copilot’s instrument panel RAD ALT DIMMING control fully clockwise.

Indication/Condition

Digital displays on pilot’s and copilot’s radar altimeters shall vary smoothly from off to bright as associated RAD ALT DIMMING control is turned clockwise.

Corrective Action

1. If Indication/Condition is not as specified, replace associated RAD ALT DIMMING control (TM 11-1520-237-23).
2. If either digital display does not go on, go to PILOT’S OR COPILOT’S RADAR ALTIMETER DIGITAL DISPLAYS DO NOT DIM, in this work package.

Step

19. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM to select bright mode.
20. Place FIRE DET TEST switch to position 1 and hold.

Indication/Condition

APU, #1 ENG EMER OFF, and #2 ENG EMER OFF T-handles shall go on bright.

Corrective Action

1. If Indication/Condition is not as specified, troubleshoot engine fire detection system (WP 0141 00).
   a. If trouble remains, replace left relay panel (WP 0825 00).

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

Step

21. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM to select dim mode.

Indication/Condition

APU, #1 ENG EMER OFF, and #2 ENG EMER OFF T-handles shall go on dimly.

Corrective Action

If Indication/Condition is not as specified, go to T-HANDLES REMAIN BRIGHT, in this work package.

Step

22. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DO NOT GO ON BRIGHT

SYMPTOM

Instrument panel and consoles indicator lights do not go on bright.

MALFUNCTION

Instrument panel and consoles indicator lights do not go on bright.

CORRECTIVE ACTION

1. **Check for 28 vdc between P902-V and ground.**
   a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **3.**
   b. If voltage is not as specified, go to **2.**
2. **Check for 28 vdc between LIGHTS ADVSY circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P902-V (WP 1747 00). Go to **3.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **3.**
3. **Procedure completed.**

PILOT'S AND COPILOT'S RADAR ALTIMETER DIGITAL DISPLAYS DO NOT GO ON BRIGHT

SYMPTOM

Pilot's and copilot's radar altimeter digital displays do not go on bright.

MALFUNCTION

Pilot's and copilot's radar altimeter digital displays do not go on bright.

CORRECTIVE ACTION

1. **Check for 28 vdc between:**
   - **P675R-7 and ground**

PILOT'S AND COPILOT'S RADAR ALTIMETER DIGITAL DISPLAYS DO NOT GO ON BRIGHT - Continued

- **P676R-6 and ground**

  a. If voltage is as specified, replace malfunctioning radar altimeter (TM 11-1520-237-23). Go to **3.**

  b. If voltage is not as specified, go to **2.**

2. **Check continuity between:**

   - **P675R-7 and P902-Y**
   - **P676R-6 and P902-n**

   a. If continuity is present, replace left relay panel (WP 0825 00). Go to **3.**

   b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **3.**

3. **Procedure completed.**

LIGHTED SWITCHES ON SELECTED PANELS, OR DISPLAYS AND ANNUNCIATORS ON AUXILIARY FUEL MANAGEMENT PANEL DO NOT DIM

SYMPTOM

Lighted switches on selected panels, or displays and annunciators on auxiliary fuel management panel do not dim.

MALFUNCTION

Lighted switches on selected panels, or displays and annunciators on auxiliary fuel management panel do not dim.

CORRECTIVE ACTION

1. **Are auxiliary fuel management panel displays and annunciators lit?**

   a. If auxiliary fuel management panel displays and annunciators are lit, go to **2.**

   b. If auxiliary fuel management panel displays and annunciators are not lit, go to **3.**

2. **With LIGHTED SWITCHES control at BRT, check for 26 vdc between P902-Z and ground.**

   a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **8.**

   b. If voltage is not as specified, go to **5.**

3. **Make sure AUX FUEL RH, UH60A UH60L No. 2 XFER CONTROL EH60A AUX FUEL QUANITY circuit breakers are pushed in. Go to 4.**

4. **Check for 26 vdc between P857-49 and ground.**

   a. If voltage is as specified, replace auxiliary fuel management panel (WP 1549 00). Go to **8.**

   b. If voltage is not as specified, go to **5.**

5. **Check for 28 vdc between P315-A and P315-C.**

   a. If voltage is as specified, go to **6.**

   b. If voltage is not as specified, go to **7.**

6. **Check continuity between P315-D and P902-Z.**

   a. If continuity is present, replace LIGHTED SWITCHES dimmer control (WP 0828 00). Go to **8.**

LIGHTED SWITCHES ON SELECTED PANELS, OR DISPLAYS AND ANNUNCIATORS ON AUXILIARY FUEL MANAGEMENT PANEL DO NOT DIM - Continued

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

7. **Check for 28 vdc between P315-A and ground.**

a. If voltage is as specified, repair/replace wiring between P315-C and ground (WP 1747 00). Go to **8.**

b. If voltage is not as specified, repair/replace wiring between P315-A and LIGHTS ADVSY circuit breaker (WP 1747 00). Go to **8.**

8. **Procedure completed.**

INDICATOR LIGHTS DO NOT DIM

SYMPTOM

Indicator lights do not dim.

MALFUNCTION

Indicator lights do not dim.

CORRECTIVE ACTION

1. **Check fuse on indicator lights dimmer.**

a. If fuse is good, go to **2.**

b. If fuse is not good, replace fuse (WP 0879 00). Go to **6.**

2. **Check for 28 vdc between P393-A and P393-B.**

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, go to **5.**

3. **With caution/advisory panel in dim mode, check continuity between P393-A and P393-C.**

a. If continuity is present, go to **4.**

b. If continuity is not present, repair/replace wiring between P393-C and J115-K (WP 1747 00). Go to **6.**

4. **Check continuity between P393-D and P902-h.**

a. If continuity is present, replace indicator lights dimmer (WP 0878 00). Go to **6.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**

5. **Check for 28 vdc between P393-B and ground.**

a. If voltage is as specified, repair/replace wiring between P393-A and ground (WP 1747 00). Go to **6.**

b. If voltage is not as specified, repair/replace wiring between P393-B and LIGHTS ADVSY circuit breaker (WP 0824 00). Go to **6.**

6. **Procedure completed.**

PILOT'S OR COPILOT'S RADAR ALTIMETER DIGITAL DISPLAYS DO NOT DIM

SYMPTOM

Pilot's or copilot's radar altimeter digital displays do not dim.

MALFUNCTION

Pilot's or copilot's radar altimeter digital displays do not dim.

CORRECTIVE ACTION

1. **With affected altimeter's RAD ALT dimming control turned fully clockwise, check for 28 vdc between:**
   - **R4-2 and R4-3**
   - **R5-2 and R5-3**

   a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **4.**

   b. If trouble remains, repair/replace wiring between (WP 1747 00), then go to **4:**
   - R4-1 and P902-W
   - R4-2 and P902-T
   - R5-1 and P902-f
   - R5-2 and P902-p

   c. If voltage is not as specified, go to **2.**
2. **Check for 28 vdc between:**
   - **R4-1 and R4-2**
   - **R5-1 and R5-2**

   a. If voltage is as specified, replace control (TM 11-1520-237-23 or TM 11-1520-249-23-3). Go to **4.**

   b. If voltage is not as specified, go to **3.**
3. **Check continuity between:**
   - **R4-1 and P902-W**
   - **R4-2 and P902-T**
   - **R5-1 and P902-f**
   - **R5-2 and P902-p**

   a. If continuity is present, replace left relay panel (WP 0825 00). Go to **4.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**
4. **Procedure completed.**

T-HANDLES REMAIN BRIGHT

SYMPTOM

T-handles remain bright.

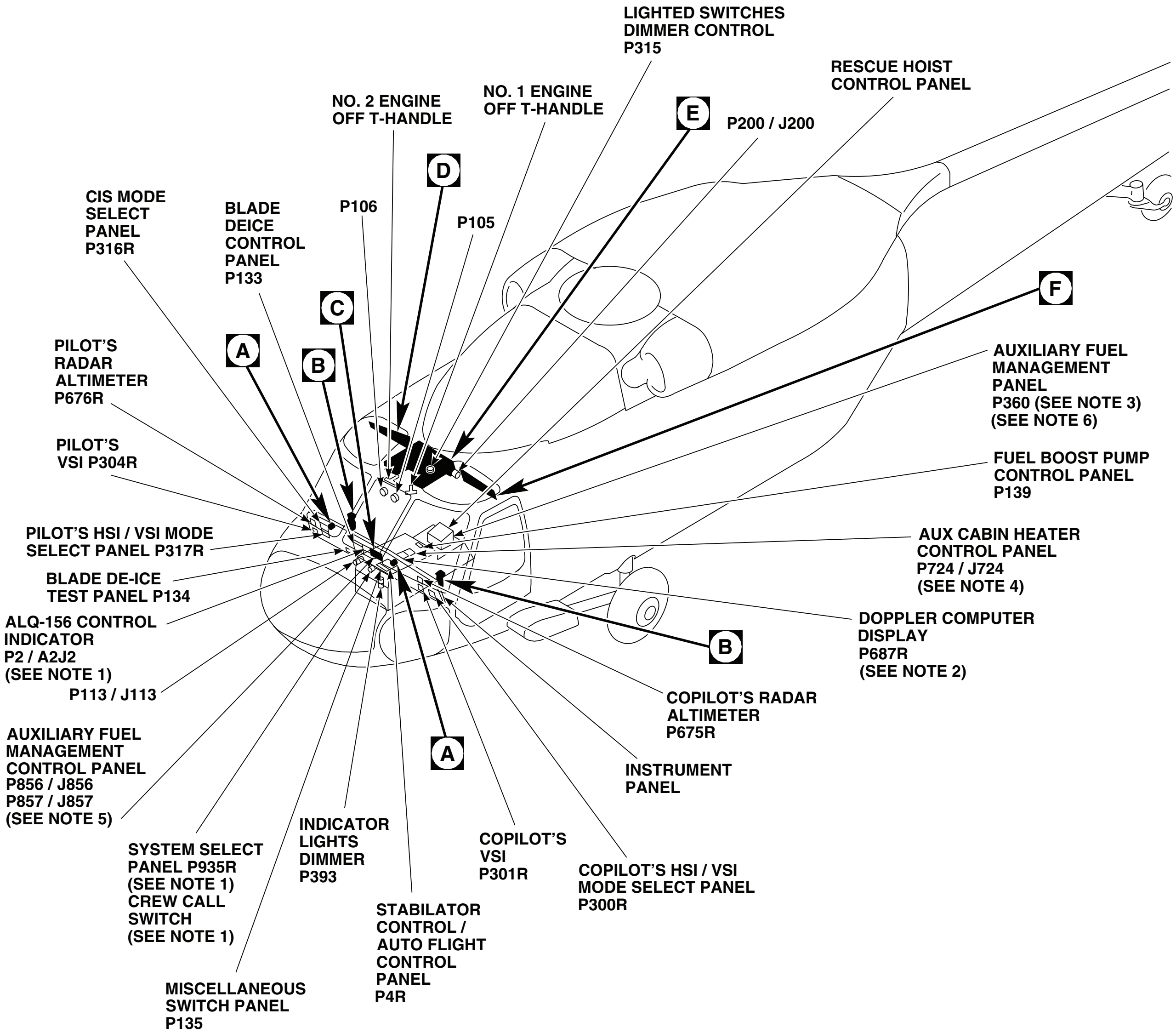
MALFUNCTION

T-handles remain bright.

CORRECTIVE ACTION

1. **Do caution/advisory capsules go dim?**
   a. If capsules are dim, go to **2.**
   b. If capsules are not dim, troubleshoot caution/advisory warning system (WP 0090 00). Go to **9.**
2. **Do instrument panel and lower console lights go dim?**
   a. If panel and lights are dim, go to **3.**
   b. If panel and lights are not dim, go to **5.**
3. **Check for 28 vdc between P902-P and ground.**
   a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **9.**
   b. If voltage is not as specified, go to **4.**
4. **Check continuity between P902-P and FIRE DET NO. 1 ENG circuit breaker terminal 1.**
   a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **9.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**
5. **Check helicopter configuration.**
   a. EMEP , go to **6.**
   b. W/O EMEP , go to **8.**
6. EMEP **Remove and check pin filtered adapter P118 (WP 0925 00).**
   a. If pin filtered adapter is good, go to **7.**
   b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9.**
7. EMEP **Install pin filtered adapter (WP 0925 00). Go to 8.**
8. **Check continuity between P902-q and P118-T.**
   a. If continuity is present, replace left relay panel (WP 0825 00). Go to **9.**
   b. If trouble remains, replace caution/advisory panel (WP 0785 00). Go to **9.**
   c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**
9. **Procedure completed.**

LOCATION



NOTES

1. EH60A
2. UH60L
3. ESSS
4. UH60A
   AUXILIARY CABIN HEATER INSTALLED
5. MWO 50-78
6. W/O MWO 50-78

AA2771_1C
SA

Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 1 of 6)

LOCATION - Continued

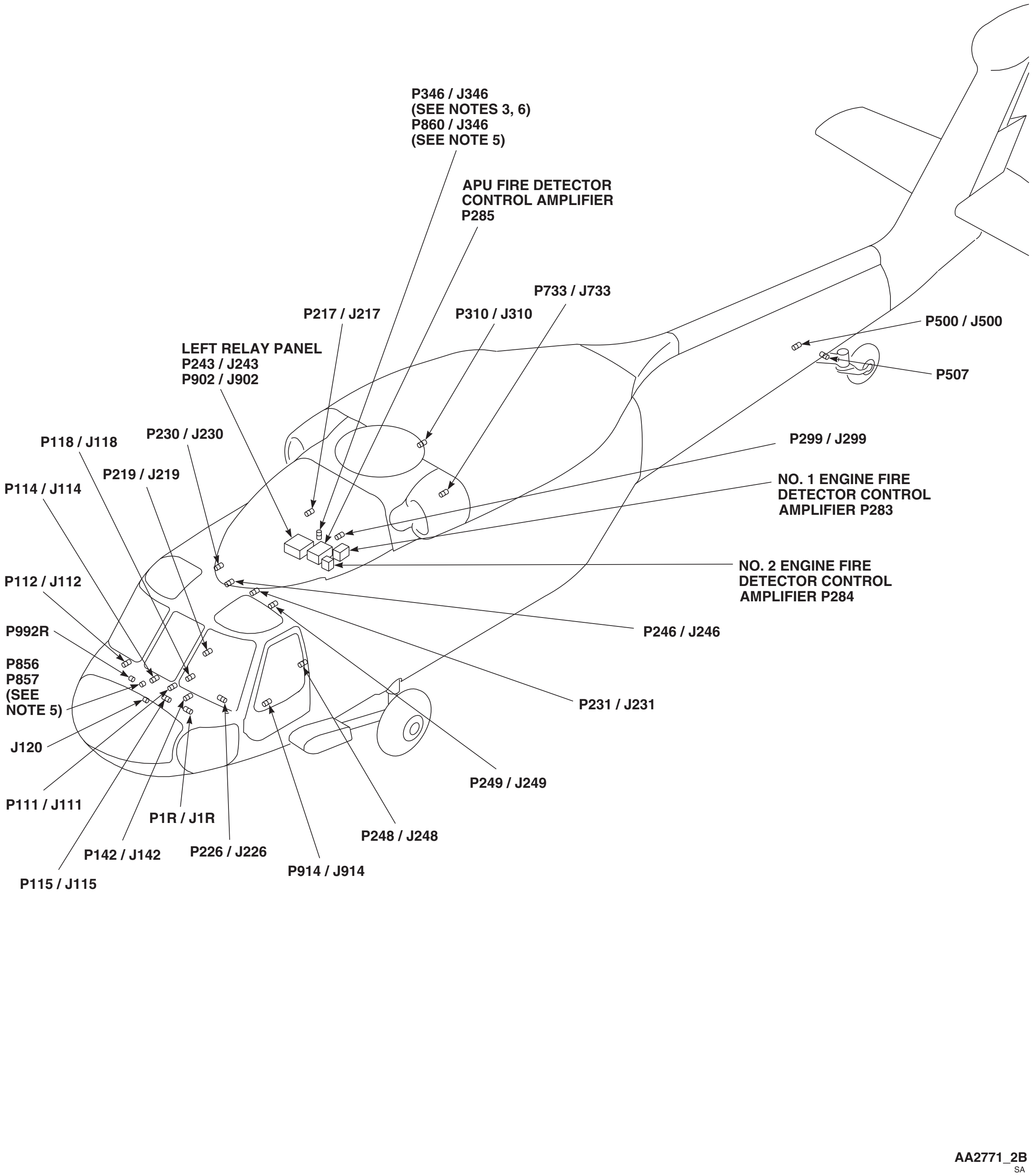


Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 2 of 6)

LOCATION - Continued

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J120	COCKPIT TUB, BL 30 RH, STA 220
P1R / J1R	COCKPIT, BL 6 LH, STA 210
P2 / A2J2 (SEE NOTE 1)	ALQ–156 CONTROL INDICATOR
P4R	STABILATOR CONTROL / AUTO FLIGHT CONTROL PANEL
P105	NO. 1 ENGINE QUADRANT
P106	NO. 2 ENGINE QUADRANT
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P112 / J112	COCKPIT, BL 6 RH, STA 197
P113 / J113	COCKPIT, BL 6 LH, STA 197
P114 / J114	COCKPIT, BL 8 RH, STA 200
P115 / J115	COCKPIT, BL 6 RH, STA 210
P118 / J118	CAUTION / ADVISORY PANEL
P133	BLADE DE–ICE CONTROL PANEL
P134	BLADE DE–ICE TEST PANEL
P135	MISCELLANEOUS SWITCH PANEL
P139	FUEL BOOST PUMP CONTROL PANEL
P142 / J142	COCKPIT, BL 3 RH, STA 210
P200 / J200	COCKPIT CEILING, BL 13 LH, STA 246
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR, BL 25 RH, STA 247
P226 / J226	COCKPIT, BL 10 LH, STA 228
P228 (SEE NOTE 2)	RESCUE HOIST CONTROL PANEL (WHEN INSTALLED)
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P231 / J231	CABIN CEILING, BL 9 LH, STA 247
P243 / J243	LEFT RELAY PANEL
P246 / J246	CABIN CEILING, BL 5 RH, STA 247
P248 / J248	CABIN, BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 23 LH
P283	NO. 1 ENGINE FIRE DETECTOR CONTROL AMPLIFIER
P284	NO. 2 ENGINE FIRE DETECTOR CONTROL AMPLIFIER

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P285	APU FIRE DETECTOR CONTROL AMPLIFIER
P299 / J299 (SEE NOTE 2)	CABIN CEILING, BL 10 LH, STA 302
P300R	COPILOT'S HSI / VSI MODE SEL PANEL
P301R	COPILOT'S VSI
P304R	PILOT'S VSI
P310 / J310	CABIN CEILING, BL 17 RH, STA 387
P315	LIGHTED SWITCHES DIMMER CONTROL
P316R	CIS MODE SELECT PANEL
P317R	PILOT'S HSI / VSI MODE SELECT PANEL
P346 / J346 (SEE NOTES 3, 6)	CABIN CEILING, BL 18 LH, STA 301
P360 (SEE NOTES 3, 6)	AUXILIARY FUEL MANAGEMENT PANEL
P393	INDICATOR LIGHTS DIMMER COCKPIT, BL 8 LH, STA 210, WL 222
P500 / J500	TAILCONE, BL 0, STA 609
P507	TAILWHEEL LOCK ACTUATOR
P675R	COPILOT'S RADAR ALTIMETER
P676R	PILOT'S RADAR ALTIMETER
P687R (SEE NOTE 2)	DOPPLER COMPUTER DISPLAY
P724 / J724 (SEE NOTE 4)	AUX CABIN HEATER CONTROL PANEL
P733 / J733	BL 24.71, STA 360, WL 265.30
P856 / J856 (SEE NOTE 5)	AUXILIARY FUEL MANAGEMENT CONTROL PANEL
P857 / J857 (SEE NOTE 5)	AUXILIARY FUEL MANAGEMENT CONTROL PANEL
P860 / J346 (SEE NOTE 5)	CABIN FUEL HARNESS
P902 / J902	LEFT RELAY PANEL
P914 / J914	COCKPIT, BL 24 LH, STA 247
P935R (SEE NOTE 1)	SYSTEM SELECT PANEL COCKPIT, BL 2.5 LH, STA 197
P992R (SEE NOTE 1)	COCKPIT BL 19.5 RH, STA 187

AA2771_3B
SA

Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 3 of 6)

LOCATION - Continued

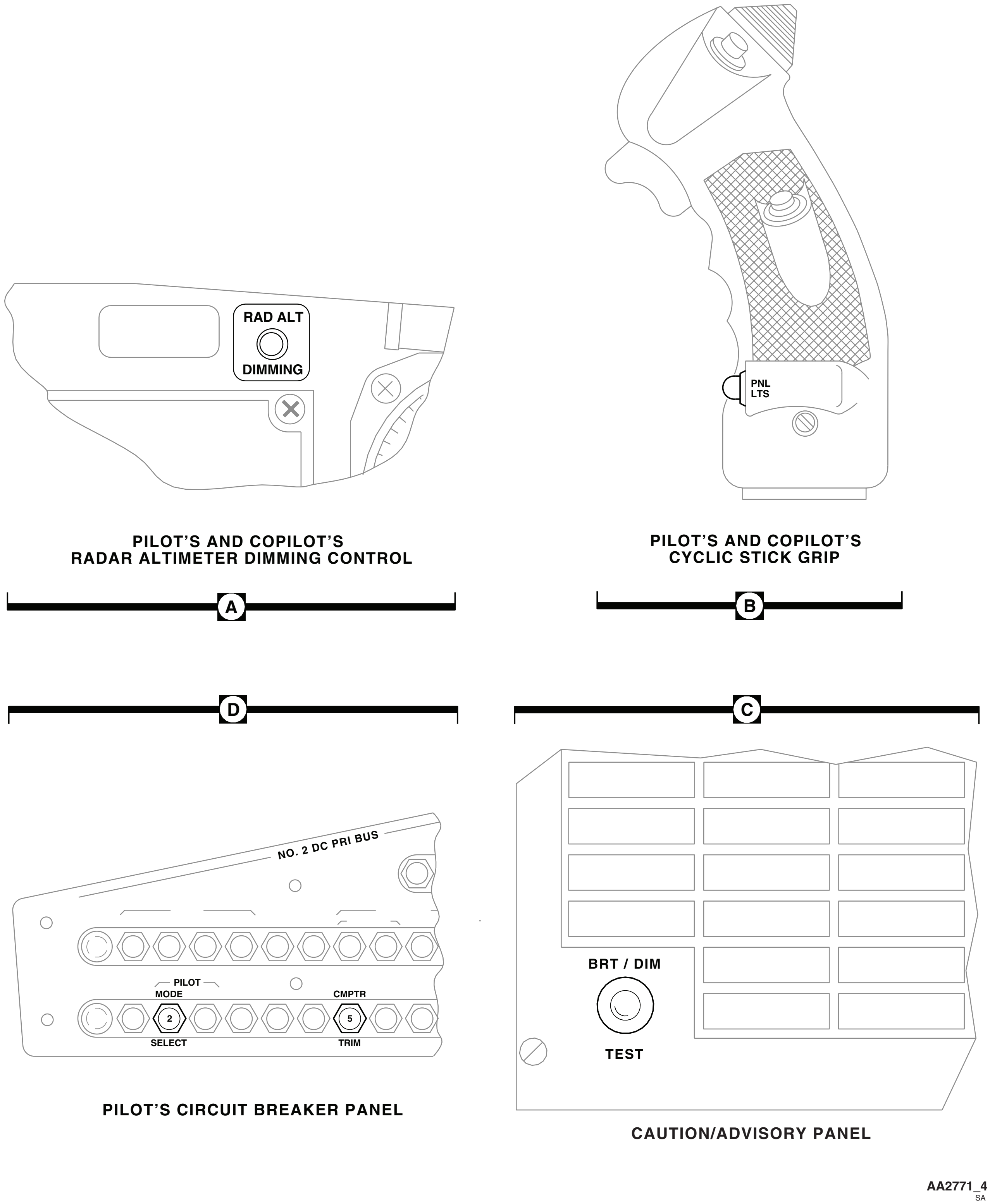


Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 4 of 6)

LOCATION - Continued

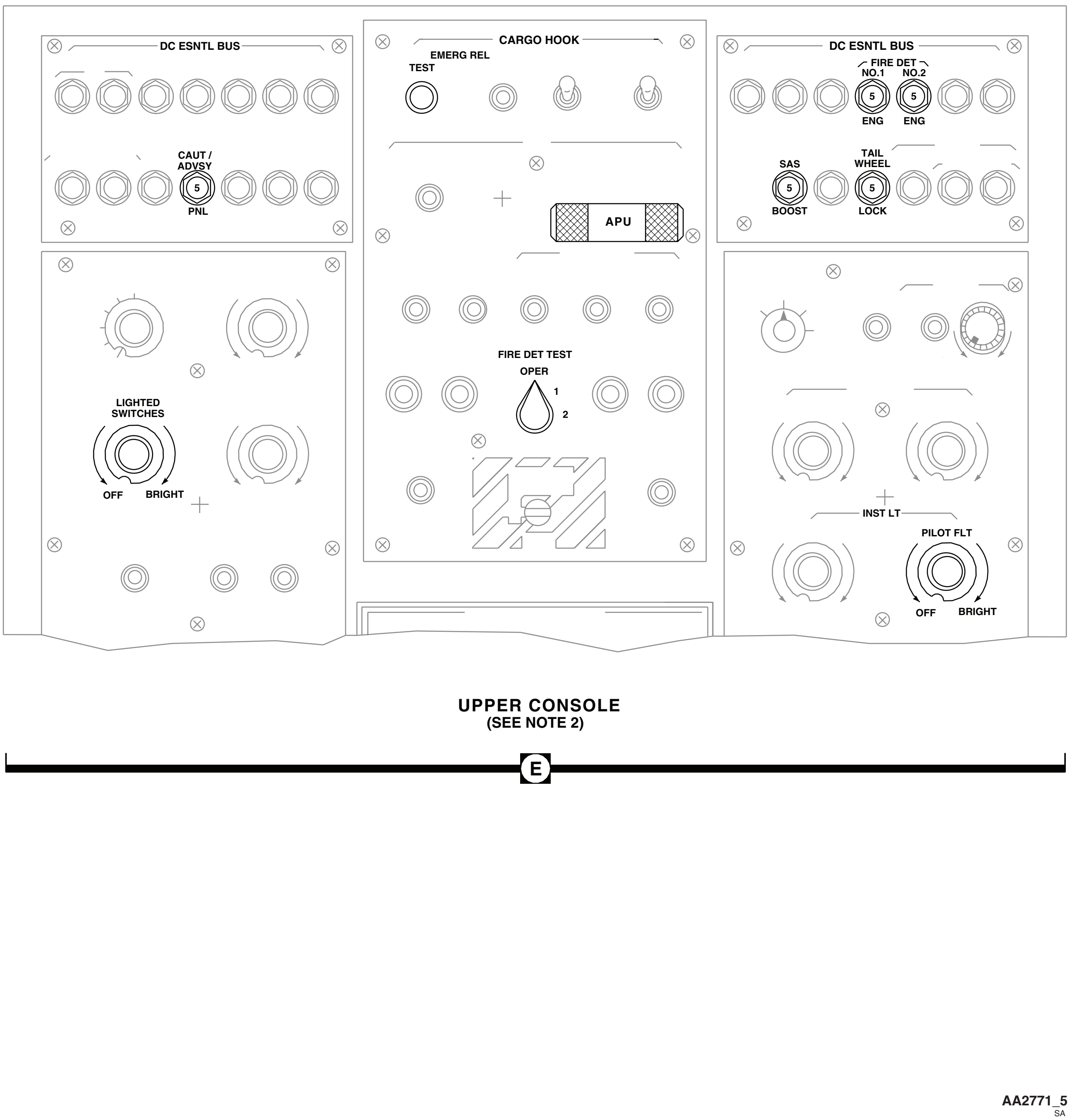


Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 5 of 6)

LOCATION - Continued

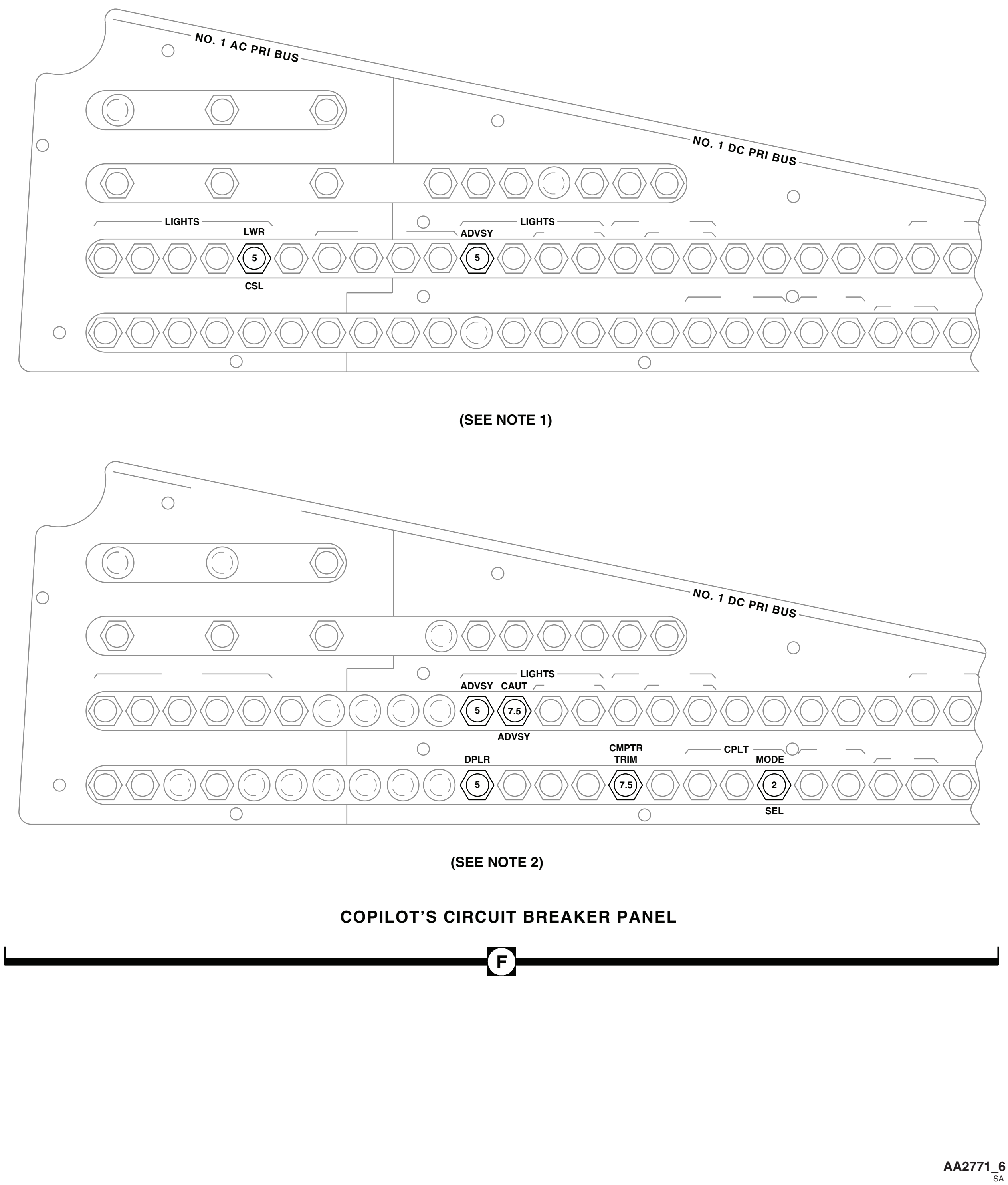
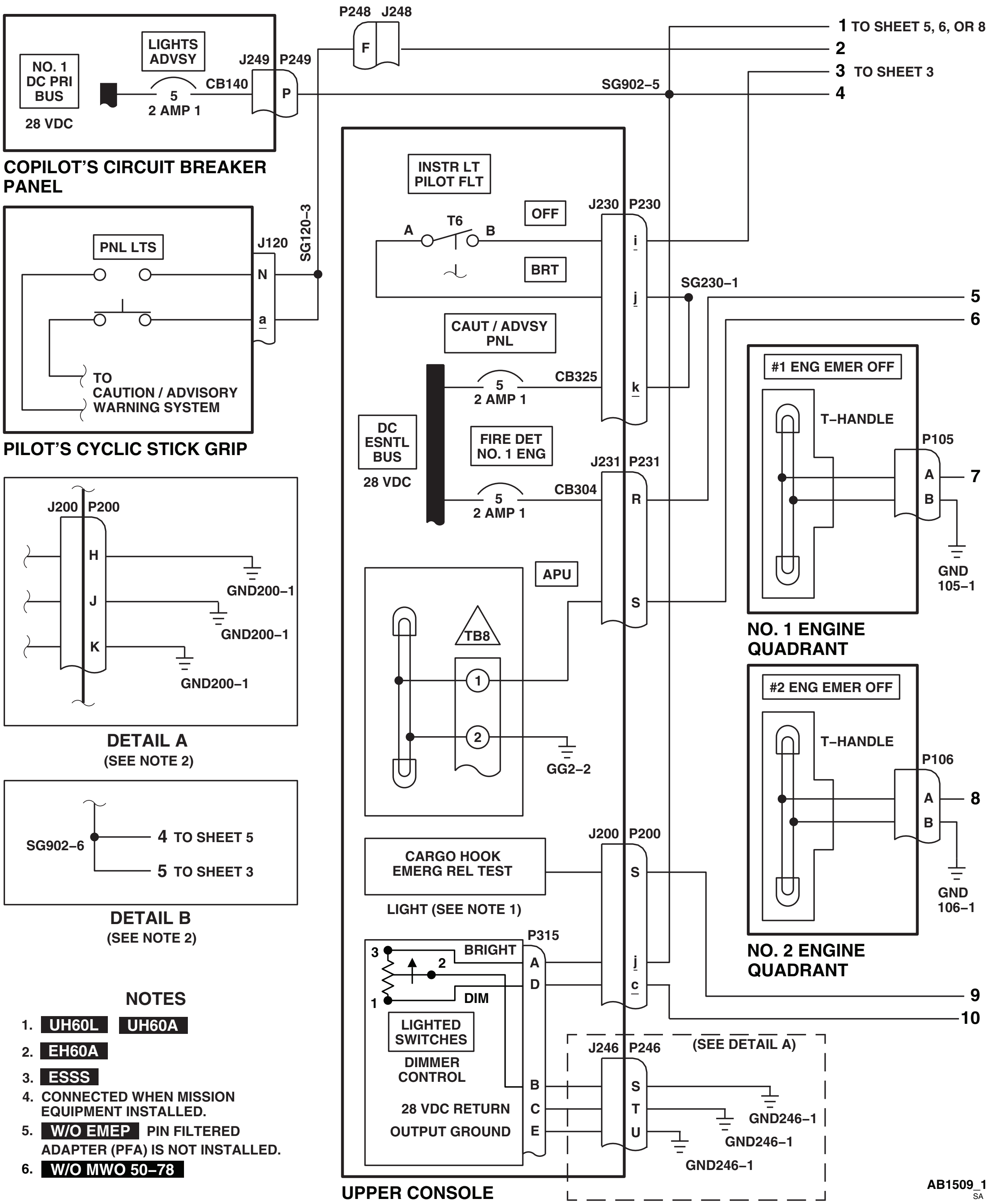


Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 6 of 6)

SCHEMATICS



NOTES

1. UH60L UH60A
2. EH60A
3. ESSS
4. CONNECTED WHEN MISSION EQUIPMENT INSTALLED.
5. W/O EMEP PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.
6. W/O MWO 50–78

Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 1 of 9)

SCHEMATICS - Continued

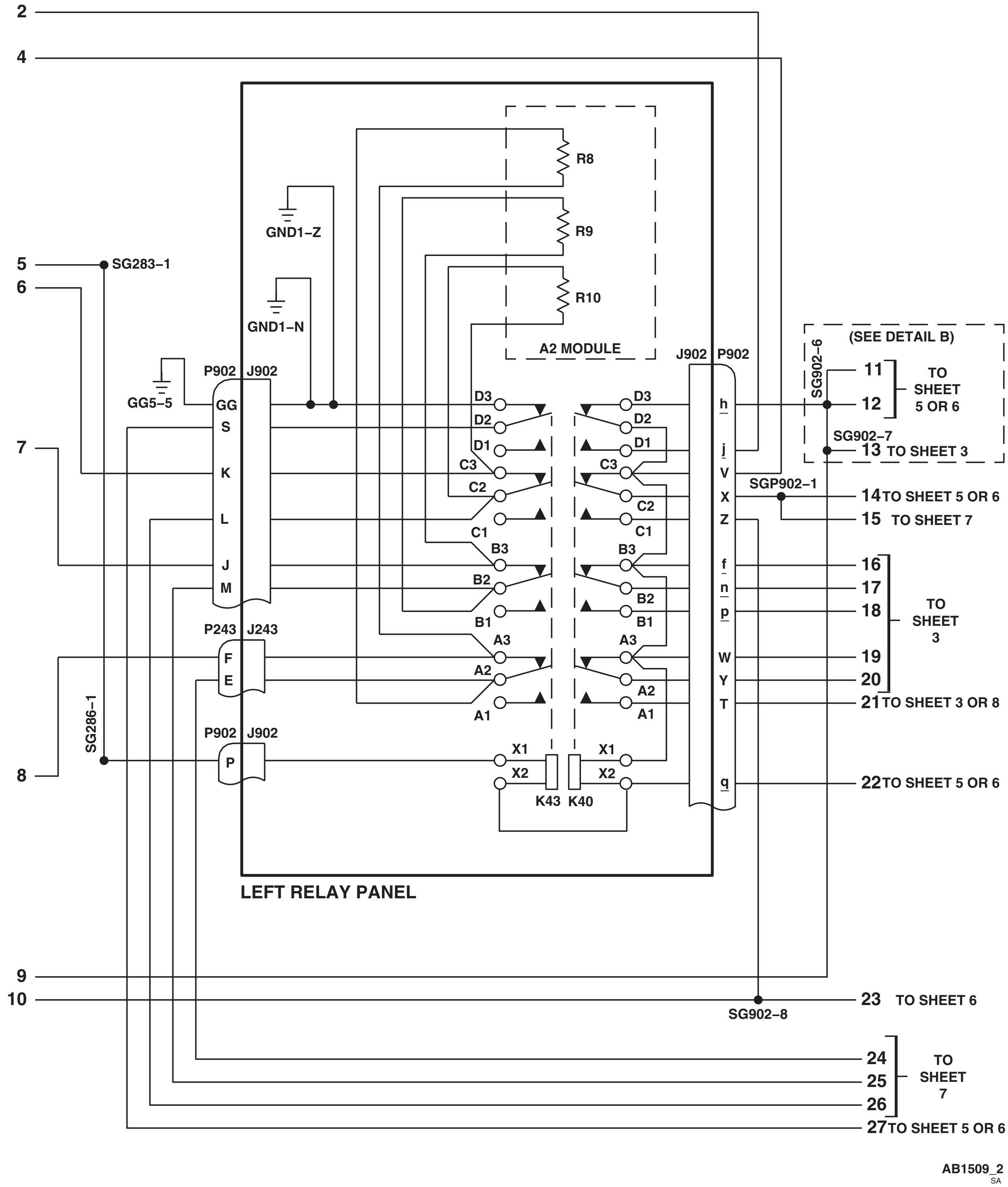


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 2 of 9)

SCHEMATICS - Continued

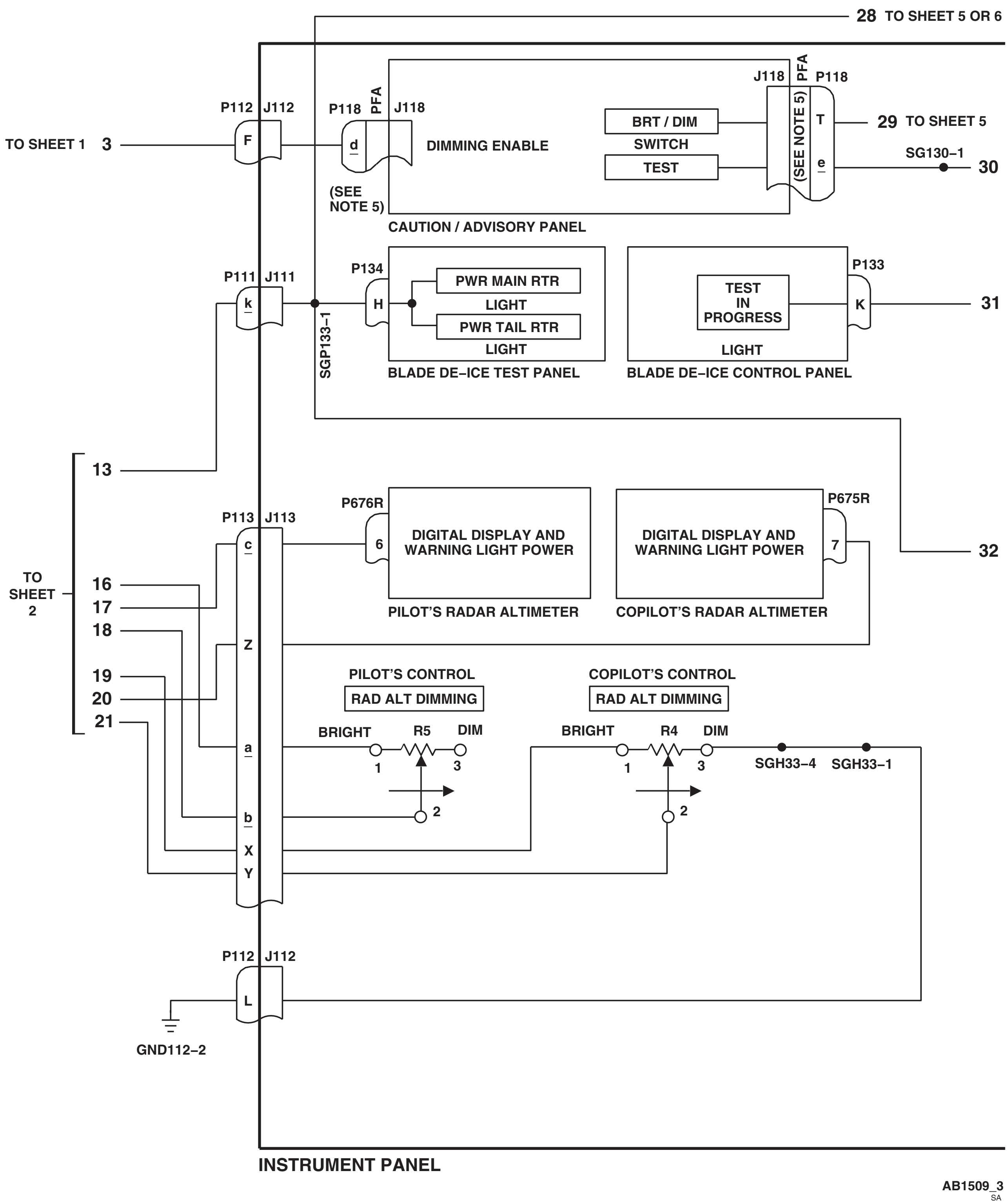


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 3 of 9)

SCHEMATICS - Continued

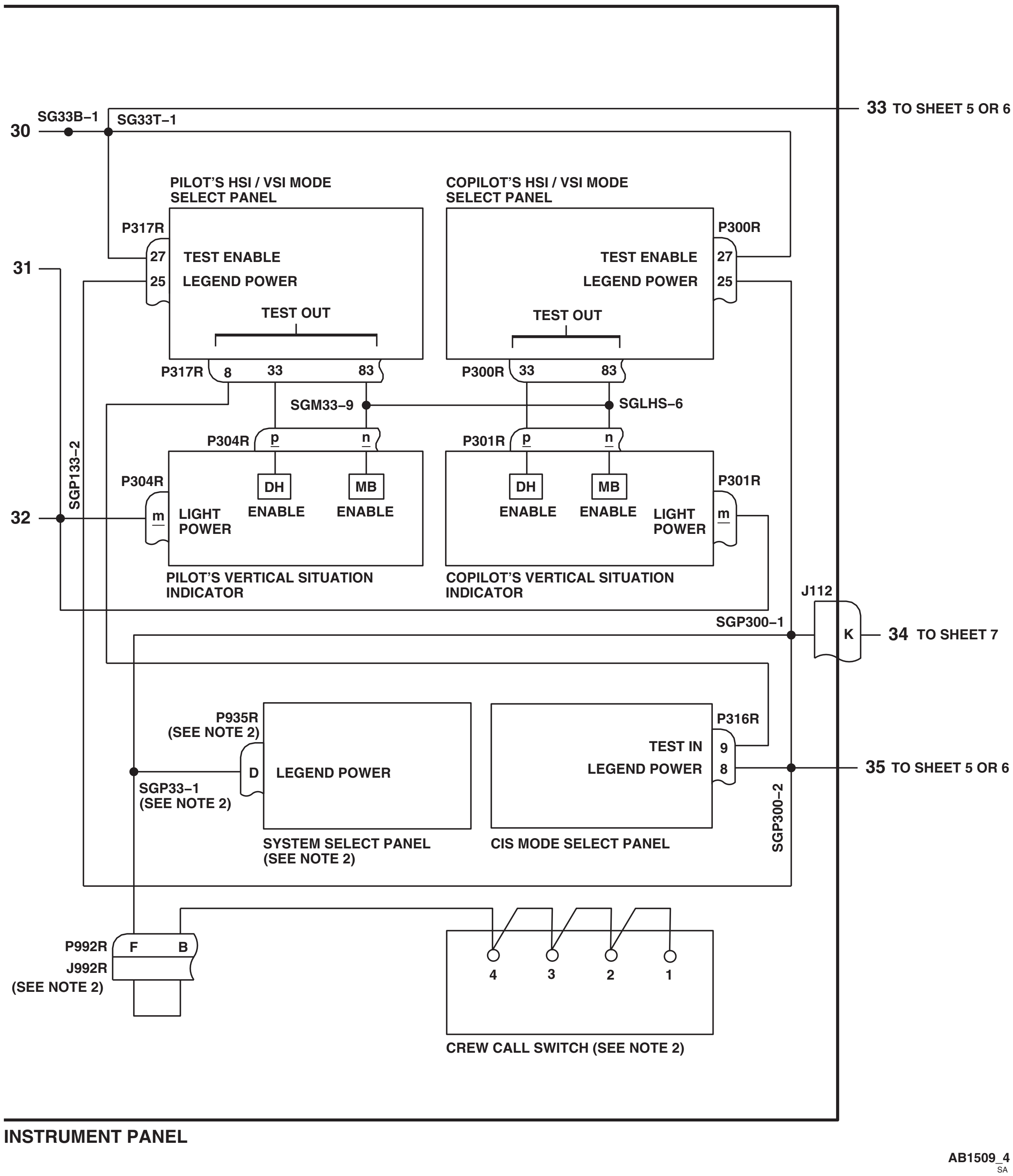


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 4 of 9)

SCHEMATICS - Continued

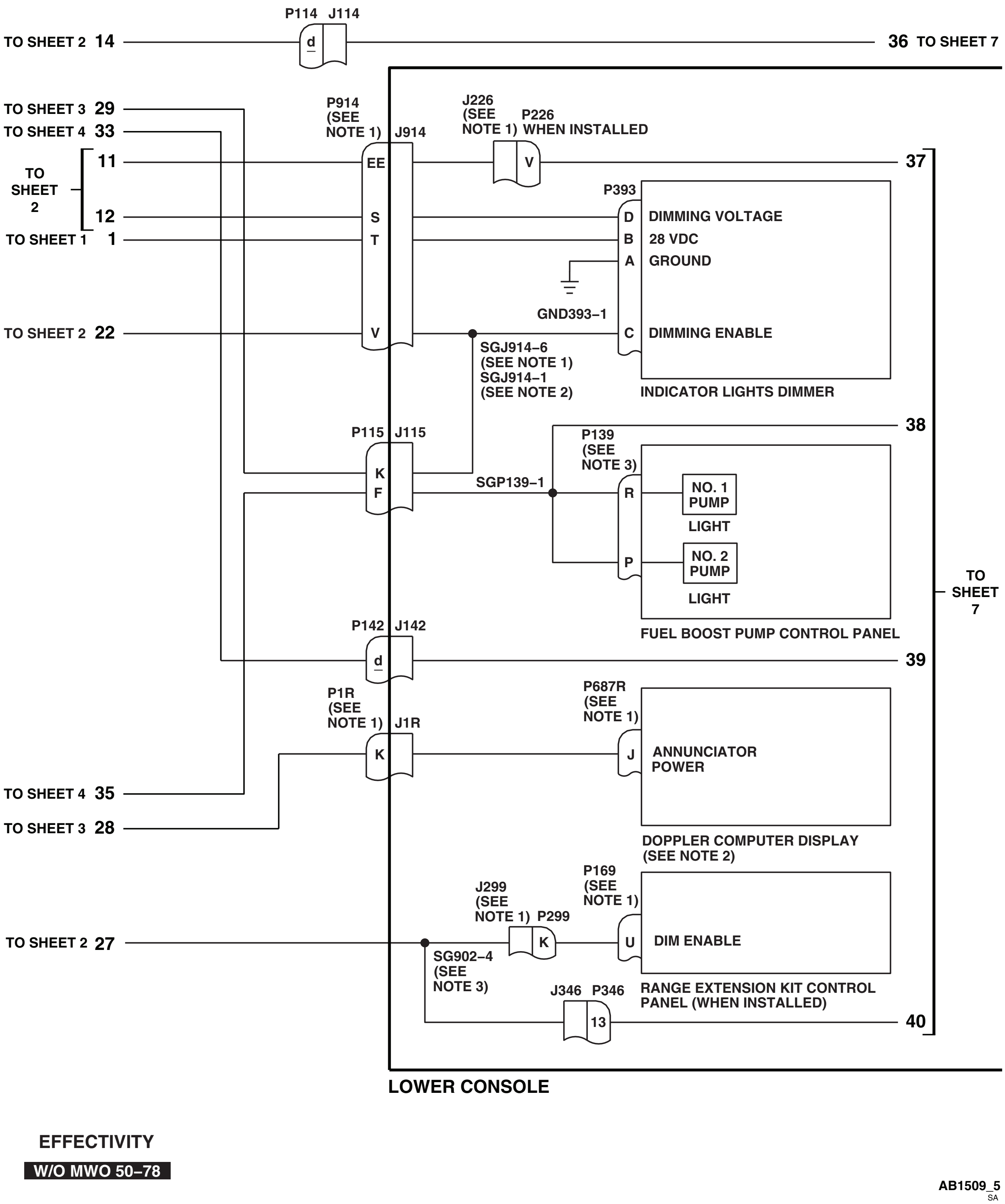


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 5 of 9)

SCHEMATICS - Continued

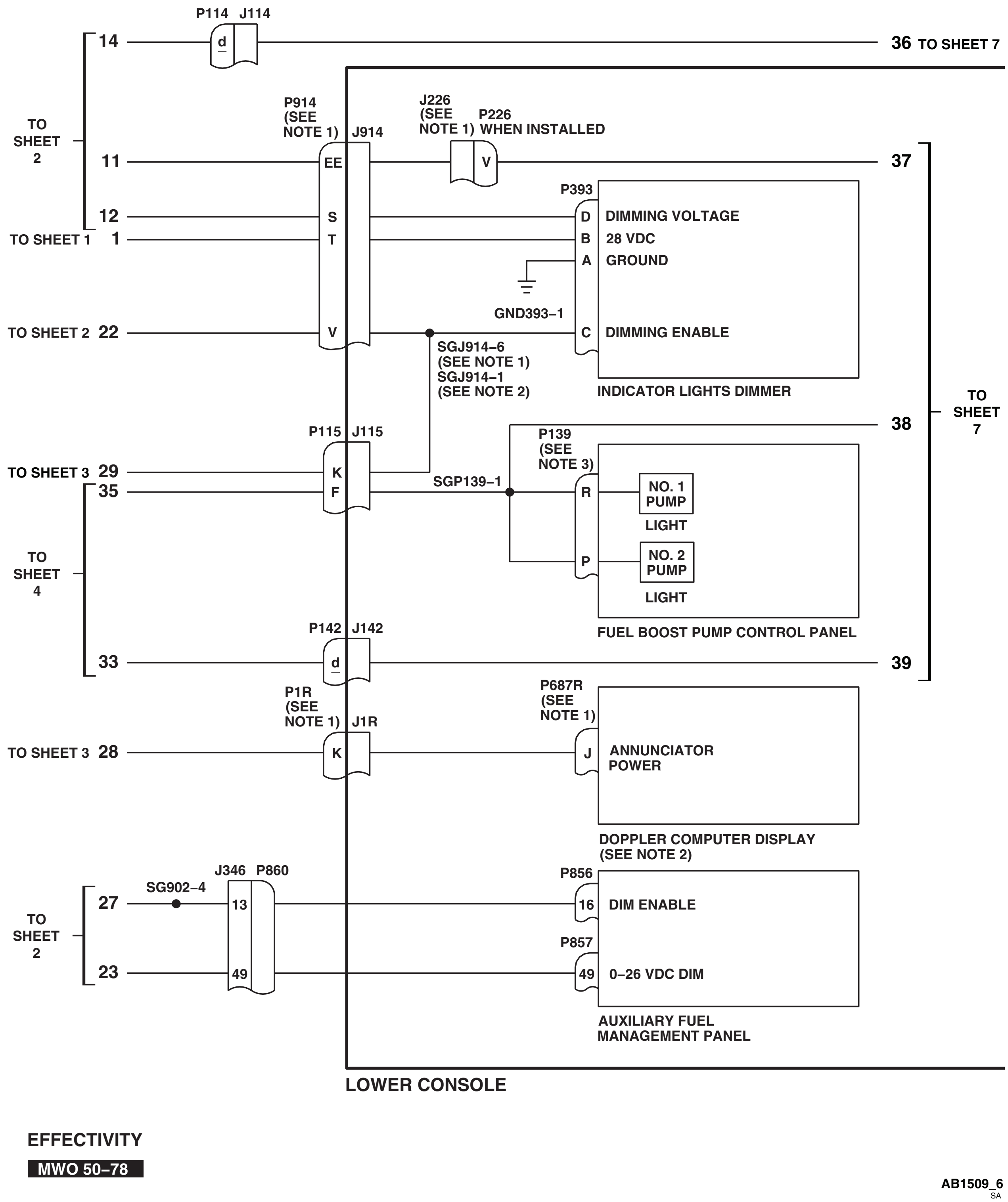


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 6 of 9)

SCHEMATICS - Continued

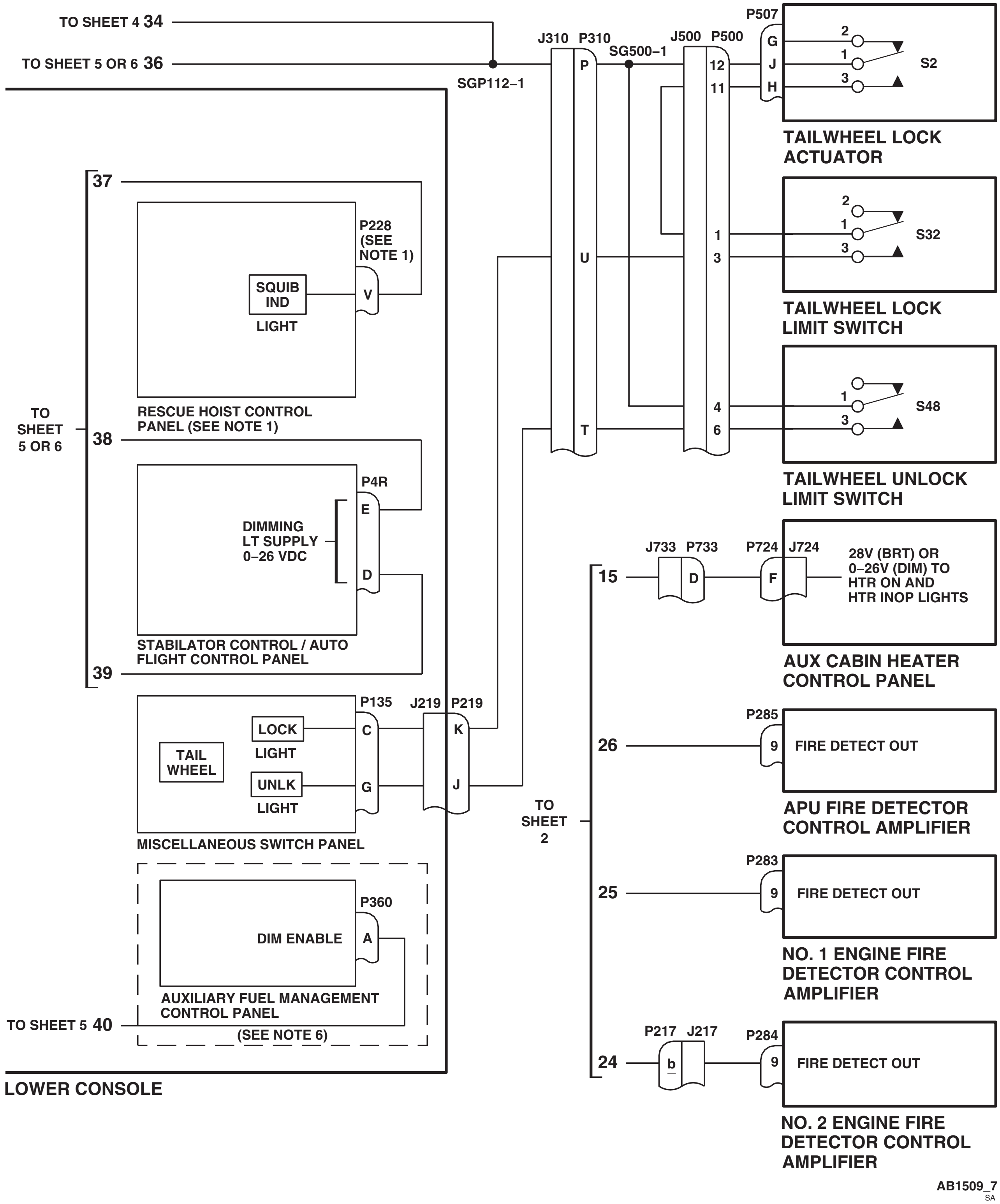


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 7 of 9)

SCHEMATICS - Continued

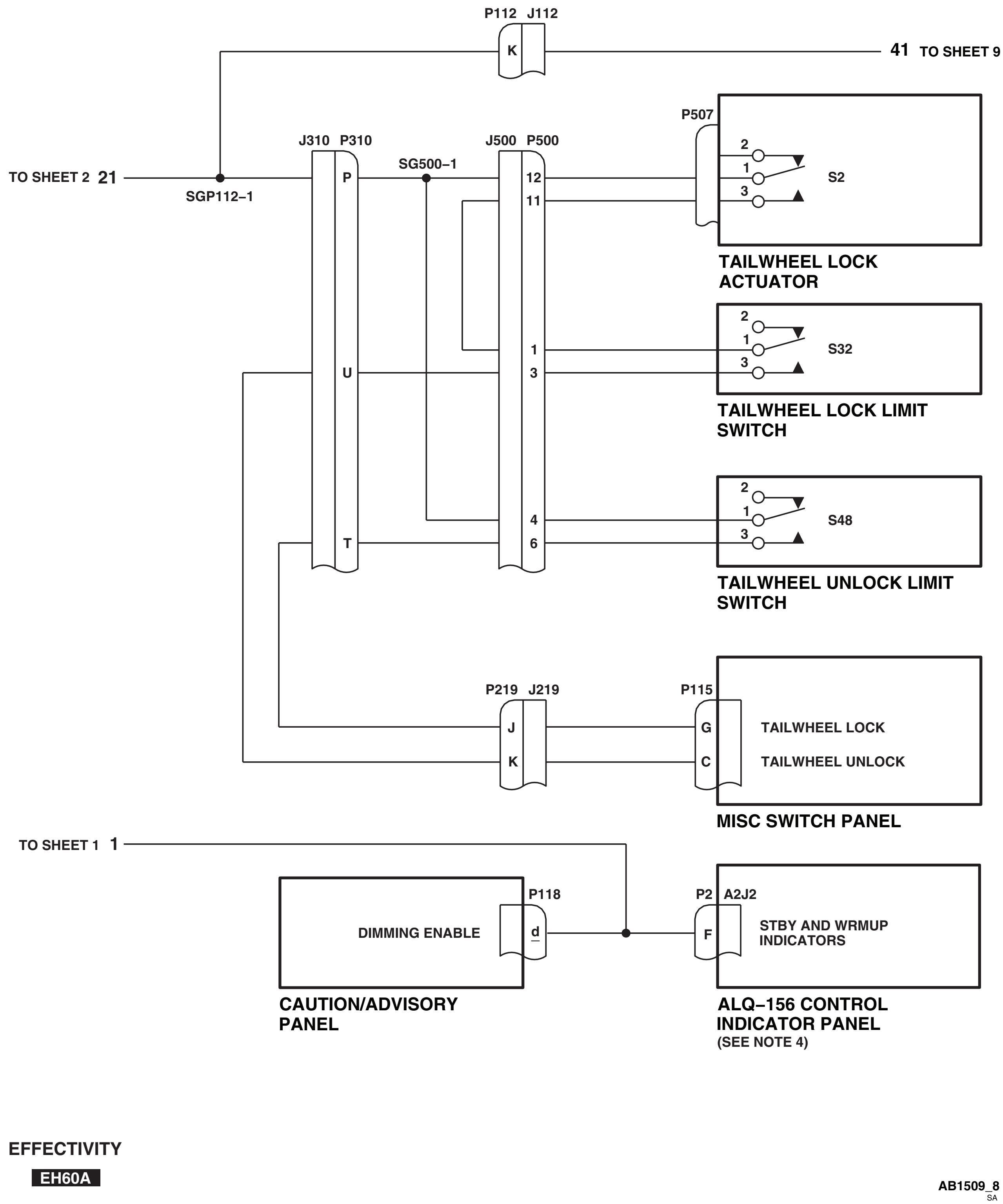


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 8 of 9)

SCHEMATICS - Continued

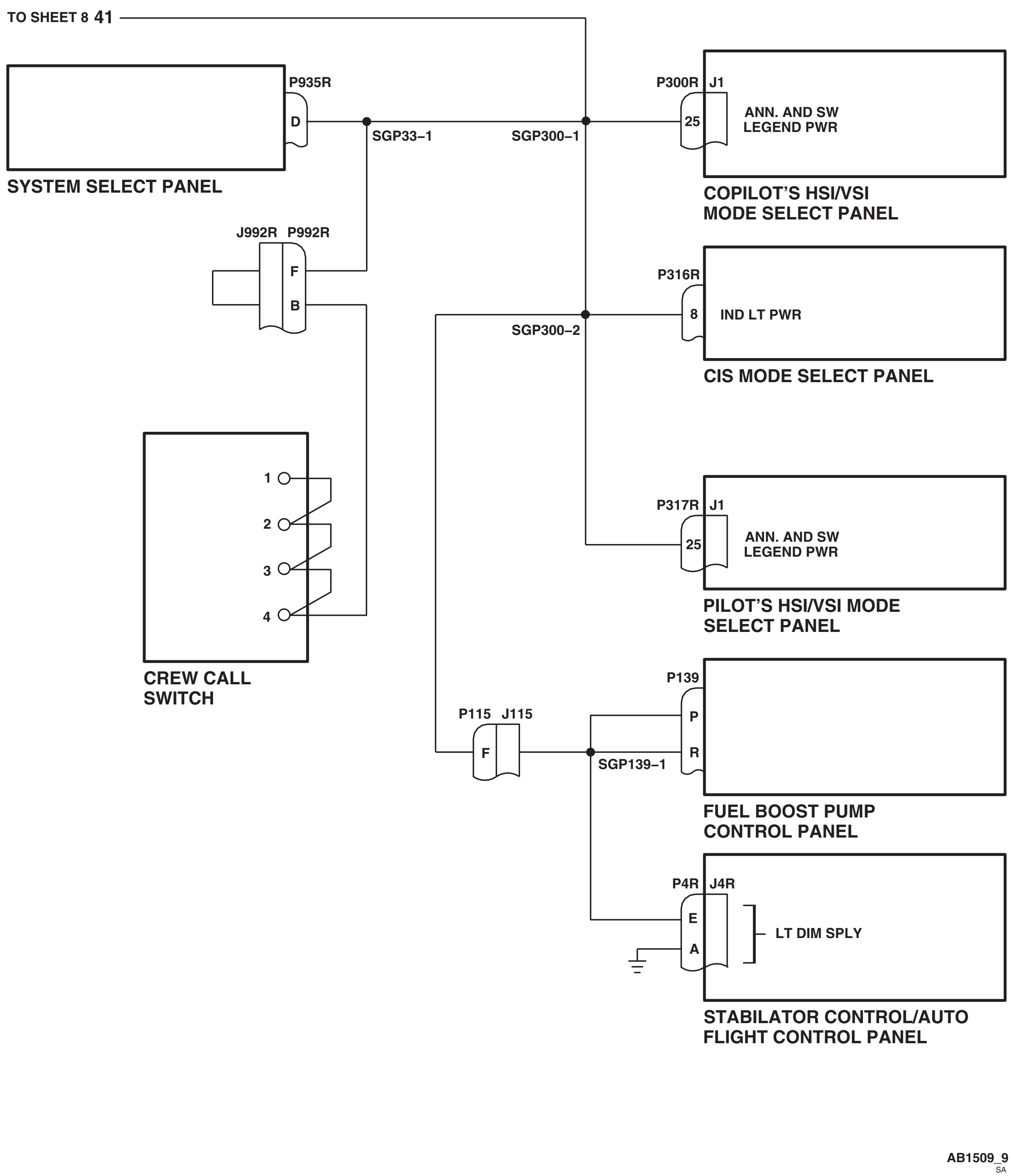


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 9 of 9)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

FORMATION LIGHTS

INITIAL SETUP:

Test Equipment
Locally-Made Infrared Light Test Set Figure 149, WP 1805 00
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
WP 0104 00
WP 0824 00
WP 0825 00
WP 0828 00
WP 0898 00
WP 1685 00
WP 1747 00
WP 1805 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

FORMATION LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Check four formation light assemblies for cracked or broken lens.

Indication/Condition

No formation light lens shall be cracked or broken.

Corrective Action

If Indication/Condition is not as specified, replace formation light assembly (WP 0898 00).

Step

2. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
LIGHTS FORM HV	Pilot's

FORMATION LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
LIGHTS FORM LV	Pilot’s
IR LTS	Pilot’s

3. Turn on electrical power.
4. Make sure NAV LTS switch is set to NORM.
5. Turn FORMATION LT control to 5.

Indication/Condition

All four formation lights shall go on.

Corrective Action

1. If Indication/Condition is not as specified, go to NO FORMATION LIGHTS GO ON, in this work package.
2. If one formation light does not go on, go to ONE FORMATION LIGHT DOES NOT GO ON, in this work package.

Step

6. Turn FORMATION LT control through positions 4, 3, 2, 1, and OFF.

Indication/Condition

Formation lights shall gradually dim and go off when control is OFF.

Corrective Action

If Indication/Condition is not as specified, replace FORMATION LT control (WP 0828 00).

Step

7. Check all four IR formation light assemblies for cracked or broken lens.

Indication/Condition

No IR formation light lens shall be cracked or broken.

Corrective Action

If Indication/Condition is not as specified, replace IR formation light assembly (WP 0898 00).

Step

8. Place locally-made IR light test set (test set) (WP 1805 00) power switch to ON.

Indication/Condition

Test set POWER ON lamp shall go on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot test set.

FORMATION LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

9. Place test set IR detector lead over POWER ON lamp.

Indication/Condition

Test set IR LIGHT ON lamp shall go on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot test set.

Step

10. Place FORMATION LT control to position 5.
11. Place NAV LTS switch to IR.
12. Place test set IR detector lead over clear lens of each IR formation light.

Indication/Condition

Test set IR LIGHT ON lamp shall go on over all four IR (bright) formation lights.

Corrective Action

1. If Indication/Condition is not as specified, go to NO IR (BRIGHT) FORMATION LIGHTS GO ON WITH FORMATION LT CONTROL IN POSITION 5, in this work package.
2. If one IR formation light does not go on, go to ONE IR (BRIGHT) FORMATION LIGHT DOES NOT GO ON, in this work package.

Step

13. Turn FORMATION LT control through position 1 through 4. At each position place IR detector lead over dark lens of IR formation light.

Indication/Condition

Test set IR LIGHT ON lamp shall go on over all four IR (dim) formation lights.

Corrective Action

1. If Indication/Condition is not a specified, check for 5 vdc between terminals C and G of FORMATION LT control.
   a. If voltage is as specified, repair/replace wiring between FORMATION LT control terminal C and the following, as required (WP 1747 00):
      - J454-3
      - J547-3
      - J711-3
      - J712-3
   b. If voltage is not as specified, replace FORMATION LT control (WP 0828 00).
2. If one IR formation light does not go on, check for 5 vdc between the following, as required:
   - J454-3 and J454-2
   - J547-3 and J547-2

FORMATION LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- J711-3 and J711-2
- J712-3 and J712-2

a. If voltage is as specified, replace light assembly as required (WP 0898 00).

b. If voltage is not as specified, repair/replace wiring between FORMATION LT control terminal C and the following, as required (WP 1747 00):

- J454-3
- J547-3
- J711-3
- J712-3

Step

14. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

NO FORMATION LIGHTS GO ON

SYMPTOM

No formation lights go on.

MALFUNCTION

No formation lights go on.

CORRECTIVE ACTION

1. Check continuity between J701-A and P243-L.

a. If continuity is present, go to **2.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

2. Check for 115 vac between P243-N and ground.

a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **10.**

b. If voltage is not as specified, go to **3.**

3. Check for 115 vac between LIGHTS FORM LV circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P243-N (WP 1747 00). Go to **10.**

b. If voltage is not as specified, go to **4.**

4. Check for 115 vac between LIGHTS FORM LV circuit breaker terminal 2 and ground.

a. If voltage is as specified, replace circuit breaker (WP 0824 00). Go to **10.**

NO FORMATION LIGHTS GO ON - Continued

b. If voltage is not as specified, go to **5.**

5. **Check for 115 vac between FORMATION LT control T9-LV and ground.**

a. If voltage is as specified, repair/replace wiring between T9-LV and LIGHTS FORM LV circuit breaker terminal 2 (WP 1747 00). Go to **10.**

b. If voltage is not as specified, go to **6.**

6. **Check for 115 vac between T9-5 and ground.**

a. If voltage is as specified, replace control (WP 0828 00). Go to **10.**

b. If voltage is not as specified, go to **7.**

7. **Check continuity between T9-G and GG2-1.**

a. If continuity is present, go to **8.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

8. **Check for 115 vac between T9-HV and ground.**

a. If voltage is as specified, replace control (WP 0828 00). Go to **10.**

b. If voltage is not as specified, go to **9.**

9. **Check for 115 vac between LIGHTS FORM HV circuit breaker terminal 1 and ground.**

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T9-HV (WP 1747 00). Go to **10.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **10.**

10. **Procedure completed.**

ONE FORMATION LIGHT DOES NOT GO ON

SYMPTOM

One formation light does not go on.

MALFUNCTION

One formation light does not go on.

CORRECTIVE ACTION

1. **Check for 115 vac between:**

- **J402-A and J402-B**
- **J506-A and J506-B**
- **J700-A and J700-B**
- **J701-A and J701-B**

a. If voltage is as specified, replace light assembly as required (WP 0898 00). Go to **3.**

b. If voltage is not as specified, go to **2.**

2. **Check continuity between:**

- **J402-B and ground**

ONE FORMATION LIGHT DOES NOT GO ON - Continued

- **J506-B and ground**
- **J700-B and ground**
- **J701-B and ground**

a. If continuity is present, repair/replace wiring between (WP 1747 00), then go to **3:**

- P243-L and J402-A
- P243-L and J506-A
- P243-L and J700-A
- P243-L and J701-A

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **3.**

3. Procedure completed.

NO IR (BRIGHT) FORMATION LIGHTS GO ON WITH FORMATION LT CONTROL IN POSITION 5

SYMPTOM

No IR (bright) formation lights go on with FORMATION LT control in position 5.

MALFUNCTION

No IR (bright) formation lights go on with FORMATION LT control in position 5.

CORRECTIVE ACTION

1. Check for 28 vdc between P243-j and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **5.**

2. Check for 10 vdc between FORMATION LT control T9-B and T9-G.

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, go to **4.**

3. Check for 10 vdc between T9-A and T9-G.

a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **8.**

b. If voltage is not as specified, replace control (WP 0828 00). Go to **8.**

4. Check continuity between T9-B and P243-g.

a. If continuity is present, replace left relay panel (WP 0825 00). Go to **8.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

5. Check for 28 vdc between NAV LTS switch S69-3 and ground.

a. If voltage is as specified, repair/replace wiring between S69-3 and P243-j (WP 1747 00). Go to **8.**

b. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between S69-2 and ground.

a. If voltage is as specified, replace NAV LTS switch (WP 0828 00). Go to **8.**

NO IR (BRIGHT) FORMATION LIGHTS GO ON WITH FORMATION LT CONTROL IN POSITION 5 - Continued

b. If voltage is not as specified, go to **7.**

7. **Check for 28 vdc between IR LTS circuit breaker terminal 1 and ground.**

a. If voltage is as specified, repair/replace wiring between S69-2 and terminal 1 of circuit breaker (WP 1747 00). Go to **8.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8.**

8. **Procedure completed.**

ONE IR (BRIGHT) FORMATION LIGHT DOES NOT GO ON

SYMPTOM

One IR (bright) formation light does not go on.

MALFUNCTION

One IR (bright) formation light does not go on.

CORRECTIVE ACTION

1. **Check for 10 vdc between:**

- **J454-1 and J454-2**
- **J547-1 and J547-2**
- **J711-1 and J711-2**
- **J712-1 and J712-2**

a. If voltage is as specified, replace light assembly as required (WP 0898 00). Go to **3.**

b. If voltage is not as specified, go to **2.**

2. **Check continuity between:**

- **J454-2 and ground**
- **J547-2 and ground**
- **J711-2 and ground**
- **J712-2 and ground**

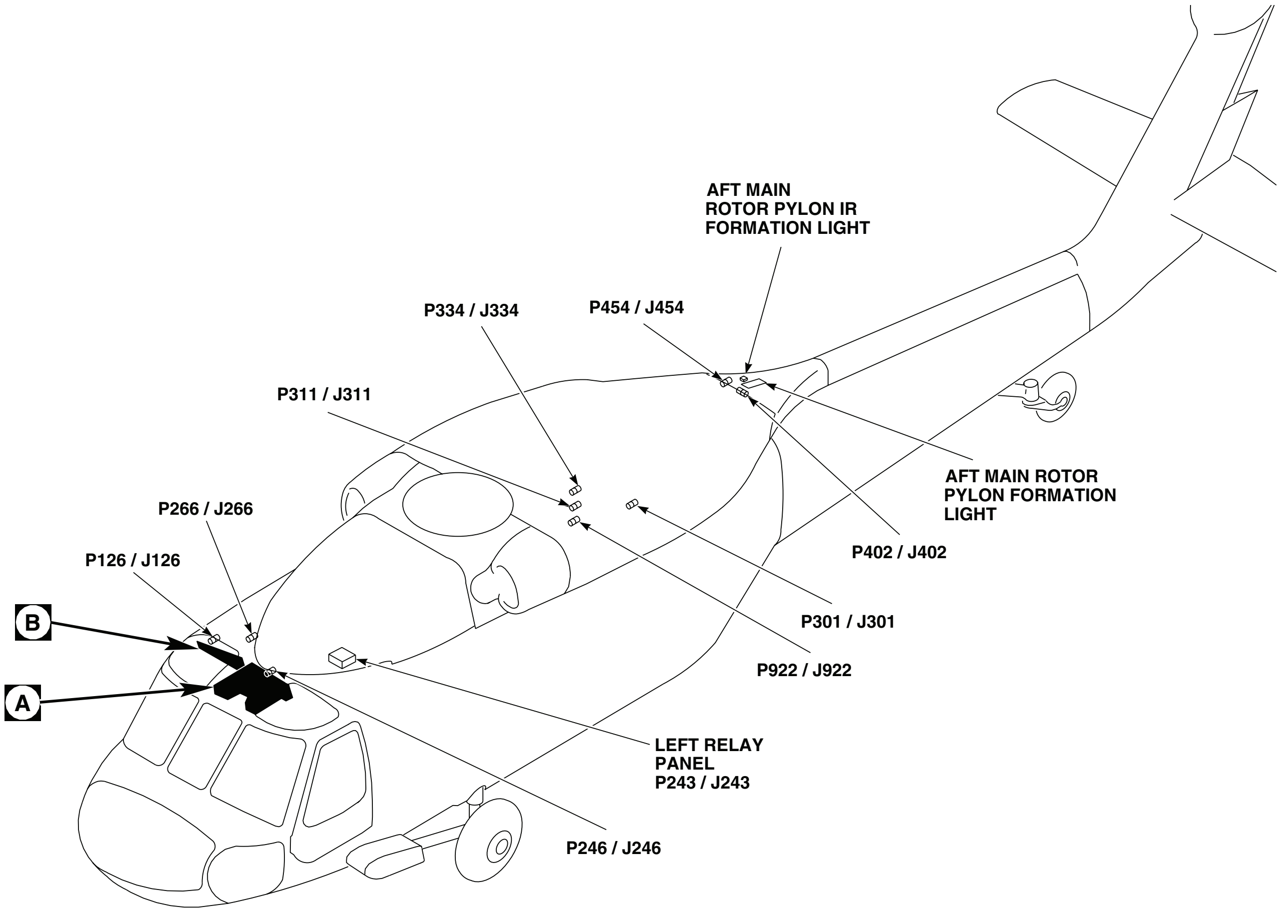
a. If continuity is present, repair/replace wiring between (WP 1747 00), then go to **3:**

- J454-1 and T9-A
- J547-1 and T9-A
- J711-1 and T9-A
- J712-1 and T9-A

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **3.**

3. **Procedure completed.**

LOCATION



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P126 / J126	COCKPIT CEILING, BL 28 RH, STA 243
P243 / J243	LEFT RELAY PANEL
P246 / J246	CABIN CEILING, BL 5 RH, STA 247
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 25 RH
P301 / J301	TRANSITION DECK, BL 12.88 LH, STA 415
P311 / J311	CABIN CEILING, BL 16.5 LH, STA 383
P334 / J334	CABIN CEILING, BL 16.5 LH, STA 394
P402 / J402	AFT MAIN ROTOR PYLON BL 3 LH, STA 482

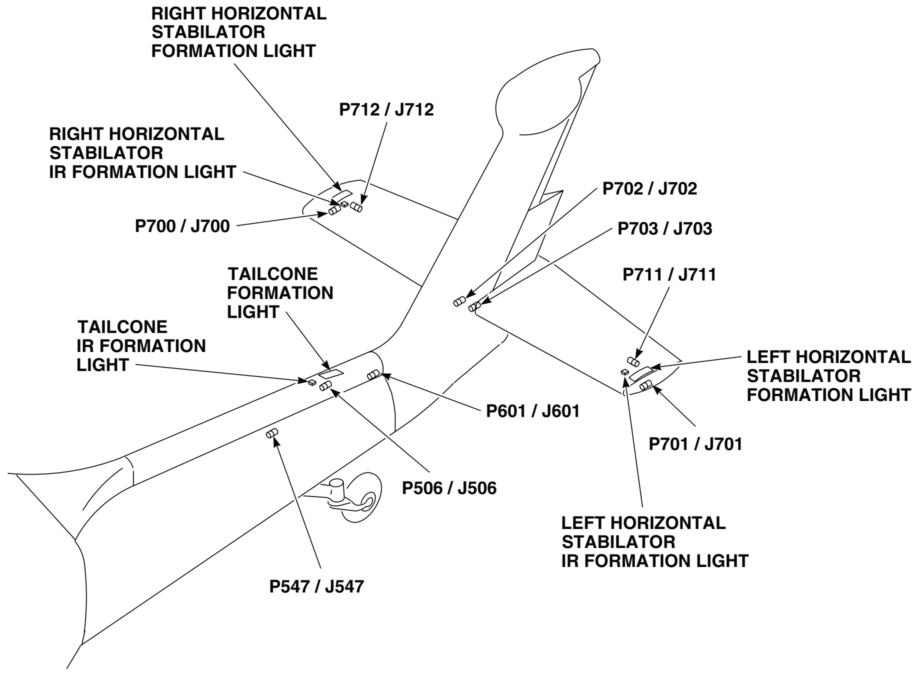
NOTE

EH60A

AA2774_1
SA

Figure 1. Formation Lights Location Diagram. (Sheet 1 of 3)

LOCATION - Continued



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P454 / J454	AFT MAIN ROTOR PYLON BL 2 RH, STA 480, WL 270
P506 / J506	TAILCONE, BL 2 RH, STA 622
P547 / J547	TAILCONE, BL 2 RH, STA 603, WL 250
P601 / J601	TAILCONE, BL 4 LH, STA 651, WL 234
P700 / J700	HORIZONTAL STABILATOR BL 81 RH, STA 709
P701 / J701	HORIZONTAL STABILATOR BL 81 LH, STA 709
P702 / J702	HORIZONTAL STABILATOR BL 5 RH, STA 694
P703 / J703	HORIZONTAL STABILATOR BL 5 LH, STA 694
P711 / J711	LEFT HORIZONTAL STABILATOR, BL 78 LH, STA 710
P712 / J712	RIGHT HORIZONTAL STABILATOR, BL 78 RH, STA 710
P922 / J922	CABIN CEILING, BL 10.7 LH, STA 380

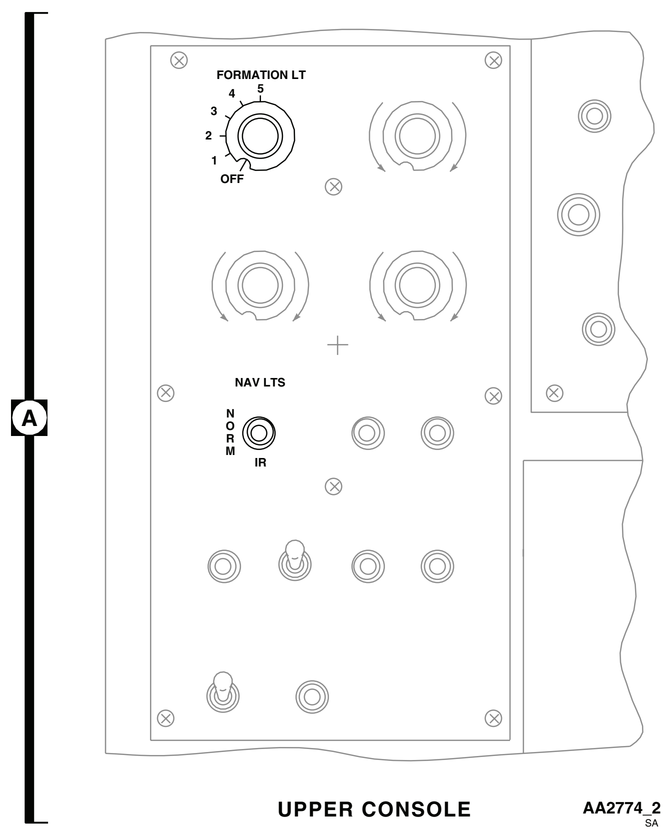


Figure 1. Formation Lights Location Diagram. (Sheet 2 of 3)

LOCATION - Continued

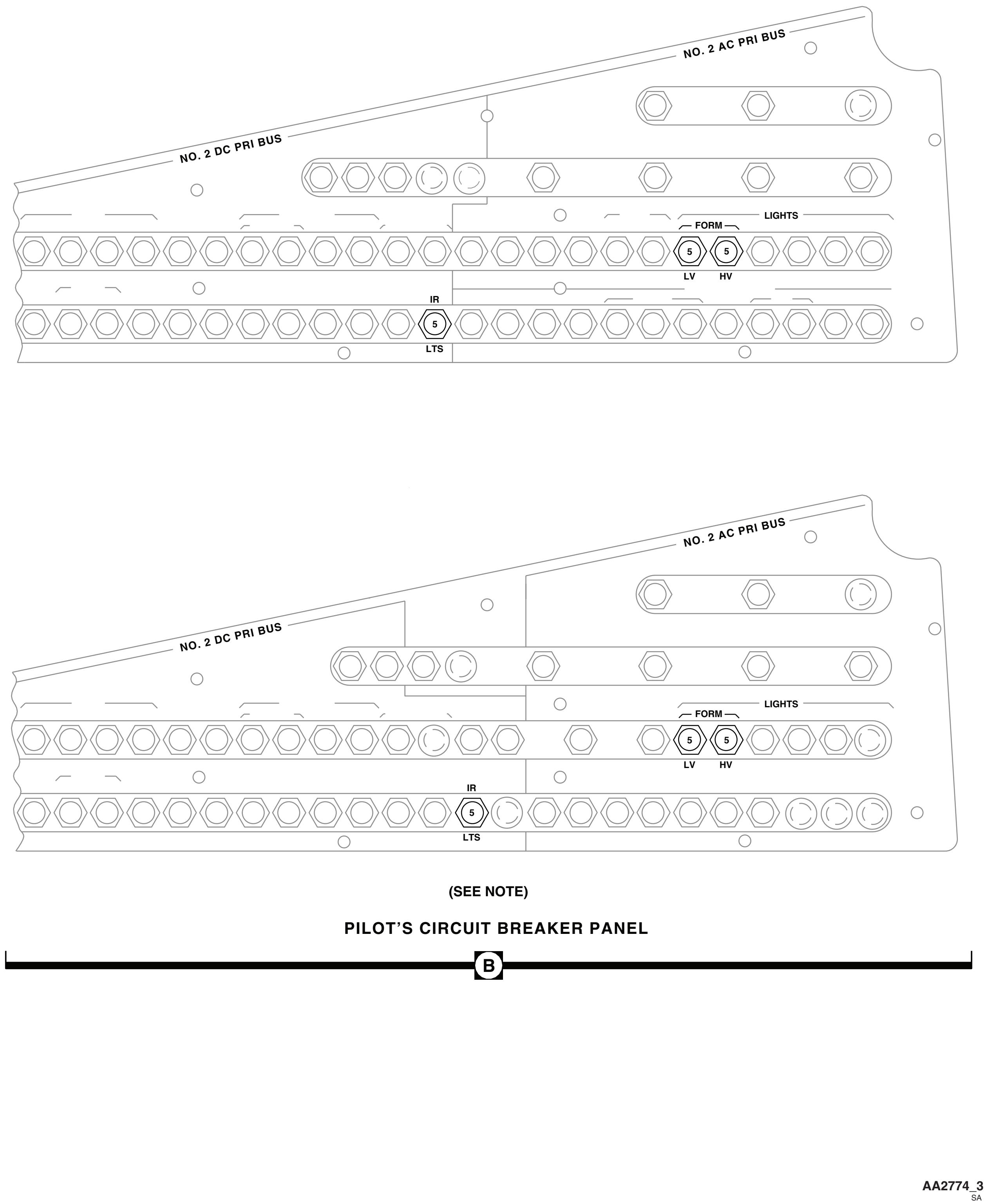


Figure 1. Formation Lights Location Diagram. (Sheet 3 of 3)

SCHEMATICS

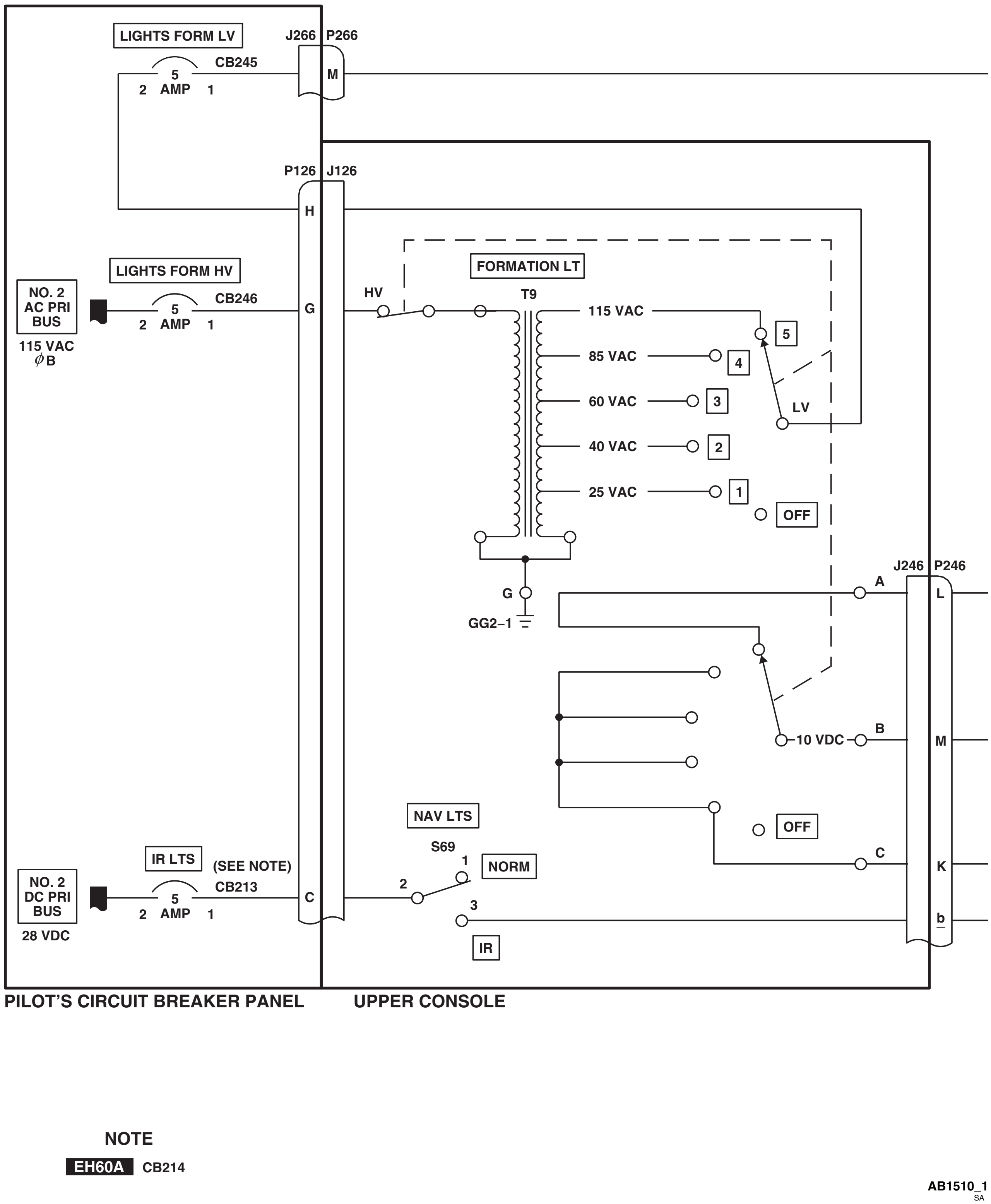


Figure 2. Formation Lights Schematic Diagram. (Sheet 1 of 4)

SCHEMATICS - Continued

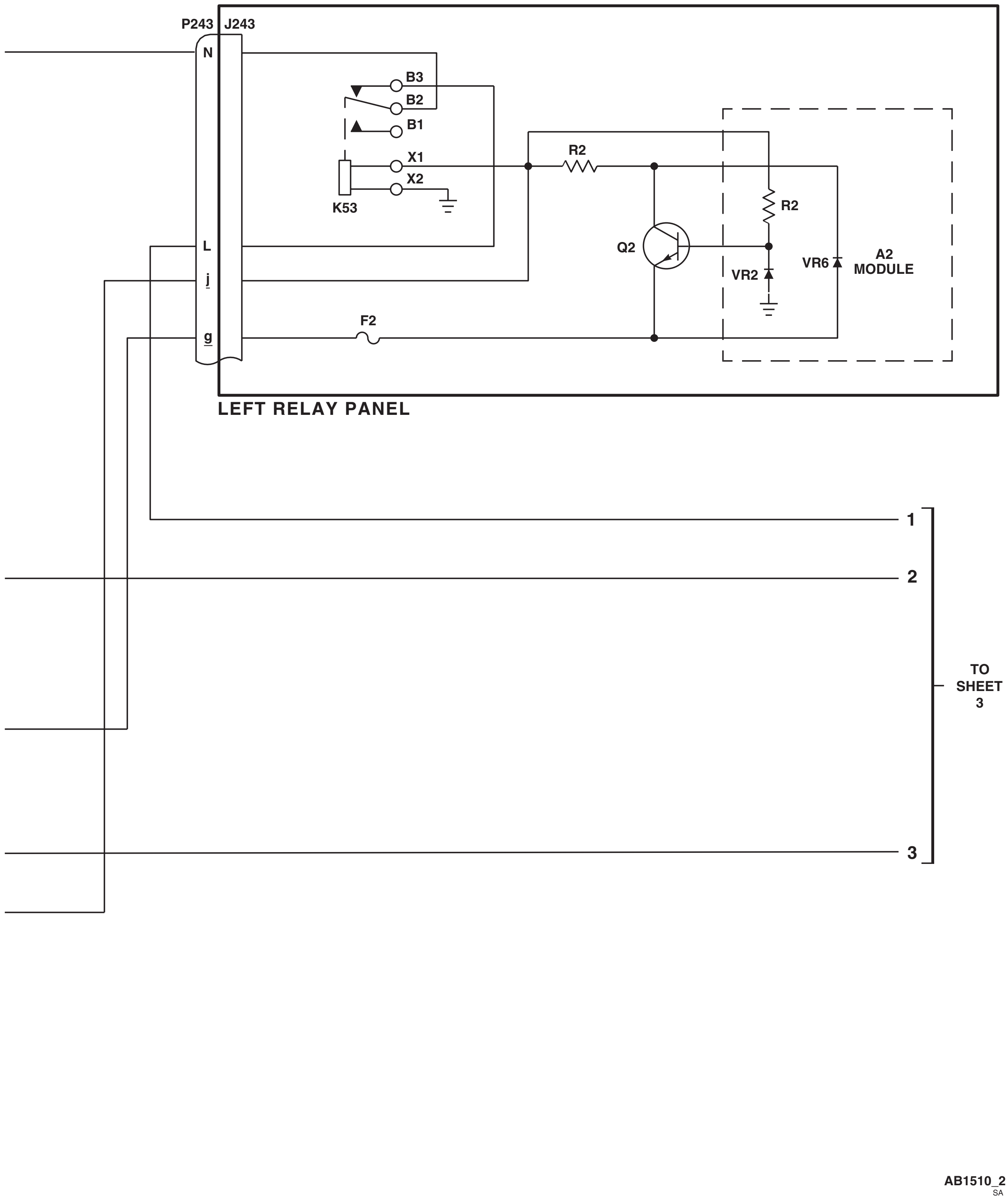


Figure 2. Formation Lights Schematic Diagram. (Sheet 2 of 4)

SCHEMATICS - Continued

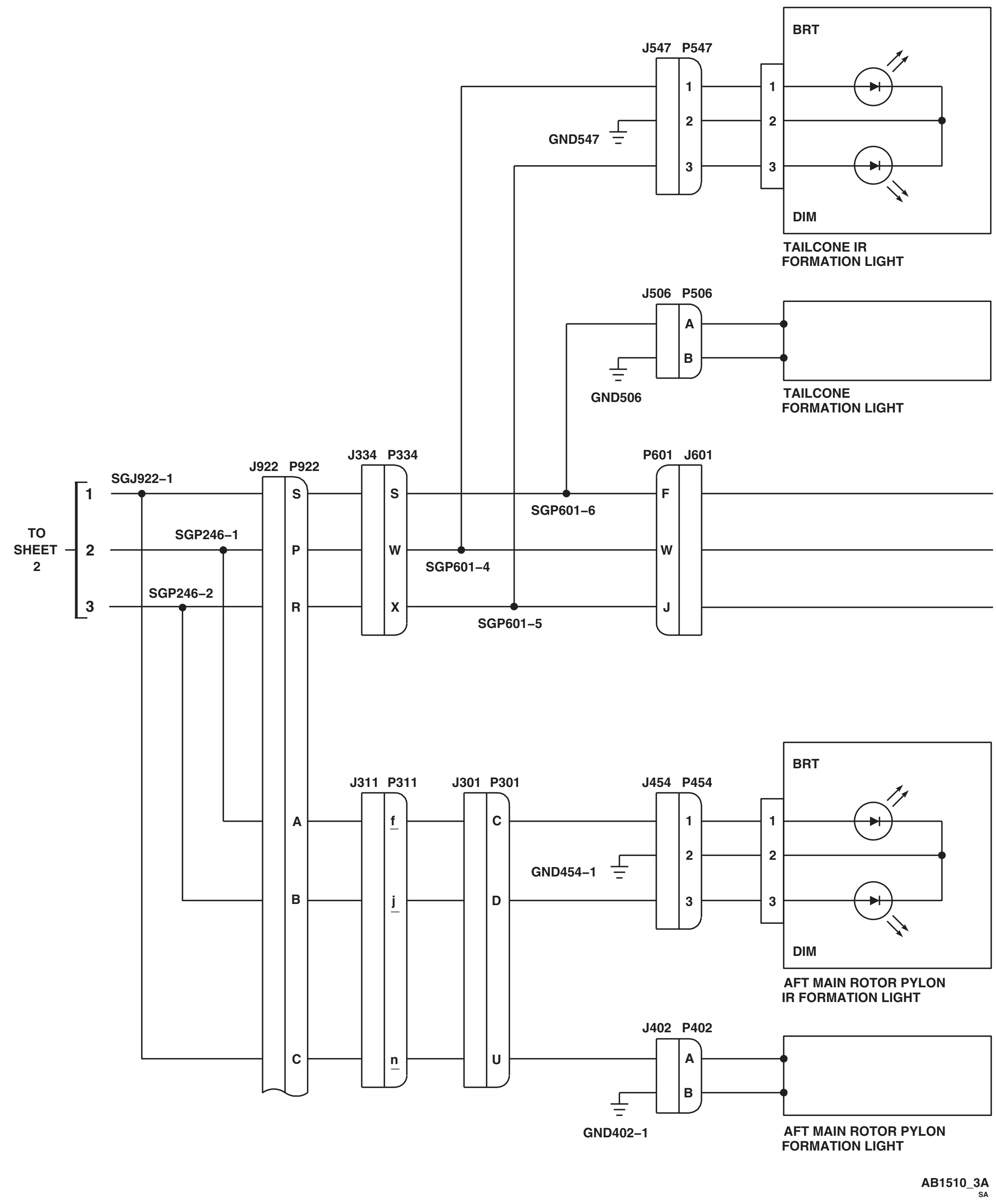


Figure 2. Formation Lights Schematic Diagram. (Sheet 3 of 4)

SCHEMATICS - Continued

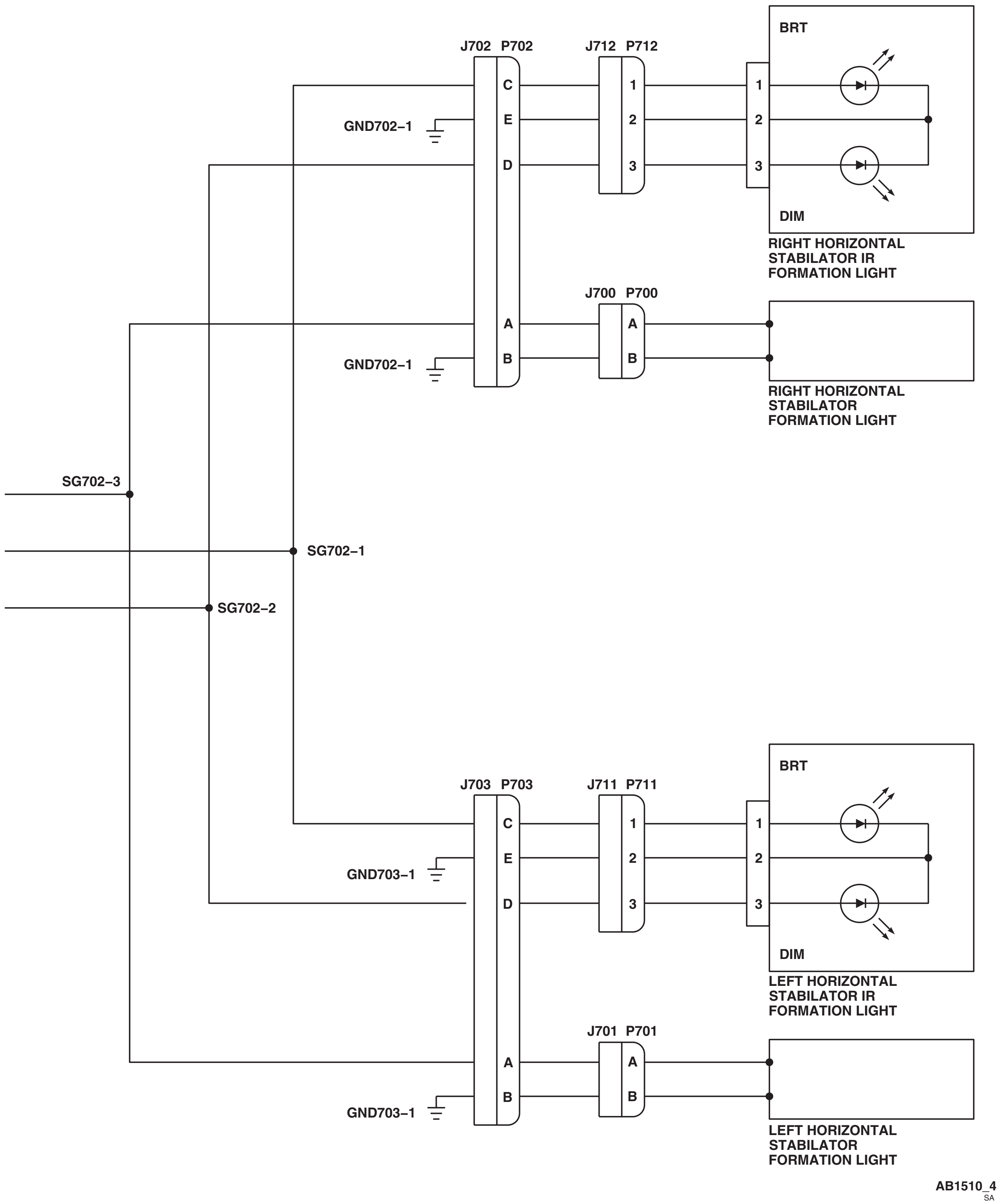


Figure 2. Formation Lights Schematic Diagram. (Sheet 4 of 4)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

CONTROLLABLE SEARCHLIGHT

This WP supersedes WP 0112 00, dated 21 March 2007.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
WP 0104 00
WP 0121 00
WP 0135 00
WP 0824 00
WP 0825 00
WP 0882 00
WP 0918 00
WP 0919 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

CONTROLLABLE SEARCHLIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

WARNING

To prevent personal injury, do not look directly into searchlight when it is turned on.

CAUTION

To avoid damage to the searchlight dimming unit, do not operate OUTPUT switch with the searchlight on.

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
LIGHTS SRCH CONTR	Upper Console

CONTROLLABLE SEARCHLIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
LIGHTS CONTR PWR	Upper console
HH-60A HH-60L MFD PILOT	Aux Circuit Breaker Panel
HH-60A HH-60L MFD COPILOT	Aux Circuit Breaker Panel

2. Turn on electrical power. HH-60A, HH-60L Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
3. Push pilot's collective grip searchlight control to EXT and hold for 5 seconds.

Indication/Condition

Searchlight shall extend from stored position.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT EXTEND WITH PILOT'S SEARCHLIGHT CONTROL PUSHED TO EXT, in this work package.

Step

4. Push pilot's collective grip searchlight control to R and release.

Indication/Condition

Controllable searchlight shall turn to right.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT TURN RIGHT WITH PILOT'S SEARCHLIGHT CONTROL PUSHED TO R, in this work package.

Step

5. Push pilot's collective grip searchlight control to L and release.

Indication/Condition

Controllable searchlight shall turn to left.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT TURN LEFT WITH PILOT'S SEARCHLIGHT CONTROL PUSHED TO L, in this work package.

Step

6. Push pilot's collective grip searchlight control to RETR and hold for 10 seconds.

Indication/Condition

Controllable searchlight shall retract.

CONTROLLABLE SEARCHLIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT RETRACT WITH PILOT'S SEARCHLIGHT CONTROL PUSHED TO RETR, in this work package.

Step

7. Push copilot's collective grip searchlight control to EXT and hold for 5 seconds.

Indication/Condition

Controllable searchlight shall extend from stored position.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT EXTEND WITH COPILOT'S SEARCHLIGHT CONTROL PUSHED TO EXT, in this work package.

Step

8. Push copilot's collective grip searchlight control to R.

Indication/Condition

Controllable searchlight shall turn to right.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT TURN RIGHT WITH COPILOT'S SEARCHLIGHT CONTROL PUSHED TO R, in this work package.

Step

9. Push copilot's collective grip searchlight control to L.

Indication/Condition

Controllable searchlight shall turn to left.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT TURN LEFT WITH COPILOT'S SEARCHLIGHT CONTROL PUSHED TO L, in this work package.

Step

10. Push copilot's collective grip searchlight control to RETR and hold for 10 seconds.

Indication/Condition

Controllable searchlight shall retract.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT RETRACT WITH COPILOT'S SEARCHLIGHT CONTROL PUSHED TO RETR, in this work package.

Step

11. If installed, remove infrared filter from the searchlight (WP 0918 00).
12. Make sure searchlight dimming unit OUTPUT switch is in NORM position.

CONTROLLABLE SEARCHLIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

13. Press pilot's collective grip SRCH LT PUSH ON/OFF switch and release.

Indication/Condition

Searchlight shall go on, and SEARCH LT ON capsule or legend shall go on.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT LIGHT WITH PILOT'S SRCH LT SWITCH ON, in this work package.

Step

14. Push pilot's collective grip SRCH LT switch to DIM and hold.

Indication/Condition

Searchlight shall decrease in brilliance and go off.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT DECREASE IN BRILLIANCE AND GO OFF WITH PILOT'S SRCH LT SWITCH HELD TO DIM, in this work package.

Step

15. Push pilot's collective grip SRCH LT switch to BRT and hold.

Indication/Condition

Searchlight shall increase in brilliance.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT INCREASE IN BRILLIANCE WITH PILOT'S SRCH LT SWITCH HELD TO BRT, in this work package.

Step

16. Press pilot's collective grip SRCH LT PUSH ON/OFF switch and release.

Indication/Condition

Searchlight shall go off and SEARCH LT ON capsule or legend shall go off.

Corrective Action

If Indication/Condition is not as specified, troubleshoot left relay panel (WP 0121 00).

Step

17. Set searchlight dimming unit OUTPUT switch to BYPASS.
18. Press pilot's collective grip SRCH LT PUSH ON/OFF switch and release.
19. Push pilot's collective grip SRCH LT switch to DIM and hold.

Indication/Condition

Searchlight shall not decrease in brilliance.

Corrective Action

If Indication/Condition is not as specified, replace controllable searchlight dimming unit (WP 0882 00).

CONTROLLABLE SEARCHLIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

20. Place dimming unit OUTPUT switch to NORM.
21. Press pilot's collective grip SRCH LT PUSH ON/OFF switch and release.
22. Press copilot's collective grip SRCH LT PUSH ON/OFF switch and release.

Indication/Condition

Searchlight shall go on, and SEARCH LT ON capsule or legend shall go on.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT LIGHT WITH COPILOT'S SRCH LT SWITCH ON, in this work package.

Step

23. Push copilot's collective grip SRCH LT PUSH ON/OFF switch to DIM and hold.

Indication/Condition

Searchlight shall decrease in brilliance and go off.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT DECREASE IN BRILLIANCE AND GO OFF WITH COPILOT'S SRCH LT SWITCH TO DIM, in this work package.

Step

24. Push copilot's collective grip SRCH LT PUSH ON/OFF switch to BRT and hold.

Indication/Condition

Searchlight shall increase in brilliance.

Corrective Action

If Indication/Condition is not as specified, go to SEARCHLIGHT DOES NOT INCREASE IN BRILLIANCE AND GO OFF WITH COPILOT'S SRCH LT SWITCH TO BRT, in this work package.

Step

25. Press copilot's collective grip SRCH LT PUSH ON/OFF switch and release.

Indication/Condition

Searchlight shall go off, and SEARCH LT ON capsule or legend shall go off.

Corrective Action

If Indication/Condition is not as specified, troubleshoot left relay panel (WP 0121 00).

Step

26. If removed, install infrared filter on the controllable searchlight (WP 0918 00).
27. Turn MFD switch OFF.
28. Turn off electrical power.

CONTROLLABLE SEARCHLIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Power shall be off.

Corrective Action

None Required

SEARCHLIGHT DOES NOT EXTEND WITH PILOT'S SEARCHLIGHT CONTROL PUSHED TO EXT

SYMPTOM

Searchlight does not extend with pilot's searchlight control pushed to EXT.

MALFUNCTION

Searchlight does not extend with pilot's searchlight control pushed to EXT.

CORRECTIVE ACTION

1. **Check for 28 vdc between J104-C and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, go to **7.**
2. **Check helicopter configuration.**
   a. W/O MOD , go to **3.**
   b. MOD , go to **5.**
3. **W/O MOD Check continuity between P104-C and P104-E.**
   a. If continuity is present, go to **4.**
   b. If continuity is not present, troubleshoot pilot's collective stick (WP 0135 00). Go to **8.**
4. **W/O MOD Check for 28 vdc between J109-A and ground.**
   a. If voltage is as specified, replace controllable searchlight (WP 0919 00). Go to **8.**
   b. If voltage is not as specified, repair/replace wiring between J104-E and J109-A (WP 1747 00). Go to **8.**
5. **MOD Check continuity between P104-C and P104-e.**
   a. If continuity is present, go to **6.**
   b. If continuity is not present, troubleshoot pilot's collective stick (WP 0135 00). Go to **8.**
6. **MOD Check for 28 vdc between J109-A and ground.**
   a. If voltage is as specified, replace controllable searchlight (WP 0918 00). Go to **8.**
   b. If voltage is not as specified, repair/replace wiring between J104-e and J109-A (WP 1747 00). Go to **8.**
7. **Check for 28 vdc between LIGHTS SRCH CONTR circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, repair/replace wiring between circuit breaker and J104-C (WP 1747 00). Go to **8.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8.**
8. **Procedure completed.**

SEARCHLIGHT DOES NOT TURN RIGHT WITH PILOT'S SEARCHLIGHT CONTROL PUSHED TO R

SYMPTOM

Searchlight does not turn right with pilot's searchlight control pushed to R.

MALFUNCTION

Searchlight does not turn right with pilot's searchlight control pushed to R.

CORRECTIVE ACTION

1. Check for 28 vdc between J109-C and ground.

a. If voltage is as specified, replace controllable searchlight (WP 0919 00). Go to **3.**

b. If voltage is not as specified, go to **2.**

2. Check continuity between P104-C and P104-D.

a. If continuity is present, repair/replace wiring between J104-D and J109-C (WP 1747 00). Go to **3.**

b. If continuity is not present, troubleshoot pilot's collective stick (WP 0135 00). Go to **3.**

3. Procedure completed.

SEARCHLIGHT DOES NOT TURN LEFT WITH PILOT'S SEARCHLIGHT CONTROL PUSHED TO L

SYMPTOM

Searchlight does not turn left with pilot's searchlight control pushed to L.

MALFUNCTION

Searchlight does not turn left with pilot's searchlight control pushed to L.

CORRECTIVE ACTION

1. Check for 28 vdc between J109-D and ground.

a. If voltage is as specified, replace controllable searchlight (WP 0919 00). Go to **3.**

b. If voltage is not as specified, go to **2.**

2. Check continuity between P104-A and P104-C.

a. If continuity is present, repair/replace wiring between J104-A and J109-D (WP 1747 00). Go to **3.**

b. If continuity is not present, troubleshoot pilot's collective stick (WP 0135 00). Go to **3.**

3. Procedure completed.

SEARCHLIGHT DOES NOT RETRACT WITH PILOT'S SEARCHLIGHT CONTROL PUSHED TO RETR

SYMPTOM

Searchlight does not retract with pilot's searchlight control pushed to RETR.

MALFUNCTION

Searchlight does not retract with pilot's searchlight control pushed to RETR.

CORRECTIVE ACTION

1. Check for 28 vdc between J109-E and ground.

SEARCHLIGHT DOES NOT RETRACT WITH PILOT'S SEARCHLIGHT CONTROL PUSHED TO RETR - Continued

a. If voltage is as specified, replace controllable searchlight (WP 0919 00). Go to **3.**

b. If voltage is not as specified, go to **2.**

2. Check continuity between P104-B and P104-C.

a. If continuity is present, repair/replace wiring between J104-B and J109-E (WP 1747 00). Go to **3.**

b. If continuity is not present, troubleshoot pilot's collective stick (WP 0135 00). Go to **3.**

3. Procedure completed.

SEARCHLIGHT DOES NOT EXTEND WITH COPILOT'S SEARCHLIGHT CONTROL PUSHED TO EXT

SYMPTOM

Searchlight does not extend with copilot's searchlight control pushed to EXT.

MALFUNCTION

Searchlight does not extend with copilot's searchlight control pushed to EXT.

CORRECTIVE ACTION

1. Check for 28 vdc between J103-C and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **7.**

2. Check helicopter configuration.

a. **W/O MOD** , go to **3.**

b. **MOD** , go to **5.**

3. W/O MOD Check continuity between P103-C and P103-E.

a. If continuity is present, go to **4.**

b. If continuity is not present, troubleshoot copilot's collective stick (WP 0135 00). Go to **8.**

4. W/O MOD Check for 28 vdc between J109-A and ground.

a. If voltage is as specified, replace controllable searchlight (WP 0919 00). Go to **8.**

b. If voltage is not as specified, repair/replace wiring between J103-E and J109-A (WP 1747 00). Go to **8.**

5. MOD Check continuity between P103-C and P103-e.

a. If continuity is present, go to **6.**

b. If continuity is not present, troubleshoot copilot's collective stick (WP 0135 00). Go to **8.**

6. MOD Check for 28 vdc between J109-A and ground.

a. If voltage is as specified, replace controllable searchlight (WP 0919 00). Go to **8.**

b. If voltage is not as specified, repair/replace wiring between J103-e and J109-A (WP 1747 00). Go to **8.**

7. Check for 28 vdc between LIGHTS SRCH CONTR circuit breaker terminal 1 and ground.

SEARCHLIGHT DOES NOT EXTEND WITH COPILOT'S SEARCHLIGHT CONTROL PUSHED TO EXT - Continued

a. If voltage is as specified, repair/replace wiring between circuit breaker and J103-C (WP 1747 00). Go to **8.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8.**

8. Procedure completed.

SEARCHLIGHT DOES NOT TURN RIGHT WITH COPILOT'S SEARCHLIGHT CONTROL PUSHED TO R

SYMPTOM

Searchlight does not turn right with copilot's searchlight control pushed to R.

MALFUNCTION

Searchlight does not turn right with copilot's searchlight control pushed to R.

CORRECTIVE ACTION

1. Check for 28 vdc between J109-C and ground.

a. If voltage is as specified, replace controllable searchlight (WP 0919 00). Go to **3.**

b. If voltage is not as specified, go to **2.**

2. Check continuity between P103-C and P103-D.

a. If continuity is present, repair/replace wiring between J103-D and J109-C (WP 1747 00). Go to **3.**

b. If continuity is not present, troubleshoot copilot's collective stick (WP 0135 00). Go to **3.**

3. Procedure completed.

SEARCHLIGHT DOES NOT TURN LEFT WITH COPILOT'S SEARCHLIGHT CONTROL PUSHED TO L

SYMPTOM

Searchlight does not turn left with copilot's searchlight control pushed to L.

MALFUNCTION

Searchlight does not turn left with copilot's searchlight control pushed to L.

CORRECTIVE ACTION

1. Check for 28 vdc between J109-D and ground.

a. If voltage is as specified, replace controllable searchlight (WP 0919 00). Go to **3.**

b. If voltage is not as specified, go to **2.**

2. Check continuity between P103-A and P103-C.

a. If continuity is present, repair/replace wiring between J103-A and J109-D (WP 1747 00). Go to **3.**

b. If continuity is not present, troubleshoot copilot's collective stick (WP 0135 00). Go to **3.**

3. Procedure completed.

SEARCHLIGHT DOES NOT RETRACT WITH COPILOT'S SEARCHLIGHT CONTROL PUSHED TO RETR

SYMPTOM

Searchlight does not retract with copilot's searchlight control pushed to RETR.

MALFUNCTION

Searchlight does not retract with copilot's searchlight control pushed to RETR.

CORRECTIVE ACTION

1. **Check for 28 vdc between J109-E and ground.**
   a. If voltage is as specified, replace controllable searchlight (WP 0919 00). Go to **3.**
   b. If voltage is not as specified, go to **2.**
2. **Check continuity between P103-B and P103-C.**
   a. If continuity is present, repair/replace wiring between J103-B and J109-E (WP 1747 00). Go to **3.**
   b. If continuity is not present, troubleshoot copilot's collective stick (WP 0135 00). Go to **3.**
3. **Procedure completed.**

SEARCHLIGHT DOES NOT LIGHT WITH PILOT'S SRCH LT SWITCH ON

SYMPTOM

Searchlight does not light with pilot's SRCH LT switch ON.

MALFUNCTION

Searchlight does not light with pilot's SRCH LT switch ON.

CORRECTIVE ACTION

1. **Check for 28 vdc between J104-K and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, go to **13.**
2. **Check for 28 vdc between J109-G and ground.**
   a. If voltage is as specified, go to **3.**
   b. If voltage is not as specified, go to **4.**
3. **Check continuity between J109-I and ground.**
   a. If continuity is present, replace controllable searchlight (WP 0919 00). Go to **14.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**
4. **Check for 28 vdc between P243-Y and ground.**
   a. If voltage is as specified, go to **5.**
   b. If voltage is not as specified, go to **12.**
5. **Check continuity between P243-W and P230-L.**
   a. If continuity is present, go to **6.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**

SEARCHLIGHT DOES NOT LIGHT WITH PILOT'S SRCH LT SWITCH ON - Continued

6. **Is SEARCH LT ON advisory capsule or legend on?**
   a. If SEARCH LT ON is on, go to **7.**
   b. If SEARCH LT ON is not on, replace left relay panel (WP 0825 00). Go to **14.**
7. **Check continuity between J109-G and P243-V.**
   a. If continuity is present, go to **8.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**
8. **Check for 28 vdc between J109-F and ground.**
   a. If voltage is as specified, replace controllable searchlight (WP 0919 00). Go to **14.**
   b. If voltage is not as specified, go to **9.**
9. **Check for 28 vdc between LIGHTS CONTR PWR circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, go to **10.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **14.**
10. **Check continuity between LIGHTS CONTR PWR circuit breaker terminal 1 and P167-B.**
    a. If continuity is present, go to **11.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**
11. **Check continuity between J109-F and P167-I.**
    a. If continuity is present, replace dimming unit (WP 0882 00). Go to **14.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**
12. **Check continuity between P104-K and P104-P.**
    a. If continuity is present, repair/replace wiring between J104-P and P243-Y (WP 1747 00). Go to **14.**
    b. If continuity is not present, troubleshoot pilot's collective stick (WP 0135 00). Go to **14.**
13. **Check for 28 vdc between LIGHTS SRCH CONTR circuit breaker terminal 1 and ground.**
    a. If voltage is as specified, repair/replace wiring between J104-K and circuit breakerterminal 1 (WP 1747 00). Go to **14.**
    b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **14.**
14. **Procedure completed.**

SEARCHLIGHT DOES NOT DECREASE IN BRILLIANCE AND GO OFF WITH PILOT'S SRCH LT SWITCH HELD TO DIM

SYMPTOM

Searchlight does not decrease in brilliance and go off with pilot's SRCH LT switch held to DIM.

MALFUNCTION

Searchlight does not decrease in brilliance and go off with pilot's SRCH LT switch held to DIM.

CORRECTIVE ACTION

1. **Make sure that the OUTPUT switch on the searchlight dimming unit is positioned to NORM. Go to 2.**
2. **Check for 28 vdc between P167-D and ground.**
   a. If voltage is as specified, go to **3.**
   b. If voltage is not as specified, go to **4.**
3. **With SRCH LT switch at DIM, check for decreasing voltage at J109-F.**
   a. If decreasing voltage is present, replace controllable searchlight (WP 0919 00). Go to **6.**
   b. If decreasing voltage is not present, replace controllable searchlight dimming unit (WP 0882 00). Go to **6.**
4. **Check for 28 vdc between J104-M and ground.**
   a. If voltage is as specified, go to **5.**
   b. If voltage is not as specified, repair/replace wiring between J104-M and P247-F (WP 1747 00). Go to **6.**
5. **With SRCH LT switch at DIM, check continuity between P104-M and P104-N.**
   a. If continuity is present, repair/replace wiring between J104-N and P167-D (WP 1747 00). Go to **6.**
   b. If continuity is not present, troubleshoot pilot's collective stick (WP 0135 00). Go to **6.**
6. **Procedure completed.**

SEARCHLIGHT DOES NOT INCREASE IN BRILLIANCE WITH PILOT'S SRCH LT SWITCH HELD TO BRT

SYMPTOM

Searchlight does not increase in brilliance with pilot's SRCH LT switch held to BRT.

MALFUNCTION

Searchlight does not increase in brilliance with pilot's SRCH LT switch held to BRT.

CORRECTIVE ACTION

1. **Make sure that the OUTPUT switch on the searchlight dimming unit is positioned to NORM. Go to 2.**
2. **Check for 28 vdc between P167-C and ground.**
   a. If voltage is as specified, go to **3.**
   b. If voltage is not as specified, go to **4.**

SEARCHLIGHT DOES NOT INCREASE IN BRILLIANCE WITH PILOT'S SRCH LT SWITCH HELD TO BRT - Continued

3. **With SRCH LT switch at BRT, check for increasing voltage at J109-F.**
   a. If increasing voltage is present, replace controllable searchlight (WP 0919 00). Go to **5.**
   b. If increasing voltage is not present, replace controllable searchlight dimming unit (WP 0882 00). Go to **5.**
4. **With SRCH LT switch at BRT, check continuity between P104-L and P104-M.**
   a. If continuity is present, repair/replace wiring between J104-L and P167-C (WP 1747 00). Go to **5.**
   b. If continuity is not present, troubleshoot pilot's collective stick (WP 0135 00). Go to **5.**
5. **Procedure completed.**

SEARCHLIGHT DOES NOT LIGHT WITH COPILOT'S SRCH LT SWITCH ON

SYMPTOM

Searchlight does not light with copilot's SRCH LT switch ON.

MALFUNCTION

Searchlight does not light with copilot's SRCH LT switch ON.

CORRECTIVE ACTION

1. **Check for 28 vdc between J103-K and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, repair/replace wiring between J103-K and P247-F (WP 1747 00). Go to **3.**
2. **Check continuity between P103-K and P103-P.**
   a. If continuity is present, repair/replace wiring between J103-P and P247-A (WP 1747 00). Go to **3.**
   b. If continuity is not present, troubleshoot copilot's collective stick (WP 0135 00). Go to **3.**
3. **Procedure completed.**

SEARCHLIGHT DOES NOT DECREASE IN BRILLIANCE AND GO OFF WITH COPILOT'S SRCH LT SWITCH TO DIM

SYMPTOM

Searchlight does not decrease in brilliance and go off with copilot's SRCH LT switch to DIM.

MALFUNCTION

Searchlight does not decrease in brilliance and go off with copilot's SRCH LT switch to DIM.

CORRECTIVE ACTION

1. **Make sure that the OUTPUT switch on the searchlight dimming unit is positioned to NORM. Go to 2.**
2. **Check for 28 vdc between J103-M and ground.**
   a. If voltage is as specified, go to **3.**

SEARCHLIGHT DOES NOT DECREASE IN BRILLIANCE AND GO OFF WITH COPILOT'S SRCH LT SWITCH TO DIM - Continued

b. If voltage is not as specified, repair/replace wiring between J103-M and P247-F (WP 1747 00). Go to **4.**

3. With SRCH LT switch at DIM, check continuity between P103-M and P103-N.

a. If continuity is present, repair/replace wiring between J103-N and P167-D (WP 1747 00). Go to **4.**

b. If continuity is not present, troubleshoot copilot's collective stick (WP 0135 00). Go to **4.**

4. Procedure completed.

SEARCHLIGHT DOES NOT INCREASE IN BRILLIANCE AND GO OFF WITH COPILOT'S SRCH LT SWITCH TO BRT

SYMPTOM

Searchlight does not increase in brilliance and go off with copilot's SRCH LT switch to BRT.

MALFUNCTION

Searchlight does not increase in brilliance and go off with copilot's SRCH LT switch to BRT.

CORRECTIVE ACTION

1. Make sure that the OUTPUT switch on the searchlight dimming unit is positioned to NORM. Go to 2.

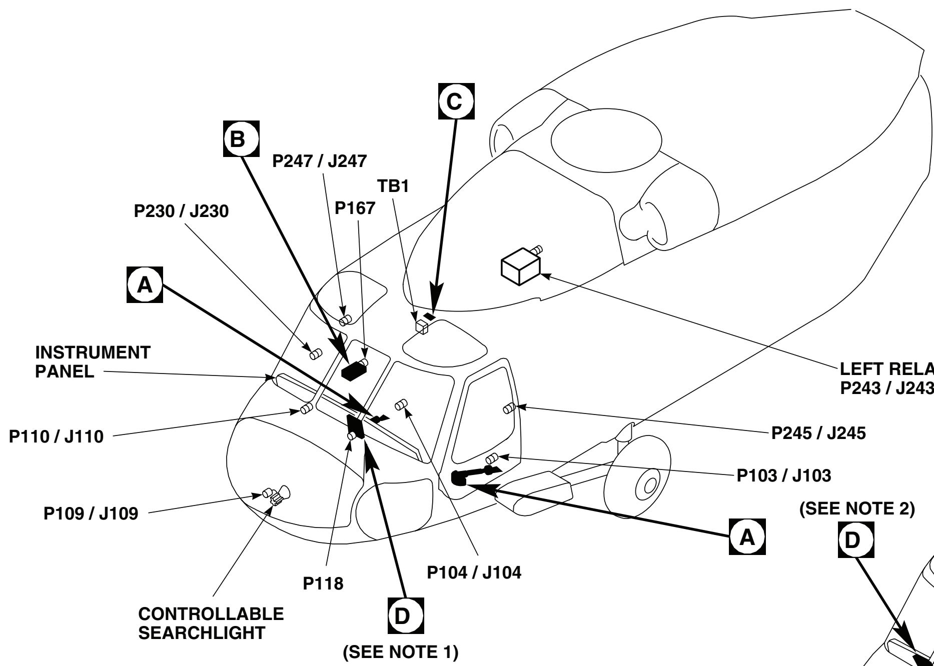
2. With SRCH LT switch at BRT, check continuity between P103-L and P103-M.

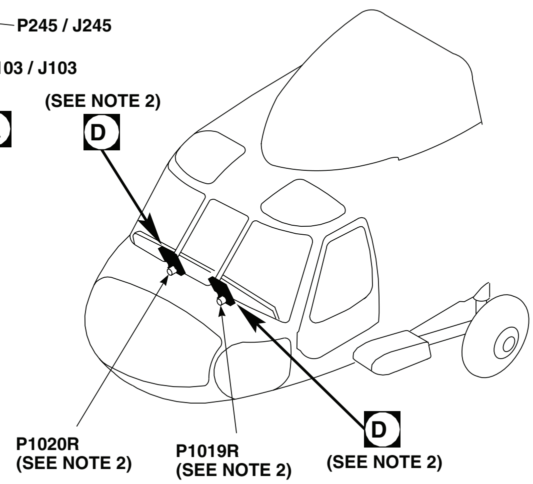
a. If continuity is present, repair/replace wiring between J103-L and P167-C (WP 1747 00). Go to **3.**

b. If continuity is not present, troubleshoot copilot's collective stick (WP 0135 00). Go to **3.**

3. Procedure completed.

LOCATION





TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P103 / J103	COCKPIT TUB BL 32 LH, STA 247
P104 / J104	COCKPIT TUB BL 15 RH, STA 247
P109 / J109	CONTROLLABLE SEARCHLIGHT
P110 / J110	INSTRUMENT PANEL STA 197, BL 7.5
P118 / J118	CAUTION / ADVISORY PANEL (SEE NOTE 1)
P167	CONTROLLABLE SEARCH–LIGHT DIMMING UNIT STA 235, BL 33
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P243 / J243	LEFT RELAY PANEL
P245 / J245	CABIN BL 40 LH, STA 248
P247 / J247	CABIN BL 40 RH, STA 248
TB1	COCKPIT CEILING BL 9 LH, STA 236
P1019R / J3	COPILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)
P1020R / J3	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)

NOTES

1. UH60A UH60L EH60A
2. HH60A HH60L

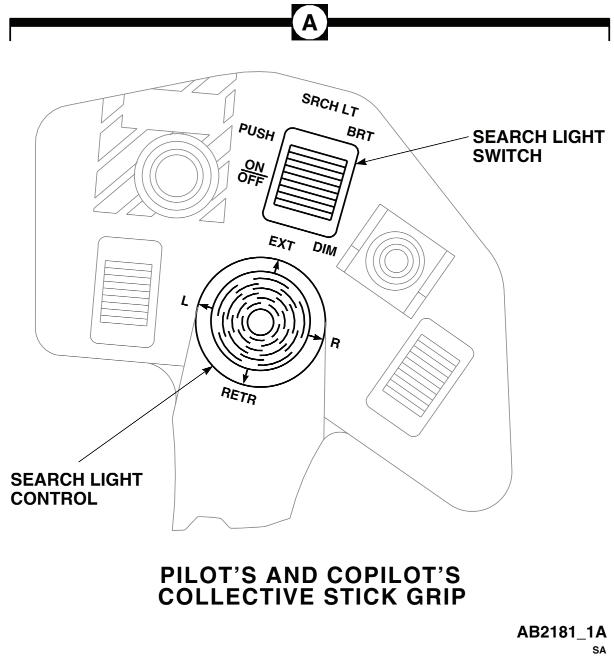


Figure 1. Controllable Searchlight Location Diagram. (Sheet 1 of 2)

LOCATION - Continued

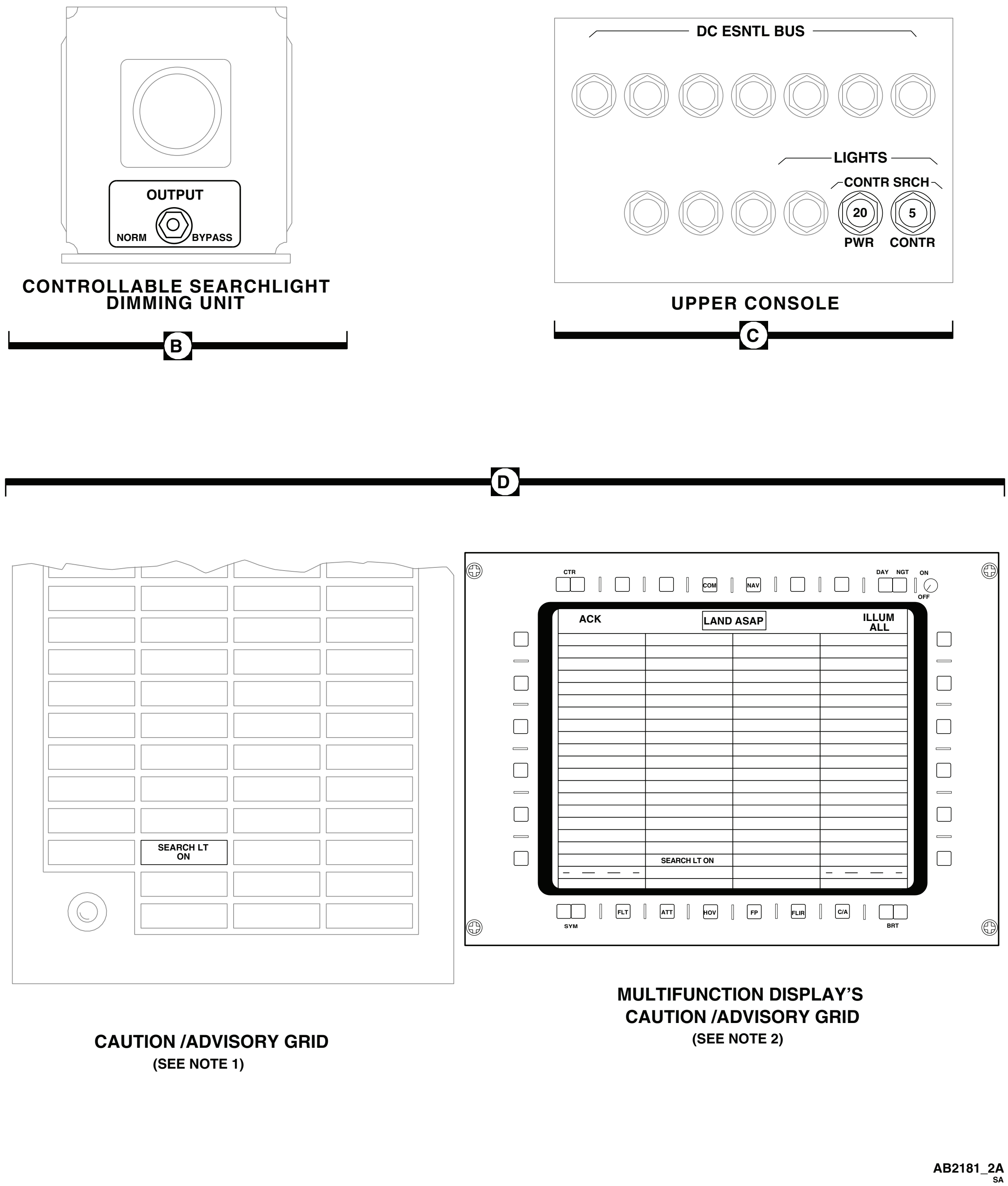


Figure 1. Controllable Searchlight Location Diagram. (Sheet 2 of 2)

SCHEMATICS

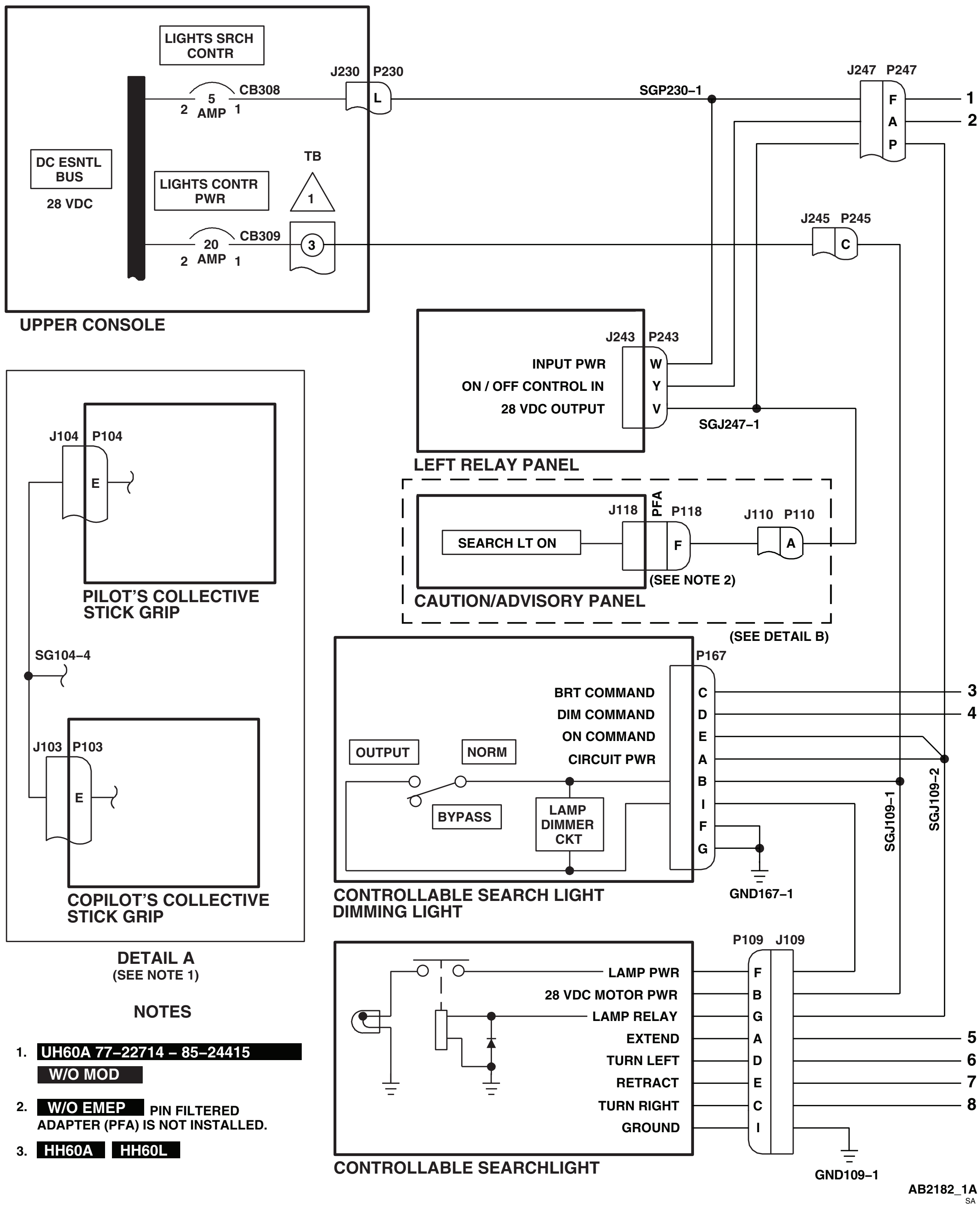


Figure 2. Controllable Searchlight Schematic Diagram. (Sheet 1 of 3)

SCHEMATICS - Continued

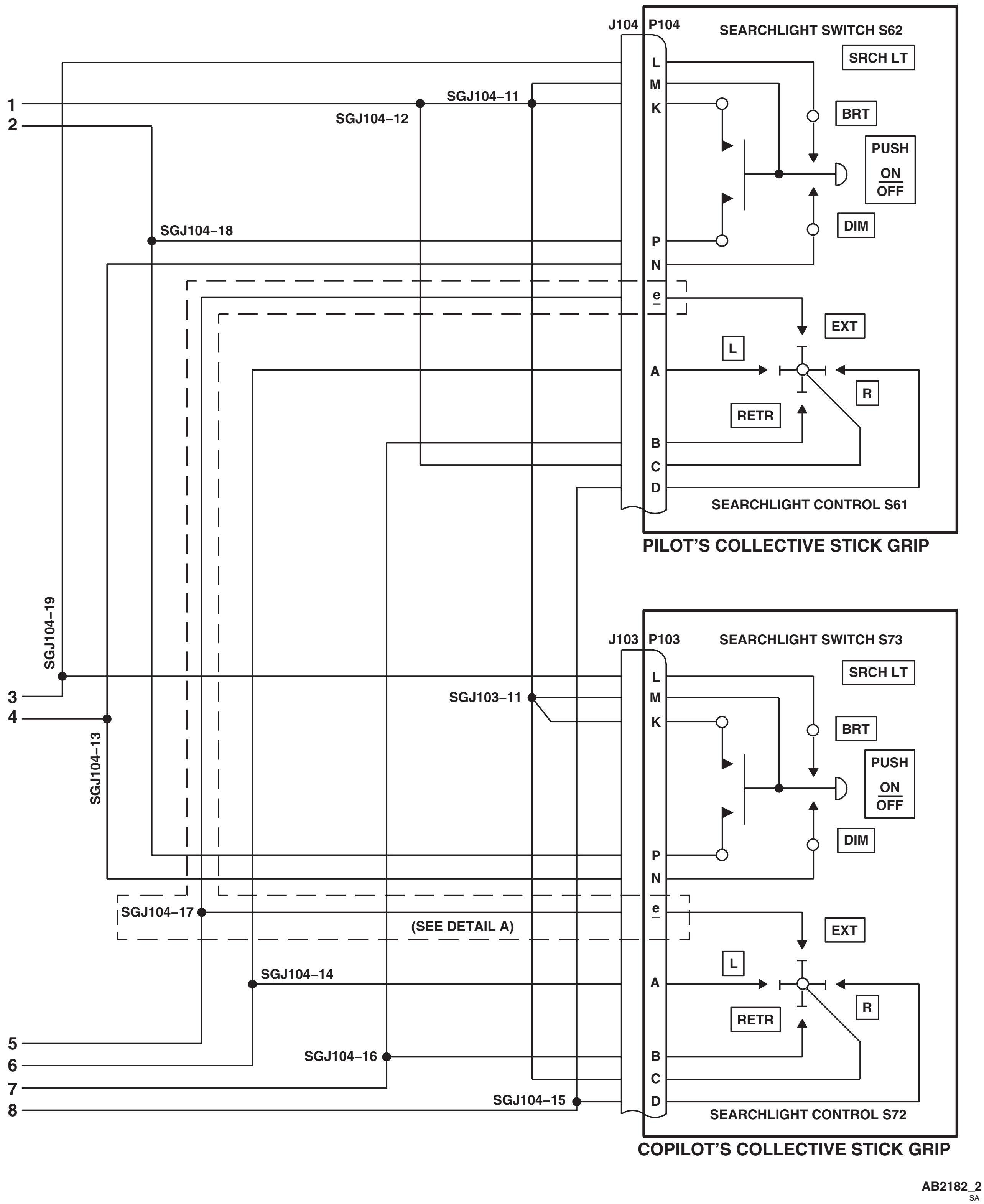


Figure 2. Controllable Searchlight Schematic Diagram. (Sheet 2 of 3)

SCHEMATICS - Continued

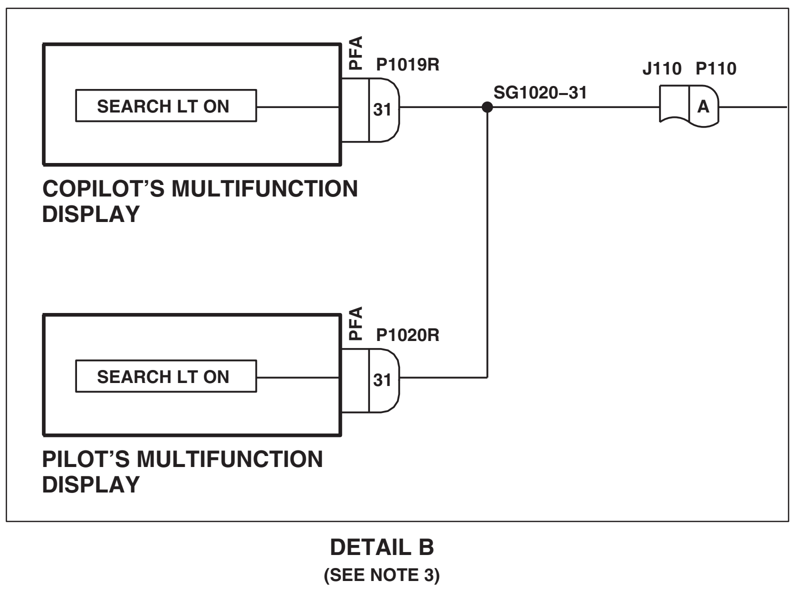


Figure 2. Controllable Searchlight Schematic Diagram. (Sheet 3 of 3)

END OF WORK PACKAGE

UNIT LEVEL
ELECTRICAL SYSTEMS
POSITION LIGHTS

This WP supersedes WP 0113 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Locally-Made Infrared Light Test Set Figure 149, WP 1805 00
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
WP 0104 00
WP 0824 00
WP 0825 00
WP 0828 00
WP 0875 00
WP 0903 00
WP 0904 00
WP 1685 00
WP 1747 00
WP 1805 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

POSITION LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- **W/O ESSS** Disconnects (P681 and P682) are jumpered so the existing three lights are operational.
- **ESSS** Position lights on the landing gear support fairings are disabled and the position lights on the horizontal stores support are operational. Therefore, three position lights are operational in either helicopter configuration.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Check position light assemblies for cracked or broken lenses.

Indication/Condition

No position light lens shall be cracked or broken.

Corrective Action

If Indication/Condition is not as specified, replace lens (WP 0904 00).

POSITION LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

2. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
POS LTS	Pilot's
IR LTS	Pilot's

3. Turn on electrical power.
4. Place NAV LTS switch to NORM.
5. Place POSITION LIGHTS DIM/OFF/BRT switch to BRT.
6. Place POSITION LIGHTS STEADY/FLASH switch to STEADY.

Indication/Condition

Three position lights shall go on.

Corrective Action

1. If Indication/Condition is not as specified, go to INCANDESCENT POSITION LIGHTS DO NOT GO ON, in this work package.
2. If one position light does not go on, go to ONE INCANDESCENT POSITION LIGHT DOES NOT GO ON, in this work package.

Step

7. Place POSITION LIGHTS DIM/OFF/BRT switch to DIM.

Indication/Condition

Position lights shall remain on but at reduced intensity.

Corrective Action

If Indication/Condition is not as specified, go to INCANDESCENT POSITION LIGHTS DO NOT LIGHT AT A REDUCED INTENSITY WITH POSITION LIGHTS DIM/OFF/BRT SWITCH AT DIM, in this work package.

Step

8. Place POSITION LIGHTS STEADY/FLASH switch to FLASH.

Indication/Condition

Position lights shall flash.

Corrective Action

If Indication/Condition is not as specified, go to INCANDESCENT POSITION LIGHTS DO NOT FLASH, in this work package.

Step

9. Place POSITION LIGHTS DIM/OFF/BRT switch to OFF.

POSITION LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Position lights shall go off.

Corrective Action

If Indication/Condition is not as specified, replace POSITION LIGHTS DIM/OFF/BRT switch S29 (WP 0828 00).

Step

10. Place NAV LIGHTS switch to IR.
11. Place locally-made IR light test set (test set) (WP 1805 00) POWER switch to ON.

Indication/Condition

POWER ON lamp shall go on.

Corrective Action

If Indication/Condition is not as specified, replace lamp, battery, or test set, as required.

Step

12. Place test set IR detector lead over POWER ON lamp.

Indication/Condition

Test set IR LIGHT ON lamp shall go on.

Corrective Action

If Indication/Condition is not as specified, replace test set.

Step

NOTE

- **W/O ESSS** Perform steps 13. through 20. for each of the three IR position lights (tail rotor pylon, and left and right landing gear support fairings).
- **ESSS** Perform steps 13. through 20. for each of the three IR position lights (tail rotor pylon, and left and right horizontal support system beams).

13. Place POSITION LIGHTS DIM/OFF/BRT switch to DIM.
14. Place POSITION LIGHTS STEADY/FLASH switch to STEADY.
15. Place test set IR detector lead over each of the IR position light dark lenses.

Indication/Condition

Test set IR LIGHT ON lamp shall go on steady over all IR position lights.

Corrective Action

1. If Indication/Condition is not as specified, go to IR LIGHTS DO NOT GO ON, in this work package.
2. If one IR position light does not go on, go to ONE IR POSITION LIGHT DOES NOT GO ON, in this work package.

POSITION LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

16. Place POSITION LIGHTS DIM/OFF/BRT switch to BRT and place test set IR detector lead over each IR position light's clear lens.

Indication/Condition

Test set IR LIGHT ON lamp shall go on over all IR position lights.

Corrective Action

1. If Indication/Condition is not as specified, check fuse F1-1A on left relay panel.
   a. If fuse is good, go to IR LIGHTS DO NOT GO ON, in this work package.
   b. If fuse is not good, replace fuse.
2. If one IR position light does not go on, go to IR POSITION LIGHT DOES NOT GO ON, in this work package.

Step

17. Place POSITION LIGHTS STEADY/FLASH switch to FLASH and place IR detector lead over each IR position light's clear lens.

Indication/Condition

Test set IR LIGHT ON lamp shall flash on and off over all IR position lights.

Corrective Action

If Indication/Condition is not as specified, go to INCANDESCENT POSITION LIGHTS DO NOT FLASH, in this work package.

Step

18. Place POSITION LIGHTS DIM/OFF/BRT switch to OFF.
19. Place test set IR detector lead over each IR position lights clear lens.

Indication/Condition

Test set IR LIGHT ON lamp shall not go on over all IR position lights.

Corrective Action

If Indication/Condition is not as specified, replace POSITION LIGHTS DIM/OFF/BRT switch (WP 0828 00).

Step

20. Place test set POWER switch to OFF.
21. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

INCANDESCENT POSITION LIGHTS DO NOT GO ON

SYMPTOM

Incandescent position lights do not go on.

MALFUNCTION

Incandescent position lights do not go on.

CORRECTIVE ACTION

1. Check for 28 vdc between position LIGHTS BRT/DIM OFF switch S29-6 and ground.

a. If voltage is as specified, repair/replace wiring between S29-6 and TB2-4 (WP 1747 00). Go to **8.**

b. If voltage is not as specified, go to **2.**

2. Check for 28 vdc between S29-5 and ground.

a. If voltage is as specified, replace switch (WP 0828 00). Go to **8.**

b. If voltage is not as specified, go to **3.**

3. Check continuity between S29-5 and P243-R.

a. If continuity is present, go to **4.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

4. Check for 28 vdc between P243-T and ground.

a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **8.**

b. If voltage is not as specified, go to **5.**

5. Check for 28 vdc between POSITION LIGHTS FLASH/STEADY switch S28-1 and ground.

a. If voltage is as specified, repair/replace wiring between S28-1 and P243-T (WP 1747 00). Go to **8.**

b. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between S28-2 and ground.

a. If voltage is as specified, replace switch (WP 0828 00). Go to **8.**

b. If voltage is not as specified, go to **7.**

7. Check for 28 vdc between POS LTS circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and S28-2 (WP 1747 00). Go to **8.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8.**

8. Procedure completed.

ONE INCANDESCENT POSITION LIGHT DOES NOT GO ON

SYMPTOM

One incandescent position light does not go on.

MALFUNCTION

One incandescent position light does not go on.

CORRECTIVE ACTION

1. **Check light lamp.**
   a. If lamp is good, go to **2.**
   b. If lamp is not good, replace lamp (WP 0904 00). Go to **4.**
2. **Check for bad ground on light assembly mounting screw.**
   a. If bad ground is present, repair ground connection (WP 1747 00). Go to **4.**
   b. If bad ground is not present, go to **3.**
3. **Check wiring between POSITION LIGHTS BRT/DIM/OFF switch S29-6 and malfunctioning light.**
   a. If wiring is good, replace position light assembly (WP 0904 00). Go to **4.**
   b. If wiring is not good, repair/replace wiring (WP 1747 00). Go to **4.**
4. **Procedure completed.**

INCANDESCENT POSITION LIGHTS DO NOT LIGHT AT A REDUCED INTENSITY WITH POSITION LIGHTS DIM/OFF/BRT SWITCH AT DIM

SYMPTOM

Incandescent position lights do not light at a reduced intensity with POSITION LIGHTS DIM/OFF/BRT switch at DIM.

MALFUNCTION

Incandescent position lights do not light at a reduced intensity with POSITION LIGHTS DIM/OFF/BRT switch at DIM.

CORRECTIVE ACTION

1. **Check for open dimming resistor.**
   a. If dimming resistor is open, replace dimming resistor R10 (WP 0875 00). Go to **3.**
   b. If dimming resistor is not open, go to **2.**
2. **Check for 28 vdc between POSITION LIGHTS BRT/DIM/OFF switch S29-4 and ground.**
   a. If voltage is as specified, repair/replace wiring between S29-4 and TB2-3 (WP 1747 00). Go to **3.**
   b. If voltage is not as specified, replace switch (WP 0828 00). Go to **3.**
3. **Procedure completed.**

INCANDESCENT POSITION LIGHTS DO NOT FLASH

SYMPTOM

Incandescent position lights do not flash.

MALFUNCTION

Incandescent position lights do not flash.

CORRECTIVE ACTION

1. **Check for 28 vdc between P232-B and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, go to **3.**
2. **Check continuity between POSITION LIGHTS FLASH/STEADY switch:**
   - **S28-1 and P232-C**
   - **S28-1 and P232-D**

   a. If continuity is present, replace flasher (WP 0903 00). Go to **4.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**
3. **Check continuity between POSITION LIGHTS FLASH/STEADY switch S28-3 and P232-B.**
   a. If continuity is present, replace switch (WP 0828 00). Go to **4.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**
4. **Procedure completed.**

IR LIGHTS DO NOT GO ON

SYMPTOM

IR lights do not go on.

MALFUNCTION

IR lights do not go on.

CORRECTIVE ACTION

1. **With NAV LTS switch at IR, check for 10 vdc between POSITION LIGHTS BRT/DIM/OFF switch S29-2 and ground.**
   a. If voltage is as specified, replace switch (WP 0828 00). Go to **7.**
   b. If voltage is not as specified, go to **2.**
2. **Check for 28 vdc between P243-j and ground.**
   a. If voltage is as specified, go to **3.**
   b. If voltage is not as specified, go to **4.**
3. **Check continuity between P243-f and S29-2.**
   a. If continuity is present, replace left relay panel (WP 0825 00). Go to **7.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7.**

IR LIGHTS DO NOT GO ON - Continued

4. **Check for 28 vdc between S69-3 and ground.**

   a. If voltage is as specified, repair/replace wiring between P243-j and S69-3 (WP 1747 00). Go to **7.**

   b. If voltage is not as specified, go to **5.**

5. **Check for 28 vdc between S69-2 and ground.**

   a. If voltage is as specified, replace switch (WP 0828 00). Go to **7.**

   b. If voltage is not as specified, go to **6.**

6. **Check for 28 vdc between IR LTS circuit breaker terminal 1 and ground.**

   a. If voltage is as specified, repair/replace wiring between S69-2 and circuit breaker (WP 1747 00). Go to **7.**

   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **7.**

7. **Procedure completed.**

ONE IR POSITION LIGHT DOES NOT GO ON

SYMPTOM

One IR position light does not go on.

MALFUNCTION

One IR position light does not go on.

CORRECTIVE ACTION

1. **Check continuity as required between:**

   - **P340-2 and ground**
   - **P341-2 and ground**
   - **P616-2 and ground**
   - **P719-2 and ground**
   - **P720-2 and ground**

   a. If continuity is present, go to **2.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3.**

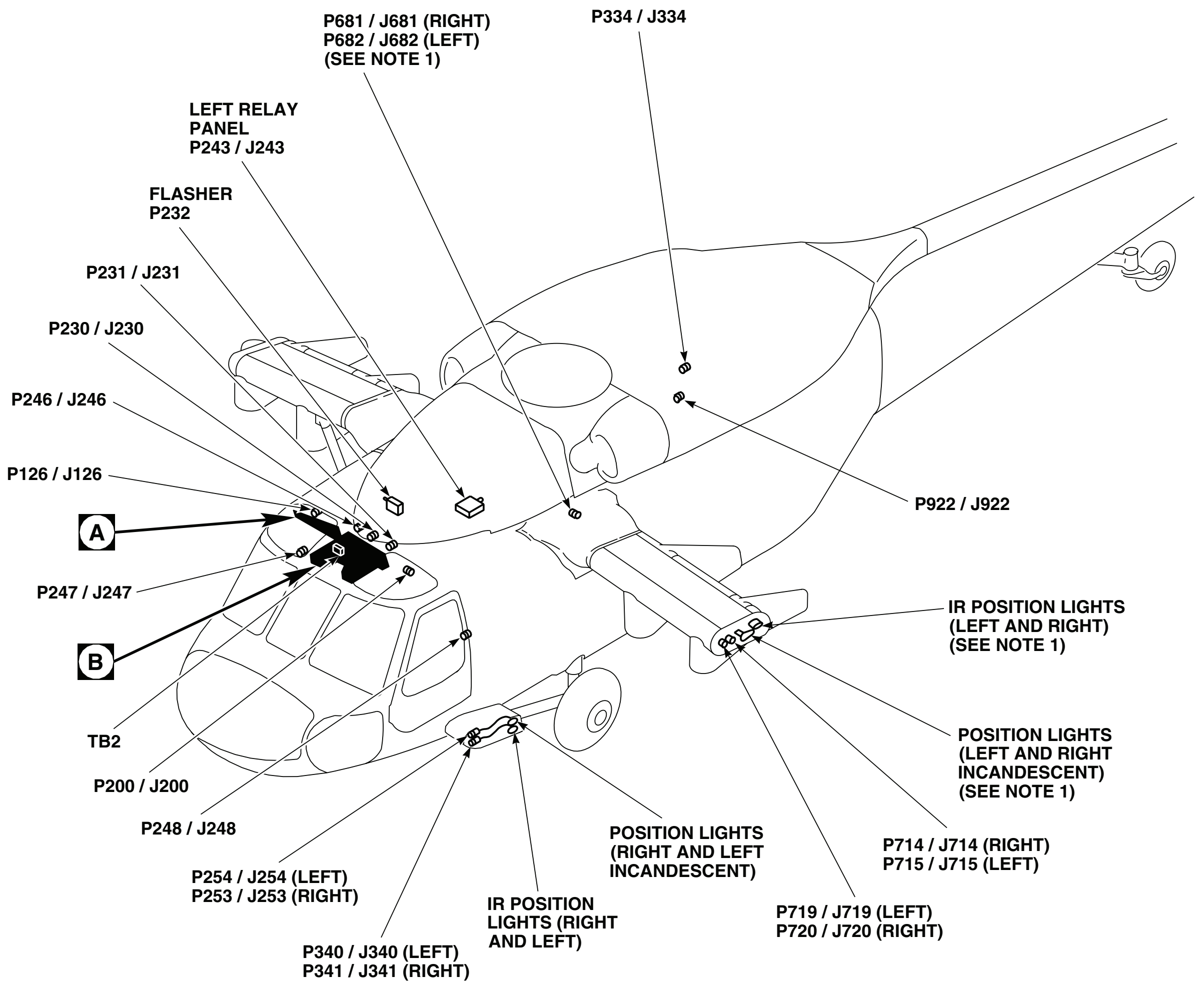
2. **Check wiring between malfunctioning IR light and P200, P246, or P231 as applicable.**

   a. If wiring is good, replace malfunctioning IR position light (WP 0904 00). Go to **3.**

   b. If wiring is not good, repair/replace wiring (WP 1747 00). Go to **3.**

3. **Procedure completed.**

LOCATION



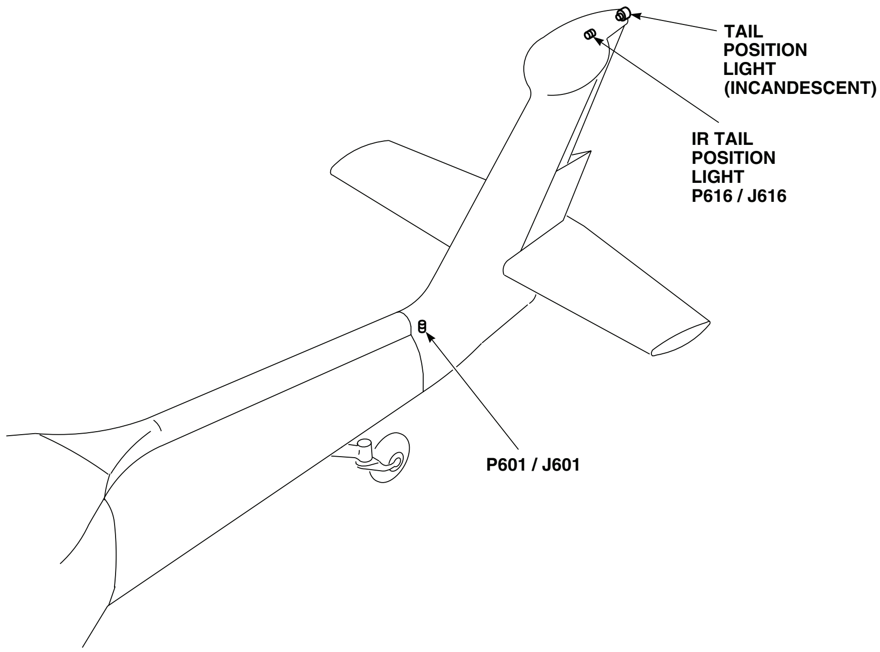
NOTES

1. **W/O ESSS** ELECTRICAL CONNECTOR J683 CONNECTED TO P681, AND J684 CONNECTED TO P682.
2. **ESSS**

AA2780_1
SA

Figure 1. Position Lights Location Diagram. (Sheet 1 of 3)

LOCATION - Continued



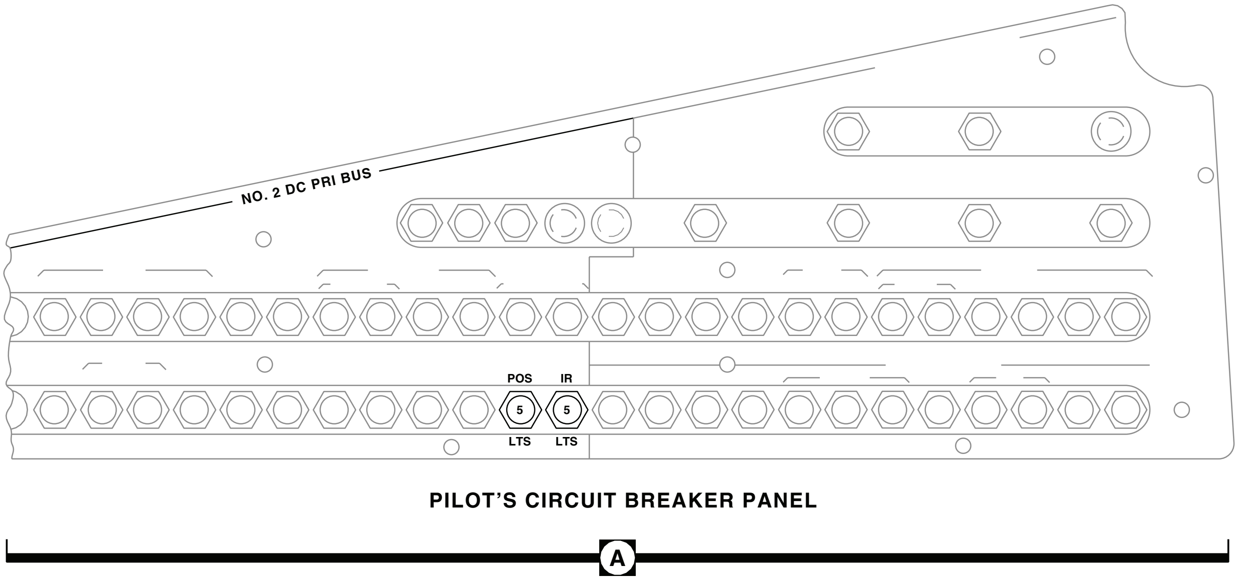
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P126 / J126	COCKPIT CEILING BL 28 RH, STA 243
P200 / J200	COCKPIT CEILING BL 13 RH, STA 246
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P231 / J231	CABIN CEILING BL 9 LH, STA 247
P232	FLASHER
P243 / J243	LEFT RELAY PANEL
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P247 / J247	CABIN BL 40 RH, STA 248
P248 / J248	CABIN BL 40 LH, STA 248
P253 / J253	LANDING GEAR SUPPORT FAIRING BL 52 RH, STA 247
P254 / J254	LANDING GEAR SUPPORT FAIRING BL 52 LH, STA 247
P334 / J334	CABIN CEILING BL 16.5 LH, STA 394

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P340 / J340	LANDING GEAR SUPPORT FAIRING BL 52 LH, STA 246
P341 / J341	LANDING GEAR SUPPORT FAIRING BL 52 RH, STA 246
P601 / J601	BL 4 LH, STA 651, WL 234
P616 / J616	TAIL CONE BL 0, STA 649
P681 / J681 (SEE NOTE 1)	HELICOPTER INTER– CONNECT BL 60 RH, STA 293
P682 / J682 (SEE NOTE 1)	HELICOPTER INTER– CONNECT BL 60 LH, STA 293
P714 / J714	HSS TIP CAP
P715 / J715	HSS TIP CAP
P719 / J719	HSS TIP CAP
P720 / J720	HSS TIP CAP
P922 / J922	CABIN CEILING BL 10.7 LH, STA 380
TB2	COCKPIT CEILING BL 9 RH, STA 236

AA2780_2
SA

Figure 1. Position Lights Location Diagram. (Sheet 2 of 3)

LOCATION - Continued



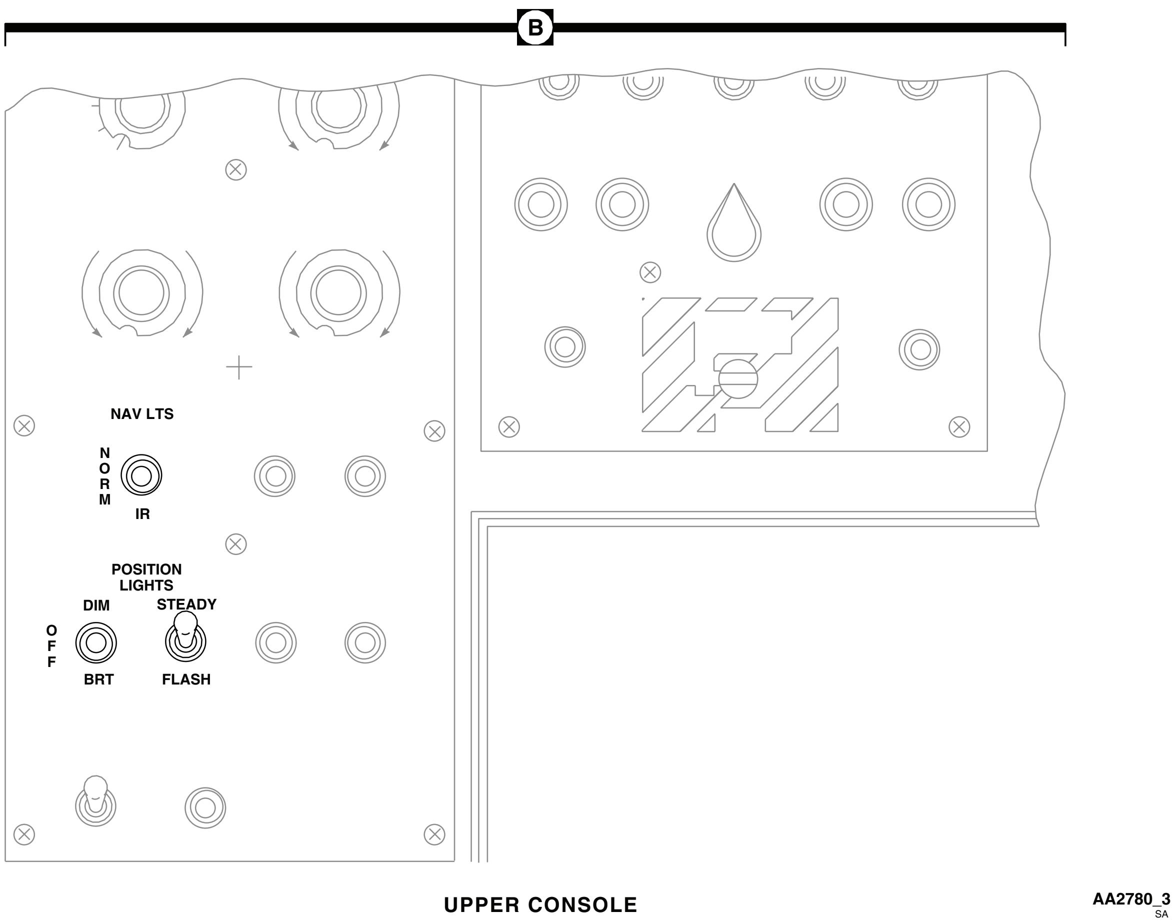


Figure 1. Position Lights Location Diagram. (Sheet 3 of 3)

SCHEMATICS

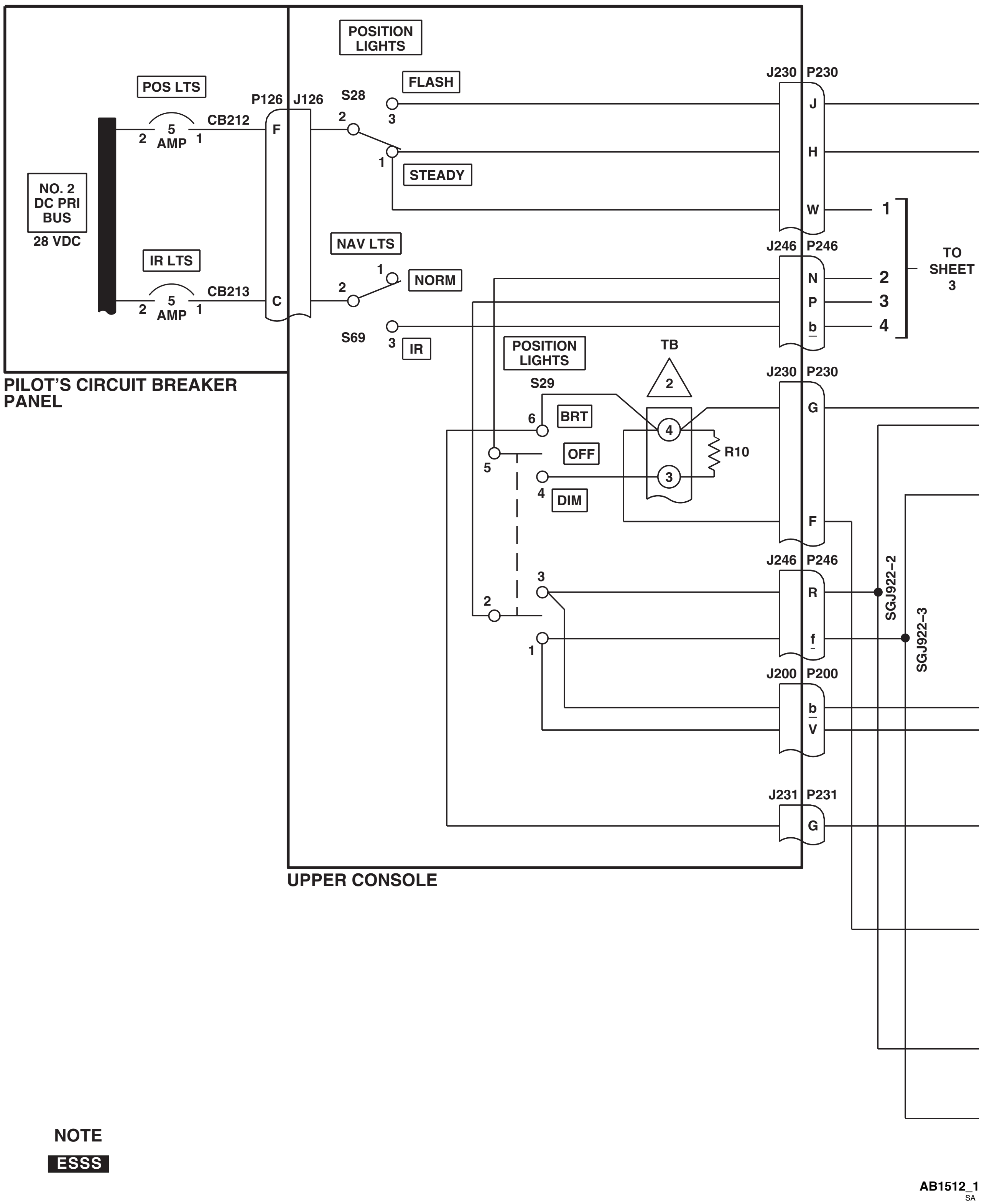


Figure 2. Position Lights Schematic Diagram. (Sheet 1 of 4)

SCHEMATICS - Continued

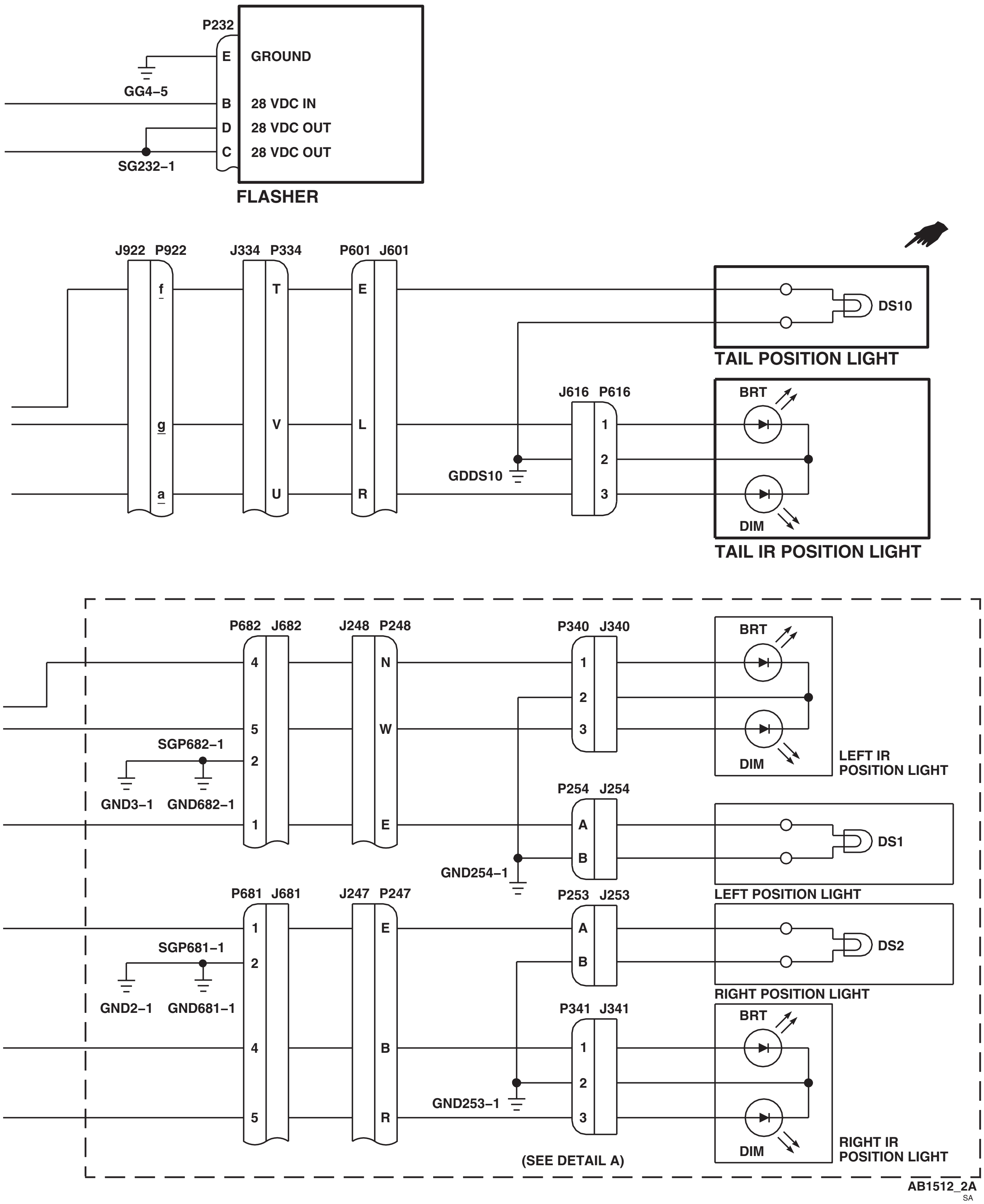


Figure 2. Position Lights Schematic Diagram. (Sheet 2 of 4)

SCHEMATICS - Continued

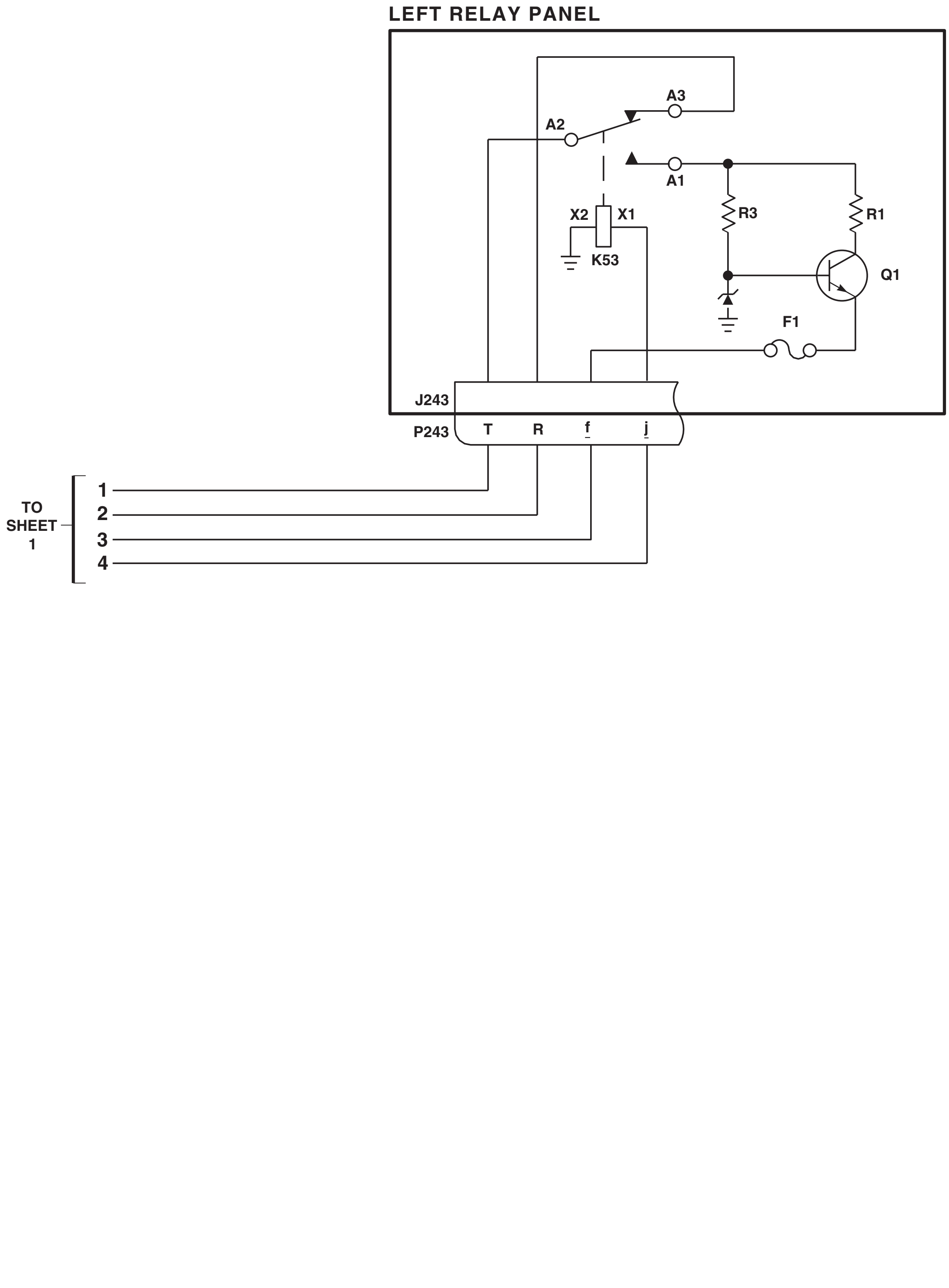


Figure 2. Position Lights Schematic Diagram. (Sheet 3 of 4)

SCHEMATICS - Continued

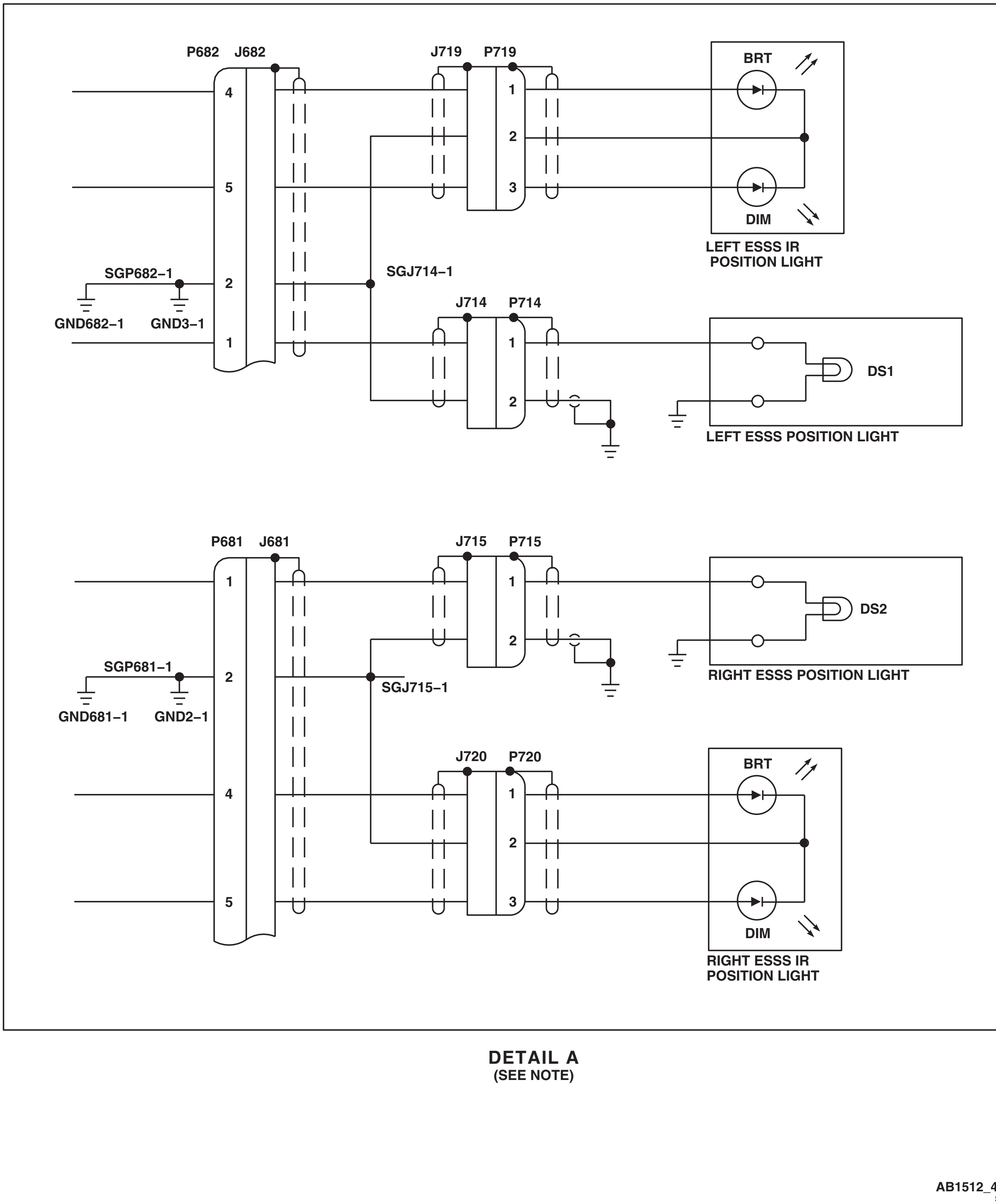


Figure 2. Position Lights Schematic Diagram. (Sheet 4 of 4)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

RETRACTABLE LANDING LIGHT

This WP supersedes WP 0114 00, dated 21 March 2007.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
WP 0090 00
WP 0100 00
WP 0104 00
WP 0135 00
WP 0824 00
WP 0825 00
WP 0916 00
WP 0917 00
WP 0925 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

RETRACTABLE LANDING LIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Slide pilot's and copilot's collective stick grip landing light switches to center position.
2. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
RETR LDG CONT	Copilot's
RETR LDG PWR	Copilot's

3. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
4. Slide pilot's collective stick grip LDG LT switch to EXT and hold.

RETRACTABLE LANDING LIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Retractable landing light shall move out to full extension.

Corrective Action

1. If Indication/Condition is not as specified, go to LANDING LIGHT DOES NOT MOVE WITH PILOT'S LDG LT SWITCH MOVED TO EXT, in this work package.
2. If retractable landing light moves but stops short of fully extended position, replace light assembly (WP 0917 00).

PROCEDURE

Step

5. Momentarily slide pilot's collective stick grip LDG LT switch to RET and release.

Indication/Condition

Retractable landing light shall retract while switch is moved to RET. Switch shall return to center when released and landing light shall stop.

Corrective Action

1. If Indication/Condition is not as specified, troubleshoot pilot's collective stick (WP 0135 00).
2. If landing light does not move, go to LANDING LIGHT DOES NOT MOVE WITH PILOT'S LDG LT SWITCH MOVED TO RET, in this work package.

Step

6. Slide pilot's collective stick grip LDG LT switch to RET and hold until landing light is fully retracted.

Indication/Condition

Retractable landing light shall retract.

Corrective Action

If Indication/Condition is not as specified, replace light assembly (WP 0917 00).

Step

7. Slide copilot's collective stick grip LDG LT switch to EXT and hold.

Indication/Condition

Landing light shall move out to full extension.

Corrective Action

1. If Indication/Condition is not as specified, with copilot's LDG LT switch held at EXT, check continuity between P103-U and P103-W.
   a. If continuity is present, repair/replace wiring between J103-W and J128-F (WP 1747 00).
   b. If continuity is not present, troubleshoot copilot's collective stick (WP 0135 00).

Step

8. Momentarily slide copilot's collective stick grip LDG LT switch to RET.

RETRACTABLE LANDING LIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Landing light shall retract while switch is held to RET. Switch shall return to center when released and landing light shall stop.

Corrective Action

1. If Indication/Condition is not as specified, with copilot's LDG LT switch held to RET, check continuity between P103-S and P103-U.
   a. If continuity is present, repair/replace wiring between J103-S and J128-E (WP 1747 00).
   b. If continuity is not present, troubleshoot copilot's collective stick (WP 0135 00).

Step

9. Momentarily press pilot's collective stick grip LDG LT switch.

Indication/Condition

a. Landing light shall go on.

b. LDG LT ON capsule or legend shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to LANDING LIGHT DOES NOT GO ON WHEN PILOT'S LDG LT SWITCH IS PRESSED AND RELEASED, in this work package.
2. If Indication/Condition b. is not as specified, check helicopter configuration.
   a. EMEP Remove and check pin filtered adapter P118, P1019R or P1020R, as applicable (WP 0925 00).
      (1) If pin filtered adapter is good, EMEP install pin filtered adapter (WP 0925 00), then W/O EMEP check for 28 vdc between P118-L, P1019R, or P1020R and ground.
         (a) If voltage is as specified, troubleshoot caution/advisory warning system or MFD/caution/advisory warning system (WP 0090 00 or WP 0100 00).
         (b) If voltage is not as specified, repair/replace wiring between P118-L, P1019R, or P1020R and P900-p (WP 1747 00).
      (2) If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00).

Step

10. Momentarily press pilot's collective stick grip LDG LT switch.

Indication/Condition

Landing light shall go off.

Corrective Action

If Indication/Condition is not as specified, go to LANDING LIGHT DOES NOT GO OFF WHEN PILOT'S LDG LT SWITCH IS PRESSED AND RELEASED, in this work package.

Step

11. Momentarily press copilot's collective stick grip LDG LT switch.

RETRACTABLE LANDING LIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Landing light and LDG LT ON capsule shall go on.

Corrective Action

If Indication/Condition is not as specified, go to LANDING LIGHT DOES NOT GO OFF WHEN COPILOT'S LDG LT SWITCH IS PRESSED AND RELEASED, in this work package.

Step

12. Momentarily press copilot's collective stick grip LDG LT switch.

Indication/Condition

Landing light shall go off.

Corrective Action

If Indication/Condition is not as specified, go to LANDING LIGHT DOES NOT GO OFF WHEN PILOT'S LDG LT SWITCH IS PRESSED AND RELEASED, in this work package.

Step

13. Slide copilot's collective stick grip landing light switch to RET until landing light is fully retracted.
14. Turn MFD switch OFF.
15. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

LANDING LIGHT DOES NOT MOVE WITH PILOT'S LDG LT SWITCH MOVED TO EXT

SYMPTOM

Landing light does not move with pilot's LDG LT switch moved to EXT.

MALFUNCTION

Landing light does not move with pilot's LDG LT switch moved to EXT.

CORRECTIVE ACTION

1. **While holding LDG LT switch to EXT, check for 28 vdc between J128-F and ground.**
   a. If voltage is as specified, replace retractable landing light (WP 0917 00). Go to **5.**
   b. If voltage is not as specified, go to **2.**
2. **Check for 28 vdc between J104-U and ground.**
   a. If voltage is as specified, go to **3.**
   b. If voltage is not as specified, go to **4.**
3. **With pilot's LDG LT switch held at EXT, check continuity between P104-U and P104-W.**

LANDING LIGHT DOES NOT MOVE WITH PILOT'S LDG LT SWITCH MOVED TO EXT - Continued

a. If continuity is present, repair/replace wiring between J104-W and J128-F (WP 1747 00). Go to **5.**

b. If continuity is not present, troubleshoot pilot's collective stick (WP 0135 00). Go to **5.**

4. Check for 28 vdc between LIGHTS RETR LDG CONT circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J104-U (WP 1747 00). Go to **5.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **5.**

5. Procedure completed.

LANDING LIGHT DOES NOT MOVE WITH PILOT'S LDG LT SWITCH MOVED TO RET

SYMPTOM

Landing light does not move with pilot's LDG LT switch moved to RET.

MALFUNCTION

Landing light does not move with pilot's LDG LT switch moved to RET.

CORRECTIVE ACTION

1. Check for 28 vdc between J128-E and ground while holding LDG LT switch to RET.

a. If voltage is as specified, replace retractable landing light (WP 0917 00). Go to **3.**

b. If voltage is not as specified, go to **2.**

2. With pilot's LDG LT switch held at RET, check continuity between P104-U and P104-S.

a. If continuity is present, repair/replace wiring between J104-S and J128-E (WP 1747 00). Go to **3.**

b. If continuity is not present, troubleshoot pilot's collective stick (WP 0135 00). Go to **3.**

3. Procedure completed.

LANDING LIGHT DOES NOT GO ON WHEN PILOT'S LDG LT SWITCH IS PRESSED AND RELEASED

SYMPTOM

Landing light does not go on when pilot's LDG LT switch is pressed and released.

MALFUNCTION

Landing light does not go on when pilot's LDG LT switch is pressed and released.

CORRECTIVE ACTION

1. Check for 28 vdc between J128-A and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **9.**

2. With LDG LT switch pushed in and held, check for 28 vdc between J128-B and ground.

a. If voltage is as specified, go to **3.**

LANDING LIGHT DOES NOT GO ON WHEN PILOT'S LDG LT SWITCH IS PRESSED AND RELEASED - Continued

b. If voltage is not as specified, go to **4.**

3. **Check if lamp is good.**

a. If lamp is good, replace landing light (WP 0917 00). Go to **10.**

b. If lamp is not good, replace lamp (WP 0916 00). Go to **10.**

4. **Check for 28 vdc between P900-n and ground.**

a. If voltage is as specified, go to **5.**

b. If voltage is not as specified, go to **8.**

5. **With pilot's LDG LT switch pressed and held in, check for 28 vdc between P900-k and ground.**

a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **10.**

b. If voltage is not as specified, go to **6.**

6. **Check continuity between J104-X and P900-k.**

a. If continuity is present, go to **7.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

7. **Check for 28 vdc between J104-Y and ground.**

a. If voltage is as specified, troubleshoot pilot's collective stick (WP 0135 00). Go to **10.**

b. If voltage is not as specified, repair/replace wiring between J104-Y and LIGHTS RETR LDG PWR circuit breaker terminal 1 (WP 1747 00). Go to **10.**

8. **Check continuity between P900-n and LIGHTS RETR LDG CONT circuit breaker terminal 1.**

a. If continuity is present, replace LIGHTS RETR LDG CONT circuit breaker (WP 0824 00). Go to **10.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

9. **Check continuity between J128-A and LIGHTS RETR LDG CONT circuit breaker terminal 1.**

a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **10.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

10. **Procedure completed.**

LANDING LIGHT DOES NOT GO OFF WHEN PILOT'S LDG LT SWITCH IS PRESSED AND RELEASED

SYMPTOM

Landing light does not go off when pilot's LDG LT switch is pressed and released.

MALFUNCTION

Landing light does not go off when pilot's LDG LT switch is pressed and released.

CORRECTIVE ACTION

1. **Check for 28 vdc between J128-B and ground.**

a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **2.**

LANDING LIGHT DOES NOT GO OFF WHEN PILOT'S LDG LT SWITCH IS PRESSED AND RELEASED - Continued

b. If voltage is not as specified, replace landing light assembly (WP 0917 00). Go to **2.**

2. Procedure completed.

LANDING LIGHT DOES NOT GO ON WHEN COPILOT'S LDG LT SWITCH IS PRESSED AND RELEASED

SYMPTOM

Landing light does not go on when copilot's LDG LT switch is pressed and released.

MALFUNCTION

Landing light does not go on when copilot's LDG LT switch is pressed and released.

CORRECTIVE ACTION

1. Check for 28 vdc between J103-Y and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, repair/replace wiring between J103-Y and P248-D (WP 1747 00). Go to **3.**

2. With copilot's LDG LT switch pressed and held in, check continuity between P103-X and P103-Y.

a. If continuity is present, repair/replace wiring between J103-X and P900-k (WP 1747 00). Go to **3.**

b. If continuity is not present, troubleshoot copilot's collective stick (WP 0135 00). Go to **3.**

3. Procedure completed.

LANDING LIGHT DOES NOT GO OFF WHEN COPILOT'S LDG LT SWITCH IS PRESSED AND RELEASED

SYMPTOM

Landing light does not go off when copilot's LDG LT switch is pressed and released.

MALFUNCTION

Landing light does not go off when copilot's LDG LT switch is pressed and released.

CORRECTIVE ACTION

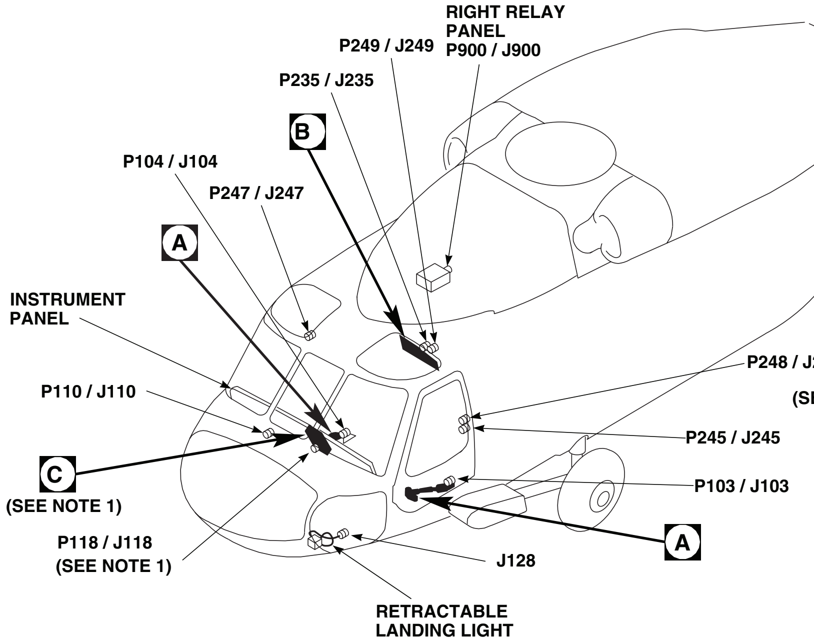
1. Check for 28 vdc between J128-B and ground.

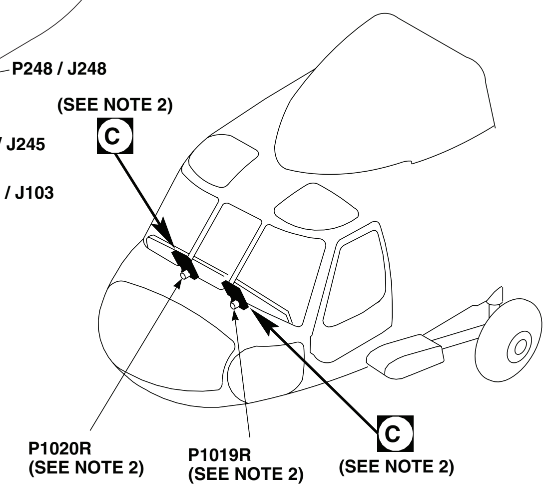
a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **2.**

b. If voltage is not as specified, replace landing light assembly (WP 0917 00). Go to **2.**

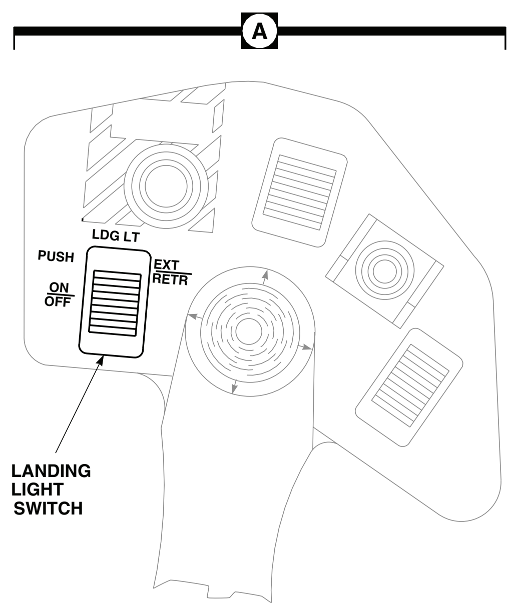
2. Procedure completed.

LOCATION





TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J128	RETRACTABLE LANDING LIGHT
P103 / J103	COCKPIT TUB BL 32 LH, STA 247
P104 / J104	COCKPIT TUB BL 15 RH, STA 247
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P118 / J118	CAUTION / ADVISORY PANEL (SEE NOTE 1)
P235 / J235	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 20 LH
P245 / J245	CABIN BL 40 LH, STA 248
P247 / J247	CABIN BL 40 RH, STA 248
P248 / J248	CABIN BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P900 / J900	RH RELAY PANEL
P1019R / J3	COPILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)
P1020R / J3	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)



NOTES

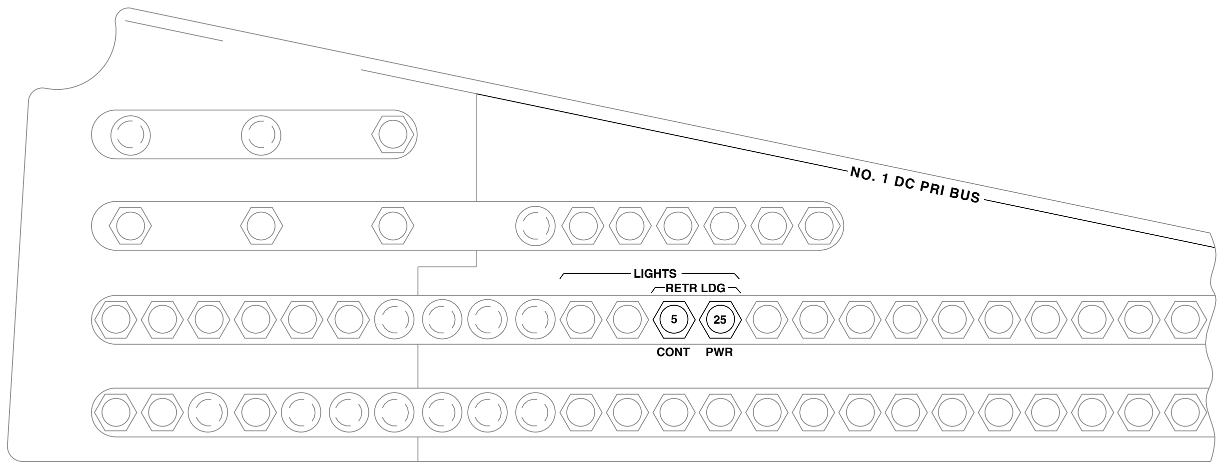
1. UH60A UH60L EH60A
2. HH60A HH60L

PILOT'S AND COPILOT'S COLLECTIVE STICK GRIP

AB2183_1A
SA

Figure 1. Retractable Landing Light Location Diagram. (Sheet 1 of 2)

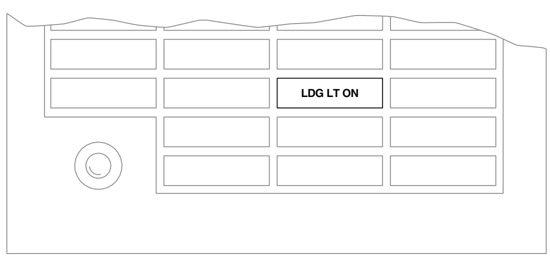
LOCATION - Continued



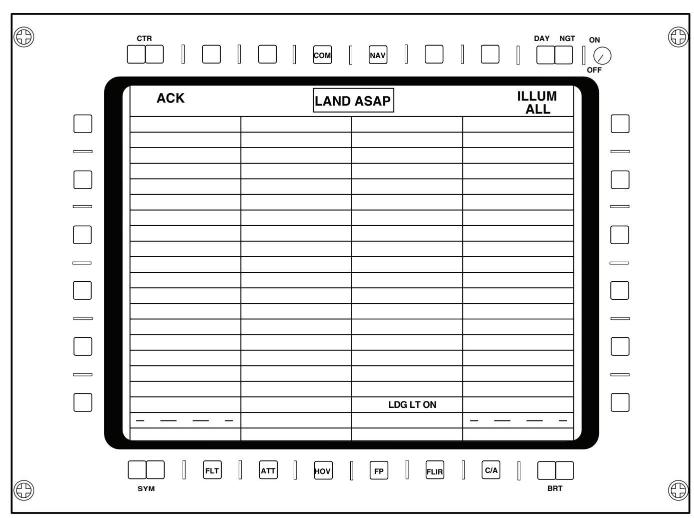
COPILOT'S CIRCUIT BREAKER PANEL

B

C



CAUTION/ADVISORY PANEL
(SEE NOTE 1)

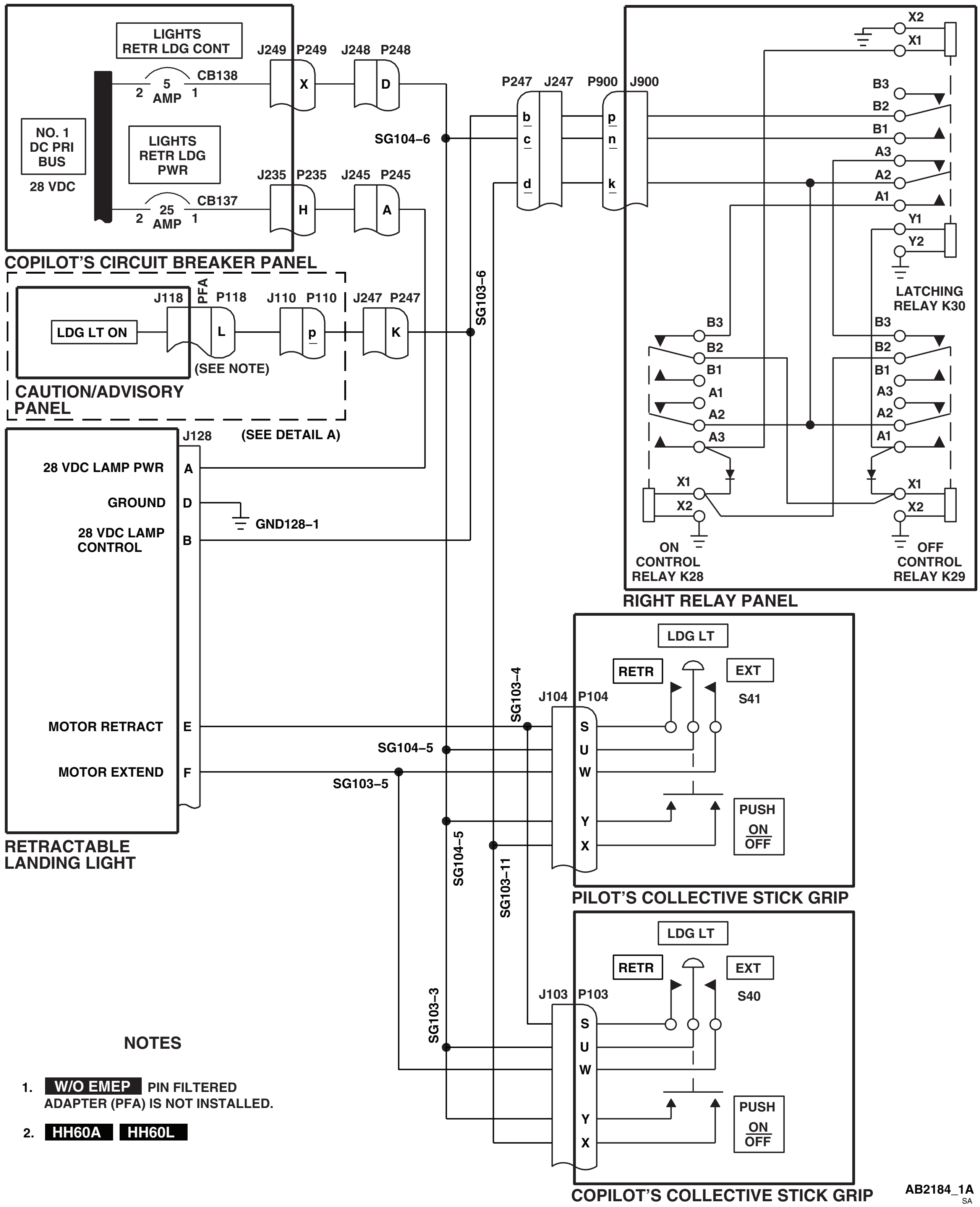


MULTIFUNCTION DISPLAY'S CAUTION/ADVISORY GRID
(SEE NOTE 2)

AB2183_2A
SA

Figure 1. Retractable Landing Light Location Diagram. (Sheet 2 of 2)

SCHEMATICS



NOTES

1. W/O EMEP PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.
2. HH60A HH60L

Figure 2. Retractable Landing Light Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

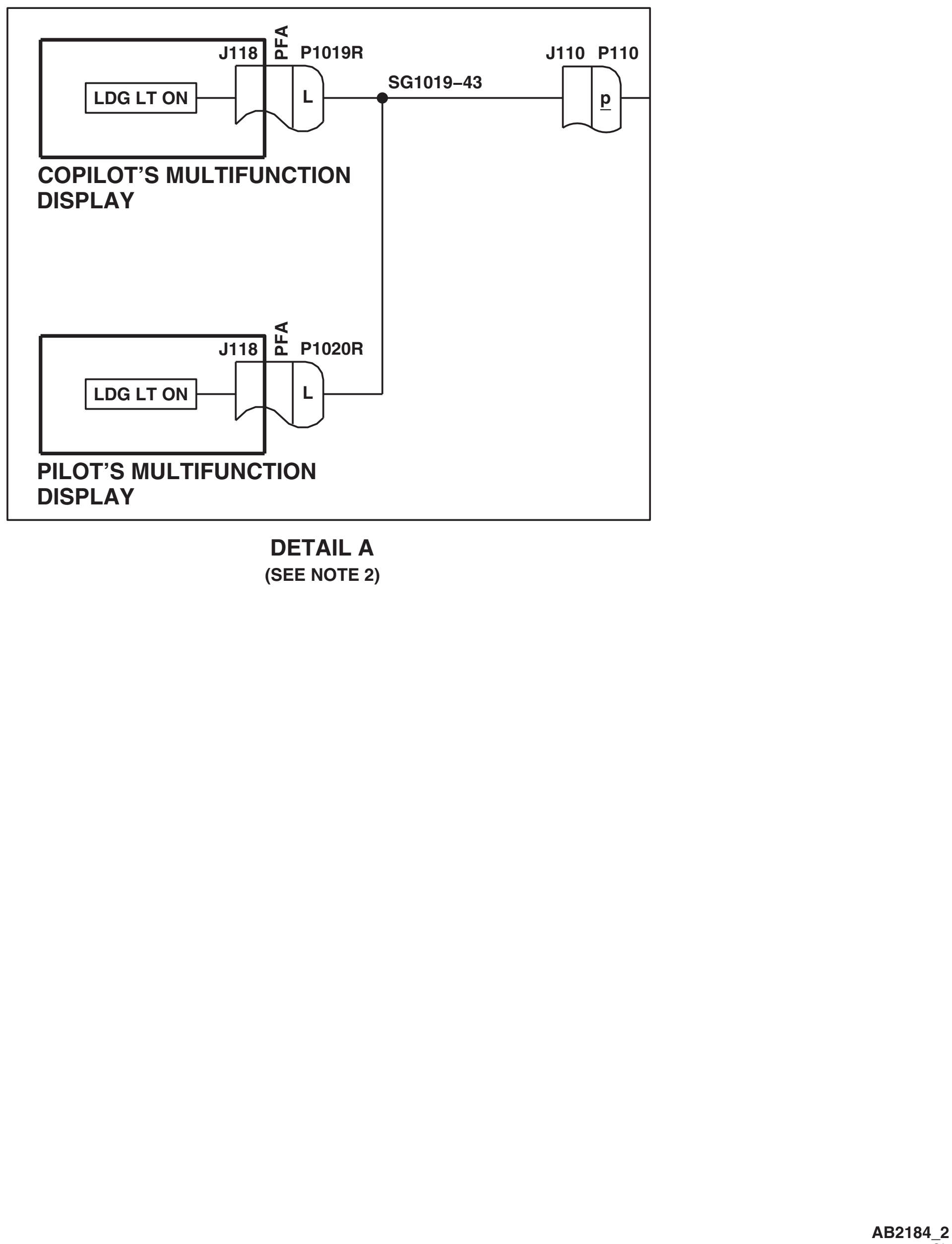


Figure 2. Retractable Landing Light Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

ANTI-COLLISION LIGHTS

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
WP 0104 00
WP 0824 00
WP 0828 00
WP 0913 00
WP 0914 00
WP 0915 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

ANTI-COLLISION LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Check anti-collision lights for cracked or broken lenses.

Indication/Condition

Anti-collision light lenses shall not be cracked or broken.

Corrective Action

If Indication/Condition is not as specified, replace lens (WP 0913 00).

Step

2. Make sure that LIGHTS ANTI COLL circuit breaker on pilot's circuit breaker panel is pushed in.
3. Turn on electrical power.
4. Place upper console ANTI COLLISION LIGHTS UPPER/BOTH/LOWER switch to BOTH.
5. Place upper console ANTI COLLISION LIGHTS DAY/OFF/NIGHT switch to DAY.

Indication/Condition

Upper and lower anti-collision lights shall alternately flash white.

ANTI-COLLISION LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition is not as specified, go to NEITHER ANTI-COLLISION LIGHT FLASHES WHITE, in this work package.
2. If one light does not flash white, go to ONE ANTI-COLLISION LIGHT DOES NOT FLASH WHITE WITH ANTI COLLISION LIGHTS SWITCH SET TO BOTH, in this work package.
3. If both day and night anti-collision lights are flashing simultaneously, replace power supply (WP 0915 00).

Step

6. Place upper console ANTI COLLISION LIGHTS DAY/OFF/NIGHT switch to NIGHT.

Indication/Condition

Upper and lower anti-collision lights shall alternately flash red.

Corrective Action

1. If Indication/Condition is not as specified, go to NEITHER ANTI-COLLISION LIGHT FLASHES RED, in this work package.
2. If one light does not flash red, go to ONE ANTI-COLLISION LIGHT DOES NOT FLASH RED WITH SWITCH SET TO BOTH, in the work package.
3. If both day and night anti-collision lights are flashing simultaneously, replace power supply (WP 0915 00).

Step

7. Place upper console ANTI COLLISION LIGHTS UPPER/BOTH/LOWER switch to UPPER.

Indication/Condition

a. Upper anti-collision light shall flash red.

b. Lower anti-collision light shall be off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to UPPER ANTI-COLLISION LIGHT DOES NOT FLASH WITH ANTI COLLISION LIGHTS SWITCH SET TO UPPER, in this work package.
2. If Indication/Condition b. is not as specified, replace ANTI COLLISION LIGHTS UPPER/BOTH/LOWER light switch S27 (WP 0828 00).

Step

8. Place upper console ANTI COLLISION LIGHTS UPPER/BOTH/LOWER switch to LOWER.

Indication/Condition

a. Lower anti-collision light shall flash red.

b. Upper anti-collision light shall be off.

Corrective Action

1. If Indication/Condition a. is not as specified, check continuity between terminals 2 and 3 of ANTI COLLISION LIGHTS UPPER/BOTH/LOWER switch.
   a. If continuity is present, repair/replace wiring to switch (WP 1747 00).
   b. If continuity is not present, replace switch (WP 0828 00).

ANTI-COLLISION LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. If Indication/Condition b. is not as specified, replace ANTI COLLISION LIGHTS UPPER/BOTH/LOWER switch S27 (WP 0828 00).

Step

9. Place upper console ANTI COLLISION UPPER/BOTH/LOWER switch to BOTH and DAY/OFF/NIGHT switch to OFF.

Indication/Condition

Upper and lower anti-collision lights shall go off.

Corrective Action

If Indication/Condition is not as specified, replace ANTI COLLISION LIGHTS DAY/OFF/NIGHT switch S26 (WP 0828 00).

Step

10. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

NEITHER ANTI-COLLISION LIGHT FLASHES WHITE

SYMPTOM

Neither anti-collision light flashes white.

MALFUNCTION

Neither anti-collision light flashes white.

CORRECTIVE ACTION

1. **Place ANTI COLLISION DAY/OFF/NIGHT switch to NIGHT. Go to 2.**
2. **Check that upper and lower anti-collision lights alternately flash red.**
   a. If lights alternately flash red, go to **3.**
   b. If lights do not alternately flash red, go to **4.**
3. **Check continuity between S26-1 and GG2-5.**
   a. If continuity is present, replace switch S26 (WP 0828 00). Go to **9.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**
4. **Check for 115 vac between P503-A and P503-C.**
   a. If voltage is as specified, go to **5.**
   b. If voltage is not as specified, go to **7.**
5. **Check continuity between P503-E and P503-B.**
   a. If continuity is present, replace power supply (WP 0915 00). Go to **9.**
   b. If continuity is not present, go to **6.**

NEITHER ANTI-COLLISION LIGHT FLASHES WHITE - Continued

6. **Check continuity between P503-E and S26-2.**
   a. If continuity is present, replace switch S26 (WP 0828 00). Go to **9.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**
7. **Check continuity between P503-C and GND503-1.**
   a. If continuity is present, go to **8.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**
8. **Check for 115 vac between LIGHTS ANTI COLL circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P503-A (WP 1747 00). Go to **9.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9.**
9. **Procedure completed.**

ONE ANTI-COLLISION LIGHT DOES NOT FLASH WHITE WITH ANTI COLLISION LIGHTS SWITCH SET TO BOTH

SYMPTOM

One anti-collision light does not flash white with ANTI COLLISION LIGHTS switch set to BOTH.

MALFUNCTION

One anti-collision light does not flash white with ANTI COLLISION LIGHTS switch set to BOTH.

CORRECTIVE ACTION

WARNING

Hazardous voltage exists on power supply connectors even with power removed. Use extreme care when disconnecting P502 and P504.

1. **Is upper anti-collision light operating?**
   a. If light is operating, go to **2.**
   b. If light is not operating, go to **9.**
2. **Check continuity between P503-B and P503-D.**
   a. If continuity is present, replace ANTI COLLISION LIGHTS UPPER/BOTH/LOWER switch (WP 0828 00). Go to **16.**
   b. If continuity is not present, go to **3.**
3. **Swap P502 and P504. Go to 4.**
4. **Is lower anti-collision light operating?**
   a. If light is operating, go to **5.**
   b. If light is not operating, go to **7.**
5. **Swap P502 and P504 to original connections. Go to 6.**
6. **Check continuity between:**
   - **P502-A and P505-A**

ONE ANTI-COLLISION LIGHT DOES NOT FLASH WHITE WITH ANTI COLLISION LIGHTS SWITCH SET TO BOTH - Continued

- **P502-B and P505-B**
- **P502-C and P505-C**
- **P502-E and P505-E**

a. If continuity is present, replace power supply (WP 0915 00). Go to **16.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **16.**

7. Swap P502 and P504 to original connections. Go to 8.

8. Replace lamp (WP 0914 00). Go to 16.

9. Check continuity between P503-D and GG2-5.

a. If continuity is present, replace ANTI COLLISION LIGHTS UPPER/BOTH/LOWER switch (WP 0828 00). Go to **16.**

b. If continuity is not present, go to **10.**

10. Swap P502 and P504. Go to 11.

11. Is upper anti-collision light operating?

a. If light is operating, go to **12.**

b. If light is not operating, go to **14.**

12. Swap P502 and P504 to original connections. Go to 13.

13. Check continuity between:

- **P504-A and P602-A**
- **P504-B and P602-B**
- **P504-C and P602-C**
- **P504-E and P602-E**

a. If continuity is present, replace power supply (WP 0915 00). Go to **16.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **16.**

14. Swap P502 and P504 to original connections. Go to 15.

15. Replace lamp (WP 0914 00). Go to 16.

16. Procedure completed.

NEITHER ANTI-COLLISION LIGHT FLASHES RED

SYMPTOM

Neither anti-collision light flashes red.

MALFUNCTION

Neither anti-collision light flashes red.

CORRECTIVE ACTION

1. Check continuity between P503-B and P503-E.

a. If continuity is present, replace power supply (WP 0915 00). Go to **4.**

NEITHER ANTI-COLLISION LIGHT FLASHES RED - Continued

b. If continuity is not present, go to **2.**

2. **Check continuity between P503-B and S27-1.**

a. If continuity is present, go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**

3. **Check continuity between S27-1 and S26-3.**

a. If continuity is present, replace switch S27 (WP 0828 00). Go to **4.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**

4. **Procedure completed.**

ONE ANTI-COLLISION LIGHT DOES NOT FLASH RED WITH SWITCH SET TO BOTH

SYMPTOM

One anti-collision light does not flash red with switch set to BOTH.

MALFUNCTION

One anti-collision light does not flash red with switch set to BOTH.

CORRECTIVE ACTION

WARNING

Hazardous voltages exist on power supply connectors even with power removed. Use extreme care when disconnecting P502 and P504.

1. **Is upper anti-collision light operating?**

a. If light is operating, go to **2.**

b. If light is not operating, go to **8.**

2. **Swap P502 and P504. Go to 3.**

3. **Is lower anti-collision light operating?**

a. If light is operating, go to **4.**

b. If light is not operating, go to **6.**

4. **Swap P502 and P504 to original connections. Go to 5.**

5. **Check continuity between P502-D and P505-D.**

a. If continuity is present, replace power supply (WP 0915 00). Go to **14.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**

6. **Swap P502 and P504 to original connections. Go to 7.**

7. **Replace lamp (WP 0914 00). Go to 14.**

8. **Swap P502 and P504. Go to 9.**

9. **Is upper anti-collision light operating?**

a. If light is operating, go to **10.**

b. If light is not operating, go to **12.**

ONE ANTI-COLLISION LIGHT DOES NOT FLASH RED WITH SWITCH SET TO BOTH - Continued

10. **Swap P502 and P504 to original connections. Go to 11.**
11. **Check continuity between P504-D and P602-D.**
    a. If continuity is present, replace power supply (WP 0915 00). Go to **14.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14.**
12. **Swap P502 and P504 to original connections. Go to 13.**
13. **Replace lamp (WP 0914 00). Go to 14.**
14. **Procedure completed.**

UPPER ANTI-COLLISION LIGHT DOES NOT FLASH WITH ANTI COLLISION LIGHTS SWITCH SET TO UPPER

SYMPTOM

Upper anti-collision light does not flash with ANTI COLLISION LIGHTS switch set to UPPER.

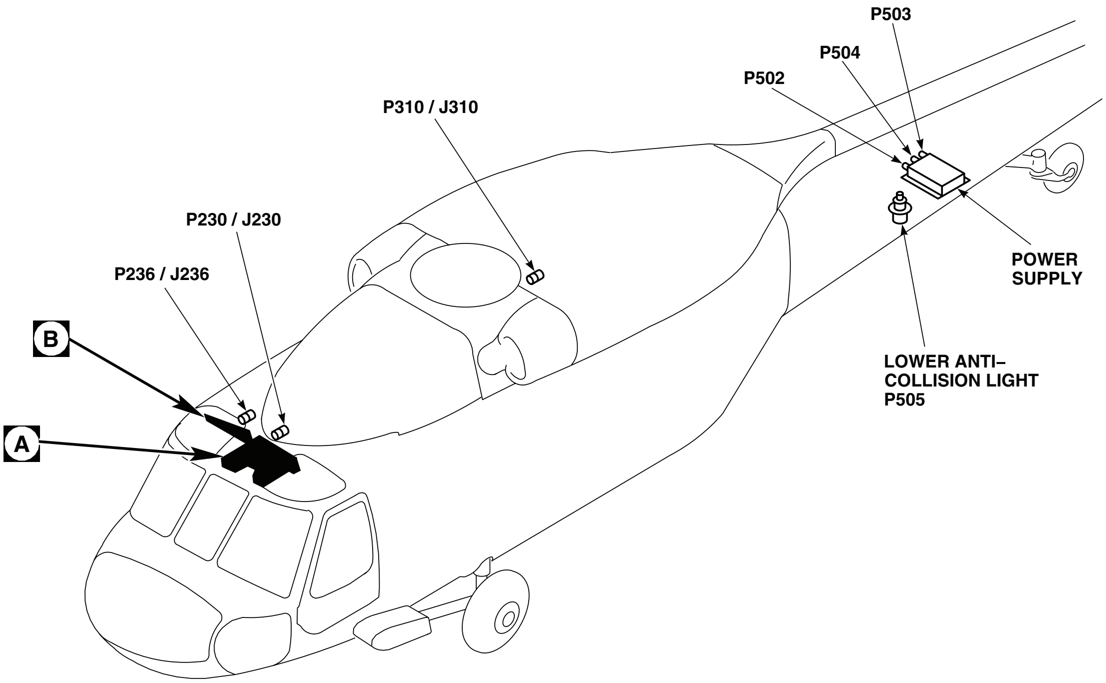
MALFUNCTION

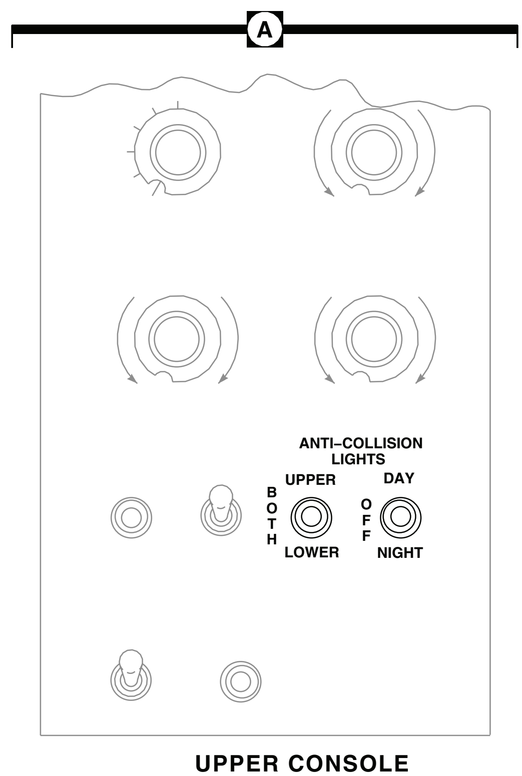
Upper anti-collision light does not flash with ANTI COLLISION LIGHTS switch set to UPPER.

CORRECTIVE ACTION

1. **Check continuity between P503-B and P503-D.**
    a. If continuity is present, replace power supply (WP 0915 00). Go to **3.**
    b. If continuity is not present, go to **2.**
2. **Check continuity between P503-D and S27-2.**
    a. If continuity is present, repair/replace wiring (WP 1747 00). Go to **3.**
    b. If continuity is not present, replace switch S27 (WP 0828 00). Go to **3.**
3. **Procedure completed.**

LOCATION



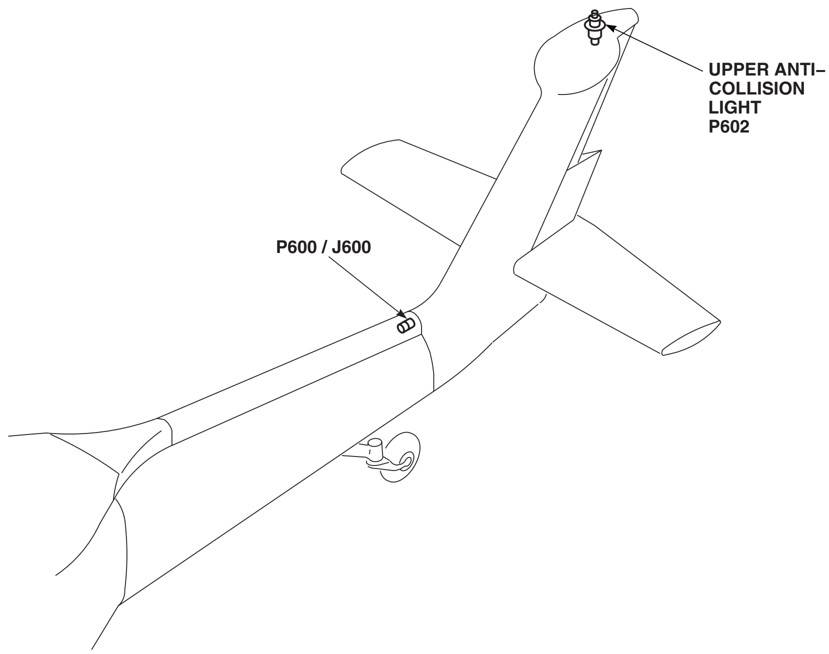


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 23 RH
P310 / J310	CABIN CEILING BL 17 RH, STA 387
P502	POWER SUPPLY
P503	POWER SUPPLY
P504	POWER SUPPLY
P505	LOWER ANTI–COLLISION LIGHT
P600 / J600	TAIL CONE BL 0, STA 650
P602	UPPER ANTI–COLLISION LIGHT

AA2785_1
SA

Figure 1. Anti-collision Lights Location Diagram. (Sheet 1 of 2)

LOCATION - Continued



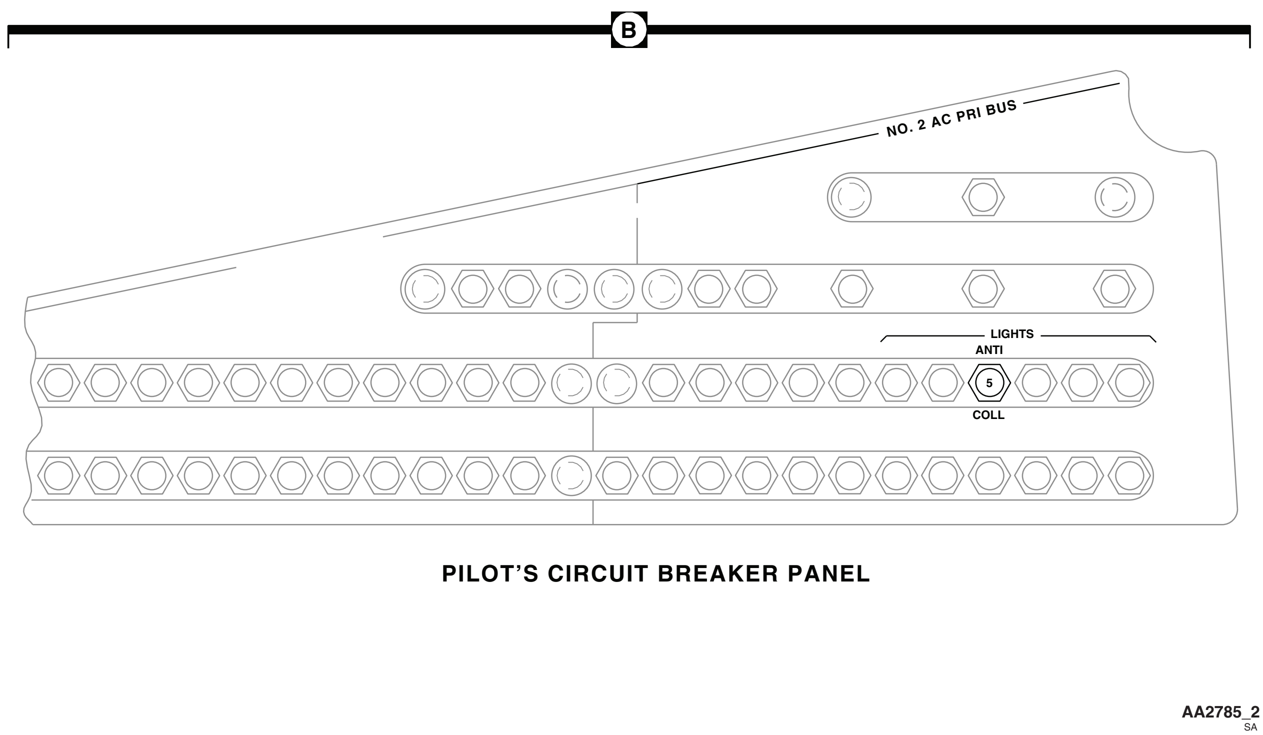


Figure 1. Anti-collision Lights Location Diagram. (Sheet 2 of 2)

SCHEMATICS

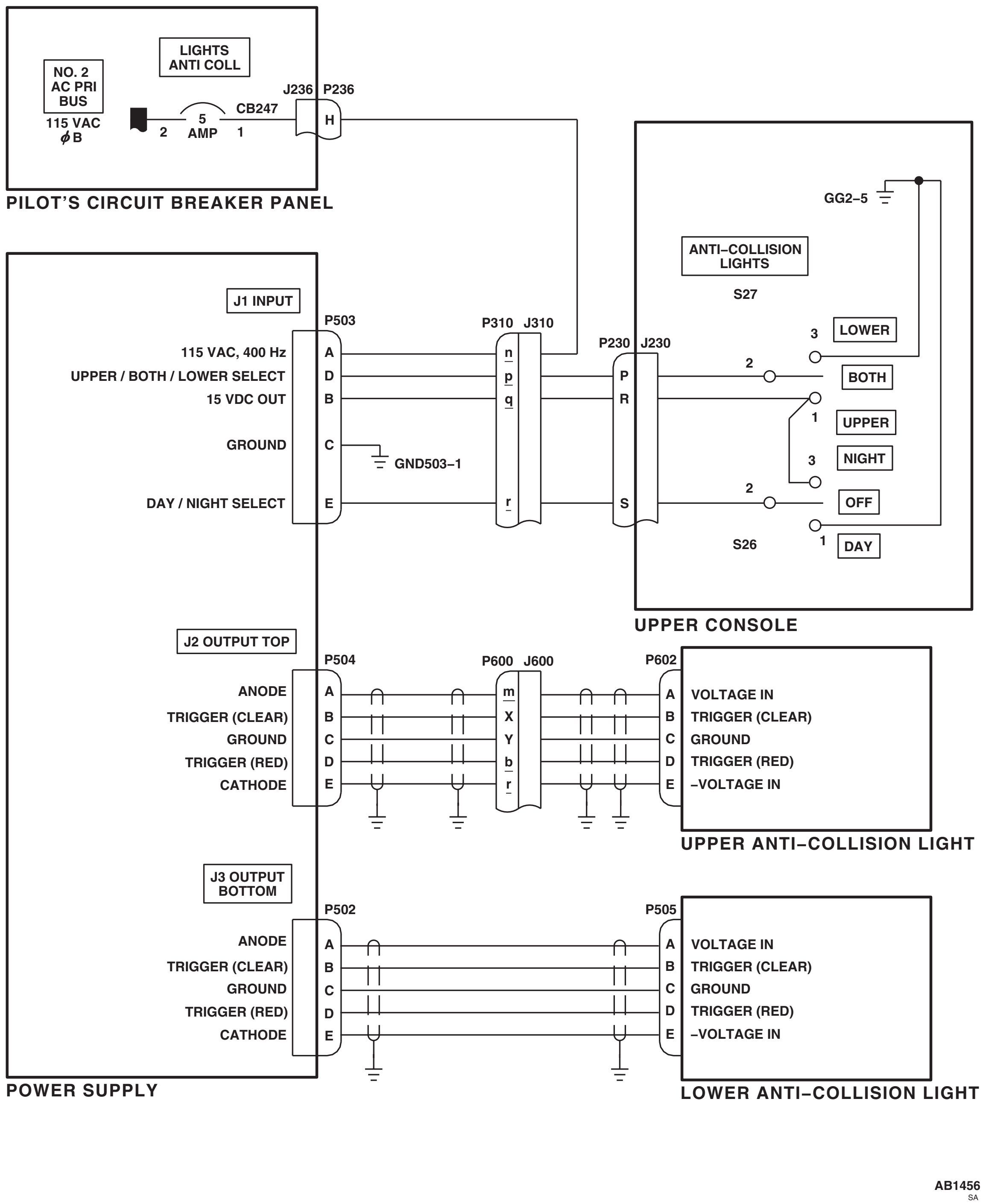


Figure 2. Anti-collision Lights Schematic Diagram.

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

CARGO HOOK LIGHTS UH-60A UH-60L HH-60A HH-60L

This WP supersedes WP 0116 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
WP 0104 00
WP 0824 00
WP 0828 00
WP 0886 00
WP 0920 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

CARGO HOOK LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE W/O MWO 50-56

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram, and 77-22714-97-26723 Figure 2 and MWO 50-56 HH-60A HH-60L UH-60L 97-26744-SUBQ Figure 3 for schematic diagrams as an aid in operational/troubleshooting.

1. Make sure LIGHTS CARGO HOOK circuit breaker on pilot's circuit breaker panel is pushed in.
2. Turn on electrical power.
3. Place upper console CARGO HOOK LT switch to ON.

Indication/Condition

Cargo hook lights shall go on.

Corrective Action

1. If Indication/Condition is not as specified, go to NO CARGO HOOK LIGHTS GO ON, in this work package.
2. If one cargo hook light does not go on, check for 115 vac between the following:
   - J389-1 and ground
   - J390-1 and ground

CARGO HOOK LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE W/O MWO 50-56 - Continued

- J391-1 and ground

a. If voltage is not as specified, repair/replace wiring as required (WP 1747 00).

b. If voltage is as specified, replace light assembly (WP 0920 00).

Step

4. Place upper console CARGO HOOK LT switch to OFF.

Indication/Condition

Cargo hook lights shall go off.

Corrective Action

If Indication/Condition is not as specified, replace CARGO HOOK LT switch (WP 0828 00).

Step

5. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

CARGO HOOK LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE MWO 50-56

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure LIGHTS CARGO HOOK circuit breaker on pilot's circuit breaker panel is pushed in.
2. Turn on electrical power.
3. Rotate HOOK LIGHT DIMMING CONTROL PANEL knob fully clockwise to BRT.
4. Place upper console CARGO HOOK LT switch to ON.

Indication/Condition

Cargo hook lights shall go on bright.

Corrective Action

If Indication/Condition is not as specified, go to NO CARGO HOOK LIGHTS GO ON WITH HOOK LIGHT CONTROL PANEL KNOB FULLY CLOCKWISE, in this work package.

Step

5. Rotate HOOK LIGHT DIMMING CONTROL PANEL knob fully counterclockwise.

CARGO HOOK LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE MWO 50-56 - Continued

Indication/Condition

Cargo hook lights shall dim.

Corrective Action

If Indication/Condition is not as specified, replace control panel (WP 0886 00).

Step

6. Place upper console CARGO HOOK LT switch to OFF.

Indication/Condition

Cargo hook lights shall go off.

Corrective Action

If Indication/Condition is not as specified, replace CARGO HOOK LT switch (WP 0828 00).

Step

7. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

NO CARGO HOOK LIGHTS GO ON W/O MWO 50-56

SYMPTOM

No cargo hook lights go on.

MALFUNCTION

No cargo hook lights go on.

CORRECTIVE ACTION

1. **Check for 115 vac between CARGO HOOK LT switch terminal 2 and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, go to **6.**
2. **Check for 115 vac between CARGO HOOK LT switch terminal 3 and ground.**
   a. If voltage is as specified, go to **3.**
   b. If voltage is not as specified, replace switch (WP 0828 00). Go to **7.**
3. **Check for 115 vac between J392-1 and J392-2.**
   a. If voltage is as specified, go to **4.**
   b. If voltage is not as specified, go to **5.**
4. **Check continuity between:**
   - **P392-1 and J389-1**

NO CARGO HOOK LIGHTS GO ON W/O MWO 50-56 - Continued

- **P392-1 and J390-1**
- **P392-1 and J391-1**

a. If continuity is present, repair/replace wiring, as required, between (WP 1747 00), then go to **7:**

- P392-2 and J389-2
- P392-2 and J390-2
- P392-2 and J391-2

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7.**

5. Check for 115 vac between J392-1 and ground.

a. If voltage is as specified, repair/replace wiring between J392-2 and ground (WP 1747 00). Go to **7.**

b. If voltage is not as specified, repair/replace wiring between J392-1 and CARGO HOOK LT switch terminal 3 (WP 1747 00). Go to **7.**

6. Check for 115 vac between LIGHTS CARGO HOOK circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and CARGO HOOK LT switch terminal 2 (WP 1747 00). Go to **7.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **7.**

7. Procedure completed.

NO CARGO HOOK LIGHTS GO ON WITH HOOK LIGHT CONTROL PANEL KNOB FULLY CLOCKWISE MWO 50-56

SYMPTOM

No cargo hook lights go on with HOOK LIGHT control panel knob fully clockwise.

MALFUNCTION

No cargo hook lights go on with HOOK LIGHT control panel knob fully clockwise.

CORRECTIVE ACTION

1. Check for 115 vac between CARGO HOOK LT switch terminal 2 and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **6.**

2. Check for 115 vac between CARGO HOOK LT switch terminal 3 and ground.

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, replace switch (WP 0828 00). Go to **8.**

3. Check for 115 vac between J392-1 and J392-2.

a. If voltage is as specified, go to **4.**

b. If voltage is not as specified, go to **5.**

4. Check for 115 vac between J389-3 and ground.

NO CARGO HOOK LIGHTS GO ON WITH HOOK LIGHT CONTROL PANEL KNOB FULLY CLOCKWISE MWO 50-56 - Continued

a. If voltage is as specified, go to **7.**

b. If voltage is not as specified, repair/replace wiring (WP 1747 00). Go to **8.**

5. **Check for 115 vac between J392-1 and ground.**

a. If voltage is as specified, repair/replace wiring between J392-2 and ground (WP 1747 00). Go to **8.**

b. If voltage is not as specified, repair/replace wiring between J392-1 and CARGO HOOK LT switch terminal 3 (WP 1747 00). Go to **8.**

6. **Check for 115 vac between LIGHTS CARGO HOOK circuit breaker terminal 1 and ground.**

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and CARGO HOOK LT switch terminal 2 (WP 1747 00). Go to **8.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8.**

7. **Check for 115 vac between:**

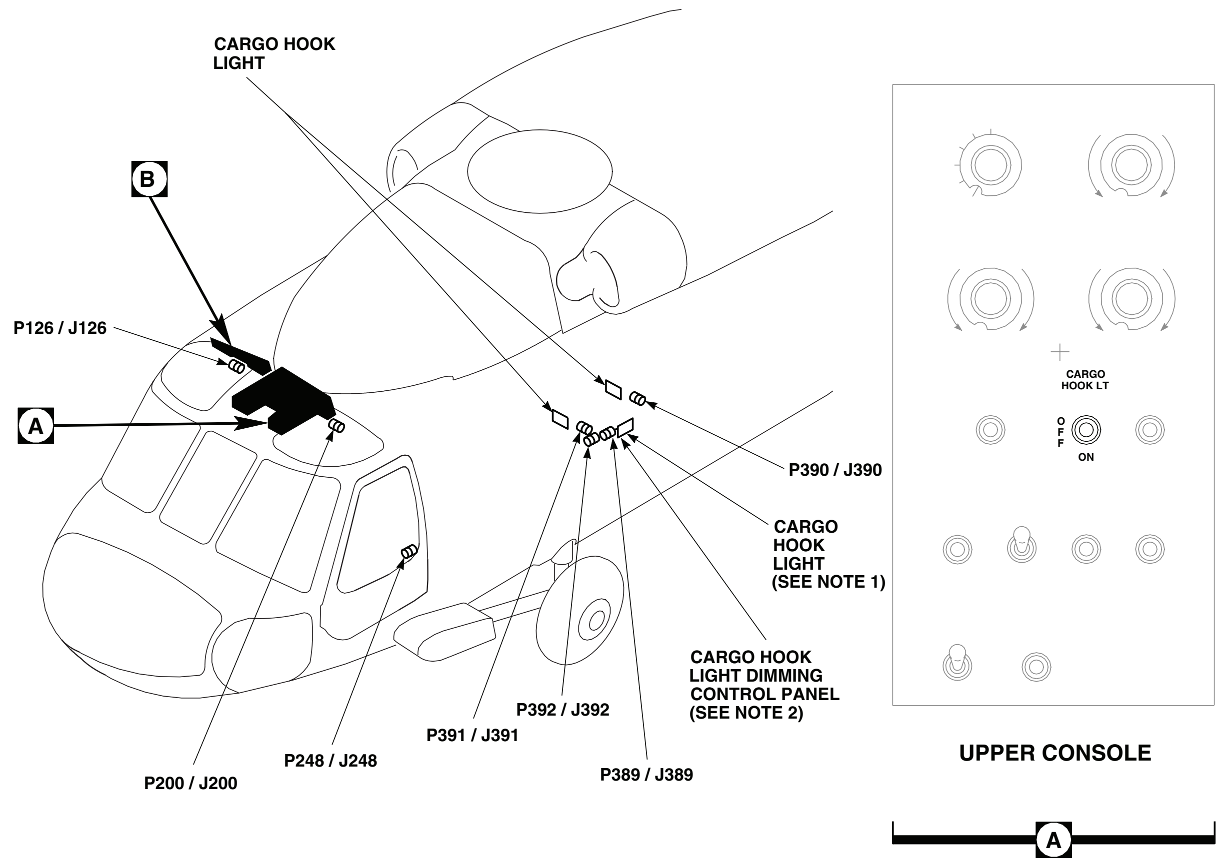
- **J390-1 and ground**
- **J391-1 and ground**

a. If voltage is as specified, replace cargo hook light (WP 0920 00). Go to **8.**

b. If voltage is not as specified, replace HOOK LIGHT DIMMING control panel (WP 0886 00). Go to **8.**

8. **Procedure completed.**

LOCATION



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P126 / J126	COCKPIT CEILING BL 28 RH, STA 243
P200 / J200	COCKPIT CEILING BL 13 LH, STA 246
P248 / J248	CABIN BL 40 LH, STA 248
P389 / J389	CARGO HOOK WELL BL 10 LH, STA 347
P390 / J390	CARGO HOOK WELL BL 5 LH, STA 363
P391 / J391	CARGO HOOK WELL BL 5 LH, STA 343
P392 / J392	CARGO HOOK WELL BL 5 LH, STA 343

NOTES

1. 77–22714 – 97–26743
2. MWO 50–56 HH–60A HH–60L UH–60L 97–26744 – SUBQ

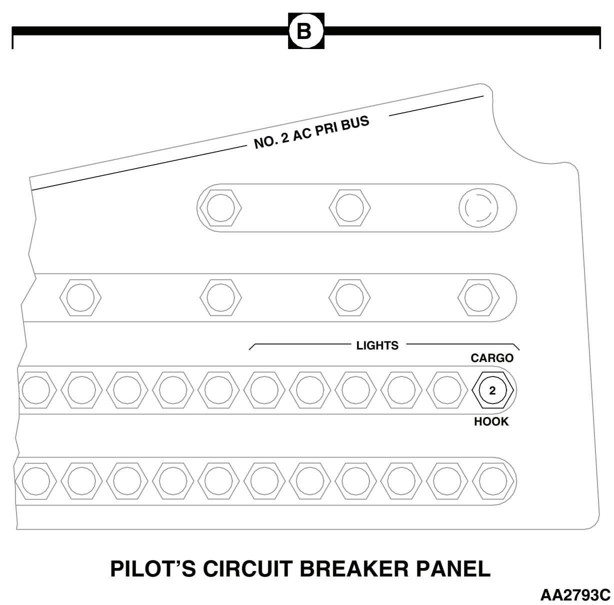
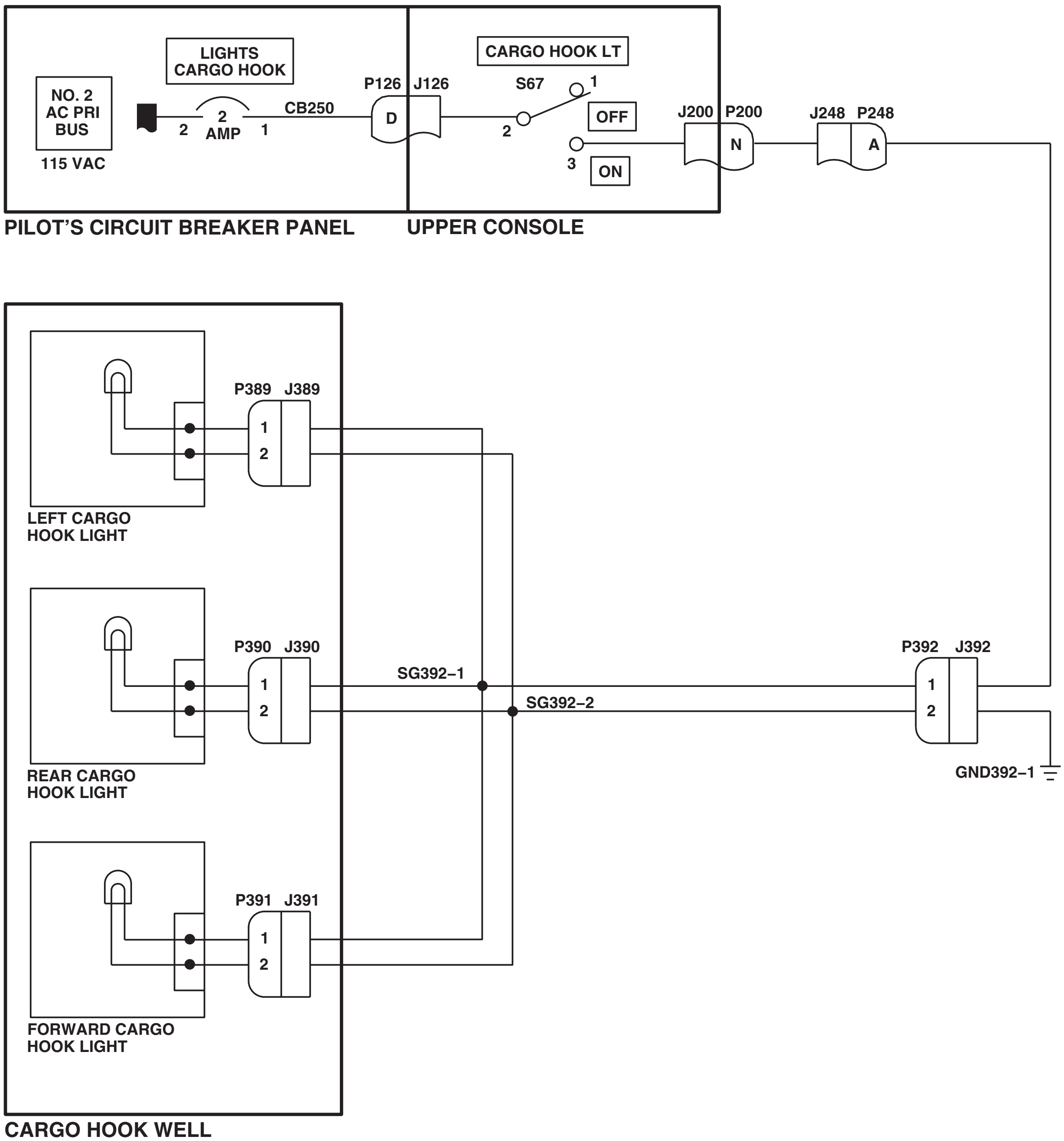


Figure 1. Cargo Hook Lights Location Diagram.

SCHEMATICS



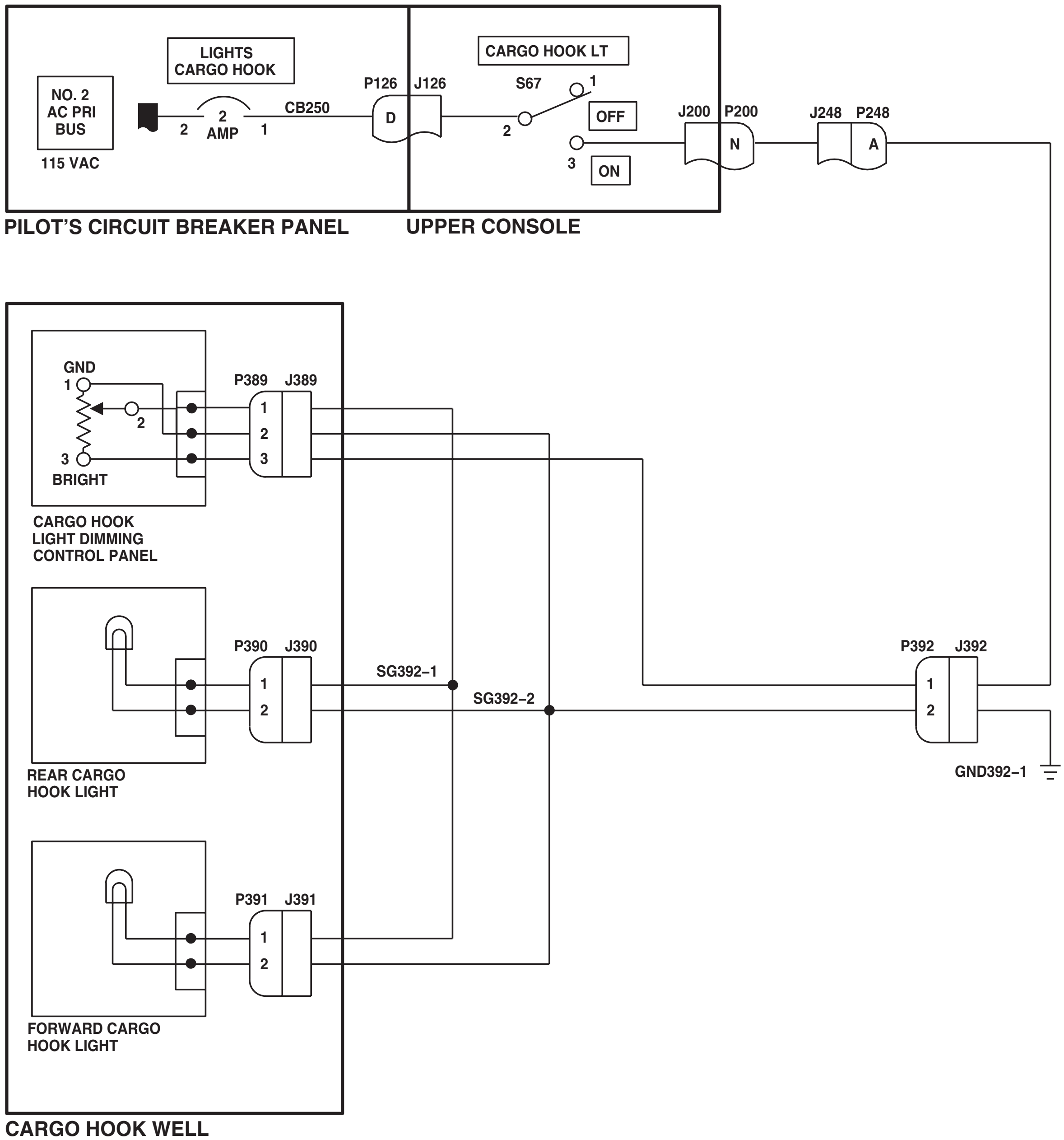
EFFECTIVITY

77–22714 – 97–26743

AB1457
SA

Figure 2. Cargo Hook Lights Schematic Diagram 77-22714-97-26723 .

SCHEMATICS - Continued



AB2329B
SA

Figure 3. Cargo Hook Lights Schematic Diagram MWO 50-56 HH-60A HH-60L UH-60L 97-26744-SUBQ .

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

ECM ANTENNA ACTUATOR ASSEMBLY EH60A

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Equipment Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)
Electrical Warfare/Intercept Aviation Systems Repairer MOS 33R (1)

References
TM 1-1520-237-10
TM 11-1520-237-23
TM 32-5865-012-10
WP 0090 00
WP 0104 00
WP 0785 00
WP 0824 00
WP 0925 00
WP 0926 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

A hover test of the ECM antenna actuator assembly and associated mission equipment is outlined in TM 32-5865-012-10. Fault isolation procedures are also provided in the manual. Faults isolated to the aircraft side of the mission interface panel, the ECM antenna switch, and the ECM antenna actuator assembly will be referred to aircraft maintenance personnel by mission equipment maintenance personnel. These faults can be isolated by this procedure with the aircraft on the ground. The procedure assumes that mission equipment is operating properly.

ECM ANTENNA ACTUATOR ASSEMBLY OPERATIONAL/TROUBLESHOOTING PROCEDURE EH60A

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Pull out Q/F EQUIP PWR circuit breaker (copilot’s circuit breaker panel).
2. Remove ECM antenna (TM 32-5865-012-10).
3. Turn on electrical power.
4. Hold caution/advisory panel BRT/DIM-TEST switch to TEST.

ECM ANTENNA ACTUATOR ASSEMBLY OPERATIONAL/TROUBLESHOOTING PROCEDURE EH60A - Continued

Indication/Condition

All capsules on caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot caution/advisory warning system (WP 0090 00).

Step

5. Push in Q/F EQUIP PWR circuit breaker (copilot's circuit breaker panel) and make sure RDR ALTM circuit breaker is pushed in.

Indication/Condition

Caution/advisory panel ANTENNA RETRACTED capsule shall go on.

Corrective Action

If Indication/Condition is not as specified, go to ANTENNA RETRACTED CAPSULE DOES NOT GO ON WHEN POWER IS APPLIED TO ECM ANTENNA SYSTEM, in this work package.

Step

6. Turn copilot's radar altimeter LO SET knob clockwise until the LO SET index is past the zero marking on the dial. Wait approximately one minute for altimeter system to stabilize.
7. Turn copilot's radar altimeter LO SET knob counterclockwise until LO warning lamp on indicator dial goes off, but OFF flag remains out of view.

Indication/Condition

Altimeter shall be on and LO SET index shall be below altitude indicated.

Corrective Action

If Indication/Condition is not as specified, go to step 17.

Step

8. Turn on ALQ-151(V)2 (TM 32-5865-012-10).

CAUTION

Damage to ECM antenna will occur if ECM ANTENNA switch is placed to EXTEND with antenna installed. Make sure ECM antenna is removed prior to ground maintenance.

9. Momentarily place instrument panel ECM ANTENNA switch to EXTEND position.

Indication/Condition

a. ECM antenna actuator assembly clevis shall pivot from fully retracted to fully extended position in approximately 20 seconds.

b. ANTENNA RETRACTED capsule shall go off.

c. ANTENNA EXTENDED capsule shall stay off.

d. ECM operator console ANTENNA EXTENDED indicator lamps shall go on.

ECM ANTENNA ACTUATOR ASSEMBLY OPERATIONAL/TROUBLESHOOTING PROCEDURE EH60A - Continued

Corrective Action

1. If ECM antenna actuator assembly clevis starts to pivot, but stops when ECM ANTENNA switch is released to OFF position, and ANTENNA EXTENDED capsule goes on, pull out RDR ALTM circuit breaker (copilot's circuit breaker panel) and check continuity between J970R-N and P674R-5.
   a. If continuity is present, replace radar altimeter (TM 11-1520-237-23).
   b. If continuity is not present, repair/replace wiring (WP 1747 00).
2. If Indication/Condition a. is not as specified, go to ECM ANTENNA ACTUATOR ASSEMBLY CLEVIS DOES NOT PIVOT FROM RETRACTED TO EXTENDED POSITION, in this work package.
3. If Indication/Condition b. is not as specified, replace ECM antenna actuator assembly (WP 0926 00).

Step

10. Momentarily place instrument panel ECM ANTENNA switch to RETRACT position.

Indication/Condition

a. ECM antenna actuator assembly clevis shall pivot from fully extended position to fully retracted position in approximately 16 seconds.

b. ANTENNA RETRACTED capsule on caution/advisory panel shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to ECM ANTENNA ACTUATOR ASSEMBLY CLEVIS DOES NOT PIVOT FROM EXTENDED TO RETRACTED POSITION, in this work package.
2. If Indication/Condition b. is not as specified, replace ECM antenna actuator assembly (WP 0926 00).

Step

CAUTION

Damage to ECM antenna will occur if ECM ANTENNA switch is placed to EXTEND with antenna installed. Make sure ECM antenna is removed prior to ground maintenance.

11. Momentarily place instrument panel ECM ANTENNA switch in EXTEND position.

Indication/Condition

a. ECM antenna actuator assembly clevis shall pivot from fully retracted to fully extended position in approximately 20 seconds.

b. ANTENNA RETRACTED capsule shall go off.

c. ANTENNA EXTENDED capsule shall stay off.

Corrective Action

If Indication/Conditions are not as specified, go to step 9.

Step

12. Turn copilot's radar altimeter LO SET knob clockwise until the LO SET index moves above the indicated altitude.

Indication/Condition

a. Copilot's radar altimeter LO warning lamp shall go on.

ECM ANTENNA ACTUATOR ASSEMBLY OPERATIONAL/TROUBLESHOOTING PROCEDURE EH60A - Continued

b. An automatic retract operation shall be initiated. The ECM antenna actuator assembly clevis shall pivot from fully extended position to fully retracted position in approximately 16 seconds.

c. ANTENNA EXTENDED capsule shall go on at beginning of retract operation.

d. ANTENNA EXTENDED capsule shall go off when fully retracted position is reached.

e. ANTENNA RETRACTED capsule shall go on when fully retracted position is reached.

Corrective Action

1. If Indication/Conditions b. and c. are not as specified, go to ECM ANTENNA ACTUATOR ASSEMBLY CLEVIS DOES NOT PIVOT FROM EXTENDED TO RETRACTED POSITION, in this work package.
2. If Indication/Condition d. or e. is not as specified, replace ECM antenna actuator assembly (WP 0926 00).

Step

13. Turn copilot's radar altimeter LO SET knob counterclockwise until the LO warning lamp goes off and the OFF flag comes into view.
14. Pull out RDR ALTM and Q/F EQUIPMENT PWR circuit breakers (copilot's circuit breaker panel).

NOTE

Operation of the radar altimeter, while aircraft is on the ground, may be erratic. The following steps test the ECM antenna actuator assembly without altimeter inputs.

15. Turn copilot's radar altimeter LO SET knob counterclockwise until OFF flag appears.
16. Pull out RDR ALTM circuit breaker (copilot's circuit breaker panel).

CAUTION

Damage to ECM antenna will occur if ECM ANTENNA switch is placed to EXTEND with antenna installed. Make sure ECM antenna is removed prior to ground maintenance.

17. Press and hold instrument panel ECM ANTENNA switch to EXTEND position.

Indication/Condition

a. ECM antenna actuator assembly clevis shall pivot from fully retracted position to fully extended position in approximately 20 seconds.

b. ANTENNA EXTENDED capsule on caution/advisory panel shall go on.

c. ANTENNA EXTENDED capsule on ECM operator console shall go on.

d. ANTENNA RETRACTED capsule on caution/advisory panel shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, pull out RDR ALTM circuit breaker (copilot's circuit breaker panel) and check continuity between J970R-V and P675R-1.
   a. If continuity is present, replace radar altimeter (TM 11-1520-237-23).
   b. If continuity is not present, repair/replace wiring (WP 1747 00).
2. If Indication/Condition b. is not as specified, go to ANTENNA EXTENDED CAPSULE DOES NOT GO ON WHEN ECM ANTENNA CLEVIS PIVOTS FROM RETRACTED TO EXTENDED POSITION, in this work package.

ECM ANTENNA ACTUATOR ASSEMBLY OPERATIONAL/TROUBLESHOOTING PROCEDURE EH60A - Continued

3. If Indication/Condition c. or d. is not as specified, replace ECM antenna actuator assembly (WP 0926 00).

Step

18. Release instrument panel ECM ANTENNA switch.
19. Momentarily place instrument panel ECM ANTENNA switch to RETRACT position.

Indication/Condition

a. ECM antenna actuator assembly clevis shall pivot from fully extended position to fully retracted position in approximately 16 seconds.

b. ANTENNA RETRACTED capsule on caution/advisory panel shall go on.

c. ANTENNA EXTENDED capsule shall go off.

Corrective Action

If Indication/Conditions are not as specified, replace ECM antenna actuator assembly (WP 0926 00).

Step

20. Pull out Q/F EQUIP PWR circuit breaker (copilot's circuit breaker panel).
21. Turn off electrical power.
22. Install ECM antenna (TM 32-5865-012-10).

Indication/Condition

ECM antenna shall be installed.

Corrective Action

None Required

ANTENNA RETRACTED CAPSULE DOES NOT GO ON WHEN POWER IS APPLIED TO ECM ANTENNA SYSTEM

SYMPTOM

ANTENNA RETRACTED capsule does not go on when power is applied to ECM antenna system.

MALFUNCTION

ANTENNA RETRACTED capsule does not go on when power is applied to ECM antenna system.

CORRECTIVE ACTION

1. Check for 28 vdc between J510-A and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **7.**

2. Check for 28 vdc between J2-S and ground.

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, replace ECM antenna actuator assembly (WP 0926 00). Go to **8.**

3. Check helicopter configuration.

a. EMEP , go to **4.**

ANTENNA RETRACTED CAPSULE DOES NOT GO ON WHEN POWER IS APPLIED TO ECM ANTENNA SYSTEM - Continued

b. **W/O EMEP** , go to **6.**

4. EMEP Remove and check pin filtered adapter P118 (WP 0925 00).

a. If pin filtered adapter is good, go to **5.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **8.**

5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.

6. Check continuity between J970R-S and P118-Y.

a. If continuity is present, replace caution/advisory panel (WP 0785 00). Go to **8.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

7. Check for 28 vdc between Q/F EQUIP PWR circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J510-A (WP 1747 00). Go to **8.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8.**

8. Procedure completed.

ECM ANTENNA ACTUATOR ASSEMBLY CLEVIS DOES NOT PIVOT FROM RETRACTED TO EXTENDED POSITION

SYMPTOM

ECM antenna actuator assembly clevis does not pivot from retracted to extended position.

MALFUNCTION

ECM antenna actuator assembly clevis does not pivot from retracted to extended position.

CORRECTIVE ACTION

1. Place ECM ANTENNA switch in the EXTEND position. Go to 2.

2. Check continuity between J970R-R and J970R-T.

a. If continuity is present, replace ECM antenna actuator assembly (WP 0926 00). Go to **4.**

b. If continuity is not present, go to **3.**

3. Check continuity between J970R-R and ECM ANTENNA switch terminal 4.

a. If continuity is present, replace ECM ANTENNA switch S97 (WP 0926 00). Go to **4.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**

4. Procedure completed.

ECM ANTENNA ACTUATOR ASSEMBLY CLEVIS DOES NOT PIVOT FROM EXTENDED TO RETRACTED POSITION

SYMPTOM

ECM antenna actuator assembly clevis does not pivot from extended to retracted position.

MALFUNCTION

ECM antenna actuator assembly clevis does not pivot from extended to retracted position.

CORRECTIVE ACTION

1. **Place ECM ANTENNA switch in the RETRACT position. Go to 2.**
2. **Check continuity between J970R-T and J970R-P.**
   a. If continuity is present, replace ECM antenna actuator assembly (WP 0926 00). Go to **4.**
   b. If continuity is not present, go to **3.**
3. **Check continuity between J970R-P and ECM ANTENNA switch terminal 6.**
   a. If continuity is present, replace ECM ANTENNA switch S97 (WP 0926 00). Go to **4.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**
4. **Procedure completed.**

ANTENNA EXTENDED CAPSULE DOES NOT GO ON WHEN ECM ANTENNA CLEVIS PIVOTS FROM RETRACTED TO EXTENDED POSITION

SYMPTOM

ANTENNA EXTENDED capsule does not go on when ECM antenna clevis pivots from retracted to extended position.

MALFUNCTION

ANTENNA EXTENDED capsule does not go on when ECM antenna clevis pivots from retracted to extended position.

CORRECTIVE ACTION

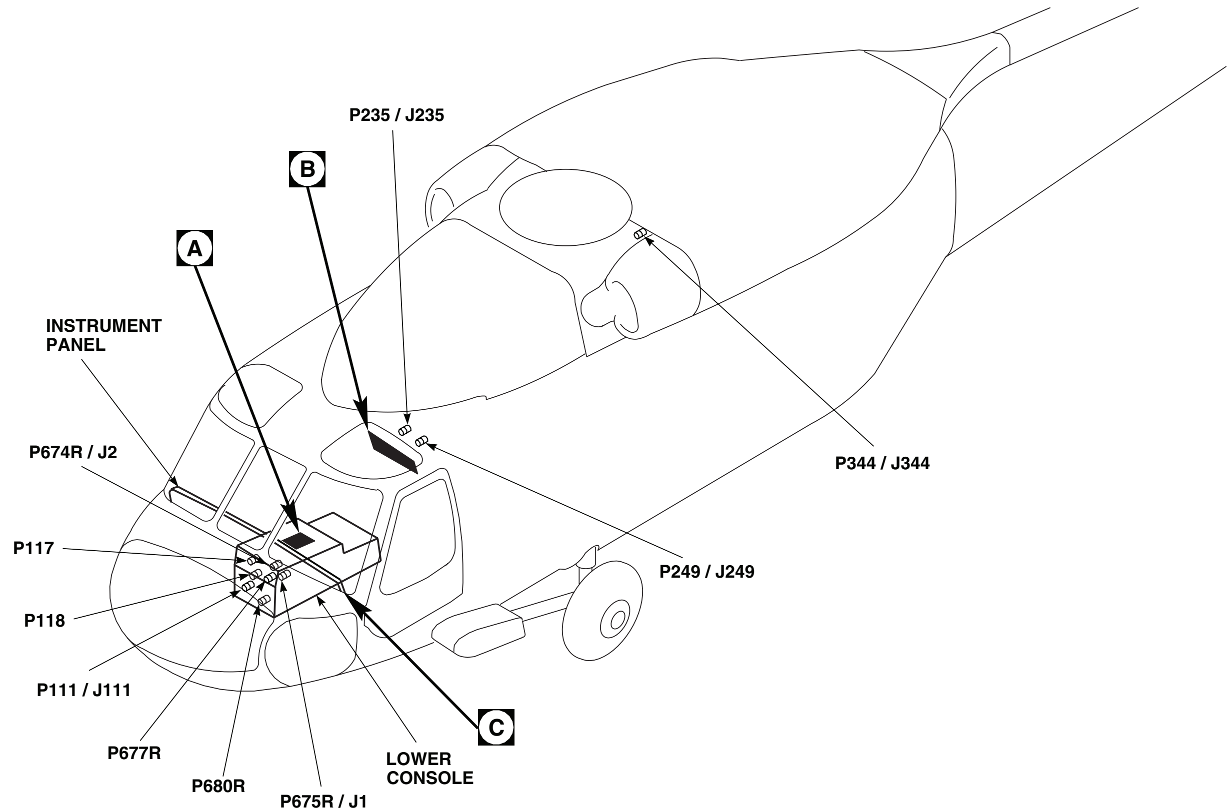
1. **Check for 28 vdc between P2-U and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, replace ECM antenna actuator assembly (WP 0926 00). Go to **6.**
2. **Check helicopter configuration.**
   a. **EMEP** , go to **3.**
   b. **W/O EMEP** , go to **5.**
3. **EMEP Remove and check pin filtered adapter P117 (WP 0925 00).**
   a. If pin filtered adapter is good, go to **4.**
   b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **6.**
4. **EMEP Install pin filtered adapter (WP 0925 00). Go to 5.**
5. **Check continuity between J970R-U and P117-38.**
   a. If continuity is present, replace caution/advisory panel (WP 0785 00). Go to **6.**

ANTENNA EXTENDED CAPSULE DOES NOT GO ON WHEN ECM ANTENNA CLEVIS PIVOTS FROM RETRACTED TO EXTENDED POSITION - Continued

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**

6. Procedure completed.

LOCATION



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P1 / J2	ECM ANTENNA RELAY ASSEMBLY
P1 / J2	ECM CONNECTOR PANEL
P1 / J3	ECM CONNECTOR PANEL
P2 / J1	ECM ANTENNA RELAY ASSEMBLY
P2 / J510	MISSION INTERFACE PANEL
P2 / J970R	MISSION INTERFACE PANEL
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P117	CAUTION / ADVISORY PANEL
P118	CAUTION / ADVISORY PANEL

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P235 / J235	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 20 LH, STA 247
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P344 / J344	CABIN CEILING, BL 3.5 RH, STA 379
P674R / J2	COPILOT'S RADAR ALTIMETER
P675R / J1	COPILOT'S RADAR ALTIMETER
W1J1 / W1	ECM ANTENNA ACTUATOR ASSEMBLY
W1P1 / A1J1	ECM ANTENNA ASSEMBLY LINEAR ACTUATOR

AA6089_1
SA

Figure 1. ECM Antenna Actuator Assembly Location Diagram. (Sheet 1 of 7)

LOCATION - Continued

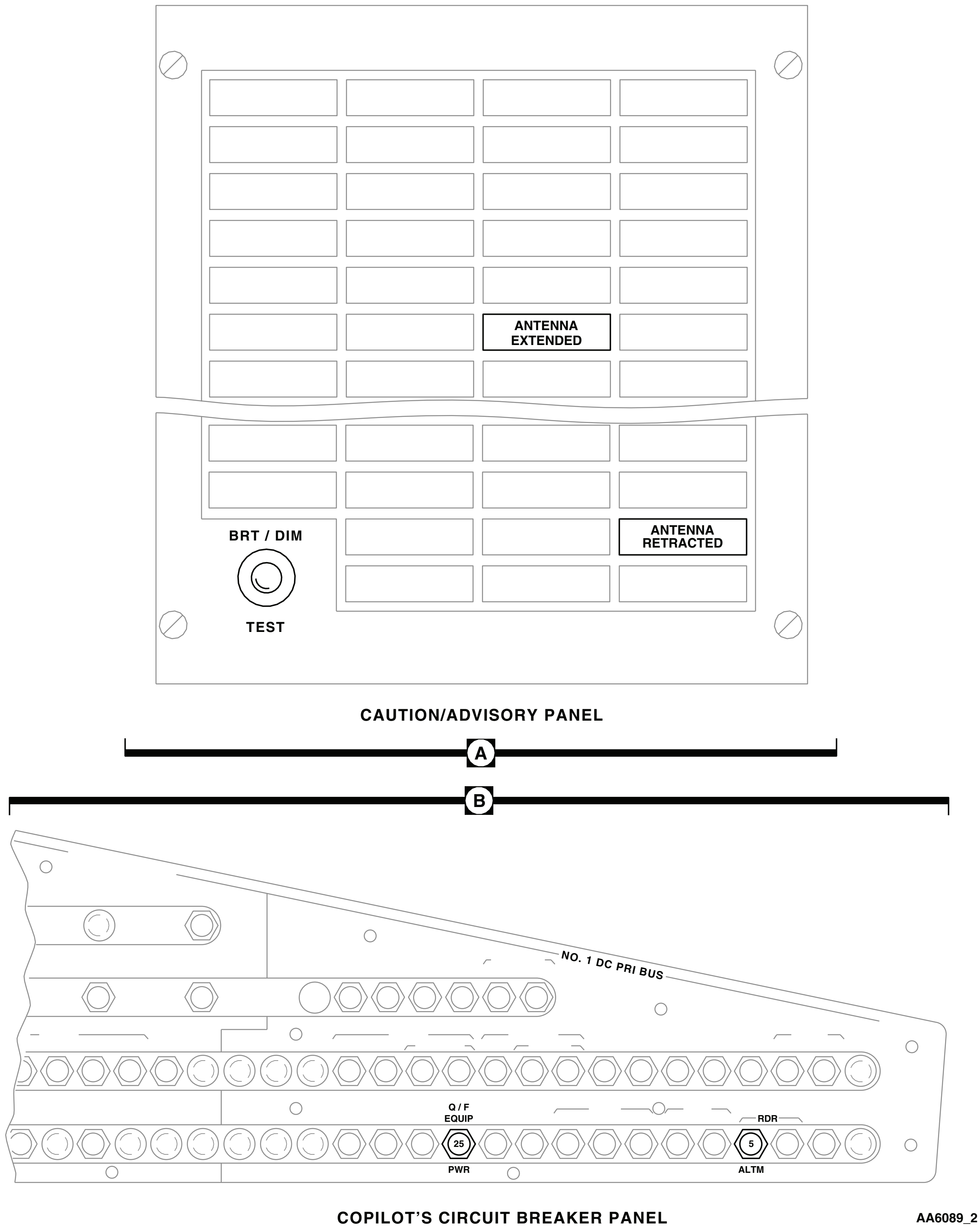


Figure 1. ECM Antenna Actuator Assembly Location Diagram. (Sheet 2 of 7)

LOCATION - Continued

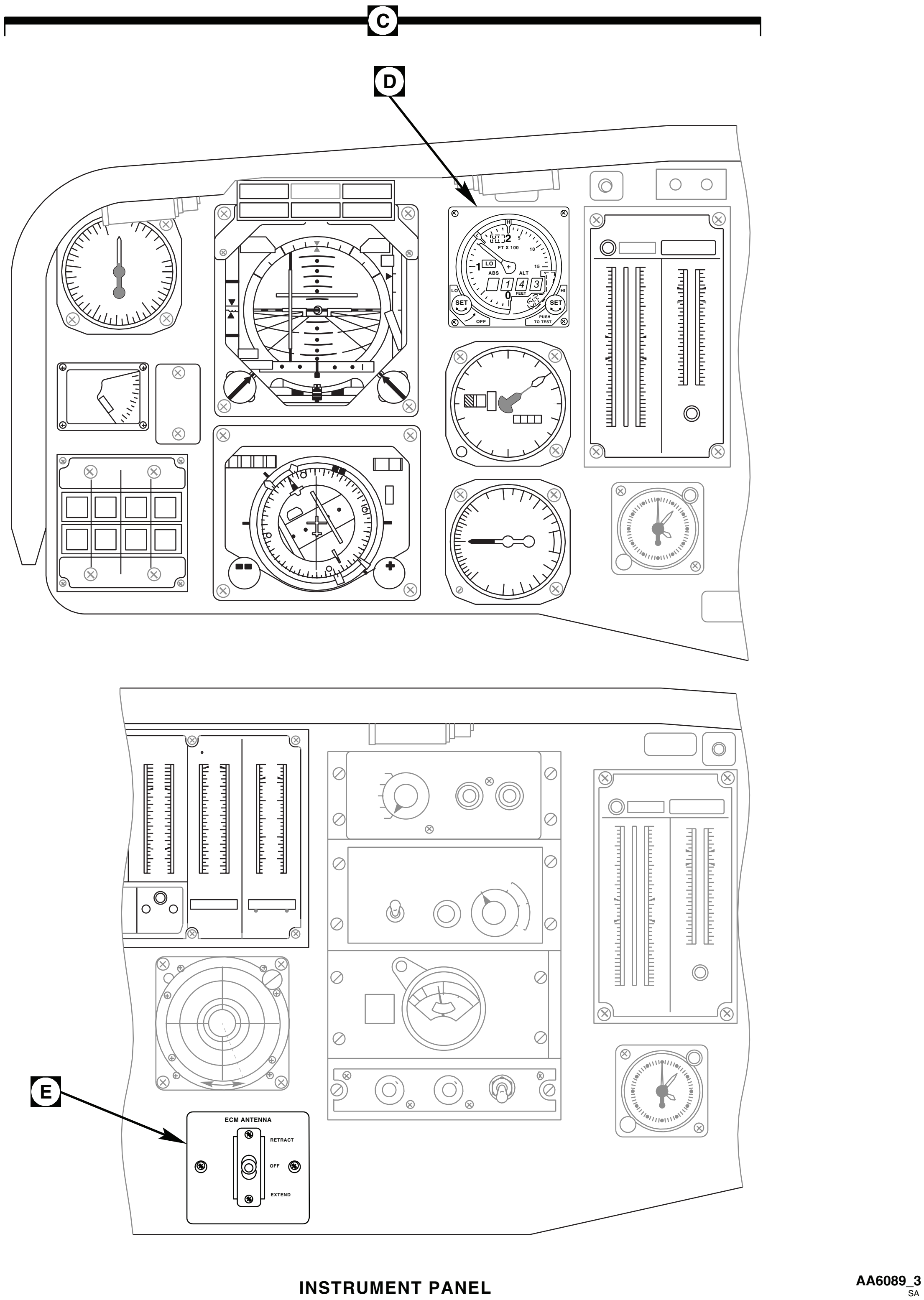
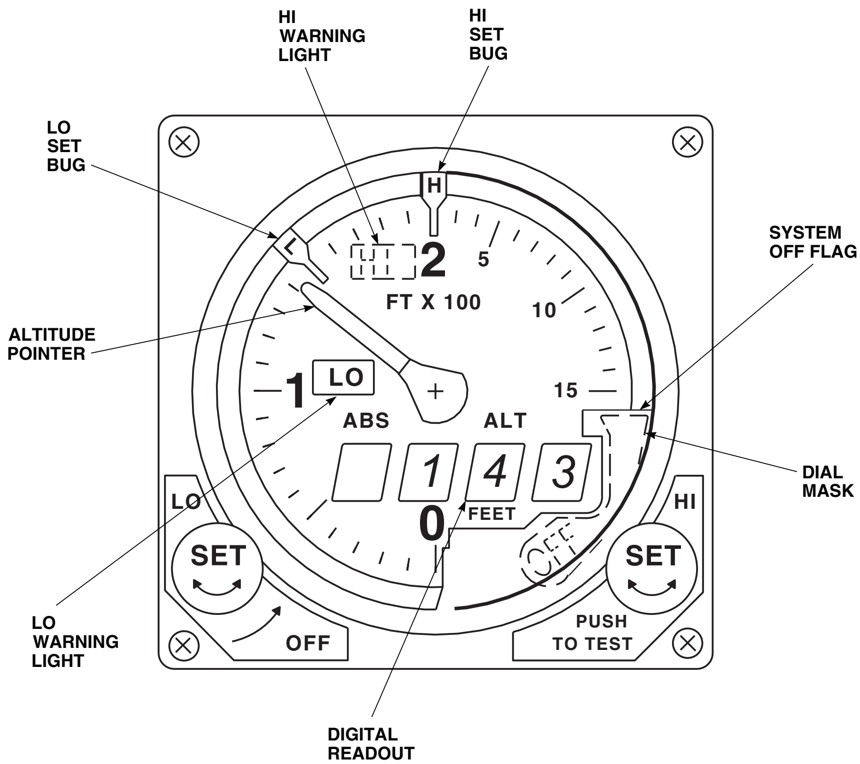


Figure 1. ECM Antenna Actuator Assembly Location Diagram. (Sheet 3 of 7)

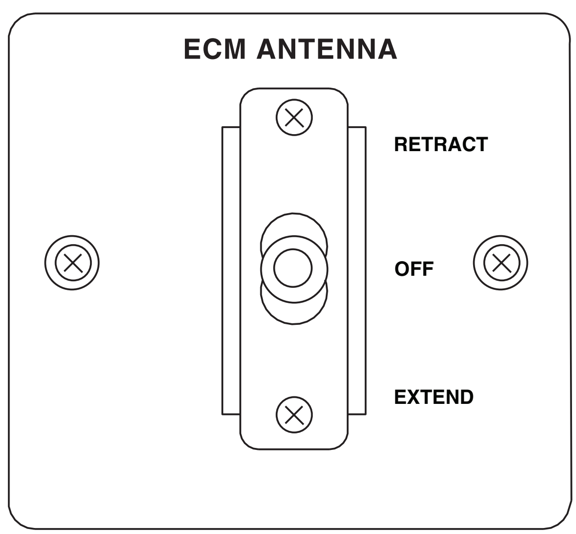
LOCATION - Continued



COPILOT'S RADAR ALTIMETER

D

E



ECM ANTENNA SWITCH

AA6089_4
SA

Figure 1. ECM Antenna Actuator Assembly Location Diagram. (Sheet 4 of 7)

LOCATION - Continued

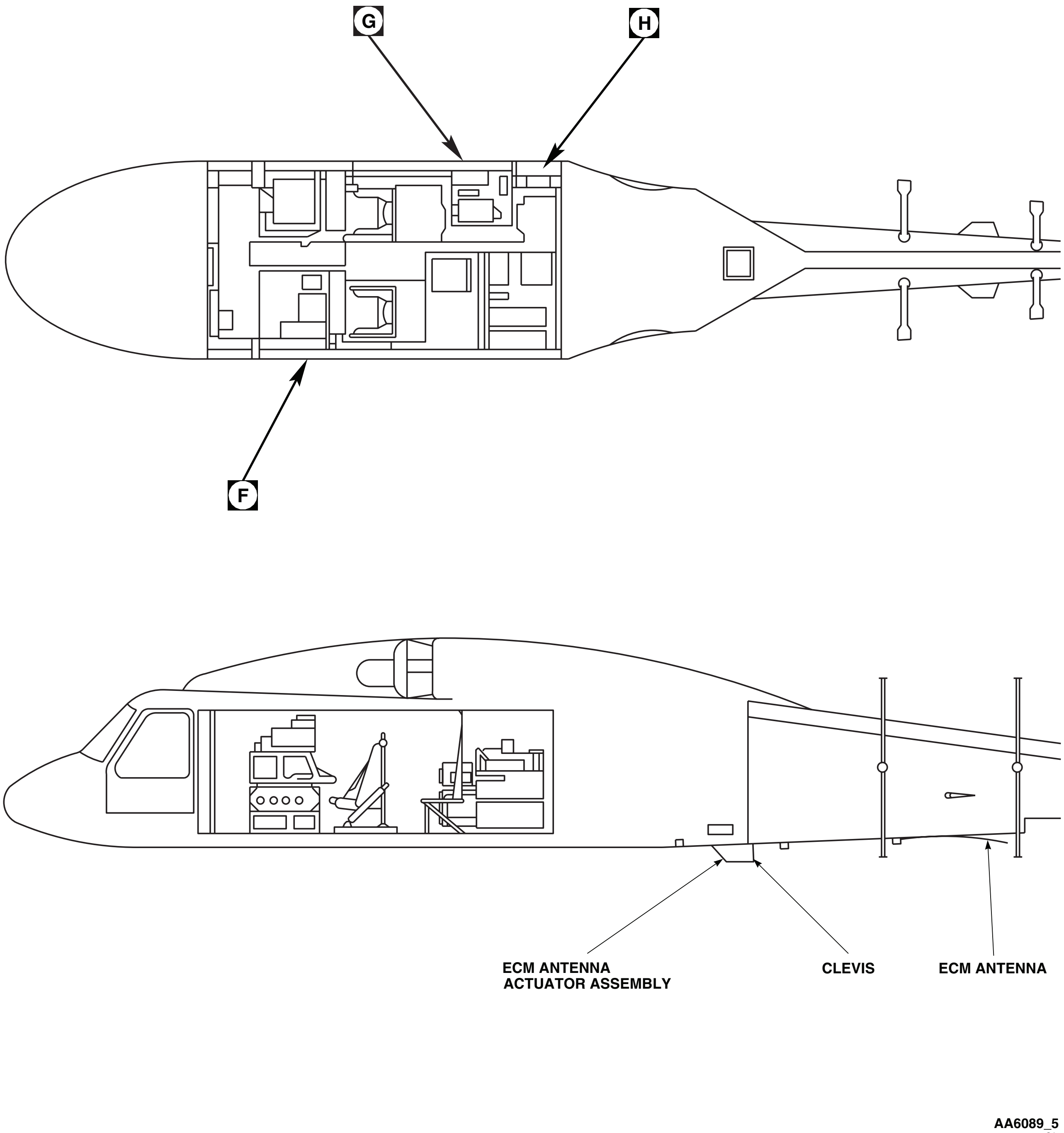


Figure 1. ECM Antenna Actuator Assembly Location Diagram. (Sheet 5 of 7)

LOCATION - Continued

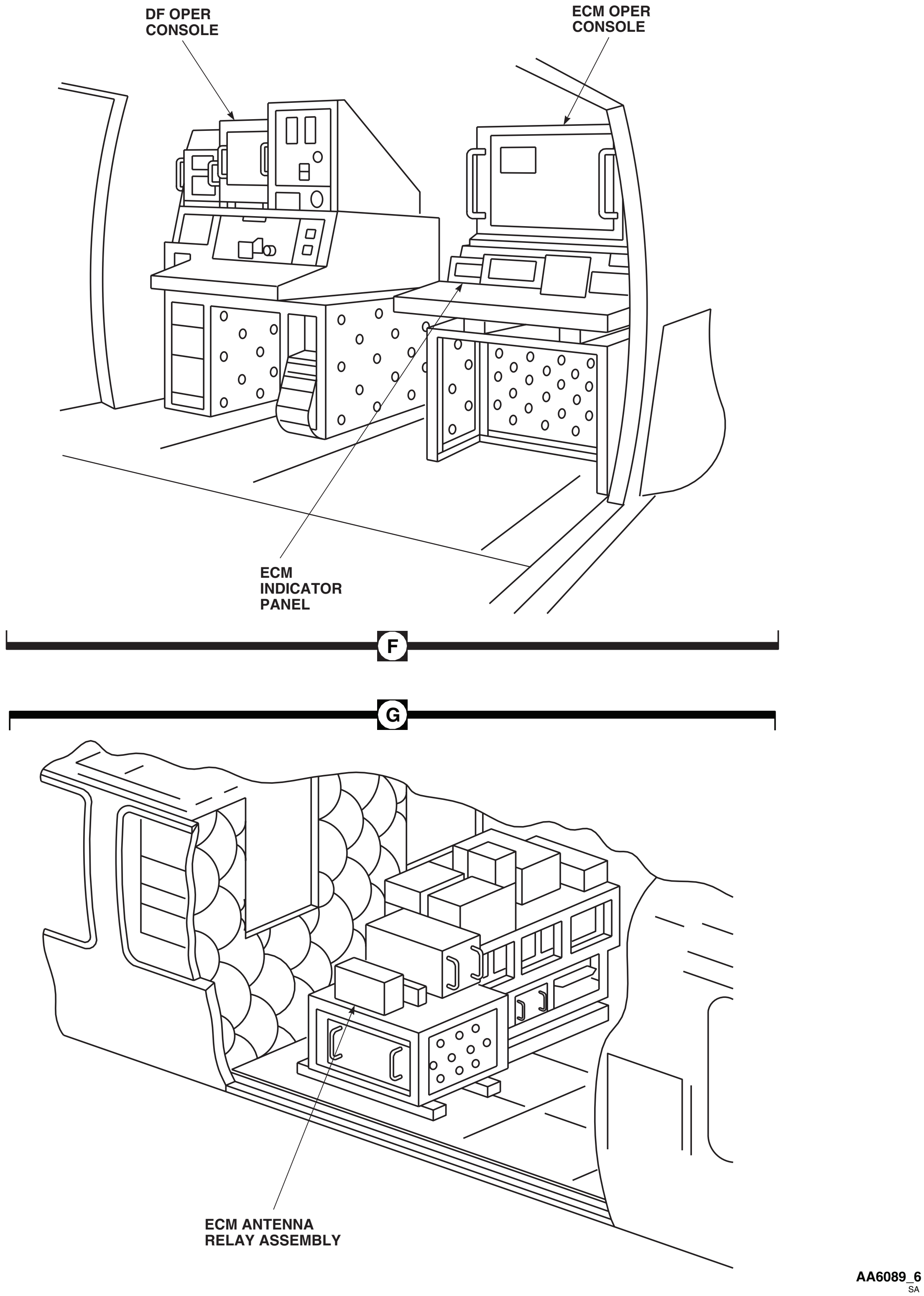


Figure 1. ECM Antenna Actuator Assembly Location Diagram. (Sheet 6 of 7)

LOCATION - Continued

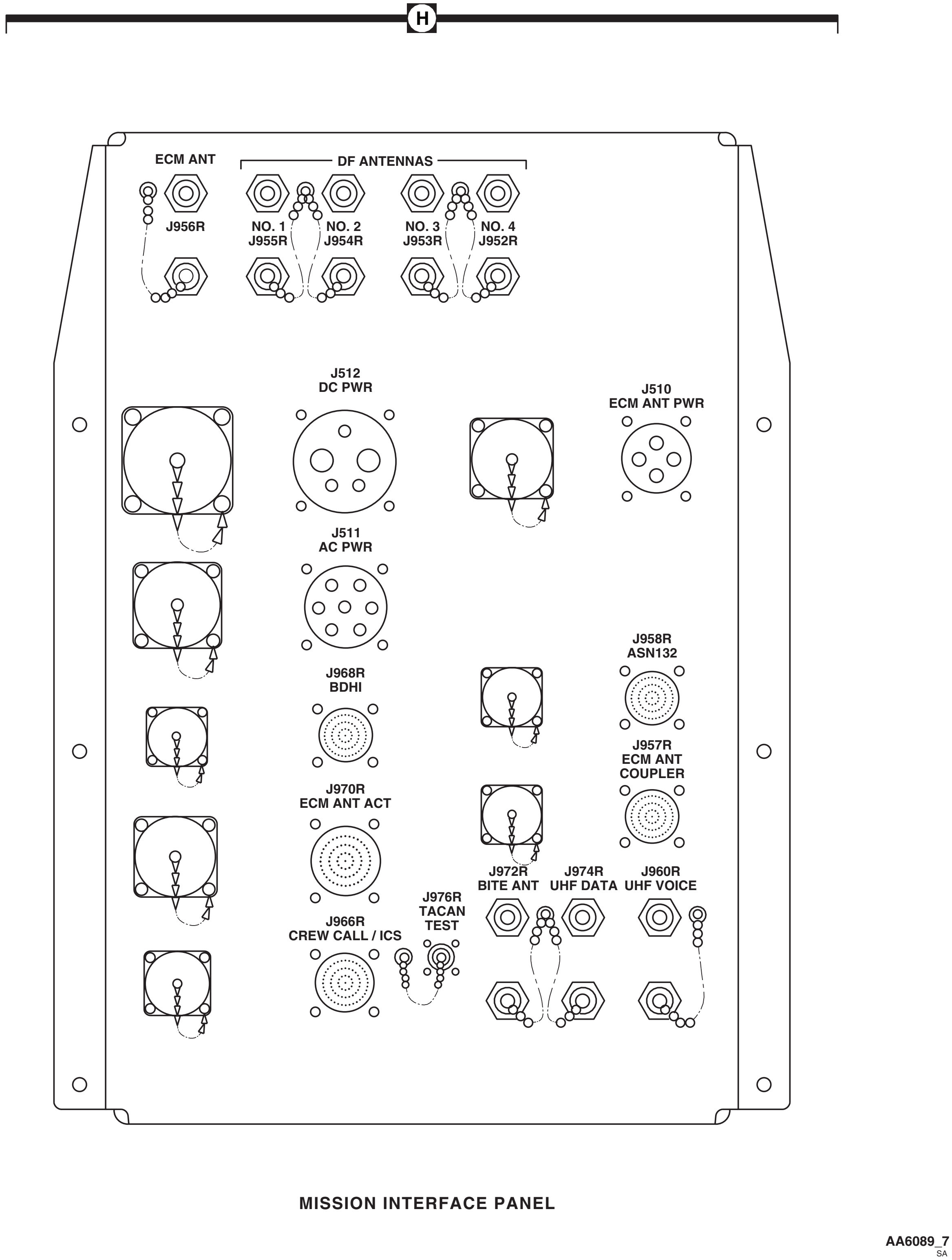
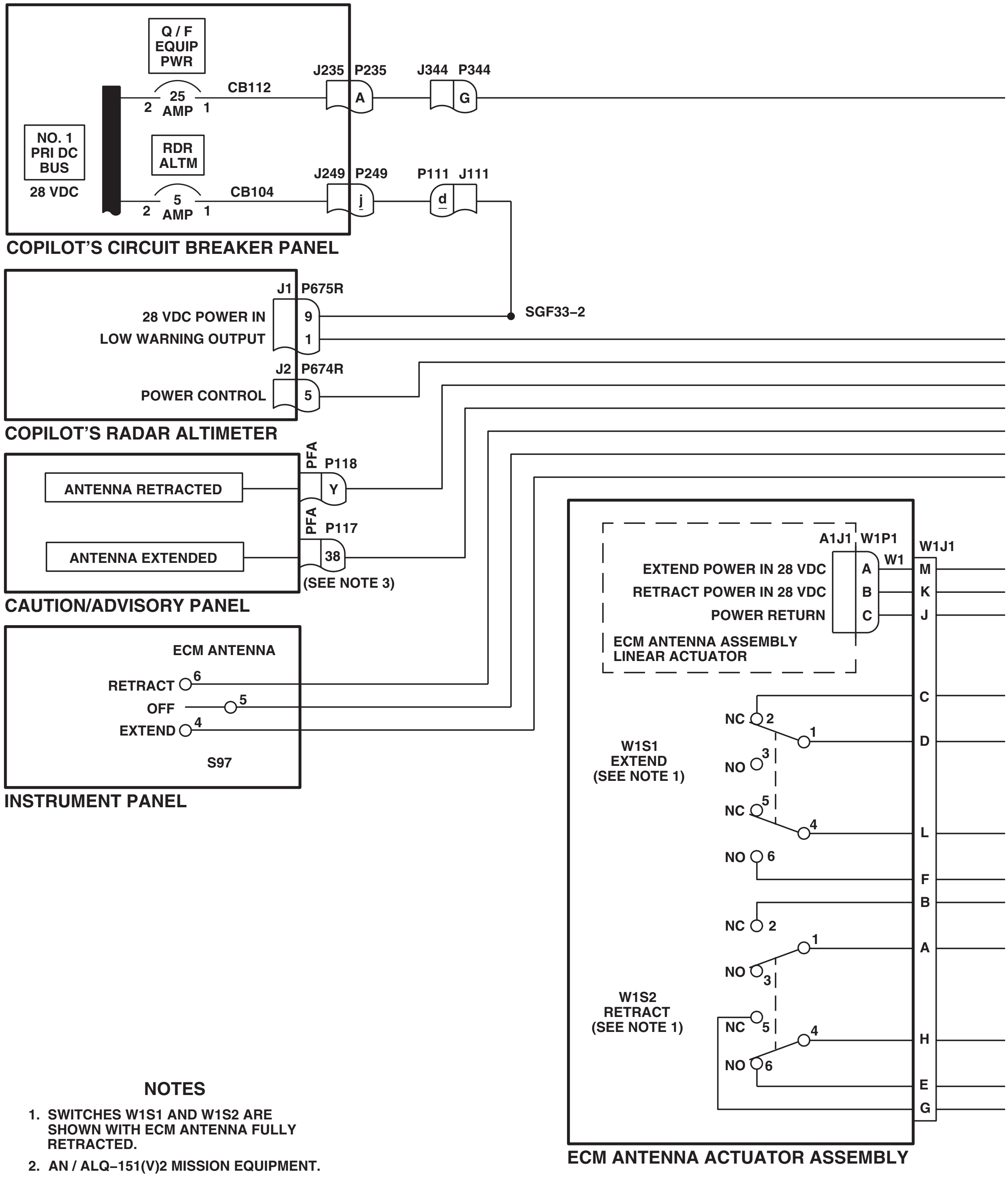


Figure 1. ECM Antenna Actuator Assembly Location Diagram. (Sheet 7 of 7)

SCHEMATICS



NOTES

1. SWITCHES W1S1 AND W1S2 ARE SHOWN WITH ECM ANTENNA FULLY RETRACTED.
2. AN / ALQ–151(V)2 MISSION EQUIPMENT.
3. W/O EMEP PIN FILTERED ADAPERS (PFA) ARE NOT INSTALLED.

AB1534_1
SA

Figure 2. ECM Antenna Actuator Assembly Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

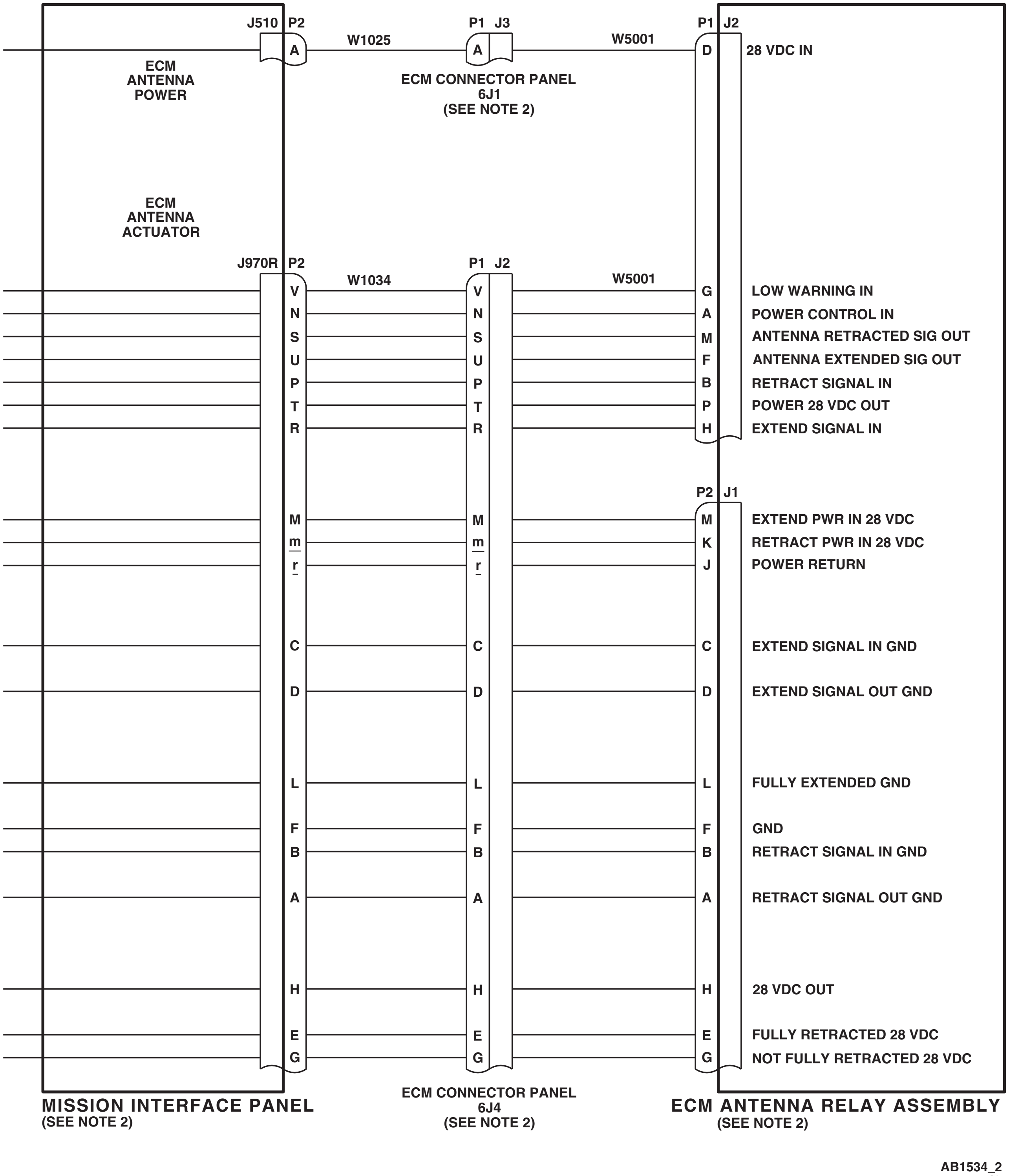


Figure 2. ECM Antenna Actuator Assembly Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

FUEL BOOST PUMP CONTROL PANEL (AVIM)

INITIAL SETUP:

Test Equipment
Multimeter, AN-PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06
Locally-Made Boost Pump Control Panel, Test Harness, Figure 153, WP 1805 00

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0867 00
WP 0868 00
WP 0953 00
WP 0954 00
WP 1747 00
WP 1805 00

Equipment Condition
Between 27 and 28 vdc, and 115 vac Bench Power Available

FUEL BOOST PUMP CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

See Figure 1 for component diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Remove cover from control panel and inspect for broken or burnt wires, security of components, and general condition.

Indication/Condition

Wires shall not be broken or burnt, and components shall be secure and in good condition.

Corrective Action

If Indication/Condition is not as specified, repair/replace as required (WP 1747 00).

Step

2. Connect locally-made boost pump control panel, test harness, P1 to fuel boost pump control panel J1.
3. With control panel NO. 1 PUMP switch placed ON, check continuity between the following jacks on test harness (WP 1805 00):
   - T and J
   - U and H
   - V and G

FUEL BOOST PUMP CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Meter shall indicate continuity.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between the following NO. 1 PUMP switch terminals:
   - 4 and 5
   - 7 and 8
   - 10 and 11

   a. If continuity is not present, replace NO. 1 PUMP switch (WP 0953 00).

   b. If continuity is present, repair/replace wiring (WP 1747 00).

Step

4. Place NO. 1 PUMP switch to OFF.
5. Check continuity between the following jacks on test harness:
   - T and J
   - U and H
   - V and G

Indication/Condition

Meter shall indicate no continuity.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between the following NO. 1 PUMP switch terminals:
   - 4 and 5
   - 7 and 8
   - 10 and 11

   a. If continuity is present, replace NO. 1 PUMP switch (WP 0953 00).

   b. If continuity is not present, repair/replace wiring (WP 1747 00).

Step

6. With control panel NO. 2 PUMP switch placed ON, check continuity between the following jacks on test harness:
   - A and F
   - B and E
   - C and D

Indication/Condition

Meter shall indicate continuity.

FUEL BOOST PUMP CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition is not as specified, check continuity between the following NO. 2 PUMP switch terminals:
   - 1 and 2
   - 4 and 5
   - 7 and 8

   a. If continuity is not present, replace NO. 2 PUMP switch (WP 0953 00).

   b. If continuity is present, repair/replace wiring (WP 1747 00).

Step

7. Place NO. 2 PUMP switch to OFF.
8. Check continuity between the following jacks on test harness:
   - A and F
   - B and E
   - C and D

Indication/Condition

Meter shall indicate no continuity.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between the following NO. 2 PUMP switch terminals:
   - 1 and 2
   - 4 and 5
   - 7 and 8

   a. If continuity is present, replace NO. 2 PUMP switch (WP 0953 00).

   b. If continuity is not present, repair/replace wiring (WP 1747 00).

Step

9. Disconnect meter.

CAUTION

To prevent possible damage to fuel boost pump control panel, verify if fuel boost pump control panel information plate uses 5 vac or 115 vac.

10. Apply 5V/115 vac across jacks L and Y on test harness.

Indication/Condition

Information plate legend shall light evenly.

FUEL BOOST PUMP CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition is not as specified, check information plate lamps and replace as required (WP 0867 00).
   a. If trouble remains, check continuity between J1-L and J2-1, and J1-Y and J2-2.
      (1) If continuity is not present, repair/replace wiring (WP 1747 00).
      (2) If continuity is present, replace information plate (WP 0867 00).

Step

11. Disconnect 5V/115 vac from test harness.
12. Apply 28 vdc between the following jacks on test harness:
    - P and Z
    - R and Z
13. On control panel, momentarily press NO. 1 PUMP PRESS TO TEST indicator.

Indication/Condition

NO. 1 PUMP indicator shall light when pressed.

Corrective Action

1. If Indication/Condition is not as specified, check indicator lamp and replace it if burnt-out (WP 0868 00).
   a. If trouble remains, check continuity between J1-R and NO. 1 PUMP indicator terminal 1, and J1-Z and NO. 1 PUMP indicator terminal 3.
      (1) If continuity is not present, repair/replace wiring (WP 1747 00).
      (2) If continuity is present, replace NO. 1 PUMP indicator (WP 0954 00).

Step

14. On control panel, momentarily press NO. 2 PUMP PRESS TO TEST indicator.

Indication/Condition

NO. 2 PUMP indicator shall light when pressed.

Corrective Action

1. If Indication/Condition is not as specified, check indicator lamp and replace it if burnt-out (WP 0954 00).
   a. If trouble remains, check continuity between J1-P and NO. 2 PUMP indicator terminal 1, and J1-Z and NO. 2 PUMP indicator terminal 3.
      (1) If continuity is not present, repair/replace wiring (WP 1747 00).
      (2) If continuity is present, replace NO. 2 PUMP indicator (WP 0954 00).

Step

15. Connect negative lead of 28 vdc power source to M test jack.

Indication/Condition

NO. 1 PUMP indicator shall light.

FUEL BOOST PUMP CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition is not as specified, check continuity between J1-M and NO. 1 PUMP indicator terminal 2.
   a. If continuity is not present, repair/replace wiring (WP 1747 00).
   b. If continuity is present, replace NO. 1 PUMP indicator (WP 0954 00).

Step

16. Connect negative lead of 28 vdc power source to N test jack.

Indication/Condition

NO. 2 PUMP indicator shall light.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between J1-N and NO. 2 PUMP indicator terminal 2.
   a. If continuity is not present, repair/replace wiring (WP 1747 00).
   b. If continuity is present, replace NO. 2 PUMP indicator (WP 0954 00).

Step

17. Disconnect 28 vdc from test harness.
18. Disconnect test harness from control panel.

Indication/Condition

Test harness from control panel shall be disconnected.

Corrective Action

None Required

LOCATION

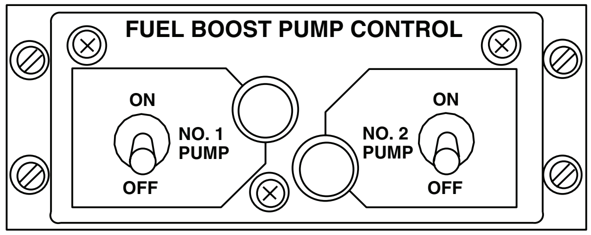


Figure 1. Fuel Boost Pump Control Panel.

SCHEMATICS

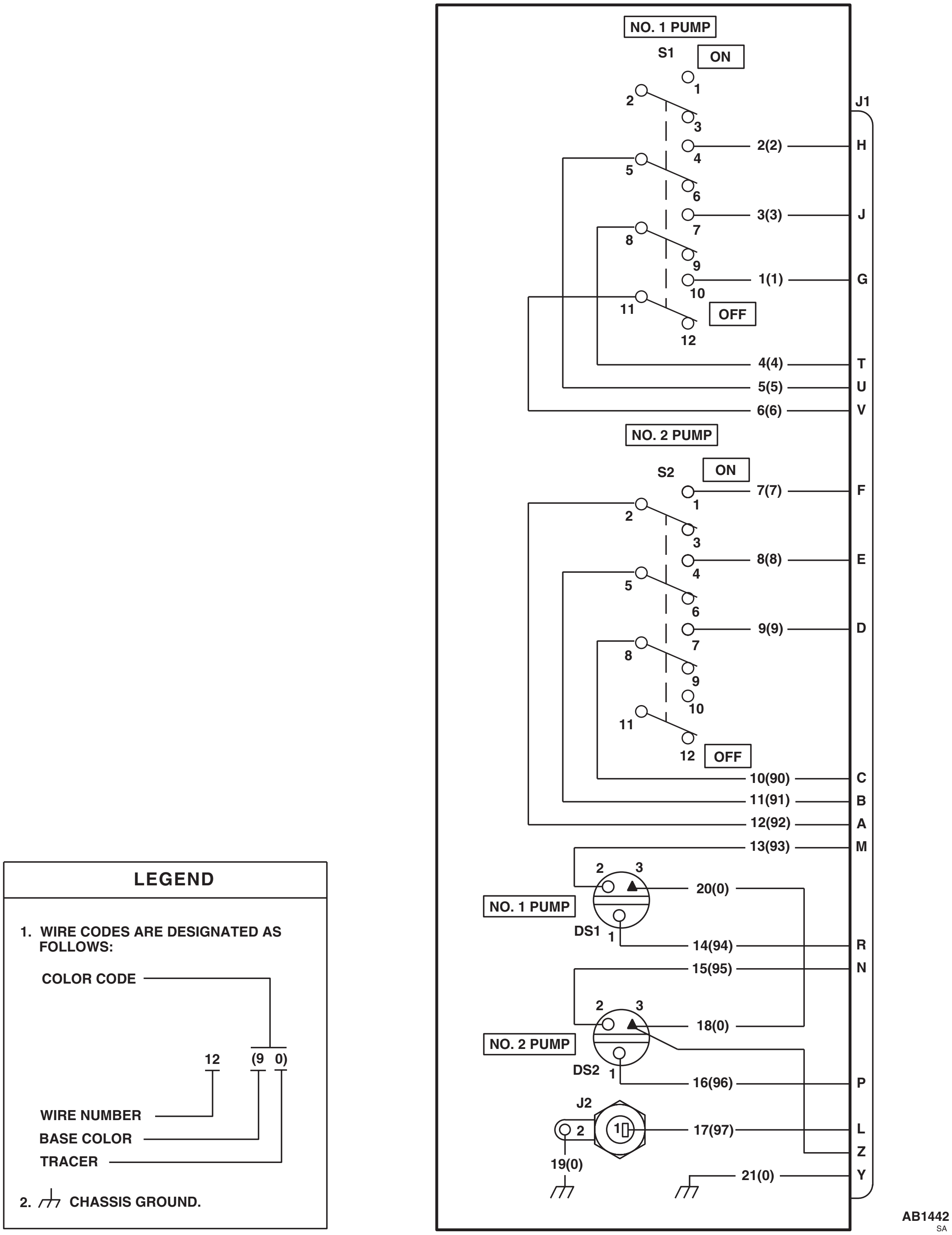


Figure 2. Fuel Boost Pump Control Panel Schematic Diagram.

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

ECM ANTENNA ACTUATOR ASSEMBLY (AVIM) EH60A

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-223

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0926 00
WP 0958 00

Equipment Condition
28 Vdc (10.0 amps) Bench Power Available

ECM ANTENNA ACTUATOR ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE EH60A

PROCEDURE

Step

NOTE

See Figure 1 for component location diagram as an aid in operational/troubleshooting.

1. Turn off bench electrical power.
2. Remove 18 screws and washers securing cover assembly to ECM antenna actuator assembly.
3. Remove seven screws and washers securing switch cover.

CAUTION

Actuator assembly is rated at 10.0 amps. Make sure connections are securely made before bench power is applied.

4. Connect 28 vdc bench test lead to W1J1-K. Connect bench power common test lead to W1J1M-J.
5. Apply bench power and allow ECM actuator assembly to move to the fully retracted position (if not already fully retracted). Actuator will be stopped by an internal limit switch.
6. Remove bench power. With ECM actuator in fully retracted position, check continuity between W1J1-C and W1J1-D, and between W1J1-H and W1J1-E.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified for W1J1-C and W1J1-D, replace limit switch W1S1 (WP 0958 00).

ECM ANTENNA ACTUATOR ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE EH60A - Continued

2. If Indication/Condition is not as specified for W1J1-E and W1J1-H, go to METER DOES NOT INDICATE CONTINUITY WITH ECM ACTUATOR ASSEMBLY IN FULLY RETRACTED POSITION, in this work package.

Step

7. Remove 28 vdc bench power test lead from W1J1-K and connect it to W1J1-M.
8. Apply bench power until retract switch arm releases W1S2 plunger.

Indication/Condition

Actuator shall move from fully retracted position.

Corrective Action

If Indication/Condition is not as specified, replace actuator (WP 0926 00).

Step

9. Remove bench power. With ECM actuator in fully retracted position, check continuity between W1J1-C and W1J1-D, and between W1J1-H and W1J1-E.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified for W1J1-C and W1J1-D, replace limit switch W1S1 (WP 0958 00).
2. If Indication/Condition is not as specified for W1J1-E and W1J1-H, go to METER DOES NOT INDICATE CONTINUITY WITH ECM ACTUATOR ASSEMBLY IN FULLY RETRACTED POSITION, in this work package.

Step

10. Remove 28 vdc bench power test lead from W1J1-M and connect it to W1J1-K.
11. Apply bench power until retract switch arm releases W1S2 plunger.

Indication/Condition

Actuator shall move from fully retracted position.

Corrective Action

If Indication/Condition is not as specified, replace actuator (WP 0926 00).

Step

12. Remove bench power. Check continuity between W1J1-A and W1J1-B, and between W1J1-G and W1J1-H.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition not as specified, replace limit switch W1S2 (WP 0958 00).

ECM ANTENNA ACTUATOR ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE EH60A - Continued

Step

13. Apply bench power until actuator moves from partially extended to fully extended position (approximately 20 seconds). Actuator will be stopped by an internal limit switch.
14. Remove bench power. With ECM actuator assembly in fully extended position, check continuity between W1J1-F and W1J1-L.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, go to METER DOES NOT INDICATE CONTINUITY WITH ECM ACTUATOR ASSEMBLY IN FULLY EXTENDED POSITION, in this work package.

Step

15. Remove 28 vdc bench power test lead from W1J1-M and connect it to W1J1-K.
16. Apply bench power until actuator moves from fully extended to fully retracted position (approximately 16 seconds). Actuator will be stopped by an internal limit switch.

Indication/Condition

Actuator shall move to retracted position.

Corrective Action

If Indication/Condition is not as specified, replace actuator (WP 0926 00).

Step

17. Remove bench power. Disconnect test leads from ECM antenna actuator assembly.

Indication/Condition

Bench power shall be removed.

Corrective Action

None Required

METER DOES NOT INDICATE CONTINUITY WITH ECM ACTUATOR ASSEMBLY IN FULLY RETRACTED POSITION

SYMPTOM

Meter does not indicate continuity with ECM actuator assembly in fully retracted position.

MALFUNCTION

Meter does not indicate continuity with ECM actuator assembly in fully retracted position.

CORRECTIVE ACTION

1. **Partially extend actuator. Go to 2.**
2. **Press W1S2 switch plunger into switch body. Go to 3.**
3. **Check continuity (short).**
   a. If continuity is present, adjust W1S2 (WP 0958 00). Go to **4.**

METER DOES NOT INDICATE CONTINUITY WITH ECM ACTUATOR ASSEMBLY IN FULLY RETRACTED POSITION - Continued

b. If continuity is not present, replace W1S2 (WP 0958 00). Go to **4.**

4. Procedure completed.

METER DOES NOT INDICATE CONTINUITY WITH ECM ACTUATOR ASSEMBLY IN FULLY EXTENDED POSITION

SYMPTOM

Meter does not indicate continuity with ECM actuator assembly in fully extended position.

MALFUNCTION

Meter does not indicate continuity with ECM actuator assembly in fully extended position.

CORRECTIVE ACTION

1. Partially retract actuator. Go to 2.

2. Press W1S1 switch plunger into switch body. Go to 3.

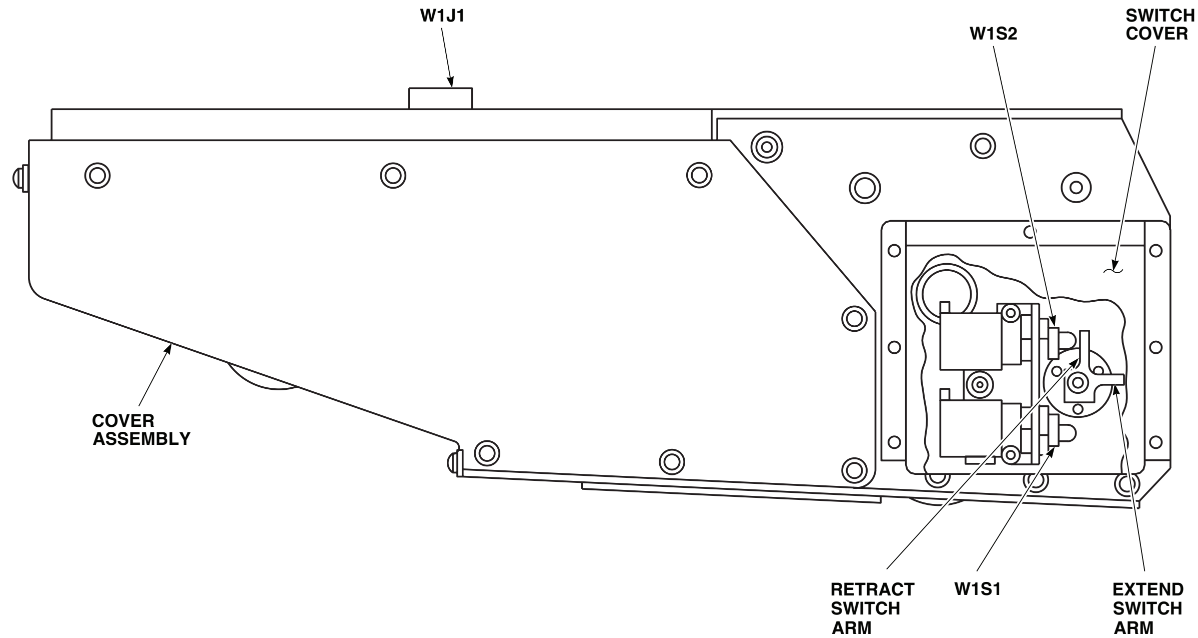
3. Check continuity (short).

a. If continuity is present, adjust W1S1 (WP 0958 00). Go to **4.**

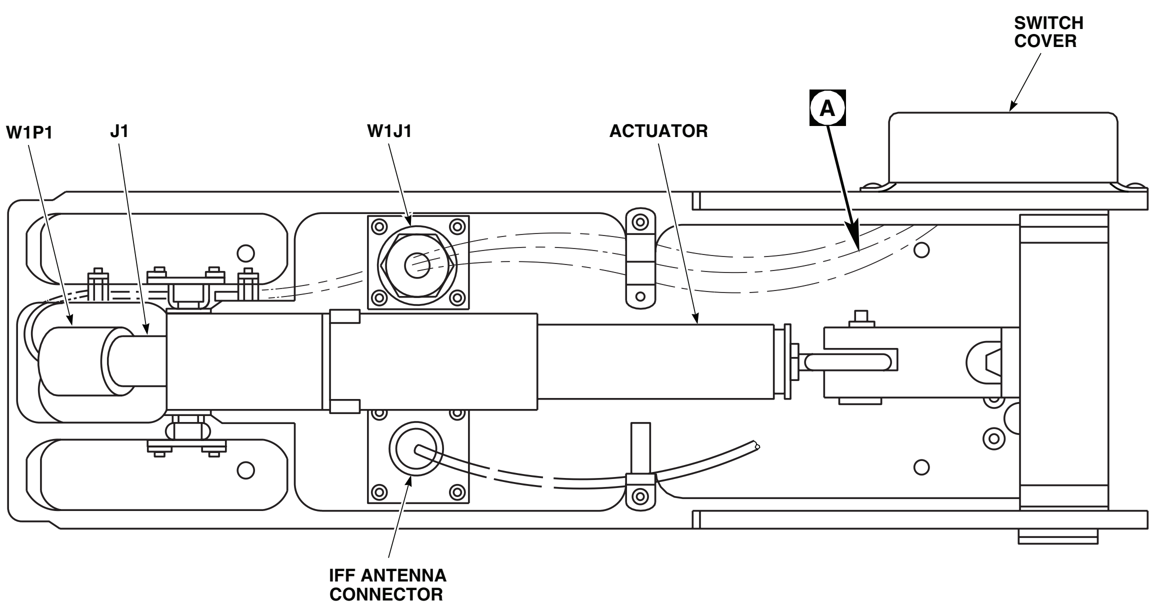
b. If continuity is not present, replace W1S1 (WP 0958 00). Go to **4.**

4. Procedure completed.

LOCATION



SIDE VIEW



BOTTOM VIEW

AA6091_1
SA

Figure 1. ECM Antenna Actuator Components Location Diagram. (Sheet 1 of 2)

LOCATION - Continued

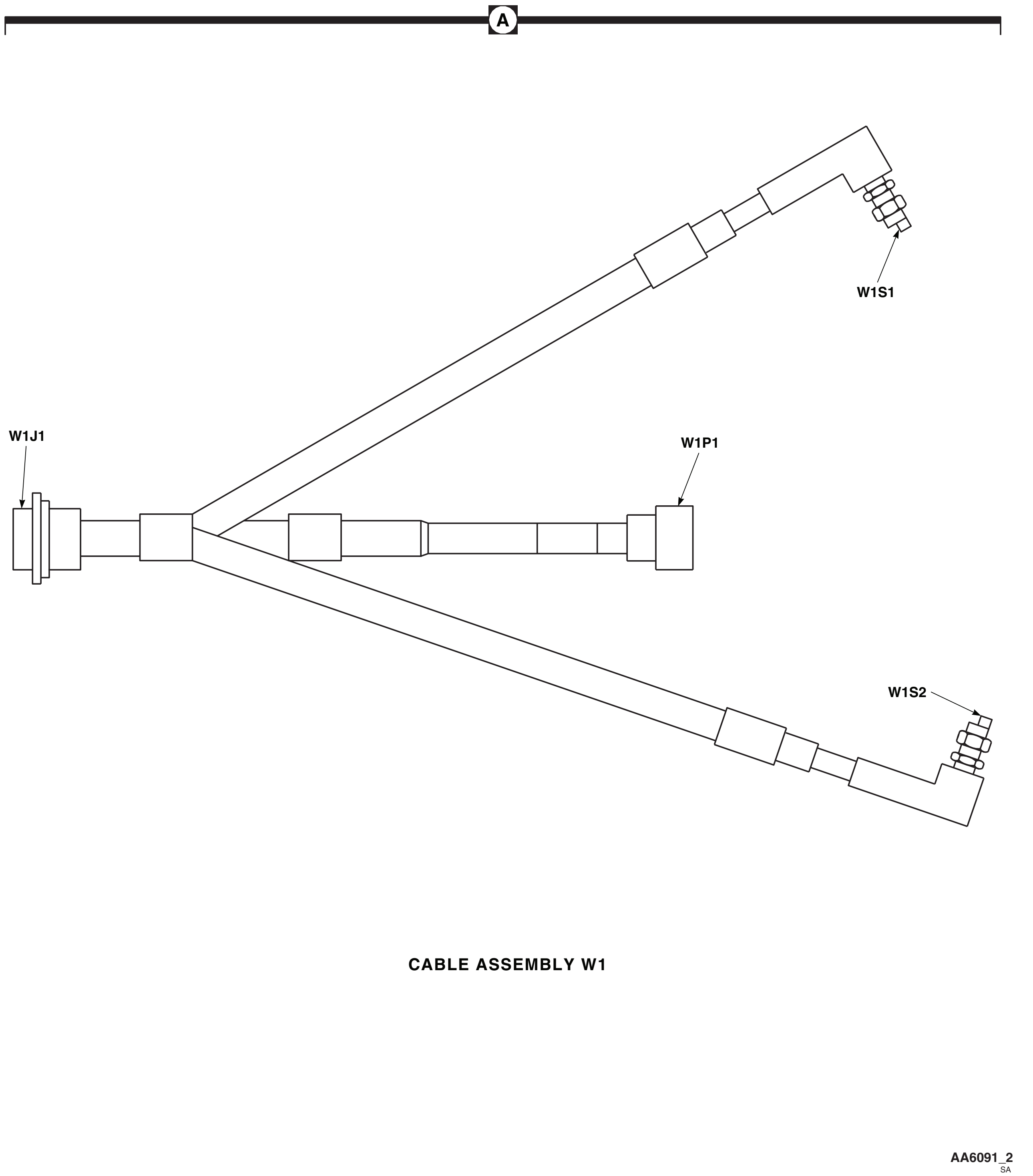


Figure 1. ECM Antenna Actuator Components Location Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

RIGHT HAND RELAY PANEL (AVIM)

INITIAL SETUP:

Tools and Special Tools

Decade Resistor, ZM57U
Electrical Repairer Toolkit, SC 5180-99-B06
Locally-Made Stores Jettison System Test Adapter (test lamp) Figure 150, WP 1805 00
Multimeter, AN/PSM-45A

Personnel Required

Aircraft Electrician MOS 15F (1)

References

WP 0848 00
WP 0937 00
WP 0944 00
WP 0945 00
WP 0947 00
WP 0948 00
WP 1747 00
WP 1805 00

Equipment Condition

Bench Power Available, 28 vdc and 115 vac, 400 Hz

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram, Figure 2 for test setup, Figure 3, **70550-02102-049, -051, AND -56** Figure 4, **70550-02102-043, -054, AND -047** Figure 5, Figure 6, **70550-02102-043, -045, -047, -049, AND -051** Figure 7, and **70550-02102-056** Figure 8 for schematic diagrams as an aid in operational/troubleshooting.

The right hand relay panel operational/troubleshooting procedure is subdivided as follows:

- T1 Current Sensor
- K20 DC Essential Bus Fail Relay
- K21 Windshield Anti-ice Lockout Relay
- K26 No. 2 Engine Start Relay
- K32 Hydraulic Accumulator Relay
- Shutdown

T1 CURRENT SENSOR OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

See Figure 2. for test setup.

1. Set decade resistors R1 and R2 to 200 ohms.
2. Apply 115 vac 400 Hz to J241-m.
3. Apply 28 vdc to J241-d.
4. Decrease resistance of R1 and R2 until each ammeter reads a minimum of 0.83 amps. Record resistance.

Indication/Condition

Test lamp (WP 1805 00) shall go off.

Corrective Action

If Indication/Condition is not as specified, go to TEST LAMP STAYS ON, in this work package.

Step

5. Increase resistance of R1 while holding R2 constant until R1 ammeter reads less than 0.67 amps.

Indication/Condition

Test lamp shall go on.

Corrective Action

If Indication/Condition is not as specified, go to TEST LAMP STAYS OFF, in this work package.

Step

6. Decrease resistance of R1 until it returns to original resistance measured in step 4.
7. Increase resistance of R2 while holding R1 constant until R2 ammeter reads less than 0.67 amps.

Indication/Condition

Test lamp shall go on.

Corrective Action

If Indication/Condition is not as specified, go to TEST LAMP STAYS OFF, in this work package.

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

K20 DC ESSENTIAL BUS FAIL RELAY OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Check continuity between J900-G and J900-H.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, repair/replace wiring (WP 1747 00).
   a. If trouble remains, replace K20 relay (WP 0944 00).

Step

2. Connect 28 vdc between J900-J and J900-M.
3. Check for no continuity between J900-G and J900-H.

Indication/Condition

No continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, repair/replace wiring (WP 1747 00).
   a. If trouble remains, replace K20 relay (WP 0944 00).

Step

4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

K21 WINDSHIELD ANTI-ICE LOCKOUT RELAY OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Check continuity between:
   - J244-F and J244-G
   - J900-E and J900-F

Indication/Condition

Continuity shall be present.

K21 WINDSHIELD ANTI-ICE LOCKOUT RELAY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

If continuity is not present, go to K21 WINDSHIELD ANTI-ICE LOCKOUT RELAY TEST FAILS, in this work package.

Step

2. For right hand relay panels 70550-02102-049, -051, and -056 only, check continuity between J900-C and J900-D.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Conditions are not as specified, go to K21 WINDSHIELD ANTI-ICE LOCKOUT RELAY TEST FAILS, in this work package.

Step

3. Connect 28 vdc between J244-E and J900-M.
4. Check for no continuity between:
   - J244-F and J244-G
   - J900-E and J900-F

Indication/Condition

No continuity shall be present.

Corrective Action

If Indication/Conditions are not as specified, go to K21 WINDSHIELD ANTI-ICE LOCKOUT RELAY TEST FAILS, in this work package.

Step

5. For right hand relay panels 70550-02102-049, -051, and -056, check for open circuit between J900-C and J900-D.

Indication/Condition

No continuity shall be present.

Corrective Action

If Indication/Conditions are not as specified, go to K21 WINSHIELD ANTI-ICE LOCKOUT RELAY TEST FAILS, in this work package.

Step

6. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

K26 NO. 2 ENGINE START RELAY OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Check continuity between:
   - J241-F and J244-J
   - J241-G and J244-K
   - J241-X and J241-Y
   - J900-L and J900-K

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Conditions are not as specified, go to K26 NO. 2 ENGINE START RELAY TEST FAILS, in this work package.

Step

2. Connect 28 vdc between J241-R and J900-M. Connect multimeter between J241-U and J900-M.

Indication/Condition

28 vdc shall be present.

Corrective Action

If Indication/Condition is not as specified, go to K26 NO. 2 ENGINE START RELAY TEST FAILS, in this work package.

Step

3. Check continuity between:
   - J241-a and J241-Z
   - J241-a and J241-b

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, go to K26 NO. 2 ENGINE START RELAY TEST FAILS, in this work package.

Step

4. Check for no continuity between:
   - J241-F and J244-J
   - J241-G and J244-K

K26 NO. 2 ENGINE START RELAY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- J241-X and J241-Y
- J900-L and J900-K

Indication/Condition

No continuity shall be present.

Corrective Action

If Indication/Conditions are not as specified, go to K26 NO. 2 ENGINE START RELAY TEST FAILS, in this work package.

Step

5. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

K32 HYDRAULIC ACCUMULATOR RELAY OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Connect 28 vdc between J241-p and J900-M.
2. Check continuity between J244-A and J244-C.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, go to K32 HYDRAULIC ACCUMULATOR RELAY TEST FAILS, in this work package.

Step

3. Connect 28 vdc between J244-a and J900-M.
4. Disconnect 28 vdc from J241-p.
5. Check continuity between J244-A and J244-C.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Conditions are not as specified, go to K32 HYDRAULIC ACCUMULATOR RELAY TEST FAILS, in this work package.

K32 HYDRAULIC ACCUMULATOR RELAY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

6. Disconnect 28 vdc from J244-a.
7. Check for no continuity between J244-A and J244-C.

Indication/Condition

No continuity shall be present.

Corrective Action

If Indication/Conditions are not as specified, go to K32 HYDRAULIC ACCUMULATOR RELAY TEST FAILS, in this work package.

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Turn off and disconnect electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

TEST LAMP STAYS ON

SYMPTOM

Test lamp stays on.

MALFUNCTION

Test lamp stays on.

CORRECTIVE ACTION

1. **Check electrical connections.**
   a. If electrical connections are good, go to **2.**
   b. If electrical connections are not good, reconnect test equipment and repeat operational test. Go to **3.**
2. **Check continuity between:**

TEST LAMP STAYS ON - Continued

- **J241-c and T1-X2**
- **J241-d and T1-A1**
- **J241-f and T1-A2**
- **J241-m and T1-X1**
- **J241-r and T1-X4**
- **T1-G and GND 1-T**

a. If continuity is present, replace T1 (WP 0937 00). Go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3.**

3. Procedure completed.

TEST LAMP STAYS OFF

SYMPTOM

Test lamp stays off.

MALFUNCTION

Test lamp stays off.

CORRECTIVE ACTION

1. Check electrical connections.

a. If electrical connections are good, go to **2.**

b. If electrical connections are not good, reconnect test equipment and repeat operational test. Go to **3.**

2. Check continuity between:

- **J241-c and T1-X2**
- **J241-d and T1-A1**
- **J241-f and T1-A2**
- **J241-m and T1-X1**
- **J241-r and T1-X4**
- **T1-G and GND 1-T**

a. If continuity is present, replace T1 (WP 0937 00). Go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3.**

3. Procedure completed.

K21 WINDSHIELD ANTI-ICE LOCKOUT RELAY TEST FAILS

SYMPTOM

K21 windshield anti-ice lockout relay test fails.

MALFUNCTION

K21 windshield anti-ice lockout relay test fails.

CORRECTIVE ACTION

1. **Check relay panel part number.**

   a. If right hand relay panel 70550-02102-043, -045, or -047, go to **2.**

   b. If right hand relay panel 70550-02102-049, -051, or -056, go to **3.**

2. **Check continuity between:**

   - **J244-E and K21-X1**
   - **GND 1-B and K21-X2**
   - **J244-F and K21-A3**
   - **J244-G and K21-A2**
   - **J900-E and K21-B3**
   - **J900-F and K21-B2**

   a. If continuity is present, replace K21 (WP 0945 00). Go to **4.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**

3. **Check continuity between:**

   - **J244-E and K21-X1**
   - **GND 1-B and K21-X2**
   - **J244-F and K21-A3**
   - **J244-G and K21-A2**
   - **J900-C and K21-C2**
   - **J900-D and K21-C3**
   - **J900-E and K21-B3**
   - **J900-F and K21-B2**

   a. If continuity is present, replace K21 (WP 0945 00). Go to **4.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**

4. **Procedure completed.**

K26 NO. 2 ENGINE START RELAY TEST FAILS

SYMPTOM

K26 No. 2 engine start relay test fails.

MALFUNCTION

K26 No. 2 engine start relay test fails.

CORRECTIVE ACTION

1. **Check continuity between:**
   - **J241-F and K26-E2**
   - **J241-G and K26-F2**
   - **J241-R and K26-X1**
   - **J241-U and K26-A2**
   - **J241-X and K26-D2**
   - **J241-Y and K26-D3**
   - **J241-Z and K26-C2**
   - **J241-a and K26-C1**
   - **J241-b and K26-C2**
   - **J241-e and GND 1-R**
   - **J244-J and K26-E3**
   - **J244-K and K26-F3**
   - **J900-K and K26-B3**
   - **J900-L and K26-B2**
   - **J900-g and K26-B1**
   - **K26-A1 and K26-X1**
   - **K26-X2 and GND 1-C**

   a. If continuity is present, replace K26 (WP 0947 00). Go to **2.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **2.**

2. **Procedure completed.**

K32 HYDRAULIC ACCUMULATOR RELAY TEST FAILS

SYMPTOM

K32 hydraulic accumulator relay test fails.

MALFUNCTION

K32 hydraulic accumulator relay test fails.

CORRECTIVE ACTION

1. **Check relay panel part numbers.**
   a. If right hand relay panel 70550-02102-043, -045, -047, -049, or -051, go to **2.**
   b. If right hand relay panel 70550-02102-056, go to **3.**
2. **Check continuity between:**
   - **J241-p and U2-18**
   - **J244-a and U2-7**
   - **E26 and U2-7**
   - **K31-A3 and U2-18**
   - **J244-A and K31-X1**
   - **J244-C and K33-B2**
   - **K31-A2 and K32-A1**
   - **K31-A3 and K32-X1**
   - **K31-B3 and E24**
   - **K31-X1 and K32-B1**
   - **K32-A2 and K33-B1**
   - **K32-B2 and K33-B2**
   - **K32-X2 and GND 1-W**
   - **E25 and K33-B1**

   a. If continuity is present, go to **4.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5.**
3. **Check continuity between:**
   - **J241-p and K31-A3**
   - **J244-a and E26**
   - **J244-A and K31-X1**
   - **J244-C and K33-B2**
   - **K31-A2 and K32-A1**
   - **K31-A3 and K32-X1**

K32 HYDRAULIC ACCUMULATOR RELAY TEST FAILS - Continued

- **K31-B3 and E24**
- **K31-X1 and K32-B1**
- **K32-A2 and K33-B1**
- **K32-B2 and K33-B2**
- **K32-X2 and GND 1-W**
- **E25 and K33-B1**

a. If continuity is present, go to **4.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5.**

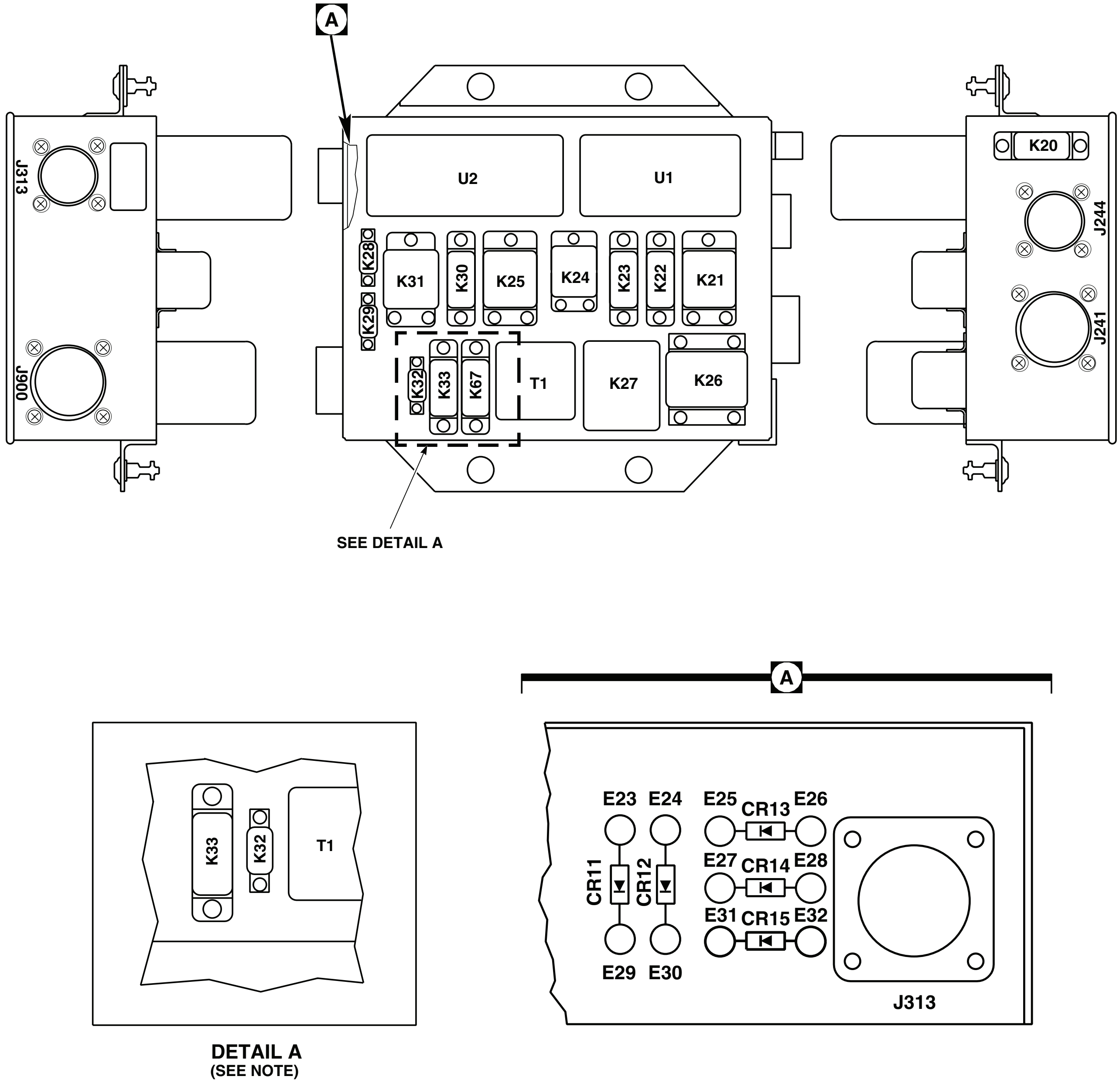
4. Check that resistance across diode CR13 is 10X greater in one direction than in the other direction.

a. If resistance is as specified, replace K32 (WP 0948 00). Go to **5.**

b. If resistance is not as specified, replace diode CR13 (WP 0848 00). Go to **5.**

5. Procedure completed.

LOCATION



NOTE

DETAIL A IS FOR RELAY PANEL PART NUMBERS 70550-02102-043 AND 70550-02102-045.

AA9147
SA

Figure 1. Right Hand Relay Panel Location Diagram.

TEST SETUP

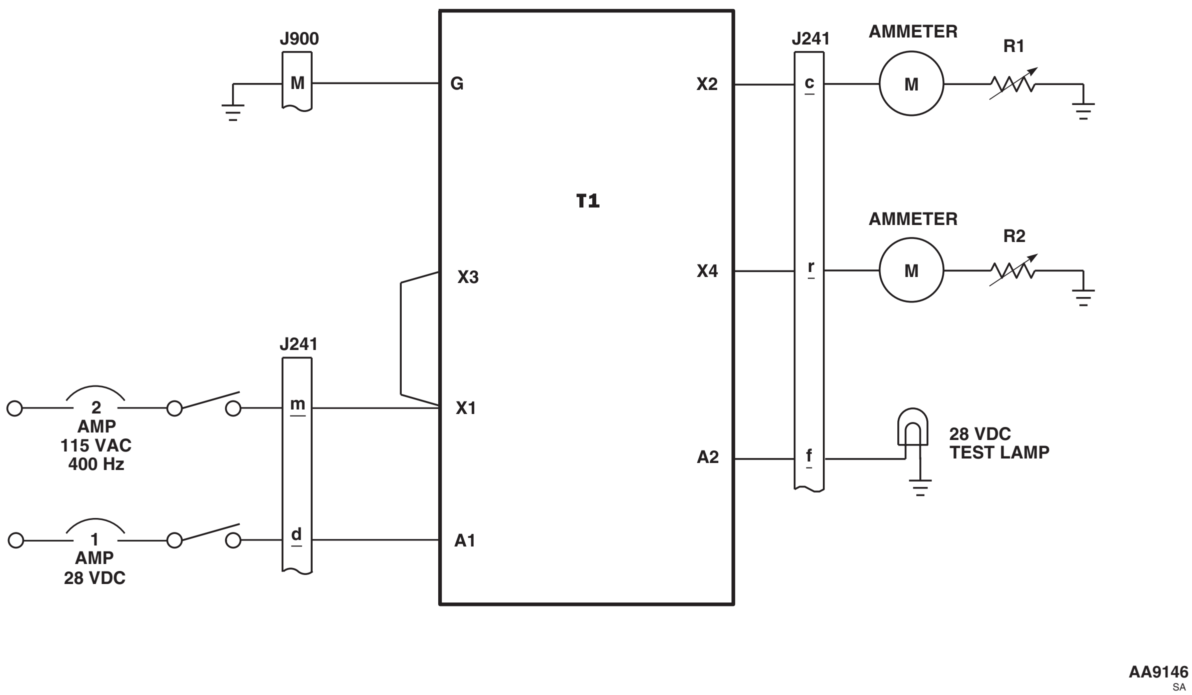


Figure 2. T1 Current Sensor Test Setup.

SCHEMATICS

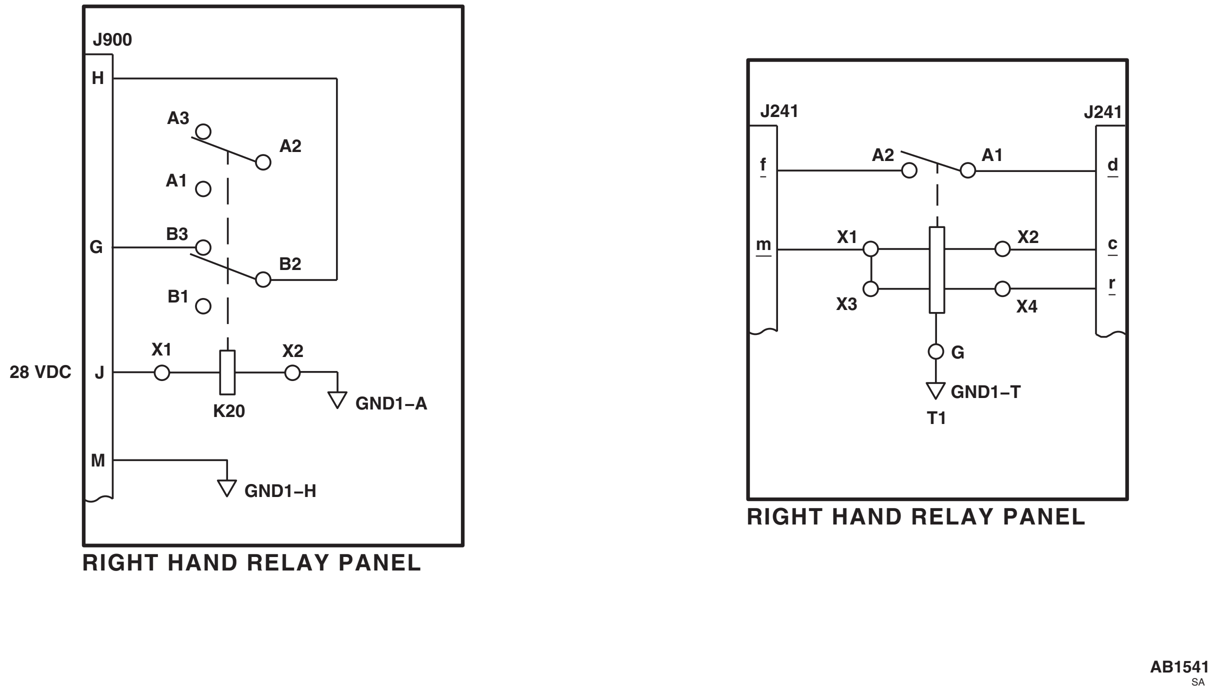
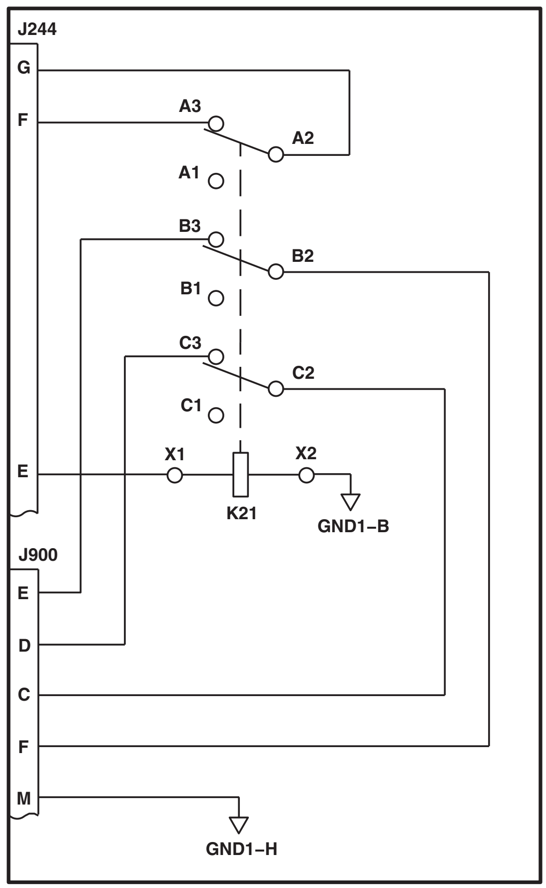


Figure 3. Right Hand Relay Panel Schematic Diagram.

SCHEMATICS - Continued



RIGHT HAND RELAY PANEL

EFFECTIVITY

RELAY PANEL 70550–02102–049, –051, AND –056.

AB1542
SA

Figure 4. Right Hand Relay Panel Schematic Diagram 70550-02102-049, -051, AND -56 .

SCHEMATICS - Continued

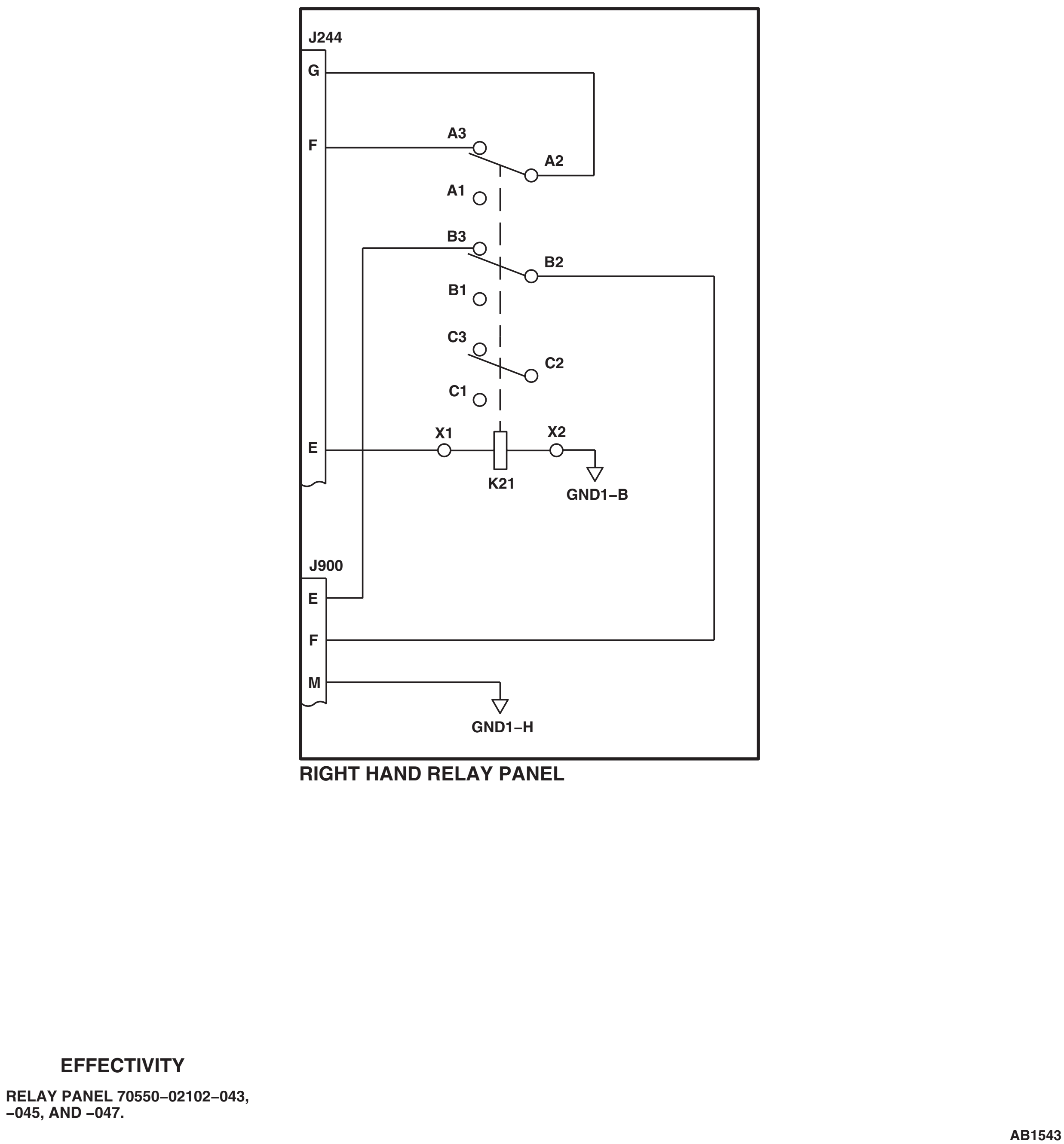


Figure 5. Right Hand Relay Panel Schematic Diagram **70550-02102-043, -045, AND -047** .

SCHEMATICS - Continued

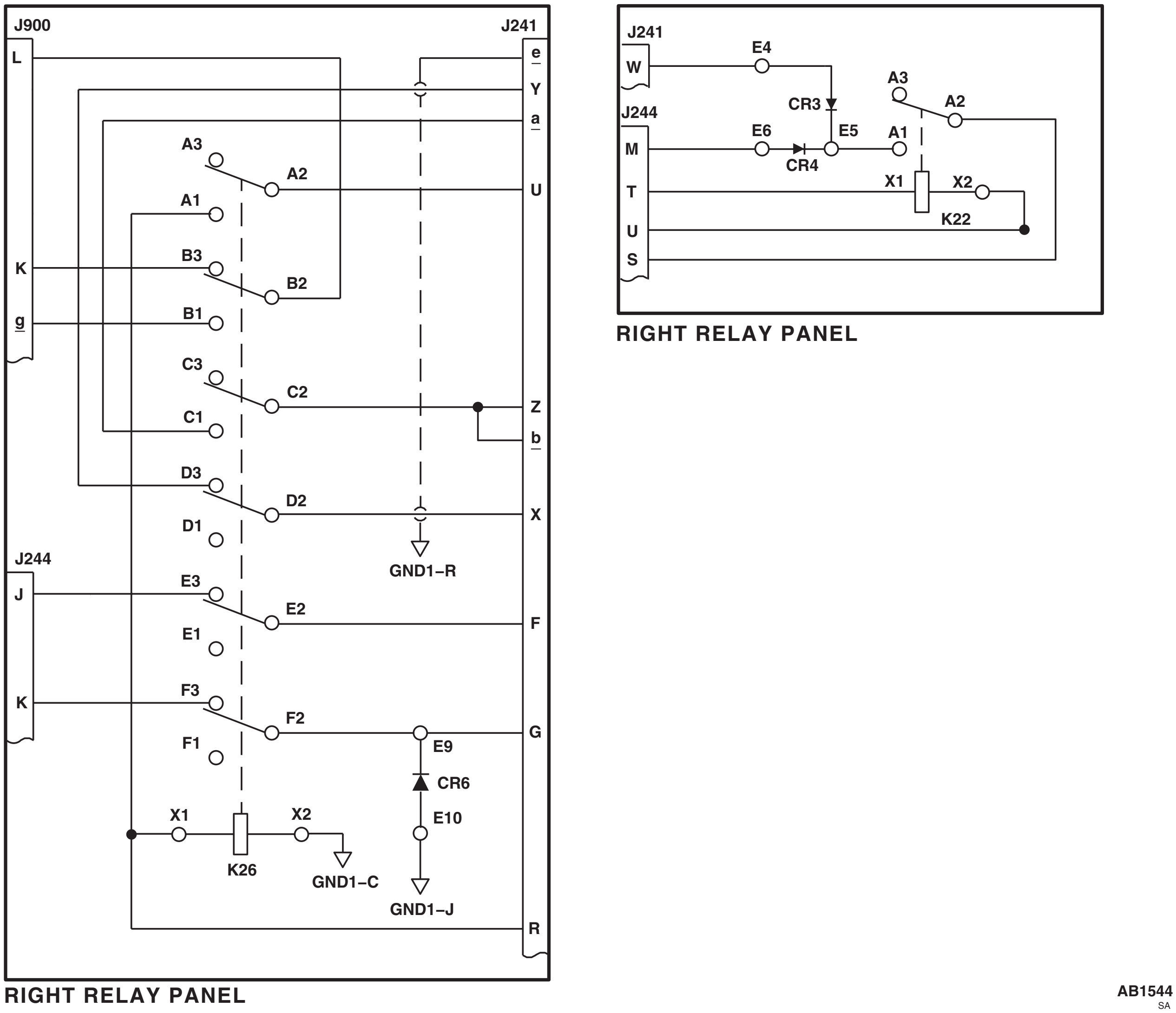
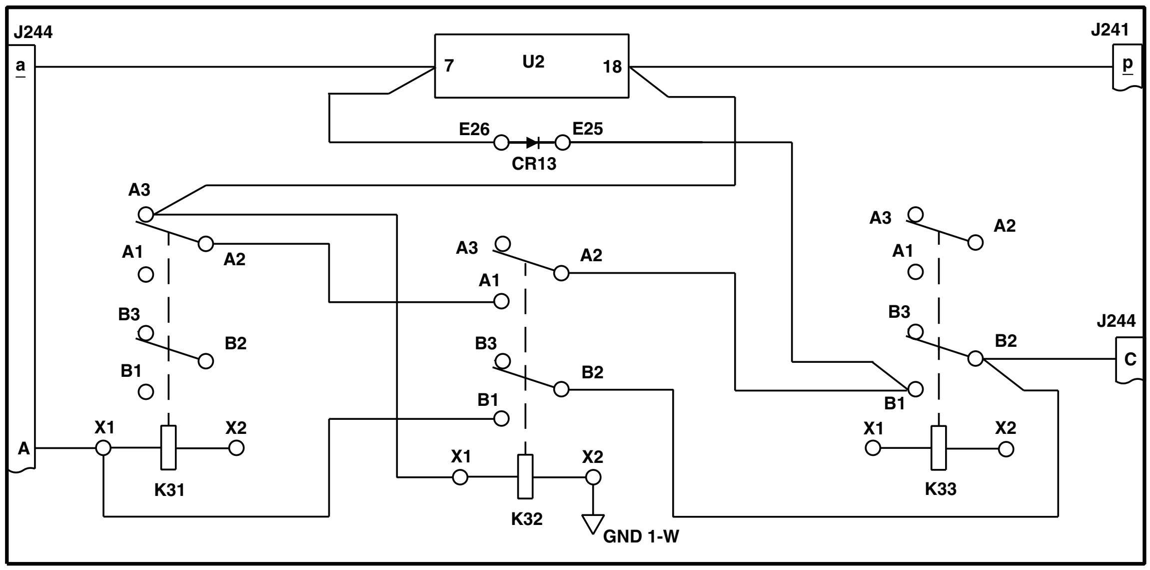


Figure 6. Right Hand Relay Panel Schematic Diagram.

SCHEMATICS - Continued



RIGHT RELAY PANEL

EFFECTIVITY

RELAY PANEL 70550-02102-043,
-045, -047, -049, AND -051.

AA9153
SA

Figure 7. Right Hand Relay Panel Schematic Diagram **70550-02102-043, -045, -047, -049, AND -051** .

SCHEMATICS - Continued

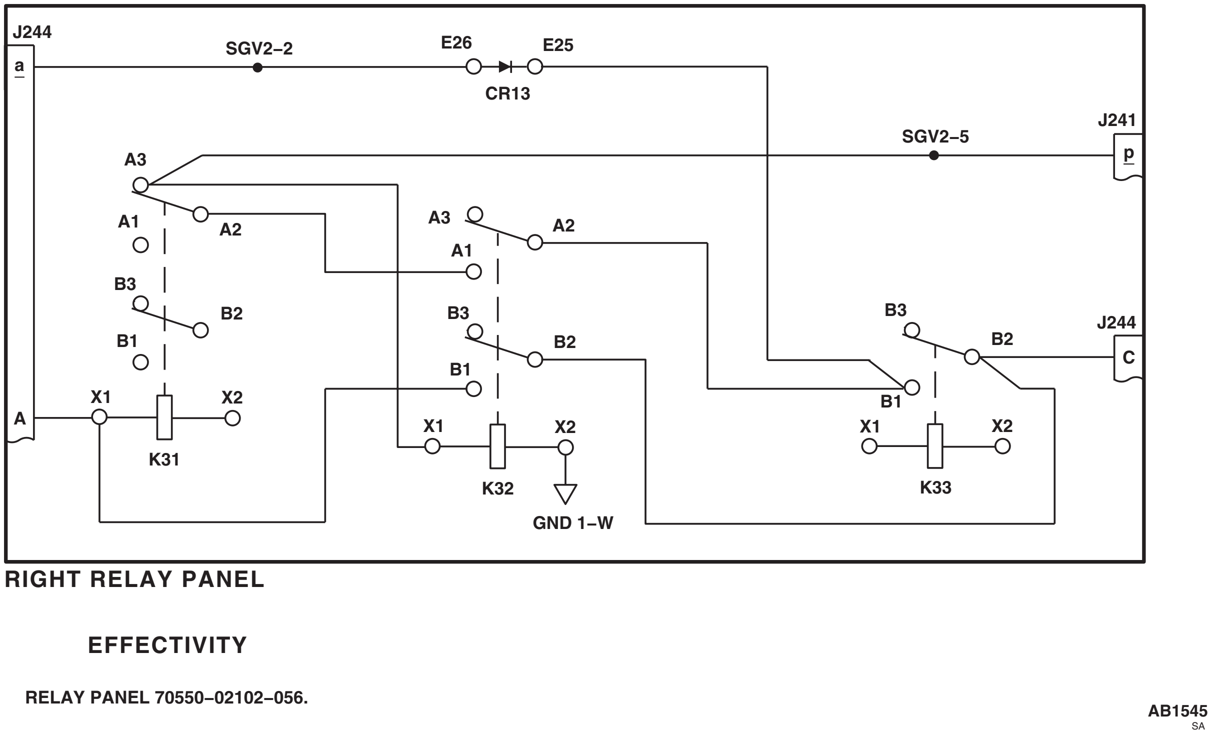


Figure 8. Right Hand Relay Panel Schematic Diagram 70550-02102-056 .

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

LEFT HAND RELAY PANEL (AVIM)

INITIAL SETUP:

Tools and Special Tools

Decade Resistor, ZM57U
Electrical Repairer Toolkit, SC 5180-99-B06
Locally-Made Stores Jettison System Test Adapter (test lamp) Figure 150, WP 1805 00
Multimeter, AN/PSM-45A

Personnel Required

Aircraft Electrician MOS 15F (1)

References

WP 0937 00
WP 0938 00
WP 0939 00
WP 0940 00
WP 0941 00
WP 0942 00
WP 1747 00
WP 1805 00

Equipment Condition

Bench Power Available, 28 vdc and 115 vac, 400 Hz

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram, Figure 2 for test setup, and Figure 3, **70550-02105-048 AND -050** Figure 4 , **70550-02105-051, -054, -57, AND -058** Figure 5 and Figure 6 for schematic diagrams as an aid in operational/troubleshooting.

The left hand relay panel operational/troubleshooting procedure is subdivided as follows:

- T1 Current Sensor
- K41 Searchlight Relay 70550-02105-048, and -050
- K41 Searchlight Relay 70550-02105-051, -054, and -057
- K45 No. 1 Engine Start Relay
- Shutdown

T1 CURRENT SENSOR OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

See Figure 2. for test setup.

1. Set decade resistors R1 and R2 to 200 ohms.

T1 CURRENT SENSOR OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. Apply 115 vac 400 Hz to J242-r.
3. Apply 28 vdc to J243-X.
4. Decrease resistance of R1 and R2 until each ammeter reads a minimum of 0.83 amps. Record resistance.

Indication/Condition

Test lamp (WP 1805 00) shall go off.

Corrective Action

If Indication/Condition is not as specified, go to TEST LAMP STAYS ON, in this work package.

Step

5. Increase resistance of R1 while holding R2 constant until ammeter reads less than 0.67 amps.

Indication/Condition

Test lamp shall go on.

Corrective Action

If Indication/Condition is not as specified, go to TEST LAMP STAYS OFF, in this work package.

Step

6. Return R1 to original resistance measured in step 4.
7. Increase resistance of R2 while holding R1 constant until ammeter reads less than 0.67 amps.

Indication/Condition

Test lamp shall go on.

Corrective Action

If Indication/Condition is not as specified, go to TEST LAMP STAYS OFF, in this work package.

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

K41 SEARCHLIGHT RELAY 70550-02105-048, AND -050 OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Check continuity between:
   - J243-d and J243-f
   - J243-e and J243-g

K41 SEARCHLIGHT RELAY 70550-02105-048, AND -050 OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- J902-f and J243-P
- P902-k and J242-i
- P902-n and J902-p
- P902-W and J902-Y

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Conditions are not as specified, go to K41 SEARCHLIGHT RELAY TEST FAILS, in this work package.

Step

2. Connect 28 vdc between P243-i and J243-P.
3. Check for no continuity between:
   - J243-d and J243-f
   - J243-e and J243-g
   - J902-f and J243-P
   - P902-k and J242-i
   - P902-n and J902-p
   - P902-W and J902-Y

Indication/Condition

No continuity shall be present.

Corrective Action

If Indication/Conditions are not as specified, go to K41 SEARCHLIGHT RELAY TEST FAILS, in this work package.

Step

4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

K41 SEARCHLIGHT RELAY 70550-02105-051, -054, AND -057 OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

K41 searchlight relay is a two coil latching relay. The de-energized state of the relay depends upon which coil was last energized.

1. Apply and remove 28 vdc between J243-Y and J243-P.
2. Measure state (continuity/no continuity) between J243-V and J243-W.
3. Apply and remove 28 vdc between J243-Y and J243-P.

Indication/Condition

State measured (continuity/no continuity) between J243-V and J243-W shall cycle upon each application of 28 vdc.

Corrective Action

If Indication/Condition is not as specified, go to K41 SEARCHLIGHT RELAY TEST FAILS, in this work package.

Step

4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

K45 NO. 1 ENGINE START RELAY OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Check continuity between:
   - J242-R and J242-S
   - J242-W and J242-X
   - J902-a and J902-c
   - J902-b and J902-d

Indication/Condition

Continuity shall be present.

Corrective Action

If continuity is not present, go to K45 NO. 1 ENGINE START RELAY TEST FAILS, in this work package.

K45 NO. 1 ENGINE START RELAY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

2. Check for no continuity between:
   - J242-T and J242-V
   - J242-P and J242-S
   - J242-Y and J242-Z
   - J242-Y and J242-U

Indication/Condition

No continuity shall be present.

Corrective Action

If continuity is present, go to K45 NO. 1 ENGINE START RELAY TEST FAILS, in this work package.

Step

3. Connect 28 vdc lamp between J242-V and J902-GG.
4. Connect 28 vdc between J242-T and J243-P.

Indication/Condition

28 vdc lamp shall light.

Corrective Action

If Indication/Condition is not as specified, go to K45 NO. 1 ENGINE START RELAY TEST FAILS, in this work package.

Step

5. Check continuity between:
   - J242-P and J242-S
   - J242-Y and J242-Z

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, go to K45 NO. 1 ENGINE START RELAY TEST FAILS, in this work package.

Step

6. Check for open circuit between:
   - J242-R and J242-S
   - J242-W and J242-X
   - J902-a and J902-c
   - J902-b and J902-d

K45 NO. 1 ENGINE START RELAY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

No continuity shall be present.

Corrective Action

If Indication/Conditions are not as specified, go to K45 NO. 1 ENGINE START RELAY TEST FAILS, in this work package.

Step

7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Turn off and disconnect electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

TEST LAMP STAYS ON

SYMPTOM

Test lamp stays on.

MALFUNCTION

Test lamp stays on.

CORRECTIVE ACTION

1. **Check electrical connections.**
   a. If electrical connections are good, go to **2.**
   b. If electrical connections are not good, reconnect test equipment and repeat operational test. Go to **3.**
2. **Check continuity between:**
   - **J242-K and T1-X2**
   - **J243-X and T1-A1**
   - **J242-b and T1-A2**

TEST LAMP STAYS ON - Continued

- **J242-r and T1-X1**
- **J242-m and T1-X4**
- **T1-G and GND 1-P**
- **T1-X1 and T1-X3**

a. If continuity is present, replace T1 (WP 0937 00). Go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3.**

3. Procedure completed.

TEST LAMP STAYS OFF

SYMPTOM

Test lamp stays off.

MALFUNCTION

Test lamp stays off.

CORRECTIVE ACTION

1. Check electrical connections.

a. If electrical connections are good, go to **2.**

b. If electrical connections are not good, reconnect test equipment and repeat operational test. Go to **3.**

2. Check continuity between:

- **J242-K and T1-X2**
- **J243-X and T1-A1**
- **J242-b and T1-A2**
- **J242-r and T1-X1**
- **J242-m and T1-X4**
- **T1-G and GND 1-P**
- **T1-X1 and T1-X3**

a. If continuity is present, replace T1 (WP 0937 00). Go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3.**

3. Procedure completed.

K41 SEARCHLIGHT RELAY TEST FAILS

SYMPTOM

K41 searchlight relay test fails.

MALFUNCTION

K41 Searchlight relay test fails.

CORRECTIVE ACTION

1. **Check continuity between:**
   - **K41-A2 and K41-A3**
   - **K41-B2 and K41-B3**
   - **K41-C2 and K41-C3**
   - **K41-D2 and K41-D3**
   - **K41-E2 and K41-E3**
   - **K41-F2 and K41-F3**

   a. If continuity is present, go to **2.**

   b. If continuity is not present, replace K41 (WP 0938 00). Go to **8.**
2. **Check for no continuity between:**
   - **K41-A1 and K41-A2**
   - **K41-B1 and K41-B2**
   - **K41-C1 and K41-C2**
   - **K41-D1 and K41-D2**
   - **K41-E1 and K41-E2**
   - **K41-F1 and K41-F2**

   a. If continuity is present, replace K41 (WP 0938 00). Go to **8.**

   b. If continuity is not present, go to **3.**
3. **Connect 28 vdc between K41-X1 and K41-X2. Check continuity between:**
   - **K41-A1 and K41-A2**
   - **K41-B1 and K41-B2**
   - **K41-C1 and K41-C2**
   - **K41-D1 and K41-D2**
   - **K41-E1 and K41-E2**
   - **K41-F1 and K41-F2**

   a. If continuity is present, go to **4.**

   b. If continuity is not present, replace K41 (WP 0938 00). Go to **8.**

K41 SEARCHLIGHT RELAY TEST FAILS - Continued

4. **Check for no continuity between:**
   - **K41-A2 and K41-A3**
   - **K41-B2 and K41-B3**
   - **K41-C2 and K41-C3**
   - **K41-D2 and K41-D3**
   - **K41-E2 and K41-E3**
   - **K41-F2 and K41-F3**

   a. If continuity is not present, go to **5.**

   b. If continuity is present, replace K41 (WP 0938 00). Go to **8.**
5. **Check continuity between:**
   - **K41-X2 and GND 1-A**
   - **K41-X1 and P243-i**
   - **K41-A2 and P243-g**
   - **K41-A3 and K40-C2**
   - **K41-B2 and P243-f**
   - **K41-B3 and K40-B2**
   - **K41-C2 and J902-n**
   - **K41-C3 and J1-13**
   - **K41-C3 and K40-A2**
   - **K41-D2 and K40-D3**
   - **K41-D3 and P902-k**
   - **K41-E2 and GND 1-B**
   - **K41-E3 and P902-f**
   - **K41-F2 and P902-Y**
   - **K41-F3 and P902-W**
   - **J243-P and GND 1-M**

   a. If continuity is present, go to **6.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**
6. **Check continuity between:**
   - **K40-A2 and K40-A3**
   - **K40-B2 and K40-B3**
   - **K40-C2 and K40-C3**

K41 SEARCHLIGHT RELAY TEST FAILS - Continued

- **K40-D2 and K40-D3**
- **K40-E2 and K40-E3**
- **K40-F2 and K40-F3**

a. If continuity is present, go to **7.**

b. If continuity is not present, replace K40 (WP 0942 00). Go to **8.**

7. **Check for no continuity between:**
   - **K40-A2 and K40-A1**
   - **K40-B2 and K40-B1**
   - **K40-C2 and K40-C1**
   - **K40-D2 and K40-D1**
   - **K40-E2 and K40-E1**
   - **K40-F2 and K40-F1**

   a. If no continuity is present, go to **8.**

   b. If continuity is present, replace K40 (WP 0942 00). Go to **8.**

8. **Procedure completed.**

K41 SEARCHLIGHT RELAY TEST FAILS

SYMPTOM

K41 searchlight relay test fails.

MALFUNCTION

K41 searchlight relay test fails.

CORRECTIVE ACTION

1. **Apply and remove 28 vdc between K41-X1 and K41-X2. Check continuity between:**
   - **K41-A1 and K41-A2**
   - **K41-B1 and K41-B2**

   a. If continuity is present, go to **2.**

   b. If continuity is not present, replace K41 (WP 0938 00). Go to **17.**

2. **Check for no continuity between:**
   - **K41-A2 and K41-A3**
   - **K41-B2 and K41-B3**

   a. If continuity is present, replace K41 (WP 0938 00). Go to **17.**

   b. If continuity is not present, go to **3.**

3. **Apply and remove 28 vdc from K41-X1 and K41-X2 and connect 28 vdc between K41-Y1 and K41-Y2. Check continuity between:**

K41 SEARCHLIGHT RELAY TEST FAILS - Continued

- **K41-A2 and K41-A3**
- **K41-B2 and K41-B3**

a. If continuity is present, go to **4.**

b. If continuity is not present, replace K41 (WP 0938 00). Go to **17.**

4. Check for no continuity between:

- **K41-A1 and K41-A2**
- **K41-B1 and K41-B2**

a. If continuity is present, replace K41 (WP 0938 00). Go to **17.**

b. If continuity is not present, go to **5.**

5. Check continuity between:

- **K41-X2 and GND**
- **K41-X1 and K51-A1**
- **K41-A1 and K51-B3**
- **K41-A2 and J243-Y**
- **K41-A2 and K52-A2**
- **K52-A2 and K51-A2**
- **K41-B2 and J243-V**
- **K41-B1 and J243-W**
- **K41-Y1 and K52-A1**
- **K41-Y2 and GND**

a. If continuity is present, go to **6.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17.**

6. Check continuity between:

- **K52-A2 and K52-A3**
- **K52-B2 and K52-B3**

a. If continuity is present, go to **7.**

b. If continuity is not present, replace K52 (WP 0940 00). Go to **17.**

7. Check for no continuity between:

- **K52-A2 and K52-A1**
- **K52-B2 and K52-B1**

a. If continuity is present, replace K52 (WP 0940 00). Go to **17.**

b. If continuity is not present, go to **8.**

8. Connect 28 vdc between K52-X1 and K52-X2. Check continuity between:

K41 SEARCHLIGHT RELAY TEST FAILS - Continued

- **K52-A2 and K52-A1**
- **K52-B2 and K52-B1**

a. If continuity is present, go to **9.**

b. If continuity is not present, replace K52 (WP 0940 00). Go to **17.**

9. Check continuity between:

- **K52-A2 and K52-A3**
- **K52-B2 and K52-B3**

a. If continuity is present, go to **10.**

b. If continuity is not present, replace K52 (WP 0940 00). Go to **17.**

10. Disconnect 28 vdc. Check continuity between:

- **K52-A1 and K41-Y1**
- **K52-A2 and K41-A2**
- **K52-A2 and K51-A2**
- **K52-B2 and K51-X1**
- **K52-B3 and K41-A3**
- **K52-X1 and K51-B2**
- **K52-X2 and GND**

a. If continuity is present, go to **11.**

b. If continuity is not present, replace K52 (WP 0940 00). Go to **17.**

11. Check continuity between:

- **K51-A2 and K51-A3**
- **K51-B2 and K51-B3**

a. If continuity is present, go to **12.**

b. If continuity is not present, replace K51 (WP 0939 00). Go to **17.**

12. Check for no continuity between:

- **K51-A2 and K51-A1**
- **K51-B2 and K51-B1**

a. If continuity is present, replace K51 (WP 0939 00). Go to **17.**

b. If continuity is not present, go to **13.**

13. Connect 28 vdc between K51-X1 and K51-X2. Check continuity between:

- **K51-A2 and K51-A1**
- **K51-B2 and K51-B1**

a. If continuity is present, go to **14.**

K41 SEARCHLIGHT RELAY TEST FAILS - Continued

b. If continuity is not present, replace K51 (WP 0939 00). Go to **17.**

14. Check for no continuity between:

- **K51-A2 and K51-A3**
- **K51-B2 and K51-B3**

a. If continuity is present, replace K51 (WP 0939 00). Go to **17.**

b. If continuity is not present, go to **15.**

15. Disconnect 28 vdc from K51. Check continuity between:

- **K51-X2 and GND**
- **K51-X1 and K52-B2**
- **K51-A1 and K41-X1**
- **K51-A2 and K52-A2**
- **K51-B3 and K41-A1**

a. If continuity is present, go to **16.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17.**

16. Check continuity between:

- **K51-A1 and J1-11**
- **K51-X1 and J1-5**
- **K52-A1 and J1-12**
- **K52-B2 and J1-5**
- **K52-X1 and J1-7**

a. If continuity is present, go to **17.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17.**

17. Procedure completed.

K45 NO. 1 ENGINE START RELAY TEST FAILS

SYMPTOM

K45 No. 1 engine start relay test fails.

MALFUNCTION

K45 No. 1 engine start relay test fails.

CORRECTIVE ACTION

1. Check continuity between:

- **K45-A2 and K45-A3**
- **K45-B2 and K45-B3**
- **K45-C2 and K45-C3**

K45 NO. 1 ENGINE START RELAY TEST FAILS - Continued

- **K45-D2 and K45-D3**
- **K45-E2 and K45-E3**
- **K45-F2 and K45-F3**

a. If continuity is present, go to **2.**

b. If continuity is not present, replace K45 (WP 0941 00). Go to **6.**

2. Check for no continuity between:

- **K45-A2 and K41-A1**
- **K45-B2 and K41-B1**
- **K45-C2 and K41-C1**
- **K45-D2 and K41-D1**
- **K45-E2 and K41-E1**
- **K45-F2 and K41-F1**

a. If continuity is present, replace K45 (WP 0941 00). Go to **6.**

b. If continuity is not present, go to **3.**

3. Connect 28 vdc between K45-X1 and K45-X2. Check continuity between:

- **K45-A2 and K45-A1**
- **K45-B2 and K45-B1**
- **K45-C2 and K45-C1**
- **K45-D2 and K45-D1**
- **K45-E2 and K45-E1**
- **K45-F2 and K45-F1**

a. If continuity is present, go to **4.**

b. If continuity is not present, replace K45 (WP 0941 00). Go to **6.**

4. Check for no continuity between:

- **K45-A2 and K45-A3**
- **K45-B2 and K45-B3**
- **K45-C2 and K45-C3**
- **K45-D2 and K45-D3**
- **K45-E2 and K45-E3**
- **K45-F2 and K45-F3**

a. If continuity is present, replace K45 (WP 0941 00). Go to **6.**

b. If continuity is not present, go to **5.**

5. Disconnect 28 vdc from K45. Check continuity between:

K45 NO. 1 ENGINE START RELAY TEST FAILS - Continued

- **K45-X2 and GND 1-E**
- **K45-X1 and K45-A1**
- **K45-A1 and J242-T**
- **K45-A2 and J242-V**
- **K45-B2 and J242-S**
- **K45-B1 and J242-P**
- **K45-B3 and J242-R**
- **K45-B3 and J1-9**
- **K45-C2 and J242-Z**
- **J242-U and J242-Z**
- **K45-D2 and J242-X**
- **K45-D3 and J242-W**
- **K45-E2 and J902-c**
- **K45-E3 and J902-a**
- **K45-F2 and J902-d**
- **K45-F3 and J902-b**

a. If continuity is present, continue operational checkout. Go to **6.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**

6. Procedure completed.

LOCATION

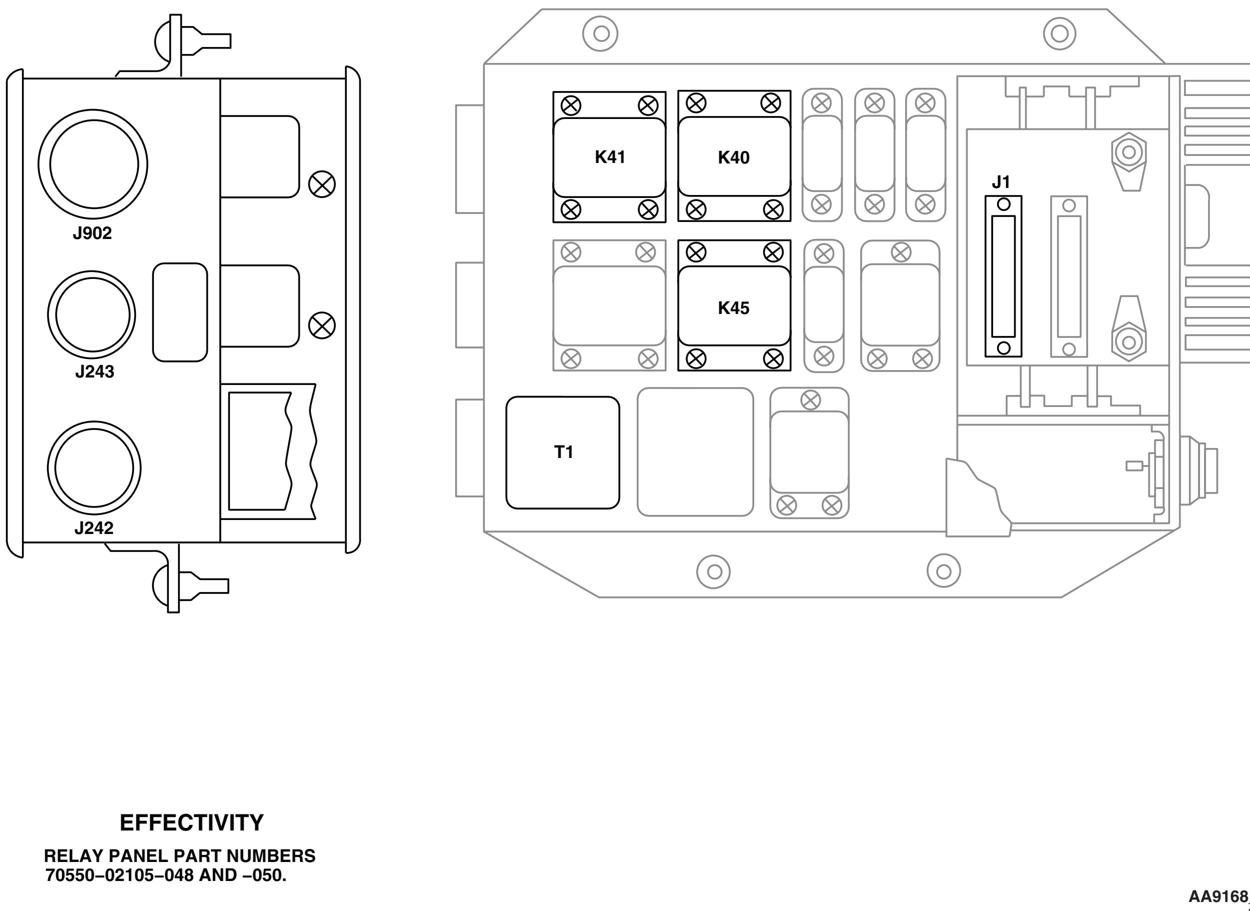


Figure 1. Left Hand Relay Panel Location Diagram. (Sheet 1 of 2)

LOCATION - Continued

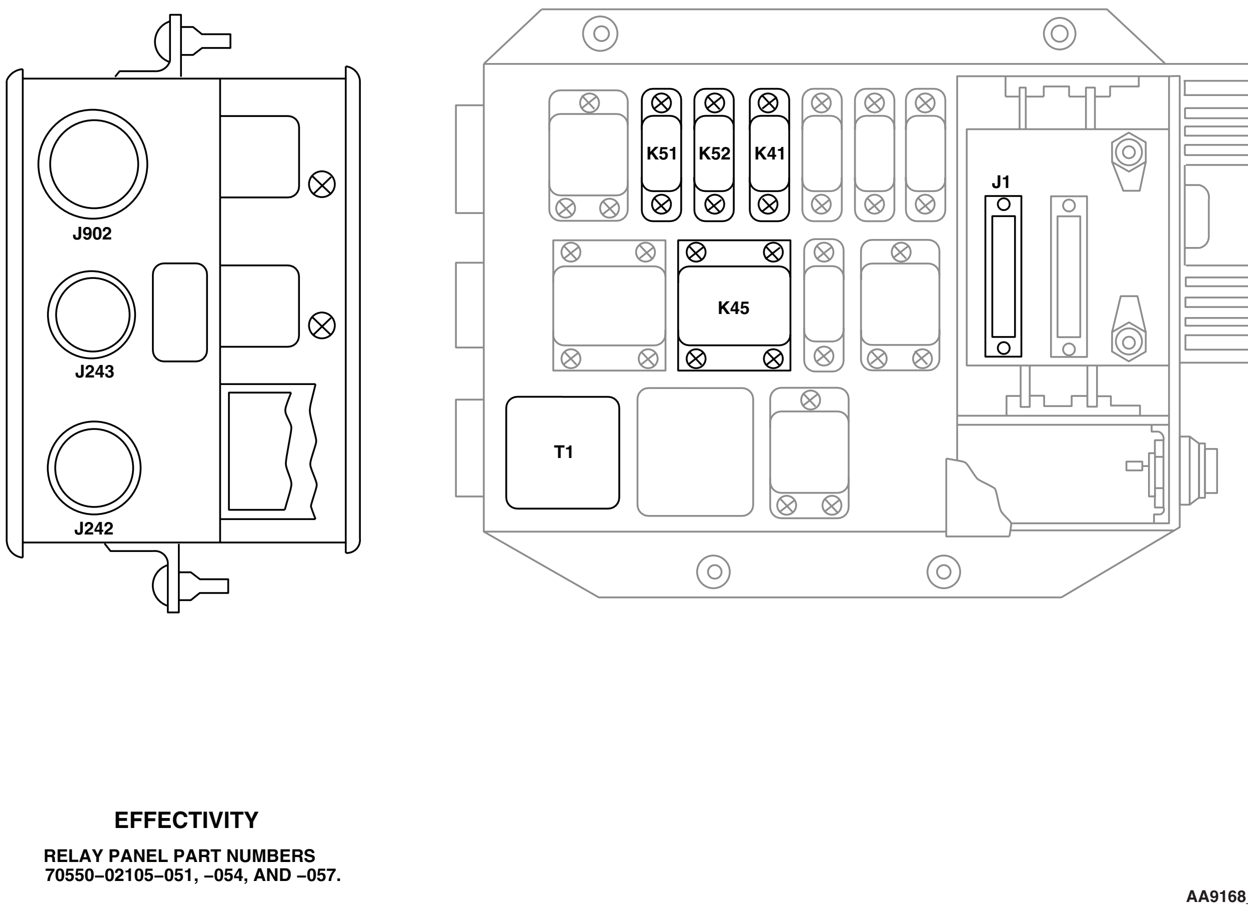


Figure 1. Left Hand Relay Panel Location Diagram. (Sheet 2 of 2)

TEST SETUP

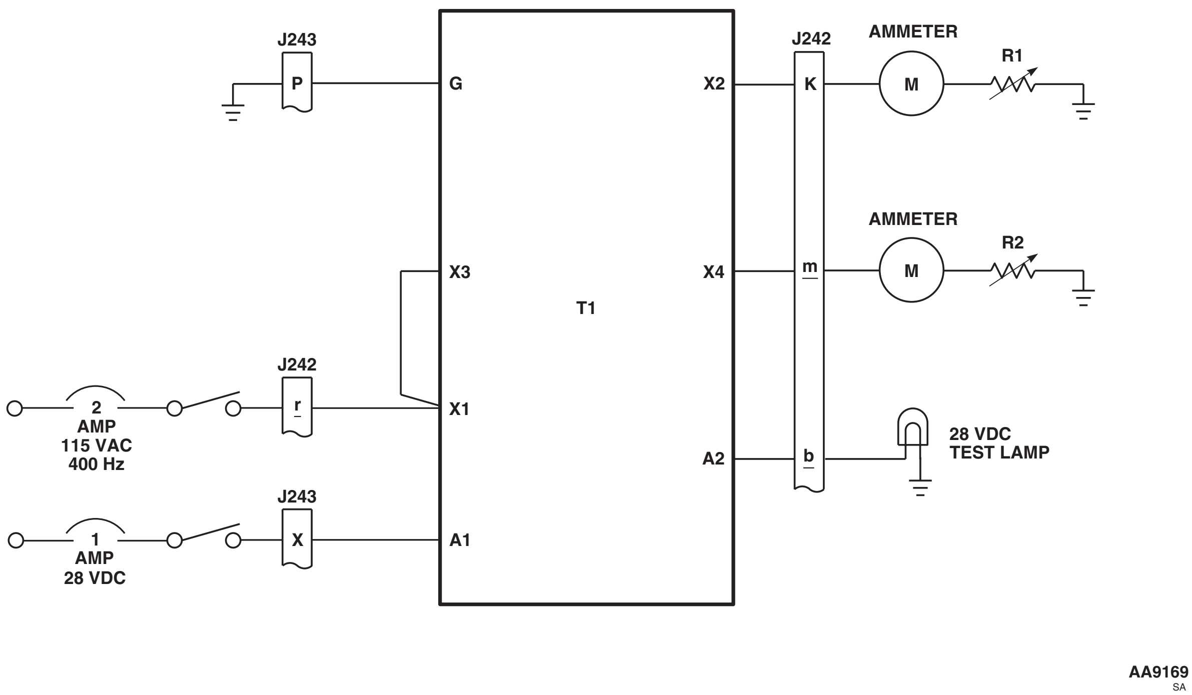


Figure 2. Left Hand Relay Panel Test Setup.

SCHEMATICS

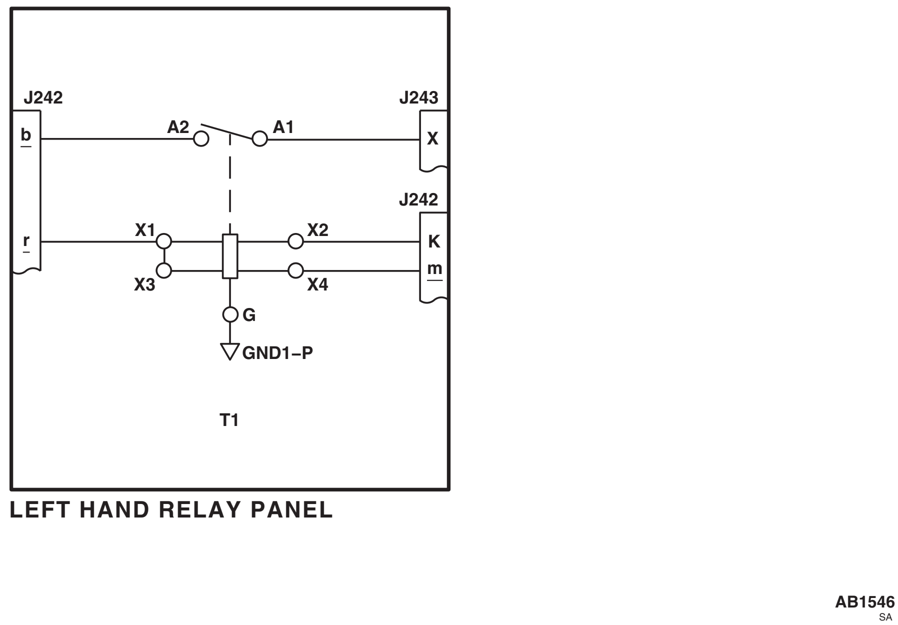
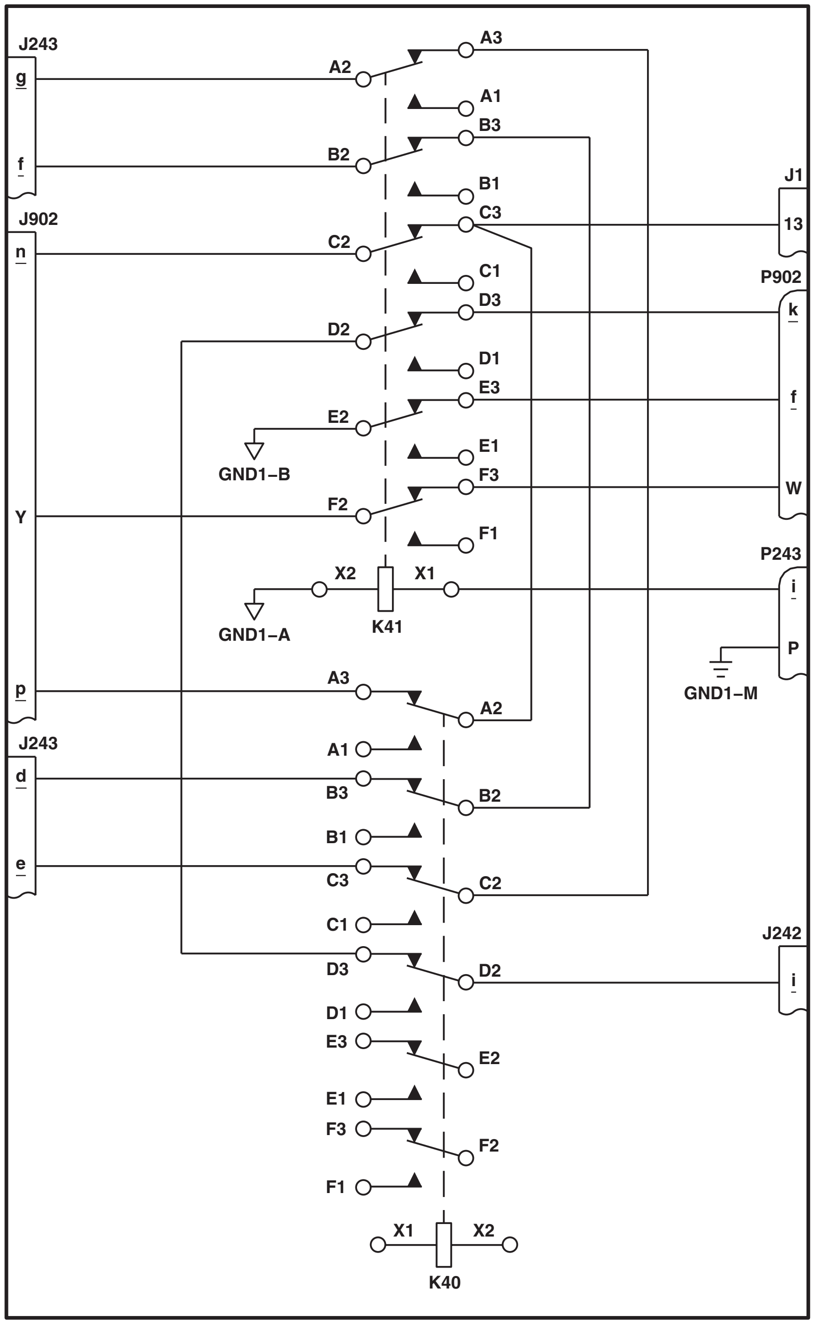


Figure 3. Left Hand Relay Panel Schematic Diagram.

SCHEMATICS - Continued



LEFT HAND RELAY PANEL

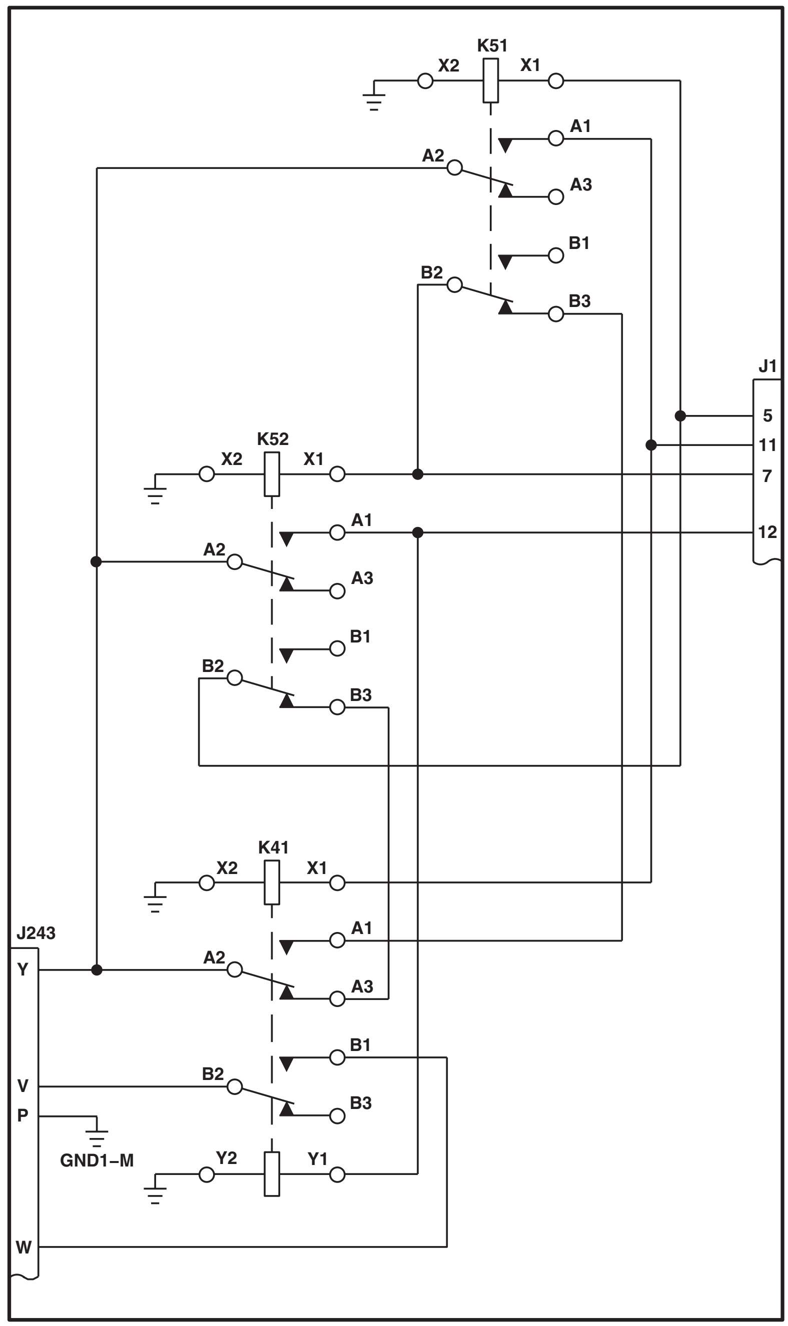
EFFECTIVITY

RELAY PANEL PART NUMBERS
70550−02105−048 AND −050.

AB1548
SA

Figure 4. Left Hand Relay Panel Schematic Diagram **70550-02105-048 AND -050** .

SCHEMATICS - Continued



LEFT RELAY PANEL

EFFECTIVITY

RELAY PANEL PART NUMBERS
70550–02105–051, –054, –057,
AND –058.

AB1547
SA

Figure 5. Left Hand Relay Panel Schematic Diagram **70550-02105-051, -054, -057, AND -058** .

SCHEMATICS - Continued

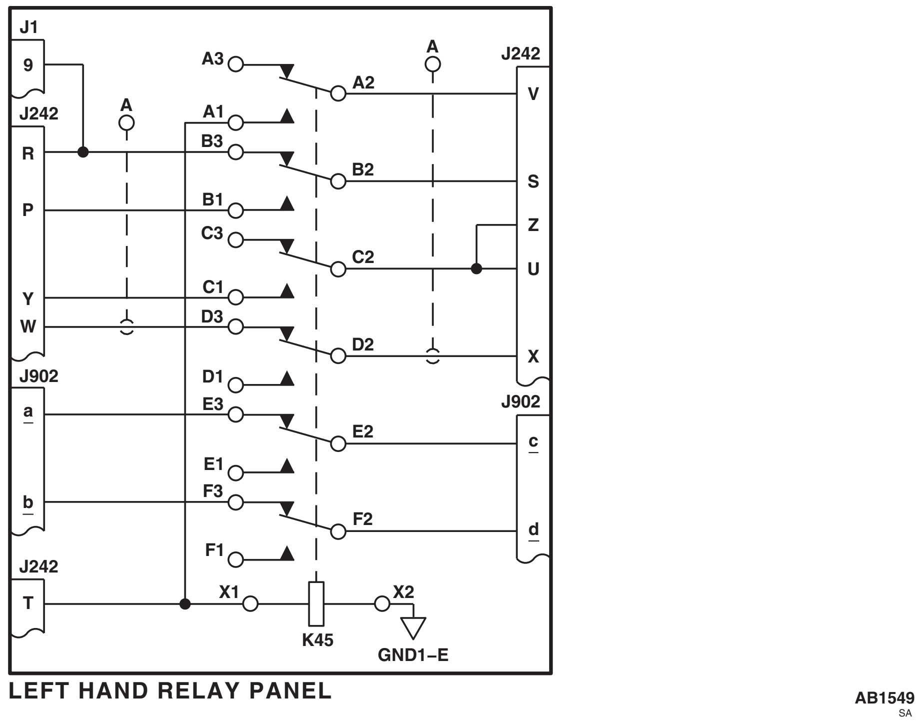


Figure 6. Left Hand Relay Panel Schematic Diagram.

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

AC ELECTRICAL SYSTEM HH-60A HH-60L

This WP supersedes WP 0122 00, dated 21 March 2007.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM- 45A
Power System Analyzer, 60B63-5-A

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer Toolkit, SC 5180-99-B06
Locally-Made Drag Beam Wedge Figure 167, WP 1805 00

Personnel Required
Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1520-237-10
WP 0100 00
WP 0104 00
WP 0123 00
WP 0145 00
WP 0443 00
WP 0799 00
WP 0824 00
WP 0825 00
WP 0828 00
WP 0833 00
WP 0834 00
WP 0835 00
WP 0836 00
WP 0837 00
WP 0838 00
WP 0839 00
WP 0840 00
WP 0843 00
WP 0844 00
WP 0845 00
WP 0925 00
WP 1227 00
WP 1685 00
WP 1747 00
WP 1805 00

Equipment Condition
External Electrical Power Available, WP 1685 00 and APU Operational, TM 1-1520-237-10

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram, Figure 2 for schematic diagram, and HH-60A Figure 3 and HH-60L Figure 4 for distribution diagrams as an aid in operational/troubleshooting.

The AC electrical system operational/troubleshooting procedure is subdivided as follows:

- APU Generator System
- No. 1 Generator System
- No. 2 Generator System
- AC External Power System
- 26 VAC Power System

SPECIAL INSTRUCTIONS - Continued

- UTIL RCPT 115 VAC 3ø 400 Hz Receptacle
- 60 Hz Converter Receptacle
- Generator and GCU Troubleshooting

CAUTION

In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.

Table 1. Temperature Requirements.

AIR TEMPERATURE F° (C°)	OPERATING TIME (MINUTES)	COOLDOWN TIME, (PUMP OFF) (MINUTES)
90° (32°) max	Unlimited	-
91° (33°) to 100° (38°)	24	72
101° (39°) to 125° (52°)	16	48

APU GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following legends on the multifunction display (MFD) will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

APU GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. Place upper console BATT switch to ON.
3. Turn pilot's and copilot's MFD switches ON.

NOTE

Pressing T6 on the MFD will display all legends for approximately 10 seconds, then only the active legends will be displayed.

4. Press MFD T6 switch.

Indication/Condition

a. AC ESS BUS OFF legends on pilot's and copilot's MFDs shall go on.

b. #1 CONV legends on pilot's and copilot's MFDs shall go on.

c. #2 CONV legends on pilot's and copilot's MFDs shall go on.

d. DC ESS BUS OFF legends on pilot's and copilot's MFDs shall be off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to AC ESS BUS OFF LEGEND DOES NOT GO ON WITHOUT POWER ON AC ESNTL BUS, in this work package.
2. If Indication/Condition b., c., or d. is not as specified, troubleshoot DC electrical system (WP 0123 00).

Step

5. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and GENERATORS APU switches are in OFF/RESET position.
6. Start APU (TM 1-1520-253-10).
7. Place upper console GENERATORS APU switch to ON.

Indication/Condition

a. APU GEN ON legends on the pilot's and copilot's MFDs shall go on.

b. #1 CONV legends on the pilot's and copilot's MFDs shall go off.

c. #2 CONV legends on the pilot's and copilot's MFDs shall go off.

d. AC ESS BUS OFF legends on the pilot's and copilot's MFDs shall go off.

e. #1 GEN and #2 GEN legends on the pilot's and copilot's MFDs shall go on.

Corrective Action

1. If Indication/Conditions a., b., and c. are not as specified, go to WITH GENERATORS APU SWITCH ON, APU GEN ON LEGEND DOES NOT GO ON AND #1 CONV AND #2 CONV LEGENDS REMAIN ON (PRIMARY AC BUSES NOT ENERGIZED), in this work package.
2. If Indication/Condition a. is not as specified, go to APU GEN ON LEGEND IS OFF WITH APU GENERATOR SUPPLYING BUSES (#1 CONV AND #2 CONV LEGENDS OFF), in this work package.
3. If Indication/Condition b. is not as specified, go to APU GENERATOR OR AC EXTERNAL POWER DOES NOT SUPPLY NO. 1 AC PRI BUS (#1 CONV LEGEND ON), in this work package.
4. If Indication/Condition c. is not as specified, check wiring from NO. 2 CONVERTER circuit breaker terminals A2, B2, and C2, to NO. 2 GEN CNTOR contactor terminals A2, B2, and C2.
   a. If continuity is not present, repair/replace wiring as required (WP 1747 00).

APU GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

b. If continuity is present, replace NO. 2 GEN CNTOR (WP 0835 00).

5. If Indication/Condition d. is not as specified, go to AC ESS BUS OFF LEGEND DOES NOT GO OFF WHEN NO. 1 AND NO. 2 AC PRIMARY BUSES ARE ENERGIZED, in this work package.
6. If Indication/Condition e. is not as specified, do NO. 1 GENERATOR SYSTEM or NO. 2 GENERATOR SYSTEM, in this work package, as required.

Step

8. Place upper console GENERATORS APU switch to OFF/RESET.
9. Turn pilot's and copilot's MFD switches OFF.
10. Shut down APU (TM 1-1520-253-10).
11. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

NO. 1 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following legends on the multifunction display (MFD) will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Place upper console BATT switch to ON.

NO. 1 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

3. Turn pilot's and copilot's MFD switches ON.

NOTE

Pressing T6 on the MFD will display all legends for approximately 10 seconds, then only the active legends will be displayed.

4. Press MFD T6 switch.

Indication/Condition

#1 GEN BRG legends on pilot's and copilot's MFDs shall not go on.

Corrective Action

If Indication/Condition is not as specified, go to #1 GEN BRG AND/OR #2 GEN BRG LEGEND GOES ON, in this work package.

Step

5. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and GENERATORS APU switches are placed to OFF/RESET.
6. Start APU (TM 1-1520-253-10).
7. Place upper console GENERATORS APU switch to ON.

Indication/Condition

a. #1 GEN legends on pilot's and copilot's MFDs shall go on.
b. #2 GEN legends on pilot's and copilot's MFDs shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #1 GEN LEGEND DOES NOT GO ON WITH NO. 1 GENERATOR OFF, in this work package.
2. If Indication/Condition b. is not as specified, do NO. 2 GENERATOR SYSTEM, in this work package.

Step

8. With qualified personnel at the controls, start No. 1 engine and operate with main rotor turning at 100% (TM 1-1520-253-10).
9. Place upper console GENERATORS NO. 1 switch to ON.

Indication/Condition

a. #1 GEN legends on pilot's and copilot's MFDs shall go off.
b. #2 GEN legends on pilot's and copilot's MFDs shall remain on.
c. #1 CONV legends on pilot's and copilot's MFDs shall go off.
d. #2 CONV legends on pilot's and copilot's MFDs shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #1 GEN LEGEND IS ON WITH GENERATORS NO. 1 SWITCH ON, in this work package.
2. If Indication/Condition b. is not as specified, do NO. 2 GENERATOR SYSTEM, in this work package.

NO. 1 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

3. If Indication/Condition c. is not as specified, go to #1 CONV LEGEND STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 1 AC PRIMARY BUS, in this work package.
4. If Indication/Condition d. is not as specified, go to #2 CONV LEGEND STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 2 AC PRIMARY BUS, in this work package.
5. If No. 1 generator is disabled due to underfrequency, go to NO. 1 AND/OR NO. 2 GENERATOR DISABLED BY UNDERFREQUENCY CONDITION DURING AUTO ROTATIVE DESCENT, in this work package.

Step

10. Place upper console GENERATORS NO. 1 switch to OFF/RESET.
11. Turn pilot's and copilot's MFD switches OFF.
12. Shut down No. 1 engine and APU (TM 1-1520-253-10).
13. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

NO. 2 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following legends on the multifunction display (MFD) will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

NO. 2 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. Place upper console BATT switch to ON.
3. Turn pilot's and copilot's MFD switches ON.

NOTE

Pressing T6 on the MFD will display all legends for approximately 10 seconds, then only the active legends will be displayed.

4. Press MFD T6 switch.

Indication/Condition

#2 GEN BRG legends on pilot's and copilot's MFDs shall not go on.

Corrective Action

If Indication/Condition is not as specified, go to #1 GEN BRG AND/OR #2 GEN BRG LEGEND GOES ON, in this work package.

Step

5. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and GENERATORS APU switches are placed to OFF/RESET.
6. Start APU (TM 1-1520-253-10).
7. Place upper console GENERATORS APU switch to ON.

Indication/Condition

a. #1 GEN legends on pilot's and copilot's MFDs shall go on.

b. #2 GEN legends on pilot's and copilot's MFDs shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, do NO. 1 GENERATOR SYSTEM, in this work package.
2. If Indication/Condition b. is not as specified, go to #2 GEN LEGEND IS OFF WITH NO. 2 GENERATOR OFF, in this work package.

Step

8. With qualified personnel at the controls, start No. 2 engine and operate with main rotor turning at 100% (TM 1-1520-253-10).
9. Place upper console GENERATORS NO. 2 switch to ON.

Indication/Condition

a. #2 GEN legends on pilot's and copilot's MFDs shall go off.

b. #1 GEN legends on pilot's and copilot's MFDs shall remain on.

c. #1 CONV and #2 CONV legends on pilot's and copilot's MFDs shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #2 GEN LEGEND IS ON WITH GENERATORS NO. 2 SWITCH ON, in this work package.
2. If Indication/Condition b. is not as specified, do NO. 1 GENERATOR SYSTEM, in this work package.
3. If Indication/Condition c. is not as specified, go to #1 CONV LEGEND STAYS ON WHEN EITHER

NO. 2 GENERATOR SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

GENERATOR IS SUPPLYING POWER TO NO. 1 AC PRIMARY BUS or #2 CONV LEGEND STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 2 AC PRIMARY BUS, in this work package, as required.

4. If No. 2 generator is disabled due to underfrequency, go to NO. 1 AND/OR NO. 2 GENERATOR DISABLED BY UNDERFREQUENCY CONDITION DURING AUTO ROTATIVE DESCENT, in this work package.

Step

10. Place upper console GENERATORS NO. 2 switch to OFF/RESET.
11. Turn pilot's and copilot's MFD switches OFF.
12. Shut down No. 2 engine and APU (TM 1-1520-253-10).
13. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

AC EXTERNAL POWER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following legends on the multifunction display (MFD) will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Place upper console BATT switch to ON.

AC EXTERNAL POWER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

3. Turn pilot's and copilot's MFD switches ON.

NOTE

Pressing T6 on the MFD will display all legends for approximately 10 seconds, then only the active legends will be displayed.

4. Press MFD T6 switch.
5. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and GENERATORS APU switches are in OFF/RESET.
6. Turn on external electrical power.

Indication/Condition

a. EXT PWR CONNECTED legends on pilot's and copilot's MFDs shall go on.

b. #1 CONV and #2 CONV legends on pilot's and copilot's MFDs shall remain on.

c. AC ESS BUS OFF legends on pilot's and copilot's MFDs shall remain on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to EXT PWR CONNECTED LEGEND IS OFF WITH A SOURCE OF EXTERNAL POWER CONNECTED, in this work package.
2. If Indication/Condition b. is not as specified, troubleshoot DC electrical system (WP 0123 00).
3. If Indication/Condition c. is not as specified, go to AC ESS BUS OFF LEGEND DOES NOT GO ON WITHOUT POWER ON AC ESNTL BUS, in this work package.

Step

7. Momentarily place upper console EXT PWR switch to RESET and then to ON.

Indication/Condition

a. #1 CONV legends on pilot's and copilot's MFDs shall go off.

b. #2 CONV legends on pilot's and copilot's MFDs shall go off.

c. #1 GEN and #2 GEN legends on pilot's and copilot's MFDs shall go on.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, go to EXTERNAL POWER DOES NOT SUPPLY AC BUSES (#1 CONV AND #2 CONV LEGENDS ON), in this work package.
2. If Indication/Condition a. is not as specified, go to APU GENERATOR OR AC EXTERNAL POWER DOES NOT SUPPLY NO. 1 AC PRI BUS (#1 CONV LEGEND ON), in this work package.
3. If Indication/Condition c. is not as specified, do NO. 1 GENERATOR SYSTEM or NO. 2 GENERATOR SYSTEM, in this work package, as required.

Step

8. Turn pilot's and copilot's MFD switches OFF.
9. Turn off external electrical power.
10. On upper console, place EXT PWR RESET/OFF/ON switch to OFF.
11. Place upper console BATT switch to OFF.

AC EXTERNAL POWER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

26 VAC POWER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following legends on the multifunction display (MFD) will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and GENERATORS APU switches are set to OFF/RESET.
3. Place upper console BATT switch to ON.
4. Turn on external electrical power.
5. Check for 26 vac between terminal 2 of 26 VAC STAB IND and 26 VAC DPLR/GPS circuit breakers and ground.

Indication/Condition

Multimeter shall indicate 26 vac.

Corrective Action

If Indication/Condition is not as specified, go to NO 26 VAC POWER ON AC ESNTL BUS, in this work package.

Step

6. Turn off external electrical power.
7. Place upper console BATT switch to OFF.

26 VAC POWER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

UTIL RCPT 115 VAC 3Ø 400 HZ RECEPTACLE OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following legends on the multifunction display (MFD) will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure UTIL RECP circuit breaker on pilot's circuit breaker panel is pushed in.
2. Place upper console BATT switch to ON.
3. Turn on external electrical power.
4. Check for 115 vac between the following:
   - J257-A and ground
   - J257-B and ground
   - J257-C and ground

Indication/Condition

Multimeter shall indicate 115 vac.

Corrective Action

1. If Indication/Condition is not as specified, check for 115 vac between terminals A2, B2, and C2 of UTIL RECP circuit breaker, and ground.
   a. If voltage is as specified, repair/replace wiring between circuit breaker and J257 as required (WP 1747 00).

UTIL RCPT 115 VAC 3Ø 400 HZ RECEPTACLE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

b. If voltage is not as specified, replace circuit breaker (WP 0824 00).

Step

5. Turn off external electrical power.
6. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch to off.

Corrective Action

None Required

60 HZ AC CONVERTER RECEPTACLE OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following legends on the multifunction display (MFD) will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Make sure 60 HZ AC CONVERTER circuit breaker on copilot's circuit breaker panel is pushed in.
2. Place upper console BATT switch to ON.
3. Turn on external electrical power.
4. Check for 115 vac between the following:
   - J276-A and ground
   - J276-B and ground
   - J276-C and ground

Indication/Condition

Multimeter shall indicate 115 vac.

60 HZ AC CONVERTER RECEPTACLE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition is not as specified, check for 115 vac between terminals A1, B1, and C1 of 60 HZ AC CONVERTER circuit breaker and ground.
   a. If voltage is as specified, repair/replace wiring between circuit breaker and P276 as required (WP 1747 00).
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00).

Step

5. Turn off external electrical power.
6. Place upper console BATT switch to OFF.

Indication/Condition

Upper console BATT switch shall be off.

Corrective Action

None Required

GENERATOR AND GCU OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds turn off electrical power and troubleshoot blade de-ice system (WP 0145 00).
- If generator control unit is removed from its mounting at any time during system troubleshooting, the generator control unit case must be grounded to airframe ground.

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following legends on the multifunction display (MFD) will go on from battery power:

  AC ESS BUS OFF

  #1 CONV

  #2 CONV

1. Place upper console GENERATORS NO. 1, GENERATORS NO. 2, or GENERATORS APU switch to OFF/RESET, as required.

GENERATOR AND GCU OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. Place power system analyzer GEN, CUR LIM, and FF toggle switches to OFF, and all other toggle switches down.
3. Turn power system analyzer TEST SELECT switch to START TEST.
4. Connect P1 of adapter cable (supplied with analyzer) to J1 AIRCRAFT connector on analyzer.

CAUTION

Always shut down APU or No. 1 engine (as applicable) before disconnecting generator control unit.

5. Disconnect P214, P215, or P216 from generator control unit and connect to P3 of analyzer adapter cable, as required.
6. Connect P2 of analyzer adapter cable to generator control unit.
7. Make sure APU CONTR INST circuit breakers on lower console circuit breaker panel are pushed in.
8. Start APU (TM 1-1520-253-10).
9. With qualified personnel at controls, start No. 1 engine and operate with main rotor turning at 100% rpm (for No. 1 or No. 2 main generator) (TM 1-1520-253-10).

CAUTION

All tests with the system analyzer must be done with the GENERATORS NO. 1, GENERATORS NO. 2, or GENERATORS APU switch set to OFF/RESET to prevent applying a simulated fault to any helicopter load.

10. Turn power system analyzer GEN/PMG VOLTS meter switch to A-B, B-C, and C-A (PMG 30V RANGE) and check pmg voltage on meter for each switch position.

Indication/Condition

Each voltage indication shall be more than 20.5 volts.

Corrective Action

If Indication/Condition is not as specified, replace generator (WP 0839 00).

Step

11. On power system analyzer, hold TEST SET switch at RESET and GEN switch to ON; then release TEST SET switch.
12. Turn power system analyzer GEN/PMG VOLTS meter switch to T-1, T-2, and T-3 (GEN 140V RANGE) and check generator output voltage on meter for each switch position.

Indication/Condition

Each voltage indication shall be between 112.5 and 117.5 vac.

Corrective Action

If Indication/Condition is not as specified, replace generator (WP 0839 00).

Step

13. Check field current on power system analyzer FIELD AMPS/REG VOLTS meter.

Indication/Condition

Field current shall be 1.0 ampere maximum.

GENERATOR AND GCU OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

If Indication/Condition is not as specified, replace generator (WP 0839 00).

Step

14. Check power system analyzer FREQUENCY meter indication.

Indication/Condition

FREQUENCY meter shall indicate between 380 and 420 Hz and ø SEQ T1-2-3 lamp shall be on.

Corrective Action

If Indication/Condition is not as specified, replace generator (WP 0839 00).

Step

15. Check power system analyzer GCR RESET, CCR RESET, CCR TRIP, SIM CONT, and TEST RELAY lamps.

Indication/Condition

GCR RESET, CCR RESET, SIM CONT, and TEST RELAY lamps shall be on and CCR TRIP lamp shall be off.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

16. Turn power system analyzer TEST SELECT switch to NL REG.
17. Place power system analyzer FIELD AMPS/REG VOLTS meter switch to READ REG BUS VOLTS and check voltage indication on meter.

Indication/Condition

Voltage indication shall be between 21 and 27 volts.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

18. Place power system analyzer FIELD AMPS/REG VOLTS meter switch to FLD CUR and GEN switch to OFF. Check GCR RESET, CCR RESET, CCR TRIP, SIM CONT, SIM WARN, and TEST RELAY lamps.

Indication/Condition

CCR TRIP, SIM WARN, and TEST RELAY lamps shall be on and GCR RESET, CCR RESET, and SIM CONT lamps shall be off.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

19. Hold power system analyzer TEST SET switch at RESET and hold GEN switch to TEST; then only release TEST SET switch.

GENERATOR AND GCU OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

20. Check power system analyzer GCR RESET, CCR RESET, CCR TRIP, SIM CONT, SIM WARN, and TEST RELAY lamps.

Indication/Condition

GCR RESET and CCR RESET lamps shall be on and CCR TRIP, SIM CONT, SIM WARN, and TEST RELAY lamps shall be off.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

21. Hold power system analyzer TEST SET switch to RESET and only release GEN switch to OFF, then place to ON; then only release TEST SET switch.
22. Turn power system analyzer GEN/PMG VOLTS meter switch to T-1, T-2, and T-3 (GEN 140V RANGE) and record generator output voltage on meter for each switch position.

Indication/Condition

Each voltage indication shall be between 112.5 and 117.5 vac.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

23. Momentarily place power system analyzer FLYWHEEL DIODE switch to TEST.

Indication/Condition

FLYWHEEL DIODE OPEN lamp shall be off.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

24. Turn power system analyzer TEST SELECT switch to FL REG.
25. Turn power system analyzer GEN/PMG VOLTS meter switch to A-B, B-C, and C-A (PMG 30V RANGE) and check pmg voltage on meter for each switch position.

Indication/Condition

Each voltage indication shall be at least 19 volts.

Corrective Action

If Indication/Condition is not as specified, replace generator (WP 0839 00).

Step

26. Turn power system analyzer GEN/PMG VOLTS meter switch to T-1, T-2, and T-3 (GEN 140V RANGE) and check generator output voltage on meter for each switch position.

Indication/Condition

Each voltage indication shall not change more than 2.5 vac from indication recorded in step 22.

GENERATOR AND GCU OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

27. Turn power system analyzer GEN/PMG VOLTS meter switch to T-1 (GEN 140V RANGE).
28. Place power system analyzer CUR LIM switch to 20 KVA.

NOTE

Voltage checks in step 29. must be done within 4 seconds of selection of each switch position, to prevent an undervoltage trip. If an undervoltage trip occurs, turn TEST SELECT switch to FL REG and place GEN switch OFF. Then, with TEST SET switch held at RESET, place GEN switch ON and repeat voltage checks.

29. Turn power system analyzer TEST SELECT switch to A ø CUR LIM, B ø CUR LIM, and C ø CUR LIM and check GEN/PMG VOLTS meter T-1 indication for each switch position.

Indication/Condition

Each voltage indication shall be between 90 and 110 volts.

Corrective Action

If Indication/Condition is not as specified, replace generator control unit (WP 0838 00).

Step

30. Place power system analyzer CUR LIM switch to 30 KVA and repeat step 29.
31. Place power system analyzer CUR LIM switch to OFF.
32. Turn power system analyzer TEST SELECT switch to A ø GEN CT, press CT CUR switch to READ, and check CT TEST CURRENT meter indication.

Indication/Condition

NOTE

It is normal for the 60B63-5A Power System Analyzer to indicate a fault when it is switched to test current transformers and contactors.

Current indication shall be between 0.70 and 1.70 mA.

Corrective Action

If Indication/Condition is not as specified, replace generator (WP 0839 00).

Step

33. Repeat step 32. with power system analyzer TEST SELECT switch in B ø GEN CT and C ø GEN CT positions.
34. Disconnect P333 from auxiliary AC junction box.
35. Turn power system analyzer TEST SELECT switch to A ø DP CT, press CT CUR switch to READ, and check CT TEST CURRENT meter indication.

Indication/Condition

Current indication shall be between 0.6 and 2.0 mA.

GENERATOR AND GCU OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

If Indication/Condition is not as specified, replace current transformer (WP 0837 00).

Step

36. Repeat step 35. with power system analyzer TEST SELECT switch at B ø DP CT and C ø DP CT.
37. Place power system analyzer GEN switch OFF and turn TEST SELECT switch to START TEST.
38. Check continuity between the following:
    - J333-C and J333-D
    - J333-E and J333-F
    - J333-G and J333-H

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, replace auxiliary AC junction box (WP 1227 00).

Step

39. Disconnect P202 from No. 2 junction box.
40. Check continuity between the following:
    - J333-C and J333-D
    - J333-E and J333-F
    - J333-G and J333-H

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, replace No. 2 junction box (WP 1227 00).

Step

41. Install P202 and P333.
42. Shut down APU and/or No. 1 engine, as applicable (TM 1-1520-253-10).
43. Disconnect P214, P215, or P216 (as applicable) from P3 of analyzer adapter cable and connect it to generator control unit.
44. Disconnect adapter cable from analyzer.

Indication/Condition

Adapter cable shall be disconnected.

Corrective Action

None Required

AC ESS BUS OFF LEGEND DOES NOT GO ON WITHOUT POWER ON AC ESNTL BUS

SYMPTOM

AC ESS BUS OFF legend does not go on without power on AC ESNTL bus.

MALFUNCTION

AC ESS BUS OFF legend does not go on without power on AC ESNTL bus.

CORRECTIVE ACTION

1. **Check for 28 vdc between P1019R-19 (copilot's)/P1020R-19 (pilot's) and ground.**
   a. If voltage is as specified, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **5.**
   b. If voltage is not as specified, go to **2.**
2. **Check for 28 vdc between K13 AC ESNTL BUS FAIL relay terminal A3 and ground.**
   a. If voltage is as specified, repair/replace wiring between K13 AC ESNTL BUS FAIL relay terminal A3 and P1019R-19 (copilot's)/P1020R-19 (pilot's) (WP 1747 00). Go to **5.**
   b. If voltage is not as specified, go to **3.**
3. **Check for 28 vdc between K13 AC ESNTL BUS FAIL relay terminal A2 and ground.**
   a. If voltage is as specified, replace relay (WP 0845 00). Go to **5.**
   b. If voltage is not as specified, go to **4.**
4. **Check for 28 vdc between ESNTL BUS AC & CONV WARN circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, repair/replace wiring between circuit breaker and relay terminal A2 (WP 1747 00). Go to **5.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **5.**
5. **Procedure completed.**

WITH GENERATORS APU SWITCH ON, APU GEN ON LEGEND DOES NOT GO ON AND #1 CONV AND #2 CONV LEGENDS REMAIN ON (PRIMARY AC BUSES NOT ENERGIZED)

SYMPTOM

With GENERATORS APU switch ON, APU GEN ON legend does not go on and #1 CONV and #2 CONV legends remain on (primary ac buses not energized).

MALFUNCTION

With GENERATORS APU switch ON, APU GEN ON legend does not go on and #1 CONV and #2 CONV legends remain on (primary ac buses not energized).

CORRECTIVE ACTION

1. **Check for 115 vac between:**
   - **J207-A and J207-D**
   - **J207-B and J207-D**
   - **J207-C and J207-D**

WITH GENERATORS APU SWITCH ON, APU GEN ON LEGEND DOES NOT GO ON AND #1 CONV AND #2 CONV LEGENDS REMAIN ON (PRIMARY AC BUSES NOT ENERGIZED) - Continued

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **24.**

2. **Remove and check pin filtered adapter from P216 (WP 0925 00).**

   a. If pin filtered adapter is good, go to **3.**

   b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **25.**

3. **Install pin filtered adapter on P216 (WP 0925 00). Go to 4.**

4. **Check for 28 vdc between GENERATORS APU switch terminal 2 and ground.**

   a. If voltage is as specified, go to **5.**

   b. If voltage is not as specified, go to **23.**

5. **Check for 28 vdc between GENERATORS APU switch terminal 5 and ground.**

   a. If voltage is as specified, go to **6.**

   b. If voltage is not as specified, go to **22.**

6. **Place GENERATORS APU switch to TEST and check that APU GEN ON legend goes on.**

   a. If APU GEN ON legend is on, go to **7.**

   b. If APU GEN ON legend is not on, go to **20.**

7. **Place GENERATORS APU switch to OFF/RESET and then ON. Go to 8.**

8. **Check for 28 vdc between GENERATORS APU switch terminal 6 and ground.**

   a. If voltage is as specified, go to **9.**

   b. If voltage is not as specified, replace GENERATORS APU switch (WP 0828 00). Go to **25.**

9. **Check for 28 vdc between P205-P and P205-N.**

   a. If voltage is as specified, go to **10.**

   b. If voltage is not as specified, go to **19.**

10. **Check for 28 vdc between P206-C and ground.**

    a. If voltage is as specified, go to **11.**

    b. If voltage is not as specified, replace K3 APU/EXT PWR CNTOR (WP 0840 00). Go to **25.**

11. **Check for 28 vdc between P241-J and ground.**

    a. If voltage is as specified, go to **12.**

    b. If voltage is not as specified, repair/replace wiring between P230-r̲ and P241-J (WP 1747 00). Go to **25.**

12. **Check for 28 vdc between P203-S and ground.**

    a. If voltage is as specified, go to **13.**

    b. If voltage is not as specified, go to **18.**

13. **Check for 28 vdc between P210-U and ground.**

    a. If voltage is as specified, go to **14.**

WITH GENERATORS APU SWITCH ON, APU GEN ON LEGEND DOES NOT GO ON AND #1 CONV AND #2 CONV LEGENDS REMAIN ON (PRIMARY AC BUSES NOT ENERGIZED) - Continued

b. If voltage is not as specified, go to **17.**

14. Check for 28 vdc between P206-P and P206-N.

a. If voltage is as specified, go to **15.**

b. If voltage is not as specified, go to **16.**

15. Check for 28 vdc between P1019R-30 (copilot's)/P1020R-30 (pilot's) and ground.

a. If voltage is as specified, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **25.**

b. If voltage is not as specified, replace K4 AC BUS TIE CNTOR (WP 0844 00). Go to **25.**

16. Check continuity between P206-N and ground.

a. If continuity is present, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **25.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **25.**

17. Check continuity between P210-U and P203-P.

a. If continuity is present, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **25.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **25.**

18. Check continuity between P203-S and P241-P.

a. If continuity is present, replace right relay panel (WP 0825 00). Go to **25.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **25.**

19. Check continuity between P205-N and ground.

a. If continuity is present, repair/replace wiring between P205-P and GENERATORS APU switch terminal 6 (WP 1747 00). Go to **25.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **25.**

20. Check for 28 vdc between GENERATORS APU switch terminal 4 and ground.

a. If voltage is as specified, go to **21.**

b. If voltage is not as specified, replace GENERATORS APU switch (WP 0828 00). Go to **25.**

21. Check for 28 vdc between P1019R-30 (copilot's)/P1020R-30 (pilot's) and ground.

a. If voltage is as specified, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **25.**

b. If voltage is not as specified, repair/replace wiring between GENERATORS APU switch terminal 4 and P1019R-30/P1020R-30 (WP 1747 00). Go to **25.**

22. Check continuity between GENERATORS APU switch terminal 5 and P216-e.

a. If continuity is present, replace generator control unit (WP 0838 00). Go to **25.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **25.**

23. Check continuity between P230-a and P216-c.

a. If continuity is present, replace generator control unit (WP 0838 00). Go to **25.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **25.**

24. Check continuity between the following for APU GENERATOR:

WITH GENERATORS APU SWITCH ON, APU GEN ON LEGEND DOES NOT GO ON AND #1 CONV AND #2 CONV LEGENDS REMAIN ON (PRIMARY AC BUSES NOT ENERGIZED) - Continued

- **Terminal T1 and J207-A**
- **Terminal T2 and J207-B**
- **Terminal T3 and J207-C**

a. If continuity is present, troubleshoot generator and GCU using system analyzer. Go to **25.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **25.**

25. Procedure completed.

APU GEN ON LEGEND IS OFF WITH APU GENERATOR SUPPLYING BUSES (#1 CONV AND #2 CONV LEGENDS OFF)

SYMPTOM

APU GEN ON legend is off with APU generator supplying buses (#1 CONV and #2 CONV legends off).

MALFUNCTION

APU GEN ON legend is off with APU generator supplying buses (#1 CONV and #2 CONV legends off).

CORRECTIVE ACTION

1. Press MFD T6 switch and check that APU GEN ON legend goes on.

a. If APU GEN ON legend is on, go to **2.**

b. If APU GEN ON legend is not on, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **7.**

2. Check for 28 vdc between P1019R-30 (copilot's)/P1020R-30 (pilot's) and ground.

a. If voltage is as specified, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **7.**

b. If voltage is not as specified, go to **3.**

3. Check for 28 vdc between P205-K and ground.

a. If voltage is as specified, go to **4.**

b. If voltage is not as specified, repair/replace wiring between P205-K and J203-T (WP 1747 00). Go to **7.**

4. Check for 28 vdc between P206-C and ground.

a. If voltage is as specified, go to **5.**

b. If voltage is not as specified, go to **6.**

5. Check continuity between P1019R-30 (copilot's)/P1020R-30 (pilot's) and P206-B.

a. If continuity is present, replace K4 AC BUS TIE CNTOR (WP 0844 00). Go to **7.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7.**

6. Check continuity between P205-L and P206-C.

a. If continuity is present, replace K3 APU/EXT PWR CNTOR (WP 0840 00). Go to **7.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7.**

7. Procedure completed.

APU GENERATOR OR AC EXTERNAL POWER DOES NOT SUPPLY NO. 1 AC PRI BUS (#1 CONV LEGEND ON)

SYMPTOM

APU generator or ac external power does not supply NO. 1 AC PRI bus (#1 CONV legend on).

MALFUNCTION

APU generator or ac external power does not supply NO. 1 AC PRI bus (#1 CONV legend on).

CORRECTIVE ACTION

1. **Check for 115 vac between the following for K1 NO. 1 GEN CNTOR:**
   - **Terminal A2 and ground**
   - **Terminal B2 and ground**
   - **Terminal C2 and ground**

   a. If voltage is as specified, repair/replace wiring between K1 NO. 1 GEN CNTOR and NO. 1 CONVERTER circuit breaker (WP 1747 00). Go to **4.**

   b. If voltage is not as specified, go to **2.**

2. **Check for 115 vac between the following for K1 NO. 1 GEN CNTOR:**
   - **Terminal A3 and ground**
   - **Terminal B3 and ground**
   - **Terminal C3 and ground**

   a. If voltage is as specified, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **4.**

   b. If voltage is not as specified, go to **3.**

3. **Check for open AC current limiters CL-1, CL-2, and CL-3.**

   a. If AC current limiters are open, replace ac current limiters as required (WP 0836 00). Go to **4.**

   b. If AC current limiters are not open, repair/replace wiring between K1 NO. 1 GEN CNTOR and K2 NO. 2 GEN CNTOR as required (WP 1747 00). Go to **4.**

4. **Procedure completed.**

AC ESS BUS OFF LEGEND DOES NOT GO OFF WHEN NO. 1 AND NO. 2 AC PRIMARY BUSES ARE ENERGIZED

SYMPTOM

AC ESS BUS OFF legend does not go off when No. 1 and No. 2 ac primary buses are energized.

MALFUNCTION

AC ESS BUS OFF legend does not go off when No. 1 and No. 2 ac primary buses are energized.

CORRECTIVE ACTION

1. **Check for 115 vac between K13 AC ESNTL BUS FAIL relay terminals X1 and X2.**

   a. If voltage is as specified, replace relay (WP 0845 00). Go to **15.**

AC ESS BUS OFF LEGEND DOES NOT GO OFF WHEN NO. 1 AND NO. 2 AC PRIMARY BUSES ARE ENERGIZED - Continued

b. If voltage is not as specified, go to **2.**

2. Check for 115 vac between K13 AC ESNTL BUS FAIL relay terminal X1 and ground.

a. If voltage is as specified, repair/replace wiring between K13 AC ESNTL BUS FAIL relay terminal X2 and ground (WP 1747 00). Go to **15.**

b. If voltage is not as specified, go to **3.**

3. Check for 115 vac between AC ESNTL BUS WARN circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between AC ESNTL BUS WARN circuit breaker terminal 1 and K13 AC ESNTL BUS FAIL relay terminal X1 (WP 1747 00). Go to **15.**

b. If voltage is not as specified, replace AC ESNTL BUS WARN circuit breaker (WP 0824 00). Go to **15.**

c. If trouble remains, go to **4.**

4. Check for 115 vac between K8 AC ESNTL BUS XFER relay terminal A2 and ground.

a. If voltage is as specified, go to **5.**

b. If voltage is not as specified, go to **13.**

5. Pull out AC ESNTL BUS SPLY circuit breaker (copilot's circuit breaker panel). Go to 6.

6. Check for 115 vac between K8 AC ESNTL BUS XFER relay terminal A2 and ground.

a. If voltage is as specified, go to **7.**

b. If voltage is not as specified, go to **9.**

7. Check for 115 vac between K8 AC ESNTL BUS XFER relay terminal A3 and ground.

a. If voltage is as specified, replace K8 AC ESNTL BUS XFER relay (WP 0845 00). Go to **15.**

b. If voltage is not as specified, go to **8.**

8. Check for 115 vac between AC ESNTL BUS SPLY circuit breaker (pilot's circuit breaker panel) terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between AC ESNTL BUS SPLY circuit breaker terminal 1 and K8 AC ESNTL BUS XFER relay terminal A3 (WP 1747 00). Go to **15.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **15.**

9. Push in AC ESNTL BUS SPLY circuit breaker (copilot's circuit breaker panel). Go to 10.

10. Check for 115 vac between K8 AC ESNTL BUS XFER terminal A2 and ground.

a. If voltage is as specified, repair/replace wiring between K8 AC ESNTL BUS XFER relay terminal A2 and AC ESNTL BUS WARN circuit breaker terminal 2 (WP 1747 00). Go to **15.**

b. If voltage is not as specified, go to **11.**

11. Check for 115 vac between K8 AC ESNTL BUS XFER relay terminal A1 and ground.

a. If voltage is as specified, repair/replace wiring between relay terminal X2 and ground (WP 1747 00). Go to **15.**

AC ESS BUS OFF LEGEND DOES NOT GO OFF WHEN NO. 1 AND NO. 2 AC PRIMARY BUSES ARE ENERGIZED - Continued

b. If voltage is not as specified, go to **12.**

12. Check for 115 vac between AC ESNTL BUS SPLY circuit breaker (copilot's circuit breaker panel) terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and K8 AC ESNTL BUS XFER relay terminal A1 (WP 1747 00). Go to **15.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **15.**

13. Check for 115 vac between K8 AC ESNTL BUS XFER relay terminal A1 and ground.

a. If voltage is as specified, repair/replace wiring between relay terminal X2 and ground (WP 1747 00). Go to **15.**

b. If voltage is not as specified, go to **14.**

14. Check for 115 vac between AC ESNTL BUS SPLY circuit breaker (copilot's circuit breaker panel) terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **15.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **15.**

15. Procedure completed.

#1 GEN BRG AND/OR #2 GEN BRG LEGEND GOES ON

SYMPTOM

#1 GEN BRG and/or #2 GEN BRG legend goes on.

MALFUNCTION

#1 GEN BRG and/or #2 GEN BRG legend goes on.

CORRECTIVE ACTION

1. Are both #1 GEN BRG and #2 GEN BRG legends on?

a. If both legends are on, troubleshoot both #1 GEN BRG and #2 GEN BRG legend circuits (WP 0100 00). Go to **10.**

b. If both legends are not on, go to **2.**

2. Is #1 GEN BRG legend on?

a. If legend is on, go to **3.**

b. If legend is not on, go to **6.**

3. Pull out, then push in NO. 1 GEN WARN circuit breaker. Go to 4.

4. Did #1 GEN BRG legend go off and remain off?

a. If legend did go off and remain off, no maintenance required. Go to **10.**

b. If legend did not go off and remain off, go to **5.**

5. Check for less than 50 ohms between P1019R-9 (copilot's)/P1020R-9 (pilot's) and ground.

a. If resistance is as specified, replace No. 1 generator (WP 0839 00). Go to **10.**

b. If resistance is not as specified, replace MFD (WP 0799 00). Go to **10.**

#1 GEN BRG AND/OR #2 GEN BRG LEGEND GOES ON - Continued

6. **Did #2 GEN BRG legend go on steady for more than 1 minute?**
   a. If legend was steady, go to **7.**
   b. If legend was not steady, no maintenance required. Go to **10.**
7. **Pull out, then push in NO. 2 GEN WARN circuit breaker. Go to 8.**
8. **Did #2 GEN BRG legend go off and remain off?**
   a. If legend did go off and remain off, no maintenance required. Go to **10.**
   b. If legend did not go off and remain off, go to **9.**
9. **Check for less than 50 ohms between P1019R-8 (copilot's)/P1020R-8 (pilot's) and ground.**
   a. If resistance is as specified, replace No. 2 generator (WP 0839 00). Go to **10.**
   b. If resistance is not as specified, replace MFD (WP 0799 00). Go to **10.**
10. **Procedure completed.**

#1 GEN LEGEND DOES NOT GO ON WITH NO. 1 GENERATOR OFF

SYMPTOM

#1 GEN legend does not go on with No. 1 generator off.

MALFUNCTION

#1 GEN legend does not go on with No. 1 generator off.

CORRECTIVE ACTION

1. **Press MFD T6 switch and check that #1 GEN legend goes on.**
   a. If #1 GEN legend goes on, go to **2.**
   b. If #1 GEN legend does not go on, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **9.**
2. **Check for 28 vdc between P1019R-17 (copilot's)/P1020-17 (pilot's) and ground.**
   a. If voltage is as specified, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **9.**
   b. If voltage is not as specified, go to **3.**
3. **Check for 28 vdc between P212-B and ground.**
   a. If voltage is as specified, go to **4.**
   b. If voltage is not as specified, go to **5.**
4. **Check continuity between P1019R-17 (copilot's)/P1020R-17 (pilot's) and P212-A.**
   a. If continuity is present, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **9.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**
5. **Check continuity between P212-B and P215-Z.**
   a. If continuity is present, go to **6.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

#1 GEN LEGEND DOES NOT GO ON WITH NO. 1 GENERATOR OFF - Continued

6. **Check continuity between P212-A and P215-f.**
   a. If continuity is present, go to **7.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**
7. **Check for 28 vdc between P215-Y and ground.**
   a. If voltage is as specified, replace No. 1 generator control unit (WP 0838 00). Go to **9.**
   b. If voltage is not as specified, go to **8.**
8. **Check for 28 vdc between NO. 1 GEN WARN circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, repair/replace wiring between circuit breaker and P215-Y (WP 1747 00). Go to **9.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9.**
9. **Procedure completed.**

#1 GEN LEGEND IS ON WITH GENERATORS NO. 1 SWITCH ON

SYMPTOM
#1 GEN legend is on with GENERATORS NO. 1 switch ON.

MALFUNCTION
#1 GEN legend is on with GENERATORS NO. 1 switch ON.

CORRECTIVE ACTION

1. **Check for 115 vac between:**
   - **J211-A and J211-D**
   - **J211-B and J211-D**
   - **J211-C and J211-D**

   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, go to **18.**
2. **Place and hold GENERATORS NO. 1 switch to TEST. Go to 3.**
3. **Check that NO. 1 GEN legend goes off.**
   a. If NO. 1 GEN legend goes off, go to **4.**
   b. If NO. 1 GEN legend does not go off, go to **8.**
4. **Place GENERATORS NO. 1 switch to ON. Go to 5.**
5. **Check for 28 vdc between P212-P and P212-N.**
   a. If voltage is as specified, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **19.**
   b. If voltage is not as specified, go to **6.**
6. **Check for 28 vdc between P212-P and ground.**
   a. If voltage is as specified, repair/replace wiring between P212-N and ground (WP 1747 00). Go to **19.**
   b. If voltage is not as specified, go to **7.**

#1 GEN LEGEND IS ON WITH GENERATORS NO. 1 SWITCH ON - Continued

7. **Check continuity between P212-P and GENERATORS NO. 1 switch terminal 6.**
   a. If continuity is present, replace switch (WP 0828 00). Go to **19.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19.**
8. **Check for 28 vdc between GENERATORS NO. 1 switch terminal 5 and ground.**
   a. If voltage is as specified, go to **9.**
   b. If voltage is not as specified, go to **13.**
9. **Check for 28 vdc between GENERATORS NO. 1 switch terminal 4 and ground.**
   a. If voltage is as specified, go to **10.**
   b. If voltage is not as specified, replace switch (WP 0828 00). Go to **19.**
10. **Remove and check pin filtered adapter from P215 (WP 0925 00).**
    a. If pin filtered adapter is good, go to **11.**
    b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **19.**
11. **Install pin filtered adapter on P215 (WP 0925 00). Go to 12.**
12. **Check continuity between GENERATORS NO. 1 switch terminal 4 and P215-d.**
    a. If continuity is present, replace No. 1 generator control unit (WP 0838 00). Go to **19.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19.**
13. **Check continuity between GENERATORS NO. 1 switch terminal 5 and P215-e.**
    a. If continuity is present, go to **14.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19.**
14. **Check for 28 vdc between GENERATORS NO. 1 switch terminal 1 and ground.**
    a. If voltage is as specified, go to **15.**
    b. If voltage is not as specified, go to **17.**
15. **Check for 28 vdc between GENERATORS NO. 1 switch terminal 1 and ground.**
    a. If voltage is as specified, go to **16.**
    b. If voltage is not as specified, replace switch (WP 0828 00). Go to **19.**
16. **Check continuity between GENERATORS NO. 1 switch terminal 1 and P215-a.**
    a. If continuity is present, replace No. 1 generator control unit (WP 0838 00). Go to **19.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19.**
17. **Check continuity between GENERATORS NO. 1 switch terminal 2 and P215-c.**
    a. If continuity is present, replace No. 1 generator control unit (WP 0838 00). Go to **19.**
    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19.**
18. **Check continuity between No. 1 generator:**
    - **Terminal T1 and J211-A**
    - **Terminal T2 and J211-B**
    - **Terminal T3 and J211-C**

#1 GEN LEGEND IS ON WITH GENERATORS NO. 1 SWITCH ON - Continued

a. If continuity is present, do GENERATOR AND GCU TROUBLESHOOTING, in this work package using system analyzer. Go to **19.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **19.**

19. Procedure completed.

#1 CONV LEGEND STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 1 AC PRIMARY BUS

SYMPTOM

#1 CONV legend stays on when either generator is supplying power to No. 1 ac primary bus.

MALFUNCTION

#1 CONV legend stays on when either generator is supplying power to No. 1 ac primary bus.

CORRECTIVE ACTION

1. Check for 115 vac between NO. 1 CONVERTER circuit breaker:

- **Terminal A2 and ground**
- **Terminal B2 and ground**
- **Terminal C2 and ground**

a. If voltage is as specified, troubleshoot DC electrical system (WP 0123 00). Go to **4.**

b. If voltage is not as specified, go to **2.**

2. Check for 115 vac between NO. 1 CONVERTER circuit breaker:

- **Terminal A1 and ground**
- **Terminal B1 and ground**
- **Terminal C1 and ground**

a. If voltage is as specified, replace circuit breaker (WP 0824 00). Go to **4.**

b. If voltage is not as specified, go to **3.**

3. Check for 115 vac between K1 NO. 1 GEN CNTOR:

- **Terminal A2 and ground**
- **Terminal B2 and ground**
- **Terminal C2 and ground**

a. If voltage is as specified, repair/replace wiring between K1 NO. 1 GEN CNTOR and NO. 1 CONVERTER circuit breaker (WP 1747 00). Go to **4.**

b. If voltage is not as specified, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **4.**

4. Procedure completed.

#2 CONV LEGEND STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 2 AC PRIMARY BUS

SYMPTOM

#2 CONV legend stays on when either generator is supplying power to No. 2 ac primary bus.

MALFUNCTION

#2 CONV legend stays on when either generator is supplying power to No. 2 ac primary bus.

CORRECTIVE ACTION

1. **Check for 115 vac between NO. 2 CONVERTER circuit breaker:**
   - **Terminal A1 and ground**
   - **Terminal B1 and ground**
   - **Terminal C1 and ground**

   a. If voltage is as specified, troubleshoot DC electrical system (WP 0123 00). Go to **6.**

   b. If voltage is not as specified, go to **2.**

2. **Check for 115 vac between NO. 2 CONVERTER circuit breaker:**
   - **Terminal A2 and ground**
   - **Terminal B2 and ground**
   - **Terminal C2 and ground**

   a. If voltage is as specified, replace circuit breaker (WP 0824 00). Go to **6.**

   b. If voltage is not as specified, go to **3.**

3. **Check for 115 vac between K2 NO. 2 GEN CNTOR:**
   - **Terminal A2 and ground**
   - **Terminal B2 and ground**
   - **Terminal C2 and ground**

   a. If voltage is as specified, repair/replace wiring between K2 NO. 2 GEN CNTOR and NO. 2 CONVERTER circuit breaker (WP 1747 00). Go to **6.**

   b. If voltage is not as specified, go to **4.**

4. **Check for 115 vac between K1 NO. 1 GEN CNTOR:**
   - **Terminal A2 and ground**
   - **Terminal B2 and ground**
   - **Terminal C2 and ground**

   a. If voltage is as specified, go to **5.**

   b. If voltage is not as specified, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **6.**

5. **Check for open current limiters CL4, CL5, and CL6.**

   a. If limiters are open, replace open AC current limiters as required (WP 0836 00). Go to **6.**

#2 CONV LEGEND STAYS ON WHEN EITHER GENERATOR IS SUPPLYING POWER TO NO. 2 AC PRIMARY BUS - Continued

b. If limiters are not open, repair/replace wiring between K1 NO. 1 GEN CNTOR and K2 NO. 2 GEN CNTOR (WP 1747 00). Go to **6.**

c. If trouble remains, replace K4 AC BUS TIE CNTOR (WP 0844 00). Go to **6.**

6. Procedure completed.

NO. 1 AND/OR NO. 2 GENERATOR DISABLED BY UNDERFREQUENCY CONDITION DURING AUTO ROTATIVE DESCENT

SYMPTOM

No. 1 and/or No. 2 generator disabled by underfrequency condition during auto rotative descent.

MALFUNCTION

No. 1 and/or No. 2 generator disabled by underfrequency condition during auto rotative descent.

CORRECTIVE ACTION

1. Are both generators disabled?

a. If both are disabled, go to **2.**

b. If both are not disabled, go to **3.**

2. Is right drag beam switch in proper adjustment?

a. If switch is in proper adjustment, replace switch (WP 0443 00). Go to **10.**

b. If switch is not in proper adjustment, adjust switch (WP 0443 00). Go to **10.**

3. Remove and check pin filtered adapter from P214 (WP 0925 00).

a. If pin filtered adapter is good, go to **4.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **10.**

4. Install pin filtered adapter on P214 (WP 0925 00). Go to 5.

5. Is No. 1 generator disabled?

a. If generator is disabled, go to **6.**

b. If generator is not disabled, go to **9.**

6. Remove and check pin filtered adapter from P215 (WP 0925 00).

a. If pin filtered adapter is good, go to **7.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **10.**

7. Install pin filtered adapter on P215 (WP 0925 00). Go to 8.

8. Check continuity between J164-C and P215-F.

a. If continuity is present, replace No. 1 generator control unit (WP 0838 00). Go to **10.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

9. Check continuity between J164-C and P214-F.

a. If continuity is present, replace No. 2 generator control unit (WP 0838 00). Go to **10.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

10. Procedure completed.

#2 GEN LEGEND IS OFF WITH NO. 2 GENERATOR OFF

SYMPTOM

#2 GEN legend is off with No. 2 generator off.

MALFUNCTION

#2 GEN legend is off with No. 2 generator off.

CORRECTIVE ACTION

1. Press MFD T6 switch and check that #2 GEN legend goes on.

a. If legend goes on, go to **2.**

b. If legend does not go on, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **9.**

2. Check for 28 vdc between P1019R-61 (copilot's)/P1020R-61 (pilot's) and ground.

a. If voltage is as specified, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **9.**

b. If voltage is not as specified, go to **3.**

3. Check for 28 vdc between P204-B and ground.

a. If voltage is as specified, go to **4.**

b. If voltage is not as specified, go to **5.**

4. Check continuity between P204-A and P1019R-61 (copilot's)/P1020R-61 (pilot's).

a. If continuity is present, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **9.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

5. Check continuity between P204-B and P214-Z.

a. If continuity is present, go to **6.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

6. Check continuity between P204-A and P214-f.

a. If continuity is present, go to **7.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

7. Check for 28 vdc between P214-Y and ground.

a. If voltage is as specified, replace No. 2 generator control unit (WP 0838 00). Go to **9.**

b. If voltage is not as specified, go to **8.**

8. Check for 28 vdc between NO. 2 GEN WARN circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **9.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9.**

9. Procedure completed.

#2 GEN LEGEND IS ON WITH GENERATORS NO. 2 SWITCH ON

SYMPTOM

#2 GEN legend is on with GENERATORS NO. 2 switch ON.

MALFUNCTION

#2 GEN legend is on with GENERATORS NO. 2 switch ON.

CORRECTIVE ACTION

1. **Check for 115 vac between:**
   - **J208-A and J208-D**
   - **J208-B and J208-D**
   - **J208-C and J208-D**

   a. If voltage is as specified, go to **2.**

   b. If voltage is not as specified, go to **17.**

2. **Place and hold GENERATORS NO. 2 switch to TEST and check that NO. 2 GEN legend goes off.**

   a. If NO. 2 GEN legend goes off, go to **3.**

   b. If NO. 2 GEN legend does not go off, go to **7.**

3. **Place GENERATORS NO. 2 switch to ON. Go to 4.**

4. **Check for 28 vdc between P204-P and P204-N.**

   a. If voltage is as specified, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **18.**

   b. If voltage is not as specified, go to **5.**

5. **Check for 28 vdc between P204-P and ground.**

   a. If voltage is as specified, repair/replace wiring between P204-N and ground (WP 1747 00). Go to **18.**

   b. If voltage is not as specified, go to **6.**

6. **Check continuity between P204-P and GENERATORS NO. 2 switch terminal 6.**

   a. If continuity is present, replace switch (WP 0828 00). Go to **18.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

7. **Remove and check pin filtered adapter from P214 (WP 0925 00).**

   a. If pin filtered adapter is good, go to **8.**

   b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **18.**

8. **Install pin filtered adapter on P214 (WP 0925 00). Go to 9.**

9. **Check for 28 vdc between GENERATORS NO. 2 switch terminal 5 and ground.**

   a. If voltage is as specified, go to **10.**

   b. If voltage is not as specified, go to **12.**

#2 GEN LEGEND IS ON WITH GENERATORS NO. 2 SWITCH ON - Continued

10. **Check for 28 vdc between GENERATORS NO. 2 switch terminal terminal 4 and ground.**

    a. If voltage is as specified, go to **11.**

    b. If voltage is not as specified, replace switch (WP 0828 00). Go to **18.**

11. **Check continuity between GENERATORS NO. 2 switch terminal 4 and P214-d.**

    a. If continuity is present, replace No. 2 generator control unit (WP 0838 00). Go to **18.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

12. **Check continuity between GENERATORS NO. 2 switch terminal 5 and P214-e.**

    a. If continuity is present, go to **13.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

13. **Check for 28 vdc between GENERATORS NO. 2 switch terminal 2 and ground.**

    a. If voltage is as specified, go to **14.**

    b. If voltage is not as specified, go to **16.**

14. **Check for 28 vdc between GENERATORS NO. 2 switch terminal 1 and ground.**

    a. If voltage is as specified, go to **15.**

    b. If voltage is not as specified, replace switch (WP 0828 00). Go to **18.**

15. **Check continuity between GENERATORS NO. 2 switch terminal 1 and P214-a.**

    a. If continuity is present, replace No. 2 generator control unit (WP 0838 00). Go to **18.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

16. **Check continuity between GENERATORS NO. 2 switch terminal 2 and P214-c.**

    a. If continuity is present, replace No. 2 generator control unit (WP 0838 00). Go to **18.**

    b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

17. **Check continuity between No. 2 generator:**

    - **Terminal T1 and J208-A**
    - **Terminal T2 and J208-B**
    - **Terminal T3 and J208-C**

    a. If continuity is present, do GENERATOR AND GCU TROUBLESHOOTING, in this work package, using system analyzer. Go to **18.**

    b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **18.**

18. **Procedure completed.**

EXT PWR CONNECTED LEGEND IS OFF WITH A SOURCE OF EXTERNAL POWER CONNECTED

SYMPTOM

EXT PWR CONNECTED legend is off with a source of external power connected.

MALFUNCTION

EXT PWR CONNECTED legend is off with a source of external power connected.

CORRECTIVE ACTION

1. **Press MFD T6 switch and check that EXT PWR CONNECTED legend goes on.**
   a. If EXT PWR CONNECTED legend goes on, go to **2.**
   b. If EXT PWR CONNECTED legend does not go on, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **6.**
2. **Check for 28 vdc between J136-F and J136-N.**
   a. If voltage is as specified, go to **3.**
   b. If voltage is not as specified, go to **4.**
3. **Check for 28 vdc between P1019R-45 (copilot's)/P1020R-45 (pilot's) and ground.**
   a. If voltage is as specified, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **6.**
   b. If voltage is not as specified, repair/replace wiring between J136-E and P1019R-45 (copilot's)/ P1020R-45 (pilot's) (WP 1747 00). Go to **6.**
4. **Check continuity between J136-N and ground.**
   a. If continuity is present, go to **5.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**
5. **Check for 28 vdc between BATT & ESNTL DC WARN EXT PWR CONTR circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J136-F (WP 1747 00). Go to **6.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6.**
6. **Procedure completed.**

EXTERNAL POWER DOES NOT SUPPLY AC BUSES (#1 CONV AND #2 CONV LEGENDS ON)

SYMPTOM

External power does not supply AC buses (#1 CONV and #2 CONV legends on).

MALFUNCTION

External power does not supply AC buses (#1 CONV and #2 CONV legends on).

CORRECTIVE ACTION

1. **Check for 115 vac between K2 NO. 2 GEN CNTOR:**
   - **Terminal A3 and ground**
   - **Terminal B3 and ground**

EXTERNAL POWER DOES NOT SUPPLY AC BUSES (#1 CONV AND #2 CONV LEGENDS ON) - Continued

- **Terminal C3 and ground**

a. If voltage is as specified, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **18.**

b. If voltage is not as specified, go to **2.**

2. Check for 115 vac between K4 AC BUS TIE CNTOR:

- **Terminal A2 and ground**
- **Terminal B2 and ground**
- **Terminal C2 and ground**

a. If voltage is as specified, repair/replace wiring between K2 NO. 2 GEN CNTOR and K4 AC BUS TIE CNTOR as required (WP 1747 00). Go to **18.**

b. If voltage is not as specified, go to **3.**

3. Check for 115 vac between K4 AC BUS TIE CNTOR:

- **Terminal A1 and ground**
- **Terminal B1 and ground**
- **Terminal C1 and ground**

a. If voltage is as specified, go to **4.**

b. If voltage is not as specified, go to **16.**

4. Check for 28 vdc between:

- **P206-P and ground**
- **P206-N and ground**

a. If voltage is as specified, replace K4 AC BUS TIE CNTOR (WP 0844 00). Go to **18.**

b. If voltage is not as specified, go to **5.**

5. Check for 28 vdc between P206-P and ground.

a. If voltage is as specified, repair/replace wiring between P206-N and ground (WP 1747 00). Go to **18.**

b. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between P212-L and ground.

a. If voltage is as specified, go to **7.**

b. If voltage is not as specified, go to **8.**

7. Check continuity between P212-M and P206-P.

a. If continuity is present, replace K1 NO. 1 GEN CNTOR (WP 0834 00). Go to **18.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

8. Check for 28 vdc between P204-M and ground.

a. If voltage is as specified, go to **9.**

b. If voltage is not as specified, go to **10.**

9. Check continuity between P204-L and P212-L.

EXTERNAL POWER DOES NOT SUPPLY AC BUSES (#1 CONV AND #2 CONV LEGENDS ON) - Continued

a. If continuity is present, replace K2 NO. 2 GEN CNTOR (WP 0835 00). Go to **18.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

10. Check for 28 vdc between P241-H and ground.

a. If voltage is as specified, go to **11.**

b. If voltage is not as specified, go to **12.**

11. Check continuity between P241-P and P204-M.

a. If continuity is present, replace right relay panel (WP 0825 00). Go to **18.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

12. Check for 28 vdc between EXT PWR switch terminal 2 and ground.

a. If voltage is as specified, repair/replace wiring between switch terminal 2 and P241-H (WP 1747 00). Go to **18.**

b. If voltage is not as specified, go to **13.**

13. Check for 28 vdc between EXT PWR switch terminal 3 and ground.

a. If voltage is as specified, replace switch (WP 0828 00). Go to **18.**

b. If voltage is not as specified, go to **14.**

14. Check continuity between EXT PWR switch terminal 3 and P213-5.

a. If continuity is present, go to **15.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18.**

15. Check continuity between K3 APU/EXT PWR CNTOR:

- **Terminal A3 and P213-3**
- **Terminal B3 and P213-2**
- **Terminal C3 and P213-1**

a. If continuity is present, replace external power monitor panel (WP 0843 00). Go to **18.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **18.**

16. Check for 115 vac between K3 APU/EXT PWR CNTOR:

- **Terminal A2 and ground**
- **Terminal B2 and ground**
- **Terminal C2 and ground**

a. If voltage is as specified, repair/replace wiring between K3 APU/EXT PWR CNTOR and K4 AC BUS TIE CNTOR as required (WP 1747 00). Go to **18.**

b. If voltage is not as specified, go to **17.**

17. Check for 115 vac between K3 APU/EXT PWR CNTOR:

- **Terminal A3 and ground**
- **Terminal B3 and ground**
- **Terminal C3 and ground**

EXTERNAL POWER DOES NOT SUPPLY AC BUSES (#1 CONV AND #2 CONV LEGENDS ON) - Continued

a. If voltage is as specified, replace K3 APU/EXT PWR CNTOR (WP 0840 00). Go to **18.**

b. If voltage is not as specified, repair/replace wiring between K3 APU/EXT PWR CNTOR and external receptacle as required (WP 1747 00). Go to **18.**

18. Procedure completed.

NO 26 VAC POWER ON AC ESNTL BUS

SYMPTOM

No 26 vac power on AC ESNTL bus.

MALFUNCTION

No 26 vac power on AC ESNTL bus.

CORRECTIVE ACTION

1. Check for 26 vac between T12 AUTO XFMR terminal HV and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **3.**

2. Check for 26 vac between T12 AUTO XFMR terminal LV and ground.

a. If voltage is as specified, repair/replace wiring between transformer terminal LV and AC ESNTL BUS 26 VAC circuit breaker (WP 1747 00). Go to **4.**

b. If voltage is not as specified, replace transformer (WP 0833 00). Go to **4.**

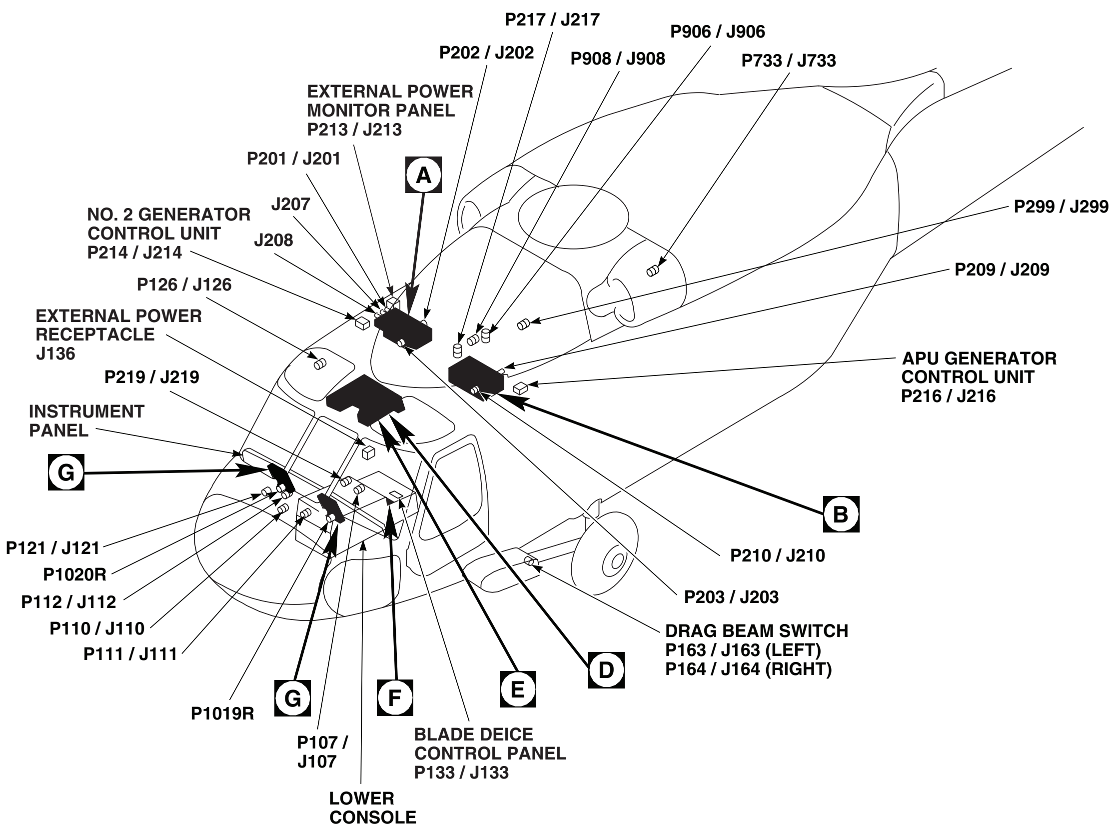
3. Check continuity between T12 AUTO XFMR terminal GND and ground.

a. If continuity is present, replace transformer (WP 0833 00). Go to **4.**

b. If continuity is not present, repair/replace wiring between T12 transformer terminal GND and ground (WP 1747 00). Go to **4.**

4. Procedure completed.

LOCATION



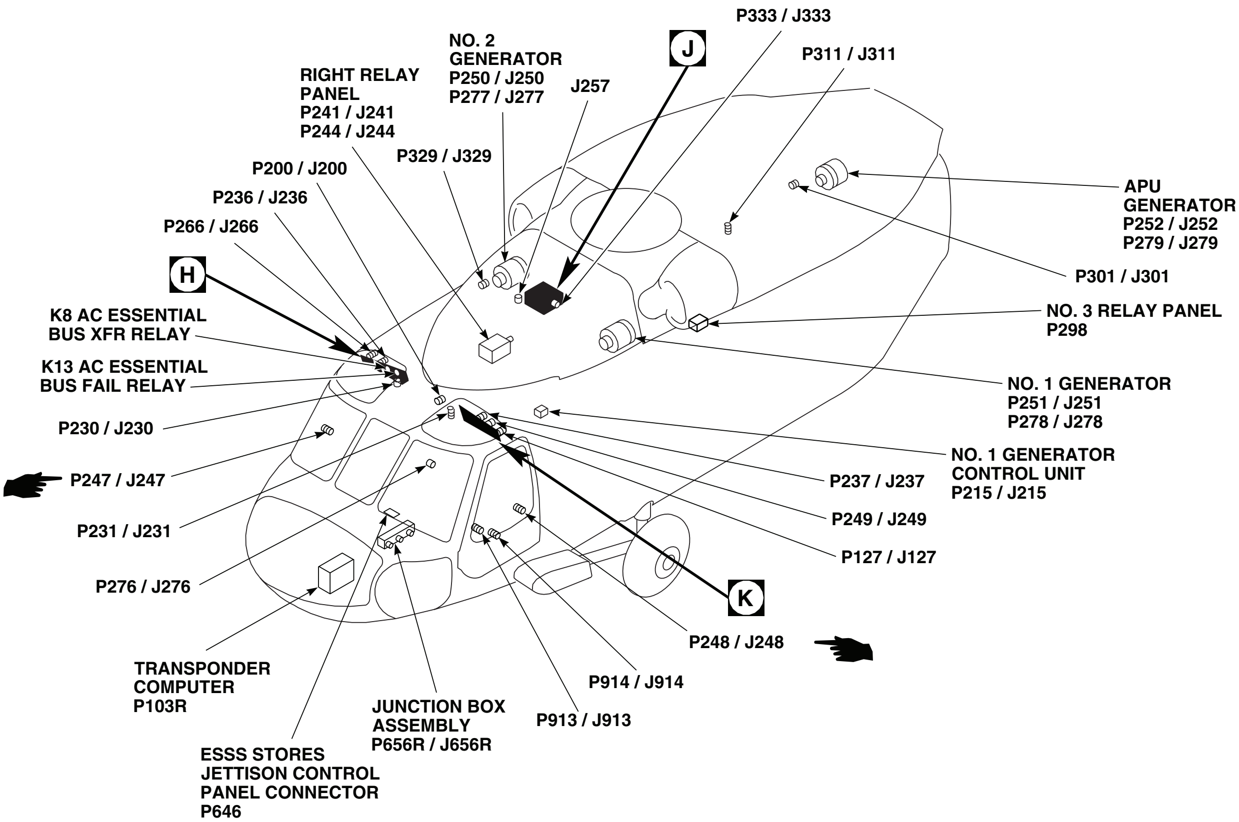
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J136	EXTERNAL POWER RECEPTACLE
J207	NO. 2 JUNCTION BOX
J208	NO. 2 JUNCTION BOX
J211	NO. 1 JUNCTION BOX
J257	UTILITY RECEPTACLE 115 V 400 Hz
P103R	IFF TRANSPONDER COMPUTER
P107 / J107	COCKPIT, BL 0 LH, STA 236

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P112 / J112	COCKPIT, BL 6 RH, STA 197
P121 / J121	COCKPIT, BL 10 RH, STA 195
P126 / J126	COCKPIT CEILING BL 28 RH, STA 243
P127 / J127	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 28 LH, STA 243
P133 / J133	BLADE DEICE CONTROL PANEL

AB0150_1
SA

Figure 1. AC Electrical System Location Diagram. (Sheet 1 of 8)

LOCATION - Continued



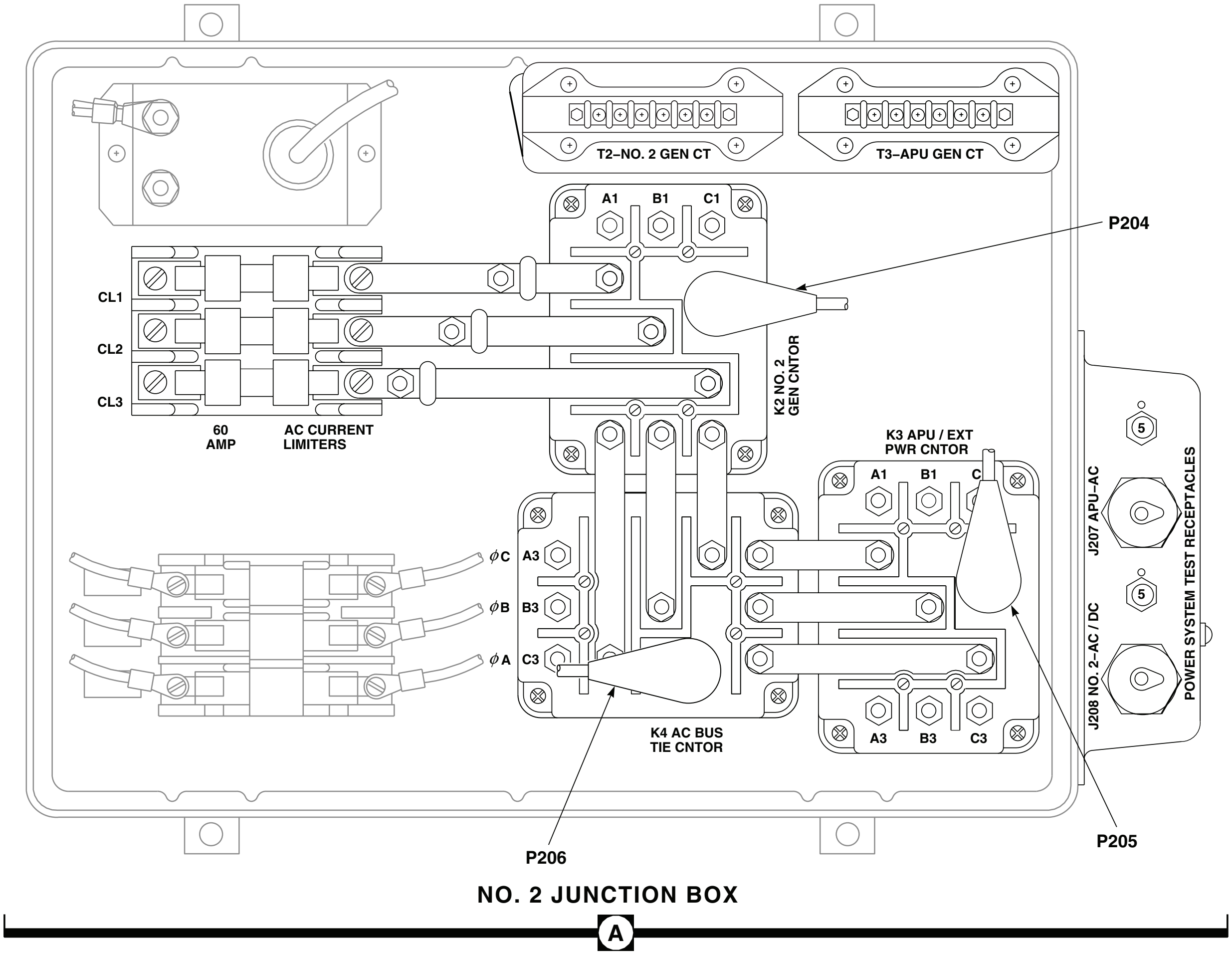
TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P163 / J163	LEFT DRAG BEAM SWITCH
P164 / J164	RIGHT DRAG BEAM SWITCH
P200 / J200	COCKPIT CEILING BL 13 LH, STA 246
P201 / J201	NO. 2 JUNCTION BOX
P202 / J202	NO. 2 JUNCTION BOX
P203 / J203	NO. 2 JUNCTION BOX
P204	NO. 2 JUNCTION BOX
P205	NO. 2 JUNCTION BOX
P206	NO. 2 JUNCTION BOX
P209 / J209	NO. 1 JUNCTION BOX
P210 / J210	NO. 1 JUNCTION BOX
P212	NO. 1 JUNCTION BOX
P213 / J213	EXTERNAL POWER MONITOR PANEL
P214 / J214	NO. 2 GENERATOR CONTROL UNIT
P215 / J215	NO. 1 GENERATOR CONTROL UNIT

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P216 / J216	APU GENERATOR CONTROL UNIT
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P231/ J231	CABIN CEILING BL 9 LH, STA 247
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 23 RH
P237 / J237	BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 25 LH
P241 / J241	RIGHT RELAY PANEL
P244 / J244	RIGHT RELAY PANEL
P247 / J247	CABIN, BL 40 RH, STA 248
P248 / J248	CABIN, BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P250 / J250	NO. 2 GENERATOR
P251 / J251	NO. 1 GENERATOR
P252 / J252	APU GENERATOR

AB0150_2B
SA

Figure 1. AC Electrical System Location Diagram. (Sheet 2 of 8)

LOCATION - Continued



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH
P276 / J276	60 Hz CONVERTER
P277 / J277	NO. 2 GENERATOR
P278 / J278	NO. 1 GENERATOR
P279 / J279	APU GENERATOR
P298	NO. 3 RELAY PANEL
P299 / J299	RANGE EXTENSION PROVISION CONTROL PANEL, STA 295, WL 265
P301 / J301	TRANSITION DECK BL 12.88 LH, STA 415
P311 / J311	CABIN CEILING BL 16.5 LH, STA 383
P329 / J329	MAIN BLADE DE–ICE JUNCTION BOX
P333 / J333	AUXILIARY AC JUNCTION BOX

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P350	ENVIRONMENTAL CONTROL UNIT DISTRIBUTION BOX
P646	ESSS STORES JETTISON CONTROL PANEL CONNECTOR
P656R / J656R	JUNCTION BOX ASSEMBLY
P721 / J721	NO. 3 RELAY PANEL
P723 / J723	NO. 3 RELAY PANEL
P733 / J733	STA 360, BL 24.71, WL 265.30
P906 / J906	CABIN CEILING, BL 0, STA 284
P908 / J908	CABIN CEILING, BL 0, STA 279
P913 / J913	COCKPIT, BL 16 LH, STA 247
P914 / J914	COCKPIT, BL 24 LH, STA 247
P1019R	COPILOT'S MFD CAUTION ADVISORY
P1020R	PILOT'S MFD CAUTION ADVISORY

AB0150_3
SA

Figure 1. AC Electrical System Location Diagram. (Sheet 3 of 8)

LOCATION - Continued

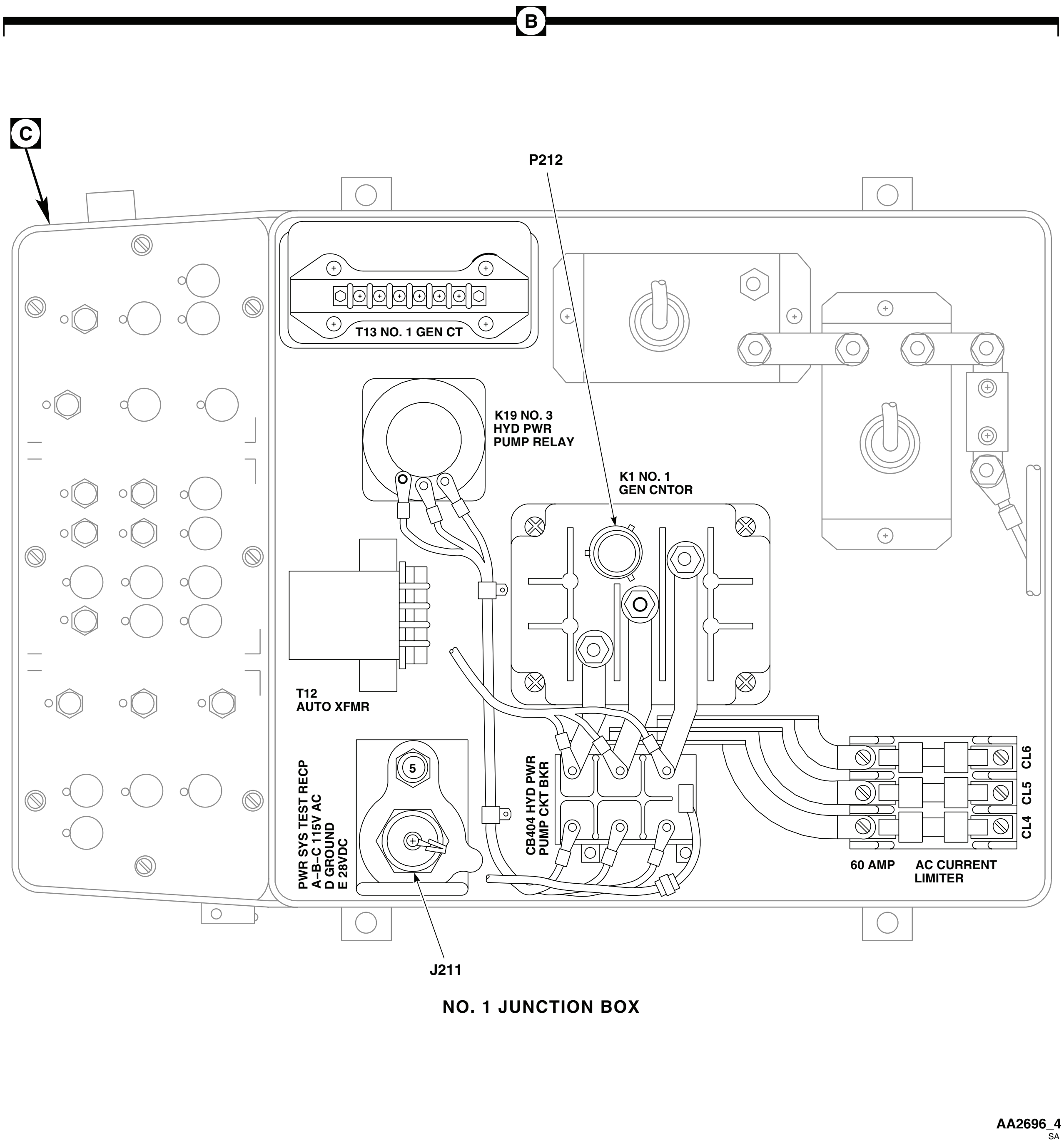


Figure 1. AC Electrical System Location Diagram. (Sheet 4 of 8)

LOCATION - Continued

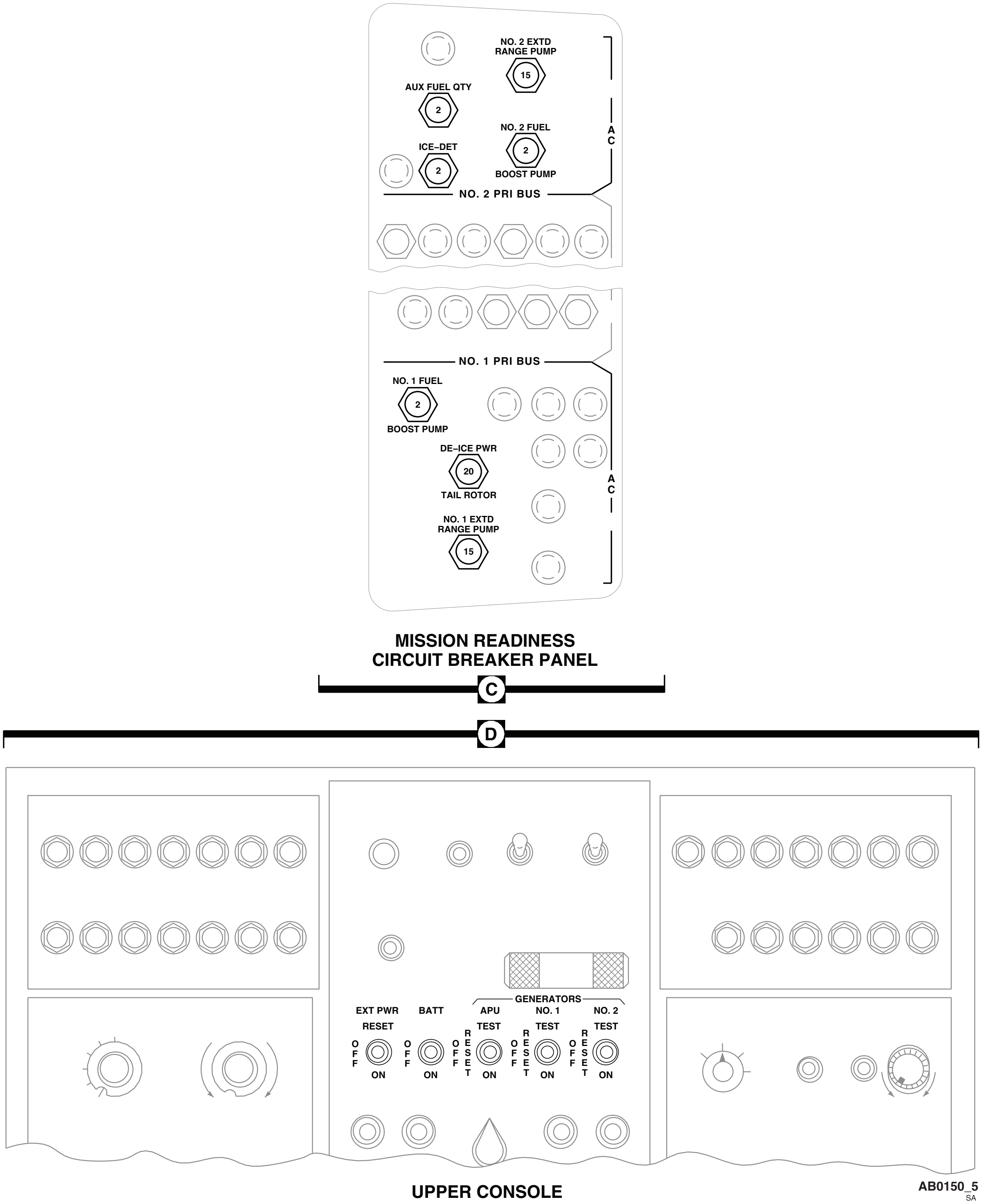


Figure 1. AC Electrical System Location Diagram. (Sheet 5 of 8)

LOCATION - Continued

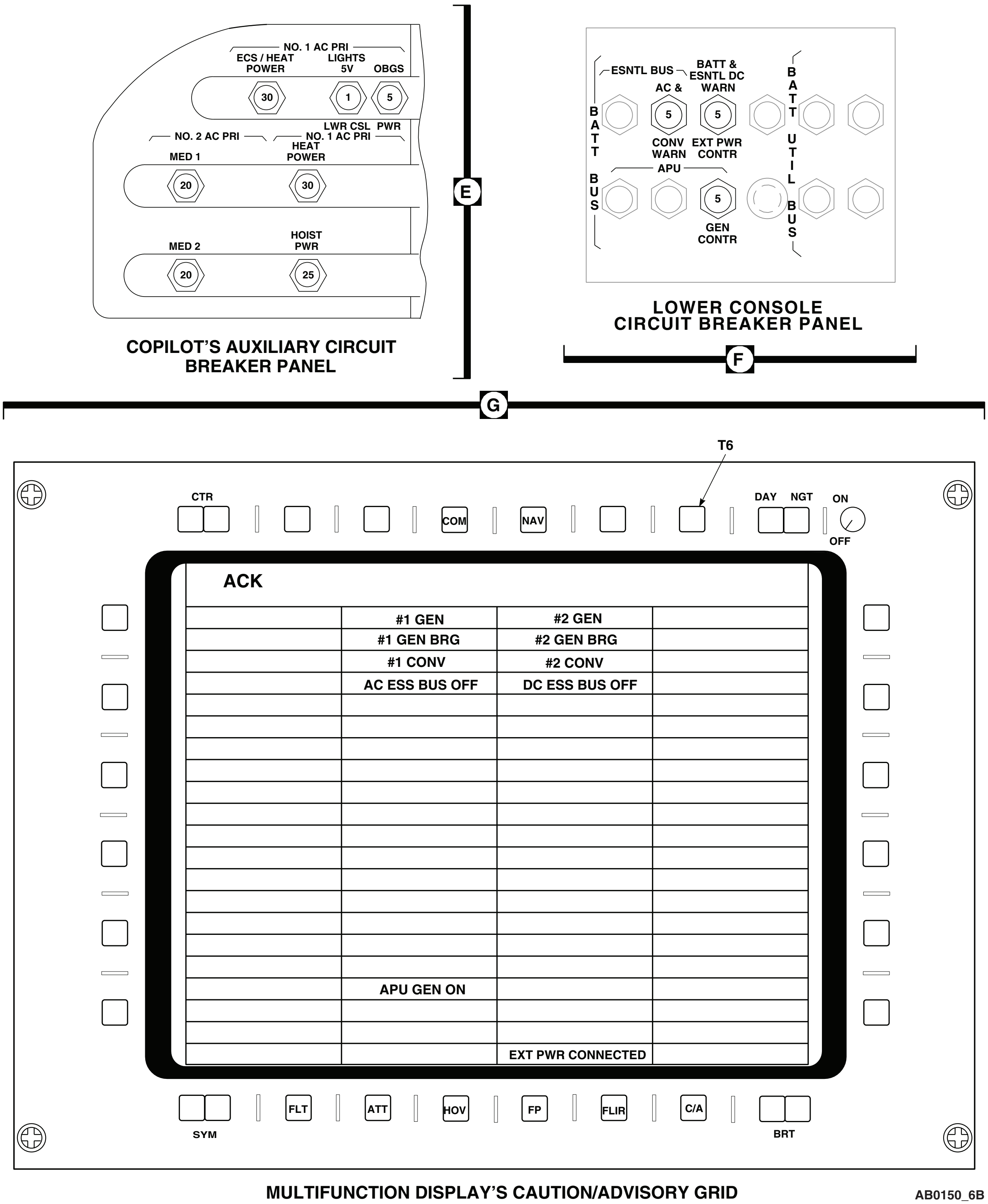


Figure 1. AC Electrical System Location Diagram. (Sheet 6 of 8)

LOCATION - Continued

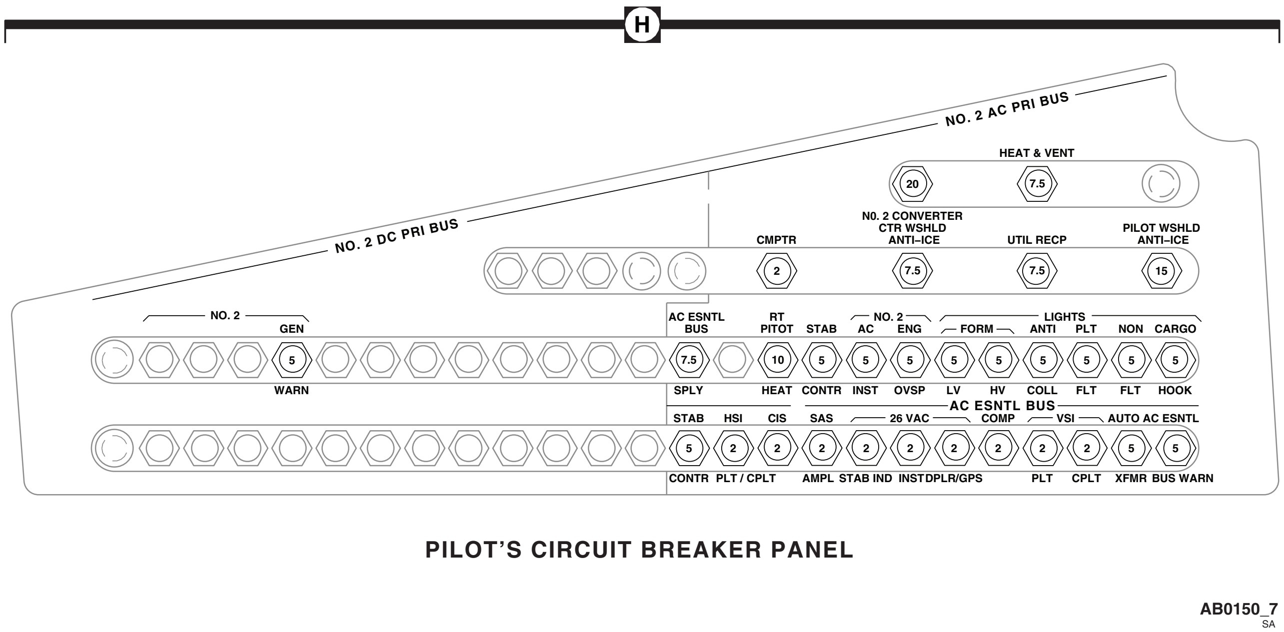
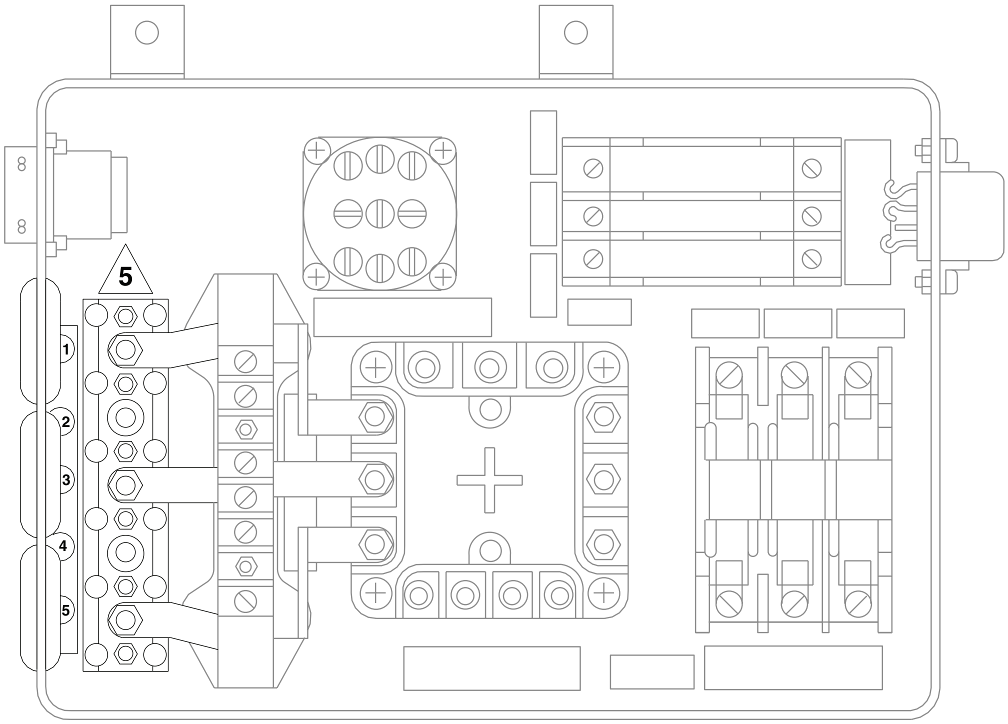


Figure 1. AC Electrical System Location Diagram. (Sheet 7 of 8)

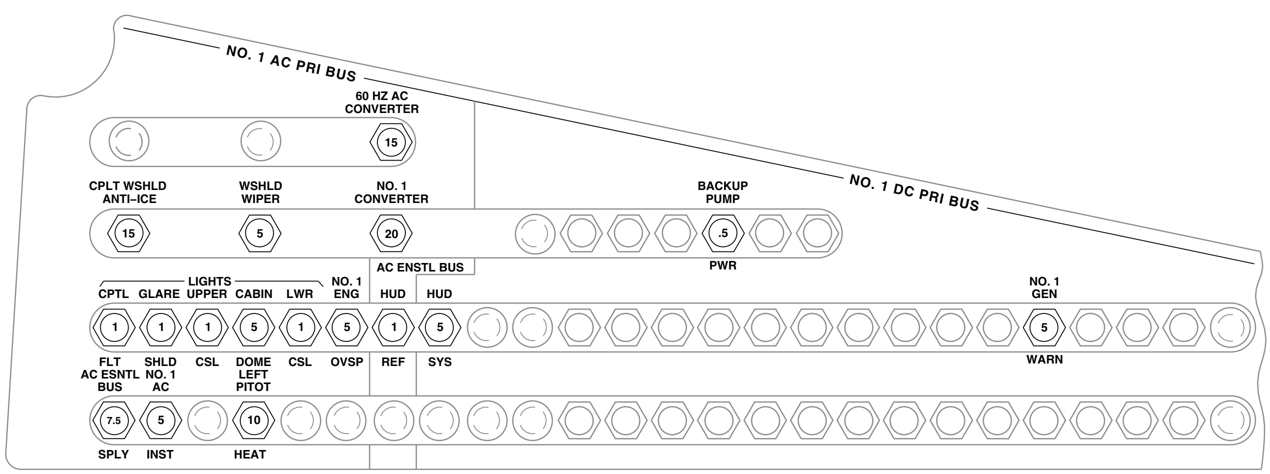
LOCATION - Continued



AUXILIARY AC JUNCTION BOX

J

K

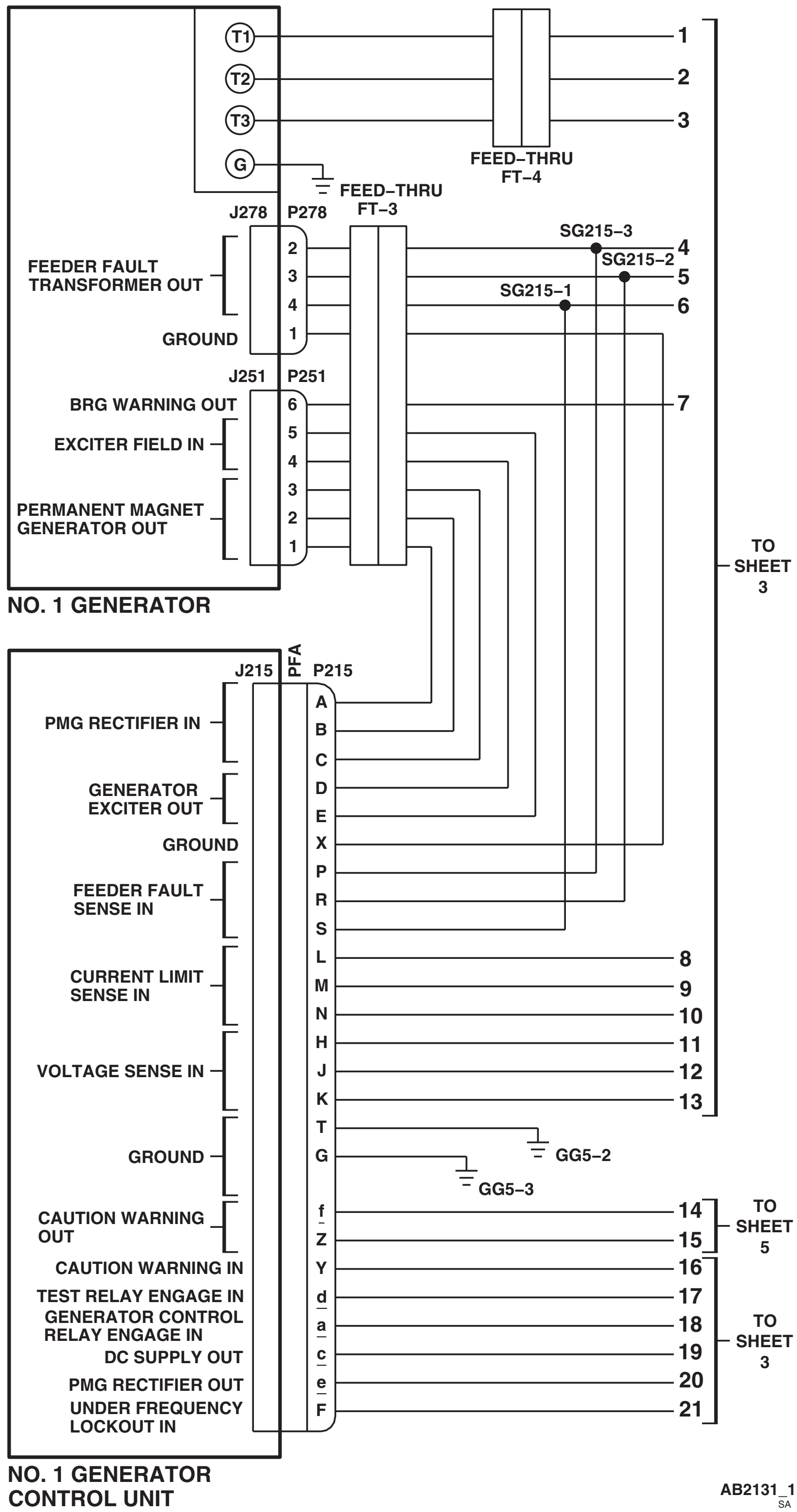


COPILOT'S CIRCUIT BREAKER PANEL

AB0150_8A
SA

Figure 1. AC Electrical System Location Diagram. (Sheet 8 of 8)

SCHEMATICS



LEGEND

POWER SOURCE

A=NO. 1 MAIN GENERATOR
B=NO. 2 MAIN GENERATOR
C=APU GENERATOR
D=EXTERNAL AC POWER
1=TURNED ON AND OPERATING
0=TURNED OFF OR FAILED

CONTACTORS/RELAYS

K1=NO. 1 GENERATOR CONTACTOR
K2=NO. 2 GENERATOR CONTACTOR
K3=APU / EXT PWR CONTACTOR
K4=AC BUS TIE CONTACTOR
K8=AC ESNTL BUS XFR RELAY
K13=AC ESNTL BUS FAIL RELAY
X=ENERGIZED
O=DE−ENERGIZED

AC POWER SOURCE				CONDITION	AC SYSTEM CONTACTORS/ RELAYS					
A	B	C	D		K1	K2	K3	K4	K8	K13
0	0	0	0	1	0	0	0	0	0	0
1	0	0	0	2	X	0	0	0	X	X
0	1	0	0	3	0	X	0	0	X	X
1	1	0	0	4	X	X	0	0	X	X
0	0	1	0	5	0	0	X	X	X	X
1	0	1	0	6	X	0	X	0	X	X
0	1	1	0	7	0	X	X	0	X	X
1	1	1	0	8	X	X	X	0	X	X
0	0	0	1	9	0	0	0	X	X	X
1	0	0	1	10	X	0	0	0	X	X
0	1	0	1	11	0	X	0	0	X	X
1	1	0	1	12	X	X	0	0	X	X
0	0	1	1	13	0	0	X	X	X	X
1	0	1	1	14	X	0	X	0	X	X
0	1	1	1	15	0	X	X	0	X	X
1	1	1	1	16	X	X	X	0	X	X

Figure 2. AC Electrical System Schematic Diagram. (Sheet 1 of 20)

SCHEMATICS - Continued

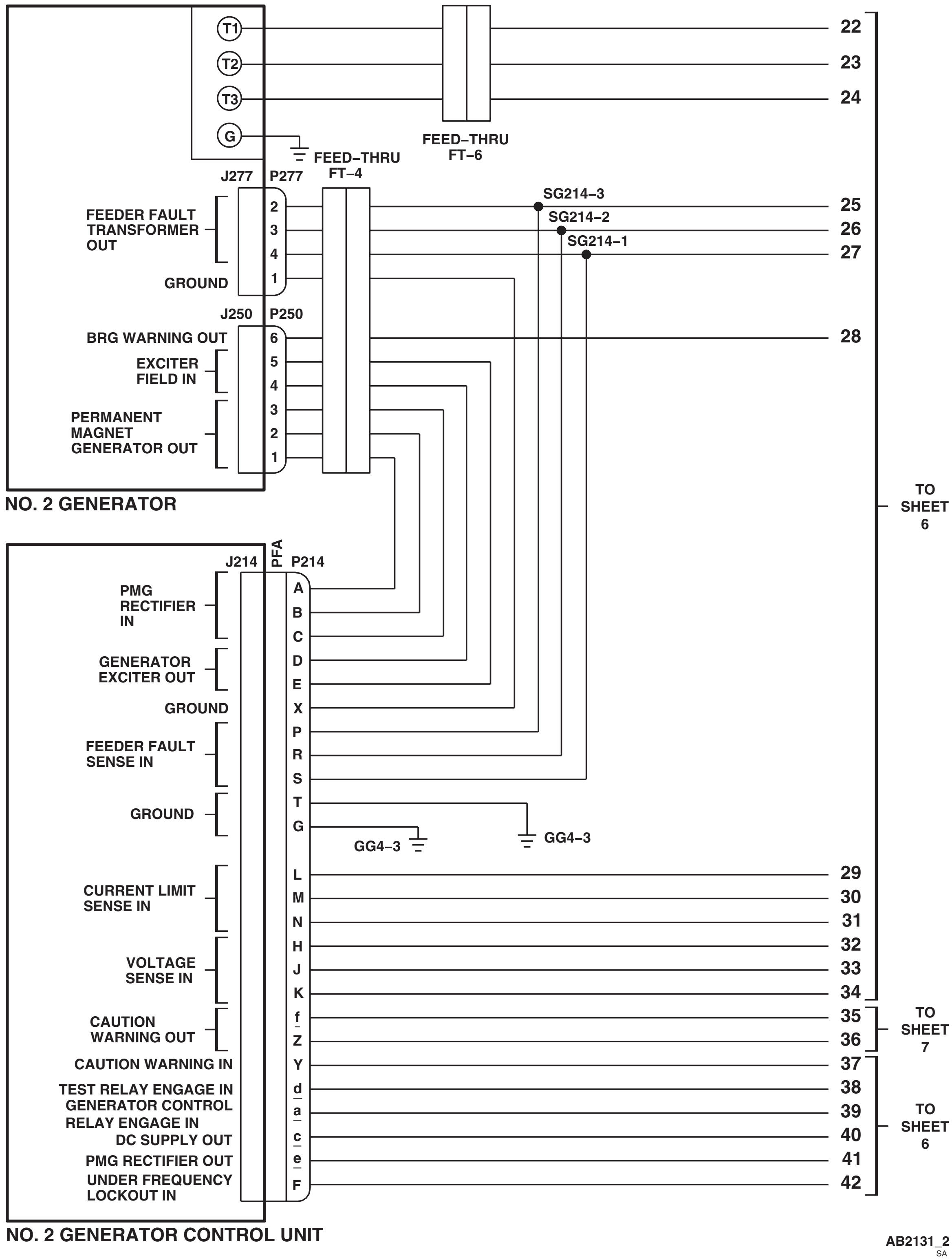


Figure 2. AC Electrical System Schematic Diagram. (Sheet 2 of 20)

SCHEMATICS - Continued

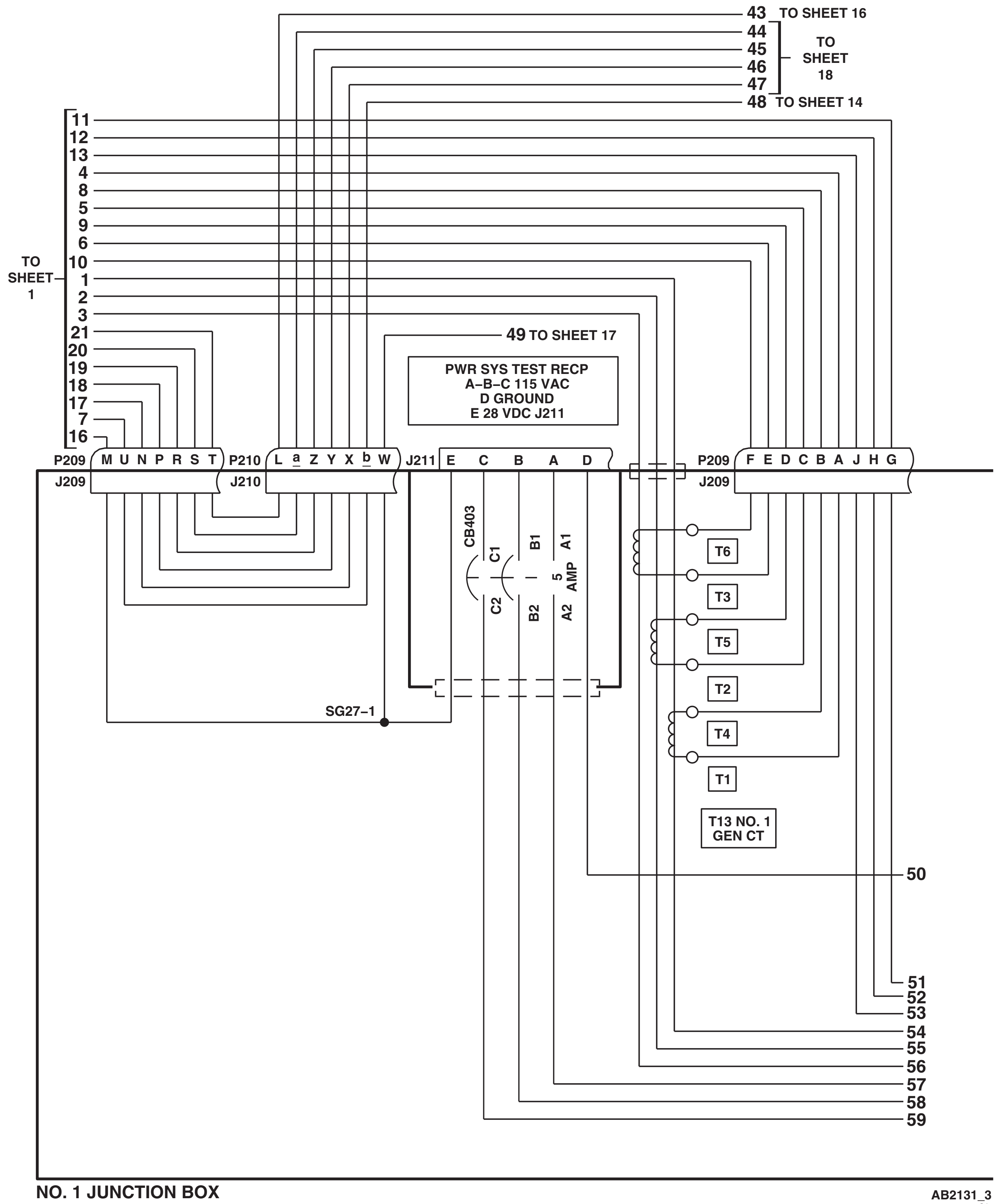


Figure 2. AC Electrical System Schematic Diagram. (Sheet 3 of 20)

SCHEMATICS - Continued

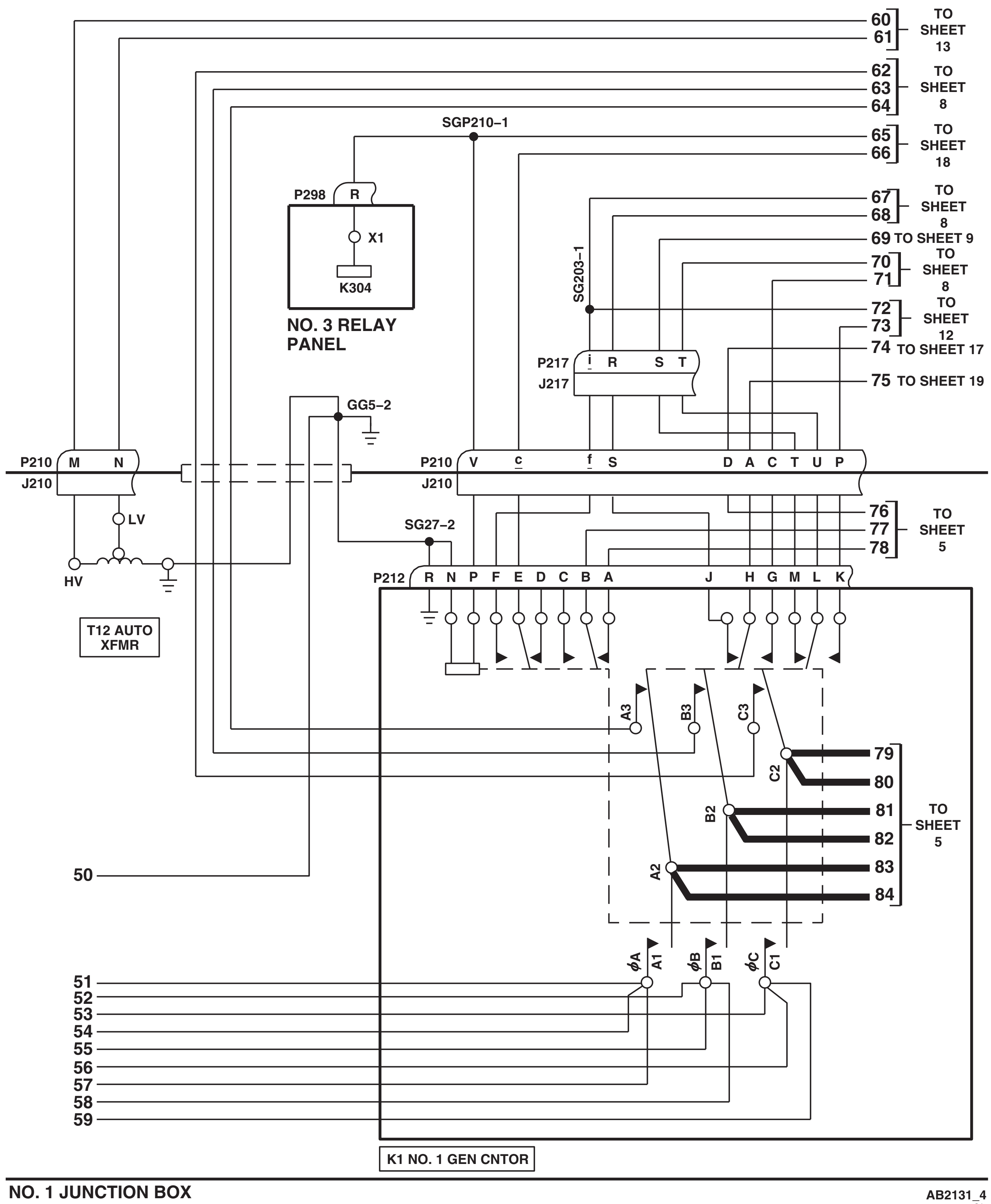


Figure 2. AC Electrical System Schematic Diagram. (Sheet 4 of 20)

SCHEMATICS - Continued

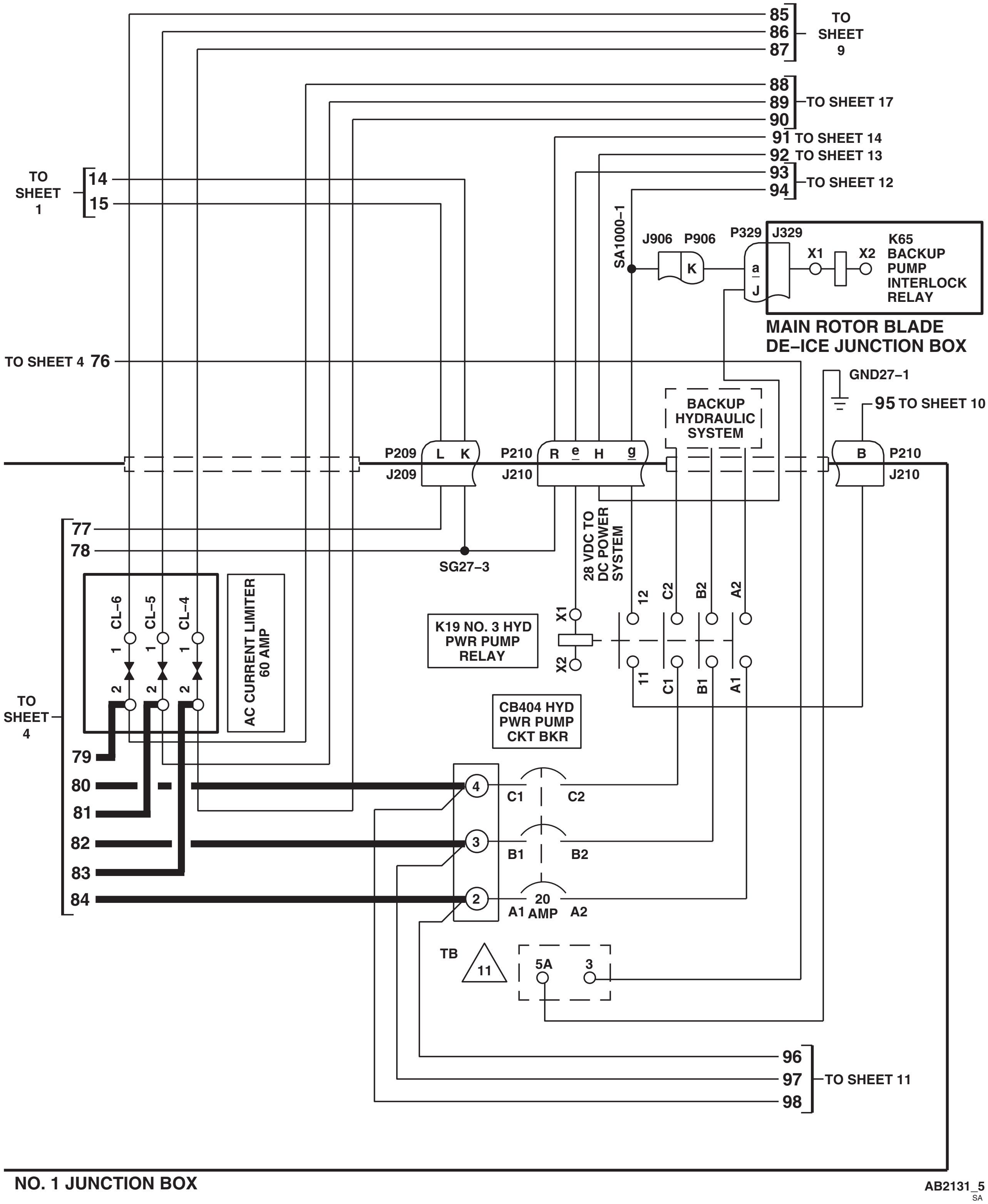


Figure 2. AC Electrical System Schematic Diagram. (Sheet 5 of 20)

SCHEMATICS - Continued

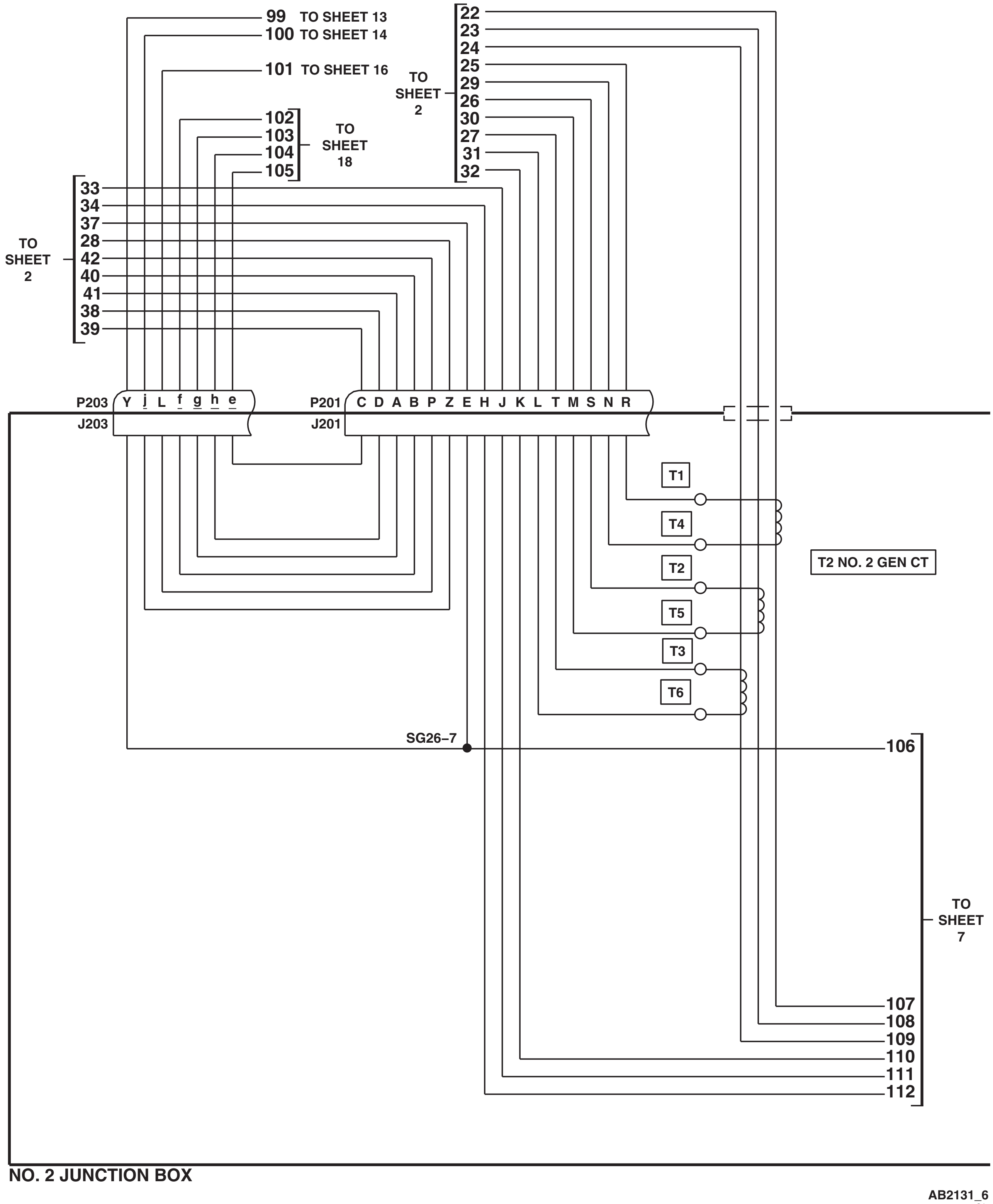


Figure 2. AC Electrical System Schematic Diagram. (Sheet 6 of 20)

SCHEMATICS - Continued

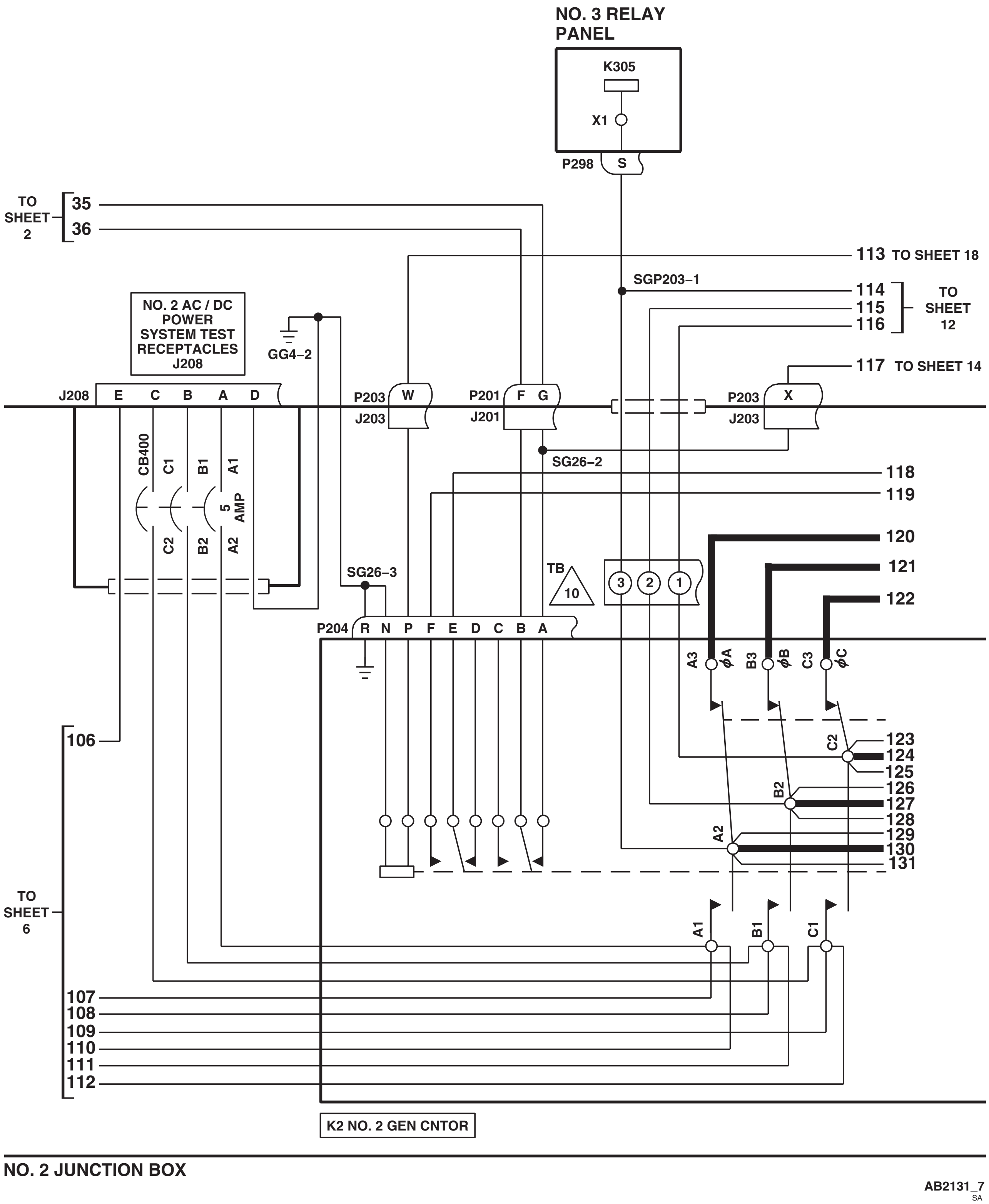


Figure 2. AC Electrical System Schematic Diagram. (Sheet 7 of 20)

SCHEMATICS - Continued

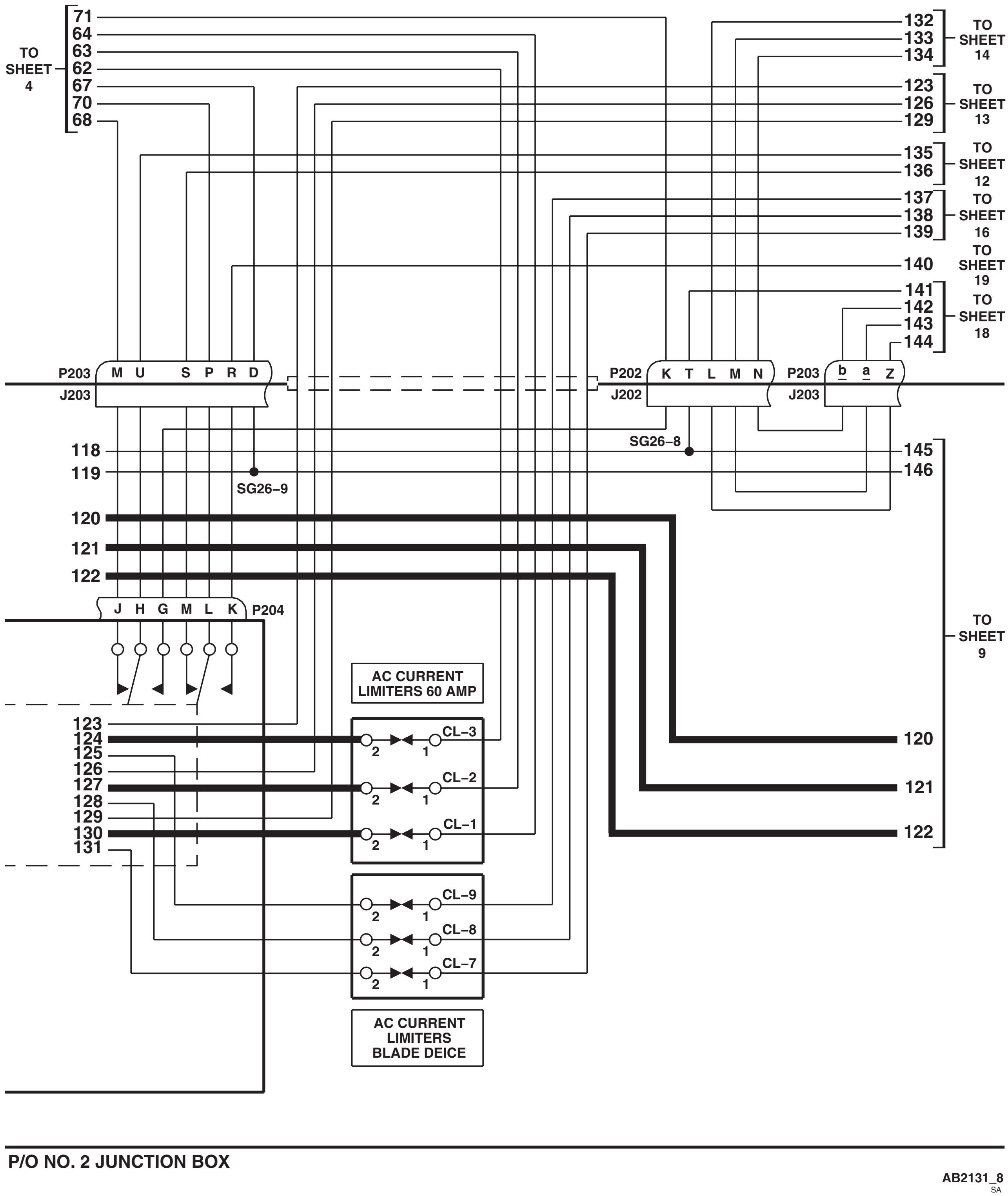


Figure 2. AC Electrical System Schematic Diagram. (Sheet 8 of 20)

SCHEMATICS - Continued

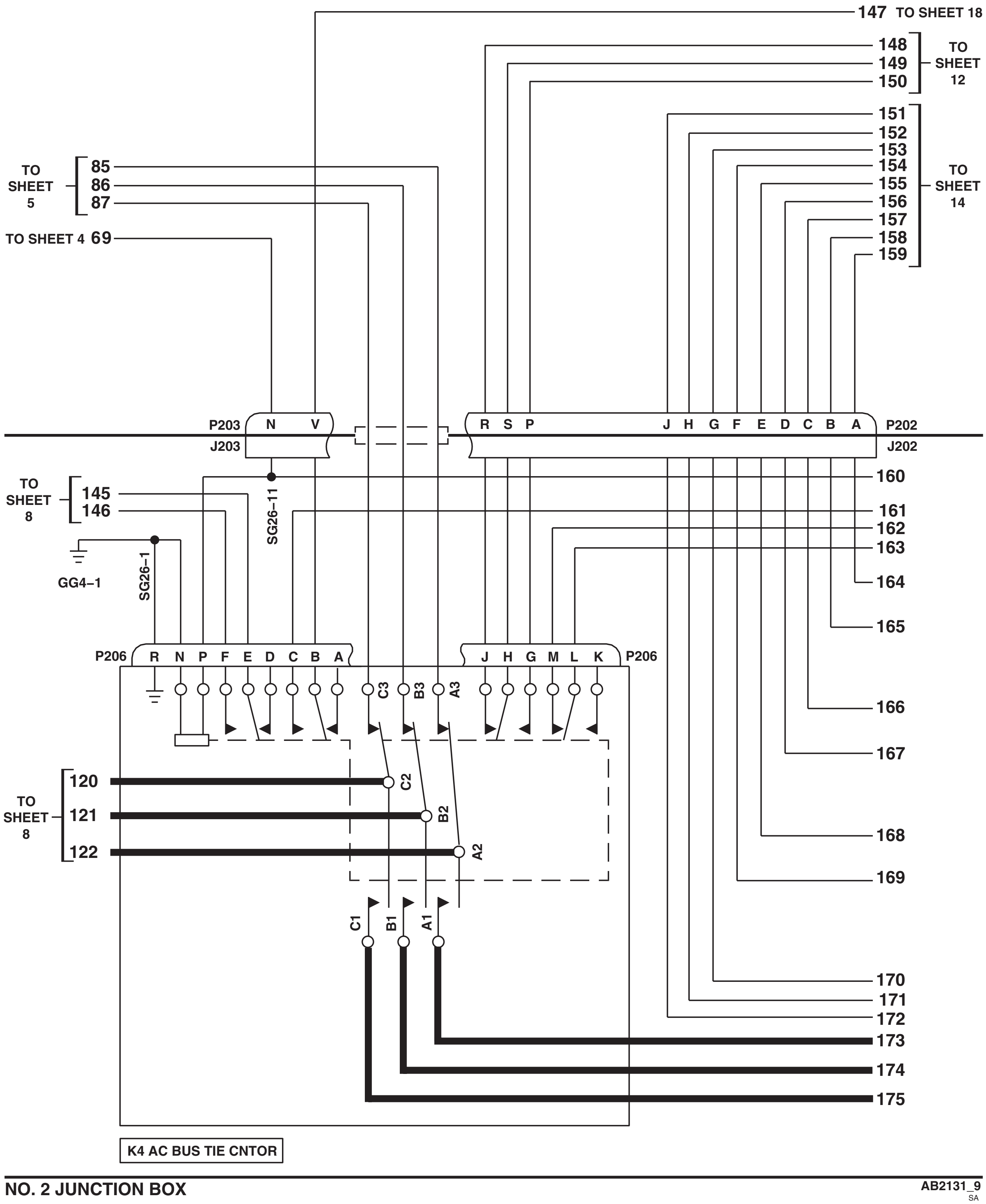


Figure 2. AC Electrical System Schematic Diagram. (Sheet 9 of 20)

SCHEMATICS - Continued

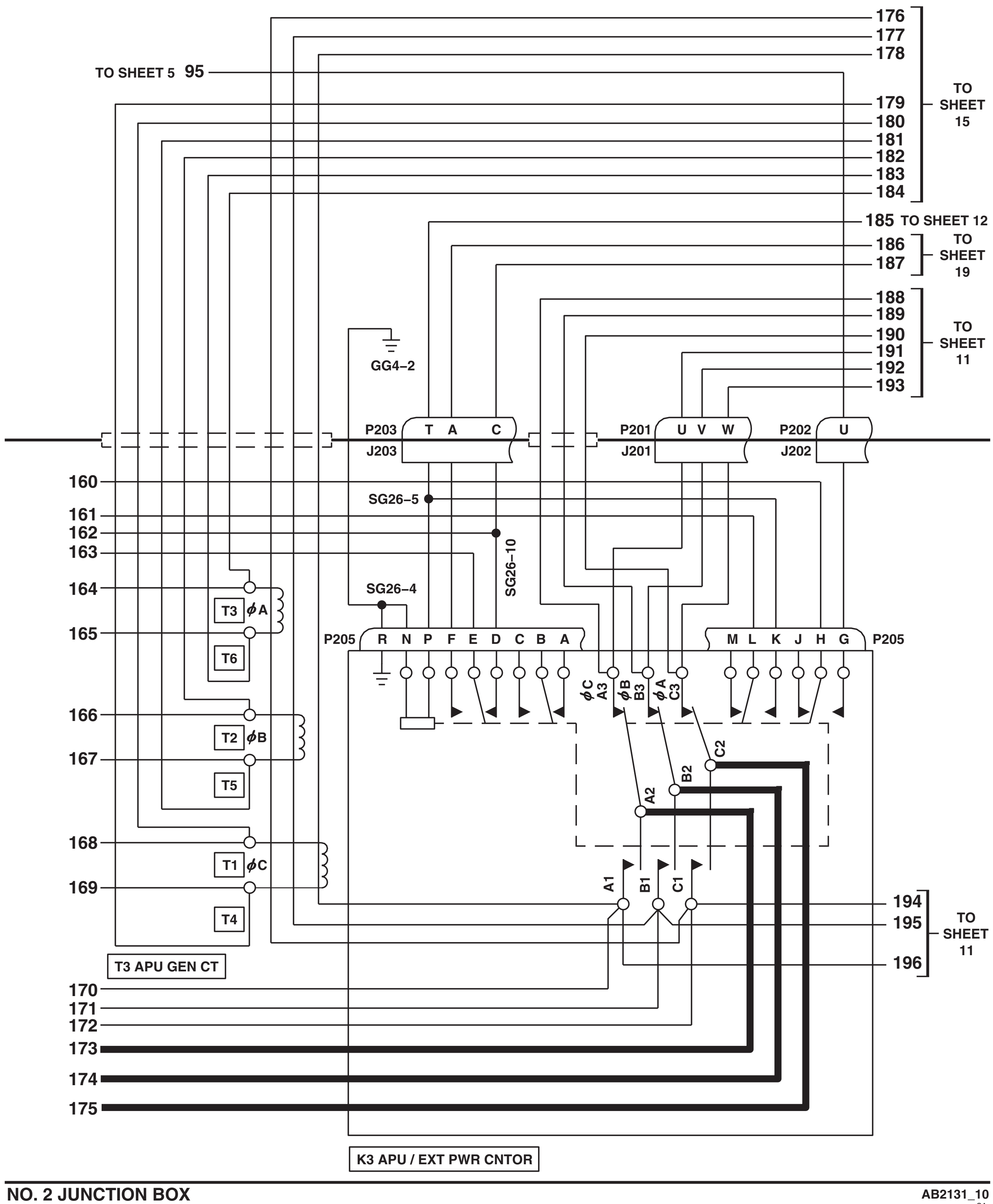


Figure 2. AC Electrical System Schematic Diagram. (Sheet 10 of 20)

SCHEMATICS - Continued

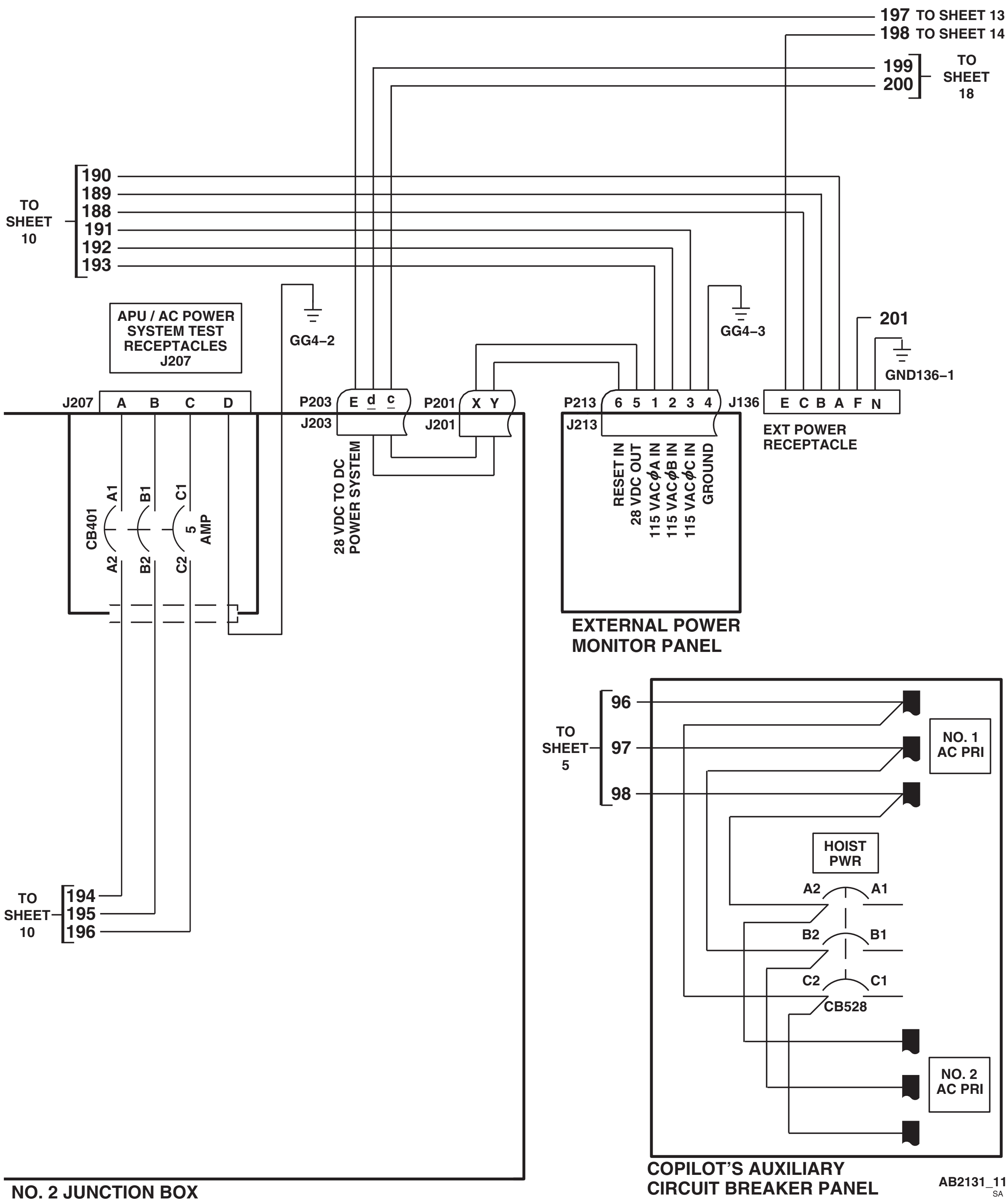


Figure 2. AC Electrical System Schematic Diagram. (Sheet 11 of 20)

SCHEMATICS - Continued

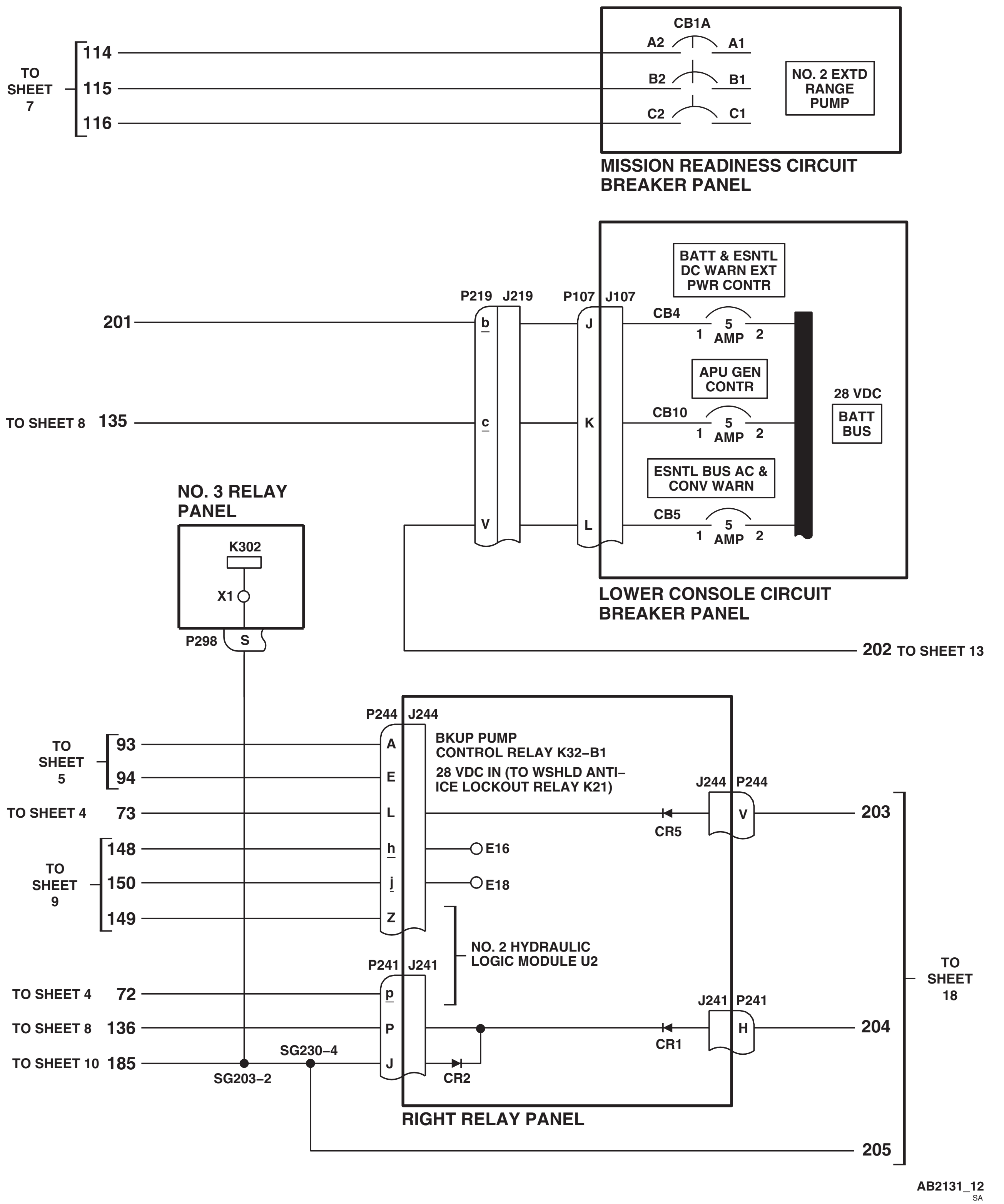


Figure 2. AC Electrical System Schematic Diagram. (Sheet 12 of 20)

SCHEMATICS - Continued

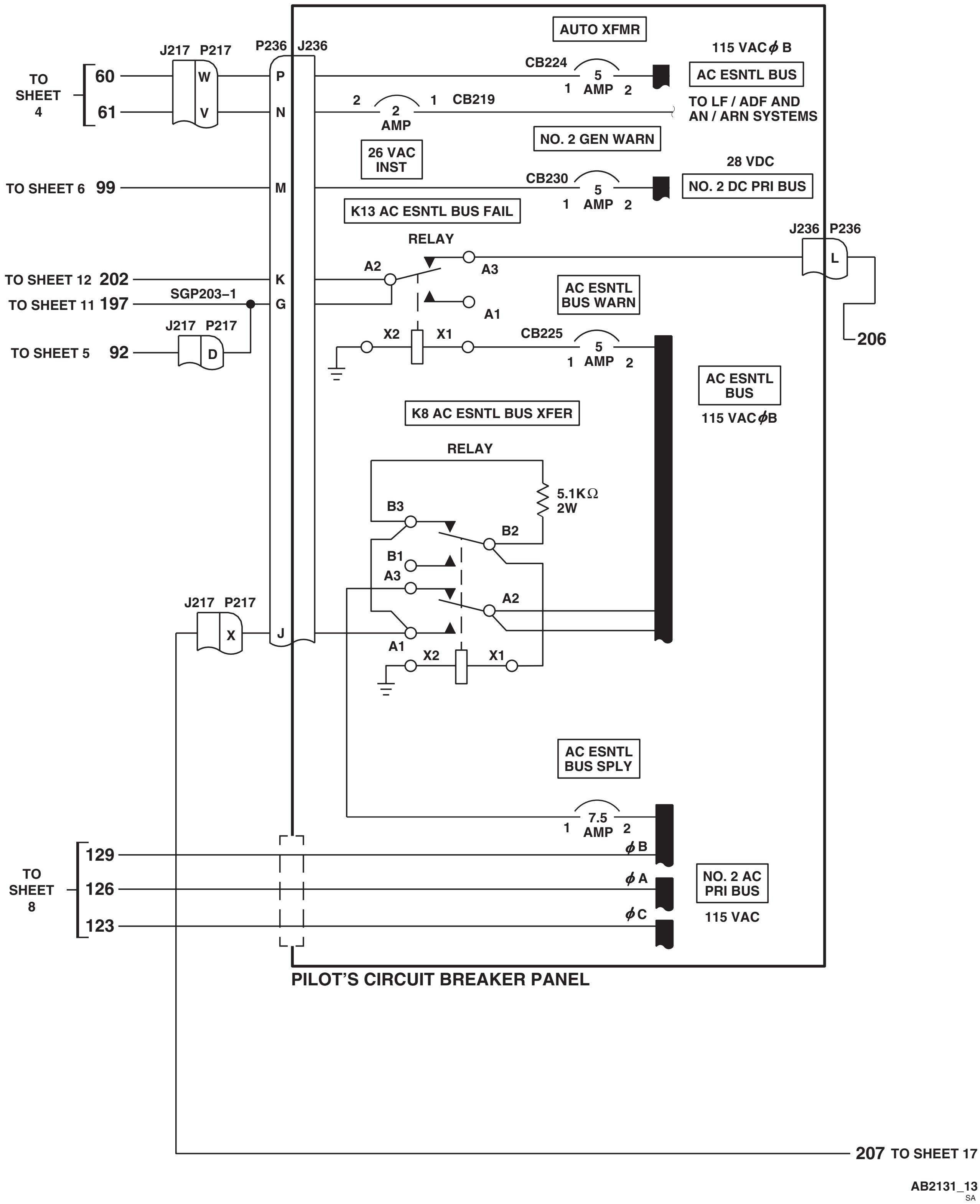


Figure 2. AC Electrical System Schematic Diagram. (Sheet 13 of 20)

SCHEMATICS - Continued

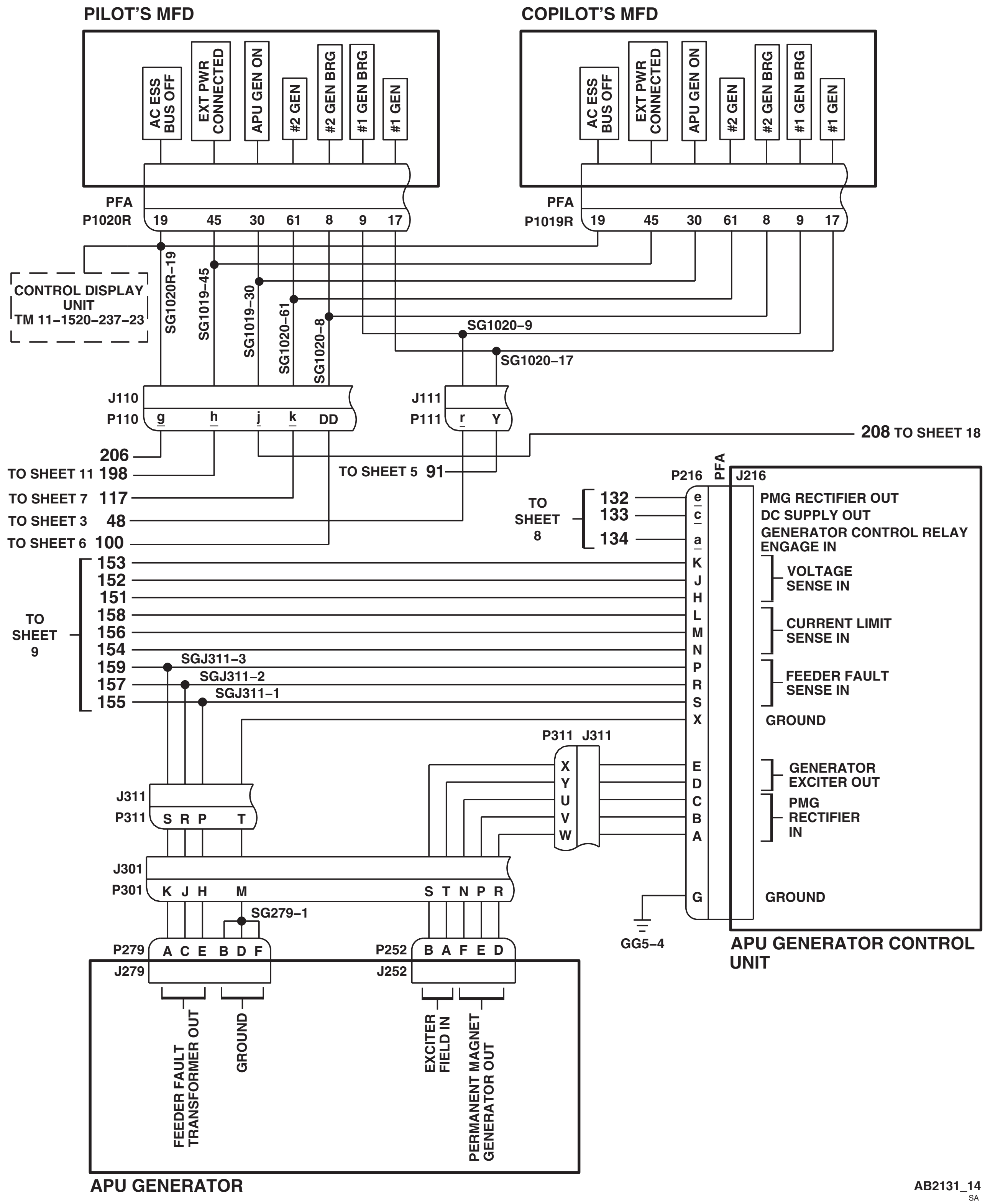


Figure 2. AC Electrical System Schematic Diagram. (Sheet 14 of 20)

SCHEMATICS - Continued

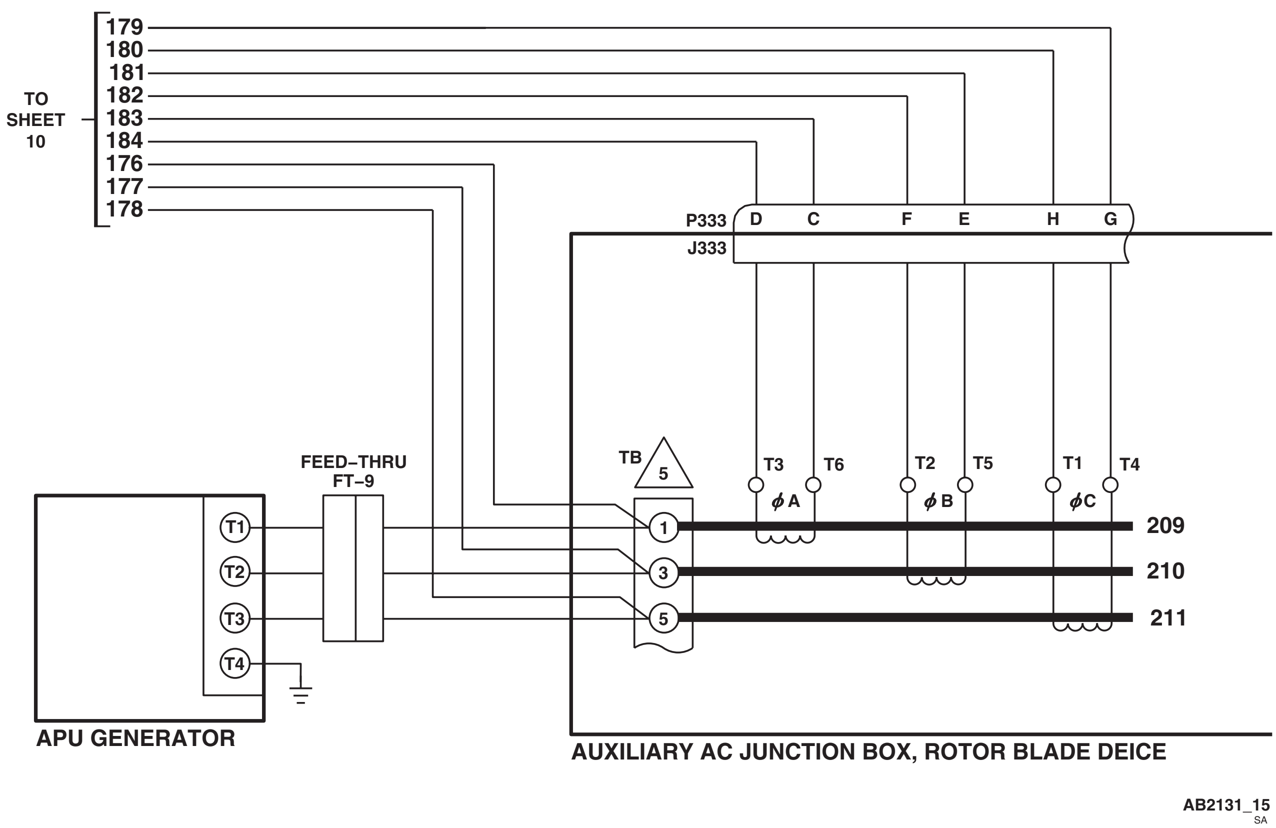


Figure 2. AC Electrical System Schematic Diagram. (Sheet 15 of 20)

SCHEMATICS - Continued

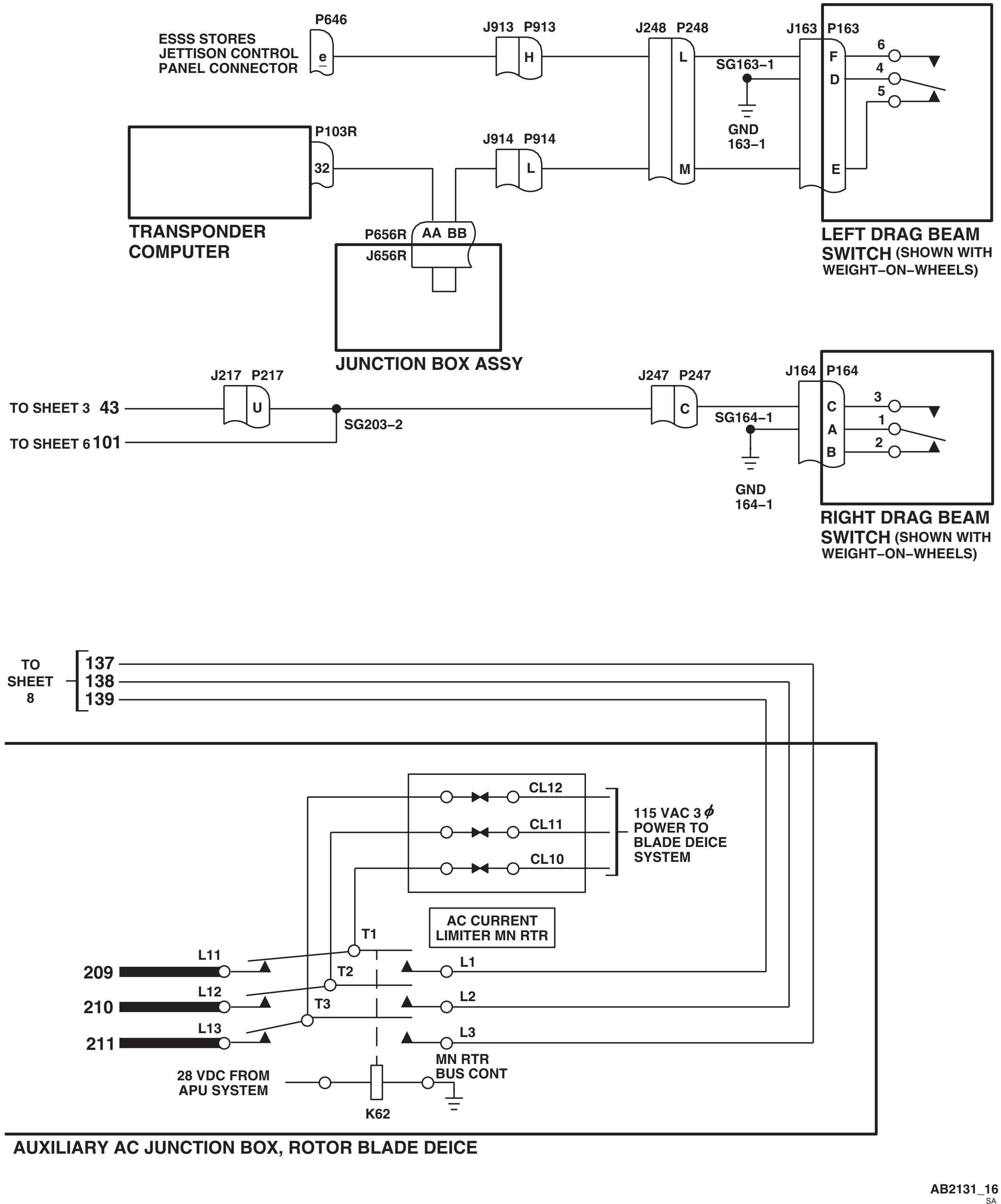


Figure 2. AC Electrical System Schematic Diagram. (Sheet 16 of 20)

SCHEMATICS - Continued

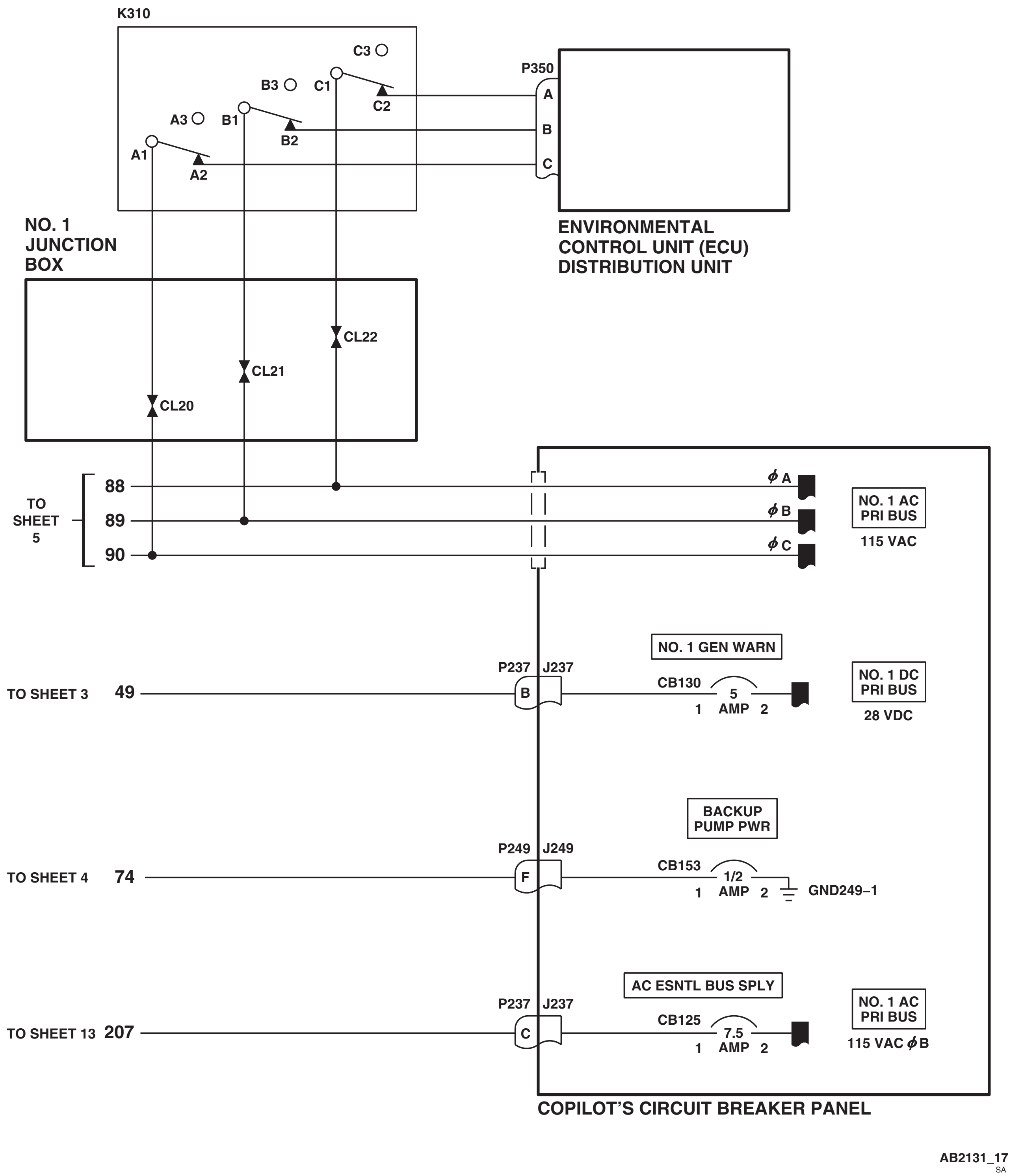


Figure 2. AC Electrical System Schematic Diagram. (Sheet 17 of 20)

SCHEMATICS - Continued

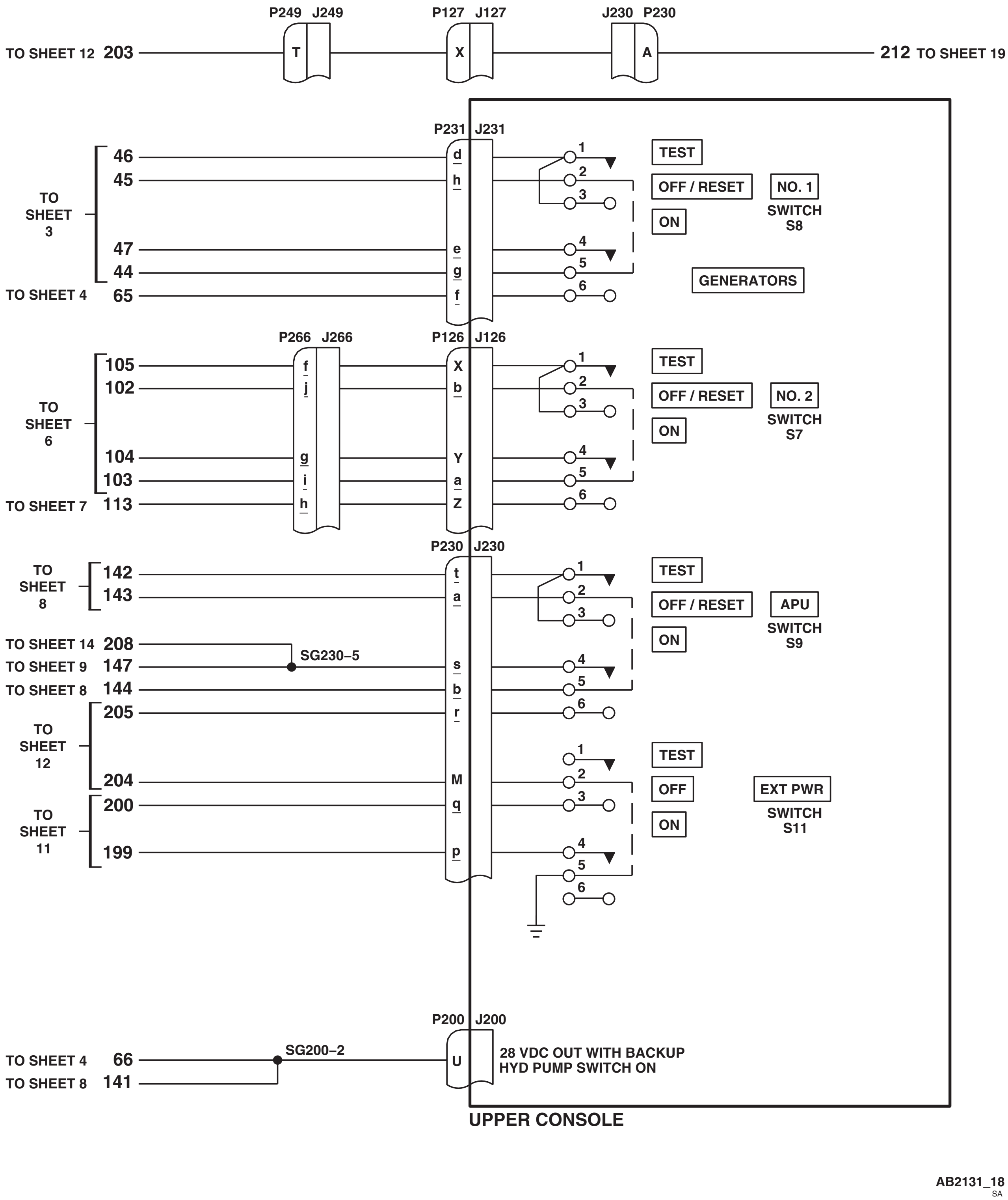


Figure 2. AC Electrical System Schematic Diagram. (Sheet 18 of 20)

SCHEMATICS - Continued

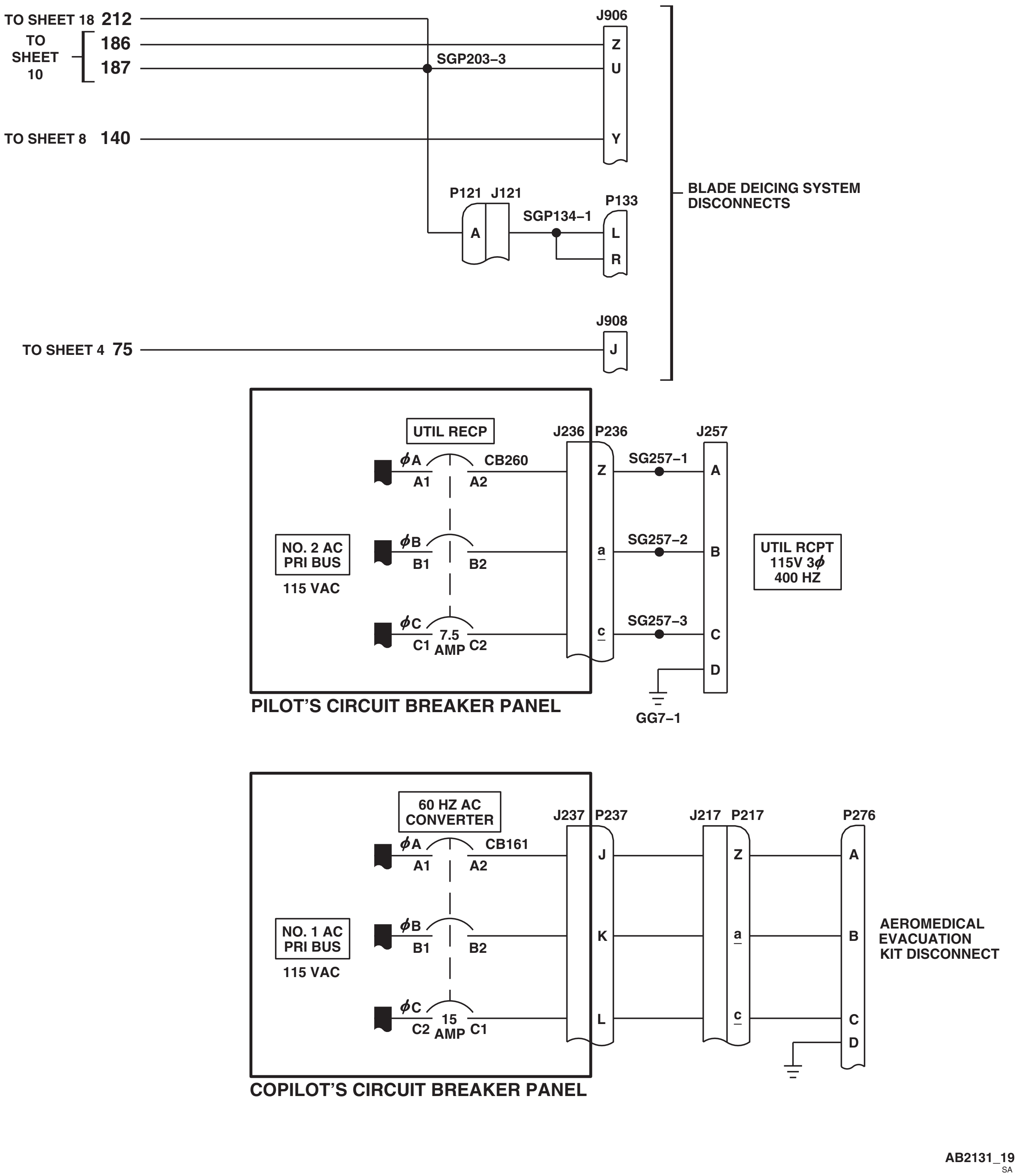


Figure 2. AC Electrical System Schematic Diagram. (Sheet 19 of 20)

SCHEMATICS - Continued

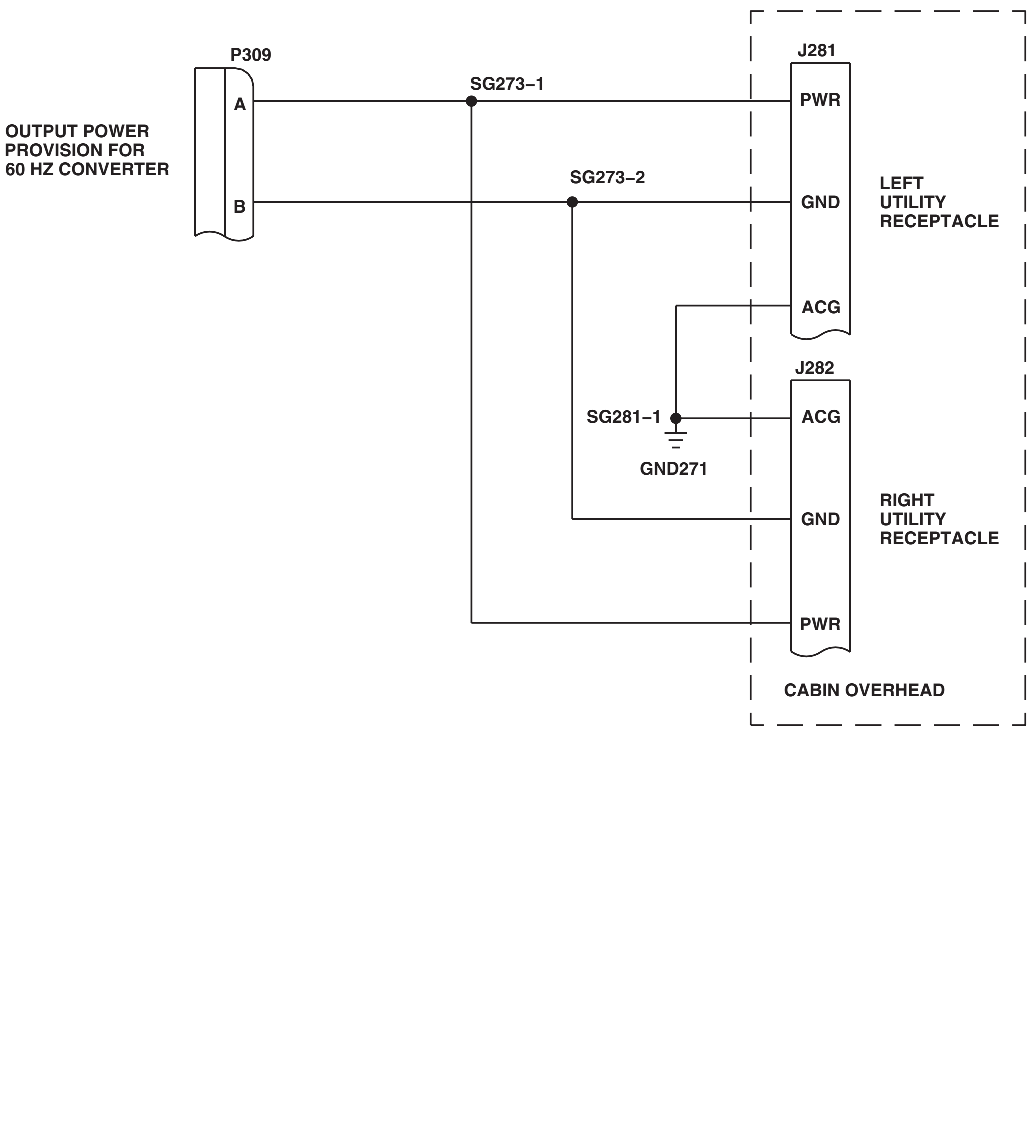


Figure 2. AC Electrical System Schematic Diagram. (Sheet 20 of 20)

SCHEMATICS - Continued

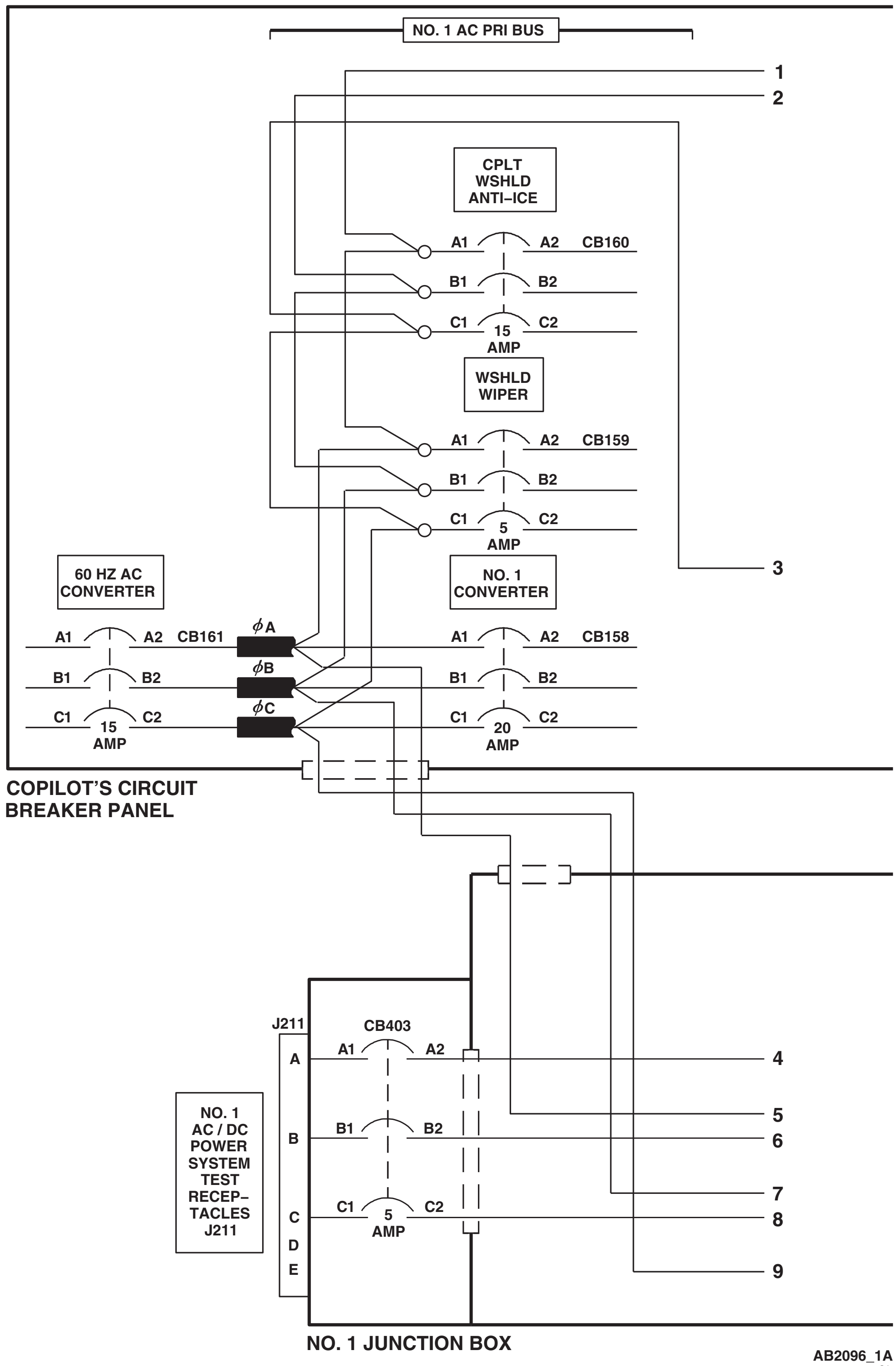


Figure 3. AC Electrical System Power Distribution Diagram HH-60A . (Sheet 1 of 6)

SCHEMATICS - Continued

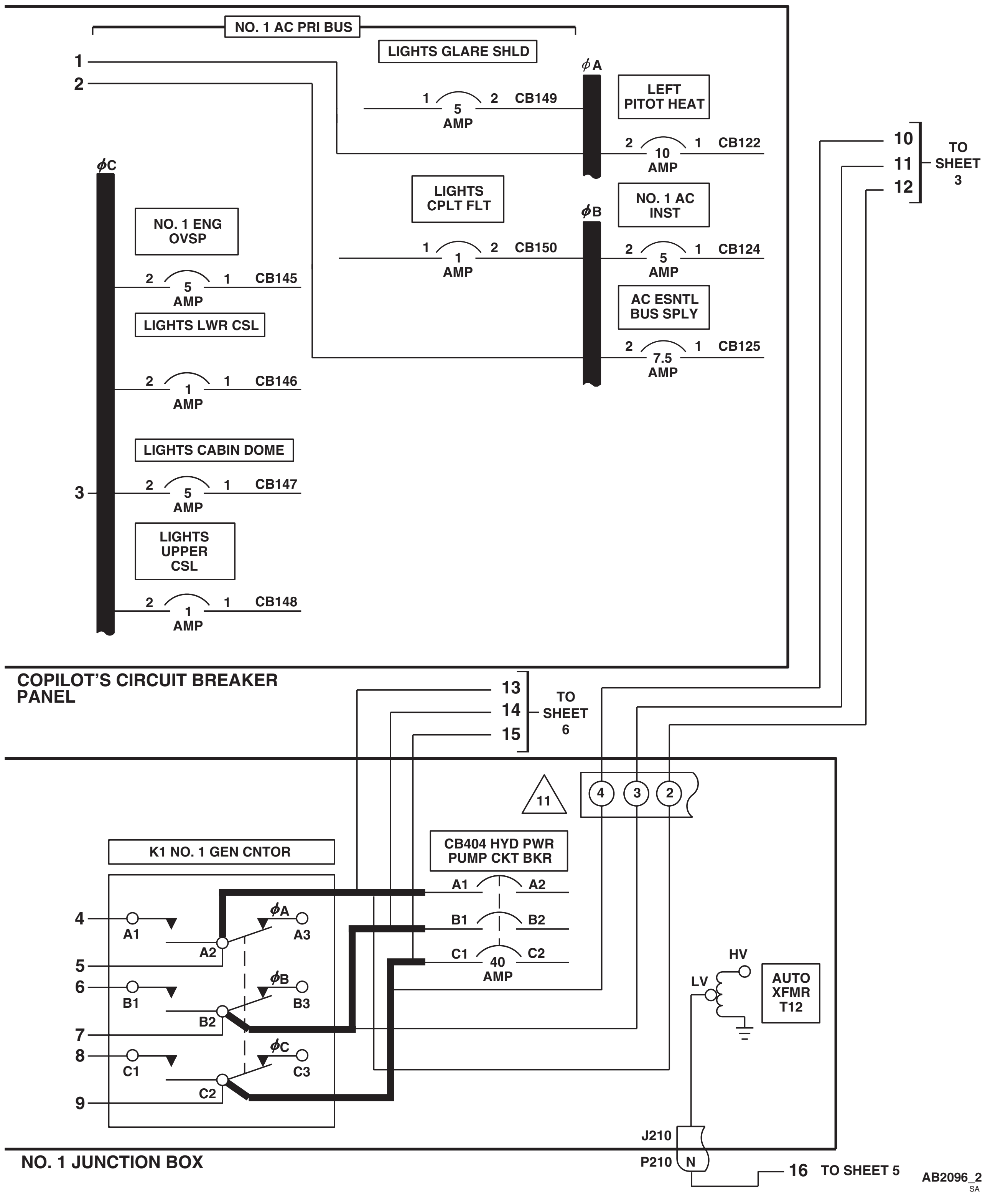


Figure 3. AC Electrical System Power Distribution Diagram **HH-60A** . (Sheet 2 of 6)

SCHEMATICS - Continued

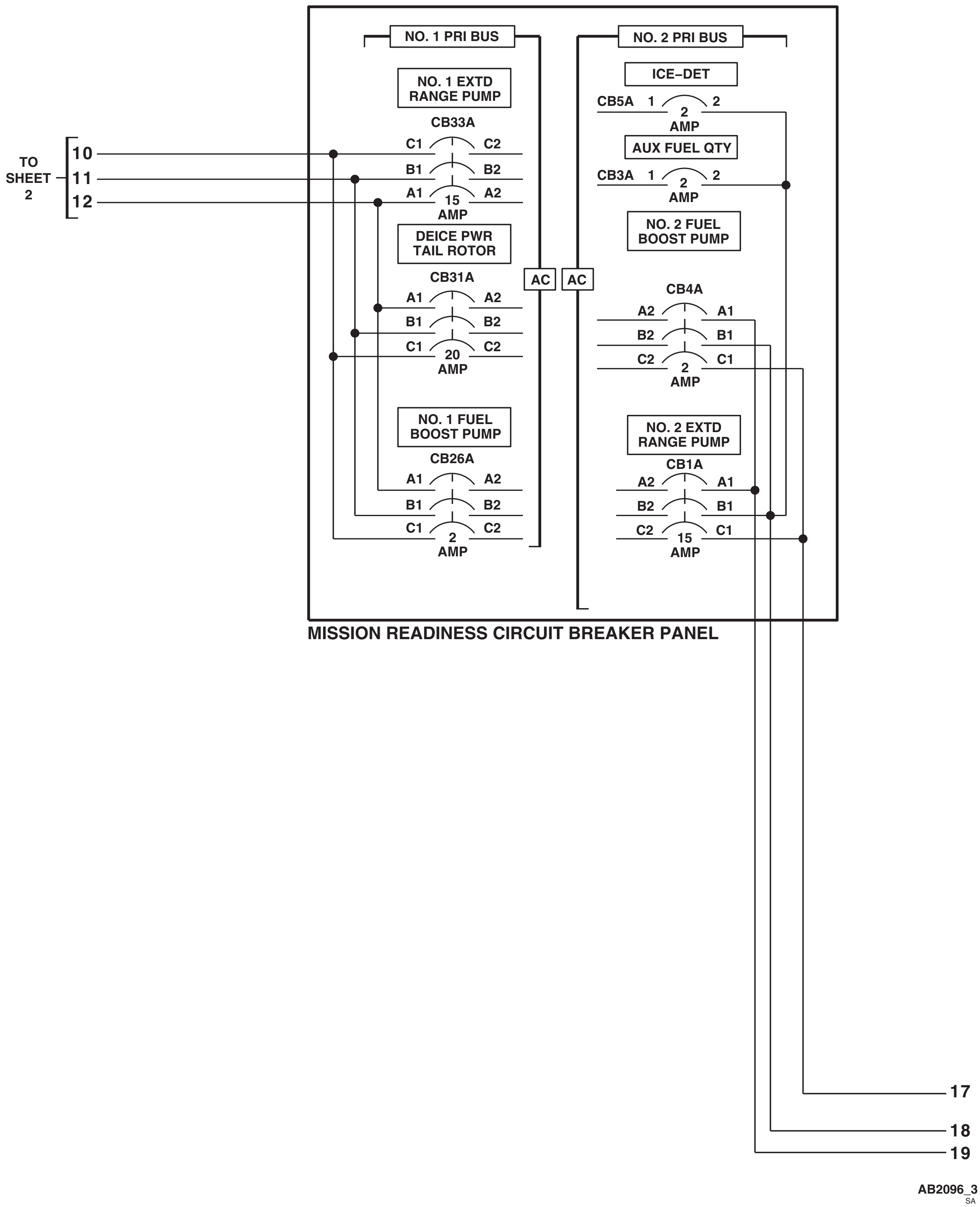


Figure 3. AC Electrical System Power Distribution Diagram **HH-60A** . (Sheet 3 of 6)

SCHEMATICS - Continued

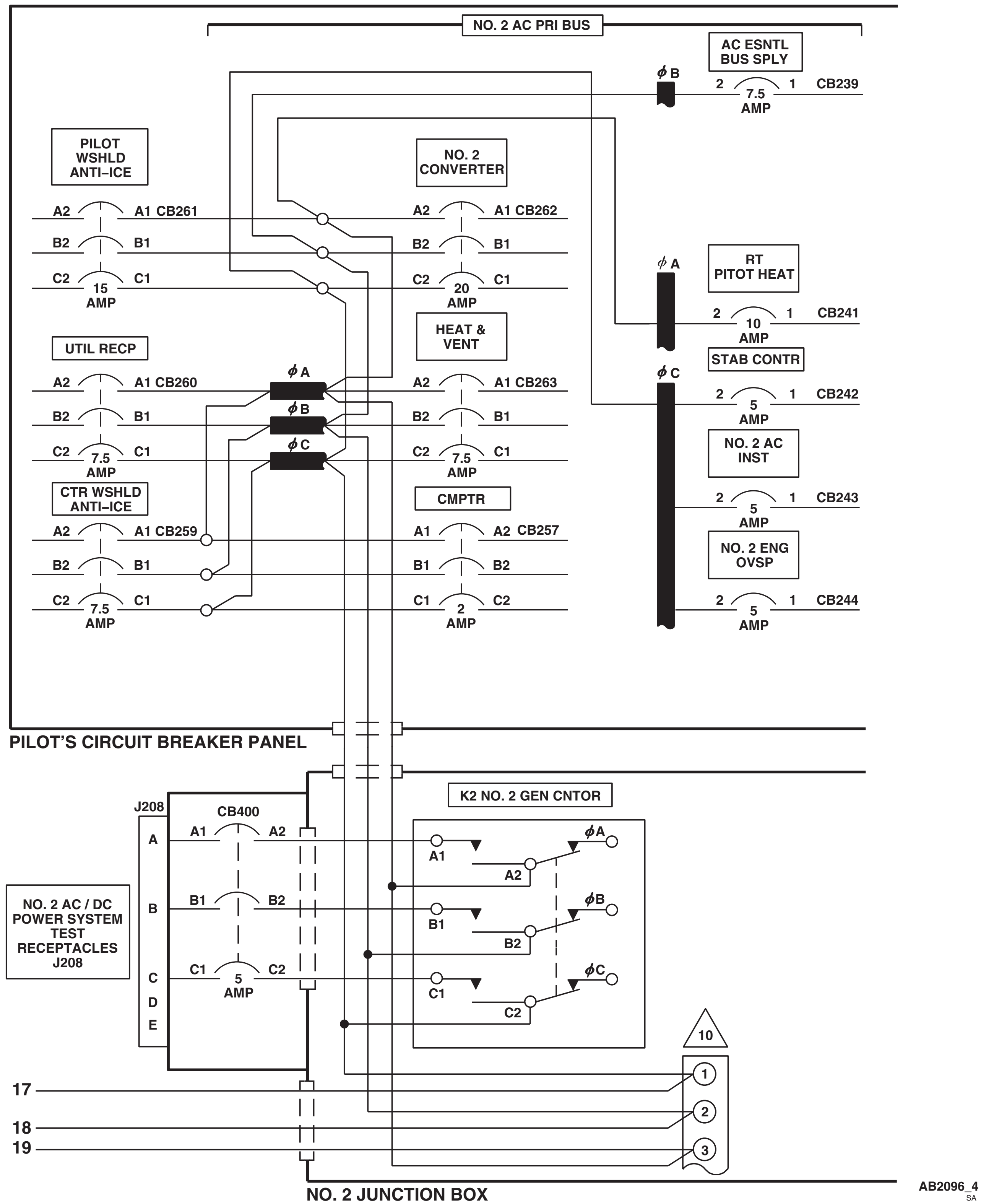


Figure 3. AC Electrical System Power Distribution Diagram **HH-60A** . (Sheet 4 of 6)

SCHEMATICS - Continued

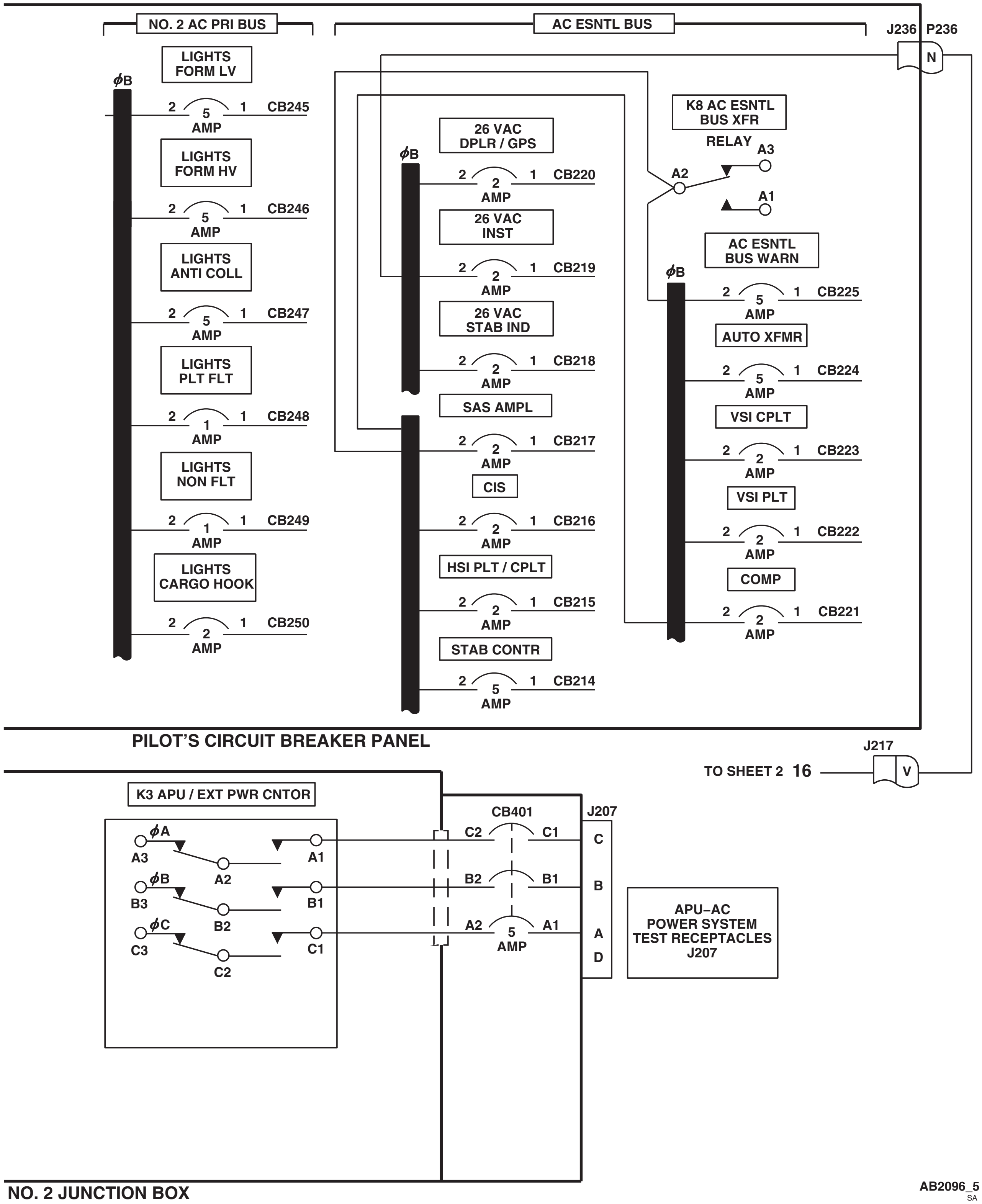


Figure 3. AC Electrical System Power Distribution Diagram **HH-60A** . (Sheet 5 of 6)

SCHEMATICS - Continued

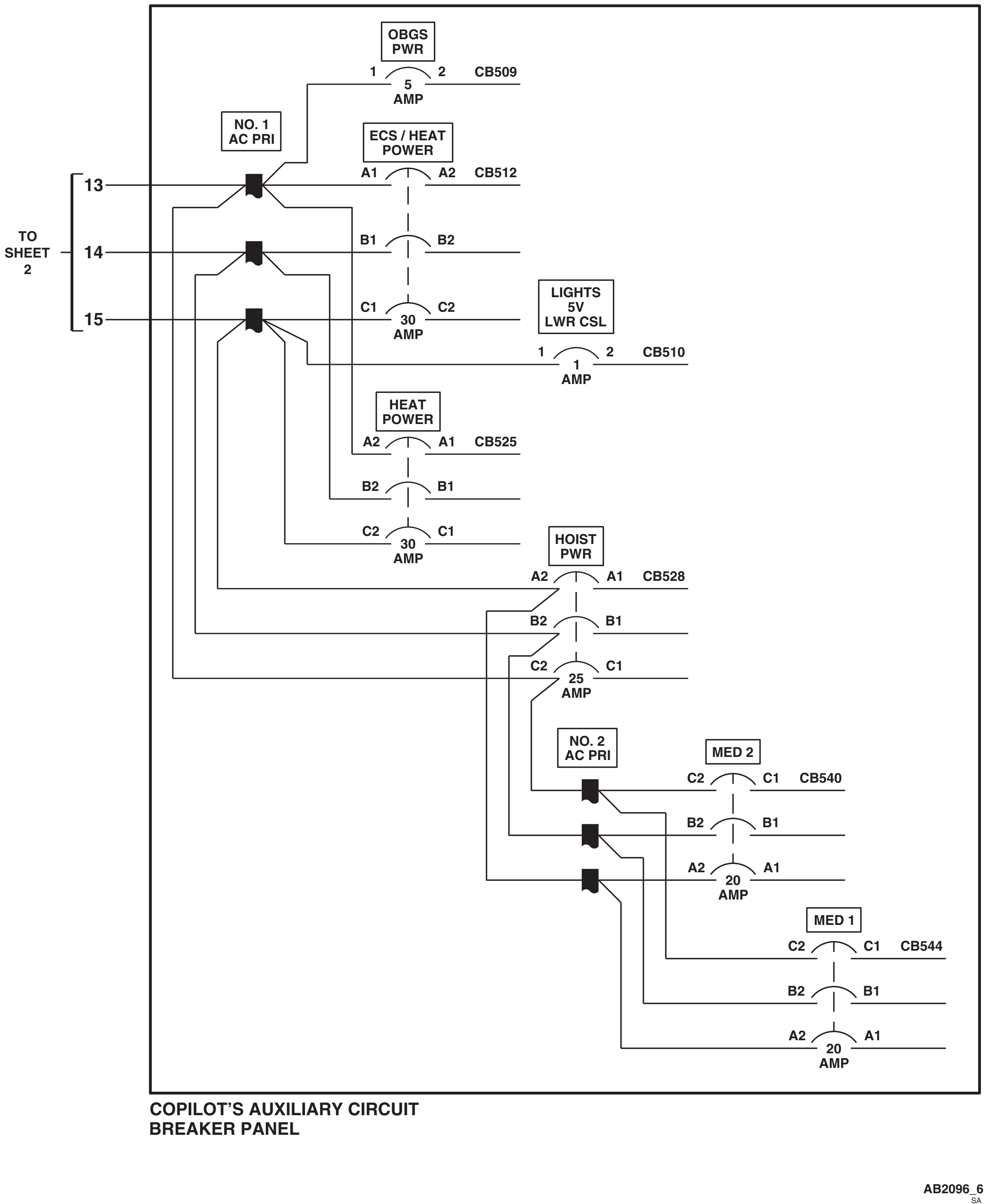


Figure 3. AC Electrical System Power Distribution Diagram **HH-60A** . (Sheet 6 of 6)

SCHEMATICS - Continued

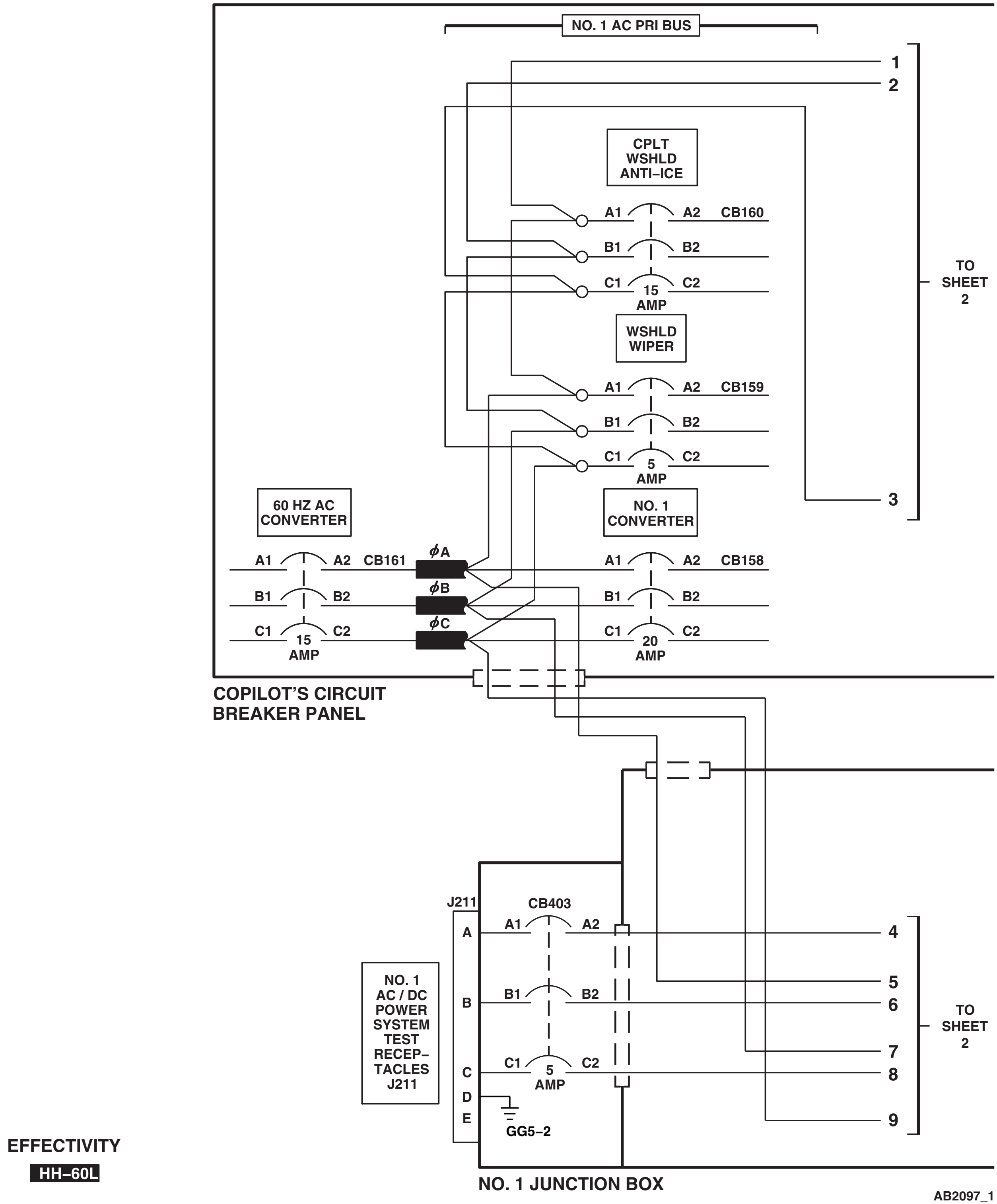


Figure 4. AC Electrical System Power Distribution Diagram HH-60L . (Sheet 1 of 6)

SCHEMATICS - Continued

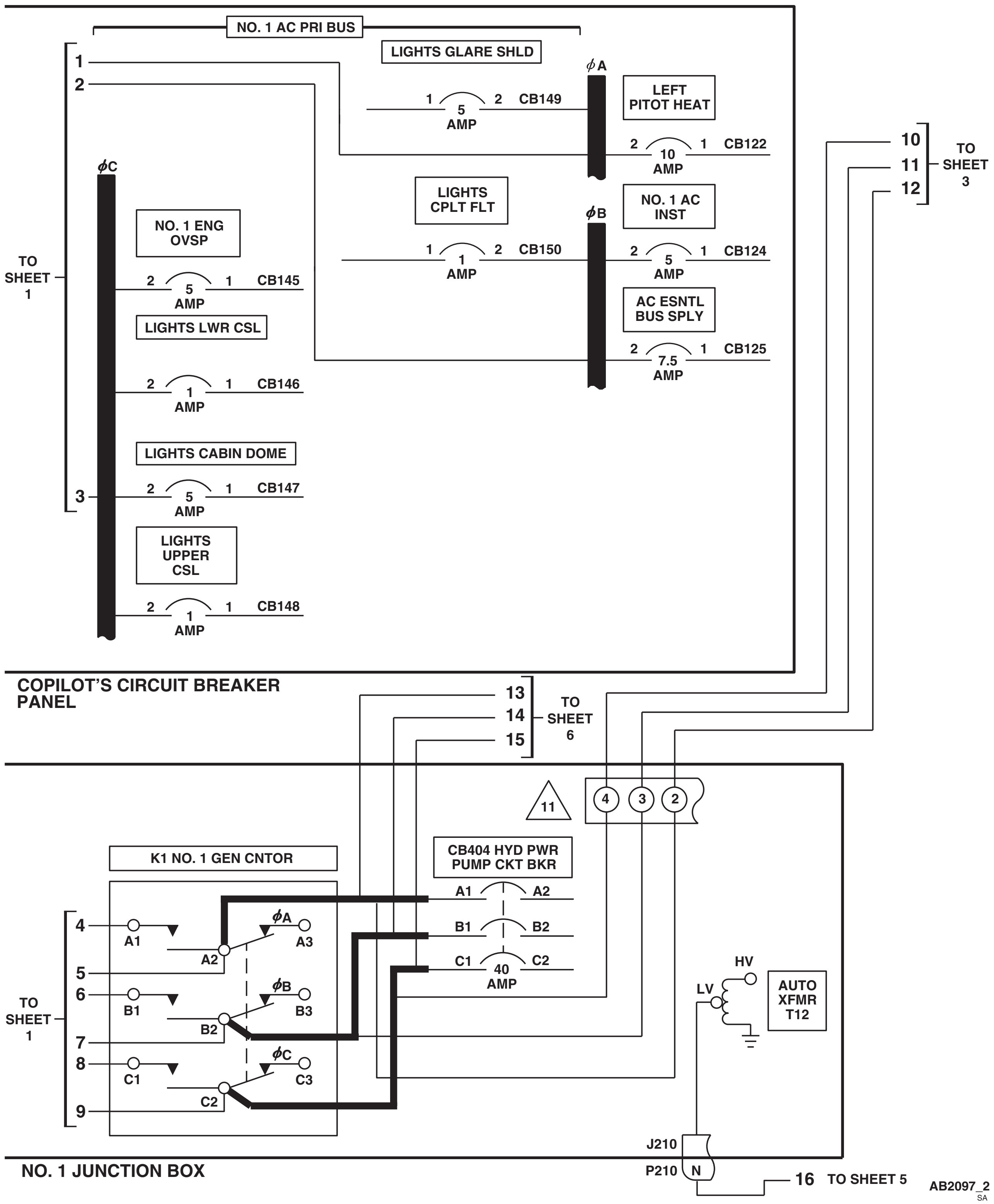


Figure 4. AC Electrical System Power Distribution Diagram HH-60L . (Sheet 2 of 6)

SCHEMATICS - Continued

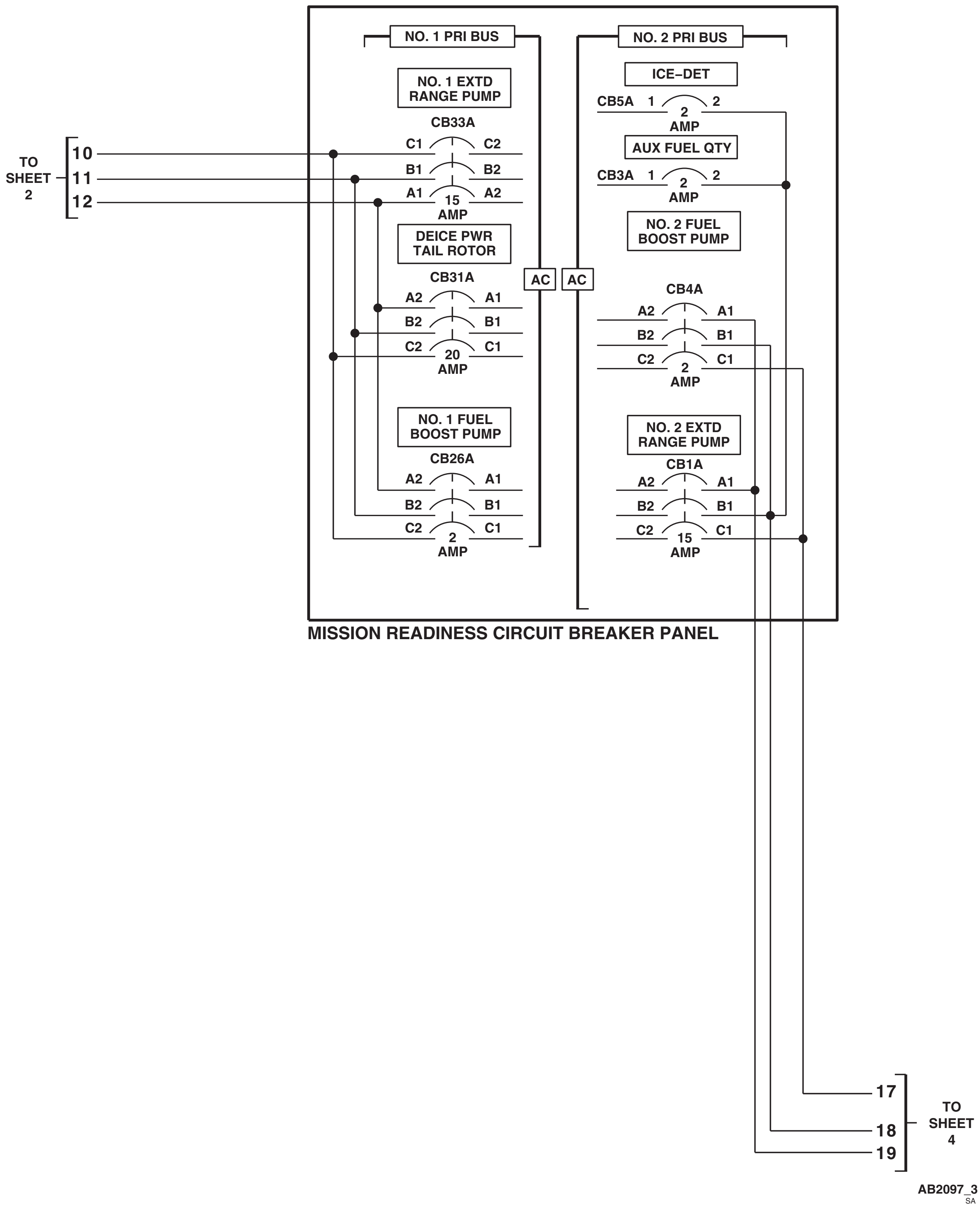


Figure 4. AC Electrical System Power Distribution Diagram **HH-60L** . (Sheet 3 of 6)

SCHEMATICS - Continued

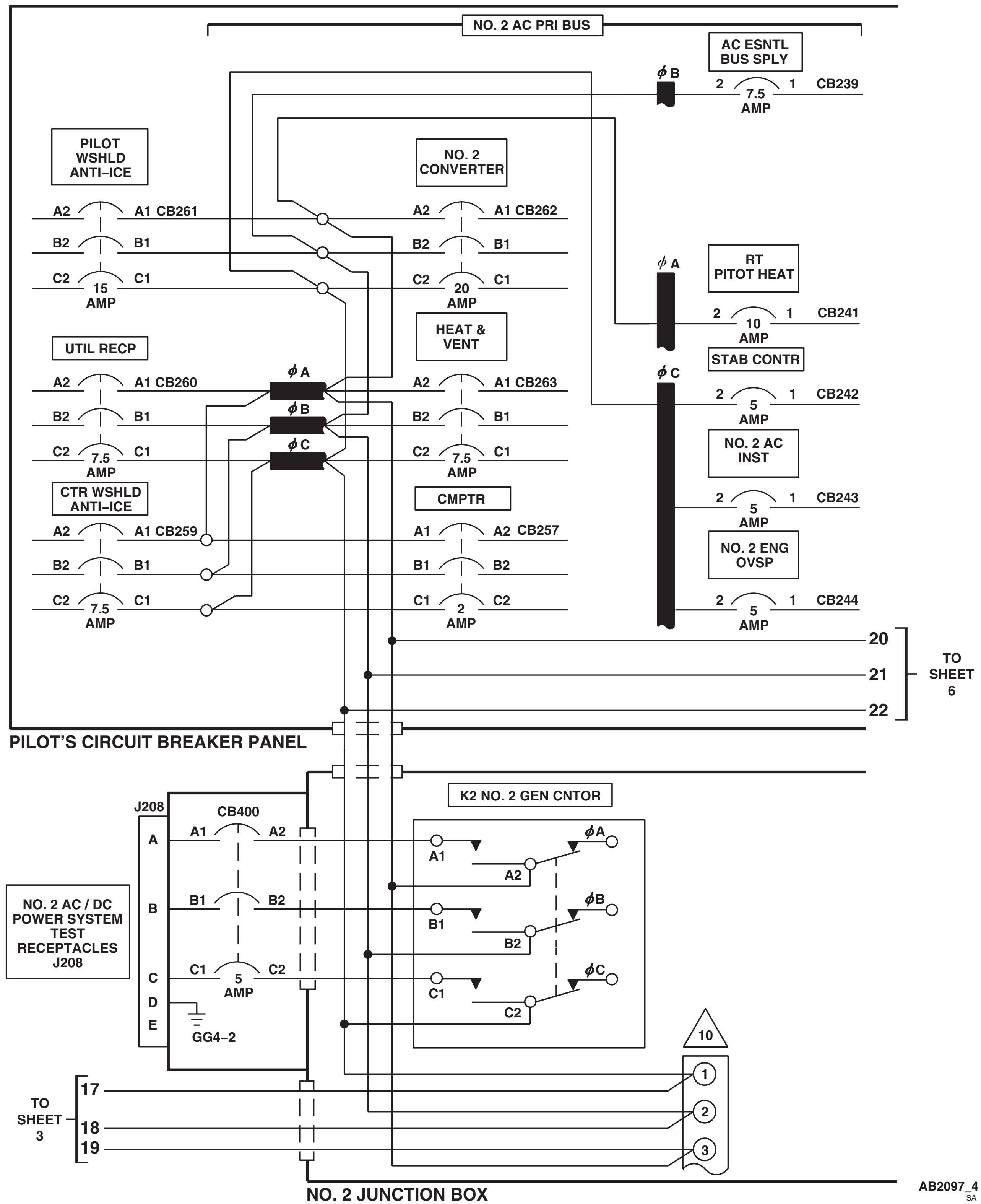


Figure 4. AC Electrical System Power Distribution Diagram **HH-60L** . (Sheet 4 of 6)

SCHEMATICS - Continued

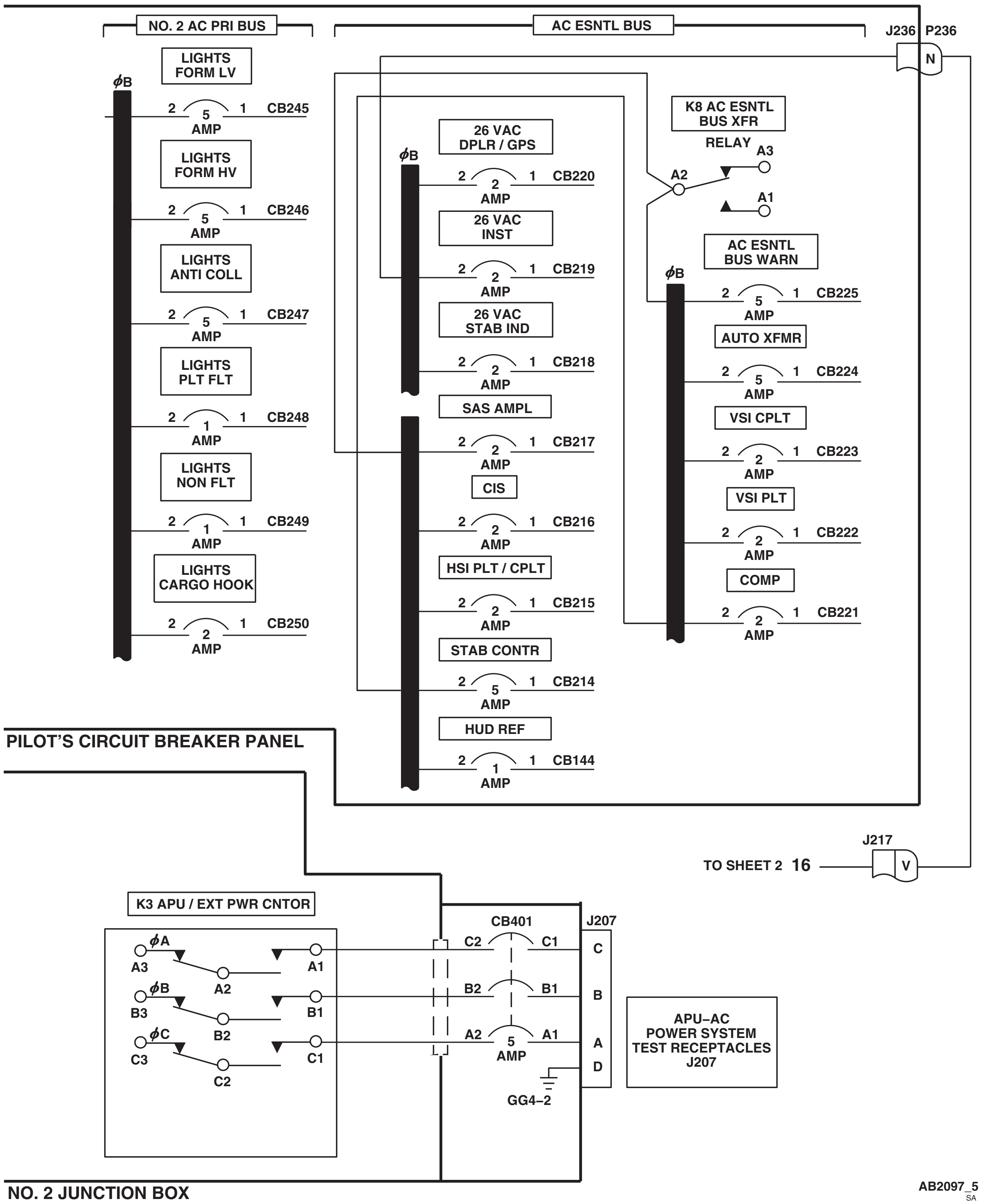


Figure 4. AC Electrical System Power Distribution Diagram HH-60L . (Sheet 5 of 6)

SCHEMATICS - Continued

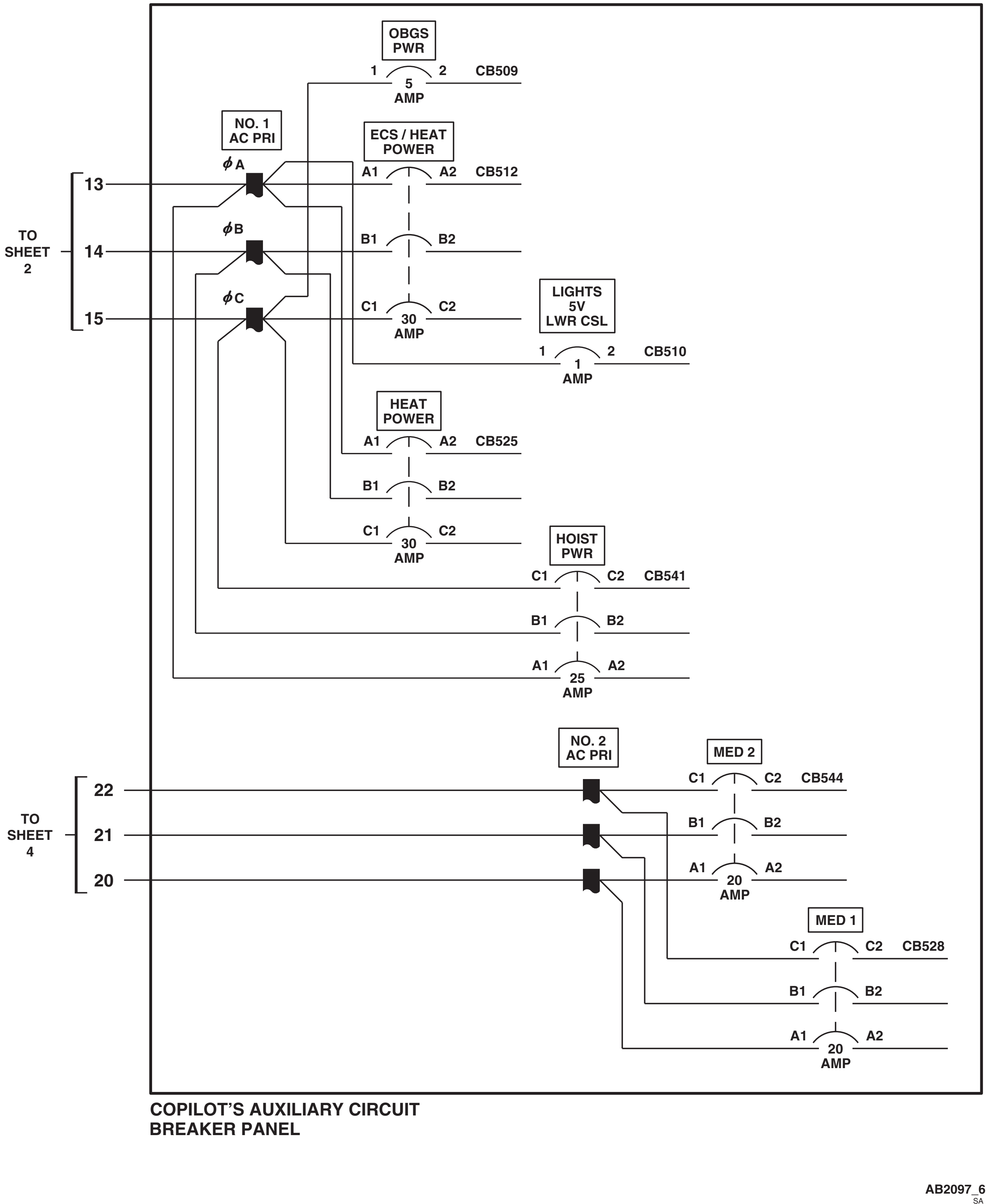


Figure 4. AC Electrical System Power Distribution Diagram **HH-60L** . (Sheet 6 of 6)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

DC ELECTRICAL SYSTEM HH-60A HH-60L

This WP supersedes WP 0123 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Battery Sensor Relay Tester Figure 258, WP 1805 00
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 10-1520-237-10
WP 0100 00
WP 0104 00
WP 0799 00
WP 0824 00
WP 0825 00
WP 0853 00
WP 0855 00
WP 0856 00
WP 0859 00
WP 0861 00
WP 0932 00
WP 0931 00
WP 0936 00
WP 1685 00
WP 1747 00
WP 1805 00

Equipment Condition
External Power Available, WP 1685 00
or APU Operational, TM 1-1520-237-10

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram, Figure 2 for schematic diagram, and HH-60A Figure 3 and HH-60L Figure 4 for distribution diagrams as an aid in operational/troubleshooting.

The dc electrical system operational/troubleshooting procedure is subdivided as follows:

- Battery Power
- External Electrical Power
- No. 1 Converter
- No. 2 Converter
- Utility DC
- Battery Low Sensing
- Shutdown

BATTERY OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- Do not start APU or connect external electrical power for battery system checkout.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. On upper console, place BATT switch to ON.
3. Turn pilot's and copilot's MFD switches ON.

Indication/Condition

a. BATT LOW CHARGE legend shall be off.

b. DC ESS BUS OFF legend shall be off.

c. SAS OFF, AC ESS BUS OFF, and BOOST SERVO OFF legends shall be on.

d. #1 CONV legend shall be on.

e. #2 CONV legend shall be on.

Corrective Action

1. If no MFD legends go on, go to NO MFD (COPILOT'S AND PILOT'S) LEGENDS GO ON WHEN BATT SWITCH IS PLACED TO ON, in this work package.
2. If Indication/Condition a. is not as specified, go to BATT LOW CHARGE LEGEND IS ON WHEN BATT SWITCH IS SET TO ON, in this work package.
3. If Indication/Condition b. is not as specified, go to DC ESS BUS OFF LEGEND IS ON WHEN BATT SWITCH IS SET TO ON, in this work package.
4. If both #1 and #2 CONV legends do not go on, go to #1 AND #2 CONV LEGENDS ARE NOT ON, in this work package.
5. If Indication/Condition d. is not as specified, go to #1 CONV LEGEND IS NOT ON, in this work package.
6. If Indication/Condition e. is not as specified, go to #2 CONV LEGEND IS NOT ON, in this work package.

Step

4. Turn pilot's and copilot's MFD switches OFF.
5. Place upper console BATT switch to OFF.
6. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

EXTERNAL ELECTRICAL POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

CAUTION

To prevent damage to the multifunction display (MFD), make sure MFD ON/OFF switch is set to OFF.

2. On upper console, make sure BATT switch is OFF.
3. Turn on external electrical power.
4. Turn pilot's and copilot's MFD switches ON.
5. Pull out ESNTL DC SENSE circuit breaker on DC ESNTL BUS (upper console circuit breaker panel).

Indication/Condition

DC ESS BUS OFF legend shall go on.

Corrective Action

If Indication/Condition is not as specified, go to DC ESS BUS OFF LEGEND DOES NOT GO ON WHEN ESNTL DC SENSE CIRCUIT BREAKER IS PULLED, in this work package.

Step

6. Push in ESNTL DC SENSE circuit breaker on DC ESNTL BUS (upper console circuit breaker panel).
7. Turn pilot's and copilot's MFD switches OFF.
8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 1 CONVERTER OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in except those noted being pulled for maintenance or safety.

CAUTION

To prevent damage to the multifunction display (MFD), make sure MFD ON/OFF switch is set to OFF.

2. Turn on external electrical power.
3. Turn pilot's and copilot's MFD switches ON.
4. Press MASTER CAUTION PRESS TO RESET capsule.
5. On upper console, make sure BATT switch is OFF.
6. Pull out NO. 2 CONVERTER circuit breaker (pilot's circuit breaker panel).

Indication/Condition

a. #1 CONV legend shall remain off.

b. #2 CONV legend shall be on.

c. DC ESS BUS OFF legend shall remain off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #1 CONV LEGEND IS ON, in this work package.
2. If Indication/Condition b. is not as specified, go to #2 CONV LEGEND IS NOT ON, in this work package.
3. If Indication/Condition c. is not as specified, go to DC ESS BUS OFF LEGEND IS ON WHEN BATT SWITCH IS SET TO ON, in this work package.

Step

7. Pull out ESNTL DC SENSE circuit breaker on DC ESNTL BUS (pilot's side upper console).

Indication/Condition

DC ESS BUS OFF legend shall go on.

Corrective Action

If Indication/Condition is not as specified, go to DC ESS BUS OFF LEGEND IS OFF WITH ESNTL SUPPLY BUS CIRCUIT BREAKER PULLED, in this work package.

Step

8. Push in ESNTL DC SENSE circuit breaker (pilot's side upper console).
9. DC ESS BUS OFF legend shall go off.

NO. 1 CONVERTER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

10. Push in NO. 2 CONVERTER circuit breaker (pilot's circuit breaker panel).
11. Turn pilot's and copilot's MFD switches OFF.
12. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 2 CONVERTER OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in except those noted being pulled for maintenance or safety.
2. Turn on external electrical power.
3. Turn pilot's and copilot's MFD switches ON.
4. Press MASTER CAUTION PRESS TO RESET capsule.
5. On upper console, make sure BATT switch is OFF.
6. Pull out NO. 1 CONVERTER circuit breaker (copilot's circuit breaker panel).

Indication/Condition

a. #2 CONV legend shall remain off.

b. #1 CONV legend shall be on.

c. DC ESS BUS OFF legend shall remain off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #2 CONV LEGEND IS ON, in this work package.
2. If Indication/Condition b. is not as specified, go to #1 CONV LEGEND IS NOT ON, in this work package.
3. If Indication/Condition c. is not as specified, go to DC ESS BUS OFF LEGEND IS ON WHEN BATT SWITCH IS SET TO ON, in this work package.

Step

7. Pull out ESNTL DC SENSE circuit breaker on DC ESNTL BUS (pilot's side upper console).

Indication/Condition

DC ESS BUS OFF legend shall go on.

Corrective Action

If Indication/Condition is not as specified, replace K9 NO. 2 DC ESNTL BUS SPLY relay (WP 0855 00).

NO. 2 CONVERTER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

8. Push in ESNTL BUS SENSE circuit breaker (pilot's side upper console).
9. DC ESS BUS OFF legend shall go off.
10. Push in NO. 1 CONVERTER circuit breaker (copilot's circuit breaker panel).
11. Turn pilot's and copilot's MFD switches OFF.
12. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

UTILITY DC OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in except those noted being pulled for maintenance or safety.
2. Turn on external electrical power.
3. Check for 28 vdc between J258-B and J258-A.

Indication/Condition

Multimeter shall indicate 28 vdc.

Corrective Action

1. If Indication/Condition is not as specified, check for 28 vdc between UTIL RECP CABIN circuit breaker terminal 1 (copilot's circuit breaker panel) and ground.
2. If voltage is not as specified, replace circuit breaker (WP 0824 00).
3. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J258-B, or between J258-A and ground, as required (WP 1747 00).

Step

4. Check for 28 vdc between J265-B and J265-A.

Indication/Condition

Multimeter shall indicate 28 vdc.

Corrective Action

1. If Indication/Condition is not as specified, check for 28 vdc between CMD CSL SET circuit breaker terminal 1 (copilot's circuit breaker panel) and ground.

UTILITY DC OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. If voltage is not as specified, replace circuit breaker (WP 0824 00).
3. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J265-B, or between J265-A and ground, as required (WP 1747 00).

Step

5. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

BATTERY LOW SENSING OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Turn off electrical power.
2. Make sure BATT switch on upper console is in OFF position.
3. Disconnect electrical connector P140 from battery.
4. Connect P140 to TP3 and TP4 of battery sensor relay tester (WP 1805 00) as follows:
   - P140 (+) to red alligator clip
   - P140 (-) to black alligator clip
   - Wire with red banana plug to TP3 on tester
   - Wire with black banana plug to TP4 on tester
5. Using battery connector P1 with pigtail wires, connect battery to tester as follows:
   - Wire with black banana plug to TP2 of tester
   - Wire with red banana plug to TP1 of tester
   - Connector P1 to battery.
6. Connect multimeter to TP3 and TP4 on tester.
7. Place BATT switch on upper console to ON position.
8. Turn pilot's and copilot's MFD switches ON.
9. On tester, push in circuit breaker CB1 and place switches S1 and S2 on.
10. On tester, turn control R1 until 23 vdc or higher is observed on multimeter.
11. On tester, turn R1 slowly clockwise until battery voltage decreases to 22.7 vdc.

BATTERY LOW SENSING OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

a. BATT LOW CHARGE legend on pilot's MFD shall go on when battery voltage is 22.7 vdc.

b. BATT LOW CHARGE legend on copilot's MFD shall go on when battery voltage is 22.7 vdc.

Corrective Action

1. If Indication/Condition a. is not as specified, go to BATT LOW CHARGE LEGEND ON PILOT'S MFD DOES NOT GO ON WHEN BATTERY VOLTAGE IS 22.7 VDC, in this work package.
2. If Indication/Condition b. is not as specified, go to BATT LOW CHARGE LEGEND ON COPILOT'S MFD DOES NOT GO ON WHEN BATTERY VOLTAGE IS 22.7 VDC, in this work package.
3. If Indication/Conditions a. and b. are not as specified, go to BATT LOW CHARGE LEGEND ON PILOT'S AND COPILOT'S MFD DOES NOT GO ON WHEN BATTERY VOLTAGE IS 22.7 VDC, in this work package.

Step

12. Place BATT switch on upper console to OFF position.
13. Disconnect all wiring from tester to electrical connector P140 and battery.
14. Turn pilot's and copilot's MFD switches OFF.
15. Reconnect P140 to battery.
16. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Make sure MFDs are turned off.
2. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

NO MFD (COPILOT'S AND PILOT'S) LEGENDS GO ON WHEN BATT SWITCH IS PLACED TO ON

SYMPTOM

No MFD (copilot's and pilot's) legends go on when BATT switch is placed to ON.

MALFUNCTION

No MFD (copilot's and pilot's) legends go on when BATT switch is placed to ON.

NO MFD (COPILOT'S AND PILOT'S) LEGENDS GO ON WHEN BATT SWITCH IS PLACED TO ON - Continued

CORRECTIVE ACTION

1. **Check for greater than 23 vdc between K9-X1 and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, go to **3.**
2. **Check continuity between K9-X2 and ground.**
   a. If continuity is present, go to **4.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20.**
3. **Check for greater than 23 vdc between BATT switch terminal 3 and ground.**
   a. If voltage is as specified, repair/replace wiring between BATT switch terminal 3 and K9-X1 (WP 1747 00). Go to **20.**
   b. If voltage is not as specified, go to **15.**
4. **Check for greater than 23 vdc between K10-A3 and ground.**
   a. If voltage is as specified, repair/replace wiring between K10-A3 and DC ESNTL Bus (WP 1747 00). Go to **20.**
   b. If voltage is not as specified, go to **5.**
5. **Check for greater than 23 vdc between K10-A4 and ground.**
   a. If voltage is as specified, replace K10 (WP 0855 00). Go to **20.**
   b. If voltage is not as specified, go to **6.**
6. **Check for greater than 23 vdc between K9-A3 and ground.**
   a. If voltage is as specified, repair/replace wiring between K9-A3 and K10-A4 (WP 1747 00). Go to **20.**
   b. If voltage is not as specified, go to **7.**
7. **Check for greater than 23 vdc between K9-A4 and ground.**
   a. If voltage is as specified, replace relay K9 (WP 0855 00). Go to **20.**
   b. If voltage is not as specified, go to **8.**
8. **Check for greater than 23 vdc between terminal 1 of ESNTL BUS DC SPLY circuit breaker and ground.**
   a. If voltage is as specified, repair/replace wiring between terminal 1 of ESNTL BUS DC SPLY circuit breaker and K9-A2 (WP 1747 00). Go to **20.**
   b. If voltage is not as specified, go to **9.**
9. **Check for greater than 23 vdc between terminal 2 of ESNTL BUS DC SPLY circuit breaker and ground.**
   a. If voltage is as specified, replace ESNTL BUS DC SPLY circuit breaker (WP 0824 00). Go to **20.**
   b. If voltage is not as specified, go to **10.**
10. **Check for greater than 23 vdc between K7-X1 and ground.**
    a. If voltage is as specified, go to **11.**

NO MFD (COPILOT'S AND PILOT'S) LEGENDS GO ON WHEN BATT SWITCH IS PLACED TO ON - Continued

b. If voltage is not as specified, repair/replace wiring between BATT switch terminal 3 and K7-X1 (WP 1747 00). Go to **20.**

11. Check continuity between:

- **K7-X2 and ground**
- **K7-A1 and K7-A3**

a. If continuity is present, go to **12.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20.**

12. Check for greater than 23 vdc between K7-A3 and ground.

a. If voltage is as specified, repair/replace wiring between K7-A3 and BATT BUS (WP 1747 00). Go to **20.**

b. If voltage is not as specified, go to **13.**

13. Check for greater than 23 vdc between K7-A2 and ground.

a. If voltage is as specified, replace K7 (WP 0856 00). Go to **20.**

b. If voltage is not as specified, Go to **14.**

14. Check continuity between:

- **P140 (+) and K7-A2**
- **P140 (-) and ground**

a. If continuity is present, replace battery (WP 0931 00). Go to **20.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20.**

15. Check for greater than 23 vdc between BATT switch terminal 2 and ground.

a. If voltage is as specified, replace BATT switch (WP 0861 00). Go to **20.**

b. If voltage is not as specified, go to **16.**

16. Check for greater than 23 vdc between BATT BUS CONTR circuit breaker terminal 1 and ground.

a. If voltage is as specified, go to **17.**

b. If voltage is not as specified, go to **19.**

17. Check continuity between terminals 7 and 9 of contactor K16.

a. If continuity is present, go to **18.**

b. If continuity is not present, replace K16 (WP 0853 00). Go to **20.**

18. Check continuity between terminals 7 and 9 of contactor K6.

a. If continuity is present, repair/replace wiring between the following as required (WP 1747 00), then go to **20:**

- BATT BUS CONTR terminal 1 and K16-9
- K16-7 and K6-7
- K6-9 and BATT switch terminal 2

NO MFD (COPILOT'S AND PILOT'S) LEGENDS GO ON WHEN BATT SWITCH IS PLACED TO ON - Continued

b. If continuity is not present, replace K16 (WP 0853 00). Go to **20.**

19. Check for greater than 23 vdc between BATT BUS CONTR circuit breaker terminal 2 and ground.

a. If voltage is as specified, replace BATT BUS CONTR circuit breaker (WP 0824 00). Go to **20.**

b. If voltage is not as specified, repair/replace wiring (WP 1747 00). Go to **20.**

20. Procedure completed.

BATT LOW CHARGE LEGEND IS ON WHEN BATT SWITCH IS SET TO ON

SYMPTOM

BATT LOW CHARGE legend is on when BATT switch is set to ON.

MALFUNCTION

BATT LOW CHARGE legend is on when BATT switch is set to ON.

CORRECTIVE ACTION

1. Check for 28 vdc between K201-X1 and K201-X2.

a. If voltage is as specified, replace K201 (WP 0936 00). Go to **4.**

b. If voltage is not as specified, go to **2.**

2. Check for 28 vdc between K201-X1 and ground.

a. If voltage is as specified, repair/repair wiring (WP 1747 00). Go to **4.**

b. If voltage is not as specified, go to **3**.

3. Check continuity between:

- **K201-X1 and BATT switch terminal 6**
- **BATT switch terminal 5 and P107-H**

a. If continuity is present, replace BATT switch (WP 0861 00). Go to **4.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**

4. Procedure completed.

DC ESS BUS OFF LEGEND IS ON WHEN BATT SWITCH IS SET TO ON

SYMPTOM

DC ESS BUS OFF legend is on when BATT switch is set to ON.

MALFUNCTION

DC ESS BUS OFF legend is on when BATT switch is set to ON.

CORRECTIVE ACTION

1. Check for greater than 23 vdc between P900-J and ground.

a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **4.**

b. If voltage is not as specified, go to **2.**

DC ESS BUS OFF LEGEND IS ON WHEN BATT SWITCH IS SET TO ON - Continued

2. **Check for greater than 23 vdc between ESNTL DC SENSE circuit breaker terminal 1 and ground.**

   a. If voltage is as specified, repair/replace wiring between ESNTL DC SENSE circuit breaker terminal 1 and P900-J (WP 1747 00). Go to **4.**

   b. If voltage is not as specified, go to **3.**

3. **Check for greater than 23 vdc between ESNTL DC SENSE circuit breaker terminal 2 and ground.**

   a. If voltage is as specified, replace circuit breaker (WP 0824 00). Go to **4.**

   b. If voltage is not as specified, repair/replace wiring between K10-A3 and DC ESNTL BUS (WP 1747 00). Go to **4.**

4. **Procedure completed.**

#1 AND #2 CONV LEGENDS ARE NOT ON

SYMPTOM

#1 and #2 CONV legends are not on.

MALFUNCTION

#1 and #2 CONV legends are not on.

CORRECTIVE ACTION

1. **Check for 28 vdc between ESNTL BUS AC & CONV WARN circuit breaker terminal 1 and ground.**

   a. If voltage is as specified, repair/replace wiring between terminal 1 of ESNTL BUS AC & CONV WARN circuit breaker and K13 AC ESNTL BUS FAIL relay terminal A2 (WP 1747 00). Go to **2.**

   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **2.**

2. **Procedure completed.**

#1 CONV LEGEND IS NOT ON

SYMPTOM

#1 CONV legend is not on.

MALFUNCTION

#1 CONV legend is not on.

CORRECTIVE ACTION

1. **Check if pilot's or copilot's #1 CONV legend is not on.**

   a. If pilot's #1 CONV legend is not on, go to **2.**

   b. If copilot's #1 CONV legend is not on, go to **5.**

2. **Check for 28 vdc between P1020R-18 and ground.**

   a. If voltage is as specified, replace pilot's MFD (WP 0799 00). Go to **6.**

   b. If voltage is not as specified, go to **3.**

#1 CONV LEGEND IS NOT ON - Continued

3. **Check for 28 vdc between K16-4 and ground.**
   a. If voltage is as specified, go to **4.**
   b. If voltage is not as specified, repair/replace wiring (WP 1747 00). Go to **6.**
4. **Check continuity between K16-6 and P1020R-18.**
   a. If continuity is present, replace K16 NO. 1 DC PRI BUS CNTOR (WP 0853 00). Go to **6.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**
5. **Check for 28 vdc between P1019R-18 and ground.**
   a. If voltage is as specified, replace copilot's MFD (WP 0799 00). Go to **6.**
   b. If voltage is not as specified, repair/replace wiring (WP 1747 00). Go to **6.**
6. **Procedure completed.**

#2 CONV LEGEND IS NOT ON

SYMPTOM

#2 CONV legend is not on.

MALFUNCTION

#2 CONV legend is not on.

CORRECTIVE ACTION

1. **Check if pilot's or copilot's #2 CONV legend is not on.**
   a. If pilot's #2 CONV legend is not on, go to **2.**
   b. If copilot's #2 CONV legend is not on, go to **5.**
2. **Check for 28 vdc between P1020R-62 and ground.**
   a. If voltage is as specified, replace pilot's MFD (WP 0799 00). Go to **6.**
   b. If voltage is not as specified, go to **3.**
3. **Check for 28 vdc between K6-1 and ground.**
   a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **6.**
   b. If voltage is not as specified, go to **4.**
4. **Check continuity between K6-3 and P1020R-62.**
   a. If continuity is present, replace K6 NO. 2 DC PRI BUS CNTOR (WP 0853 00). Go to **6.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**
5. **Check for 28 vdc between P1019R-62 and ground.**
   a. If voltage is as specified, replace copilot's MFD (WP 0799 00). Go to **6.**
   b. If voltage is not as specified, repair/replace wiring (WP 1747 00). Go to **6.**
6. **Procedure completed.**

DC ESS BUS OFF LEGEND DOES NOT GO ON WHEN ESNTL DC SENSE CIRCUIT BREAKER IS PULLED

SYMPTOM

DC ESS BUS OFF legend does not go on when ESNTL DC SENSE circuit breaker is pulled.

MALFUNCTION

DC ESS BUS OFF legend does not go on when ESNTL DC SENSE circuit breaker is pulled.

CORRECTIVE ACTION

1. **Check for 28 vdc between these points:**
   - **P1019R-34 and ground**
   - **P1020R-34 and ground**

   a. If voltage is as specified, troubleshoot MFD/caution/advisory system (WP 0100 00). Go to **6.**

   b. If voltage is not as specified, go to **2.**

2. **Check for 28 vdc between P900-G and ground.**

   a. If voltage is as specified, go to **3.**

   b. If voltage is not as specified, repair/replace wiring between (WP 1747 00), then go to **6:**
   - P900-G and P1019R-34
   - P900-G and P1020R-34

3. **Check for 28 vdc between P900-H and ground.**

   a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **6.**

   b. If voltage is not as specified, go to **4.**

4. **Check for 28 vdc between BATT & ESNTL DC WARN EXT PWR CONTR circuit breaker terminal 1 and ground.**

   a. If voltage is as specified, repair/replace wiring between P900-H and BATT & ESNTL DC WARN EXT PWR CONTR circuit breaker terminal 1 (WP 1747 00). Go to **6.**

   b. If voltage is not as specified, go to **5**.

5. **Check for 28 vdc between BATT & ESNTL DC WARN EXT PWR CONTR circuit breaker terminal 2 and ground.**

   a. If voltage is as specified, replace BATT & ESNTL DC WARN EXT PWR CONTR circuit breaker (WP 0824 00). Go to **6.**

   b. If voltage is not as specified, repair/replace wiring between battery bus and BATT & ESNTL DC WARN EXT PWR CONTR circuit breaker (WP 1747 00). Go to **6.**

6. **Procedure completed.**

#1 CONV LEGEND IS ON

SYMPTOM

#1 CONV legend is on.

MALFUNCTION

#1 CONV legend is on.

CORRECTIVE ACTION

1. **Check for 28 vdc between P239-A and P239-B.**

   a. If voltage is as specified, replace K16 NO. 1 DC PRI BUS CNTOR (WP 0853 00). Go to **8.**

   b. If voltage is not as specified, go to **2.**

2. **Check for 28 vdc between P239-A and ground.**

   a. If voltage is as specified, repair/replace wiring between P239-B and ground (WP 1747 00). Go to **8.**

   b. If voltage is not as specified, go to **3.**

3. **Check for 28 vdc between K16 NO. 1 DC PRI BUS CNTOR terminal A1 and ground.**

   a. If voltage is as specified, repair/replace wiring between contactor terminal A1 and P239-A (WP 1747 00). Go to **8.**

   b. If voltage is not as specified, go to **4.**

4. **Check continuity between No. 1 converter (+) terminal and K16 NO. 1 DC PRI BUS CNTOR terminal A1.**

   a. If continuity is present, go to **5.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

5. **Check for 115 vac between:**

   - **P903-A and P903-D**
   - **P903-B and P903-D**
   - **P903-C and P903-D**

   a. If voltage is as specified, replace No. 1 converter (WP 0859 00). Go to **8.**

   b. If voltage is not as specified, go to **6.**

6. **Check continuity between P903-D and ground.**

   a. If continuity is present, go to **7.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

7. **Check for 115 vac between NO. 1 CONVERTER circuit breaker:**

   - **Terminal A1 and ground**
   - **Terminal B1 and ground**
   - **Terminal C1 and ground**

#1 CONV LEGEND IS ON - Continued

a. If voltage is as specified, repair/replace wiring between circuit breaker and No. 1 converter (WP 1747 00). Go to **8.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8.**

8. Procedure completed.

DC ESS BUS OFF LEGEND IS OFF WITH ESNTL SUPPLY BUS CIRCUIT BREAKER PULLED

SYMPTOM

DC ESS BUS OFF legend is off with ESNTL SUPPLY BUS circuit breaker pulled.

MALFUNCTION

DC ESS BUS OFF legend is off with ESNTL SUPPLY BUS circuit breaker pulled.

CORRECTIVE ACTION

1. Check for 28 vdc between K10 NO. 1 DC ESNTL BUS SPLY relay terminals X1 and X2.

a. If voltage is as specified, replace relay (WP 0855 00). Go to **5.**

b. If voltage is not as specified, go to **2.**

2. Check for 28 vdc between K10 NO. 1 DC ESNTL BUS SPLY relay terminal X1 and ground.

a. If voltage is as specified, repair/replace wiring between relay terminal X2 and ground (WP 1747 00). Go to **5.**

b. If voltage is not as specified, go to **3.**

3. Check for 28 vdc between K10 NO. 1 DC ESNTL BUS SPLY relay terminal A2 and ground.

a. If voltage is as specified, repair/replace wiring between relay terminal A2 and X1 (WP 1747 00). Go to **5.**

b. If voltage is not as specified, go to **4.**

4. Check for 28 vdc between NO. 1 DC ESNTL BUS SPLY circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and K10 NO. 1 DC ESNTL BUS SPLY relay terminal A2 (WP 1747 00). Go to **5.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **5.**

5. Procedure completed.

#2 CONV LEGEND IS ON

SYMPTOM

#2 CONV legend is on.

MALFUNCTION

#2 CONV legend is on.

CORRECTIVE ACTION

1. Check for 28 vdc between P238-A and P238-B.

#2 CONV LEGEND IS ON - Continued

a. If voltage is as specified, replace K6 NO. 2 DC PRI BUS CNTOR (WP 0853 00). Go to **8.**

b. If voltage is not as specified, go to **2.**

2. Check for 28 vdc between P238-A and ground.

a. If voltage is as specified, repair/replace wiring between P238-B and ground (WP 1747 00). Go to **8.**

b. If voltage is not as specified, go to **3.**

3. Check for 28 vdc between K6 NO. 2 DC PRI BUS CNTOR terminal A2 and ground.

a. If voltage is as specified, repair/replace wiring between K6 NO. 2 DC PRI BUS CNTOR terminal A2 and P238-A (WP 1747 00). Go to **8.**

b. If voltage is not as specified, go to **4.**

4. Check continuity between K6 NO. 2 DC PRI BUS CNTOR terminal A2 and No. 2 converter.

a. If continuity is present, go to **5.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

5. Check for 115 vac between:

- **P904-A and P904-D**
- **P904-B and P904-D**
- **P904-C and P904-D**

a. If voltage is as specified, replace converter (WP 0859 00). Go to **8.**

b. If voltage is not as specified, go to **6.**

6. Check continuity between P904-D and ground.

a. If continuity is present, go to **7.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8.**

7. Check for 115 vac between NO. 2 CONVERTER circuit breaker:

- **Terminal A1 and ground**
- **Terminal B1 and ground**
- **Terminal C1 and ground**

a. If voltage is as specified, repair/replace wiring between circuit breaker and No. 2 converter (WP 1747 00). Go to **8.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8.**

8. Procedure completed.

BATT LOW CHARGE LEGEND ON PILOT'S MFD DOES NOT GO ON WHEN BATTERY VOLTAGE IS 22.7 VDC

SYMPTOM

BATT LOW CHARGE legend on pilot's MFD does not go on when battery voltage is 22.7 vdc.

MALFUNCTION

BATT LOW CHARGE legend on pilot's MFD does not go on when battery voltage is 22.7 vdc.

CORRECTIVE ACTION

1. Check for 28 vdc between P1020R-20 and ground.

a. If voltage is as specified, troubleshoot MFD/caution/advisory warning system (WP 0855 00). Go to **2.**

b. If voltage is not as specified, repair/replace wiring between P1020R-20 and K201-B3 (WP 1747 00). Go to **2.**

2. Procedure completed.

BATT LOW CHARGE LEGEND ON COPILOT'S MFD DOES NOT GO ON WHEN BATTERY VOLTAGE IS 22.7 VDC

SYMPTOM

BATT LOW CHARGE legend on copilot's MFD does not go on when battery voltage is 22.7 vdc.

MALFUNCTION

BATT LOW CHARGE legend on copilot's MFD does not go on when battery voltage is 22.7 vdc.

CORRECTIVE ACTION

1. Check for 28 vdc between P1019R-20 and ground.

a. If voltage is as specified, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **2.**

b. If voltage is not as specified, repair/replace wiring between P1019R-20 and K201-B3 (WP 1747 00). Go to **2.**

2. Procedure completed.

BATT LOW CHARGE LEGEND ON PILOT'S AND COPILOT'S MFD DOES NOT GO ON WHEN BATTERY VOLTAGE IS 22.7 VDC

SYMPTOM

BATT LOW CHARGE legend on pilot's and copilot's MFD does not go on when battery voltage is 22.7 vdc.

MALFUNCTION

BATT LOW CHARGE legend on pilot's and copilot's MFD does not go on when battery voltage is 22.7 vdc.

CORRECTIVE ACTION

1. Check continuity between:

- **P1020R-20 and K201-B3**
- **P1019R-20 and K201-B3**

BATT LOW CHARGE LEGEND ON PILOT'S AND COPILOT'S MFD DOES NOT GO ON WHEN BATTERY VOLTAGE IS 22.7 VDC - Continued

- **K201-X2 and ground**

a. If continuity is present, go to **2.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**

2. With BATT switch ON, check for 23 vdc or greater between K201-X1 and ground.

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, go to **4.**

3. Check for 23 vdc or greater between K201-Y1 and ground.

a. If voltage is as specified, replace relay K201 (WP 0936 00). Go to **6.**

b. If voltage is not as specified, repair/replace wiring between (WP 1747 00), then go to **6:**

- K201-Y1 and BATT switch terminal 6
- K201-X1 and BATT switch terminal 6

4. Check for 23 vdc or greater between BATT switch terminal 5 and ground.

a. If voltage is as specified, replace BATT switch (WP 0861 00). Go to **6.**

b. If voltage is not as specified, go to **5.**

5. Check continuity between:

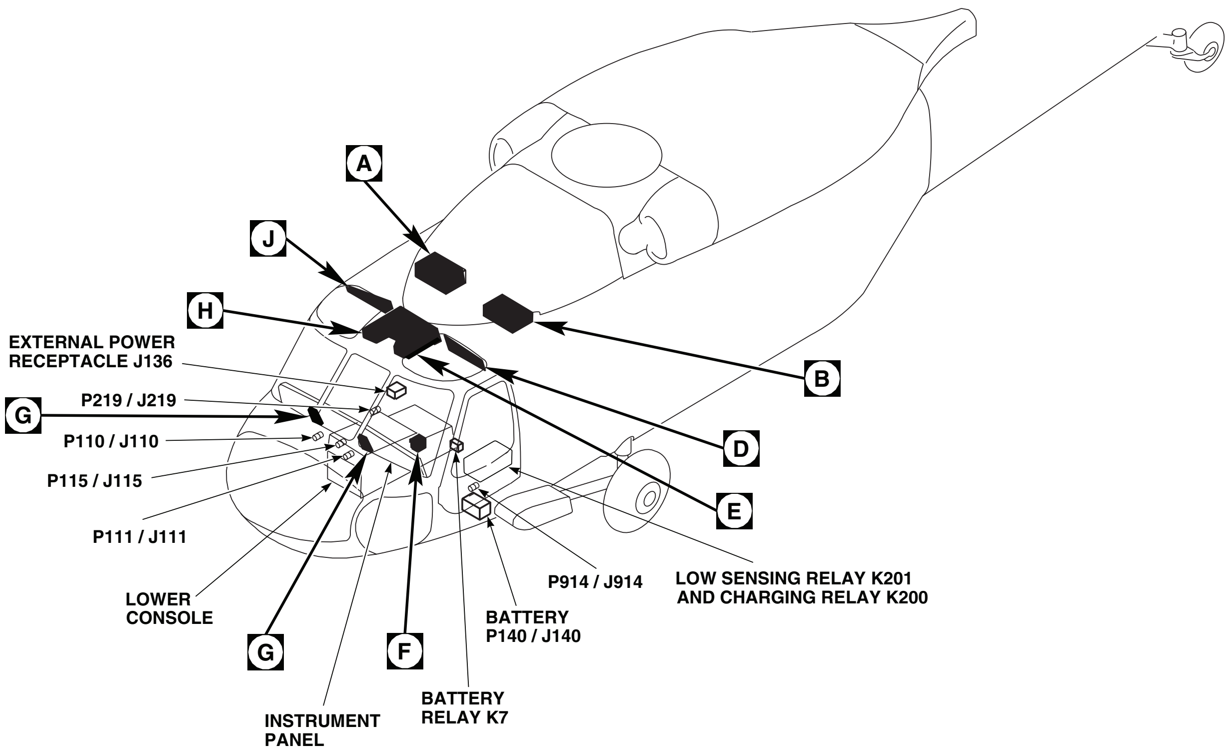
- **BATT switch terminal 5 and P140 (+)**
- **P140 (-) and ground**

a. If continuity is present, replace battery (WP 0931 00). Go to **6.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**

6. Procedure completed.

LOCATION

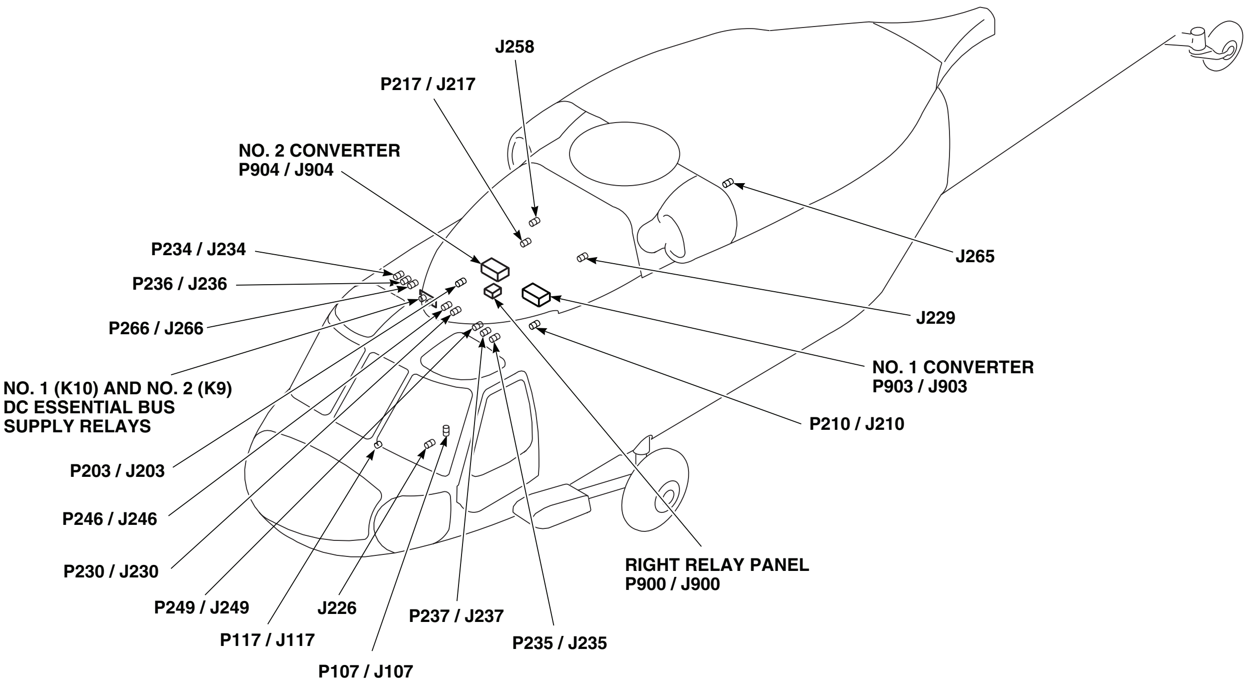


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J136	EXTERNAL POWER RECEPTACLE
J226	RESCUE HOIST KIT DISCONNECTS
J229	RESCUE HOIST KIT DISCONNECTS
J258	UTILITY RECEPTACLE +28 VDC
J265	CMD CSL POWER RECPT +28 VDC
P107 / J107	COCKPIT BL 9 LH, STA 236
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P115 / J115	COCKPIT BL 6 RH, STA 210

AB0184_1
SA

Figure 1. DC Electrical System Location Diagram. (Sheet 1 of 6)

LOCATION - Continued



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P140 / J140	BATTERY
P203 / J203	NO. 2 JUNCTION BOX
P210 / J210	NO. 1 JUNCTION BOX
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P234 / J234	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 27 RH
P235 / J235	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 20 LH
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 23 RH

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P237 / J237	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 25 LH
P238	NO. 2 JUNCTION BOX
P239	NO. 1 JUNCTION BOX
P240	NO. 1 JUNCTION BOX
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH
P900 / J900	RIGHT RELAY PANEL
P903 / J903	NO. 1 CONVERTER
P904 / J904	NO. 2 CONVERTER
P914 / J914	COCKPIT BL 24 LH, STA 247
P1019R / J1019R	COPILOT MFD
P1020R / J1020R	PILOT MFD
TLTB24	NEAR BATTERY CHARGING RELAY

AB0184_2
SA

Figure 1. DC Electrical System Location Diagram. (Sheet 2 of 6)

LOCATION - Continued

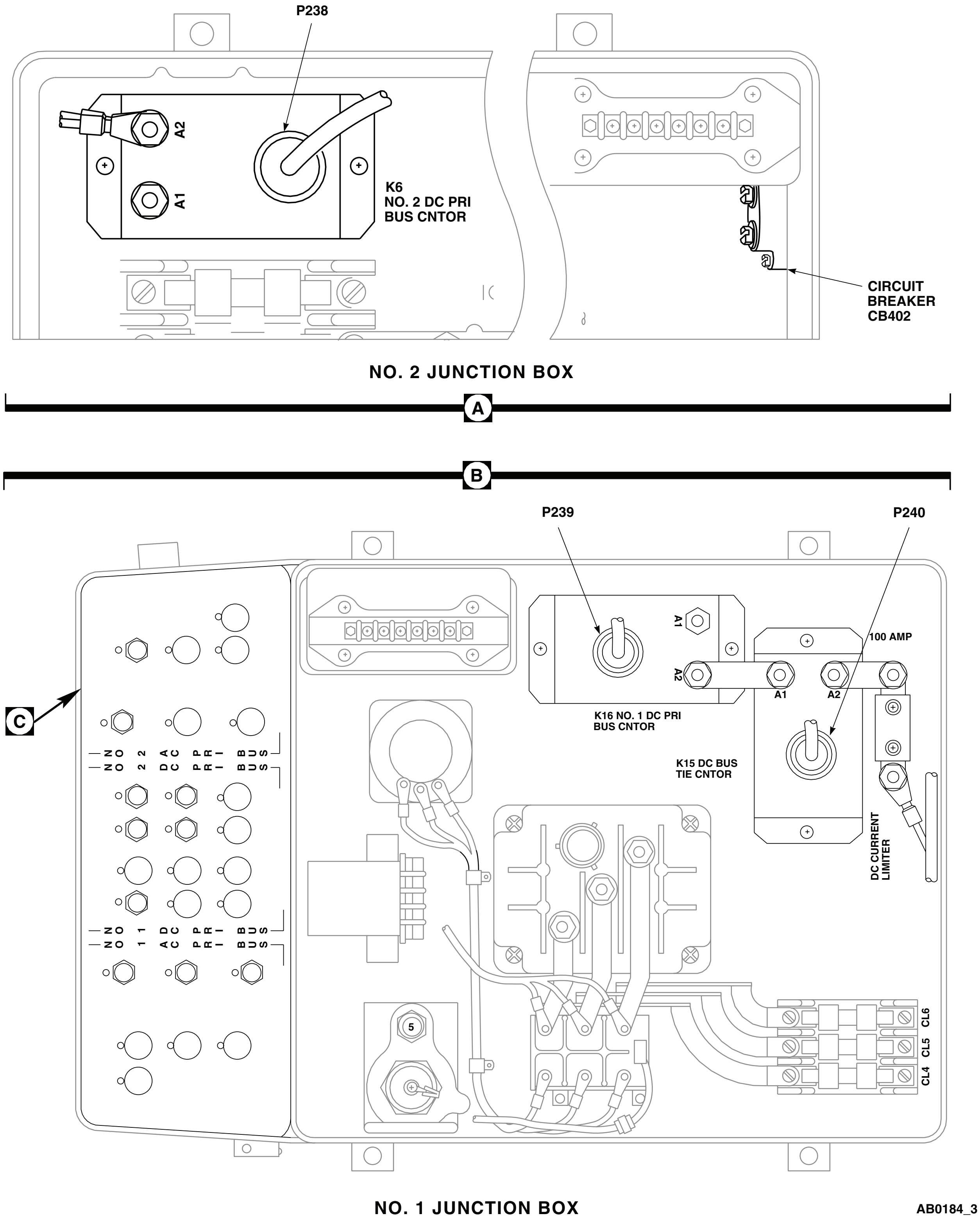


Figure 1. DC Electrical System Location Diagram. (Sheet 3 of 6)

LOCATION - Continued

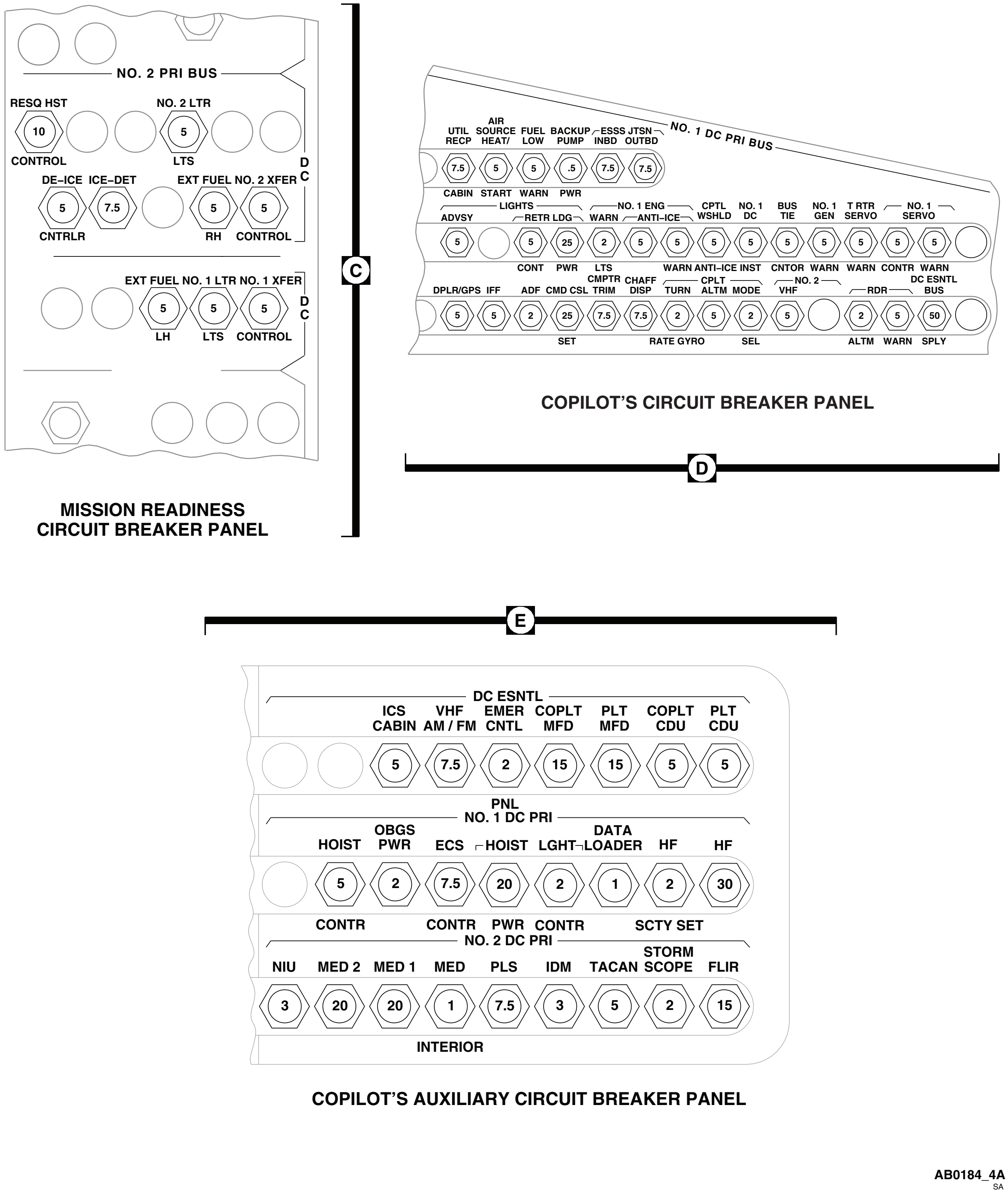
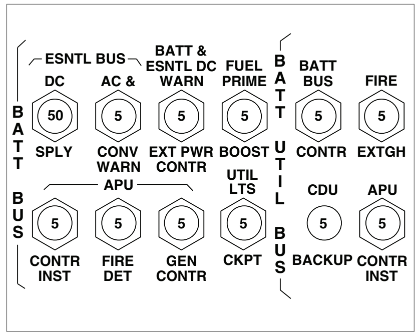


Figure 1. DC Electrical System Location Diagram. (Sheet 4 of 6)

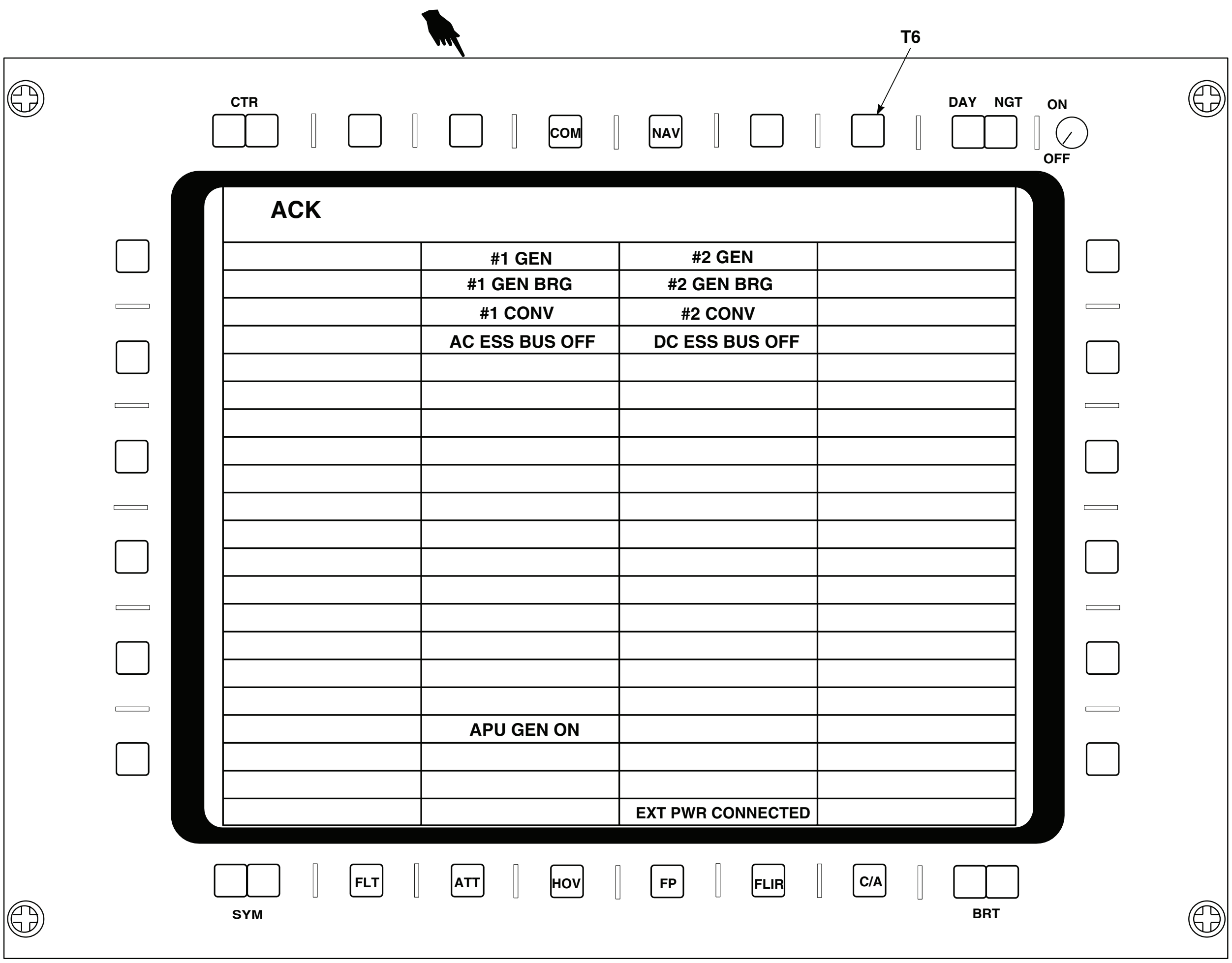
LOCATION - Continued



LOWER CONSOLE
CIRCUIT BREAKER PANEL

F

G

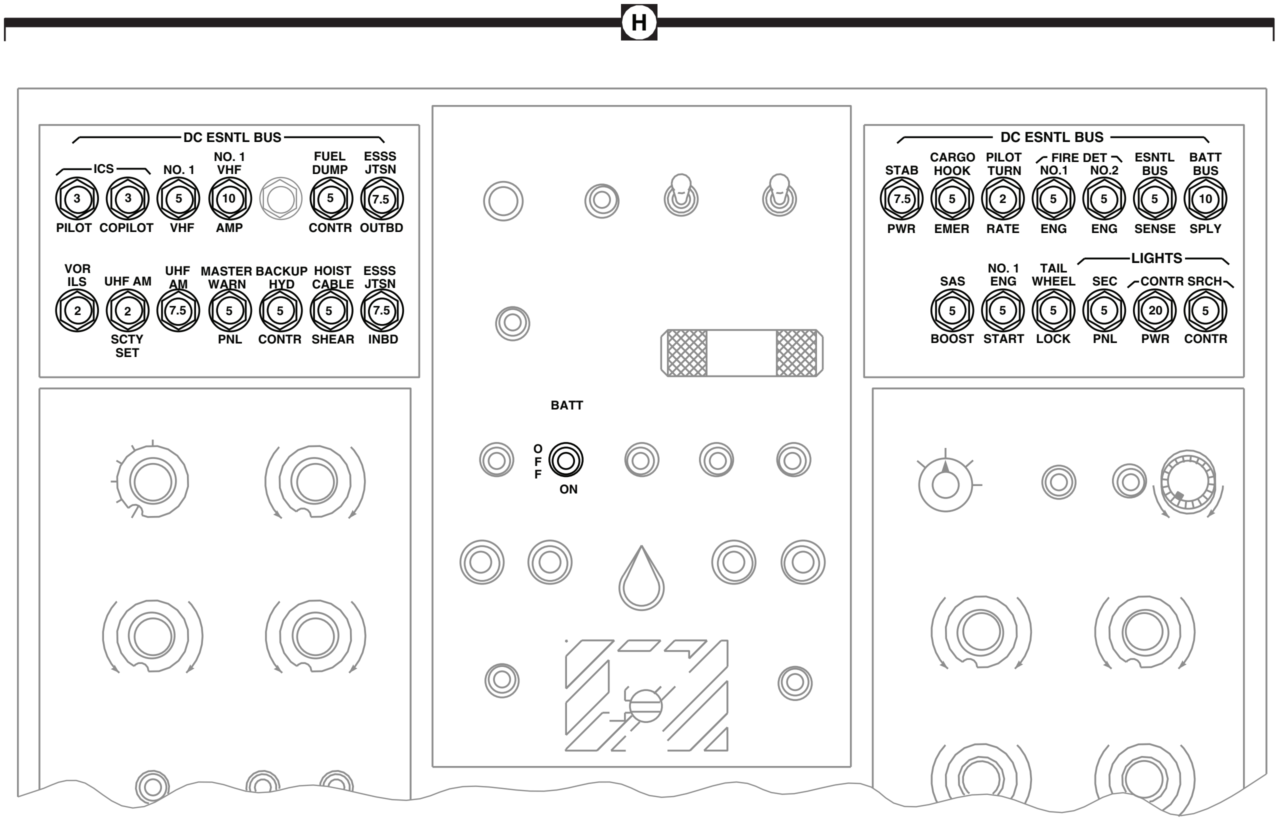


MULTIFUNCTION DISPLAY'S CAUTION/ADVISORY GRID

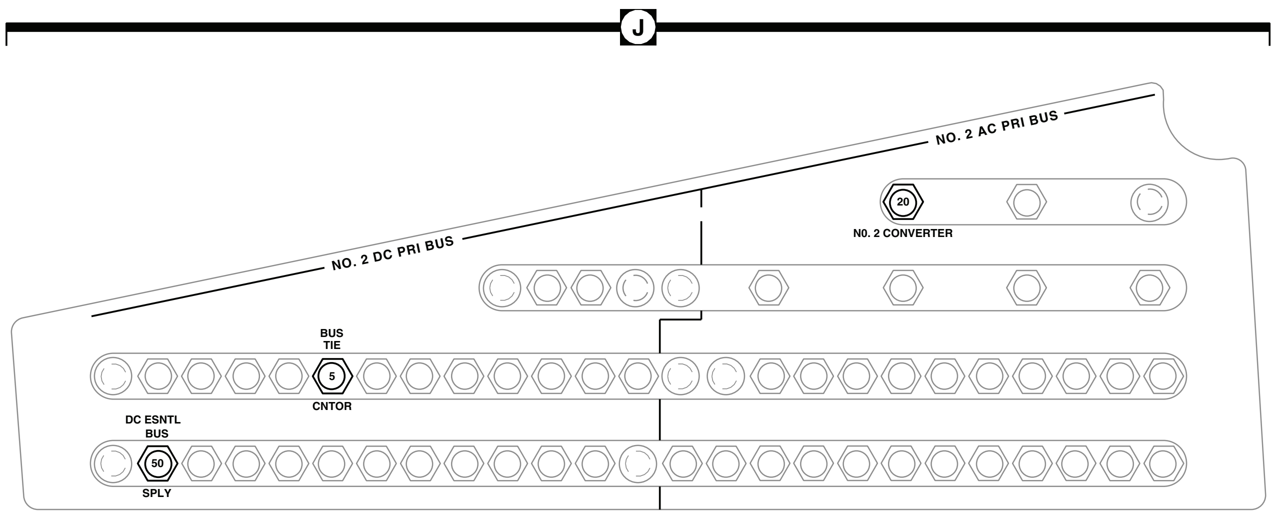
AB0184_5B
SA

Figure 1. DC Electrical System Location Diagram. (Sheet 5 of 6)

LOCATION - Continued



UPPER CONSOLE

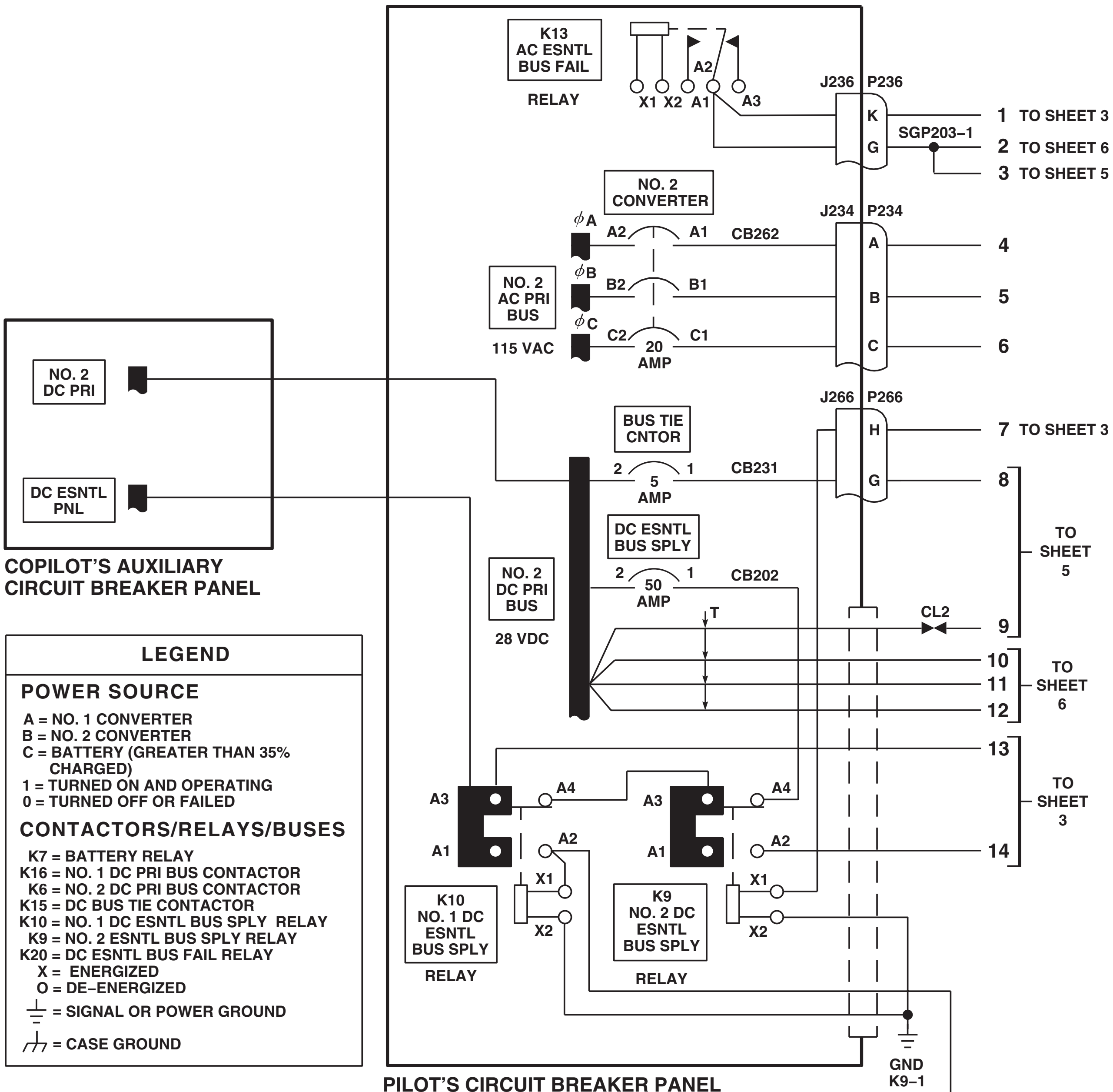


PILOT'S CIRCUIT BREAKER PANEL

AB0184_6
SA

Figure 1. DC Electrical System Location Diagram. (Sheet 6 of 6)

SCHEMATICS



DC POWER SOURCE			CONDITION	DC SYSTEM CONTACTORS / RELAYS							DC BUSES				
A	B	C		K7	K16	K6	K15	K10	K9	K20	NO. 1 DC PRI	NO. 2 DC PRI	DC ESNTL	BATT UTIL	BATT
0	0	0	1	O	O	O	O	O	O	O	O	O	O	O	O
1	0	0	2	O	X	O	X	X	O	X	X	X	X	X	X
0	1	0	3	O	O	X	X	X	O	X	X	X	X	X	X
1	1	0	4	O	X	X	O	X	O	X	X	X	X	X	X
0	0	1	5	X	O	O	O	O	X	X	O	O	X	X	X
1	0	1	6	O	X	O	X	X	O	X	X	X	X	X	X
0	1	1	7	O	O	X	X	X	O	X	X	X	X	X	X
1	1	1	8	O	X	X	O	X	O	X	X	X	X	X	X

AB2132_1
SA

Figure 2. DC Electrical System Schematic Diagram. (Sheet 1 of 6)

SCHEMATICS - Continued

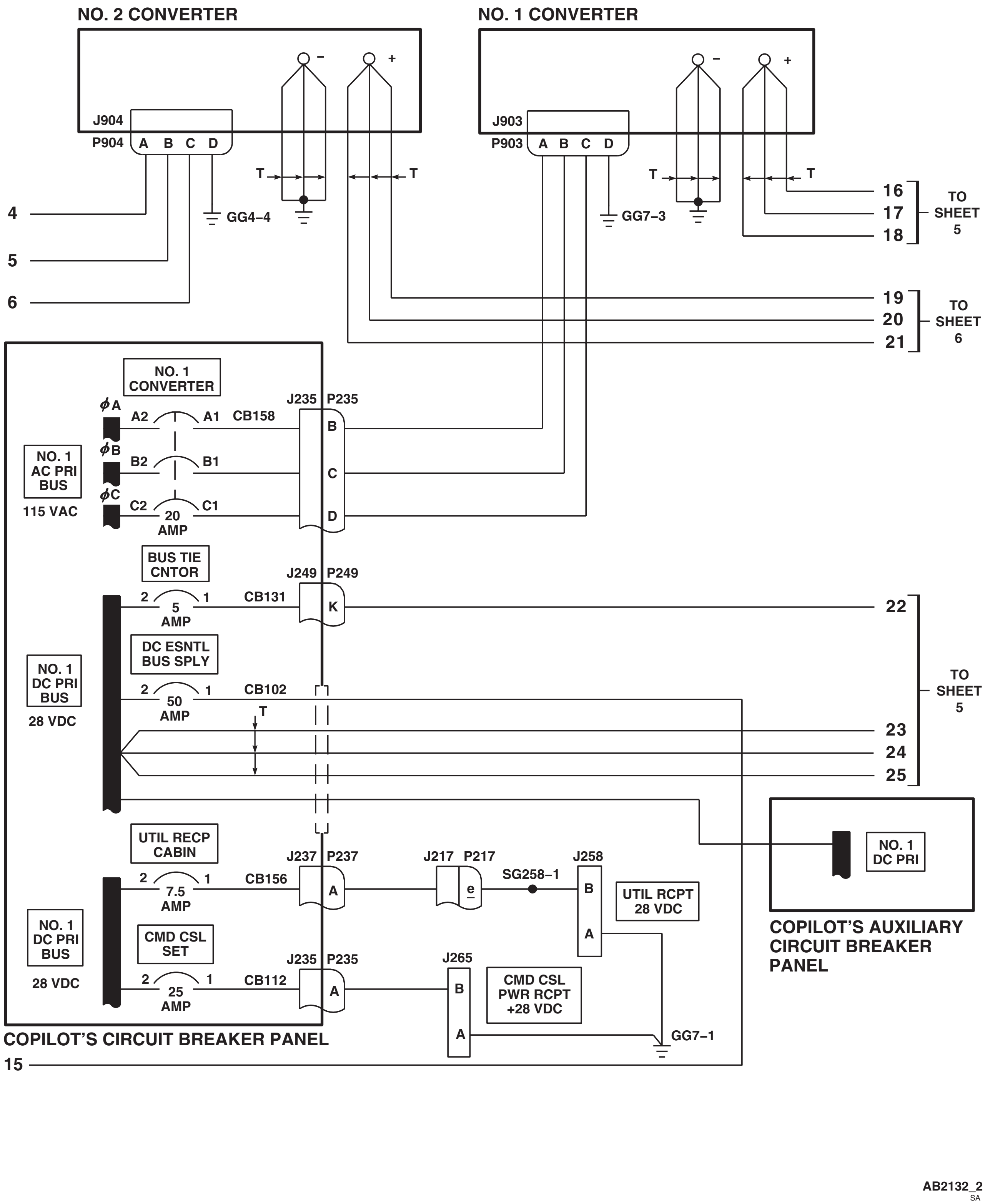


Figure 2. DC Electrical System Schematic Diagram. (Sheet 2 of 6)

SCHEMATICS - Continued

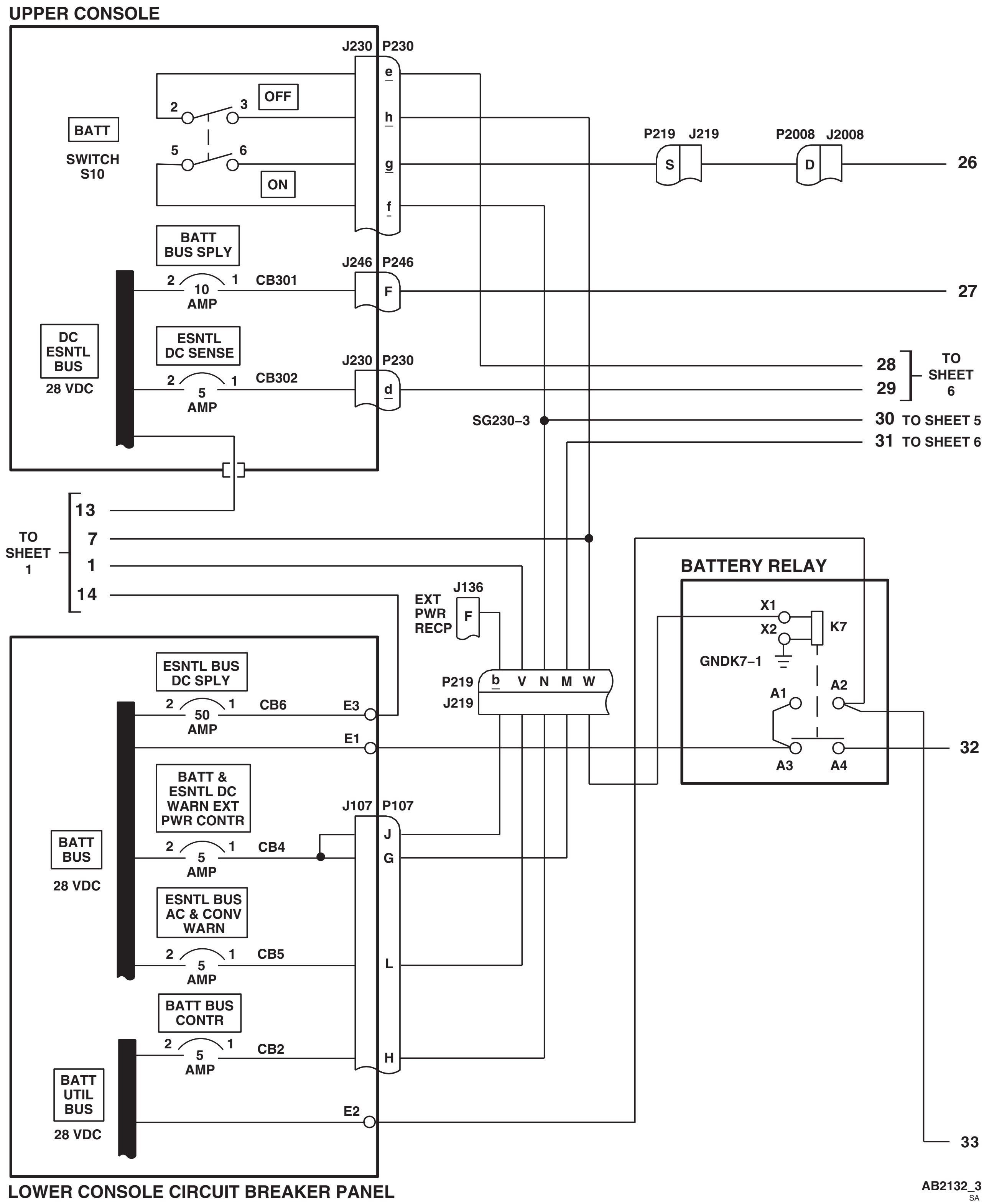


Figure 2. DC Electrical System Schematic Diagram. (Sheet 3 of 6)

SCHEMATICS - Continued

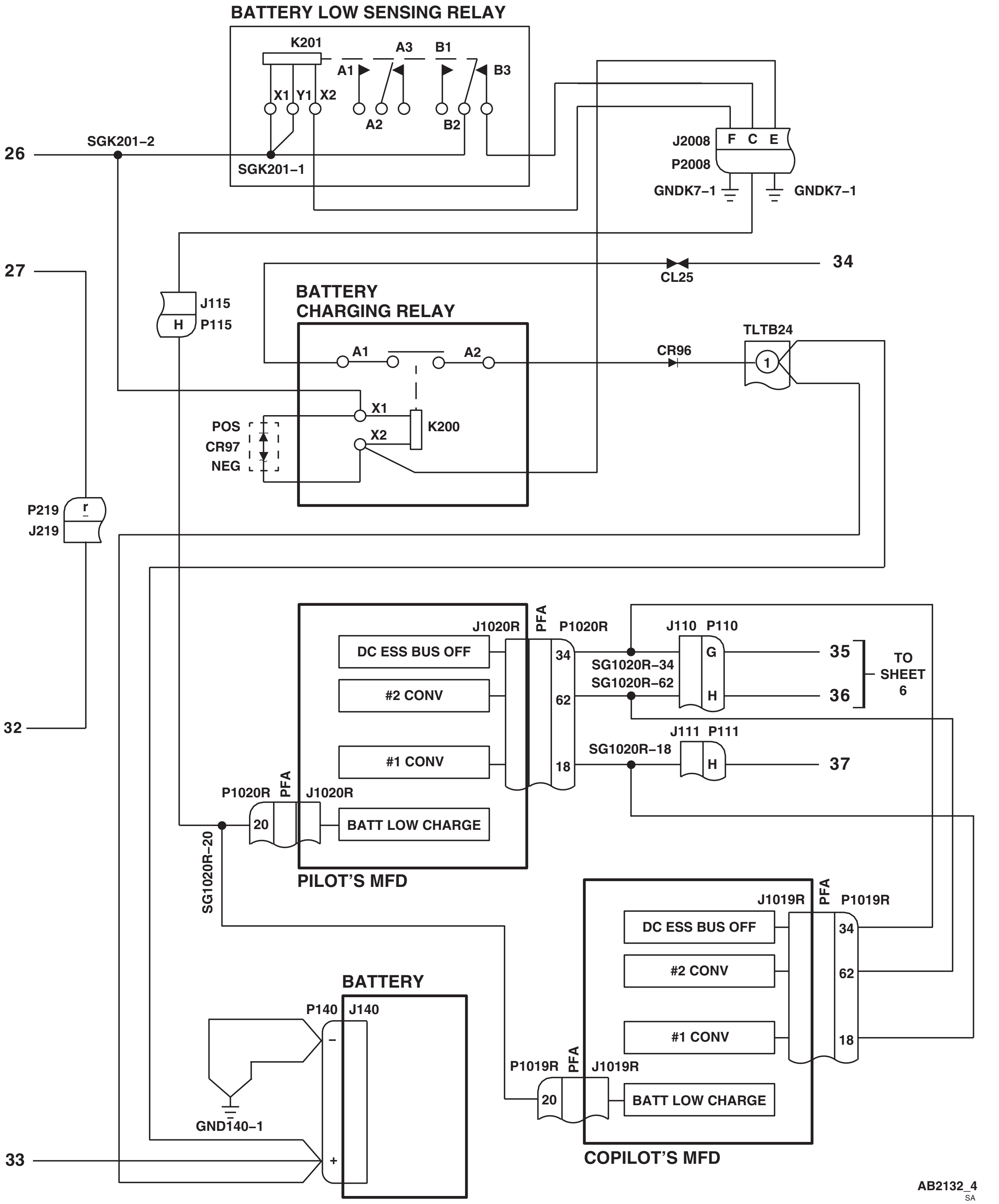


Figure 2. DC Electrical System Schematic Diagram. (Sheet 4 of 6)

SCHEMATICS - Continued

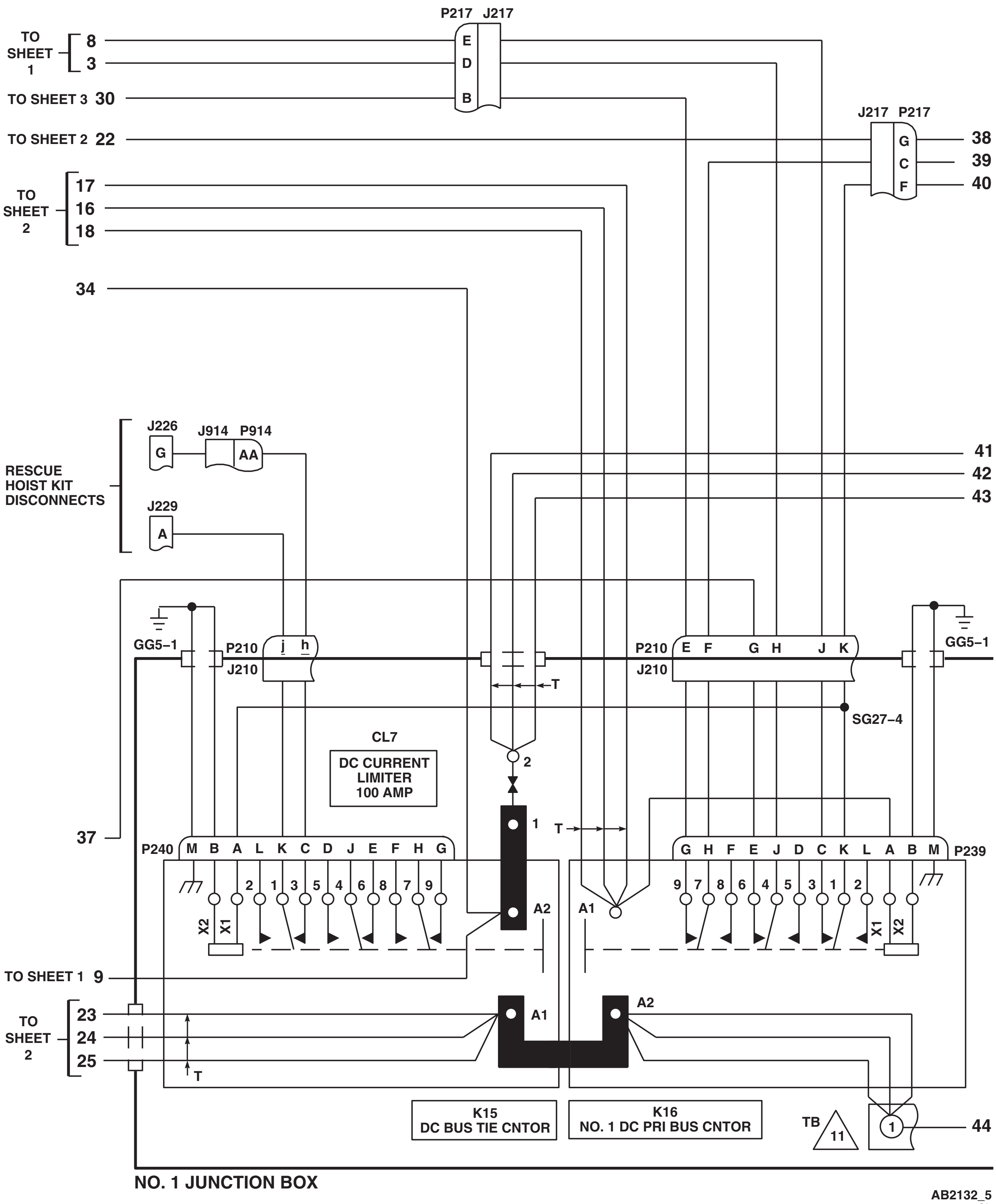


Figure 2. DC Electrical System Schematic Diagram. (Sheet 5 of 6)

SCHEMATICS - Continued

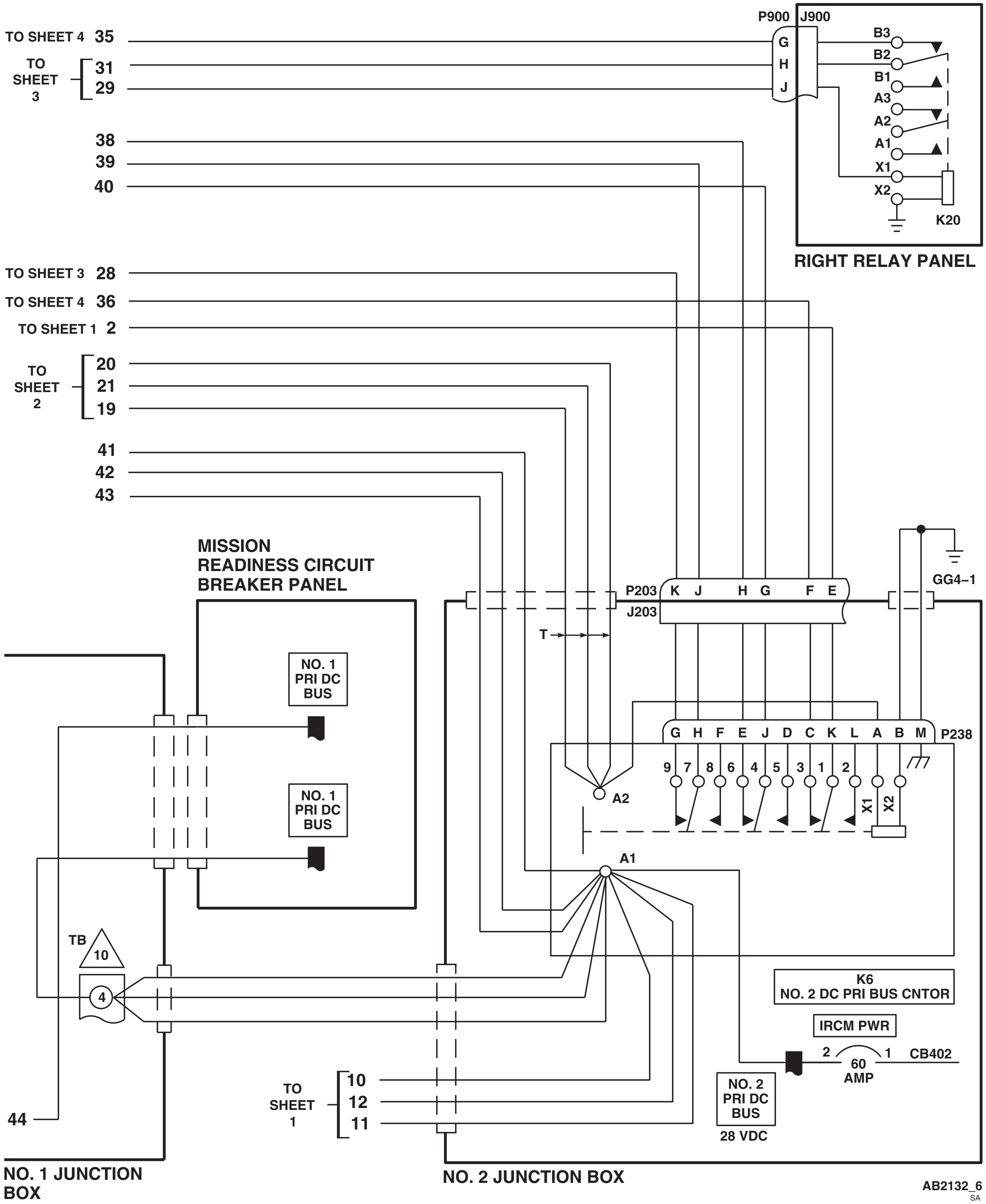


Figure 2. DC Electrical System Schematic Diagram. (Sheet 6 of 6)

SCHEMATICS - Continued

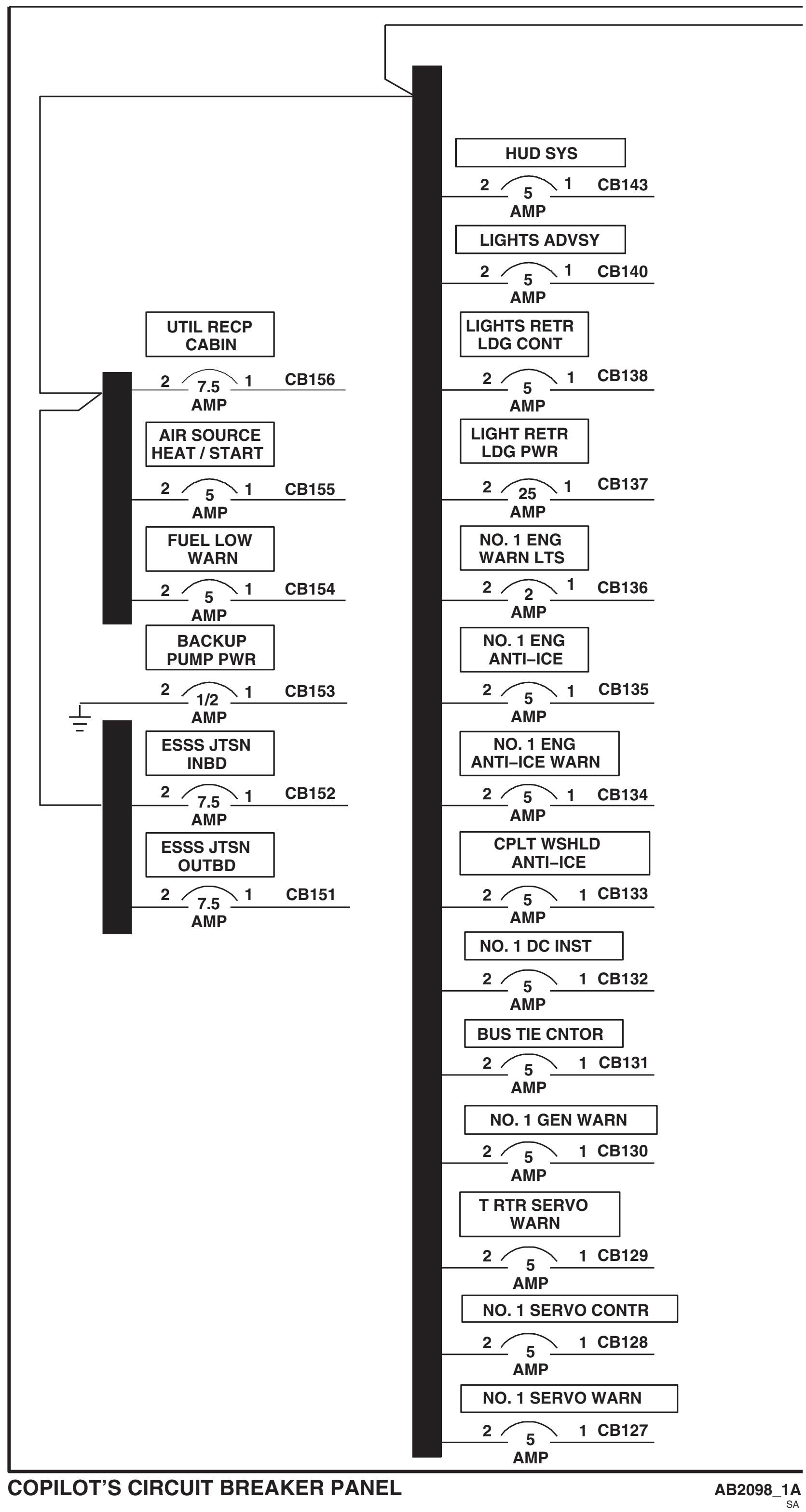


Figure 3. DC Electrical System Power Distribution Diagram **HH-60A** . (Sheet 1 of 8)

SCHEMATICS - Continued

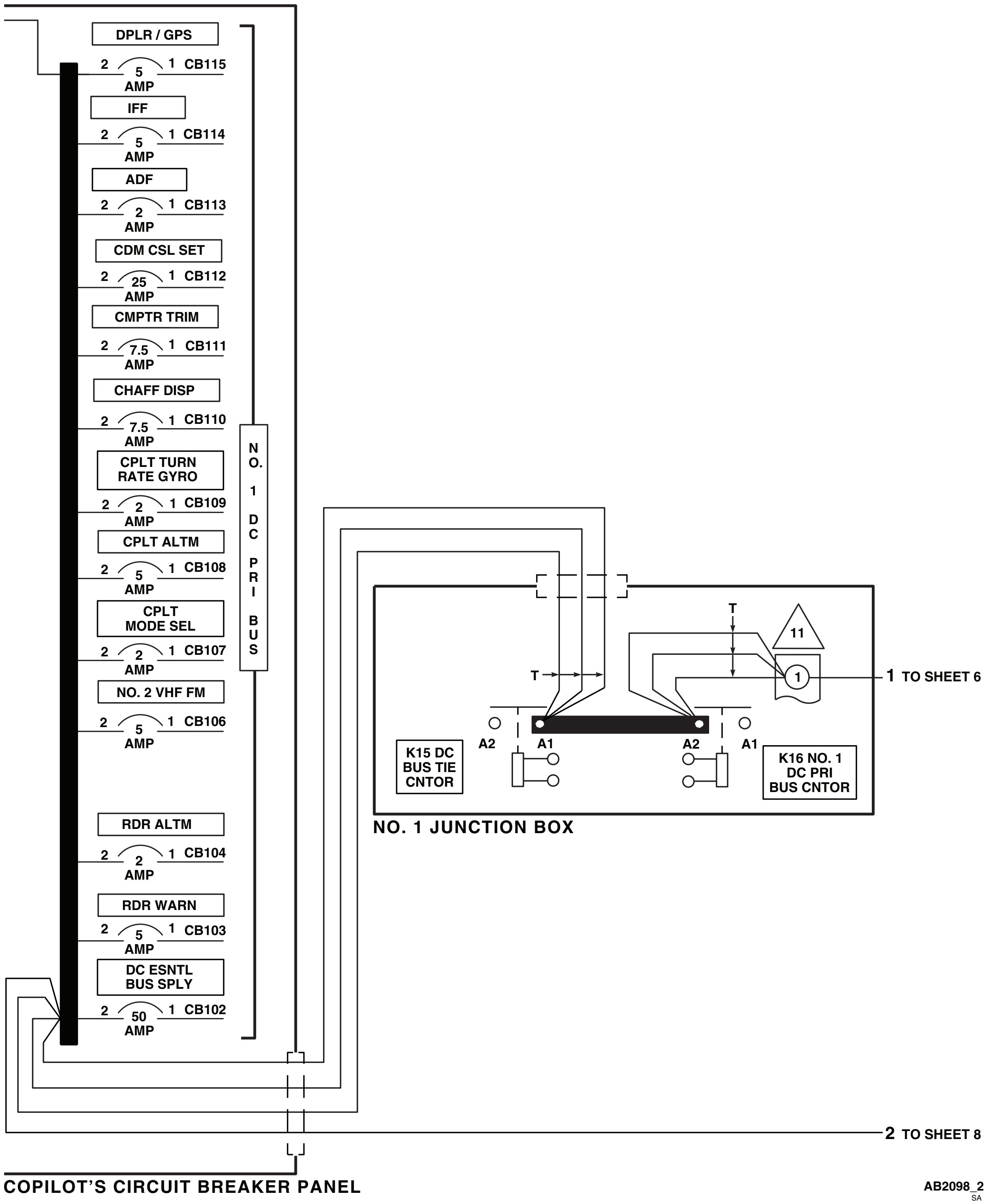


Figure 3. DC Electrical System Power Distribution Diagram HH-60A . (Sheet 2 of 8)

SCHEMATICS - Continued

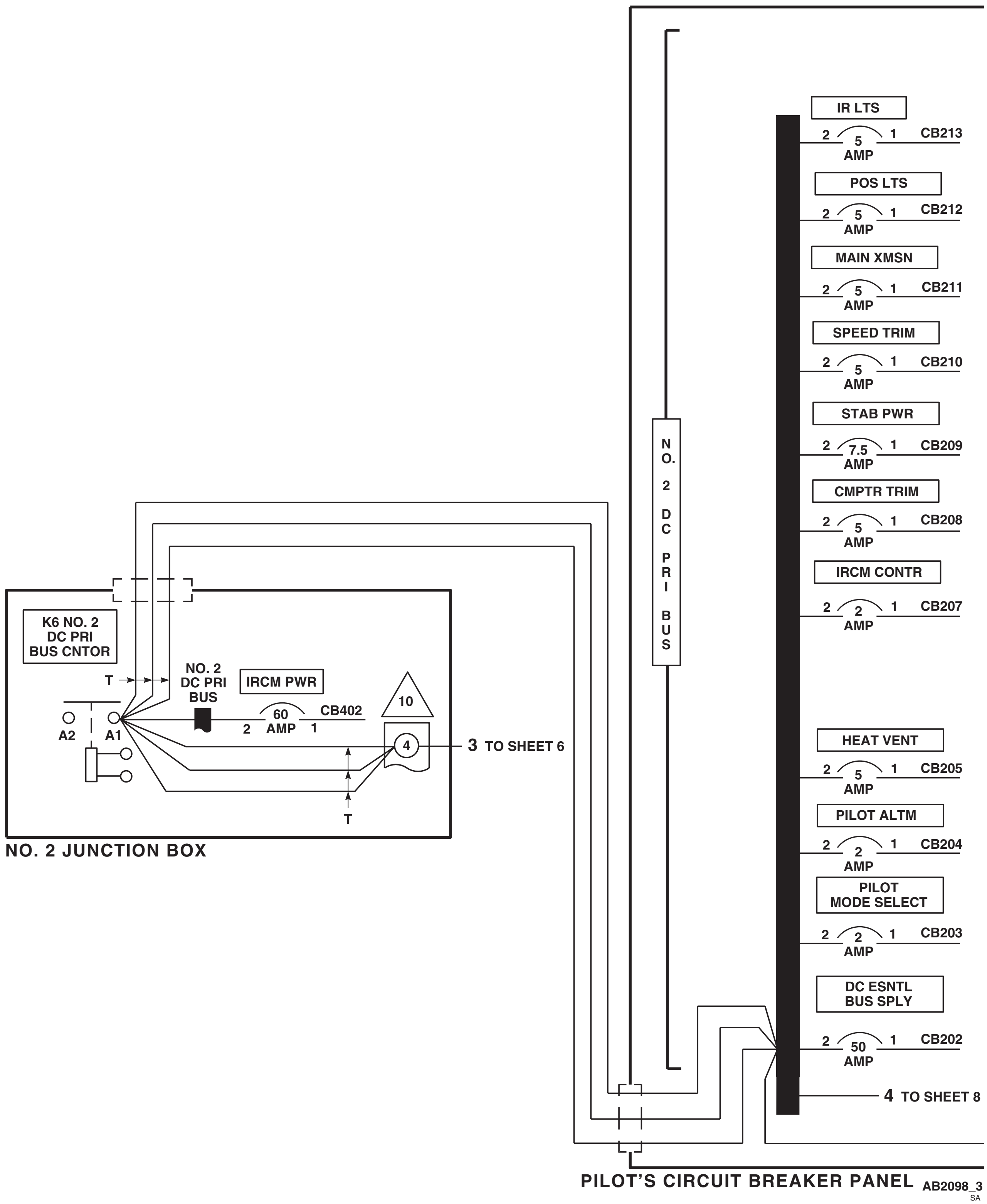


Figure 3. DC Electrical System Power Distribution Diagram HH-60A . (Sheet 3 of 8)

SCHEMATICS - Continued

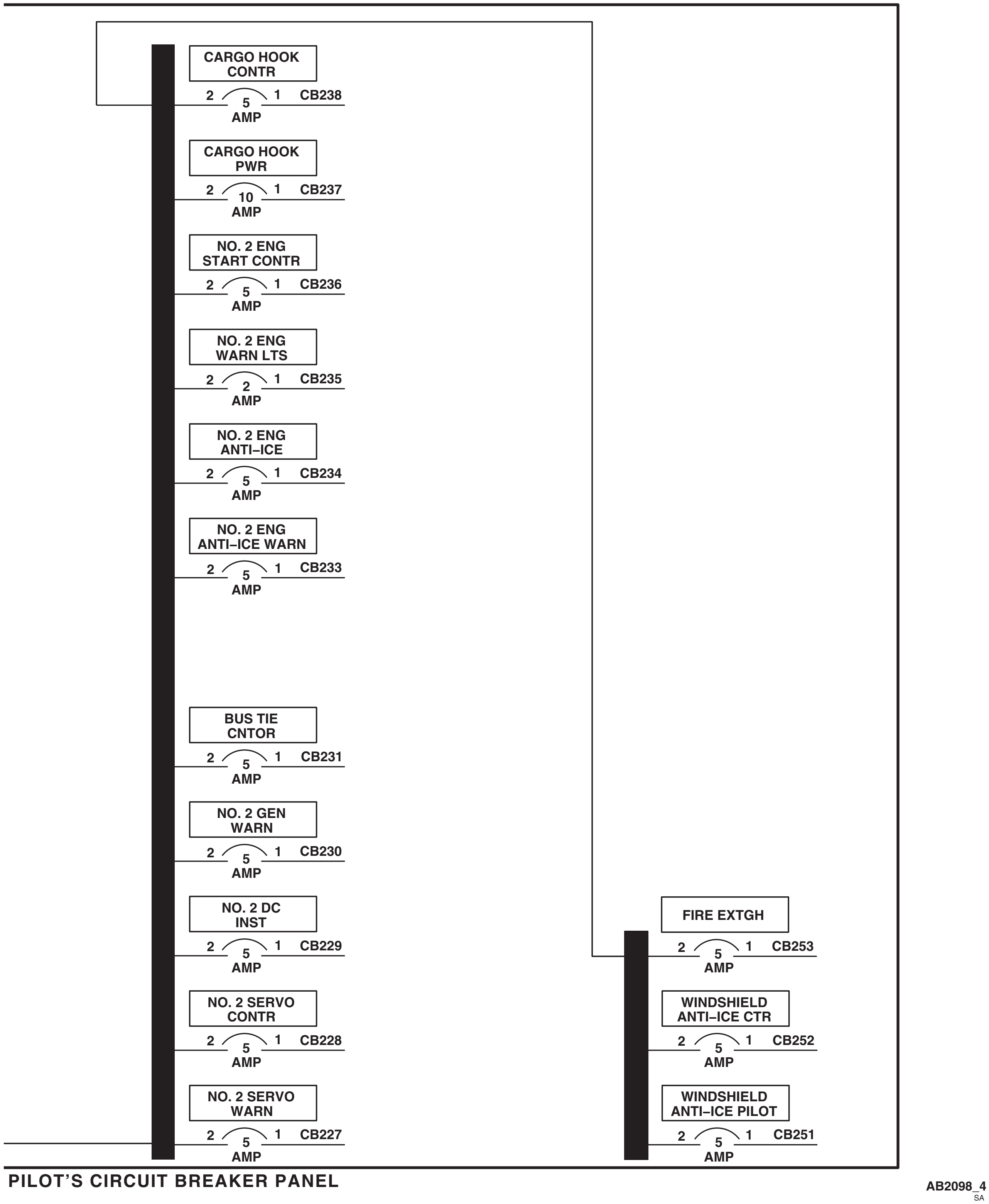


Figure 3. DC Electrical System Power Distribution Diagram **HH-60A** . (Sheet 4 of 8)

SCHEMATICS - Continued

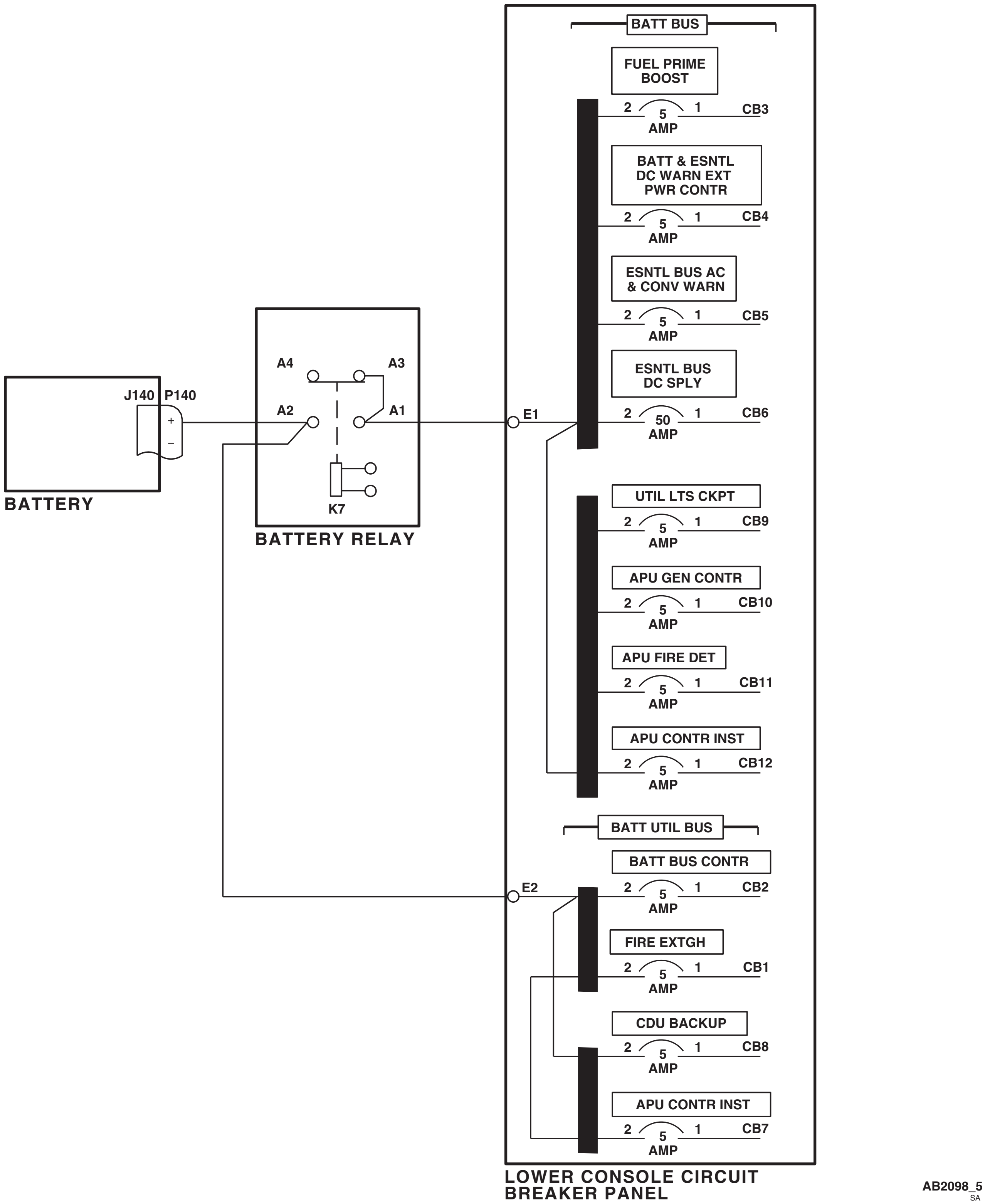


Figure 3. DC Electrical System Power Distribution Diagram HH-60A . (Sheet 5 of 8)

SCHEMATICS - Continued

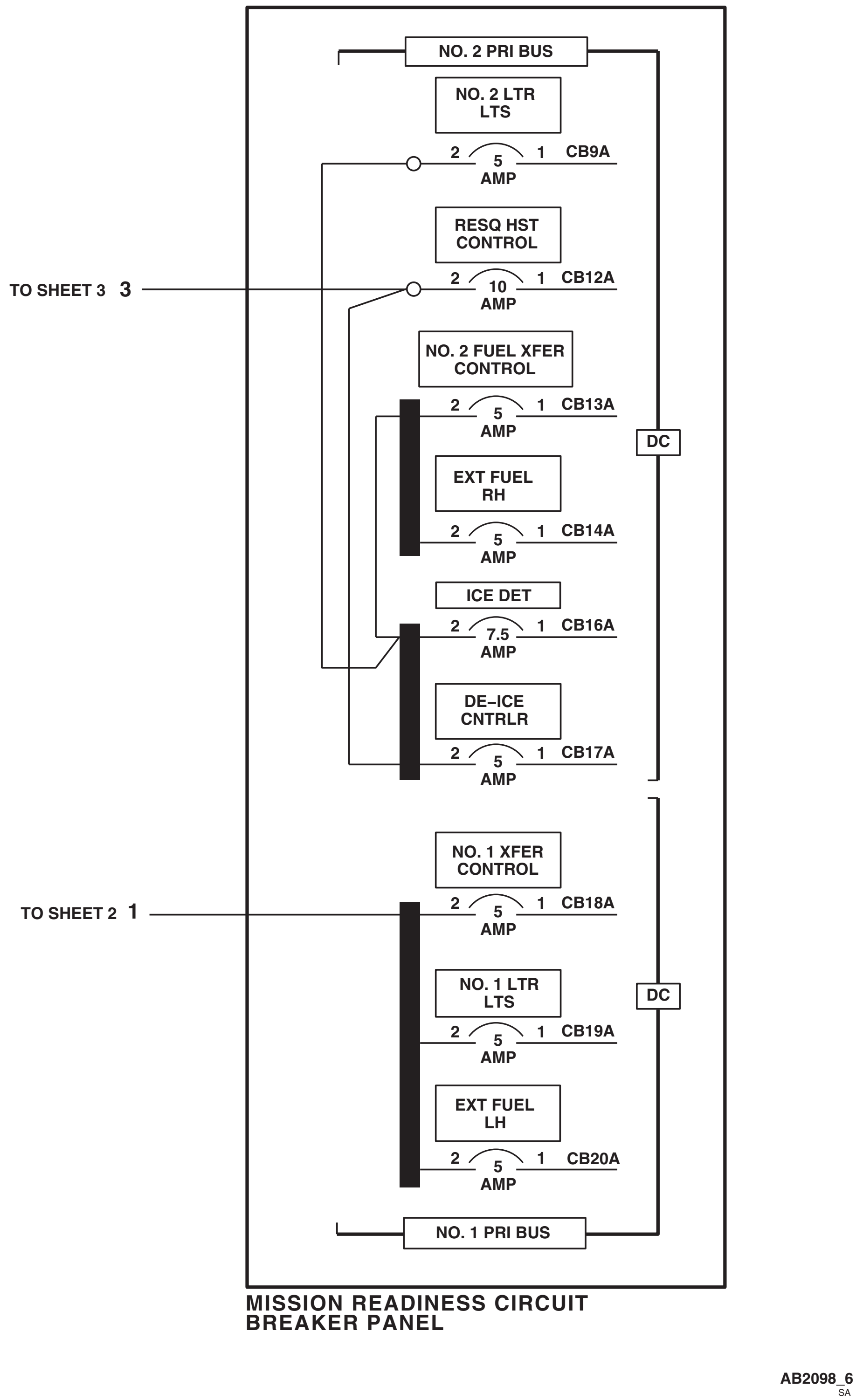


Figure 3. DC Electrical System Power Distribution Diagram **HH-60A** . (Sheet 6 of 8)

SCHEMATICS - Continued

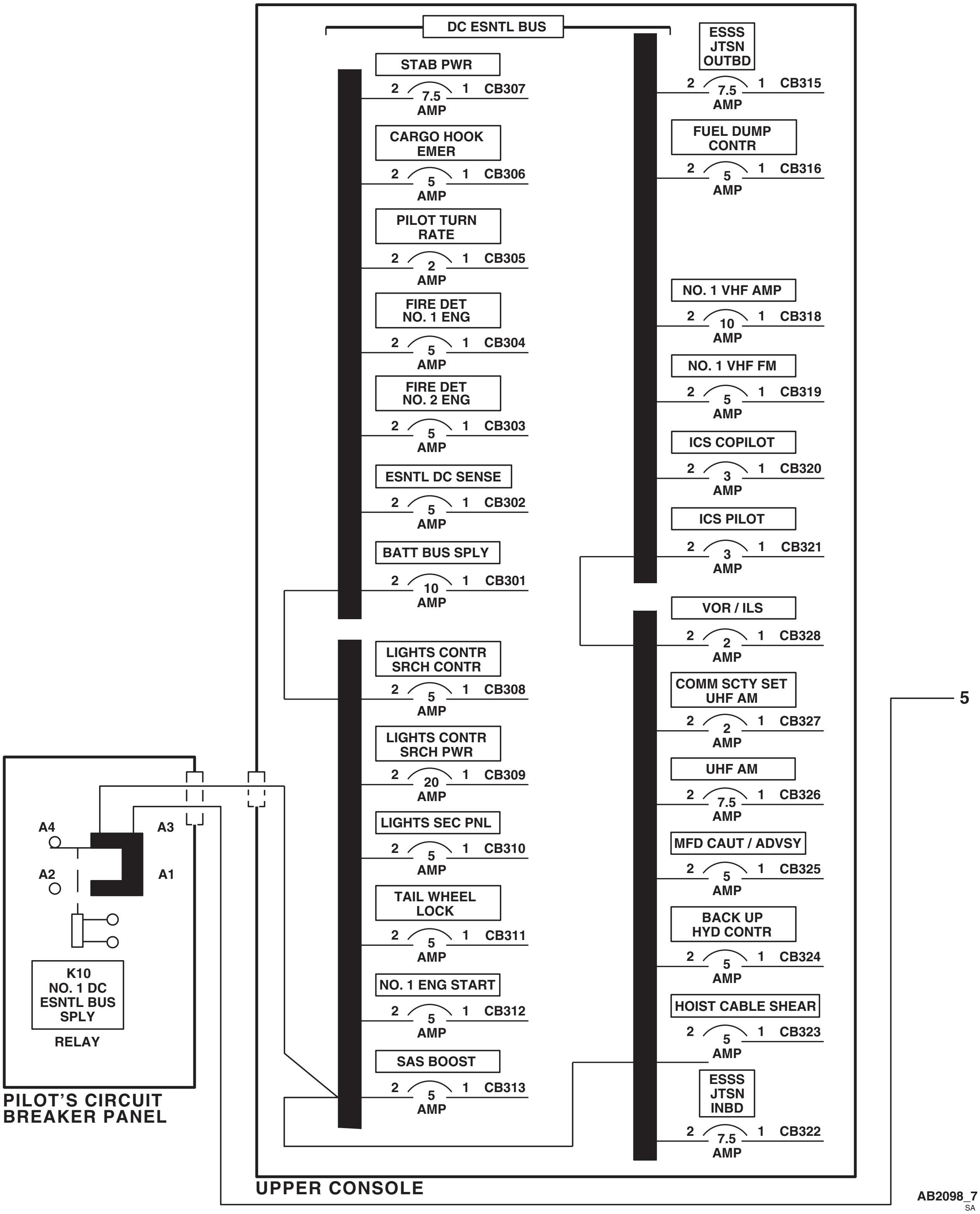


Figure 3. DC Electrical System Power Distribution Diagram **HH-60A** . (Sheet 7 of 8)

SCHEMATICS - Continued

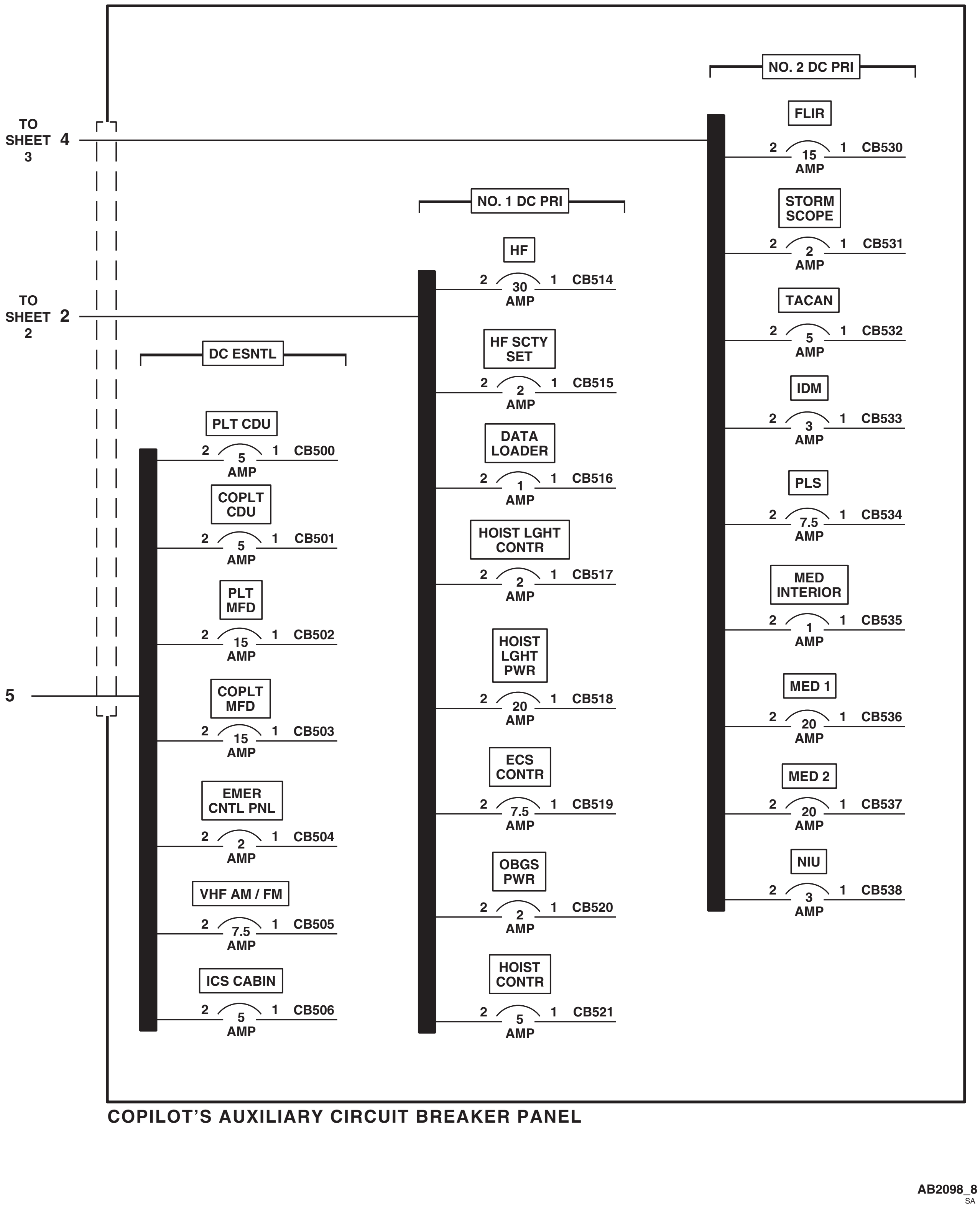
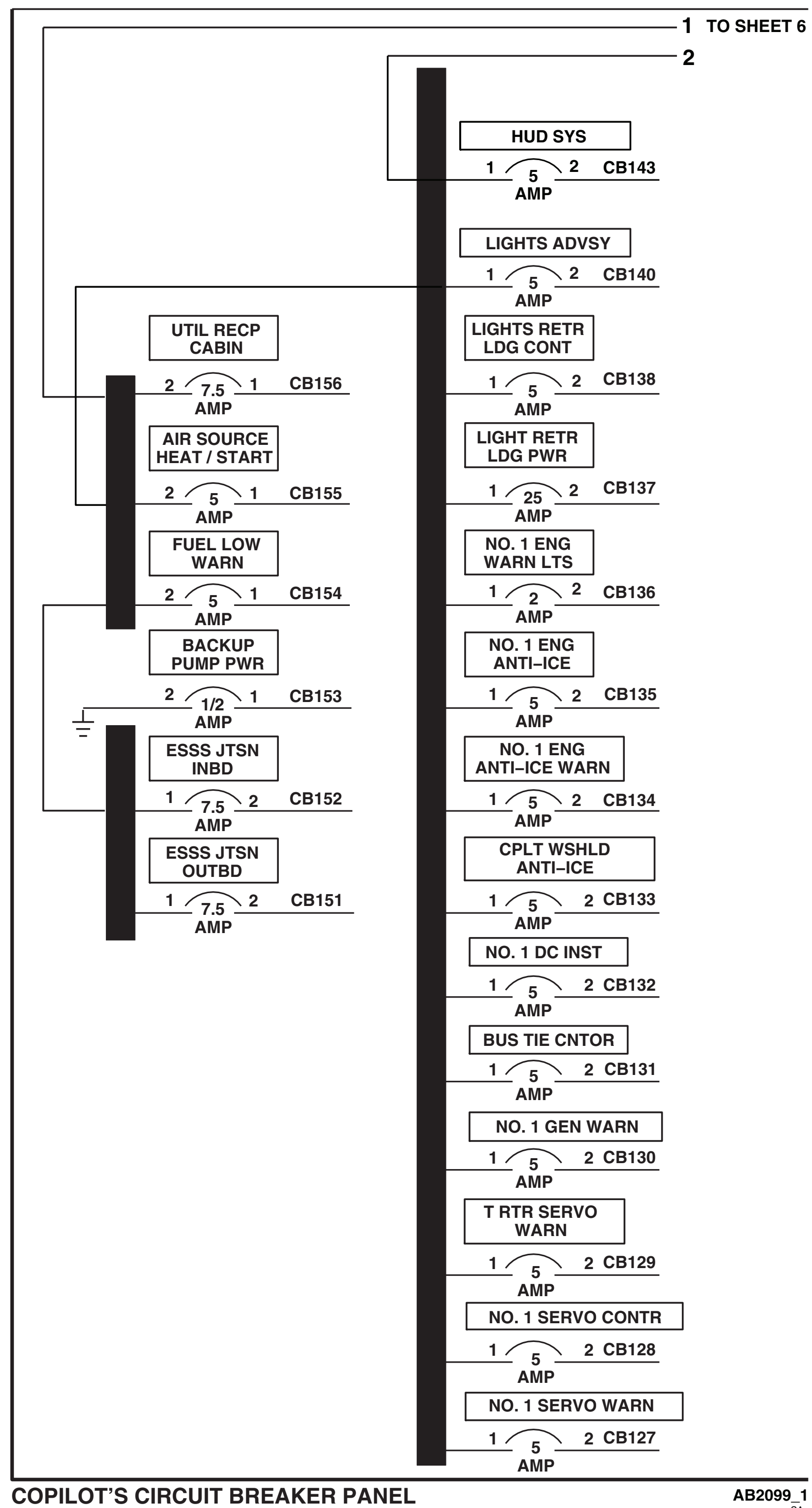


Figure 3. DC Electrical System Power Distribution Diagram HH-60A . (Sheet 8 of 8)

SCHEMATICS - Continued



NOTE

HH–60L

Figure 4. DC Electrical System Power Distribution Diagram HH-60L . (Sheet 1 of 8)

SCHEMATICS - Continued

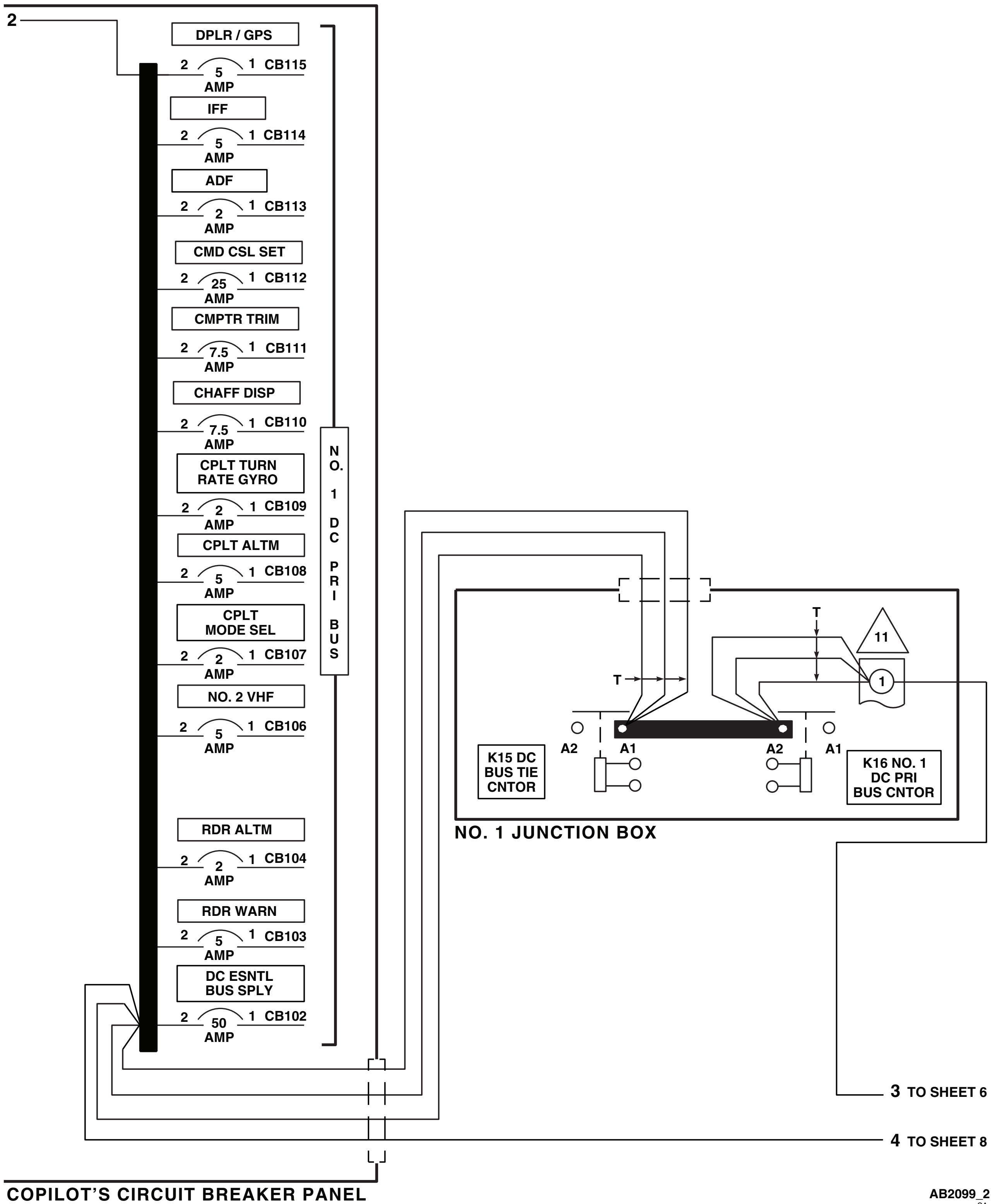


Figure 4. DC Electrical System Power Distribution Diagram **HH-60L** . (Sheet 2 of 8)

SCHEMATICS - Continued

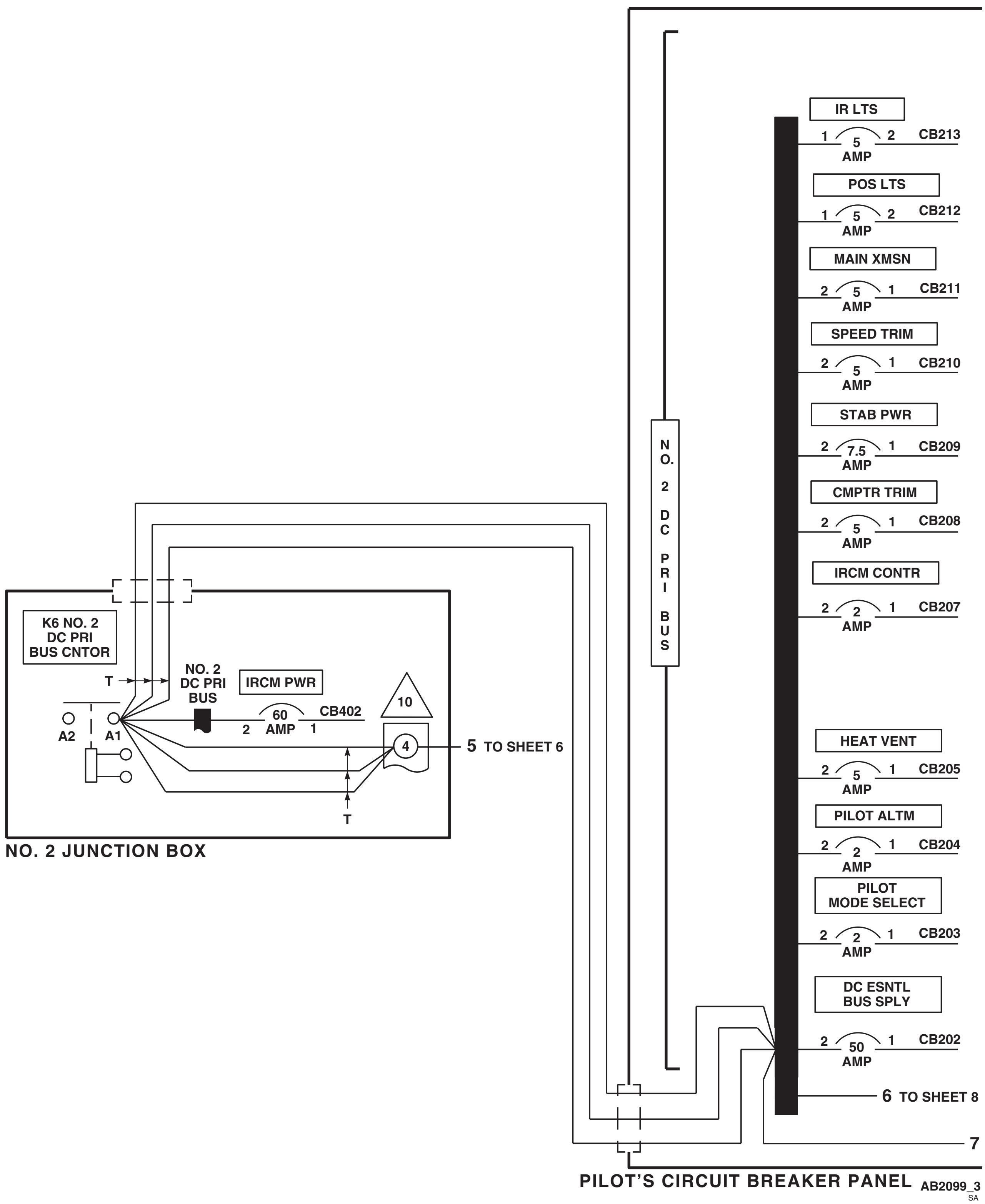


Figure 4. DC Electrical System Power Distribution Diagram HH-60L . (Sheet 3 of 8)

SCHEMATICS - Continued

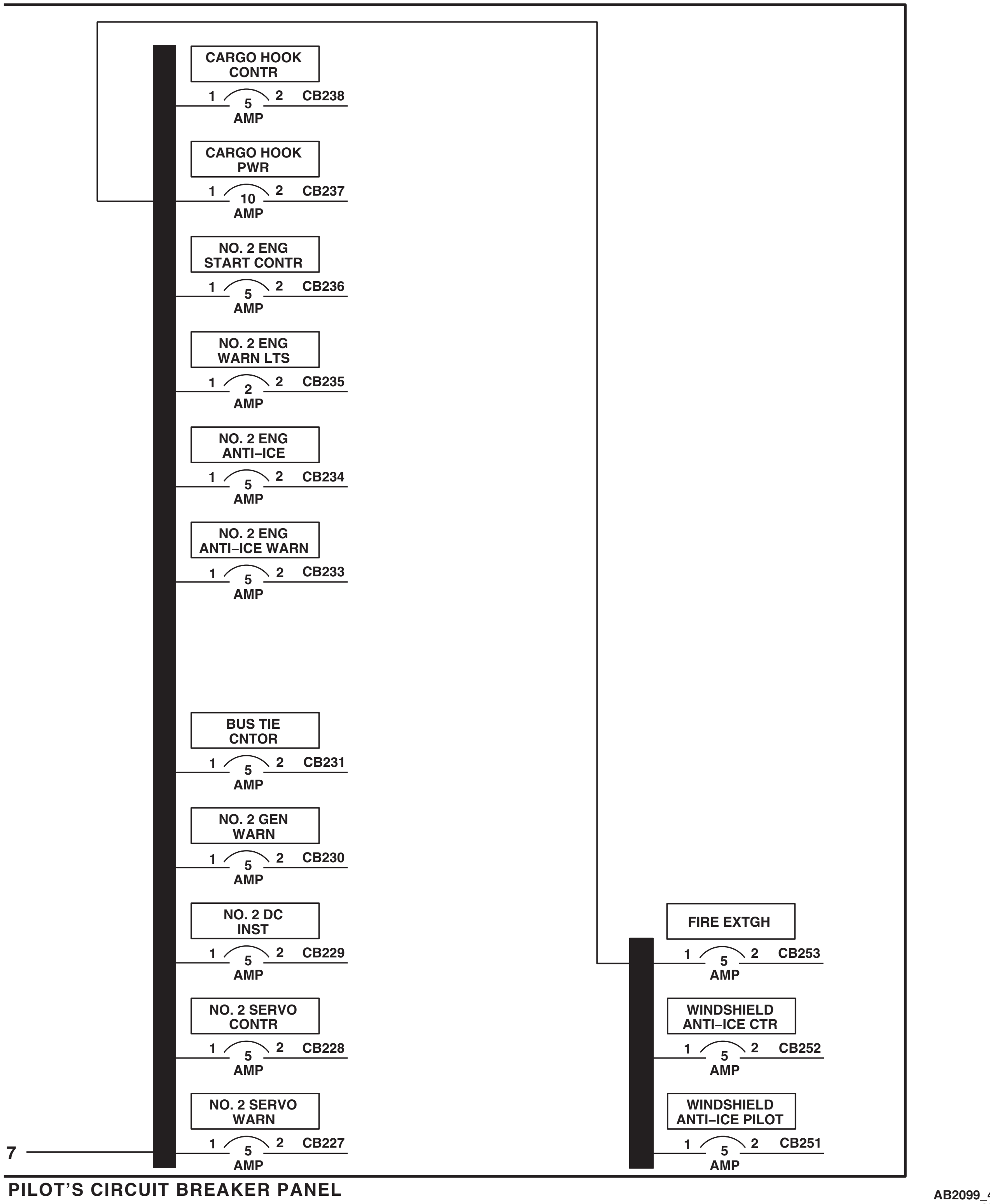


Figure 4. DC Electrical System Power Distribution Diagram HH-60L . (Sheet 4 of 8)

SCHEMATICS - Continued

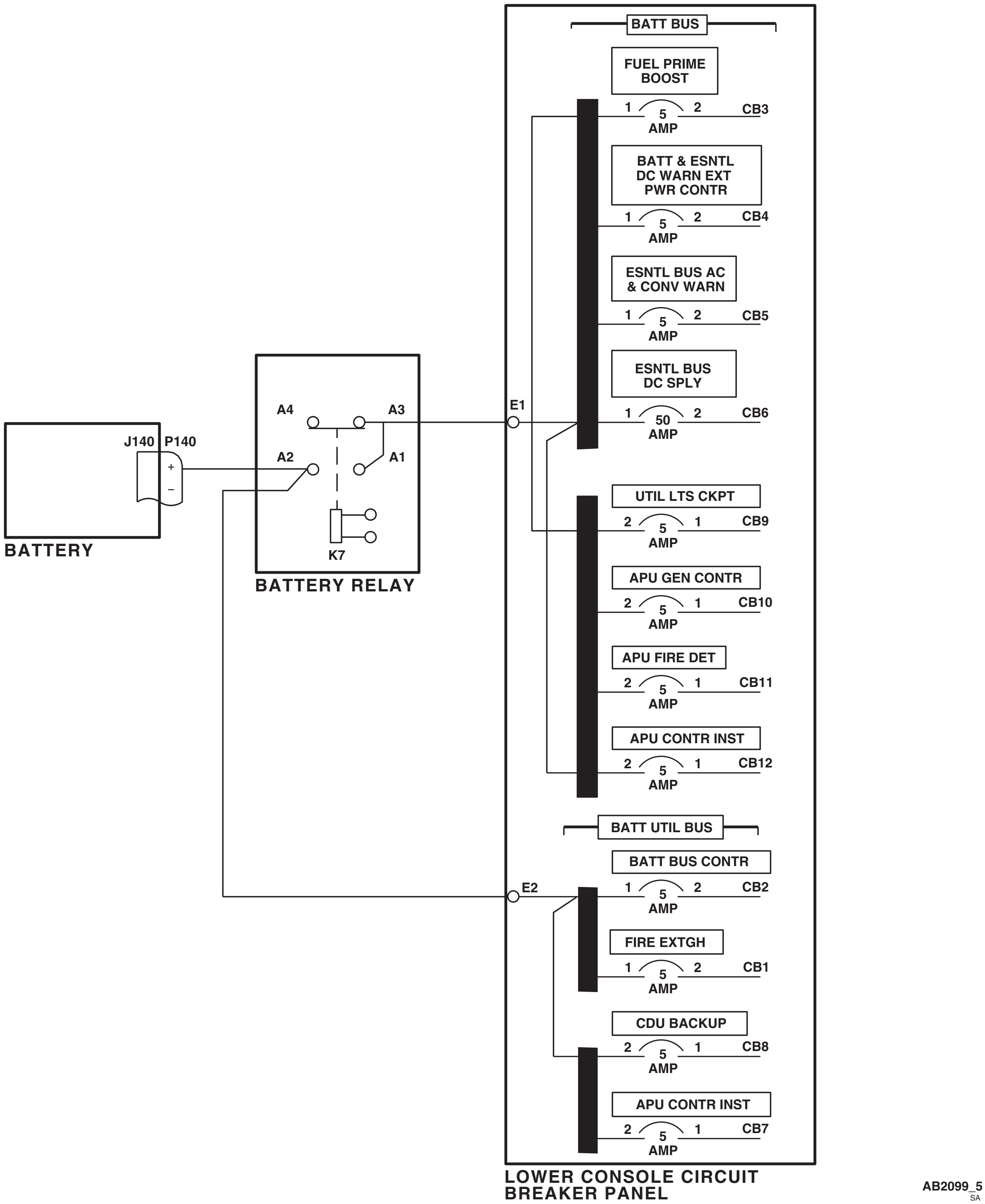


Figure 4. DC Electrical System Power Distribution Diagram **HH-60L** . (Sheet 5 of 8)

SCHEMATICS - Continued

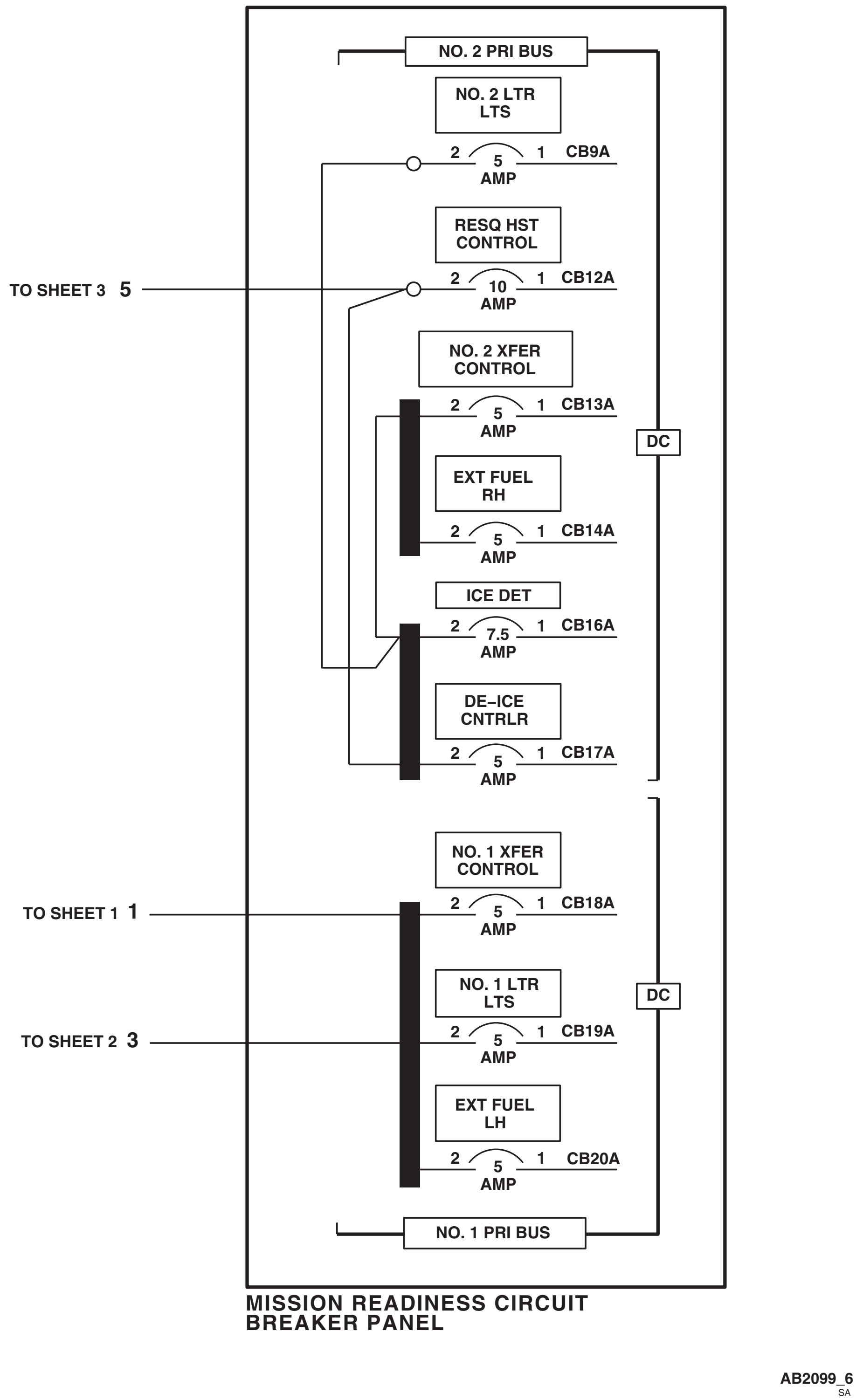


Figure 4. DC Electrical System Power Distribution Diagram HH-60L . (Sheet 6 of 8)

SCHEMATICS - Continued

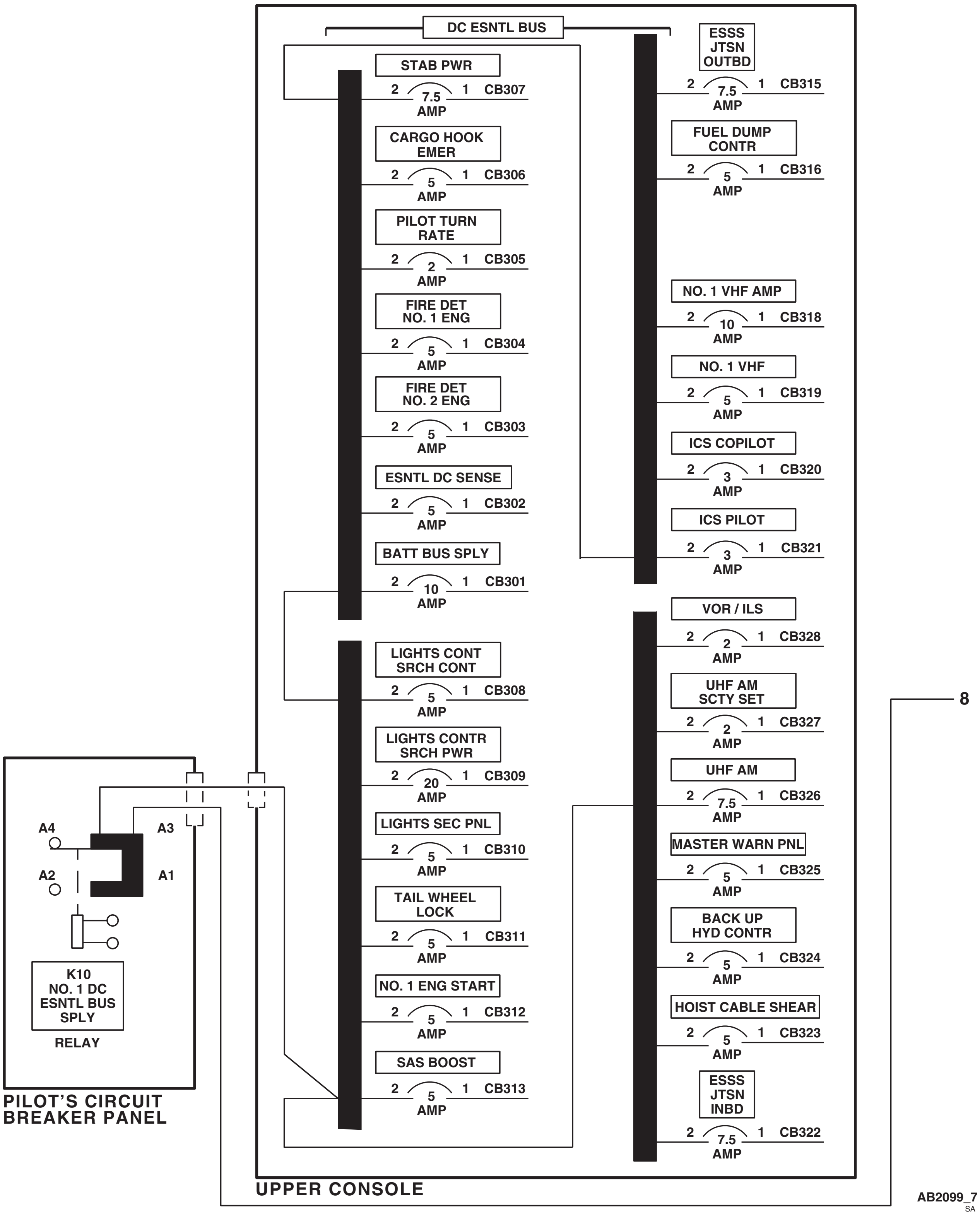


Figure 4. DC Electrical System Power Distribution Diagram HH-60L . (Sheet 7 of 8)

SCHEMATICS - Continued

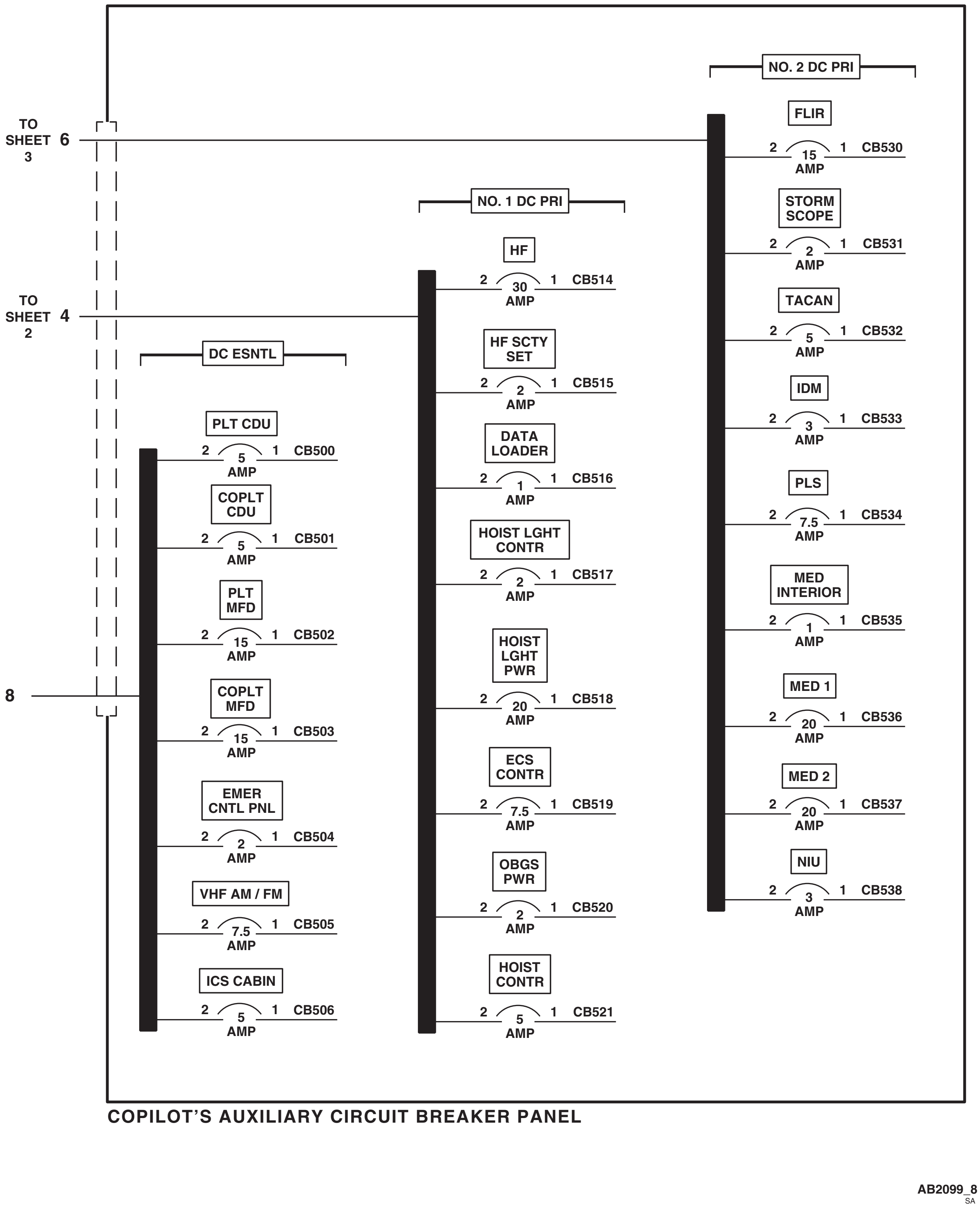


Figure 4. DC Electrical System Power Distribution Diagram HH-60L . (Sheet 8 of 8)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

UPPER AND LOWER CONSOLE LIGHTS HH-60A HH-60L

This WP supersedes WP 0124 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
WP 0104 00
WP 0108 00
WP 0824 00
WP 0875 00
WP 0935 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

UPPER AND LOWER CONSOLE LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Turn on electrical power.
3. On upper console, turn CONSOLE LT LOWER BRT/OFF control to BRT.

Indication/Condition

Lighting (0-115 vac) on these control panels shall come on:

- Stabilator control/auto flight control panel
- Fuel boost pump control panel
- Personnel locator system (PLS) control panel
- Compass system control panel
- Blade de-ice control panel
- Blade de-ice test panel

UPPER AND LOWER CONSOLE LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- Miscellaneous switch panel
- ESSS stores jettison panel
- Pilot's ICS control panel
- Copilot's ICS control panel
- MED 2 ICS control panel
- MED 1 ICS control panel
- Crew chief's ICS control panel
- Auxiliary fuel management control panel

Corrective Action

1. If none of this lighting comes on, go to LOWER CONSOLE 0-115 VAC LIGHTING FAILS, in this work package.
2. If lower console lighting flickers when control is turned clockwise, replace CONSOLE LT LOWER dimming control T11 (WP 0875 00).
3. If lighting on PLS, Compass System, Blade De-ice Control, Blade De-ice Test, Miscellaneous Switch, ESSS Stores, Pilot's ICS, Copilot's ICS, MED 2 ICS, MED 1 ICS, and Crew Chief's ICS control panels fails, check for 115 vac between TJ115A-H and TJ115A-G and ground with dimming control in BRT position.
   a. If voltage is as specified, replace TJ115A (WP 1747 00).
   b. If voltage is not as specified, repair/replace wiring between TJ115A-H and T11-LV and between TJ115A-G and T11-LV (WP 1747 00).
   c. If any individual 0-115 vac panel lighting fails, go to LOWER CONSOLE 0-115 VAC AUXILIARY LIGHTING CHECKS, in this work package.

Step

4. On upper console, turn CONSOLE LT LOWER BRT/OFF control to OFF.

Indication/Condition

Lower console lighting shall go from bright to dim, then off as control is turned counterclockwise.

Corrective Action

If Indication/Condition is not as specified, replace CONSOLE LT LOWER dimmer control T11 (WP 0875 00).

Step

5. On upper console, turn CONSOLE LT UPPER BRT/OFF control to BRT.

Indication/Condition

The following lighting shall go on bright as control is turned clockwise:

- Upper console panels
- Cockpit flood and secondary lights panel
- Engine controls quadrant panels

UPPER AND LOWER CONSOLE LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition is not as specified, go to UPPER CONSOLE LIGHTING DOES NOT GO ON, in this work package.
2. If upper console lighting flickers when control is turned clockwise, replace CONSOLE LT UPPER dimmer control T3 (WP 0875 00).
3. If individual panel lighting fails, go to UPPER CONSOLE PANEL LIGHTING CHECKS, in this work package.

Step

6. On upper console, turn CONSOLE LT UPPER BRT/OFF control to OFF.

Indication/Condition

The following lighting shall go from bright to dim, then off as control is turned counterclockwise:

- Upper console panels
- Cockpit flood and secondary lights panel
- Engine controls quadrant panels

Corrective Action

If Indication/Condition is not as specified, replace CONSOLE LT UPPER dimmer control T3 (WP 0875 00).

Step

7. On instrument panel, turn LOWER CSL AUX DIMMER control fully clockwise.

Indication/Condition

Lighting (0-5 Vac) on these panels shall come on:

- Ice rate meter
- Copilot’s CDU
- Emergency control panel
- Pilot’s rescue hoist control panel
- On-board oxygen generating (OBOGS) status panel
- Environmental control unit (ECU)
- Auxiliary switch panel
- Forward looking infrared (FLIR) control panel (if installed)
- VHF AM/FM radio
- Crew’s rescue hoist control panel
- Pilot’s CDU

Corrective Action

1. If none of the 0-5 vac lighting comes on, go to LOWER CONSOLE 0-5 VAC LIGHTING FAILS, in this work package.

UPPER AND LOWER CONSOLE LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. If any individual panel lighting fails, go to LOWER CONSOLE 0-5 VAC AUXILIARY LIGHTING CHECKS, in this work package.

Step

8. Turn LOWER CSL AUX DIMMER control fully counterclockwise.

Indication/Condition

Auxiliary 0-5 vac lighting shall go off.

Corrective Action

If Indication/Condition is not as specified, replace LOWER CSL AUX DIMMER control T15 (WP 0935 00).

Step

9. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

LOWER CONSOLE 0-115 VAC LIGHTING FAILS

SYMPTOM

Lower console 0-115 vac lighting fails.

MALFUNCTION

Lower console 0-115 vac lighting fails.

CORRECTIVE ACTION

1. **Check continuity between CONSOLE LT LOWER dimming control T11-G and ground.**
   a. If continuity is present, go to **2.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5.**
2. **Check for 115 vac between T11-HV and T11-G.**
   a. If voltage is as specified, go to **3.**
   b. If voltage is not as specified, go to **4.**
3. **Check for 115 vac between T11-LV and ground with dimming control in BRT position.**
   a. If voltage is as specified, repair/replace wiring between P217-g and T11-LV (WP 1747 00). Go to **5.**
   b. If voltage is not as specified, replace dimming control T11 (WP 0875 00). Go to **5.**
4. **Check for 115 vac between of LIGHTS LWR CSL circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, repair/replace wiring between T11-HV and circuit breaker terminal 1 (WP 1747 00). Go to **5.**

LOWER CONSOLE 0-115 VAC LIGHTING FAILS - Continued

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **5.**

5. Procedure completed.

Table 1. Lower Console 0-115 Vac Auxiliary Lighting Checks.

COMPONENT	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
STABILATOR CONTROL /AUTO FLIGHT CONTROL PANEL	P4R-e and T11-LV, P4R-A and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
FUEL BOOST PUMP CONTROL PANEL	P139-L and T11-LV, P139-Z and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
PLS CONTROL PANEL	P145R-C and TJ115A-Z, P145R-M and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
COMPASS SYSTEM CONTROL PANEL	P134R-U and TJ115A-A, P134R-V and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
BLADE DE-ICE CONTROL PANEL	P133-A and TJ115A-L, P133-E and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
BLADE DE-ICE TEST PANEL	P134-b and TJ115A-J, P134-c and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
MISCELLANEOUS SWITCH PANEL	P659R-G and TJ115A-B, P659R-H and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
ESSS STORES JET-TISON PANEL	P645-V and TJ115A-K, P645-W and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
ICS CONTROL PANEL, PILOT'S	P124R-J and TJ115A-E, P124R-U and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
ICS CONTROL PANEL, COPILOT'S	P175R-J and TJ115A-S, P175R-U and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
ICS CONTROL PANEL, MED 2	P220R-J and TJ115A-N, P220R-U and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
ICS CONTROL PANEL, MED 1	P204R-J and TJ115A-P, P204R-U and ground	Replace Panel	Repair/replace wiring (WP 1747 00).

LOWER CONSOLE 0-115 VAC LIGHTING FAILS - Continued

Table 1. Lower Console 0-115 Vac Auxiliary Lighting Checks. - Continued

COMPONENT	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
ICS CONTROL PANEL, CREW CHIEF'S	P207R-J and TJ115A-R, P207R-U and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
AUXILIARY FUEL MANAGEMENT PANEL	P360-T and T11-LV, P360-S and ground	Replace Panel	Repair/replace wiring (WP 1747 00).

UPPER CONSOLE LIGHTING DOES NOT GO ON

SYMPTOM

Upper console lighting does not go on.

MALFUNCTION

Upper console lighting does not go on.

CORRECTIVE ACTION

1. **Check continuity between CONSOLE LT UPPER dimming control T3-G and ground.**
   a. If continuity is present, go to **2.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**
2. **Check for 115 vac between T3-HV and T3-G.**
   a. If voltage is as specified, replace dimmer control T3 (WP 0875 00). Go to **4.**
   b. If voltage is not as specified, go to **3.**
3. **Check for 115 vac between LIGHTS UPPER CSL circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T3-HV (WP 1747 00). Go to **4.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **4.**
4. **Procedure completed.**

Table 2. Upper Console Panel Lighting Checks.

INFORMATION PANEL	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
ENGINE CONTROL LEFT QUADRANT	J394-1 and T3-LV, J394-2 and ground	Replace Panel	Repair/replace wiring (WP 1747 00).

UPPER CONSOLE LIGHTING DOES NOT GO ON - Continued

Table 2. Upper Console Panel Lighting Checks. - Continued

INFORMATION PANEL	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
ENGINE CONTROL CENTER QUADRANT	J395-1 and T3-LV, J395-2 and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
ENGINE CONTROL RIGHT QUADRANT	J396-1 and T3-LV, J396-2 and ground	Replace Panel	Repair/replace wiring (WP 1747 00).
COCKPIT FLOOD AND SECONDARY LIGHTS	P288-E and T3-LV, P288-D and ground	Troubleshoot Cockpit Flood And Secondary Lights (WP 0108 00)	Repair/replace wiring (WP 1747 00).
UPPER CONSOLE LEFT SIDE	1J2-LV and T3-LV	Replace Panel	Repair/replace wiring (WP 1747 00).
UPPER CONSOLE CENTER	1J1-LV and 1J3-LV, 1J1-LV and 1J2-LV	Replace Panel	Repair/replace wiring (WP 1747 00).
UPPER CONSOLE RIGHT SIDE	1J3-LV and T3-LV, 1J3-GND and TB2-2	Replace Panel	Repair/replace wiring (WP 1747 00).
UPPER CONSOLE ENG SPEED TRIM	1J4-LV and 1J3-LV, 1J4-GND and T3-G	Replace Panel	Repair/replace wiring (WP 1747 00).

LOWER CONSOLE 0-5 VAC LIGHTING FAILS

SYMPTOM

Lower console 0-5 vac lighting fails.

MALFUNCTION

Lower console 0-5 vac lighting fails.

CORRECTIVE ACTION

1. **Check for 115 vac between T15-HV and ground.**

   a. If voltage is as specified, go to **2.**

   b. If voltage is not as specified, go to **5.**

2. **Check continuity between T15-G and ground.**

   a. If continuity is present, go to **3.**

   b. If continuity is not present, repair/replace wiring to ground (WP 1747 00). Go to **6.**

3. **With LOWER CSL AUX DIMMER control fully clockwise, check for 5 vac between T15-LV and ground.**

   a. If voltage is as specified, go to **4.**

LOWER CONSOLE 0-5 VAC LIGHTING FAILS - Continued

b. If voltage is not as specified, replace LOWER CSL AUX DIMMER control T15 (WP 0935 00). Go to **6.**

4. Check continuity between:

- **T15-LV and TJ5VA-A**
- **T15-LV and TJ5VA-B**
- **T15-LV and TJ5VA-C**
- **T15-LV and TJ5VA-D**
- **T15-LV and TJ5VB-A**
- **T15-LV and TJ5VB-B**
- **T15-LV and TJ5VB-C**
- **T15-LV and TJ5VB-D**

a. If continuity is present, replace TJ5VA or TJ5VB as required (WP 1747 00). Go to **6.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**

5. Check for 115 vac between LIGHTS 5V LWR CSL circuit breaker terminal 2 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker and T15-HV (WP 1747 00). Go to **6.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6.**

6. Procedure completed.

Table 3. Lower Console 0-5 Vac Auxiliary Lighting Checks.

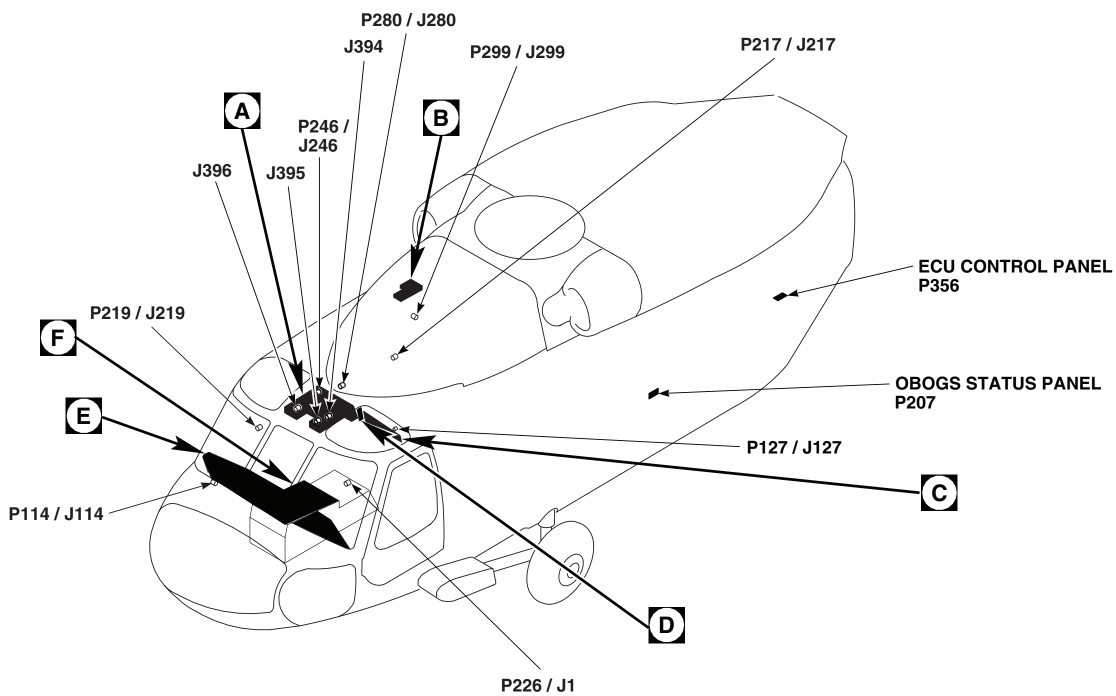
COMPONENT	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
ICE RATE METER	P132-1 and TJ5VA-G, P132-2 and ground	Replace meter	Repair/replace wiring (WP 1747 00).
CDU, PILOT'S	P1012R-1 and TJ5VB-E, P1012R-2 and ground	Replace CDU	Repair/replace wiring (WP 1747 00).
CDU, COPILOT'S	P1011R-1 and TJ5VA-E, P1011R-2 and ground	Replace CDU	Repair/replace wiring (WP 1747 00).
EMERGENCY CONTROL PANEL	P140R-1 and TJ5VA-F, P140R-2 and ground	Replace panel	Repair/replace wiring (WP 1747 00).

LOWER CONSOLE 0-5 VAC LIGHTING FAILS - Continued

Table 3. Lower Console 0-5 Vac Auxiliary Lighting Checks. - Continued

COMPONENT	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
PILOT'S RESCUE HOIST CONTROL PANEL	P218-C and TJ5VA-H, P218-D and ground	Replace panel	Repair/replace wiring (WP 1747 00).
OBOGS STATUS PANEL	P207-2 and TJ5VA-K, P207-1 and ground	Replace panel	Repair/replace wiring (WP 1747 00).
ECU	P356-N and TJ5VB-L, P356-P and ground	Replace panel	Repair/replace wiring (WP 1747 00).
AUXILIARY SWITCH PANEL	P170-N and TJ5VB-K, P170-P and ground	Replace panel	Repair/replace wiring (WP 1747 00).
FLIR CONTROL PANEL	P214R-9 and TJ5VB-G, P214R-10 and ground	Replace panel	Repair/replace wiring (WP 1747 00).
VHF AM/FM RADIO	P110R-n and TJ5VB-F, P110R-b and ground	Replace radio	Repair/replace wiring (WP 1747 00).
CREW'S RESCUE HOIST CONTROL PANEL	P226-h and TJ5VB-J, P226-i and ground	Replace panel	Repair/replace wiring (WP 1747 00).

LOCATION

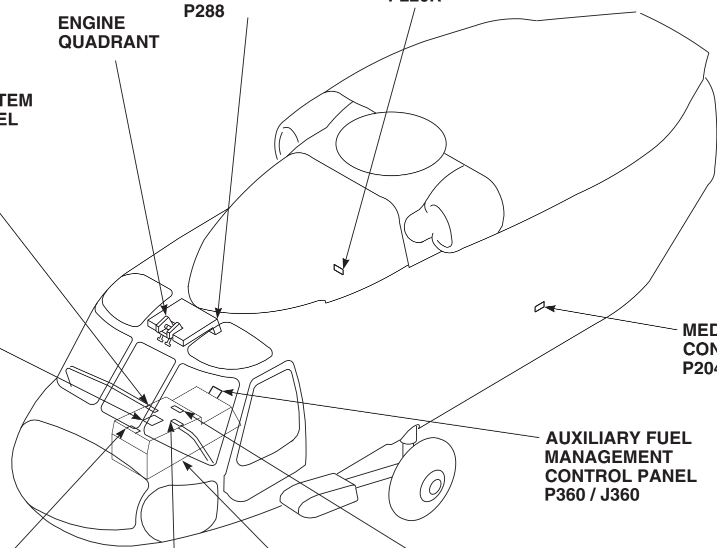


TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
J394	COCKPIT CEILING, BL 9 LH, STA 233
J395	COCKPIT CEILING, BL 4 LH, STA 229
J396	COCKPIT CEILING, BL 4 RH, STA 229
P4R	STABILATOR CONTROLS / AUTO FLIGHT CONTROL PANEL
P110R / J1	VHF AM / FM RADIO
P114 / J114	COCKPIT, BL 8 RH, STA 200
P124R	PILOT'S ICS CONTROL PANEL
P127 / J127	BEHIND COPILOT'S CIRCUIT BREAKER PANEL, L 28 LH, STA 243
P129 / J129	AUXILIARY CIRCUIT BREAKER PANEL
P132	ICE RATE METER
P133	BLADE DE–ICE CONTROL PANEL
P134	BLADE DE–ICE TEST PANEL
P134R	COMPASS SYSTEM CONTROL PANEL
P139	FUEL BOOST PUMP CONTROL PANEL

AB0110_1A
SA

Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 1 of 6)

LOCATION - Continued



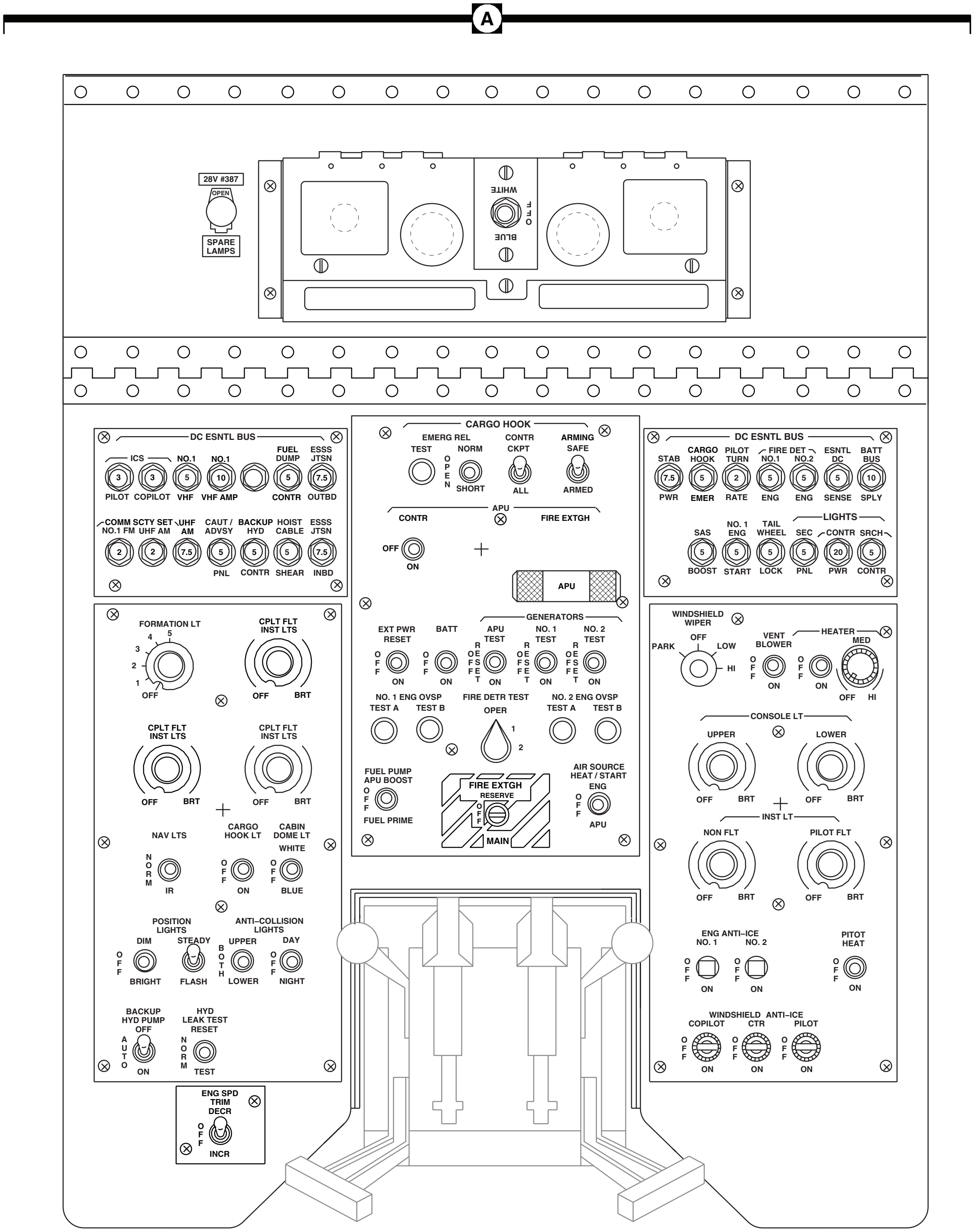
TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P140R	EMERGENCY CONTROL PANEL
P145R	PERSONNEL LOCATOR SYSTEM (PLS) CONTROL PANEL
P170	AUXILIARY SWITCH PANEL
P175R	COPILOT'S ICS CONTROL PANEL
P204R	MED1 ICS CONTROL PANEL
P207	OBOGS STATUS PANEL
P207R	CREW CHIEF'S ICS CONTROL PANEL
P214R	FLIR CONTROL PANEL
P217 / J217	INTERCONNECT, CABIN CEILING
P218	PILOT'S RESCUE HOIST CONTROL PANEL
P219 / J219	CABIN FLOOR, BL 25 RH, STA 247
P220R	MED2 ICS CONTROL PANEL

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P226 / J1	CREW'S RESCUE HOIST CONTROL PANEL
P246 / J246	CABIN CEILING, BL 5 RH , STA 247
P280 / J280	CABIN CEILING, BL 6 LH, STA 247
P288	COCKPIT FLOOD AND SECONDARY LIGHTS PANEL
P299 / J299	CABIN CEILING, BL 10 LH, STA 302
P356	ECU CONTROL PANEL
P360 / J360	AUXILIARY FUEL MANAGEMENT CONTROL PANEL
P645	ESSS STORES JETTISON PANEL CONNECTOR
P659R	MISCELLANEOUS SWITCH PANEL
P1011R	COPILOT'S CDU
P1012R	PILOT'S CDU

AB0110_2A
SA

Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 2 of 6)

LOCATION - Continued



UPPER CONSOLE

AB0110_3
SA

Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 3 of 6)

LOCATION - Continued

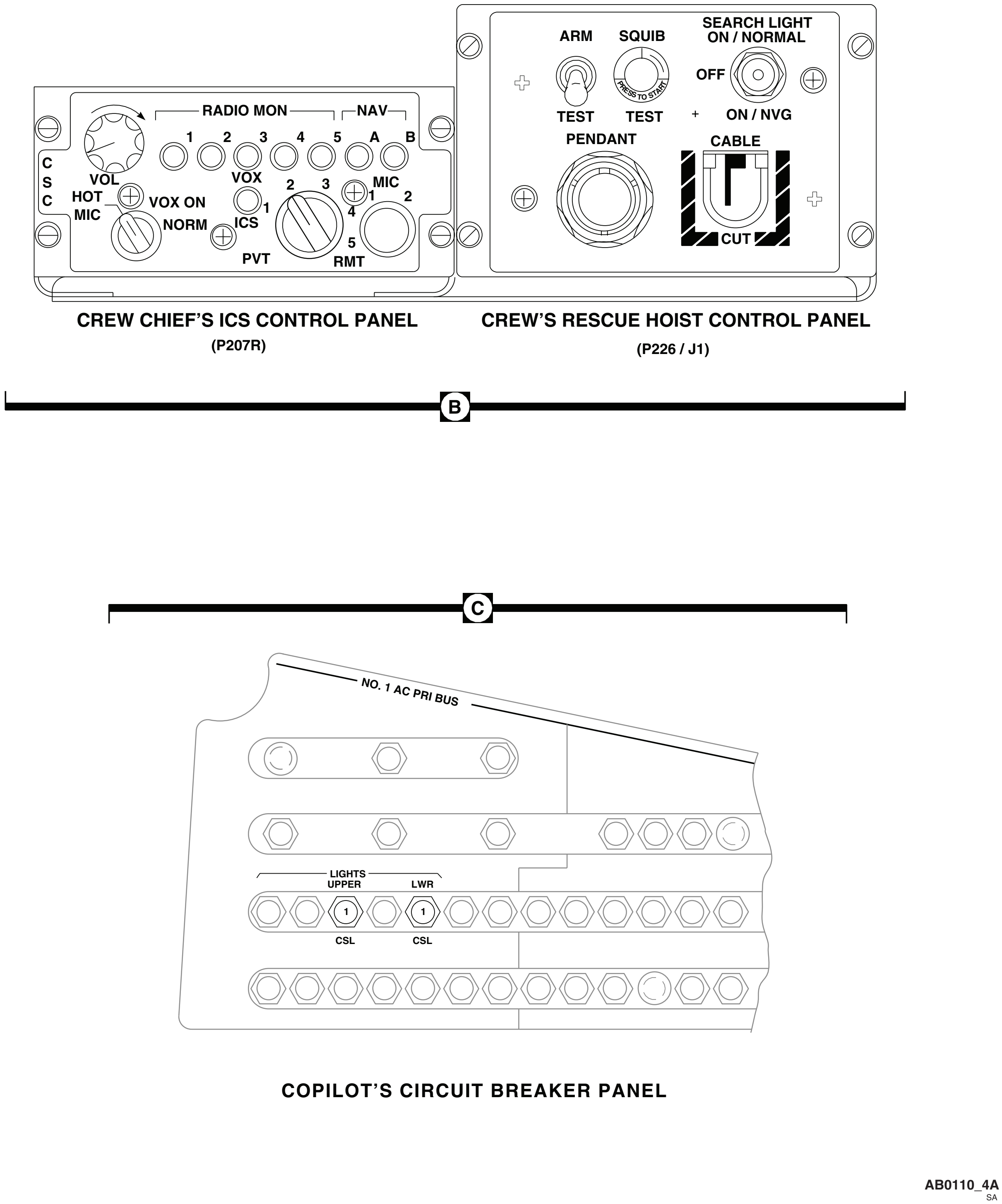


Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 4 of 6)

LOCATION - Continued

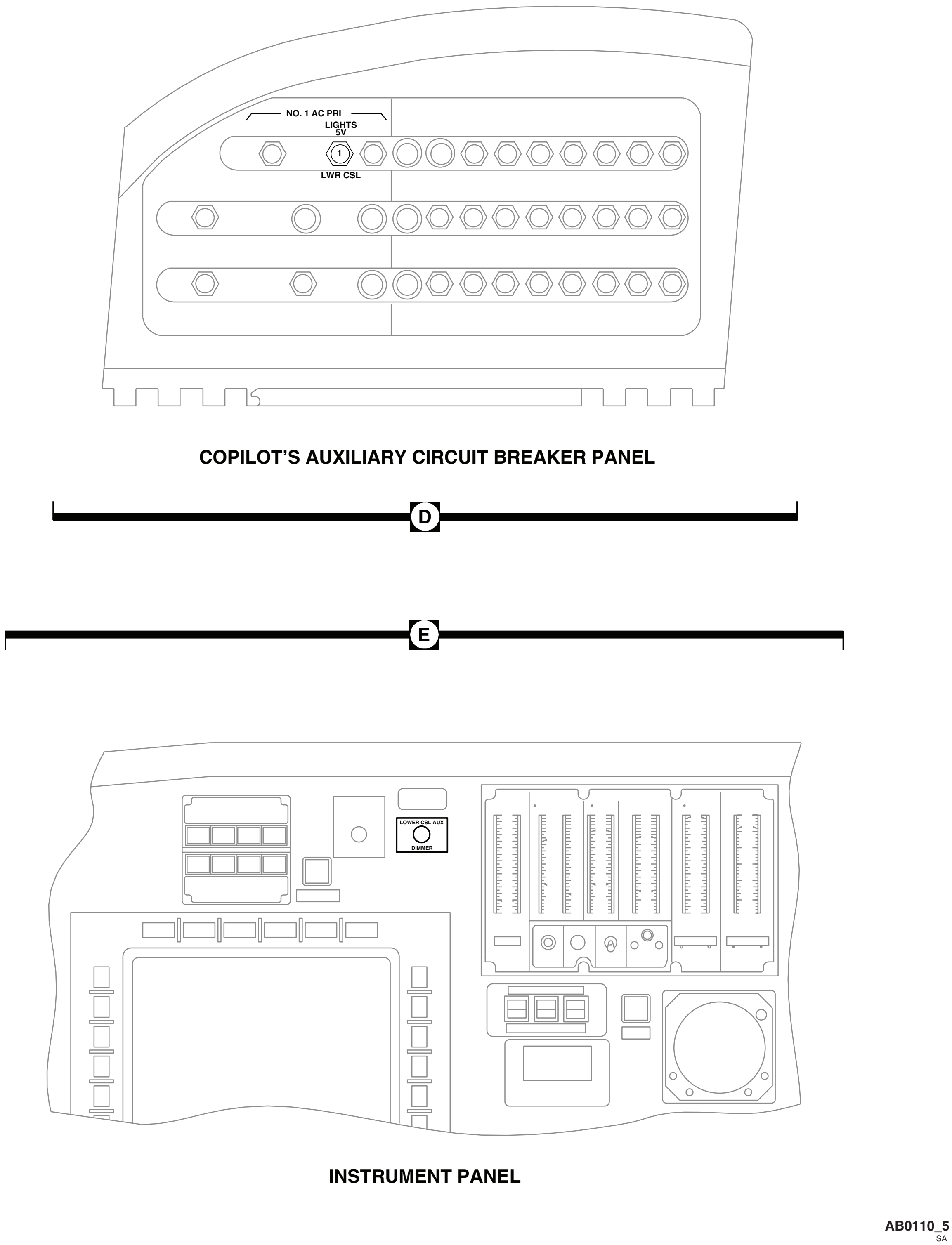


Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 5 of 6)

LOCATION - Continued

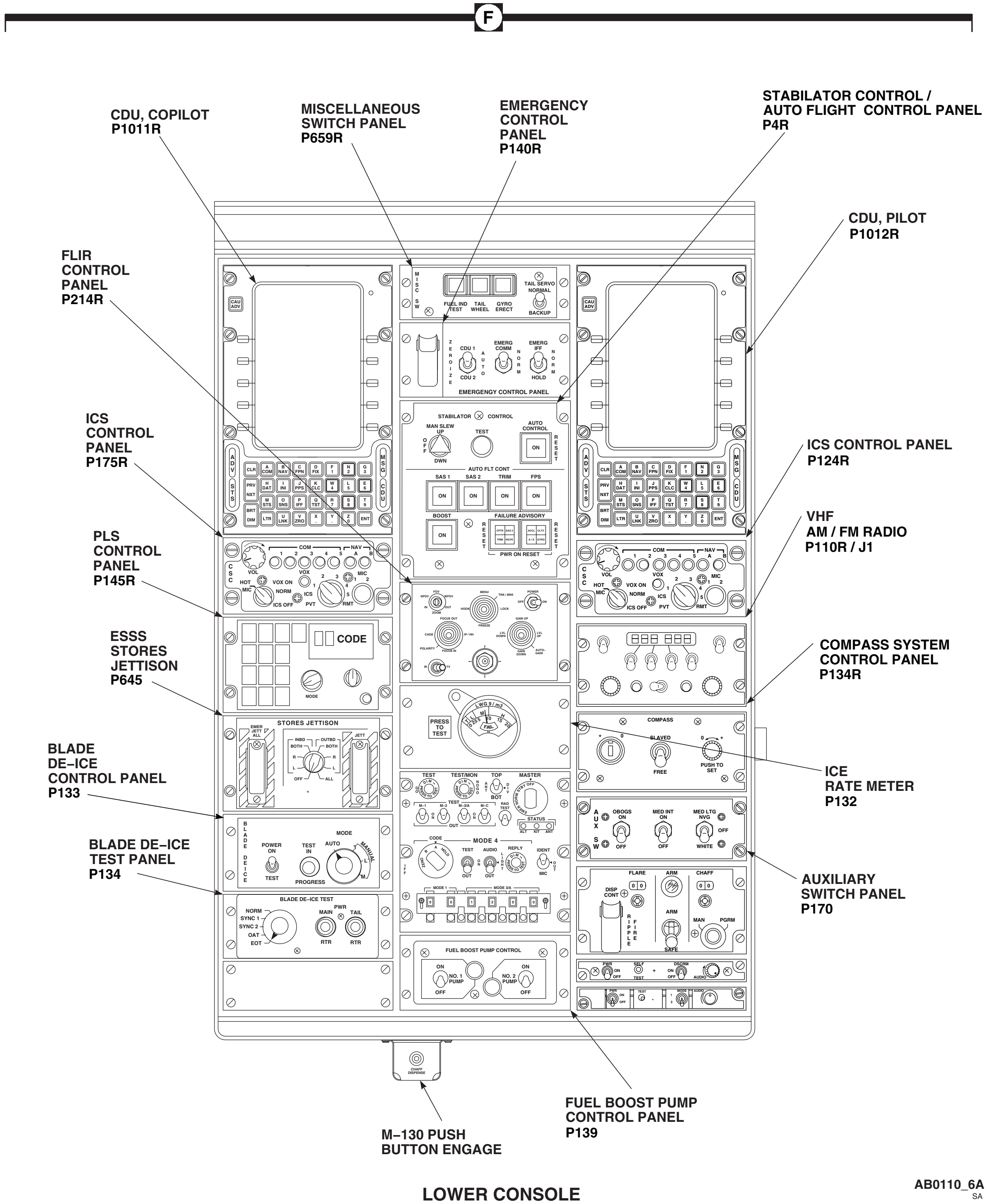


Figure 1. Upper and Lower Console Lights Location Diagram. (Sheet 6 of 6)

SCHEMATICS

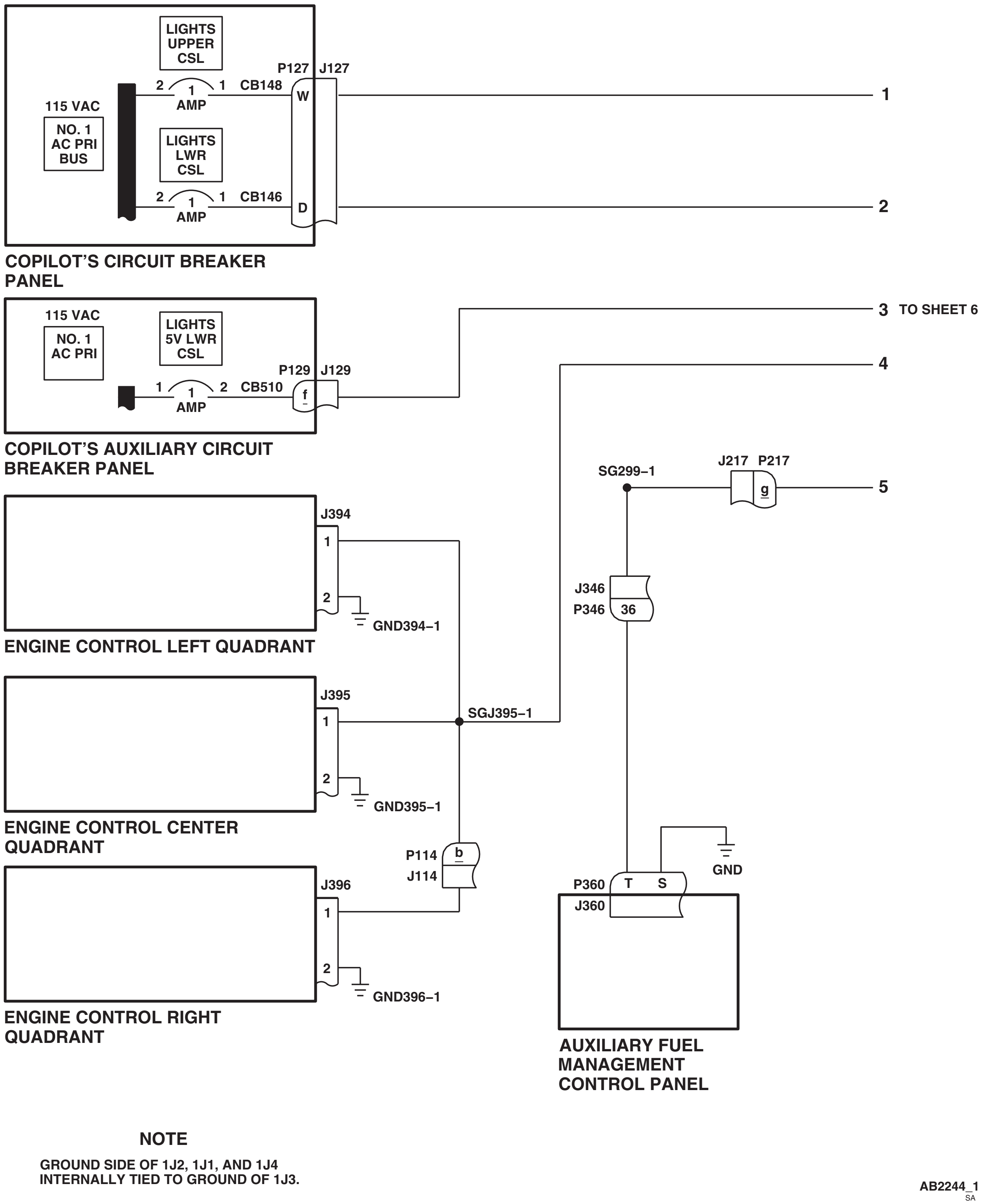


Figure 2. Upper and Lower Console Lights Schematic Diagram. (Sheet 1 of 6)

SCHEMATICS - Continued

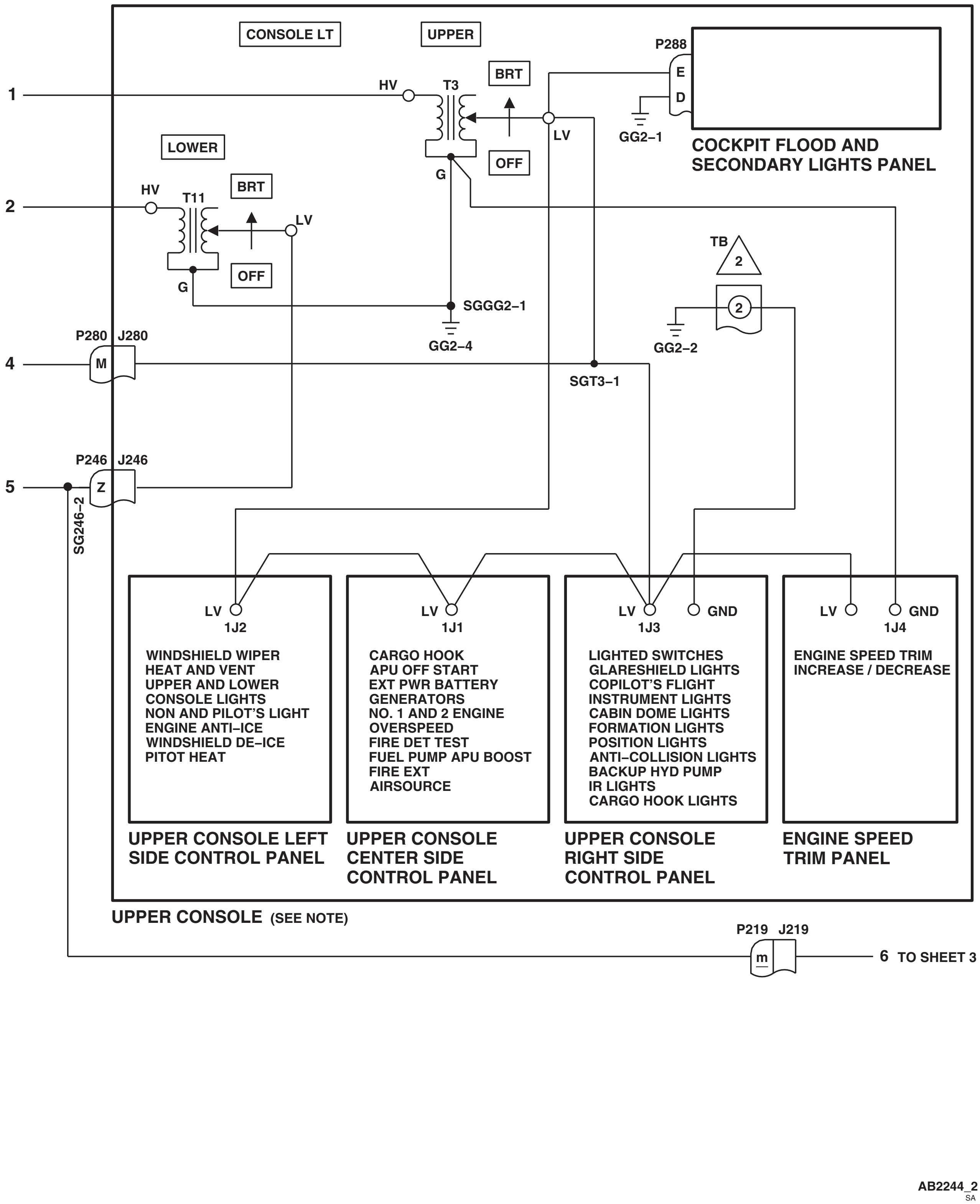


Figure 2. Upper and Lower Console Lights Schematic Diagram. (Sheet 2 of 6)

SCHEMATICS - Continued

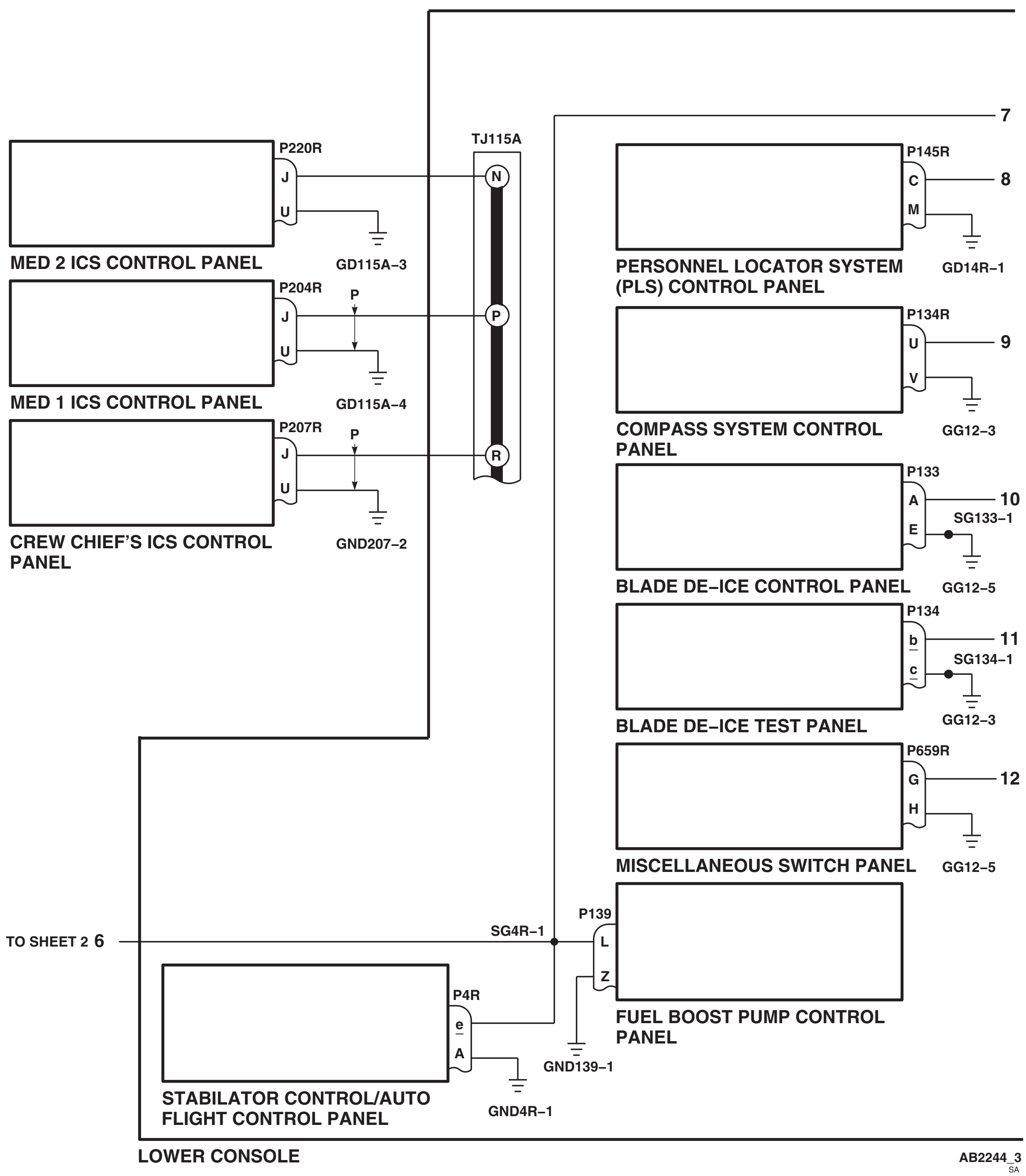


Figure 2. Upper and Lower Console Lights Schematic Diagram. (Sheet 3 of 6)

SCHEMATICS - Continued

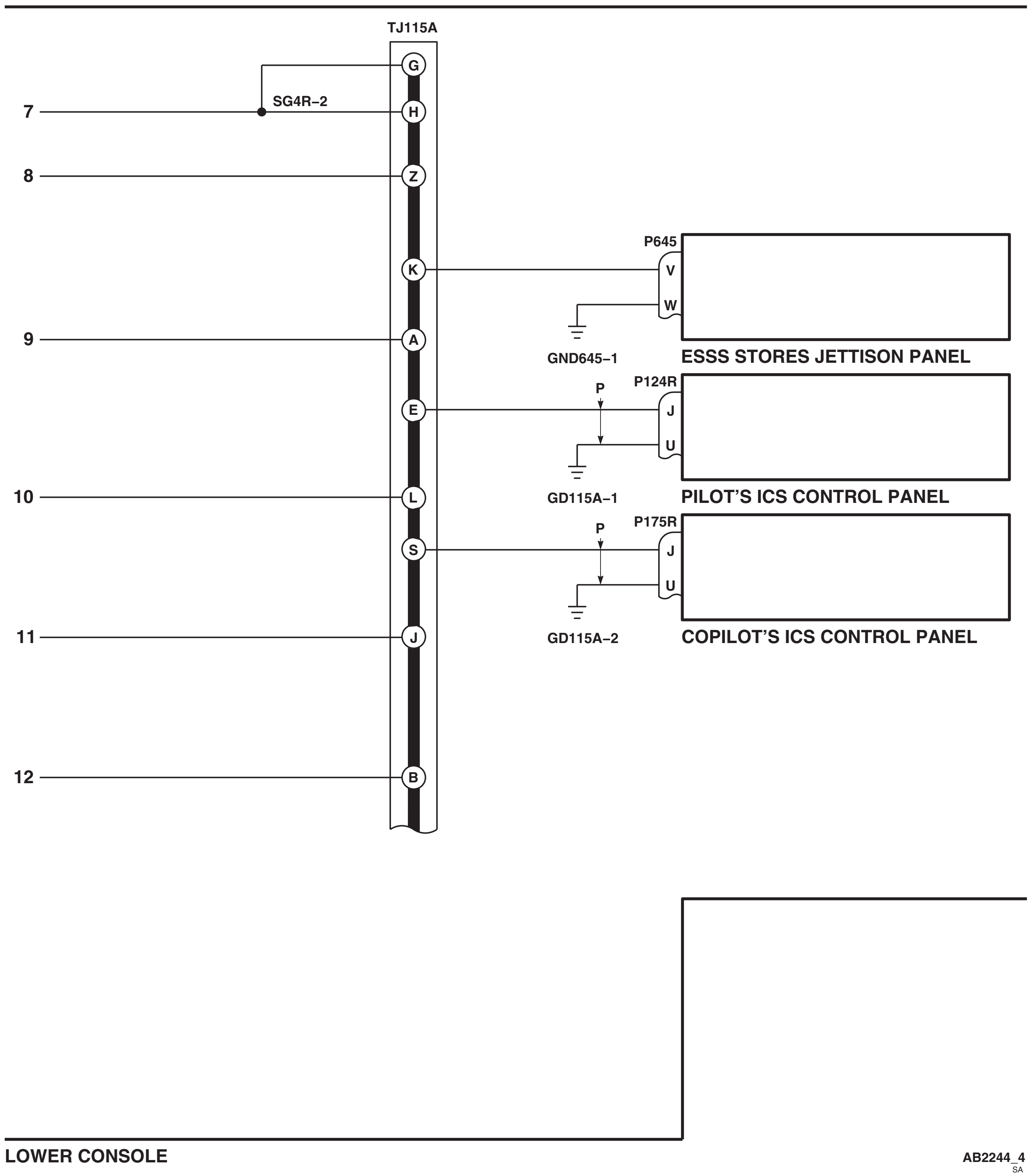


Figure 2. Upper and Lower Console Lights Schematic Diagram. (Sheet 4 of 6)

SCHEMATICS - Continued

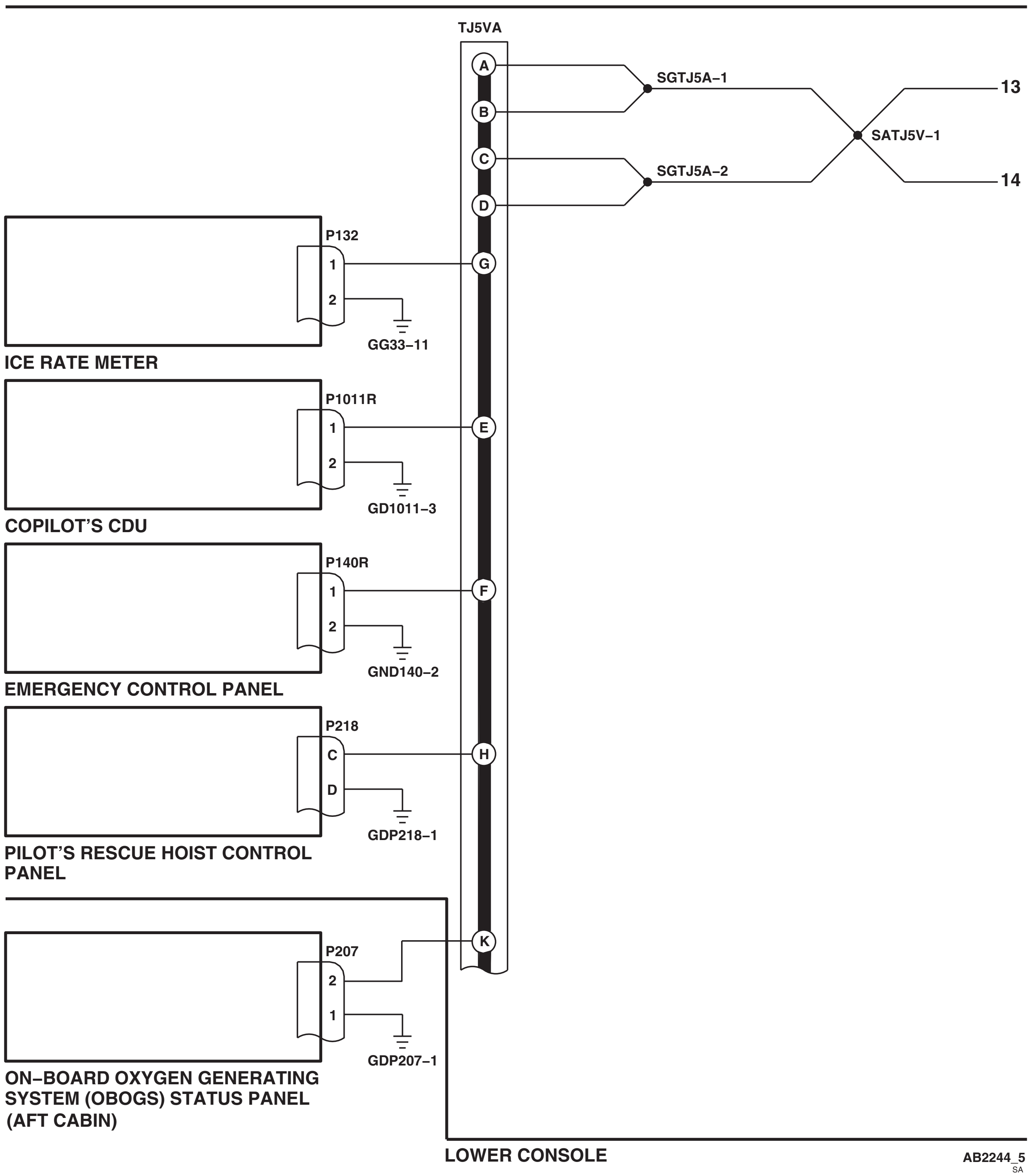


Figure 2. Upper and Lower Console Lights Schematic Diagram. (Sheet 5 of 6)

SCHEMATICS - Continued

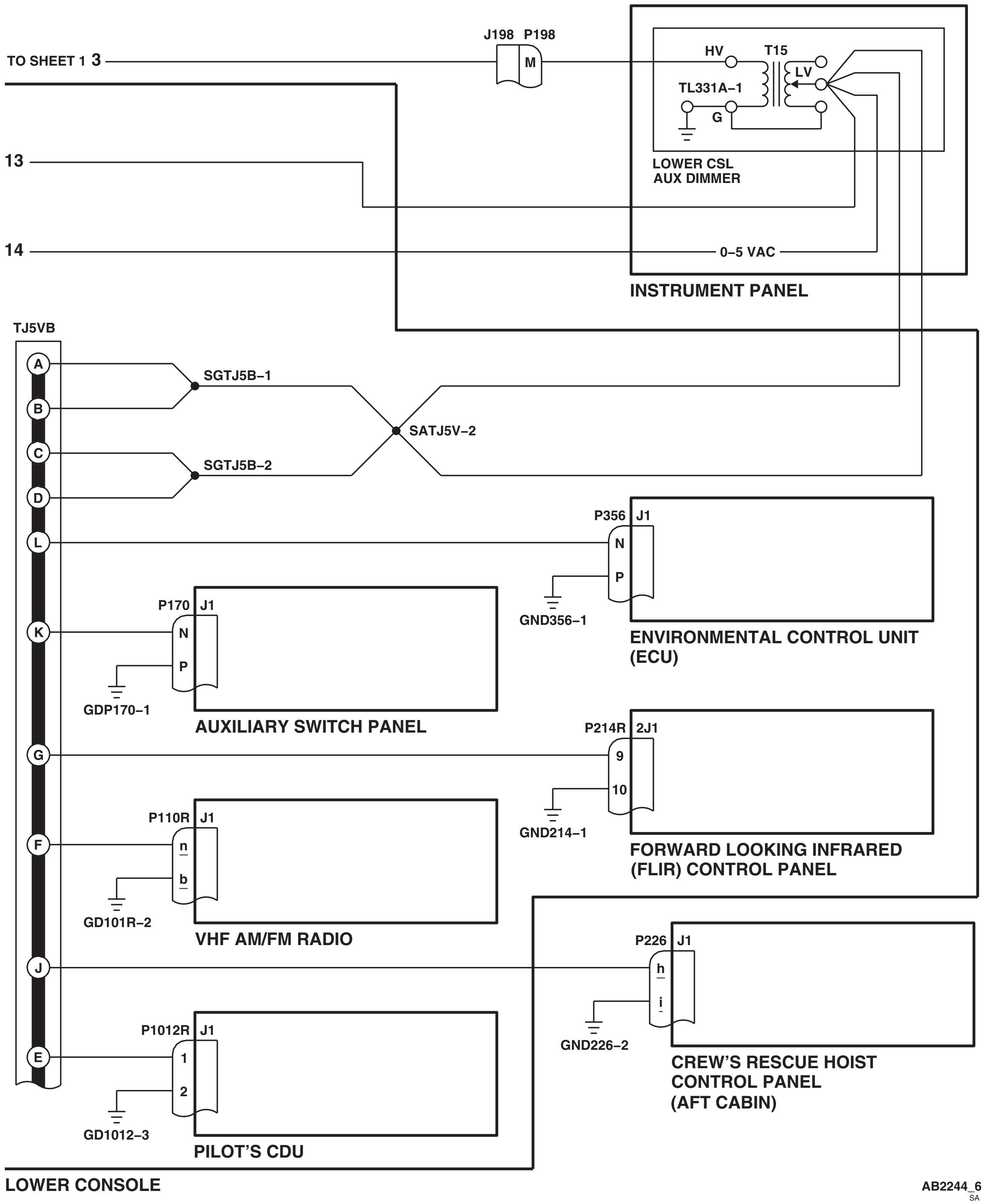


Figure 2. Upper and Lower Console Lights Schematic Diagram. (Sheet 6 of 6)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

INSTRUMENT PANEL LIGHTS HH-60A HH-60L

This WP supersedes WP 0125 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

References
TM 1-1520-237-10
WP 0089 00
WP 0104 00
WP 0824 00
WP 0875 00
WP 0881 00
WP 0902 00
WP 1066 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

INSTRUMENT PANEL LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

To prevent damage to lighting, do not measure voltage or continuity across transformers T5, T6, and T7.

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Turn on electrical power.
3. Turn CPLT FLT INST LTS control to OFF.
4. Turn INST LT PILOT FLT control to OFF.
5. Turn INST LT NON FLT control to BRT.

Indication/Condition

a. On instrument panel, stabilator placard and stabilator indicator bezel shall go on.

b. Pilot’s and copilot’s collective stick grip information panel lighting shall go on.

c. Central Display Unit (CDU) faceplate lights shall go on bright.

INSTRUMENT PANEL LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, go to STABILATOR PLACARD, STABILATOR INDICATOR BEZEL, AND PILOT'S AND COPILOT'S COLLECTIVE STICK GRIP INFORMATION PANEL LIGHTING FAILS TO COME ON, in this work package.
2. If stabilator placard does not go on, go to STABILATOR PLACARD LIGHTING FAILS TO COME ON, in this work package.
3. If stabilator indicator lighted bezel does not go on, go to STABILATOR INDICATOR LIGHTED BEZEL FAILS TO COME ON, in this work package.
4. If pilot's collective stick grip information panel lighting does not go on, go to PILOT'S COLLECTIVE STICK GRIP INFORMATION PANEL LIGHTING FAILS TO COME ON, in this work package.
5. If copilot's collective stick grip information panel lighting does not go on, go to COPILOT'S COLLECTIVE STICK GRIP INFORMATION PANEL LIGHTING FAILS TO COME ON, in this work package.
6. If CDU faceplate lights do not go on bright, troubleshoot instrument display system (WP 0089 00).

Step

6. Turn INST LT NON FLT control to OFF.

Indication/Condition

Stabilator placard, stabilator indicator bezel, pilot's and copilot's collective grip lighting, and CDU faceplate lights shall dim and go off.

Corrective Action

If Indication/Condition is not as specified, replace INST LT NON FLT dimmer control T7 (WP 0875 00).

Step

7. Turn pilot's Multifunction Display (MFD) switch ON.
8. Turn INST LT PILOT FLT control to BRT.

Indication/Condition

a. Lighted bezels on the following pilot's instruments shall go on:

- Airspeed indicator
- Altimeter encoder
- Radar altimeter
- Vertical velocity indicator (VVI)
- Vertical situation indicator (VSI)
- Horizontal situation indicator (HSI)

b. Lights on the following pilot's instruments shall go on:

- MODE SEL panel
- MFD
- Digital clock

c. Pilot's display unit (PDU) faceplate lights shall go on bright.

INSTRUMENT PANEL LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If none of the bezels or lights go on, go to NONE OF THE BEZELS OR LIGHTS GO ON, in this work package.
2. If an individual bezel or light does not go on, go to INDIVIDUAL BEZEL OR LIGHT DOES NOT GO ON, in this work package.
3. If pilot's MODE SEL panel lighting fails, go to PILOT'S MODE SEL PANEL LIGHTING FAILS, in this work package.
4. If pilot's PDU faceplate lights do not go on bright, troubleshoot instrument display system (WP 0089 00).

Step

9. Turn INST LT PILOT FLT control to OFF.

Indication/Condition

All pilot flight instrument panel lights shall dim and go off.

Corrective Action

If Indication/Condition is not as specified, replace INST LT PILOT FLT dimmer control T6 (WP 0875 00).

Step

10. Turn pilot's MFD switch OFF.
11. Turn copilot's MFD switch ON.
12. Turn CPLT FLT INST LTS control to BRT.

Indication/Condition

a. Lighted bezels on the following copilot's instruments shall go on:

- Airspeed indicator
- Barometric altimeter
- Radar altimeter
- VVI
- VSI
- HSI

b. Lights on the following copilot's instruments shall go on:

- MODE SEL panel
- MFD
- Digital clock

c. Copilot's display unit (PDU) faceplate lights shall go on bright.

Corrective Action

1. If none of the bezels or lights go on, go to NONE OF THE BEZELS OR LIGHTS GO ON, in this work package.

INSTRUMENT PANEL LIGHTS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. If an individual bezel or light does not come on, go to INDIVIDUAL BEZEL OR LIGHT DOES NOT GO ON, in this work package.
3. If copilot's MODE SEL panel lighting fails, go to COPILOT'S MODE SEL PANEL LIGHTING FAILS, in this work package.
4. If copilot's PDU faceplate lights do not go on bright, troubleshoot instrument display system (WP 0089 00).

Step

13. Turn CPLT FLT INST LTS control to OFF.

Indication/Condition

All copilot flight instrument panel lights shall dim and go off.

Corrective Action

If Indication/Condition is not as specified, replace CPLT FLT INST LTS dimmer control T5 (WP 0875 00).

Step

14. Turn copilot's MFD switch OFF.
15. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

STABILATOR PLACARD, STABILATOR INDICATOR BEZEL, AND PILOT'S AND COPILOT'S COLLECTIVE STICK GRIP INFORMATION PANEL LIGHTING FAILS TO COME ON

SYMPTOM

Stabilator placard, stabilator indicator bezel, and pilot's and copilot's collective stick grip information panel lighting fails to come on.

MALFUNCTION

Stabilator placard, stabilator indicator bezel, and pilot's and copilot's collective stick grip information panel lighting fails to come on.

CORRECTIVE ACTION

1. **Check for 115 vac between T7-HV and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, go to **3.**
2. **Check continuity between T7-G and ground.**
   a. If continuity is present replace dimmer control T7 (WP 0875 00). Go to **4.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**
3. **Check for 115 vac between terminal 1 of CB249 and ground.**
   a. If voltage is as specified, repair/replace wiring between LIGHTS NON FLT circuit breaker terminal 1 and T7-HV (WP 1747 00). Go to **4.**

STABILATOR PLACARD, STABILATOR INDICATOR BEZEL, AND PILOT'S AND COPILOT'S COLLECTIVE STICK GRIP INFORMATION PANEL LIGHTING FAILS TO COME ON - Continued

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **4.**

4. Procedure completed.

STABILATOR PLACARD LIGHTING FAILS TO COME ON

SYMPTOM

Stabilator placard lighting fails to come on.

MALFUNCTION

Stabilator placard lighting fails to come on.

CORRECTIVE ACTION

1. Check for 115 vac between PWR terminal and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, repair/replace wiring between PWR terminal and T7-LV (WP 1747 00). Go to **3.**

2. Check continuity between GND terminal and ground.

a. If continuity is present, replace placard (WP 0902 00). Go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3.**

3. Procedure completed.

STABILATOR INDICATOR LIGHTED BEZEL FAILS TO COME ON

SYMPTOM

Stabilator indicator lighted bezel fails to come on.

MALFUNCTION

Stabilator indicator lighted bezel fails to come on.

CORRECTIVE ACTION

1. Check for 5 vac between P196-A and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **3.**

2. Check continuity between P196-A and ground.

a. If continuity is present, replace bezel (WP 0902 00). Go to **5.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5.**

3. Check for 115 vac between P388-C and ground.

a. If voltage is as specified, go to **4.**

b. If voltage is not as specified, repair/replace wiring between P388-C and T7-LV (WP 1747 00). Go to **5.**

4. Check continuity between:

- **P388-A and P196-A**
- **P388-B and ground**

STABILATOR INDICATOR LIGHTED BEZEL FAILS TO COME ON - Continued

- **P388-D and ground**
- **P388-E and ground**

a. If continuity is present, replace nonflight instrument lights transformer (WP 0881 00). Go to **5.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5.**

5. Procedure completed.

PILOT'S COLLECTIVE STICK GRIP INFORMATION PANEL LIGHTING FAILS TO COME ON

SYMPTOM

Pilot's collective stick grip information panel lighting fails to come on.

MALFUNCTION

Pilot's collective stick grip information panel lighting fails to come on.

CORRECTIVE ACTION

1. Check for 115 vac between J104-T and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, repair/replace wiring between J104-T and T7-LV (WP 1747 00). Go to **3.**

2. Check continuity between J104-V and ground.

a. If continuity is present, replace information panel (WP 1066 00). Go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3.**

3. Procedure completed.

COPILOT'S COLLECTIVE STICK GRIP INFORMATION PANEL LIGHTING FAILS TO COME ON

SYMPTOM

Copilot's collective stick grip information panel lighting fails to come on.

MALFUNCTION

Copilot's collective stick grip information panel lighting fails to come on.

CORRECTIVE ACTION

1. Check for 115 vac between J103-T and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, repair/replace wiring between J103-T and T7-LV (WP 1747 00). Go to **3.**

2. Check continuity between J103-V and ground.

a. If continuity is present, replace information panel (WP 1066 00). Go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3.**

3. Procedure completed.

NONE OF THE BEZELS OR LIGHTS GO ON

SYMPTOM

None of the bezels or lights go on.

MALFUNCTION

None of the bezels or lights go on.

CORRECTIVE ACTION

1. Check for 115 vac between T6-HV and ground.

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **3.**

2. Check continuity between T6-G and ground.

a. If continuity is present, replace dimmer control T6 (WP 0875 00). Go to **4.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**

3. Check for 115 vac between terminal 1 of CB248 and ground.

a. If voltage is as specified, repair/replace wiring between LIGHTS PLT FLT circuit breaker terminal 1 and T6-HV (WP 1747 00). Go to **4.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **4.**

4. Procedure completed.

Table 1. Individual Bezel or Light Does Not Go On.

CONTROL PANEL OR COMPONENT	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
PILOT'S DIGITAL CLOCK	J112-m and P186-3, J112-L and P186-4	Replace bezel	Repair/replace wiring (WP 1747 00).
PILOT'S AIRSPEED INDICATOR	J112-m and P187-A, J112-L and P187-B	Replace bezel	Repair/replace wiring (WP 1747 00).
PILOT'S MFD	J112-m and P1018R-21, J112-L and P1018R-22	Replace panel	Repair/replace wiring (WP 1747 00).
PILOT'S ALTIMETER/ ENCODER	J112-r and P183-A, J112-L and P183-B	Replace bezel	Repair/replace wiring (WP 1747 00).
PILOT'S RADAR ALTIMETER	J112-r and P182-A, J112-L and P182-B	Replace bezel	Repair/replace wiring (WP 1747 00).
PILOT'S VVI	J112-r and P181-A, J112-L and P181-B	Replace bezel	Repair/replace wiring (WP 1747 00).

NONE OF THE BEZELS OR LIGHTS GO ON - Continued

Table 1. Individual Bezel or Light Does Not Go On. - Continued

CONTROL PANEL OR COMPONENT	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
PILOT'S HSI	J112-r̲ and JJ1-2, J112-L̄ and JJ2-2	Replace bezel	Repair/replace wiring (WP 1747 00).
PILOT'S VSI	J112-r̲ and P184-A, J112-L̄ and P184-B	Replace bezel	Repair/replace wiring (WP 1747 00).

PILOT'S MODE SEL PANEL LIGHTING FAILS

SYMPTOM

Pilot's MODE SEL panel lighting fails.

MALFUNCTION

Pilot's MODE SEL panel lighting fails.

CORRECTIVE ACTION

1. **Check for 115 vac between P317R-36 and ground.**
   a. If voltage is as specified, replace MODE SEL Panel (TM 11-1520-237-23). Go to **5.**
   b. If voltage is not as specified, go to **2.**
2. **Check continuity between P317R-36 and P386-C.**
   a. If continuity is present, go to **3**.
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5.**
3. **Check for 5 vac between P386-A and ground.**
   a. If voltage is as specified, go to **4.**
   b. If voltage is not as specified, repair/replace wiring between P386-A and T6-LV (WP 1747 00). Go to **5.**
4. **Check continuity between:**
   - **P386-B and ground**
   - **P386-D and ground**
   - **P386-E and ground**

   a. If continuity is present, replace pilot's flight instrument lights transformer (WP 0881 00). Go to **5.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5.**
5. **Procedure completed.**

NONE OF THE BEZELS OR LIGHTS GO ON

SYMPTOM

None of the bezels or lights go on.

MALFUNCTION

None of the bezels or lights go on.

CORRECTIVE ACTION

1. **Check for 115 vac between T5-HV and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, go to **3.**
2. **Check continuity between T5-G and ground.**
   a. If continuity is present, replace dimmer control T5 (WP 0875 00). Go to **4.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**
3. **Check for 115 vac between terminal 1 of circuit breaker CB150 and ground.**
   a. If voltage is as specified, repair/replace wiring between LIGHTS CPLT FLT circuit breaker terminal 1 and T5-HV (WP 1747 00). Go to **4.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **4.**
4. **Procedure completed.**

Table 2. Individual Bezel or Light Does Not Go On.

CONTROL PANEL OR COMPONENT	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
COPILOT'S DIGITAL CLOCK	J113-r and P195-3, J113-H̄ and P195-4	Replace bezel	Repair/replace wiring (WP 1747 00).
COPILOT'S AIR-SPEED INDICATOR	J113-m and P189-A, J113-H̄ and P189-B	Replace bezel	Repair/replace wiring (WP 1747 00).
COPILOT'S MFD	J113-r and P1017R-21, J113-H̄ and P1017R-22	Replace panel	Repair/replace wiring (WP 1747 00).
COPILOT'S BAROMETRIC ALTIMETER	J113-r and P193-A, J113-H̄ and P193-B	Replace bezel	Repair/replace wiring (WP 1747 00).
COPILOT'S RADAR ALTIMETER	J112-r and P192-A, J113-H̄ and P192-B	Replace bezel	Repair/replace wiring (WP 1747 00).
COPILOT'S VVI	J113-r and P194-A, J113-H̄ and P194-B	Replace bezel	Repair/replace wiring (WP 1747 00).

NONE OF THE BEZELS OR LIGHTS GO ON - Continued

Table 2. Individual Bezel or Light Does Not Go On. - Continued

CONTROL PANEL OR COMPONENT	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
COPILOT'S HSI	J113-$\underline{r}$ and JJ3-2, J113-$\overline{H}$ and JJ4-2	Replace bezel	Repair/replace wiring (WP 1747 00).
COPILOT'S VSI	J113-$\underline{m}$ and P190-A, J113-$\overline{H}$ and P190-B	Replace bezel	Repair/replace wiring (WP 1747 00).

COPILOT'S MODE SEL PANEL LIGHTING FAILS

SYMPTOM

Copilot's MODE SEL panel lighting fails.

MALFUNCTION

Copilot's MODE SEL panel lighting fails.

CORRECTIVE ACTION

1. Check for 115 vac between P300R-36 and ground.

a. If voltage is as specified, replace MODE SEL Panel (TM 11-1520-237-23). Go to **5.**

b. If voltage is not as specified, go to **2.**

2. Check continuity between P300R-36 and P387-C.

a. If continuity is present, go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5.**

3. Check for 5 vac between P387-A and ground.

a. If voltage is as specified, go to **4.**

b. If voltage is not as specified, repair/replace wiring between P387-A and T5-LV (WP 1747 00). Go to **5.**

4. Check continuity between:

- **P387-B and ground**
- **P387-D and ground**
- **P387-E and ground**

a. If continuity is present, replace copilot's flight instrument lights transformer (WP 0881 00). Go to **5.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5.**

5. Procedure completed.

LOCATION

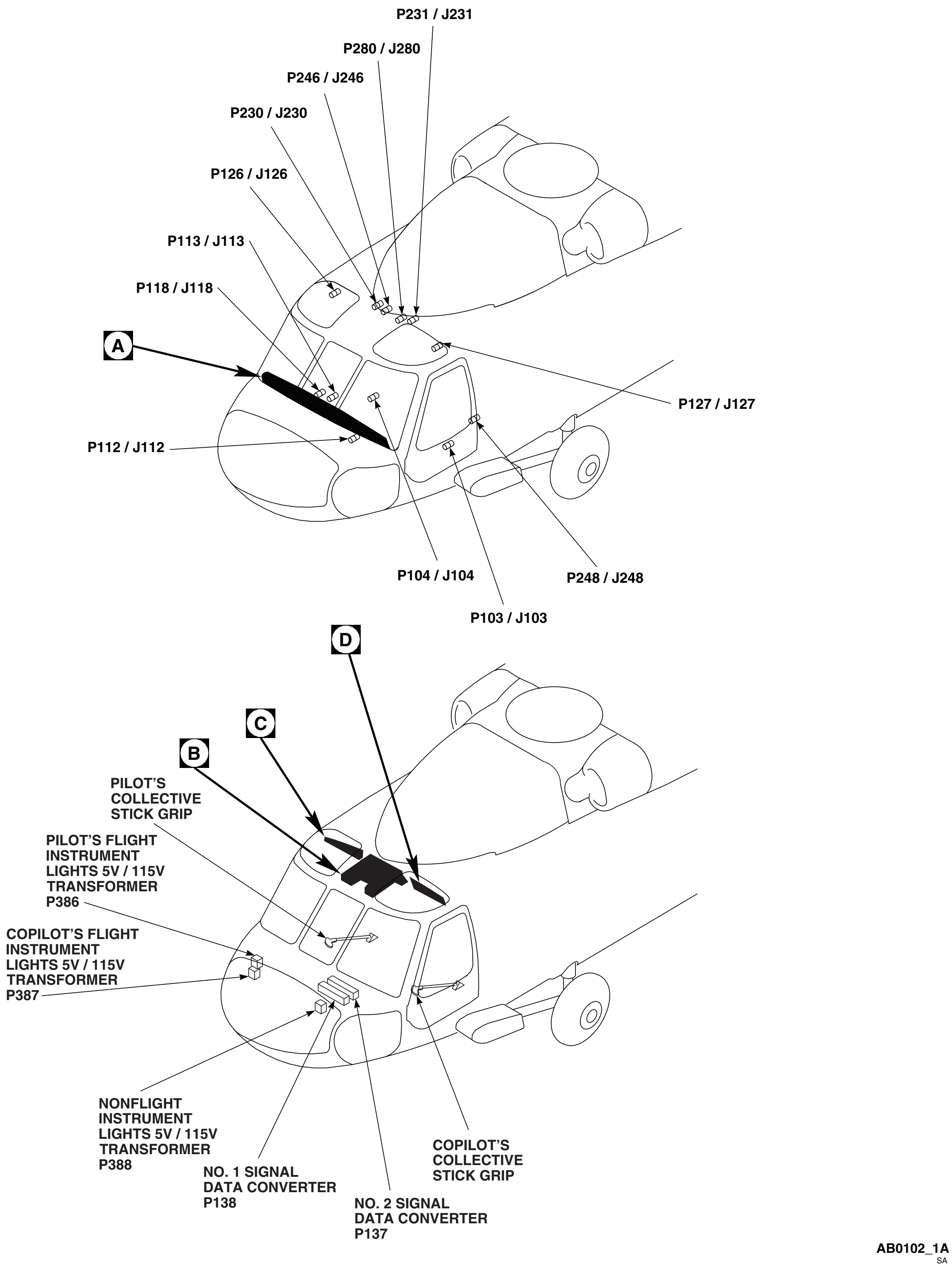


Figure 1. Instrument Panel Lights Location Diagram. (Sheet 1 of 5)

LOCATION - Continued

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
JJ1 / JJ2	PILOT'S HORIZONTAL SITUATION INDICATOR LIGHTED BEZEL ASSEMBLY
JJ3 / JJ4	COPILOT'S HORIZONTAL SITUATION INDICATOR LIGHTED BEZEL ASSEMBLY
P103 / J103	COCKPIT TUB, BL 32 LH, STA 247
P104 / J104	COCKPIT TUB, BL 15 RH, STA 247
P112 / J112	COCKPIT, BL 6 RH, STA 197
P113 / J113	COCKPIT, BL 6 LH, STA 197
P118 / J118	CAUTION / ADVISORY PANEL
P126 / J126	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 28 RH, STA 243
P127 / J127	BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 28 LH, STA 243
P132	ICING RATE METER
P133	BLADE DE–ICE CONTROL PANEL
P134	BLADE DE–ICE TEST PANEL
P137	NO. 2 SIGNAL DATA CONVERTER
P138	NO. 1 SIGNAL DATA CONVERTER
P149	NO. 1 SIGNAL DATA CONVERTER
P150	NO. 2 SIGNAL DATA CONVERTER
P157	COPILOT'S CDU
P158	PILOT'S CDU
P181	PILOT'S VERTICAL VELOCITY INDICATOR (VVI) LIGHTED BEZEL ASSEMBLY
P182	PILOT'S RADAR ALTIMETER INDICATOR LIGHTED BEZEL ASSEMBLY
P183	PILOT'S ALTIMETER ENCODER INDICATOR LIGHTED BEZEL ASSEMBLY
P184	PILOT'S VERTICAL SITUATION INDICATOR LIGHTED BEZEL ASSEMBLY

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P186	PILOT'S DIGITAL CLOCK
P187	PILOT'S AIRSPEED INDICATOR LIGHTED BEZEL ASSEMBLY
P189	COPILOT'S AIRSPEED INDICATOR LIGHTED BEZEL ASSEMBLY
P190	COPILOT'S VERTICAL SITUATION INDICATOR LIGHTED BEZEL ASSEMBLY
P192	COPILOT'S RADAR ALTIMETER LIGHTED BEZEL ASSEMBLY
P193	COPILOT'S BAROMETRIC ALTIMETER LIGHTED BEZEL ASSEMBLY
P194	COPILOT'S VERTICAL VELOCITY INDICATOR (VVI) LIGHTED BEZEL ASSEMBLY
P195	COPILOT'S DIGITAL CLOCK
P196	STABILATOR INDICATOR LIGHTED BEZEL ASSEMBLY
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P231 / J231	CABIN CEILING, BL 9 LH, STA 247
P246 / J246	CABIN CEILING, BL 5 RH, STA 247
P248 / J248	CABIN, BL 40 LH, STA 248
P280 / J280	CABIN CEILING, BL 6 LH, STA 247
P300R	COPILOT'S HSI / VSI MODE SELECT PANEL
P317R	PILOT'S HSI / VSI MODE SELECT PANEL
P386	PILOT'S FLIGHT INSTRUMENT LIGHTS 5V / 115V TRANSFORMER
P387	COPILOT'S FLIGHT INSTRUMENT LIGHTS 5V / 115V TRANSFORMER
P388	NONFLIGHT INSTRUMENT LIGHTS 5V / 115V TRANSFORMER
P1017R	COPILOT'S MFD
P1018R	PILOT'S MFD

AB0102_2A
SA

Figure 1. Instrument Panel Lights Location Diagram. (Sheet 2 of 5)

LOCATION - Continued

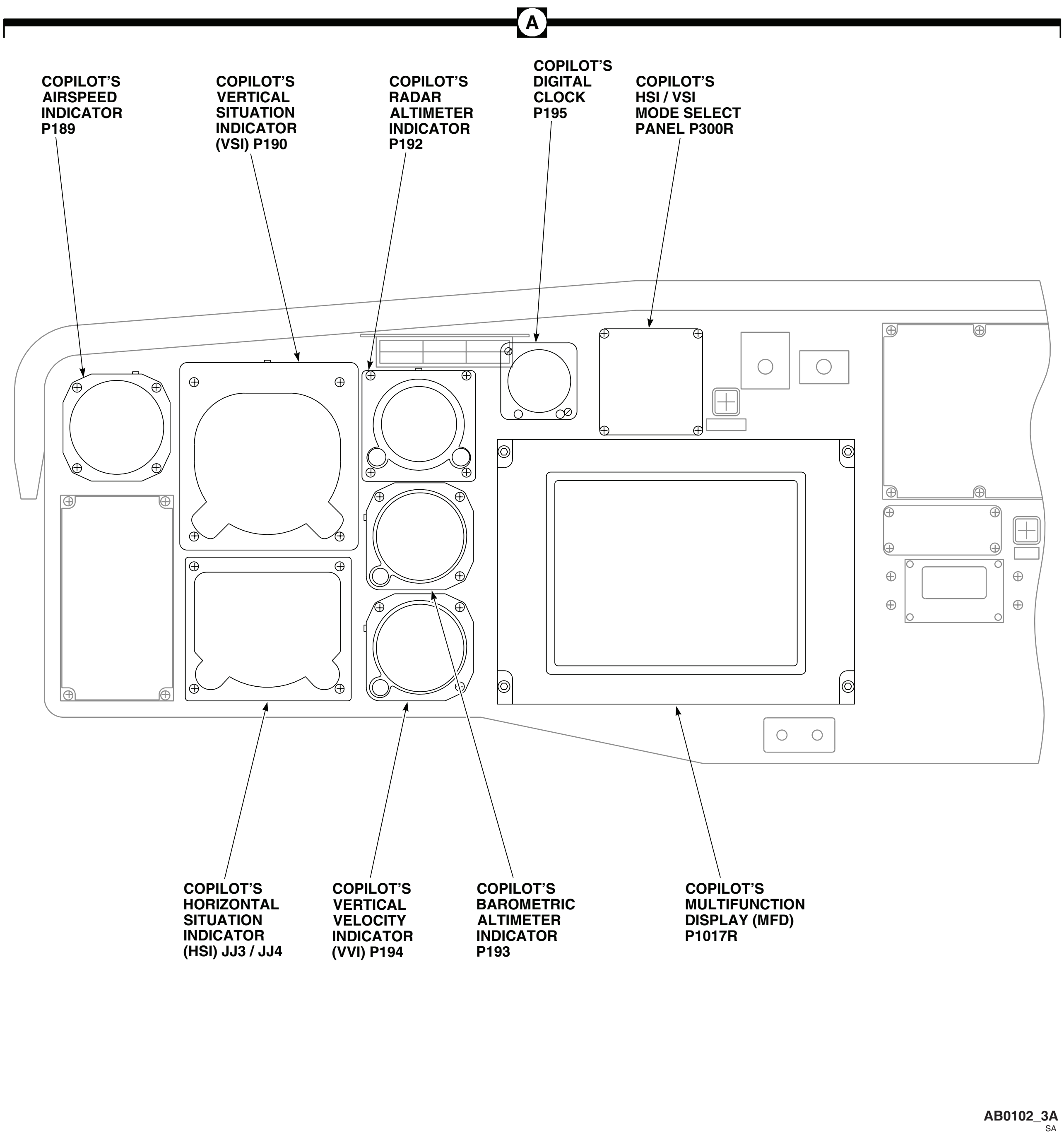


Figure 1. Instrument Panel Lights Location Diagram. (Sheet 3 of 5)

LOCATION - Continued

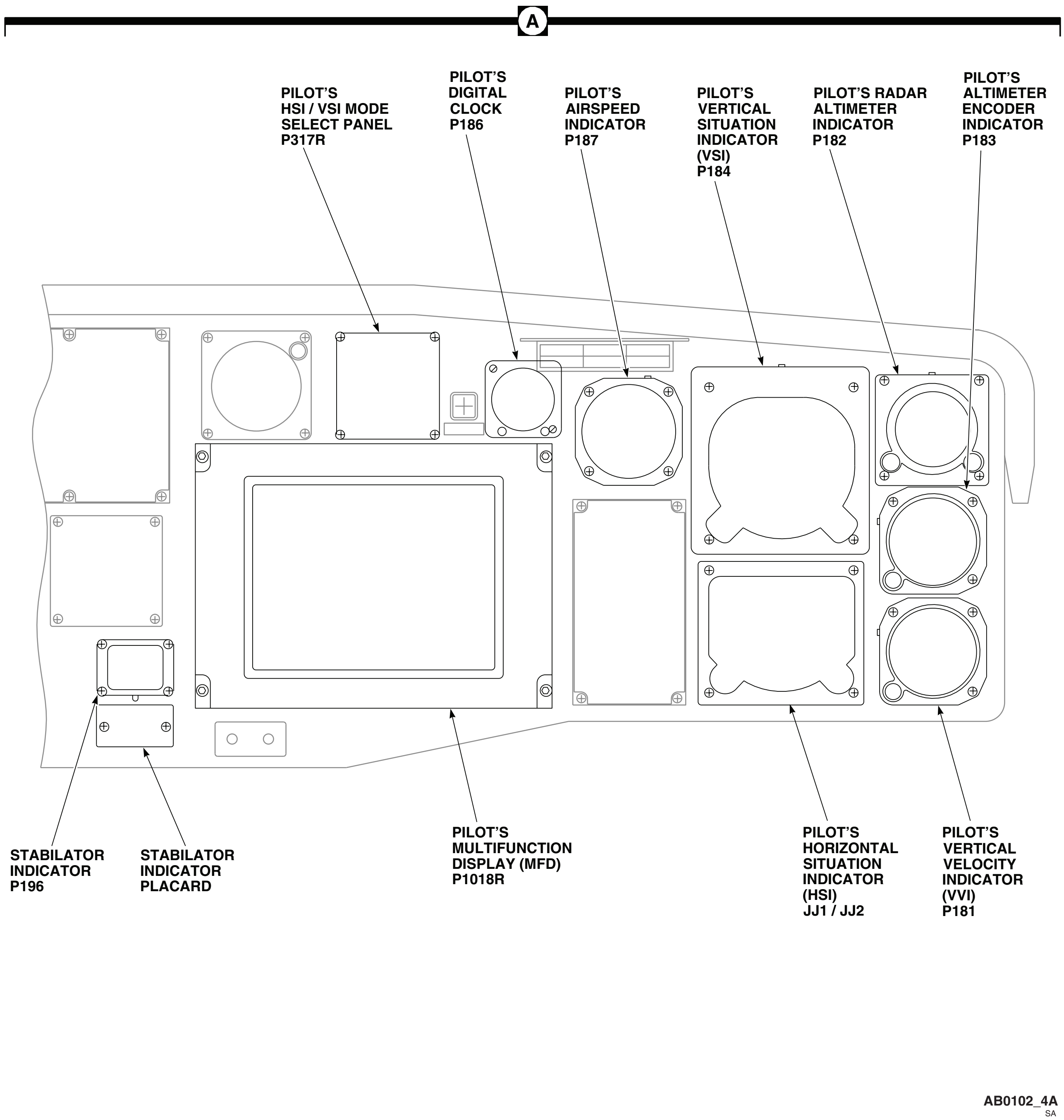


Figure 1. Instrument Panel Lights Location Diagram. (Sheet 4 of 5)

LOCATION - Continued

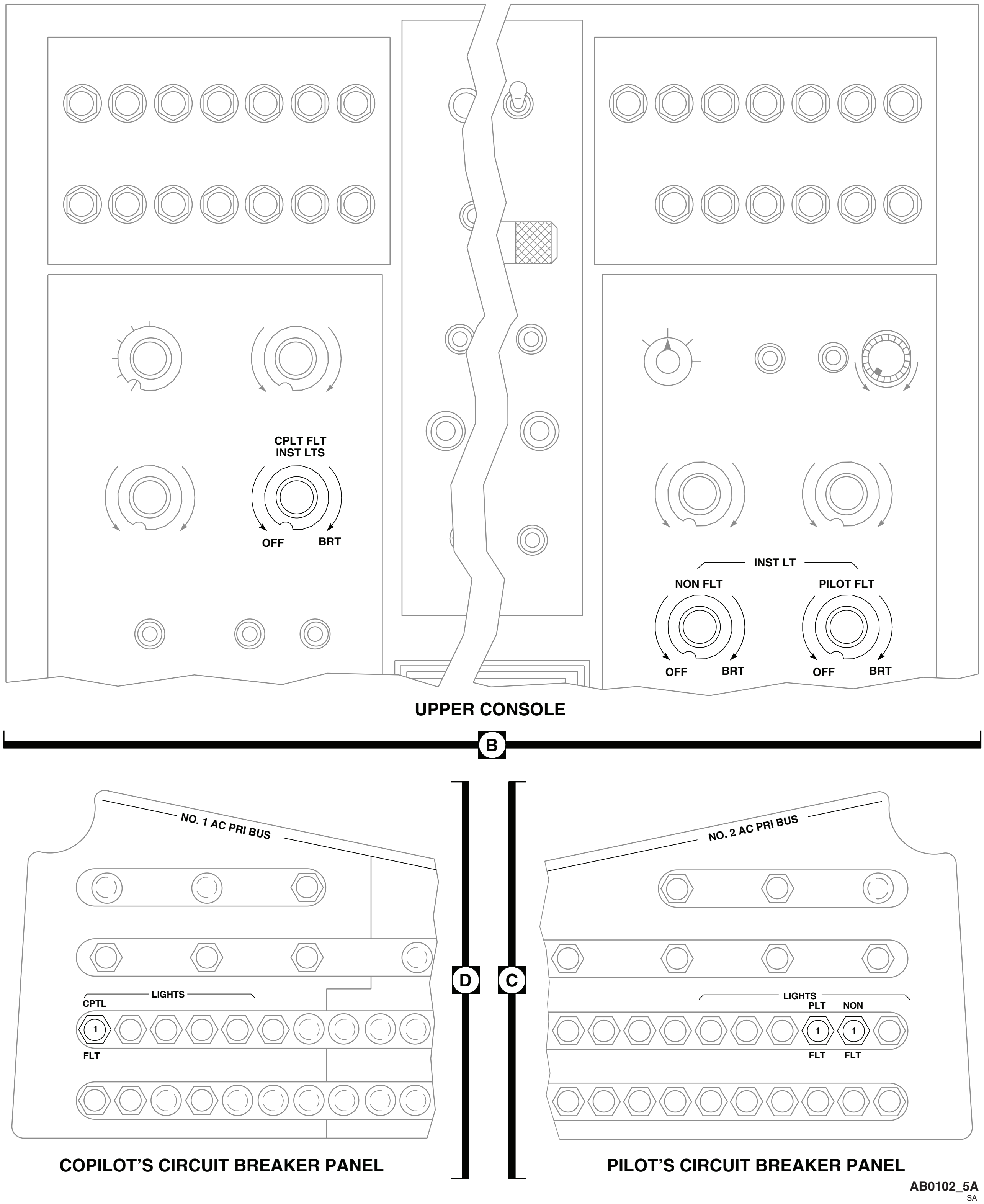


Figure 1. Instrument Panel Lights Location Diagram. (Sheet 5 of 5)

SCHEMATICS

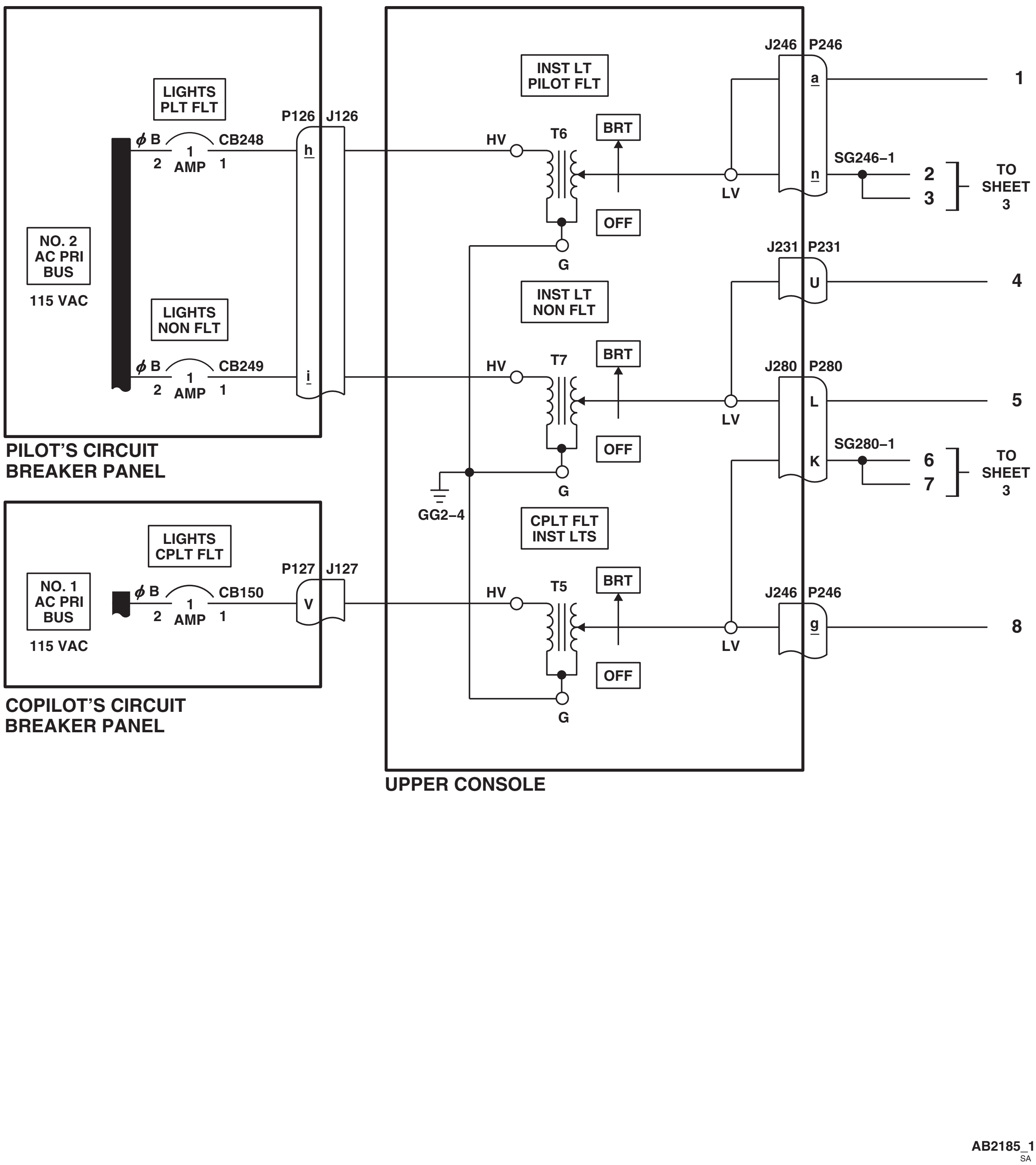


Figure 2. Instrument Panel Lights Schematic Diagram. (Sheet 1 of 3)

SCHEMATICS - Continued

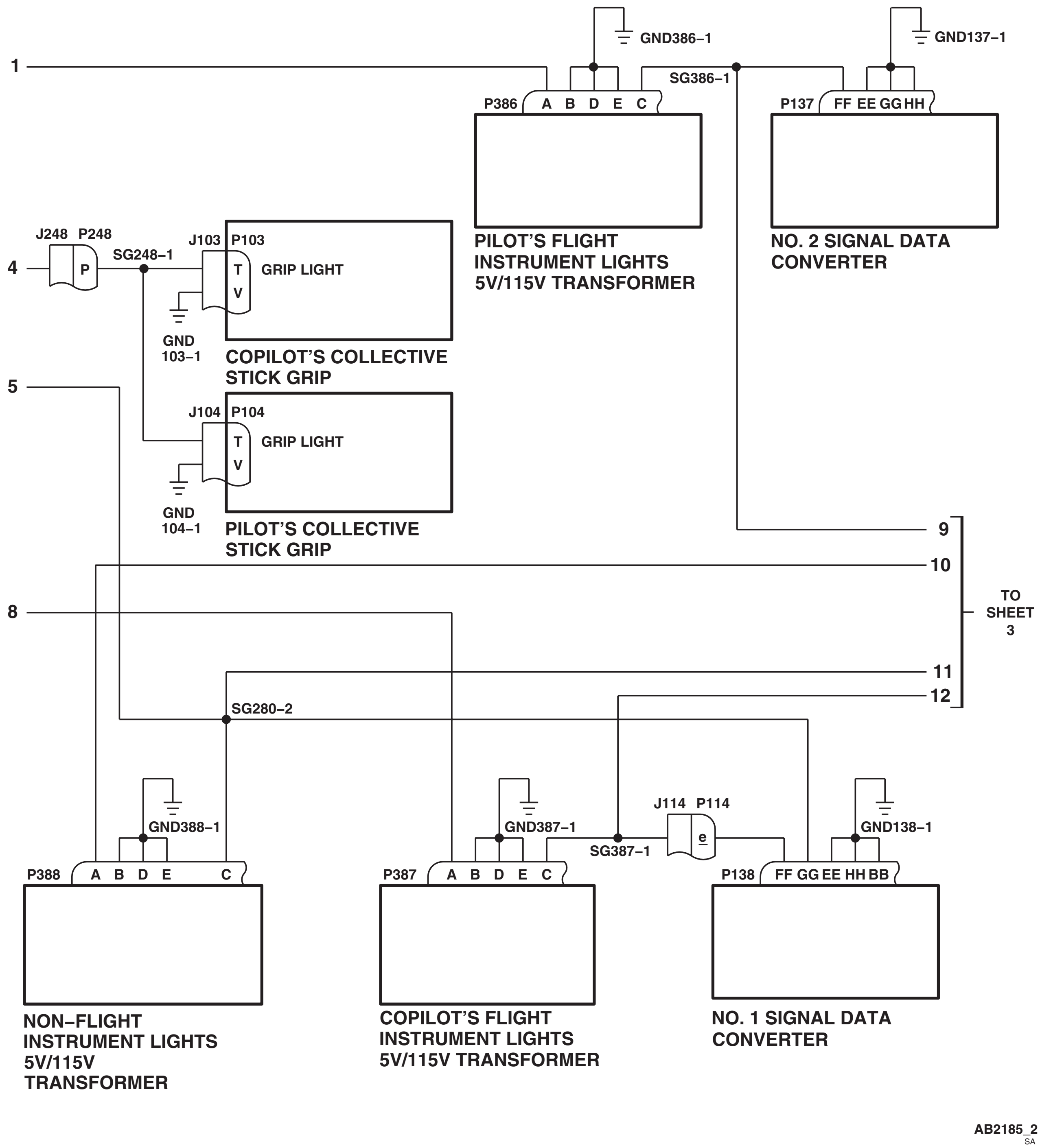


Figure 2. Instrument Panel Lights Schematic Diagram. (Sheet 2 of 3)

SCHEMATICS - Continued

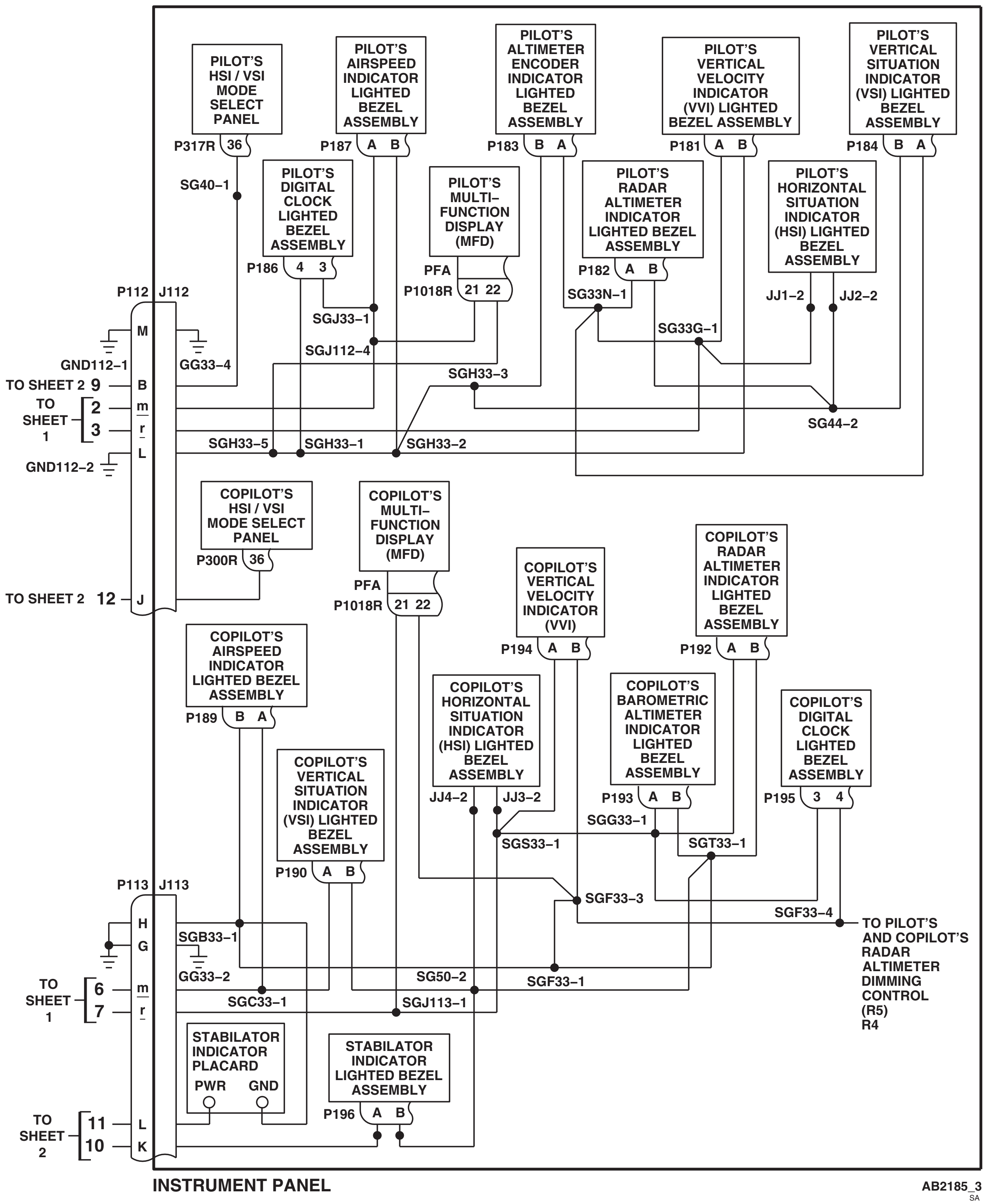


Figure 2. Instrument Panel Lights Schematic Diagram. (Sheet 3 of 3)

END OF WORK PACKAGE

UNIT LEVEL

ELECTRICAL SYSTEMS

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING HH-60A HH-60L

This WP supersedes WP 0126 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
TM 11-1520-237-23
TM 11-1520-253-13
WP 0071 00
WP 0104 00
WP 0146 00
WP 0153 00
WP 0824 00
WP 0825 00
WP 0868 00
WP 0876 00
WP 0878 00
WP 0879 00
WP 0883 00
WP 1224 00
WP 1241 00
WP 1340 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING OPERATIONAL/ TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. On upper console, turn INST LT PILOT FLT and LIGHTED SWITCHES controls fully counterclockwise to OFF.
3. On instrument panel, turn pilot's and copilot's RAD ALT NVG DIMMING controls fully counterclockwise.
4. Turn on electrical power.
5. On pilot's and copilot's radar altimeters, turn LO SET control to on.
6. On miscellaneous switch panel, press TAIL WHEEL switch to the LOCK position.

Indication/Condition

a. On instrument panel, these indicators shall come on to full brightness:

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

- Switch legends on CIS mode select panel.
- Switch legends on pilot's mode select panel.
- Switch legends on copilot's mode select panel.
- GA, DH, or MB indicator on pilot's and copilot's VSI.
- Pilot's BRG 1/DIST (DOP/GPS or TACAN legend).
- Copilot's BRG 1/DIST (DOP/GPS or TACAN legend).
- VTAC (VOR or TACAN legend).
- Pilot's radar altimeter digital display and warning light.
- Copilot's radar altimeter digital display and warning light.

b. On lower console, these indicators shall come on to full brightness:

- On stabilator controls/auto flight control panel, press BOOST switch, ON legend shall come on.
- TAIL WHEEL LOCK legend on miscellaneous switch panel.

NOTE

PWR ON indicator light shall only go on if the OBOGS ON-OFF switch on auxiliary switch panel is set to ON.

c. On OBOGS oxygen status panel, PWR ON indicator shall come on to full brightness.

d. On ECS control panel, AC ON and HEAT ON indicators shall be on.

Corrective Action

1. If none of these indicators come on, go to INSTRUMENT PANEL AND CONSOLE INDICATOR LIGHTS DO NOT GO ON BRIGHT, in this work package.
2. If any individual indicator does not go on, go to INDICATOR LIGHTS DO NOT DIM, in this work package.

Step

7. Check these press-to-test indicators:

a. Lower console:

(1) Blade de-ice control panel TEST IN PROGRESS indicator.

(2) Blade de-ice test panel PWR/MAIN RTR and PWR/TAIL RTR indicators.

(3) Fuel boost pump control panel NO. 1 PUMP and NO. 2 PUMP indicators.

b. Upper console CARGO HOOK EMER/TEST indicator.

NOTE

The SQUIB TEST indicator shall only go on if the HOIST POWER switch on the pilot's hoist control panel is set to ON.

c. Crew's hoist control panel SQUIB TEST indicator.

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Each indicator shall come on to full brightness when pressed.

Corrective Action

1. If any indicator fails to come on when pressed, replace indicator lamp (WP 1224 00) (WP 1241 00) (WP 0868 00) (WP 1340 00).
2. If trouble remains, go to PRESS-TO-TEST INDICATOR CHECKS, in this work package.
3. If CARGO HOOK EMER/TEST indicator fails to come on, troubleshoot cargo hook system (WP 0146 00).
4. If SQUIB TEST indicator fails, troubleshoot rescue hoist (WP 0153 00).

Step

8. Open nose compartment door.
9. Place doppler computer display MODE switch to LAMP TEST.

Indication/Condition

a. MEM and MAL failure indicators shall go on bright.
b. Digital displays shall vary according to DIM control.

Corrective Action

1. If Indication/Condition a. is not as specified, repair/replace wiring (WP 1747 00) or replace doppler computer display (TM 11-1520-237-23), as required.
2. If one indicator in Indication/Condition a. is not as specified, replace lamp (TM 11-1520-237-23).
   a. If trouble remains, replace doppler computer display (TM 11-1520-237-23).

Step

10. Place doppler computer display MODE switch to OFF.
11. On upper console, turn INST LT PILOT FLT control at least 1/4 turn clockwise from OFF.
12. On instrument panel, set INDICATOR LTS switch to BRT/DIM and release to select dim lighting mode.

Indication/Condition

The following indicators shall go dim:

a. Pilot’s BRG 1/DIST (DOP/GPS or TACAN).
b. Copilot’s BRG 1/DIST (DOP/GPS or TACAN).
c. GA, DH, or MB indicator on pilot’s VSI.
d. GA, DH, or MB indicator on copilot’s VSI.
e. Blade de-ice control panel TEST IN PROGRESS when pressed.
f. Blade de-ice test panel PWR/MAIN RTR and PWR/TAIL RTR when pressed.
g. OBOGS oxygen status panel PWR ON.
h. Crew’s hoist control panel SQUIB TEST when pressed.
i. CARGO HOOK EMER/TEST when pressed.

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If indicators do not dim, go to INDICATOR LIGHTS DO NOT DIM, in this work package.

Step

13. On upper console, slowly turn LIGHTED SWITCHES control clockwise from OFF to BRT.

Indication/Condition

a. On instrument panel, these indicators shall vary smoothly from off to bright:

- Switch legends on CIS mode select panel
- Switch legends on pilot's mode select panel
- Switch legends on copilot's mode select panel
- Pilot's BRG 1/DIST
- Copilot's BRG 1/DIST
- VTAC

b. On lower console, these indicators shall vary smoothly from off to bright:

- BOOST ON legend on stabilator controls/auto flight control panel.
- TAIL WHEEL LOCK legends on miscellaneous switch panel.

Corrective Action

If Indication/Condition is not as specified, go to LIGHTED SWITCHES FUNCTION INOPERATIVE, in this work package.

Step

14. On instrument panel, slowly turn pilot's RAD ALT NVG DIMMING control from fully counterclockwise to fully clockwise.

Indication/Condition

Altimeter's digital display and warning lights shall go from off to dim to bright.

Corrective Action

None Required

Step

15. On instrument panel, slowly turn copilot's RAD ALT NVG DIMMING control from fully counterclockwise to fully clockwise.

Indication/Condition

Altimeter's digital display and warning lights shall go from off to dim to bright.

Corrective Action

If Indication/Condition is not as specified, go to PILOT'S AND COPILOT'S RADAR ALTIMETER DIGITAL DISPLAY AND WARNING LIGHTS FAIL TO DIM, in this work package.

INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

Step

16. If bright lighting mode is desired, set INDICATOR LTS switch to BRT/DIM and release.
17. Turn off external electrical power.
18. Close nose compartment door.

Indication/Condition

Nose compartment door shall be closed.

Corrective Action

None Required

INSTRUMENT PANEL AND CONSOLE INDICATOR LIGHTS DO NOT GO ON BRIGHT

SYMPTOM

Instrument panel and console indicator lights do not go on bright.

MALFUNCTION

Instrument panel and console indicator lights do not go on bright.

CORRECTIVE ACTION

1. **Check for 28 vdc between P902-V and ground.**
   a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **3.**
   b. If voltage is not as specified, go to **2.**
2. **Check for 28 vdc between LIGHTS ADVSY circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P902-V (WP 1747 00). Go to **3.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **3.**
3. **Procedure completed.**

Table 1. Indicator Light Checks, Bright Mode.

CONTROL PANEL OR COMPONENT	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
CIS MODE SELECT SWITCH LEGENDS	P316R-8 and P902-X	Replace panel	Repair/replace wiring (WP 1747 00).
PILOT'S MODE SELECT SWITCH LEGENDS	P317R-25 and P902-X	Replace panel	Repair/replace wiring (WP 1747 00).
COPILOT'S MODE SELECT SWITCH LEGENDS	P300R-25 and P902-X	Replace panel	Repair/replace wiring (WP 1747 00).

INSTRUMENT PANEL AND CONSOLE INDICATOR LIGHTS DO NOT GO ON BRIGHT - Continued

Table 1. Indicator Light Checks, Bright Mode. - Continued

CONTROL PANEL OR COMPONENT	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
PILOT'S AND COPILOT'S VSI GA, DH, MB FAIL	P304R-m and P902-h and P301R-m and P902-h	Replace VSI	Repair/replace wiring (WP 1747 00).
PILOT'S BRG 1/DIST DGNS TCN LEGEND	S121-1 and P902-X, S121-2 and P902-X, S121-3 and P902-X, S121-4 and P902-X	Replace switch	Repair/replace wiring (WP 1747 00).
COPILOT'S BRG 1/DIST DGNS TCN LEGEND	S122-1 and P902-X, S122-2 and P902-X, S122-3 and P902-X, S122-4 and P902-X	Replace switch	Repair/replace wiring (WP 1747 00).
VTAC	S123-1 and P902-X, S123-2 and P902-X, S123-3 and P902-X, S123-4 and P902-X	Replace switch	Repair/replace wiring (WP 1747 00).
PILOT'S RADAR ALTIMETER DIGITAL DISPLAY AND WARNING LIGHT	P676R-6 and P902-n	Replace altimeter	Repair/replace wiring (WP 1747 00).
COPILOT'S RADAR ALTIMETER DIGITAL DISPLAY AND WARNING LIGHT	P675R-7 and P902-Y	Replace altimeter	Repair/replace wiring (WP 1747 00).
FLIGHT CONTROL	P4R-E and P902-X	Replace panel	Repair/replace wiring (WP 1747 00).
MISCELLANEOUS SWITCH PANEL TAIL WHEEL LOCK LEGEND	P135-C and P902-X	Replace panel	Troubleshoot Tail Wheel Lock System (WP 0071 00)
OBOGS OXYGEN STATUS POWER ON	P207-12 and P902-h	Troubleshoot OBOGS (TM 1-1520- 253-13&P)	Repair/replace wiring (WP 1747 00).
ECS AC ON AND HEAT ON	P356-G and P902-X, P356-J and P902-X	Replace panel	Repair/replace wiring (WP 1747 00).

INSTRUMENT PANEL AND CONSOLE INDICATOR LIGHTS DO NOT GO ON BRIGHT - Continued

Table 2. Press-To-Test Indicator Checks.

INDICATOR	CHECK CONTINUITY BETWEEN THESE POINTS	IF YES	IF NO
Blade De-ice Control Panel TEST IN PROGRESS	P133-K and P902-h	Replace blade de-ice control panel	Repair/replace wiring (WP 1747 00).
Blade De-ice Test Panel PWR/MAIN RTR and PWR/TAIL RTR	P134-H and P902-h	Replace blade de-ice test panel	Repair/replace wiring (WP 1747 00).
Fuel Boost Pump Control Panel NO. 1 PUMP and NO. 2 PUMP	P139-P and P902-X, P139-R and P902-X	Replace fuel boost pump control panel	Repair/replace wiring (WP 1747 00).

INDICATOR LIGHTS DO NOT DIM

SYMPTOM

Indicator lights do not dim.

MALFUNCTION

Indicator lights do not dim.

CORRECTIVE ACTION

1. Check light dimmer fuse.

a. If fuse is good, go to **2.**

b. If fuse is not good, replace it (WP 0879 00). Go to **6.**

2. Check for 28 vdc between P393-B and ground.

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, repair/replace wiring between P393-B and LIGHTS ADVSY circuit breaker (WP 0824 00). Go to **6.**

3. Check continuity between P393-A and ground.

a. If continuity is present, go to **4.**

b. If continuity is not present, repair/replace wiring between P393-A and ground (WP 1747 00). Go to **6.**

4. Check continuity between P393-C and P1000-D.

a. If continuity is present, go to **5.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**

5. Check continuity between P393-D and P902-h.

a. If continuity is present, replace light dimmer (WP 0878 00). Go to **6.**

INDICATOR LIGHTS DO NOT DIM - Continued

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**

6. Procedure completed.

LIGHTED SWITCHES FUNCTION INOPERATIVE

SYMPTOM

Lighted switches function inoperative.

MALFUNCTION

Lighted switches function inoperative.

CORRECTIVE ACTION

1. With LIGHTED SWITCHES control at BRT, check for 26 vdc between P902-Z and ground.

a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **5.**

b. If voltage is not as specified, go to **2.**

2. Check for 28 vdc between P315-A and ground.

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, repair/replace wiring between P249-P and P315-A (WP 1747 00). Go to **5.**

3. Check continuity between:

- **P315-B and ground**
- **P315-C and ground**
- **P315-E and ground**

a. If continuity is present, go to **4.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5.**

4. Check continuity between P315-D and P902-Z.

a. If continuity present, replace LIGHTED SWITCHES dimmer control (WP 0876 00). Go to **5.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5.**

5. Procedure completed.

PILOT'S AND COPILOT'S RADAR ALTIMETER DIGITAL DISPLAY AND WARNING LIGHTS FAIL TO DIM

SYMPTOM

Pilot's and copilot's radar altimeter digital display and warning lights fail to dim.

MALFUNCTION

Pilot's and copilot's radar altimeter digital display and warning lights fail to dim.

CORRECTIVE ACTION

1. With pilot's and copilot's radar altimeter controls rotated fully counterclockwise, check for 28 vdc between these points:

- **P676R-6 and ground**

PILOT'S AND COPILOT'S RADAR ALTIMETER DIGITAL DISPLAY AND WARNING LIGHTS FAIL TO DIM - Continued

- **P675R-7 and ground**

a. If voltage is as specified, go to **2.**

b. If voltage is not as specified, go to **3.**

2. Check for 28 vdc between:

- **R4-1 and ground**
- **R5-1 and ground**

a. If voltage is as specified, go to **4.**

b. If voltage is not as specified, repair/replace wiring between these points (WP 1747 00), then go to **6:**

- R4-1 and P902-f
- R5-1 and P902-W

3. Check continuity between:

- **P676R-6 and P902-n**
- **P675R-7 and P902-Y**

a. If continuity is present, replace left relay panel (WP 0825 00). Go to **6.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**

4. Check continuity between:

- **R4-2 and P902-T**
- **R5-2 and P902-p**

a. If continuity is present, go to **5.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**

5. Check continuity between:

- **R4-3 and ground**
- **R5-3 and ground**

a. If continuity is present, replace malfunctioning control (WP 0883 00). Go to **6.**

b. If continuity is not present, repair/replace wiring between these points (WP 1747 00), then go to **6:**

- R4-3 and J112-L
- R5-3 and J113-H
- P112-L and ground
- P113-H and ground

6. Procedure completed.

LOCATION

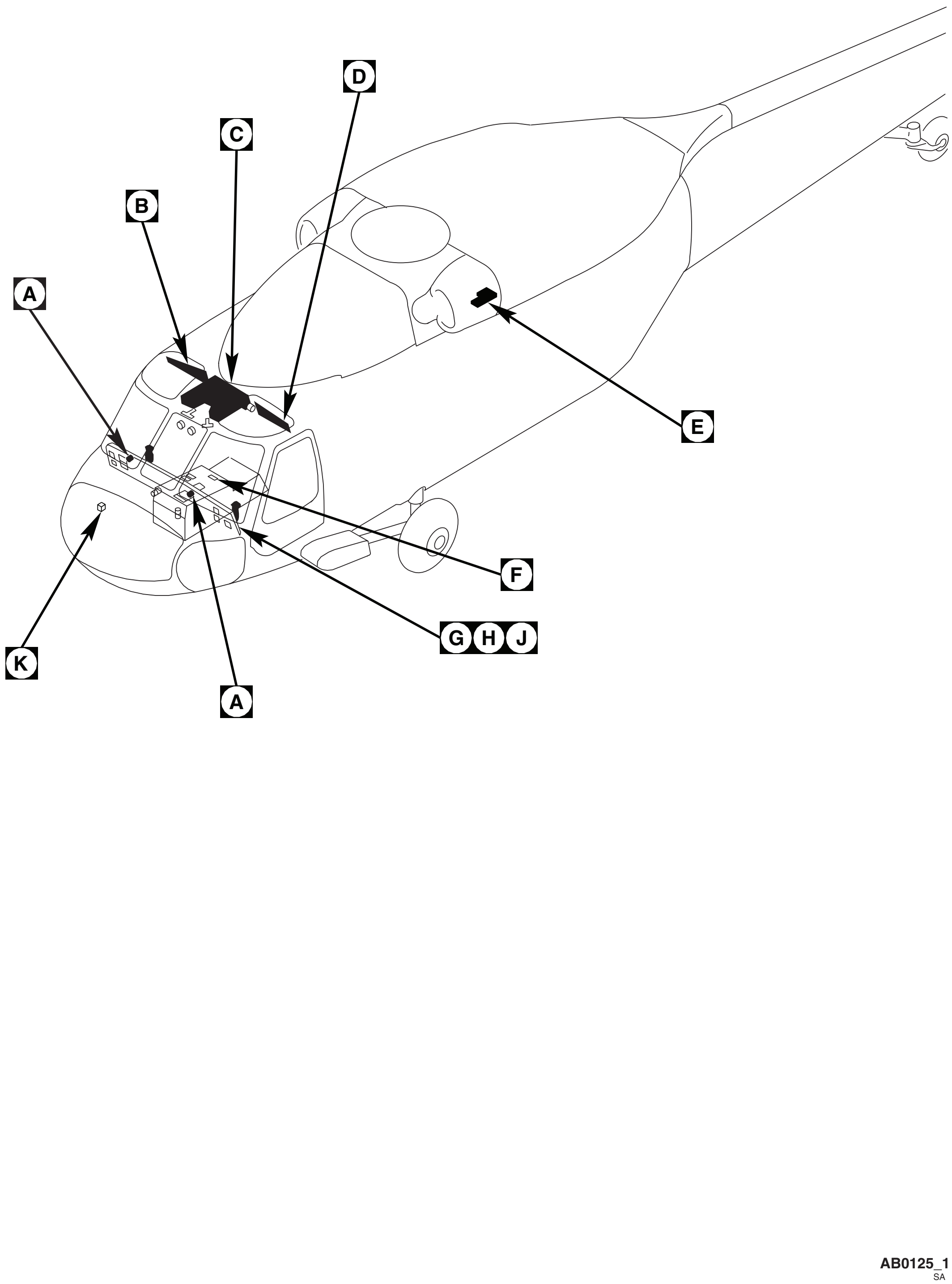


Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 1 of 9)

LOCATION - Continued

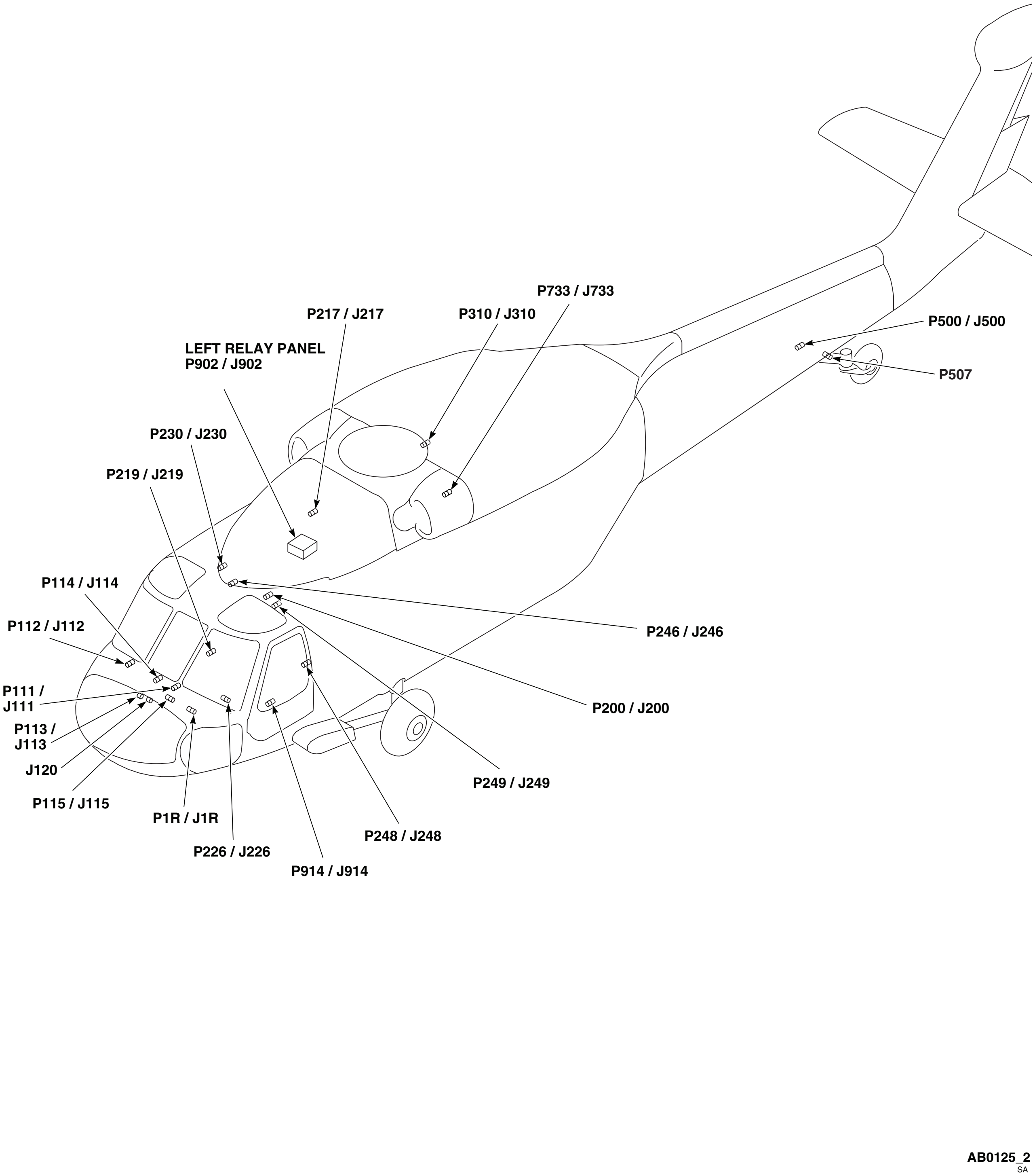


Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 2 of 9)

LOCATION - Continued

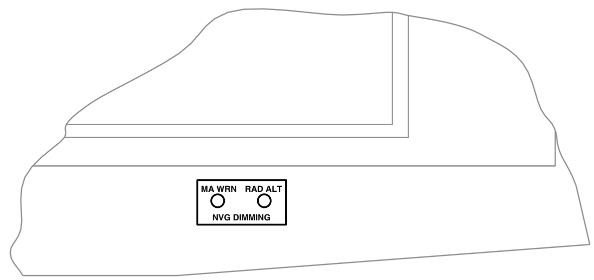
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J346	CABIN CEILING, BL 18 LH, STA 301
P1R / J1R	COCKPIT, BL 6 LH, STA 210
P4R	STABILATOR CONTROL / AUTO FLIGHT CONTROL PANEL
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P112 / J112	COCKPIT, BL 6 RH, STA 197
P113 / J113	COCKPIT, BL 6 LH, STA 197
P114 / J114	COCKPIT, BL 8 RH, STA 200
P115 / J115	COCKPIT, BL 6 RH, STA 210
P133	BLADE DE–ICE CONTROL PANEL
P134	BLADE DE–ICE TEST PANEL
P135	MISCELLANEOUS SWITCH PANEL
P139	FUEL BOOST PUMP CONTROL PANEL
P198 / J198	INSTRUMENT PANEL DISCONNECT
P199 / J199	INSTRUMENT PANEL DISCONNECT
P200 / J200	COCKPIT CEILING, BL 13 LH, STA 246
P207 / J207	OBOGS STATUS PANEL
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR, BL 25 RH, STA 247
P226 / J226	CREW'S HOIST CONTROL PANEL
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P246 / J246	CABIN CEILING, BL 5 RH, STA 247
P248 / J248	CABIN, BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 23 LH

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P298 / J298	NO. 3 RELAY PANEL
P300R	COPILOT'S HSI / VSI MODE SEL PANEL
P301R	COPILOT'S VSI
P304R	PILOT'S VSI
P310 / J310	CABIN CEILING, BL 17 RH, STA 387
P315	LIGHTED SWITCHES DIMMER CONTROL
P316R	CIS MODE SELECT PANEL
P317R	PILOT'S HSI / VSI MODE SELECT PANEL
P356	ECS CONTROL PANEL
P360	AUXILIARY FUEL MANAGEMENT PANEL
P393	INDICATOR LIGHTS DIMMER COCKPIT, BL 8 LH, STA 210, WL 222
P500 / J500	TAILCONE, BL 0, STA 609
P507	TAILWHEEL LOCK ACTUATOR
P675R	COPILOT'S RADAR ALTIMETER
P676R	PILOT'S RADAR ALTIMETER
P687R / 3J1	DOPPLER COMPUTER DISPLAY
P733 / J733	BL 24.71, STA 360, WL 265.30
P902 / J902	LEFT RELAY PANEL
P914 / J914	COCKPIT, BL 24 LH, STA 247
P1000	MASTER WARNING DIMMER
S121	BRG 1 / DIST DGNS TCN
S122	BRG 1 / DIST DGNS TCN
S123	VTAC VOR TCN

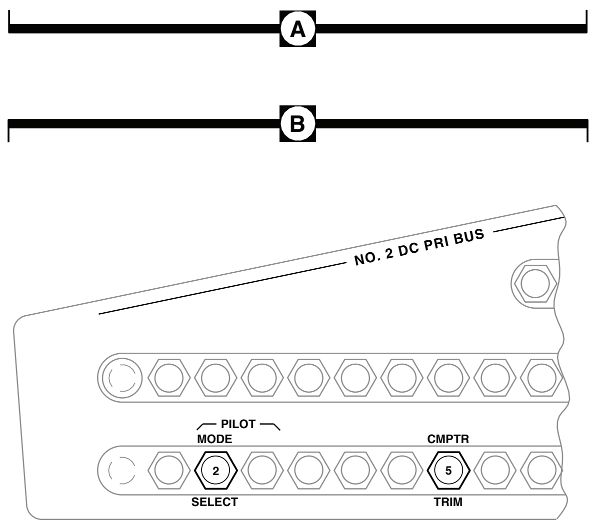
AB0125_3
SA

Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 3 of 9)

LOCATION - Continued



PILOT'S AND COPILOT'S RADAR ALTIMETER DIMMING CONTROL



PILOT'S CIRCUIT BREAKER PANEL

AB0125_4
SA

Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 4 of 9)

LOCATION - Continued

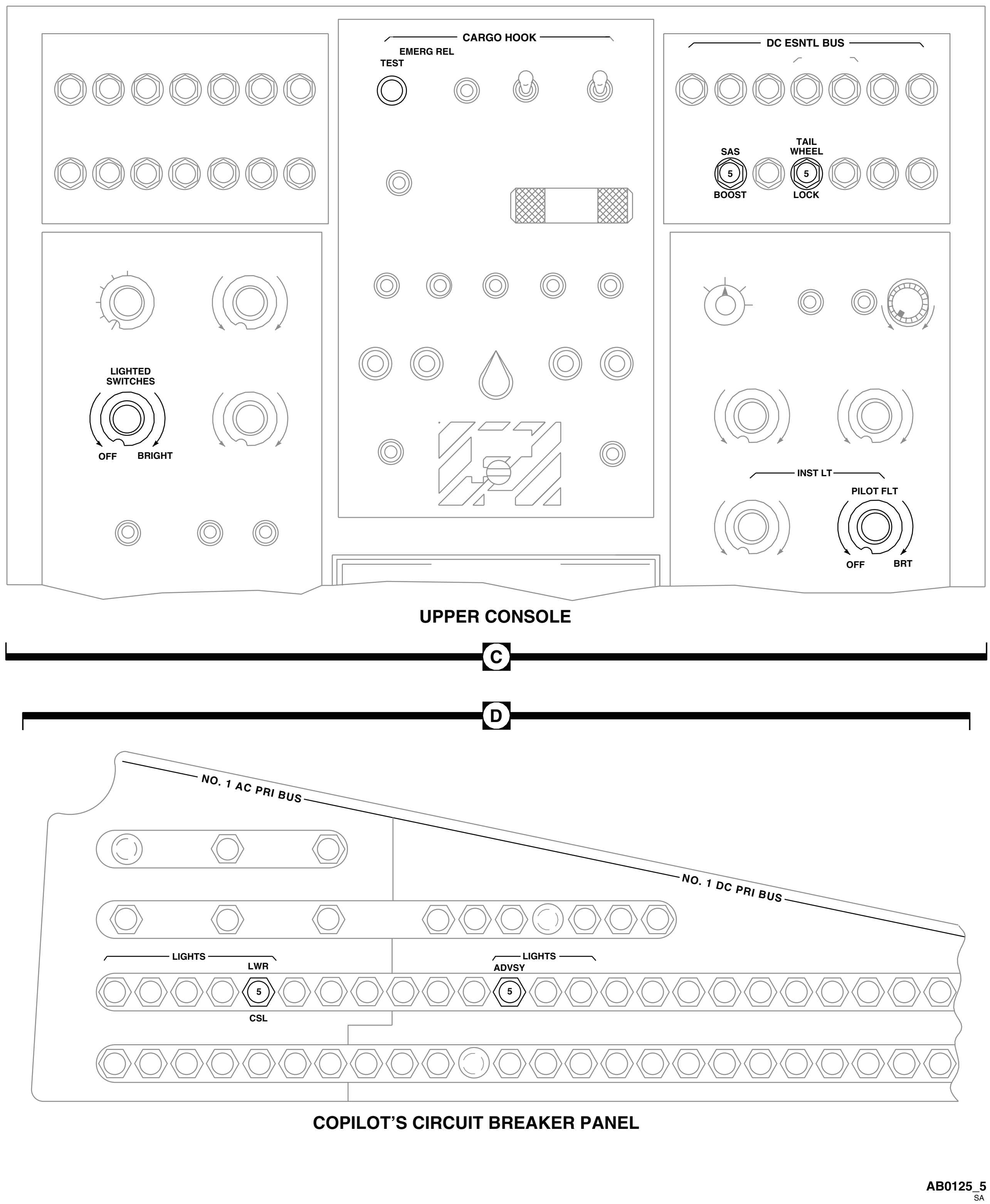
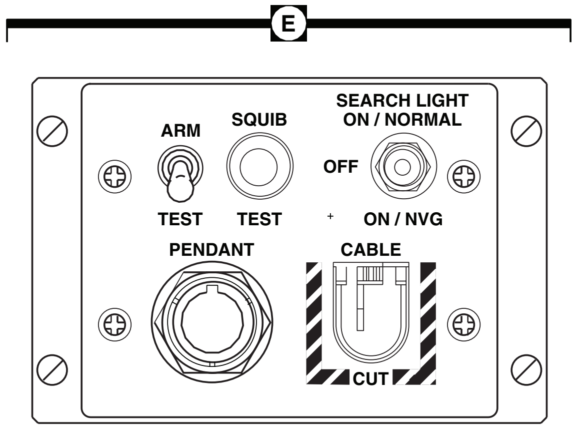


Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 5 of 9)

LOCATION - Continued

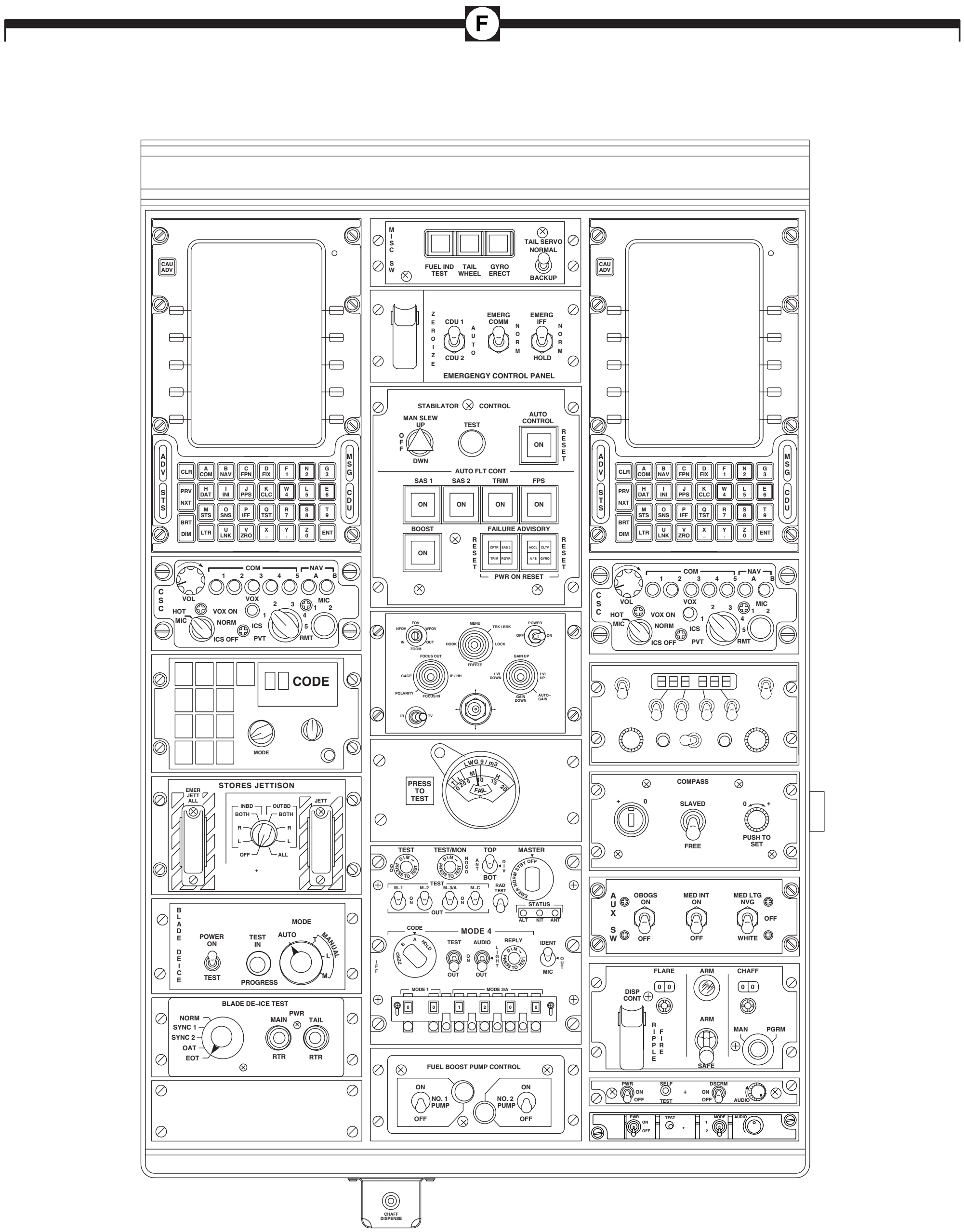


CREW'S HOIST CONTROL PANEL

AB0125_6
SA

Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 6 of 9)

LOCATION - Continued

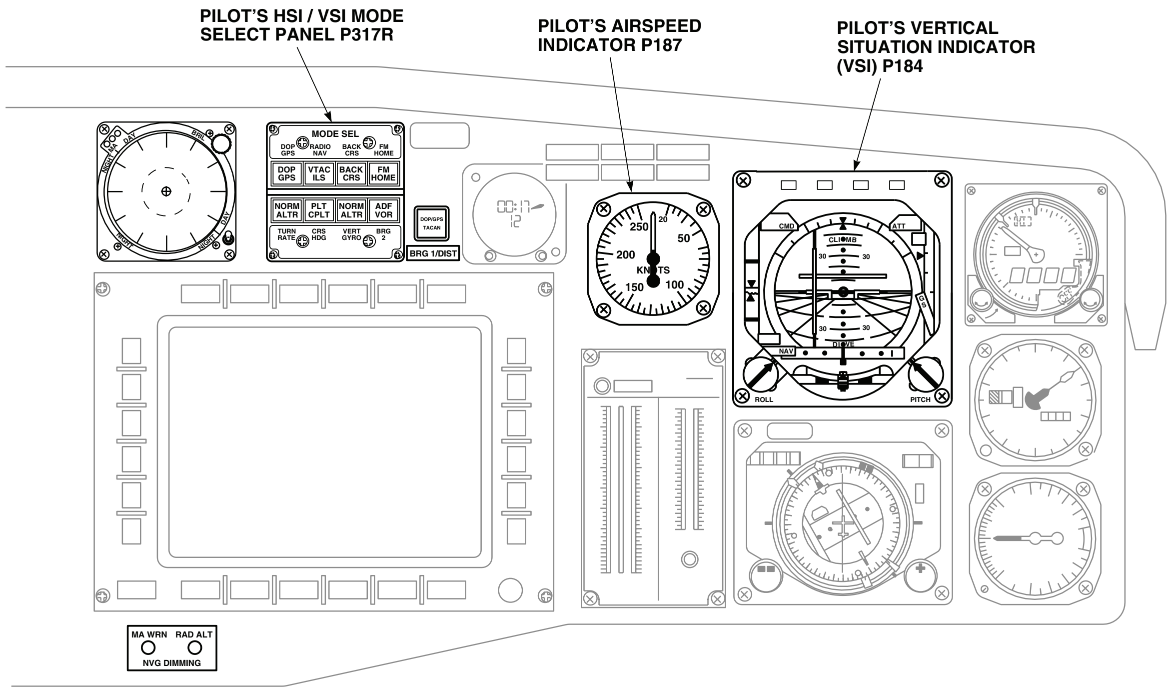


LOWER CONSOLE

AB0125_7A
SA

Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 7 of 9)

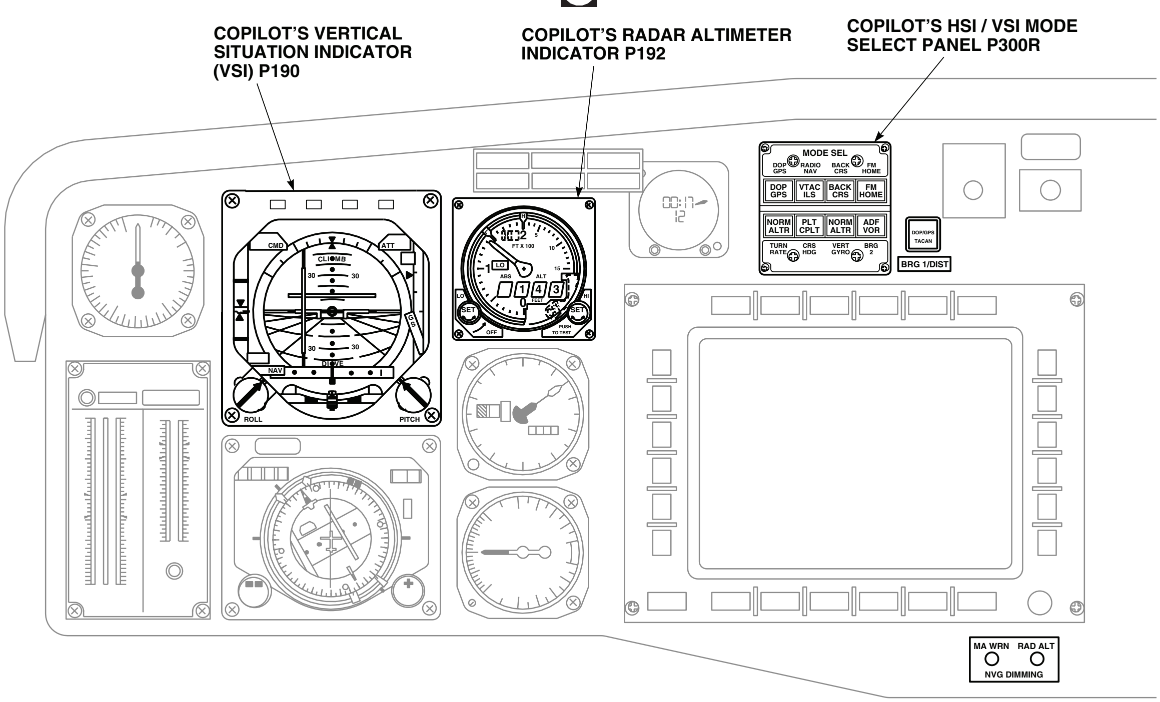
LOCATION - Continued



INSTRUMENT PANEL PILOT'S SIDE

G

H

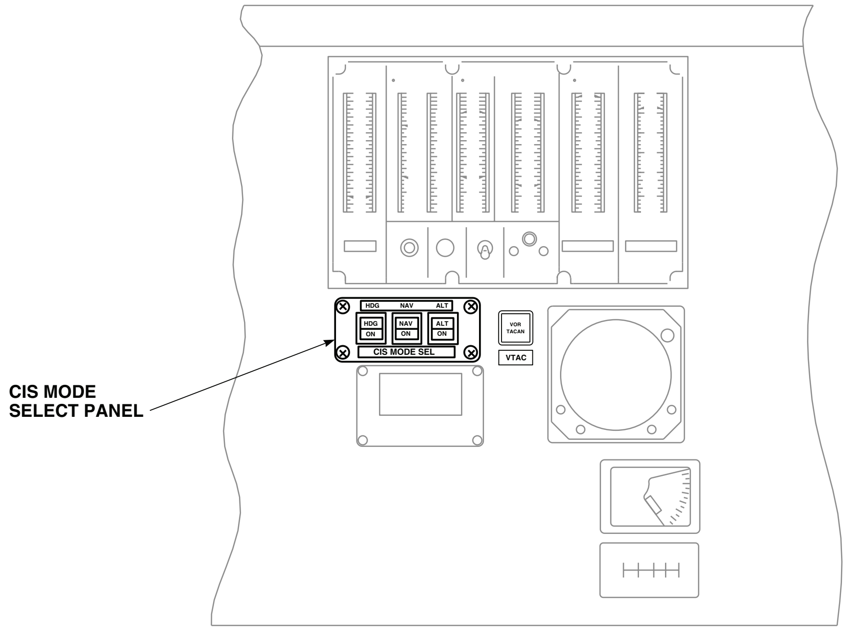


INSTRUMENT PANEL COPILOT'S SIDE

AB0125_8A
SA

Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 8 of 9)

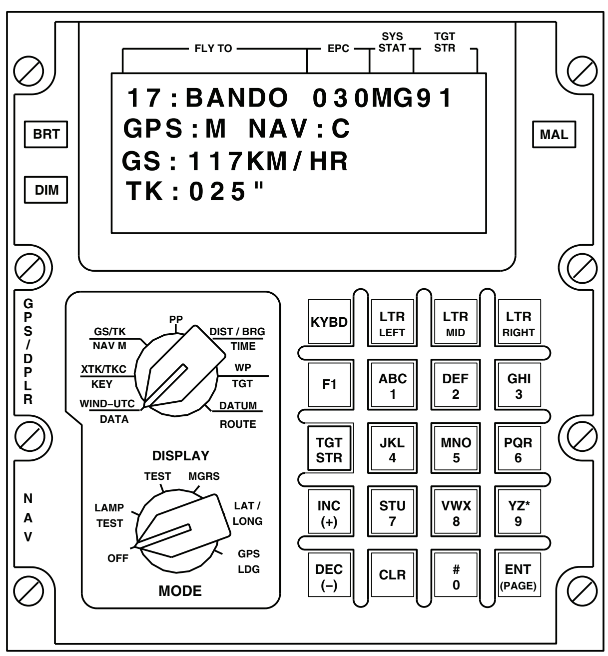
LOCATION - Continued



INSTRUMENT PANEL (CENTER)

J

K



DOPPLER COMPUTER–DISPLAY UNIT

AB0125_9A
SA

Figure 1. Instrument Panel and Consoles Indicator Lights Dimming Location Diagram. (Sheet 9 of 9)

SCHEMATICS

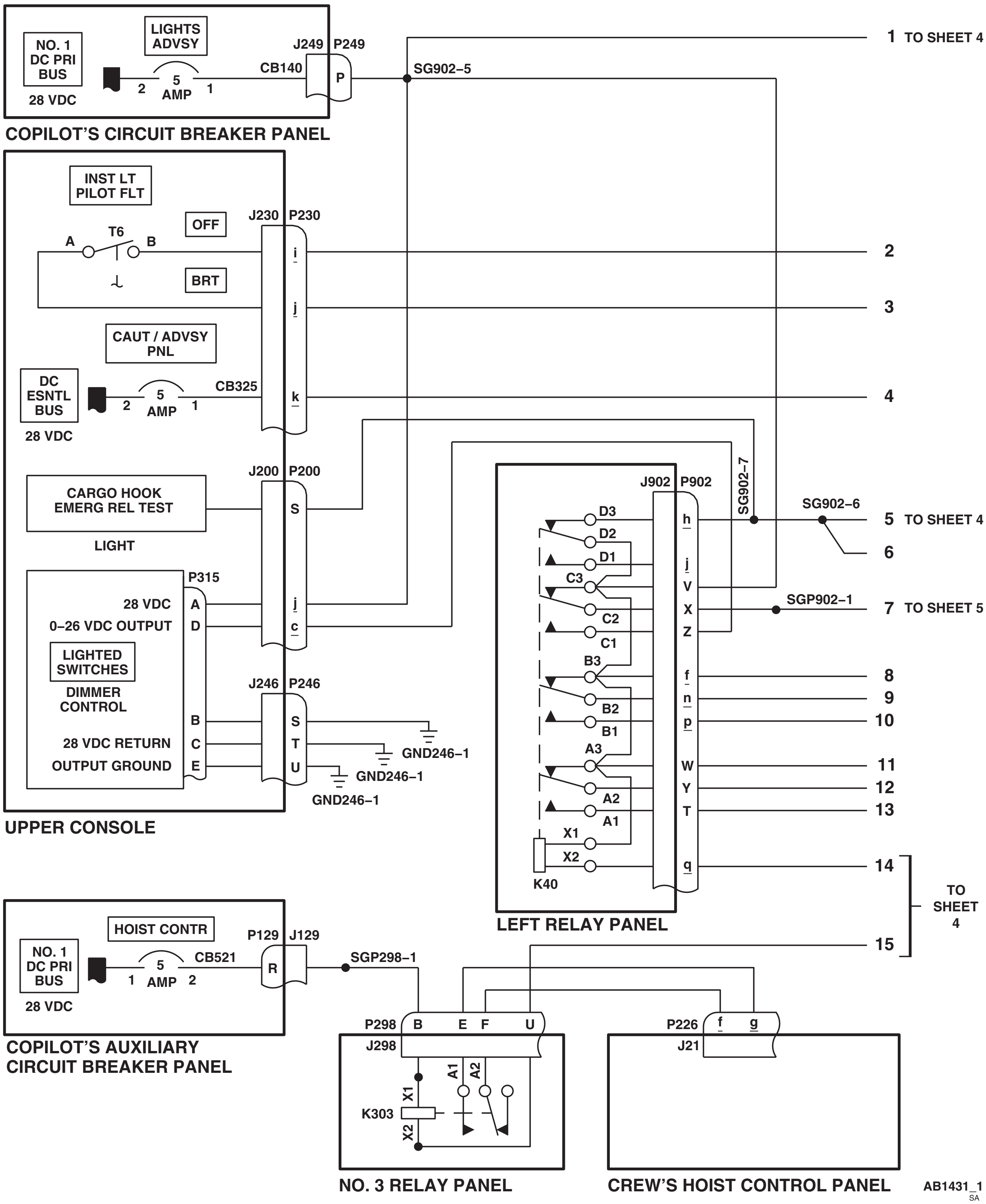


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 1 of 5)

SCHEMATICS - Continued

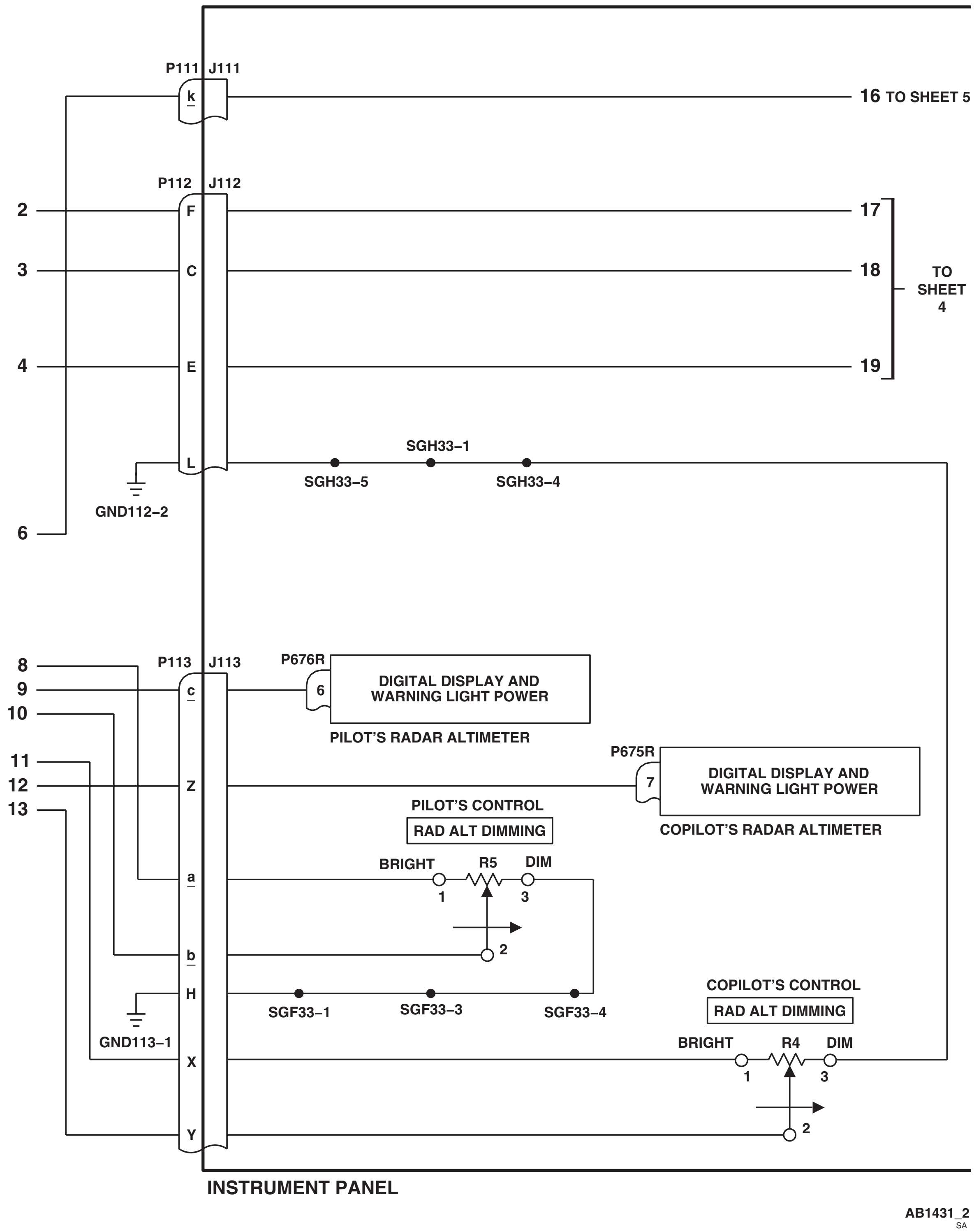


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 2 of 5)

SCHEMATICS - Continued

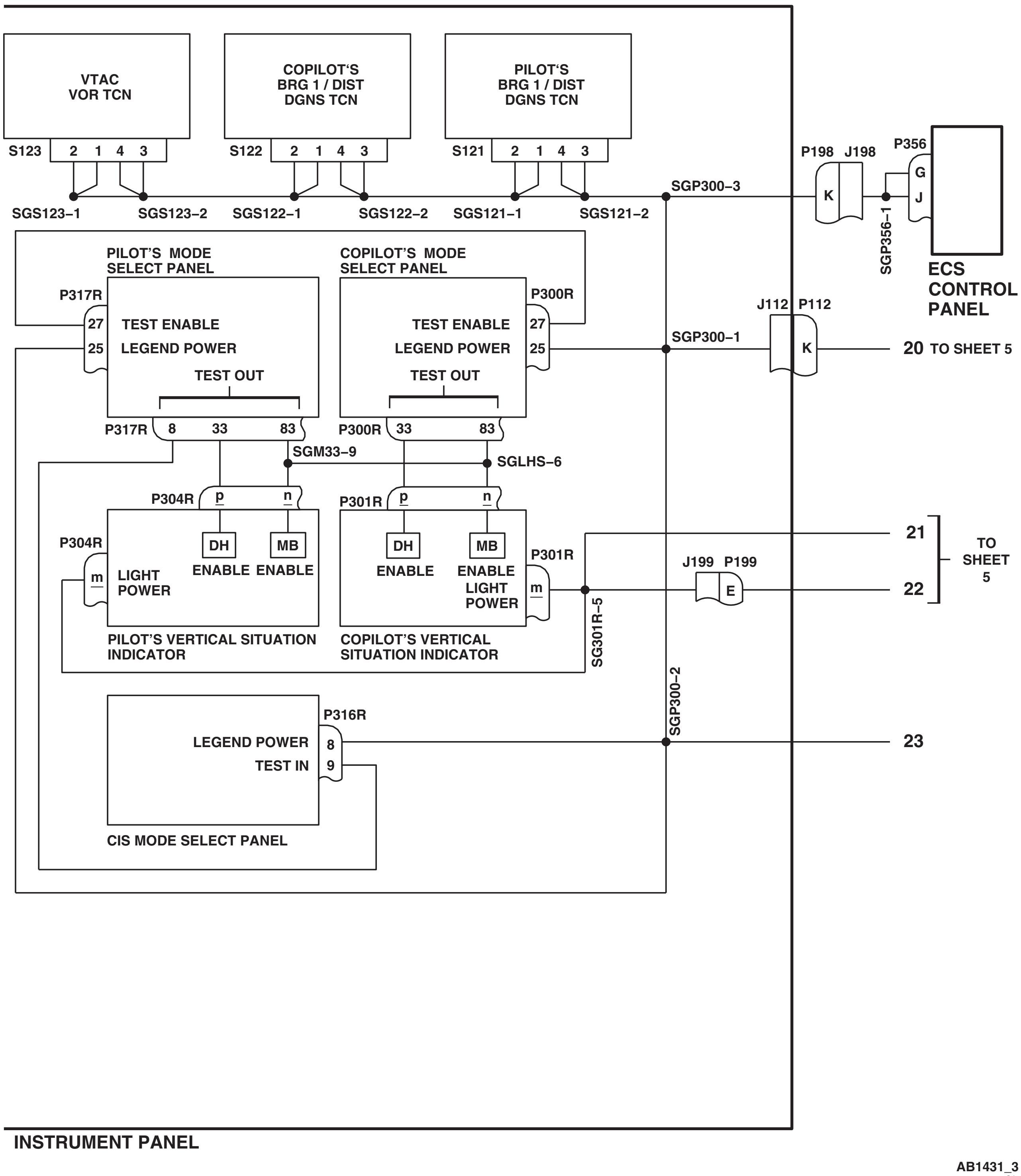


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 3 of 5)

SCHEMATICS - Continued

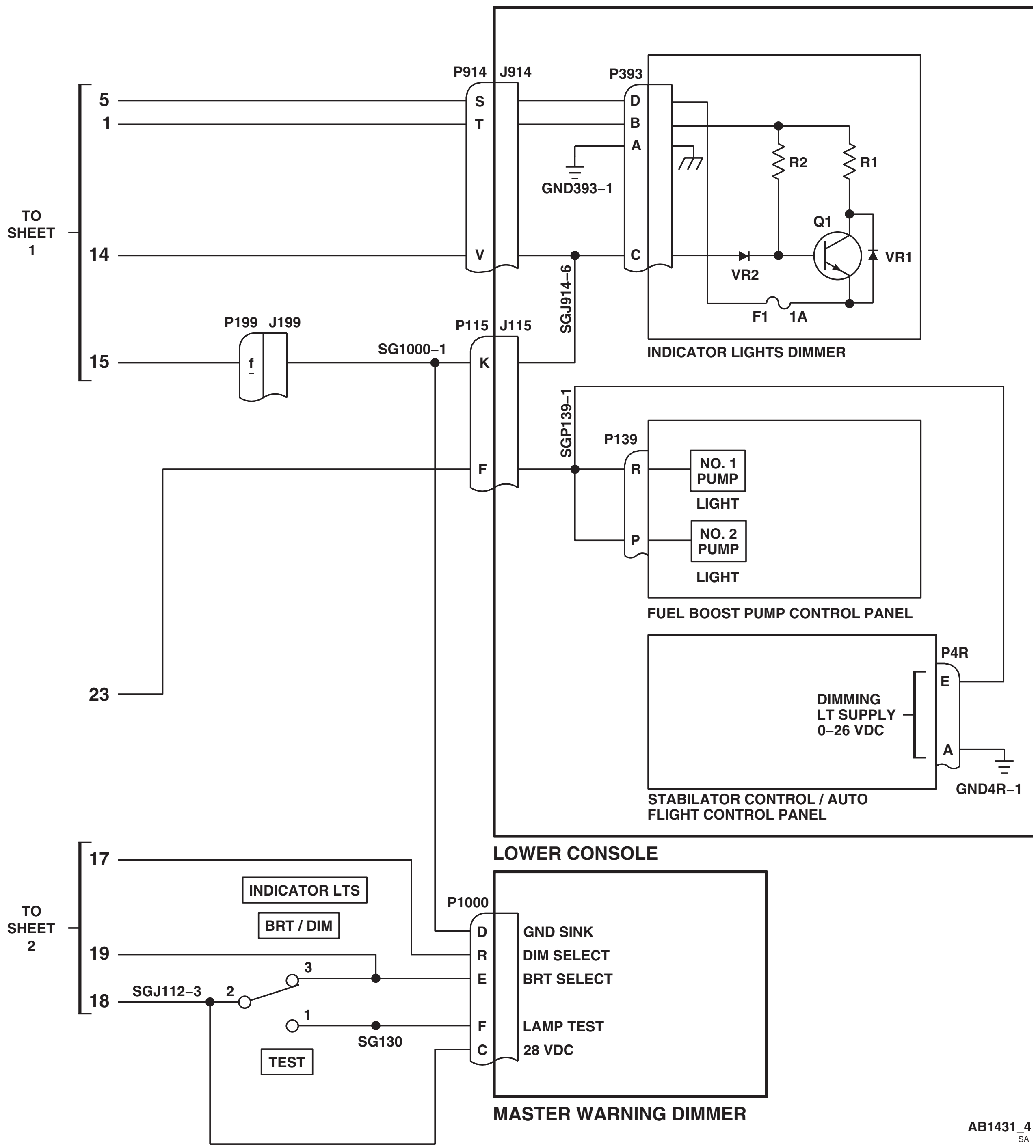


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 4 of 5)

SCHEMATICS - Continued

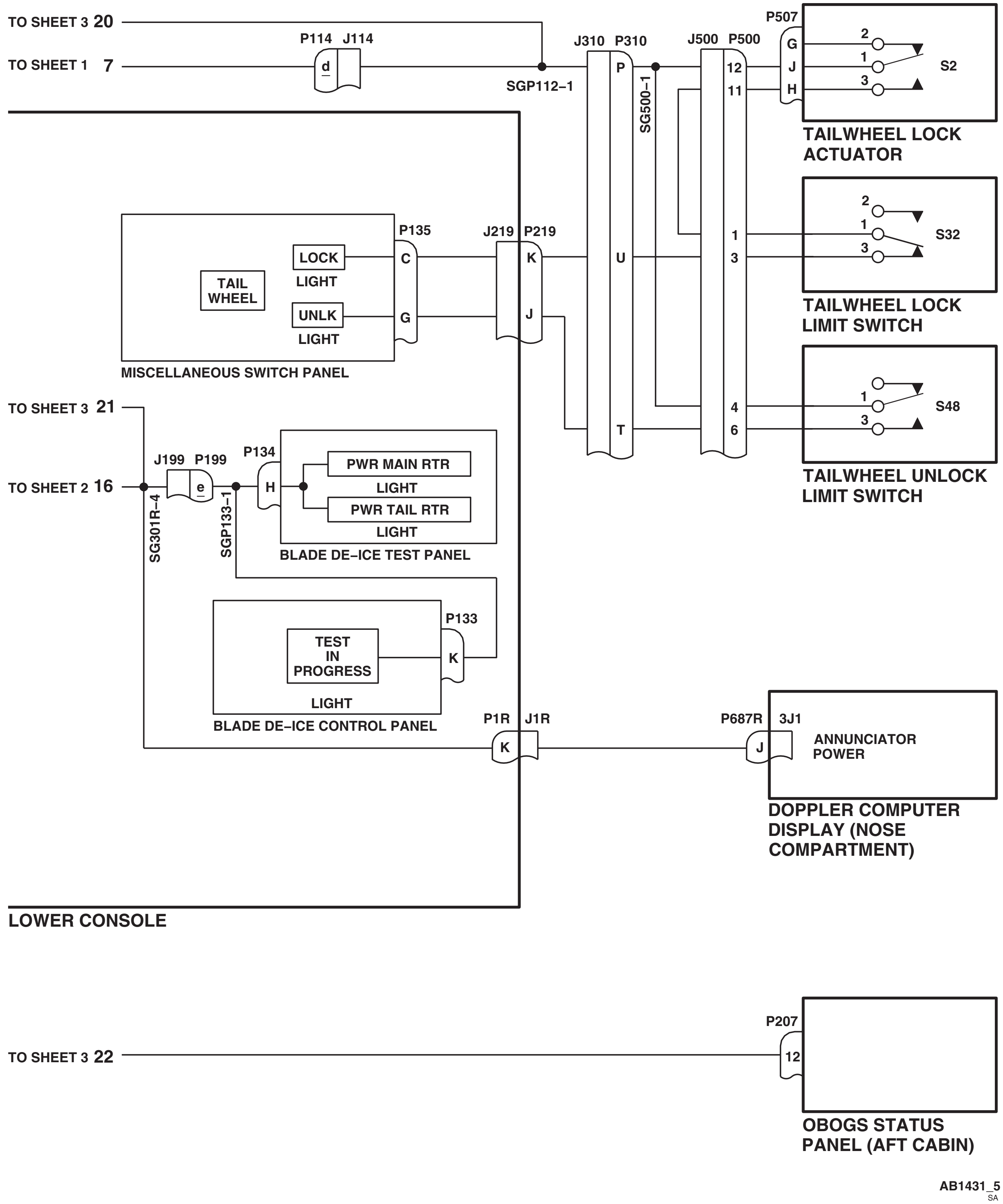


Figure 2. Instrument Panel and Consoles Indicator Lights Dimming Schematic Diagram. (Sheet 5 of 5)

END OF WORK PACKAGE

UNIT LEVEL

FUEL SYSTEM

FUEL LOW LEVEL WARNING SYSTEM

This WP supersedes WP 0127 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1520-237-10
WP 0090 00
WP 0100 00
WP 0104 00
WP 0785 00
WP 0786 00
WP 0799 00
WP 0824 00
WP 0830 00
WP 0994 00
WP 0996 00
WP 1612 00
WP 1685 00
WP 1747 00

Equipment Condition
External Power Available, WP 1685 00
or APU Operational, TM 1-1520-237-10
No. 1 and No. 2 Fuel Tanks Each Filled With **200 pounds** of Fuel, WP 1612 00

FUEL LOW LEVEL WARNING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- If #1 or #2 FUEL LOW capsules on caution/advisory panel or **HH-60A HH-60L** #1 and #2 FUEL legends on MFDs are ever on steadily (not flashing), replace caution/advisory panel (WP 0785 00) or **HH-60A HH-60L** MFDs (WP 0799 00) and then continue with operational/troubleshooting procedure.
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure FUEL LOW WARN circuit breaker on copilot's circuit breaker panel is pushed in.
2. Turn on electrical power. **HH-60A HH-60L** Turn pilot's and copilot's multifunction display (MFD) switches ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
3. Press miscellaneous switch panel FUEL IND TEST button.

FUEL LOW LEVEL WARNING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

#1 and #2 FUEL LOW capsules on caution/advisory panel or **HH-60A HH-60L** #1 and #2 FUEL legends on MFDs ■ and master caution capsules, on master warning panels, shall flash.

Corrective Action

1. If Indication/Condition is not as specified, check for 28 vdc between P255-J and ground, and between P255-K and ground.
   a. If voltage is as specified, replace low level warning conditioner (WP 0996 00).
   b. If voltage is not as specified, replace miscellaneous switch panel (WP 0830 00).
      (1) If trouble remains, repair/replace wiring (WP 1747 00).

Step

4. Release miscellaneous switch panel FUEL IND TEST button.
5. As you defuel No. 1 tank to **150 pounds**, observe #1 FUEL LOW capsule on caution/advisory panel or **HH-60A HH-60L** #1 FUEL legends on MFDs ■ . Record poundage when **HH-60A HH-60L** #1 FUEL LOW capsule or #1 FUEL legends on MFDs ■ begin to flash.

Indication/Condition

a. #1 FUEL LOW capsule o **HH-60A HH-60L** #1 FUEL legends on MFDs ■ r shall flash with less than **170 to 180 pounds** of fuel in No. 1 tank.

b. #1 FUEL LOW capsule or **HH-60A HH-60L** #1 FUEL legends on MFDs ■ shall be off with more than **170 to 180 pounds** of fuel in No. 1 tank.

Corrective Action

1. If Indication/Condition a. is not as specified, go to NO FUEL LOW INDICATION FROM ONE TANK WITH FUEL LEVEL BELOW 170 TO 180 POUNDS. SECOND TANK NORMAL, in this work package.
2. If Indication/Condition b. is not as specified, go to FUEL LOW INDICATION ON ONE TANK WITH FUEL QUANTITY GREATER THAN 170 TO 180 POUNDS. SECOND TANK NORMAL, in this work package.

Step

6. As you defuel No. 2 tank to **150 pounds**, observe #2 FUEL LOW capsule on caution/advisory panel or **HH-60L** #2 FUEL legends on MFDs ■ . Record poundage when **HH-60A HH-60L** #2 FUEL LOW capsule or #2 FUEL legends begins to flash on MFDs. ■

Indication/Condition

a. #2 FUEL LOW capsule or **HH-60A HH-60L** #2 FUEL legends on MFDs ■ shall flash with less than **170 to 180 pounds** of fuel in No. 2 tank.

b. #2 FUEL LOW capsule or **HH-60A HH-60L** #2 FUEL legends on MFDs ■ shall be off with more than **170 to 180 pounds** of fuel in No. 2 tank.

Corrective Action

1. If Indication/Condition a. is not as specified, go to NO FUEL LOW INDICATION FROM ONE TANK WITH FUEL LEVEL BELOW 170 TO 180 POUNDS. SECOND TANK NORMAL, in this work package.

FUEL LOW LEVEL WARNING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. If Indication/Condition b. is not as specified, go to FUEL LOW INDICATION ON ONE TANK WITH FUEL QUANTITY GREATER THAN 170 TO 180 POUNDS. SECOND TANK NORMAL, in this work package.

Step

7. Check poundage recorded in steps 5. and 6.

Indication/Condition

#1 FUEL LOW and #2 FUEL LOW capsules or HH-60A HH-60L #1 and #2 FUEL legends on MFDs shall flash between **170 to 180 pounds**.

Corrective Action

If Indication/Condition is not as specified, go to NO FUEL LOW INDICATION FROM BOTH TANKS WITH FUEL BELOW 170 TO 180 POUNDS, in this work package.

Step

8. Turn MFD switches OFF.
9. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

NO FUEL LOW INDICATION FROM ONE TANK WITH FUEL LEVEL BELOW 170 TO 180 POUNDS. SECOND TANK NORMAL

SYMPTOM

No fuel low indication from one tank with fuel level below 170 to 180 pounds. Second tank normal.

MALFUNCTION

No fuel low indication from one tank with fuel level below 170 to 180 pounds. Second tank normal.

CORRECTIVE ACTION

1. **Press FUEL IND TEST button on miscellaneous switch panel and observe that FUEL LOW capsules or HH-60A HH-60L FUEL legends on MFDs flash.**
   a. If capsules or legends flash, go to **2.**
   b. If capsules or legends do not flash, go to **4.**
2. **Check resistance between:**
   - **P255-A and P255-B 650-1500 Ohms**
   - **P255-A and P255-C 360-440 Ohms**
   - **P255-E and P255-F 650-1500 Ohms**
   - **P255-E and P255-G 360-440 Ohms**

   a. If resistance is as specified, replace low level warning conditioner (WP 0996 00). Go to **9.**

NO FUEL LOW INDICATION FROM ONE TANK WITH FUEL LEVEL BELOW 170 TO 180 POUNDS. SECOND TANK NORMAL - Continued

b. If resistance is not as specified, go to **3.**

3. Check continuity between:

- **P255-A and J307-C**
- **P255-B and J307-B**
- **P255-C and J307-A**
- **P255-E and J308-C**
- **P255-F and J308-B**
- **P255-G and J308-A**

a. If continuity is present, replace malfunctioning fuel quantity probe (WP 0994 00). Go to **9.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**

4. Hold caution/advisory panel BRT/DIM-TEST switch to TEST or HH-60A HH-60L press T6 switch on MFD, and check that FUEL LOW capsule or HH-60A HH-60L FUEL legends on MFDs go on.

a. If capsule(s) or legends go on, go to **5.**

b. If capsule(s) or legends do not go on, go to **8.**

5. Check for 28 vdc between:

- **P255-N (No. 1 tank) and ground**
- **P255-S (No. 2 tank) and ground**

a. If voltage is as specified, go to **6.**

b. If voltage is not as specified, repair/replace wiring as required (WP 1747 00). Go to **9.**

6. Install an insulated jumper wire as required between, then go to 7:

- **P255-N and P255-P (No. 1 tank)**
- **P255-N and P255-T (No. 2 tank)**

7. Check that FUEL LOW capsules or HH-60A HH-60L FUEL capsules on MFDs flashes.

a. If capsule(s) or legends flash, replace low level warning conditioner (WP 0996 00). Go to **9.**

b. If capsule(s) or legends do not flash, repair/replace wiring as required between low level warning conditioner and caution/advisory panel or MFDs (WP 1747 00). Go to **9.**

8. Check if FUEL LOW capsule lamps are good. HH-60A HH-60L Check if FUEL legends on MFDs are good.

a. If lamp(s) or legends are good, troubleshoot caution/advisory warning system (WP 0090 00) or HH-60A HH-60L MFDs (WP 0100 00). Go to **9.**

b. If lamp(s) or legends are not good, replace lamp (WP 0786 00) or MFD (WP 0799 00). Go to **9.**

9. Procedure completed.

FUEL LOW INDICATION ON ONE TANK WITH FUEL QUANTITY GREATER THAN 170 TO 180 POUNDS. SECOND TANK NORMAL

SYMPTOM

Fuel low indication on one tank with fuel quantity greater than 170 to 180 pounds. Second tank normal.

MALFUNCTION

Fuel low indication on one tank with fuel quantity greater than 170 to 180 pounds. Second tank normal.

CORRECTIVE ACTION

1. Check thermistor resistance between:

- **P255-A and P255-B (650-1500 Ohms)**
- **P255-A and P255-C (360-440 Ohms)**
- **P255-E and P255-F (650-1500 Ohms)**
- **P255-E and P255-G (360-440 Ohms)**

a. If resistance is as specified, replace low level warning conditioner (WP 0996 00). Go to **3.**

b. If trouble remains, replace miscellaneous switch panel (WP 0830 00). Go to **3.**

c. If resistance is not as specified, go to **2.**

2. Check continuity between:

- **P255-A and J307-C**
- **P255-B and J307-B**
- **P255-C and J307-A**
- **P255-E and J308-C**
- **P255-F and J308-B**
- **P255-G and J308-A**

a. If continuity is present, replace malfunctioning fuel quantity probe or wiring harness (WP 0994 00). Go to **3.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **3.**

3. Procedure completed.

NO FUEL LOW INDICATION FROM BOTH TANKS WITH FUEL BELOW 170 TO 180 POUNDS

SYMPTOM

No fuel low indication from both tanks with fuel below 170 to 180 pounds.

MALFUNCTION

No fuel low indication from both tanks with fuel below 170 to 180 pounds.

CORRECTIVE ACTION

1. Hold caution/advisory panel BRT/DIM-TEST switch to TEST or HH-60A HH-60L press T6 switch on MFD, and check that FUEL LOW capsule or HH-60A HH-60L FUEL legends on MFDs

NO FUEL LOW INDICATION FROM BOTH TANKS WITH FUEL BELOW 170 TO 180 POUNDS - Continued

a. If capsule(s) or legends go on, go to **2.**

b. If capsule(s) or legends do not go on, go to **4.**

2. Check for 28 vdc between:

- **P255-N and ground**
- **P255-S and ground**

a. If voltage is as specified, replace low level warning conditioner (WP 0996 00). Go to **5.**

b. If voltage is not as specified, go to **3.**

3. Check continuity between FUEL LOW WARN circuit breaker terminal 1 and J255-N or J255-S.

a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **5.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **5.**

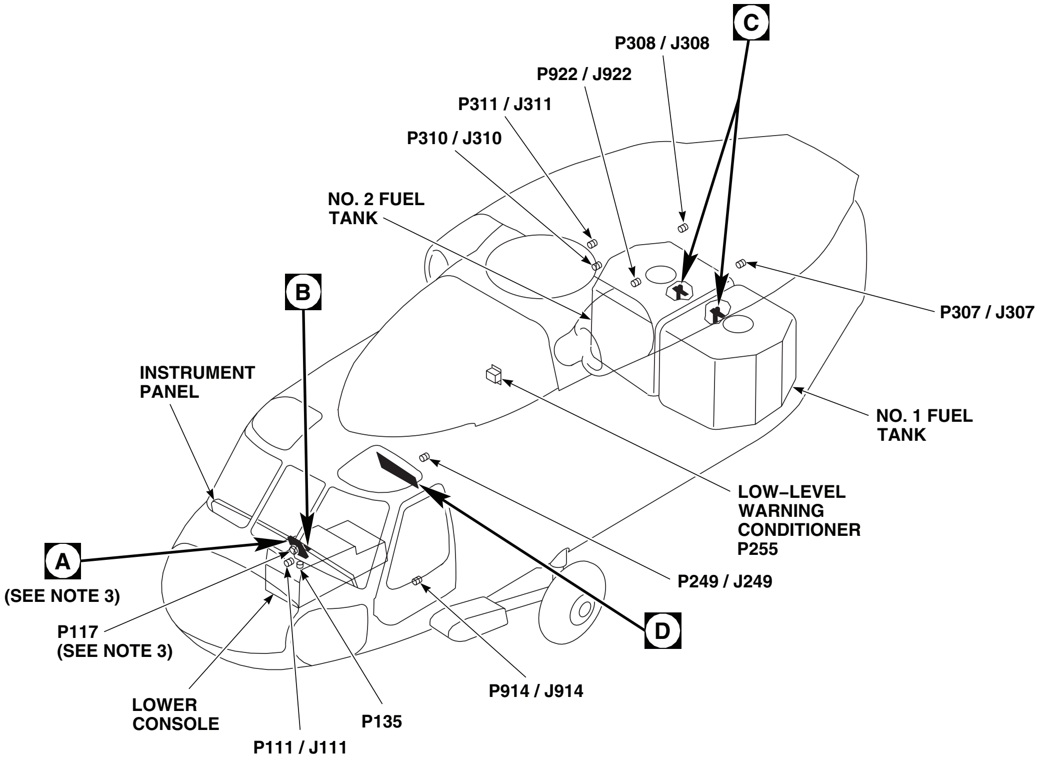
4. Check if FUEL LOW capsule lamp or **HH-60A HH-60L** FUEL legends on MFDs **are good.**

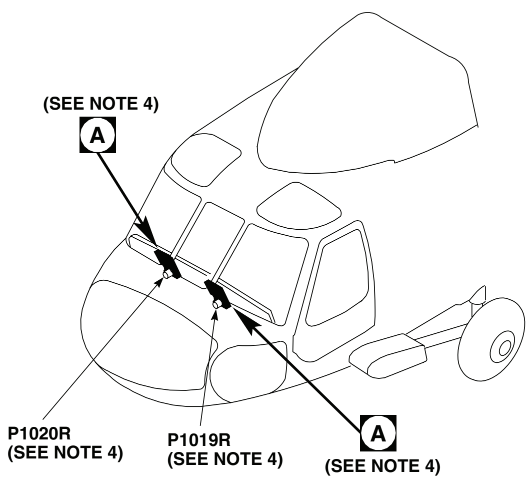
a. If lamp(s) or legends are good, troubleshoot caution/advisory warning system (WP 0090 00) or **HH-60A HH-60L** MFDs (WP 0100 00). Go to **5.**

b. If lamp(s) or legends are not good, replace lamp(s) (WP 0786 00) or MFD (WP 0799 00). Go to **5.**

5. Procedure completed.

LOCATION





TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P117 / J117	CAUTION / ADVISORY PANEL (SEE NOTE 3)
P135	MISCELLANEOUS SWITCH PANEL
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P255	LOW–LEVEL WARNING CONDITIONER
P307 / J307	CABIN CEILING, BL 15 LH, STA 430
P308 / J308	CABIN CEILING, BL 15 RH, STA 430
P310 / J310	CABIN CEILING, BL 17 RH, STA 387
P311 / J311	CABIN CEILING, BL 16.5 LH, STA 383
P914 / J914	COCKPIT, BL 24 LH, STA 247
P922 / J922	CABIN CEILING, BL 10.7 LH, STA 380
P1019R / J3	COPILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 4)
P1020R / J3	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 4)

NOTES

1. UH–60A UH–60L HH–60A HH–60L
2. EH–60A
3. UH–60A UH–60L EH–60A
4. HH–60A HH–60L

AA2672_1A
SA

Figure 1. Fuel Low Level Warning System Location Diagram. (Sheet 1 of 3)

LOCATION - Continued

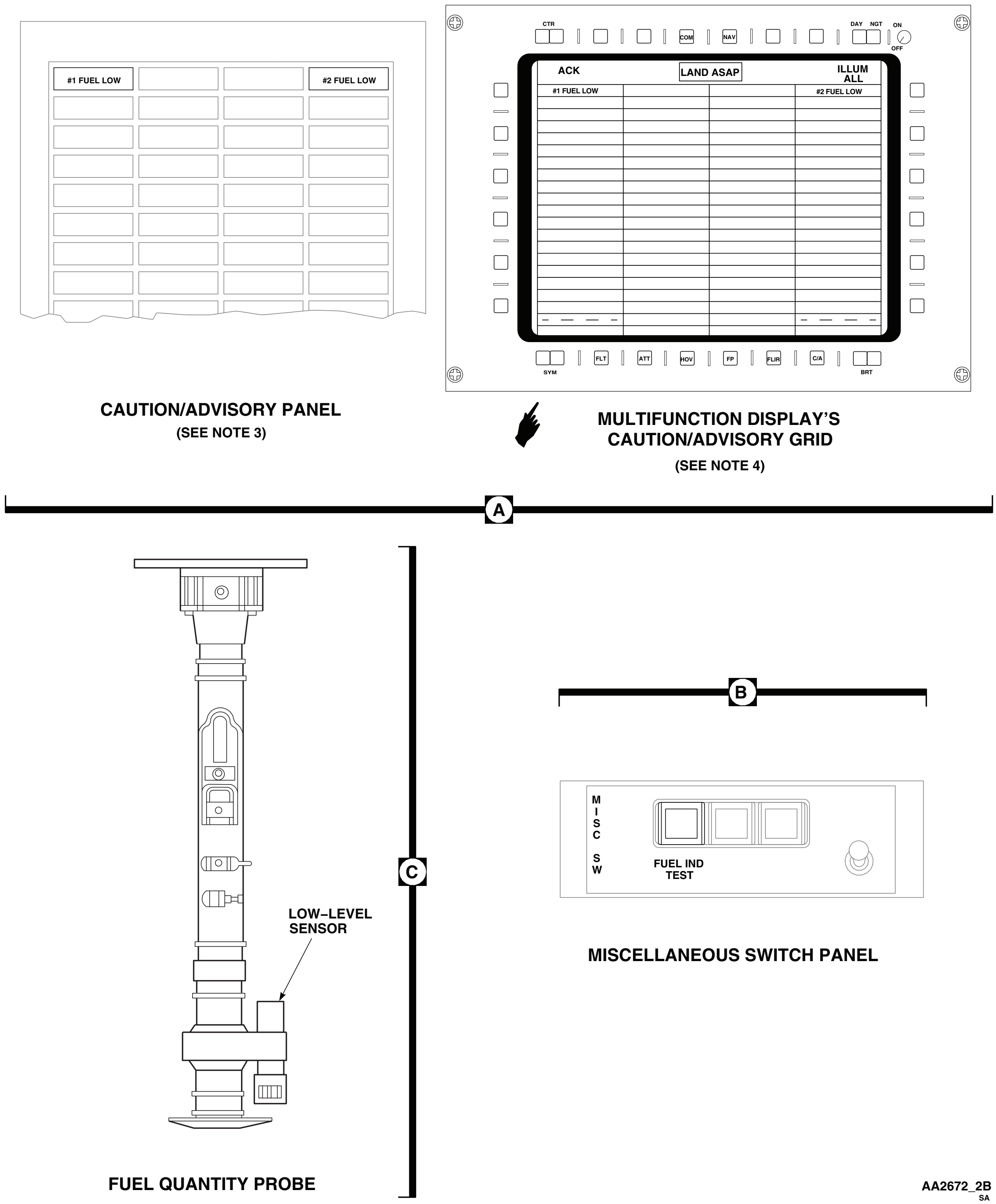


Figure 1. Fuel Low Level Warning System Location Diagram. (Sheet 2 of 3)

LOCATION - Continued

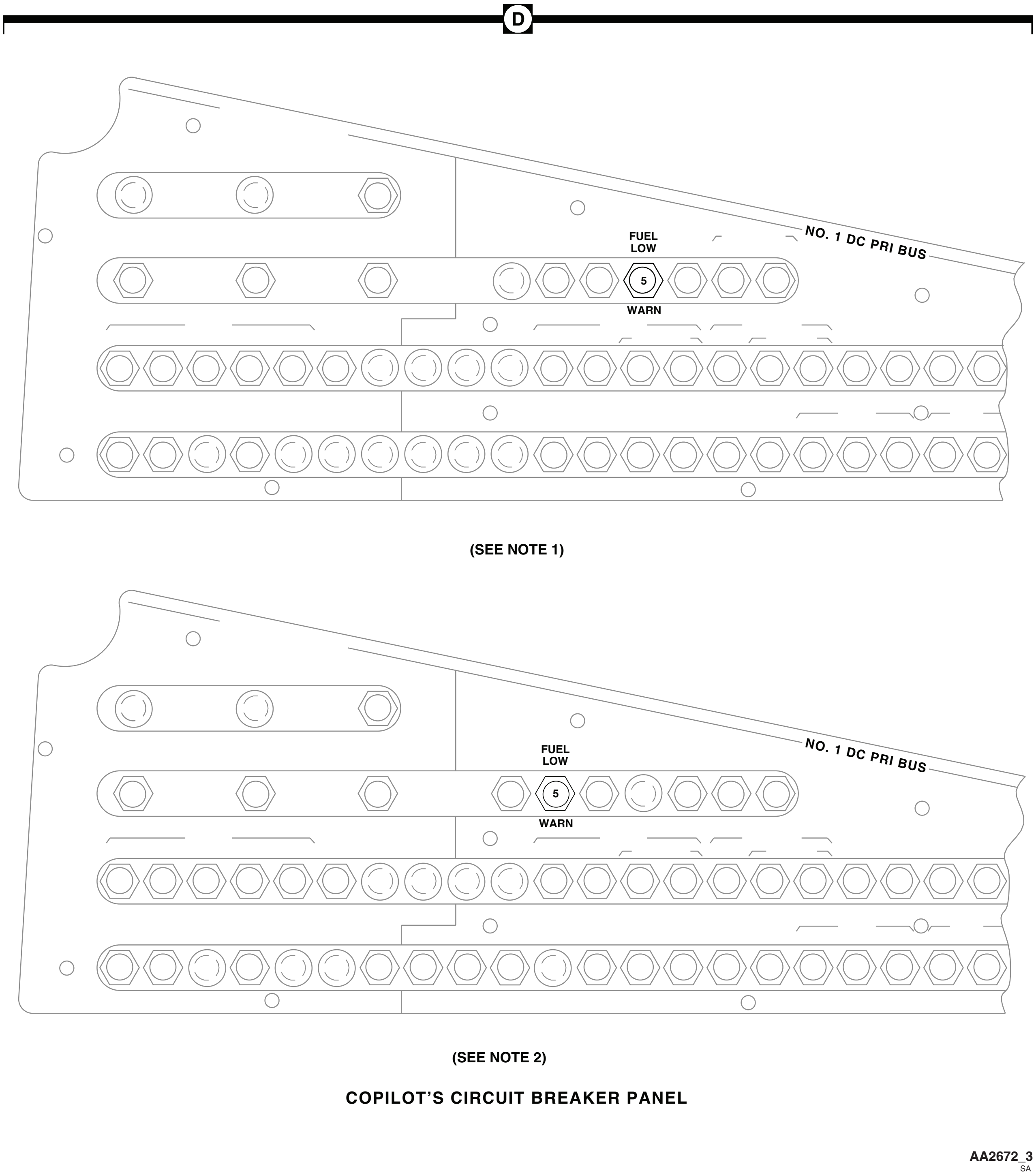


Figure 1. Fuel Low Level Warning System Location Diagram. (Sheet 3 of 3)

SCHEMATICS

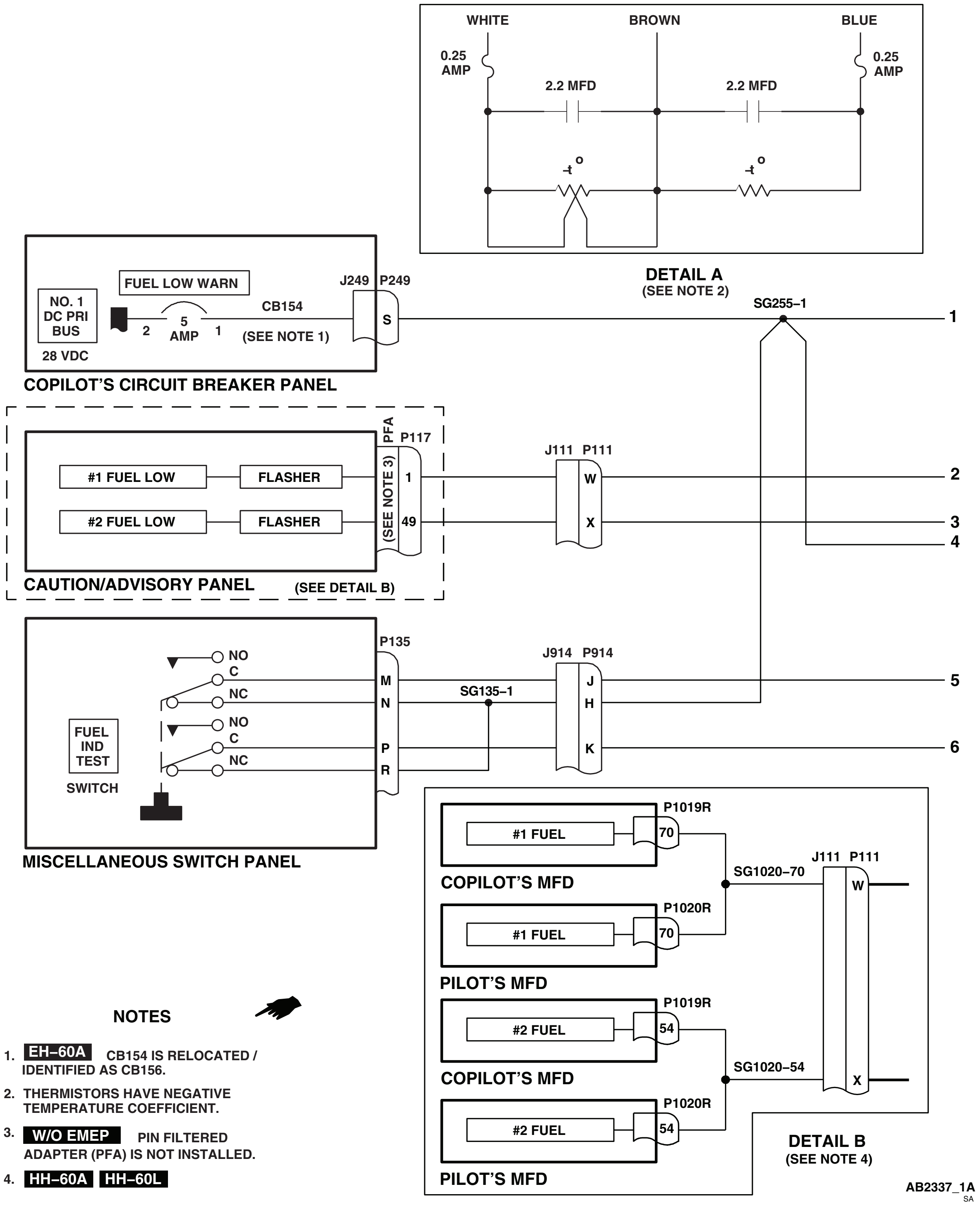


Figure 2. Fuel Low Level Warning System Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

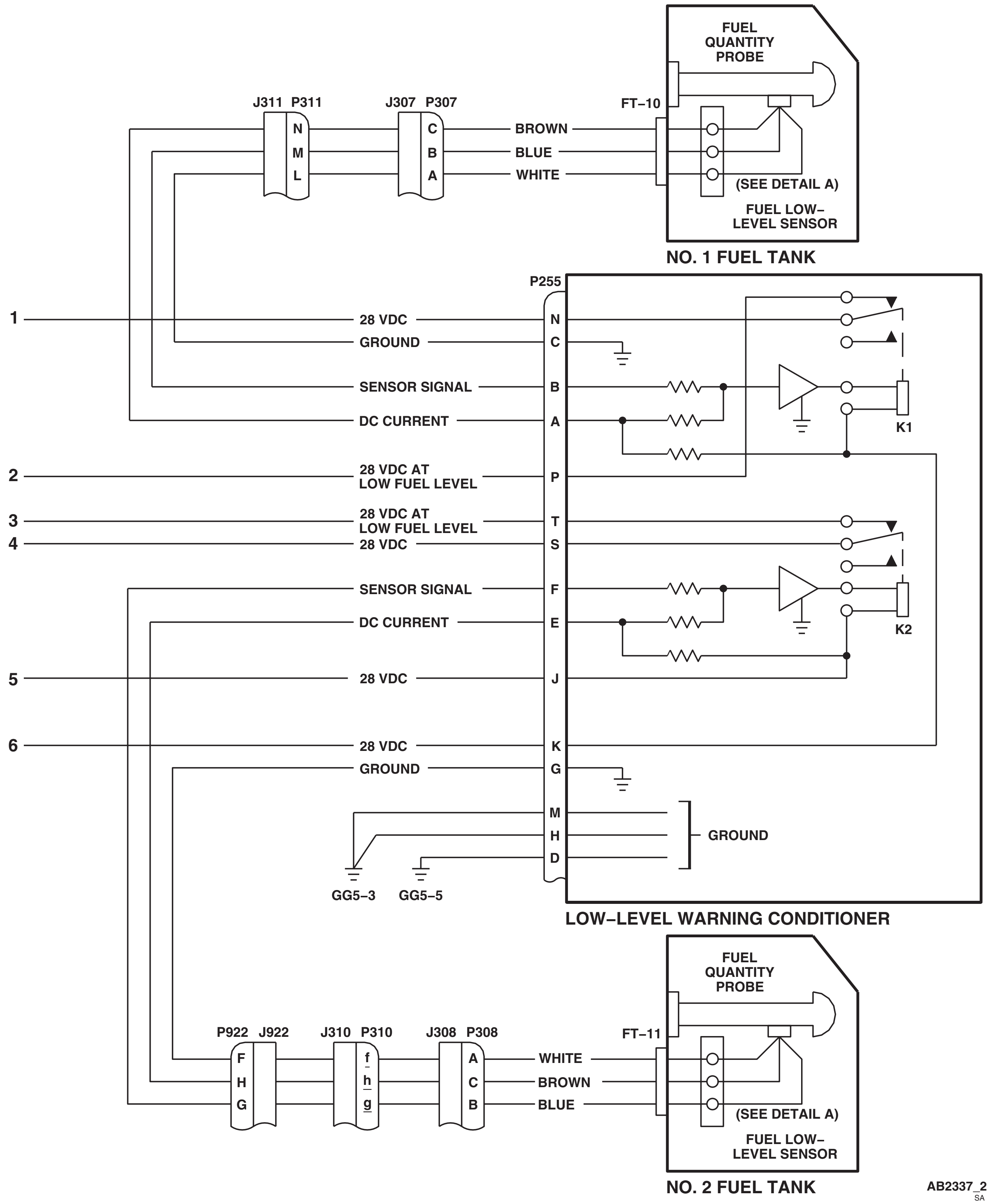


Figure 2. Fuel Low Level Warning System Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

FUEL SYSTEM

SUBMERGED FUEL BOOST PUMP SYSTEM

This WP supersedes WP 0128 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer Toolkit, SC 5180-99-B06

Material/Parts
Suitable Container

Personnel Required
Aircraft Electrician MOS 15F (1)
Tactical Transport Helicopter Repair MOS 15T (1)

References
TM 1-1520-237-10
TM 1-2840-248-23
WP 0104 00
WP 0110 00
WP 0509 00
WP 0510 00
WP 0824 00
WP 0866 00
WP 0868 00
WP 0973 00
WP 0974 00
WP 0975 00
WP 0984 00
WP 0986 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10
No. 1 and No. 2 Engines Primed, TM 1-1520-237-10
No. 1 and No. 2 Fuel Tanks Serviced, TM 1-1520-237-10

SUBMERGED FUEL BOOST PUMP SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
UH-60A UH-60L HH-60A HH-60L	
NO. 1 ENG WARN LTS	Copilot's

SUBMERGED FUEL BOOST PUMP SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
NO. 2 ENG WARN LTS	Pilot's
LIGHTS ADVSY	Pilot's
NO. 1 FUEL BOOST PUMP	Mission Readiness
NO. 2 FUEL BOOST PUMP	Mission Readiness
EH-60A	
NO. 1 ENG WARN LTS	Copilot's
NO. 1 FUEL BOOST PUMP	Copilot's
NO. 2 ENG WARN LTS	Pilot's
NO. 2 FUEL BOOST PUMP	Pilot's
LIGHTS ADVSY	Pilot's

2. Turn on electrical power. **HH-60A HH-60L** Turn pilot's and copilot's multifunction display (MFD) switches ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
3. Place engine controls quadrant NO. 1 ENG FUEL SYS selector handle to DIR.
4. Place lower console NO. 1 PUMP switch to ON.

Indication/Condition

a. NO. 1 PUMP pressure light shall go on.

b. #1 FUEL PRESS (caution/advisory panel capsule or MFD legends) shall go off.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, go to NO. 1 PUMP LIGHT DID NOT GO ON AND #1 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF, in this work package.
2. If Indication/Condition a. is not as specified, go to #1 FUEL PRESS CAPSULE OR LEGENDS GO OFF, BUT NO. 1 PUMP LIGHT DOES NOT GO ON, in this work package.
3. If Indication/Condition b. is not as specified, go to NO. 1 PUMP LIGHT GOES ON, BUT #1 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF, in this work package.

Step

5. Move engine controls quadrant NO. 1 ENG FUEL SYS selector handle to OFF.
6. Place lower console NO. 1 PUMP switch to OFF.

Indication/Condition

NO. 1 PUMP pressure light shall go off.

SUBMERGED FUEL BOOST PUMP SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

If Indication/Condition is not as specified, go to NO. 1 PUMP LIGHT DOES NOT GO OFF WHEN NO. 1 PUMP SWITCH IS PLACED TO OFF, in this work package.

Step

7. Place engine controls quadrant NO. 2 ENG FUEL SYS selector handle to DIR.
8. Place lower console NO. 2 PUMP switch to ON.

Indication/Condition

a. NO. 2 PUMP pressure light shall go on.

b. #2 FUEL PRESS (caution/advisory panel capsule or MFD legends) shall go off.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, go to NO. 2 PUMP LIGHT DID NOT GO ON AND #2 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF, in this work package.
2. If Indication/Condition a. is not as specified, go to #2 FUEL PRESS CAPSULE OR LEGENDS GO OFF, BUT NO. 2 PUMP LIGHT DOES NOT GO ON, in this work package.
3. If Indication/Condition b. is not as specified, go to NO. 2 PUMP LIGHT GOES ON, BUT #2 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF, in this work package.

Step

9. Place engine controls quadrant NO. 2 ENG FUEL SYS selector handle to OFF.
10. Place lower console NO. 2 PUMP switch to OFF.

Indication/Condition

NO. 2 PUMP pressure light shall go off.

Corrective Action

If Indication/Condition is not as specified, go to NO. 2 PUMP PRESSURE LIGHT DOES NOT GO OFF WHEN NO. 2 PUMP SWITCH IS PLACED OFF, in this work package.

Step

11. Place engine controls quadrant NO. 1 ENG FUEL SYS selector handle to XFD.
12. Place lower console NO. 2 PUMP switch to ON.

Indication/Condition

a. NO. 2 PUMP pressure light shall go on.

b. #1 FUEL PRESS (caution/advisory panel capsule or MFD legends) shall go off.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, go to NO. 2 PUMP LIGHT DID NOT GO ON AND #1 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF, in this work package.
2. If Indication/Condition a. is not as specified, go to #1 FUEL PRESS CAPSULE OR LEGENDS GO OFF, BUT NO. 2 PUMP LIGHT DOES NOT GO ON, in this work package.

SUBMERGED FUEL BOOST PUMP SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

3. If Indication/Condition b. is not as specified, go to NO. 2 PUMP LIGHT GOES ON, BUT #1 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF, in this work package.

Step

13. Move engine controls quadrant NO. 1 ENG FUEL SYS selector handle to OFF.
14. Place lower console NO. 2 PUMP switch to OFF.
15. Place engine controls quadrant NO. 2 ENG FUEL SYS selector handle to XFD.
16. Place lower console NO. 1 PUMP switch to ON.

Indication/Condition

a. NO. 1 PUMP pressure light shall go on.

b. #2 FUEL PRESS (caution/advisory panel capsule or MFD legends) shall go off.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, go to NO. 1 PUMP LIGHT DID NOT GO ON AND #2 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF, in this work package.
2. If Indication/Condition a. is not as specified, go to #2 FUEL PRESS CAPSULE OR LEGENDS GO OFF, BUT NO. 1 PUMP LIGHT DOES NOT GO ON, in this work package.
3. If Indication/Condition b. is not as specified, go to NO. 1 PUMP LIGHT GOES ON, BUT #2 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF, in this work package.

Step

17. Move engine controls quadrant NO. 2 ENG FUEL SYS selector handle to OFF.
18. Place lower console NO. 1 PUMP switch to OFF.
19. Turn MFD switches OFF.
20. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

NO. 1 PUMP LIGHT DID NOT GO ON AND #1 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF

SYMPTOM

NO. 1 PUMP light did not go on and #1 FUEL PRESS capsule or legends did not go off.

MALFUNCTION

NO. 1 PUMP light did not go on and #1 FUEL PRESS capsule or legends did not go off.

CORRECTIVE ACTION

1. With NO. 1 PUMP switch at ON, check for 115 vac between:

- **J322-A and J322-D**
- **J322-B and J322-D**

NO. 1 PUMP LIGHT DID NOT GO ON AND #1 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF - Continued

- **J322-C and J322-D**

a. If voltage is as specified, replace No. 1 boost pump (WP 0973 00). Go to **6.**

b. If voltage is not as specified, go to **2.**

2. Check continuity between J322-D and ground.

a. If continuity is present, go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**

3. Check continuity between:

- **P139-G and J322-A**
- **P139-H and J322-B**
- **P139-J and J322-C**

a. If continuity is present, go to **4.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**

4. Check for 115 vac between:

- **P139-V and ground**
- **P139-U and ground**
- **P139-T and ground**

a. If voltage is as specified, replace fuel boost pump control panel (WP 0866 00). Go to **6.**

b. If voltage is not as specified, go to **5.**

5. Check for 115 vac between NO. 1 FUEL BOOST PUMP circuit breaker:

- **Terminal A1 and ground**
- **Terminal B1 and ground**
- **Terminal C1 and ground**

a. If voltage is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **6:**

- Terminal A1 andP139-V
- Terminal B1 andP139-U
- Terminal C1 andP139-T

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6.**

6. Procedure completed.

#1 FUEL PRESS CAPSULE OR LEGENDS GO OFF, BUT NO. 1 PUMP LIGHT DOES NOT GO ON

SYMPTOM

#1 FUEL PRESS capsule or legends go off, but NO. 1 PUMP light does not go on.

MALFUNCTION

#1 FUEL PRESS capsule or legends go off, but NO. 1 PUMP light does not go on.

CORRECTIVE ACTION

1. **Press face of NO. 1 PUMP light. Check that light goes on.**
   a. If light goes on, go to **2.**
   b. If light does not go on, go to **5.**
2. **With NO. 1 PUMP switch at ON, check continuity between DS1 contacts 2 and 1.**
   a. If continuity is present, go to **3.**
   b. If continuity is not present, replace pressure switch (WP 0975 00). Go to **9.**
3. **Check continuity between P321-2 and ground.**
   a. If continuity is present, go to **4.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**
4. **Check continuity between P139-M and P321-1.**
   a. If continuity is present, replace fuel boost pump control panel (WP 0866 00). Go to **9.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**
5. **Check NO. 1 PUMP switch lamp.**
   a. If lamp is good, go to **6.**
   b. If lamp is not good, replace lamp (WP 0868 00). Go to **9.**
6. **Make sure caution/advisory panel or HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch is in BRT mode. Go to 7.**
7. **Check for 28 vdc between P139-R and ground.**
   a. If voltage is as specified, replace fuel boost pump control panel (WP 0866 00). Go to **9.**
   b. If voltage is not as specified, go to **8.**
8. **Check continuity between P139-R and P902-X.**
   a. If continuity is present, troubleshoot instrument panel and consoles indicator lights dimming (WP 0110 00). Go to **9.**
   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9.**
9. **Procedure completed.**

NO. 1 PUMP LIGHT GOES ON, BUT #1 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF

SYMPTOM

NO. 1 PUMP light goes on, but #1 FUEL PRESS capsule or legends did not go off.

MALFUNCTION

NO. 1 PUMP light goes on, but #1 FUEL PRESS capsule or legends did not go off.

CORRECTIVE ACTION

1. **Place suitable container beneath left side of helicopter fuselage at No. 1 engine overboard drain. Go to2.**
2. **Make sure NO. 1 ENG WARN LTS circuit breaker is pushed in. Go to 3.**
3. **Move engine controls quadrant NO. 1 ENG FUEL SYS selector handle to DIR. Go to 4.**
4. **Place ENG POWER CONT lever to LOCKOUT. Go to 5.**
5. **Check for fuel flow overboard.**
   a. If fuel flow is present, go to **6.**
   b. If fuel flow is not present, go to **8.**
6. **Return engine controls quadrant NO. 1 ENG FUEL SYS selector handle and ENG POWER CONT lever to OFF. Go to 7.**
7. **Replace fuel pressure sensor (TM 1-2840-248-23). Go to 10.**
8. **Return engine controls quadrant NO. 1 ENG FUEL SYS selector handle and ENG POWER CONT lever to OFF. Go to 9.**
9. **Replace check valve (WP 0974 00). Go to 10.**
10. **Procedure completed.**

NO. 1 PUMP LIGHT DOES NOT GO OFF WHEN NO. 1 PUMP SWITCH IS PLACED TO OFF

SYMPTOM

NO. 1 PUMP light does not go off when NO. 1 PUMP switch is placed to off.

MALFUNCTION

NO. 1 PUMP light does not go off when NO. 1 PUMP switch is placed to off.

CORRECTIVE ACTION

1. **With NO. 1 PUMP switch at OFF, place suitable container beneath left side of helicopter fuselage at No. 1 engine overboard drain. Go to 2.**
2. **Make sure NO. 1 ENG WARN LTS circuit breaker is pushed in. Go to 3.**
3. **Move engine controls quadrant NO. 1 ENG FUEL SYS selector handle to DIR. Go to 4.**
4. **Place ENG POWER CONT lever to LOCKOUT. Fuel should not flow overboard.**
   a. If fuel does not flow overboard, go to **5.**
   b. If fuel does flow overboard, go to **9.**

NO. 1 PUMP LIGHT DOES NOT GO OFF WHEN NO. 1 PUMP SWITCH IS PLACED TO OFF - Continued

5. **Return engine controls quadrant NO. 1 ENG FUEL SYS selector handle to OFF. Go to 6.**
6. **Check continuity between pressure switch terminals 2 and 3.**
   a. If continuity is present, go to **7.**
   b. If continuity is not present, replace pressure switch (WP 0975 00). Go to **11.**
7. **Connect P321 to pressure switch. Go to 8.**
8. **Check continuity between P139-M and ground.**
   a. If continuity is present, repair/replace wiring (WP 1747 00). Go to **11.**
   b. If continuity is not present, replace fuel boost pump control panel (WP 0866 00). Go to **11.**
9. **Return engine controls quadrant NO. 1 ENG FUEL SYS selector handle to OFF. Go to 10.**
10. **Replace fuel boost pump control panel (WP 0866 00). Go to 11.**
11. **Procedure completed.**

NO. 2 PUMP LIGHT DID NOT GO ON AND #2 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF

SYMPTOM

NO. 2 PUMP light did not go on and #2 FUEL PRESS capsule or legends did not go off.

MALFUNCTION

NO. 2 PUMP light did not go on and #2 FUEL PRESS capsule or legends did not go off.

CORRECTIVE ACTION

1. **With NO. 2 PUMP switch at ON, check for 115 vac between:**
   - **J323-A and J323-D**
   - **J323-B and J323-D**
   - **J323-C and J323-D**

   a. If voltage is as specified, replace No. 2 boost pump (WP 0973 00). Go to **6.**
   b. If voltage is not as specified, go to **2.**
2. **Check continuity between J323-D and ground.**
   a. If continuity is present, go to **3.**
   b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**
3. **Check continuity between:**
   - **J323-A and P139-D**
   - **J323-B and P139-E**
   - **J323-C and P139-F**

   a. If continuity is present, go to **4.**
   b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**

NO. 2 PUMP LIGHT DID NOT GO ON AND #2 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF - Continued

4. **Check for 115 vac between:**
   - **P139-A and ground**
   - **P139-B and ground**
   - **P139-C and ground**

   a. If voltage is as specified, replace fuel boost pump control panel (WP 0866 00). Go to **6.**

   b. If voltage is not as specified, go to **5.**

5. **Check for 115 vac between NO. 2 FUEL BOOST PUMP circuit breaker:**
   - **Terminal A1 and ground**
   - **Terminal B1 and ground**
   - **Terminal C1 and ground**

   a. If voltage is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **6:**
      - Terminal A1 and P139-C
      - Terminal B1 and P139-B
      - Terminal C1 and P139-A

   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6.**

6. **Procedure completed.**

#2 FUEL PRESS CAPSULE OR LEGENDS GO OFF, BUT NO. 2 PUMP LIGHT DOES NOT GO ON

SYMPTOM

#2 FUEL PRESS capsule or legends go off, but NO. 2 PUMP light does not go on.

MALFUNCTION

#2 FUEL PRESS capsule or legends go off, but NO. 2 PUMP light does not go on.

CORRECTIVE ACTION

1. **With NO. 2 PUMP switch at ON, press face of NO. 2 PUMP light. Go to 2.**
2. **Check that NO. 2 PUMP light goes on.**

   a. If light is on, go to **3.**

   b. If light is not on, go to **6.**

3. **Check continuity between NO. 2 PUMP switch contacts 2 and 1.**

   a. If continuity is present, go to **4.**

   b. If continuity is not present, replace pressure switch (WP 0975 00). Go to **10.**

4. **Check continuity between P320-2 and ground.**

   a. If continuity is present, go to **5.**

   b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

5. **Check continuity between P139-N and P320-1.**

#2 FUEL PRESS CAPSULE OR LEGENDS GO OFF, BUT NO. 2 PUMP LIGHT DOES NOT GO ON - Continued

a. If continuity is present, replace fuel boost pump control panel (WP 0866 00). Go to **10.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

6. **Check NO. 2 PUMP switch lamp.**

a. If lamp is good, go to **7.**

b. If lamp is not good, replace lamp (WP 0868 00). Go to **10.**

7. **Make sure caution/advisory panel or HH-60A HH-60L instrument panel INDICATOR LTS [redacted] BRT/DIM-TEST switch is in BRT mode. Go to 8.**

8. **Check for 28 vdc between P139-P and ground.**

a. If voltage is as specified, replace fuel boost pump control panel (WP 0866 00). Go to **10.**

b. If voltage is not as specified, go to **9.**

9. **Check continuity between P139-P and P902-X.**

a. If continuity is present, troubleshoot instrument panel and consoles indicator lights dimming (WP 0110 00). Go to **10.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

10. **Procedure completed.**

NO. 2 PUMP LIGHT GOES ON, BUT #2 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF

SYMPTOM

NO. 2 PUMP light goes on, but #2 FUEL PRESS capsule or legends did not go off.

MALFUNCTION

NO. 2 PUMP light goes on, but #2 FUEL PRESS capsule or legends did not go off.

CORRECTIVE ACTION

1. **Place suitable container beneath right side of helicopter fuselage at No. 2 engine overboard drain. Go to 2.**

2. **Make sure NO. 2 ENG WARN LTS circuit breaker is pushed in. Go to 3.**

3. **Move engine controls quadrant NO. 2 ENG FUEL SYS selector handle to DIR. Go to 4.**

4. **Place ENG POWER CONT lever to LOCKOUT. Go to 5.**

5. **Check for fuel flow overboard.**

a. If fuel flows overboard, go to **6.**

b. If fuel does not flow overboard, go to **8.**

6. **Return engine controls quadrant NO. 2 ENG FUEL SYS selector handle and ENG POWER CONT lever to OFF. Go to 7.**

7. **Replace pressure switch (WP 0975 00). Go to 10.**

8. **Return engine controls quadrant NO. 2 ENG FUEL SYS selector handle and ENG POWER CONT lever to OFF. Go to 9.**

9. **Replace check valve (WP 0974 00). Go to 10.**

NO. 2 PUMP LIGHT GOES ON, BUT #2 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF - Continued

10. **Procedure completed.**

NO. 2 PUMP PRESSURE LIGHT DOES NOT GO OFF WHEN NO. 2 PUMP SWITCH IS PLACED OFF

SYMPTOM

NO. 2 PUMP pressure light does not go off when NO. 2 PUMP switch is placed OFF.

MALFUNCTION

NO. 2 PUMP pressure light does not go off when NO. 2 PUMP switch is placed OFF.

CORRECTIVE ACTION

1. **With NO. 2 PUMP switch at OFF, place suitable container beneath right side of helicopter fuselage at No. 2 engine overboard drain. Go to 2.**
2. **Make sure NO. 2 ENG WARN LTS circuit breaker is pushed in. Go to 3.**
3. **Move engine controls quadrant NO. 2 ENG FUEL SYS selector handle to DIR. Go to 4.**
4. **Place ENG POWER CONT lever to LOCKOUT. Go to 5.**
5. **Check that fuel does not flow overboard.**
   a. If fuel does not flow overboard, go to **6.**
   b. If fuel flows overboard, go to **10.**
6. **Return engine controls quadrant NO. 2 ENG FUEL SYS fuel selector handle to OFF. Go to 7.**
7. **Check continuity between No. 2 boost pump pressure switch terminals 2 and 3.**
   a. If continuity is present, go to **8.**
   b. If continuity is not present, replace pressure switch (WP 0975 00). Go to **12.**
8. **Connect P320 to pressure switch. Go to 9.**
9. **Check continuity between P139-N and ground.**
   a. If continuity is present, repair/replace wiring (WP 1747 00). Go to **12.**
   b. If continuity is not present, replace fuel boost pump control panel (WP 0866 00). Go to **12.**
10. **Return engine controls quadrant NO. 2 ENG FUEL SYS fuel selector handle to OFF. Go to 11.**
11. **Replace fuel boost pump control panel (WP 0866 00). Go to 12.**
12. **Procedure completed.**

NO. 2 PUMP LIGHT DID NOT GO ON AND #1 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF

SYMPTOM

NO. 2 PUMP light did not go on and #1 FUEL PRESS capsule or legends did not go off.

MALFUNCTION

NO. 2 PUMP light did not go on and #1 FUEL PRESS capsule or legends did not go off.

CORRECTIVE ACTION

1. **With lower console NO. 2 PUMP switch at ON, check for 115 vac between:**
   - **J323-A and J323-D**
   - **J323-B and J323-D**
   - **J323-C and J323-D**

   a. If voltage is as specified, replace No. 2 boost pump (WP 0973 00). Go to **6.**

   b. If voltage is not as specified, go to **2.**

2. **Check continuity between J323-D and ground.**

   a. If continuity is present, go to **3.**

   b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**

3. **Check continuity between:**
   - **J323-A and P139-D**
   - **J323-B and P139-E**
   - **J323-C and P139-F**

   a. If continuity is present, go to **4.**

   b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**

4. **Check for 115 vac between:**
   - **P139-A and ground**
   - **P139-B and ground**
   - **P139-B and ground**

   a. If voltage is as specified, replace fuel boost pump control panel (WP 0866 00). Go to **6.**

   b. If voltage is not as specified, go to **5.**

5. **Check for 115 vac between NO. 2 FUEL BOOST PUMP circuit breaker:**
   - **Terminal A1 and ground**
   - **Terminal B1 and ground**
   - **Terminal C1 and ground**

   a. If voltage is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **6:**
      - Terminal A1 and P139-C

NO. 2 PUMP LIGHT DID NOT GO ON AND #1 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF - Continued

- Terminal B1 and P139-B
- Terminal C1 and P139-A

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6.**

6. Procedure completed.

#1 FUEL PRESS CAPSULE OR LEGENDS GO OFF, BUT NO. 2 PUMP LIGHT DOES NOT GO ON

SYMPTOM

#1 FUEL PRESS capsule or legends go off, but NO. 2 PUMP light does not go on.

MALFUNCTION

#1 FUEL PRESS capsule or legends go off, but NO. 2 PUMP light does not go on.

CORRECTIVE ACTION

1. With lower console NO. 2 PUMP switch at ON, press face of NO. 2 PUMP light. Go to 2.

2. Check that NO. 2 PUMP light goes on.

a. If light is on, go to **3.**

b. If light is not on, go to **7.**

3. Disconnect P320 and connect jumper between P320-1 and P320-2. Go to 4.

4. Check that NO. 2 PUMP light goes on.

a. If light is on, replace No. 2 boost pump pressure switch (WP 0975 00). Go to **11.**

b. If light is not on, go to **5.**

5. Check continuity between P320-2 and ground.

a. If continuity is present, go to **6.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **11.**

6. Check continuity between P139-N and P320-1.

a. If continuity is present, replace fuel boost pump control panel (WP 0866 00). Go to **11.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **11.**

7. Check NO. 2 PUMP switch lamp.

a. If lamp is good, go to **8.**

b. If lamp is not good, replace lamp (WP 0868 00). Go to **11.**

8. Make sure caution/advisory panel or HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch is in BRT mode. Go to 9.

9. Check for 28 vdc between P139-P and ground.

a. If voltage is as specified, replace fuel boost pump control panel (WP 0866 00). Go to **11.**

b. If voltage is not as specified, go to **10.**

10. Check continuity between P139-P and P902-X.

#1 FUEL PRESS CAPSULE OR LEGENDS GO OFF, BUT NO. 2 PUMP LIGHT DOES NOT GO ON - Continued

a. If continuity is present, troubleshoot instrument panel and consoles indicator lights dimming (WP 0110 00). Go to **11.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **11.**

11. Procedure completed.

NO. 2 PUMP LIGHT GOES ON, BUT #1 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF

SYMPTOM

NO. 2 PUMP light goes on, but #1 FUEL PRESS capsule or legends did not go off.

MALFUNCTION

NO. 2 PUMP light goes on, but #1 FUEL PRESS capsule or legends did not go off.

CORRECTIVE ACTION

1. Place suitable container beneath left side of helicopter fuselage at No. 1 engine overboard drain. Go to 2.

2. Make sure NO. 1 ENG WARN LTS circuit breaker is pushed in. Go to 3.

3. Move engine controls quadrant NO. 1 ENG FUEL SYS selector handle to XFD. Go to 4.

4. Place lower console NO. 2 PUMP switch to ON and place ENG POWER CONT lever to LOCKOUT. Go to5.

5. Check for fuel flow overboard.

a. If fuel flow is present, go to **6.**

b. If fuel flow is not present, go to **8.**

6. Return engine controls quadrant NO. 1 ENG FUEL SYS selector handle, ENG POWER CONT lever, and lower console NO. 2 PUMP switch to OFF. Go to 7.

7. Replace pressure switch on No. 1 engine (WP 0975 00). Go to 13.

8. Return engine controls quadrant NO. 1 ENG FUEL SYS selector handle and lower console NO. 2 PUMP switch to OFF. Go to 9.

9. Place engine controls quadrant NO. 1 ENG FUEL SYS selector handle in DIR and XFD positions and observe that push/pull cable actuates fuel selector valve.

a. If fuel selector valve is actuated, go to **10.**

b. If fuel selector valve is not actuated, replace push/pull cable (WP 0509 00). Go to **13.**

c. If trouble remains, replace control box (WP 0510 00). Go to **13.**

10. Using empty 1 gallon container to catch fuel, disconnect fuel hose from breakaway tee valve. Go to 11.

11. Momentarily place lower console NO. 2 PUMP switch to ON. Go to 12.

12. Observe that fuel flows from fuel line.

a. If fuel does not flow from fuel line, replace check valve (WP 0984 00). Go to **13.**

b. If fuel does flow from fuel line, replace fuel selector valve (WP 0986 00). Go to **13.**

13. Procedure completed.

NO. 1 PUMP LIGHT DID NOT GO ON AND #2 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF

SYMPTOM

NO. 1 PUMP light did not go on and #2 FUEL PRESS capsule or legends did not go off.

MALFUNCTION

NO. 1 PUMP light did not go on and #2 FUEL PRESS capsule or legends did not go off.

CORRECTIVE ACTION

1. With NO. 1 PUMP switch ON, check for 115 vac between:

- **J322-A and J322-D**
- **J322-B and J322-D**
- **J322-C and J322-D**

a. If voltage is as specified, replace No. 1 boost pump (WP 0973 00). Go to **6.**

b. If voltage is not as specified, go to **2.**

2. Check continuity between J322-D and ground.

a. If continuity is present, go to **3.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**

3. Check continuity between:

- **P139-G and J322-A**
- **P139-H and J322-B**
- **P139-J and J322-C**

a. If continuity is present, go to **4.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**

4. Check for 115 vac between:

- **P139-V and ground**
- **P139-U and ground**
- **P139-T and ground**

a. If voltage is as specified, replace fuel boost pump control panel (WP 0866 00). Go to **6.**

b. If voltage is not as specified, go to **5.**

5. Check for 115 vac between NO. 1 FUEL BOOST PUMP circuit breaker:

- **Terminal A1 and ground**
- **Terminal B1 and ground**
- **Terminal C1 and ground**

a. If voltage is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **6:**

- Terminal A1 and P139-V

NO. 1 PUMP LIGHT DID NOT GO ON AND #2 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF - Continued

- Terminal B1 and P139-U
- Terminal C1 and P139-T

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6.**

6. Procedure completed.

#2 FUEL PRESS CAPSULE OR LEGENDS GO OFF, BUT NO. 1 PUMP LIGHT DOES NOT GO ON

SYMPTOM

#2 FUEL PRESS capsule or legends go off, but NO. 1 PUMP light does not go on.

MALFUNCTION

#2 FUEL PRESS capsule or legends go off, but NO. 1 PUMP light does not go on.

CORRECTIVE ACTION

1. Press face of NO. 1 PUMP light. Check that light goes on.

a. If light goes on, go to **2.**

b. If light does not go on, go to **6.**

2. Disconnect P321 and connect jumper between P321-2 and P321-1. Go to 3.

3. Check that NO. 1 PUMP light goes on.

a. If light goes on, go to **4.**

b. If light does not go on, replace No. 1 boost pump pressure switch (WP 0975 00). Go to **10.**

4. Check continuity between P321-2 and ground.

a. If continuity is present, go to **5.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

5. Check continuity between P139-M and P321-1.

a. If continuity is present, replace fuel boost pump control panel (WP 0866 00). Go to **10.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

6. Check NO. 1 PUMP switch lamp.

a. If lamp is good, go to **7.**

b. If lamp is not good, replace lamp (WP 0868 00). Go to **10.**

7. Make sure caution/advisory panel or HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch is in BRT mode. Go to 8.

8. Check for 28 vdc between P139-R and ground.

a. If voltage is as specified, replace fuel boost pump control panel (WP 0866 00). Go to **10.**

b. If voltage is not as specified, go to **9.**

9. Check continuity between P139-R and P902-X.

a. If continuity is present, troubleshoot instrument panel and consoles indicator lights dimming (WP 0110 00). Go to **10.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10.**

#2 FUEL PRESS CAPSULE OR LEGENDS GO OFF, BUT NO. 1 PUMP LIGHT DOES NOT GO ON - Continued

10. **Procedure completed.**

NO. 1 PUMP LIGHT GOES ON, BUT #2 FUEL PRESS CAPSULE OR LEGENDS DID NOT GO OFF

SYMPTOM

NO. 1 PUMP light goes on, but #2 FUEL PRESS capsule or legends did not go off.

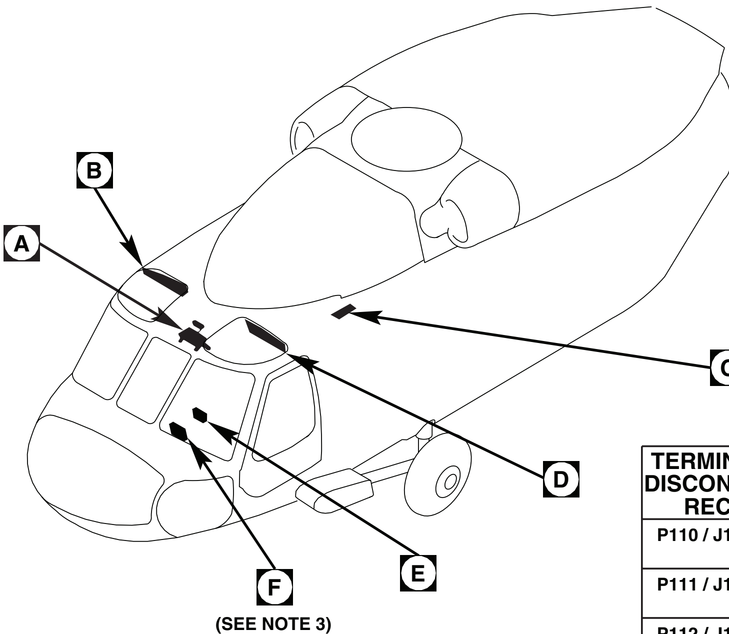
MALFUNCTION

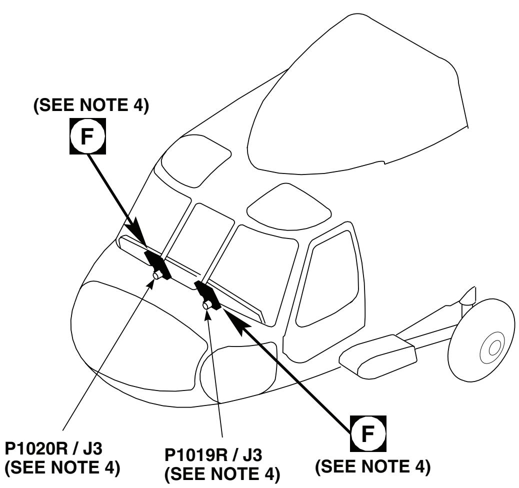
NO. 1 PUMP light goes on, but #2 FUEL PRESS capsule or legends did not go off.

CORRECTIVE ACTION

1. **Place suitable container beneath left side of helicopter fuselage at No. 2 engine overboard drain. Go to 2.**
2. **Make sure NO. 2 ENG WARN LTS circuit breaker is pushed in. Go to 3.**
3. **Move engine controls quadrant NO. 2 ENG FUEL SYS fuel selector handle to XFD. Go to 4.**
4. **Place lower console NO. 1 PUMP switch to ON and place ENG POWER CONT lever to LOCKOUT. Go to 5.**
5. **Check for fuel flow overboard.**
   a. If fuel flow is present, go to **6.**
   b. If fuel flow is not present, go to **8.**
6. **Return engine controls quadrant NO. 2 ENG FUEL SYS fuel selector handle, ENG POWER CONT lever, and lower console NO. 1 PUMP switch to OFF. Go to 7.**
7. **Replace pressure switch (WP 0975 00). Go to 13.**
8. **Return engine controls quadrant NO 2 ENG FUEL SYS selector handle and lower console NO. 1 PUMP switch to OFF. Go to 9.**
9. **Place engine controls quadrant NO. 2 ENG FUEL SYS selector handle in DIR and XFD positions and observe that push/pull cable actuates fuel selector valve.**
   a. If fuel selector valve is actuated, go to **10.**
   b. If fuel selector valve is not actuated, replace push/pull cable (WP 0509 00). Go to **13.**
   c. If trouble remains, replace control box (WP 0510 00). Go to **13.**
10. **Using empty 1 gallon container to catch fuel, disconnect fuel hose from breakaway tee valve. Go to 11.**
11. **Momentarily place lower console NO. 1 PUMP switch to ON. Go to 12.**
12. **Observe that fuel flows from fuel line.**
    a. If fuel does not flow from fuel line, replace check valve (WP 0984 00). Go to **13.**
    b. If fuel does flow from fuel line, replace fuel selector valve (WP 0986 00). Go to **13.**
13. **Procedure completed.**

LOCATION





TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P112 / J112	COCKPIT BL 6 RH, STA 197
P113 / J113	COCKPIT BL 6 LH, STA 197
P114 / J114	COCKPIT BL 8 RH, STA 200
P115 / J115	COCKPIT BL 6 RH, STA 210
P117	CAUTION / ADVISORY PANEL (SEE NOTE 3)
P139 / J1	FUEL BOOST PUMP CONTROL PANEL
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P223 / J223	JUNCTION BOX MAIN ROTOR PYLON DECK
P224 / J224	JUNCTION BOX MAIN ROTOR PYLON DECK
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 23 RH
P237 / J237	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 25 LH
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH
P311 / J311	CABIN CEILING BL 16.5 LH, STA 383
P316 / J316	CABIN CEILING BL 17 RH, STA 387
P320	TOP OF PLATE NO. 2 FUEL TANK
P321	TOP OF PLATE NO. 1 FUEL TANK

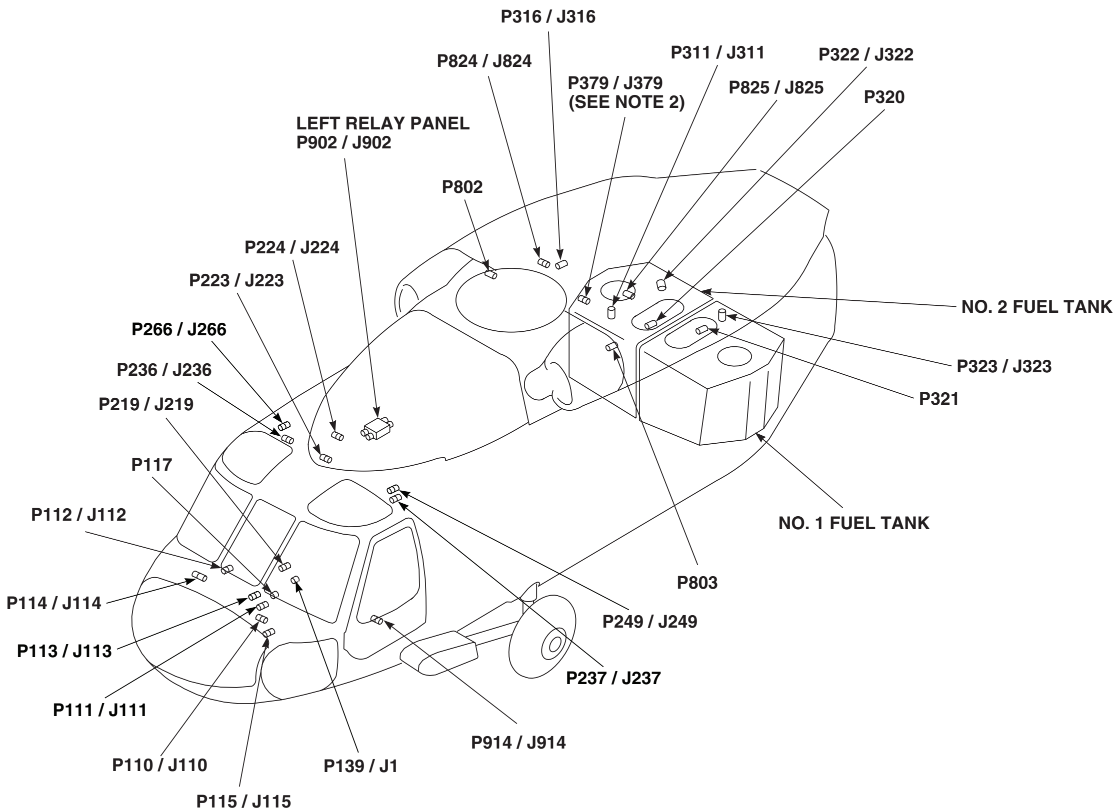
NOTES

1. UH60A UH60L HH60A HH60L
2. EH60A
3. UH60A UH60L EH60A
4. HH60A HH60L

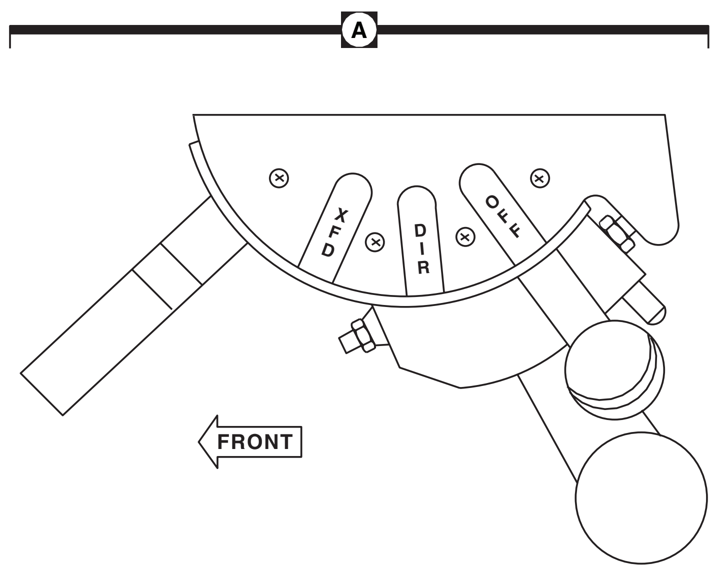
AA2693_1A
SA

Figure 1. Submerged Fuel Boost Pump System Location Diagram. (Sheet 1 of 6)

LOCATION - Continued



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P322 / J322	ABOVE NO. 1 FUEL TANK BL 7, WL 269
P323 / J323	ABOVE NO. 2 FUEL TANK BL 7, WL 269
P379 / J379 (SEE NOTE 2)	CABIN CEILING BL 5 RH, STA 379, WL 269
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P803	NO. 1 ENGINE INTERFACE DISCONNECT
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 356.25
P825 / J825	FIREWALL DISCONNECT BL 14 LH, STA 358.25
P902 / J902	LEFT RELAY PANEL
P914 / J914	COCKPIT BL 24 LH, STA 247
P1019R / J3	MULTIFUNCTION DISPLAY (SEE NOTE 4)
P1020R / J3	MULTIFUNCTION DISPLAY (SEE NOTE 4)



ENGINE CONTROLS QUADRANT

AA2693_2A
SA

Figure 1. Submerged Fuel Boost Pump System Location Diagram. (Sheet 2 of 6)

LOCATION - Continued

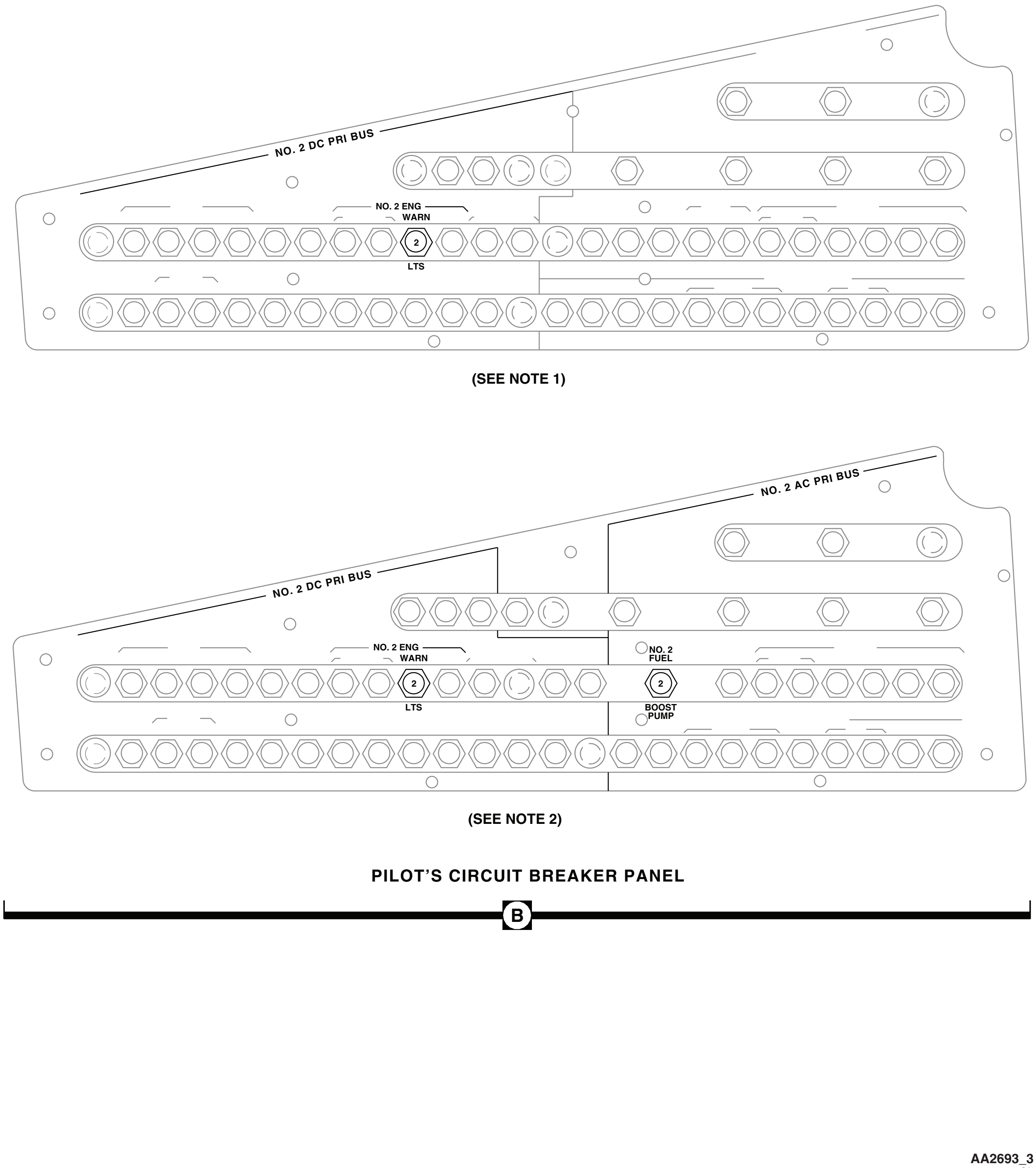


Figure 1. Submerged Fuel Boost Pump System Location Diagram. (Sheet 3 of 6)

LOCATION - Continued

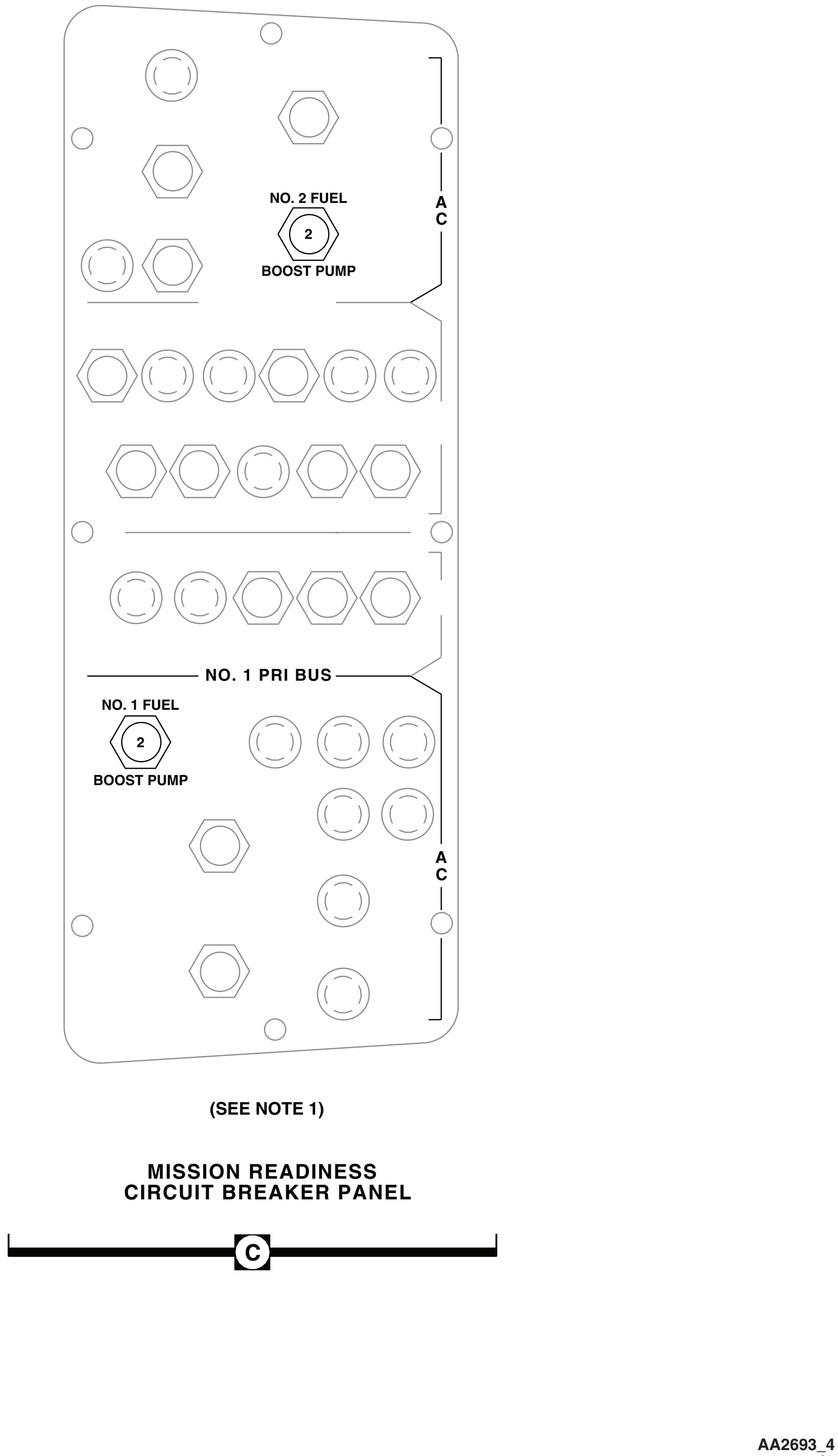
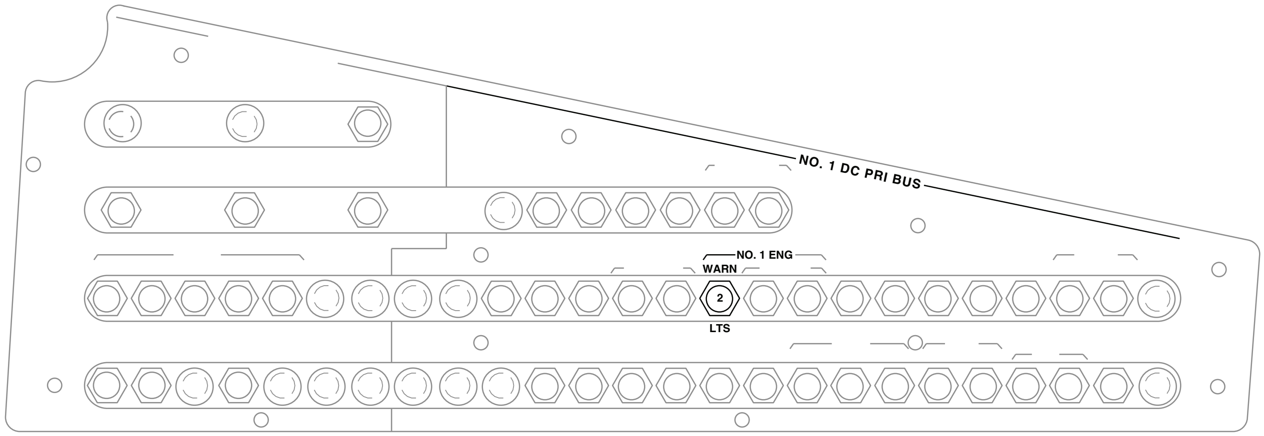
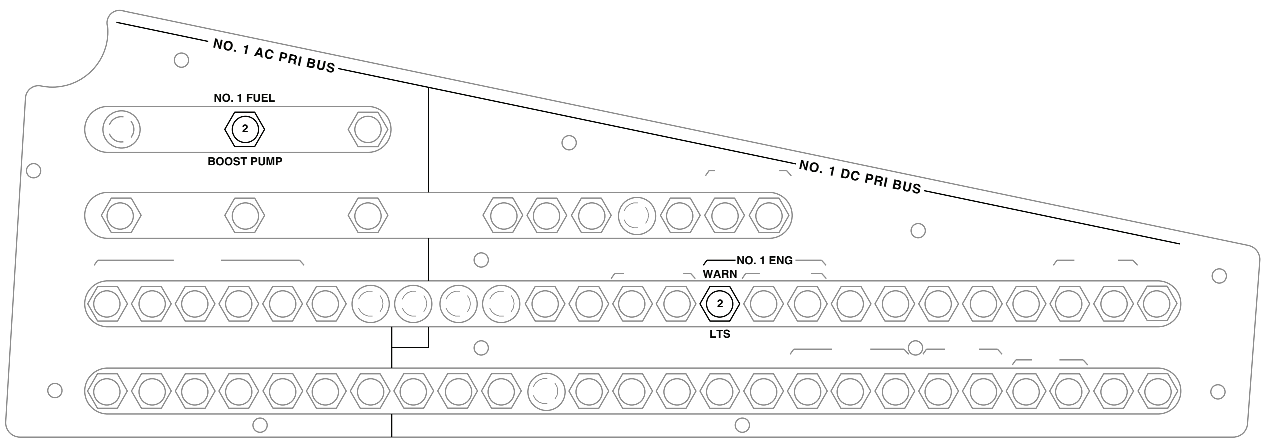


Figure 1. Submerged Fuel Boost Pump System Location Diagram. (Sheet 4 of 6)

LOCATION - Continued



(SEE NOTE 1)



(SEE NOTE 2)

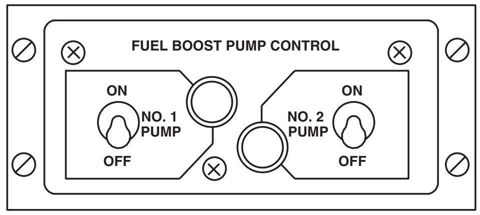
COPILOT'S CIRCUIT BREAKER PANEL

D

AA2693_5
SA

Figure 1. Submerged Fuel Boost Pump System Location Diagram. (Sheet 5 of 6)

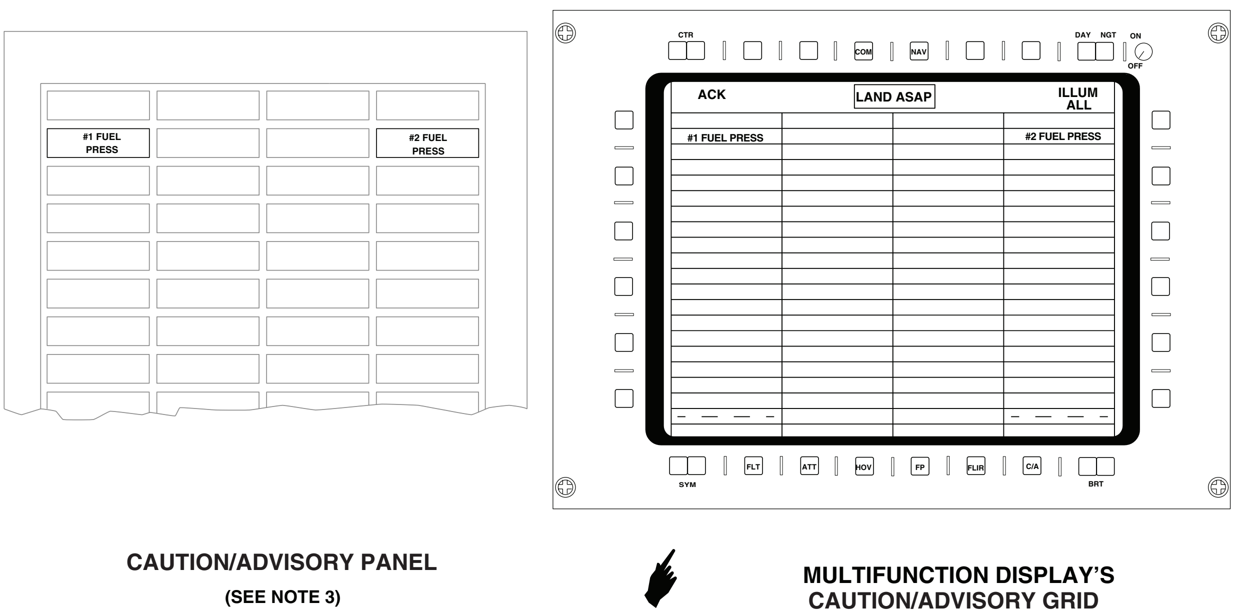
LOCATION - Continued



FUEL BOOST PUMP CONTROL PANEL

E

F



CAUTION/ADVISORY PANEL

(SEE NOTE 3)

MULTIFUNCTION DISPLAY'S CAUTION/ADVISORY GRID

(SEE NOTE 4)

AA2693_6B
SA

Figure 1. Submerged Fuel Boost Pump System Location Diagram. (Sheet 6 of 6)

SCHEMATICS

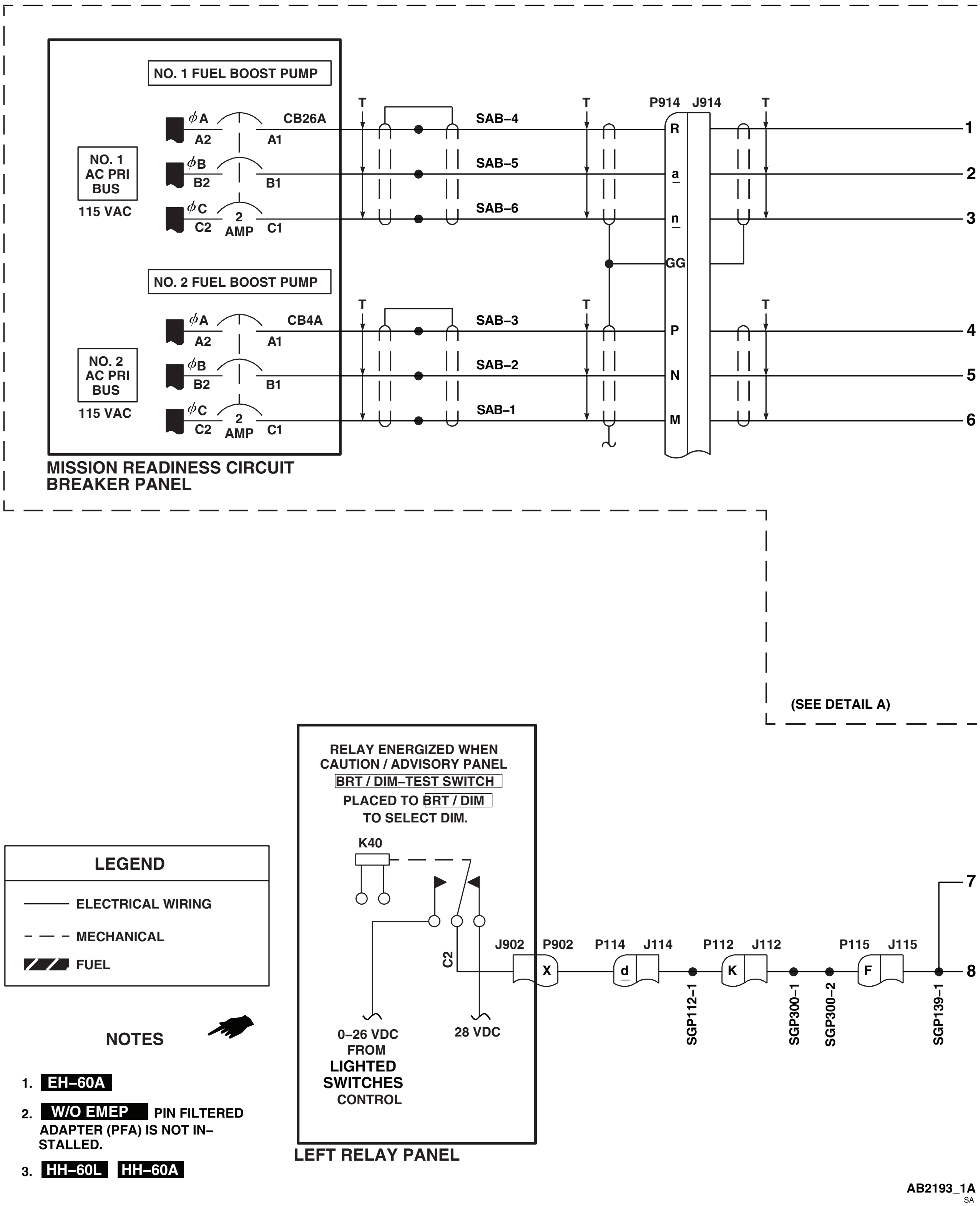


Figure 2. Submerged Fuel Boost Pump System Schematic Diagram. (Sheet 1 of 7)

SCHEMATICS - Continued

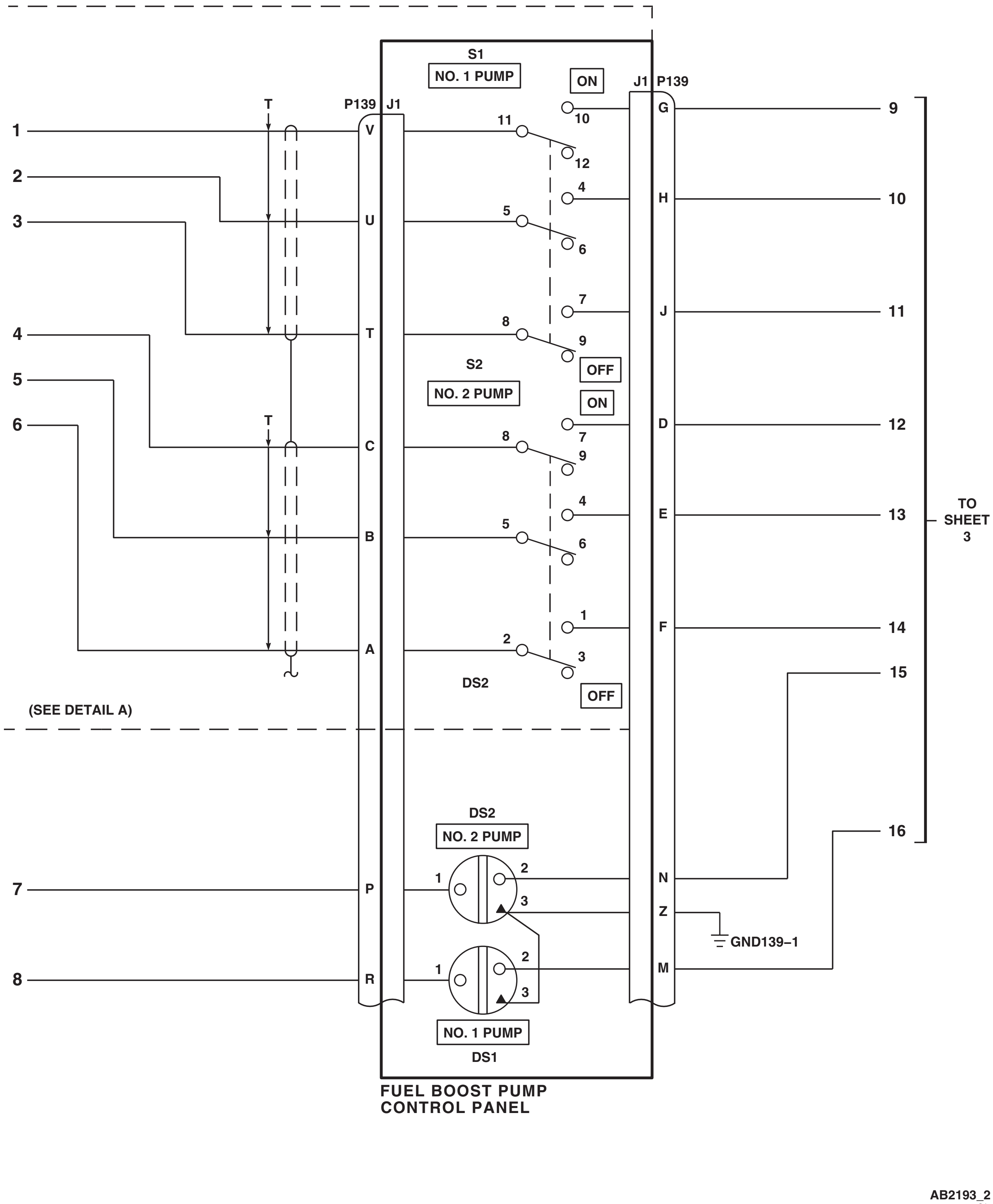


Figure 2. Submerged Fuel Boost Pump System Schematic Diagram. (Sheet 2 of 7)

SCHEMATICS - Continued

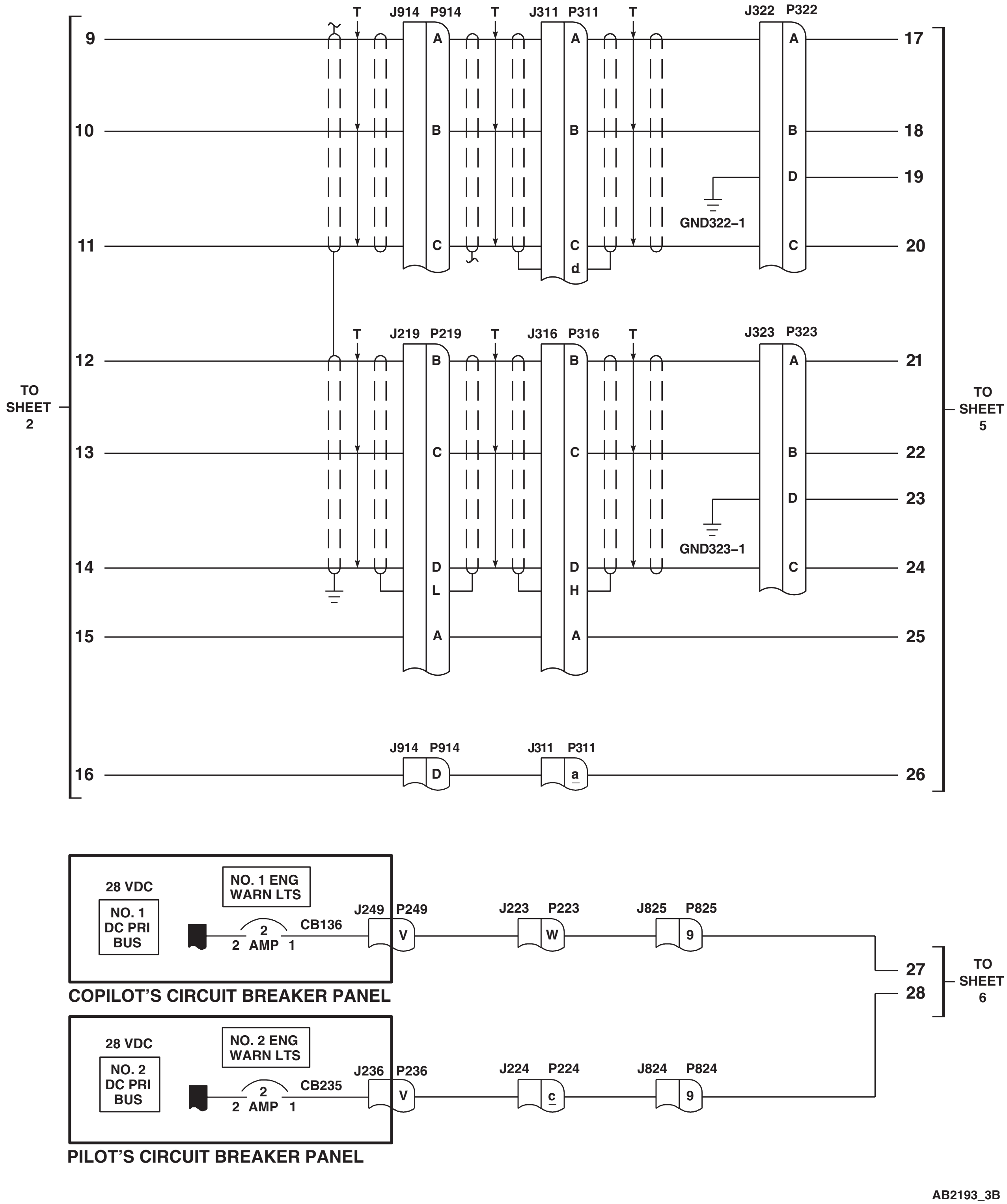
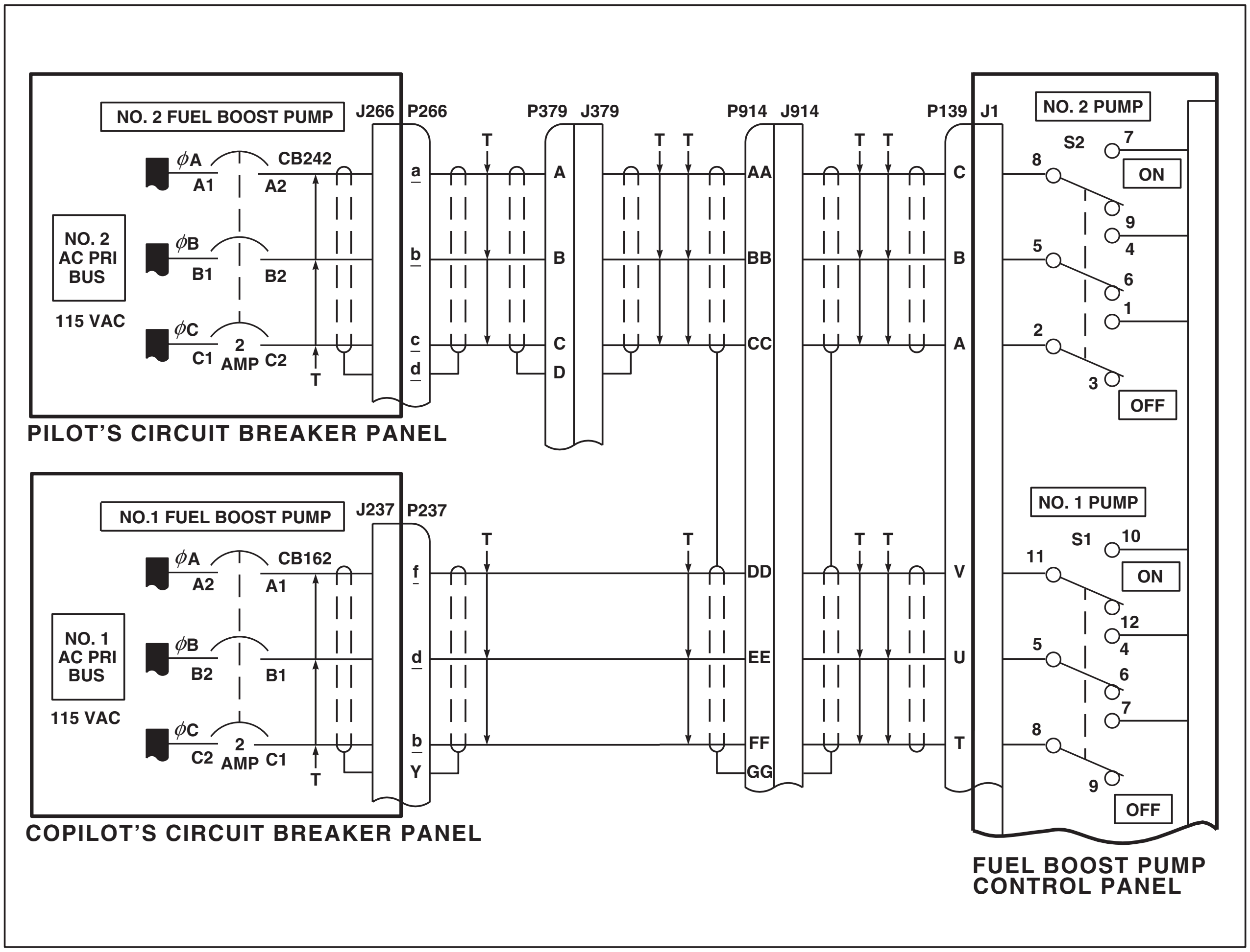


Figure 2. Submerged Fuel Boost Pump System Schematic Diagram. (Sheet 3 of 7)

SCHEMATICS - Continued



DETAIL A
(SEE NOTE 1)

Figure 2. Submerged Fuel Boost Pump System Schematic Diagram. (Sheet 4 of 7)

SCHEMATICS - Continued

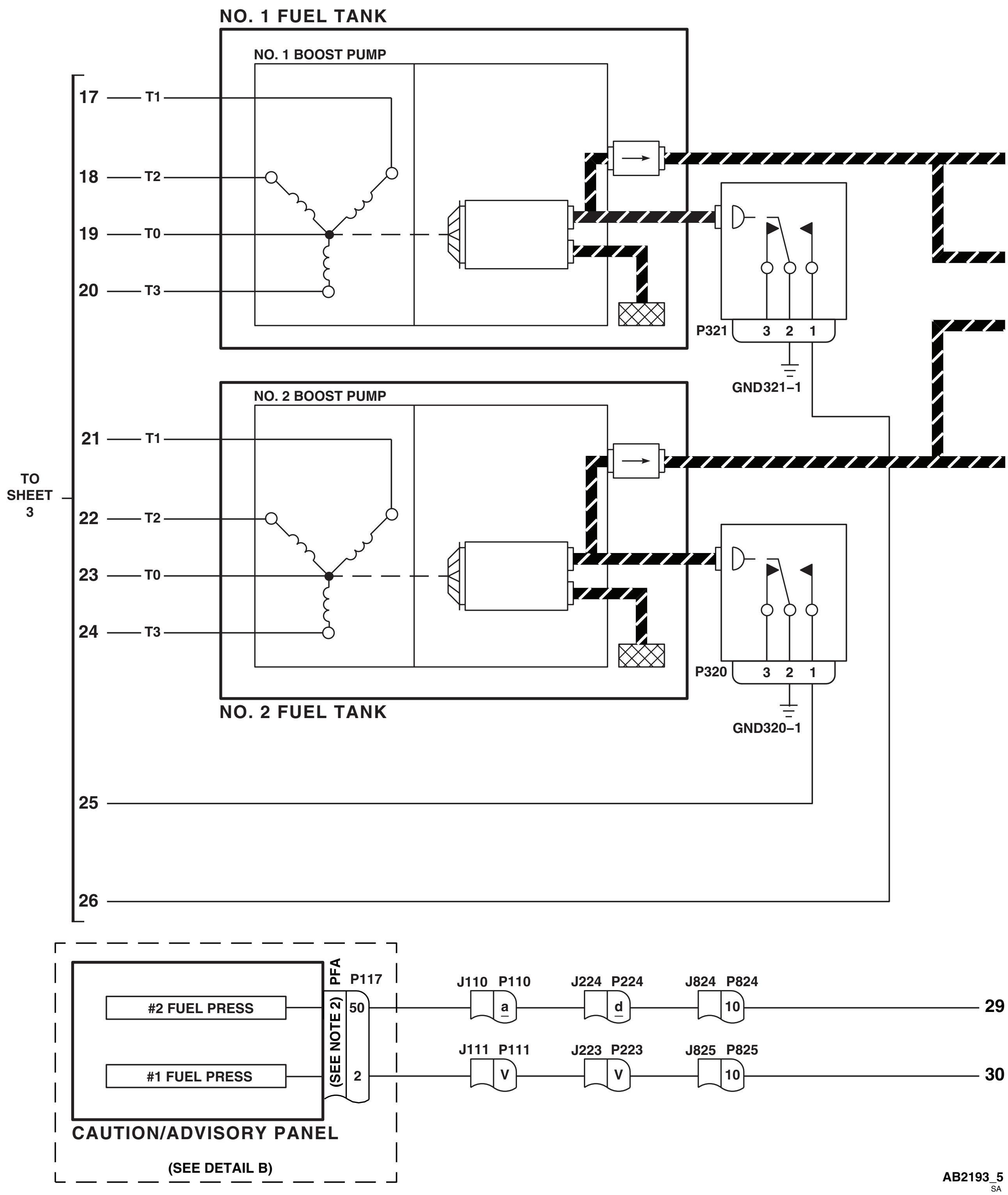


Figure 2. Submerged Fuel Boost Pump System Schematic Diagram. (Sheet 5 of 7)

SCHEMATICS - Continued

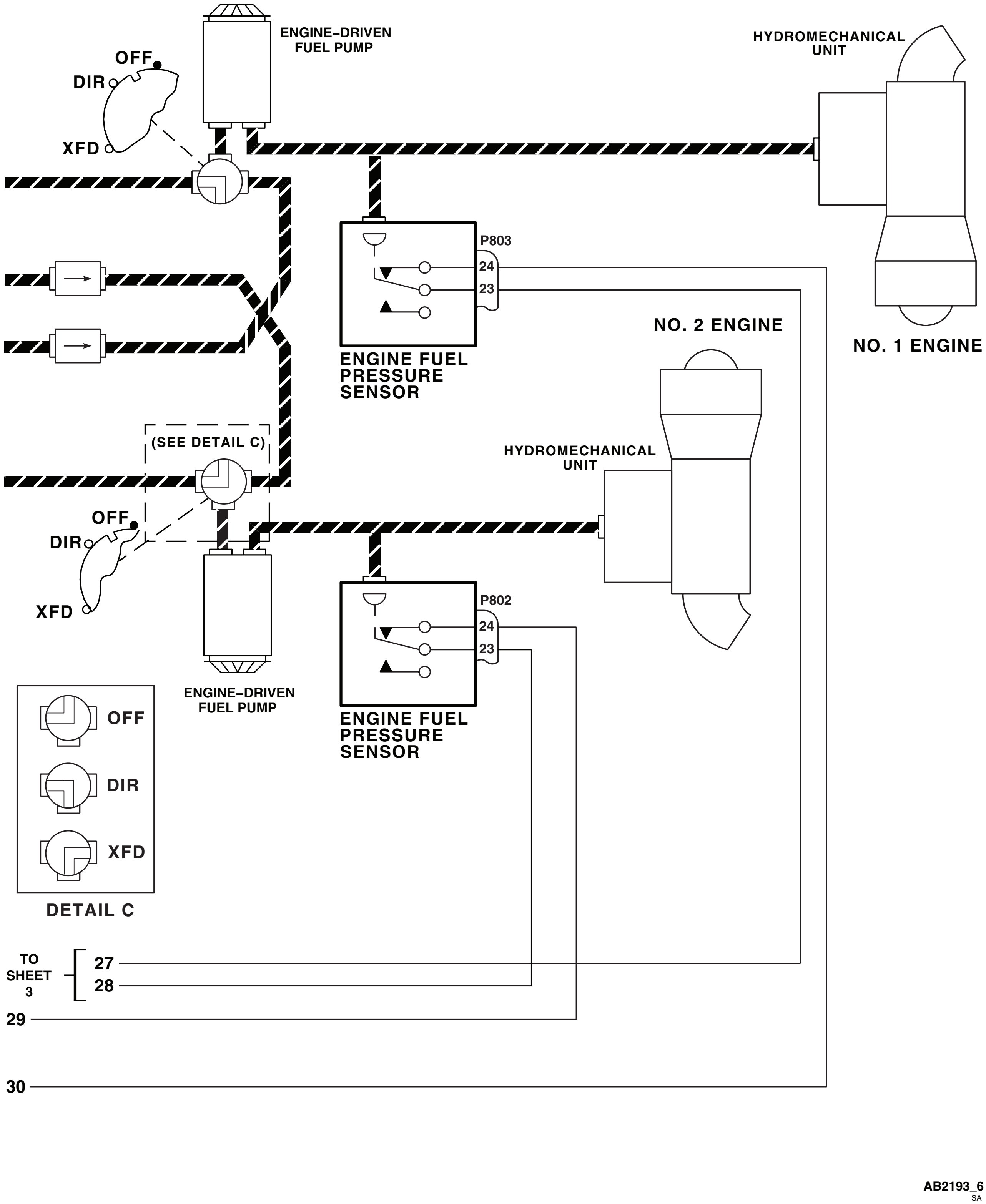


Figure 2. Submerged Fuel Boost Pump System Schematic Diagram. (Sheet 6 of 7)

SCHEMATICS - Continued

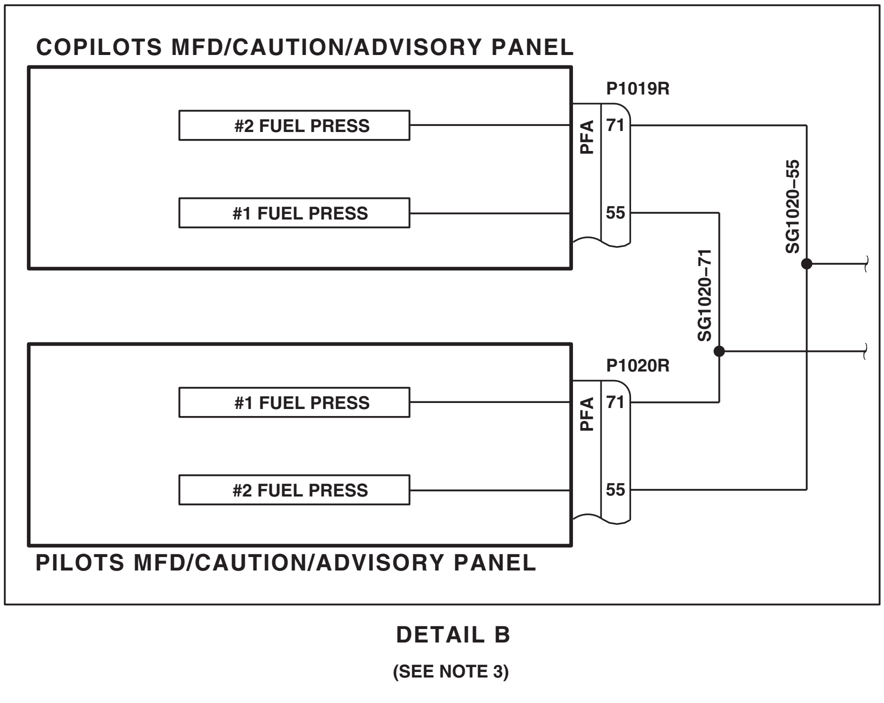


Figure 2. Submerged Fuel Boost Pump System Schematic Diagram. (Sheet 7 of 7)

END OF WORK PACKAGE

UNIT LEVEL

FUEL SYSTEM

FUEL PRIME BOOST SYSTEM

This WP supersedes WP 0129 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-223

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer Toolkit, SC 5180-99-B06

Material/Parts
Suitable Container

Personnel Required
Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1520-237-10
WP 0079 00
WP 0090 00
WP 0100 00
WP 0104 00
WP 0506 00
WP 0509 00
WP 0785 00
WP 0824 00
WP 0825 00
WP 0828 00
WP 0869 00
WP 0925 00
WP 0976 00
WP 0981 00
WP 0986 00
WP 0988 00
WP 1203 00
WP 1368 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10
No. 1 Fuel Tank Serviced, TM 1-1520-237-10

FUEL PRIME BOOST SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

WARNING

To prevent injury to personnel or damage to equipment, use normal fire precautions while doing procedures.

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
FUEL PRIME BOOST	Lower Console
CAUT/ADVSY PNL	Upper Console
LIGHTS CAUT ADVSY	Copilot’s

2. Turn on electrical power. **HH-60A HH-60L** Turn pilot’s or copilot’s multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).

NOTE

Make sure APU T-handle (on upper console panel) is pushed in or fuel prime/boost pump will not operate.

3. Place upper console BATT switch to ON.

Indication/Condition

PRIME FUEL BOOST PUMP ON capsule or legends shall not go on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot engine start and ignition system (WP 0079 00).

Step

4. Hold caution/advisory panel BRT/DIM-TEST switch or **HH-60A HH-60L** press T6 switch on MFD .

Indication/Condition

a. All capsules on caution/advisory or **HH-60A HH-60L** legends on MFDs , pilot’s and copilot’s master warning panels, and all switch legends on pilot’s and copilot’s HSI/VSI mode select panels shall go on.

FUEL PRIME BOOST SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

b. #1 FUEL LOW, #2 FUEL LOW, and LOW ROTOR RPM capsules shall flash.

Corrective Action

If Indication/Conditions are not as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

Step

5. Release BRT/DIM-TEST switch.
6. Place a rag under nut on APU fuel control enclosure. Loosen nut enough to allow fuel to leak.
7. Place upper console FUEL PUMP switch to APU BOOST.

Indication/Condition

a. PRIME BOOST PUMP ON capsule or legends shall go on.

b. Boost pump operation shall be heard.

c. Fuel shall leak from nut.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, go to PRIME BOOST PUMP ON CAPSULE OR LEGENDS NOT ON, AND BOOST PUMP NOT OPERATING, in this work package.
2. If Indication/Condition a. is not as specified, check helicopter configuration, go to **EMEP** or **W/O EMEP** .
   a. **EMEP** Remove and check pin filtered adapter P118 (WP 0925 00).
      (1) **EMEP** If pin filtered adapter is good, install pin filtered adapter (WP 0925 00).
      (2) **EMEP** If pin filtered adapter is not good, replace it (WP 0925 00).
   b. **EMEP W/O EMEP** Check for 28 vdc between P118-K and ground.
      (1) **EMEP W/O EMEP** If voltage is as specified, replace caution/advisory panel (WP 0785 00).
      (2) **EMEP W/O EMEP** If voltage is not as specified, repair/replace wiring between P118-K and S101-C (WP 1747 00).
3. If Indication/Condition b. is not as specified, go to PRIME BOOST PUMP ON CAPSULE OR LEGENDS ON, BOOST PUMP NOT OPERATING, in this work package.
4. If Indication/Condition c. is not as specified, go to PRIME BOOST PUMP ON CAPSULE OR LEGENDS ON AND PUMP OPERATING, BUT NO FUEL COMING OUT OF NUT, in this work package.

Step

8. Tighten nut on APU fuel control enclosure and wipe fuel from nut.
9. Place upper console FUEL PUMP switch to OFF.
10. Move and hold control quadrant NO. 1 ENG POWER CONT lever in cockpit to LOCKOUT position.
11. Move control quadrant NO. 1 ENG FUEL SYS selector lever to DIR detent.

Indication/Condition

NO. 1 ENG FUEL SYS selector lever shall move without difficulty.

FUEL PRIME BOOST SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

If Indication/Condition is not as specified, go to NO. 1 Or NO. 2 ENG FUEL SYS SELECTOR LEVER DOES NOT MOVE OR IS DIFFICULT TO MOVE, in this work package.

Step

12. Place suitable container beneath left side of helicopter fuselage at No. 1 engine overboard drain.
13. Place upper console FUEL PUMP switch to FUEL PRIME until fuel is coming from overboard drain for 3 minutes, then place switch to OFF.

Indication/Condition

a. PRIME BOOST PUMP ON capsule or legends shall go on.

b. Fuel from No. 1 engine overboard drain line shall drain into container.

c. Operation of boost pump shall be heard.

Corrective Action

If Indication/Conditions are not as specified, go to NO FUEL COMING OUT OF NO. 1 ENGINE FUEL DRAIN, in this work package.

Step

14. Place control quadrant NO. 1 ENG FUEL SYS selector lever to OFF.
15. Pull control quadrant NO. 1 ENG POWER CONT lever back to OFF.
16. Move and hold control quadrant NO. 2 ENG POWER CONT lever in cockpit to LOCKOUT position.
17. Move control quadrant NO. 2 ENG FUEL SYS selector lever in DIR detent.

Indication/Condition

NO. 2 ENG FUEL SYS selector lever shall move without difficulty.

Corrective Action

If Indication/Condition is not as specified, go to NO. 1 OR NO. 2 ENG FUEL SYS SELECTOR LEVER DOES NOT MOVE OR IS DIFFICULT TO MOVE, in this work package.

Step

18. Place suitable container beneath right side of helicopter fuselage at No. 2 engine overboard drain.
19. Place upper console FUEL PUMP switch to FUEL PRIME until fuel is coming from overboard drain for 3 minutes, then place switch to OFF.

Indication/Condition

a. PRIME BOOST PUMP ON capsule or legends shall go on.

b. Fuel from No. 2 engine overboard drain line shall drain into container.

c. Operation of boost pump shall be heard.

Corrective Action

If Indication/Conditions are not as specified, go to NO FUEL COMING OUT OF NO. 2 ENGINE FUEL DRAIN, in this work package.

FUEL PRIME BOOST SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

20. Place control quadrant No. 2 ENG FUEL SYS selector lever to OFF.
21. Pull control quadrant No. 2 ENG POWER CONT lever back to OFF.
22. Press control quadrant No. 1 or No. 2 engine start button.

Indication/Condition

PRIME BOOST PUMP ON capsule or legends shall go on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot engine start and ignition system (WP 0079 00).

Step

23. Turn MFD switch OFF.
24. Turn off electrical power.
25. Remove container from under helicopter.

Indication/Condition

Container shall be removed.

Corrective Action

None Required

PRIME BOOST PUMP ON CAPSULE OR LEGENDS NOT ON, AND BOOST PUMP NOT OPERATING

SYMPTOM

PRIME BOOST PUMP ON capsule or legends not on, and boost pump not operating.

MALFUNCTION

PRIME BOOST PUMP ON capsule or legends not on, and boost pump not operating.

CORRECTIVE ACTION

1. **Check for 28 vdc between APU T-handle switch S101-C and ground.**
   a. If voltage is as specified, repair/replace wiring between switch and J231-Y (WP 1747 00). Go to **7.**
   b. If voltage is not as specified, go to **2.**
2. **Check for 28 vdc between APU T-handle switch terminal NC and ground.**
   a. If voltage is as specified, replace switch (WP 1203 00). Go to **7.**
   b. If voltage is not as specified, go to **3.**
3. **Check for 28 vdc between FUEL PUMP switch terminal 4 and ground.**
   a. If voltage is as specified, repair/replace wiring between FUEL PUMP switch and APU T-handle switch S101 (WP 1747 00). Go to **7.**
   b. If voltage is not as specified, go to **4.**
4. **Check for 28 vdc between FUEL PUMP switch terminal 5 and ground.**

PRIME BOOST PUMP ON CAPSULE OR LEGENDS NOT ON, AND BOOST PUMP NOT OPERATING - Continued

a. If voltage is as specified, replace switch (WP 0828 00). Go to **7.**

b. If voltage is not as specified, go to **5.**

5. Check for 28 vdc between FUEL PUMP switch terminal 2 and ground.

a. If voltage is as specified, repair/replace wiring between switch terminals 2 and 5 (WP 1747 00). Go to **7.**

b. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between FUEL PRIME BOOST circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between circuit breaker and switch (WP 1747 00). Go to **7.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **7.**

7. Procedure completed.

PRIME BOOST PUMP ON CAPSULE OR LEGENDS ON, BOOST PUMP NOT OPERATING

SYMPTOM

PRIME BOOST PUMP ON capsule or legends on, boost pump not operating.

MALFUNCTION

PRIME BOOST PUMP ON capsule or legends on, boost pump not operating.

CORRECTIVE ACTION

1. With FUEL PUMP switch placed to APU BOOST, check for 28 vdc between P303-A and P303-B.

a. If voltage is as specified, replace prime/boost pump (WP 0981 00). Go to **3.**

b. If voltage is not as specified, go to **2.**

2. Check for 28 vdc between P303-A and ground.

a. If voltage is as specified, repair/replace wiring between P303-B and GND311-1 (WP 1747 00). Go to **3.**

b. If voltage is not as specified, repair/replace wiring between P303-A and APU T-handle switch S101-C (WP 1747 00). Go to **3.**

3. Procedure completed.

PRIME BOOST PUMP ON CAPSULE OR LEGENDS ON AND PUMP OPERATING, BUT NO FUEL COMING OUT OF NUT

SYMPTOM

PRIME BOOST PUMP ON capsule or legends on and pump operating, but no fuel coming out of nut.

MALFUNCTION

PRIME BOOST PUMP ON capsule or legends on and pump operating, but no fuel coming out of nut.

CORRECTIVE ACTION

1. Check breakaway valve for visible yellow line in center of valve.

PRIME BOOST PUMP ON CAPSULE OR LEGENDS ON AND PUMP OPERATING, BUT NO FUEL COMING OUT OF NUT - Continued

a. If yellow line is present, replace breakaway valve (WP 0988 00). Go to **6.**

b. If yellow line is not present, go to **2.**

2. With FUEL PUMP switch placed to APU BOOST, check for 28 vdc between P306-A and P306-C.

a. If voltage is as specified, replace APU fuel shutoff valve (WP 1368 00). Go to **6.**

b. If voltage is not as specified, go to **3.**

3. Check for 28 vdc between P306-A and ground.

a. If voltage is as specified, repair/replace wiring between P306-B and GND310-1 (WP 1747 00). Go to **6.**

b. If voltage is not as specified, go to **4.**

4. Check continuity between P306-A and P241-S.

a. If continuity is present, go to **5.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**

5. Check continuity between P241-T and APU T-handle switch S101-C.

a. If continuity is present, replace right relay panel (WP 0825 00). Go to **6.**

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6.**

6. Procedure completed.

NO. 1 OR NO. 2 ENG FUEL SYS SELECTOR LEVER DOES NOT MOVE OR IS DIFFICULT TO MOVE

SYMPTOM

NO. 1 or NO. 2 ENG FUEL SYS selector lever does not move or is difficult to move.

MALFUNCTION

NO. 1 or NO. 2 ENG FUEL SYS selector lever does not move or is difficult to move.

CORRECTIVE ACTION

1. Inspect inside of engine controls quadrant for loose items that may cause interference. Go to 2.

2. Disconnect push/pull cable from control box. Go to 3.

3. Cycle ENG FUEL SYS selector lever back and forth. Movement of lever shall be smooth.

a. If movement is smooth, replace control box/fuel selector valve (WP 0986 00). Go to **4.**

b. If movement is not smooth, replace fuel selector push/pull cable (WP 0509 00). Go to **4.**

4. Procedure completed.

NO FUEL COMING OUT OF NO. 1 ENGINE FUEL DRAIN

SYMPTOM

No fuel coming out of No. 1 engine fuel drain.

MALFUNCTION

No fuel coming out of No. 1 engine fuel drain.

CORRECTIVE ACTION

1. Visually check drain tube for obvious obstructions.

a. If obstructions are present, remove obstructions. Go to **10.**

b. If obstructions are not present, go to **2.**

2. Position ENG POWR CONT lever to LOCKOUT. As lever is advanced to LOCKOUT, check to hear a click in hydromechanical unit (HMU).

a. If click is heard in HMU, go to **3.**

b. If click is not heard in HMU, go to **9.**

3. Check push/pull cable to fuel selector valve for proper rigging.

a. If push/pull cable is properly rigged, go to **4.**

b. If push/pull cable is not properly rigged, correctly rig push/pull cable (WP 0509 00). Go to **10.**

4. Check for 28 vdc between P304-A and P304-B.

a. If voltage is as specified, replace No. 1 engine prime shutoff valve (WP 0976 00). Go to **10.**

b. If voltage is not as specified, go to **5.**

5. Check for 28 vdc between P304-A and ground.

a. If voltage is as specified, repair/replace wiring between P304-B and GND310-3 (WP 1747 00). Go to **10.**

b. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between TB1-8 and ground.

a. If voltage is as specified, repair/replace wiring between TB1-8 and P304-A (WP 1747 00). Go to **10.**

b. If voltage is not as specified, go to **7.**

7. Check for 28 vdc between TB2-9 and ground.

a. If voltage is as specified, replace diode board (WP 0869 00). Go to **10.**

b. If voltage is not as specified, go to **8.**

8. Check for 28 vdc between FUEL PUMP switch terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between TB2-9 and switch terminal 1 (WP 1747 00). Go to **10.**

b. If voltage is not as specified, replace switch (WP 0828 00). Go to **10.**

9. Check engine control for restrictions/proper rigging at LOCKOUT (WP 0506 00).

NO FUEL COMING OUT OF NO. 1 ENGINE FUEL DRAIN - Continued

a. If engine controls are properly rigged, replace HMU (TM 1-2840-248-23). Go to **10.**

10. Procedure completed.

NO FUEL COMING OUT OF NO. 2 ENGINE FUEL DRAIN

SYMPTOM

No fuel coming out of No. 2 engine fuel drain.

MALFUNCTION

No fuel coming out of No. 2 engine fuel drain.

CORRECTIVE ACTION

1. Visually check drain tube for obvious obstructions.

a. If obstructions are present, remove obstructions. Go to **8.**

b. If obstructions are not present, go to **2.**

2. Position ENG POWR CONT lever to LOCKOUT. As lever is advanced to LOCKOUT, check to hear click in HMU.

a. If click is heard in HMU, go to **3.**

b. If click is not heard in HMU, go to **7.**

3. Check push/pull cable to fuel selector valve for proper rigging.

a. If push/pull cable is properly rigged, go to **4.**

b. If push/pull cable is not properly rigged, correctly rig push/pull cable (WP 0509 00). Go to **8.**

4. Check for 28 vdc between P305-A and P305-B.

a. If voltage is as specified, replace No. 2 engine prime shutoff valve (WP 0976 00). Go to **8.**

b. If voltage is not as specified, go to **5.**

5. Check for 28 vdc between P305-A and ground.

a. If voltage is as specified, repair/replace wiring between P305-B and GND310-2 (WP 1747 00). Go to **8.**

b. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between TB2-8 and ground.

a. If voltage is as specified, repair/replace wiring between TB2-8 and P305-A (WP 1747 00). Go to **8.**

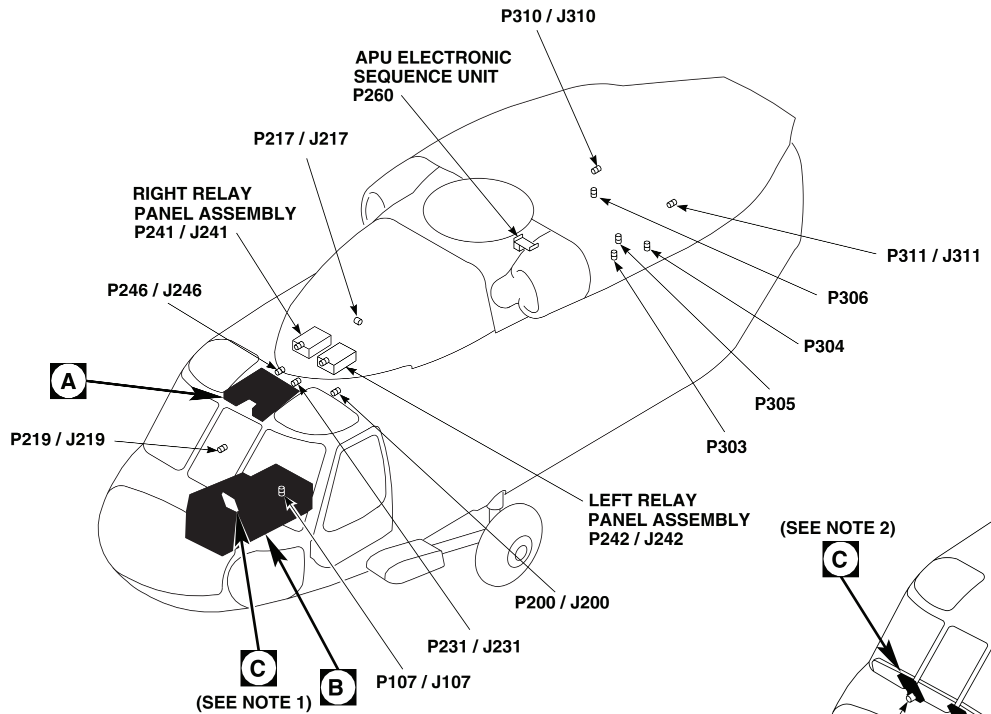
b. If voltage is not as specified, replace diode board (WP 0869 00). Go to **8.**

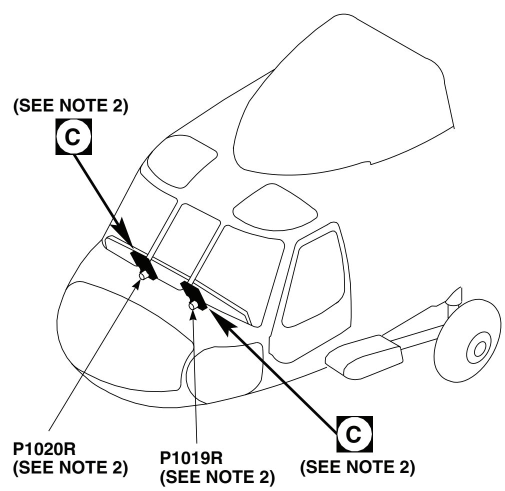
7. Check engine control for restrictions/proper rigging at LOCKOUT (WP 0506 00).

a. If engine controls are properly rigged, replace HMU (TM 1-2840-248-23). Go to **8.**

8. Procedure completed.

LOCATION





TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P107 / J107	COCKPIT BL 9 LH, STA 236
P111 / J111	COCKPIT B. 7.5 LH, STA 197
P118 / J118	CAUTION / ADVISORY PANEL (SEE NOTE 1)
P200 / J200	CABIN CEILING BL 13 LH, STA 246
P217 / J217	CABIN CEILING BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P231 / J231	CABIN CEILING BL 9 LH, STA 247
P241 / J241	RIGHT RELAY PANEL ASSEMBLY
P242	LEFT RELAY PANEL ASSEMBLY
P246 / J246	CABIN CEILING BL 5 RH, STA 247

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P260	APU ELECTRONIC SEQUENCE UNIT
P303	PRIME BOOST PUMP
P304	NO. 1 ENGINE PRIME SHUTOFF VALVE
P305	NO. 2 ENGINE PRIME SHUTOFF VALVE
P306	APU FUEL SHUTOFF VALVE
P310 / J310	CABIN CEILING BR 17 LH, STA 387
P311 / J311	CABIN CEILING BL 165 LH, STA 343
TB1	COCKPIT CEILING BL 9 LH, STA 236
TB2	COCKPIT CEILING BL 9 RH, STA 236
P1019R / J3	COPILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)
P1020R / J3	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)

NOTES

1. UH–60A UH–60L EH–60A
2. HH–60A HH–60L

AA2763_1B
SA

Figure 1. Fuel Prime Boost System Location Diagram. (Sheet 1 of 3)

LOCATION - Continued

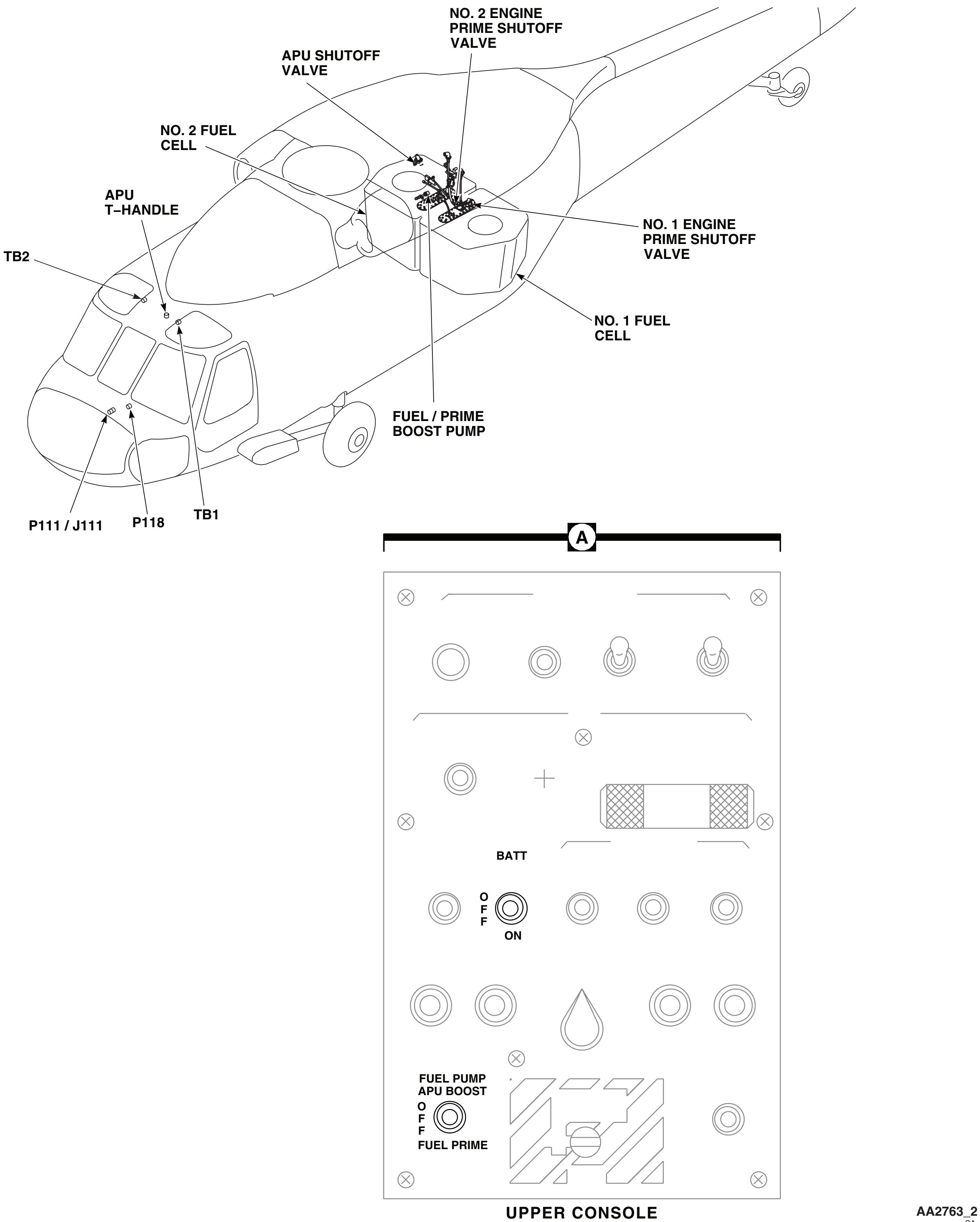
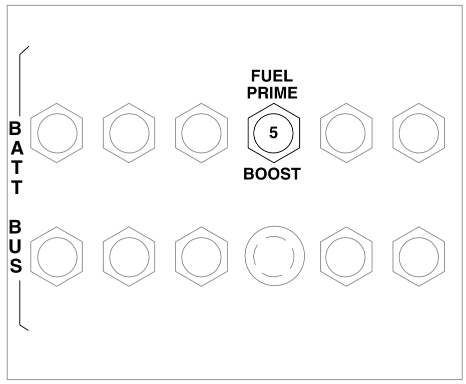


Figure 1. Fuel Prime Boost System Location Diagram. (Sheet 2 of 3)

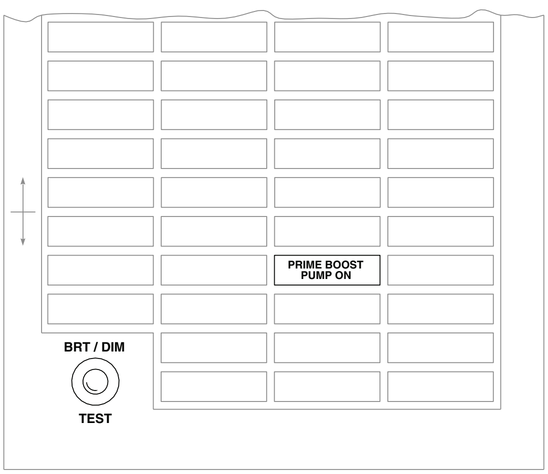
LOCATION - Continued



LOWER CONSOLE

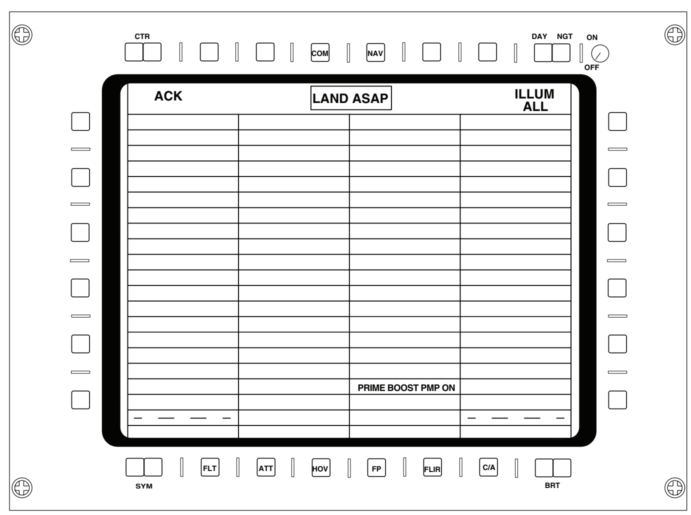
B

C



CAUTION/ADVISORY PANEL

(SEE NOTE 1)



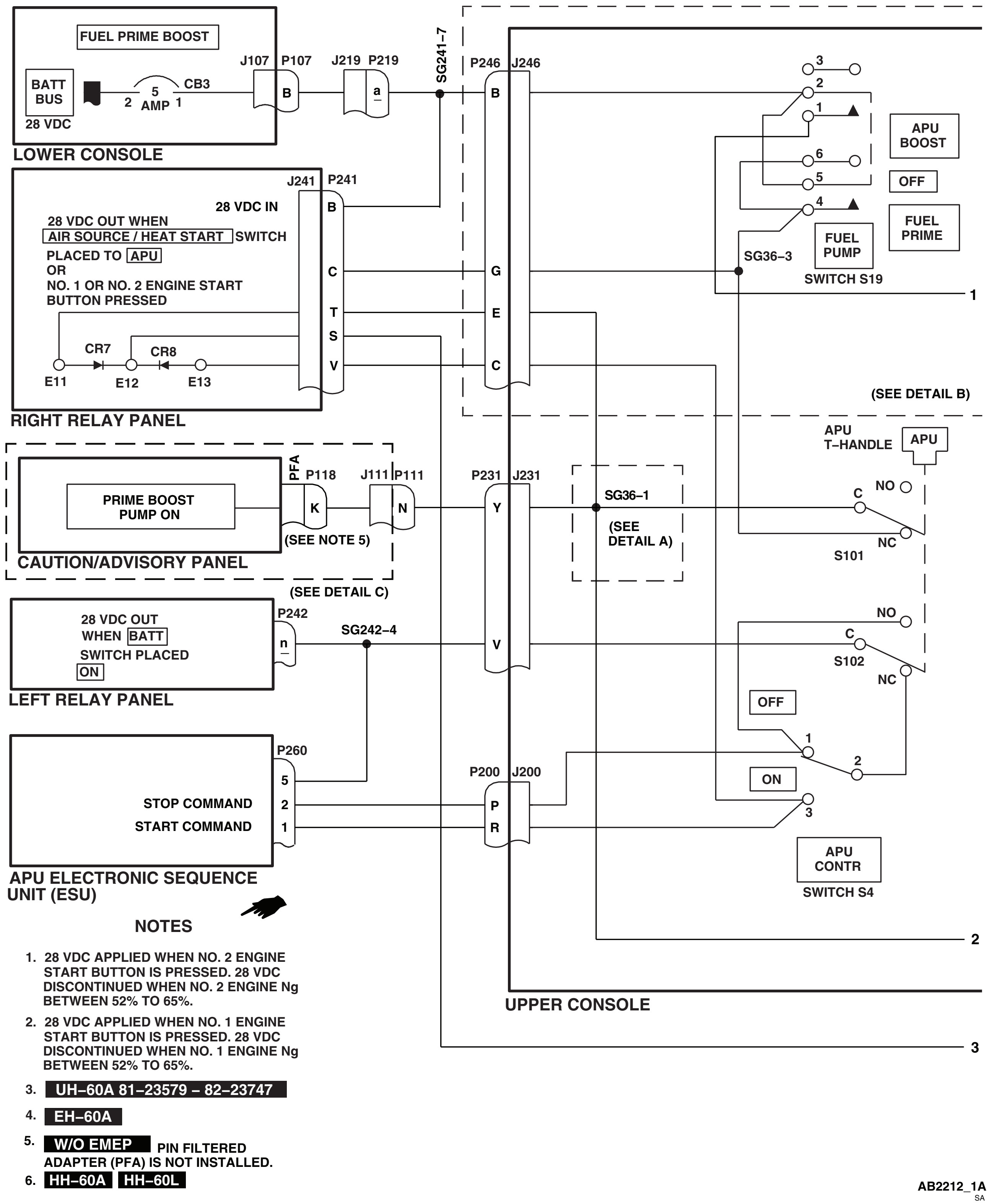
MULTIFUNCTION DISPLAY'S CAUTION/ADVISORY GRID

(SEE NOTE 2)

AA2763_3B
SA

Figure 1. Fuel Prime Boost System Location Diagram. (Sheet 3 of 3)

SCHEMATICS



NOTES

1. 28 VDC APPLIED WHEN NO. 2 ENGINE START BUTTON IS PRESSED. 28 VDC DISCONTINUED WHEN NO. 2 ENGINE Ng BETWEEN 52% TO 65%.
2. 28 VDC APPLIED WHEN NO. 1 ENGINE START BUTTON IS PRESSED. 28 VDC DISCONTINUED WHEN NO. 1 ENGINE Ng BETWEEN 52% TO 65%.
3. UH-60A 81-23579 – 82-23747
4. EH-60A
5. W/O EMEP PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.
6. HH-60A HH-60L

Figure 2. Fuel Prime Boost System Schematic Diagram. (Sheet 1 of 5)

SCHEMATICS - Continued

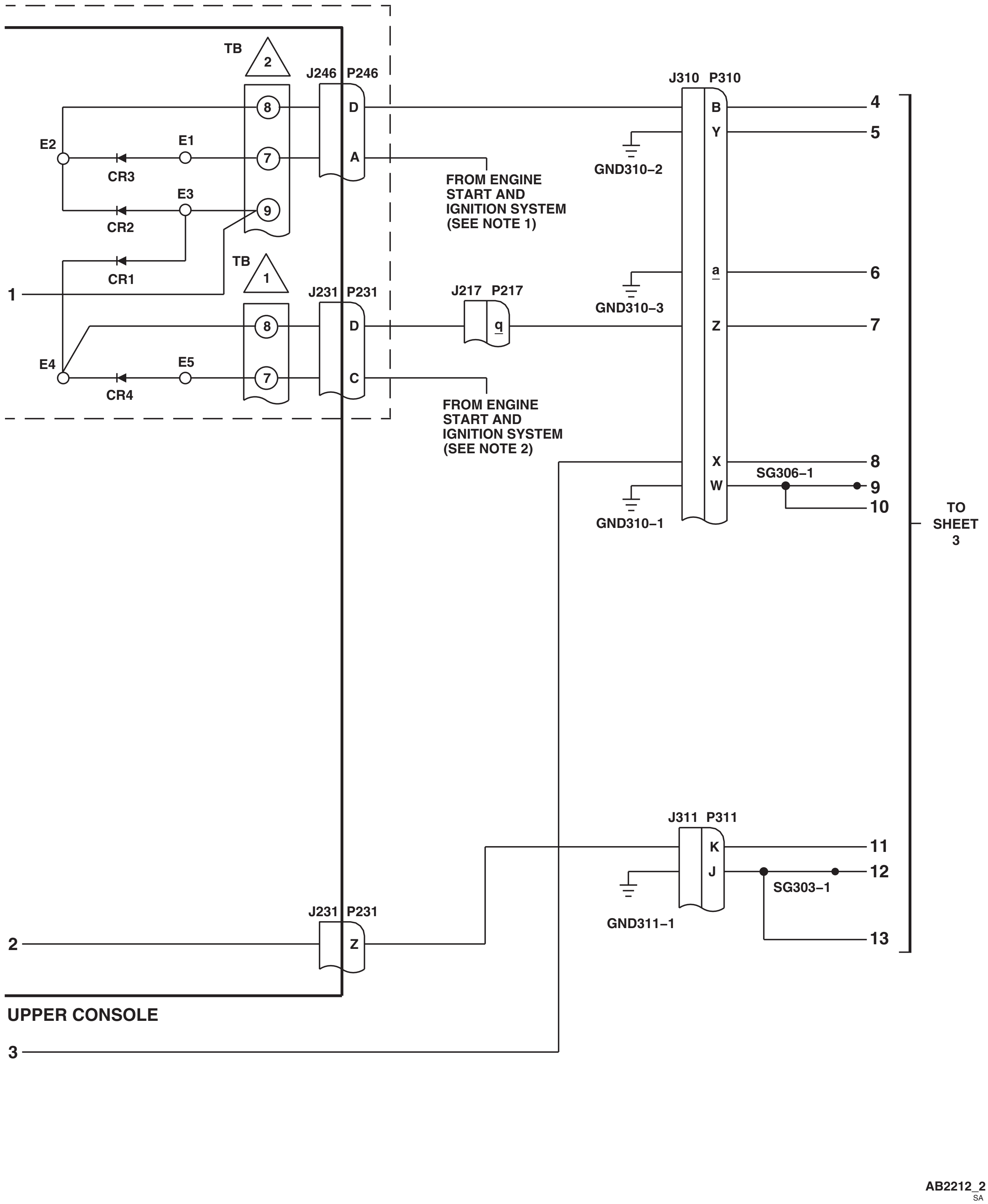


Figure 2. Fuel Prime Boost System Schematic Diagram. (Sheet 2 of 5)

SCHEMATICS - Continued

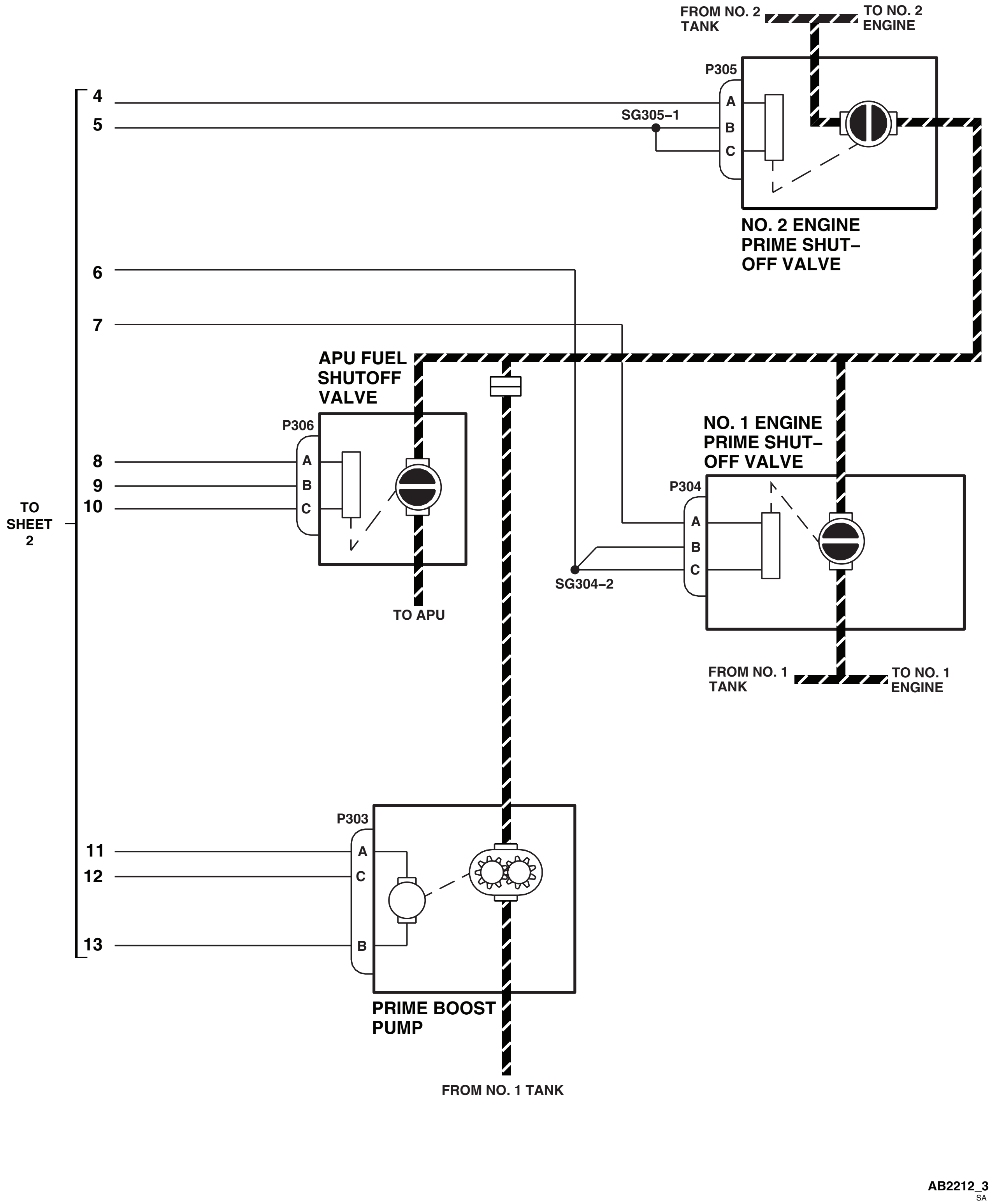
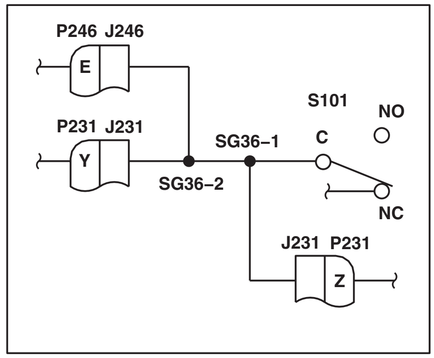
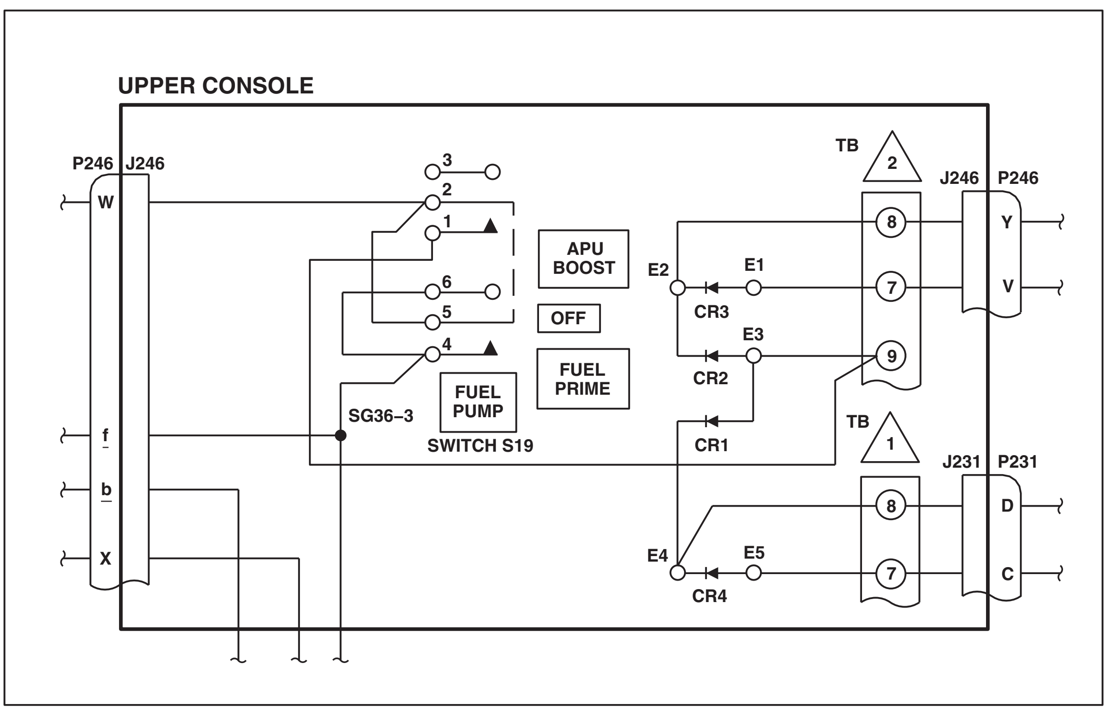


Figure 2. Fuel Prime Boost System Schematic Diagram. (Sheet 3 of 5)

SCHEMATICS - Continued



DETAIL A
(SEE NOTE 3)



DETAIL B
(SEE NOTE 4)

AB2212_4
SA

Figure 2. Fuel Prime Boost System Schematic Diagram. (Sheet 4 of 5)

SCHEMATICS - Continued

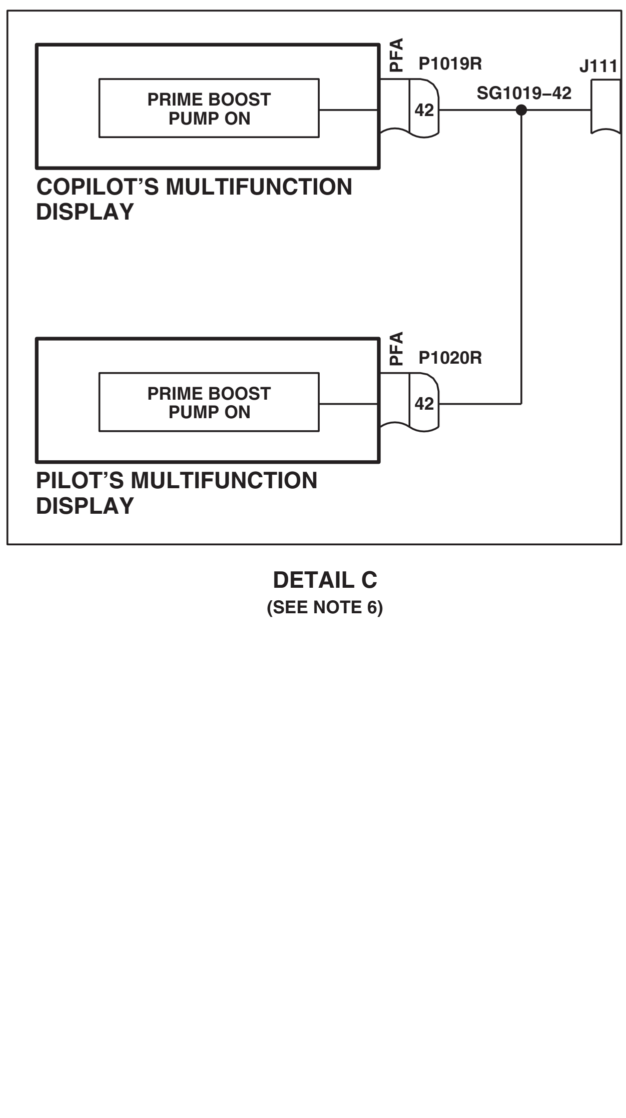


Figure 2. Fuel Prime Boost System Schematic Diagram. (Sheet 5 of 5)

END OF WORK PACKAGE

UNIT LEVEL

FUEL SYSTEM

PRESSURE REFUEL/DEFUEL SYSTEM

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1520-237-10
WP 0963 00
WP 0965 00
WP 0966 00
WP 0970 00
WP 0978 00
WP 0979 00
WP 1612 00
WP 1615 00
WP 1685 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

The pressure refuel/defuel system operational/troubleshooting procedure is subdivided as follows:

- Pressure Fueling
- Pressure Defueling

PRESSURE FUELING OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

WARNING

Use extreme care when removing gravity fill port cap; fuel may come out with pressure.

1. Turn on electrical power.

NOTE

- Monitor fuel quantity indication (on main instrument panel) during pressure fueling to verify that fuel is flowing into tank.
- When pressure fueling is completed, remove gravity fill port cap to verify proper high level shutoff valve operation.

2. Pressure-fuel both tanks (WP 1612 00).

PRESSURE FUELING OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

a. When the tanks are full, the high-level shutoff valve shall cause the pressure-fueling valve to close.

b. Fuel flow from the refueling truck shall stop. At this time there shall be a noticeable jump from the fueling hose.

c. Fuel shall not come from the vent overboard tube at the bottom of the helicopter.

d. When the gravity fill port cap is removed, fuel shall not spill out.

Corrective Action

1. For minor spills from gravity fill cap, first check if helicopter is in ground level attitude, then check installation of high level sensor.
2. If Indication/Condition c. or d. is not as specified, replace both high-level shutoff valves in overfilled fuel tank (WP 0965 00).
   a. If trouble remains, replace refuel/defuel valve in bottom of tank (WP 0966 00).
3. If only tank #2 fills, check for hanging floats in the high level sensor of tank #1. Replace pressure fueling valve if floats are not hung up (WP 0966 00).
4. If only tank #1 fills, check for hanging floats in the high level sensor of tank #2. Additionally, inspect the breakaway interconnect valve to see if it has activated and sealed off tank #2 (WP 0963 00).
   a. If valve is inoperable, replace valve (WP 0979 00).

Step

3. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

PRESSURE DEFUELING OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Turn on electrical power.

NOTE

Monitor fuel quantity indication (on main instrument panel) during defueling to verify that fuel is leaving tank.

2. Pressure defuel both tanks (WP 1615 00).

Indication/Condition

Both tanks shall empty at about the same time.

PRESSURE DEFUELING OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition is not as specified (as shown on fuel quantity indicator), do the following checks on tank with fuel remaining:
   a. Visually check breakaway vent valve at top of tank. If a yellow paint stripe can be seen, replace vent valve (WP 0978 00).
   b. If vent valve is all right, replace low-level shutoff valve (WP 0970 00).
   c. If tank still cannot be properly pressure-defueled, replace pressure fuel/defuel valve (WP 0966 00).
   d. If discrepancy is isolated to tank #2, inspect the breakaway interconnect valve to see if it has activated and sealed off tank #2 (WP 0963 00).

Step

3. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

LOCATION

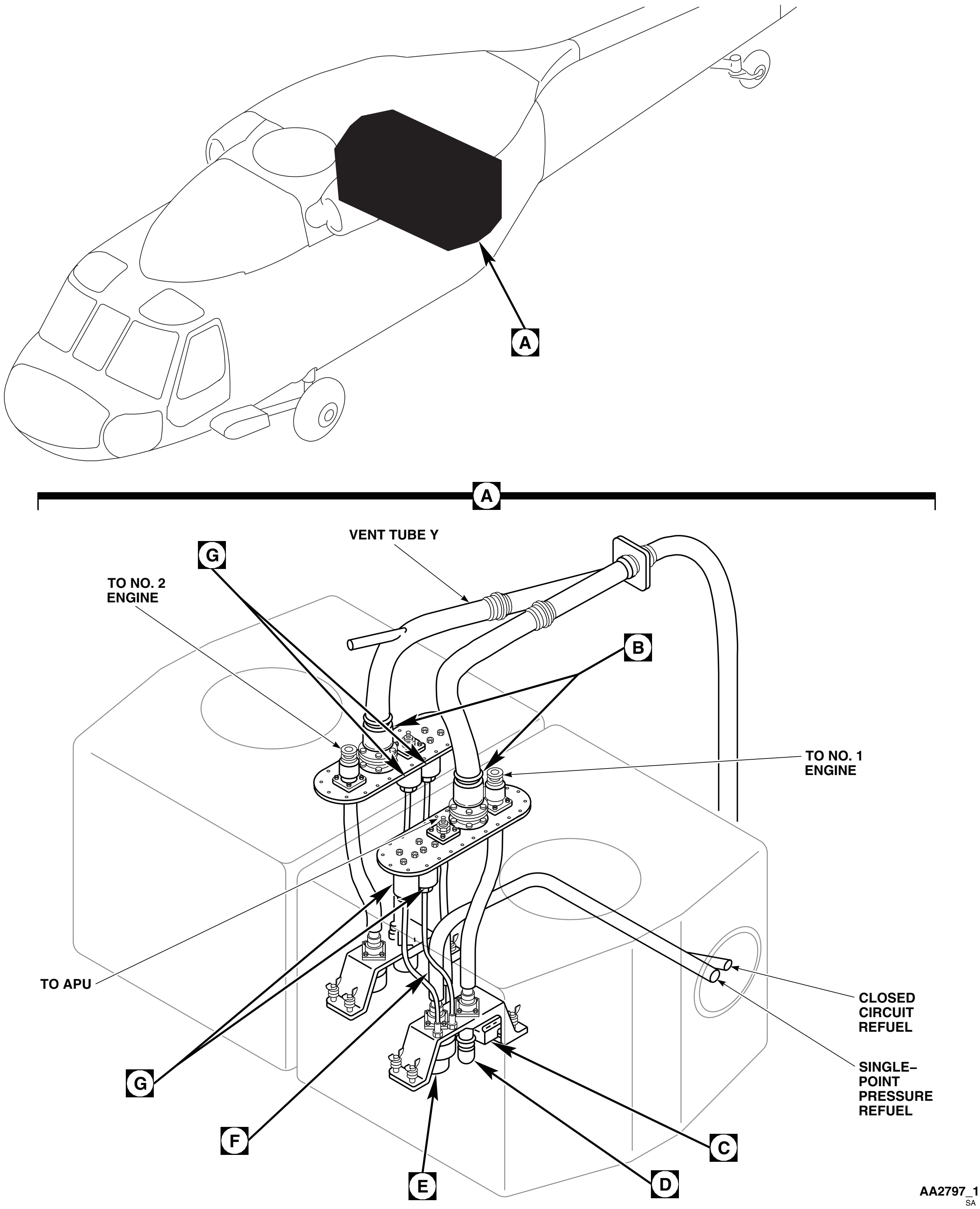


Figure 1. Pressure Refuel/Defuel System Location Diagram. (Sheet 1 of 3)

LOCATION - Continued

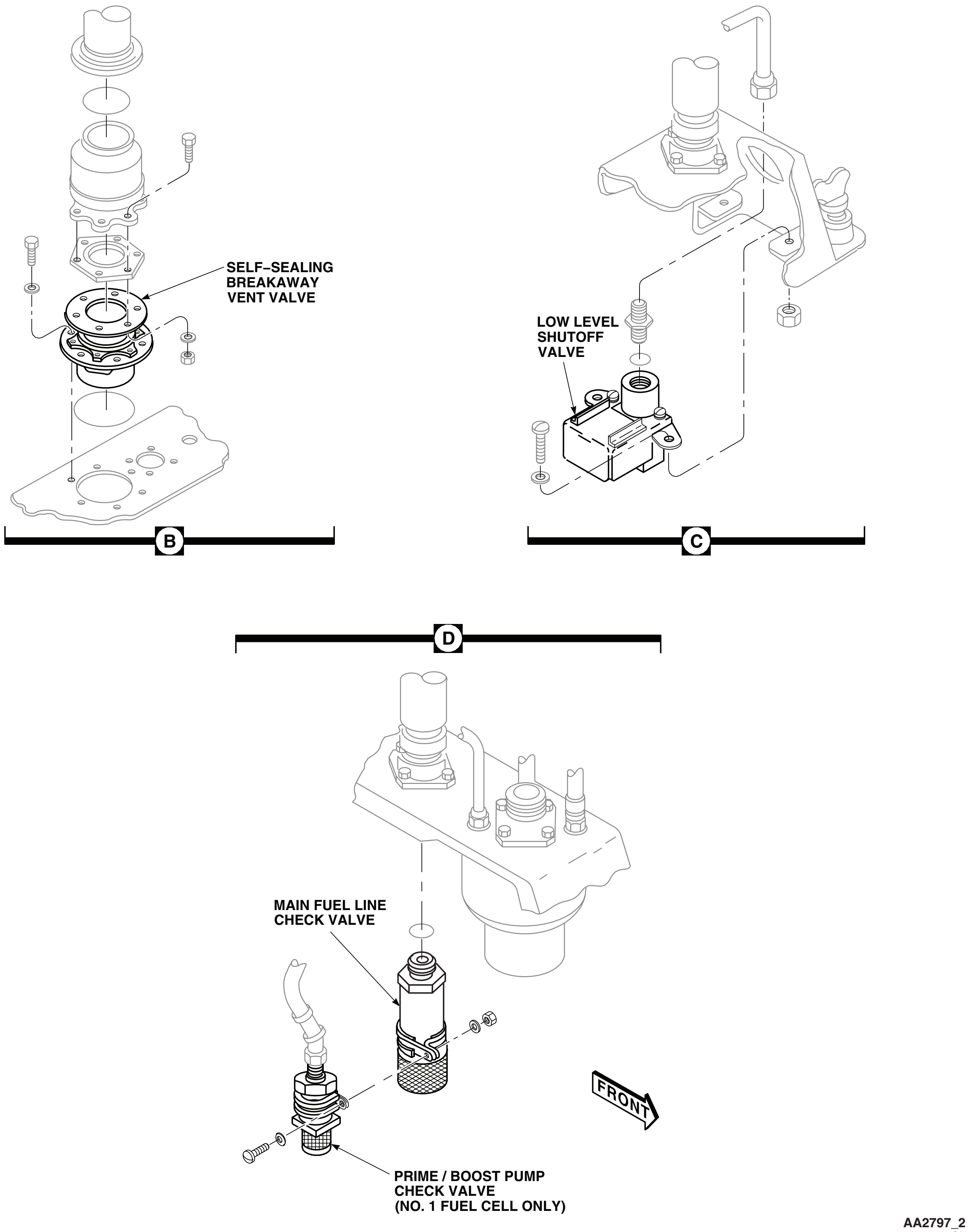


Figure 1. Pressure Refuel/Defuel System Location Diagram. (Sheet 2 of 3)

LOCATION - Continued

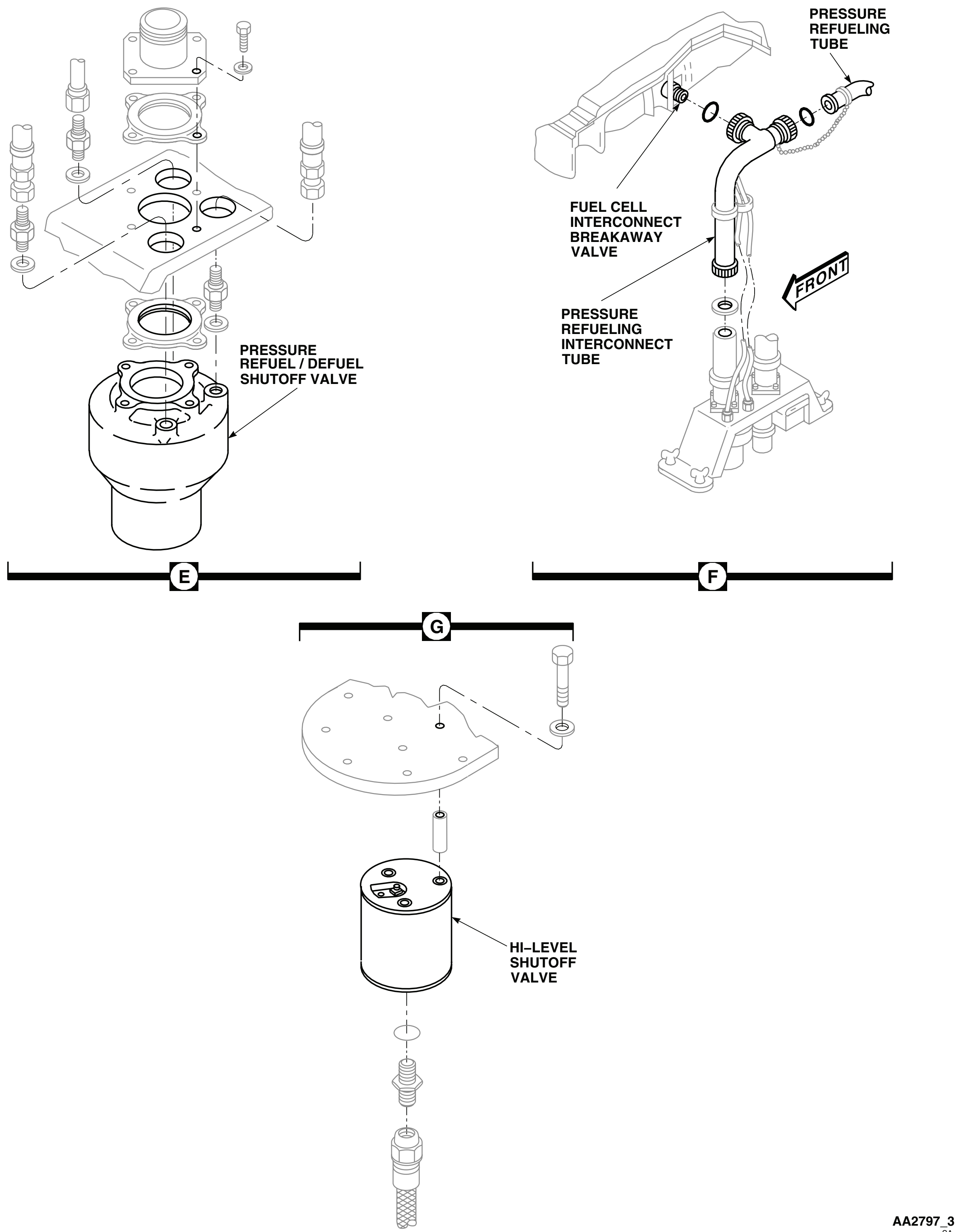


Figure 1. Pressure Refuel/Defuel System Location Diagram. (Sheet 3 of 3)

SCHEMATICS

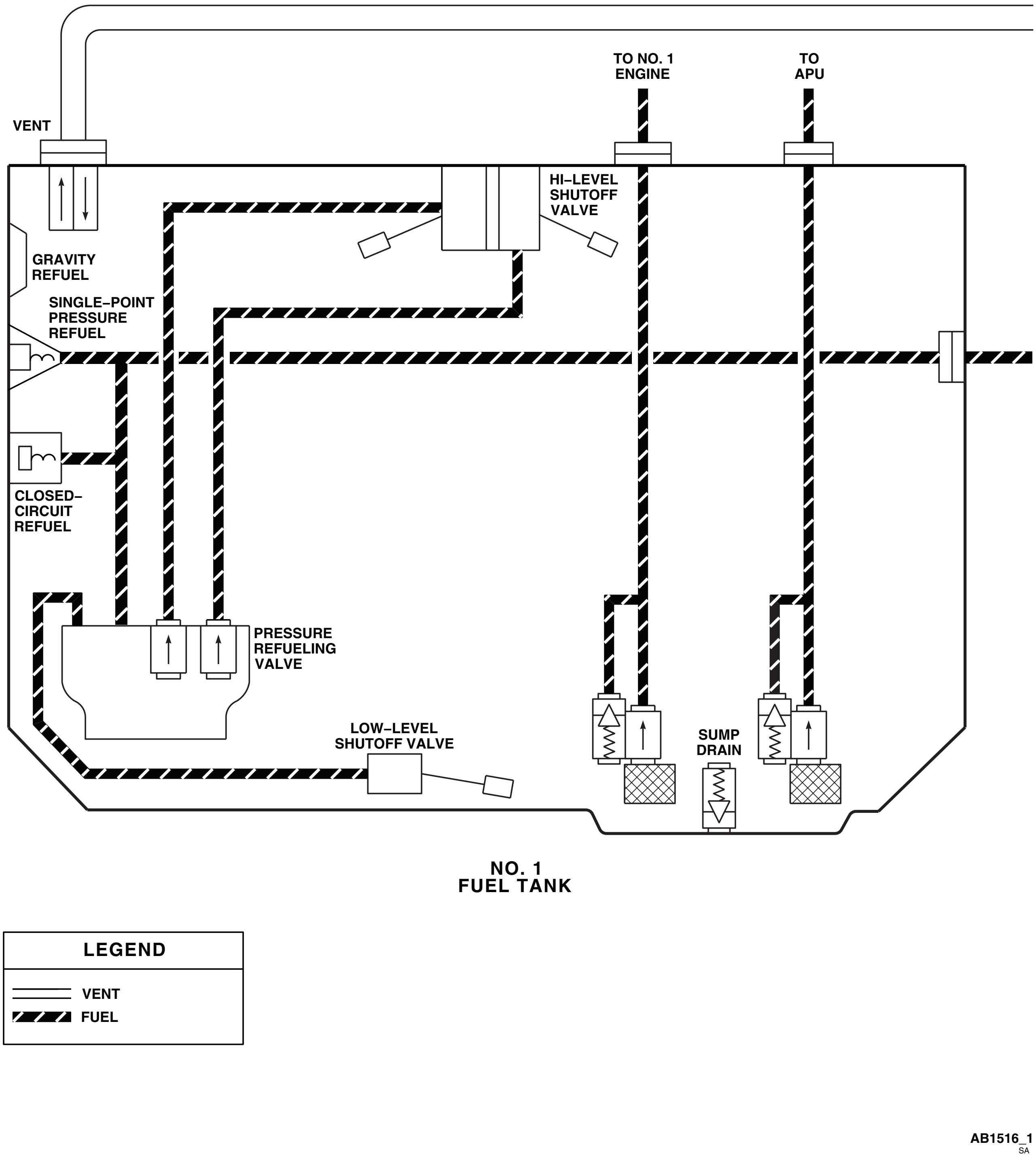


Figure 2. Pressure Refuel/Defuel System Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

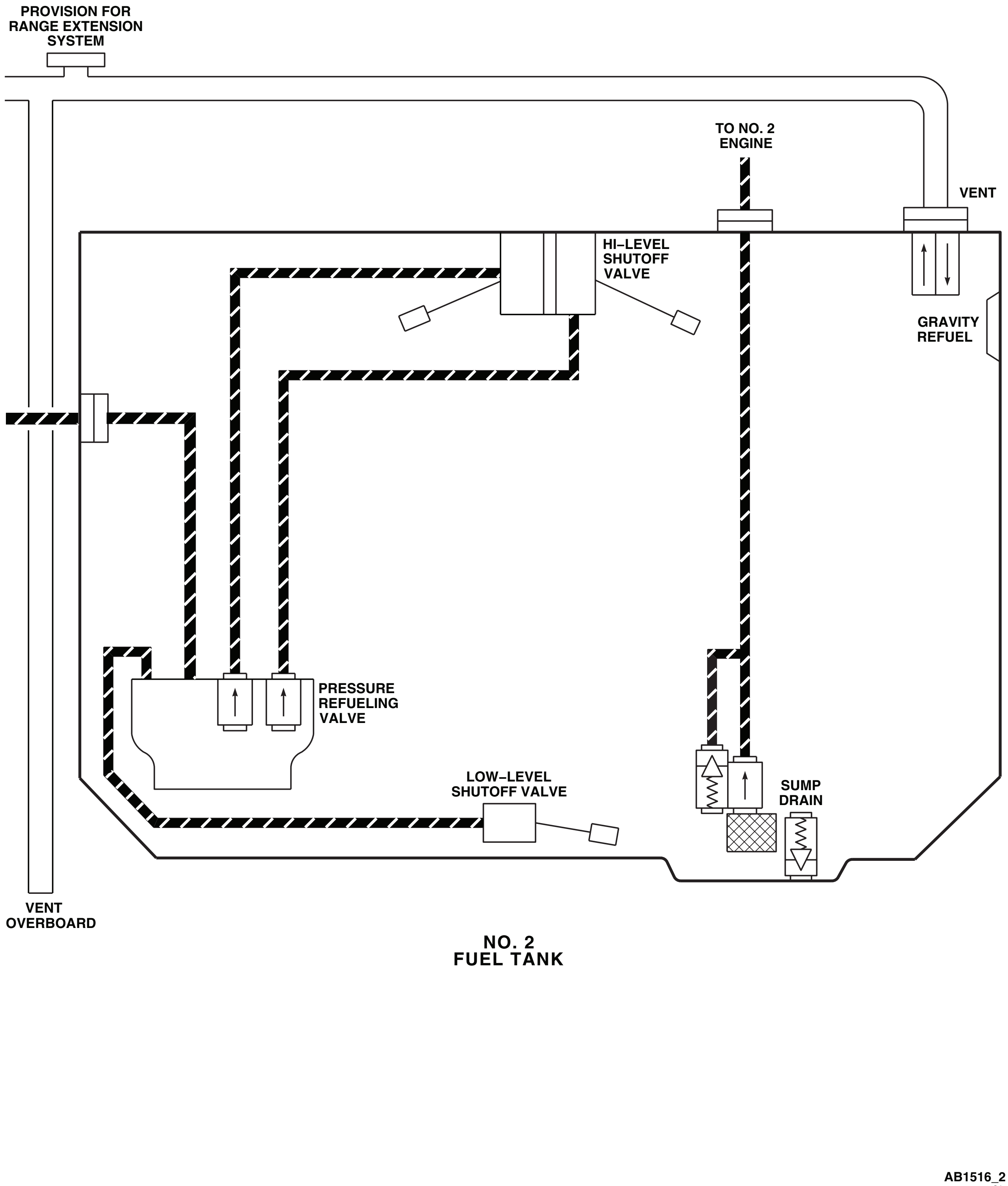


Figure 2. Pressure Refuel/Defuel System Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

FUEL SYSTEM

FUEL QUANTITY SYSTEM USING TEST SET TF-579

This WP supersedes WP 0131 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Fuel Quantity Test Set, TF-579
Multimeter, AN/PSM-45A

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06
Fuel Quantity System Harness Adapter, 70700-20602-042

Personnel Required

Aircraft Electrician MOS 15F (1)

References

WP 0089 00
WP 0104 00
WP 0824 00
WP 0830 00
WP 0925 00
WP 0994 00
WP 0995 00
WP 0997 00
WP 1001 00
WP 1685 00
WP 1747 00

Equipment Condition

External Electrical Power Available, WP 1685 00
Fuel Tanks Empty and Purged (Dry method only), WP 1001 00

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram, Figure 2 for schematic diagram, and **DRY METHOD** Figure 3 ■ and **WET METHOD** Figure 4 ■ for test setup diagrams as an aid in operational/troubleshooting.

The fuel quantity system operational/troubleshooting procedure is subdivided as follows:

- Dry Method (Preferred)
- Wet Method (Alternate)

DRY METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- On fuel quantity test set, to use PROBE 25-250 MMF control, loosen clamp at its lower right to unlock control, obtain desired indication, and then tighten clamp to lock control.

1. On fuel quantity test set (test set) front panel, remove protective cap from 115 vac receptacle and connect right-angle connector of power cable assembly (supplied with test set) to receptacle.
2. Connect other end of power cable to UTILITY RECEPTACLE 115/200 VAC receptacle in cabin ceiling.
3. Connect test set GRD and harness adapter box CASE GND binding post to any convenient point on helicopter ground bus.
4. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance.
5. Turn on external electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
6. Place test set ON-OFF switch to ON and allow 5 minutes for warm-up.
7. Calibrate test set using procedures contained with unit.
8. Connect harness adapter TANK TEST cables to test set TANK UNIT connections. Spread test harness connectors P1 and J1 apart and do not allow either to touch any metal surface.
9. Place harness adapter FUNCTION switch to TEST and TANK SELECT switch to TANK 1.
10. Place test set FUNCTION SELECTOR switch to TANK UNIT TEST UNSH and CAP-RES/CHECK switch to CAP.
11. Turn test set CAPACITANCE INDICATOR RANGE SELECTOR switch to lowest multiplier setting that gives an on-scale reading. Record dial indication; this is a measure of tank 1 harness adapter lead capacitance.
12. Place harness adapter TANK SELECT switch to TANK 2.
13. Turn test set CAPACITANCE INDICATOR RANGE SELECTOR switch to lowest multiplier setting that gives an on-scale reading. Record dial indication; this is a measure of tank 2 harness adapter lead capacitance.
14. Disconnect TANK TEST cables and connect harness adapter IND TEST cables to test set TANK UNIT connections.
15. Place harness adapter TANK SELECT switch to TANK 1.
16. Turn test set CAPACITANCE INDICATOR RANGE SELECTOR switch to lowest multiplier setting that gives an on-scale reading. Record dial indication; this is a measure of indicator 1 harness adapter lead capacitance.
17. Place harness adapter TANK SELECT switch to TANK 2.
18. Turn test set CAPACITANCE INDICATOR RANGE SELECTOR switch to lowest multiplier setting that gives an on-scale reading. Record dial indication; this is a measure of indicator 2 harness adapter lead capacitance.
19. Pull out NO. 1 AC INST circuit breaker (copilot's circuit breaker panel).
20. Disconnect P905 from fuel quantity signal conditioner connector.

DRY METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

21. Connect harness adapter J1 to P905 and harness adapter P1 to fuel quantity signal conditioner (Figure 3).
22. Connect harness adapter IND TEST cables to test set TEST IND connections and harness adapter TANK TEST cables to test set TANK UNIT connections.
23. Place harness adapter TANK SELECT switch to TANK 1 and FUNCTIONS switch to TEST.
24. Place test set FUNCTION SELECTOR switch to TANK UNIT UNSH and record capacitance indication.
25. Calculate and record capacitance of probe 1 as follows: subtract indication recorded in step 11. from indication recorded in step 24.

Indication/Condition

Capacitance of probe shall be 69.8 to 71.4 pF.

Corrective Action

If Indication/Condition is not as specified, go to PROBE CAPACITANCE IS NOT 69.8 TO 71.4 PF, in this work package.

Step

26. Calculate accurate tank 1 empty capacitance as follows: subtract indicator 1 harness adapter lead capacitance recorded in step 16. from probe 1 capacitance recorded in step 25.
27. Turn two test set PROBE MMF switches and PROBE 25-250 MMF control to get result obtained in step 26. on CAPACITANCE INDICATOR dial. Set test set as close to value as possible.
28. Set test set FUNCTION SELECTOR switch to TEST IND TEST.
29. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
30. Loosen retaining screws on signal conditioner adjustment cover plate and slide plate back to expose adjustment potentiometers.
31. Connect multimeter to OUTPUT VDC test points on test harness adapter.
32. Using a jeweler's screwdriver, adjust TK1 E potentiometer on signal conditioner to obtain between -0.005 and 0.005 vdc indication on multimeter. When adjusting control, start with control fully clockwise and then turn counterclockwise until first null indication is obtained.

Indication/Condition

a. Multimeter shall indicate between -0.005 and 0.005 vdc.

b. On central display unit, FUEL QTY 1 vertical indicator shall indicate between 0 and 50 pounds of fuel.

Corrective Action

1. If Indication/Condition a. is not as specified, go to MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 1 TANK, in this work package.
2. If Indication/Condition b. is not as specified, go to FUEL QTY 1 VERTICAL INDICATOR NOT AS SPECIFIED, in this work package.

Step

33. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
34. Place test set FUNCTION SELECTOR switch to TEST IND PROBE SET.
35. Calculate full capacitance 1 as follows: add 41.1 pF to indication recorded in step 25., then subtract indication recorded in step 16.

DRY METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

36. Turn two test set PROBE MMF switches and PROBE 25-250 MMF control to get result obtained in step 35. on CAPACITANCE INDICATOR dial. Set test set as close to value as possible.
37. Place test set FUNCTION SELECTOR switch to TEST IND TEST.
38. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
39. Using a jeweler's screwdriver, adjust TK1 F potentiometer until central display unit FUEL QTY 1 vertical indicator indicates between 1150 and 1200 pounds of fuel.

Indication/Condition

a. Multimeter shall indicate between 5.97 and 6.03 vdc.

b. On central display unit, FUEL QTY 1 vertical indicator shall indicate between 1150 and 1200 pounds of fuel.

Corrective Action

1. If Indication/Condition a. is not as specified, go to MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 1 TANK, in this work package.
2. If Indication/Condition b. is not as specified, go to FUEL QTY 1 VERTICAL INDICATOR NOT AS SPECIFIED, in this work package.

Step

40. Disconnect multimeter from test harness adapter.
41. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
42. Place harness adapter TANK SELECT switch to TANK 2.
43. Place test set FUNCTION SELECTOR switch to TANK UNIT UNSH and record capacitance indication.
44. Calculate and record capacitance of probe 2 as follows: subtract indication recorded in step 13. from indication recorded in step 43.

Indication/Condition

Capacitance of probe shall be 69.8 to 71.4 pF.

Corrective Action

If Indication/Condition is not as specified, go to PROBE CAPACITANCE IS NOT 69.8 TO 71.4 PF, in this work package.

Step

45. Calculate the accurate tank 2 empty capacitance as follows: subtract indicator 2 harness adapter lead capacitance recorded in step 18. from probe 2 capacitance recorded in step 44.
46. Turn two test set PROBE MMF switches and PROBE 25-250 MMF control to get result obtained in step 45. on CAPACITANCE INDICATOR dial. Set test set as close to value as possible.
47. Place test set FUNCTION SELECTOR switch to TEST IND TEST.
48. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
49. Connect digital multimeter to OUTPUT VDC test points on test harness adapter.
50. Using a jeweler's screwdriver, adjust TK2 E potentiometer on signal conditioner to obtain between -0.005 and 0.005 vdc indication on multimeter. When adjusting this control, start with control fully clockwise and then turn counterclockwise until first null indication is obtained.

DRY METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

a. Multimeter shall indicate between -0.005 and 0.005 vdc.

b. On central display unit, FUEL QTY 2 vertical indicator shall indicate between 0 and 50 pounds of fuel.

Corrective Action

1. If Indication/Condition a. is not as specified, go to MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 2 TANK, in this work package.
2. If Indication/Condition b. is not as specified, go to FUEL QTY 2 VERTICAL INDICATOR NOT AS SPECIFIED, in this work package.

Step

51. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
52. Place test set FUNCTION SELECTOR switch to TEST IND PROBE SET.
53. Calculate full capacitance 2 as follows: add 41.1 pF to indication recorded in step 45., then subtract indication recorded in step 18.
54. Turn two test set PROBE MMF switches and PROBE 25-250 MMF control to get result obtained in step 53. on CAPACITANCE INDICATOR dial. Set test set as close to value as possible.
55. Place test set FUNCTION SELECTOR switch to TEST IND TEST.
56. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
57. Using a jeweler's screwdriver, adjust TK2 F potentiometer until central display unit FUEL QTY 2 vertical indicator indicates between 1150 and 1200 pounds of fuel.

Indication/Condition

a. Multimeter shall indicate between 5.97 and 6.03 vdc.

b. FUEL QTY 2 vertical indicator shall indicate between 1150 and 1200 pounds of fuel.

Corrective Action

1. If Indication/Condition a. is not as specified, go to MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 2 TANK, in this work package.
2. If Indication/Condition b. is not as specified, go to FUEL QTY 2 VERTICAL INDICATOR NOT AS SPECIFIED, in this work package.

Step

58. Disconnect multimeter from test harness adapter.
59. Momentarily press lower console MISC SW panel FUEL IND TEST switch.

Indication/Condition

a. Fuel value on central display unit FUEL QTY 2 shall decrease while switch is pressed.

b. Switch shall light while pressed.

Corrective Action

If Indication/Conditions are not as specified, go to NO FUEL IND TEST FUNCTION, in this work package.

DRY METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

60. Place harness adapter TANK SELECT switch to TANK 1.
61. Momentarily press lower console MISC SW panel FUEL IND TEST switch.

Indication/Condition

a. Fuel value on central display unit FUEL QTY 1 shall decrease while switch is pressed.

b. Switch shall light while pressed.

Corrective Action

If Indication/Conditions are not as specified, go to NO FUEL IND TEST FUNCTION, in this work package.

Step

62. Pull out NO. 1 AC INST circuit breaker on copilot’s circuit breaker panel.
63. Disconnect harness adapter J1 from P905 and harness adapter P1 from fuel quantity signal conditioner.
64. Connect P905 to fuel quantity signal conditioner.
65. Push in NO. 1 AC INST circuit breaker on copilot’s circuit breaker panel.

Indication/Condition

Central display unit FUEL QTY 1 and 2 shall indicate between 0 and 50 pounds of fuel.

Corrective Action

If Indication/Condition is not as specified, replace fuel quantity probe (WP 0994 00).

Step

66. **HH-60A HH-60L** Turn MFD switch OFF.
67. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

WET METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- On fuel quantity test set, to use PROBE 25-250 MMF control, loosen clamp at its lower right to unlock control, obtain desired indication, and then tighten clamp to lock control.

1. On test set front panel, remove protective cap from 115 vac receptacle and connect right-angle connector of power cable assembly (supplied with test set) to receptacle.
2. Connect other end of power cable to UTILITY RECEPTACLE 115/200 VAC receptacle in cabin ceiling.
3. Connect test set GRD and harness adapter box CASE GND binding post to any convenient point on helicopter ground bus.
4. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance.
5. Turn on electrical power.
6. Place test set ON-OFF switch to ON and allow 5 minutes for warm-up.
7. Calibrate test set using procedures contained with unit.
8. Connect harness adapter IND TEST cables to test set TANK UNIT connections. Spread test harness connectors P1 and J1 apart and do not allow either to touch any metal surface.
9. Place harness adapter FUNCTION switch to TEST and TANK SELECT switch to TANK 1.
10. Place test set FUNCTION SELECTOR switch to TANK UNIT TEST UNSH and CAP-RES/CHECK switch to CAP.
11. Turn test set CAPACITANCE INDICATOR RANGE SELECTOR switch to lowest multiplier setting that gives an on-scale reading. Record dial indication; this is a measure of No. 1 indicator lead capacitance.
12. Calculate and record No. 1 tank empty and full capacitance as follows:
    - Empty Capacitance = 70.6 pF - step 11. indication
    - Full Capacitance = 111.7 pF - step 11. indication
13. Place harness adapter TANK SELECT switch to TANK 2.
14. Turn test set CAPACITANCE INDICATOR RANGE SELECTOR switch to lowest multiplier setting that gives an on-scale reading. Record dial indication; this is a measure of No. 2 indicator lead capacitance.
15. Calculate and record No. 2 tank empty and full capacitance as follows:
    - Empty Capacitance = 70.6 pF - step 14. indication
    - Full Capacitance = 111.7 pF - step 14. indication
16. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
17. Disconnect P905 from fuel quantity signal conditioner connector.
18. Connect harness adapter J1 to P905 and harness adapter P1 to fuel quantity signal conditioner connector (Figure 4).

WET METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

19. Connect harness adapter IND TEST cables to test set TEST IND connections.
20. Place test set FUNCTION SELECTOR switch to TEST IND PROBE SET.
21. Place harness adapter TANK SELECTOR switch to TANK 1 and FUNCTION switch to TEST.
22. Turn two test set PROBE MMF switches and PROBE 25-250 MMF control to empty capacitance value obtained in step 12. on CAPACITANCE INDICATOR dial. Set test set as close to value as possible.
23. Place test set FUNCTION SELECTOR switch to TEST IND TEST.
24. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
25. Loosen retaining screws on signal conditioner adjustment cover plate and slide plate back to expose adjustment potentiometers.
26. Connect digital multimeter to OUTPUT VDC test points on test harness adapter.
27. Using a jeweler's screwdriver, adjust TK1 E potentiometer until central display unit FUEL QTY 1 vertical indicator indicates between 0 and 50 pounds of fuel.

Indication/Condition

a. Multimeter shall indicate between -0.005 and 0.005 vdc.

b. On central display unit, FUEL QTY 1 vertical indicator shall indicate between 0 and 50 pounds of fuel.

Corrective Action

1. If Indication/Condition a. is not as specified, go to MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 1 TANK, in this work package.
2. If Indication/Condition b. is not as specified, go to FUEL QTY 1 VERTICAL INDICATOR NOT AS SPECIFIED, in this work package.

Step

28. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
29. Place harness adapter TANK SELECT switch to TANK 2 and FUNCTION switch to TEST.
30. Place test set FUNCTION SELECTOR switch to TEST IND PROBE SET.
31. Turn two test set PROBE MMF switches and PROBE 25-250 MMF control empty capacitance value obtained in step 15. on CAPACITANCE INDICATOR dial. Set test set as close to value as possible.
32. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
33. Place test set FUNCTION SELECTOR switch to TEST IND TEST.
34. Using a jeweler's screwdriver, adjust TK2 E potentiometer until central display unit FUEL QTY 2 vertical indicator indicates between 0 and 50 pounds of fuel.

Indication/Condition

a. Multimeter shall indicate between -0.005 and 0.005 vdc.

b. On central display unit, FUEL QTY 2 vertical indicator shall indicate between 0 and 50 pounds of fuel.

Corrective Action

1. If Indication/Condition a. is not as specified, go to MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 2 TANK, in this work package.

WET METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. If Indication/Condition b. is not as specified, go to FUEL QTY 2 VERTICAL INDICATOR NOT AS SPECIFIED, in this work package.

Step

35. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
36. Place test set FUNCTION SELECTOR switch to TEST IND PROBE SET.
37. Place harness adapter TANK SELECT switch to TANK 1.
38. Turn two test set PROBE MMF switches and PROBE 25-250 MMF control to full capacitance value obtained in step 12. on CAPACITANCE INDICATOR dial. Set test set as close to value as possible.
39. Place test set FUNCTION SELECTOR switch to TEST IND TEST.
40. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
41. Using a jeweler's screwdriver, adjust signal conditioner TK1 F potentiometer until central display unit FUEL QTY 1 vertical indicator indicates between 1150 and 1200 pounds of fuel.

Indication/Condition

a. Multimeter shall indicate between 5.97 and 6.03 vdc.

b. On central display unit, FUEL QTY 1 vertical indicator shall indicate between 1150 and 1200 pounds of fuel.

Corrective Action

1. If Indication/Condition a. is not as specified, go to MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 1 TANK, in this work package.
2. If Indication/Condition b. is not as specified, go to FUEL QTY 1 VERTICAL INDICATOR NOT AS SPECIFIED, in this work package.

Step

42. Momentarily press lower console MISC SW panel FUEL IND TEST switch.

Indication/Condition

a. Fuel value on central display unit FUEL QTY 1 shall decrease while switch is pressed.

b. Switch shall light while pressed.

Corrective Action

If Indication/Conditions are not as specified, go to NO FUEL IND TEST FUNCTION, in this work package.

Step

43. Place test set FUNCTION SELECTOR switch to TEST IND PROBE SET.
44. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
45. Place harness adapter TANK SELECT switch to TANK 2.
46. Turn two test set PROBE MMF switches and PROBE 25-250 MMF control to full capacitance value obtained in step 15. on CAPACITANCE INDICATOR dial. Set test set as close to value as possible.
47. Place test set FUNCTION SELECTOR switch to TEST IND TEST.
48. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.

WET METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

49. Using a jeweler's screwdriver, adjust TK2 F potentiometer until central display unit FUEL QTY 2 vertical indicator indicates between 1150 and 1200 pounds of fuel.

Indication/Condition

a. Multimeter shall indicate between 5.97 and 6.03 vdc.

b. On central display unit, FUEL QTY 2 vertical indicator shall indicate between 1150 and 1200 pounds of fuel.

Corrective Action

1. If Indication/Condition a. is not as specified, go to MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 2 TANK, in this work package.
2. If Indication/Condition b. is not as specified, go to FUEL QTY 2 VERTICAL INDICATOR NOT AS SPECIFIED, in this work package.

Step

50. Momentarily press lower console MISC SW panel FUEL IND TEST switch.

Indication/Condition

a. Fuel value on central display unit FUEL QTY 2 shall decrease while switch is pressed.

b. Switch shall light while pressed.

Corrective Action

If Indication/Conditions are not as specified, go to NO FUEL IND TEST FUNCTION, in this work package.

Step

51. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
52. Disconnect multimeter from test harness.
53. Disconnect harness adapter J1 from P905 and harness adapter P1 from fuel quantity signal conditioner.
54. Connect P905 to fuel quantity signal conditioner.
55. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

PROBE CAPACITANCE IS NOT 69.8 TO 71.4 PF

SYMPTOM

Probe capacitance is not 69.8 to 71.4 pf.

MALFUNCTION

Probe capacitance is not 69.8 to 71.4 pf.

CORRECTIVE ACTION

1. Check continuity between:

PROBE CAPACITANCE IS NOT 69.8 TO 71.4 PF - Continued

- **P905-M and J307-D**
- **P905-N and J308-D**
- **P905-B and J307-G**
- **P905-D and J308-G**

a. If continuity is present, replace fuel quantity probe (WP 0994 00). Go to **3.**

b. If trouble remains, replace fuel tank wiring harness (WP 0995 00). Go to **3.**

c. If continuity is not present, go to **2.**

2. Check continuity between:

- **P135-L and P135-K**
- **P135-H and P135-J**

a. If continuity is present, replace miscellaneous switch panel (WP 0830 00). Go to **3.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **3.**

3. Procedure completed.

MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 1 TANK

SYMPTOM

Multimeter does not indicate specified voltage for No. 1 tank.

MALFUNCTION

Multimeter does not indicate specified voltage for No. 1 tank.

CORRECTIVE ACTION

1. Check for 115 vac between P905-H and ground.

a. If voltage is as specified go to **2.**

b. If voltage is not as specified, go to **6.**

2. Check helicopter configuration.

a. **EMEP** , go to **3.**

b. **W/O EMEP** , go to **5.**

3. EMEP Remove and check pin filtered adapter P138 (WP 0925 00).

a. If pin filtered adapter is good, go to **4.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **7.**

4. EMEP Install pin filtered adapter (WP 0925 00). Go to 5.

5. Check continuity between:

- **P138-G and P905-L**
- **P138-H and P905-K**
- **P905-F and ground**
- **P905-G and ground**

MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 1 TANK - Continued

- **P905-J and ground**

a. If continuity is present, replace fuel quantity signal conditioner (WP 0997 00). Go to **7.**

b. If trouble remains, replace fuel quantity probe or wiring harness (WP 0994 00). Go to **7.**

c. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7.**

6. Check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between terminal 1 and P905-H (WP 1747 00). Go to **7.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **7.**

7. Procedure completed.

FUEL QTY 1 VERTICAL INDICATOR NOT AS SPECIFIED

SYMPTOM

FUEL QTY 1 vertical indicator not as specified.

MALFUNCTION

FUEL QTY 1 vertical indicator not as specified.

CORRECTIVE ACTION

1. Check helicopter configuration.

a. **EMEP** , go to **2.**

b. **W/O EMEP** , go to **4.**

2. EMEP Remove and check pin filtered adapter P138 (WP 0925 00).

a. If pin filtered adapter is good, go to **3.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **6.**

3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.

4. Check continuity between:

- **P905-L and P138-G**
- **P905-K and P138-H**
- **P905-M and P135-H**
- **P135-J and J307-D**
- **J307-G and P905-B**

a. If continuity is present go to **5.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**

5. Check continuity between P905-M and J307-D.

a. If continuity is present, troubleshoot instrument display system (WP 0089 00). Go to **6.**

b. If trouble remains, replace fuel quantity signal conditioner (WP 0997 00). Go to **6.**

c. If continuity is not present, replace miscellaneous switch panel (WP 0830 00). Go to **6.**

FUEL QTY 1 VERTICAL INDICATOR NOT AS SPECIFIED - Continued

6. **Procedure completed.**

MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 2 TANK

SYMPTOM

Multimeter does not indicate specified voltage for No. 2 tank.

MALFUNCTION

Multimeter does not indicate specified voltage for No. 2 tank.

CORRECTIVE ACTION

1. **Check for 115 vac between P905-H and ground.**
   a. If voltage is as specified, go to **2.**
   b. If voltage is not as specified, go to **6.**
2. **Check helicopter configuration.**
   a. EMEP , go to **3.**
   b. W/O EMEP , go to **5.**
3. **EMEP Remove and check pin filtered adapter P137 (WP 0925 00).**
   a. If pin filtered adapter is good, go to **4.**
   b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **7.**
4. **EMEP Install pin filtered adapter (WP 0925 00). Go to 5.**
5. **Check continuity between:**
   - **P137-G and P905-U**
   - **P137-H and P905-V**
   - **P905-F and ground**
   - **P905-G and ground**
   - **P905-J and ground**

   a. If continuity is present, replace fuel quantity signal conditioner (WP 0997 00). Go to **7.**
   b. If trouble remains, replace fuel quantity probe or wiring harness (WP 0994 00). Go to **7.**
   c. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7.**
6. **Check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.**
   a. If voltage is as specified, repair/replace wiring between terminal 1 and P905-H (WP 1747 00). Go to **7.**
   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **7.**
7. **Procedure completed.**

FUEL QTY 2 VERTICAL INDICATOR NOT AS SPECIFIED

SYMPTOM

FUEL QTY 2 vertical indicator not as specified.

MALFUNCTION

FUEL QTY 2 vertical indicator not as specified.

CORRECTIVE ACTION

1. Check helicopter configuration.

a. EMEP , go to **2.**

b. W/O EMEP , go to **4.**

2. EMEP Remove and check pin filtered adapter P137 (WP 0925 00).

a. If pin filtered adapter is good, go to **3.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **6.**

3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.

4. Check continuity between:

- **P905-U and P137-G**
- **P905-V and P137-H**
- **P905-N and P135-L**
- **P135-K and J308-D**
- **J308-G and P905-D**

a. If continuity is present, go to **5.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**

5. Check continuity between P905-N and J308-D.

a. If continuity is present, troubleshoot instrument display system (WP 0089 00). Go to **6.**

b. If trouble remains, replace fuel quantity signal conditioner (WP 0997 00). Go to **6.**

c. If continuity is not present, replace miscellaneous switch panel (WP 0830 00). Go to **6.**

6. Procedure completed.

NO FUEL IND TEST FUNCTION

SYMPTOM

No FUEL IND TEST function.

MALFUNCTION

No FUEL IND TEST function.

CORRECTIVE ACTION

1. With FUEL IND TEST switch pressed, check for open circuit between:

- **J135-K and J135-L**

NO FUEL IND TEST FUNCTION - Continued

- **J135-H and J135-J**

a. If open circuit is present, replace fuel quantity signal conditioner (WP 0997 00). Go to **2.**

b. If trouble remains, troubleshoot instrument display system (WP 0089 00). Go to **2.**

c. If open circuit is not present, replace miscellaneous switch panel (WP 0830 00). Go to **2.**

2. Procedure completed.

DIGITAL DISPLAY DOES NOT INDICATE GREATER THAN 1 MEGOHM

SYMPTOM

Digital display does not indicate greater than 1 megohm.

MALFUNCTION

Digital display does not indicate greater than 1 megohm.

CORRECTIVE ACTION

1. With multimeter, check for greater than 1 megohm between:

- **P905-B and P905-A**
- **P905-B and P905-P**
- **P905-D and P905-E**
- **P905-D and P905-R**
- **P307-G and P307-F**
- **P307-G and P307-H**
- **P307-G and P307-K**
- **P307-G and P307-E**
- **P308-G and P308-F**
- **P308-G and P308-H**
- **P308-G and P308-K**
- **P308-G and P308-E**

a. If multimeter indicates greater than 1 megohm, replace fuel quantity probe (WP 0994 00). Go to **2.**

b. If multimeter does not indicate greater than 1 megohm, repair/replace wiring (WP 1747 00). Go to **2.**

2. Procedure completed.

DIGITAL DISPLAY DOES NOT INDICATE 0 (ZERO) MEGOHM

SYMPTOM

Digital display does not indicate 0 (zero) megohm.

MALFUNCTION

Digital display does not indicate 0 (zero) megohm.

CORRECTIVE ACTION

1. With multimeter, check for 0 megohm between:

- **P905-A and ground**
- **P905-E and ground**
- **P905-F and ground**
- **P905-G and ground**
- **P905-J and ground**
- **P905-P and ground**
- **P905-R and ground**

a. If multimeter indicates 0 megohm, replace fuel quantity signal conditioner unit (WP 0997 00). Go to **2.**

b. If multimeter does not indicate 0 megohm, repair/replace wiring (WP 1747 00). Go to **2.**

2. Procedure completed.

Table 1. Worksheet 1-A. Dry Method (Preferred).

NUMBER	ACTION	TANK #1	TANK #2
1.	Measure empty tank plus cable capacitance.	pF	pF
2.	Measure ″T″ cable capacitance.	pF	pF
3.	Subtract line 2 from line 1. Actual empty tank capacitance equals (must be between 69.8 pF and 71.4 pF).	pF	pF
4.	Measure LOZ-HIZ insulation resistance (must be over 20 megohms).	meg	meg

DIGITAL DISPLAY DOES NOT INDICATE 0 (ZERO) MEGOHM - Continued

Table 1. Worksheet 1-A. Dry Method (Preferred). - Continued

NUMBER	ACTION	TANK #1	TANK #2
5.	Measure insulation resistance of: (must be over 1 megohm) LOZ-SHLD HIZ-SHLD LOZ-GND HIZ-GND	meg meg meg meg	meg meg meg meg
6.	Measure insulation resistance of SHLD-GND (must be 0 megohm).	meg	meg
	Nominal empty tank capacitance is:	40.0 pF	40.0 pF
7.	Subtract line 2 cable capacitance ″simulated added″ capacitance equals	pF	pF
8.	Switch FUNCT switch to AC ONLY Adjust TK E potentiometer to read between -0.005 and 0.005 vdc	vdc	vdc
9.	Vertical scale indicators read Total fuel indicator reads lb	lb	lb
10.	Switch FUNCT switch to SIM TU ONLY		
11.	Adjust TK F potentiometer to read between 5.95 and 6.05 vdc Vertical scale indicators read Total fuel indicator reads lb	vdc lb	vdc lb
12.	Readjust TK E potentiometer, if necessary, to read between -0.005 and 0.005 vdc	vdc	vdc

DIGITAL DISPLAY DOES NOT INDICATE 0 (ZERO) MEGOHM - Continued

Table 2. Worksheet 1-B. Wet Method (Alternate).

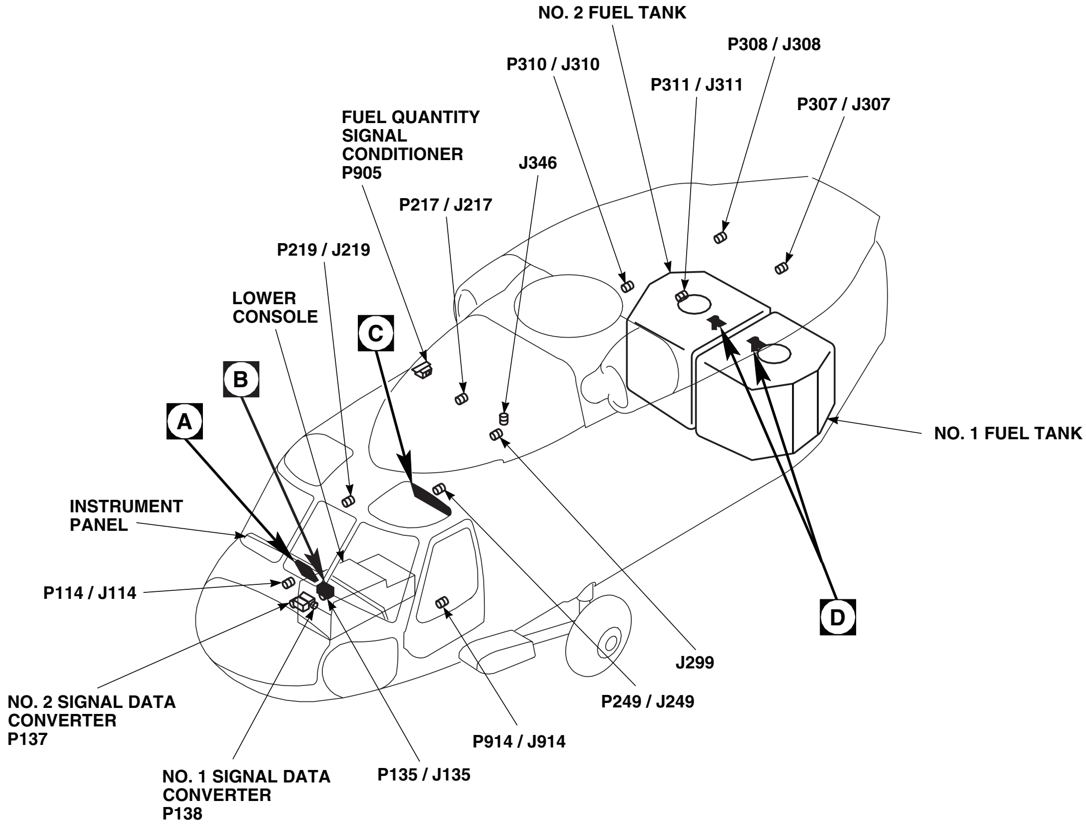
NUMBER	ACTION	TANK #1	TANK #2
1.	Nominal empty tank capacitance equals	70.6 pF	70.6 pF
2.	Measure ″T″ cable capacitance	pF	pF
3.	Subtract line 2 from line 1 Simulated empty tank capacitance equals	pF	pF
4.	Measure LOZ-HIZ insulation resistance (must be over 20 megohms).	meg	meg
5.	Measure insulation resistance of: (must be over 1 megohms)		
	LOZ-SHLD	meg	meg
	HIZ-SHLD	meg	meg
	LOZ-GND	meg	meg
	HIZ-GND	meg	meg
6.	Measure insulation resistance of SHLD-GND (must read 0 megohms).	meg	meg
7.	Adjust TK E potentiometer to read between -0.005 and 0.005 vdc	vdc	vdc
8.	Vertical scale indicators read	lb	lb
9.	Total fuel indicator reads lb	lb	lb
10.	Adjust TK F potentiometer to read between 5.95 and 6.05 vdc	vdc	vdc
11.	Vertical scale indicators read	lb	lb

DIGITAL DISPLAY DOES NOT INDICATE 0 (ZERO) MEGOHM - Continued

Table 2. Worksheet 1-B. Wet Method (Alternate). - Continued

NUMBER	ACTION	TANK #1	TANK #2
12.	Total fuel indicator reads lb	lb	lb
13.	Readjust TK E potentiometer to read between -0.005 and 0.005 vdc	vdc	vdc

LOCATION



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
J299	CABIN CEILING BL 10 LH, STA 302
J346 (SEE NOTE)	CABIN CEILING BL 18 LH, STA 301
P114 / J114	COCKPIT BL 8 RH, STA 200
P135 / J135	MISCELLANEOUS SWITCH PANEL
P137	INSTRUMENT DISPLAY SYSTEM NO. 2 SIGNAL DATA CONVERTER

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P138	INSTRUMENT DISPLAY SYSTEM NO. 1 SIGNAL DATA CONVERTER
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P307 / J307	TRANSITION DECK BL 15 LH, STA 430
P308 / J308	TRANSITION DECK BL 15 RH, STA 430
P310 / J310	CABIN CEILING BL 17 RH, STA 387
P311 / J311	CABIN CEILING BL 16 LH, STA 383
P905	FUEL QUANTITY SIGNAL CONDITIONER
P914 / J914	COCKPIT BL 24 LH, STA 247

NOTE

UH–60A 82–23748 – SUBQ
EH–60A UH–60L HH–60A
HH–60L

AA2789_1C
SA

Figure 1. Fuel Quantity System Location Diagram. (Sheet 1 of 2)

LOCATION - Continued

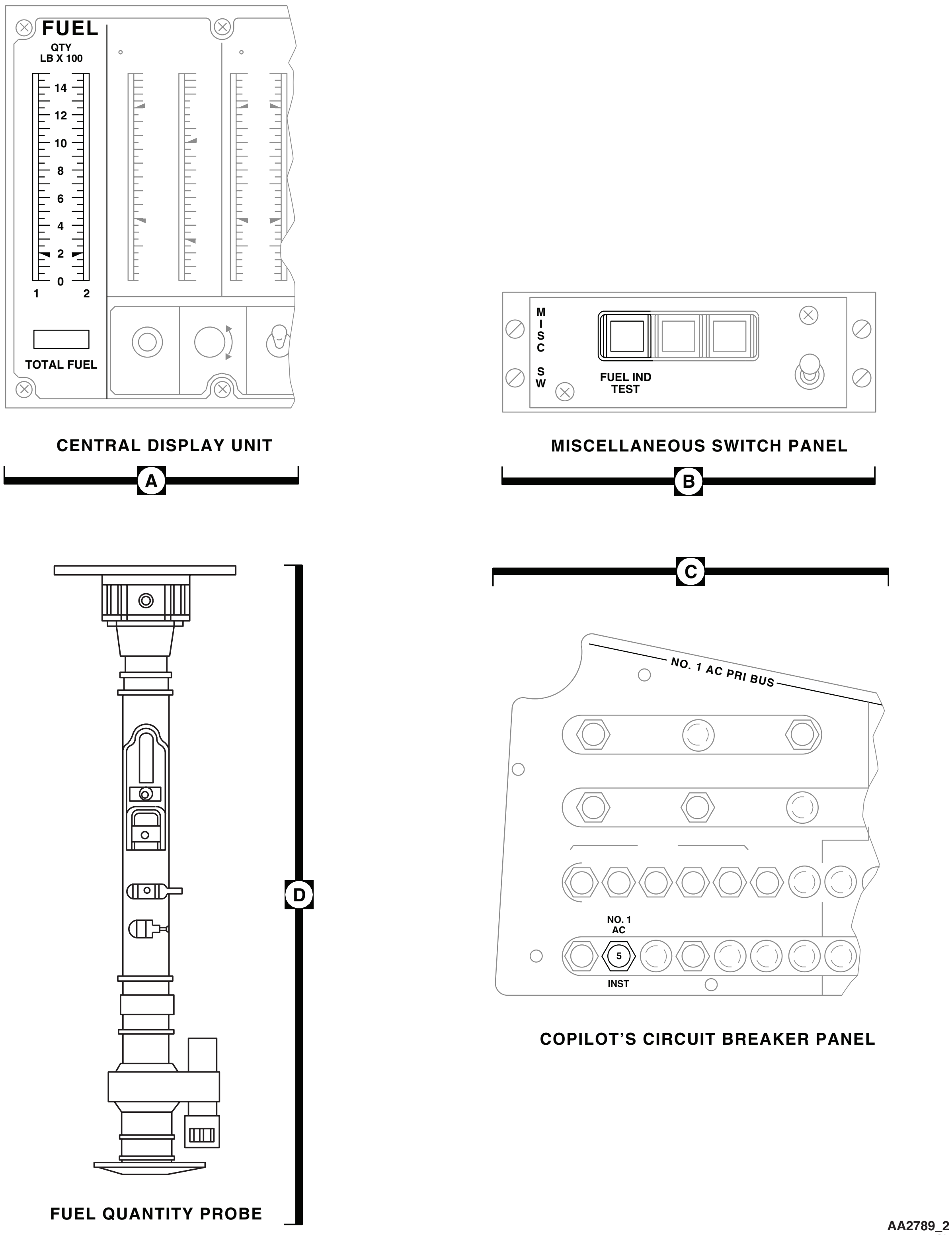


Figure 1. Fuel Quantity System Location Diagram. (Sheet 2 of 2)

SCHEMATICS

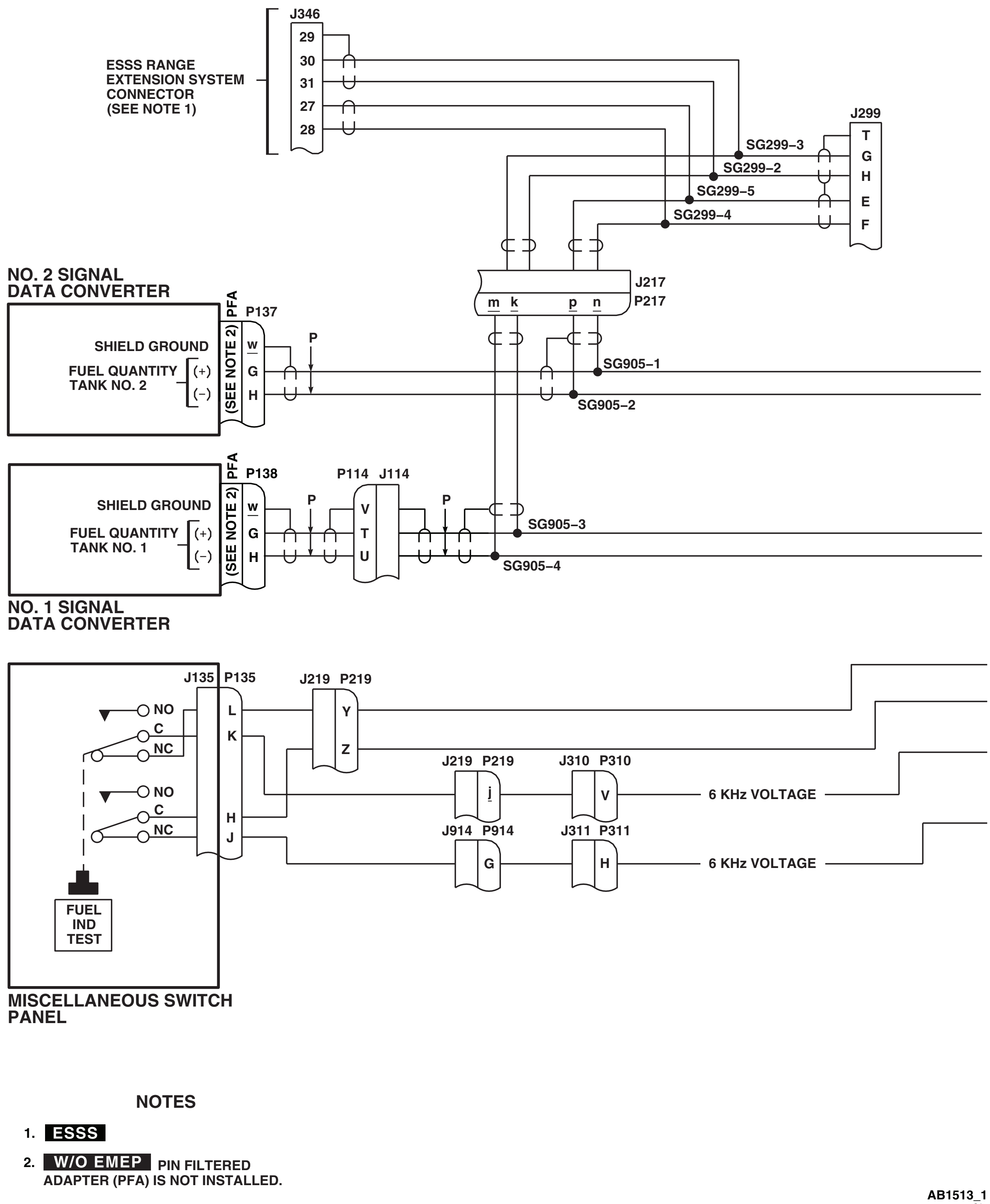


Figure 2. Fuel Quantity System Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

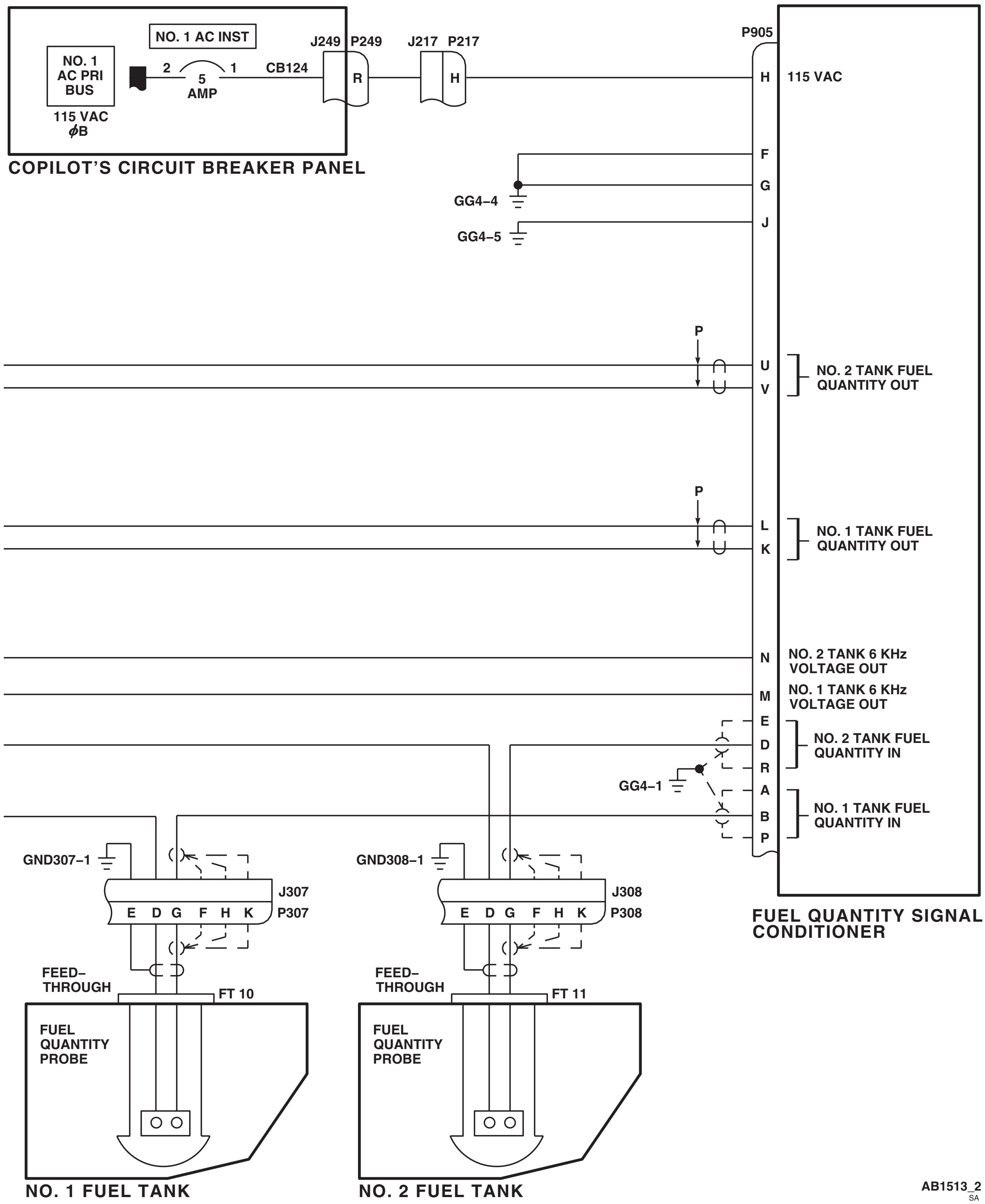


Figure 2. Fuel Quantity System Schematic Diagram. (Sheet 2 of 2)

TEST SETUP

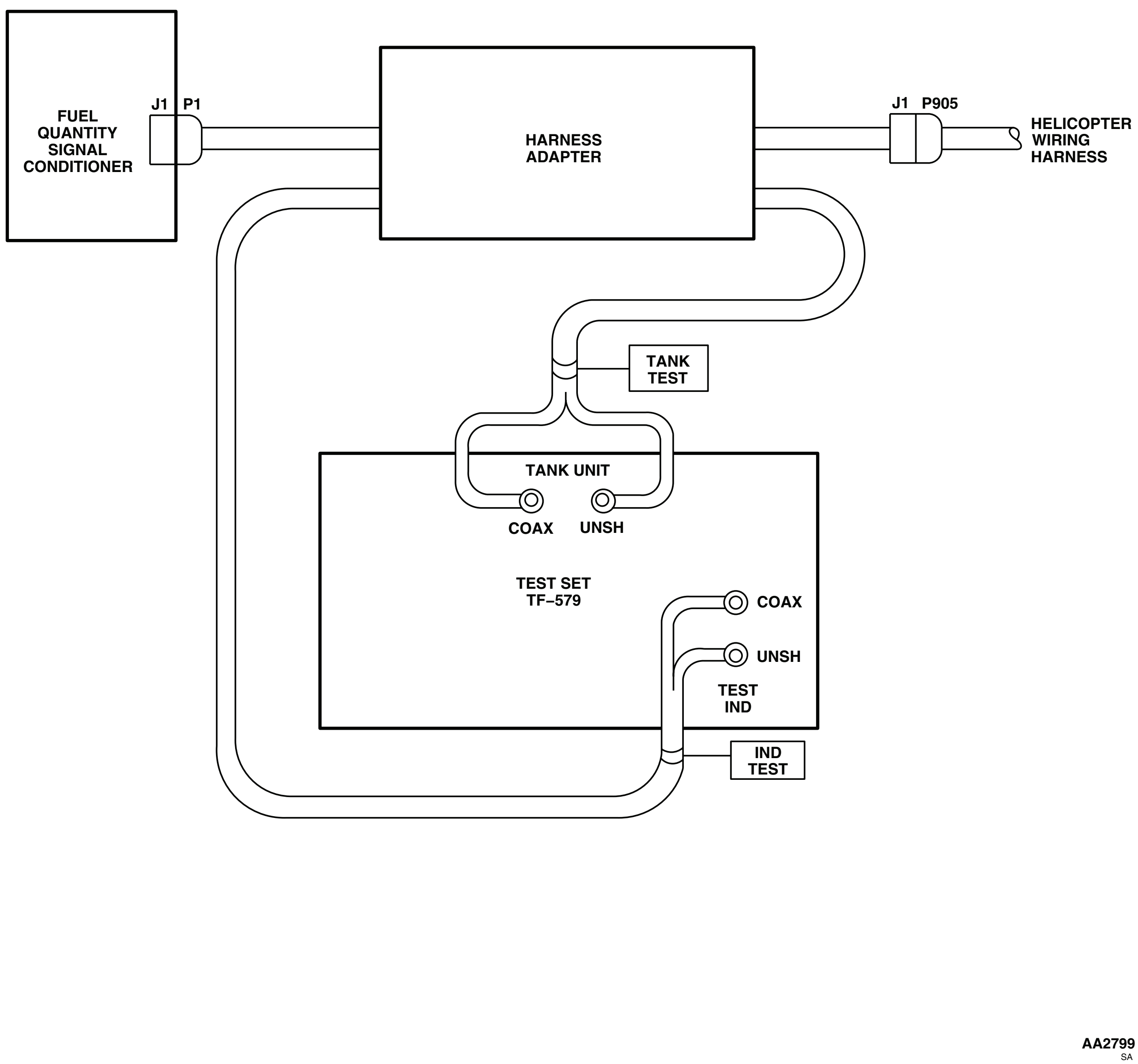


Figure 3. Fuel Quantity System Test Setup Diagram **DRY METHOD** .

TEST SETUP - Continued

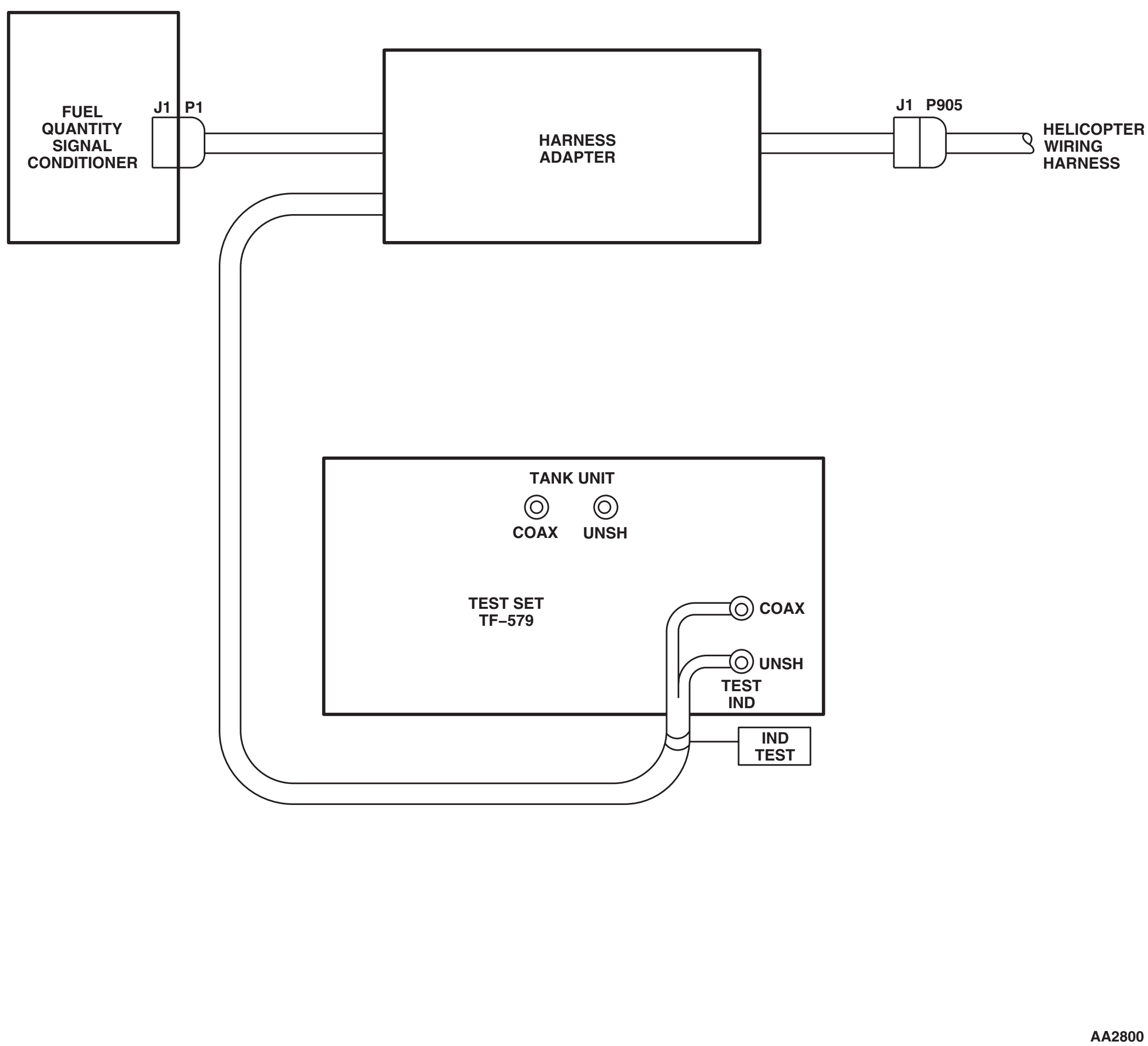


Figure 4. Fuel Quantity System Test Setup Diagram WET METHOD .

END OF WORK PACKAGE

UNIT LEVEL

FUEL SYSTEM

FUEL QUANTITY SYSTEM USING TEST SET PSD60-1AF

This WP supersedes WP 0132 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Fuel Quantity Test Set, PSD60-1AF
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06
Fuel Quantity System ″T″ Cable, PSD, AF-113

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0089 00
WP 0104 00
WP 0824 00
WP 0830 00
WP 0925 00
WP 0994 00
WP 0995 00
WP 0997 00
WP 1615 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00
Fuel Tanks Empty and Purged (Dry method only), WP 1001 00

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram, Figure 2 for schematic diagram, and Figure 3 for test setup diagram as an aid in operational/troubleshooting.

The fuel quantity system operational/troubleshooting procedure is subdivided as follows:

- Dry Method (Preferred)
- Wet Method (Alternate)

DRY METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- On the fuel quantity test set, to use PROBE 25-250 MMF control, loosen clamp at its lower right to unlock control, obtain desired indication, and then tighten clamp to lock control.

1. Defuel tank to be calibrated. Completely drain all fuel from tank. Failure to do so invalidates calibration (WP 1615 00).
2. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance.
3. Turn on external electrical power. **HH-60A** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
4. Pull out (open) the NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
5. Lift to unlock test set ON-OFF switch and place to ON. If words ″LO BAT″ appear in upper left-hand corner of digital display, replace batteries.
6. Allow test set to warm up for at least 3 minutes.
7. Connect PSDAF-113 ″T″ cable LOZ lead for tank being tested to test set TANK UNITS LOZ terminal J-4.
8. Connect PSDAF-113 ″T″ cable HIZ lead for tank being tested to test set TANK UNITS HIZ terminal J-5.

NOTE

Do not connect ″CP″ lead to test set for this test.

9. Turn test set FUNCT switch to MEASURE EXT.
10. Turn test set SELECT switch to TU.
11. Place the TANKS IN/OUT switch on ″T″ cable to IN.
12. Read capacitance of TU test leads.

Indication/Condition

Test set digital readout shall be 0.5 pF or less.

Corrective Action

If reading is over 0.5 pF, turn in cable for repair.

Step

13. Record capacitance on Worksheet 1-A., line 2.
14. Locate fuel quantity signal conditioner unit (SCU) on right side of cabin, over crewchief/gunner's window, and disconnect P905 from SCU.
15. Connect J-1 of ″T″ cable to P905.

DRY METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

16. Read capacitance of empty tank and ″T″ cable. Record capacitance on Worksheet 1-A., line 1.
17. Subtract capacitance recorded in step 13. from reading displayed in step 16.

Indication/Condition

Empty tank capacitance calculated shall be between 69.1 pF and 72.1 pF.

Corrective Action

If Indication/Condition is not as specified, go to PROBE CAPACITANCE IS NOT 69.1 TO 72.1 PF, in this work package.

Step

18. Record calculated empty tank capacitance on Worksheet 1-A., line 3.
19. Turn SELECT switch on test set to LOZ-HIZ.

NOTE

The PSD60-1AF test set reads ″over-range″ above 10,000 megohms by flashing three colons and displaying four zeros.

20. Allow reading to stabilize for 30 seconds. Verify that test set digital display reads over 20 megohms.
21. Turn SELECT switch to the following positions and verify that test set digital display reads over 1 megohm in each position.
    - LOZ-SHLD
    - HIZ-SHLD
    - LOZ-GND
    - HIZ-GND

Indication/Condition

Digital display shall indicate over 1 megohm in each position.

Corrective Action

If Indication/Condition is not as specified, go to DIGITAL DISPLAY DOES NOT INDICATE GREATER THAN 1 MEGOHM, in this work package.

Step

22. Turn SELECT switch to SHLD-GND position.

Indication/Condition

Digital readout on test set indicates 0 megohms.

Corrective Action

If Indication/Condition is not as specified, go to DIGITAL DISPLAY DOES NOT INDICATE 0 (ZERO) MEGOHM, in this work package.

Step

23. Disconnect ″T″ cable LOZ and HIZ leads from test set TANK UNITS terminals.

DRY METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

24. Connect cable LOZ lead to test set INDICATOR LOZ terminal J-1.
25. Connect cable HIZ lead to test set INDICATOR HIZ terminal J-2.
26. Turn test set FUNCT switch to MEASURE INT.
27. Turn SELECT switch to TU.
28. Set TU simulator decade D-1 to read ″002″.
29. Subtract cable capacitance measured in step 13. from 40.0 pF.
30. Adjust TU vernier V-1 on test set until digital readout indicates calculated capacitance value in step 29.
31. Turn FUNCT switch to AIRCRAFT ONLY.
32. Connect ″T″ cable P-1 to SCU.
33. Select 0-10 VDC scale on digital multimeter.
34. Connect digital voltmeter leads to METER jacks on ″T″ cable. Verify correct polarity. Turn meter ON.
35. Switch the DC OUT switch on PSDAF-113 ″T″ cable to tank being calibrated.
36. Push in (close) NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
37. Loosen screws holding SCU cover plate. Slide plate back to expose potentiometers.
38. Identify ″TK E″ potentiometer for tank being calibrated.
39. While observing DC voltage and polarity on meter, adjust ″TK E″ potentiometer with nonmagnetic screwdriver, counterclockwise.
    - If polarity on multimeter is positive (+) and adjustment decreases the voltage, continue turning ″TK E″ potentiometer counterclockwise until multimeter indicates -0.05 vdc, then turn potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.
    - If polarity on multimeter is positive (+) and adjustment increases the voltage, turn ″TK E″ potentiometer clockwise until multimeter indicates negative (-) polarity, continue turning potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.
    - If polarity on multimeter is negative (-), turn ″TK E″ potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.

Indication/Condition

Multimeter shall indicate -0.005 to 0.005 vdc.

Corrective Action

If Indication/Condition is not as specified, go to MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 1 TANK or MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 2 TANK, in this work package, for tank being tested.

Step

40. Observe fuel quantity indicators on instrument panel.

Indication/Condition

Fuel quantity indicators shall read as follows:

- vertical scale 0 lb

DRY METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- total fuel 0-20 lb

Corrective Action

If Indication/Condition is not as specified, go to FUEL QTY 1 VERTICAL INDICATOR NOT AS SPECIFIED or FUEL QTY 2 VERTICAL INDICATOR NOT AS SPECIFIED, in this work package, for tank being tested.

Step

41. Connect test set CHASSIS jack J-7 to airframe ground block near signal conditioner unit using ground lead supplied with test set.
42. Turn the test set FUNCT switch to SIM TU ONLY.
43. Adjust the ″TK F″ potentiometer on the SCU for tank being calibrated to read between 5.95 to 6.05 vdc on multimeter.

Indication/Condition

Multimeter shall indicate 5.95 to 6.05 vdc.

Corrective Action

If Indication/Condition is not as specified, go to MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 1 TANK or MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 2 TANK, in this work package, for tank being tested.

Step

44. Observe fuel quantity indicators on instrument panel.

Indication/Condition

Fuel quantity indicators shall read as follows:

- vertical scale 1200 lb
- total fuel 1180-1220 lb

Corrective Action

If Indication/Condition is not as specified, go to FUEL QTY 1 VERTICAL INDICATOR NOT AS SPECIFIED or FUEL QTY 2 VERTICAL INDICATOR NOT AS SPECIFIED, in this work package, for tank being tested.

Step

45. Turn FUNCT switch back to AC ONLY position. Adjust ″TK E″ potentiometer for tank being calibrated, if necessary, to read between -0.005 to 0.005 vdc on multimeter.
46. Repeat steps 7. through 45. for other tank if it is being calibrated.
47. Pull (open) NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
48. Disconnect PSDAF-113 ″T″ cable from SCU, from P905, and from test set.
49. Reconnect P905 to SCU.
50. Turn MFD switch OFF.
51. Turn off electrical power.
52. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
53. Turn test set OFF. Store test set grounding cable in lid of test set.

DRY METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Test set shall be off.

Corrective Action

None Required

WET METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- On the fuel quantity test set, to use PROBE 25-250 MMF control, loosen clamp at its lower right to unlock control, obtain desired indication, and then tighten clamp to lock control.

1. Connect test set CHASSIS jack J-7 to airframe ground block near signal conditioner unit using ground lead supplied with test set.
2. Pull out (open) the NO. 1 AC INST circuit breaker on copilot's AC CB panel.
3. Lift to unlock test set ON-OFF switch and turn to ON. If words ″LO BAT″ appear in upper left-hand corner of digital display, replace batteries.
4. Allow test set to warm up for at least 3 minutes.
5. Switch the TANKS IN/OUT switch on ″T″ cable to IN.
6. Switch DC OUT switch on ″T″ cable to tank being calibrated.
7. Connect PSDAF-113 ″T″ cable LOZ lead for tank being tested to test set TANK UNITS LOZ terminal J-4.
8. Connect PSDAF-113 ″T″ cable HIZ lead for tank being tested to test set TANK UNITS HIZ terminal J-5.
9. Turn test set FUNCT switch to MEASURE EXT.
10. Turn test set SELECT switch to TU.
11. Read capacitance of TU test leads.

Indication/Condition

Test set digital readout shall be 0.5 pF or less.

Corrective Action

If reading is over 0.5 pF, turn in cable for repair.

Step

12. Record capacitance on Worksheet 1-B., line 1.
13. Locate fuel quantity signal conditioner unit (SCU) on right side of cabin, over crew chief/gunner's window.
14. Disconnect P905 from SCU.
15. Connect J-1 of ″T″ cable to P905.

WET METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

16. Turn SELECT switch on test set to LOZ-HIZ.

NOTE

The PSD60-1AF test set reads ″over-range″ above 10,000 megohms by flashing three colons and displaying four zeros.

17. Allow reading to stabilize for 30 seconds. Verify that test set digital display reads over 20 megohms.
18. Turn SELECT switch to the following positions and verify that test set digital display reads over 1 megohm in each position.
    - LOZ-HIZ
    - HIZ-SHLD
    - LOZ-GND
    - HIZ-GND

Indication/Condition

Digital display shall indicate over 1 megohm in each position.

Corrective Action

If Indication/Condition is not as specified, go to DIGITAL DISPLAY DOES NOT INDICATE GREATER THAN 1 MEGOHM, in this work package.

Step

19. Turn SELECT switch to SHLD-GND position.

Indication/Condition

Digital readout on test set indicates 0 megohms.

Corrective Action

If Indication/Condition is not as specified, go to DIGITAL DISPLAY DOES NOT INDICATE 0 (ZERO) MEGOHM, in this work package.

Step

20. Disconnect ″T″ cable LOZ and HIZ leads from test set TANK UNITS terminals.
21. Connect cable LOZ lead to test set INDICATOR LOZ terminal J-1.
22. Connect cable HIZ lead to test set INDICATOR HIZ terminal J-2.
23. Connect COMP lead for same tank to INDICATOR COMP terminal.
24. On ″T″ cable, switch TANK IN/OUT switch to OUT position.
25. Subtract capacitance of adapter harness IND TEST leads, recorded in step 12. from 70.6 pF. Record on Worksheet 1-B., line 2.
26. Turn test set FUNCT switch to MEASURE INT.
27. Turn SELECT switch to TU.
28. Set TU SIMULATOR decade D-1 to read ″005″.
29. Adjust TU vernier V-1 on test set until digital display reads the simulated capacitance value calculated in step 25.

WET METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

30. Turn SELECT switch to COMP.
31. Set the COMP SIMULATOR D-2 on test set to read ″02″.
32. Adjust the COMP vernier V-2 until digital display reads 40.0 pF.
33. Turn FUNCT switch to the SIM TU ONLY position.
34. Connect ″T″ cable P-1 to SCU.
35. Select 0-10 VDC scale on digital multimeter.
36. Connect digital voltmeter leads to METER jacks on ″T″ cable. Verify correct polarity. Turn meter ON.
37. Turn on electrical power.
38. Push in (close) NO. 1 AC INST circuit breaker on copilot’s circuit breaker panel.
39. Loosen screws holding SCU cover plate. Slide plate back to expose potentiometers.
40. While observing DC voltage and polarity on meter, adjust ″TK E″ potentiometer with nonmagnetic screwdriver, counterclockwise.
    - If polarity on multimeter is positive (+) and adjustment decreases the voltage, continue turning ″TK E″ potentiometer counterclockwise until multimeter indicates -0.05 vdc, then turn potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.
    - If polarity on multimeter is positive (+) and adjustment increases the voltage, turn ″TK E″ potentiometer clockwise until multimeter indicates negative (-) polarity, continue turning potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.
    - If polarity on multimeter is negative (-), turn ″TK E″ potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.

Indication/Condition

Multimeter shall indicate -0.005 to 0.005 vdc.

Corrective Action

If Indication/Condition is not as specified, go to MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 1 TANK or MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 2 TANK, in this work package, for tank being tested.

Step

41. Observe fuel quantity indicators on instrument panel for tank being calibrated.

Indication/Condition

Fuel quantity indicators shall read as follows:

- vertical scale 0 lb
- total fuel 0-20 lb

Corrective Action

If Indication/Condition is not as specified, go to FUEL QTY 1 VERTICAL INDICATOR NOT AS SPECIFIED or FUEL QTY 2 VERTICAL INDICATOR NOT AS SPECIFIED, in this work package, for tank being tested.

WET METHOD OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

42. Turn the test set FUNCT switch to TU & COMP.
43. Adjust the ″TK F″ potentiometer on the SCU for tank being calibrated to read between 5.95 to 6.05 vdc on multimeter.

Indication/Condition

Multimeter shall indicate 5.95 to 6.05 vdc.

Corrective Action

If Indication/Condition is not as specified, go to MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 1 TANK or MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 2 TANK, in this work package, for tank being tested.

Step

44. Observe fuel quantity indicators on instrument panel.

Indication/Condition

Fuel quantity indicators shall read as follows:

- vertical scale 1200 lb
- total fuel 1180-1220 lb

Corrective Action

If Indication/Condition is not as specified, go to FUEL QTY 1 VERTICAL INDICATOR NOT AS SPECIFIED or FUEL QTY 2 VERTICAL INDICATOR NOT AS SPECIFIED, in this work package, for tank being tested.

Step

45. Turn FUNCT switch back to SIM TU ONLY position. Adjust ″TK E″ potentiometer for tank being calibrated, if necessary, to read between -0.005 to 0.005 vdc on multimeter.
46. Repeat steps 5. through 98. above for other tank if it is being calibrated.
47. Pull (open) NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
48. Disconnect PSDAF-113 ″T″ cable from SCU, from P905, and from test set.
49. Reconnect P905 to SCU.
50. Turn off external electrical power.
51. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
52. Turn test set OFF. Store test set grounding cable in lid of test set.

Indication/Condition

Test set shall be off.

Corrective Action

None Required

PROBE CAPACITANCE IS NOT 69.1 TO 72.1 PF

SYMPTOM

Probe capacitance is not 69.1 to 72.1 pf.

MALFUNCTION

Probe capacitance is not 69.1 to 72.1 pf.

CORRECTIVE ACTION

1. Check continuity between:

- **P905-M and J307-D**
- **P905-N and J308-D**
- **P905-B and J307-G**
- **P905-D and J308-G**

a. If continuity is present, replace fuel quantity probe (WP 0994 00). Go to **3.**

b. If trouble remains, replace fuel tank wiring harness (WP 0995 00). Go to **3.**

c. If continuity is not present, go to **2.**

2. Check continuity between:

- **P135-L and P135-K**
- **P135-H and P135-J**

a. If continuity is present, replace miscellaneous switch panel (WP 0830 00). Go to **3.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **3.**

3. Procedure completed.

MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 1 TANK

SYMPTOM

Multimeter does not indicate specified voltage for No. 1 tank.

MALFUNCTION

Multimeter does not indicate specified voltage for No. 1 tank.

CORRECTIVE ACTION

1. Check for 115 vac between P905-H and ground.

a. If voltage is as specified go to **2.**

b. If voltage is not as specified, go to **6.**

2. Check helicopter configuration.

a. **EMEP** , go to **3.**

b. **W/O EMEP** , go to **5.**

3. EMEP Remove and check pin filtered adapter P138 (WP 0925 00).

a. If pin filtered adapter is good, go to **4.**

MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 1 TANK - Continued

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **7.**

4. **EMEP Install pin filtered adapter (WP 0925 00). Go to 5.**
5. **Check continuity between:**
   - **P138-G and P905-L**
   - **P138-H and P905-K**
   - **P905-F and ground**
   - **P905-G and ground**
   - **P905-J and ground**

   a. If continuity is present, replace fuel quantity signal conditioner (WP 0997 00). Go to **7.**

   b. If trouble remains, replace fuel quantity probe or wiring harness (WP 0994 00). Go to **7.**

   c. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7.**
6. **Check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.**

   a. If voltage is as specified, repair/replace wiring between terminal 1 and P905-H (WP 1747 00). Go to **7.**

   b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **7.**
7. **Procedure completed.**

FUEL QTY 1 VERTICAL INDICATOR NOT AS SPECIFIED

SYMPTOM

FUEL QTY 1 vertical indicator not as specified.

MALFUNCTION

FUEL QTY 1 vertical indicator not as specified.

CORRECTIVE ACTION

1. **Check helicopter configuration.**

   a. EMEP , go to **2.**

   b. W/O EMEP , go to **4.**
2. **EMEP Remove and check pin filtered adapterP138 (WP 0925 00).**

   a. If pin filtered adapter is good, go to **3.**

   b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **6.**
3. **EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**
4. **Check continuity between:**
   - **P905-L and P138-G**
   - **P905-K and P138-H**
   - **P905-M and P135-H**
   - **P135-J and J307-D**

FUEL QTY 1 VERTICAL INDICATOR NOT AS SPECIFIED - Continued

- **J307-G and P905-B**

a. If continuity is present go to **5.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**

5. Check continuity between P905-M and J307-D.

a. If continuity is present, troubleshoot instrument display system (WP 0089 00). Go to **6.**

b. If trouble remains, replace fuel quantity signal conditioner (WP 0997 00). Go to **6.**

c. If continuity is not present, replace miscellaneous switch panel (WP 0830 00). Go to **6.**

6. Procedure completed.

MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 2 TANK

SYMPTOM

Multimeter does not indicate specified voltage for No. 2 tank.

MALFUNCTION

Multimeter does not indicate specified voltage for No. 2 tank.

CORRECTIVE ACTION

1. Check for 115 vac between P905-H and ground.

a. If voltage is as specified go to **2.**

b. If voltage is not as specified, go to **6.**

2. Check helicopter configuration.

a. **EMEP** , go to **3.**

b. **W/O EMEP** , go to **5.**

3. EMEP Remove and check pin filtered adapter P137 (WP 0925 00).

a. If pin filtered adapter is good, go to **4.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **7.**

4. EMEP Install pin filtered adapter (WP 0925 00). Go to 5.

5. Check continuity between:

- **P137-G and P905-U**
- **P137-H and P905-V**
- **P905-F and ground**
- **P905-G and ground**
- **P905-J and ground**

a. If continuity is present, replace fuel quantity signal conditioner (WP 0997 00). Go to **7.**

b. If trouble remains, replace fuel quantity probe or wiring harness (WP 0994 00). Go to **7.**

c. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7.**

6. Check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.

MULTIMETER DOES NOT INDICATE SPECIFIED VOLTAGE FOR NO. 2 TANK - Continued

a. If voltage is as specified, repair/replace wiring between terminal 1 and P905-H (WP 1747 00). Go to **7.**

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **7.**

7. Procedure completed.

FUEL QTY 2 VERTICAL INDICATOR NOT AS SPECIFIED

SYMPTOM

FUEL QTY 2 vertical indicator not as specified.

MALFUNCTION

FUEL QTY 2 vertical indicator not as specified.

CORRECTIVE ACTION

1. Check helicopter configuration.

a. **EMEP** , go to **2.**

b. **W/O EMEP** , go to **4.**

2. EMEP Remove and check pin filtered adapter P137 (WP 0925 00).

a. If pin filtered adapter is good, go to **3.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **6.**

3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.

4. Check continuity between:

- **P905-U and P137-G**
- **P905-V and P137-H**
- **P905-N and P135-L**
- **P135-K and J308-D**
- **J308-G and P905-D**

a. If continuity is present, go to **5.**

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6.**

5. Check continuity between P905-N and J308-D.

a. If continuity is present, troubleshoot instrument display system (WP 0089 00). Go to **6.**

b. If trouble remains, replace fuel quantity signal conditioner (WP 0997 00). Go to **6.**

c. If continuity is not present, replace miscellaneous switch panel (WP 0830 00). Go to **6.**

6. Procedure completed.

NO FUEL IND TEST FUNCTION

SYMPTOM
No FUEL IND TEST function.

MALFUNCTION
No FUEL IND TEST function.

CORRECTIVE ACTION

1. With FUEL IND TEST switch pressed, check for open circuit between:

- **J135-K and J135-L**
- **J135-H and J135-J**

a. If open circuit is present, replace fuel quantity signal conditioner (WP 0997 00). Go to **2.**

b. If trouble remains, troubleshoot instrument display system (WP 0089 00). Go to **2.**

c. If open circuit is not present, replace miscellaneous switch panel (WP 0830 00). Go to **2.**

2. Procedure completed.

DIGITAL DISPLAY DOES NOT INDICATE GREATER THAN 1 MEGOHM

SYMPTOM
Digital display does not indicate greater than 1 megohm.

MALFUNCTION
Digital display does not indicate greater than 1 megohm.

CORRECTIVE ACTION

1. With multimeter, check for greater than 1 megohm between:

- **P905-B and P905-A**
- **P905-B and P905-P**
- **P905-D and P905-E**
- **P905-D and P905-R**
- **P307-G and P307-F**
- **P307-G and P307-H**
- **P307-G and P307-K**
- **P307-G and P307-E**
- **P308-G and P308-F**
- **P308-G and P308-H**
- **P308-G and P308-K**
- **P308-G and P308-E**

DIGITAL DISPLAY DOES NOT INDICATE GREATER THAN 1 MEGOHM - Continued

a. If multimeter indicates greater than 1 megohm, replace fuel quantity probe (WP 0994 00). Go to **2.**

b. If multimeter does not indicate greater than 1 megohm, repair/replace wiring (WP 1747 00). Go to **2.**

2. Procedure completed.

DIGITAL DISPLAY DOES NOT INDICATE 0 (ZERO) MEGOHM

SYMPTOM

Digital display does not indicate 0 (zero) megohm.

MALFUNCTION

Digital display does not indicate 0 (zero) megohm.

CORRECTIVE ACTION

1. With multimeter, check for 0 megohm between:

- **P905-A and ground**
- **P905-E and ground**
- **P905-F and ground**
- **P905-G and ground**
- **P905-J and ground**
- **P905-P and ground**
- **P905-R and ground**

a. If multimeter indicates 0 megohm, replace fuel quantity signal conditioner unit (WP 0997 00). Go to **2.**

b. If multimeter does not indicate 0 megohm, repair/replace wiring (WP 1747 00). Go to **2.**

2. Procedure completed.

Table 1. Worksheet 1-A. Dry Method (Preferred).

NUMBER	ACTION	TANK #1	TANK #2
1.	Measure empty tank plus cable capacitance.	pF	pF
2.	Measure ″T″ cable capacitance.	pF	pF
3.	Subtract line 2 from line 1. Actual empty tank capacitance equals. (Must be between 69.1 pF and 72.1 pF)	pF	pF

DIGITAL DISPLAY DOES NOT INDICATE 0 (ZERO) MEGOHM - Continued

Table 1. Worksheet 1-A. Dry Method (Preferred). - Continued

NUMBER	ACTION	TANK #1	TANK #2
4.	Measure LOZ-HIZ insulation resistance. (Must be over 20 megohms)	meg	meg
5.	Measure insulation resistance of: (Must be over 1 megohm)		
	LOZ-SHLD	meg	meg
	HIZ-SHLD	meg	meg
	LOZ-GND	meg	meg
	HIZ-GND	meg	meg
6.	Measure insulation resistance of SHLD-GND (Must be 0 megohm)	meg	meg
	Nominal empty tank capacitance is:	40.0 pF	40.0 pF
7.	Subtract line 2 cable capacitance ″simulated added″ capacitance equals	pF	pF
8.	Switch FUNCT switch to AC ONLY Adjust TK E potentiometer to read between -0.005 and 0.005 vdc	vdc	vdc
9.	Vertical scale indicators read Total fuel indicator reads lb	lb	lb
10.	Switch FUNCT switch to SIM TU ONLY		
11.	Adjust TK F potentiometer to read between 5.95 and 6.05 vdc Vertical scale indicators read Total fuel indicator reads lb	vdc lb	vdc lb

DIGITAL DISPLAY DOES NOT INDICATE 0 (ZERO) MEGOHM - Continued

Table 1. Worksheet 1-A. Dry Method (Preferred). - Continued

NUMBER	ACTION	TANK #1	TANK #2
12.	Readjust TK E potentiometer, if necessary, to read between -0.005 and 0.005 vdc	vdc	vdc

Table 2. Worksheet 1-B. Wet Method (Alternate).

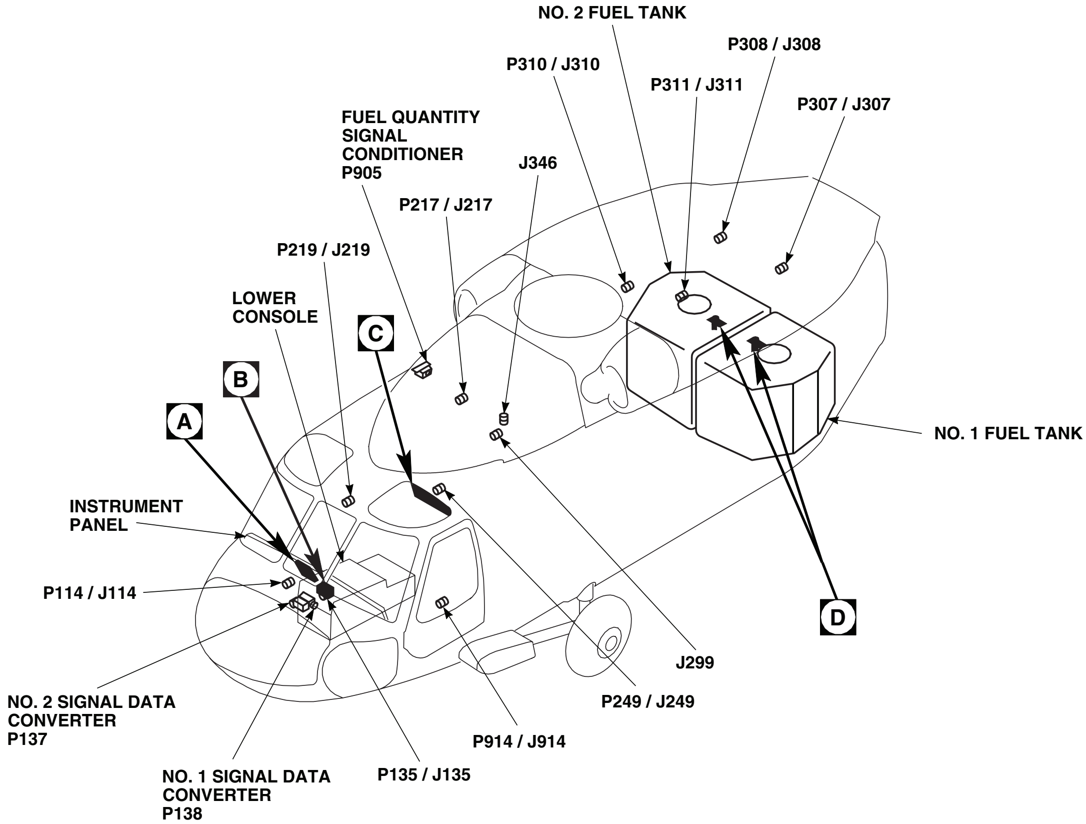
NUMBER	ACTION	TANK #1	TANK #2
1.	Nominal empty tank capacitance equals	70.6 pF	70.6 pF
2.	Measure "T" cable capacitance	pF	pF
3.	Subtract line 2 from line 1 Simulated empty tank capacitance equals	pF	pF
4.	Measure LOZ-HIZ insulation resistance (must be over 20 megohms)	meg	meg
5.	Measure insulation resistance of: (must be over 1 megohms)		
	LOZ-SHLD	meg	meg
	HIZ-SHLD	meg	meg
	LOZ-GND	meg	meg
	HIZ-GND	meg	meg
6.	Measure insulation resistance of SHLD-GND (must read 0 megohms)	meg	meg
7.	Adjust TK E potentiometer to read between -0.005 and 0.005 vdc	vdc	vdc
8.	Vertical scale indicators read	lb	lb

DIGITAL DISPLAY DOES NOT INDICATE 0 (ZERO) MEGOHM - Continued

Table 2. Worksheet 1-B. Wet Method (Alternate). - Continued

NUMBER	ACTION	TANK #1	TANK #2
9.	Total fuel indicator reads lb	lb	lb
10.	Adjust TK F potentiometer to read between 5.95 and 6.05 vdc	vdc	vdc
11.	Vertical scale indicators read	lb	lb
12.	Total fuel indicator reads lb	lb	lb
13.	Readjust TK E potentiometer to read between -0.005 and 0.005 vdc	vdc	vdc

LOCATION



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
J299	CABIN CEILING BL 10 LH, STA 302
J346 (SEE NOTE)	CABIN CEILING BL 18 LH, STA 301
P114 / J114	COCKPIT BL 8 RH, STA 200
P135 / J135	MISCELLANEOUS SWITCH PANEL
P137	INSTRUMENT DISPLAY SYSTEM NO. 2 SIGNAL DATA CONVERTER

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P138	INSTRUMENT DISPLAY SYSTEM NO. 1 SIGNAL DATA CONVERTER
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P307 / J307	TRANSITION DECK BL 15 LH, STA 430
P308 / J308	TRANSITION DECK BL 15 RH, STA 430
P310 / J310	CABIN CEILING BL 17 RH, STA 387
P311 / J311	CABIN CEILING BL 16 LH, STA 383
P905	FUEL QUANTITY SIGNAL CONDITIONER
P914 / J914	COCKPIT BL 24 LH, STA 247

NOTE

UH–60A 82–23748 – SUBQ
EH–60A UH–60L HH–60A
HH–60L

AA2789_1C
SA

Figure 1. Fuel Quantity System Location Diagram. (Sheet 1 of 2)

LOCATION - Continued

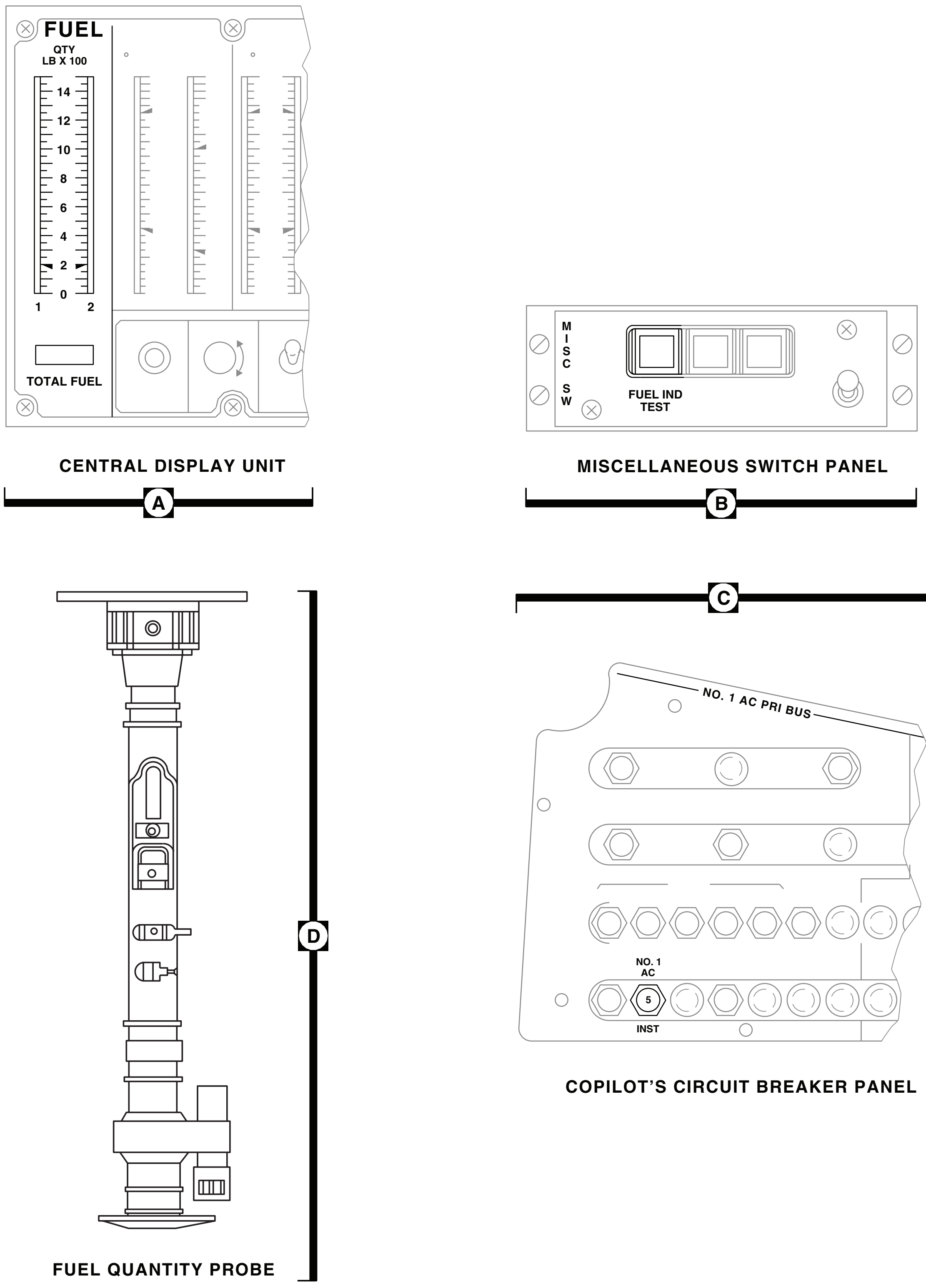
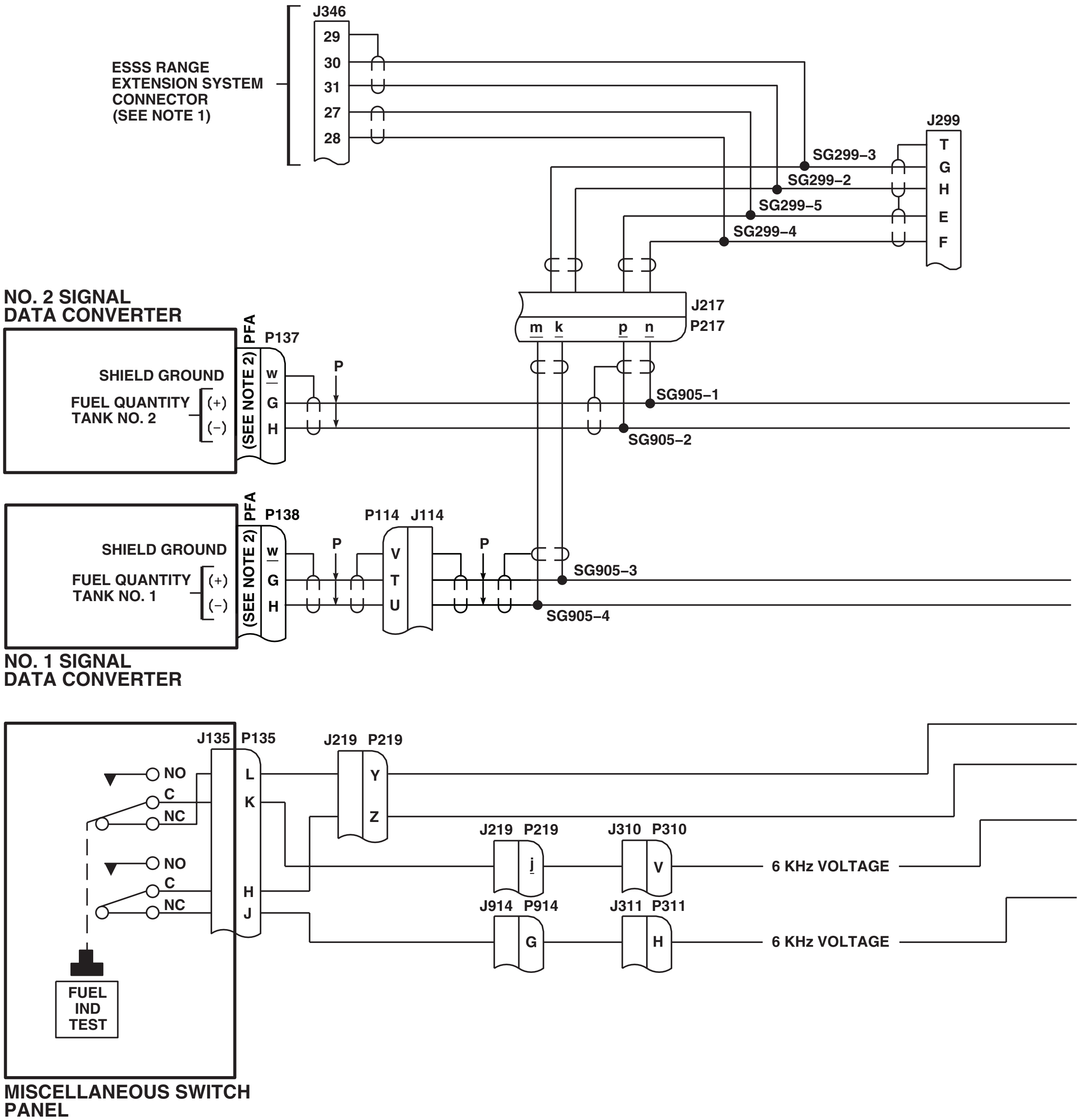


Figure 1. Fuel Quantity System Location Diagram. (Sheet 2 of 2)

SCHEMATICS



NOTES

1. ESSS
2. W/O EMEP PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.

AB1513_1
SA

Figure 2. Fuel Quantity System Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

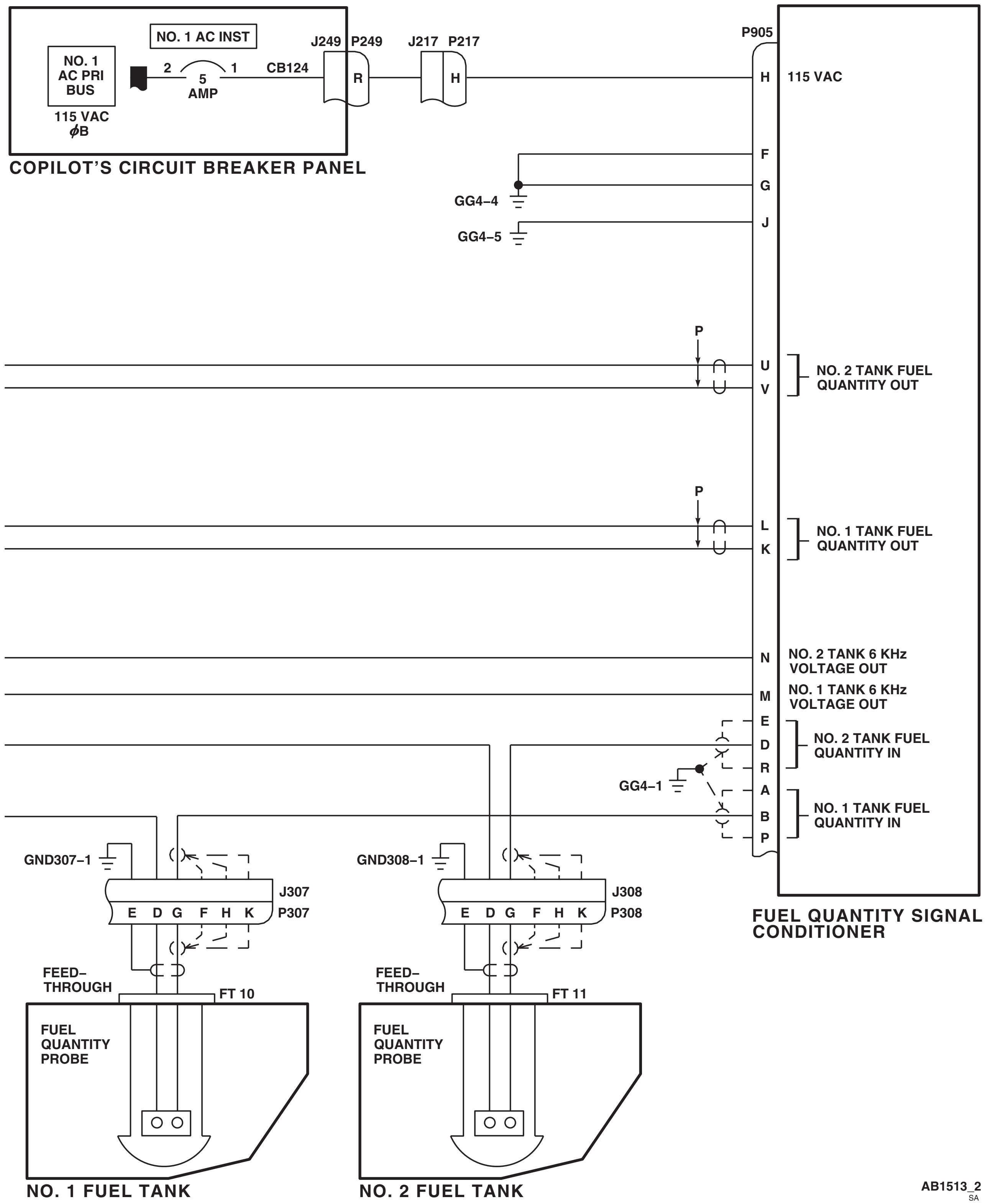


Figure 2. Fuel Quantity System Schematic Diagram. (Sheet 2 of 2)

TEST SETUP

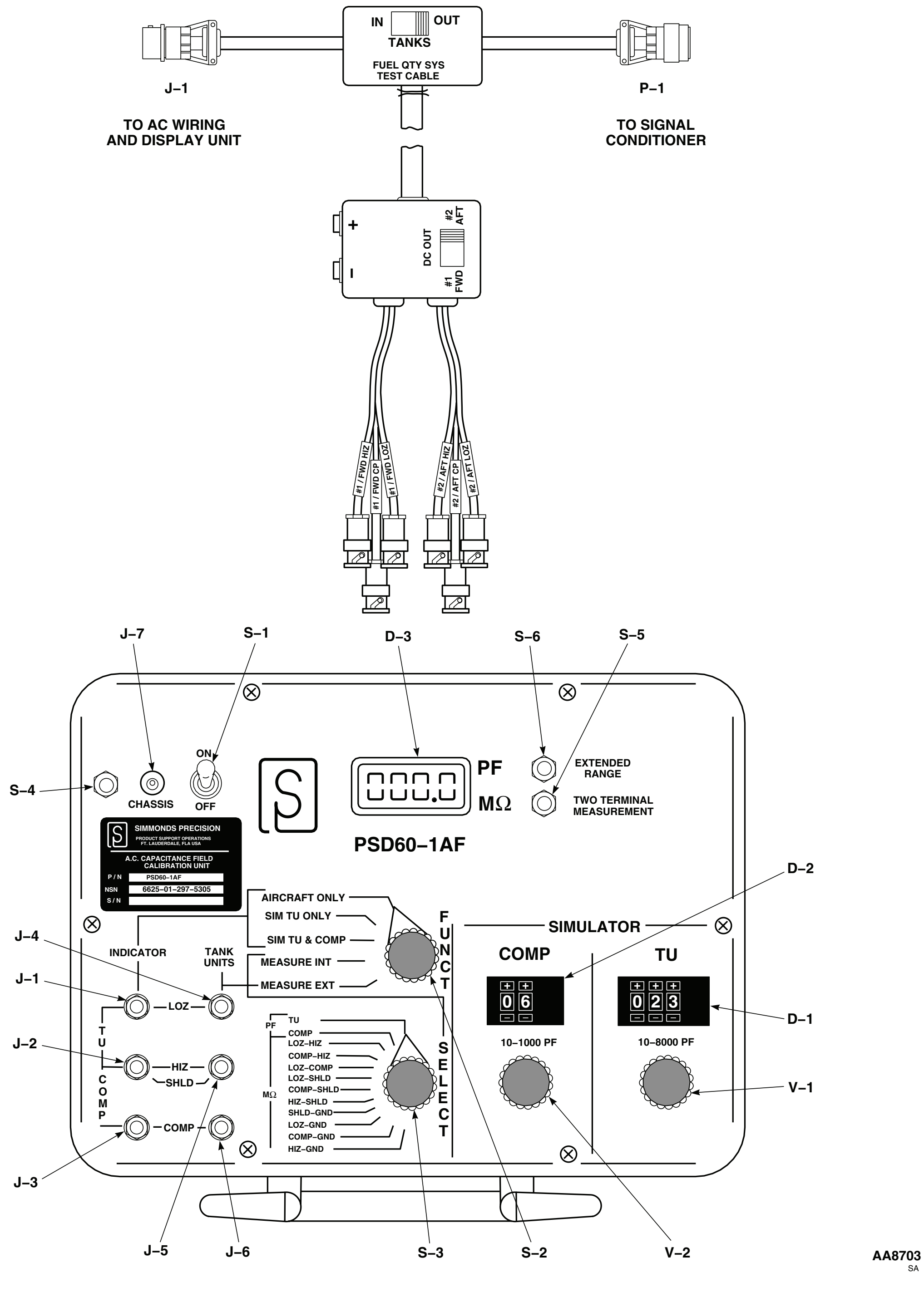


Figure 3. Fuel Quantity System Test (PSD60-1AF) Setup Diagram.

END OF WORK PACKAGE

UNIT LEVEL

FUEL SYSTEM

FUEL QUANTITY SYSTEM HARNESS ADAPTER (AVIM)

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 58F (1)

References
WP 1592 00

FUEL QUANTITY SYSTEM HARNESS ADAPTER (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

See Figure 1 for harness adapter diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Check continuity between the following points:
   - J1-C and P1-C
   - J1-G and P1-G
   - J1-H and P1-H
   - J1-J and P1-J
   - J1-F and P1-F
   - J1-E and J1-R
   - J1-P and J1-A
   - J1-L and P1-L
   - J1-K and P1-K
   - J1-U and P1-U
   - J1-V and P1-V
   - P1-E and P1-R
   - P1-P and P1-A
   - J1-F and CASE GND
   - P1-F and CASE GND

FUEL QUANTITY SYSTEM HARNESS ADAPTER (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, repair/replace wiring.

Step

2. Place TANK SELECT switch to TANK 1 and FUNCTION switch to TEST.
3. Check continuity between P5 and J1-B.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, go to NO CONTINUITY BETWEEN P5 AND J1-B, in this work package.

Step

4. Check continuity between P4 and J1-M.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, go to NO CONTINUITY BETWEEN P4 AND J1-M, in this work package.

Step

5. Check continuity between P3 and P1-B.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, go to NO CONTINUITY BETWEEN P3 AND P1-B, in this work package.

Step

6. Check continuity between P2 and P1-M.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, go to NO CONTINUITY BETWEEN P2 AND P1-M, in this work package.

FUEL QUANTITY SYSTEM HARNESS ADAPTER (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

7. Place TANK SELECT switch to TANK 2.
8. Check continuity between P5 and J1-D.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, go to NO CONTINUITY BETWEEN P5 AND J1-D, in this work package.

Step

9. Check continuity between P4 and J1-N.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, go to NO CONTINUITY BETWEEN P4 AND J1-N, in this work package.

Step

10. Check continuity between P3 and P1-D.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, go to NO CONTINUITY BETWEEN P3 AND P1-D, in this work package.

Step

11. Check continuity between P2 and P1-N.

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, go to NO CONTINUITY BETWEEN P2 AND P1-N, in this work package.

Step

12. Place FUNCTION switch to NORM.
13. Check continuity between J1-B and P1-B.

Indication/Condition

Continuity shall be present.

FUEL QUANTITY SYSTEM HARNESS ADAPTER (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition is not as specified, check continuity between S2-A1 and S2-C1.
   a. If continuity is present, replace FUNCTION switch (WP 1592 00).
   b. If continuity is not present, repair/replace wiring.

Step

14. Check continuity between J1-M and P1-M.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between S2-B1 and S2-D1.
   a. If continuity is present, replace FUNCTION switch (WP 1592 00).
   b. If continuity is not present, repair/replace wiring.

Step

15. Check continuity between J1-D and P1-D.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between S2-E1 and S2-G1.
   a. If continuity is present, replace FUNCTION switch (WP 1592 00).
   b. If continuity is not present, repair/replace wiring.

Step

16. Check continuity between J1-N and P1-N.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between S2-F1 and S2-H1.
   a. If continuity is present, replace FUNCTION switch (WP 1592 00).
   b. If continuity is not present, repair/replace wiring.

Step

17. Check continuity between OUTPUT VDC (+) and J1-U.

Indication/Condition

Continuity shall be present.

FUEL QUANTITY SYSTEM HARNESS ADAPTER (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition is not as specified, check continuity between S1-G and S1-G2.
   a. If continuity is present, repair/replace wiring.
   b. If continuity is not present, replace TANK SELECT switch (WP 1592 00).

Step

18. Check continuity between OUTPUT VDC (-) and J1-V.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between S1-H and S1-H2.
   a. If continuity is present, repair/replace wiring.
   b. If continuity is not present, replace TANK SELECT switch (WP 1592 00).

Step

19. Place TANK SELECT switch to TANK 1.
20. Check continuity between OUTPUT VDC (+) and J1-L.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between S1-G and S1-G1.
   a. If continuity is present, repair/replace wiring.
   b. If continuity is not present, replace TANK SELECT switch (WP 1592 00).

Step

21. Check continuity between OUTPUT VDC (-) and J1-K.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between S1-H and S1-H1.
   a. If continuity is present, repair/replace wiring.
   b. If continuity is not present, replace TANK SELECT switch (WP 1592 00).

Step

22. Check continuity between the following points:
    - P5 shield and J1-E
    - P5 shield and J1-P

FUEL QUANTITY SYSTEM HARNESS ADAPTER (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- P3 shield and P1-E
- P3 shield and P1-P

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, repair/replace wiring.

NO CONTINUITY BETWEEN P5 AND J1-B

SYMPTOM

No continuity between P5 and J1-B.

MALFUNCTION

No continuity between P5 and J1-B.

CORRECTIVE ACTION

1. **Check continuity between:**
   - **P5 and S1-A**
   - **S1-A1 and S2-A2**
   - **S2-A and J1-B**

   a. If continuity is present, go to **2.**

   b. If continuity is not present, repair/replace wiring. Go to **3.**
2. **Check continuity between S1-A and S1-A1.**

   a. If continuity is present, replace FUNCTION switch (WP 1592 00). Go to **3.**

   b. If continuity is not present, replace TANK SELECT switch (WP 1592 00). Go to **3.**
3. **Procedure completed.**

NO CONTINUITY BETWEEN P4 AND J1-M

SYMPTOM

No continuity between P4 and J1-M.

MALFUNCTION

No continuity between P4 and J1-M.

CORRECTIVE ACTION

1. **Check continuity between:**
   - **P4 and S1-B**
   - **S1-B1 and S2-B2**
   - **S2-B and J1-M**

   a. If continuity is present, go to **2.**

NO CONTINUITY BETWEEN P4 AND J1-M - Continued

b. If continuity is not present, repair/replace wiring. Go to **3.**

2. Check continuity between S1-B and S1-B2.

a. If continuity is present, replace FUNCTION switch (WP 1592 00). Go to **3.**

b. If continuity is not present, replace TANK SELECT switch (WP 1592 00). Go to **3.**

3. Procedure completed.

NO CONTINUITY BETWEEN P3 AND P1-B

SYMPTOM

No continuity between P3 and P1-B.

MALFUNCTION

No continuity between P3 and P1-B.

CORRECTIVE ACTION

1. Check continuity between:

- **P3 and S1-C**
- **S1-C1 and S2-C2**
- **S2-C and P1-B**

a. If continuity is present, go to **2.**

b. If continuity is not present, repair/replace wiring. Go to **3.**

2. Check continuity between S1-C and S1-C1.

a. If continuity is present, replace FUNCTION switch (WP 1592 00). Go to **3.**

b. If continuity is not present, replace TANK SELECT switch (WP 1592 00). Go to **3.**

3. Procedure completed.

NO CONTINUITY BETWEEN P2 AND P1-M

SYMPTOM

No continuity between P2 and P1-M.

MALFUNCTION

No continuity between P2 and P1-M.

CORRECTIVE ACTION

1. Check continuity between:

- **P2 and S1-D**
- **S1-D1 and S2-D2**
- **S2-D and P1-M**

a. If continuity is present, go to **2.**

b. If continuity is not present, repair/replace wiring. Go to **3.**

NO CONTINUITY BETWEEN P2 AND P1-M - Continued

2. **Check continuity between S1-D and S1-D1.**
   a. If continuity is present, replace FUNCTION switch (WP 1592 00). Go to **3.**
   b. If continuity is not present, replace TANK SELECT switch (WP 1592 00). Go to **3.**
3. **Procedure completed.**

NO CONTINUITY BETWEEN P5 AND J1-D

SYMPTOM

No continuity between P5 and J1-D.

MALFUNCTION

No continuity between P5 and J1-D.

CORRECTIVE ACTION

1. **Check continuity between:**
   - **S1-A2 and S2-E2**
   - **S2-E and J1-D**

   a. If continuity is present, go to **2.**
   b. If continuity is not present, repair/replace wiring. Go to **3.**
2. **Check continuity between S1 and S1-A1.**
   a. If continuity is present, replace FUNCTION switch (WP 1592 00). Go to **3.**
   b. If continuity is not present, replace TANK SELECT switch (WP 1592 00). Go to **3.**
3. **Procedure completed.**

NO CONTINUITY BETWEEN P4 AND J1-N

SYMPTOM

No continuity between P4 and J1-N.

MALFUNCTION

No continuity between P4 and J1-N.

CORRECTIVE ACTION

1. **Check continuity between:**
   - **S1-B2 and S2-F2**
   - **S2-F and J1-N**

   a. If continuity is present, go to **2.**
   b. If continuity is not present, repair/replace wiring. Go to **3.**
2. **Check continuity between S1-B and S1-B2.**
   a. If continuity is present, replace FUNCTION switch (WP 1592 00). Go to **3.**
   b. If continuity is not present, replace TANK SELECT switch (WP 1592 00). Go to **3.**
3. **Procedure completed.**

NO CONTINUITY BETWEEN P3 AND P1-D

SYMPTOM

No continuity between P3 and P1-D.

MALFUNCTION

No continuity between P3 and P1-D.

CORRECTIVE ACTION

1. **Check continuity between:**
   - **S1-C2 and S2-G2**
   - **S2-G and P1-D**

   a. If continuity is present, go to **2.**

   b. If continuity is not present, repair/replace wiring. Go to **3.**

2. **Check continuity between S1-C and S1-C2.**

   a. If continuity is present, replace FUNCTION switch (WP 1592 00). Go to **3.**

   b. If continuity is not present, replace TANK SELECT switch (WP 1592 00). Go to **3.**

3. **Procedure completed.**

NO CONTINUITY BETWEEN P2 AND P1-N

SYMPTOM

No continuity between P2 and P1-N.

MALFUNCTION

No continuity between P2 and P1-N.

CORRECTIVE ACTION

1. **Check continuity between:**
   - **S1-D2 and S2-H2**
   - **S2-H and P1-N**

   a. If continuity is present, go to **2.**

   b. If continuity is not present, repair/replace wiring. Go to **3.**

2. **Check continuity between S1-D and S1-D2.**

   a. If continuity is present, replace FUNCTION switch (WP 1592 00). Go to **3.**

   b. If continuity is not present, replace TANK SELECT switch (WP 1592 00). Go to **3.**

3. **Procedure completed.**

LOCATION

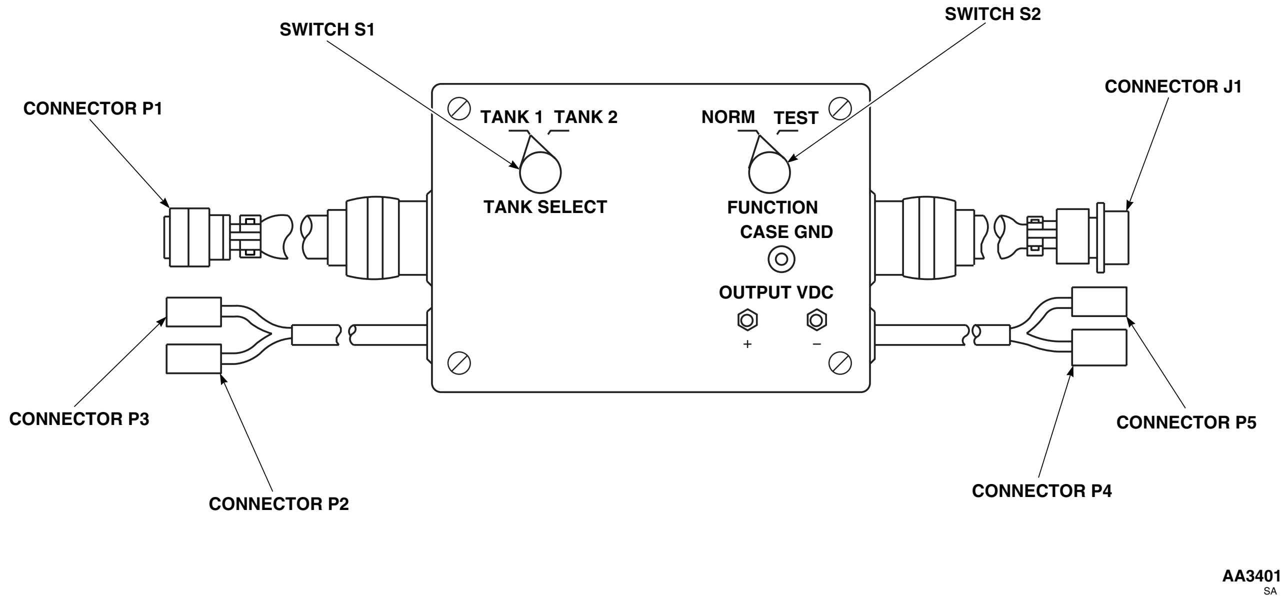


Figure 1. Fuel Quantity System Harness Adapter.

SCHEMATICS

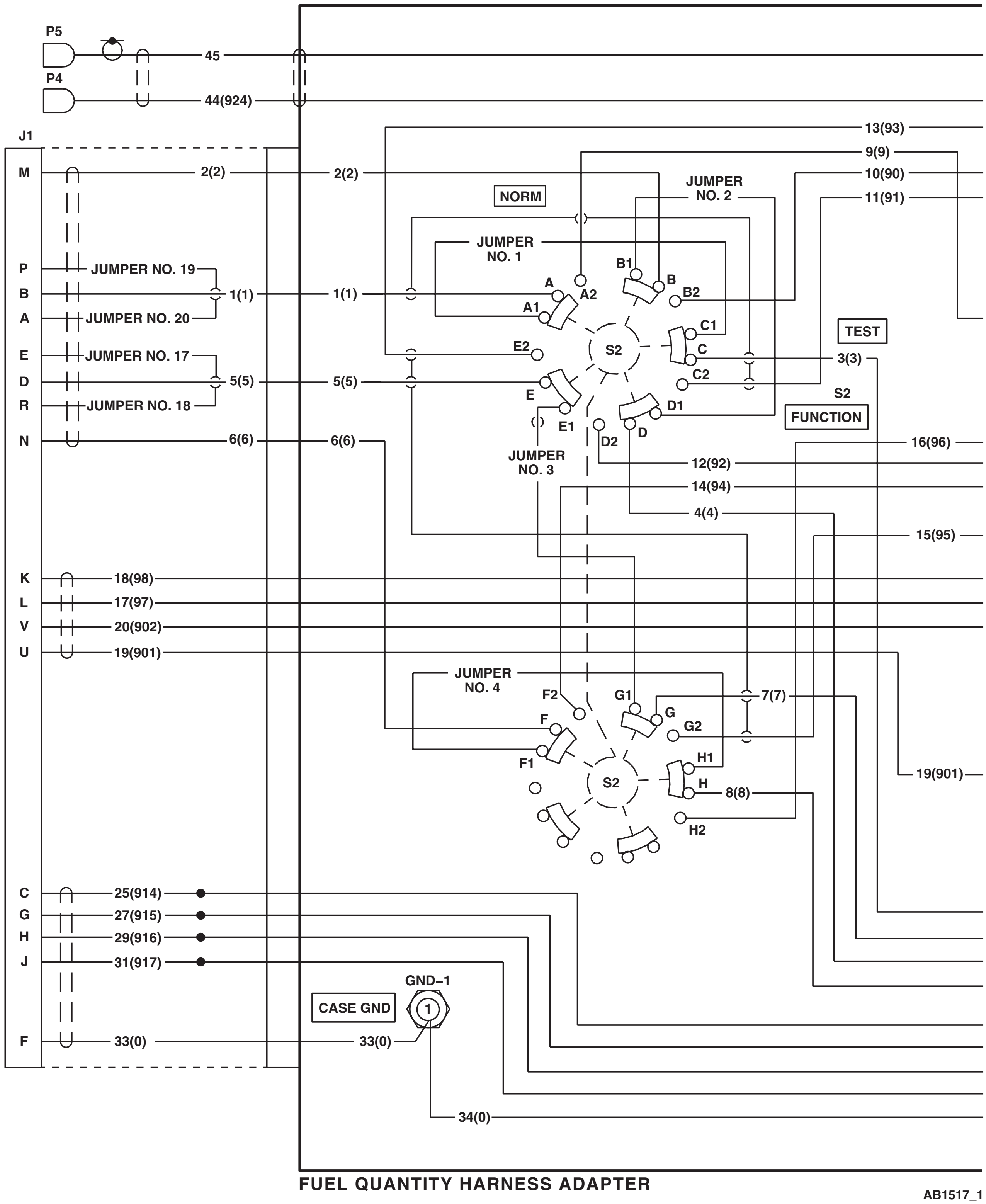


Figure 2. Fuel Quantity System Harness Adapter Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

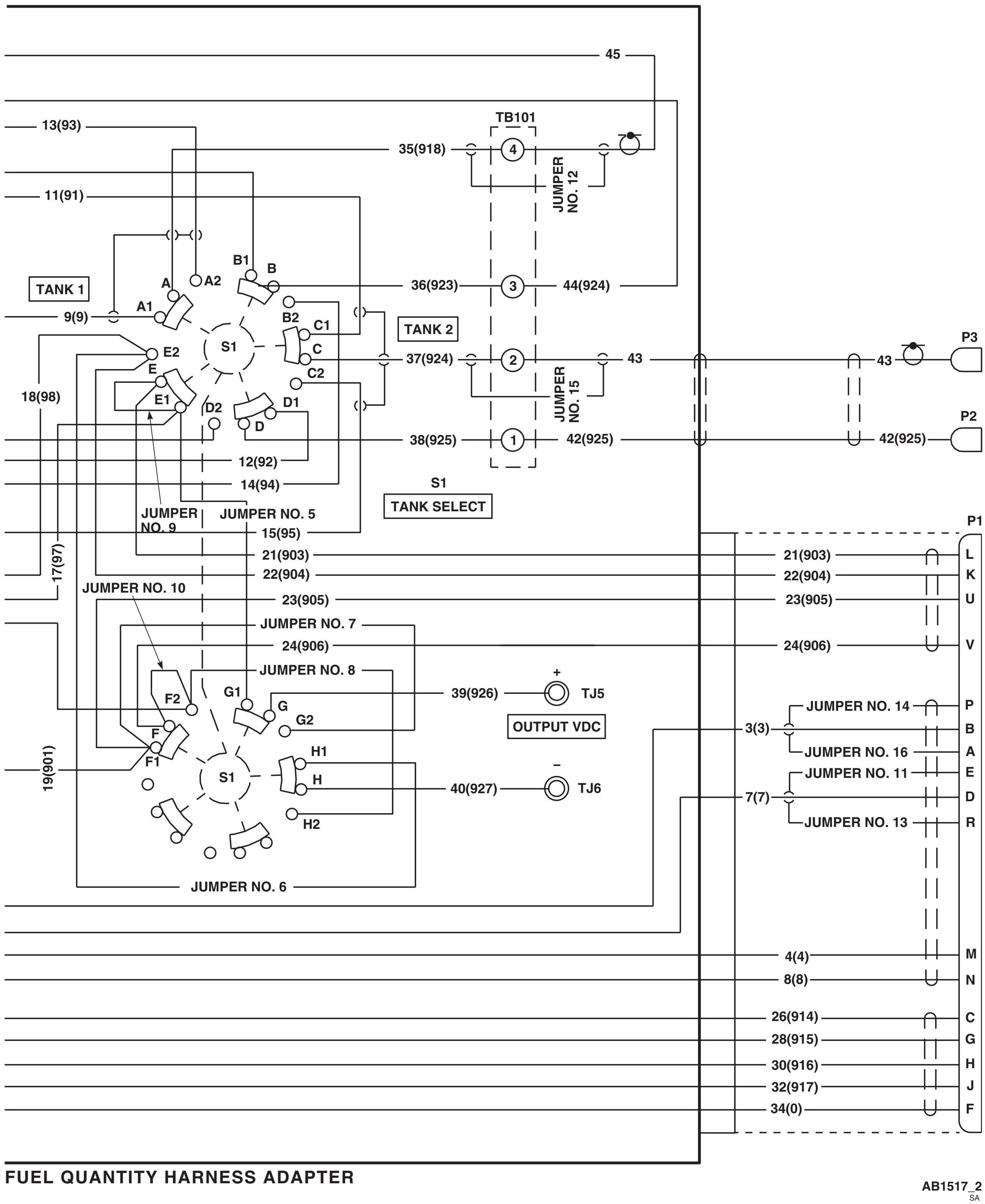


Figure 2. Fuel Quantity System Harness Adapter Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

FLIGHT CONTROLS

CYCLIC STICK ASSEMBLY

This WP supersedes WP 0134 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06
Torque Wrench, A-A-1274

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 11-1520-237-23
TM 11-1520-249-23-2
WP 1052 00
WP 1053 00
WP 1054 00
WP 1055 00
WP 1057 00
WP 1059 00
WP 1747 00

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram and Figure 2, **UH-60L UH-60A HH-60L EH-60A 87-26047- SUBQ UH-60A 88-24067 - SUBQ** Figure 3, and **EH-60A 84-24017 - 87-24666 UH-60A 77-22714 - 88-24046** Figure 4 for schematic diagrams as an aid in operational/troubleshooting.

The cyclic stick assembly operational/troubleshooting procedure is subdivided as follows:

- Grip Assembly Electrical
- Slew-Up Switch Electrical
- Grip Assembly Mechanical Inspection
- Slew-Up Switch Mechanical Inspection

GRIP ASSEMBLY ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Check continuity between following pins while setting switches listed in Table 1:

GRIP ASSEMBLY ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 1. Grip Assembly Resistance Check.

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN) CONNECTOR PIN - SWITCH TERMINAL
TRIM REL	*Push	A - D	Infinity	**Replace switch
TRIM REL	Release	A - D	0	A - 1 D - 2
CARGO REL	*Push	E - F	0	E - CARGO REL. (E-20) F - CARGO REL. (F- 20)
CARGO REL	Release	E - F	Infinity	**Replace switch
RADIO-ICS	*RADIO	G - H	0	G - RADIO - ICS (G-20) H - RADIO - ICS (H- 20)
RADIO-ICS	*RADIO	J - K	Infinity	**Replace switch
RADIO-ICS	*ICS	J - K	0	J - RADIO - ICS (J-20) K - RADIO - ICS (K- 20)
RADIO-ICS	*ICS	G - H	Infinity	**Replace switch
STICK TRIM	Center (off)	T - U	Infinity	**Replace switch
STICK TRIM	Center (off)	T - V	Infinity	**Replace switch
STICK TRIM	Center (off)	T - W	Infinity	**Replace switch
STICK TRIM	Center (off)	T - X	Infinity	**Replace switch
STICK TRIM	*L	T - U	0	T - 16 U - 15

GRIP ASSEMBLY ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 1. Grip Assembly Resistance Check. - Continued

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN) CONNECTOR PIN - SWITCH TERMINAL
STICK TRIM	*R	T - X	0	T - 6 X - 5
STICK TRIM	*FWD	T - V	0	T - 2 V - 1
STICK TRIM	*AFT	T - W	0	T - 12 W - 11
GA	*Push	Y - Z	Infinity	**Replace switch
GA	Release	Y - Z	0	Y - 1 Z - 2
PNL LTS	Push (on)	N - P	0	N - 3 P - 4
PNL LTS	Push (on)	a - b	Infinity	**Replace switch
PNL LTS	Push (off)	a - b	0	a - 1 b - 2
PNL LTS	Push (off)	N - P	Infinity	**Replace switch

*Indicates momentary contact. Switch must be held at this position.

** WP 1053 00, WP 1054 00, WP 1055 00

Indication/Condition

Meter shall indicate corresponding resistance.

Corrective Action

1. If result is not as specified, do corresponding corrective action.

GRIP ASSEMBLY ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

a. If a meter indication of infinity is specified, and continuity is found, replace switch (WP 1053 00, WP 1054 00, WP 1055 00).

b. If a meter indication of 0 is specified, and continuity is not found, repair/replace wiring (WP 1747 00).

SLEW-UP SWITCH ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. **EH-60A 87-24667 - SUBQ UH-60A 88-26047 - SUBQ UH-60L HH-60A HH-60L** Check continuity between the following pins while operating switches as listed in Table 2:

Table 2. Slew-Up Resistance Check
(EH-60A 89-24667-SUBQ UH-60A 88-26047-SUBQ UH-60L HH-60A HH-60L).

SLEW-UP SWITCH POSITION	CHECK RESISTANCE BETWEEN PINS (P7R OR P8R)	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN)
*Push	A - G	0	A to 1 NO 1 NO to 1 Wiper 1 Wiper to 2 Wiper 2 Wiper to G
Release	A - G	Infinity	**Replace switch
*Push	G - H	0	G to 2 NO 2 NO to 2 Wiper
Release	G - H	Infinity	**Replace switch
*Push	D - E	0	D to 3 NO 3 NO to 3 Wiper 3 Wiper to 4 Wiper 4 Wiper to E
Release	D - E	Infinity	**Replace switch

SLEW-UP SWITCH ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 2. Slew-Up Resistance Check
(EH-60A 89-24667-SUBQ UH-60A 88-26047-SUBQ UH-60L HH-60A HH-60L). - Continued

SLEW-UP SWITCH POSITION	CHECK RESISTANCE BETWEEN PINS (P7R OR P8R)	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN)
*Push	E - C	0	C to 4 NO 4 NO to 4 Wiper
Release	E - C	Infinity	**Replace switch
*Indicates momentary contact. Switch must be held at this position.			
** TM 11-1520-237-23 or EH-60A **TM 11-1520-249-23-2**			

Indication/Condition

Meter shall indicate corresponding resistance.

Corrective Action

1. If result is not as specified, do corresponding corrective action.

   a. If a meter indication of infinity is specified, and continuity is found, replace switch (TM 11-1520-237-23 or EH-60A TM 11-1520-249-23-2).

   b. If a meter indication of 0 is specified, and continuity is not found, repair/replace wiring (WP 1747 00).

Step

2. EH-60A 84-24017 - 88-26046 UH-60A 77-22714 - 87-24666 Check continuity between the following pins while operating switch as listed in Table 3:

SLEW-UP SWITCH ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 3. Slew-Up Resistance Check (EH-60A 84-24017 - 88-26046 UH-60A 77-22714-87-24666).

SLEW-UP SWITCH POSITION	CHECK RESISTANCE BETWEEN PINS (P7R OR P8R)	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN)
*Push	A - B	0	A to 1 NO 1 NO to 1 Wiper 1 Wiper to 2 Wiper 2 Wiper to B
Release	A - B	Infinity	**Replace switch
*Push	B - C	0	C to 2 NO 2 NO to 2 Wiper
Release	B - C	Infinity	**Replace switch
*Push	D - E	0	D to 3 NO 3 NO to 3 Wiper 3 Wiper to 4 Wiper 4 Wiper to E
Release	D - E	Infinity	**Replace switch
*Push	E - F	0	F to 4 NO 4 NO to 4 Wiper
Release	E - F	Infinity	**Replace switch

*Indicates momentary contact. Switch must be held at this position.

**TM 11-1520-237-23 or EH-60A TM 11-1520-249-23-2

Indication/Condition

Meter shall indicate corresponding resistance.

SLEW-UP SWITCH ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If result is not as specified, do corresponding corrective action.
   a. If a meter indication of infinity is specified, and continuity is found, replace switch (TM 11-1520-237-23 or **EH-60A** TM 11-1520-249-23-2).
   b. If a meter indication of 0 is specified, and continuity is not found, repair/replace wiring (WP 1747 00).

GRIP ASSEMBLY MECHANICAL INSPECTION OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Visually check grip for cracks.

Indication/Condition

Grip shall not be cracked.

Corrective Action

If result is not as specified, replace grip assembly (WP 1052 00).

Step

2. Visually check tube assembly for cracks.

Indication/Condition

Tube assembly shall not be cracked.

Corrective Action

If result is not as specified, replace tube assembly (WP 1059 00).

Step

3. Visually check grip for loose or cracked controls.

Indication/Condition

Grip controls shall not be loose or cracked.

Corrective Action

If result is not as specified, replace grip assembly (WP 1052 00).

Step

4. If tag attached to assembly indicates rough or sloppy stick movement, do the following:
   - Inspect socket assembly bearings for ratcheting or play.
   - If ratcheting or play is found, replace malfunctioning bearing(s) and/or socket assembly (WP 1057 00).
   - Check each bearing for binding by turning with torque wrench. Bearing shall turn with 2 inch-ounce torque or less.
   - If bearing does not turn, replace malfunctioning bearing(s) and/or socket assembly (WP 1057 00).

GRIP ASSEMBLY MECHANICAL INSPECTION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Movement shall not be rough or sloppy.

Corrective Action

None Required

SLEW-UP SWITCH MECHANICAL INSPECTION OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Inspect spring for proper tension by determining if trigger returns to static stop (full open position) after pressing trigger.

Indication/Condition

Trigger shall return to full open position.

Corrective Action

If result is not as specified, replace spring (TM 11-1520-237-23 or **EH-60A** TM 11-1520-249-23-2).

Step

2. Inspect trigger guide and switch button for proper alignment.

Indication/Condition

Trigger guide shall remain on switch button and cause no binding.

Corrective Action

If result is not as specified, replace trigger assembly (TM 11-1520-237-23 or **EH-60A** TM 11-1520-249-23-2).

Step

3. Inspect trigger assembly for looseness or play.

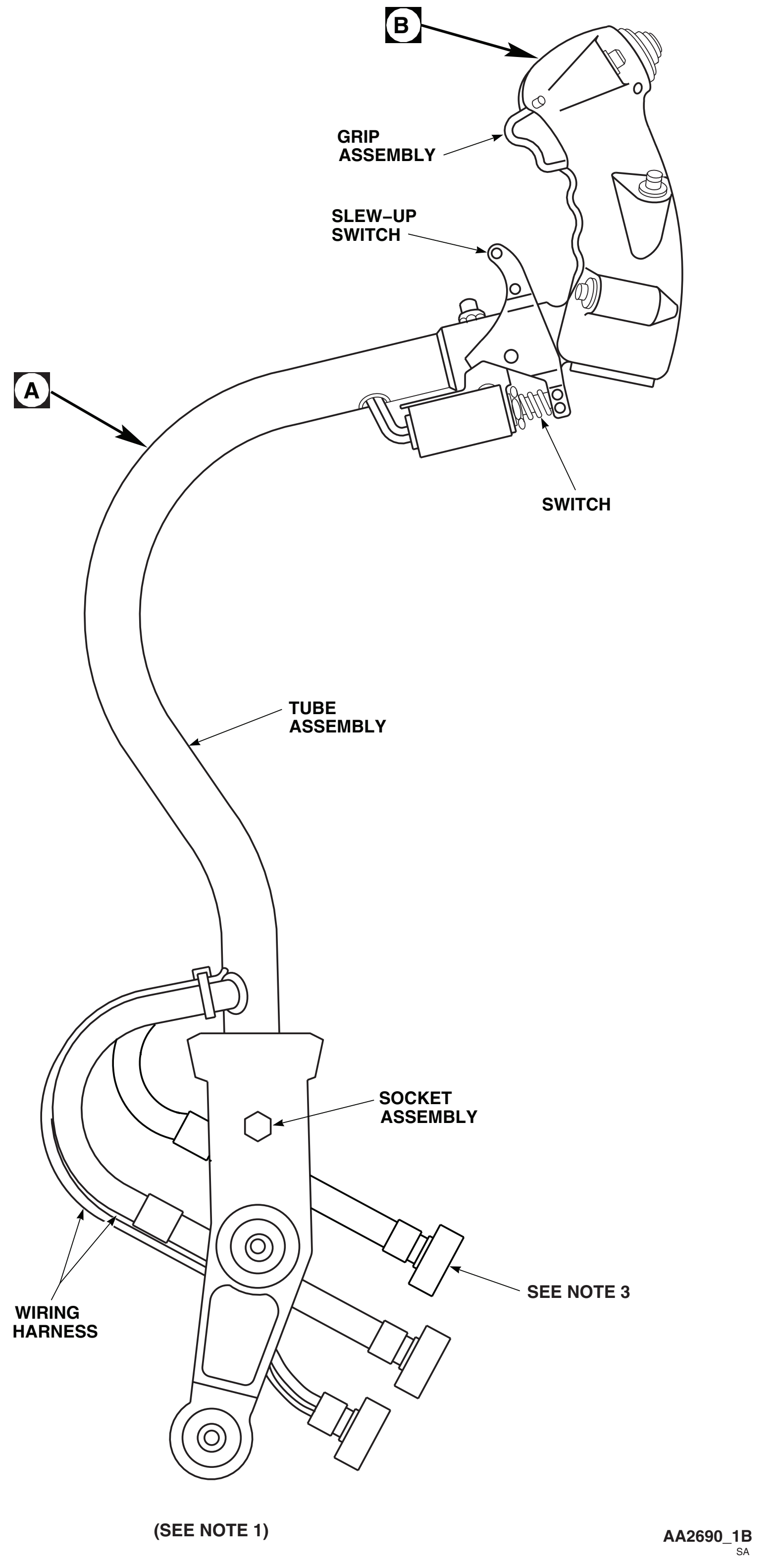
Indication/Condition

No looseness or play shall be found.

Corrective Action

If result is not as specified, replace trigger assembly (TM 11-1520-237-23 or **EH-60A** TM 11-1520-249-23-2).

LOCATION



NOTES

1. UH60L HH60A HH60L
   UH60A 88–24067 – SUBQ
   EH60A 87–26047 – SUBQ
2. UH60A 77–22714 – 88–24046
   EH60A 84–24017 – 87–24666
3. HFIRE/VOL

Figure 1. Cyclic Stick Assembly. (Sheet 1 of 2)

LOCATION - Continued

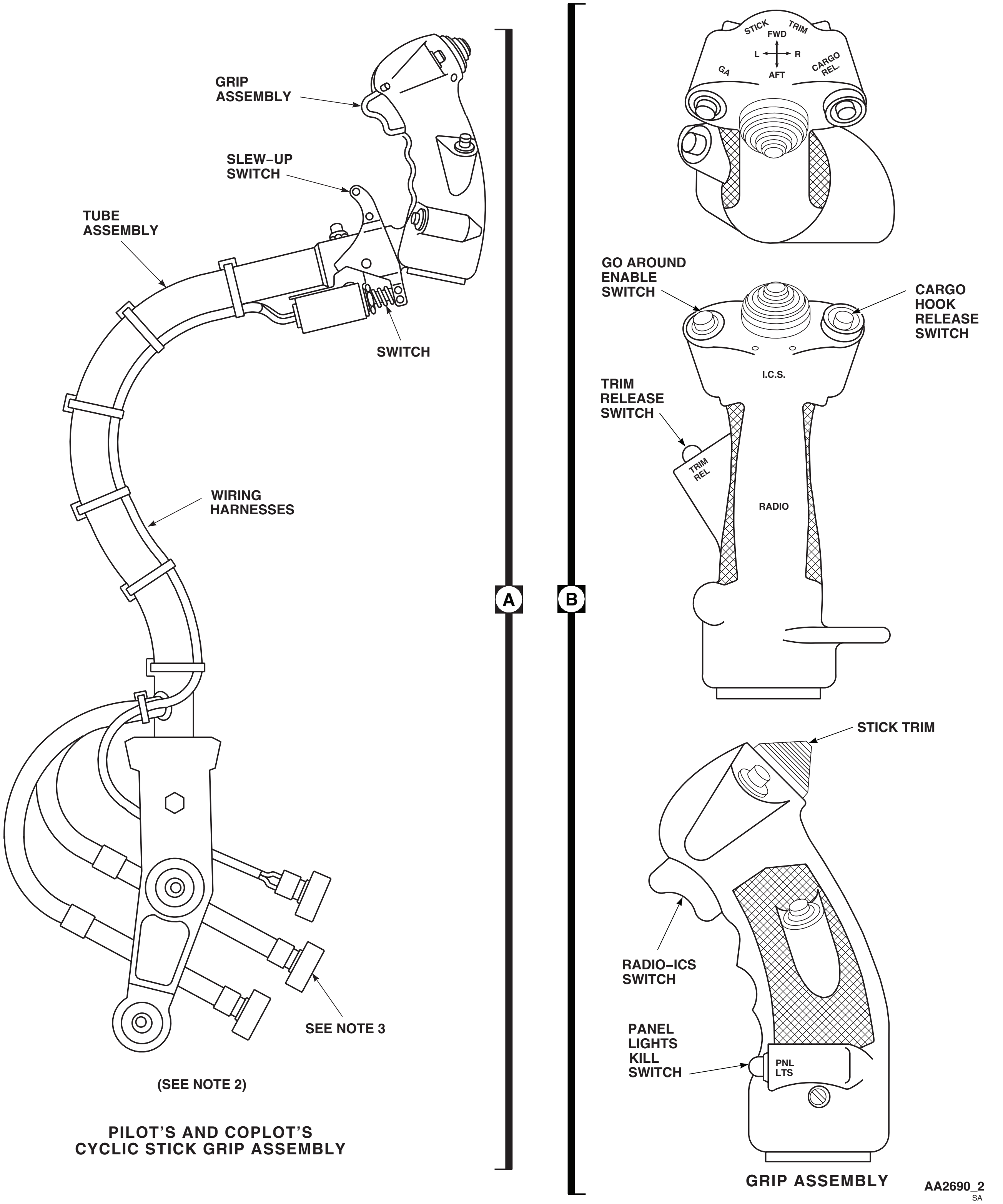
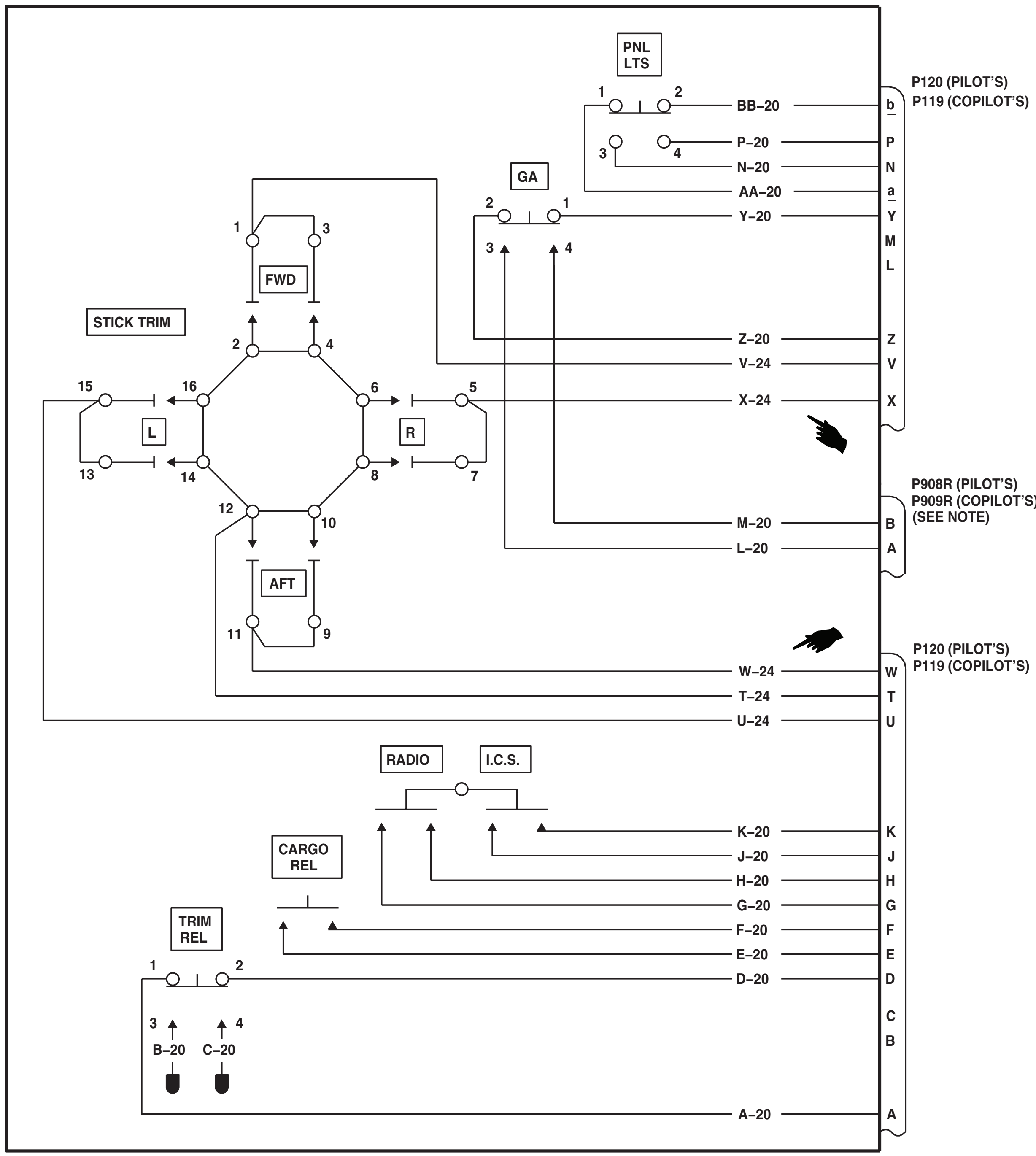


Figure 1. Cyclic Stick Assembly. (Sheet 2 of 2)

SCHEMATICS



NOTE

HFIRE/VOL

AB1450A
SA

Figure 2. Cyclic Stick Grip Assembly Schematic Diagram.

SCHEMATICS - Continued

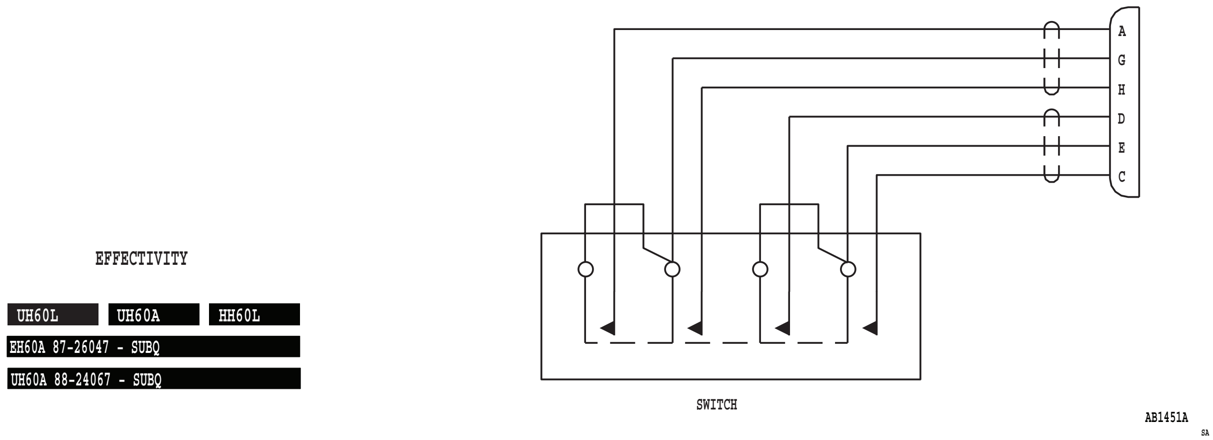


Figure 3. Cyclic Stick Slew-Up Switch Schematic Diagram

UH-60L UH-60A HH-60L EH-60A 87-26047- SUBQ UH-60A 88-24067 - SUBQ .

SCHEMATICS - Continued

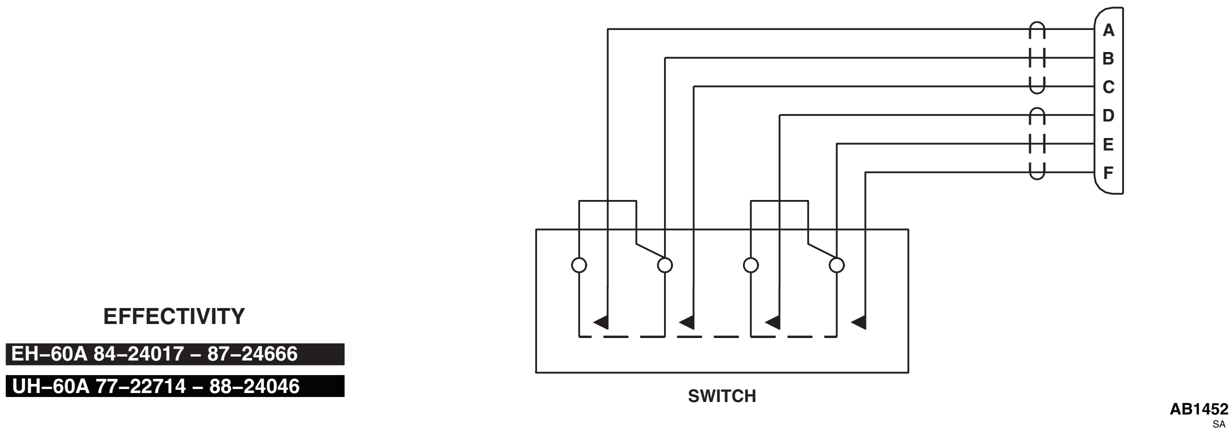


Figure 4. Cyclic Stick Slew-Up Switch Schematic Diagram **EH-60A 84-24017 - 87-24666 UH-60A 77-22714 - 88-24046** .

END OF WORK PACKAGE

UNIT LEVEL

FLIGHT CONTROLS

COLLECTIVE STICK ASSEMBLY

This WP supersedes WP 0135 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 1-1520-237-10
WP 0104 00
WP 0110 00
WP 1060 00
WP 1065 00
WP 1066 00
WP 1067 00
WP 1068 00
WP 1069 00
WP 1070 00
WP 1071 00
WP 1072 00
WP 1075 00
WP 1076 00
WP 1077 00
WP 1685 00
WP 1747 00

Equipment Condition
External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for collective stick assembly diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

The collective stick assembly operational/troubleshooting procedure is subdivided as follows:

- Electrical
- Mechanical Inspection, Pilot
- Mechanical Inspection, Copilot

ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- MOD represents UH-60A helicopters prior to Serial No. 85-24415 modified by MWO 1-1520-237-55-7 (MWO 1-1520-237-55-7 upgrades electrical harnesses of above helicopters with UH-60L electrical harnesses).
- W/O MOD represents UH-60A helicopters prior to Serial No. 85-24415 not modified by MWO 1-1520-237-55-7.
- HUD represents UH-60A, UH-60L, and HH-60A helicopters prior to Serial No. 96-26723 modified by MWO 1-1520-237-50-62. UH-60L Serial Nos. 96-26723 - SUBQ and HH-60L helicopters have product line incorporation of the ANVIS Heads Up Display Set.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Turn on electrical power.
3. Adjust both upper console INSTR LT PILOT FLT and CPLT FLT INSTR LTS to BRT.

Indication/Condition

Collective sticks shall be evenly lit.

Corrective Action

If Indication/Condition is not as specified, go to LIGHTED PANEL IS NOT LIT CORRECTLY, in this work package.

Step

4. Turn off electrical power.
5. Do CONTINUITY CHECK (UH-60A 85-24415 - SUBQ MOD UH-60L EH-60A HH-60A HH-60L) or CONTINUITY CHECK (UH-60A 77-22714 - 85-24414 W/O MOD):

Table 1. Continuity Check (UH-60A 85-24415 - SUBQ MOD UH-60L EH-60A HH-60A HH-60L).

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN)
LDG LT	*EXT	U - W	0	U - 1 and W - 3
LDG LT	*EXT	U - S	Infinity	Replace switch (WP 1068 00)
LDG LT	Center (off)	U - W	Infinity	Replace switch (WP 1068 00)

ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 1. Continuity Check (UH-60A 85-24415 - SUBQ MOD UH-60L EH-60A HH-60A HH-60L). - Continued

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN)
LDG LT	Center (off)	U - S	Infinity	Replace switch (WP 1068 00)
LDG LT	*RETRACT	U - S	0	S - 2
LDG LT	*RETRACT	U - W	Infinity	Replace switch (WP 1068 00)
LDG LT	*Pressed	X - Y	0	X - 5 and Y - 4
LDG LT	Not Pressed	X - Y	Infinity	Replace switch (WP 1068 00)
SRCH LT	Center (off)	M - N	Infinity	Replace switch (WP 1067 00)
SRCH LT	Center (off)	L - M	Infinity	Replace switch (WP 1067 00)
SRCH LT	*DIM	M - N	0	M - 1 and N - 2
SRCH LT	*DIM	L - M	Infinity	Replace switch (WP 1067 00)
SRCH LT	*BRT	L - M	0	L - 3 and M - 1
SRCH LT	*BRT	M - N	Infinity	Replace switch (WP 1067 00)
SRCH LT	*Pressed	K - P	0	K - 5 and P - 4
SRCH LT	Not Pressed	K - P	Infinity	Replace switch (WP 1067 00)
SVO OFF	1ST STG	f - J	0	J - 6 and f - 5
SVO OFF	1ST STG	c - n	0	n - 3 and c - 2
SVO OFF	2ND STG	f - p	0	p - 4
SVO OFF	2ND STG	c - d	0	d - 1
EMERG HOOK REL	Push (on)	F - H	0	H - 1 and F - 2
EMERG HOOK REL	Push (off)	F - H	Infinity	Replace switch (WP 1068 00)
EH-60A UH-60A UH-60L ENG RPM	Center (off)	h - g	0	h - 5 and g - 6
EH-60A UH-60A UH-60L ENG RPM	Center (off)	h - i	Infinity	Replace switch (WP 1069 00)

ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 1. Continuity Check (UH-60A 85-24415 - SUBQ MOD UH-60L EH-60A HH-60A HH-60L). - Continued

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN)
EH-60A UH-60A UH-60L ENG RPM	Center (off)	k - j	0	k - 2 and j - 1
EH-60A UH-60A UH-60L ENG RPM	Center (off)	k - m	Infinity	Replace switch (WP 1069 00)
EH-60A UH-60A UH-60L ENG RPM	*INC	k - m	0	m - 3
EH-60A UH-60A UH-60L ENG RPM	*INC	g - h	0	Replace switch (WP 1069 00)
EH-60A UH-60A UH-60L ENG RPM	*DECR	h - i	0	i - 4
EH-60A UH-60A UH-60L ENG RPM	*DECR	j - k	0	Replace switch (WP 1069 00)
HH-60A HH-60L RAD SEL	Center (off)	j - h	Infinity	Replace switch (WP 1070 00)
HH-60A HH-60L RAD SEL	Center (off)	j - i	Infinity	Replace switch (WP 1070 00)
HH-60A HH-60L RAD SEL	Center (off)	j- k	Infinity	Replace switch (WP 1070 00)
HH-60A HH-60L RAD SEL	Center (off)	j- m	Infinity	Replace switch (WP 1070 00)
HH-60A HH-60L RAD SEL	UP	j - m	0	m - D
HH-60A HH-60L RAD SEL	DN	j - i	0	i - B
HH-60A HH-60L RAD SEL	L	j - h	0	h - C
HH-60A HH-60L RAD SEL	R	j - k	0	k - A
Searchlight control	Center (off)	C - A	Infinity	Replace switch (WP 1071 00)

ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 1. Continuity Check (UH-60A 85-24415 - SUBQ MOD UH-60L EH-60A HH-60A HH-60L). - Continued

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN)
Searchlight control	Center (off)	C - B	Infinity	Replace switch (WP 1071 00)
Searchlight control	Center (off)	C - D	Infinity	Replace switch (WP 1071 00)
Searchlight control	Center (off)	C - e	Infinity	Replace switch (WP 1071 00)
Searchlight control	*EXT	C - e	0	e - P and C - E
Searchlight control	*RETR	C - B	0	B - B
Searchlight control	*L	C - A	0	A - A
Searchlight control	*R	C - D	0	D - C
HUD control	Center (off)	Z - q	Infinity	Replace switch (WP 1072 00)
HUD control	Center (off)	Z - r	Infinity	Replace switch (WP 1072 00)
HUD control	Center (off)	Z - t	Infinity	Replace switch (WP 1072 00)
HUD control	Center (off)	Z - s	Infinity	Replace switch (WP 1072 00)
HUD control	*BRT	Z - s	0	s - C and Z - E
HUD control	*DIM	Z - r	0	r - A
HUD control	*MODE	Z - q	0	q - D
HUD control	*DCLT	Z - t	0	t - B

*Indicates momentary contact. Switch must be held at this position.

Table 2. Continuity Check (UH-60A 77-22714 - 85-24414 W/O MOD).

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN)
LDG LT	*EXT	U - W	0	U - 1 and W - 3
LDG LT	*EXT	U - S	Infinity	Replace switch (WP 1068 00)

ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 2. Continuity Check (UH-60A 77-22714 - 85-24414 W/O MOD). - Continued

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN)
LDG LT	Center (off)	U - W	Infinity	Replace switch (WP 1068 00)
LDG LT	Center (off)	U - S	Infinity	Replace switch (WP 1068 00)
LDG LT	*RETRACT	U - S	0	S - 2
LDG LT	*RETRACT	U - W	Infinity	Replace switch (WP 1068 00)
LDG LT	*Pressed	X - Y	0	X - 5 and Y - 4
LDG LT	Not Pressed	X - Y	Infinity	Replace switch (WP 1068 00)
SRCH LT	Center (off)	M - N	Infinity	Replace switch (WP 1067 00)
SRCH LT	Center (off)	L - M	Infinity	Replace switch (WP 1067 00)
SRCH LT	*DIM	M - N	0	M - 1 and N - 2
SRCH LT	*DIM	L - M	Infinity	Replace switch (WP 1067 00)
SRCH LT	*BRT	L - M	0	L - 3 and M - 1
SRCH LT	*BRT	M - N	Infinity	Replace switch (WP 1067 00)
SRCH LT	*Pressed	K - P	0	K - 5 and P - 4
SRCH LT	Not Pressed	K - P	Infinity	Replace switch (WP 1067 00)
SVO OFF	1ST STG	f - J	0	J - 6 and f - 5
SVO OFF	1ST STG	c - n	0	n - 3 and c - 2
SVO OFF	2ND STG	f - p	0	p - 4
SVO OFF	2ND STG	c - d	0	d - 1
EMERG HOOK REL	Push (on)	F - G	0	F - 1 and G - 2
EMERG HOOK REL	Push (off)	F - G	Infinity	Replace switch (WP 1068 00)

ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 2. Continuity Check (UH-60A 77-22714 - 85-24414 W/O MOD). - Continued

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN)
ENG RPM	Center (off)	h - g	0	h - 5 and g - 6
ENG RPM	Center (off)	h - i	Infinity	Replace switch (WP 1069 00)
ENG RPM	Center (off)	k - j	0	k - 2 and j - 1
ENG RPM	Center (off)	k - m	Infinity	Replace switch (WP 1069 00)
ENG RPM	*INC	k - m	0	m - 3
ENG RPM	*INC	g - h	0	Replace switch (WP 1069 00)
ENG RPM	*DECR	h - i	0	i - 4
ENG RPM	*DECR	j - k	0	Replace switch (WP 1069 00)
Searchlight control	Center (off)	C - A	Infinity	Replace switch (WP 1071 00)
Searchlight control	Center (off)	C - B	Infinity	Replace switch (WP 1071 00)
Searchlight control	Center (off)	C - D	Infinity	Replace switch (WP 1071 00)
Searchlight control	Center (off)	C - E	Infinity	Replace switch (WP 1071 00)
Searchlight control	*EXT	C - E	0	E - P and C - H
Searchlight control	*RETR	C - B	0	B - B
Searchlight control	*L	C - A	0	A - A

ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 2. Continuity Check (UH-60A 77-22714 - 85-24414 W/O MOD). - Continued

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (OHM)	CORRECTIVE ACTION (CHECK CONTINUITY BETWEEN)
Searchlight control	*R	C - D	0	D - C
HUD control	Center (off)	Z - q	Infinity	Replace switch (WP 1072 00)
HUD control	Center (off)	Z - r	Infinity	Replace switch (WP 1072 00)
HUD control	Center (off)	Z - t	Infinity	Replace switch (WP 1072 00)
HUD control	Center (off)	Z - s	Infinity	Replace switch (WP 1072 00)
HUD control	*BRT	Z - s	0	s - C and Z - E
HUD control	*DIM	Z - r	0	r - A
HUD control	*MODE	Z - q	0	q - D
HUD control	*DCLT	Z - t	0	t - B

*Indicates momentary contact. Switch must be held at this position.

Indication/Condition

Meter shall indicate corresponding resistance.

Corrective Action

1. If Indication/Condition is not as specified, do corresponding CORRECTIVE ACTION.
   a. If a continuity check is specified and continuity is present, replace switch (WP 1067 00 through WP 1072 00).
   b. If continuity is not present, repair/replace wiring (WP 1747 00).

MECHANICAL INSPECTION, PILOT OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- **MOD** represents UH-60A helicopters prior to Serial No. 85-24415 modified by MWO 1-1520-237-55-7 (MWO 1-1520-237-55-7 upgrades electrical harnesses of above helicopters with UH-60L electrical harnesses).
- **W/O MOD** represents UH-60A helicopters prior to Serial No. 85-24415 not modified by MWO 1-1520-237-55-7.
- **HUD** represents UH-60A, UH-60L, and HH-60A helicopters prior to Serial No. 96-26723 modified by MWO 1-1520-237-50-62. UH-60L Serial Nos. 96-26723 - SUBQ and HH-60L helicopters have product line incorporation of the ANVIS Heads Up Display Set.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Visually check grip for cracks.

Indication/Condition

Grip shall not be cracked.

Corrective Action

If Indication/Condition is not as specified, replace grip assembly (WP 1065 00).

Step

2. Visually check friction lock boot for tears or cracks.

Indication/Condition

Boot shall not be torn or cracked.

Corrective Action

If Indication/Condition is not as specified, replace boot (WP 1060 00).

Step

3. Check stick friction operation.

Indication/Condition

Friction grip shall work and stick shall move smoothly.

Corrective Action

If Indication/Condition is not as specified, go to FRICTION GRIP DOES NOT WORK AND/OR STICK DOES NOT MOVE SMOOTHLY, in this work package.

MECHANICAL INSPECTION, COPILOT OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- **MOD** represents UH-60A helicopters prior to Serial No. 85-24415 modified by MWO 1-1520-237-55-7 (MWO 1-1520-237-55-7 upgrades electrical harnesses of above helicopters with UH-60L electrical harnesses).
- **W/O MOD** represents UH-60A helicopters prior to Serial No. 85-24415 not modified by MWO 1-1520-237-55-7.
- **HUD** represents UH-60A, UH-60L, and HH-60A helicopters prior to Serial No. 96-26723 modified by MWO 1-1520-237-50-62. UH-60L Serial Nos. 96-26723 - SUBQ and HH-60L helicopters have product line incorporation of the ANVIS Heads Up Display Set.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Visually check grip for cracks.

Indication/Condition

Grip shall not be cracked.

Corrective Action

If Indication/Condition is not as specified, replace grip assembly (WP 1065 00).

Step

2. Check assembly for rough or sloppy stick movement.

Indication/Condition

Assembly movement shall not be rough or sloppy.

Corrective Action

1. If Indication/Condition is not as specified, do the following:
   a. Inspect socket assembly bearings for ratcheting or play.
      (1) If ratcheting or play is found, replace malfunctioning bearing(s) and/or socket assembly (WP 1077 00 or WP 1076 00).
   b. Check each bearing for binding by turning with torque wrench. Bearing shall turn with 8 inch-ounce torque or less.
      (1) If bearing binds, replace malfunctioning bearing(s) and/or socket assembly (WP 1077 00 or WP 1076 00).

LIGHTED PANEL IS NOT LIT CORRECTLY

SYMPTOM

Lighted panel is not lit correctly.

MALFUNCTION

Lighted panel is not lit correctly.

CORRECTIVE ACTION

1. Is lighted panel not lit?

a. If panel is not lit, go to **2.**

b. If panel is lit, go to **4.**

2. Check for 115 vac between:

- **J103-T and J103-V (Copilot's)**
- **J104-T and J104-V (Pilot's)**

a. If voltage is as specified, go to **3.**

b. If voltage is not as specified, troubleshoot instrument panel and consoles indicator lights dimming (WP 0110 00). Go to **5.**

3. Check wiring between:

- **P103-T and panel lights**
- **P103-V and panel lights**
- **P104-T and panel lights**
- **P104-V and panel lights**

a. If wiring is good, replace lighted panel assembly (WP 1066 00). Go to **5.**

b. If wiring is not good, repair/replace wiring (WP 1747 00). Go to **5.**

4. Check for burnt-out lamps.

a. If lamps are good, replace lighted panel assembly (WP 1066 00). Go to **5.**

b. If lamps are not good, replace lamps as required (WP 1066 00). Go to **5.**

5. Procedure completed.

FRICTION GRIP DOES NOT WORK AND/OR STICK DOES NOT MOVE SMOOTHLY

SYMPTOM

Friction grip does not work and/or stick does not move smoothly.

MALFUNCTION

Friction grip does not work and/or stick does not move smoothly.

CORRECTIVE ACTION

1. Does friction grip cause friction malfunction?

a. If grip causes malfunction, go to **2.**

FRICTION GRIP DOES NOT WORK AND/OR STICK DOES NOT MOVE SMOOTHLY - Continued

b. If grip does not cause malfunction, go to **3.**

2. Is over-tightening of grip possible?

a. If over-tightening is possible, replace friction lockring and stop assembly (WP 1075 00). Go to **6.**

b. If over-tightening is not possible, replace friction lock (WP 1075 00). Go to **6.**

3. Inspect socket assembly bearings for ratcheting or play.

a. If bearings have ratcheting or play, replace malfunctioning bearings and/or socket assembly (WP 1077 00 or WP 1076 00). Go to **6.**

b. If bearings do not have ratcheting or play, go to **4.**

4. Check each bearing for binding by turning with torque wrench. Go to 5.

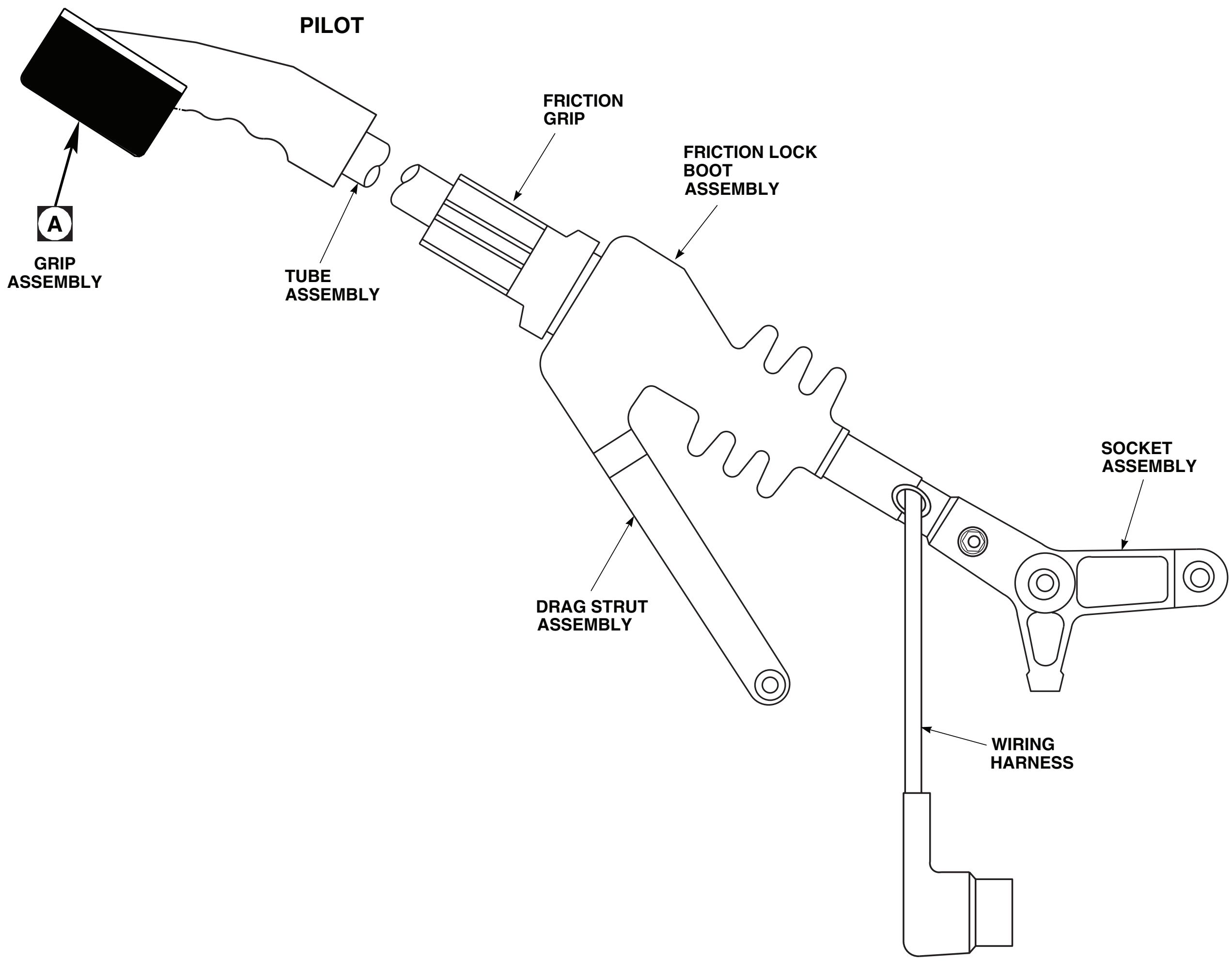
5. Do they turn with less than 8 inch-ounces torque?

a. If bearing turns with less than 8 inch-ounces torque, replace friction lock (WP 1075 00). Go to **6.**

b. If bearing does not turn with less than 8 inch-ounces torque, replace malfunctioning bearings and/or socket assembly (WP 1077 00 or WP 1076 00). Go to **6.**

6. Procedure completed.

LOCATION



NOTES

1. HUD
2. HH–60A HH–60L ENGINE SPEED TRIM SWITCH LOCATED ON UPPER CONSOLE.
3. HH–60A HH–60L

AA2287_1B
SA

Figure 1. Collective Stick Assembly. (Sheet 1 of 3)

LOCATION - Continued

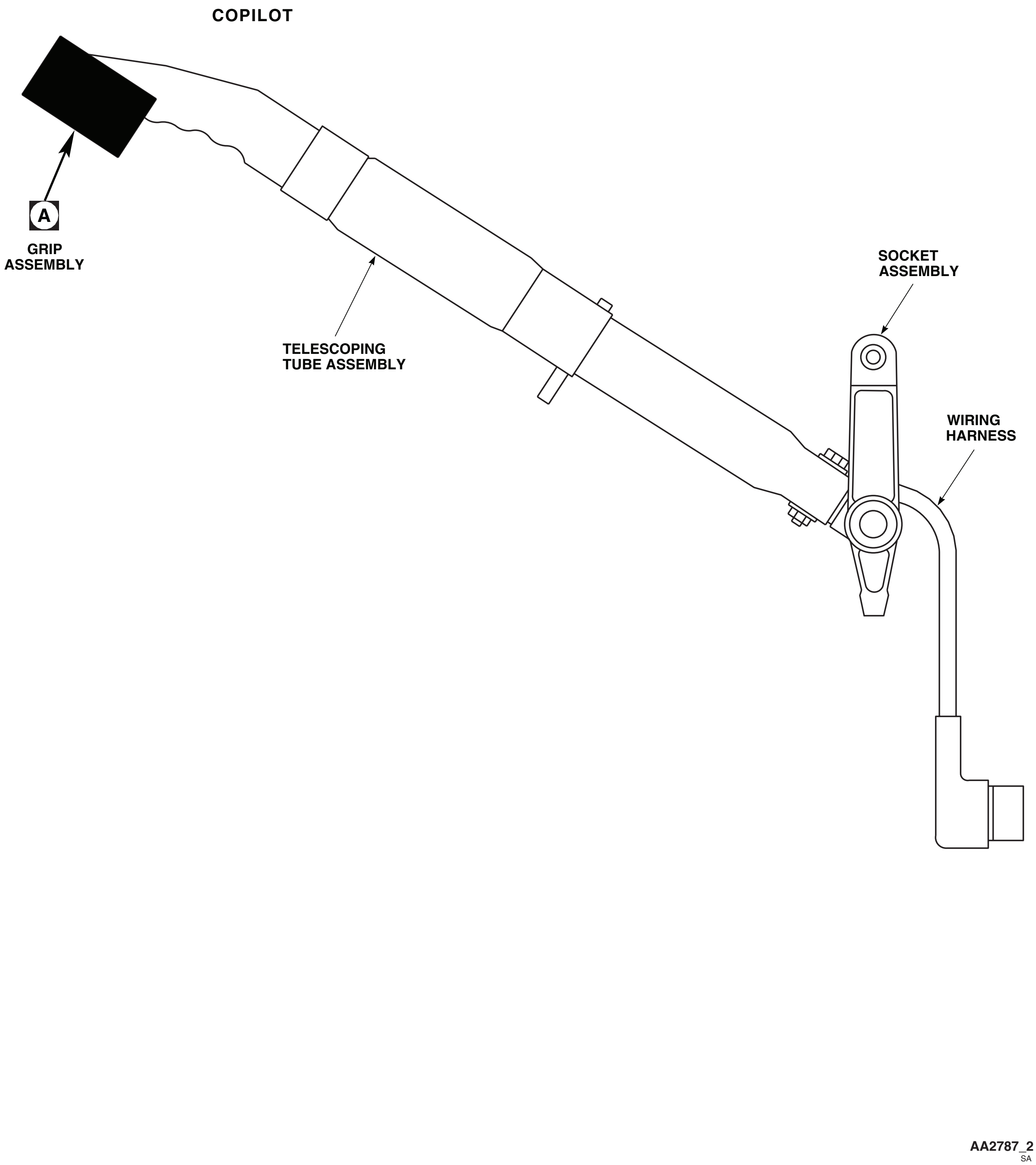


Figure 1. Collective Stick Assembly. (Sheet 2 of 3)

LOCATION - Continued

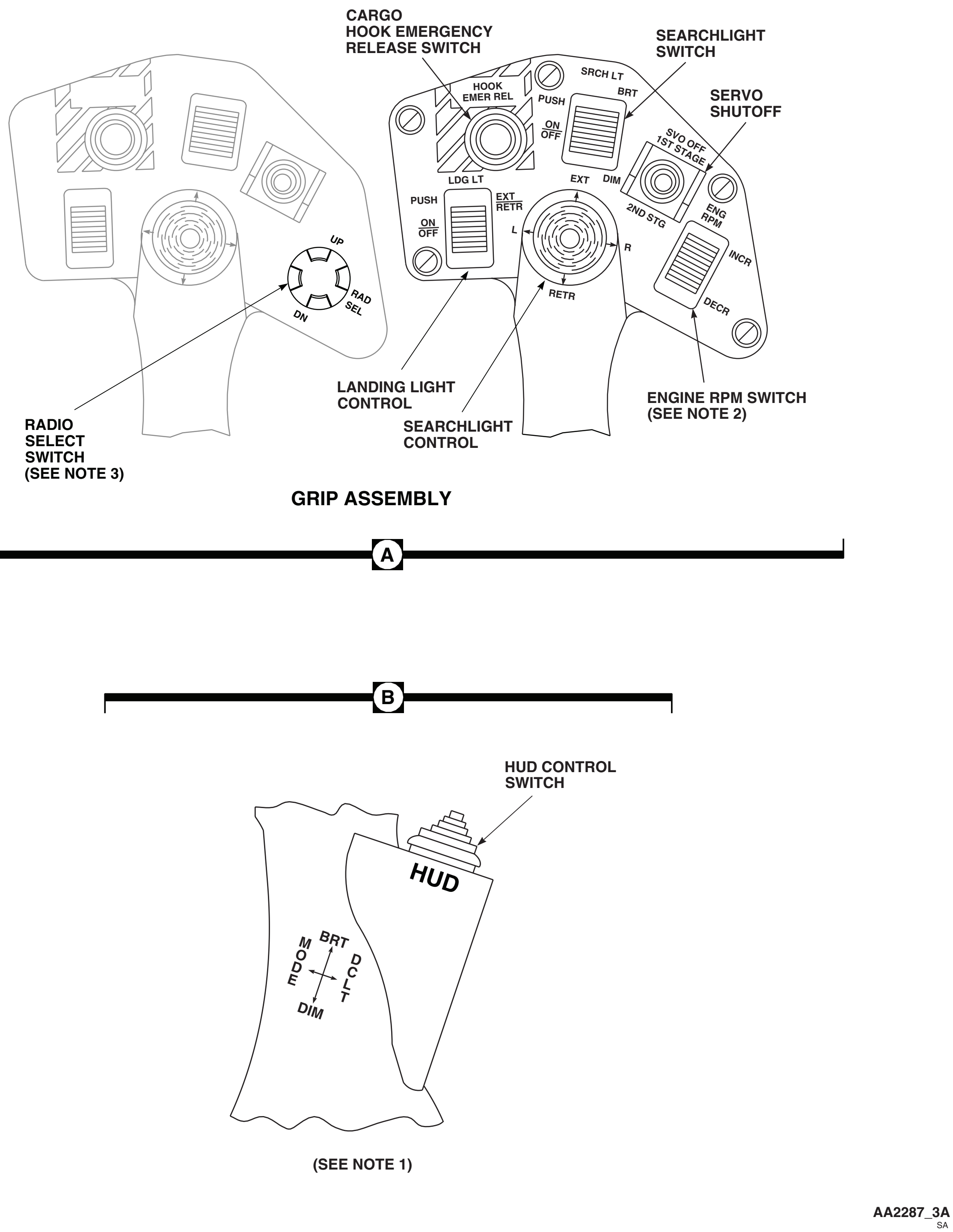
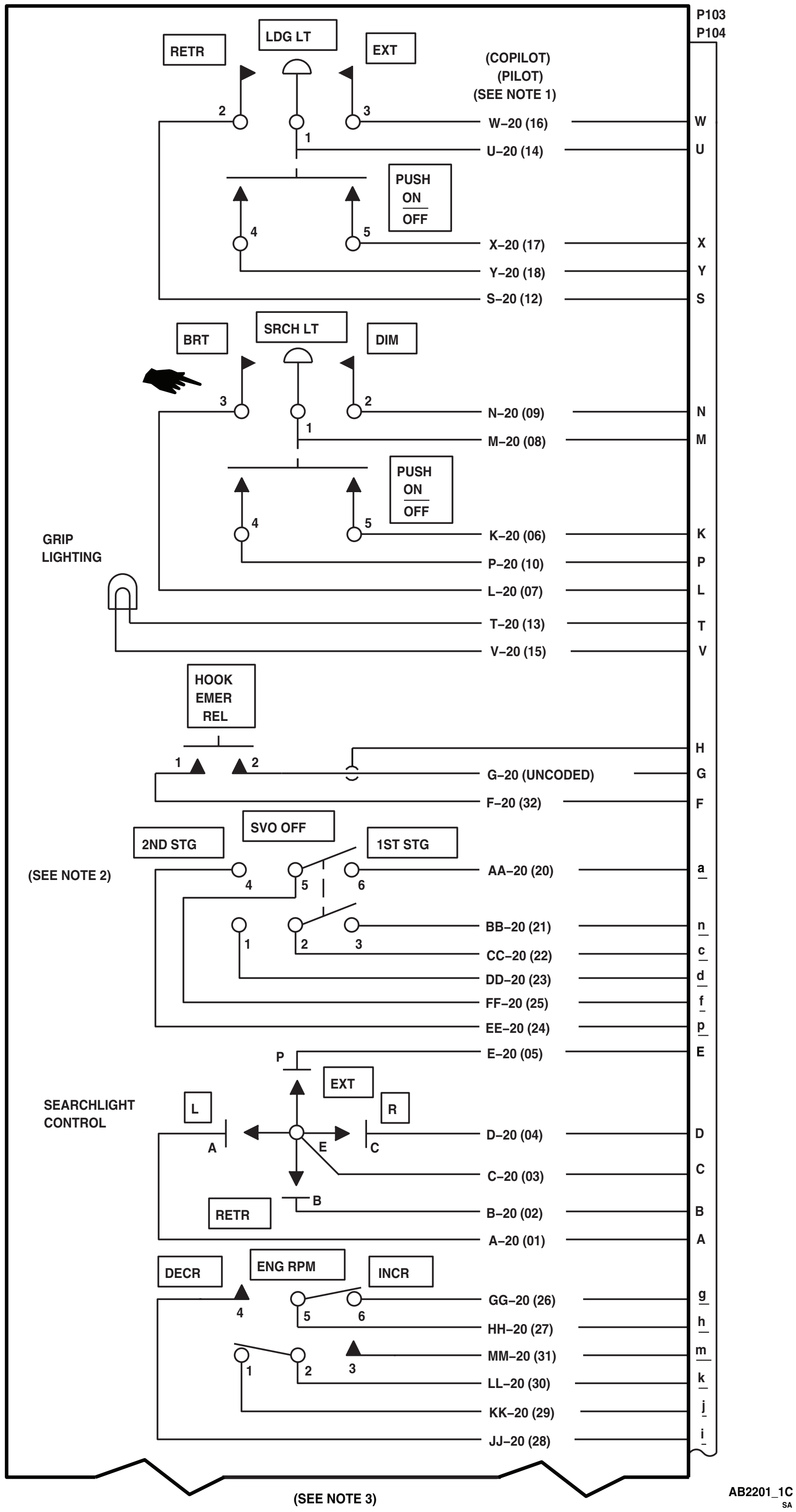


Figure 1. Collective Stick Assembly. (Sheet 3 of 3)

SCHEMATICS



NOTES

1. PILOT STICK ASSEMBLY WIRE NUMBERS SHOWN WITHOUT PARENTHESES. CO–PILOT STICK ASSEMBLY WIRE NUMBERS SHOWN IN PARENTHESES.
2. SVO OFF SWITCH SHOWN IN OFF POSITION. WITH SWITCH IN 2ND STG, CONNECTIONS BETWEEN 5 AND 4, AND 2 AND 1 ARE MADE. WITH SWITCH IN 1ST STG, CONNECTIONS BETWEEN 5 AND 6, AND 2 AND 3 ARE MADE.
3. **UH–60A 77–22714 – 85–24414**
   W/O MOD CHECK PINS G, b, a, Z, AND E. ALL SHOULD BE SHORTED TO EACH OTHER, IAW FIGURE 11–2–2–2, SHEET 2. IF NOT, IAW FIGURE 11–2–2–2, SHEET 1.
4. **UH–60A 85–24415 – SUBQ**
   MOD **UH60L** **EH60A**
5. **HUD** JUMPER WIRE FROM SG–1 TO Z IS REMOVED.
6. **HUD** APPLICABLE FOR PILOT AND CO–PILOT STICK ASSEMBLIES.
7. **HH–60A** **HH–60L** ENGINE SPEED TRIM SWITCH LOCATED ON UPPER CONSOLE.
8. **HH–60A** **HH–60L**

Figure 2. Collective Stick Assembly Schematic Diagram. (Sheet 1 of 3)

SCHEMATICS - Continued

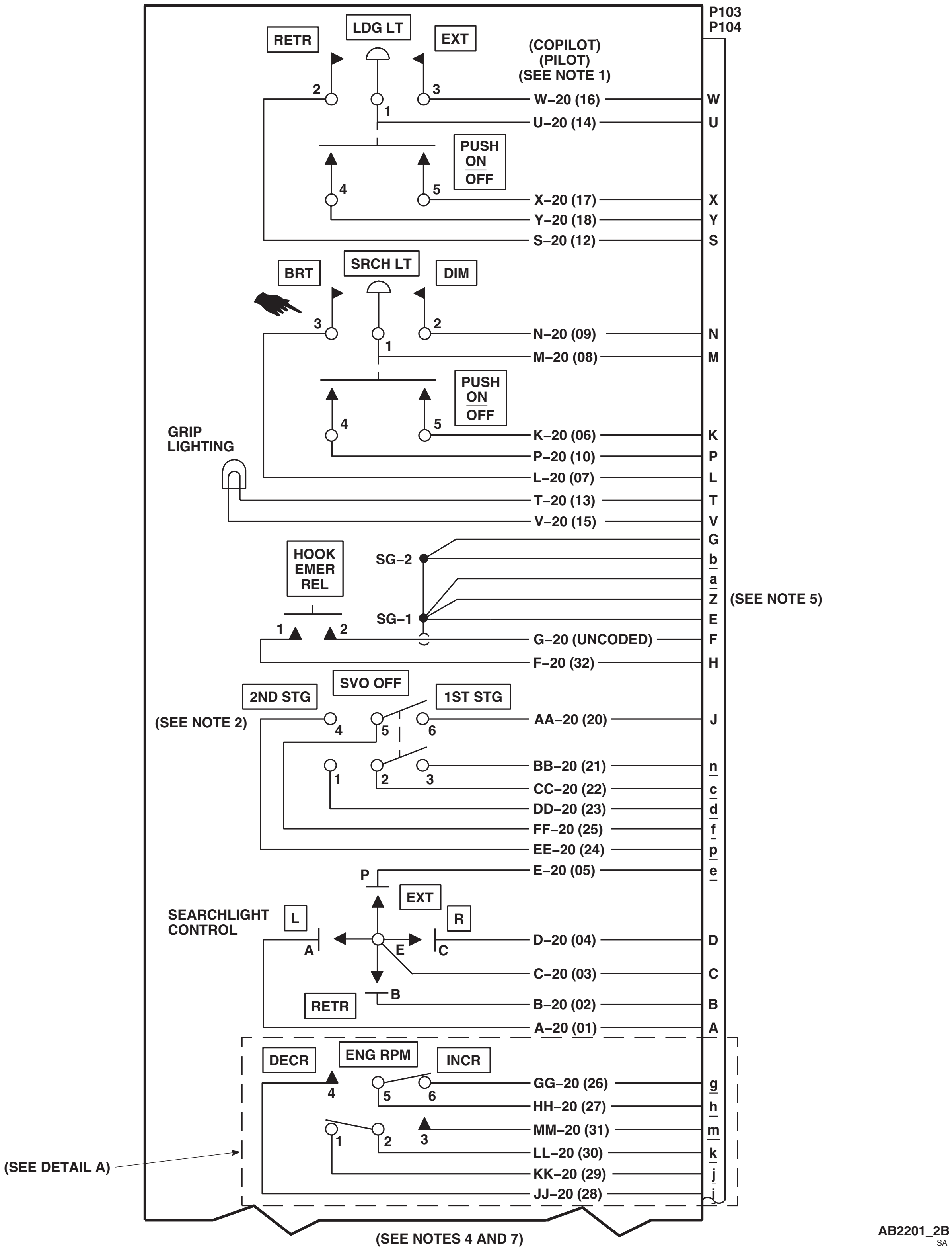
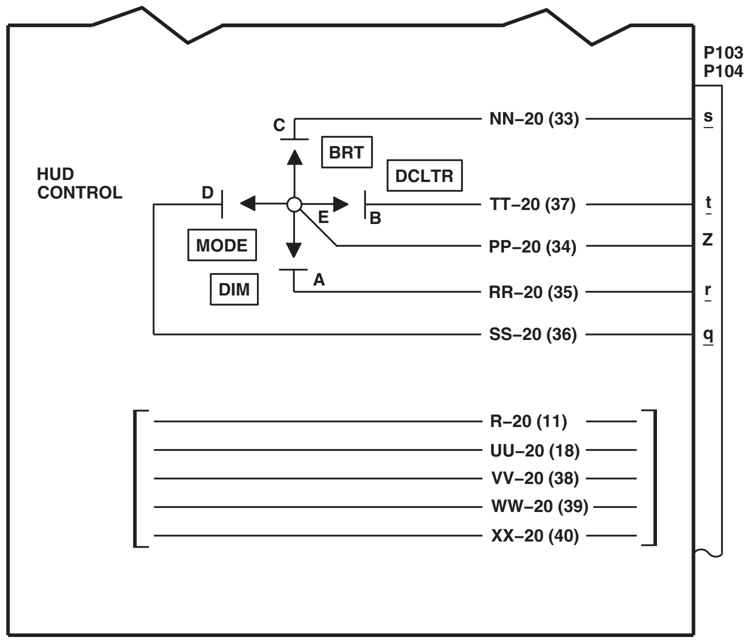
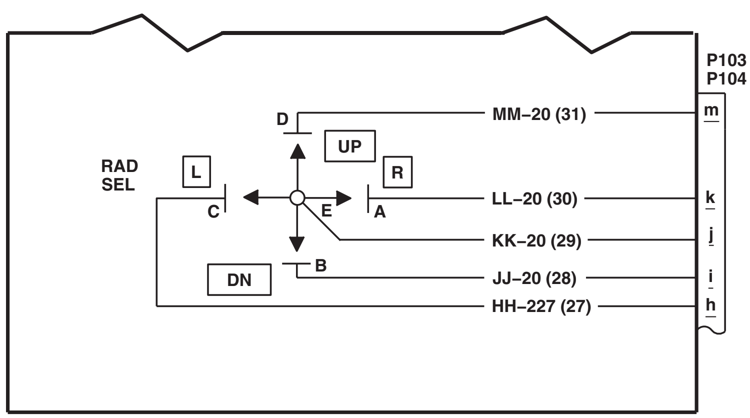


Figure 2. Collective Stick Assembly Schematic Diagram. (Sheet 2 of 3)

SCHEMATICS - Continued



(SEE NOTE 6)



DETAIL A
(SEE NOTE 8)

AB2201_3
SA

Figure 2. Collective Stick Assembly Schematic Diagram. (Sheet 3 of 3)

END OF WORK PACKAGE

UNIT LEVEL

FLIGHT CONTROLS

FLIGHT CONTROLS (MECHANICAL)

This WP supersedes WP 0136 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment
Multimeter, AN/PSM-45A

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1520-237-10
TM 11-1520-237-23
TM 11-1520-249-23-2
WP 0086 00
WP 0378 00
WP 0785 00
WP 0799 00
WP 0824 00
WP 0925 00
WP 1008 00
WP 1010 00
WP 1014 00
WP 1018 00
WP 1022 00
WP 1030 00
WP 1035 00
WP 1037 00
WP 1040 00
WP 1041 00
WP 1042 00
WP 1046 00
WP 1047 00
WP 1058 00
WP 1060 00
WP 1078 00
WP 1080 00
WP 1083 00
WP 1085 00
WP 1087 00
WP 1089 00
WP 1091 00
WP 1094 00
WP 1095 00
WP 1096 00
WP 1097 00
WP 1098 00
WP 1099 00
WP 1100 00
WP 1104 00
WP 1105 00
WP 1107 00
WP 1108 00
WP 1110 00
WP 1114 00
WP 1116 00
WP 1117 00
WP 1122 00
WP 1125 00
WP 1127 00
WP 1128 00
WP 1129 00
WP 1131 00
WP 1132 00
WP 1135 00
WP 1136 00
WP 1137 00
WP 1144 00
WP 1147 00
WP 1148 00
WP 1152 00
WP 1154 00
WP 1685 00
WP 1747 00

Equipment Condition
All Flight Control Components Installed
Backup Hydraulic Pump Operational,
TM 1-1520-237-10
External Electrical Power Available, WP 1685 00
or APU Operational, TM 1-1520-237-10

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram, Figure 2 for schematic diagram, and Figure 3 for inspection diagram as an aid in operational/troubleshooting.

The flight controls operational/troubleshooting procedure is subdivided as follows:

- Binding
- In-Flight Attitude Oscillations
- Flight Control Noise Checkout

CAUTION

In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.

Table 1. Temperature Requirements.

FAT F° (C°)	OPERATING TIME (MINUTES)	COOLDOWN TIME (PUMP OFF) (MINUTES)
90° (32°) max	Unlimited	-
91° (33°) to 100° (38°)	24	72
101° (39°) to 125° (52°)	16	48

BINDING OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- Flight control system will be damaged if cockpit controls are moved while control rods are disconnected. Loose control rods will strike other components. Make sure disconnected control rods are secured and flight controls are not moved when performing maintenance in this area.
- Damage to swashplate will result if primary servo input control rod is removed while hydraulic power is applied to the helicopter. Do not remove primary servo input control rods with hydraulic power on.
- To prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.

NOTE

- If any unusual noises are heard and no binding is felt while moving pilot or copilot pedals, collective sticks or cyclic sticks perform Flight Control Noise Checkout procedure.
- If any one of the following capsules on the caution/advisory panel or MFD legends go on during the operational/troubleshooting procedure, go to WP 0086 00:

  #1 PRI SERVO PRESS

  #2 PRI SERVO PRESS

  #1 TAIL RTR SERVO
- If any one of the following capsules on the caution/advisory panel or MFD legends go on during the operational/troubleshooting procedure, go to TM 11-1520-237-23 or TM 11-1520-249-23-2.

  SAS OFF

  BOOST SERVO OFF

1. Visually check complete flight control system to make sure all flight control rods are connected.
2. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
3. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
4. Make sure BOOST, TRIM, and FPS buttons on stabilator control/auto flight control panel are disengaged (capsules not on).
5. Place upper console BACKUP HYD PUMP switch to ON.
6. Move cyclic stick forward, back, right, left, and back to center position.

BINDING OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Cyclic stick shall move freely in all directions with no restrictions.

Corrective Action

1. If cyclic stick is difficult to move forward, but moves back freely (or the reverse), visually check for objects interfering with flight controls.
2. If cyclic stick is difficult to move forward and back, go to CYCLIC STICK IS DIFFICULT TO MOVE FORWARD AND AFT, in this work package.
3. If cyclic stick is difficult to move left and right, go to CYCLIC STICK IS DIFFICULT TO MOVE LEFT AND RIGHT, in this work package.

Step

7. Push on left and right pedals through their full range.

Indication/Condition

Pedals shall move freely with no restrictions.

Corrective Action

1. If TAIL ROTOR QUADRANT capsule or **HH-60A HH-60L** TAIL RTR QUADRANT legend ■ goes on, go to **HH-60A HH-60L** TAIL ROTOR QUADRANT CAPSULE OR HH-60A HH-60L TAIL RTR QUADRANT LEGEND IS ON ■ , in this work package.
2. If there is too much free play in pedals, go to EXCESSIVE FREE PLAY IN TAIL ROTOR PEDALS, in this work package.
3. If pedals are difficult to move, go to PEDALS ARE DIFFICULT TO MOVE, in this work package.

Step

8. Completely unlock friction lock on collective stick by turning friction grip until friction lock moves forward.

NOTE

It is normal for pedals to move when collective stick is moved.

9. Move collective stick up and down through full range.

Indication/Condition

Collective stick shall move freely with no restrictions.

Corrective Action

1. If collective stick requires much greater force to raise than to lower, go to COLLECTIVE STICK REQUIRES GREATER FORCE TO MOVE IN ONE DIRECTION THAN THE OTHER, in this work package.
2. If collective stick is difficult to move up and down, go to COLLECTIVE STICK IS DIFFICULT TO MOVE UP AND DOWN, in this work package.

Step

10. Install an insulated jumper between J611-C and J611-A.

BINDING OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

TAIL ROTOR QUADRANT capsule on caution/advisory panel or **HH-60A HH-60L** TAIL RTR QUADRANT legend on MFD shall go on.

Corrective Action

If Indication/Condition is not as specified, go to **HH-60A HH-60L** TAIL ROTOR QUADRANT CAPSULE OR HH-60A HH-60L TAIL RTR QUADRANT LEGEND DOES NOT GO ON , in this work package.

Step

11. Install an insulated jumper between J611-C and J611 -B.

Indication/Condition

TAIL ROTOR QUADRANT capsule on caution/advisory panel or **HH-60A HH-60L** TAIL RTR QUADRANT legend on MFD shall go on.

Corrective Action

If Indication/Condition is not as specified, go to **HH-60A HH-60L** TAIL ROTOR QUADRANT CAPSULE OR HH60A HH-60L TAIL RTR QUADRANT LEGEND DOES NOT GO ON , in this work package.

Step

12. Remove jumper wire and reconnect electrical connector.
13. Place upper console BACKUP HYD PUMP switch to OFF.
14. **HH-60A HH-60L** Turn MFD switch OFF.
15. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

IN-FLIGHT ATTITUDE OSCILLATIONS OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- Flight control system will be damaged if cockpit controls are moved while control rods are disconnected. Loose control rods will strike other components. Make sure disconnected control rods are secured and flight controls are not moved when performing maintenance in this area.
- Damage to swashplate will result if primary servo input control rod is removed while hydraulic power is applied to the helicopter. Do not remove primary servo input control rods with hydraulic power on.
- To prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.

NOTE

- If any unusual noises are heard and no binding is felt while moving pilot or copilot pedals, collective sticks or cyclic sticks perform Flight Control Noise Checkout procedure.
- In-flight attitude oscillations are caused by trouble in the flight control systems, from the controls in the cockpit to the inputs to the swashplate. Since these oscillations may feel like a lateral, longitudinal, or vertical movement (yaw, wallow, or porpoising) to the pilot, or any combination of these, the condition may be difficult to diagnose. The type of oscillations dealt with in this procedure are low frequency, (less than 1/rev) stable amplitude movements in any axis of the helicopter, in forward flight or in a hover.
- If any one of the following capsules on the caution/advisory panel or MFD legends go on during the operational/troubleshooting procedure, go to WP 0086 00:

  #1 PRI SERVO PRESS

  #2 PRI SERVO PRESS

  #1 TAIL RTR SERVO
- If any one of the following capsules on the caution/advisory panel or MFD legends go on during the operational/troubleshooting procedure, go to TM 11-1520-237-23 or TM 11-1520-249-23-2:

  SAS OFF

  BOOST SERVO OFF
- **UH-60A 77-22714 - 78-23015 W/O MOD** In forward flight at 145 kts, a pitch oscillation may occur. This oscillation does not indicate a malfunction.
- **UH-60A 78-23016 - SUBQ MOD UH-60A EH-60A HH-60A HH-60L** This oscillation has been eliminated.

IN-FLIGHT ATTITUDE OSCILLATIONS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- During maintenance test flight, it is necessary to troubleshoot AFCS, to isolate problem area of flight controls (TM 11-1520-237-23 or TM 11-1520-249-23-2).

1. Perform maintenance test flight to determine type of attitude oscillation and troubleshoot AFCS as follows (TM 1-1520-237-MTF).

NOTE

- If type of attitude oscillation cannot be determined, troubleshoot AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).
- If type of attitude oscillation has most effect on pitch, roll, collective, or yaw, check that system first.

2. Fly helicopter straight-and-level in forward flight.
3. Disengage AFCS TRIM.

Indication/Condition

Oscillation shall persist.

Corrective Action

If Indication/Condition is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step

4. Engage AFCS TRIM.
5. Disengage SAS 1.

Indication/Condition

Oscillation shall persist.

Corrective Action

If Indication/Condition is not as specified, troubleshoot analog SAS units (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step

6. Disengage SAS 2.

Indication/Condition

Oscillation shall persist.

Corrective Action

If Indication/Condition is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step

7. If oscillation has a noticeable pitch component, disengage stabilator AUTO CONTROL.

Indication/Condition

Oscillation shall persist.

IN-FLIGHT ATTITUDE OSCILLATIONS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

If Indication/Condition is not as specified, troubleshoot stabilator units (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step

8. Check primary servo valve crank assembly for play (WP 1042 00).

Indication/Condition

Primary servo valve crank assembly shall not have play.

Corrective Action

If Indication/Condition is not as specified, replace primary servo (WP 1091 00).

Step

9. Check all flight control system components, connections and hardware, between primary servos and swashplate, for security, damage and corrosion.

Indication/Condition

Flight control system components between primary servos and swashplate shall not have damage, corrosion or loose hardware beyond limits.

Corrective Action

If Indication/Condition is not as specified, repair/replace flight control system components, as required.

Step

10. Check all flight control system components, connections and hardware, between primary servos and pilots' controls, for security, damage and corrosion.

Indication/Condition

Flight control system components between primary servos and pilots' controls shall not have damage, corrosion or loose hardware beyond limits.

Corrective Action

If Indication/Condition is not as specified, tighten and repair/replace hardware as required.

Step

11. If no deficiencies were found in flight control system being inspected, inspect all other flight control systems.
12. Do a rigging check (WP 1037 00).
13. Do a maintenance test flight (TM 1-1520-237-MTF).

Indication/Condition

Maintenance test flight shall be done.

Corrective Action

None Required

FLIGHT CONTROL NOISE CHECKOUT OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- Flight control system will be damaged if cockpit controls are moved while control rods are disconnected. Loose control rods will strike other components. Make sure disconnected control rods are secured and flight controls are not moved when performing maintenance in this area.
- Damage to swashplate will result if primary servo input control rod is removed while hydraulic power is applied to the helicopter. Do not remove primary servo input control rods with hydraulic power on.
- To prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.

NOTE

- If any unusual noises are heard and no binding is felt while moving pilot or copilot pedals, collective sticks or cyclic sticks perform Flight Control Noise Checkout procedure.
- In flight control systems, some slight noise can be expected as the system ages. Some of these noises are normal and do not indicate a problem. Other noises indicate that a problem exists which requires investigation and correction. The following procedures will help localize and correct known sources of unusual noises.
- If any one of the following capsules on the caution/advisory panel or MFD legends go on during the operational/troubleshooting procedure, go to WP 0086 00:

  #1 PRI SERVO PRESS

  #2 PRI SERVO PRESS

  #1 TAIL RTR SERVO
- If any one of the following capsules on the caution/advisory panel or MFD legends go on during the operational/troubleshooting procedure, go to TM 11-1520-237-23 or TM 11-1520-249-23-2:

  SAS OFF

  BOOST SERVO OFF
- During the following checkout procedure, two personnel will be required: one to move flight controls and another to listen for location of unusual noises.

1. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
2. Place upper console BACKUP HYD PUMP switch to ON.
3. On stabilator control/auto flight control panel, press BOOST ON switch (capsule on).

FLIGHT CONTROL NOISE CHECKOUT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

4. While slowly moving pilot cyclic stick fully left then fully right, carefully listen for any unusual noises in the lateral cyclic control channel.

Indication/Condition

No unusual noises shall be heard while moving pilot cyclic stick from left to right.

Corrective Action

If noise is heard while moving pilot cyclic stick from left to right, go to NOISE IS HEARD WHILE MOVING PILOT CYCLIC STICK FROM LEFT TO RIGHT, in this work package.

Step

5. While slowly moving pilot cyclic stick fully forward then fully aft, carefully listen for any unusual noises in the pitch cyclic control channel.

Indication/Condition

No unusual noises shall be heard while moving pilot cyclic stick from forward to aft.

Corrective Action

If noise is heard while moving pilot cyclic stick from forward to aft, go to NOISE IS HEARD WHILE MOVING PILOT CYCLIC STICK FROM FORWARD TO AFT, in this work package.

Step

NOTE

If noises in the collective control channel are suspected to be coming from forward of the boost actuators, disengage stabilator control/auto flight control panel BOOST ON switch (capsule off) and repeat step 6.

6. While slowly moving pilot collective stick fully up then fully down, carefully listen for any unusual noises in the collective control channel.

Indication/Condition

No unusual noises shall be heard while moving collective stick up then down.

Corrective Action

If noise is heard while moving collective stick up then down, go to NOISE IS HEARD WHILE MOVING COLLECTIVE STICK UP AND DOWN, in this work package.

Step

NOTE

- In order to reproduce unusual noises in the yaw control channel, momentary movement of the collective stick from the down position to the up position may be required.
- If noises in the yaw control channel are suspected to be coming from forward of the boost actuators, disengage stabilator control/auto flight control panel BOOST ON switch (capsule off) and repeat step 7.

7. While slowly pushing on pilot left and right pedals, carefully listen for any unusual noises in the yaw control channel.

FLIGHT CONTROL NOISE CHECKOUT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

No unusual noises shall be heard while moving pilot left and right pedals.

Corrective Action

If noise is heard while moving pilot left and right pedals, go to NOISE IS HEARD WHILE PUSHING ON LEFT AND RIGHT PEDAL, in this work package.

Step

8. On stabilator control/auto flight control panel ensure SAS ON is disengaged (capsule off).
9. On stabilator control/auto flight control panel ensure BOOST ON is disengaged (capsule off).
10. Place upper console BACKUP HYD PUMP switch to OFF.
11. **HH-60A HH-60L** Turn MFD switch OFF.
12. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

CYCLIC STICK IS DIFFICULT TO MOVE FORWARD AND AFT

SYMPTOM

Cyclic stick is difficult to move forward and aft.

MALFUNCTION

Cyclic stick is difficult to move forward and aft.

CORRECTIVE ACTION

1. Visually inspect mixer assembly for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace mixer assembly (WP 1089 00). Go to **10.**

b. If corrosion, damage, or an interfering object is not present, go to **2.**

2. Visually inspect pitch trim assembly input and output linkage for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace pitch trim assembly (WP 1083 00). Go to **10.**

b. If corrosion, damage, or an interfering object is not present, go to **3.**

3. Visually inspect forward primary servo input and output linkage for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace forward primary servo (WP 1091 00). Go to **10.**

b. If corrosion, damage, or an interfering object is not present, go to **4.**

CYCLIC STICK IS DIFFICULT TO MOVE FORWARD AND AFT - Continued

4. **Visually inspect primary servo input and output linkage for corrosion, damage, or an interfering object.**

   a. If corrosion, damage, or an interfering object is present, remove interfering object or replace rear primary servo linkage (WP 1091 00). Go to **10.**

   b. If corrosion, damage, or an interfering object is not present, go to **5.**

5. **Visually inspect primary servo for corrosion, damage, or an interfering object.**

   a. If corrosion, damage, or an interfering object is present, remove interfering object or replace rear primary servo (WP 1091 00). Go to **10.**

   b. If corrosion, damage, or an interfering object is not present, go to **6.**

6. **Remove cabin soundproofing, visually inspect cyclic stick balance spring for corrosion, damage, or an interfering object.**

   a. If corrosion, damage, or an interfering object is present, remove interfering object or replace cyclic stick balance spring (WP 1099 00). Go to **10.**

   b. If corrosion, damage, or an interfering object is not present, go to **7.**

7. **Check adjustment of cyclic stick balance spring.**

   a. If cyclic stick balance spring is good, go to **8.**

   b. If cyclic stick balance spring is not good, adjust cyclic stick balance spring (WP 1040 00). Go to **10.**

8. **Visually inspect pitch torque shaft and levers for corrosion, damage, or an interfering object.**

   a. If corrosion, damage, or an interfering object is present, remove interfering object or replace pitch torque shaft and levers (WP 1095 00). Go to **10.**

   b. If corrosion, damage, or an interfering object is not present, go to **9.**

9. **Visually inspect all flight control rods, bearings, and bellcranks for corrosion, damage, or an interfering object.**

   a. If corrosion, damage, or an interfering object is present, remove interfering object or replace control rods, bearings, and bellcranks as necessary (WP 1100 00). Go to **10.**

   b. If corrosion, damage, or an interfering object is not present, rig and adjust flight control system (WP 1035 00). Go to **10.**

10. **Procedure completed.**

CYCLIC STICK IS DIFFICULT TO MOVE LEFT AND RIGHT

SYMPTOM

Cyclic stick is difficult to move left and right.

MALFUNCTION

Cyclic stick is difficult to move left and right.

CORRECTIVE ACTION

1. **Visually inspect mixer assembly for corrosion, damage, or an interfering object.**

CYCLIC STICK IS DIFFICULT TO MOVE LEFT AND RIGHT - Continued

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace mixer assembly (WP 1089 00). Go to **8.**

b. If corrosion, damage, or an interfering object is not present, go to **2.**

2. Visually inspect roll trim assembly output linkage (under boot) for evidence of scoring.

a. If scoring is present, replace roll trim assembly (WP 1087 00). Go to **8.**

b. If scoring is not present, go to **3.**

3. Visually inspect roll trim assembly for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace roll trim assembly (WP 1087 00). Go to **8.**

b. If corrosion, damage, or an interfering object is not present, go to **4.**

4. Visually inspect lateral primary servo for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace lateral primary servo (WP 1091 00). Go to **8.**

b. If corrosion, damage, or an interfering object is not present, go to **5.**

5. Visually inspect roll torque shaft and levers for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace roll torque shaft and levers. Go to **8.**

b. If corrosion, damage, or an interfering object is not present, go to **6.**

6. With BACKUP PUMP switch ON, slowly move cyclic stick forward, back, left, and right, while visually inspecting flight control area for binding against wire bundles.

a. If binding is present, reposition wire bundles as necessary. Go to **8.**

b. If binding is not present, go to **7.**

7. Visually inspect all flight control rods, bearings, and bellcranks for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace control rods, bearings, and bellcranks as necessary. Go to **8.**

b. If corrosion, damage, or an interfering object is not present, rig and adjust flight control system (WP 1035 00). Go to **8.**

8. Procedure completed.

TAIL ROTOR QUADRANT CAPSULE OR HH-60A HH-60L TAIL RTR QUADRANT LEGEND IS ON

SYMPTOM

TAIL ROTOR QUADRANT capsule or HH-60A HH-60L **TAIL RTR QUADRANT legend** is on.

MALFUNCTION

TAIL ROTOR QUADRANT capsule or HH-60A HH-60L **TAIL RTR QUADRANT legend** is on.

CORRECTIVE ACTION

1. **Visually inspect tail rotor cables for broken or disconnected cables.**
   a. If broken or disconnected cables are present, repair/replace broken or disconnected cables (WP 1144 00). Go to **9.**
   b. If broken or disconnected cables are not present, go to **2.**
2. **Disconnect J611. Go to 3.**
3. **Does TAIL ROTOR QUADRANT capsule or HH-60A HH-60L TAIL RTR QUADRANT legend go off?**
   a. If capsule is off, go to **4.**
   b. If capsule is not off, go to **5.**
4. **Check adjustment of tail rotor quadrant warning switches.**
   a. If warning switches are good, replace tail pylon cable (harness) (WP 1147 00). Go to **9.**
   b. If warning switches are not good, adjust rotor quadrant warning switches (WP 1152 00). Go to **9.**
5. **Check helicopter configuration.**
   a. EMEP , go to **6.**
   b. W/O EMEP , go to **8.**
6. **EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).**
   a. If pin filtered adapter is good, go to **7.**
   b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9.**
7. **EMEP Install pin filtered adapter (WP 0925 00). Go to 8.**
8. **Check for 28 vdc between P117-9 or HH-60A HH-60L P1019R-77 or P1020R-77 and ground with J611 disconnected.**
   a. If voltage is as specified, repair/replace short circuit between J611-A, J611- B, and J611-C (WP 1747 00). Go to **9.**
   b. If voltage is not as specified, replace caution advisory panel (WP 0785 00) or multifunction display (WP 0799 00). Go to **9.**
9. **Procedure completed.**

EXCESSIVE FREE PLAY IN TAIL ROTOR PEDALS

SYMPTOM

Excessive free play in tail rotor pedals.

MALFUNCTION

Excessive free play in tail rotor pedals.

CORRECTIVE ACTION

1. **Inspect pedal assembly mounting hardware for incorrect torque or wear.**
   a. If pedal assembly hardware is incorrect, torque or replace hardware as required. Go to **6.**
   b. If pedal assembly hardware is correct, go to **2.**
2. **Inspect pedal pivot shaft and directional control rods for looseness or wear.**
   a. If pedal pivot shaft and directional control rods are loose or worn, tighten or replace hardware or components as required. Go to **6.**
   b. If pedal pivot shaft and directional control rods are not loose or worn, go to **3.**
3. **Inspect pedal adjuster for looseness or wear.**
   a. If pedal adjuster is loose or worn, tighten or replace hardware of components as required (WP 1047 00). Go to **6.**
   b. If pedal adjuster is not loose or worn, go to **4.**
4. **Inspect both cable disconnect fittings in tail cone. Go to 5.**
5. **Make sure retainers are bottomed out in quick disconnect fittings. Go to 6.**
6. **Procedure completed.**

PEDALS ARE DIFFICULT TO MOVE

SYMPTOM

Pedals are difficult to move.

MALFUNCTION

Pedals are difficult to move.

CORRECTIVE ACTION

1. **Visually inspect pedal floor cover for binding (around pedal shaft on cockpit floor).**
   a. If binding is present, replace pedal floor covers (WP 1046 00). Go to **10.**
   b. If binding is not present, go to **2.**
2. **Visually inspect mixer assembly for corrosion, damage, or an interfering object.**
   a. If corrosion, damage, or an interfering object is present, remove interfering object or replace mixer assembly (WP 1089 00). Go to **10.**
   b. If corrosion, damage, or an interfering object is not present, go to **3.**
3. **Visually inspect tail rotor system for incorrectly routed, kinked, or broken cables, damaged cable pulleys, or an object interfering with cable movement.**

PEDALS ARE DIFFICULT TO MOVE - Continued

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace cables or pulleys (WP 1144 00). Go to **10.**

b. If corrosion, damage, or an interfering object is not present, go to **4.**

4. **Visually inspect plastic tubes and cables on tail pylon for binding.**

a. If binding is present, repair/replace tail rotor cables or tubes (WP 1144 00). Go to **10.**

b. If binding is not present, go to **5.**

5. **Visually inspect yaw boost assembly input and output linkage for corrosion, damage, or an interfering object.**

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace yaw boost assembly (WP 1080 00). Go to **10.**

b. If corrosion, damage, or an interfering object is not present, go to **6.**

6. **Check cables for proper tension.**

a. If proper tension is present, go to **7.**

b. If proper tension is not present, adjust cable tension. Go to **10.**

7. **Visually inspect tail rotor servo for corrosion, damage, or an interfering object.**

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace tail rotor servo (WP 1148 00). Go to **10.**

b. If corrosion, damage, or an interfering object is not present, go to **8.**

8. **Visually inspect directional (yaw) trim servo for corrosion, damage, or an interfering object.**

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace directional (yaw) trim servo (WP 1078 00). Go to **10.**

b. If corrosion, damage, or an interfering object is not present, go to **9.**

9. **Visually inspect flight control rods, bearings, and bellcranks for corrosion, damage, or an interfering object.**

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace control rods, bearings, and bellcranks as necessary. Go to **10.**

b. If corrosion, damage, or an interfering object is not present, rig and adjust flight control system (WP 1035 00). Go to **10.**

10. **Procedure completed.**

COLLECTIVE STICK REQUIRES GREATER FORCE TO MOVE IN ONE DIRECTION THAN THE OTHER

SYMPTOM

Collective stick requires greater force to move in one direction than the other.

MALFUNCTION

Collective stick requires greater force to move in one direction than the other.

CORRECTIVE ACTION

1. **Remove soundproofing, visually inspect collective balance spring for corrosion, damage, or an interfering object.**

COLLECTIVE STICK REQUIRES GREATER FORCE TO MOVE IN ONE DIRECTION THAN THE OTHER - Continued

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace collective balance spring (WP 1099 00). Go to **5.**

b. If corrosion, damage, or an interfering object is not present, go to **2.**

2. Check adjustment of collective balance spring.

a. If collective balance spring is good, go to **3.**

b. If collective balance spring is not good, adjust collective stick balance spring (WP 1099 00). Go to **5.**

3. Visually inspect boot at base of collective stick.

a. If boot at base of collective stick is good, go to **4.**

b. If boot at base of collective stick is not good, replace collective stick boot (WP 1060 00). Go to **5.**

4. Visually inspect all flight control rods, bearings, and bellcranks for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace control rods, bearings, and bellcranks as necessary. Go to **5.**

b. If corrosion, damage, or an interfering object is not present, rig and adjust flight control system (WP 1035 00). Go to **5.**

5. Procedure completed.

COLLECTIVE STICK IS DIFFICULT TO MOVE UP AND DOWN

SYMPTOM

Collective stick is difficult to move up and down.

MALFUNCTION

Collective stick is difficult to move up and down.

CORRECTIVE ACTION

1. Visually inspect boot at base of collective stick.

a. If boot at base of collective stick is good, go to **2.**

b. If boot at base of collective stick is not good, replace collective stick boot (WP 1060 00). Go to **11.**

2. Remove soundproofing and visually inspect collective stick balance spring for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace collective stick balance spring (WP 1099 00). Go to **11.**

b. If corrosion, damage, or an interfering object is not present, go to **3.**

3. Check adjustment of collective balance spring.

a. If collective balance spring is good, go to **4.**

b. If collective balance spring is not good, adjust collective stick balance spring (WP 1041 00). Go to **11.**

4. Visually inspect mixer assembly for corrosion, damage, or an interfering object.

COLLECTIVE STICK IS DIFFICULT TO MOVE UP AND DOWN - Continued

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace mixer assembly (WP 1089 00). Go to **11.**

b. If corrosion, damage, or an interfering object is not present, go to **5.**

5. Visually inspect collective boost assembly for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace collective boost assembly (WP 1085 00). Go to **11.**

b. If corrosion, damage, or an interfering object is not present, go to **6.**

6. Visually inspect collective trim servo for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace collective trim servo (WP 1116 00). Go to **11.**

b. If corrosion, damage, or an interfering object is not present, go to **7.**

7. Visually inspect forward primary servo input and output linkage for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace forward primary servo (WP 1091 00). Go to **11.**

b. If corrosion, damage, or an interfering object is not present, go to **8.**

8. Visually inspect aft primary servo input and output linkage for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace rear primary servo (WP 1091 00). Go to **11.**

b. If corrosion, damage, or an interfering object is not present, go to **9.**

9. Visually inspect lateral primary servo for corrosion, damage, or an interfering object.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace lateral primary servo (WP 1091 00). Go to **11.**

b. If corrosion, damage, or an interfering object is not present, go to **10.**

10. Visually inspect tail rotor system for incorrectly routed, kinked, or broken cables, damaged cable pulleys or an object interfering with cable movement.

a. If corrosion, damage, or an interfering object is present, remove interfering object or replace cables or pulleys (WP 1144 00). Go to **11.**

b. If corrosion, damage, or an interfering object is not present, rig and adjust flight control system. Go to **11.**

11. Procedure completed.

TAIL ROTOR QUADRANT CAPSULE OR HH-60A HH-60L TAIL RTR QUADRANT LEGEND ■ DOES NOT GO ON

SYMPTOM

TAIL ROTOR QUADRANT capsule or HH-60A HH-60L TAIL RTR QUADRANT legend ■ does not go on.

MALFUNCTION

TAIL ROTOR QUADRANT capsule or HH-60A HH-60L TAIL RTR QUADRANT legend ■ does not go on.

CORRECTIVE ACTION

1. Install an insulated jumper between J611-A and J611-C. Go to 2.

2. Install an insulated jumper between J611-B and J611-C. Go to 3.

3. Check helicopter configuration.

a. **EMEP** , go to **4.**

b. **W/O EMEP** , go to **6.**

4. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R ■ (WP 0925 00).

a. If pin filtered adapter is good, go to **5.**

b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **11.**

5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.

6. Check for 28 vdc between P117-9 or HH-60A HH-60L P1019R-77 or P1020R-77 ■ and ground.

a. If voltage is as specified, replace caution advisory panel (WP 0785 00) or multifunction display (WP 0799 00). Go to **11.**

b. If voltage is not as specified, go to **7.**

7. Remove insulated jumpers. Go to 8.

8. Check for 28 vdc between:

- **J611-A and ground**
- **J611-B and ground**

a. If voltage is as specified, go to **9.**

b. If voltage is not as specified, go to **10.**

9. Check continuity between J611-C and P117-9 or HH-60A HH-60L P1019R-77 or P1020R-77 ■ .

a. If voltage is as specified, replace quadrant switches (WP 1154 00). Go to **11.**

b. If voltage is not as specified, repair/replace wiring (WP 1747 00). Go to **11.**

10. Check continuity between T RTR SERVO WARN circuit breaker terminal 1 and J611-B.

a. If voltage is as specified, replace circuit breaker (WP 0824 00). Go to **11.**

TAIL ROTOR QUADRANT CAPSULE OR HH-60A HH-60L TAIL RTR QUADRANT LEGEND DOES NOT GO ON - Continued

b. If voltage is not as specified, repair/replace wiring (WP 1747 00). Go to **11.**

11. Procedure completed.

NOISE IS HEARD WHILE MOVING PILOT CYCLIC STICK FROM LEFT TO RIGHT

SYMPTOM

Noise is heard while moving pilot cyclic stick from left to right.

MALFUNCTION

Noise is heard while moving pilot cyclic stick from left to right.

CORRECTIVE ACTION

1. Is noise coming from forward of vertical torque shaft?

a. If noise is coming from forward of vertical torque shaft, go to **7.**

b. If noise is not coming from forward of vertical torque shaft, go to **2.**

2. Is noise coming from control shaft in upper broom closet?

a. If noise is coming from control shaft in upper broom closet, go to **10.**

b. If noise is not coming from control shaft in upper broom closet, go to **3.**

3. Is noise coming from vertical torque shaft?

a. If noise is coming from vertical torque shaft, go to **16.**

b. If noise is not coming from vertical torque shaft, go to **4.**

4. Is noise coming from aft of vertical torque shaft?

a. If noise is coming from aft of vertical torque shaft, go to **22.**

b. If noise is not coming from aft of vertical torque shaft, go to **5.**

5. Is noise coming from mixer assembly?

a. If noise is coming from mixer assembly, go to **25.**

b. If noise is not coming from mixer assembly, go to **6.**

6. Is noise coming from aft of mixer assembly but forward of stationary swashplate?

a. If noise is coming from aft of mixer assembly but forward of stationary swashplate, go to **28.**

b. If noise is not coming from aft of mixer assembly but forward of stationary swashplate, perform foward, aft, and lateral flight control channel inspection (WP 1030 00). Go to **31.**

7. Visually inspect area forward of vertical torque shafts for missing or loose mounting hardware.

a. If missing or loose mounting hardware is present, repair/replace as required. Go to **31.**

b. If missing or loose mounting hardware is not present, go to **8.**

8. Inspect the following components forward of the vertical torque tubes for excessive radial play in bearings:

- **Pilot and copilot cyclic sticks (WP 1010 00)**
- **Cockpit lateral bellcrank (WP 1131 00)**

NOISE IS HEARD WHILE MOVING PILOT CYCLIC STICK FROM LEFT TO RIGHT - Continued

- **Cabin lateral bellcrank (WP 1132 00)**

a. If excessive radial play is present in cyclic stick, cockpit lateral bellcrank, or cabin lateral bellcrank bearings, repair/replace cyclic stick bearings or lateral bellcranks as required (WP 1058 00, WP 1131 00, or WP 1132 00). Go to **31.**

b. If excessive radial play in bearings is not present, go to **9.**

9. Visually inspect pilot cyclic lateral control rods, copilot cyclic lateral control rods, and cabin lateral control rod forward of vertical torque shafts for loose swaged inserts (Figure 3).

a. If swaged inserts on control rods are loose, replace pilot's directional and cyclic control rods, copilot's directional and cyclic control rods, or cabin control rods as required (WP 1105 00, WP 1108 00, or WP 1110 00). Go to **31.**

b. If swaged inserts on control rods are not loose, perform forward, aft and lateral flight control inspection, and bearing friction wear inspection (WP 1030 00). Go to **31.**

10. Visually inspect lateral control rods leading to upper broom closet for loose or missing mounting hardware.

a. If missing or loose mounting hardware is present, repair/replace as required. Go to **31.**

b. If missing or loose mounting hardware is not present, go to **11.**

11. Inspect midsection bellcranks and shafts for excessive radial play in bearings (WP 1100 00).

a. If excessive radial play in bearings is present, repair/replace midsection bellcranks and shaft (WP 1100 00). Go to **31.**

b. If excessive radial play in bearings is not present, go to **12.**

12. Inspect pilot and copilot lateral control rods leading to upper broom closet for loose swaged inserts (Figure 3).

a. If lateral control rod swaged inserts are loose, repair/replace lateral control rods leading to upper broom closet (WP 1110 00). Go to **31.**

b. If lateral control rod swaged insert are not loose, go to **13.**

13. Inspect midsection bellcranks split spacers for proper gap (WP 1100 00).

a. If split spacer gap is within tolerance, go to **14.**

b. If split spacer gaps are not within tolerance, repair/replace split spacers as required (WP 1100 00). Go to **31.**

14. Check helicopter configuration.

a. **UH-60A 79-23269 - SUBQ** , perform forward, aft, and lateral flight control channel inspection (WP 1030 00). Go to **31.**

b. **UH-60A 77-22714 - 79-23268** , go to **15.**

15. Inspect right side riveted rigging pin adapter located between pitch and roll bellcranks for looseness.

a. If riveted rigging pin adapter is loose, replace midsection bellcranks and shaft (WP 1100 00). Go to **31.**

NOISE IS HEARD WHILE MOVING PILOT CYCLIC STICK FROM LEFT TO RIGHT - Continued

b. If riveted rigging pin adapter is not loose, perform forward, aft, and lateral flight control channel inspection (WP 1030 00). Go to **31.**

16. Visually inspect lateral vertical torque shaft for loose or missing hardware.

a. If missing or loose hardware is present, repair/replace as required. Go to **31.**

b. If missing or loose hardware is not present, go to **17.**

17. Inspect lateral vertical torque shafts for excessive radial play in bearings.

a. If excessive radial play in bearings is present, replace vertical torque shafts as required (WP 1110 00). Go to **31.**

b. If excessive radial play in bearings is not present, go to **18.**

18. Inspect swaged insert on lateral vertical torque shafts for loose swaged inserts (Figure 3).

a. If vertical torque shafts swaged inserts are loose, replace swaged insert vertical torque shafts as required (WP 1110 00). Go to **31.**

b. If vertical torque shafts swaged inserts are not loose, go to **19.**

19. Inspect riveted type lateral vertical torque shafts for loose rod ends (Figure 3).

a. If riveted type vertical torque shafts rod ends are loose, replace riveted type vertical torque shafts as required (WP 1110 00). Go to **31.**

b. If riveted type vertical torque shafts rod ends are not loose, go to **20.**

20. Inspect lateral torque shaft and levers for loose or missing tapered pins (WP 1096 00).

a. If loose or missing tapered pins are present, replace tapered pins are required (WP 1098 00). Go to **31.**

b. If loose or missing tapered pins are not present, go to **21.**

21. Inspect airframe around roll trim actuator, and lateral vertical torque tubes for loose or missing rivets.

a. If rivets are working loose in airframe near torque shaft or roll trim actuator, replace rivets or repair airframe structure as required (WP 0378 00). Go to **31.**

b. If rivets are not working loose in airframe near torque shaft or roll trim actuator, perform forward, aft, and lateral flight control channel inspection (WP 1030 00). Go to **31.**

22. Visually inspect area aft of lateral vertical torque shafts for loose or missing hardware.

a. If missing or loose hardware is present, repair/replace as required. Go to **31.**

b. If missing or loose hardware is not present, go to **23.**

23. Inspect roll trim actuator assembly for excessive radial play in bearings.

a. If excessive radial play in bearings is present, replace roll trim assembly (WP 1087 00). Go to **31.**

b. If excessive radial play in bearings is not present, go to **24.**

24. Inspect riveted type pushrods for loose rod ends (Figure 3).

NOISE IS HEARD WHILE MOVING PILOT CYCLIC STICK FROM LEFT TO RIGHT - Continued

a. If riveted type pushrod ends are loose, replace riveted type pushrods as required (WP 1114 00 or WP 1117 00). Go to **31.**

b. If riveted type pushrod ends are not loose, go to **25.**

25. Visually inspect mixer assembly for loose or missing hardware (WP 1014 00).

a. If missing or loose hardware is present, repair/replace as required (WP 1089 00). Go to **31.**

b. If loose or missing hardware is not present, go to **26.**

26. Inspect mixer assembly bearing for excessive radial play (WP 1089 00).

a. If excessive radial play in bearings is present, repair mixer (WP 1089 00). Go to **31.**

b. If excessive radial play in bearings is not present, go to **27.**

27. Inspect mixer assembly area for loose or missing taper pins.

a. If loose or missing taper pins are present, repair/replace taper pins as required (WP 1098 00). Go to **31.**

b. If loose or missing taper pins are not present, perform forward, aft, and lateral flight control channel inspection (WP 1030 00). Go to **31.**

28. Visually inspect area aft of mixer assembly but forward of stationary swash plate for loose or missing hardware.

a. If loose or missing hardware is present, repair/replace as required. Go to **31.**

b. If loose or missing hardware is not present, go to **29.**

29. Inspect lateral swashplate link uniball bearings for excessive radial play and freedom of movement (WP 1018 00).

a. If excessive radial play or sticky bearings are present, replace lateral swashplate link (WP 1129 00). Go to **31.**

b. If excessive radial play or sticky bearings are not present, go to **30.**

30. Inspect the following components for worn journal bearings:

- **Lateral primary servo (WP 1091 00)**
- **Mixer to lateral primary servo control rod (WP 1122 00)**
- **Lateral control rod (WP 1127 00)**
- **Lateral bellcrank (WP 1136 00)**
- **Left and right tie rods (WP 1022 00)**

a. If worn journal bearings are present, replace lateral primary servo, mixer to lateral primary servo control rod, lateral control rod, lateral bellcrank, or left and right tie rod as required (WP 1091 00, WP 1127 00, WP 1122 00, WP 1136 00, or WP 1022 00). Go to **31.**

b. If worn journal bearings are not present, perform forward, aft, and lateral flight control channel inspection (WP 1030 00). Go to **31.**

31. Procedure completed.

NOISE IS HEARD WHILE MOVING PILOT CYCLIC STICK FROM FORWARD TO AFT

SYMPTOM

Noise is heard while moving pilot cyclic stick from forward to aft.

MALFUNCTION

Noise is heard while moving pilot cyclic stick from forward to aft.

CORRECTIVE ACTION

1. Is noise coming from forward of vertical torque shaft?

a. If noise is coming from forward of vertical torque shaft, go to **7.**

b. If noise is not coming from forward of vertical torque shaft, go to **2.**

2. Is noise coming from control shaft in upper broom closet?

a. If noise is coming from control shaft in upper broom closet, go to **10.**

b. If noise is not coming from control shaft in upper broom closet, go to **3.**

3. Is noise coming from vertical torque shaft?

a. If noise is coming from vertical torque shaft, go to **16.**

b. If noise is not coming from vertical torque shaft, go to **4.**

4. Is noise coming from aft of vertical torque shaft?

a. If noise is coming from aft of vertical torque shaft, go to **22.**

b. If noise is not coming from aft of vertical torque shaft, go to **5.**

5. Is noise coming from mixer assembly?

a. If noise is coming from mixer assembly, go to **25.**

b. If noise is not coming from mixer assembly, go to **6.**

6. Is noise coming from aft of mixer assembly but forward of stationary swashplate?

a. If noise is coming from aft of mixer assembly but forward of stationary swashplate, go to **28.**

b. If noise is not coming from aft of mixer assembly but forward of stationary swashplate perform forward, aft, and lateral flight control channel inspection (WP 1030 00). Go to **31.**

7. Visually inspect area forward of pitch vertical torque shafts for missing or loose mounting hardware.

a. If missing or loose mounting hardware is present, repair/replace as required. Go to **31.**

b. If missing or loose mounting hardware is not present, go to **8.**

8. Inspect the following components forward of the pitch vertical torque tubes for excessive radial play in bearings:

- **Pilot and copilot cyclic sticks (WP 1010 00)**
- **Cockpit forward and aft bellcrank (WP 1131 00)**
- **Cabin longitudinal bellcrank (WP 1132 00)**

a. If excessive radial play is present in cyclic stick, cockpit forward and aft bellcrank, or cabin

NOISE IS HEARD WHILE MOVING PILOT CYCLIC STICK FROM FORWARD TO AFT - Continued

longitudinal bellcrank bearings, repair/replace cyclic stick bearings, cockpit forward and aft bellcrank, or cabin longitudinal bellcrank as required (WP 1058 00, WP 1131 00, or WP 1132 00). Go to **31.**

b. If excessive radial play in bearings is not present, go to **9.**

9. Visually inspect pilot cyclic pitch control rods, copilot cyclic pitch control rods, and cabin pitch control rod forward of vertical torque shafts for loose swaged inserts (Figure 3).

a. If swaged inserts on control rods are loose, replace pilot's directional and cyclic control rods, copilot's directional and cyclic control rods, or cabin control rods as required (WP 1105 00, WP 1108 00, or WP 1110 00). Go to **31.**

b. If swaged inserts on control rods are not loose, perform forward, aft and lateral flight control inspection (WP 1030 00). Go to **31.**

10. Visually inspect pitch control rods leading to upper broom closet for loose or missing mounting hardware.

a. If loose or missing mounting hardware is present, repair/replace as required. Go to **31.**

b. If loose or missing mounting hardware is not present, go to **11.**

11. Inspect midsection bellcranks and shafts for excessive radial play in bearings (WP 1100 00).

a. If excessive radial play in bearings is present, repair/replace midsection bellcranks and shaft (WP 1100 00). Go to **31.**

b. If excessive radial play in bearings is not present, go to **12.**

12. Inspect pilot and copilot pitch control rods leading to upper broom closet for loose swaged inserts (Figure 3).

a. If pitch control rods swaged inserts are loose, repair/replace pitch control rods leading to upper broom closet are required (WP 1110 00). Go to **31.**

b. If pitch control rod swaged inserts are not loose, go to **13.**

13. Inspect midsection bellcranks split spacers for proper gap (WP 1100 00).

a. If split spacer gaps are within tolerance, go to **14.**

b. If split spacer gaps are not within tolerance, repair/replace split spacers as required (WP 1100 00). Go to **31.**

14. Check helicopter configuration.

a. **UH-60A 79-23269 - SUBQ** , perform forward, aft, and lateral flight control channel inspection (WP 1030 00). Go to **31.**

b. **UH-60A 77-22714 - 79-23268** , go to **15.**

15. Inspect right side riveted rigging pin adapter located between pitch and roll bellcranks for looseness.

a. If riveted rigging pin adapter is loose, replace midsection bellcranks and shaft (WP 1100 00). Go to **31.**

b. If riveted rigging pin adapter is not loose, perform forward, aft, and lateral flight control channel inspection (WP 1030 00). Go to **31.**

NOISE IS HEARD WHILE MOVING PILOT CYCLIC STICK FROM FORWARD TO AFT - Continued

16. **Visually inspect pitch vertical torque shafts for loose or missing hardware.**
    a. If missing or loose hardware is present, repair/replace as required. Go to **31.**
    b. If missing or loose hardware is not present, go to **17.**
17. **Inspect pitch vertical torque shafts for excessive radial play in bearings.**
    a. If excessive radial play in bearings is present, replace pitch vertical torque shafts as required (WP 1110 00). Go to **31.**
    b. If excessive radial play in bearings is not present, go to **18.**
18. **Inspect swaged inserts on pitch vertical torque shafts for loose swaged inserts (Figure 3).**
    a. If swaged inserts on pitch vertical torque shafts are loose, replace swaged inserts on pitch vertical torque shafts as required (WP 1110 00). Go to **31.**
    b. If swaged inserts on pitch vertical torque shafts are not loose, go to **19.**
19. **Inspect riveted type pitch vertical torque shafts for loose rod ends (Figure 3).**
    a. If riveted type pitch vertical torque shafts rod ends are loose, replace riveted type pitch vertical torque shafts as required (WP 1110 00). Go to **31.**
    b. If riveted type pitch vertical torque shafts rod ends are not loose, go to **20.**
20. **Inspect pitch torque shaft and levers for loose or missing tapered pins (WP 1095 00).**
    a. If loose or missing tapered pins are present, replace tapered pins are required (WP 1098 00). Go to **31.**
    b. If loose or missing tapered pins are not present, go to **21.**
21. **Inspect airframe around pitch trim actuator, and pitch vertical torque tubes for loose or missing rivets.**
    a. If rivets are working loose in airframe near torque shaft or pitch trim actuator, replace rivets or repair airframe structure as required (WP 0378 00). Go to **31.**
    b. If rivets are not working loose in airframe near torque shaft or pitch trim actuator, perform forward, aft, and lateral flight control channel inspection (WP 1030 00). Go to **31.**
22. **Visually inspect area aft of pitch vertical torque shafts for loose or missing hardware.**
    a. If missing or loose hardware is present, repair/replace as required. Go to **31.**
    b. If missing or loose hardware is not present, go to **23.**
23. **Inspect pitch trim actuator assembly for excessive radial play in bearings.**
    a. If excessive radial play in bearings is present, replace pitch trim assembly (WP 1083 00). Go to **31.**
    b. If excessive radial play in bearings is not present, go to **24.**
24. **Inspect riveted type pushrods for loose rod ends (Figure 3).**
    a. If riveted type pushrod ends are loose, replace riveted type pushrods as required (WP 1112 00 or WP 1116 00). Go to **31.**
    b. If riveted type pushrod ends are not loose, go to **25.**

NOISE IS HEARD WHILE MOVING PILOT CYCLIC STICK FROM FORWARD TO AFT - Continued

25. Visually inspect mixer assembly for loose or missing hardware (WP 1014 00).

a. If missing or loose hardware is present, repair/replace as required (WP 1089 00). Go to **31.**

b. If loose or missing hardware is not present, go to **26.**

26. Inspect mixer assembly bearing for excessive radial play (WP 1089 00).

a. If excessive radial play in bearings is present, repair mixer (WP 1089 00). Go to **31.**

b. If excessive radial play in bearings is not present, go to **27.**

27. Inspect mixer assembly area for loose or missing taper pins.

a. If loose or missing taper pins are present, repair/replace taper pins as required (WP 1098 00). Go to **31.**

b. If loose or missing taper pins are not present, perform forward, aft, and lateral flight control channel inspection (WP 1030 00). Go to **31.**

28. Visually inspect area aft of mixer assembly but forward of stationary swash plate for loose or missing hardware.

a. If loose or missing hardware is present, repair/replace as required. Go to **31.**

b. If loose or missing hardware is not present, go to **29.**

29. Inspect forward and aft swashplate link uniball bearings for excessive radial play and freedom of movement (WP 1018 00).

a. If excessive radial play or sticky bearings are present, replace forward and aft swashplate link as required (WP 1129 00). Go to **31.**

b. If excessive radial play or sticky bearings are not present, go to **30.**

30. Inspect the following components for worn journal bearings:

- **Forward and aft primary servos (WP 1091 00)**
- **Mixer to aft primary servo control rod**
- **Aft control rod (WP 1126 00)**
- **Long control rod (WP 1128 00)**
- **Aft bellcrank (WP 1137 00)**

a. If worn journal bearings are present, replace forward primary servo, aft primary servo, mixer to aft primary servo control rod, Aft control rod, long control rod, aft bellcrank as required (WP 1091 00, WP 1126 00, WP 1128 00, or WP 1137 00). Go to **31.**

b. If worn journal bearings are not present, perform forward, aft, and lateral flight control channel inspection (WP 1030 00) Go to **31.**

31. Procedure completed.

NOISE IS HEARD WHILE MOVING COLLECTIVE STICK UP AND DOWN

SYMPTOM

Noise is heard while moving collective stick up and down.

MALFUNCTION

Noise is heard while moving collective stick up and down.

CORRECTIVE ACTION

1. Is noise coming from forward of vertical torque shaft?

a. If noise is coming from forward of vertical torque shaft, go to **7.**

b. If noise is not coming from forward of vertical torque shaft, go to **2.**

2. Is noise coming from collective control shaft in upper broom closet?

a. If noise is coming from control shaft in upper broom closet, go to **10.**

b. If noise is not coming from control shaft in upper broom closet, go to **3.**

3. Is noise coming from vertical torque shaft?

a. If noise is coming from vertical torque shaft, go to **17.**

b. If noise is not coming from vertical torque shaft, go to **4.**

4. Is noise coming from aft of vertical torque shaft?

a. If noise is coming from aft of vertical torque shaft, go to **23.**

b. If noise is not coming from aft of vertical torque shaft, go to **5.**

5. Is noise coming from mixer assembly?

a. If noise is coming from mixer assembly, go to **26.**

b. If noise is not coming from mixer assembly, go to **6.**

6. Is noise coming from aft of mixer assembly but forward of stationary swashplate?

a. If noise is coming from aft of mixer assembly but forward of stationary swashplate, go to **29.**

b. If noise is not coming from aft of mixer assembly but forward of stationary swashplate, perform bearing friction wear inspection (WP 1008 00). Go to **32.**

7. Visually inspect area forward of collective vertical torque shafts for missing or loose mounting hardware.

a. If missing or loose mounting hardware is present, repair/replace as required. Go to **32.**

b. If missing or loose mounting hardware is not present, go to **8.**

8. Inspect the following components forward of the collective vertical torque tubes for excessive radial play in bearings:

- **Pilot and copilot collective sticks (WP 1011 00)**
- **Cabin collective bellcrank (WP 1132 00)**

a. If excessive radial play is present in collective sticks, or cabin collective bellcrank bearings, repair/replace collective stick bearings, or cabin collective bellcrank as required (WP 1058 00 or WP 1131 00). Go to **32.**

NOISE IS HEARD WHILE MOVING COLLECTIVE STICK UP AND DOWN - Continued

b. If excessive radial play in bearings is not present, go to **9.**

9. **Visually inspect pilot collective control rods, copilot collective control rods, and cabin collective control rod forward of vertical torque shafts for loose swaged inserts (Figure 3).**

   a. If swaged inserts on control rods are loose, replace pilot's collective control rods, copilot's collective control rods, or cabin collective control rods as required (WP 1106 00, WP 1109 00, or WP 1110 00). Go to **32.**

   b. If swaged inserts on control rods are not loose, perform bearing friction wear inspection (WP 1008 00). Go to **32.**

10. **Visually inspect collective control rods leading to upper broom closet for loose or missing mounting hardware.**

    a. If loose or missing mounting hardware is present, repair/replace as required. Go to **32.**

    b. If loose or missing mounting hardware is not present, go to **11.**

NOTE

The popping sound generated between the shaft and forward attachment fitting is considered benign and acceptable.

11. **Check for a small amount of motion between the shaft and forward attachment fitting by cycling high collective left pedal to low collective right pedal.**

    a. If a large amount of motion is present, replace collective torque shafts (WP 1094 00). Go to **32.**

    b. If a large amount of motion is not present, go to **12.**

12. **Inspect midsection bellcranks and shafts for excessive radial play in bearings (WP 1100 00).**

    a. If excessive radial play in bearings is present, repair/replace midsection bellcranks and shaft (WP 1100 00). Go to **32.**

    b. If excessive radial play in bearings is not present, go to **13.**

13. **Inspect pilot and copilot collective control rods leading to upper broom closet for loose swaged inserts (Figure 3).**

    a. If collective control rods swaged inserts are loose, repair/replace collective control rods leading to upper broom closet as required (WP 1110 00). Go to **32.**

    b. If collective control rod swaged inserts are not loose, go to **14.**

14. **Inspect midsection bellcranks split spacers for proper gap (WP 1100 00).**

    a. If split spacer gap is within tolerance, go to **15.**

    b. If split spacer gaps are not within tolerance, repair/replace split spacers as required (WP 1100 00). Go to **32.**

15. **Check helicopter configuration.**

    a. **UH-60A 79-23269 - SUBQ** , perform bearing friction wear inspection (WP 1008 00). Go to **32.**

    b. **UH-60A 77-22714 - 79-23268** , go to **16.**

NOISE IS HEARD WHILE MOVING COLLECTIVE STICK UP AND DOWN - Continued

16. **Inspect right side riveted rigging pin adapter located between pitch and roll bellcranks for looseness.**

    a. If riveted rigging pin adapter is loose, replace midsection bellcranks and shaft (WP 1100 00). Go to **32.**

    b. If riveted rigging pin adapter is not loose, perform bearing friction wear inspection (WP 1008 00). Go to **32.**

17. **Visually inspect collective vertical torque shafts for loose or missing hardware.**

    a. If missing or loose hardware is present, repair/replace as required. Go to **32.**

    b. If missing or loose hardware is not present, go to **18.**

18. **Inspect collective vertical torque shafts for excessive radial play in bearings.**

    a. If excessive radial play in bearings is present, replace collective vertical torque shafts as required (WP 1097 00). Go to **32.**

    b. If excessive radial play in bearings is not present, go to **19.**

19. **Inspect swaged insert on collective vertical torque shafts for loose swaged inserts (Figure 3).**

    a. If swaged inserts on collective vertical torque shafts are loose, replace swaged insert collective vertical torque shafts as required (WP 1110 00. Go to **32.**

    b. If swaged inserts on collective vertical torque shafts are not loose, go to **20.**

20. **Inspect riveted type collective vertical torque shafts for loose rod ends (Figure 3).**

    a. If riveted type collective vertical torque shafts rod ends are loose, replace riveted type collective vertical torque shafts as required (WP 1110 00). Go to **32.**

    b. If riveted type collective vertical torque shafts rod ends are not loose, go to **21.**

21. **Inspect collective torque shaft and levers for loose or missing tapered pins (WP 1094 00).**

    a. If loose or missing tapered pins are present, replace tapered pins are required (WP 1098 00). Go to **32.**

    b. If loose or missing tapered pins are not present, go to **22.**

22. **Inspect airframe around collective vertical torque tubes for loose or missing rivets.**

    a. If rivets are working loose in airframe near torque shaft, replace rivets or repair airframe structure as required (WP 0378 00). Go to **32.**

    b. If rivets are not working loose in airframe near torque shaft, perform bearing friction wear inspection (WP 1008 00). Go to **32.**

23. **Visually inspect area aft of collective vertical torque shafts for loose or missing hardware.**

    a. If missing or loose hardware is present, repair/replace as required. Go to **32.**

    b. If missing or loose hardware is not present, go to **24.**

24. **Inspect collective boost assembly for excessive radial play in bearings.**

    a. If excessive radial play in bearings is present, replace collective boost assembly (WP 1080 00). Go to **32.**

NOISE IS HEARD WHILE MOVING COLLECTIVE STICK UP AND DOWN - Continued

b. If excessive radial play in bearings is not present, go to **25.**

25. Inspect riveted type pushrods for loose rod ends (Figure 3).

a. If riveted type pushrod ends are loose, replace riveted type pushrods as required (WP 1112 00 or WP 1116 00). Go to **32.**

b. If riveted type pushrod ends are not loose, go to **26.**

26. Visually inspect mixer assembly for loose or missing hardware (WP 1014 00).

a. If missing or loose hardware is present, repair/replace as required (WP 1089 00). Go to **32.**

b. If loose or missing hardware is not present, go to **27.**

27. Inspect mixer assembly bearing for excessive radial play (WP 1089 00).

a. If excessive radial play in bearings is present, repair mixer (WP 1089 00). Go to **32.**

b. If excessive radial play in bearings is not present, go to **28.**

28. Inspect mixer assembly area for loose or missing taper pins.

a. If loose or missing taper pins are present, repair/replace taper pins as required (WP 1098 00). Go to **32.**

b. If loose or missing taper pins are not present, perform bearing friction wear inspection (WP 1008 00). Go to **32.**

29. Visually inspect area aft of mixer assembly but forward of stationary swash plate for loose or missing hardware.

a. If loose or missing hardware is present, repair/replace as required. Go to **32.**

b. If loose or missing hardware is not present, go to **30.**

30. Inspect forward and aft swashplate link uniball bearings for excessive radial play and freedom of movement (WP 1018 00).

a. If excessive radial play or sticky bearings are present, replace forward and aft swashplate link as required (WP 1129 00). Go to **32.**

b. If excessive radial play or sticky bearings are not present, go to **31.**

31. Inspect the following components for worn journal bearings:

- **Forward primary servo (WP 1091 00)**
- **Mixer to forward primary servo control rod (WP 1120 00)**
- **Forward control rod (WP 1125 00)**
- **Forward bellcrank (WP 1135 00)**

a. If worn journal bearings are present, replace forward primary servo, mixer to forward primary servo control rod, forward control rod, forward bellcrank as required (WP 1091 00, WP 1120 00, WP 1125 00, or WP 1135 00). Go to **32.**

b. If worn journal bearings are not present, perform bearing friction wear inspection (WP 1008 00). Go to **32.**

32. Procedure completed.

NOISE IS HEARD WHILE PUSHING ON LEFT AND RIGHT PEDAL

SYMPTOM

Noise is heard while pushing on left and right pedal.

MALFUNCTION

Noise is heard while pushing on left and right pedal.

CORRECTIVE ACTION

1. Is noise coming from vicinity of pedals?

a. If noise in the vicinity of pedals, go to **8.**

b. If noise is not coming from vicinity of pedals, go to **2.**

2. Is noise between pedals and vertical torque shafts

a. If noise is coming from between pedals and torque shafts, go to **14.**

b. If noise is not coming from between pedals and torque shafts, go to **3.**

3. Is noise coming from control shafts in upper broom closet.

a. If noise is coming from control shafts in upper broom closet, go to **17.**

b. If noise is not coming from control shafts in upper broom closet, go to **4.**

4. Is noise coming from vertical torque shaft?

a. If noise is coming from vertical torque shaft, go to **25.**

b. If noise is not coming from vertical torque shaft, go to **5.**

5. Is noise coming from aft of vertical torque shafts?

a. If noise is aft of vertical torque shafts, go to **31.**

b. If noise is not coming from aft of vertical torque shafts, go to **6.**

6. Is noise coming from mixer assembly?

a. If noise is coming from mixer assembly, go to **34.**

b. If noise is not coming from mixer assembly, go to **7.**

7. Is noise coming from mixer assembly and forward quadrant.

a. If noise is coming from mixer assembly and forward quadrant, go to **37.**

b. If noise is not coming from mixer assembly and forward quadrant, perform bearing friction wear inspection (WP 1008 00). Go to **40.**

8. Visually inspect pilot and copilot pedal adjusters for missing or loose mounting hardware.

a. If missing or loose mounting hardware is present, repair/replace as required. Go to **40.**

b. If missing or loose mounting hardware is not present, go to **9.**

9. Inspect pilot and copilot pedal adjuster bearings for excessive radial play (WP 1047 00).

a. If excessive radial play is present in pilot or copilot pedal adjusters bearings, repair/replace as required (WP 1058 00). Go to **40.**

NOISE IS HEARD WHILE PUSHING ON LEFT AND RIGHT PEDAL - Continued

b. If excessive radial play is not present in pilot or copilot pedal adjuster bearings, go to **10.**

10. Inspect pilot and copilot pedal and adjuster control rods for loose swaged inserts (Figure 3).

a. If swaged inserts on pilot or copilot pedal and adjuster control rods are loose, replace control rods as required (WP 1105 00). Go to **40.**

b. If swaged inserts on pilot or copilot pedal and adjuster control rods are not loose, go to **11.**

11. Inspect pilot and copilot pedal and adjuster control rods for elongation of tapered pin holes on housing.

a. If tapered pin holes are elongated, replace pedal adjuster assembly (WP 1104 00 or WP 1107 00). Go to **40.**

b. If tapered pin holes are not elongated, go to **12.**

12. Inspect pilot and copilot pedal adjuster taper pins for security.

a. If loose tapered pins are present, repair/replace tapered pins as required (WP 1098 00). Go to **40.**

b. If loose tapered pins are not present, go to **13.**

13. Inspect pilot and copilot pedal adjusters for proper shimming.

a. If pilot and copilot pedal adjuster are properly shimmed, preform bearing friction wear inspection (WP 1008 00). Go to **40.**

b. If pilot or copilot pedal adjusters are not properly shimmed, repair/replace shims as required (WP 1058 00). Go to **40.**

14. Visually inspect area between pilot and copilot pedals and upper broom closet for missing or loose hardware.

a. If missing or loose hardware is present, repair/replace hardware as required. Go to **40.**

b. If missing or loose hardware is not present, go to **15.**

15. Inspect the following components in pilot and copilot yaw channel for excessive radial play in bearings:

- **Yaw lever (WP 1131 00)**
- **Yaw bellcrank (WP 1132 00)**

a. If pilot yaw lever, pilot yaw bellcrank, copilot yaw lever, or copilot yaw bellcrank bearings have excessive radial play, repair/replace component as required (WP 1131 00, WP 1132 00). Go to **40.**

b. If pilot yaw lever, pilot yaw bellcrank, copilot yaw lever, or copilot yaw bellcrank bearings do not have excessive radial play, go to **16.**

16. Inspect pilot and copilot yaw directional control rods in cockpit for loose swaged inserts (Figure 3).

a. If pilot or copilot yaw directional control rods, in cockpit, swaged inserts are loose, repair/replace control rods as required (WP 1110 00). Go to **40.**

b. If control rod swaged inserts are not loose, perform bearing friction wear inspection (WP 1008 00). Go to **40.**

17. Visually inspect upper broom closet for loose or missing hardware.

NOISE IS HEARD WHILE PUSHING ON LEFT AND RIGHT PEDAL - Continued

a. If missing or loose hardware is present, repair/replace hardware as required. Go to **40.**

b. If missing or loose hardware is not present, go to **18.**

NOTE

The popping sound generated between the bellcrank shaft and forward attachment fitting is considered benign and acceptable.

18. Check for a small amount of motion between the bellcrank shaft and forward attachment fitting by cycling the tail rotor pedals.

a. If a large amount of motion is present, replace yaw torque shaft (WP 1097 00). Go to **40.**

b. If a large amount of motion is not present, go to **19.**

19. With upper console BACKUP HYD PUMP switch OFF, check midsection bellcranks and shafts for excessive radial play in bearings (WP 1100 00).

a. If excessive radial play in bearings is present, repair/replace midsection bellcranks and shaft (WP 1100 00). Go to **40.**

b. If excessive radial play in bearings is not present, go to **20.**

20. Inspect pilot and copilot yaw control rods leading to upper broom closet for loose swaged inserts (Figure 3).

a. If yaw control rods swaged inserts are loose, repair/replace control rods leading to upper broom closet as required (WP 1110 00). Go to **40.**

b. If yaw control rod swaged inserts are not loose, go to **21.**

21. Inspect midsection bellcranks split spacers for proper gap (WP 1100 00).

a. If split spacer gaps are within tolerance, go to **22.**

b. If split spacer gaps are not within tolerance, repair/replace split spacers as required (WP 1100 00). Go to **40.**

22. Inspect airframe around bellcrank attachment fittings for loose or missing rivets.

a. If rivets are working loose in airframe near bellcrank attachment fittings, repair airframe structure as required (WP 0378 00). Go to **40.**

b. If rivets are not working loose in airframe near bellcrank attachment fittings, go to **23.**

23. Check helicopter configuration.

a. **UH-60A 79-23269 - SUBQ** , perform bearing friction wear inspection (WP 1008 00). Go to **40.**

b. **UH-60A 77-22714 - 79-23268** , go to **24.**

24. Inspect right side riveted rigging pin adapter located between pitch and roll bellcranks for looseness.

a. If riveted rigging pin adapter is loose, replace midsection bellcranks and shaft (WP 1100 00). Go to **40.**

b. If riveted rigging pin adapter is not loose, perform bearing friction wear inspection (WP 1008 00). Go to **40.**

25. Visually inspect pilot and copilot yaw vertical torque shafts for loose or missing hardware.

NOISE IS HEARD WHILE PUSHING ON LEFT AND RIGHT PEDAL - Continued

a. If missing or loose hardware is present, repair/replace as required. Go to **40.**

b. If missing or loose hardware is not present, go to **26.**

26. With upper console BACKUP HYD PUMP switch OFF, check yaw torque shaft and levers for excessive radial play in bearings (WP 1097 00).

a. If excessive radial play in bearings is present, replace yaw torque shaft and levers (WP 1097 00). Go to **40.**

b. If excessive radial play in bearings is not present, go to **27.**

27. Inspect swaged inserts on yaw vertical torque shafts for loose swaged inserts (Figure 3).

a. If swaged inserts on yaw vertical torque shafts are loose, replace yaw vertical torque shafts as required (WP 1110 00). Go to **40.**

b. If swaged inserts on yaw vertical torque shafts are not loose, go to **28.**

28. Inspect riveted type yaw vertical torque shafts for loose rod ends (Figure 3).

a. If riveted type yaw vertical torque shafts rod ends are loose, replace riveted type yaw vertical torque shafts as required (WP 1110 00). Go to **40.**

b. If riveted type collective vertical torque shafts rod ends are not loose, go to **29.**

29. Inspect yaw torque shaft and levers for loose or missing tapered pins (WP 1097 00).

a. If loose or missing tapered pins are present, replace tapered pins are required (WP 1098 00). Go to **40.**

b. If loose or missing tapered pins are not present, go to **30.**

30. Inspect airframe around yaw trim actuator, and yaw vertical torque tubes for loose or missing rivets.

a. If rivets are working loose in airframe near torque tubes or yaw trim actuator, replace rivets or repair airframe structure as required (WP 0378 00). Go to **40.**

b. If rivets are not working loose in airframe near torque tubes or yaw trim actuator, perform bearing friction wear inspection (WP 1008 00). Go to **40.**

31. Visually inspect area aft of yaw vertical torque shafts for loose or missing hardware.

a. If missing or loose hardware is present, repair/replace as required. Go to **40.**

b. If missing or loose hardware is not present, go to **32.**

32. Inspect yaw boost assembly for excessive radial play in bearings.

a. If excessive radial play in bearings is present, replace yaw boost assembly (WP 1080 00). Go to **40.**

b. If excessive radial play in bearings is not present, go to **33.**

33. Inspect riveted type pushrods for loose rod ends (Figure 3).

a. If riveted type pushrod ends are loose, replace riveted type pushrods as required (WP 1112 00 or WP 1116 00). Go to **40.**

NOISE IS HEARD WHILE PUSHING ON LEFT AND RIGHT PEDAL - Continued

b. If riveted type pushrod ends are not loose, perform bearing friction wear inspection (WP 1008 00). Go to **40.**

34. Visually inspect mixer assembly for loose or missing hardware.

a. If missing or loose hardware is present, repair/replace as required (WP 1089 00). Go to **40.**

b. If loose or missing hardware is not present, go to **35.**

35. Inspect mixer assembly bearing for excessive radial play.

a. If excessive radial play in bearings is present, repair mixer (WP 1089 00). Go to **40.**

b. If excessive radial play in bearings is not present, go to **36.**

36. Inspect mixer assembly area for loose or missing taper pins.

a. If loose or missing taper pins are present, repair/replace taper pins as required (WP 1098 00). Go to **40.**

b. If loose or missing taper pins are not present, perform bearing friction wear inspection (WP 1008 00). Go to **40.**

37. Visually inspect area between mixer and forward quadrant for loose or missing hardware.

a. If loose or missing hardware is present, repair/replace as required. Go to **40.**

b. If loose or missing hardware is not present, go to **38.**

38. Inspect forward quadrant and support bearings for excessive radial play.

a. If excessive radial play in bearings is present, replace forward quadrant and supports, as required (WP 1102 00). Go to **40.**

b. If excessive radial play in bearings is not present, go to **39.**

39. Inspect the following components for loose rod ends:

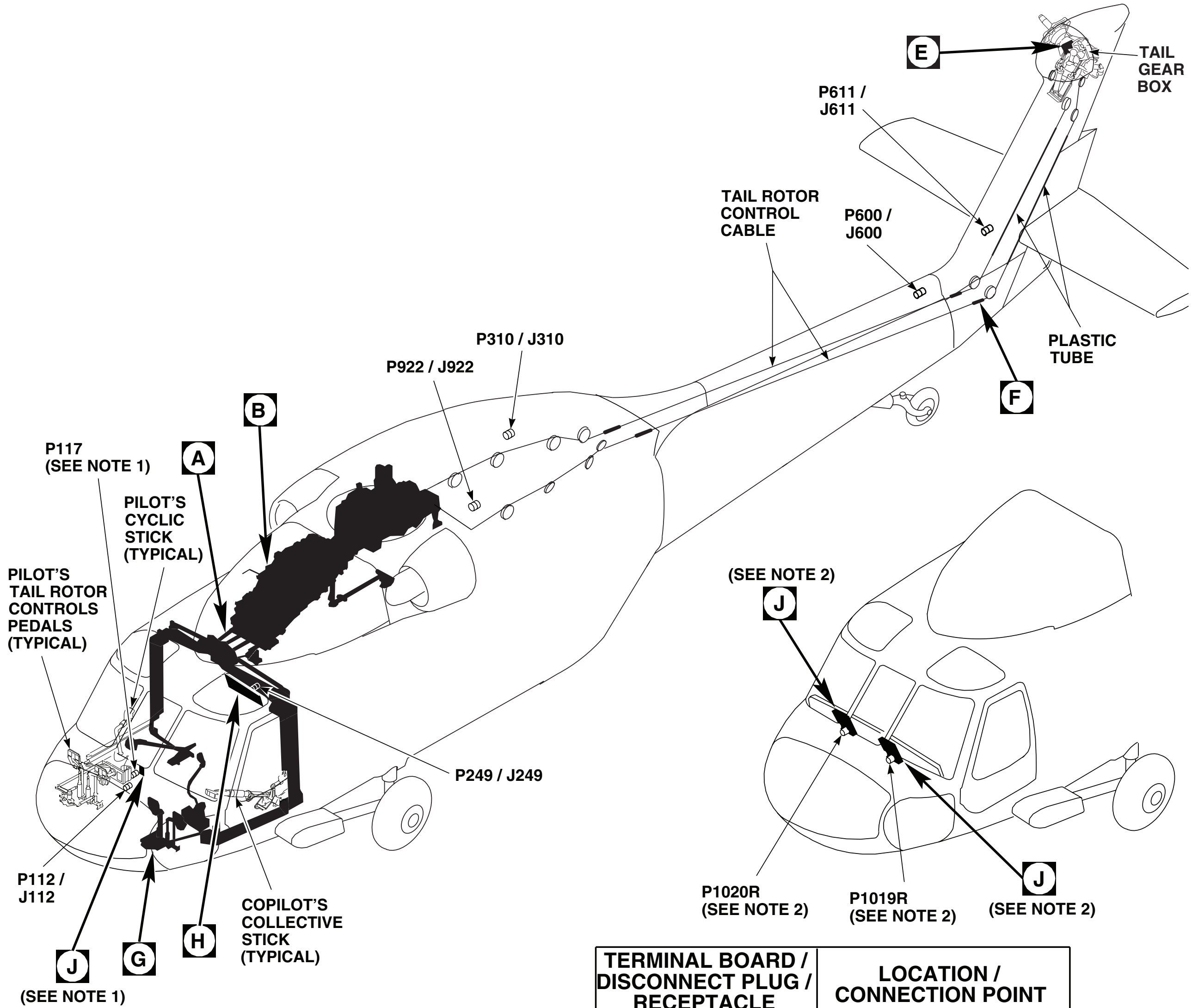
- **Mixer to yaw lever pushrod (Figure 3)**
- **Yaw lever to forward quadrant control rod**

a. If loose control rod ends are present, replace control rod as required (WP 1123 00, WP 1124 00). Go to **40.**

b. If loose rod ends are not present, perform bearing friction wear inspection (WP 1008 00). Go to **40.**

40. Procedure completed.

LOCATION



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P112 / J112	COCKPIT BL 7.5 LH, STA 197
P117 / J117	CAUTION / ADVISORY PANEL (SEE NOTE 1)
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P310 / J310	CABIN CEILING, BL 17 RH, STA 387
P600 / J600	TAIL CONE, BL 0, STA 650
P611 / J611	TAIL RTR QUAD BL 0, WL P100, STA P215
P922 / J922	CABIN CEILING BL 10.7 LH, STA 380
P1019R / J3	COPILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)
P1020R / J3	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)

NOTES

1. UH–60A UH–60L EH–60A
2. HH–60A HH–60L

AA3403_1B
SA

Figure 1. Flight Controls (Mechanical) System Location Diagram. (Sheet 1 of 5)

LOCATION - Continued

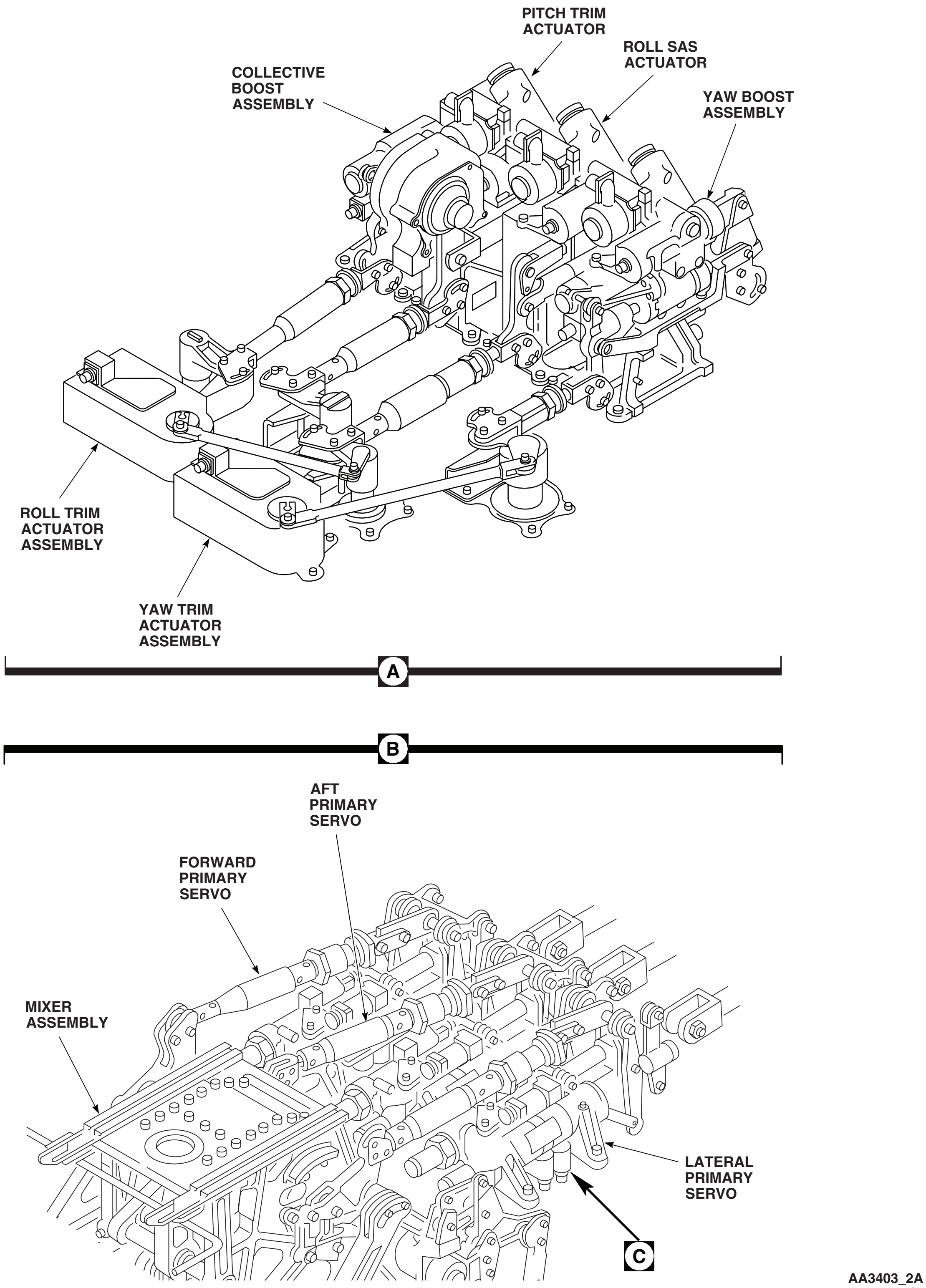


Figure 1. Flight Controls (Mechanical) System Location Diagram. (Sheet 2 of 5)

LOCATION - Continued

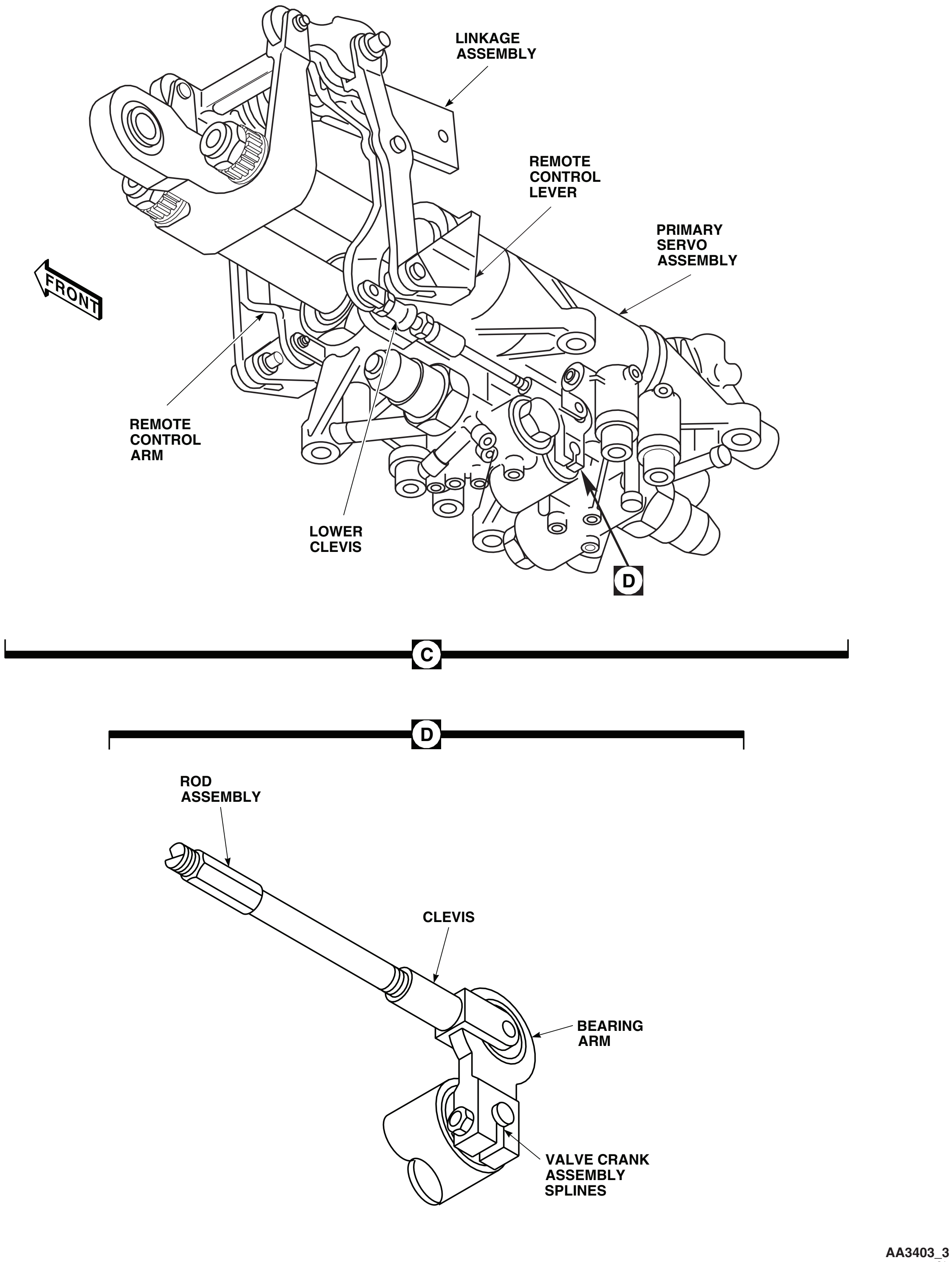


Figure 1. Flight Controls (Mechanical) System Location Diagram. (Sheet 3 of 5)

LOCATION - Continued

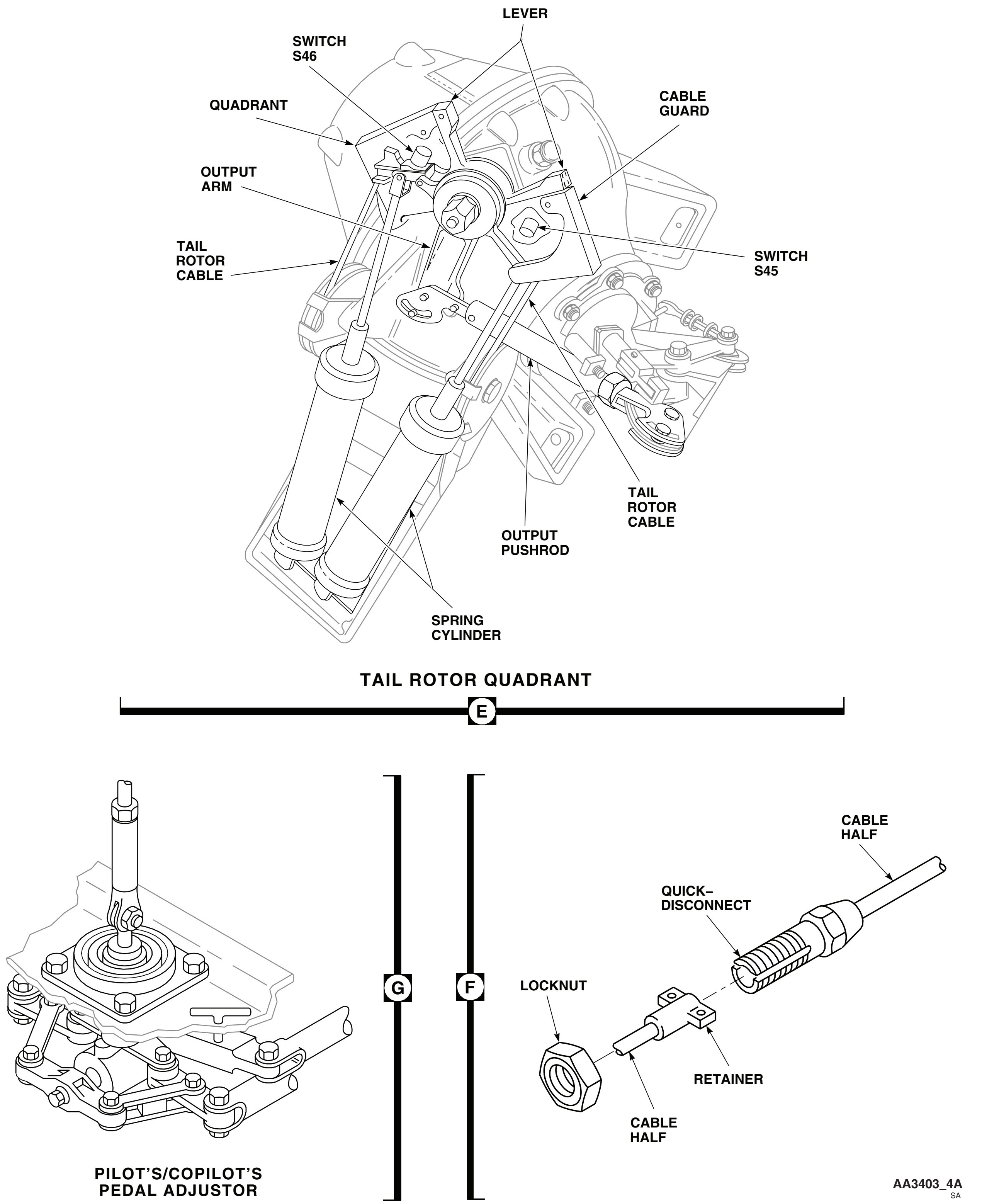


Figure 1. Flight Controls (Mechanical) System Location Diagram. (Sheet 4 of 5)

LOCATION - Continued

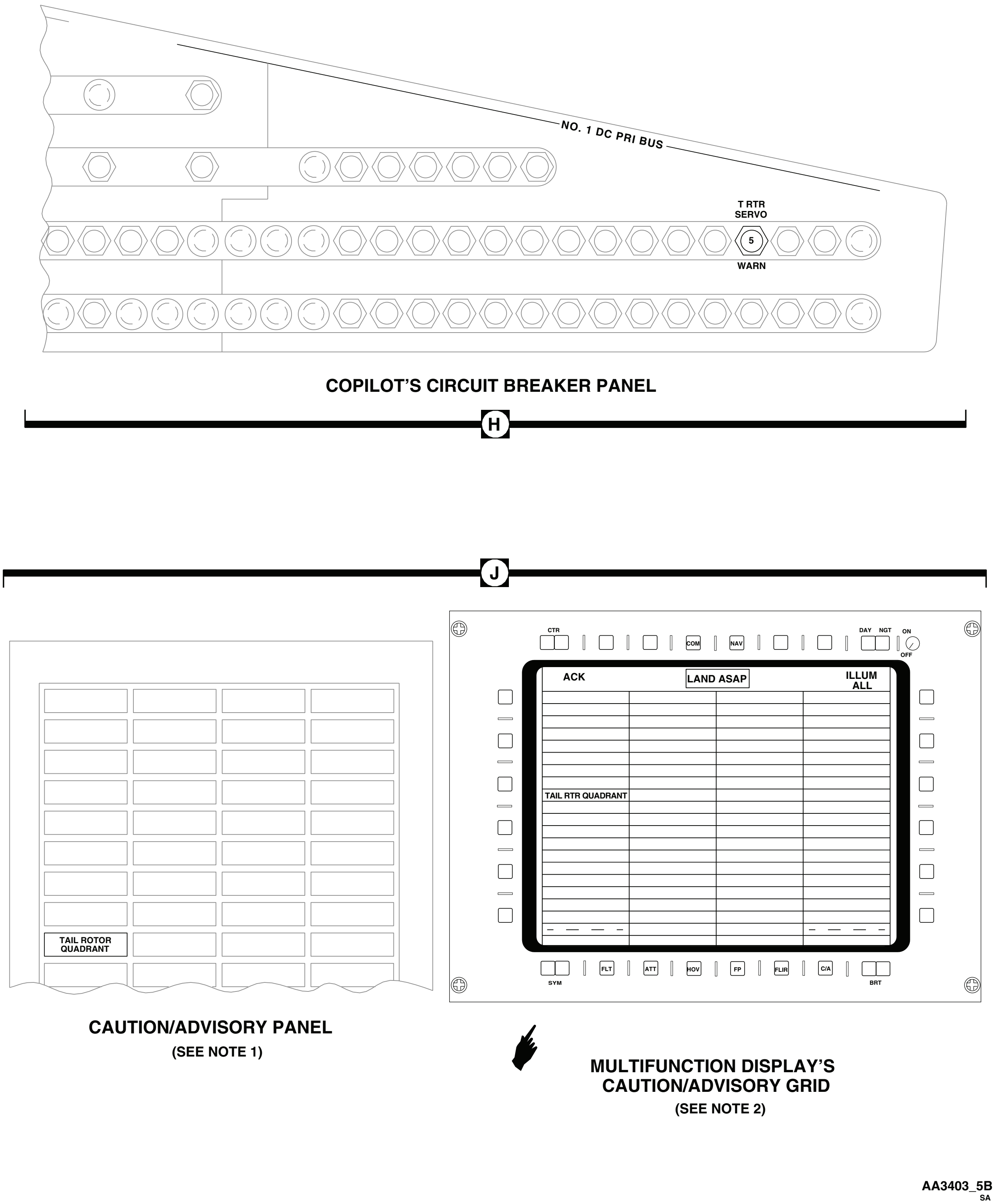
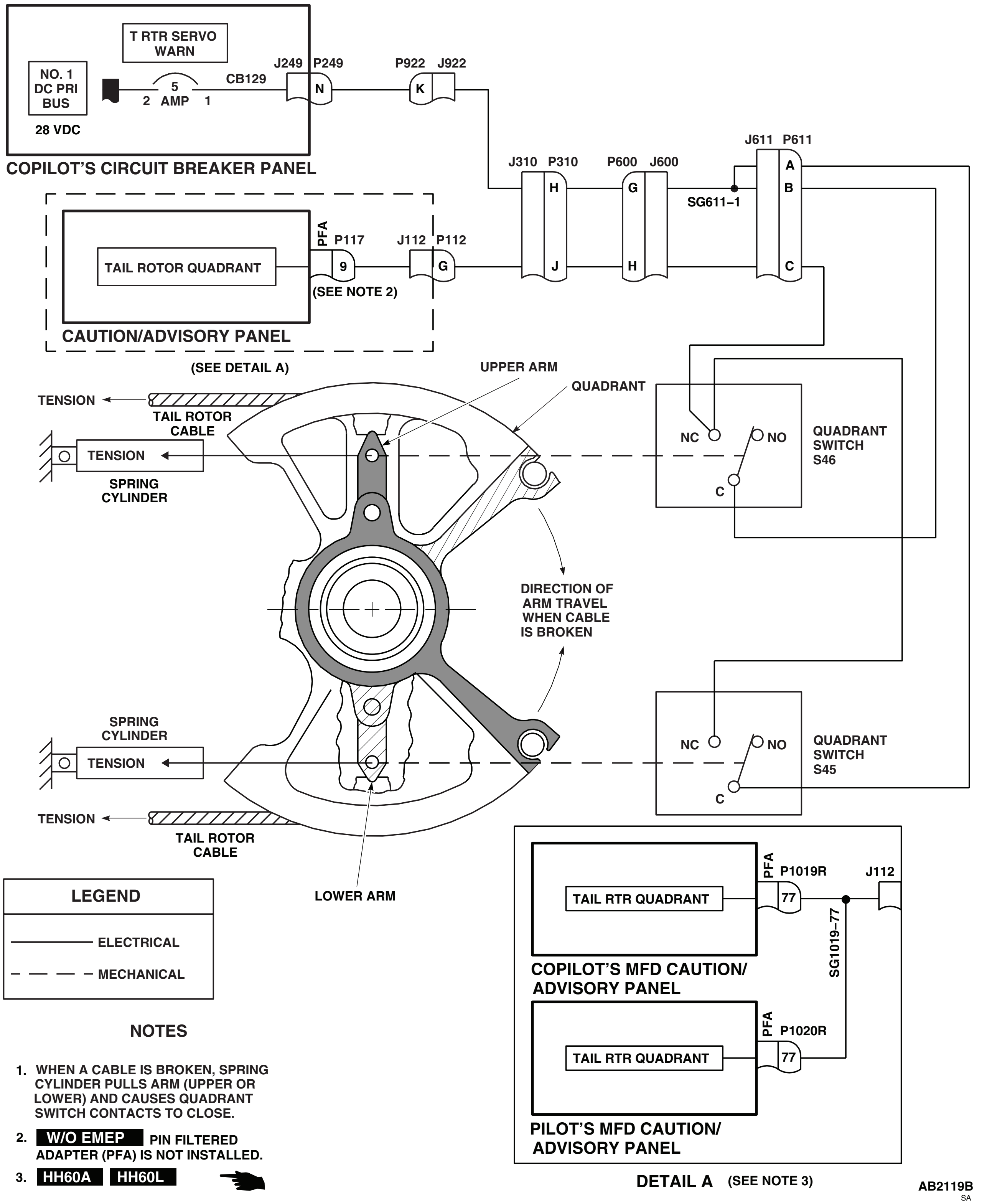


Figure 1. Flight Controls (Mechanical) System Location Diagram. (Sheet 5 of 5)

SCHEMATICS

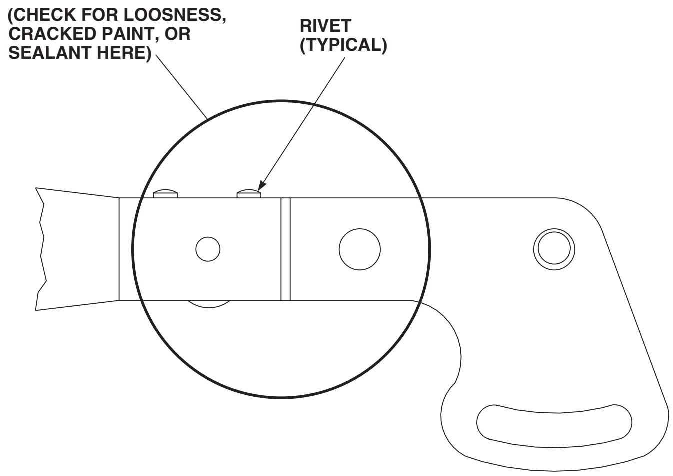


NOTES

1. WHEN A CABLE IS BROKEN, SPRING CYLINDER PULLS ARM (UPPER OR LOWER) AND CAUSES QUADRANT SWITCH CONTACTS TO CLOSE.
2. W/O EMEP PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.
3. HH60A HH60L

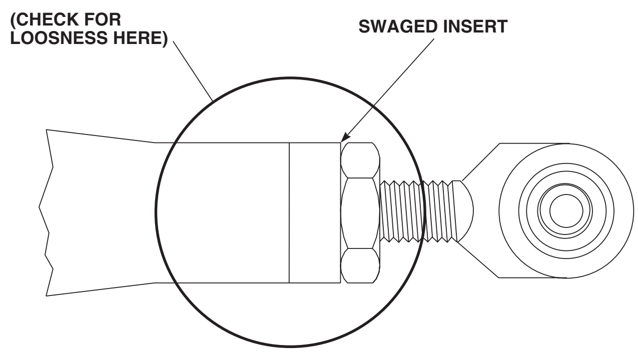
Figure 2. Tail Rotor Quadrant Warning System Schematic Diagram.

SCHEMATICS - Continued



RIVETED TYPE PUSHROD

(SEE NOTE 1)



SWAGED INSERT TYPE PUSHROD

(SEE NOTE 2)

NOTES

1. USUALLY FOUND IN THE YAW SYSTEM, BETWEEN THE YAW BOOST SERVO AND THE YAW SYSTEM STOP IN THE CABIN OVERHEAD.
2. USUALLY FOUND IN THE YAW SYSTEM, MOST COMMONLY THE FOUR RODS CONNECTING THE PEDALS TO THE PEDAL ADJUSTORS.

AB1298
SA

Figure 3. Control Rod Inspection Diagram.

END OF WORK PACKAGE

UNIT LEVEL

FLIGHT CONTROLS

ROLL SAS ASSEMBLY (AVIM)

INITIAL SETUP:

Test Equipment
Dial Indicator, A-A-2348
Nulling Fixture Assembly, SAS Actuator, 70700-20675-041

Tools and Special Tools
Hydraulic Repairer Toolkit, SC 5180-99-CL-A03
Magnet, GGG-F-0036
Torque Wrench, 0 - 30 in. lbs., A-A-2411
Torque Wrench, 30 - 150 in. lbs., GGG-W-686

Material/Parts
Wire, Nonelectrical (Lockwire), Item 355, WP 1803 00

Personnel Required
Aircraft Pneudraulics Repairer MOS 15H (1)

References
TM 1-1500-204-23
WP 1166 00
WP 1167 00
WP 1803 00

Equipment Condition
Hydraulic (3000 psi) Bench Power Available
28 vdc and 115 vac, Single-Phase, 60 Hz Bench Power Available

ROLL SAS ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

To prevent leakage between the nulling fixture and servo, make sure manifold and servo self-sealing couplings are clean.

NOTE

- If a servovalve or actuator has to be replaced, repeat operational/troubleshooting procedure.
- See Figure 1 for Roll SAS Servo Assembly diagram and Figure 2 for test setup diagram as an aid in operational/troubleshooting.

1. Position roll assembly on nulling fixture base. Carefully slide roll assembly toward manifold and engage self-sealing couplings.
2. Install three bolts securing roll assembly to base. Tighten bolts.
3. Place all test box pilot-assist/nulling switches to OFF and place TRIM COIL SELECT switch to BOTH.

ROLL SAS ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CAUTION

When using the 70700-20682-042 cable, make sure that the connector marked SAS ACTUATOR is correctly connected. If the incorrect connector is used, servovalve may become damaged.

4. Connect test box cables to pilot-assist module and roll assembly on nulling fixture as follows:

CABLE CONNECTOR	CONNECT TO
J466R	Pilot-assist module (Trim turn-on valve)
J467R	Pilot-assist module (SAS pressure switch)
J468R	Pilot-assist module (SAS shutoff valve)
J469R	Pilot-assist module (Boost shutoff valve)
J3	Roll assembly (Servovalve) (P/N 70700-20682-042 cable assembly)

5. Connect test box power cables to electrical power source.
6. Connect hydraulic lines from hydraulic power source to PRESSURE and RETURN ports on nulling fixture.
7. Turn on hydraulic power source and set pressure for 3000 psi at a maximum of 6 gpm.
8. Place test box AC and DC POWER switches to ON.

Indication/Condition

POWER ON indicators shall light and SAS PRESS ON lights shall go on.

Corrective Action

If Indication/Condition is not as specified, make sure test box AC and DC circuit breakers are pushed in.

Step

9. Place test box SAS switch to ON and then to OFF.

Indication/Condition

SAS PRESS ON and OFF lights shall cycle from on to off.

Corrective Action

If Indication/Condition is not as specified, replace test box.

Step

10. Repeat step 9. several times and leave switch at ON (SAS PRESS ON light on).
11. Vary test box CURRENT CONTROL until VALVE CURRENT meter indicates 0.
12. Vary test box CURRENT CONTROL from 0 to 4 mA, back to 0, then to -4 mA, and back to 0 several times, to stroke SAS piston. Return CURRENT CONTROL to 0.

ROLL SAS ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CAUTION

If bleed screw is not backed out properly, packing on bleed screw will be damaged.

13. Decrease hydraulic pressure to 500 psi. Back out bleed screw until bleed hole is visible and a good flow is established.
14. Allow fluid to flow from valve until fluid is free of air. Decrease pressure to 0 psi. Close bleed valve. **TORQUE TO 19 - 21 INCH-POUNDS** and lockwire (Item 355, WP 1803 00) (Figure 1., TM 1-1500-204-23).
15. Increase hydraulic pressure to 3000 psi and watch SAS assembly output piston for any movement.

Indication/Condition

Piston shall not move.

Corrective Action

If Indication/Condition is not as specified, there is still air trapped in servo. Repeat steps 8. through 14.

Step

16. Install dial indicator against SAS actuator link. Zero indicator.
17. Cycle test box SAS switch several times from ON to OFF. Watch actuator piston for any jump or movement.

Indication/Condition

Piston jump shall be below 0.005 inches to be nulled.

Corrective Action

NOTE

- Repeat piston jump check after torquing jamnut in substep (2). Readjust eccentric screw if necessary.
- Substep 1 verifies if SAS output rod moves properly when a specific current is applied to servovalve.

1. If Indication/Condition is not as specified, do the following:
   a. Remove cover from inside of jamnut using magnet.
   b. Loosen jamnut 1/4 turn.
   c. Insert screwdriver through jamnut and turn eccentric screw as needed to reduce piston jump below 0.005 inch. Cycle SAS switch to check for jump.
      (1) If piston jump is out of limits, remove servovalve (4 screws) and use 1/16-inch brass rod to center SAS actuator housing before installing servovalve with 4 screws. Repeat operational/troubleshooting procedure.
      (2) If piston jump is below 0.005 inch, **TORQUE JAMNUT 47 - 53 INCH-POUNDS**. Install cover on inside of jamnut and lockwire (Figure 1., TM 1-1500-204-23).
         (a) On test box, vary CURRENT CONTROL and do VALVE CURRENT check as shown in Figure 2.
            1 If SAS output rod movement does not correspond to milliamp input from test box and if nulling check of servo is acceptable, replace SAS module (WP 1167 00) and/or electrohydraulic servo valve (WP 1166 00).

ROLL SAS ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

18. Turn off hydraulic pressure and electrical power.
19. Remove dial indicator.
20. Disconnect hydraulic lines from nulling fixture.
21. Remove test box cable from power source.
22. Remove roll assembly from nulling fixture by removing three bolts.

Indication/Condition

Roll assembly shall be removed.

Corrective Action

None Required

LOCATION

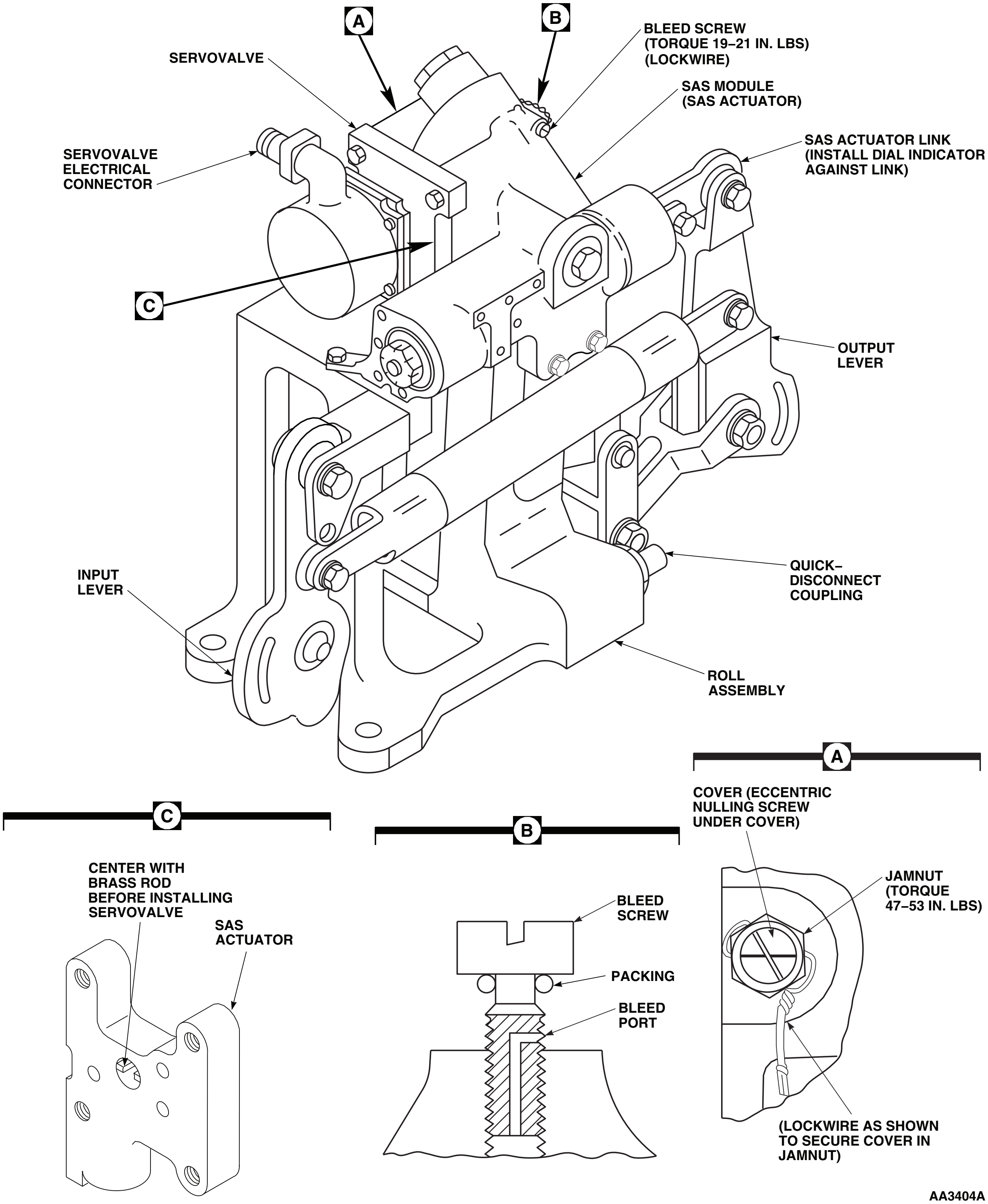
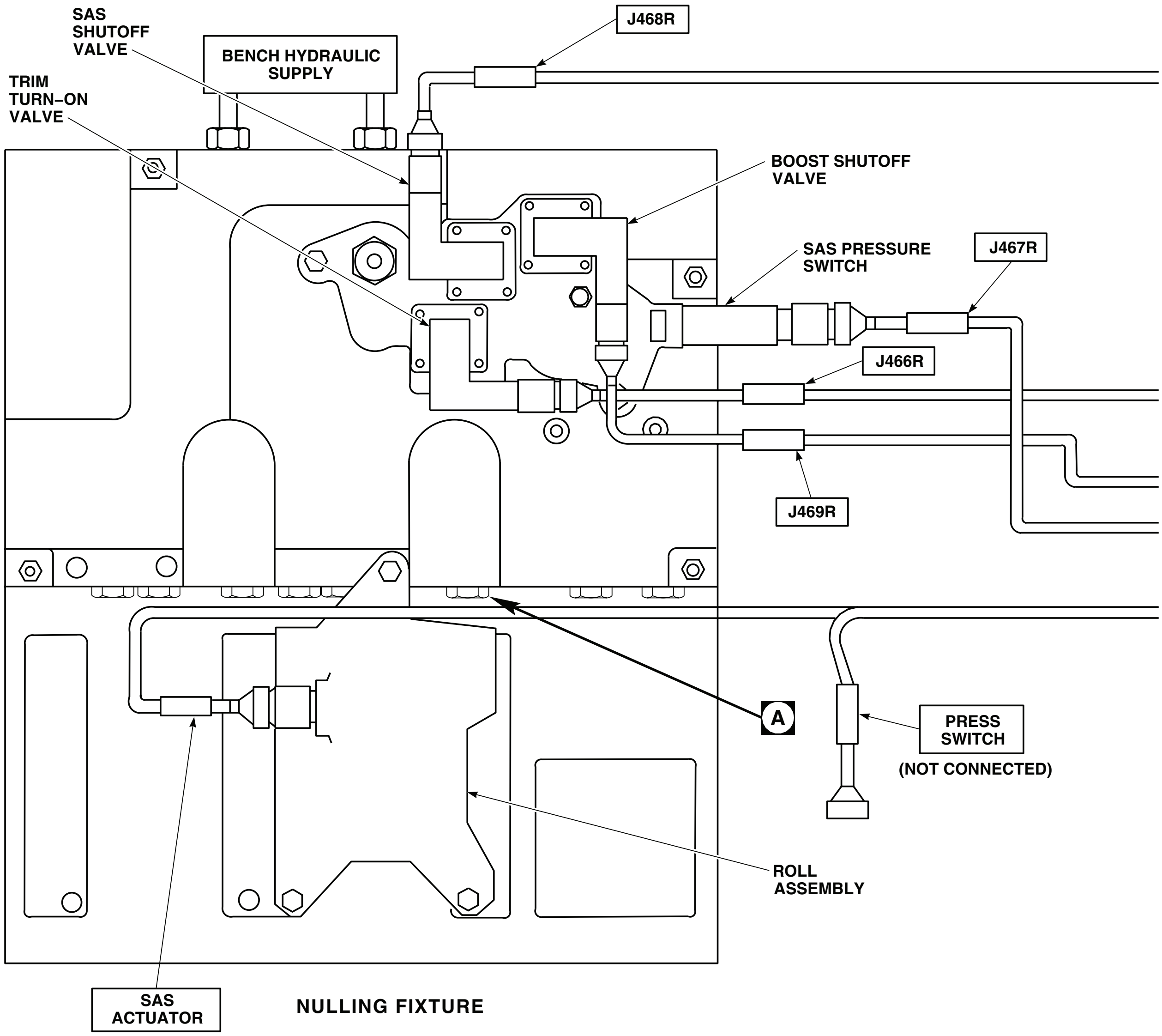


Figure 1. Roll SAS Servo Assembly.

TEST SETUP



VALVE CURRENT CHART

TRIM COIL SELECT SWITCH IN BOTH POSITION

SIGNAL MILLI–AMPS (MA)	DISPLACEMENT MEASURED FROM NULL (INCHES)
0	0 (NULLED)
–3.70	0.090 TO 0.110 (EXTENDED FROM NULL)
–6.60	0.155 TO 0.200 (EXTENDED FROM NULL)
+3.70	0.090 TO 0.110 (RETRACTED FROM NULL)
+6.60	0.155 TO 0.200 (RETRACTED FROM NULL)

TRIM COIL SELECT SWITCH IN 1 OR 2 POSITION

SIGNAL MILLI–AMPS (MA)	DISPLACEMENT MEASURED FROM NULL (INCHES)
0	0 (NULLED)
–1.85	0.045 TO 0.055 (EXTENDED FROM NULL)
–3.30	0.0775 TO 0.100 (EXTENDED FROM NULL)
+1.85	0.045 TO 0.055 (RETRACTED FROM NULL)
+3.30	0.0775 TO 0.100 (RETRACTED FROM NULL)

AA3405_1
SA

Figure 2. Roll SAS Servo Assembly Test Setup Diagram. (Sheet 1 of 3)

TEST SETUP - Continued

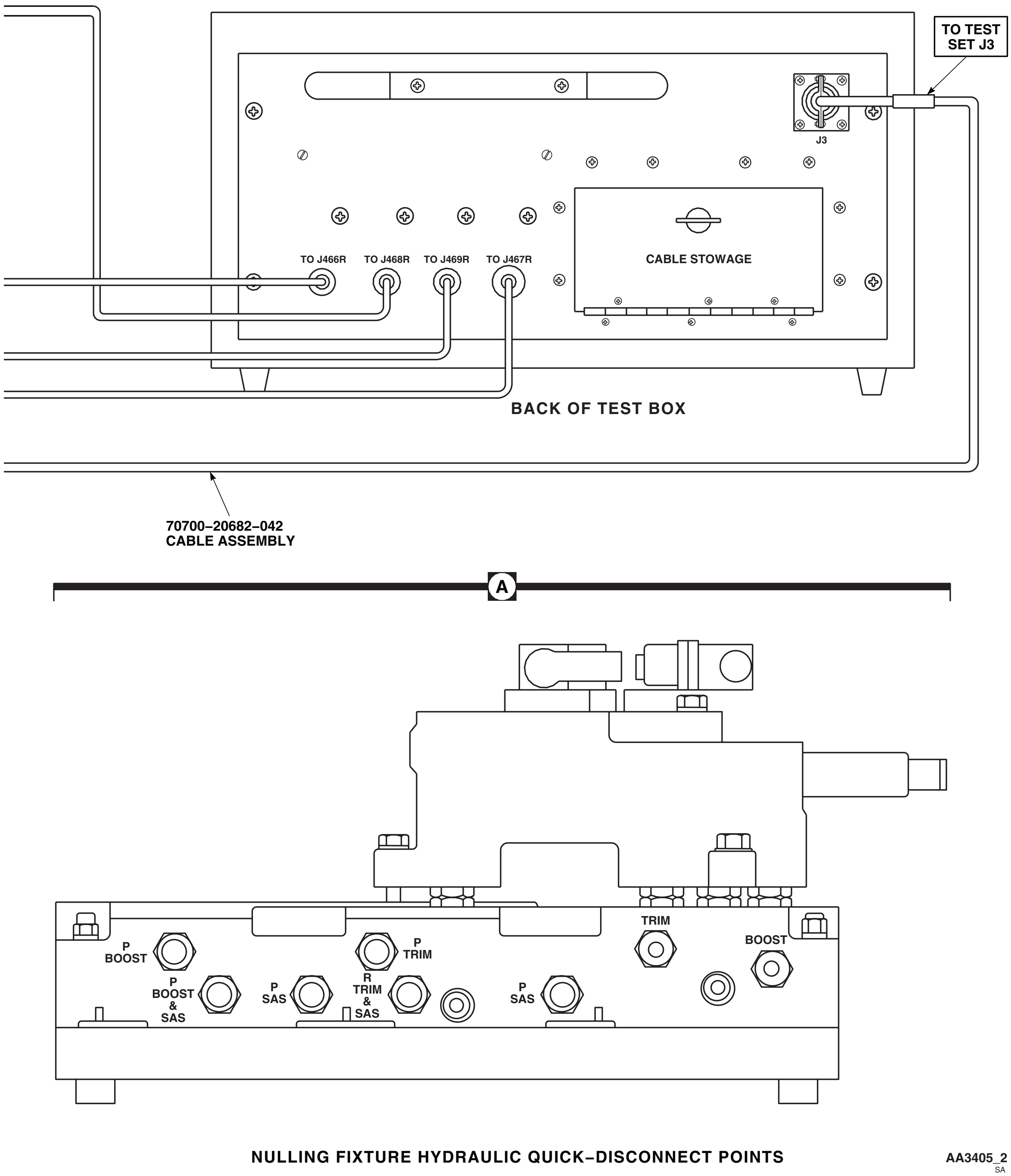


Figure 2. Roll SAS Servo Assembly Test Setup Diagram. (Sheet 2 of 3)

TEST SETUP - Continued

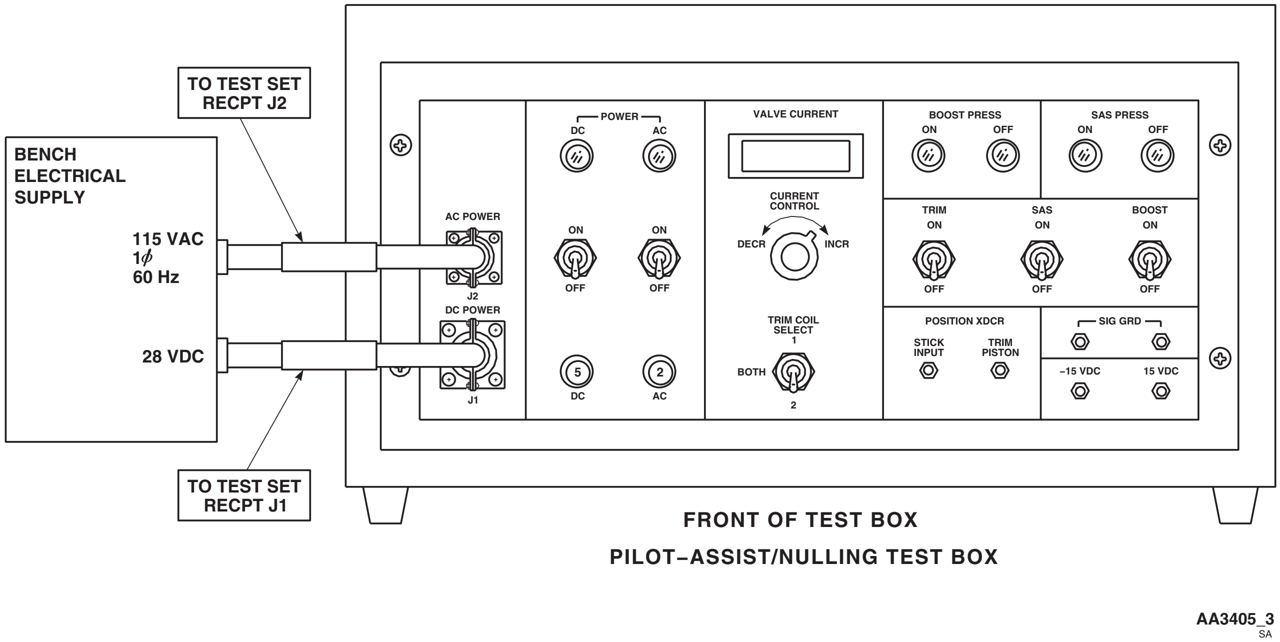


Figure 2. Roll SAS Servo Assembly Test Setup Diagram. (Sheet 3 of 3)

END OF WORK PACKAGE

UNIT LEVEL

FLIGHT CONTROLS

YAW BOOST ASSEMBLY (AVIM)

INITIAL SETUP:

Test Equipment
Dial Indicator, A-A-2348
Nulling Fixture Assembly, SAS Assembly, 70700-20675-041

Tools and Special Tools
Hydraulic Repairer Toolkit, SC 5180-99-CL-A03
Magnet, GGG-F-0036
Torque Wrench, 0-30 in. lbs., A-A-2411
Torque Wrench, 30-150 in. lbs., GGG-W-686

Material/Parts
Wire, Nonelectrical (Lockwire), Item 355, WP 1803 00

Personnel Required
Aircraft Pneudraulics Repairer MOS 15H (1)

References
TM 1-1500-204-23
WP 1162 00
WP 1163 00
WP 1803 00

Equipment Condition
Hydraulic (3000 psi) Bench Power Available
28 vdc and 115 vac, Single-Phase, 60 Hz Bench Power Available

YAW BOOST ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

To prevent leakage between the nulling fixture and servo, make sure manifold and servo self-sealing couplings are clean.

NOTE

- If a SAS module or actuator has to be replaced, repeat operational/troubleshooting procedure.
- See Figure 1 for Yaw Boost Assembly diagram and Figure 2 for test setup diagram as an aid in operational/troubleshooting.

1. Position yaw boost assembly on nulling fixture base. Carefully slide yaw boost assembly toward manifold and engage self-sealing couplings.
2. Install four bolts securing yaw boost assembly to base. Tighten bolts.

YAW BOOST ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CAUTION

When using the 70700-20682-042 cable, make sure that the connectors marked PRESS SWITCH and SAS ACTUATOR are correctly connected. If an incorrect connection is made, SERVO VALVE may become damaged.

3. Connect test box cables to pilot assist module and yaw boost assembly on nulling fixture as follows:

CABLE CONNECTOR	CONNECT TO
J466R	Pilot-assist module (Trim turn-on valve)
J467R	Pilot-assist module (SAS pressure switch)
J468R	Pilot-assist module (SAS shutoff valve)
J469R	Pilot-assist module (Boost shutoff valve)
J3	Yaw boost assembly (Servo valve and pressure switch)

4. Connect test box power cables to electrical power source.
5. Connect hydraulic lines from hydraulic power source to PRESSURE and RETURN ports on nulling fixture.
6. Turn on hydraulic power source and set pressure for 3000 psi at a maximum of 6 gpm.
7. Place test box AC and DC POWER switches to ON.

Indication/Condition

POWER ON indicators shall light and BOOST PRESS OFF and SAS PRESS OFF lights shall be on.

Corrective Action

If Indication/Condition is not as specified, make sure test box AC and DC circuit breakers are pushed in.

Step

8. Place test box BOOST switch to ON and then to OFF several times (BOOST PRESS ON and OFF lights cycle), then leave ON. Move servo input lever back and forth several times to cycle output piston.
9. Place test box SAS switch to ON and then to OFF.

Indication/Condition

SAS PRESS ON and OFF lights shall cycle from on to off.

Corrective Action

If Indication/Condition is not as specified, replace test box.

Step

10. Repeat step 9. several times and leave switch at ON (SAS PRESS ON light on).
11. Vary test box CURRENT CONTROL until VALVE CURRENT meter indicates 0.
12. Vary test box CURRENT CONTROL from 0 to 5.0 , back to 0, then to -5.0 , and back to 0 several times to stroke the SAS piston. Return CURRENT CONTROL to 0.

YAW BOOST ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CAUTION

If bleed screw is not backed out properly, packing on bleed screw will be damaged.

13. Decrease hydraulic pressure to 500 psi. Back out bleed screw until bleed hole is visible and a good flow is established.
14. Allow fluid to flow from valve until fluid is free of air. Decrease pressure to 0 psi. Close bleed valve. **TORQUE TO 19 - 21 INCH-POUNDS** and lockwire (WP 1803 00) (Figure 1, TM 1-1500-204-23).
15. Increase hydraulic pressure to 3000 psi and watch SAS assembly output piston for any movement.

Indication/Condition

Piston shall not move.

Corrective Action

If Indication/Condition is not as specified, there is still air trapped in servo. Repeat steps 8. through 14.

Step

16. Install dial indicator against SAS actuator link. Zero indicator.
17. Cycle test box SAS switch several times from ON to OFF. Watch actuator piston for any jump or movement.

Indication/Condition

Actuator piston jump shall be less than 0.005 inches to be nulled.

Corrective Action

NOTE

- Repeat piston jump check after torquing jamnut in substep (2). Readjust eccentric screw if necessary.
- Substep 1 verifies if SAS output rod moves properly when a specific current is applied to servovalve.

1. If Indication/Condition is not as specified, do the following:
   a. Remove cover from inside of jamnut using magnet.
   b. Loosen jamnut 1/4 turn.
   c. Insert screwdriver through jamnut and turn eccentric screw as needed to reduce piston jump below 0.005 inch. Cycle SAS switch and check for jump.
      (1) If piston jump is out of limits, remove servovalve (4 screws) and use 1/16-inch brass rod to center SAS actuator housing before installing servovalve with 4 screws. Repeat operational/troubleshooting procedure.
         (a) If trouble continues, replace SAS module (WP 1163 00) and repeat operational/troubleshooting procedure.
      (2) If piston jump is below 0.005 inch, **TORQUE JAMNUT TO 47 - 53 INCH-POUNDS**. Install cover on inside of jamnut and lockwire (Figure 1, TM 1-1500-204-23).
         (a) On test box, vary CURRENT CONTROL and do VALVE CURRENT check as shown in Figure 2.

YAW BOOST ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

1 If SAS output rod movement does not correspond to milliamp input from test box and if nulling check of servo is acceptable, replace SAS module (WP 1163 00) and/or electrohydraulic servo valve (WP 1162 00).

Step

18. Turn off hydraulic pressure and electrical power.
19. Remove dial indicator.
20. Disconnect hydraulic lines from nulling fixture.
21. Remove test box cable from power source.
22. Remove yaw boost assembly from nulling fixture by removing four bolts.

Indication/Condition

Yaw boost assembly shall be removed.

Corrective Action

None Required

LOCATION

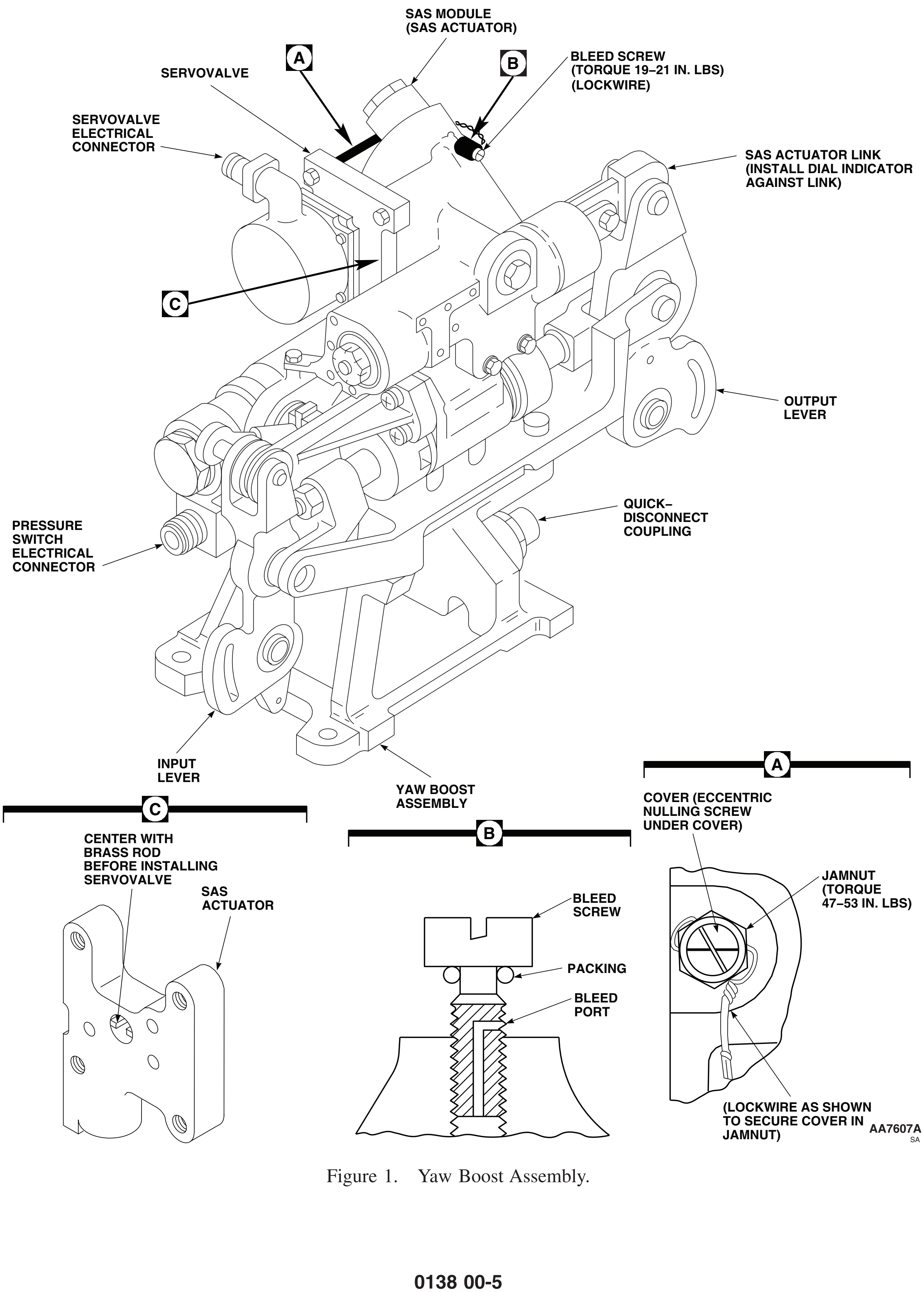
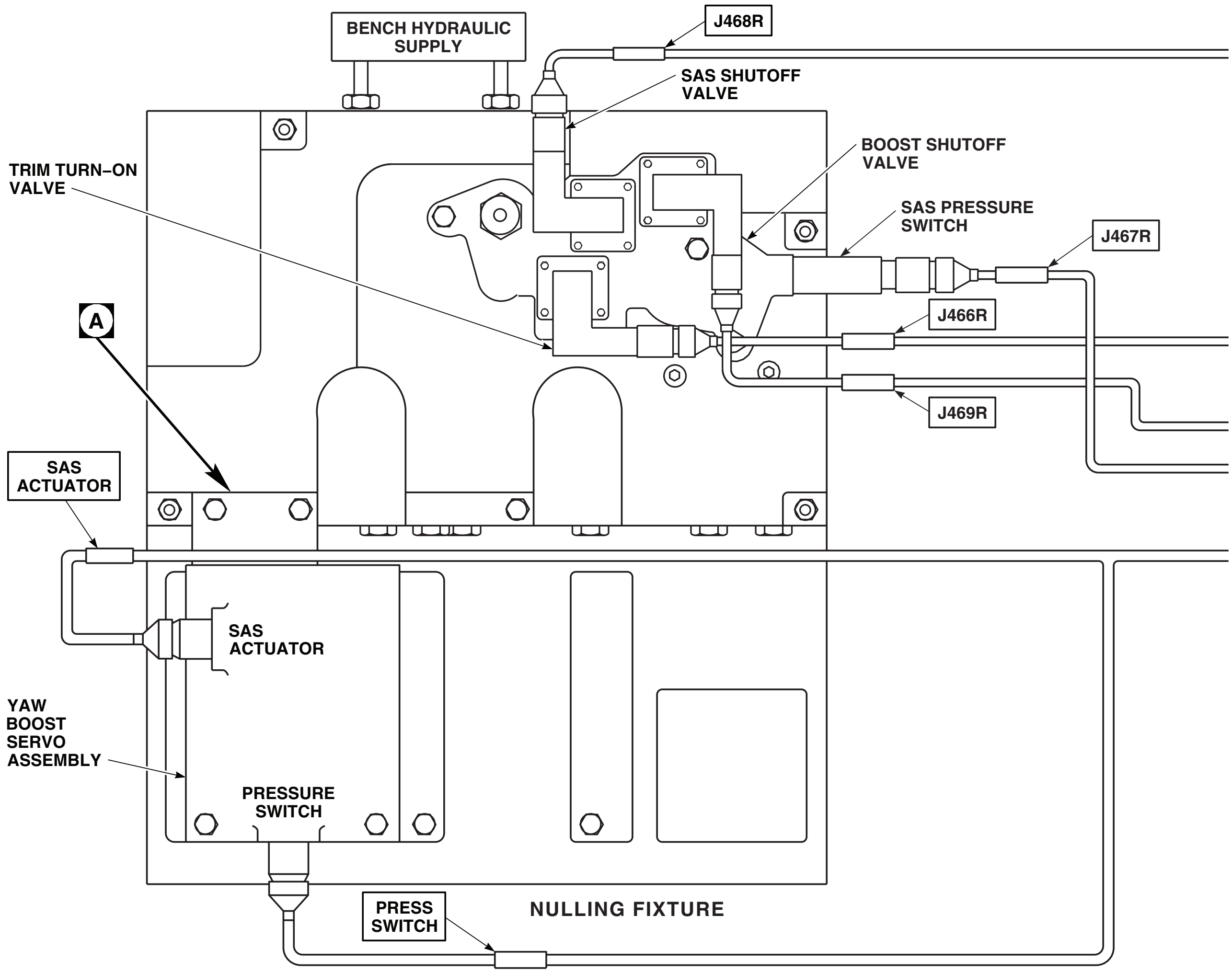


Figure 1. Yaw Boost Assembly.

TEST SETUP



VALVE CURRENT CHART

TRIM COIL SELECT SWITCH IN BOTH POSITION

SIGNAL MILLI–AMPS (MA)	DISPLACEMENT MEASURED FROM NULL (INCHES)
0	0 (NULLED)
–3.70	0.090 TO 0.110 (EXTENDED FROM NULL)
–6.60	0.155 TO 0.200 (EXTENDED FROM NULL)
+3.70	0.090 TO 0.110 (RETRACTED FROM NULL)
+6.60	0.155 TO 0.200 (RETRACTED FROM NULL)

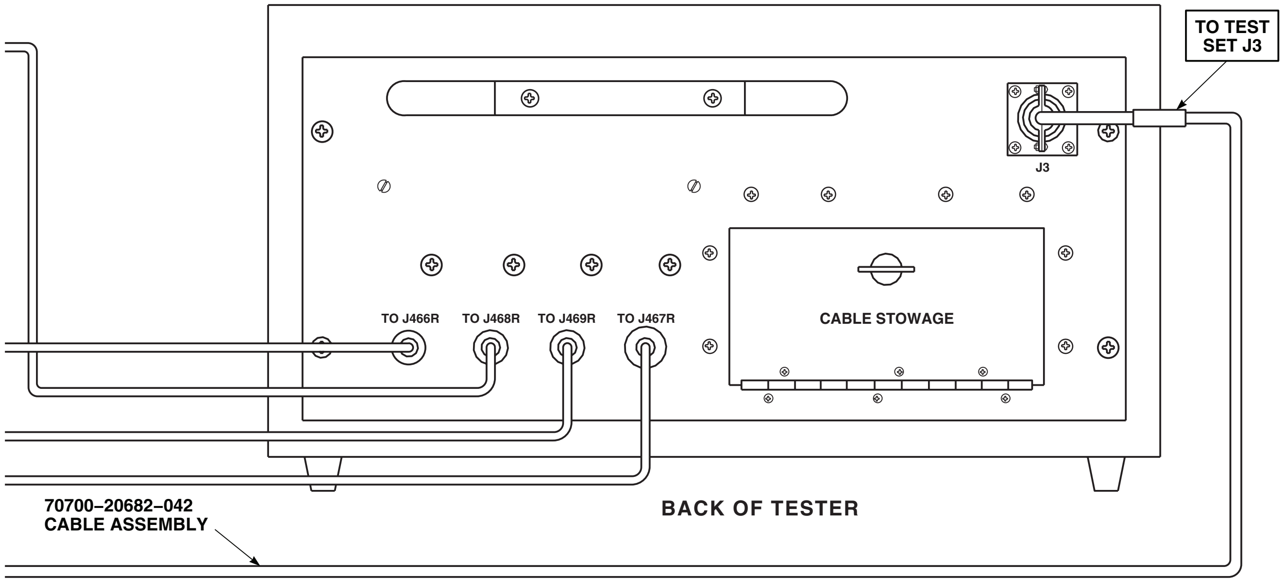
TRIM COIL SELECT SWITCH IN 1 OR 2 POSITION

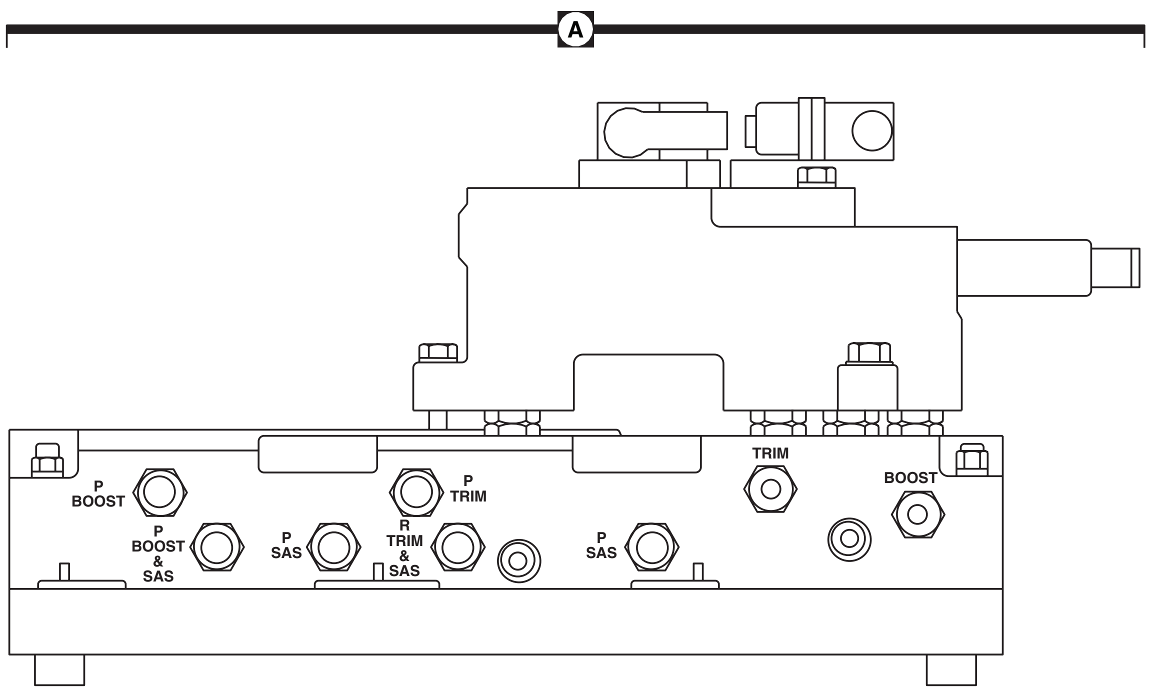
SIGNAL MILLI–AMPS (MA)	DISPLACEMENT MEASURED FROM NULL (INCHES)
0	0 (NULLED)
–1.85	0.045 TO 0.055 (EXTENDED FROM NULL)
–3.30	0.0775 TO 0.100 (EXTENDED FROM NULL)
+1.85	0.045 TO 0.055 (RETRACTED FROM NULL)
+3.30	0.0775 TO 0.100 (RETRACTED FROM NULL)

AA3406–1
SA

Figure 2. Yaw Boost Assembly Test Setup. (Sheet 1 of 3)

TEST SETUP - Continued





NULLING FIXTURE HYDRAULIC QUICK–DISCONNECT POINTS

AA3406_2
SA

Figure 2. Yaw Boost Assembly Test Setup. (Sheet 2 of 3)

TEST SETUP - Continued

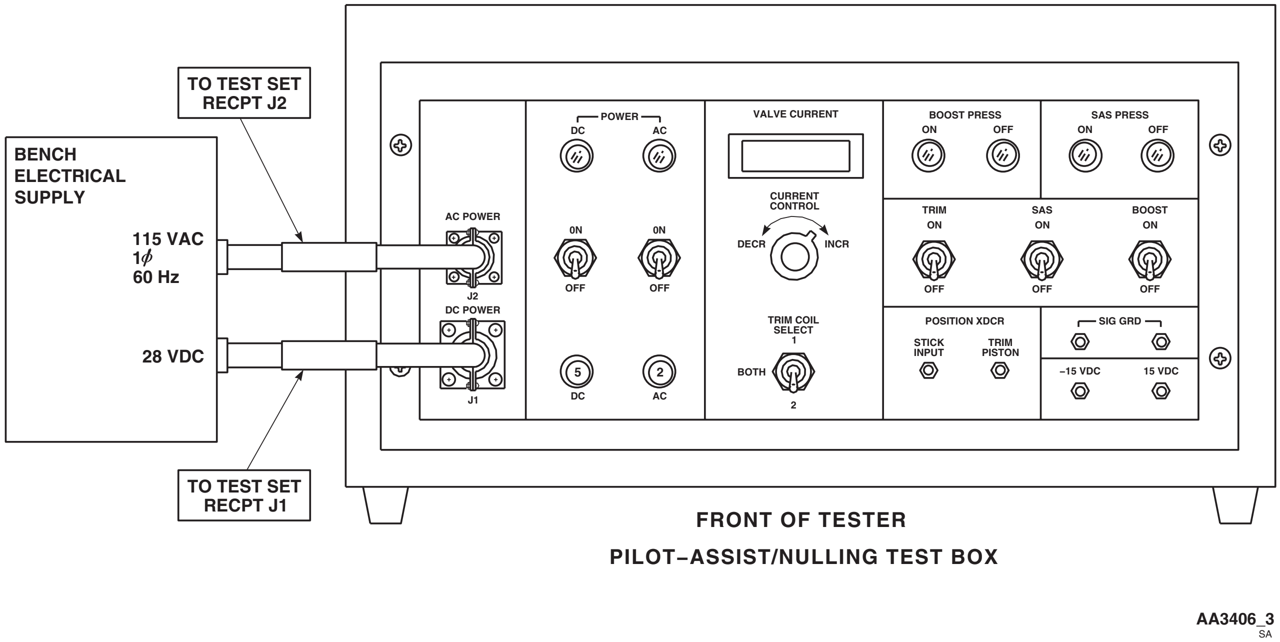


Figure 2. Yaw Boost Assembly Test Setup. (Sheet 3 of 3)

END OF WORK PACKAGE

UNIT LEVEL

FLIGHT CONTROLS

PITCH TRIM ACTUATOR (AVIM)

INITIAL SETUP:

Test Equipment
Dial Indicator, A-A-2348
Nulling Fixture Assembly, SAS Actuator, 70700-20675-041

Tools and Special Tools
Hydraulic Repairer Toolkit, SC 5180-99-CL-A03
Magnet, GGG-G-0036
Torque Wrench, A-A-2411
Torque Wrench, GGG-W-686

Material/Parts
Wire, Nonelectrical (Lockwire), Item 355, WP 1803 00

Personnel Required
Aircraft Pneudraulics Repairer MOS 15H (1)

References
TM 1-1500-204-23
WP 1164 00
WP 1165 00
WP 1803 00

Equipment Condition
Hydraulic (3000 psi) Bench Power Available
28 vdc and 115 vac, Single-Phase, 60 Hz Bench Power Available

PITCH TRIM ACTUATOR (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- To prevent leakage between the nulling fixture and servo, make sure manifold and servo self-sealing couplings are clean.
- To prevent damage to test equipment, do not apply electrical power until hydraulic pressure is applied.

NOTE

- If a servovalve and actuator has to be replaced, repeat operational/troubleshooting procedure.
- See Figure 1 for Pitch Trim Actuator diagram and Figure 2 for test setup diagram as an aid in operational/troubleshooting.

1. Position pitch trim actuator on nulling fixture base. Carefully slide pitch trim actuator toward manifold and engage self-sealing couplings.
2. Install three bolts securing pitch trim actuator to base. Tighten bolts.

PITCH TRIM ACTUATOR (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

3. Connect test box cables to pilot-assist module and pitch trim actuator on nulling fixture as follows:

CABLE CONNECTOR	CONNECT TO
J466R	Pilot-assist module (Trim shutoff valve)
J467R	Pilot-assist module (SAS pressure switch)
J468R	Pilot-assist module (SAS shutoff valve)
J469R	Pilot-assist module (Boost shutoff valve)
J3	Pitch trim actuator (Servo valve) (70700-20682-041 cable assembly)

4. Connect test box power cables to electrical power source.
5. Connect hydraulic lines from hydraulic power source to PRESSURE and RETURN ports on nulling fixture.
6. Turn on hydraulic power source and set pressure for 3000 psi at a maximum of 6 gpm.
7. Place AC and DC test box POWER switches to ON.

Indication/Condition

POWER ON indicators shall light and SAS PRESS OFF light shall go on.

Corrective Action

If Indication/Condition is not as specified, make sure test box AC and DC circuit breakers are pushed in.

Step

8. Place SAS switch to ON and then to OFF several times; then leave ON. Move servo input lever back and forth several times to cycle output piston. SAS PRESS ON light shall be on.
9. Turn TRIM switch ON.
10. Vary CURRENT CONTROL until VALVE CURRENT meter indicates 0.
11. Vary CURRENT CONTROL from 0 to 5.0 , back to 0, then to -5.0 , and back to 0 several times to stroke trim piston. Return CURRENT CONTROL to 0.

CAUTION

When using the 70700-20682-042 cable assembly, make sure that the connector marked SAS ACTUATOR is correctly connected. If the incorrect connector is used, servo valve may become damaged.

12. Disconnect cable assembly, 70700-20682-041, from test box and trim servovalve. Connect cable assembly, 70700-20682-042 to test set and SAS servovalve.
13. Cycle SAS switch several times ON and OFF. After cycling switch leave in ON position.
14. Vary CURRENT CONTROL from 0 to 5.0 , back to 0, then to -5.0 , and back to 0 several times, to stroke SAS position.

PITCH TRIM ACTUATOR (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CAUTION

If bleed screw is not backed out properly, packing on bleed screw will be damaged.

15. Decrease hydraulic pressure to 500 psi. Back out bleed screw until bleed hole is visible and a good flow is established.
16. Allow fluid to flow from valve until fluid is free of air. Decrease pressure to 0 psi. Close bleed valve. **TORQUE TO 19 - 21 INCH-POUNDS** and lockwire (WP 1803 00) (Figure 1., TM 1-1500-204-23).
17. Increase hydraulic pressure to 3000 psi and watch SAS assembly output piston for any movement.

Indication/Condition

Piston shall not move.

Corrective Action

If piston moves there is still air trapped in servo. Repeat steps 7. through 17.

Step

18. Install dial indicator against SAS actuator link.
19. Adjust indicator to read zero.
20. Cycle SAS switch several times from ON to OFF. Watch actuator piston for any jump or movement.

Indication/Condition

Actuator piston jump shall be less than 0.005 inches to be nulled.

Corrective Action

NOTE

- Repeat piston jump check after torquing jamnut in substep (2). Readjust eccentric screw if necessary.
- Substep 1 verifies if SAS output rod moves properly when a specific current is applied to servovalve.

1. If Indication/Condition is not as specified, do the following:
   a. Remove cover from inside of jamnut using magnet.
   b. Loosen jamnut 1/4 turn.
   c. Insert screwdriver through jamnut and turn eccentric screw as needed to reduce piston jump below 0.005 inch. Cycle SAS switch and check for jump.
      (1) If piston jump is out of limits, remove servovalve (4 screws) and use 1/16-inch brass rod to center SAS actuator housing before installing servovalve with 4 screws. Repeat operational/troubleshooting procedure.
      (2) If piston jump is below 0.005 inch, **TORQUE JAMNUT TO 47 - 53 INCH-POUNDS.** Install cover on inside of jamnut and lockwire (Figure 1. , TM 1-1500-204-23).
         (a) On test box, vary CURRENT CONTROL and do VALVE CURRENT check as shown in Figure 2.

PITCH TRIM ACTUATOR (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

1 If SAS output rod movement does not correspond to milliamp input from test box and if nulling check of servo is acceptable, replace SAS module (WP 1165 00) and/or electrohydraulic servo valve (WP 1164 00).

Step

21. Turn off hydraulic pressure and electrical power.
22. Remove dial indicator.
23. Disconnect hydraulic lines from nulling fixture.
24. Remove test box cable from power source.
25. Remove pitch trim actuator from nulling fixture by removing three bolts.

Indication/Condition

Pitch trim actuator shall be removed.

Corrective Action

None Required

LOCATION

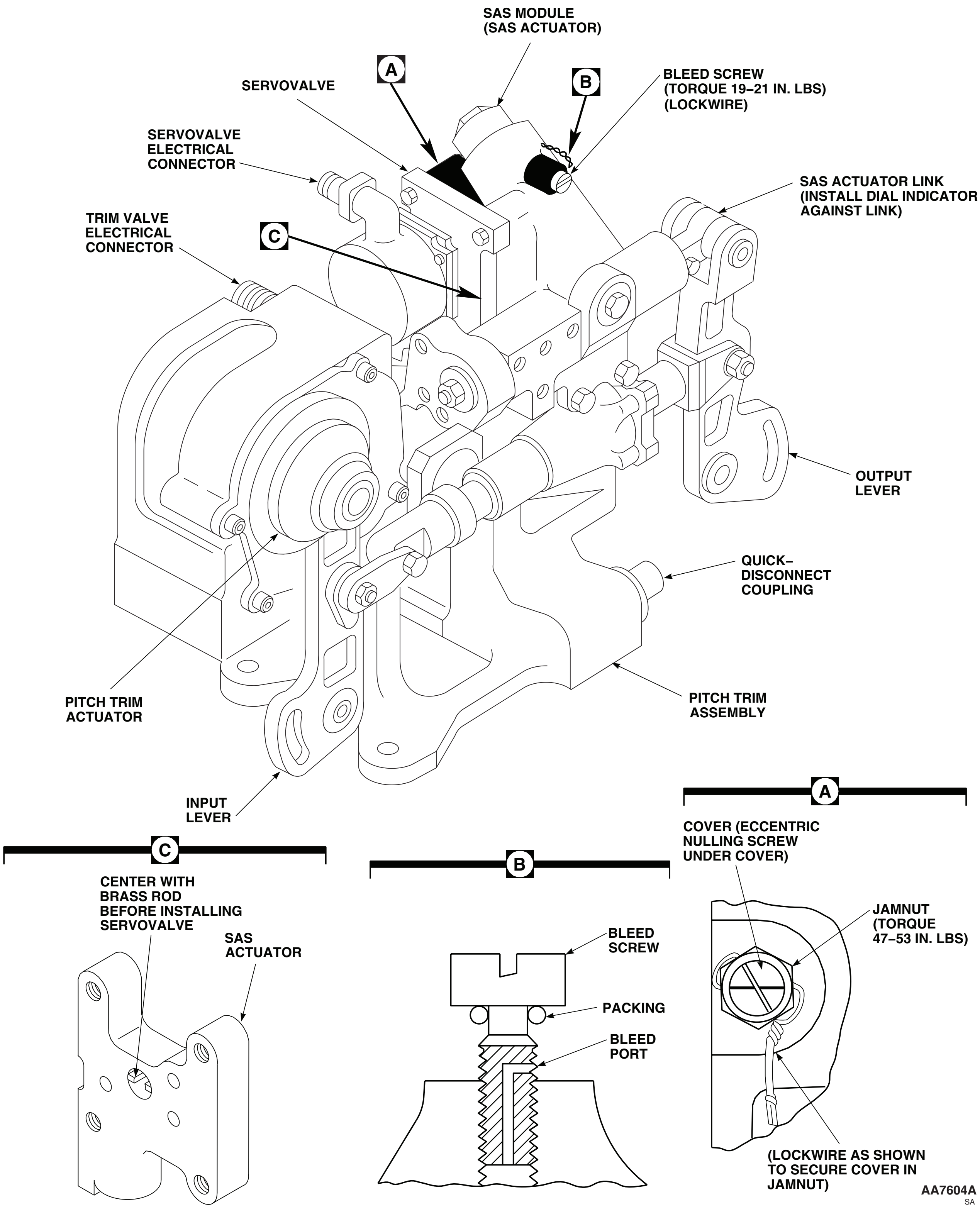
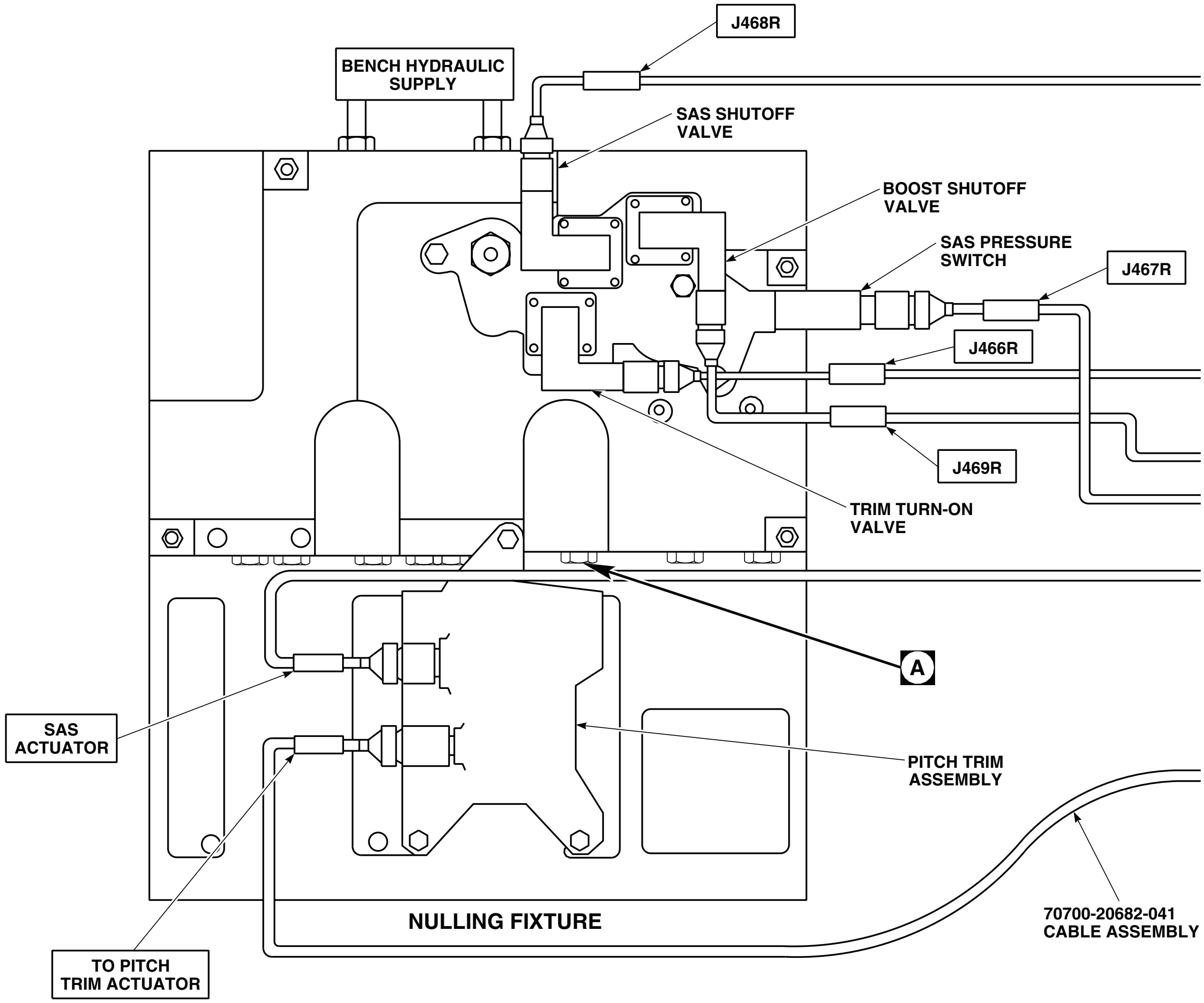


Figure 1. Pitch Trim Actuator.

TEST SETUP



VALVE CURRENT CHART

TRIM COIL SELECT SWITCH IN BOTH POSITION

SIGNAL MILLI-AMPS (MA)	DISPLACEMENT MEASURED FROM NULL (INCHES)
0	0 (NULLED)
-3.70	0.090 TO 0.110 (EXTENDED FROM NULL)
-6.60	0.155 TO 0.200 (EXTENDED FROM NULL)
+3.70	0.090 TO 0.110 (RETRACTED FROM NULL)
+6.60	0.155 TO 0.200 (RETRACTED FROM NULL)

TRIM COIL SELECT SWITCH IN 1 OR 2 POSITION

SIGNAL MILLI-AMPS (MA)	DISPLACEMENT MEASURED FROM NULL (INCHES)
0	0 (NULLED)
-1.85	0.045 TO 0.055 (EXTENDED FROM NULL)
-3.30	0.0775 TO 0.100 (EXTENDED FROM NULL)
+1.85	0.045 TO 0.055 (RETRACTED FROM NULL)
+3.30	0.0775 TO 0.100 (RETRACTED FROM NULL)

NOTE

CONNECT CONNECTOR AS SPECIFIED IN OPERATIONAL CHECK.

AA3407_1
SA

Figure 2. Pitch Trim Actuator Test Setup Diagram. (Sheet 1 of 3)

TEST SETUP - Continued

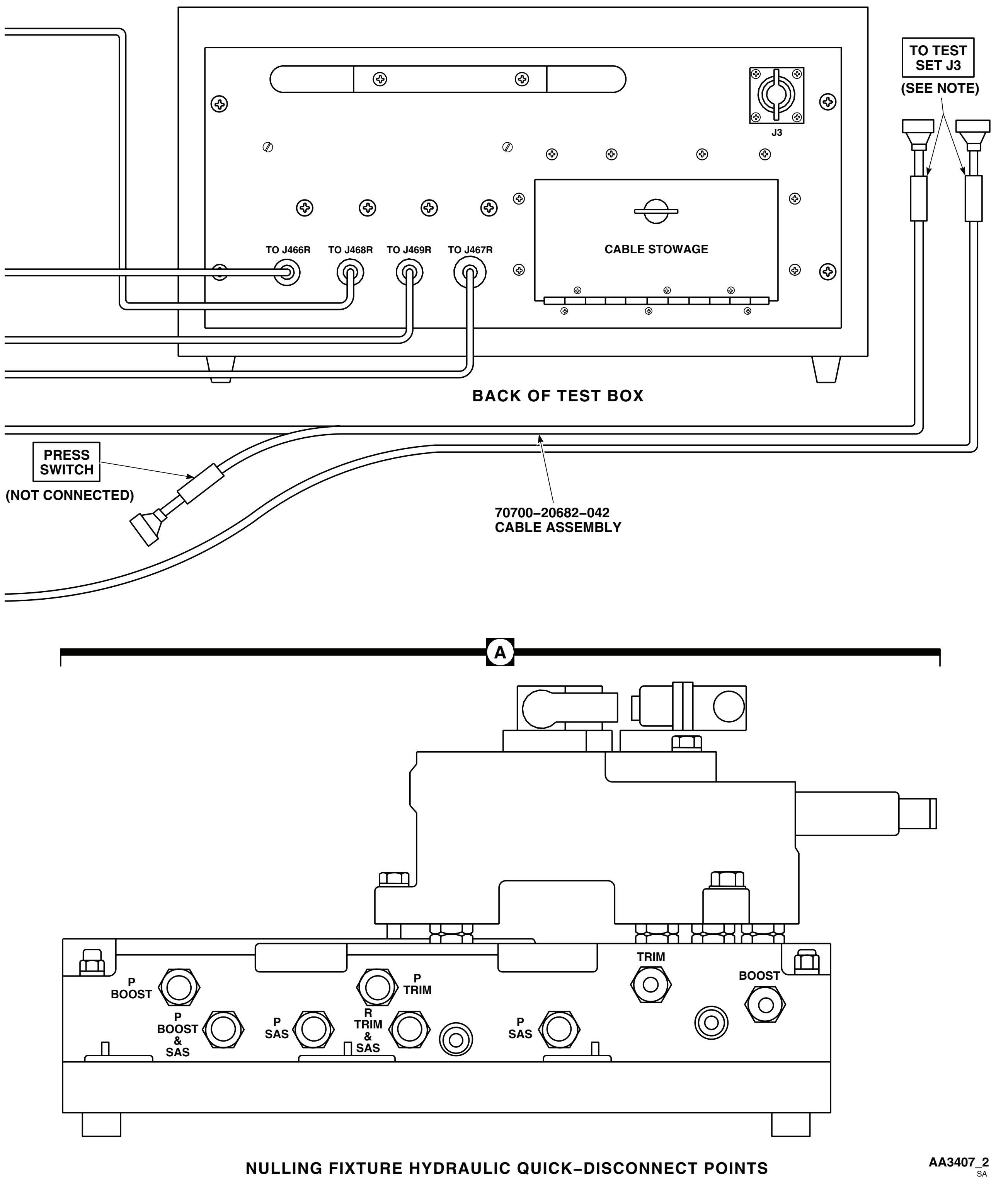


Figure 2. Pitch Trim Actuator Test Setup Diagram. (Sheet 2 of 3)

TEST SETUP - Continued

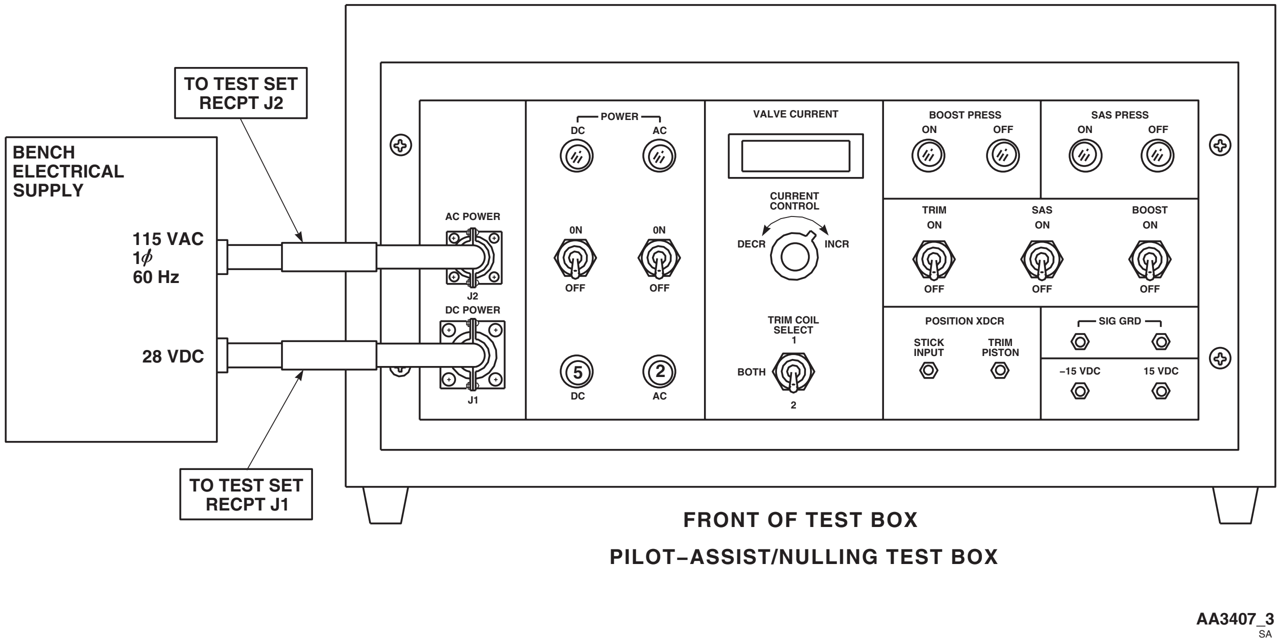


Figure 2. Pitch Trim Actuator Test Setup Diagram. (Sheet 3 of 3)

END OF WORK PACKAGE

UNIT LEVEL

FLIGHT CONTROLS

PILOT-ASSIST MODULE (AVIM)

INITIAL SETUP:

Test Equipment
Nulling Fixture Assembly, SAS Actuator, 70700-20675-041

Tools and Special Tools
Hydraulic Repairer's Toolkit, SC 5180-99-C
Torque Wrench, 150 - 750 in. lbs., GGG-W-686

Personnel Required
Aircraft Pneudraulics Repairer MOS 15H (1)

References
TM 1-1500-204-23
WP 0729 00
WP 0748 00
WP 1169 00
WP 1170 00

Equipment Condition
Hydraulic Power Source (3000 psi) Available
28 vdc Electrical Power Available

PILOT-ASSIST MODULE (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

To prevent leakage between the nulling fixture and pilot-assist module, make sure manifold and servo self-sealing couplings are clean.

NOTE

- If a three-way valve or pressure reducer has to be replaced, repeat operational/troubleshooting procedure.
- See Figure 1 for Pilot-assist module diagram and Figure 2 for test setup diagram as an aid in operational/troubleshooting.

1. Remove existing pilot-assist module from nulling fixture by removing three bolts and electrical connectors. Lift module from nulling fixture.
2. Install pilot-assist module to be tested on nulling fixture, making sure quick-disconnects are lined up. Secure module with three bolts. **TORQUE BOLTS TO 180 - 200 INCH-POUNDS (Figure 2., TM 1-1500-204-23).**
3. Install a test block on each servo position on nulling fixture.
4. Install a pressure gage in TRIM and BOOST pressure ports of test fixture.
5. Place all test box pilot-assist/nulling switches on to OFF and place TRIM COIL SELECT switch to BOTH.

PILOT-ASSIST MODULE (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

6. Connect test box cables to pilot-assist module assembly on nulling fixture as follows:

CABLE CONNECTOR	CONNECT TO
J466R	Pilot-assist module (Trim turn-on valve)
J467R	Pilot-assist module (SAS pressure switch)
J468R	Pilot-assist module (SAS shutoff valve)
J469R	Pilot-assist module (Boost shutoff valve)

7. Connect test box power cables to electrical power source.
8. Connect hydraulic lines from hydraulic power source to PRESSURE and RETURN ports on nulling fixture.
9. Turn on hydraulic power source and set pressure to 3000 psi at 6 gpm.
10. Place test box AC and DC POWER switches to ON.

Indication/Condition

a. Pressure gage connected to TRIM pressure port shall indicate 0 psi, and pressure gage connected to BOOST pressure port shall indicate 0 psi.

b. POWER ON indicator shall light.

c. SAS PRESS OFF light shall be on.

Corrective Action

1. If Indication/Condition a. is not as specified, replace boost three-way valve (WP 1169 00).
2. If Indication/Conditions b. and c. are not as specified, make sure that test box AC and DC circuit breakers are pushed in.
3. If Indication/Condition c. is not as specified, check SAS pressure switch and replace if necessary (WP 0748 00).

Step

11. Place test box TRIM switch to ON.

Indication/Condition

Pressure gage connected to TRIM pressure port shall indicate 950 to 1050 psi.

Corrective Action

1. If gage remains at 0 psi, replace trim three-way valve (WP 1169 00).
2. If gage indicates more than 1050 or less than 950 psi, replace pressure reducer (WP 1170 00).

Step

12. Place test box TRIM switch to OFF.

Indication/Condition

a. TRIM pressure gage shall indicate 0 psi.

PILOT-ASSIST MODULE (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

b. Button on pressure reducer shall not pop.

Corrective Action

1. If Indication/Condition a. is not as specified, replace trim three-way valve (WP 1169 00).
2. If Indication/Condition b. is not as specified and will not remain reset, replace pressure reducer (WP 1170 00).

Step

13. Place test box SAS switch to ON.

Indication/Condition

SAS PRESS ON light shall be on.

Corrective Action

1. If SAS PRESS OFF light stays lit, check SAS pressure switch and replace if necessary (WP 0729 00).
   a. If trouble remains, replace SAS three-way valve (WP 1169 00).

Step

14. Place test box SAS switch to OFF.

Indication/Condition

SAS PRESS ON light shall go off.

Corrective Action

1. If SAS PRESS OFF light is not on, check SAS pressure switch and replace if necessary (WP 0729 00).
   a. If trouble remains, replace SAS three-way valve (WP 1169 00).

Step

15. Place test box BOOST switch to ON.

Indication/Condition

Pressure gage connected to BOOST pressure port shall indicate 3000 psi.

Corrective Action

If Indication/Condition is not as specified, replace boost three-way valve (WP 1169 00).

Step

16. Place test box BOOST switch to OFF.

Indication/Condition

BOOST pressure gage shall indicate 0 psi.

Corrective Action

If Indication/Condition is not as specified, replace boost three-way valve (WP 1169 00).

Step

17. Place test box DC POWER switch to OFF.

PILOT-ASSIST MODULE (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

18. Turn off hydraulic pressure and electrical power.
19. Disconnect hydraulic lines from nulling fixture.
20. Remove test box cable from power source.
21. Disconnect electrical cables from pilot-assist module.
22. Remove pilot-assist module from nulling fixture by removing three bolts and lifting module.
23. Install previously attached pilot-assist module onto nulling fixture. Install three bolts. Tighten bolts.

Indication/Condition

Module shall be installed.

Corrective Action

None Required

LOCATION

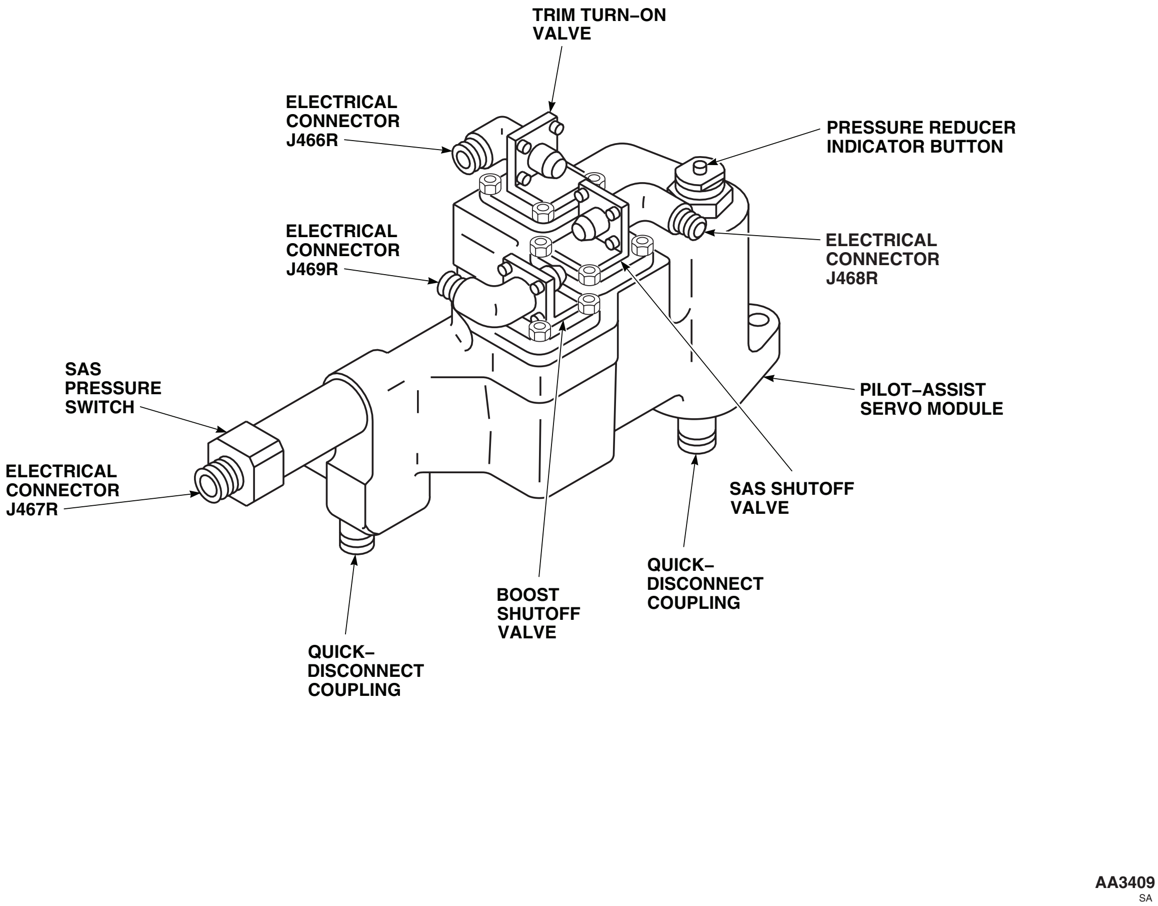


Figure 1. Pilot-assist Module.

TEST SETUP

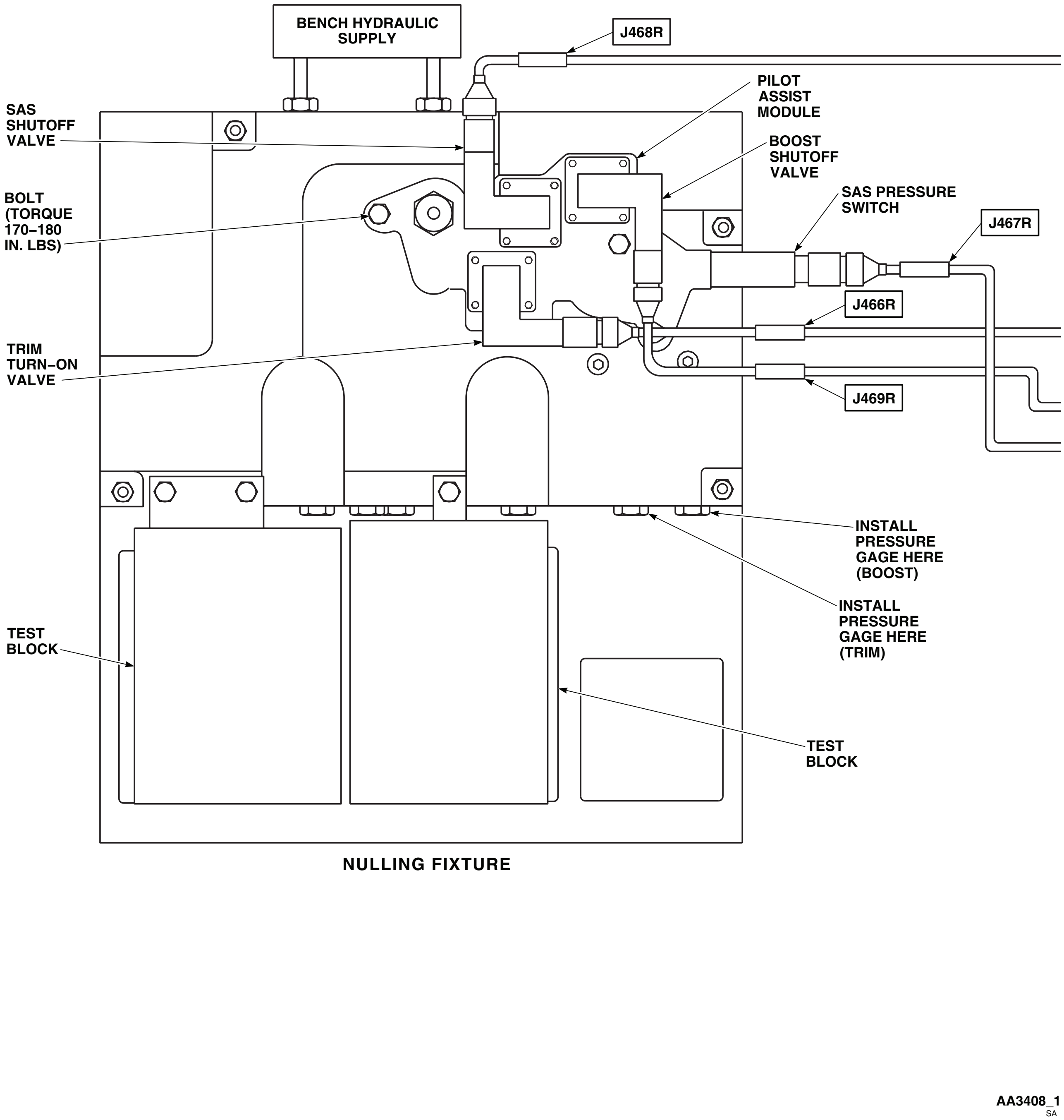


Figure 2. Pilot-assist Module Test Setup. (Sheet 1 of 2)

TEST SETUP - Continued

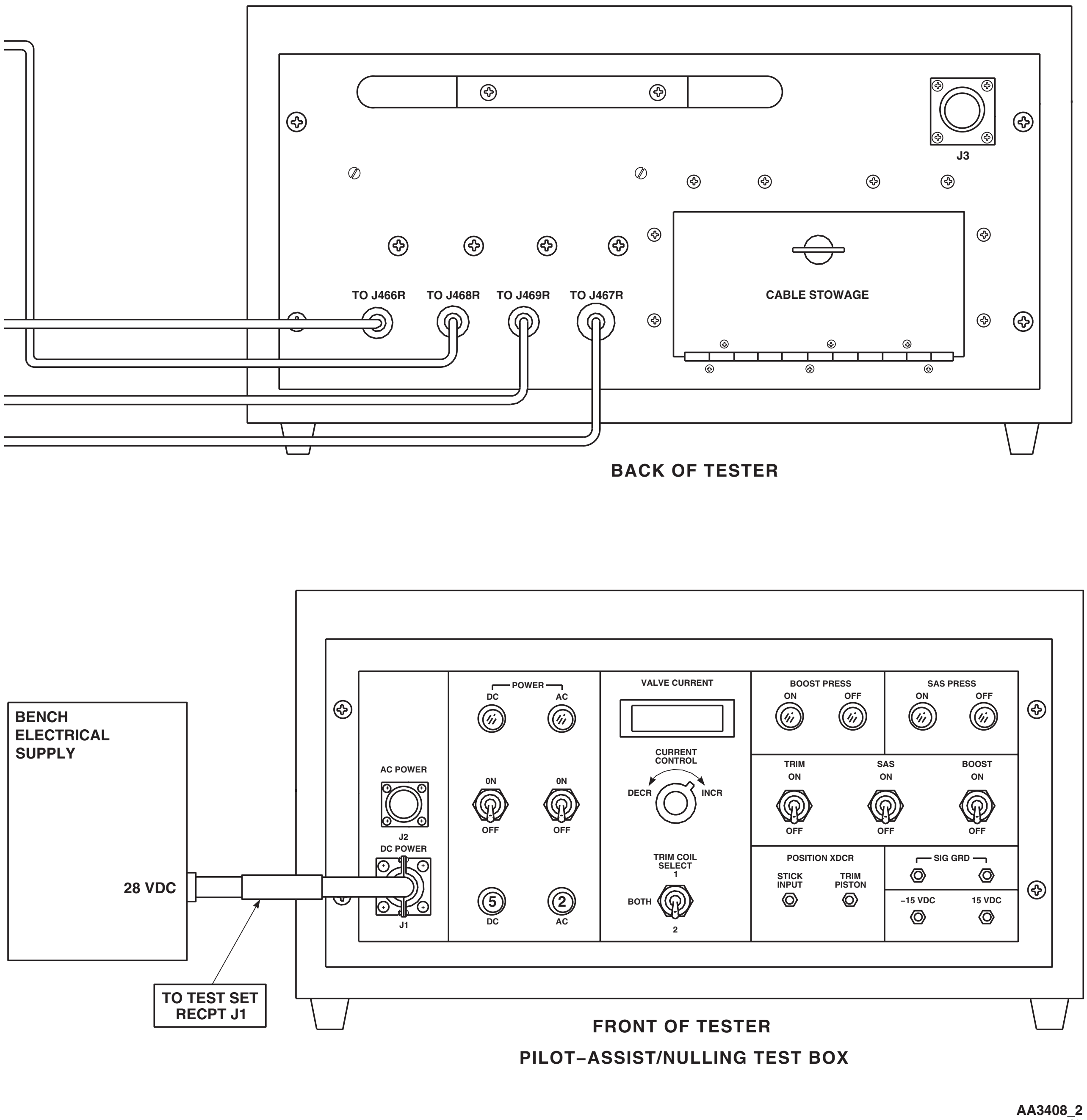


Figure 2. Pilot-assist Module Test Setup. (Sheet 2 of 2)

END OF WORK PACKAGE

By Order of the Secretary of the Army:

PETER J. SCHOOMAKER
General, United States Army
Chief of Staff

Official:

Joyce E. Morrow

JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army
0608613

DISTRIBUTION:
To be distributed in accordance with Initial Distribution Number (IDN) 311286, requirements for TM 1-1520-237-23-3.

These are the instructions for sending an electronic 2028

The following format must be used if submitting an electronic 2028. The subject line must be exactly the same and all fields must be included; however only the following fields are mandatory: 1, 3, 4, 5, 6, 7, 8, 9, 10, 13, 15, 16, 17, and 27.

From: "Whomever" <whomever@wherever.army.mil>
To: 2028@redstone.army.mil

Subject: DA Form 2028

1. From: Joe Smith
2. Unit: home
3. Address: 4300 Park
4. City: Hometown
5. St: MO
6. Zip: 77777
7. Date Sent: 19--OCT–93
8. Pub no: 55--2840--229--23
9. Pub Title: TM
10. Publication Date: 04--JUL--85
11. Change Number: 7
12. Submitter Rank: MSG
13. Submitter FName: Joe
14. Submitter MName: T
15. Submitter LName: Smith
16. Submitter Phone: 123–123–1234
17. Problem: 1
18. Page: 2
19. Paragraph: 3
20. Line: 4
21. NSN: 5
22. Reference: 6
23. Figure: 7
24. Table: 8
25. Item: 9
26. Total: 123
27. Text:

This is the text for the problem below line 27.

RECOMMENDED CHANGES TO PUBLICATIONS AND BLANK FORMS For use of this form, see AR 25-30; the proponent agency is ODISC4.	Use Par reverse) for Repair Parts and Special Tool Lists (RPSTL) and Supply Catalogs/ Supply Manuals (SC/SM)	DATE 8/30/02

TO: (Forward to proponent of publication or form)(Include ZIP Code)	FROM: (Activity and location)(Include ZIP Code)
Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, 35898	MSG, Jane Q. Doe 1234 Any Street Nowhere Town, AL 34565

PART 1 – ALL PUBLICATIONS (EXCEPT RPSTL AND SC/SM) AND BLANK FORMS

PUBLICATION/FORM NUMBER	DATE	TITLE
TM 9-1005-433-24	16 Sep 2002	Organizational, Direct Support, And General Support Maintenance Manual for Machine Gun, .50 Caliber M3P and M3P Machine Gun Electrical Test Set Used On Avenger Air Defense Weapon System

ITEM NO.	PAGE NO.	PARA-GRAPH	LINE NO. *	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASON
1	W PG 3		2			Test or Corrective Action column should identify a different WP number.

EXAMPLE

* Reference to line numbers within the paragraph or subparagraph.

TYPED NAME, GRADE OR TITLE	TELEPHONE EXCHANGE/ AUTOVON, PLUS EXTENSION	SIGNATURE
MSG, Jane Q. Doe, SFC	788--1234	

DA FORM 2028, FEB 74 REPLACES DA FORM 2028, 1 DEC 68, WHICH WILL BE USED. USAPA V3.01

TO: (Forward direct to addressee listed in publication)	FROM: (Activity and location) (Include ZIP Code)	DATE
Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM–MMC–MA–NP Redstone Arsenal, 35898	MSG, Jane Q. Doe 1234 Any Street Nowhere Town, AL 34565	8/30/02

PART II – REPAIR PARTS AND SPECIAL TOOL LISTS AND SUPPLY CATALOGS/SUPPLY MANUALS

PUBLICATION NUMBER	DATE	TITLE

PAGE NO.	COLM NO.	LINE NO.	NATIONAL STOCK NUMBER	REFERENCE NO.	FIGURE NO.	ITEM NO.	TOTAL NO. OF MAJOR ITEMS SUPPORTED	RECOMMENDED ACTION

PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.)

TYPED NAME, GRADE OR TITLE	TELEPHONE EXCHANGE/AUTOVON, PLUS EXTENSION	SIGNATURE
MSG, Jane Q. Doe, SFC	788–1234	

RECOMMENDED CHANGES TO PUBLICATIONS AND BLANK FORMS For use of this form, see AR 25-30; the proponent agency is ODISC4.	Use Par reverse) for Repair Parts and Special Tool Lists (RPSTL) and Supply Catalogs/ Supply Manuals (SC/SM)	DATE

TO: (Forward to proponent of publication or form)(Include ZIP Code) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, AL 35898 - 5000	FROM: (Activity and location)(Include ZIP Code)

PART 1 – ALL PUBLICATIONS (EXCEPT RPSTL AND SC/SM) AND BLANK FORMS

PUBLICATION/FORM NUMBER	DATE	TITLE

ITEM NO.	PAGE NO.	PARA-GRAPH	LINE NO. *	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASON

* Reference to line numbers within the paragraph or subparagraph.

TYPED NAME, GRADE OR TITLE	TELEPHONE EXCHANGE/ AUTOVON, PLUS EXTENSION	SIGNATURE

TO: (Forward direct to addressee listed in publication) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, AL 35898	FROM: (Activity and location) (Include ZIP Code)	DATE

PART II – REPAIR PARTS AND SPECIAL TOOL LISTS AND SUPPLY CATALOGS/SUPPLY MANUALS

PUBLICATION NUMBER	DATE	TITLE

PAGE NO.	COLM NO.	LINE NO.	NATIONAL STOCK NUMBER	REFERENCE NO.	FIGURE NO.	ITEM NO.	TOTAL NO. OF MAJOR ITEMS SUPPORTED	RECOMMENDED ACTION

PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.)

TYPED NAME, GRADE OR TITLE	TELEPHONE EXCHANGE/AUTOVON, PLUS EXTENSION	SIGNATURE

The Metric System and Equivalents

Linear Measure

1 centimeter = 10 millimeters = .39 inch
1 decimeter = 10 centimeters = 3.94 inches
1 meter = 10 decimeters = 39.37 inches
1 dekameter = 10 meters = 32.8 feet
1 hectometer = 10 dekameters = 328.08 feet
1 kilometer = 10 hectometers = 3,280.8 feet

Weights

1 centigram = 10 milligrams = .15 grain
1 decigram = 10 centigrams = 1.54 grains
1 gram = 10 decigram = .035 ounce
1 decagram = 10 grams = .35 ounce
1 hectogram = 10 decagrams = 3.52 ounces
1 kilogram = 10 hectograms = 2.2 pounds
1 quintal = 100 kilograms = 220.46 pounds
1 metric ton = 10 quintals = 1.1 short tons

Liquid Measure

1 centiliter = 10 milliters = .34 fl. ounce
1 deciliter = 10 centiliters = 3.38 fl. ounces
1 liter = 10 deciliters = 33.81 fl. ounces
1 dekaliter = 10 liters = 2.64 gallons
1 hectoliter = 10 dekaliters = 26.42 gallons
1 kiloliter = 10 hectoliters = 264.18 gallons

Square Measure

1 sq. centimeter = 100 sq. millimeters = .155 sq. inch
1 sq. decimeter = 100 sq. centimeters = 15.5 sq. inches
1 sq. meter (centare) = 100 sq. decimeters = 10.76 sq. feet
1 sq. dekameter (are) = 100 sq. meters = 1,076.4 sq. feet
1 sq. hectometer (hectare) = 100 sq. dekameters = 2.47 acres
1 sq. kilometer = 100 sq. hectometers = .386 sq. mile

Cubic Measure

1 cu. centimeter = 1000 cu. millimeters = .06 cu. inch
1 cu. decimeter = 1000 cu. centimeters = 61.02 cu. inches
1 cu. meter = 1000 cu. decimeters = 35.31 cu. feet

Approximate Conversion Factors

To change	*To*	*Multiply by*	*To change*	*To*	*Multiply by*
inches	centimeters	2.540	ounce-inches	Newton-meters	.007062
feet	meters	.305	centimeters	inches	.394
yards	meters	.914	meters	feet	3.280
miles	kilometers	1.609	meters	yards	1.094
square inches	square centimeters	6.451	kilometers	miles	.621
square feet	square meters	.093	square centimeters	square inches	.155
square yards	square meters	.836	square meters	square feet	10.764
square miles	square kilometers	2.590	square meters	square yards	1.196
acres	square hectometers	.405	square kilometers	square miles	.386
cubic feet	cubic meters	.028	square hectometers	acres	2.471
cubic yards	cubic meters	.765	cubic meters	cubic feet	35.315
fluid ounces	milliliters	29,573	cubic meters	cubic yards	1.308
pints	liters	.473	milliliters	fluid ounces	.034
quarts	liters	.946	liters	pints	2.113
gallons	liters	3.785	liters	quarts	1.057
ounces	grams	28.349	liters	gallons	.264
pounds	kilograms	.454	grams	ounces	.035
short tons	metric tons	.907	kilograms	pounds	2.205
pound-feet	Newton-meters	1.356	metric tons	short tons	1.102
pound-inches	Newton-meters	.11296			

Temperature (Exact)

°F	Fahrenheit temperature	5/9 (after subtracting 32)	Celsius temperature	°C